slippery chicken

Generated with ROBODoc Version 4.99.44 (Oct 31 2021)

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Contents

1 sc/all.lsp

[Modules]

NAME:

```
all

File:             all.lsp

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Load all the lisp files associated with slippery-chicken.
                  No public interface envisaged (so no robodoc entries).

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    5th December 2000

$$ Last modified:  15:59:36 Thu Mar 28 2024 CET
```

2 sc/clm-akoustik-piano-spectra.lsp

[Modules]

NAME:

```
globals

File:             clm-piano-spectra.lsp

Class Hierarchy:  None: no classes defined.

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          For reading in from spectra.lsp: From Akoustic Piano's
                  samples library: "/Akoustik Piano
                  Library/Samples/ConcertGrand/Samples/". We can use this
                  data for the calculation of chord dissonance values in
                  the chord class's calculate-dissonance method.
```

```
Author:           Michael Edwards: m@michael-edwards.org

Creation date:    5th August 2015

$$ Last modified: 12:30:03 Tue Feb  2 2016 GMT

SVN ID: $Id: clm-piano-spectra.lsp 963 2010-04-08 20:58:32Z medward2 $
```

3 sc/clm-piano-spectra.lsp

[Modules]

NAME:

```
globals

File:             clm-piano-spectra.lsp

Class Hierarchy:  None: no classes defined.

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          For reading in from spectra.lsp: Modified data from CLM's
                  spectr.clm which itself was created from data provided
                  courtesy of J.A. Moorer. We'll use this data for the
                  calculation of chord dissonance values in the chord
                  class's calculate-dissonance method.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    5th August 2015

$$ Last modified: 16:22:57 Wed Aug  5 2015 BST

SVN ID: $Id: clm-piano-spectra.lsp 963 2010-04-08 20:58:32Z medward2 $
```

4 sc/clm-violin-ensemble-spectra.lsp

[Modules]

NAME:

```
globals

File:             violin-ensemble-spectra.lsp

Class Hierarchy:  None: no classes defined.

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          For reading in from spectra.lsp: From Kontakt's violin
                  ensemble samples library: Kontakt 3 Library/Orchestral/Z
                  - Samples/01 Violin ensemble - 14/VI-14_mV_sus_mf". We can
                  use this data for the calculation of chord dissonance
                  values in the chord class's calculate-dissonance method.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    5th August 2015

$$ Last modified:  19:36:36 Thu Jun  8 2017 BST

SVN ID: $Id: clm-piano-spectra.lsp 963 2010-04-08 20:58:32Z medward2 $
```

5 sc/cm

[Modules]

NAME:

```
cm

File:             cm.lsp

Class Hierarchy:  none (no classes defined)

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Definition of common-music related and other functions
                  like transposition of notes/chords, enharmonic
                  equivalents etc.

Author:           Michael Edwards: m@michael-edwards.org
```

```
Creation date:    1st March 2001

$$ Last modified:  09:57:54 Wed Jun 18 2025 CEST

SVN ID: $Id$
```

5.1 cm/degree-to-note

[cm] [Functions]

DESCRIPTION:

```
Get the specified scale degree number as a note-name pitch symbol within
the current scale. An optional argument allows the user to specify that the
scale degree number should be used to get the note-name pitch from a
different scale.
```

ARGUMENTS:

```
An integer that is a scale degree number.
```

OPTIONAL ARGUMENTS:

```
- The scale from which the note-name pitch symbol associated with the
  specified scale degree is to be drawn.
```

RETURN VALUE:

```
A note-name pitch symbol.
```

EXAMPLE:

```
(degree-to-note 127 'chromatic-scale)

=> G9

(degree-to-note 127 'twelfth-tone)

=> ATS0

(degree-to-note 127 'quarter-tone)

=> EQF4
```

SYNOPSIS:

```
(defun degree-to-note (degree &optional (scale cm::*scale*))
```

5.2 cm/degrees-per-octave

[cm] [Functions]

DESCRIPTION:

```
Return the number of scale degrees in the span of one octave within the
current tuning system.
```

ARGUMENTS:

```
- No arguments.
```

RETURN VALUE:

```
- An integer that is the number of scale degrees in each octave.
```

EXAMPLE:

```
(in-scale :chromatic)
(degrees-per-octave)

=> 12

(in-scale :quarter-tone)
(degrees-per-octave)

=> 24
```

SYNOPSIS:

```
(defun degrees-per-octave (&optional (scale cm::*scale*))
```

5.3 cm/degrees-per-semitone

[cm] [Functions]

DESCRIPTION:

```
Get the number of scale degrees per equal-tempered semitone in the current
tuning scale.
```

ARGUMENTS:

```
- No arguments
```

OPTIONAL ARGUMENTS:

```
- The scale for which the number of degrees per semitone is to be
  retrieved.
```

RETURN VALUE:

```
An integer.
```

EXAMPLE:

```
(degrees-per-semitone 'chromatic-scale)

=> 1

(degrees-per-semitone 'twelfth-tone)

=> 6

(degrees-per-semitone 'quarter-tone)

=> 2
```

SYNOPSIS:

```
(defun degrees-per-semitone (&optional (scale cm::*scale*))
```

5.4 cm/degrees-to-notes

[cm] [Functions]

DESCRIPTION:

```
NB: If the specified scale-degree number within the current scale would
    result in pitch outside of the maximum MIDI pitch range for that tuning
    (chromatic: C-1 to B10; quarter-tone: C-1 to BQS10; twelfth-tone: C-1
    to CTF11), the function will return an error.
```

ARGUMENTS:

```
An integer that is a scale degree number in the current tuning.
```

RETURN VALUE:

```
A list of note-name pitch symbols.
```

EXAMPLE:

```
(in-scale :chromatic)
(degrees-to-notes '(0 143 116 127 38))

=> (C-1 B10 AF8 G9 D2)

(in-scale :twelfth-tone)
(degrees-to-notes '(0 144 116 127 38 287 863))

=> (C-1 C1 GSS0 ATS0 FSSS-1 CTF3 CTF11)

(in-scale :quarter-tone)
(degrees-to-notes '(0 144 116 127 38 287))

=> (C-1 C5 BF3 EQF4 G0 BQS10)
```

SYNOPSIS:

```
(defun degrees-to-notes (degrees)
```

5.5 cm/event-list-to-midi-file

[cm] [Functions]

DESCRIPTION:

```
Write the events in a list to a midi-file. events-update-time is a related
function useful for preparind ad-hoc events for midi-file-writing.
```

ARGUMENTS:

```
- A list of events objects
- the path to the midi-file
- the starting tempo (integer: BPM)
- a time-offset for the events (seconds)
```

OPTIONAL ARGUMENTS:

```
- whether to overwrite events' amplitude slots and use a single
velocity/amplitude value given here (0-1.0 (float) or 0-127 (integer)
```

RETURN VALUE:

```
The MIDI file path
```

SYNOPSIS:

```
(defun event-list-to-midi-file (event-list
                                &key (midi-file "/tmp/tmp.mid")
                                  (start-tempo 120) (time-offset 0)
                                  (auto-open (get-sc-config
                                              'midi-play-auto-open))
                                  force-velocity)
```

5.6 cm/freq-to-degree

[cm] [Functions]

DESCRIPTION:

```
Get the scale degree of the specified frequency in Hertz within the current
scale.

NB: This method will return fractional scale degrees.
```

ARGUMENTS:

```
A frequency in Hertz.
```

OPTIONAL ARGUMENTS:

```
- The scale in which to find the corresponding scale degree.
```

RETURN VALUE:

```
A scale degree number. This may be a decimal number.
```

EXAMPLE:

```
(freq-to-degree 423 'chromatic-scale)

=> 68.317856

(freq-to-degree 423 'twelfth-tone)
```

```
=> 409.9071

(freq-to-degree 423 'quarter-tone)

=> 136.63571
```

SYNOPSIS:

```
(defun freq-to-degree (freq &optional (scale cm::*scale*))
```

5.7 cm/freq-to-midi

[cm] [Functions]

DATE:

```
10th July 2016, Essen Werden
```

DESCRIPTION:

```
convert a frequency in Hertz to a MIDI note number (possibly floating
point, indicating microtonality).
```

ARGUMENTS:

```
a frequency in Hertz
```

RETURN VALUE:

```
a floating point value representing the MIDI note number of the given frequency
```

EXAMPLE:

```
(freq-to-midi 260) --> 59.892094
(freq-to-midi (midi-to-freq 60)) --> 60.0
```

SYNOPSIS:

```
(defun freq-to-midi (freq)
```

5.8 cm/freq-to-note

[cm] [Functions]

DESCRIPTION:

Get the note-name pitch equivalent of the specified frequency, rounded to the nearest scale degree of the current scale.

ARGUMENTS:

A number that is a frequency in Hertz.

OPTIONAL ARGUMENTS:

- The scale in which the note-name pitch equivalent is to be sought (Common Music scale object or symbol). If a symbol, then 'chromatic-scale, 'twelfth-tone, or 'quarter-tone only at present.

RETURN VALUE:

A note-name pitch symbol.

EXAMPLE:

```
(freq-to-note 423 'chromatic-scale)

=> AF4

(freq-to-note 423 'twelfth-tone)

=> GSSS4

(freq-to-note 423 'quarter-tone)

=> AQF4
```

SYNOPSIS:

```
(defun freq-to-note (freq &optional (scale cm::*scale*))
```

5.9 cm/get-pitch-bend

[cm] [Functions]

DESCRIPTION:

Get the MIDI pitch-bend value necessary for application to a MIDI pitch in order to achieve the specified frequency.

```
NB: This will always return a positive value between 0.0 and 1.0, as
    slippery-chicken always applies pitch-bends upwards from the nearest
    chromatic note below the specified frequency.

NB: This value will be the same in all tuning scales.
```

ARGUMENTS:

```
A frequency in Hertz.
```

RETURN VALUE:

```
A two-digit decimal number that is the pitch-bend value required to achieve
the specified frequency in MIDI.
```

EXAMPLE:

```
(get-pitch-bend 423)

=> 0.32
```

SYNOPSIS:

```
(defun get-pitch-bend (freq)
```

5.10 cm/in-scale

[cm] [Functions]

DESCRIPTION:

```
Set the global scale (tuning) for the current slippery-chicken
environment. Current options are :chromatic, :quarter-tone or
:twelfth-tone. See the file cm-load.lsp for specifications and the html
manual page "More about note-names and scales" for more details on use.
```

ARGUMENTS:

```
- A scale (tuning) designation.
```

RETURN VALUE:

```
the (new) Common Music scale object
```

EXAMPLE:

```
(in-scale :chromatic)

=> #<tuning "chromatic-scale">

(in-scale :quarter-tone)

=> #<tuning "quarter-tone">

(in-scale :twelfth-tone)

=> #<tuning "twelfth-tone">
```

SYNOPSIS:

```
(defun in-scale (scale)
```

5.11 cm/midi-file-high-low

[cm] [Functions]

DATE:

```
30-Dec-2010
```

DESCRIPTION:

```
Print the highest and lowest pitch in a specified MIDI file as a MIDI note
number.

NB: This is a Common Music function and as such must be called with the
    package qualifier cm:: if used within slippery chicken.
```

ARGUMENTS:

```
- The path (including the name) to the MIDI file.
```

OPTIONAL ARGUMENTS:

```
- An integer or NIL to indicate which track in the specified MIDI file is
  to be accessed. If NIL, all tracks will be accessed. NB: CM (and
  therefore slippery-chicken too) generates some MIDI files by writing each
  channel to a different track, so the "track" would seem synonymous with
  "channel" here.
```

RETURN VALUE:

```
Two integer values (using the values function) that are the highest and
lowest pitches in the specified MIDI file.
```

EXAMPLE:

```
(cm::midi-file-high-low "/tmp/multi-ps.mid")

=> 72, 60
```

SYNOPSIS:

```
(defun midi-file-high-low (file &optional track)
```

5.12 cm/midi-file-one-note

[cm] [Functions]

DESCRIPTION:

```
Write all midi notes in the file out to a new one-channel file using the
single pitch <note> and channel number <channel>.
```

ARGUMENTS:

```
- A string that is the file path, including file-name and extension.
- A note-name symbol or MIDI-note integer that is the pitch to write.
- An integer that is the channel to which the output should be written
  (1-based)
```

OPTIONAL ARGUMENTS:

```
- An integer that is the an existing MIDI channel in the original file. If
  this argument is given, only notes on this channel of the original file
  will be written (1-based).
```

RETURN VALUE:

```
The path to the new file.
```

EXAMPLE:

```
(cm::midi-file-one-note "/tmp/multi-ps.mid" 'c4 1)
```

SYNOPSIS:

```
(defun midi-file-one-note (file note channel &optional old-channel)
```

5.13 cm/midi-file-to-events

[cm] [Functions]

DATE:

```
28/6/16, Edinburgh
```

DESCRIPTION:

```
Read in a MIDI file and convert notes to event objects.
NB This won't (yet) import microtones as indicated by pitch-bend messages
```

ARGUMENTS:

```
the path to the midi file
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :track. The track number to read. Default = NIL which means read all
tracks.
- :tempo. The tempo of the track in crotchets (quarter notes) per
minute. This should override the midi-file's tempo. See also
test-midi-file-to-events in sc-test-suite.lsp
```

RETURN VALUE:

```
a list of event objects
```

SYNOPSIS:

```
(defun midi-file-to-events (file &key track tempo)
```

5.14 cm/midi-to-degree

[cm] [Functions]

DESCRIPTION:

```
Convert the specified MIDI note number to the degree number of the current
scale.
```

ARGUMENTS:

- A MIDI note number.

RETURN VALUE:

- An integer that is the scale-degree equivalent of the specified MIDI note number in the current scale.

EXAMPLE:

```
(in-scale :chromatic)
(midi-to-degree 64)

=> 64

(in-scale :twelfth-tone)
(midi-to-degree 64)

=> 384

(in-scale :quarter-tone)
(midi-to-degree 64)

=> 128
```

SYNOPSIS:

```
(defun midi-to-degree (midi-note &optional (scale cm::*scale*))
```

5.15 cm/midi-to-freq

[cm] [Functions]

DESCRIPTION:

Get the frequency equivalent in Hertz to the specified MIDI note number.

ARGUMENTS:

- A number (can be a decimal) that is a MIDI note number.

RETURN VALUE:

A decimal number that is a frequency in Hertz.

EXAMPLE:

```
(midi-to-freq 67)

=> 391.99542

(midi-to-freq 67.9)

=> 412.91272
```

SYNOPSIS:

```
(defun midi-to-freq (midi-note)
```

5.16 cm/midi-to-note

[cm] [Functions]

DESCRIPTION:

```
Get the note-name pitch symbol equivalent of the specified MIDI note
number.
```

ARGUMENTS:

```
- a MIDI note number: integer or float. If float, appropriate microtones
will be returned if the current scale is microtonal (e.g. (in-scale
:quarter-tone))
```

RETURN VALUE:

```
A note-name pitch symbol.
```

EXAMPLE:

```
(midi-to-note 67)
=> G4
(midi-to-note 60.5)
=>CQS4
```

SYNOPSIS:

```
(defun midi-to-note (midi-note &optional (scale cm::*scale*))
```

5.17 cm/midi2qlist

[cm] [Functions]

DATE:

```
10th November 2016, Edinburgh
```

DESCRIPTION:

```
Convert a midi-file to qlist text format, for sequencing in PD or MaxMSP.
If you don't want specific tracks from the file, just pass two arguments so
<tracks> remains nil.
```

ARGUMENTS:

```
- the path to the midi file you want to convert (string)
- the path to the qlist text file you'd like to write (string). If this is
  nil then we'll write to the same patch as the midi file with the extension
  .txt added. NB If this file exists it will be overwritten, no questions
  asked.
- (&rest) the numbers of the tracks you'd like to convert, starting from 1.
```

RETURN VALUE:

```
The number of notes written (integer)
```

SYNOPSIS:

```
(defun midi2qlist (midi-file qlist-file &rest tracks)
```

5.18 cm/note-to-degree

[cm] [Functions]

DESCRIPTION:

```
Get the scale degree number of the specified note-name pitch symbol within
the current scale.
```

ARGUMENTS:

```
- A note-name pitch symbol.
```

OPTIONAL ARGUMENTS:

```
 - The scale in which to find the scale-degree of the specified pitch.
```

RETURN VALUE:

```
 An integer that is a scale degree in the current scale.
```

EXAMPLE:

```
(note-to-degree 'AF4 'chromatic-scale)

=> 68

(note-to-degree 'AF4 'twelfth-tone)

=> 408

(note-to-degree 'AF4 'quarter-tone)

=> 136
```

SYNOPSIS:

```
(defun note-to-degree (note &optional (scale cm::*scale*))
```

5.19 cm/note-to-freq

[cm] [Functions]

DESCRIPTION:

```
 Get the frequency in Hertz of the specified note-name pitch symbol.
```

ARGUMENTS:

```
 - A note-name pitch symbol.
```

RETURN VALUE:

```
 A frequency in Hertz.
```

EXAMPLE:

```
(in-scale :chromatic)
(note-to-freq 'AF4)
```

```
=> 415.3047

(in-scale :twelfth-tone)
(note-to-freq 'GSSS4)

=> 423.37845

(in-scale :quarter-tone)
(note-to-freq 'AQF4)

=> 427.47403
```

SYNOPSIS:

```
(defun note-to-freq (note)
```

5.20 cm/note-to-midi

[cm] [Functions]

DESCRIPTION:

```
Get the MIDI note number equivalent for a chromatic note-name pitch
symbol.
```

ARGUMENTS:

```
- A chromatic note-name pitch symbol.
```

RETURN VALUE:

```
An integer.
```

EXAMPLE:

```
(note-to-midi 'g4)

=> 67
```

SYNOPSIS:

```
(defun note-to-midi (midi-note)
```

5.21 cm/parse-midi-file

[cm] [Functions]

DESCRIPTION:

```
Return Common Music MIDI events in the specified file as a list.

NB: This is a Common Music function and as such must be called with the
    package qualifier cm:: if used within slippery chicken.
```

ARGUMENTS:

```
- The path (including the file name) to the MIDI file.
```

OPTIONAL ARGUMENTS:

```
- An integer or NIL to indicate which track in the specified MIDI file is
  to be accessed. If NIL, all tracks will be accessed. NB: CM (and
  therefore slippery-chicken too) generates some MIDI files by writing each
  channel to a different track, so the "track" would seem synonymous with
  "channel" here.
```

RETURN VALUE:

```
The CM data for the MIDI events in the specified file, and the number of
events.
```

EXAMPLE:

```
(cm::parse-midi-file "/tmp/multi-ps.mid")

=>
(#i(midi-tempo-change time 0.0 usecs 357142)
 #i(midi-time-signature time 0.0 numerator 2 denominator 4 clocks 24 32nds 8)
 #i(midi time 0.0 keynum 36 duration 0.357142 amplitude 0.09448819 channel 15)
 #i(midi-tempo-change time 0.0 usecs 357142)
 #i(midi-time-signature time 0.0 numerator 2 denominator 4 clocks 24 32nds 8)
 #i(midi-tempo-change time 0.0 usecs 357142)
 #i(midi time 0.178571 keynum 66 duration 0.178571 amplitude 0.09448819 channel 15)
 #i(midi time 0.357142 keynum 68 duration 0.0892855 amplitude 0.09448819 channel 15)
 #i(midi time 0.357142 keynum 40 duration 0.357142 amplitude 0.09448819 channel 15)
 #i(midi time 0.6249985 keynum 66 duration 0.0892855 amplitude 0.09448819 channel 15)
 #i(midi-time-signature time 0.714284 numerator 3 denominator 4 clocks 24 32nds 8)
```

SYNOPSIS:

```
(defun parse-midi-file (file &optional track tempo)
```

5.22 cm/set-diapason

[cm] [Functions]

DATE:

March 11th 2021

DESCRIPTION:

Set the frequency of the note A4 (the diapason). By default, in slippery chicken (and common music) this is 440 Hertz. If this function is called, it will change the diapason for all currently defined scales, and thus will affect all calculations in any scale that converts to or from frequency.

NB Explicit calling of this function is discouraged. Use instead something like this: (set-sc-config 'diapason 442)

ARGUMENTS:

- the new frequency of the note A4 (middle A: i.e. the note played by the oboe when a western orchestra tunes). (For internal use, if this is NIL, the frequency of the lowest A will be returned.)

RETURN VALUE:

The frequency in Hertz of the lowest A in the (new) tuning.

EXAMPLE: SYNOPSIS:

```
(let ((low-a 440/64))                   ; 6.875Hz (lowest A)
  (defun set-diapason (hertz)
```

6 sc/cm-cm

[Modules]

NAME:

cm

File: cm-cm.lsp

Class Hierarchy: none (no classes defined)

```
Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Definition of common-music functions etc. that need to be
                  in the cm package

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    26th March 2020 (abstracted out of cm.lsp)

$$ Last modified:  13:50:33 Tue May 20 2025 CEST

SVN ID: $Id$
```

7 sc/cm-load

[Modules]

NAME:

```
cm-load

File:             cm-load.lsp

Class Hierarchy:  none (no classes defined)

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Definition of the common-music quarter-tone scale and
                  twelfth-tone scale which should be loaded and not
                  compiled.  The quarter tone scale is our default
                  No public interface envisaged (so no robodoc entries).

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    7th February 2003

$$ Last modified:  10:21:48 Fri Mar 12 2021 CET

SVN ID: $Id$
```

8 sc/cmn

[Modules]

NAME:

```
cmn

File:             cmn.lsp

Class Hierarchy:  None: no classes defined.

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Interface from complete-set to Bill's CMN package for
                  displaying of sets in musical notation.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    11th February 2002

$$ Last modified:  14:01:36 Wed Dec 17 2025 CET

SVN ID: $Id$
```

9 sc/cmn-glyphs

[Modules]

NAME:

```
cmn-glyphs

File:             cmn-glyphs.lsp

Class Hierarchy:  none, no classes defined

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Definition of various postscript glyphs (accidentals
                  etc.) for cmn.
```

```
Author:           Michael Edwards: m@michael-edwards.org

Creation date:    10th November 2002

$$ Last modified:  17:54:01 Mon May 28 2018 CEST

SVN ID: $Id$
```

10 sc/export-symbols.lsp

[Modules]

NAME:

```
export-symbols.lsp

File:             slippery-chicken.asd

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Export all symbols defined in slippery chicken.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    March 15th 2024

$$ Last modified:  17:58:03 Tue Apr 16 2024 CEST
```

11 sc/get-spectrum

[Modules]

NAME:

```
get-spectrum

File:             get-spectrum.lsp

Class Hierarchy:  none (no classes defined)

Version:          1.1.0
```

```
Project:          slippery chicken (algorithmic composition)

Purpose:          Used mainly by spectra.lsp, get-spectrum returns, as a
                  list, the most prominent frequencies in a sound file.
                  The list is ordered from the most prominent to the least
                  prominent frequency (:order-by 'amp) or the highest to
                  lowest frequency (:order-by 'freq) and the number of
                  elements in the list is set by the <num-partials>
                  parameter.  The instrument spec-an is called by
                  get-spectrum and is therefore not meant to be used as a
                  public function (it is written as an instrument to take
                  advantage of CLM's run-loop efficiency).  The results of
                  analysis are stored in
                  *slippery-chicken-get-spectrum-last-result* so that if
                  you request the same analysis repeatedly, the analysis
                  will not need to be reperformed each time.  Be careful
                  though, only the input file, the ordering, the number of
                  partials requested and the analysis start time are used
                  to detect whether analysis has to be reperformed or not.
                  If you want to force reanalysis, do (setf
                  *slippery-chicken-get-spectrum-last-result* nil) before
                  the call to get-spectrum.

Creation date:    This is very old code but it was added to slippery
                  chicken on August 5th 2015

Author:           Michael Edwards: m@michael-edwards.org

$$ Last modified:  10:17:28 Fri Apr 19 2024 CEST

SVN ID: $Id: get-spectrum.lsp 5359 2015-07-24 20:53:22Z medward2 $
```

12 sc/globals.lsp

[Modules]

NAME:

```
globals

File:             globals.lsp

Class Hierarchy:  None: no classes defined.
```

```
Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Definition of the user-changeable configuration data and
                  globals for internal programme use. See top of cmn.lsp and
                  osc-sc-bsd.lsp for a few more globals relevant to
                  functionality in their respective packages.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    30th May 2013

$$ Last modified:  16:20:27 Sat Jan 31 2026 CET

SVN ID: $Id: sclist.lsp 963 2010-04-08 20:58:32Z medward2 $
```

13 sc/instruments

[Modules]

NAME:

```
instrument

File:             instruments.lsp

Class Hierarchy:  none (no classes defined)

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Definition of various standard instruments and other
                  data/functions useful to slippery-chicken users.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    30th December 2010

$$ Last modified:  12:07:34 Sat May 10 2025 CEST

SVN ID: $Id$
```

13.1 instruments/+slippery-chicken-standard-instrument-palette+

[instruments] [Global Parameters]

DESCRIPTION:

```
A palette of standard instruments (by no means exhaustive...) for use
directly in projects or for combining with user palettes e.g.

(combine
 +slippery-chicken-standard-instrument-palette+
 (make-instrument-palette
  'esoteric-stuff
   '((toy-piano
      (:staff-name "toy piano" ....
```

SYNOPSIS:

```
(defparameter +slippery-chicken-standard-instrument-palette+
  (make-instrument-palette
   'slippery-chicken-standard-instrument-palette
   ;; SAR Fri Jan 20 11:43:32 GMT 2012: Re-ordering these to Adler's "standard"
   ;; score order for easier look-up
   '((piccolo
      (:staff-name "piccolo" :staff-short-name "picc"
       :lowest-written d4 :highest-written c7 :transposition-semitones 12
       :missing-notes nil
       :largest-fast-leap 19
       :starting-clef treble
       :chords nil
       :microtones t
       :midi-program 73))
     (flute
      (:staff-name "flute" :staff-short-name "fl"
       :lowest-written c4 :highest-written d7
       :missing-notes (cqs4 dqf4)
       :largest-fast-leap 19
       :starting-clef treble
       :chords nil
       :microtones t
       :midi-program 74))
     (alto-flute
      (:staff-name "alto flute" :staff-short-name "alt fl"
       :lowest-written c4 :highest-written c7 :transposition-semitones -5
       :missing-notes (cqs4 dqf4)
       :largest-fast-leap 17
```

```
  :starting-clef treble
  :chords nil
  :microtones t
  :midi-program 74))
;; SAR Fri Jan 20 11:46:45 GMT 2012: Modified bass flute range to that
;; stated by Adler.
(bass-flute
 (:staff-name "bass flute" :staff-short-name "bass fl"
  :lowest-written c4 :highest-written c7 :transposition-semitones -12
  :missing-notes (cqs4 dqf4)
  :largest-fast-leap 15
  :clefs-in-c (treble bass) :starting-clef treble
  :chords nil
  :microtones t
  :midi-program 74))
;; MDE Wed Oct 20 11:48:45 2021, Heidhausen
(sopranino-recorder
 (:lowest-written f4 :highest-written g6 :chords nil
  :staff-name "sopranino recorder" :staff-short-name "sop.no rec."
  :midi-program 75 :microtones nil :largest-fast-leap 12
  :transposition-semitones 12  :starting-clef treble))
(soprano-recorder
 (:lowest-written c4 :highest-written d6 :chords nil
  :staff-name "soprano recorder" :staff-short-name "sop. rec."
  :midi-program 75 :microtones nil :largest-fast-leap 12
  :transposition-semitones 12 :starting-clef treble))
(alto-recorder
 (:lowest-written f4 :highest-written g6 :chords nil
  :staff-name "alto recorder" :staff-short-name "alt. rec."
  :midi-program 75 :microtones nil :largest-fast-leap 12
  :starting-clef treble))
(tenor-recorder
 (:lowest-written c4 :highest-written d6 :chords nil
  :staff-name "tenor recorder" :staff-short-name "ten. rec."
  :midi-program 75 :microtones nil :largest-fast-leap 12
  :starting-clef treble))
;; this is a special instrument that is not fully chromatic and has a
;; tuning of a4 = 466 Hz. I wrote a piece for 41 of these for Ulrike
;; Volkhardt and colleagues in 2021/2
(consort-tenor-recorder
 (:lowest-written c4 :highest-written a5 :chords nil
  :staff-name "consort tenor recorder" :staff-short-name "cons. ten. rec."
  :midi-program 75 :microtones nil :largest-fast-leap 12
  :missing-notes (cs4 ds4) :starting-clef treble))
(bass-recorder
 (:lowest-written f2 :highest-written f4 :chords nil
```

```
  :staff-name "bass recorder" :staff-short-name "bs. rec."
  :midi-program 75 :microtones nil :largest-fast-leap 12
  :transposition-semitones 12 :starting-clef bass))
;; SAR Fri Jan 20 12:01:37 GMT 2012: Added oboe. Conservative range taken
;; from the Adler
(oboe
 (:staff-name "oboe" :staff-short-name "ob"
  :lowest-written bf3 :highest-written a6
  :largest-fast-leap 12
  :starting-clef treble
  :chords nil :microtones t
  :missing-notes (bqf3 bqs3 cqs4 dqf4)
  :midi-program 69))
;; MDE Tue Feb 20 19:43:50 2018
(cor-anglais
 (:staff-name "cor anglais" :staff-short-name "c.a."
  :lowest-written bf3 :highest-written a6
  :transposition-semitones -7
  :largest-fast-leap 12
  :starting-clef treble
  :chords nil :microtones t
  :missing-notes (bqf3 bqs3 cqs4 dqf4)
  :midi-program 70))
(oboe-d-amore
 (:staff-name "oboe d'amore" :staff-short-name "ob d'am"
  :lowest-written bf3 :highest-written a6
  :transposition-semitones -3
  :largest-fast-leap 12
  :starting-clef treble
  :chords nil :microtones t
  :missing-notes (bqf3 bqs3 cqs4 dqf4)
  :midi-program 70))
(e-flat-clarinet
 (:staff-name "E-flat clarinet" :staff-short-name "E-flat cl"
  :lowest-written e3 :highest-written a6 :transposition-semitones 3
  :missing-notes (aqs4 bqf4 bqs4 cqs5 dqf5 gqf3 fqs3 fqf3)
  :largest-fast-leap 15
  :starting-clef treble
  :chords nil
  :microtones t
  :midi-program 72))
(b-flat-clarinet
 (:staff-name "B-flat clarinet" :staff-short-name "B-flat cl"
  :lowest-written e3 :highest-written a6 :transposition-semitones -2
  :missing-notes (aqs4 bqf4 bqs4 cqs5 dqf5 gqf3 fqs3 fqf3)
  :largest-fast-leap 15
```

```
  :starting-clef treble
  :chords nil
  :microtones t
  :midi-program 72))
(a-clarinet
 (:staff-name "A clarinet" :staff-short-name "A cl"
  :lowest-written e3 :highest-written a6 :transposition-semitones -3
  :missing-notes (aqs4 bqf4 bqs4 cqs5 dqf5 gqf3 fqs3 fqf3)
  :largest-fast-leap 15
  :starting-clef treble
  :chords nil
  :microtones t
  :midi-program 72))
(bass-clarinet
 (:staff-name "bass clarinet" :staff-short-name "bass cl"
  :lowest-written c3 :highest-written g6 :transposition-semitones -14
  :missing-notes (aqs4 bqf4 bqs4 cqs5 dqf5 gqf3 fqs3 fqf3 eqf3 dqs3 dqf3
                       cqs3)
  :largest-fast-leap 13
  :prefers-notes low
  :clefs (treble) :clefs-in-c (treble bass) :starting-clef treble
  :chords nil
  :microtones t
  :midi-program 72))
(sopranino-sax
 (:staff-name "sopranino saxophone" :staff-short-name "spino sax"
  :lowest-written bf3 :highest-written fs6 :transposition-semitones 3
  :missing-notes (gqs4 gqs5)
  :largest-fast-leap 15
  :starting-clef treble
  :chords nil
  :microtones t
  :midi-program 65))
(soprano-sax
 (:staff-name "soprano saxophone" :staff-short-name "sop sax"
  :lowest-written bf3 :highest-written fs6 :transposition-semitones -2
  :missing-notes (gqs4 gqs5)
  :largest-fast-leap 15
  :starting-clef treble
  :chords nil
  :microtones t
  :midi-program 65))
(alto-sax
 (:staff-name "alto saxophone" :staff-short-name "alt sax"
  ;; altissimo extra....by hand...
  :lowest-written bf3 :highest-written fs6 :transposition-semitones -9
```

```
  :missing-notes (gqs4 gqs5)
  :largest-fast-leap 15
  :starting-clef treble
  :chords nil
  :microtones t
  :midi-program 66))
(tenor-sax
 (:staff-name "tenor sax" :staff-short-name "ten sax"
  :lowest-written bf3 :highest-written fs6 :transposition-semitones -14
  :missing-notes (gqs4 gqs5)
  :largest-fast-leap 13
  :starting-clef treble :clefs-in-c (treble bass)
  :chords nil
  :microtones t
  :midi-program 67))
(baritone-sax
 (:staff-name "baritone sax" :staff-short-name "bar sax"
  :lowest-written bf3 :highest-written fs6 :transposition-semitones -21
  :missing-notes (gqs4 gqs5)
  :largest-fast-leap 11
  :clefs-in-c (treble bass) :starting-clef treble
  :chords nil
  :microtones t
  :midi-program 68))
(bassoon
 (:staff-name "bassoon" :staff-short-name "bsn"
  ;; of course it can go higher but best not to algorithmically select
  ;; these
  :lowest-written bf1 :highest-written c5
  ;; Wolfgang Ruediger says all 1/4 tones are OK above low E
  :missing-notes (bqf1 bqs1 cqs2 dqf2 dqs2 eqf2)
  :largest-fast-leap 13
  :clefs (bass tenor) :starting-clef bass
  :chords nil :microtones t :midi-program 71))
(contra-bassoon
 (:staff-name "contrabassoon" :staff-short-name "cbsn"
  ;; some pieces go up to c5 but let's not
  :lowest-written bf1 :highest-written a4
  :transposition-semitones -12
  ;; Wolfgang Ruediger says all 1/4 tones are OK above low E for the
  ;; bassoon, so let's assume for now that these work here too.
  :missing-notes (bqf1 bqs1 cqs2 dqf2 dqs2 eqf2)
  :largest-fast-leap 13
  :clefs (bass tenor) :starting-clef bass
  :chords nil :microtones t :midi-program 71))
(french-horn
```

```
 (:staff-name "french horn" :staff-short-name "hn"
  :lowest-written f2 :highest-written c6 :transposition-semitones -7
  :largest-fast-leap 9
  ;; :clefs (treble bass) :starting-clef treble
  :clefs (treble)
  :chords nil
  :microtones t
  :midi-program 61))
;; MDE Fri Mar 5 08:42:59 2021, Heidhausen -- for typical orchestral
;; scorings of 1+3 (high), 2+4 (low)
(french-horn-high
 (:staff-name "french horn" :staff-short-name "hn"
  :lowest-written g3 :highest-written c6 :transposition-semitones -7
  :largest-fast-leap 9
  :clefs (treble)
  :chords nil
  :microtones t
  :midi-program 61))
(french-horn-low
 (:staff-name "french horn" :staff-short-name "hn"
  :lowest-written f2 :highest-written g5 :transposition-semitones -7
  :largest-fast-leap 9
  :clefs (treble)
  :chords nil
  :microtones t
  :midi-program 61))
(c-trumpet
 (:staff-name "trumpet in c" :staff-short-name "c tpt"
  :lowest-written fs3 :highest-written c6
  :largest-fast-leap 9
  :clefs (treble) :starting-clef treble
  :chords nil
  :microtones t
  :midi-program 57))
;; SAR Fri Jan 20 12:09:41 GMT 2012: Added b-flat-trumpet from Adler
;; MDE Mon Feb 20 20:02:55 2012 -- modified to keep in line with clarinet
(b-flat-trumpet
 (:staff-name "B-flat trumpet" :staff-short-name "b-flat tpt"
  ;; the -flat should be converted in CMN and LilyPond to the flat sign
  :lowest-written fs3 :highest-written d6 :transposition-semitones -2
  :largest-fast-leap 9
  :starting-clef treble
  :chords nil :midi-program 57))
;; SAR Fri Jan 20 12:17:24 GMT 2012: Added tenor trombone from Adler
(tenor-trombone
 (:staff-name "trombone" :staff-short-name "tbn"
```

```
       :lowest-written e2 :highest-written bf4
       :largest-fast-leap 7
       :microtones t
       :clefs (bass tenor) :starting-clef bass
       :chords nil :midi-program 58))
     ;; MDE Thu Nov  8 19:32:19 2018 --
     (bass-trombone
      (:staff-name "bass trombone" :staff-short-name "btbn"
       ;; these are of course not true for every instrument/player
       :lowest-written e1 :highest-written g4
       :largest-fast-leap 7
       :microtones t
       :clefs (bass tenor) :starting-clef bass
       :chords nil :midi-program 58))
;;; SAR Fri Jul 13 12:35:35 BST 2012
     (tuba
      (:staff-name "tuba" :staff-short-name "tba"
       :lowest-written d1 :highest-written g4
       :largest-fast-leap 5
       :clefs (bass) :starting-clef bass
       :chords nil :midi-program 59))
     (marimba
      (:staff-name "marimba" :staff-short-name "mba"
       :lowest-written c3 :highest-written c7
       :starting-clef treble :clefs (treble) ; (treble bass)
       :chords t :chord-function mallet-chord-fun
       :microtones nil :midi-program 13))
     (vibraphone
      (:staff-name "vibraphone" :staff-short-name "vib"
       :lowest-written f3 :highest-written f6
       :starting-clef treble
       :chords t :chord-function mallet-chord-fun
       :microtones nil :midi-program 12))
     (accordion
      (:staff-name "accordion" :staff-short-name "acc"
       :lowest-written e1 :highest-written bf7
       :clefs (treble bass double-treble double-bass) :starting-clef treble
       ;; clearly a better function should be provided at some point
       :chords t :chord-function piano-chord-fun
       :microtones nil :midi-program 22))
     ;; MDE Mon Mar 24 20:21:08 2014 -- following three added from data given
     ;; by Zach Howell (thanks).
     (glockenspiel
      (:staff-name "glockenspiel" :staff-short-name "glk"
       :lowest-written f3 :highest-written c6
       :transposition-semitones +24
```

```
  :starting-clef treble
  :chords nil :microtones nil :missing-notes nil
  :midi-program 10))
(xylophone
 (:staff-name "xylophone" :staff-short-name "xyl"
  :lowest-written f3 :highest-written c7
  :transposition-semitones +12
  :starting-clef treble
  :chords nil :microtones nil :missing-notes nil
  :midi-program 14))
(celesta
 (:staff-name "celesta" :staff-short-name "cel"
  :lowest-written c3 :highest-written c7
  :transposition-semitones +12
  :starting-clef treble
  :chords t :microtones nil :missing-notes nil
  :midi-program 9))
(crotales
 (:staff-name "crotales" :staff-short-name "crt"
  :lowest-written c4 :highest-written c6
  :starting-clef treble
  :transposition-semitones 24
  :chords t :chord-function mallet-chord-fun
  ;; no crotales in GM set so use glockenspiel
  :microtones nil :midi-program 10))
(piano
 (:staff-name "piano" :staff-short-name "pno"
  :lowest-written a0 :highest-written c8
  :largest-fast-leap 9
  :clefs (treble bass double-treble double-bass) :starting-clef treble
  :chords t :chord-function piano-chord-fun
  :microtones nil
  :midi-program 1))
;; We generally treat the piano as two instruments (LH, RH), generating
;; lines separately. So this is the same as the piano instrument but has
;; no staff-name and starts with bass clef. Use set-limits to change the
;; range of the two hands, as they're both set to be full piano range
;; here.
(piano-lh
 (:lowest-written a0 :highest-written c8
  ;; MDE Tue Aug 21 17:47:07 2012 -- to avoid the NIL ins name in CMN
  :staff-name "" :staff-short-name ""
  :largest-fast-leap 9
  :chords t :chord-function piano-chord-fun
  :clefs (treble bass double-treble double-bass) :starting-clef bass
  :microtones nil
```

```
  :midi-program 1))
;; MDE Wed Sep 12 12:51:00 2018 -- the 5 octave instrument...range varies
;; according to model of course
(harpsichord
 (:staff-name "harpsichord" :staff-short-name "hpd"
  :lowest-written f1 :highest-written f6
  :largest-fast-leap 9
  :clefs (treble bass) :starting-clef treble
  :chords t :chord-function piano-chord-fun
  :microtones nil
  :midi-program 7))
(harp
 (:staff-name "harp" :staff-short-name "hp"
  :lowest-written b0 :highest-written gs7
  :largest-fast-leap 12
  :clefs (treble bass) :starting-clef treble
  :chords t :chord-function piano-chord-fun
  :microtones nil :harmonics t
  :open-strings nil :nodes ((12 2))
  :midi-program 47))
;; MDE Fri Jul 24 20:18:16 2015 -- range typical for modern organ with
;; 61-key manuals and 32-key pedalboard
(organ
 (:lowest-written c2 :highest-written c7 :staff-name "organ"
  :staff-short-name "org" :largest-fast-leap 9 :chords t
  :chord-function piano-chord-fun :clefs (treble bass)
  :starting-clef treble :microtones nil :midi-program 20))
(organ-pedals
 (:lowest-written c2 :highest-written g4 :staff-name "organ pedals"
  :staff-short-name "ped" :largest-fast-leap 6 :chords nil
  :clefs (bass) :starting-clef bass :microtones nil :midi-program 20))
(percussion
 (:staff-name "percussion" :staff-short-name "perc" :staff-lines 5
  :lowest-written d4 :highest-written g5
  :starting-clef percussion
  :midi-program 1))
(tambourine
 (:staff-name "tambourine" :staff-short-name "tmb" :staff-lines 1
  ;; MDE Fri Nov 13 16:05:46 2015 -- put on b4 not c4
  ;; :lowest-written c4 :highest-written c4
  :lowest-written b4 :highest-written b4
  :starting-clef percussion
  :midi-program 1))
(guitar
 (:staff-name "guitar" :staff-short-name "gtr"
  :lowest-written e3 :highest-written b6 :transposition-semitones -12
```

```
  :largest-fast-leap 31
  :starting-clef treble
  :chords t :chord-function guitar-chord-selection-fun
  :microtones nil :harmonics t
  :open-strings (e5 b4 g4 d4 a3 e3) ; written!
  :open-string-marks (c1 c2 c3 c4 c5 c6)
  :midi-program 25))
;; MDE Wed Oct  9 12:21:22 2013
(mandolin
 (:staff-name "mandolin" :staff-short-name "mln"
  :lowest-written g3 :highest-written c7
  :largest-fast-leap 25
  :starting-clef treble
  ;; mandolin has same tuning as the violin
  :open-strings (e5 a4 d4 g3)
  :open-string-marks (c1 c2 c3 c4) ; assuming guitar numbering strategy
  :chords t :chord-function violin-chord-selection-fun
  ;; there is no GM programme for mandolin so use either steel string
  ;; guitar (26) or banjo 106 perhaps
  :microtones nil :midi-program 26))
(soprano                          ; voice
 (:staff-name "soprano" :staff-short-name "sop"
  :lowest-written c4 :highest-written c6
  :starting-clef treble :midi-program 54))
(mezzo                            ; voice
 (:staff-name "mezzo-soprano" :staff-short-name "mez"
  :lowest-written a3 :highest-written a5
  :starting-clef treble :midi-program 54))
(alto                             ; voice
 (:staff-name "alto" :staff-short-name "alt"
  :lowest-written f3 :highest-written f5
  :starting-clef treble :midi-program 54))
(countertenor                     ; voice
 (:staff-name "countertenor" :staff-short-name "ctr"
  :lowest-written e3 :highest-written e5
  :starting-clef treble-8vb :midi-program 54))
(tenor                            ; voice
 (:staff-name "tenor" :staff-short-name "ten"
  :lowest-written c3 :highest-written c5
  :starting-clef treble-8vb :midi-program 54))
(baritone                         ; voice
 (:staff-name "baritone" :staff-short-name "bar"
  :lowest-written a2 :highest-written a4
  :starting-clef bass :midi-program 54))
(bass                             ; voice
 (:staff-name "bass" :staff-short-name "bas"
```

```
  :lowest-written e2 :highest-written e4
  :starting-clef bass :midi-program 54))
(violin
 (:staff-name "violin" :staff-short-name "vln"
  :lowest-written g3 :highest-written c7
  :largest-fast-leap 13
  :starting-clef treble
  :chords t :chord-function violin-chord-selection-fun
  :microtones t :harmonics t
  :open-strings (e5 a4 d4 g3)
  :open-string-marks (i ii iii iv)
  :midi-program 41))
(viola
 (:staff-name "viola" :staff-short-name "vla"
  :lowest-written c3 :highest-written f6
  :largest-fast-leap 13
  :clefs (alto treble) :starting-clef alto
  :chords t :chord-function viola-chord-selection-fun
  :microtones t :harmonics t
  :open-strings (a4 d4 g3 c3)
  :open-string-marks (i ii iii iv)
  :midi-program 42))
(viola-d-amore
 (:staff-name "viola d'amore" :staff-short-name "vla d'am"
  :lowest-written a2 :highest-written f7
  :largest-fast-leap 13
  :clefs (alto treble) :starting-clef alto
  :chords t :chord-function nil
  :microtones t :harmonics t
  :midi-program 41))
(cello
 (:staff-name "cello" :staff-short-name "vc"
  ;; of course it can go higher but best not to algorithmically select
  ;; these
  :lowest-written c2 :highest-written a5
  :largest-fast-leap 12
  :clefs (bass tenor treble) :starting-clef bass
  :chords t :chord-function cello-chord-selection-fun
  :microtones t :harmonics t
  :open-strings (a3 d3 g2 c2)
  :open-string-marks (i ii iii iv)
  :midi-program 43))
(double-bass
 (:staff-name "double bass" :staff-short-name "db"
  :lowest-written e2 :highest-written g5 :transposition-semitones -12
  :prefers-notes low
```

```
    :largest-fast-leap 8 ; 10
    :clefs (bass tenor treble) :starting-clef bass
    :chords nil
    :microtones t :harmonics t
    :open-strings (g3 d3 a2 e2)      ; written!
    :open-string-marks (i ii iii iv)
    :midi-program 44))
  (bass-guitar
   (:staff-name "bass guitar" :staff-short-name "b. gtr"
    :lowest-written e2 :highest-written g4 :transposition-semitones -12
    :prefers-notes low
    :largest-fast-leap 10
    :clefs (bass treble) :starting-clef bass
    :chords t
    :microtones nil :harmonics t
    :open-strings (g3 d3 a2 e2)      ; written!
    :open-string-marks (c1 c2 c3 c4)
    :midi-program 33))
  ;; SAR Thu Apr 12 18:19:21 BST 2012: Added "computer" part for "silent"
  ;; parts in case the user would like to create rhythmically independent
  ;; computer parts.
  ;; MDE Jul 2012 -- changed to reflect more clefs (and removed percussion)
  (computer
   (:staff-name "computer" :staff-short-name "comp"
    :lowest-sounding C-1 :highest-sounding bf8
    :clefs (treble bass double-treble double-bass)
    ;; MDE Thu Oct  8 17:13:35 2020, Heidhausen -- FX 4 (atmosphere)
    :midi-program 100
    ;; MDE Sat Aug  5 18:42:03 2017 --
    :microtones t
    :starting-clef treble)))))
```

13.2 instruments/cello-chord-selection-fun

[instruments] [Functions]

DESCRIPTION:

```
Create a double-stop chord object using the core string-chord-selection-fun
and a value of 'G2 for the open III string.
```

SYNOPSIS:

```
(let ((vc-III (make-pitch 'g2)))
  (defun cello-chord-selection-fun (curve-num index pitch-list pitch-seq
                                    instrument set)
```

13.3 instruments/chord-fun-aux

[instruments] [Functions]

DESCRIPTION:

```
An auxiliary function that allows users to create moderately tailored chord
functions by setting values for the number of notes in the current set to
skip, the number of desired notes in the resulting chord, and the maximum
span of the resulting chord in semitones.

This function must be called within a call to the defun macro to create a
new chord function, as demonstrated below.
```

ARGUMENTS:

```
The first six arguments -- curve-num, index, pitch-list, pitch-seq,
instrument, and set -- are inherited and not required to be directly
accessed by the user.

- skip: An integer that is the step by which the function skips through the
  subset of currently available pitches. A value of 2, for example, will
  instruct the method to build chords from every second pitch in that
  subset.
- An integer that is the number of pitches that should be in each resulting
  chord. If the list of pitches available to an instrument is too short to
  make a chord with x notes, a chord with fewer pitches may be made
  instead.
- An integer that is the largest interval in semitones allowed between the
  bottom and top notes of the chord. If a chord made with the specified
  number of notes surpasses this span, a chord with fewer pitches may be
  made instead.
```

EXAMPLE:

```
(defun new-chord-function (curve-num index pitch-list pitch-seq instrument set)
  (chord-fun-aux curve-num index pitch-list pitch-seq instrument set 4 3 14))

=> NEW-CHORD-FUNCTION
```

SYNOPSIS:

```
(defun chord-fun-aux (curve-num index pitch-list pitch-seq instrument set
                      skip num-notes max-span)
```

13.4 instruments/chord-fun1

[instruments] [Functions]

DESCRIPTION:

```
Generate three-note chords where possible, using every second pitch from
the list of pitches currently available to the given instrument from the
current set, and ensuring that none of the chords it makes span more than
an octave.

In all functions that take this format curve-num is the
```

SYNOPSIS:

```
(defun chord-fun1 (curve-num index pitch-list pitch-seq instrument set)
```

13.5 instruments/chord-fun2

[instruments] [Functions]

DESCRIPTION:

```
Generates 4-note chords where possible, using every third pitch from the
list of pitches currently available to the given instrument from the
current set, with (almost) no limit on the total span of the chord.
```

SYNOPSIS:

```
(defun chord-fun2 (curve-num index pitch-list pitch-seq instrument set)
```

13.6 instruments/get-standard-ins

[instruments] [Functions]

DATE:

```
August 29th 2018, Heidhausen
```

DESCRIPTION:

```
Convenience function for get-data from the standard instrument palette
```

ARGUMENTS:

```
- the ID of the instrument (symbol)
```

OPTIONAL ARGUMENTS:

```
- either NIL (no error/warning) or a function (object) to call if the
  instrument isn't found. Default = #'error
```

RETURN VALUE:

```
the instrument object
```

SYNOPSIS:

```
(defun get-standard-ins (ins &optional (on-fail #'error))
```

13.7 instruments/guitar-chord-selection-fun

[instruments] [Functions]

DESCRIPTION:

```
Create chord objects with differing numbers of pitches, drawing the pitches
from set-palette object subsets with the ID 'guitar.

This function was written for the composition "Cheat Sheet", in which the
pitch sets were defined explicitly such that all of the pitches available
to the guitar at any moment were playable as a guitar chord. As such, this
function always assumes that the pitch-list it is drawing from contains
pitches that are already playable as a guitar chord. It also adds the
fingering as mark above each chord when outputting to CMN, which may or may
not be desirable.
```

SYNOPSIS:

```
(defun guitar-chord-selection-fun (curve-num index pitch-list pitch-seq
                                   instrument set)
```

13.8 instruments/natural-harmonic

[instruments] [Functions]

DATE:

```
December 24th 2013
```

DESCRIPTION:

```
Determine whether a pitch can be played as a natural harmonic on a (string)
instrument (with the guitar as default).

MDE Tue Jan 8 18:22:09 2019 -- This function is retained for historical
reasons but has been superseded by the instrument method natural-harmonic?
along with the associated instrument slots open-strings, open-string-marks,
and nodes (which altogether are more flexible and efficient than this
function alone)
```

ARGUMENTS:

```
- the pitch (symbol or pitch object)
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :tuning. a list of the fundamentals of the open strings, as pitch objects
  or symbols.  These should descend from the highest string.  Default:
  guitar tuning.
- :highest-partial.  Integer. What we consider the highest harmonic possible
  (counting the fundamental as 1).
- :tolerance.  The deviation in cents that we can accept for the frequency
  comparison.  Default = 10.
- :debug. Print data for debugging/testing purposes.  Default = NIL.
```

RETURN VALUE:

```
The string number and partial number as a list if possible as a harmonic,
or NIL if not.
```

EXAMPLE:

```
(NATURAL-HARMONIC 'b5) ; octave harmonic of B string
=> (2 2)
SC> (NATURAL-HARMONIC 'b6) ; octave + 5th of high E string
=> (1 3)
```

SYNOPSIS:

```
(defun natural-harmonic (pitch &key (tuning '(e5 b4 g4 d4 a3 e3))
                         (highest-partial 6) (tolerance 15) debug)
```

13.9 instruments/piano-chord-fun

[instruments] [Functions]

DESCRIPTION:

```
Generate four-note chords, where possible, from consecutive notes in the
current set, with the number enclosed in parentheses in the pitch-seq being
the top note of that chord, where possible.
```

SYNOPSIS:

```
(defun piano-chord-fun (curve-num index pitch-list pitch-seq instrument set)
```

13.10 instruments/piano-chord-fun3

[instruments] [Functions]

DATE:

```
August 2nd 2019, Heidhausen
```

DESCRIPTION:

```
Almost exactly the same as piano-chord-fun but generally get just three
notes rather than four. If we want to force a maximum of three notes, it
will need to be done with subseq or similar. I find this 'usually three
notes' preferable though as some four-note chords are returned, for variety.
```

SYNOPSIS:

```
(defun piano-chord-fun3 (curve-num index pitch-list pitch-seq instrument set)
```

13.11 instruments/play-all

[instruments] [Functions]

DESCRIPTION:

```
A very simple chord function to simply play all the notes in the set, no
matter which instrument should play them. Useful perhaps for computer
voices. In any case there's no checking of instrument range as we ignore
the pitch-list here.
```

SYNOPSIS:

```
(defun play-all (curve-num index pitch-list pitch-seq instrument set)
```

13.12 instruments/salzedo-p

[instruments] [Functions]

DESCRIPTION:

```
Determine whether a given input is a valid salzedo-list, which is a 7-number
list reflecting the pedal positions of a concert harp according to the
Salzedo-notation. The order of the pedals in the list is (D C B E F G A),
where -1 indicates lowering the respective note a half step, 0 means no
alteration, 1 means raising the note a half. Thus, '(0 1 0 0 0 -1 0)
implies the following alteration: D C# B E F Gb A, or (in a more common
order: C# D E F Gb A B.
```

ARGUMENTS:

```
The object to be tested.
```

RETURN VALUE:

```
T if the tested object is a salzedo-list, otherwise NIL.
```

EXAMPLE:

```
(salzedo-p '(-1 1 0 1 0 1 1))
=> T
```

SYNOPSIS:

```
(defun salzedo-p (object)
```

13.13 instruments/salzedo-to-symbol

[instruments] [Functions]

DESCRIPTION:

```
This function converts a salzedo harp pedaling list (cf.
harp-salzedo-to-tl-set) into a symbol which could be further used e.g. as an
id for subsets or sets.
During the processing, the values in the salzedo-list will be replaced
with the following characters:
-1 => F   (lower a semitone/flat)
 0 => N   (natural)
 1 => S   (raise a semitone/sharp)
```

ARGUMENTS:

```
The salzedo list (e.g. '(0 -1 1 1 0 -1 1)) to be parsed into a symbol.
```

OPTIONAL ARGUMENTS:

```
keyword-arguments:
- :prefix. The prefix to be added to the generated symbol. Default = 'harp-.
- :suffix. The suffix to be added to the generated symbol. Default = NIL.
```

RETURN VALUE:

```
A symbol generated from the given salzedo-list.
```

EXAMPLE:

```
(salzedo-to-symbol '(-1 1 0 1 1 -1 0) :prefix 'harp-)
;; => 'HARP-FSNSSFN
```

SYNOPSIS:

```
(defun salzedo-to-symbol (salzedo &key
                                    (prefix 'harp-)
                                    suffix)
```

13.14 instruments/set-standard-range

[instruments] [Functions]

DATE:

```
September 24th 2018, Heidhausen
```

DESCRIPTION:

```
Set the lowest and highest pitches of an instrument in the standard
instrument palette. Use this function to correctly set the highest sounding
and written slots of transposing and non-transposing instruments alike.
```

ARGUMENTS:

```
- the instrument id (symbol)
- the lowest pitch (symbol or pitch object). This can also be NIL, whereupon
  no change will be made to the current slot's value.
- the highest pitch (symbol or pitch object). Can also be NIL.
```

OPTIONAL ARGUMENTS:

- T or NIL to indicated whether the pitches given are to be interpreted as written or sounding pitches. In either case, both written and sounding pitches will be set for the instrument. Default = NIL = written.

RETURN VALUE:

The instrument object for which the changes were made.

SYNOPSIS:

```
(defun set-standard-range (ins low high &optional sounding)
```

13.15 instruments/string-chord-selection-fun

[instruments] [Functions]

DESCRIPTION:

This is the core function for creating instances of double-stop chords for strings, ensuring that the highest note of the double-stop is not lower than the open III string. The pitch of the open III string is passed as an argument in the chord-selection functions for the individual stringed instruments.

This function uses the best-string-diad function. If no double-stops instances can be created using best-string-diad, two-note chords will be created using the default-chord-function. If neither of these are possible, a chord of a single pitch will be returned instead.

SYNOPSIS:

```
(defun string-chord-selection-fun (curve-num index pitch-list pitch-seq
                                   instrument set string-III)
```

13.16 instruments/symbol-to-salzedo

[instruments] [Functions]

DESCRIPTION:

This method converts a salzedo symbol (e.g. created via salzedo-to-symbol) back to a salzedo-list (cf. ibid). It removes the prefix and suffix (if

```
given) from the symbol and translates each of the seven (!) characters of
the remaining symbol as follows:
F => -1  (lower a semitone/flat)
N =>  0  (natural)
S =>  1  (raise a semitone/sharp)
```

ARGUMENTS:

```
A symbol.
```

OPTIONAL ARGUMENTS:

```
keyword-arguments:
- :prefix. A symbol which will be removed from the start of the input
  symbol and be ignored while processing. Default = 'harp-.
- :suffix. A symbol which will be removed from the end of the input symbol
  and be ignored while processing. Default = NIL.
```

RETURN VALUE:

```
A salzedo-list (cf. salzedo-to-symbol).
```

EXAMPLE:

```
(symbol-to-salzedo 'HARP-FSNSSFN-ONE :prefix 'harp- :suffix '-ONE)
;; => (-1 1 0 1 1 -1 0)
|#
(defun symbol-to-salzedo (symbol &key
                                  (prefix 'harp-)
                                  suffix)
```

13.17 instruments/viola-chord-selection-fun

[instruments] [Functions]

DESCRIPTION:

```
Create a double-stop chord object using the core string-chord-selection-fun
and a value of 'G3 for the open III string.
```

SYNOPSIS:

```
(let ((vla-III (make-pitch 'g3)))
  (defun viola-chord-selection-fun (curve-num index pitch-list pitch-seq
                                    instrument set)
```

13.18 instruments/violin-chord-selection-fun

[instruments] [Functions]

DESCRIPTION:

```
Create a double-stop chord object using the core string-chord-selection-fun
and a value of 'D4 for the open III string.
```

SYNOPSIS:

```
(let ((vln-III (make-pitch 'd4)))
  (defun violin-chord-selection-fun (curve-num index pitch-list pitch-seq
                                     instrument set)
```

14 sc/lilypond

[Modules]

14.1 lilypond/lp-get-mark

[lilypond] [Functions]

DESCRIPTION:

```
lp-get-mark:
Translation function for LilyPond marks (dynamics, accents, etc.).  Not
generally called by the user but the list of symbols that can be used will
be useful.  If <silent> then non-existing marks will not produce
warnings/errors (but we'll return nil).
MDE Thu Sep 15 10:50:47 2016 -- default to the
config setting
```

SYNOPSIS:

```
          (a "-> ")                    ; accent
          (lhp "-+ ")
          ;; see p229 of lilypond.pdf: need to define this command in file
          ;; this is done for us in lilypond.ly, which will be included if we
          ;; call write-lp-data-for-all with :use-custom-markup T
          (bartok "^\\snappizzicato ")
          (pizz "^\"pizz.\" ")
          (ord "^\"ord.\" ")
          (pizzp "^\"(pizz.)\" ")
```

```
(clb "^\"clb\"")
(cl "^\"cl\" ")
(col-legno "^\"col legno\" ")
(clt "^\"clt\" ")
(arco "^\"arco\" ")
(batt "^\"batt.\" ")
(spe "^\"spe\" ")
(sp "^\"sul pont.\" ")
(st "^\"sul tasto\" ")
(mv "^\"molto vib.\" ")
(sv "^\"senza vib.\" ")
(poco-crini "^\"poco crini\" ")
(flaut '(wds "^\"flautando\" "))
(det '(wds "^\"detache\" "))
(s "-. ")
(ss "-! ") ; MDE Thu Dec 2 15:16:27 2021, Heidhausen
;; MDE Mon Dec 15 12:25:29 2025, Heidhausen -- something close to
;; dorico's 'stress'
(stress "-! ") ; MDE Thu Dec 2 15:16:27 2021, Heidhausen
(nail (no-lp-mark 'nail))
(stopped (no-lp-mark 'stopped))
(as "->-. ")
(at "->-- ")
(ts "-_ ")
(te "-- ")
;; so unmeasured is implicit
(t3 (format nil ":~a " (* 32 (expt 2 num-flags))))
(flag "\\flageolet ")
(niente "^\markup { niente } ")
(pppp "\\pppp ")
(ppp "\\ppp ")
(pp "\\pp ")
(p "\\p ")
(mp "\\mp ")
(mf "\\mf ")
(f "\\f ")
(ff "\\ff ")
(fff "\\fff ")
(ffff "\\ffff ")
;; MDE Sat Aug 11 15:51:16 2012 -- dynamics in parentheses
(ffff-p "\\parenFFFF ")
(fff-p "\\parenFFF ")
(ff-p "\\parenFF ")
(f-p "\\parenF ")
(mf-p "\\parenMF ")
(mp-p "\\parenMP ")
```

```
(p-p "\\parenP ")
(pp-p "\\parenPP ")
(ppp-p "\\parenPPP ")
(pppp-p "\\parenPPPP ")
(sfz "\\sfz ")
(downbow "\\downbow ")
(upbow "\\upbow ")
(open "\\open ")
(I "^\\markup { \\teeny \"I\" } ")
(II "^\\markup { \\teeny \"II\" } ")
(III "^\\markup { \\teeny \"III\" } ")
(IV "^\\markup { \\teeny \"IV\" } ")
;; MDE Thu Dec 26 14:14:34 2013 -- guitar string numbers
(c1 "\\1 ")
(c2 "\\2 ")
(c3 "\\3 ")
(c4 "\\4 ")
(c5 "\\5 ")
(c6 "\\6 ")
(beg-sl "( ")
(end-sl  ") ")
;; MDE Fri Apr  6 21:57:59 2012 -- apparently LP can't have nested
;; slurs but it does have phrase marks:
(beg-phrase "\\( ")
(end-phrase "\\) ")
;; there's no start gliss / end gliss in lilypond
(beg-gliss "\\glissando ")
(end-gliss "")
;; 13.4.11
(beg-8va "\\ottava #1 ")
(end-8va "\\ottava #0 ")
(beg-8vb "\\ottava #-1 ")
(end-8vb "\\ottava #0 ")
;; MDE Mon Jan 18 11:43:43 2016
(beg-15ma "\\ottava #2 ")
(end-15ma "\\ottava #0 ")
(beg-15mb "\\ottava #-2 ")
(end-15mb "\\ottava #0 ")
;; NB note heads used to be added via (add-mark-before) but can now
;; be added by add-mark and friends (16.6.20) so if adding new heads
;; , make sure to add the mark symbol to the move-elements call in
;; event::separate-marks-before
;;
;; (circled-x "\\once \\override NoteHead #'style = #'xcircle ")
(circled-x "\\tweak #'style #'xcircle ")
;; MDE Thu May 29 10:25:02 2025, Heidhausen -- I'm surprised that
```

```
;; lilypond doesn't have a circled note head (e.g. for rim shots) ---
;; or did I miss something at
;; https://lilypond.org/doc/v2.23/Documentation/notation/note-heads ?
(circled (no-lp-mark 'circled))
;; (x-head "\\once \\override NoteHead #'style = #'cross ")
(x-head "\\xNote ")
;; (triangle "\\once \\override NoteHead #'style = #'triangle ")
(triangle "\\tweak #'style #'triangle ")
;; (triangle-up "\\once \\override NoteHead #'style = #'do ")
(triangle-up "\\tweak #'style #'do ")
(airy-head (no-lp-mark 'airy-head))
;; this has to be added to the event _before_ the one which needs to
;; start with these noteheads.
(improvOn "\\improvisationOn ")
(improvOff "\\improvisationOff ")
;; MDE Sat Nov  9 20:21:19 2013 -- in CMN it's :breath-in: a
;; triangle on its side (pointing left)
;; (wedge "\\once \\override NoteHead #'style = #'fa ")
(wedge "\\tweak #'style #'fa ")
;; (square "\\once \\override NoteHead #'style = #'la ")
(square "\\tweak #'style #'la ")
;; (mensural "\\once \\override NoteHead #'style = #'slash ")
;;(flag-head "\\once \\override NoteHead #'style = #'harmonic-mixed
;;")
;; MDE Mon Apr 30 20:46:06 2012 -- see event::get-lp-data for how
;; this is handled
(flag-head "\\harmonic ") ; diamond
;; MDE Mon Apr 30 20:46:31 2012 -- flag-heads by default don't
;; display dots so we need to add-mark-before to get these to
;; display or turn them off again
(flag-dots-on "\\set harmonicDots = ##t ")
(flag-dots-off "\\set harmonicDots = ##f ")
;; circle head but stem extends through it like a vertical slash
(none (no-lp-mark 'none))
(trill-f (no-lp-mark 'trill-f))
(trill-n (no-lp-mark 'trill-n))
(trill-s (no-lp-mark 'trill-s))
(beg-trill-a "\\pitchedTrill ") ; must be before note
;; we'll also need e.g. (trill-note g5) to give the note in ()
(end-trill-a "\\stopTrillSpan ") ; after note
;; (no-lp-mark 'square))
(slash (no-lp-mark 'slash))
;; MDE Sat Dec 28 11:37:22 2013 -- up and down arrows on arpeggio
;; lines will need more complex treatment (need a note-before mark
;; :/ )
(arp "\\arpeggio ")
```

```
(arrow-up (no-lp-mark 'arrow-up))
(arrow-down (no-lp-mark 'arrow-down))
(cresc-beg "\\< ")
(cresc-end "\\! ")
(dim-beg "\\> ")
(dim-end "\\! ")
(<< "<< ")
(>> ">> ")
;; NB this override has to come exactly before the note/dynamic it
;; applies to. MDE Tue Jun 16 23:38:57 2020, Heidhausen -- again this
;; override should be OK (rather than tweak) as it's below the staff
;; rather than a property of a note (in a chord)
;; (hairpin0 "\\once \\override Hairpin #'circled-tip = ##t ")
;; MDE Thu May 11 15:26:00 2023, Heidhausen -- version 2.21 changes
(hairpin0 "\\once \\override Hairpin.circled-tip = ##t ")
;; (dim0-beg "\\once \\override Hairpin #'circled-tip = ##t \\> ")
(pause "\\fermata ")
(long-pause "\\longfermata ")
(short-pause
 "^\\markup { \\musicglyph #\"scripts.ushortfermata\" } ")
;; MDE Thu Apr  5 16:17:11 2012 -- these need the graphics files in
;; lilypond-graphics.zip to be in the same directory as the
;; generated lilypond files
(aeolian-light "^\\aeolianLight ")
(aeolian-dark "^\\aeolianDark ")
;; this one uses the graphic for close bracket
(bracket-end "^\\bracketEnd ")
(mphonic "^\\mphonic ")
(mphonic-arr "^\\mphonicArr ")
(mphonic-cons "^\\mphonicCons ")
(mphonic-diss "^\\mphonicDiss ")
(mphonic-cluster "^\\mphonicCluster ")
(sing "^\\sing ")
(high-sine "^\\high-sine ")
(noise "^\\noise ")
(focus "^\\focus ")
(balance "^\\balance ")
(alternate "^\\alternate ")
(sing-arr "^\\singArr ")
(arrow-up-down "^\\arrowUpDown ")
;; end lilypond-graphics.zip files
;; these must have been set up with the event::add-arrow method
(start-arrow "\\startTextSpan ")
(end-arrow "\\stopTextSpan ")
(harm "^\\flageolet ")
;; 2.3.11
```

```
;; write sost. pedal as text (usually held for long time so brackets
;; not a good idea)
;; MDE Wed May 25 12:40:57 2016 -- update: do use the LP
;; sost. brackets after all
(sost "\\sostenutoOn ")
(sost-up "\\sostenutoOff ")
(sost^ "\\sostenutoOff\\sostenutoOn ")
(ped "\\sustainOn ")
(ped^ "\\sustainOff\\sustainOn ")
(ped-up "\\sustainOff ")
(uc "\\unaCorda ")
(tc "\\treCorde ")
```

14.2 lilypond/lp-salzedo-mark

[lilypond] [Functions]

DESCRIPTION:

```
Returns the LilyPond markup for a harp salzedo pedal-mark given as a salzedo
list (cf. harp-salzedo-to-tl-set).
```

ARGUMENTS:

```
- The salzedo list. E.g. '(0 -1 1 0 1 -1 1) for d, cf, bs, e, fs, gf, as
```

EXAMPLE:

```
(lp-salzedo-mark '(1 -1 1 0 1 0 1))
;; => "_\\markup { \\harp-pedal \"v^v|-v-v\" }"
```

SYNOPSIS:

```
(defun lp-salzedo-mark (salzedo)
```

14.3 lilypond/salzedo-to-ly-string

[lilypond] [Functions]

DESCRIPTION:

```
This method converts a salzedo list to a string to be used with LilyPond's
\harp-pedal markup. Cf.
https://lilypond.org/doc/v2.23/Documentation/notation/harp
```

```
N.B.: Steinberg's Dorico notation software uses the same syntax for entering
harp pedal diagrams.
```

ARGUMENTS:

```
A salzedo list (cf. harp-salzedo-to-tl-set).
```

RETURN VALUE:

```
The LilyPond salzedo-string.
```

EXAMPLE:

```
(salzedo-to-ly-string '(0 1 0 1 -1 1 0))
=>
"-v-|v^v-"
```

SYNOPSIS:

```
(defun salzedo-to-ly-string (salzedo)
```

15 sc/music-xml

[Modules]

15.1 music-xml/xml-get-mark

[music-xml] [Functions]

DESCRIPTION:

```
xml-get-mark:
Translation function for music-xml marks (dynamics, accents, etc.). Not
generally called by the user but the list of symbols that can be used will
be useful. If <silent> then non-existing marks will not produce
warnings/errors (but we'll return nil).
default to the config setting
(print silent)
```

SYNOPSIS:

```
;; if returns a string then xml-articulation will be called,
;; otherwise (dir ...) (tech ...)
;; also possible: (art "accent" "below") or a list of
;; strings, in which case there'll be 2+ xml-articulation
(a "accent")       ; accent
(s "staccato")
(ss "staccatissimo") ; MDE Thu Dec 2 15:16:27 2021, Heidhausen
(as '("accent" "staccato"))
(at '("accent" "tenuto"))
(ts "detached-legato")
(te "tenuto")
;; MDE Mon Dec 15 12:21:47 2025, Heidhausen -- like a cmn
;; wedge (staccatissimo, like above) but at an angle
(stress "stress")
;; tech means xml-technical will be called; see bottom of
;; this function
(lhp '(tech "stopped" a))
(bartok '(tech "snap-pizzicato" a))
;; todo: could also add the <sound> tag here to change
;; midi-programme (see MozartTrio.xml for e.g.)
;; dir means xml-direction will be called
(pizz '(wds "pizz." a))
(ord '(wds "ord."))
(pizzp '(wds "(pizz.)"))
(clb '(wds "clb"))
(cl '(wds "cl"))
(col-legno '(wds "col legno"))
(clt '(wds "clt"))
(arco '(wds "arco"))
(batt '(wds "batt"))
(spe '(wds "spe"))
(sp '(wds "sul pont."))
(st '(wds "sul tasto"))
(mv '(wds "molto vib"))
(sv '(wds "senza vib"))
(flaut '(wds "flautando"))
(det '(wds "detache"))
(poco-crini '(wds "poco crini"))
(nail '(tech "fingernails" a))
(stopped (no-xml-mark 'stopped))
;; so unmeasured tremolo is implicit
(t3 '(orn "tremolo" nil 3))
(flag '(tech "harmonic" a))
;; MDE Fri Dec  7 18:11:29 2018 -- dorico still doesn't handle
;; the little circles to start/end a hairpin
(niente '(wds "niente"))
```

```
(pppp  '(dyn "pppp"))
(ppp  '(dyn "ppp"))
(pp  '(dyn "pp"))
(p  '(dyn "p"))
(mp  '(dyn "mp"))
(mf  '(dyn "mf"))
(f  '(dyn "f"))
(ff  '(dyn "ff"))
(fff  '(dyn "fff"))
(ffff  '(dyn "ffff"))
;; todo: find out how to put dynamics in parentheses; for now
;; just put in the dynamic without
(pppp-p '(dyn "pppp"))
(ppp-p '(dyn "ppp"))
(pp-p '(dyn "pp"))
(p-p '(dyn "p"))
(mp-p '(dyn "mp"))
(mf-p '(dyn "mf"))
(f-p '(dyn "f"))
(ff-p '(dyn "ff"))
(fff-p '(dyn "fff"))
(ffff-p '(dyn "ffff"))
(sfz '(dyn "sfz"))
(downbow '(tech "down-bow" a))
(upbow '(tech "up-bow" a))
(open '(tech "fingering" a 0))
(I '(wds "I"))
(II '(wds "II"))
(III '(wds "III"))
(IV '(wds "IV"))
(V '(wds "V"))
(VI '(wds "VI"))
(VII '(wds "VII"))
(VIII '(wds "VIII"))
;; guitar string numbers; todo: find a way of circling them
(c1 "1")
(c2 "2")
(c3 "3")
(c4 "4")
(c5 "5")
(c6 "6")
;; music xml doesn't seem to have separate slur and phrase
;; marks so we'll simply use the slur tag with number=1 for
;; slurs and number=2 for phrases
(beg-sl '(not "slur" "number=\"1\" type=\"start\""))
(end-sl '(not "slur" "number=\"1\" type=\"stop\""))
```

```
(beg-phrase '(not "slur" "number=\"2\" type=\"start\""))
(end-phrase '(not "slur" "number=\"2\" type=\"stop\""))
(beg-gliss '(not "glissando" "number=\"1\" type=\"start\" line-type=\"solid\""))
(end-gliss '(not "glissando" "number=\"1\" type=\"stop\" line-type=\"solid\""))
(beg-8va '(dir "octave-shift"
           ;; it really is down, probably because what's
           ;; notated is an 8ve lower than sounding
           "dash-length=\"7.5\" size=\"8\" space-length=\"7.5\" type=\"down\""))
(end-8va '(dir "octave-shift" "size=\"8\" type=\"stop\""))
(beg-8vb '(dir "octave-shift"
           "dash-length=\"7.5\" size=\"-8\" space-length=\"7.5\" type=\"up\""))
(end-8vb '(dir "octave-shift" "size=\"-8\" type=\"stop\""))
(beg-15ma '(dir "octave-shift"
            "dash-length=\"7.5\" size=\"15\" space-length=\"7.5\" type=\"down\""))
(end-15ma '(dir "octave-shift" "size=\"15\" type=\"stop\""))
(beg-15mb '(dir "octave-shift"
            "dash-length=\"7.5\" size=\"-15\" space-length=\"7.5\" type=\"up\""))
(end-15mb '(dir "octave-shift" "size=\"-15\" type=\"stop\""))
;; noteheads
(circled-x '(hd "circle-x"))
;; MDE Thu May 29 10:20:50 2025, Heidhausen -- e.g. rim shots
(circled '(hd "circled"))
(x-head '(hd "x"))
(triangle '(hd "triangle"))
(triangle-up '(hd "inverted triangle"))
(airy-head (no-xml-mark 'airy-head))
(none '(hd "none"))
;; the On and Off versions were for lilypond; here we only
;; need the On version (or preferably, just improv but for
;; compatibility...)
(improvOn '(hd "slash"))
(improvOff nil)
(wedge '(hd "fa"))
(square '(hd "la"))
;; todo: look at xml doc for harmonic and sort out a way of
;; representing artificial harms in xml
(flag-head '(hd "diamond"))
(flag-dots-on (no-xml-mark 'flag-dots-on))
(flag-dots-off (no-xml-mark 'flag-dots-off))
(trill-f (no-xml-mark 'trill-f))
(trill-n (no-xml-mark 'trill-n))
(trill-s (no-xml-mark 'trill-s))
(beg-trill-a (xml-beg-trill stream))
(end-trill-a (xml-end-trill stream))
(slash (no-xml-mark 'slash))
(arp '(not "arpeggiate" "default-x=\"-12\"  number=\"1\""))
```

```
(arrow-up (no-xml-mark 'arrow-up))
(arrow-down (no-xml-mark 'arrow-down))
(cresc-beg '(dir "wedge" nil
             "spread=\"0\" type=\"crescendo\"" b))
(cresc-end '(dir "wedge" nil
             "spread=\"15\" type=\"stop\"" b))
(dim-beg '(dir "wedge" nil
           "spread=\"15\" type=\"diminuendo\"" b))
(dim-end '(dir "wedge" nil
           "spread=\"0\" type=\"stop\"" b))
(<< (no-xml-mark '<<))
(>> (no-xml-mark '>>))
;; we could faff around with lexical variables and manipulate
;; the niente property of "wedge" but as this would be
;; post-gen anyway we can do it in the notation software
(hairpin0 (no-xml-mark 'hairpin0))
(pause (xml-pause stream))
;; seems there is a fermata-shape element but damned if I can
;; find out where it could go and be interpreted correctly.
;; MDE Mon Mar 27 13:09:50 2017 -- see xml-pause above
(long-pause (xml-pause stream)); "lunga"))
(short-pause (xml-pause stream)); "breva"))
(aeolian-light (no-xml-mark 'aeolianLight))
(aeolian-dark (no-xml-mark 'aeolianDark))
;; this one uses the graphic for close bracket
(bracket-end (no-xml-mark 'bracketEnd))
(mphonic (no-xml-mark 'mphonic))
(mphonic-arr (no-xml-mark 'mphoniArr))
(mphonic-cons (no-xml-mark 'mphonicCons))
(mphonic-diss (no-xml-mark 'mphonicDiss))
(mphonic-cluster (no-xml-mark 'mphonicCluster))
(sing (no-xml-mark 'sing))
(high-sine (no-xml-mark 'high-sine))
(noise (no-xml-mark 'noise))
(focus (no-xml-mark 'focus))
(balance (no-xml-mark 'balance))
(alternate (no-xml-mark 'alternate))
(sing-arr (no-xml-mark 'singArr))
(arrow-up-down (no-xml-mark 'arrowUpDown))
;; no need to fiddle with these: can do in notation software,
;; no?
(start-arrow (no-xml-mark 'start-arrow))
(end-arrow (no-xml-mark 'end-arrow))
(harm '(tech "harmonic" a "<natural />"))
(sost '(wds "sost."))
(sost-up '(wds "sost.*"))
```

```
(sost^ (no-xml-mark 'sost^)) ;"\\sostenutoOff\\sostenutoOn ")
(ped '(dir "pedal" nil "line=\"yes\" type=\"start\""))
(ped^ '(dir "pedal" nil "line=\"yes\" type=\"change\""))
(ped-up '(dir "pedal" nil "line=\"yes\" type=\"stop\""))
(uc '(wds "una corda" nil b))
(tc '(wds "tre corde" nil b))
```

15.2 music-xml/xml-salzedo

[music-xml] [Functions]

DESCRIPTION:

```
This function returns the MusicXML-markup for a harp salzedo pedalling
indication from a given salzedo-list (cf. harp-salzedo-to-tl-set).

N.B.: Though the MusicXML standard provides the facility to define
harp pedal indications (via <harp-pedals>), most music notation programs,
as of now (2023-12-19), do not support this specific element. Thus, the
harp pedalling can be included via a <words>-tag. This can be achieved by
setting the respective config variable accordingly via
(set-sc-config 'xml-salzedo-as-text t).
 The pedal instruction is then given with the respective characters as
defined in the SMuFL Harp techniques table
(https://w3c.github.io/smufl/latest/tables/harp-techniques.html).
For correct representation, the font-family attribute is set to the
music font. As some notation programs are not capable of dealing with
generic font-families (e.g. "music"), a list of common music/engraving
font-families is given as the standard value to the font-family argument.

If (get-sc-config 'xml-salzedo-as-text) = NIL, the salzedo mark will be
output according to the MusicXML-standard (via the <harp-pedals> element).
```

ARGUMENTS:

```
- The output stream.
- A salzedo list.
```

OPTIONAL ARGUMENTS:

```
keyword-arguments:
- :as-text. When T, the pedal mark will be exported as a <words> text mark,
  instead of the MusicXML <harp-pedals> element (see above).
  Default = The value of (get-sc-config 'xml-salzedo-as-text).
- :placement. The location of the text mark. Default = 'b.
```

```
- :font-size. The font size. Default = 14.
- :font-family. The font family used to display the mark. This should be
  SMuFL compatible. Default = The value of
  (get-sc-config 'xml-engraving-font).
```

RETURN VALUE:

```
The MusicXML-markup for the pedal indication written to the stream.
```

SYNOPSIS:

```
(defun xml-salzedo (stream salzedo &key
                                    (as-text
                                     (get-sc-config 'xml-salzedo-as-text))
                                    (placement 'b)
                                    (font-size 14)
                                    (font-family
                                     (get-sc-config 'xml-engraving-font)))
```

16 sc/package

[Modules]

17 sc/permutations

[Modules]

NAME:

```
permutations

File:             permutations.lsp

Class Hierarchy:  none, no classes defined.

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Various permutation functions.

Author:           Michael Edwards: m@michael-edwards.org
```

```
Creation date:    10th November 2002

$$ Last modified:  09:43:01 Wed Jun 24 2026 CEST

SVN ID: $Id$
```

17.1 permutations/permutations

[permutations] [Functions]

DESCRIPTION:

```
Systematically produce a list of all possible permutations of a set of
consecutive integers beginning with zero. The function's only argument,
<level>, is an integer that determines how many consecutive integers from 0
are to be used for the process.

This is a more efficient permutation algorithm, but the results will always
be in a certain order, with the same number at the end until that
permutation is exhausted, then the number below that etc.
```

ARGUMENTS:

```
An integer that indicates how many consecutive integers from 0 are to be
used for the process.
```

OPTIONAL ARGUMENTS:

```
- allow-file: whether to allow the results to be written to a file rather
than returned to the interpreter. This will only happen if the length of
the list is > 8. The advantage is that you can then read in the file rather
than slowly regenerate the results if you need them again.
```

RETURN VALUE:

```
A list of sequences (lists), each of which is a permutation of the
original.
```

EXAMPLE:

```
;; Produce a list consisting of all permutations that can be made of 4
;; consecutive integers starting with 0 (i.e., (0 1 2 3))
(permutations 4)

=>
```

```
((0 1 2 3) (1 0 2 3) (0 2 1 3) (2 0 1 3) (1 2 0 3) (2 1 0 3) (0 1 3 2)
 (1 0 3 2) (0 3 1 2) (3 0 1 2) (1 3 0 2) (3 1 0 2) (0 2 3 1) (2 0 3 1)
 (0 3 2 1) (3 0 2 1) (2 3 0 1) (3 2 0 1) (1 2 3 0) (2 1 3 0) (1 3 2 0)
 (3 1 2 0) (2 3 1 0) (3 2 1 0))
```

SYNOPSIS:

```
(defun permutations (level &optional (allow-file t))
```

17.1.1 permutations/avoid-common-elements

[permutations] [Functions]

DATE:

```
December 6th 2019, Heidhausen
```

DESCRIPTION:

```
Taking a list of sublists, move the sublists around to try and avoid
consecutive sublists having common elements. Note that even if it's not
possible to completely solve this problem, a re-ordered list will be
returned along with a warning detailing where things were no longer
possible. No further, more complex attempts are made at that point. NB In
that case shuffling before calling this function might help
```

ARGUMENTS:

```
- the list of sublists (of any element type)
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :test. The function to be passed to (intersection) for testing equivalence
  amongst list elements. Default = #'eq
- :accept. The number of common elements to accept amongst sublists. Default
  = 0
- :warn. Issue a warning if we can't complete? Default = T
- :all-on-fail. Return the unprocessed (failing) elements if we can't
  complete? If NIL then we'll return less sublists than given but there will
  be no common elements, at least. Default = T.
```

RETURN VALUE:

```
a list: the first argument with re-ordered sublists.
```

EXAMPLE:

```
(avoid-common-elements
 '((1 2 3) (4 5 6) (5 4 6) (4 6 1) (2 3 4) (2 3 1) (1 3 2)))
-->
WARNING: permutations::avoid-common-elements: did 4 but can't complete.
Current: (1 3 2)
Rest: ((4 6 1) (2 3 4))
((1 2 3) (4 5 6) (2 3 1) (5 4 6) (1 3 2) (4 6 1) (2 3 4))

(avoid-common-elements
 '((1 2 3) (4 5 6) (5 4 6) (3 2 1) (2 3 1) (6 5 4) (3 1 2)))
-->
((1 2 3) (4 5 6) (3 2 1) (5 4 6) (2 3 1) (6 5 4) (3 1 2))
```

SYNOPSIS:

```
(defun avoid-common-elements (lists &key (test #'eq) (accept 0) (warn t)
                                      (all-on-fail t))
```

17.1.2 permutations/inefficient-permutations

[permutations] [Functions]

DESCRIPTION:

```
Return a shuffled, non-systematic list of all possible permutations of a
set of consecutive integers beginning with zero.

The function's first argument, <level>, is an integer that determines how
many consecutive integers from 0 are to be used for the process. An
optional keyword argument <max> allows the user to specify the maximum
number of permutations to return.

This function differs from the "permutations" function in that it's result
is not ordered systematically.

The function simply returns a list of <max> permutations of the numbers
less than <level>; it does not permutate a given list.

The function is inefficient in so far as it simply shuffles the numbers and
so always has to check whether the new list already contains the shuffled
before storing it.

The order of the permutations returned will always be the same unless <fix>
```

is set to NIL.

Keyword argument <skip> allows the user to skip a number of permutations,
which is only sensible if :fix is set to T.

ARGUMENTS:

An integer that indicates how many consecutive integers from 0 are to be
used for the process.

OPTIONAL ARGUMENTS:

keyword arguments:
- :max. An integer that indicates the maximum number of permutations to be
  returned.
- :skip. An integer that indicates a number of permutations to skip.
- :fix. T or NIL to indicate whether the given sequence should always be
  shuffled with the same (fixed) random seed (thus always producing the
  same result). T = fixed seed. Default = T.
- :if-not-enough. A function object (or NIL) to call when :max was
  requested but we can't return that many results. Default = #'error.

RETURN VALUE:

A list.

EXAMPLE:

```
;; Creating a shuffled, non-systematic list of all permutations of consecutive
;; integers 0 to 4
(inefficient-permutations 4)

=> ((2 3 0 1) (3 1 2 0) (2 0 3 1) (1 0 2 3) (1 2 3 0) (0 2 3 1) (2 1 0 3)
    (0 1 2 3) (2 3 1 0) (1 2 0 3) (3 0 1 2) (3 1 0 2) (1 3 2 0) (1 0 3 2)
    (2 0 1 3) (3 2 1 0) (2 1 3 0) (3 2 0 1) (1 3 0 2) (0 2 1 3) (3 0 2 1)
    (0 1 3 2) (0 3 2 1) (0 3 1 2))

;; Using 0 to 4 again, but limiting the number of results returned to a maximum
;; of 7
(inefficient-permutations 4 :max 7)

=> ((2 3 0 1) (3 1 2 0) (2 0 3 1) (1 0 2 3) (1 2 3 0) (0 2 3 1) (2 1 0 3))

;; The same call will return the same "random" results each time by default
(loop repeat 4 do (print (inefficient-permutations 3 :max 5)))
```

```
=>
((2 0 1) (2 1 0) (0 2 1) (1 0 2) (1 2 0))
((2 0 1) (2 1 0) (0 2 1) (1 0 2) (1 2 0))
((2 0 1) (2 1 0) (0 2 1) (1 0 2) (1 2 0))
((2 0 1) (2 1 0) (0 2 1) (1 0 2) (1 2 0))

;; Setting the :fix argument to NIL will result in different returns
(loop repeat 4 do (print (inefficient-permutations 3 :max 5 :fix nil)))

=>
((1 0 2) (0 1 2) (1 2 0) (2 1 0) (0 2 1))
((1 2 0) (2 0 1) (2 1 0) (1 0 2) (0 1 2))
((0 1 2) (1 0 2) (2 0 1) (1 2 0) (2 1 0))
((0 2 1) (1 2 0) (0 1 2) (2 0 1) (1 0 2))
```

SYNOPSIS:

```
(defun inefficient-permutations (level &key (max nil) (skip 0) (fix t)
                                            (if-not-enough #'error))
```

17.1.3 permutations/inefficiently-permutate

[permutations] [Functions]

DESCRIPTION:

Return a shuffled, non-systematically ordered list of all possible permutations of an original list of elements of any type. An optional keyword argument <max> allows the user to specify the maximum number of permutations to return.

As opposed to the function "permutate", inefficiently-permutate returns the elements of the specified <list> as a flat list, unless the keyword argument <sublists> is set to T, whereupon the function returns the result as a list of lists, each one being a permutation of <list>.

The function is inefficient in so far as it simply shuffles the numbers and so always has to check whether the new list already contains the shuffled sublist before storing it.

The order of the permutations returned will always be the same unless <fix> is set to NIL.

ARGUMENTS:

```
- A list.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :max. An integer that indicates the maximum number of permutations to be
  returned.
- :skip. An integer that indicates a number of permutations to skip.
- :fix. T or NIL to indicate whether the given sequence should always be
  shuffled with the same (fixed) random seed (thus always producing the
  same result). T = fixed seed. Default = T.
- :sublists. T or NIL to indicate whether the returned result should be
  flattened into a one-dimensional list or should be left as a list of
  lists. T = leave as list of lists. Default = NIL.
- :clone. T or NIL to indicate whether objects in the list should be cloned
  as they are permutated (so that they are unique objects rather than
  shared data space).  Useful perhaps if e.g. you're cloning chords which
  will then have their own marks. etc.  If T then the list must contain
  slippery-chicken named-objects or types subclassed from them (as is every
  slippery-chicken class).
- :if-not-enough. A function object (or NIL) to call when :max was
  requested but we can't return that many results.  Default = #'error.
```

RETURN VALUE:

```
A list.
```

EXAMPLE:

```
;; By default the function returns a flattened list of all possible
;; permutations in a shuffled (random) order
(inefficiently-permutate '(a b c))

=> (C A B C B A A C B B A C B C A A B C)

;; The length of the list returned can be potentially shortened using the :max
;; keyword argument. Note here that the value given here refers to the number
;; of permutations before the list is flattened, not to the number of
;; individual items in the flattened list.
(inefficiently-permutate '(a b c) :max 3)

=> (C A B C B A A C B)

;; By default the function is set to using a fixed random seed, causing it to
;; return the same result each time
```

```
(loop repeat 4 do (print (inefficiently-permutate '(a b c))))

=>
(C A B C B A A C B B A C B C A A B C)
(C A B C B A A C B B A C B C A A B C)
(C A B C B A A C B B A C B C A A B C)
(C A B C B A A C B B A C B C A A B C)

;; Setting the :fix keyword argument to NIL allows the function to produce
;; different output each time
(loop repeat 4 do (print (inefficiently-permutate '(a b c) :fix nil)))

=>
(B A C A C B B C A A B C C B A C A B)
(A C B B A C C B A C A B B C A A B C)
(A C B B A C B C A A B C C A B C B A)
(B A C A B C C A B C B A B C A A C B)

;; Setting the :sublists keyword argument to T causes the function to return a
;; list of lists instead
(inefficiently-permutate '(a b c) :sublists t)

=> ((C A B) (C B A) (A C B) (B A C) (B C A) (A B C))
```

SYNOPSIS:

```
(defun inefficiently-permutate (list &key (max nil) (skip 0) (fix t)
                                clone (sublists nil) (if-not-enough #'error))
```

17.1.4 permutations/move-repeats

[permutations] [Functions]

DESCRIPTION:

```
Move, when possible, any elements within a given list that are repeated
consecutively.

When two consecutive elements repeat, such as the c in '(a b c c b a),
the function moves the repeated element to the next place in the given
list that won't produce a repetition. When no such place can be found in
the remainder of the list, the offending element is moved to the end of the
given list and a warning is printed.

This function can be applied to simple lists and lists with sublists.
```

However, due to this function being designed for--but not limited to--use with the results of permutations, if the list has sublists, then instead of repeating sublists being moved, the last element of a sublist is checked for repetition with the first element of the next sublist. See the first example below.

NB: This function only moves elements further along the list; it won't place them earlier than their original position. Thus:

(move-repeats '(3 3 1))

will return (3 1 3), while

(move-repeats '(1 3 3))

will leave the list untouched and print a warning.

ARGUMENTS:

- A list.

OPTIONAL ARGUMENTS:

- A function that serves as the comparison test. Default = #'eq.

RETURN VALUE:

A list.

EXAMPLE:

```
;;; Used with a list of lists.  Note that the repeating C, end of sublist 1,
;;; beginning of sublist 2, is moved, not the whole repeating sublist (c a b).
(move-repeats '((a b c) (c a b) (c a b) (d e f) (a b c) (g h i)))

=> ((A B C) (D E F) (C A B) (C A B) (A B C) (G H I))

;;; Works with simple lists too:
(move-repeats '(1 2 3 3 4 5 6 7 8 8 9 10))

=> (1 2 3 4 3 5 6 7 8 9 8 10)

;; Moves the offending element to the end of the list and prints a warning when
;; no solution can be found
(move-repeats '((a b c d) (d c b a) (b c a d) (c a b d)))
```

```
=> ((A B C D) (B C A D) (C A B D) (D C B A))
WARNING:
   move-repeats: can't find non-repeating place!
   present element: (D C B A), elements left: 1
```

SYNOPSIS:

```
(defun move-repeats (list &key (warn t) (test #'eq))
```

17.1.5 permutations/multi-shuffle

[permutations] [Functions]

DESCRIPTION:

```
Applies the shuffle function a specified number of times to a specified
list.

NB: As with the plain shuffle function, the order of the permutations
    returned will always be the same unless the keyword argument
    :fix is set to NIL.
```

ARGUMENTS:

```
- A sequence.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :start. A zero-based index integer indicating the first element of a
  subsequence to be shuffled. Default = 0.
- :end. A zero-based index integer indicating the last element of a
  subsequence to be shuffled. Default = the length of the given sequence.
- :copy. T or NIL to indicate whether the given sequence should be copied
  before it is modified or should be destructively shuffled.
  T = copy. Default = T.
- :fix. T or NIL to indicate whether the given sequence should always be
  shuffled with the same (fixed) random seed (thus always producing the
  same result). T = fixed seed. Default = T.
- :reset. T or NIL to indicate whether the random state should be reset
  before the function is performed.  T = reset. Default = T.
```

RETURN VALUE:

```
 - A sequence.
```

EXAMPLE:

```
;; Simple multi-shuffle with default keywords.
(multi-shuffle '(a b c d e f g) 3)

=> (B A C E D G F)

;; Always returns the same result by default.
(loop repeat 4 do (print (multi-shuffle '(a b c d e f g) 3)))

=>
(B A C E D G F)
(B A C E D G F)
(B A C E D G F)
(B A C E D G F)

;; Set keyword argument :fix to NIL to return different results each time
(loop repeat 4 do (print (multi-shuffle '(a b c d e f g) 3 :fix nil)))

=>
(G C F B D E A)
(A G F B D C E)
(A B D G C F E)
(G C A D E F B)

;; Set keyword arguments :start and :end to shuffle just a subsequence of the
;; given sequence
(loop repeat 4
   do (print (multi-shuffle '(a b c d e f g) 3
                            :fix nil
                            :start 2
                            :end 5)))

=>
(A B D E C F G)
(A B E C D F G)
(A B E D C F G)
(A B D C E F G)
```

SYNOPSIS:

```
(defun multi-shuffle (seq num-shuffles &key
                      (start 0)
                      (end (length seq))
```

```
                    (copy t)
                    (fix t)
                    (reset t))
```

17.1.6 permutations/multi-shuffle-with-perms

[permutations] [Functions]

DESCRIPTION:

```
Return one permutation from a shuffled list of permutations of the
specified list. The second argument determines how many shuffled
permutations will be in the list from which the resulting permutation is
selected. Similar to the "multi-shuffle" function, but uses the function
"inefficient-permutations" as part of the process.

The <num-shuffles> argument allows the user to always return the same
specific permutation.

NB: This function always uses a fixed random seed and has no optional
    arguments to allow the user to alter that setting.
```

ARGUMENTS:

```
- A list.
- An integer that is the number of consecutive shuffles to be collected in
  the list from which the resulting permutation is selected.
```

RETURN VALUE:

```
- A list that is a single permutation of the specified list.
```

EXAMPLE:

```
;; Returns a permutation of a shuffled version of the specified list
(let ((l '(0 1 2 3 4)))
  (multi-shuffle-with-perms l 7))

=> (3 1 4 2 0)

;; Always returns the same result
(loop repeat 4 do (print (multi-shuffle-with-perms '(0 1 2 3 4) 7)))

=>
(3 1 4 2 0)
```

```
(3 1 4 2 0)
(3 1 4 2 0)
(3 1 4 2 0)

;; Different <num-shuffles> values return different permutations
(loop for i from 0 to 5
   do (print (multi-shuffle-with-perms '(0 1 2 3 4) i)))

=>
(0 1 2 3 4)
(1 4 2 0 3)
(0 3 1 4 2)
(4 0 2 1 3)
(1 2 3 4 0)
(2 1 3 0 4)
```

SYNOPSIS:

```
(defun multi-shuffle-with-perms (seq num-shuffles)
```

17.1.7 permutations/permutate

[permutations] [Functions]

DESCRIPTION:

```
Systematically produce a list of all possible permutations of an original
list of elements of any type.

NB: Such lists can quickly become very long, so slippery-chicken
    automatically defaults to outputting the resulting list to a file and
    printing a warning when the results exceed a certain length.
```

ARGUMENTS:

```
- A list with elements of any type.
```

OPTIONAL ARGUMENTS:

```
- allow-file: whether to allow the results to be written to a file rather
than returned to the interpreter. This will only happen if the length of
the list is > 8. The advantage is that you can then read in the file rather
than slowly regenerate the results if you need them again.
```

RETURN VALUE:

```
A list of lists that are all possible permutations of the original,
specified list.

Interrupts with an error if the method is passed anything but a list.
```

EXAMPLE:

```
;; Simple usage
(permutate '(a b c))

=> ((A B C) (B A C) (A C B) (C A B) (B C A) (C B A))

;; When the list is more than 8 elements long, the resulting permutations are
;; written to a file due to the very high number of results
(permutate '(1 2 3 4 5 6 7 8 9))

=>
WARNING: permutations::permutations: This call will return 362880
results so they are being written to the file
'/tmp/permutations.txt'.
```

SYNOPSIS:

```
(defun permutate (list &optional (allow-file t))
```

17.1.8 permutations/random-rep

[permutations] [Functions]

DESCRIPTION:

```
Return a random non-negative number that is less than the specified
value. An optional argument allows for the random state to be reset.
```

ARGUMENTS:

```
- A number.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether the random state should be reset before the
  function is performed. T = reset. Default = NIL.
```

RETURN VALUE:

A number.

EXAMPLE:

```
;; By default returns a different value each time
(loop repeat 10 do (print (random-rep 5)))

=>
1
3
4
4
3
4
2
0
2
0

;; Setting the optional argument to T resets the random state before
;; performing the function
(loop repeat 10 do (print (random-rep 5 t)))

=>
3
3
3
3
3
3
3
3
3
3
```

SYNOPSIS:

```
(defun random-rep (below &optional (reset nil))
```

17.1.9 permutations/shuffle

[permutations] [Functions]

DESCRIPTION:

Create a random ordering of a given sequence or a subsequence of a given

```
sequence.  By default we used fixed-seed randomness so we can guarantee the
same results each time (perhaps counter-intuitively).  So the order of the
permutations returned will always be the same unless keyword argument :fix
is set to NIL.

NB: This function is a modified form of Common Music's shuffle function.
```

ARGUMENTS:

```
- A sequence (list, vector (string)).
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :start. A zero-based index integer indicating the first element of a
  subsequence to be shuffled. Default = 0.
- :end. A zero-based index integer indicating the last element of a
  subsequence to be shuffled. Default = the length of the given sequence.
- :copy. T or NIL to indicate whether the given sequence should be copied
  before it is modified or should be destructively shuffled.
  T = copy. Default = T.
- :fix. T or NIL to indicate whether the given sequence should always be
  shuffled with the same (fixed) random seed (thus always producing the
  same result). T = fixed seed. Default = T.
- :reset. T or NIL to indicate whether the random state should be reset
  before the function is performed.  T = reset. Default = T.
```

RETURN VALUE:

```
A list.
```

EXAMPLE:

```
;; Simple shuffle with default keywords.
(shuffle '(1 2 3 4 5 6 7))

=> (5 4 3 6 7 1 2)

;; Always returns the same result by default.
(loop repeat 4 do (print (shuffle '(1 2 3 4 5 6 7))))

=>
(5 4 3 6 7 1 2)
(5 4 3 6 7 1 2)
(5 4 3 6 7 1 2)
```

```
(5 4 3 6 7 1 2)

;; Set keyword argument :fix to NIL to return different results each time
(loop repeat 4 do (print (shuffle '(1 2 3 4 5 6 7) :fix nil)))

=>
(1 2 6 3 5 4 7)
(1 3 5 2 7 4 6)
(4 7 2 5 1 6 3)
(1 5 3 7 4 2 6)

;; Set the keyword argument :reset to t only at the beginning so we get the
;; same result that time but different (but repeatable) results thereafter.
(loop repeat 3 do
     (print 'start)
     (loop for i below 4
        do (print (shuffle '(1 2 3 4 5 6 7) :reset (zerop i)))))

=>
START
(5 4 3 6 7 1 2)
(4 6 5 2 3 1 7)
(3 4 1 6 5 7 2)
(3 2 7 4 1 6 5)
START
(5 4 3 6 7 1 2)
(4 6 5 2 3 1 7)
(3 4 1 6 5 7 2)
(3 2 7 4 1 6 5)
START
(5 4 3 6 7 1 2)
(4 6 5 2 3 1 7)
(3 4 1 6 5 7 2)
(3 2 7 4 1 6 5)


;; Set keyword arguments :start and :end to shuffle just a subsequence of the
;; given sequence
(loop repeat 4
   do (print (shuffle '(1 2 3 4 5 6 7)
                      :fix nil
                      :start 2
                      :end 5)))

=>
(1 2 5 4 3 6 7)
```

```
(1 2 3 5 4 6 7)
(1 2 4 5 3 6 7)
(1 2 3 4 5 6 7)
```

SYNOPSIS:

```
(defun shuffle (seq &key
                (start 0)
                (end (length seq))
                (copy t)
                (fix t)
                (reset t)
                &aux (width (- end start)))
```

18 sc/samp5

[Modules]

NAME:

```
samp5

File:             samp5.lsp

Class Hierarchy:  none, no classes defined

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          clm instrument for sample processing; called by
                  slippery-chicken::clm-play

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    12th June 2004

$$ Last modified:  12:29:13 Fri May 16 2025 CEST

SVN ID: $Id$
```

19 sc/sine

[Modules]

NAME:

```
samp5

File:             sine.lsp

Class Hierarchy:  none, no classes defined

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          clm instrument for simple sine wave generation.  This is
                  used mainly as an example to show how user instruments
                  can be used in clm-play but it may also be useful for a
                  quick sinewave rendition of a piece.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    12th June 2004

$$ Last modified:  15:28:03 Wed Nov  6 2024 CET

SVN ID: $Id: sine.lsp 4223 2013-10-29 10:57:09Z medward2 $
```

20 sc/slippery-chicken-edit

[Modules]

NAME:

```
slippery-chicken-edit

File:             slippery-chicken-edit.lsp

Class Hierarchy:  named-object -> slippery-chicken

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Post-generation editing methods for the slippery-chicken
                  class.

Author:           Michael Edwards: m@michael-edwards.org
```

```
Creation date:    April 7th 2012

$$ Last modified:  11:33:15 Tue Feb  3 2026 CET

SVN ID: $Id$
```

20.1 slippery-chicken-edit/add-arrow-to-events

[slippery-chicken-edit] [Methods]

DATE:

```
April 9th 2012
```

DESCRIPTION:

```
Adds an arrow above the specified notes of a slippery-chicken object,
coupled with text to be printed in the score at the start and end of the
arrow. Can be used, for example, for transitions from one playing state to
another.

If no text is desired, this must be indicated by a space in quotes (" ")
rather than empty quotes ("").

See also the add-arrow method in the event class.
```

ARGUMENTS:

```
- A slippery-chicken object.
- A text string for the beginning of the arrow.
- A text string for the end of the arrow.
- A list that is the starting event reference, in the form (bar-number
  event-number). Event numbers count from 1 and include rests and tied
  notes.
- A list that is the end event reference, in the form (bar-number
  event-number).
- The ID of the player to whose part the arrow should be attached.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether or not to print a warning when trying to
  attach an arrow and accompanying marks to a rest.
  T = print warning. Default = NIL.
```

RETURN VALUE:

```
T
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :title "mini"
        :ensemble '(((pno (piano :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c4 d4 f4 g4 a4 c5 d5 f5 g5 a5 c6)))
                       (2 ((cs4 ds4 fs4 gs4 as4 cs5 ds5 fs5 gs5 as5))))
        :set-map '((1 (1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q q))
                                :pitch-seq-palette ((1 (2))))))
        :rthm-seq-map '((1 ((pno (1 1 1 1 1 1))))))))
  (add-arrow-to-events mini "here" "there" '(1 1) '(5 1) 'pno)
  (write-lp-data-for-all mini))
```

SYNOPSIS:

```
(defmethod add-arrow-to-events ((sc slippery-chicken) start-text end-text
                                event1-ref event2-ref player
                                &optional warn-rest)
```

20.2 slippery-chicken-edit/add-auxiliary-notes

[slippery-chicken-edit] [Methods]

DATE:

```
May 26th 2016, Edinburgh
```

DESCRIPTION:

```
Analyses phrase by phrase to find the most frequent X notes then changes
some of them (according to an activity-levels object) to auxiliary notes
(up a semitone by default). Chords are ignored.
```

ARGUMENTS:

```
- the slippery-chicken object
```

OPTIONAL ARGUMENTS:

```
keyword arguments
- :players. A single player (symbol) or list of players to process. If NIL
   then all players will be processed.
- :start-bar. An integer indicating the bar in which the processes should
   start. Default = NIL which means we'll start at the beginning.
- :end-bar. An integer indicating the bar in which the processes should
   stop. Default = NIL which means we'll process through to the end
- :verbose. T or NIL to print information about the pitch changes during
   the process. Default = NIL.
- :ignore. A symbol or pitch-object (or list thereof) to represent pitches
   which should not be processed. Default = NIL = process (potentially) all
   pitches.
- :activity-level. Whether to change the note is decided by an
  activity-levels object. This argument determines the 'level' argument used
  in that class's active method. Default = 5 = 50% of the notes will be
  changed. NB The activity-levels object will be reinitialised for each
  phrase so the 50/50 spread might not be obvious.
- :num-notes. The number of notes to process per phrase. We calculate the
  most used pitches in every phrase then use this number of pitches to add
  auxiliaries to. Default = 3.
- :intervals. The intervals that the original pitches will be transposed by
  to create the auxiliary notes. This can be a single value in semitones or
  a list of values. The list does not have to have the same length as
  :num-notes as it will be cycled through when necessary. Default = 1.
```

RETURN VALUE:

```
T
```

SYNOPSIS:

```
(defmethod add-auxiliary-notes ((sc slippery-chicken)
                                &key players start-bar end-bar
                                  verbose ignore (activity-level 5)
                                  (num-notes 3) (intervals 1))
```

20.3 slippery-chicken-edit/add-clef

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Attach a specified clef symbol to a specified clef object within a given
slippery-chicken object.
```

ARGUMENTS:

```
- A slippery-chicken object.
- The ID of the player to whose part the clef symbol is to be added.

NB: The optional arguments are actually required.
```

OPTIONAL ARGUMENTS:

```
- An integer that is the bar number in which the clef symbol is to be
  placed.
- An integer that is the event number within the given bar to which the
  clef symbol is to be attached.
- A symbol that is the clef type to be attached. See the documentation for
  the make-instrument function of the instrument class for a list of
  possible clef types.
```

RETURN VALUE:

```
Returns the new value of the MARKS-BEFORE slot of the given event object.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :title "mini"
        :ensemble '(((vn (violin :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c4 d4 f4 g4 a4 c5 d5 f5 g5 a5 c6)))
                       (2 ((cs4 ds4 fs4 gs4 as4 cs5 ds5 fs5 gs5 as5))))
        :set-map '((1 (1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                                :pitch-seq-palette ((1 2 3 4)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1 1)))))))
  (add-clef mini 'vn 3 2 'alto))

  => ((CLEF ALTO))
```

SYNOPSIS:

```
(defmethod add-clef ((sc slippery-chicken) player &optional
                     bar-num event-num clef)
```

20.4 slippery-chicken-edit/add-event-to-bar

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Add an event object to a specified bar either at the end of that bar or at
a specified position within that bar.

It's the user's responsibility to make sure the bar is full and
update-slots is called after all post-generation editing such as this.

NB Since January 2026 a list of events can be passed rather than just a
single object.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An event object or list thereof
- An integer that is the bar number or a list that is the reference to the
  bar in the form '(section sequence bar), where sequence and bar are
  numbers counting from 1)
- The ID of the player to whose part the event should be added.
```

OPTIONAL ARGUMENTS:

```
keyword argument:
- :position. NIL or an integer indicating the position in the bar (0-based)
   where the event should be added. If NIL, the new event is placed at the
   end of the bar. Default = NIL.
```

RETURN VALUE:

```
T
```

EXAMPLE:

```
;;; Adding two events to separate bars, once using a bar number with
;;; :position's default to NIL, and once using a bar number reference list with
;;; :position specified as 2. Print the bars after adding to see the changes.

(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
```

```
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c4 d4 f4 g4 a4 c5 d5 f5 g5 a5 c6)))
                       (2 ((cs4 ds4 fs4 gs4 as4 cs5 ds5 fs5 gs5 as5 cs6))))
        :set-map '((1 (1 1 1 1 1 1))
                   (2 (2 2 2 2 2 2)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                                :pitch-seq-palette ((1 2 3 4))))
                           (2 ((((2 4) e s s q))
                                :pitch-seq-palette ((1 2 3 4)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1 1))))
                        (2 ((vn (2 2 2 2 2 2)))))))
  (add-event-to-bar mini (make-event 'cs4 'e) 2 'vn)
  (print-simple (first (get-bar mini 2)))
  (add-event-to-bar mini (make-event 'c4 'q) '(2 2 1) 'vn :position 2)
  (print-simple (first (get-bar mini '(2 2 1)))))

=>
(2 4): C4 Q, D4 E, F4 S, G4 S, CS4 E
(2 4): CS4 E, DS4 S, C4 Q, FS4 S, GS4 Q
```

SYNOPSIS:

```
(defmethod add-event-to-bar ((sc slippery-chicken) event bar-num-or-ref player
                             &key (position nil))
```

20.5 slippery-chicken-edit/add-half-beat-rest

[slippery-chicken-edit] [Methods]

DATE:

```
August 26th 2016, Edinburgh
```

DESCRIPTION:

```
Calls the method of the same name on each rthm-seq-bar object at the given
bar number. Adds a half beat rest at the end of the bar. This is a
destructive method. It will change the time signature, so a 2/4 bar
becomes a 5/8 bar, with an 1/8th rest at the end. This is done for all the
players in the ensemble (so that scores can be correctly written)
```

ARGUMENTS:

```
- a slippery-chicken object
```

OPTIONAL ARGUMENTS:

```
- the bar number to modify. Default = 1.
```

RETURN VALUE:

```
T
```

SYNOPSIS:

```
(defmethod add-half-beat-rest ((sc slippery-chicken) &optional (bar-num 1))
```

20.6 slippery-chicken-edit/add-mark-all-players

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Add a specified mark to a specified event in the parts of all players. The
event can either be specified as a 1-based integer, in which case the mark
will be attached to the same event in all parts, or as a list of integers,
in which the mark is attached to different events in the same bar for each
player, passing from the top of the ensemble downwards.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the bar number or a list of integers that is a
  reference to the bar number in the form (section sequence bar).
- An integer that is the event to which to attach the specified mark in all
  parts, or a list of integers that are the individual events to which to
  attach the mark in the consecutive players.
- The mark to be added.
```

RETURN VALUE:

```
Always returns T.
```

EXAMPLE:

```
;;; Apply the method twice: Once using an integer to attach the mark to the
;;; same event in all players, and once using a list to attach the mark to
;;; different events in the consecutive players. Print the corresponding marks
;;; slots to see the results.
```

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (hn (french-horn :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((cl (1 1 1 1 1))
                            (hn (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (2 ((cl (1 1 1 1 1))
                            (hn (1 1 1 1 1))
                            (vc (1 1 1 1 1)))))))))
  (add-mark-all-players mini 3 1 'ppp)
  (add-mark-all-players mini '(2 2 1) '(1 2 3) 'fff)
  (loop for i in '(cl hn vc)
     do (print (marks (get-event mini 3 1 i))))
  (loop for i in '(cl hn vc)
     for e in '(1 2 3)
     do (print (marks (get-event mini '(2 2 1) e i)))))

=>
(PPP)
(PPP)
(PPP)
(FFF)
(FFF)
(FFF)
```

SYNOPSIS:

```
(defmethod add-mark-all-players ((sc slippery-chicken)
                                 bar-num event-num mark)
```

20.7 slippery-chicken-edit/add-mark-before-note

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
 Add the specified mark to the MARKS-BEFORE slot of the specified note
```

```
 object within the given slippery-chicken object.

 NB: This method counts notes, not events; i.e., rests are not counted.
```

ARGUMENTS:

```
 - A slippery-chicken object.
 - An integer that is the bar number in which the mark is to be added.
 - An integer that is the NOTE number to which the mark is to be added (not
   the event number; i.e., rests are not counted).
 - The ID of the player to which the mark is to be added.
 - The mark to be added.
```

RETURN VALUE:

```
 Returns the new value of the MARKS-BEFORE slot of the given event object.
```

EXAMPLE:

```
;;; The method adds the mark to the specified note, not event. Add the mark to
;;; note 2, print the MARKS-BEFORE slots of events 2 (which is a rest) and 3.
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c4 d4 f4 g4 a4 c5 d5 f5 g5 a5 c6))))
        :set-map '((1 (1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q (e) s s))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1 1)))))))
  (add-mark-before-note mini 3 2 'vn 'ppp)
  (print (marks-before (get-event mini 3 2 'vn)))
  (print (marks-before (get-event mini 3 3 'vn))))

=>
NIL
(PPP)
```

SYNOPSIS:

```
(defmethod add-mark-before-note ((sc slippery-chicken)
                                 bar-num note-num player mark)
```

20.8 slippery-chicken-edit/add-mark-to-event

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Add the specified mark to the MARKS slot of the specified event within the
given slippery-chicken object.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the bar number to which the mark is to be added.
- An integer that is the event number in the specified bar to which the
  mark is to be added.
- The ID of the player to which to add the mark.
- The mark to add.
```

RETURN VALUE:

```
Returns T.
```

EXAMPLE:

```
;;; Add a mark to an event object then read the value of the MARKS slot of that
;;; event to see the result
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c4 d4 f4 g4 a4 c5 d5 f5 g5 a5 c6))))
        :set-map '((1 (1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q (e) s s))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1 1))))))))
  (add-mark-to-event mini 3 2 'vn 'ppp)
  (marks (get-event mini 3 2 'vn)))

=> (PPP)
```

SYNOPSIS:

```
(defmethod add-mark-to-event ((sc slippery-chicken) bar-num event-num player
                              mark)
```

20.9 slippery-chicken-edit/add-mark-to-note

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Add the specified mark to the specified note of a given slippery-chicken
object.

NB: This method counts notes, not events; i.e., not rests.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the bar number to which to add the mark
- An integer that is the note number two which to add the mark. This is
  1-based, and counts notes not events; i.e., not rests. If a list, then it
  is of the form (note-num chord-note), where chord-note is 1-based and
  counts from the lowest note up.
- The ID of the player to whose part the mark is to be added.
- The mark to add.
```

RETURN VALUE:

```
Returns T.
```

EXAMPLE:

```
;;; Add a mark to a note in a bar with a rest. Print the corresponding event
;;; object to see the result.
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c4 d4 f4 g4 a4 c5 d5 f5 g5 a5 c6))))
        :set-map '((1 (1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q (e) s s))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1 1))))))))
  (add-mark-to-note mini 3 2 'vn 'ppp)
  (print (marks (get-event mini 3 2 'vn)))
  (print (marks (get-event mini 3 3 'vn))))

=>
NIL
(PPP)
```

SYNOPSIS:

```
(defmethod add-mark-to-note ((sc slippery-chicken)
                             bar-num note-num player mark)
```

20.10 slippery-chicken-edit/add-marks-sh

[slippery-chicken-edit] [Methods]

DATE:

27-Jun-2011

DESCRIPTION:

Add marks in a somewhat more free list form, with the option of implementing a user-defined shorthand.

ARGUMENTS:

- A slippery-chicken object.
- A list of lists containing the players, bar and note refs, and marks to be added. The first element of each contained list will be the ID of the player to whose part the marks are to be added followed by a pattern of <mark bar-number note-number> triplets, or if a mark is to be added repeatedly then <mark bar note bar note... >. A mark can be a string or a symbol.

OPTIONAL ARGUMENTS:

keyword arguments:
- For marks given as symbols, the user can supply a shorthand table that will expand an abbreviation, such as sp, to the full mark name, such as short-pause. This table takes the form of a simple Lisp association list, e.g.: '((al aeolian-light)
  (ad aeolian-dark)
  (wt "WT")
  (h harm))
- :warn. T or NIL to indicate whether to print a warning for unrecognized marks. T = print warning. Default = T.
- :verbose. T or NIL to indicate whether the method is to print verbose feedback about each mark added to the Listener. T = print feedback. Default = NIL.

RETURN VALUE:

Returns NIL.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (va (viola :midi-channel 2))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c4 d4 f4 g4 a4 c5 d5 f5))))
        :set-map '((1 (1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1 1))
                            (va (1 1 1 1 1 1)))))))
  (add-marks-sh mini
                '((vn a 1 1 1 2 3 1 s 2 1 2 2 2 5)
                  (va pizz 1 3 2 3 sp 3 1))
                :shorthand '((sp short-pause))
                :verbose t))

=> NIL
```

SYNOPSIS:

```
(defmethod add-marks-sh ((sc slippery-chicken) player-data
                         &key shorthand (warn t) verbose)
```

20.11 slippery-chicken-edit/add-marks-to-note

[slippery-chicken-edit] [Methods]

DESCRIPTION:

Add one or more specified marks to a specified note within a given slippery-chicken object.

NB: This method counts notes, not events; i.e., rests are not counted.

ARGUMENTS:

- A slippery-chicken object.
- An integer that is the bar number to which the mark or marks should to be added.

```
 - An integer that is the note within the specified bar to which the mark or
   marks should be added.
 - The ID of the player to whose part the mark or marks should be added.
 - The mark or marks to add.
```

RETURN VALUE:

```
 Returns T.
```

EXAMPLE:

```
;;; Add several marks to one note, then print the corresponding MARKS slot to
;;; see the difference.
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (va (viola :midi-channel 2))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c4 d4 f4 g4 a4 c5 d5 f5))))
        :set-map '((1 (1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e (e) e e (e) e e e))
                                :pitch-seq-palette ((1 2 3 4 5 6)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1 1))
                            (va (1 1 1 1 1 1))))))))
  (add-marks-to-note mini 2 3 'va 'a 's 'lhp 'pizz)
  (print (marks (get-note mini 2 3 'va))))

=> (PIZZ LHP S A)
```

SYNOPSIS:

```
(defmethod add-marks-to-note ((sc slippery-chicken) bar-num note-num
                              player &rest marks)
```

20.12 slippery-chicken-edit/add-marks-to-notes

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
 Add the specified mark or marks to a consecutive sequence of multiple notes
 within the given slippery-chicken object.
```

ARGUMENTS:

- A slippery-chicken object.
- An integer or a list consisting of two numbers to indicate the start bar/note. If this is an integer, all notes in this bar will receive the specified mark or marks. If this is a two-number list, the first number determines the bar, the second the note within that bar.
- An integer or a list consisting of two numbers to indicate the end bar/note. If this is an integer, all notes in this bar will receive the specified mark or marks. If this is a two-number list, the first number determines the bar, the second the note within that bar.
- The ID of the player or players to whose parts the mark or marks should be attached. This can be a single symbol or a list.
- T or NIL to indicate whether the mark should be added to the MARKS slot or the MARKS-BEFORE slot of the given events objects.
- The mark or marks to be added.

RETURN VALUE:

Returns T.

EXAMPLE:

```
;;; This example calls the method twice: Once using the single-integer
;;; indication for full bars, with one instrument and one mark; and once using
;;; the bar/note reference lists for more specific placement, a list of several
;;; players that should all receive the marks, and multiple marks to add.
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (va (viola :midi-channel 2))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c4 d4 f4 g4 a4 c5 d5 f5))))
        :set-map '((1 (1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1 1))
                            (va (1 1 1 1 1 1)))))))
  (add-marks-to-notes mini 2 3 'vn nil 'lhp)
  (add-marks-to-notes mini '(1 3) '(2 2) '(vn va) nil 's 'a))

=> T
```

SYNOPSIS:

```
(defmethod add-marks-to-notes ((sc slippery-chicken) start end players before
                               &rest marks)
```

20.13 slippery-chicken-edit/add-ornament

[slippery-chicken-edit] [Methods]

DATE:

15th January 2026

DESCRIPTION:

Add ornaments (trill, mordent, grace note(s), glissandi) to notes in the piece.

ARGUMENTS:

- the slippery-chicken object
- the bar number in which we want to add the mark. If nil, and an event is passed as the next argument, then this will be taken from the event.
- the bar's note number (i.e. attacked note number not event), counting from 1. For convenience, e.g. if processing lists of events, this can be an event object itself rather than a number.
- the player (symbol) whose part should have the ornament. If an event is passed, then it will be taken from the event.
- the type of ornament: a symbol such as trill, mordent, grace, gliss, trill

OPTIONAL ARGUMENTS:

- data for the ornament. Currently only relevant to grace notes. In this case this could either be a single symbol/pitch or list of the pitches (symbols, objects) to add or an integer. In the latter case, the specified number of notes will be taken from the pitches of the current and previous bar and grace notes nearest to the main note will be automatically created.
- the maximum interval in semitones for a mordent. If no nearer notes are available near the note for the ornament, then it will be skipped.
- the maximum distance from the main note that a grace note can be

RETURN VALUE:

T or NIL for whether the ornament was added or not

SYNOPSIS:

```
(defmethod add-ornament ((sc slippery-chicken) bar-num note player
                         type &optional (data 1) (mordent-max 5) (warn t)
                         (grace-max 13))
```

20.14 slippery-chicken-edit/add-pitches-to-chord

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Add specified pitches to an existing chord object.
```

ARGUMENTS:

```
- The slippery-chicken object which contains the given chord object.
- The ID of the player whose part is to be affected.
- An integer that is the number of the bar that contains the chord object
  that is to be modified.
- An integer that is the number of the note that is the chord object to be
  modified.
- The pitches to be added. These can be pitch objects or any data that can
  be passed to make-pitch, or indeed lists of these, as they will be
  flattened.
```

RETURN VALUE:

```
The chord object that has been changed.
```

EXAMPLE:

```
(let* ((ip-clone (clone +slippery-chicken-standard-instrument-palette+)))
  (set-slot 'chord-function 'chord-fun1 'guitar ip-clone)
  (let* ((mini
          (make-slippery-chicken
           '+mini+
           :instrument-palette ip-clone
           :ensemble '(((gtr (guitar :midi-channel 1))))
           :set-palette '((1 ((e3 f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5 d5 e5 f5
                               g5 a5 b5 c6 d6 e6))))
           :set-map '((1 (1)))
           :rthm-seq-palette
           '((1 ((((4 4) e e e e e e e e))
                 :pitch-seq-palette ((1 (2) 3 (4) 5 (6) 7 (8))))))
           :rthm-seq-map '((1 ((gtr (1))))))))
    (print (get-pitch-symbols
            (pitch-or-chord (get-event mini 1 2 'gtr))))
    (add-pitches-to-chord mini 'gtr 1 2 'cs4 'ds4)
    (print (get-pitch-symbols
            (pitch-or-chord (get-event mini 1 2 'gtr))))))
```

```
=>
(E3 G3 B3)
(E3 G3 B3 CS4 DS4)
```

SYNOPSIS:

```
(defmethod add-pitches-to-chord ((sc slippery-chicken) player bar-num note-num
                                 &rest pitches)
```

20.15 slippery-chicken-edit/add-player

[slippery-chicken-edit] [Methods]

DATE:

```
October 30th 2018, Heidhausen
```

DESCRIPTION:

```
Add a player to the ensemble and piece slots/objects of the given
slippery-chicken object. This clones the section/sub-section structure of
the slippery-chicken object, creating all the necessary data structures for
empty bars in the right meter, tempo etc.

NB As this method may be called several times successively, it's the
caller's duty to call (update-slots sc) in order to have timing and other
data updated correctly.
```

ARGUMENTS:

```
- the slippery-chicken object
- a player object or symbol ID for the new player
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :instrument. a symbol ID for an existing instrument in the
  instrument-palette (the next argument). This is actually required unless
  the default of 'computer is acceptable or a player object is passed as
  second argument.
- :instrument-palette. an instrument-palette object in which the instrument
  exists. Default is the standard palette.
- :midi-channel. the midi-channel for the new player
- microtones-midi-channel. the microtones-midi-channel for the new player.
  Default = -1 = same as :midi-channel.
```

RETURN VALUE:

```
the new player object from the ensemble slot of the slippery-chicken object
```

SYNOPSIS:

```
(defmethod add-player ((sc slippery-chicken) player
                       &key (instrument 'computer)
                         (instrument-palette
                          +slippery-chicken-standard-instrument-palette+)
                         ;; MDE Tue Jul 14 19:08:15 2020, Heidhausen
                         (midi-channel 1)
                         (microtones-midi-channel -1))
```

20.16 slippery-chicken-edit/add-tuplet-bracket-to-bar

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Add a tuplet bracket (with number) to a specified bar in a slippery-chicken
object. This method adds only one tuplet bracket of one tuplet type
(triplet, quintuplet etc.) at a time.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the number of the bar to which the tuplet bracket is
  to be added.
- The ID of the player to whose part the tuplet bracket is to be added.
- The bracket info defining the tuplet bracket to be added. This takes the
  form of a three-element list specifying tuplet value, number of the event
  (zero-based) on which the bracket is to begin, and number of the event on
  which the bracket is to end, e.g. '(3 0 2).
```

OPTIONAL ARGUMENTS:

```
T or NIL to indicate whether all existing tuplet brackets in the given bar
are to be deleted first. T = delete. Default = NIL>
```

RETURN VALUE:

```
T
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+sc-object+
        :ensemble '(((va (viola :midi-channel 2))))
        :set-palette '((1 ((c3 d3 e3 f3 g3 a3 b3 c4 d4))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((3 4) (te) - te te - { 3 te ts+ts te }
                                  - fs fs fs fs fs -))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8 9 8)))))
        :rthm-seq-map '((1 ((va (1 1 1))))))))
  (add-tuplet-bracket-to-bar mini 1 'va '(3 0 2))
  (add-tuplet-bracket-to-bar mini 2 'va '(5 7 11))
  (add-tuplet-bracket-to-bar mini 3 'va '(3 3 4) t)
  (add-tuplet-bracket-to-bar mini 3 'va '(3 5 6)))

=> T
```

SYNOPSIS:

```
(defmethod add-tuplet-bracket-to-bar ((sc slippery-chicken) bar-num player
                                      bracket-info
                                      &optional (delete-all-tuplets-first nil))
```

20.17 slippery-chicken-edit/add-tuplet-brackets-to-beats

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Add the specified tuplet brackets (and numbers) to the specified event
objects in the specified bars within the given slippery-chicken object.
```

ARGUMENTS:

```
- A slippery-chicken object.
- The ID of the player to whose part the tuplet brackets are to be added.
- A list of 4-element sublists that is the bracket info. Each sublist must
  consist of: the number of the bar to which the bracket is to be added;
  the number that is the tuplet type (3 = triplet, 5 = quintuplet etc.);
  the zero-based number of the event where the bracket is to begin; the
  zero-based number that is the number of the event where the bracket is to
  end; e.g. '((2 3 0 5) (3 3 0 3) (5 5 0 4))
```

OPTIONAL ARGUMENTS:

```
T or NIL to indicate whether all existing tuplet bracket info in the given
bars is to first be deleted. T = delete. Default = NIL.
```

RETURN VALUE:

```
NIL.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+sc-object+
        :ensemble '(((va (viola :midi-channel 2))))
        :set-palette '((1 ((c3 e3 g3 c4))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((3 4) - te te te - - fs fs fs fs fs -
                                 - 28 28 28 28 28 28 28 -))
                               :pitch-seq-palette ((1 2 3 4 1 2 3 4 1 2 3 4 1
                                                    2 3)))))
        :rthm-seq-map '((1 ((va (1 1 1))))))))
  (add-tuplet-brackets-to-beats mini 'va '((1 3 0 2) (2 5 3 7) (3 7 8 14))))

=> NIL
```

SYNOPSIS:

```
(defmethod add-tuplet-brackets-to-beats
    ((sc slippery-chicken) player bracket-info
     &optional (delete-all-tuplets-first nil))
```

20.18 slippery-chicken-edit/auto-accidentals

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Automatically determine which notes in each bar need accidentals and which
don't.

This method also places cautionary accidentals (in parentheses) based on
how many notes back the last occurrence of that note/accidental combination
appeared in the bar. The first optional argument to the method allows the
user to specify how many notes back to look.

NB: As both cmn-display and write-lp-data-for-all call respell-notes by
```

```
default, that option must be set to NIL for this method to be
effective (see below).
```

ARGUMENTS:

```
- A slippery-chicken object.
```

OPTIONAL ARGUMENTS:

```
- An integer that is the number of notes back to look when placing
  cautionary accidentals in parentheses. If the last occurrence of a given
  repeated note/accidental combination was farther back than this number,
  the accidental will be placed in the score in parentheses.
```

RETURN VALUE:

```
Returns NIL.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :set-palette '((1 ((fs4 gs4 as4))))
        :set-map '((1 (1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 2 1 2 3 2)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1)))))))
  (auto-accidentals mini 4)
  (cmn-display mini :respell-notes nil))

=> NIL
```

SYNOPSIS:

```
(defmethod auto-accidentals ((sc slippery-chicken) &optional
                             (cautionary-distance 3)
                             ignore1 ignore2)
```

20.19 slippery-chicken-edit/auto-add-ornaments

[slippery-chicken-edit] [Methods]

DATE:

17th January 2026

DESCRIPTION:

Automatically add ornaments to a piece using a cycle-repeats object and the data passed to :cycle-repeats. Ornaments are only added to attacked notes with a minimum duration passed as :min-dur. See the add-ornament method (and expand where desired) to see the meaning of mordent, grace, etc.

ARGUMENTS:

- the slippery-chicken object
- the player (or list of players) to add ornaments to (symbol). If nil then all players will be processed.

OPTIONAL ARGUMENTS:

keyword arguments:
- :ale-env. An activity-levels-envlope to determine how often a note will get an ornament. Default will be a constant level of 4 which means c. 40% of notes (deterministically, not randomly).
- :ale-cycle-length: if nil then it will be set to the number of (attacked) notes of sufficient duration which the player has in the whole piece or in the given bars
- :cycle-repeats (object). Determines which ornaments will be added and their number of repetitions.
- :start-bar and :end-bar default to 1 and the last bar of the piece unless specific bar numbers are passed.
- :min-dur. The minimum duration of an event (compound-duration i.e. including ties)

RETURN VALUE:

the number of ornaments added (integer)

SYNOPSIS:

```
(defmethod auto-add-ornaments ((sc slippery-chicken) players
                               &key start-bar end-bar
                               (ale-env '(0 4 100 4)) ale-cycle-length
                               (min-dur 0.250) ; seconds
                               (update t)
                               (warn t)
                               (mordent-max 5) ; semitones
                               ;; NB grace is a sublist as it needs an
```

```
;; argument (how many notes or a list of
;; specific pitch symbols)
(cycle-repeats '((mordent 2) ((grace 3) 1)
                 (mordent 3) ((grace 4) 1)
                 (mordent 1) (gliss 5)
                 ((grace 3) 2))))
```

20.20 slippery-chicken-edit/auto-beam

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Automatically places indications for beam start- and end-points (1 and 0)
in the BEAMS slot of the corresponding event objects.

By default, this method determines the start and end indications for beams
on the basis of the beat found in the time signature, but the user can
specify a different beat basis using the first optional argument.
```

ARGUMENTS:

```
- A slippery-chicken object.
```

OPTIONAL ARGUMENTS:

```
- NIL, an integer that is a power-of-two rhythmic duration, or the
  alphabetic representation of such a rhythm to specify the beat basis for
  setting beams (e.g. 4 or 'h).
- T or NIL to indicate whether the method is to check whether an exact beat
  of rhythms can be found for each beat of the bar. If T, a warning will be
  printed when an exact beat cannot be found for each beat of the bar.
  Default = T.
- players, a single symbol or list of players to process. Default = NIL =
  process all players in the ensemble
```

RETURN VALUE:

```
Returns NIL.
```

EXAMPLE:

```
;; Auto-beam the events of the given slippery-chicken object on the basis of a ;
;; half note:                          ;
(let ((mini
```

```
      (make-slippery-chicken
       '+mini+
       :ensemble '(((vn (violin :midi-channel 1))))
       :tempo-map '((1 (q 60)))
       :set-palette '((1 ((fs4 gs4 as4))))
       :set-map '((1 (1 1 1 1)))
       :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                               :pitch-seq-palette ((1 2 3 2 1 2 3 2)))))
       :rthm-seq-map '((1 ((vn (1 1 1 1)))))))
 (auto-beam mini 'h))

 => NIL
```

SYNOPSIS:

```
(defmethod auto-beam ((sc slippery-chicken) &optional (beat nil) (check-dur t)
                                                      players)
```

20.21 slippery-chicken-edit/auto-clefs

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Automatically create clef changes in the specified player's or players'
part or parts by adding the appropriate clef symbol to the MARKS-BEFORE
slot of the corresponding event object.

This routine will only place clef symbols that are present in the given
instrument object's CLEFS slot.

This method is called automatically by cmn-display and
write-lp-data-for-all, with the delete-clefs option set to T.

NB: While this routine generally does a good job of putting the proper
    clefs in place, it will get confused if the pitches in a given player's
    part jump from very high to very low (e.g. over the complete range of
    the piano).
```

ARGUMENTS:

```
- A slippery-chicken object.
```

OPTIONAL ARGUMENTS:

```
 keyword arguments:
 - :verbose. T or NIL to indicate whether the method is to print feedback
   about its operations to the Listener. T = print feedback. Default = NIL.
 - :in-c. T or NIL to indicate whether the pitches processed are to be
   handled as sounding or written pitches. T = sounding. Default = T.
 - :players. A list containing the IDs of the players whose parts are to be
   to have clefs automatically added.
 - :delete-clefs. T or NIL to indicate whether the method should first
   delete all clef symbols from the MARKS-BEFORE slots of all event objects
   it is processing before setting the automatic clef changes.
 - :delete-marks-before. T or NIL to indicate whether the MARKS-BEFORE slot
   of all event objects processed should first be set to NIL.
   T = set to NIL. Default = NIL.
```

RETURN VALUE:

```
 Returns T
```

EXAMPLE:

```
;;; Straightforward usage applied to just the VC player ;
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c2 e2 d4 e4 f4 g4 a4 f5))))
        :set-map '((1 (1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1))
                            (vc (1 1 1 1))))))))
  (auto-clefs mini :players '(vc)))

  => T
```

SYNOPSIS:

```
(defmethod auto-clefs ((sc slippery-chicken)
                       &key verbose in-c players
                         (delete-clefs t)
                         (delete-marks-before nil))
```

20.22 slippery-chicken-edit/auto-slur

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Automatically add slurs to note groups in the specified measure range of a
given player's part.

This method places slurs above all consecutive notes between rests. If a
value is specified for :end-bar and the last event in the end bar is not a
rest, the final sequence of attacked notes in that bar will not be
slurred.

NB: Slurs will automatically stop at repeated pitches.  Staccato marks will
not stop the auto-slurring process but staccatos can be removed (see
below).
```

ARGUMENTS:

```
- A slippery-chicken object.
- A player ID or list of player IDs for the parts in which the slurs are to
  be placed.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :start-bar. An integer that is the first bar in which to automatically
  place slurs.
- :end-bar. An integer that is the last bar in which to automatically place
  slurs.
- :rm-slurs-first. T or NIL to indicate whether to first remove existing
  slurs from the specified region. NB: If you already have slur marks
  attached to events, setting this to NIL can produce unwanted results
  caused by orphaned beg-slur or end-slur marks.  T = remove existing slurs
  first. Default = T.
- :rm-staccatos. T or NIL to indicate whether to first remove existing
  staccato, tenuto, and accented staccato marks from the specified
  region. T = remove staccatos. Default = NIL.
- :over-accents. T or NIL. Default = T.
- :verbose. T or NIL to indicate whether to print feedback from the process
  to the Lisp listener. T = print. Default = NIL.
```

RETURN VALUE:

```
A list of sublists, each of which contains the start and end event, plus
the number of notes under the slur, for each slur added.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c3 d3 e3 f3 g4 a3 b3
                            c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) - e e - (s) e.
                                  - s s e - - s (s) s s -))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8 9))
                                :marks (a 4))))
        :rthm-seq-map '((1 ((vn (1 1 1))))))))
  (auto-slur mini 'vn
             :start-bar 1
             :end-bar 2))
```

SYNOPSIS:

```
(defmethod auto-slur ((sc slippery-chicken) players
                      &key start-bar end-bar
                      (rm-slurs-first t)
                      (rm-staccatos t)
                      ;; 5.4.11
                      (over-accents t)
                      verbose)
```

20.23 slippery-chicken-edit/bars-to-sc

[slippery-chicken-edit] [Functions]

DESCRIPTION:

```
Take a list of rthm-seq-bars and add them to a new or existing
slippery-chicken object. It works in three different ways:
   1. Create a new slippery chicken object with one player and one section.
   2. Add new parts to an already existing section in an already exisiting
      slippery chicken object.
   3. Add a new section to an already exisiting part in and already
      exisiting object.

In the second case, we assume you are creating a slippery chicken object
part by part with this function. It is not currently possible to
```

add a part like this in the middle of the score--the new part will be added
to the end of the last group of the ensemble (bottom of score) so make sure
to add parts in the order you want them.

In the third case, you can add a new section to an exisitng part, but you
cannot add a new section to a part that does not exist.

NB Bear in mind that if you want to use midi-play, then the events in the
   bars will need to have their midi-channel set (e.g. via make-event).
   It's also the caller's responsibility to ensure that the parts added have
   the same time-signature structure as any existing part.

ARGUMENTS:

- A list of rthm-seq-bars

OPTIONAL ARGUMENTS:

keyword arguments:
- :sc. Either an existing slippery-chicken object or nil if one should be
  created automatically. If nil, the following three arguments must be
  specified, otherwise they will be ignored. Default = NIL.
- :sc-name. The name (symbol) for the slippery-chicken object to be
  created. This will become a global variable. Default = '*auto*.
- :player. The name (symbol) of the player to create. Default =
  'player-one. (Remember that Lilypond has problems with player names
   with numbers in them :/)
- :instrument. The id (symbol) of an already existing instrument in the
   instrument-palette. Default = 'flute.
- :update. Whether to call update-slots on the new slippery-chicken
  object (to update timing info. etc.). Default = T.
- :section-id. The section id. Default = 1.
- :tempo. The tempo in BPM for the whole slippery-chicken object (piece).
- :midi-channels. A 2-element list specifying the midi-channels for the
  chromatic and microtonal notes of the player. This will not change
  channels of the events in the bars but is assumed to correspond to those.

RETURN VALUE:

A slippery-chicken object.

SYNOPSIS:

```
(defun bars-to-sc (bars &key sc (sc-name '*auto*) (player 'player-one)
                           (midi-channels '(1 2))
```

```
                        (instrument-palette
                         +slippery-chicken-standard-instrument-palette+)
                        (tempo 60) (instrument 'flute) (section-id 1)
                        (update t))
```

20.24 slippery-chicken-edit/change-pitch

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Change the pitch of a specified event to a new specified pitch. The new
pitch is not required to be a member of the current set.

NB The new pitch is the sounding pitch if a transposing instrument, unless
optional argument is set to T.

NB In the case of ties, this doesn't update the following 'tied-to' notes
but see the documentation for the change-pitches method below (or call
check-ties, which fixes them too).
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the bar number in which the pitch is to be changed.
- An integer that is the number of the note in the specified bar whose
  pitch is to be changed (1-based).
- The ID of the player for whom the pitch is to be changed.
- A note-name symbol that is the new pitch.
```

OPTIONAL ARGUMENTS:

```
T or NIL to indicate whether the written or sounding pitch should be
changed.  Default = NIL (sounding).
```

RETURN VALUE:

```
Returns T.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vc (cello :midi-channel 1))))
```

```
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((d3 e3 f3 g3 a3 b3 c4 e4))))
        :set-map '((1 (1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8)))))
        :rthm-seq-map '((1 ((vc (1 1 1 1))))))))
  (change-pitch mini 1 3 'vc 'fs3))

=> T
```

SYNOPSIS:

```
(defmethod change-pitch ((sc slippery-chicken) bar-num note-num player
                         new-pitch &optional written)
```

20.25 slippery-chicken-edit/change-pitches

[slippery-chicken-edit] [Methods]

DESCRIPTION:

Change the pitches of the specified event objects for a given player to the specified new pitches.

If the new pitches are passed as a simple flat list, the method will just change the pitch of each consecutive attacked event object (with NIL indicating no change), moving from bar to bar as necessary, until all of the specified new pitches are used up. Also, if a flat list is passed, each new pitch specified will be applied to each consecutive attacked note; i.e., ties don't count as new pitches.

Tied-to events are left with the old pitch information, which is of course a potential problem. When generating scores though, we usually call respell-notes, which calls check-ties, which corrects spellings of tied-to notes and therefore in effect changes those notes too. So generally, we don't have to worry about this, but if you explicitly tell slippery chicken not to respell notes, you'll need to call check-ties with the first optional argument as T.

Also see the documentation in the bar-holder class for the method of the same name.

NB As various methods (e.g. transposition, change-pitch(es)) in various classes (slippery-chicken, bar-holder, rthm-seq-bar, rthm-seq) may change pitch information and midi-channels stored in events, chords, pitches,

etc. it is recommended to call the slippery-chicken reset-midi-channels method before calling midi-play, if any of those methods have been used. This way midi channels for pitches belonging to particular players will use channels originally allocated in the ensemble for chromatic and microtonal pitches, no matter where they originated or how they were changed.

ARGUMENTS:

- A slippery-chicken object.
- The ID of the player whose part is to be modified.
- An integer that is the number of the first bar whose pitches are to be modified.
- A list note-name symbols and NILs, or a list of lists of note-name symbols and NILs, which are the new pitches. If a simple flat list, see the comment in the description above. If a list of lists, each sub-list will represent a full bar; e.g., (change-pitches bh 'vla 5 '((g3 gs4) nil (nil nil aqf5))) will change the pitches in bars 5 and 7 (for the player 'vla), whereas bar six, indicated by nil, wouldn't be changed; similarly the first two notes of bar 7, being nil, will also not be changed, but note 3 will.

OPTIONAL ARGUMENTS:

keyword arguments:
- :use-last-octave. T or NIL to indicate whether or not each consecutive new pitch listed will automatically take the most recent octave number specified; e.g. '((a3 b g cs4)). T = use last octave number. Default = T.
- :marks. A list of marks to be added to the events objects. This option can only be used in conjunction with the simple flat list of pitches. In this case the list of pitches and list of marks must be the same length and correspond to each other item by item. Sub-lists can be used to add several marks to a single event. NB: See cmn.lsp::get-cmn-marks for the list of recognised marks. If NIL, no marks will be added. Default = NIL.
- :written. T or NIL to indicate whether these are the written or sounding notes for a transposing instrument. Default = NIL.
- :warn. If there are more pitches in the given list than there are events in the slippery-chicken structure, issue a warning, unless NIL. Default = T.

RETURN VALUE:

If the new pitches are passed as a simple flat list, the method returns the number of the bar in which the pitches were changed;
otherwise returns T.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vc (cello :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((d3 e3 f3 g3 a3 b3 c4 e4))))
        :set-map '((1 (1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8)))))
        :rthm-seq-map '((1 ((vc (1 1 1 1 1 1))))))))
  (change-pitches mini 'vc 2 '((fs3 gs3 as3)))
  (change-pitches mini 'vc 3 '((nil nil fs3 gs as ds fs gs)
                               nil
                               (cs4 ds fs))))

=> T
```

SYNOPSIS:

```
(defmethod change-pitches ((sc slippery-chicken) player start-bar new-pitches
                           &key (use-last-octave t) marks written
                             (warn t))
```

20.26 slippery-chicken-edit/change-time-sig

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Force a change of the time-sig associated with a specified bar.

NB: This method does not check to see if the rhythms in the bar add up to a
    complete bar in the new time-sig.

Also see rthm-seq-bar (setf time-sig).
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the number of the bar whose time signature should be
  changed or a list that is a reference to the bar whose time signature is
  to be changed in the format (section sequence bar).
- The new signature in the format (number-of-beats beat-unit).
```

RETURN VALUE:

```
Returns T.
```

EXAMPLE:

```
;;; Changing two time signatures; once using the integer bar reference, the
;;; second time using the list reference to the bar number.
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vc (cello :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((d3 e3 f3 g3 a3 b3 c4 e4))))
        :set-map '((1 (1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8)))))
        :rthm-seq-map '((1 ((vc (1 1 1 1))))))))
  (change-time-sig mini 2 '(3 8))
  (change-time-sig mini '(1 1 1) '(5 8)))

=> T
```

SYNOPSIS:

```
(defmethod change-time-sig ((sc slippery-chicken) bar-num-or-ref new-time-sig)
```

20.27 slippery-chicken-edit/consolidate-all-notes

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
A convenience method which just calls the consolidate-notes method from the
rthm-seq-bar class for all the bars specified in the arguments.
```

ARGUMENTS:

```
- the slippery-chicken object
- the first bar number in which consolidation should take place
- the last bar number in which consolidation should take place
  (inclusive). If NIL then the process will run until the last bar.
- A list of the IDs of the players to whose parts the consolidation should
  be applied. Can also be a single symbol.
```

RETURN VALUE:

```
A list of the rthm-seq-bar objects that were modified. See map-over-bars
for more details.
```

SYNOPSIS:

```
(defmethod consolidate-all-notes ((sc slippery-chicken)
                                  &optional (start-bar 1) end-bar players
                                  ;; MDE Fri Sep 21 09:08:09 2018 -- added
                                  ;; consolidate-notes' optional args
                                    check-dur beat (auto-beam t))
```

20.28 slippery-chicken-edit/consolidate-all-rests

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
A convenience method which just calls the consolidate-rests method from the
rthm-seq-bar class for all the bars specified in the arguments.
```

ARGUMENTS:

```
- the slippery-chicken object
- the first bar number in which consolidation should take place
- the last bar number in which consolidation should take place
  (inclusive).  If NIL then the process will run until the last bar.
- A list of the IDs of the players to whose parts the consolidation should
  be applied. Can also be a single symbol.
```

OPTIONAL ARGUMENTS:

```
T or NIL to indicate whether the method should print a warning to the Lisp
listener if it is mathematically unable to consolidate the rests. T = print
warning. Default = NIL.
```

RETURN VALUE:

```
- A list of the rthm-seq-bar objects that were modified.  See map-over-bars
for more details.
```

SYNOPSIS:

```
(defmethod consolidate-all-rests ((sc slippery-chicken)
                                  &optional (start-bar 1) end-bar players warn
                                    beat)
```

20.29 slippery-chicken-edit/delete-bars

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Delete a sequence of consecutive bars from the given slippery-chicken
object.

NB This might delete rehearsal letters, instrument changes (and maybe other
things) attached to a bar/event.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the first bar to delete.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :num-bars. An integer that is the number of consecutive bars, including
  the start-bar, to delete. This argument cannot be used simultaneously
  with :end-bar
- :end-bar. An integer that is the number of the last of the consecutive
  bars to delete. This argument cannot be used simultaneously with
  :num-bars.
- :print. Print feedback of the process to the Listener, including a
  print-simple of the bars deleted.
```

RETURN VALUE:

```
Returns T.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vc (cello :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((d3 e3 f3 g3 a3 b3 c4 e4))))
        :set-map '((1 (1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8)))))
        :rthm-seq-map '((1 ((vc (1 1 1 1))))))))
```

```
(delete-bars mini 2 :end-bar 3)
(delete-bars mini 2 :num-bars 1))

=> T
```

SYNOPSIS:

```
(defmethod delete-bars ((sc slippery-chicken) start-bar
                        &key num-bars end-bar print)
```

20.30 slippery-chicken-edit/delete-clefs

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Delete the clef symbol held in the MARKS-BEFORE slot of the specified event
object within the given slippery-chicken object.
```

ARGUMENTS:

```
NB: The optional arguments are actually required.
- A slippery-chicken object.
- The ID of the player from whose part the clef symbol is to be deleted.
- An integer that is the number of the bar from which the clef symbol is to
  be deleted.
- An integer that is the number of the event object within the specified
  from whose MARKS-BEFORE slot the clef symbol is to be deleted. This is a
  1-based index but counts rests and ties.
```

RETURN VALUE:

```
Returns NIL.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vc (cello :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c2 e2 d4 e4 f4 g4 a4 f5))))
        :set-map '((1 (1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8)))))
```

```
      :rthm-seq-map '((1 ((vc (1 1 1 1))))))))
  (auto-clefs mini)
  (delete-clefs mini 'vc 1 3))

  => NIL
```

SYNOPSIS:

```
(defmethod delete-clefs ((sc slippery-chicken) &optional
                         player bar-num event-num)
```

20.31 slippery-chicken-edit/delete-events

[slippery-chicken-edit] [Methods]

DATE:

```
21-Jul-2011 (Pula)
```

DESCRIPTION:

```
Turn notes into rests by setting the IS-REST slots of the specified
consecutive event objects within the given slippery-chicken object to T.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the number of the first bar for which the notes are to
  be changed to rests.
- An integer that is the index of the first event object within the
  specified start bar for which the IS-REST slot is to be changed to
  T. This number is 1-based and counts rests and ties.
- An integer that is the number of the last bar for which the notes are to
  be changed to rests.
- An integer that is the index of the last event object within the
  specified end bar for which the IS-REST slot is to be changed to T. This
  number is 1-based and counts rests and ties. If NIL, apply the change to
  all events in the given bar.
```

OPTIONAL ARGUMENTS:

```
- A list of the IDs of the players whose parts are to be modified. If NIL,
  apply the method to the parts of all players.
- T or NIL to indicate whether to consolidate resulting consecutive rests
  into one longer rest each. T = consolidate. Default = T.
```

RETURN VALUE:

```
Returns T.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vc (cello :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c2 e2 d4 e4 f4 g4 a4 f5))))
        :set-map '((1 (1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8)))))
        :rthm-seq-map '((1 ((vc (1 1 1 1))))))))
  (delete-events mini 2 2 3 nil 'vc))

  => T
```

SYNOPSIS:

```
(defmethod delete-events ((sc slippery-chicken) start-bar start-event end-bar
                          end-event &optional players (consolidate-rests t))
```

20.32 slippery-chicken-edit/delete-rehearsal-letter

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Delete the rehearsal letter from a specified bar of on or more specified
players' parts by setting the REHEARSAL-LETTER slot of the corresponding
rthm-seq-bar object to NIL.

NB: This deletes the given rehearsal letter without resetting and
    re-ordering the remaining rehearsal letters.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the number of the bar from which the rehearsal letter
  is to be deleted. NB: The rehearsal letter for a given bar is internally
  actually attached to the previous bar. The number given here is the
  number from the user's perspective, but the change will be reflected in
  the bar with the number specified -1.
```

OPTIONAL ARGUMENTS:

```
- A list consisting of the IDs of the players from whose parts the
  rehearsal letter is to be deleted.
```

RETURN VALUE:

```
Returns the deleted rehearsal letter from the last player which had one
attached.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vc (cello :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c2 e2 d4 e4 f4 g4 a4 f5))))
        :set-map '((1 (1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8)))))
        :rthm-seq-map '((1 ((vc (1 1 1 1 1 1)))))
        :rehearsal-letters '(2 4 6))))
  (delete-rehearsal-letter mini 2 '(vc)))

  => NIL
```

SYNOPSIS:

```
(defmethod delete-rehearsal-letter ((sc slippery-chicken) bar-num
                                    &optional players)
```

20.33 slippery-chicken-edit/delete-slur

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Delete a slur mark that starts on a specified note within a specified bar
of a specified player's part by deleting the BEG-SL and END-SL marks from
the corresponding event objects.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the number of the bar from which the slur is to be
  deleted.
- An integer that is the number of the note on which the slur to be deleted
  starts within the given bar. This number counts tied-notes but not
  rests.
- The ID of the player from whose part the slur is to be deleted.
```

RETURN VALUE:

```
Returns NIL.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vc (cello :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c2 e2 d4 e4 f4 g4 a4 f5))))
        :set-map '((1 (1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8))
                                :marks (slur 1 8))))
        :rthm-seq-map '((1 ((vc (1 1 1 1 1 1))))))))
  (delete-slur mini 1 1 'vc)
  (delete-slur mini 3 1 'vc))

  => NIL
```

SYNOPSIS:

```
(defmethod delete-slur ((sc slippery-chicken) bar-num note-num player)
```

20.34 slippery-chicken-edit/delete-tempi

[slippery-chicken-edit] [Methods]

DATE:

```
June 26th 2022, Karlsruhe
```

DESCRIPTION:

```
Delete all tempo-change slot tempo objects from all events in the given
```

players. NB Call update-slots after using this and similar methods for the last time, so that event timing reflects tempo changes.

ARGUMENTS:

-

OPTIONAL ARGUMENTS:

- the start bar. Default = NIL = 1
- the end bar. Default = NIL = last bar of piece
- the player(s). Either a list of or single symbol(s). Default = NIL = all players

RETURN VALUE:

- the newly modified slippery-chicken object

SYNOPSIS:

```
(defmethod delete-tempi ((sc slippery-chicken) &optional start-bar end-bar
                                                         players)
```

20.35 slippery-chicken-edit/double-events

[slippery-chicken-edit] [Methods]

DATE:

20-Jul-2011 (Pula)

DESCRIPTION:

Copy the specified events from one player to the corresponding bars of one or more other players.

NB: Although partial bars can be copied from the source player, the entire bars of the target players are always overwritten, resulting in rests in those segments of the target players' bars that do not contain the copied material. This method thus best lends itself to copying into target players parts that have rests in the corresponding bars.

ARGUMENTS:

- A slippery-chicken object.
- The ID of the player from whose part the events are to be copied.
- The ID or a list of IDs of the player or players into whose parts the copied events are to be placed.
- An integer that is the number of the first bar from which the events are to be copied.
- An integer that is the number of the first event to be copied from the specified start bar. This number is 1-based and counts rests and ties.
- An integer that is the number of the last bar from which the events are to be copied.
- NIL or an integer that is the number of the last event to be copied from the specified end bar. This number is 1-based and counts rests and ties. If NIL, all events from the given bar will be copied.

OPTIONAL ARGUMENTS:

keyword arguments:
- :transposition. A positive or negative number that is the number of semitones by which the copied material is to be first transposed. This number can be a decimal number, in which case the resulting pitches will be rounded to the nearest microtone (if the current tuning environment is capable of microtones).
- :consolidate-rests. T or NIL to indicate whether resulting consecutive rests should be consolidated each into one longer rest. T = consolidate. Default = T.
- :update. T or NIL to indicate whether to update the slots of the given slippery-chicken object after copying. T = update. Default = T.
- :auto-beam. T or NIL to indicate whether to call auto-beam on the doubled notes. Default = T.
- :pitches. T or NIL to indicate whether pitch information should be copied over. This would usually be T but some methods might just want to duplicate events as rests (e.g. orchestrate). Default = T.

RETURN VALUE:

Returns T

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((bsn (bassoon :midi-channel 1))
                     (tbn (tenor-trombone :midi-channel 2))
                     (vlc (cello :midi-channel 3))))
```

```
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c2 e2 d4 e4 f4 g4 a4 f5))))
        :set-map '((1 (1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) (w)))))
                            (2 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8)))))
        :rthm-seq-map '((1 ((bsn (1 1 1 1 1 1))
                            (tbn (1 1 1 1 1 1))
                            (vlc (2 2 2 2 2 2)))))))
  (double-events mini 'vlc '(bsn tbn) 2 3 4 2)
  (double-events mini 'vlc 'bsn 5 1 5 nil :transposition 3.5))

  => T
```

SYNOPSIS:

```
(defmethod double-events ((sc slippery-chicken) master-player doubling-players
                          start-bar start-event end-bar end-event
                          &key transposition (consolidate-rests t) (update t)
                            (pitches t) (auto-beam t))
```

20.36 slippery-chicken-edit/double-events-scaled

[slippery-chicken-edit] [Methods]

DATE:

```
15.5.25
```

DESCRIPTION:

```
Double the events of one player and then rhythmically scale (augment or
diminute) them. Add them to a new part, where time signature and rhythmic
properties allow. Note that
```

ARGUMENTS:

```
- the slippery-chicken object
- the scaler to apply to the rhythms, e.g. 2 would be half speed
- the player whose events will be copied before scaling
- the symbol associated with the new player in the ensemble
- the instrument symbol (from the :instrument-palette) for the new player
```

OPTIONAL ARGUMENTS:

keyword arguments:
- :transposition. The semitone transposition for the master-player's pitches. Either this or the :pitches argument may be given, not both.
- :pitches. A list of pitch/chord objects or symbols/symbols lists (chords) that will be used to replace the pitches/chords of the events of the master player. If this is shorter than the number of events that we generate, then they will be reused circularly. If this is not passed, then the pitches of the master-player will be used, optionally transposed (see above).
- :auto-beam. Whether to call auto-beam on the slippery-chicken object after the process is finished.
- :update-slots. Whether to call update-slots on the slippery-chicken object after the process is finished. This is left to the user as it is computationally expensive and thus might be put off to a later stage.
- :verbose. Print useful information as the process takes place.
- :midi-channel. the midi-channel for the new player
- microtones-midi-channel. the microtones-midi-channel for the new player. Default = -1 = same as :midi-channel.
- :instrument. a symbol ID for an existing instrument in the instrument-palette (the next argument). This is actually required unless the default of 'computer is acceptable or a player object is passed as second argument.
- :instrument-palette. an instrument-palette object in which the instrument exists. Default is the standard palette.
- :start-bar (:start-event, :end-bar, :end-event). The start/end points for that determine the range that we first of all copy, before scaling and begin pasting from :new-start-bar. Defaults are the whole piece.
- :new-start-bar. The bar at which we start pasting in the scaled events (it's not possible to specify a starting event) . Default = 1.
- :new-end-bar. The bar at which we end (inclusive) pasting in the scaled events (it's not possible to specify a stopping event) . Default = NIL = end of the piece.
- :bar-ok-fun. A function to call in every bar to decide whether to double events to this bar (e.g. you might want to skip over bars in which other players aren't playing). Requires and will be passed two arguments: the rthm-seq-bar and slippery-chicken objects. Default returns T = every bar.

RETURN VALUE:

the sc object with the new player added

SYNOPSIS:

```
(defmethod double-events-scaled
    ((sc slippery-chicken) rthm-scaler master-player
     new-player new-instrument
```

```
&key transposition (new-start-bar 1) new-end-bar (start-bar 1)
start-event end-bar end-event pitches (auto-beam t) (midi-channel 1)
(microtones-midi-channel -1) (update-slots t) verbose
(bar-ok-fun #'(lambda (bar sc) (declare (ignore bar sc)) t))
(instrument-palette +slippery-chicken-standard-instrument-palette+))
```

20.37 slippery-chicken-edit/double-player-inverted

[slippery-chicken-edit] [Methods]

DATE:

```
July 14th 2020, Heidhausen
```

DESCRIPTION:

```
Add a new player to the ensemble/piece, double the events of an existing
player, then use the invert method for each bars within the given range to
turn rests into notes and vice-versa. To choose new notes we use curves to
limit high and low, using the sets (from the map) used to generate the
existing voices.
```

ARGUMENTS:

```
- the slippery-chicken object
- the ID of a new player (symbol)
- the instrument for the new player (must be in the instrument-palette)
- the ID of the existing player whose events are inverted (symbol)
- a curve for the maximum upper notes
- a curve for the maximum lower notes. Both these curves can have any
  arbitrary x range but the y values should use MIDI note numbers or note
  symbols. Note that the x range will be stretched over the number of bars
  in the piece, not the given bar range, so that the bar number can be used
  for interpolation.
- a curve to decide whether to actually play a note in the new part or
  not. This can also have any arbitrary x range and will also be stretched
  over the number of bars in the piece, but the y range should be from 0 to
  10 as the interpolated value for the notes in each bar is passed to an
  activity-levels object to determine whether to play or not.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
:start-bar :start-event :end-bar :end-event. These are passed to
double-events. See that method for details but note that the defaults for
```

```
the end-* arguments are NIL which in turn default to the last bar/event.
:reset-midi-channels. T or NIL to reset all events in the slippery-chicken
object to the original midi channels of the parent player. Default = T.
```

RETURN VALUE:

```
the (modified) slippery-chicken object
```

SYNOPSIS:

```
(defmethod double-player-inverted ((sc slippery-chicken) new-player new-ins
                                   existing-player upper-curve lower-curve
                                   &key (start-bar 1) (start-event 1)
                                     (activity-curve '(0 10 100 10))
                                     end-bar end-event
                                     (reset-midi-channels t) (update-slots t)
                                     (midi-channel 1)
                                     (microtones-midi-channel -1))
```

20.38 slippery-chicken-edit/enharmonic-spellings

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Change the pitch of specified event objects to their enharmonic
equivalents.

This takes as its second argument a list of lists, each of which consists
of the ID of the player whose part is to be altered and a series of
bar-number/event-number pairs, where (2 3) indicates that the pitch of the
third event of the second bar is to be changed to its enharmonic
equivalent.

Pitches within chords are specified by following the bar number with a
2-item list consisting of the event number and the number of the pitch
within the chord, counting from low to high, where (2 (2 4)) indicates that
the fourth pitch from the bottom of the chord located in the second event
object of bar 2 should be changed to its enharmonic equivalent.

An optional T can be included to indicate that the written pitch is to be
changed, but not the sounding pitch, as in (cl (3 4 t)).

NB: In order for this method to work, the :respell-notes option of
    cmn-display and write-lp-data-for-all must be set to NIL.
```

ARGUMENTS:

```
- A slippery-chicken object.
- The list of changes to be made, in the format '((player changes...)),
  e.g.:

  '((cl (3 3 t) (3 4 t))
    (pn (2 (2 4)))
    (vc (1 1) (1 3) (1 4) (1 6)))
```

RETURN VALUE:

```
Returns T.
```

EXAMPLE: SYNOPSIS:

```
(defmethod enharmonic-spellings ((sc slippery-chicken) corrections)
```

20.39 slippery-chicken-edit/enharmonics

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Change the sharp/flat pitches of a specified region of a specified player's
part to their enharmonic equivalent.

NB: This method only affects pitches that already have sharp/flat
    accidentals. It does not affect "white-key" notes (e.g. C-natural =
    B-sharp etc.)

NB: As the cmn-display and write-lp-data-for-all methods call
   :respell-notes by default, this option must be explicitly set to NIL for
   this method to be effective.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer or a 2-item list of integers that indicates the first bar in
  which the enharmonics are to be changed. If an integer, the method will
  be applied to all sharp/flat pitches in the bar of that number. If a
  2-item list of integers, these represent '(bar-number note-number). The
  note number is 1-based and counts ties. If nil, it defaults to 1.
- An integer or a 2-item list of integers that indicates the last bar in
  which the enharmonics are to be changed. If an integer, the method will
```

```
   be applied to all sharp/flat pitches in the bar of that number. If a
   2-item list of integers, these represent '(bar-number note-number). The
   note number is 1-based and counts ties. If nil it defaults to (num-bars sc)
 - The ID of the player or players whose part is to be changed. If nil, the
   method will be applied to all players.
```

OPTIONAL ARGUMENTS:

```
 keyword arguments
 - :written. T or NIL to indicate whether to change written-only pitches or
   sounding-only pitches. T = written-only. Default = T.
 - :pitches. NIL or a list of note-name symbols. If NIL, all sharp/flat
   pitches in the specified region will be changed to their enharmonic
   equivalents. If a list of one or more note-name symbols, only those
   pitches will be affected.
 - :force-naturals. T or NIL to indicate whether to force "natural" note
   names that contain no F or S in their name to convert to their enharmonic
   equivalent (ie, B3 = CF4).  NB double-flats/sharps are not implemented so
   this will only work on F/E  B/C.
```

RETURN VALUE:

```
 Returns T.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (pn (piano :midi-channel 2))
                     (vn (violin :midi-channel 3))))
        :set-palette '((1 ((cs4 ds4 e4 fs4 gs4 as4 b4 cs5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) - e e e e - - e e e e -))
                                :pitch-seq-palette ((1 (2) 3 4 (5) 6 (7) 8)))))
        :rthm-seq-map '((1 ((cl (1 1 1 1 1))
                            (pn (1 1 1 1 1))
                            (vn (1 1 1 1 1))))))))
  (enharmonics mini 1 2 'vn)
  (enharmonics mini 2 3 'pn :pitches '(cs4 ds4))
  (enharmonics mini 3 4 'cl :written nil)
  (enharmonics mini nil nil nil))

=> T
```

SYNOPSIS:

```
(defmethod enharmonics ((sc slippery-chicken) start end players
                        &key (written t) pitches force-naturals)
```

20.40 slippery-chicken-edit/fast-microtone-to-chromatic

[slippery-chicken-edit] [Methods]

DATE:

4 September 2019, London

DESCRIPTION:

Round microtonal pitches to their nearest chromatic pitch if the time between successive events is less than or equal to a specified threshold in seconds. This method is intended to make playing fast pasages in microtonal music easier for human players.

ARGUMENTS:

- The slippery chiocken object containing the pitches to be tested.
- The player or players to be tested

OPTIONAL ARGUMENTS:

- start-bar: the first bar to start testing in. Default = 1
- end-bar: the last bar to test in. Default = (num-bars sc)
- threshold: the time in seconds between successive events that determines whether or not a pitch should be rounded to its nearest chromatic. Default = (fast-leap-threshold sc).

RETURN VALUE:

A list containing the number of rounded pitches per instrument.

EXAMPLE:

```
(let ((mini (make-slippery-chicken
                 '+mini+
                 :title "Your Title Here"
                 :composer "Your Name Here"
                 :ensemble '(((flt (flute :midi-channel 1 :microtones-midi-channel 2))))
```

```
                :staff-groupings '(1)
                :tempo-map '((1 (q 60)))
                :set-palette '((set1 ((fqs2 b2 dqs4 aqf4 dqs5 e5 a5 d6))))
                :set-map '((1 (set1 set1 set1 set1 set1 set1)))
                :rthm-seq-palette
                '((seq1 ((((4 4) q - s s s s -  - e e - q))
                         :pitch-seq-palette (1 2 3 4 5 6 7 8))))
                :rthm-seq-map
                '((1 ((flt (seq1 seq1 seq1 seq1 seq1 seq1))))))))
      (fast-microtone-to-chromatic mini nil :threshold 10)
      (cmn-display mini))
```

SYNOPSIS:

```
(defmethod fast-microtone-to-chromatic ((sc slippery-chicken) players
                                        &key
                                          (start-bar 1)
                                          end-bar
                                          ;; threshold in secs
                                          threshold)
```

20.41 slippery-chicken-edit/force-artificial-harmonics

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
For string scoring purposes only: Transpose the pitch of the given event
object down two octaves and add the harmonic symbol at the perfect fourth.

If this results in a fingered pitch (or even a touched perfect fourth) that
is out of the range of the instrument, a warning will be printed to the
Listener, the pitch will not be transposed, and the harmonic diamond will
not be added.

NB This method assumes the player doesn't change instrument during the bars
to be processed.
```

ARGUMENTS:

```
- A slippery-chicken object.
- The ID of the player whose part is to be changed.
- An integer that is the number of the first bar in which artificial
  harmonics are to be created.
- An integer that is the number of the first event in that bar that is to
```

```
   be changed into an artificial harmonic.
 - An integer that is the number of the last bar in which artificial
   harmonics are to be created.
```

OPTIONAL ARGUMENTS:

```
 - An integer that is the number of the first event in that bar that is to
   be changed into an artificial harmonic. If no end-event is specified, all
   event objects in the last bar will be changed to artificial harmonics.
 - T or NIL to trigger warnings when forcing would take a fingered pitch out
   of the instrument's range, or when we encounter chords or rests
 - T or NIL to force natural-harmonics to become artificial. Default = T.
```

RETURN VALUE:

```
 Returns T.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c4 f4 b4 e5 a5 d6 g7 c8))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8)))))
        :rthm-seq-map '((1 ((vn (1 1 1))))))))
  (force-artificial-harmonics mini 'vn 2 3 3 2))

=> T
```

SYNOPSIS:

```
(defmethod force-artificial-harmonics ((sc slippery-chicken) player start-bar
                                       start-event end-bar
                                       &optional end-event (warn t)
                                         (naturals t))
```

20.42 slippery-chicken-edit/force-harmonics

[slippery-chicken-edit] [Methods]

DATE: DESCRIPTION: ARGUMENTS: OPTIONAL ARGUMENTS: RETURN VALUE: EXAMPLE: SYNOPSIS:

```
(defmethod force-harmonics ((sc slippery-chicken) player start-bar
                            &key (start-event 1) end-bar end-event warn
                              (tolerance 15))
```

20.43 slippery-chicken-edit/force-in-range

[slippery-chicken-edit] [Methods]

DATE:

```
October 19th 2018, Heidhausen
```

DESCRIPTION:

```
Force all pitches (and chords) to be within the range of the instruments who
play them. Of course this is usually a part of the slippery-chicken pitch
selection algorithm but sometimes we change pitches (algorithmically in
particular) and this might bring them out of range. Notes that are out of
range will be transposed up and down the required number of octaves. This
applies to single pitches in chords also. Instrument changes will be
respected on a bar-by-bar basis. See also the three force-in-range methods
in instrument.lsp
```

ARGUMENTS:

```
- the slippery-chicken object
- a symbol or list of symbols for the players to be processed. Default = NIL
  = all players.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :start-bar. The bar to start at. Default = NIL = 1
- :end-bar. The bar to end at. Default = NIL = last bar in piece/object
- any other keyword arguments to the force-in-range instrument method
```

RETURN VALUE:

```
the modified slippery-chicken object
```

SYNOPSIS:

```
(defmethod force-in-range ((sc slippery-chicken) players
                           &rest keyargs &key &allow-other-keys)
```

20.44 slippery-chicken-edit/force-natural-harmonics

[slippery-chicken-edit] [Methods]

DATE: DESCRIPTION: ARGUMENTS: OPTIONAL ARGUMENTS: RETURN VALUE: EXAMPLE: SYNOPSIS:

```
(defmethod force-natural-harmonics ((sc slippery-chicken) player start-bar
                                    &optional start-event end-bar end-event
                                      (warn t) (tolerance 15))
```

20.45 slippery-chicken-edit/force-rest-bars

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Delete all notes from the specified bars and replace them with full-bar
rests.

NB: The start-bar and end-bar index numbers are inclusive
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the number of the first bar to change to a full bar of
  rest.
- An integer that is the number of the last bar to change to a full bar of
  rest. If NIL then we process all bars.
- A list containing the IDs of the players in whose parts the full-bar
  rests are to be forced. If NIL then all players will be processed.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate that only bars containing exclusively rests should be
  processed.
```

RETURN VALUE:

```
Returns the number of bars processed (integer).
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
```

```
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (va (viola :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c4 e4 g4 b4 d5 f5 a5 c6))))
        :set-map '((1 (1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1 1))
                            (va (1 1 1 1 1 1))
                            (vc (1 1 1 1 1 1)))))))
  (force-rest-bars mini 3 5 '(vn vc)))

=> NIL
```

SYNOPSIS:

```
(defmethod force-rest-bars ((sc slippery-chicken) start-bar end-bar players
                            &optional only-if-all-rests)
```

20.46 slippery-chicken-edit/make-hammer-friendly

[slippery-chicken-edit] [Methods]

DATE:

October 4th 2019

DESCRIPTION:

Yamaha Disklavier pianos can't play a note immediately after a note-off. The hammer needs a little time to return to its 'off' position before it can be used to restrike a note. Tests show this time to be around 60 milliseconds. of course this will be instrument-dependent, so a little experimentation might be needed.

This method will go through each event and decrement those events which repeat a note on the same MIDI channel which start < 'min-time' after the end-time of the previous event.

Do not call update-slots after this method as you will get 'bar not full' errors (due to hard-changed durations)

ARGUMENTS:

```
- the slippery-chicken object
- either a single symbol or a list of symbols for the players to be
  processed. If nil then all players will be processed.
```

OPTIONAL ARGUMENTS:

```
keyword arguments
- :start-bar. The bar at which to begin processing (integer). Default = 1.
- :end-bar. The bar at which to end processing (integer). Default = NIL =
  last bar
- :min-gap. The minimum time the hammer needs to return, in
  milliseconds. Default = 60.
- :min-dur. The minimum duration of a note in milliseconds, after reducing
  duration. Default = 50 milliseconds.
```

RETURN VALUE:

```
A list of the number of notes affected per player.
```

SYNOPSIS:

```
(defmethod make-hammer-friendly ((sc slippery-chicken) players
                                 &key
                                 (warn t)
                                 (start-bar 1)
                                 end-bar
                                 (min-gap 60)
                                 (min-dur 50))
```

20.47 slippery-chicken-edit/map-over-bars

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Apply the specified method/function to the bars (all rthm-seq-bar objects)
of one or more players' parts in the given slippery-chicken object.
```

ARGUMENTS:

```
- A slippery-chicken object.
- A number that is the first bar to which the function should be
  applied. Default = NIL in which case 1 will be used.
- A number that is the last bar to which the function should be
  applied. Default = NIL in which case all bars will be processed.
```

- A list of the IDs of the players to whose parts the function should be applied. Can also be a single symbol. If NIL then all players will be processed.
- The method or function itself. This can be a user-defined function or the name of an existing method or function. It should take at least one argument, a rthm-seq-bar, and any other arguments as supplied.

OPTIONAL ARGUMENTS:

- Any additional argument values the specified method/function may take or require. See the thin method below for an example that uses additional arguments.

RETURN VALUE:

- A list of the rthm-seq-bar objects that were modified. NB This might be a long list, and, depending on your Lisp implementation, formatting of the bars might cause Lisp to appear to 'hang'.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((sax ((alto-sax tenor-sax) :midi-channel 1))))
        :instrument-change-map '((1 ((sax ((1 alto-sax) (3 tenor-sax)))))
                                 (2 ((sax ((2 alto-sax) (5 tenor-sax)))))
                                 (3 ((sax ((3 alto-sax) (4 tenor-sax))))))
        :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h e (s) (s) e+s+s))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((sax (1 1 1 1 1))))
                        (2 ((sax (1 1 1 1 1))))
                        (3 ((sax (1 1 1 1 1))))))))
  (print (map-over-bars mini 1 nil nil #'consolidate-notes nil 'q)))

=>
(
RTHM-SEQ-BAR: time-sig: 2 (4 4), time-sig-given: T, bar-num: 1,
[...]
RTHM-SEQ-BAR: time-sig: 2 (4 4), time-sig-given: T, bar-num: 2,
[...]
```

```
RTHM-SEQ-BAR: time-sig: 2 (4 4), time-sig-given: T, bar-num: 3,
[...]
RTHM-SEQ-BAR: time-sig: 2 (4 4), time-sig-given: T, bar-num: 4,
[...]
)
```

SYNOPSIS:

```
(defmethod map-over-bars ((sc slippery-chicken) start-bar end-bar players
                          function &rest further-args)
```

20.48 slippery-chicken-edit/map-over-events

[slippery-chicken-edit] [Methods]

DATE:

1 September 2019, London

DESCRIPTION:

A simple method that mimics the behaviour of map-over-bars but on events instead. At its heart is a next-event loop. Use this method to apply an arbitrary function to every event within a specific range of bars.

ARGUMENTS:

- A slippery chicken object
- A number that is the first bar to which the function should be applied. Default = NIL in which case 1 will be used.
- A number that is the last bar to which the function should be applied. Default = NIL in which case all bars will be processed.
- A list of the IDs of the players to whose parts the function should be applied. Can also be a single symbol. If NIL then all players will be processed.
- The method or function itself. This can be a user-defined function or the name of an existing method or function. It should take at least one argument, an event, and any other arguments as supplied.

OPTIONAL ARGUMENTS:

- Any additional argument values the specified method/function may take or require. See the thin method below for an example that uses additional arguments.

RETURN VALUE:

```
- A list containing the number of events changed per instrument
```

EXAMPLE:

```
(let* ((mini
          (make-slippery-chicken
           '+mini+
           :ensemble '(((sax ((alto-sax tenor-sax) :midi-channel 1))))
           :instrument-change-map '((1 ((sax ((1 alto-sax) (3 tenor-sax)))))
                                    (2 ((sax ((2 alto-sax) (5 tenor-sax)))))
                                    (3 ((sax ((3 alto-sax) (4 tenor-sax))))))
           :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
           :set-map '((1 (1 1 1 1 1))
                      (2 (1 1 1 1 1))
                      (3 (1 1 1 1 1)))
           :rthm-seq-palette '((1 ((((4 4) h e (s) (s) e+s+s))
                                   :pitch-seq-palette ((1 2 3)))))
           :rthm-seq-map '((1 ((sax (1 1 1 1 1))))
                          (2 ((sax (1 1 1 1 1))))
                          (3 ((sax (1 1 1 1 1)))))))
        (print (is-rest (get-event mini (num-bars mini) 2 'sax)))
        (print (map-over-events mini 1 nil nil #'force-rest))
        (is-rest (get-event mini (num-bars mini) 3 'sax)))

=> NIL
   (105)
   T
```

SYNOPSIS:

```
(defmethod map-over-events ((sc slippery-chicken) start-bar end-bar players
                            function &rest further-args)
```

20.49 slippery-chicken-edit/map-over-events-aux

[slippery-chicken-edit] [Functions]

DATE:

```
3 September 2019, London
```

DESCRIPTION:

Auxilliary function for map-over-notes and map-over-events. Saves
duplicating code.

ARGUMENTS:

- A slippery chicken object
- A number that is the first bar to which the function should be
  applied. Default = NIL in which case 1 will be used.
- A number that is the last bar to which the function should be
  applied. Default = NIL in which case all bars will be processed.
- A list of the IDs of the players to whose parts the function should be
  applied. Can also be a single symbol. If NIL then all players will be
  processed.
- T or NIL to indicate whether or not rests will be ignored.
  T = attacked notes only (i.e. no rests), NIL = include rests.
- The method or function itself. This can be a user-defined function or the
  name of an existing method or function. It should take at least one
  argument, an event, and any other arguments as supplied.
- Any additional argument values the specified method/function may
  take or require. See the thin method below for an example that uses
  additional arguments.

RETURN VALUE:

- A list containing the number of events changed per instrument

SYNOPSIS:

```
(defun map-over-events-aux (sc start-bar end-bar players attacked-notes-only
                            function further-args)
```

20.50 slippery-chicken-edit/map-over-notes

[slippery-chicken-edit] [Methods]

DATE:

1 September 2019, London

DESCRIPTION:

A simple method that mimics the behaviour of map-over-bars but on notes
instead. At its heart is a next-event loop. Use this method to apply an
arbitrary function to every note within a specific range of bars.
NB This works on NOTES only, rests will be skipped!

ARGUMENTS:

- A slippery chicken object
- A number that is the first bar to which the function should be applied. Default = NIL in which case 1 will be used.
- A number that is the last bar to which the function should be applied. Default = NIL in which case all bars will be processed.
- A list of the IDs of the players to whose parts the function should be applied. Can also be a single symbol. If NIL then all players will be processed.
- The method or function itself. This can be a user-defined function or the name of an existing method or function. It should take at least one argument, a note event (can be a single pitch or a chord), and any other arguments as supplied.

OPTIONAL ARGUMENTS:

- Any additional argument values the specified method/function may take or require. See the thin method below for an example that uses additional arguments.

RETURN VALUE:

- A list containing the number of notes changed per instrument

EXAMPLE:

```
(let* ((mini
          (make-slippery-chicken
           '+mini+
           :ensemble '(((sax ((alto-sax tenor-sax) :midi-channel 1))))
           :instrument-change-map '((1 ((sax ((1 alto-sax) (3 tenor-sax)))))
                                    (2 ((sax ((2 alto-sax) (5 tenor-sax)))))
                                    (3 ((sax ((3 alto-sax) (4 tenor-sax))))))
           :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
           :set-map '((1 (1 1 1 1 1))
                      (2 (1 1 1 1 1))
                      (3 (1 1 1 1 1)))
           :rthm-seq-palette '((1 ((((4 4) h e (s) (s) e+s+s))
                                   :pitch-seq-palette ((1 2 3)))))
           :rthm-seq-map '((1 ((sax (1 1 1 1 1))))
                           (2 ((sax (1 1 1 1 1))))
                           (3 ((sax (1 1 1 1 1)))))))))
       (print (is-chord (get-note mini (num-bars mini) 3 'sax)))
       (print (map-over-notes mini 1 nil nil #'add-pitches 'c4 'd4))
```

```
        (is-chord (get-note mini (num-bars mini) 3 'sax)))

=> NIL
   (45)
   3
```

SYNOPSIS:

```
(defmethod map-over-notes ((sc slippery-chicken) start-bar end-bar players
                           function &rest further-args)
```

20.51 slippery-chicken-edit/map-over-sequenzes

[slippery-chicken-edit] [Methods]

DATE:

```
February 2nd 2018, Heidhausen
```

DESCRIPTION:

```
Apply a function to each sequenz object in a slippery-chicken piece. Note
that as a sequenz is a subclass of rthm-seq (and bar-holder) and that
rthm-seq has its rthm-seq-bars in the bars slot, we've got access to all
kinds of data which we can switch on in the function passed here,
e.g. player-section-ref, rsp-id
```

ARGUMENTS:

```
- the slippery-chicken object
- the player(s) to process (single or list of symbols). If nil then all
  players will be processed.
- the function to apply to each sequenz. This should accept a sequenz object
  as its first argument.
```

OPTIONAL ARGUMENTS:

```
&rest: further arguments to be passed to the given function
```

RETURN VALUE:

```
T
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((hn (french-horn :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5))))
                           (2 ((((4 4) h h))
                               :pitch-seq-palette ((1 2)))))
        :rthm-seq-map '((1 ((hn (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (2 ((hn (2 2 2 2 2))
                            (vc (2 2 2 2 2))))
                        (3 ((hn (1 1 1 1 1))
                            (vc (1 1 1 1 1)))))))))
  (map-over-sequenzes mini 'hn
                      #'(lambda (seq) (print (player-section-ref seq)))))
=>
(1 HN)
(1 HN)
(1 HN)
(1 HN)
(1 HN)
(2 HN)
(2 HN)
(2 HN)
(2 HN)
(2 HN)
(3 HN)
(3 HN)
(3 HN)
(3 HN)
(3 HN)
T
```

SYNOPSIS:

```
(defmethod map-over-sequenzes ((sc slippery-chicken) players function
                               &rest further-args)
```

20.52 slippery-chicken-edit/move-clef

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Move a specified clef from a specified event object to another.

NB: As the :auto-clefs option of cmn-display and write-lp-data-for all
    first deletes all clefs before automatically placing them, this
    argument must be set to NIL. The auto-clefs method can be called
    outside of the cmn-display or write-lp-data-for-all methods instead.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the number of the bar in which the given clef is
  located.
- An integer that is the number of the event object in the given bar to
  which the given clef is attached.
- An integer that is the number of the bar to which the given clef is
  to be moved (this can be the same bar).
- An integer that is the number of the event object in the new bar to
  which the given clef is to attached.
- The ID of the player in whose part the clef is to be moved.
```

RETURN VALUE:

```
Returns the value of the MARKS-BEFORE slot of the event object to which the
clef is moved.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vc (cello :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c2 e2 d4 e4 f4 g4 a4 f5))))
        :set-map '((1 (1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8)))))
        :rthm-seq-map '((1 ((vc (1 1 1 1))))))))
  (auto-clefs mini)
  (move-clef mini 1 6 1 8 'vc)
  (cmn-display mini :auto-clefs nil))
```

SYNOPSIS:

```
(defmethod move-clef ((sc slippery-chicken) from-bar from-event
                      to-bar to-event player)
```

20.53 slippery-chicken-edit/move-events

[slippery-chicken-edit] [Methods]

DATE:

20-Jul-2011 (Pula)

DESCRIPTION:

Move a specified sequence of consecutive event objects from one player to another, deleting the events from the source player.

NB: Although partial bars can be moved from the source player, the entire bars of the target players are always overwritten, resulting in rests in those segments of the target players' bars that do not contain the moved material. This method thus best lends itself to moving into target players parts that have rests in the corresponding bars.

ARGUMENTS:

- A slippery-chicken object.
- The ID of the source player.
- The ID of the target player.
- A number that is the first bar from which events are to be moved.
- A number that is the first event within the start-bar that is to be moved.
- A number that is the last bar from which events are to be moved.
- A number that is the last event within the end-bar that is to be moved.

OPTIONAL ARGUMENTS:

keyword arguments:
- :transposition. A positive or negative number that is the number of semitones by which the copied material is to be first transposed. This number can be a decimal number, in which case the resulting pitches will be rounded to the nearest microtone (if the current tuning environment is capable of microtones).
- :consolidate-rests. T or NIL to indicate whether resulting consecutive

```
   rests should be consolidated each into one longer rest.
   T = consolidate. Default = T.
```

RETURN VALUE:

```
 Returns T.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((bn (bassoon :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c2 e2 d4 e4 f4 g4 a4 f5))))
        :set-map '((1 (1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8))))
                           (2 ((((4 4) (w))))))
        :rthm-seq-map '((1 ((bn (1 1 1 1))
                            (vc (2 2 2 2))))))))
  (move-events mini 'bn 'vc 2 3 3 2)
  (move-events mini 'bn 'vc 4 1 4 2 :transposition 4.5))

  => T
```

SYNOPSIS:

```
(defmethod move-events ((sc slippery-chicken) from-player to-player
                        start-bar start-event end-bar end-event
                        &key transposition (consolidate-rests t))
```

20.54 slippery-chicken-edit/move-rehearsal-letter

[slippery-chicken-edit] [Methods]

DATE:

```
 March 15th 2017, Edinburgh
```

DESCRIPTION:

```
 Move a rehearsal letter for those players who have one at <from-bar> to
 <to-bar>
```

ARGUMENTS:

```
- a slippery-chicken object
- the bar number (integer) to move the letter from
- the bar number (integer) to move the letter to
```

RETURN VALUE:

```
The last rehearsal-letter that was seen (in the lowest player in the
ensemble)
```

SYNOPSIS:

```
(defmethod move-rehearsal-letter ((sc slippery-chicken) from-bar to-bar)
```

20.55 slippery-chicken-edit/note-add-bracket-offset

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
For CMN only: Adjust the position, lengths, and angles of the tuplet
bracket attached to a specified event object.

NB: The bracket data is stored in the BRACKET slot of the first event
    object of a given tuplet figure.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the number of the bar in which the tuplet bracket is
  located.
- An integer that is the event to which the tuplet bracket is
  attached. Tuplet brackets are attached to the first event object of a
  given tuplet figure.
- The ID of the player in whose part the tuplet bracket is located.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
NB: At least one of these arguments must be set in order to create a
    change.
- :dx. A positive or negative decimal number to indicate the horizontal
  offset of the entire bracket.
```

- :dy. A positive or negative decimal number to indicate the vertical offset of the entire bracket.
- :dx0. A positive or negative decimal number to indicate the horizontal offset of the left corner of the bracket.
- :dy0.A positive or negative decimal number to indicate the vertical offset of the left corner of the bracket.
- :dx1. A positive or negative decimal number to indicate the horizontal offset of the right corner of the bracket.
- :dy1. A positive or negative decimal number to indicate the vertical offset of the right corner of the bracket.
- :index. An integer that indicates which bracket of a nested bracket on the same event is to be affected. 0 = outermost bracket, 1 = first nested bracket, etc. Default = 0.

RETURN VALUE:

Returns a list of the bracket start/end indicator and the tuplet value followed by the offset values passed to the keyword arguments.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vc (cello :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((f3 g3 a3 b3))))
        :set-map '((1 (1)))
        :rthm-seq-palette '((1 ((((2 4) { 3 te te te } q ))
                                :pitch-seq-palette ((1 2 3 4)))))
        :rthm-seq-map '((1 ((vc (1))))))))
  (note-add-bracket-offset mini 1 1 'vc
                           :dx -.1 :dy -.3
                           :dx0 -.1 :dy0 -.4
                           :dx1 .3 :dy1 -.1))

=> (1 3 -0.1 -0.3 -0.1 -0.4 0.3 -0.1)
```

SYNOPSIS:

```
(defmethod note-add-bracket-offset ((sc slippery-chicken)
                                    bar-num note-num player
                                    &key (dx nil) (dy nil)
                                    (dx0 nil) (dy0 nil)
                                    (dx1 nil) (dy1 nil)
                                    (index 0))
```

20.56 slippery-chicken-edit/octavise-repeated-notes

[slippery-chicken-edit] [Methods]

DATE:

```
July 13th 2020, Heidhausen (though taken from earlier projects)
```

DESCRIPTION:

```
Transpose fast repeated notes by given intervals, typically, as in piano
music, by an octave. Note that if there are several fast repeated notes then
there will be multiple transpositions e.g. (c4 c4 c4 c4 c4) would become by
default (c4 c5 c4 c5 c4).
```

ARGUMENTS:

```
- the slippery-chicken object
- the player (symbol) or players (list) to process
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :start-bar. The bar to start processing at (integer). Default = 1.
- :end-bar. The bar to stop processing at (inclusive). Default = NIL = last
  bar of the piece.
- :intervals. A list of semitone intervals. Default = '(12) = only transpose
  up one octave. A list of any length may be used for variation; it will be
  used cyclically.
- :threshold. The maximum time difference in seconds between events that
  will result in the 'octavisation'. Default = the slippery-chicken
  object's fast-leap-threshold (0.125 seconds by default).
- :verbose. Print bar numbers where processing takes place. Default = NIL.
- :whole-chords. T or NIL to transpose whole chords or just the repeated
  notes in chords (could result in unplayable chords if NIL). Default = T =
  transpose whole chord.
- :check-ties. Whether to call the check-ties method, which as a side-effect
  makes sure tied notes are of the same pitch. Default = T.
```

RETURN VALUE:

```
the (modified) slippery-chicken object.
```

SYNOPSIS:

```
(defmethod octavise-repeated-notes ((sc slippery-chicken) players
                                    &key (start-bar 1) end-bar verbose
                                      (intervals '(12)) (whole-chords t)
                                      (check-ties t)
                                      (threshold (fast-leap-threshold sc)))
```

20.57 slippery-chicken-edit/orchestrate

[slippery-chicken-edit] [Methods]

DATE:

November 2nd 2018, Heidhausen

DESCRIPTION:

'Orchestrate' the chords and single pitches in an existing slippery-chicken object. This involves mapping the existing players' pitch events onto a group of new players (a 'combo'). The procedure is fairly complex and by no means perfect (in fact it may exit with an error, especially if your requirements are :strict (see below)), but it works like this:

First, we assume that your existing music is more vertical/harmonic than horizontal/linear/melodic, particularly because the mapping process treats each chord separately and potentially maps each chord onto different players---the choice of whether to change ensemble is decided by an activity-levels object by default (see :combo-change-fun below) but also by the number of notes in any given chord and whether it's possible to play any subsequent chord on the same instruments. The method also assumes that each 'player' in the existing slippery-chicken object has a sequence of one or more chords which can be separated into 'phrases' using the get-phrases method; that the new ensemble uses player IDs that don't exist in the original ensemble (though any player can use the same instruments, of course); and that using the player IDs from the new ensemble we provide lists of sublists of possible 'combos' that should be used to play the existing chords. These combos should range from 2 players to however many we'll need to play the existing chords. We pass these combos to the method and/or allow them to be created by the 'lotsa-combos' ensemble method (see :combos below)

Whether a chord is playable by a given combo is determined by the range of the chord and the instruments of any given combo, along with those instruments' abilities to play chords and/or microtones (if the chord is microtonal). When trying to determine whether a given combo can play a

chord, we permutate the possible instrument orderings to see if the pitches are in range. If :artificial-harmonics is T we also try for such on notes normally considered too high for any given string instrument. If we can't find a combo for any given chord we'll keep track of the best match and if our :relax argument allows, use that instead of backing out completely.

One of the complexities we face, after determining that a chord is playable by a particular combo, is whether the players are free (i.e. not already playing) at the time the chord needs to be played. We take ties into account of course, but one limitiation, presently, is that in order be deemed 'free' each instrument of the combo must have a complete bar free for the start and end of any given chord.

As we select the current instrument using the instrument-change-map slot of the slippery-chicken object (which may have been changed after the original data was generated) this method works with players who double on two or more instruments.

NB If all goes well, each note in a chord will be played by one instrument. Of course orchestration often involves 'doubling' where each note is played by more than one instrument. To achieve this, call orchestrate multiple times, perhaps increasing the :relax value on subsequent calls so that the method doesn't exit when it can't find a 2nd, 3rd, or 4th (...) combo to play a given chord.

ARGUMENTS:

- the slippery-chicken object whose existing players and event data will be used to orchestrate onto new players
- a new ensemble, the players/instruments of which will be used for the new orchestration. (This shouldn't be a nested ensemble but that's a rarity.) This can either be an ensemble object or a list structure that can be made into one (as used for most calls to make-slippery-chicken).
- a list of player IDs indicating the original music to be re-orchestrated.

OPTIONAL ARGUMENTS:

keyword arguments:
- :combos. A list of lists indicating the player groupings to be used to orchestrate the events. Default = NIL whereupon the <new-ensemble> will be used to generate groupings via the 'lotsa-combos' ensemble class method.
- :add-more-combos. T or NIL to indicate whether the :combos slot should be expanded via the 'lotsa-combos' method. If T then whatever is passed to :combos will nevertheless be retained after the expansion. Default = NIL.
- :sub-combos. T or NIL to indicate whether the :combos slot should be expanded via the sub-groups function. This is quite different to

:add-more-combos as that approach uses permutations of all the instruments in the ensemble whereas this approach uses permutations of sub-sets of the given :combos (only) instead.
- :start-bar. The bar number to start orchestrating. Default = NIL which will in turn default to 1
- :end-bar. The bar number to stop orchestrating (inclusive). Default = NIL which will in turn default to the last bar of the slippery-chicken object.
- :auto-beam. T or NIL to indicate whether to call auto-beam in the new parts. Default = T.
- :chords. Whether individual instruments capable of playing chords should attempt to do so. This may determine whether a given combo is determined to be capable of playing one of the original slippery-chicken object's chords, and also whether under some circumstances some notes will be doubled across different instruments. Default = T.
- :combo-change-fun. A function which should return T or NIL to indicate whether we should change combo on any given event. This should take two arguments (the combo as a list of player IDs and an assoc-list of all possible combos (as created by the organise-combos method using the <combos> argument given here) plus two further optional arguments which are free to define. The function should also provide "reset" functionality or do nothing if one of the two required arguments is NIL. Default = combo-change? (defined in event.lsp).
- :verbose. T or NIL to turn on the printing of orchestration decisions as the algorithm is run.
- :artificial-harmonics. T or NIL to allow the selection of string artificial harmonics in order to reach high notes outwith normal ranges.
- :sticky-stats. T or NIL to indicate whether the statistics collected and returned by the method should be reset or retained between calls. Default = NIL = reset.
- :relax. An integer between 0 and 3 inclusive to indicate how strict the algorithm should when allocating chord pitches to instruments. The values have the following meanings with degrees of relaxedness ascending:
  0: each pitch in the chord has to be playable by at least one instrument in the combo; no instrument can sit this chord out but instruments can play chords that include notes found in other instruments.
  1. instruments may sit out a chord if a chord-playing instrument can fill in the notes (all notes of chord must still be played)
  2. notes can be left out and instruments can sit out a chord
  3. whole chords can be left out

RETURN VALUE:

Five statistics values representing the number of chords where: 1. all pitches & every player used; 2. all pitches but some players tacit; 3. some pitches and instruments missing; 4. chords completely left out; 5. total chords attempted

EXAMPLE:

```
(let* ((mini (make-slippery-chicken
              '+mini+
              :ensemble '(((flt (flute :midi-channel 1))))
              :staff-groupings '(1)
              :tempo-map '((1 (q 60)))
              :set-palette '((set1 ((fs2 b2 d4 a4 d5 e5 a5 d6)))
                             (set2 ((b2 fs3 d4 e4 a4 d5 e5 a5 d6))))
              :set-map '((1 (set1 set1 set2 set1 set1 set2)))
              :rthm-seq-palette
              '((seq1 ((((4 4) q q q q))
                       :pitch-seq-palette (1 2 1 3)))
                (seq2 ((((4 4) (e) e q h))
                       :pitch-seq-palette (1 2 3))))
              :rthm-seq-map '((1 ((flt (seq1 seq1 seq2 seq1 seq1 seq2)))))))
       (new-ens (make-ensemble 'new-ens
                               '((vln (violin :midi-channel 2))
                                 (clr (b-flat-clarinet :midi-channel 3))))))
  (orchestrate mini new-ens 'flt)
  (cmn-display mini))

=> T
```

SYNOPSIS:

```
(let (successes)                          ; for stats
  (defmethod orchestrate ((sc slippery-chicken) new-ensemble original-players
                          &key combos add-more-combos sub-combos start-bar
                            end-bar verbose
                            (artificial-harmonics t) (relax 0) (auto-clefs t)
                            (auto-beam t) (combo-change-fun #'combo-change?)
                            (chords t) sticky-stats)
```

20.58 slippery-chicken-edit/pause-last

[slippery-chicken-edit] [Methods]

DATE:

```
23 September 2016, London
```

DESCRIPTION:

```
Add a pause mark to the last note of every (any) player's part. It defaults
```

```
to the last bar but this can be changed with the :bar-num keyword arg. You
also have the option to change the bar line type in the same bar using the
:bar-line keyword arg. See (change-bar-line-type) for details.
```

ARGUMENTS:

```
- the slippery chicken object
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :players. The player (symbol) or list of players that will be
  effected. Default = NIL = all players.
- :bar-num. The bar number to add the pause mark. Default = last bar of
  piece.
- :bar-line. The bar line type of the bar we'll add a pause to (symbol). Can
  be one of 'normal 'double-bar 'final-double 'begin-repeat
  'begin-end-repeat or 'end-repeat. Default = NIL = no change.
- :pause. The type of pause to add (symbol). Can be one of 'pause
  'long-pause or 'short-pause. Default = 'pause
```

RETURN VALUE:

```
- the number of pause marks added
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((pno (piano :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c4 d4 f4 g4 a4 c5 d5 f5 g5 a5 c6))))
        :set-map '((1 (1 1 1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q q))
                                :pitch-seq-palette ((1 (2))))))
        :rthm-seq-map '((1 ((pno (1 1 1 1 1 1 1 1))))))))
  (pause-last mini))

=> 1
```

SYNOPSIS:

```
(defmethod pause-last ((sc slippery-chicken)
                       &key
```

```
;; MDE Tue Feb 26 09:07:43 2019 -- type of pause added
;; can vary
(pause 'pause)
;; MDE Sat Feb 23 12:52:15 2019 -- making players a key
;; arg instead of required
players
;; we might want the pause mark somehere else
(bar-num (num-bars sc))
bar-line)
```

20.59 slippery-chicken-edit/process-events-by-time

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Apply the given function to all event objects within the given measure
range in order of their chronological occurrence.  The function can take
one argument only: the current event object.  NB If the time of the event
is needed it can be accessed in the given function via the event's
start-time slot.
```

ARGUMENTS:

```
- A slippery-chicken object.
- A function (or variable to which a function has been assigned).
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :start-bar. An integer that is the first bar in which the function is to
  be applied to event objects. Default = 1.
- :end-bar. NIL or an integer that is the last bar in which the function is
  to be applied to event objects. If NIL, the last bar of the
  slippery-chicken object is used. Default = NIL.
```

RETURN VALUE:

```
T
```

EXAMPLE:

```
(let ((marks (make-cscl '(a s as te ts at))))
  (defun add-random-marks (event)
    (unless (is-rest event)
```

```
      (setf (marks event) (list (get-next marks))))))

(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (va (viola :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((3 4) s (e.)  (s) s (e) (e) s (s)))
                                :pitch-seq-palette ((1 2 3))))
                            (2 ((((3 4) (s) s (e) (e) s (s) s (e.)))
                                :pitch-seq-palette ((1 2 3))))
                            (3 ((((3 4) (e) s (s) s (e.)  (s) s (e)))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((vn (1 2 3))
                            (va (2 3 1))
                            (vc (3 1 2))))))))
  (process-events-by-time mini #'add-random-marks))
```

SYNOPSIS:

```
(defmethod process-events-by-time ((sc slippery-chicken) function
                                   &key (start-bar 1) end-bar)
```

20.60 slippery-chicken-edit/re-bar

[slippery-chicken-edit] [Methods]

DESCRIPTION:

Arrange the events of specified consecutive bars in a given slippery-chicken object into new bars of a different time signature. If the number of beats in the specified series of events does not fit evenly into full measures of the the specified time signature, the method will do its best to create occasional bars of a different time-signature that are as close as possible to the desired length.

This method will only combine existing short bars into longer ones; it won't split up longer bars and recombine them.

NB: This method should not be confused with the rebar method (which is used internally, not directly by the user).

NB: as this reorganises fundamental structures within the slippery-chicken object, it may cause methods like clm-play to fail with some arguments which seem reasonable to the user. So use immediately before creating the score and after creating other outputs.

ARGUMENTS:

- A slippery-chicken object.

OPTIONAL ARGUMENTS:

keyword arguments
- :start-bar. An integer that is the number of the first bar whose events are to be re-barred.
- :end-bar. An integer that is the number of the last bar whose events are to be re-barred.
- :min-time-sig. A time signature in the form of a 2-item list containing the number of beats and the beat unit; e.g. '(3 4). This is a target time signature from which the method may occasionally deviate if the number of events does not fit evenly into full bars of the specified time signature.
- :verbose. T or NIL to indicate whether to print feedback on the re-barring process to the Listener. T = print feedback. Default = NIL.
- :check-ties. T or NIL to indicate whether to force the method to ensure that all ties have a beginning and ending. T = check. Default = T.
- :auto-beam. T, NIL, or an integer. If T, the method will automatically attach beam indications to the corresponding events according to the beat unit of the time signature. If an integer, the method will beam in accordance with a beat unit that is equal to that integer. If NIL, the method will not automatically place beams. Default = T.
- :update-slots. T or NIL to indicate whether to update all slots of the given slippery-chicken object after applying the method. This is an internal argument and will generally not be needed by the user.

RETURN VALUE:

Returns T.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
```

```
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c2 e2 d4 e4 f4 g4 a4 f5))))
        :set-map '((1 (1 1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                                :pitch-seq-palette ((1 2 3 4)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1 1 1)))))))
  (re-bar mini :start-bar 2 :end-bar 5 :min-time-sig '(4 4) :auto-beam 4))

  => T
```

SYNOPSIS:

```
(defmethod re-bar ((sc slippery-chicken)
                   &key start-bar
                   end-bar
                   ;; the following is just a list like '(3 8) '(5 8)
                   min-time-sig
                   verbose
                   ;; MDE Thu Feb  9 10:36:02 2012 -- seems if we don't
                   ;; update-slots then the new bar structure isn't displayed
                   (update-slots t)
                   (check-ties t)
                   ;; could also be a beat rhythmic unit
                   (auto-beam t))
```

20.61 slippery-chicken-edit/rehearsal-letters-at-sections

[slippery-chicken-edit] [Methods]

DATE:

```
August 19th 2016, Edinburgh
```

DESCRIPTION:

```
Place a rehearsal letter at the beginning of every section (and subsection)
in the piece.
```

ARGUMENTS:

```
- a slippery chicken object
```

RETURN VALUE:

```
A list of the bar numbers where rehearsal letters will appear.
```

SYNOPSIS:

```
(defmethod rehearsal-letters-at-sections ((sc slippery-chicken))
```

20.62 slippery-chicken-edit/remove-extraneous-dynamics

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
A post-generation editing method: If two or more consecutive event objects
have the same dynamic, remove that dynamic marking from all but the first
of these.
```

ARGUMENTS:

```
- A slippery-chicken object.
```

RETURN VALUE:

```
Returns T.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c2 e2 d4 e4 f4 g4 a4 f5))))
        :set-map '((1 (1 1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                                :pitch-seq-palette ((1 2 3 4))
                                :marks (f 1 f 2 f 3 f 4))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1 1 1))))))))
  (remove-extraneous-dynamics mini))

  => T
```

SYNOPSIS:

```
(defmethod remove-extraneous-dynamics ((sc slippery-chicken))
```

20.63 slippery-chicken-edit/replace-events

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Replace one or more consecutive existing event objects with new event
objects. All references are 1-based. This method can be applied to only one
bar at a time.

One or more new event objects can be specified as a replacement for one
single original event object.

It's the user's responsibility to make sure the bar is full and
update-slots is called after all post-generation editing.
```

ARGUMENTS:

```
- A slippery-chicken object.
- The ID of the player whose part is to be modified.
- An integer that is the number of the bar in which the change is to be
  made; or a reference to the bar in the format '(section sequence bar).
- An integer that is the number of the first event object in the given bar
  to replace.
- An integer that is the total number of consecutive original event objects
  to replace.
- A list of the new event objects, each in turn specified as a 2-item list
  in the format (pitch rhythm), e.g. '((c4 e)). Rests are indicated with
  NIL or 'r, e.g. (nil s) (r h). Chords are indicated by enclosing the
  pitches of the chord in a list, e.g. ((c4 e4) e).
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether to automatically re-beam the given bar after
  replacing the events. T = beam. Default = NIL.
- A list of integers to indicate tuplet bracket placement, in the format
 '(tuplet-value start-event end-event). These numbers are 0-based and
 inclusive and count rests.
```

RETURN VALUE:

```
Returns T.
```

EXAMPLE:

```
(let ((mini
```

```
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c2 e2 d4 e4 f4 g4 a4 f5))))
        :set-map '((1 (1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q (e) s s))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1))))))))
  (replace-events mini 'vn 1 2 1 '((nil s) ((ds5 fs5) s)) t)
  (replace-events mini 'vn 2 2 1 '((cs5 e)))
  (replace-events mini 'vn '(1 3 1) 3 1 '((df4 s)))
  (replace-events mini 'vn 4 1 1 '((ds4 te) (r te) (b3 te)) t '(3 0 2)))

=> T
```

SYNOPSIS:

```
(defmethod replace-events ((sc slippery-chicken) player bar-num start-event
                           replace-num-events new-events
                           &optional (auto-beam nil) tuplet-brackets)
```

20.64 slippery-chicken-edit/replace-mark

[slippery-chicken-edit] [Methods]

DATE:

17th December 2025

DESCRIPTION:

Replace a mark in all the events of a piece. Applies to both the marks and marks-before slots. NB works with marks that are strings, symbols, lists, numbers or anything that is #'string= or #'equal

ARGUMENTS:

- the slippery-chicken object
- the mark to replace

OPTIONAL ARGUMENTS:

- the start- and end-bars, and players. if these are nil they default to 1, last bar of piece, and all players

RETURN VALUE:

- A list of the rthm-seq-bar objects that were modified.

SYNOPSIS:

```
(defmethod replace-mark ((sc slippery-chicken) what with &optional
                         start-bar end-bar players)
```

20.65 slippery-chicken-edit/replace-multi-bar-events

[slippery-chicken-edit] [Methods]

DESCRIPTION:

Replace specified consecutive event objects across several bars.

The new rhythms provided must produce full bars for all bars specified; i.e., if only a quarter note is provided as the new event for a 2/4 bar, the method will not automatically fill up the remainder of the bar. In that case you'll get a warning and the procedure will exit. This means if you don't know how many bars you want to replace events for, you can pass a large number and ignore the warning and providing you've filled the bars, all should be well.

ARGUMENTS:

- A slippery-chicken object.
- The ID of the player whose part is to be modified.
- An integer that is the number of the first bar in which event objects are to be replaced. This can be an absolute bar number or a list in the form '(section sequence bar); or with subsections then e.g. '((3 1) 4 2)).
- An integer that is the number of bars in which event objects will be replaced.
- The list of new event objects. The new event objects can be passed as complete event objects; as a list of 2-item lists that are note-name/rhythm pairs, e.g: '((c4 q) (d4 e)); or as a list with two sub-lists, the first being just the sequence of rhythms and the second being just the sequence of pitches, e.g: '((q e) (c4 d4)). For the latter, :interleaved must be set to NIL. (see :interleaved below). Pitch data is the usual cs4 or (cs4 cd3) for chords, and NIL or 'r indicate a rest. NB: All pitches are sounding pitches; written pitches will be created for transposing instruments where necessary.

OPTIONAL ARGUMENTS:

keyword arguments:
- :interleaved. T or NIL to indicate whether the new event data is to be processed as a list of note-name/rhythm pairs (or existing event objects), or if it is to be processed as a list with two sub-lists, the first containing the sequence of rhythms and the second containing the sequence of pitches (see above). T = interleaved, i.e. already existing event objects or a list of note-name/rhythm pairs. NIL = separate lists for rhythms and pitches. Default = T.
  If this argument is T, the list of 2-element lists (note-name/rhythm pairs) is passed to make-events, but such a list can contain no ties. If the argument is set to NIL, the rhythm and pitch data is passed as two separate lists to make-events2 where + can be used to indicate ties.
- :consolidate-rests. T or NIL to indicate whether shorter rests should automatically be consolidated into a single longer rest.
  T = consolidate. Default = T.
  NB: slippery chicken will always consolidate full bars of rest into measure-rests, regardless of the value of this argument.
- :beat. NIL or an integer (rhythm symbol) that indicates which beat basis will be used when consolidating rests. If NIL, the beat of the time signature will be used (e.g. quarter in 4/4). Default = NIL.
- :auto-beam. T or NIL to indicate whether to automatically beam the new events. T = automatically beam. Default = T.
- :tuplet-bracket. NIL or an integer to indicate whether to automatically add tuplet (e.g. triplet/quintuplet) brackets to the new events where applicable. If this is an integer, all tuplets in the given bar will be given a tuplet bracket with that integer as the tuplet number. NB: This option does not allow for setting tuplets of different numbers for the same bar. To do that, set :tuplet-bracket to NIL and add the tuplet-brackets manually. NIL = place no brackets. Default = NIL.

RETURN VALUE:

The number of new events used to replace the old ones.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :set-palette '((1 ((d4 e4 f4 g4))))
        :set-map '((1 (1 1 1 1 1 1))
                   (2 (1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                                :pitch-seq-palette ((1 2 3 4))))
                            (2 ((((2 4) e s s q)
```

```
                                (s s e +e e))
                               :pitch-seq-palette ((1 2 3 4 3 2 4 1)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1 1))))
                       (2 ((vn (2 2 2 2 2 2))))))))
  (replace-multi-bar-events mini 'vn 2 3
                            '((cs5 h) ((ds5 fs5) h) (nil h)))
  (replace-multi-bar-events mini 'vn '(2 2 2) '3
                            '((h h h) (cs5 (ds5 fs5) nil))
                            :interleaved nil)
  (replace-multi-bar-events mini 'vn 1 1
                            '((nil e) (nil e) (nil e) (cs4 e))
                            :consolidate-rests t)
  (replace-multi-bar-events mini 'vn 8 1
                            '((nil q) (b3 e) (cs4 s) (ds4 s))
                            :auto-beam t))

=> 4
```

SYNOPSIS:

```
(defmethod replace-multi-bar-events ((sc slippery-chicken)
                                     player start-bar num-bars new-events
                                     &key
                                       ;; 24.3.11: see above.
                                       (interleaved t)
                                       ;; MDE Mon Apr 23 12:36:08 2012 --
                                       ;; changed default to nil
                                       (consolidate-rests nil)
                                       ;; for consolidate rests
                                       (beat nil)
                                       ;; MDE Mon Apr 23 12:36:08 2012 --
                                       ;; changed default to nil
                                       (auto-beam nil)
                                       ;; MDE Fri Aug 29 10:18:29 2014
                                       (warn t)
                                       ;; MDE Mon Sep  1 16:41:02 2014
                                       (delete-beams t)
                                       (delete-tuplets t)
                                       ;; MDE Sat Jan 17 17:52:07 2015 --
                                       ;; should we update the player slot of
                                       ;; each event?
                                       (write-player nil)
                                       ;; MDE Thu Dec  2 23:22:18 2021
                                       (update-slots t)
                                       ;; 31.3.11: if this is t, then rthms > a
                                       ;; beat will case an error
```

```
                                    (auto-beam-check-dur t)
                                    (tuplet-bracket nil))
```

20.66 slippery-chicken-edit/replace-tempo-map

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Replace the tempo data for a given slippery-chicken object with new
specified tempo indications.

Calls not only the setf method - which converts bar references like
(section-num sequence-num bar-num) to numbers and makes a tempo-map object,
but also updates all event objects to reflect new start times etc.
```

ARGUMENTS:

```
- A slippery-chicken object
- A list that is the new tempo-map.
```

RETURN VALUE:

```
T
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((pno (piano :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c4 d4 f4 g4 a4 c5 d5 f5 g5 a5 c6))))
        :set-map '((1 (1 1 1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q q))
                                :pitch-seq-palette ((1 (2))))))
        :rthm-seq-map '((1 ((pno (1 1 1 1 1 1 1 1))))))))
  (replace-tempo-map mini '((1 (q 60 "Andante")) ((1 3 1) (e 80)))))

=> T
```

SYNOPSIS:

```
(defmethod replace-tempo-map ((sc slippery-chicken) tm)
```

20.67 slippery-chicken-edit/reset-midi-channels

[slippery-chicken-edit] [Methods]

DATE:

June 9th 2020

DESCRIPTION:

Reset players' event objects to use the midi-channels given in the ensemble rather than those inherited created elsewhere.

ARGUMENTS:

- the slippery-chicken object to process

OPTIONAL ARGUMENTS:

- a symbol or list of symbols representing the players we want to process (i.e. the same player symbols found in the ensemble slot, not instruments). Default = NIL = all players
- the start-bar in which events should begin to be processed. Default = 1
- the end bar (inclusive) in which the last events should be processed. Default = NIL = the last bar of the slippery-chicken object.

RETURN VALUE:

T

SYNOPSIS:

20.68 slippery-chicken-edit/respell-bars

[slippery-chicken-edit] [Methods]

DESCRIPTION:

Look for enharmonically equivalent pitches in the same bar and try to unify their spelling. The method applies this process to every bar in the given slippery-chicken object.

Also see rthm-seq-bar/respell-bar and slippery-chicken/respell-notes.

ARGUMENTS:

```
 - A slippery-chicken object.
```

RETURN VALUE:

```
 Returns NIL.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((cs4 ds4 df5 ef5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                                :pitch-seq-palette ((1 2 3 4)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1))))))))
  (respell-bars mini))

=> T
```

SYNOPSIS:

```
(defmethod respell-bars ((sc slippery-chicken))
```

20.69 slippery-chicken-edit/respell-notes

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
 Pass through the entire given slippery-chicken object and change some of
 the pitch objects to their enharmonic equivalents to produce more sensible
 spellings of consecutive pitches in the score.

 An optional argument takes a list specifying which pitches to change in the
 same format found in the method enharmonic-spellings; i.e.
 '((player (bar note-num))). These notes are changed after the respelling
 routine has run.

 NB: If a list of corrections is specified, the :respell-notes argument of
     any subsequent call to cmn-display or write-lp-data-for-all must be set
     NIL, otherwise the modified pitches may be overwritten.  Also, although
     this algorithm corrects tied notes when respelling, notes referenced in
```

the corrections list will not be followed through to any subsequent ties.

ARGUMENTS:

- A slippery-chicken object.

OPTIONAL ARGUMENTS:

- A list of specific notes whose pitches are to be enharmonically flipped, in the format, e.g. '((vn (1 1) (1 4)) (vc (2 3) (3 3)))

RETURN VALUE:

Returns T.

EXAMPLE:

```
;; An example using respell-notes for the whole slippery-chicken object.
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((cs4 ds4 df5 ef5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                                :pitch-seq-palette ((1 2 3 4)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1))))))))
  (respell-notes mini))

;; An example specifying which pitches are to be enharmonically changed.
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((cs4 ds4 df5 ef5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                                :pitch-seq-palette ((1 2 3 4)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1))))))))
  (respell-notes mini '((vn (1 1) (1 4))))
  (cmn-display mini :respell-notes nil))

=> T
```

SYNOPSIS:

```
(defmethod respell-notes ((sc slippery-chicken) &optional corrections)
```

20.70 slippery-chicken-edit/respell-notes-for-player

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Pass through the pitches of a specified player's part and change some of
the pitches to their enharmonic equivalents in order to produce more
sensible spellings of consecutive notes.

This is just a very simple attempt to better spell notes by comparing each
note to the previous two and making it the same accidental type. It
doesn't look further back or ahead as of yet.

If the optional argument is set to T, then look at the written notes
instead of the sounding notes.

NB: Since both the cmn-display and write-lp-data-for-all methods
    automatically call respell-notes for all players of an entire
    sc-object, their :respell-notes argument may need to be set to NIL for
    this method to produce the desired results.
```

ARGUMENTS:

```
- A slippery-chicken object.
- The ID of the player whose pitches are to be modified.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether to change written pitches only or sounding
  pitches only. T = change written pitches only. Default = NIL.
```

RETURN VALUE:

```
Returns NIL.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
```

```
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (vn (violin :midi-channel 2))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((b3 cs4 b4 cs5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                                :pitch-seq-palette ((1 2 3 4)))))
        :rthm-seq-map '((1 ((cl (1 1 1 1 1))
                            (vn (1 1 1 1 1))))))))
  (respell-notes-for-player mini 'cl t)
  (cmn-display mini :respell-notes nil :in-c nil))
```

```
=> T
```

SYNOPSIS:

```
(defmethod respell-notes-for-player ((sc slippery-chicken) player
                                     &optional written)
```

20.71 slippery-chicken-edit/rest-to-note

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Change a specified event object from a rest into a note by supplying a
pitch or chord (as objects or symbols).

Marks to be attached to the new note can be supplied as a symbol or a list
as an optional argument.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the number of the bar in which the rest is to be
  changed to a note.
- An integer that is the number of the rest in the given bar that is to be
  changed. This number counts rests only, not sounding notes or events.
- The ID of the player whose part is to be changed.
- A note-name symbol that is to be the pitch of the new note, or a list of
  note-name symbols that will make up a chord.
```

OPTIONAL ARGUMENTS:

```
 - A mark or list of marks to be attached to the new note.
```

RETURN VALUE:

```
 Returns the new event object created.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((cs4 ds4 fs4))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q (e) s s))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1))))))))
  (rest-to-note mini 2 1 'vn 'gs5)
  (rest-to-note mini 3 1 'vn '(gs5 b5))
  (rest-to-note mini 4 1 'vn '(gs4 b4) 'ppp)
  (rest-to-note mini 5 1 'vn '(gs4 b4) '(fff pizz)))

=>
EVENT: start-time: 9.000, end-time: 9.500,
       duration-in-tempo: 0.500,
       compound-duration-in-tempo: 0.500,
       amplitude: 0.900
       bar-num: 5, marks-before: NIL,
       tempo-change: NIL
       instrument-change: NIL
       display-tempo: NIL, start-time-qtrs: 9.000,
       midi-time-sig: NIL, midi-program-changes: NIL,
       8va: 0
       pitch-or-chord:
CHORD: auto-sort: T, marks: NIL, micro-tone: NIL
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (
[...]
```

SYNOPSIS:

```
(defmethod rest-to-note ((sc slippery-chicken) bar-num rest-num player new-note
                         &rest marks)
```

20.72 slippery-chicken-edit/rm-marks-from-note

[slippery-chicken-edit] [Methods]

DESCRIPTION:

Remove one or more specific marks from the MARKS slot of a specified event object.

ARGUMENTS:

- A slippery-chicken object.
- An integer that is the number of the bar from which the marks are to be removed.
- An integer that is the number of the note in that bar from which the marks are to be removed.
- The ID of the player from whose part the marks are to be removed.

OPTIONAL ARGUMENTS:

- A specific mark or list of specific marks that are to be removed. If this argument is not specified, no marks will be removed.

RETURN VALUE:

Returns T.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((cs4 ds4 fs4))))
        :set-map '((1 (1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                                :pitch-seq-palette ((1 2 3 4))
                                :marks (a 2 s 2 fff 2 pizz 2))))
        :rthm-seq-map '((1 ((vn (1 1 1 1)))))))
  (rm-marks-from-note mini 2 2 'vn 'pizz)
  (rm-marks-from-note mini 3 2 'vn '(pizz fff))
  (rm-marks-from-note mini 3 2 'vn))

=> T
```

SYNOPSIS:

```
(defmethod rm-marks-from-note ((sc slippery-chicken) bar-num note-num
                               player &rest marks)
```

20.73 slippery-chicken-edit/rm-marks-from-notes

[slippery-chicken-edit] [Methods]

DESCRIPTION:

Remove only the specified marks from the MARKS slots of specified events in the parts of specified players. If the <players> argument is set to NIL, remove the mark or marks from all players.

ARGUMENTS:

- A slippery-chicken object.
- An integer or a 2-item list of integers indicating the first bar and note from which to remove marks. If an integer, this is the bar number and the mark will be removed from all notes in the bar. If a 2-item list, this is a reference to the bar number and number of the first note in the bar from which to start removing marks, in the form e.g. '(3 1).
- An integer or a 2-item list of integers indicating the last bar and note from which to remove marks. If an integer, this is the bar number and the mark will be removed from all notes in the bar. If this is a 2-item list, this is a reference to the bar number and number of the first note in the bar from which to start removing marks, in the form e.g. '(3 1).
- The ID or a list of IDs of the players from whose parts the marks are to be removed.

OPTIONAL ARGUMENTS:

NB: The <marks> argument is a required argument for this method.
- The mark or a list of the marks to remove. This method will only remove specified marks.

RETURN VALUE:

Returns T.

EXAMPLE:

```
(let ((mini
```

```
       (make-slippery-chicken
        '+mini+
        :ensemble '(((fl (flute :midi-channel 1))
                     (hn (french-horn :midi-channel 2))
                     (vn (violin :midi-channel 3))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((cs4 ds4 fs4))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                                :pitch-seq-palette ((1 2 3 4))
                                :marks (a 2 s 2 fff 2))))
        :rthm-seq-map '((1 ((fl (1 1 1 1 1))
                            (hn (1 1 1 1 1))
                            (vn (1 1 1 1 1)))))))
  (rm-marks-from-notes mini 1 2 'fl 'fff)
  (rm-marks-from-notes mini '(1 2) '(2 1) 'hn '(fff a))
  (rm-marks-from-notes mini 3 '(4 3) '(hn vn) '(fff s a))
  (rm-marks-from-notes mini 5 5 nil 'fff))

=> T
```

SYNOPSIS:

```
(defmethod rm-marks-from-notes ((sc slippery-chicken) start end
                                players &rest marks)
```

20.74 slippery-chicken-edit/rm-pitches-from-chord

[slippery-chicken-edit] [Methods]

DESCRIPTION:

Remove the specified pitches from an existing chord object.

ARGUMENTS:

- The slippery-chicken object which contains the given chord object.
- The ID of the player whose part is to be affected.
- An integer that is the number of the bar that contains the chord object that is to be modified.
- An integer that is the number of the note that is the chord object to be modified.
- The pitches to be removed. These can be pitch objects or any data that can be passed to make-pitch, or indeed lists of these, as they will be flattened.

RETURN VALUE:

The chord object that has been changed.

EXAMPLE:

```
(let* ((ip-clone (clone +slippery-chicken-standard-instrument-palette+)))
  (set-slot 'chord-function 'chord-fun2 'guitar ip-clone)
  (let* ((mini
          (make-slippery-chicken
           '+mini+
           :instrument-palette ip-clone
           :ensemble '(((gtr (guitar :midi-channel 1))))
           :set-palette '((1 ((e3 f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5 d5 e5 f5
                               g5 a5 b5 c6 d6 e6))))
           :set-map '((1 (1)))
           :rthm-seq-palette
           '((1 ((((4 4) e e e e e e e e))
                 :pitch-seq-palette ((1 (2) 3 (4) 5 (6) 7 (8))))))
           :rthm-seq-map '((1 ((gtr (1))))))))
    (print (get-pitch-symbols
            (pitch-or-chord (get-event mini 1 2 'gtr))))
    (rm-pitches-from-chord mini 'gtr 1 2 'a3 'd4)
    (print (get-pitch-symbols
            (pitch-or-chord (get-event mini 1 2 'gtr))))))

=>
(E3 A3 D4 G4)
(E3 G4)
```

SYNOPSIS:

```
(defmethod rm-pitches-from-chord ((sc slippery-chicken) player bar-num
                                  note-num &rest pitches)
```

20.75 slippery-chicken-edit/rm-repeated-pitches

[slippery-chicken-edit] [Methods]

DATE:

April 23rd 2016, Edinburgh

DESCRIPTION:

Sometimes, even pitch curves without repeated notes result in repeated notes in the generated piece. Often it's the removal of large fast leaps in instrumental parts that ends up creating repeated notes. This method will do its best to replace repeated notes, even in chords, with other notes from the current set and within the range of the current instrument and any set-limits. If a chord is required, then the current instrument's chord function will be used to create the replacement chord.

NB If you're interested in instrument/players statistics like tessitura you'll have to call update-slots after you're done calling this and related methods.

ARGUMENTS:

- the slippery-chicken object
- the player (symbol)

OPTIONAL ARGUMENTS:

- the start bar (integer). Default = 1.
- the end bar (integer). Default = NIL which means we'll process up to the end of the piece.

RETURN VALUE:

The slippery-chicken object with all possible repeated notes removed.

SYNOPSIS:

```
(defmethod rm-repeated-pitches ((sc slippery-chicken) player
                                &optional (start-bar 1) end-bar)
```

20.76 slippery-chicken-edit/rm-slurs

[slippery-chicken-edit] [Methods]

DESCRIPTION:

Remove the specified slurs from the MARKS slots of specified events in the parts of specified players. If the <players> argument is set to NIL, remove the specified slurs from all players.

ARGUMENTS:

```
 - A slippery-chicken object.
 - An integer or a 2-item list of integers indicating the first bar and note
   from which to remove slurs. If an integer, this is the bar number and the
   slurs will be removed from all notes in the bar. If a 2-item list, this is
   a reference to the bar number and number of the first note in the bar
   from which to start removing slurs, in the form e.g. '(3 1).
 - An integer or a 2-item list of integers indicating the last bar and note
   from which to remove slurs. If an integer, this is the bar number and the
   slurs will be removed from all notes in the bar. If this is a 2-item list,
   this is a reference to the bar number and number of the first note in the
   bar from which to start removing slurs, in the form e.g. '(3 1).
 - The ID or a list of IDs of the players from whose parts the marks are to
   be removed.
```

RETURN VALUE:

```
 Returns T.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((fl (flute :midi-channel 1))
                     (hn (french-horn :midi-channel 2))
                     (vn (violin :midi-channel 3))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c4 d4 e4 fs4 gs4 as4 c5 d5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 8))
                                :marks (slur 1 2 slur 3 4 slur 5 6 slur 7 8))))
        :rthm-seq-map '((1 ((fl (1 1 1 1 1))
                            (hn (1 1 1 1 1))
                            (vn (1 1 1 1 1))))))))
  (rm-slurs mini 1 2 'fl)
  (rm-slurs mini '(1 3) '(2 1) 'hn)
  (rm-slurs mini 3 '(4 3) '(hn vn))
  (rm-slurs mini 5 5 nil))

=> T
```

SYNOPSIS:

```
(defmethod rm-slurs ((sc slippery-chicken) start end players)
```

20.77 slippery-chicken-edit/round-to-nearest

[slippery-chicken-edit] [Methods]

DATE:

October 18th 2018, Heidhausen

DESCRIPTION:

For the given bars and players, round the events' pitch data to the nearest in the current or given scale. For example, a slippery-chicken object could be generated in the :quarter-tone scale, then the scale changed to :chromatic (or the keyword argument :scale 'chromatic-scale given), this method then called, and all pitches would be rounded to the nearest chromatic pitch. See the pitch class method for more details.

ARGUMENTS:

the slippery-chicken argument

OPTIONAL ARGUMENTS:

keyword arguments:
- :start-bar. A number that is the first bar which should be processed. Default = NIL in which case 1 will be used.
- :end-bar. A number that is the last bar which should be processed. Default = NIL in which case all bars will be processed.
- :players. A list of the IDs of the players to whose parts the function should be applied. Can also be a single symbol. If NIL then all players will be processed.
- :scale. The scale to use when rounding. (Common Music tuning object or symbol). If a symbol, then 'chromatic-scale, 'twelfth-tone, or 'quarter-tone only at present. Default is the current scale as set by (in-scale :...).

RETURN VALUE:

the modified slippery-chicken object

SYNOPSIS:

```
(defmethod round-to-nearest ((sc slippery-chicken)
                             &key start-bar end-bar players (scale cm::*scale*))
```

20.78 slippery-chicken-edit/sc-delete-beams

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Delete beam indications from specified notes. If only a bar number is
specified, this method deletes all beams in the bar.

NB: If specifying start and end notes, the start notes specified must be
    the first note of a beamed group of notes (i.e. the BEAMS slot of the
    corresponding event object must be 1), and the end note must be the
    last note of a beamed group of notes (i.e., the BEAMS slot of the
    corresponding event object must be 0), otherwise errors may
    occur. Also, if specifying one of these arguments, both must be
    specified.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the number of the bar in which the beams are to be
  deleted.
- The ID of the player from whose part the beams are to be deleted.
```

OPTIONAL ARGUMENTS:

```
- An integer that is the number of the note that currently holds the
  start-beam information (i.e., the BEAMS slot is 1). This number is
  1-based and counts ties.
- An integer that is the number of the note that currently holds the
  end-beam information (i.e., the BEAMS slot is 0). This number is 1-based
  and counts ties.
```

RETURN VALUE:

```
If deleting all beams in a bar, returns T, otherwise returns NIL.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vc (cello :midi-channel 1))))
        :set-palette '((1 ((d3 e3 f3 g3 a3 b3 c4 e4))))
        :set-map '((1 (1 1 1 1)))
```

```
       :rthm-seq-palette '((1 ((((4 4) - e e - - e e - - e e - - e e -))
                              :pitch-seq-palette ((1 2 3 4 5 6 7 8)))))
       :rthm-seq-map '((1 ((vc (1 1 1 1))))))))
  (sc-delete-beams mini 2 'vc)
  (sc-delete-beams mini 3 'vc 3 4))

  => NIL
```

SYNOPSIS:

```
(defmethod sc-delete-beams ((sc slippery-chicken) bar-num player
                            &optional start-note end-note)
```

20.79 slippery-chicken-edit/sc-delete-marks

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Delete all marks from the MARKS slot of a given note event object and
set the slot to NIL.

NB: This method counts notes, not rests, and is 1-based.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the number of the bar in which the marks are to be
  deleted.
- An integer that is the number of the note from which the marks are to be
  deleted.
- The ID of the player from whose part the marks are to be deleted.
```

RETURN VALUE:

```
Returns T.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :set-palette '((1 ((cs4 ds4 fs4))))
        :set-map '((1 (1 1 1 1)))
```

```
        :rthm-seq-palette '((1 ((((2 4) q (e) s s))
                               :pitch-seq-palette ((1 2 3))
                               :marks (a 2 s 2 fff 2 pizz 2))))
        :rthm-seq-map '((1 ((vn (1 1 1 1))))))))
  (sc-delete-marks mini 2 2 'vn))

=> T
```

SYNOPSIS:

```
(defmethod sc-delete-marks ((sc slippery-chicken) bar-num note-num player)
```

20.80 slippery-chicken-edit/sc-delete-marks-before

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Deletes all data from the MARKS-BEFORE slot of a specified event object and
replaces it with NIL.

NB: In addition to clef symbol data, the MARKS-BEFORE slot also stores part
    of the required data for trills and arrows. Deleting just the
    MARKS-BEFORE components of those markings may result in unwanted
    results.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the number of the bar in which the event object is to
  be  modified.
- An integer that is the number of the note within the given bar for which
  the MARKS-BEFORE slot is to be set to NIL.
- The ID of the player whose part is to be affected.
```

RETURN VALUE:

```
Returns NIL.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vc (cello :midi-channel 1))))
```

```
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((d3 e3 f3 g3 a3 b3 c4 e4))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                               :pitch-seq-palette ((1 2 3 4)))))
        :rthm-seq-map '((1 ((vc (1 1 1)))))))
  (add-mark-before-note mini 2 3 'vc 'fff)
  (add-mark-before-note mini 2 3 'vc 's)
  (add-mark-before-note mini 2 3 'vc 'lhp)
  (sc-delete-marks-before mini 2 3 'vc))

=> NIL
```

SYNOPSIS:

```
(defmethod sc-delete-marks-before ((sc slippery-chicken)
                                   bar-num note-num player)
```

20.81 slippery-chicken-edit/sc-delete-marks-from-event

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Delete all data from the MARKS slot of the specified event object and
replace it with NIL.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the number of the bar from which the marks are to be
  deleted.
- An integer that is the number of the event within the given bar from
  which the marks are to be deleted.
- The ID of the player from whose part the marks are to be deleted.
```

RETURN VALUE:

```
Returns NIL.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
```

```
        :ensemble '(((vc (cello :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((d3 e3 f3 g3 a3 b3 c4 e4))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                               :pitch-seq-palette ((1 2 3 4))
                               :marks (a 1 4 lhp 4 s 3 4 slur 1 2))))
        :rthm-seq-map '((1 ((vc (1 1 1))))))))
  (sc-delete-marks-from-event mini 2 4 'vc))

=> NIL
```

SYNOPSIS:

```
(defmethod sc-delete-marks-from-event ((sc slippery-chicken)
                                       bar-num event-num player)
```

20.82 slippery-chicken-edit/sc-force-rest

[slippery-chicken-edit] [Methods]

DATE:

23-Jul-2011 (Pula)

DESCRIPTION:

Change the specified event object to a rest. If events tied from this event should automatically be forced to rests also, use the sc-force-rest2 method instead.

ARGUMENTS:

- A slippery-chicken object.
- An integer that is the number of the bar in which the rest is to be forced.
- An integer that is the number of the event within that bar which is to be changed into a rest. This number is 1-based and counts tied notes but not rests.
- The ID of the player whose part is to be modified.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether the specified bar should be automatically beamed after the change has been made. NB: In general, calling auto-beam

```
   is a good idea (esp. when deleting notes under an existing beam);
   however, auto-beam may fail when addressing bars that contain notes
   longer than one beat. T = automatically beam. Default = NIL.
```

RETURN VALUE:

```
 The new rthm-seq-bar object.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vc (cello :midi-channel 1))))
        :set-palette '((1 ((a3 b3 c4 e4))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                                :pitch-seq-palette ((1 2 3 4)))))
        :rthm-seq-map '((1 ((vc (1 1 1))))))))
  (sc-force-rest mini 2 3 'vc)
  (sc-force-rest mini 3 3 'vc t))

  =>

  RTHM-SEQ-BAR: time-sig: 3 (2 4), time-sig-given: T, bar-num: 3,
  old-bar-nums: NIL, write-bar-num: NIL, start-time: 4.000,
  start-time-qtrs: 4.0, is-rest-bar: NIL, multi-bar-rest: NIL,
  show-rest: T, notes-needed: 3,
  tuplets: NIL, nudge-factor: 0.35, beams: ((1 2)),
  current-time-sig: 3, write-time-sig: NIL, num-rests: 1,
  num-rhythms: 4, num-score-notes: 3, parent-start-end: NIL,
  missing-duration: NIL, bar-line-type: 2,
  player-section-ref: (1 VC), nth-seq: 2, nth-bar: 0,
  rehearsal-letter: NIL, all-time-sigs: (too long to print)
  sounding-duration: 1.750,
  rhythms: (
[...]
```

SYNOPSIS:

```
(defmethod sc-force-rest ((sc slippery-chicken) bar-num note-num player
                          &optional auto-beam)
```

20.83 slippery-chicken-edit/sc-force-rest2

[slippery-chicken-edit] [Methods]

DESCRIPTION:

Turn events into rests, doing the same with any following tied events.

NB As it is foreseen that this method may be called many times iteratively, there is no call to check-ties, auto-beam, consolidate-rests, or update-instrument-slots (for statistics)--it is advised that these methods are called once the last call to this method has been made. gen-stats is however called for each affected bar, so the number of rests vs. notes should be consistent with the new data.

NB This fill trigger an error if attempted with grace-notes

ARGUMENTS:

- A slippery-chicken object
- The bar number (integer)
- The event number in the bar (integer, counting from 1)
- The player name (symbol)

OPTIONAL ARGUMENTS:

- T or NIL to delete all marks when a note is turned into a rest. Default = NIL.
- A function object to be called on error (could be #'error (default), #'warn, #'print or simply NIL for no error)

RETURN VALUE:

The number of events turned into rests.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vc (cello :midi-channel 1))))
        :set-palette '((1 ((a3 b3 c4 e4))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette
        '((1 ((((4 4) - e.. 32 - h.) (+w) (+w) ((w)) ((h) (e) q e)
               (+q - +s e. - +h) (+w) (+w) ((w))))))
        :rthm-seq-map '((1 ((vc (1 1 1))))))))
  (sc-force-rest2 mini 1 3 'vc))
=> 3
```

SYNOPSIS:

```
(defmethod sc-force-rest2 ((sc slippery-chicken) bar-num event-num player
                           &optional delete-marks (on-error #'error))
```

20.84 slippery-chicken-edit/sc-move-dynamic

[slippery-chicken-edit] [Methods]

DESCRIPTION:

Move the dynamic attached to a specified event object to another specified event object.

By default the dynamics are moved between events within the same bar. An optional argument allows for dynamics to be moved to events in a different bar.

ARGUMENTS:

- A slippery-chicken object.
- An integer that is the number of the bar in which to move the dynamic.
- The ID of the player in whose part the dynamic is located.
- An integer that is the number of the event object from which the dynamic is to be moved. This number is 1-based and counts both rests and ties.
- An integer that is the number of the event object to which the dynamic is to be moved. This number is 1-based and counts both rests and ties.

OPTIONAL ARGUMENTS:

- An integer that is the number of the bar to which the dynamic should be moved. If this is not specified, the dynamic will be moved to the specified event within the same bar.

RETURN VALUE:

Returns T.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vc (cello :midi-channel 1))))
```

```
        :set-palette '((1 ((a3 b3 c4 e4))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                                :pitch-seq-palette ((1 2 3 4))
                                :marks (fff 1))))
        :rthm-seq-map '((1 ((vc (1 1 1))))))))
  (sc-move-dynamic mini 1 'vc 1 3)
  (sc-move-dynamic mini 2 'vc 1 4 3))

  => T
```

SYNOPSIS:

```
(defmethod sc-move-dynamic ((sc slippery-chicken) bar-num player
                            ;; event numbers 1-based but counting rests and ties
                            from to &optional to-bar)
```

20.85 slippery-chicken-edit/sc-remove-dynamic

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Remove all dynamics from the MARKS slot of one or more specified event
objects.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the number of the bar from which the dynamics are to
  be removed.
- The ID of the player from whose part the dynamics are to be removed.
- An integer or a list of integers that are the numbers of the events from
  which the dynamics are to be removed. Event numbers include ties and
  rests.
```

RETURN VALUE:

```
Returns the last dynamic removed.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
```

```
        :ensemble '(((vc (cello :midi-channel 1))))
        :set-palette '((1 ((a3 b3 c4 e4))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                                :pitch-seq-palette ((1 2 3 4))
                                :marks (fff 1 ppp 3))))
        :rthm-seq-map '((1 ((vc (1 1 1)))))))
  (sc-remove-dynamic mini 2 'vc 1)
  (sc-remove-dynamic mini 3 'vc '(1 3)))

  => PPP
```

SYNOPSIS:

```
(defmethod sc-remove-dynamic ((sc slippery-chicken) bar-num player
                              &rest event-nums)
```

20.86 slippery-chicken-edit/sc-remove-dynamics

[slippery-chicken-edit] [Methods]

DATE:

16-Mar-2011

DESCRIPTION:

Remove all dynamic marks from the MARKS slots of all consecutive event objects (or just rests, if preferred) within a specified region of bars.

ARGUMENTS:

- A slippery-chicken object.
- An integer or a list of two integers. If a single integer, this is the number of the first bar from which the dynamics will be removed, and all dynamics will be removed from the full bar. If this is a list of two integers, they are the numbers of the first bar and first note within that bar from which the dynamics will be removed, in the form '(bar-num note-num). Note numbers are 1-based and count ties but not rests.
- An integer or a list of two integers. If a single integer, this is the number of the last bar from which the dynamics will be removed, and all dynamics will be removed from the full bar. If this is a list of two integers, they are the numbers of the last bar and last note within that bar from which the dynamics will be removed, in the form '(bar-num note-num). Note numbers are 1-based and count ties but not rests.

```
 - A single ID or a list of IDs of the players from whose parts the dynamics
   are to be removed. If NIL then all players will be processed. Default =
   NIL = all players.
```

OPTIONAL ARGUMENTS:

```
 - T or NIL to indicate that dynamics should be removed only from
   rests. Default = NIL.
```

RETURN VALUE:

```
 Returns T.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (va (viola :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :set-palette '((1 ((d3 e3 f3 g3 a3 b3 c4 e4 f4 g4 a4 b4))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                                :pitch-seq-palette ((1 2 3 4))
                                :marks (fff 1 ppp 3))))
        :rthm-seq-map '((1 ((vn (1 1 1))
                            (va (1 1 1))
                            (vc (1 1 1))))))))
  (sc-remove-dynamics mini '(1 2) '(2 2) 'vn)
  (sc-remove-dynamics mini 2 3 '(va vc)))

  => T
```

SYNOPSIS:

```
(defmethod sc-remove-dynamics ((sc slippery-chicken) start end players
                               &optional just-rests)
```

20.87 slippery-chicken-edit/set-cautionary-accidental

[slippery-chicken-edit] [Methods]

DATE:

28-Sep-2011

DESCRIPTION:

A post-generation editing method: Place a cautionary accidental (sharp/flat/natural sign in parentheses) before a specified note.

NB: Adding cautionary accidentals to pitches within chords is currently only possible in LilyPond output. Adding cautionary accidentals to single pitches is possible in both CMN and LilyPond.

NB: Since the cmn-display and write-lp-data-for-all methods call respell-notes by default, that option must be explicitly set to NIL within the calls to those methods in order for this method to be effective.

ARGUMENTS:

- A slippery-chicken object.
- An integer that is the number of the bar in which to add the cautionary accidental.
- An integer or a 2-item list of integers that is the number of the note within that bar to which to add the cautionary accidental. This number is 1-based and counts ties. If a 2-item list such, this indicates that the pitch is within a chord; e.g., '(1 2) indicates that a cautionary accidental should be added to the 2nd pitch up from the bottom of the chord located at the 1st note position in the bar.
- The ID of the player to whose part the cautionary accidental is to be added.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether to add the cautionary accidental to only the written pitch or only the sounding pitch. T = written only. Default = NIL.

RETURN VALUE:

Returns T.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
```

```
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (pn (piano :midi-channel 2))))
        :set-palette '((1 ((ds3 e3 fs3 af3 bf3 c4 ef4 fs4))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                                :pitch-seq-palette ((1 2 (3) 4))
                                :marks (fff 1 ppp 3))))
        :rthm-seq-map '((1 ((cl (1 1 1))
                            (pn (1 1 1)))))))
  (respell-notes mini)
  (set-cautionary-accidental mini 3 2 'cl t)
  (set-cautionary-accidental mini 2 1 'pn)
  (set-cautionary-accidental mini 2 2 'pn)
  (set-cautionary-accidental mini 3 '(3 3) 'pn)
  (write-lp-data-for-all mini :respell-notes nil))

=> T
```

SYNOPSIS:

```
(defmethod set-cautionary-accidental ((sc slippery-chicken) bar-num note-num
                                      player &optional written)
```

20.88 slippery-chicken-edit/set-limits-by-section

[slippery-chicken-edit] [Methods]

DATE:

```
February 20th 2018, Heidhausen
```

DESCRIPTION:

```
For each section/subsection of a piece, set the highest or lowest pitch,
i.e. create the curve for one player in the set-limits-high or
set-limits-low slot.

N.B.1 As well as returning the generated list, the respective slot value
(<which>) is also changed.

N.B.2 As the internal structure of the slippery-chicken object is modified
here, and this has implications for pitch-selection, i.e. constraints for an
already-existing slippery-chicken object (made, probably, by
make-slippery-chicken), the piece will have to be regenerated by explicitly
calling (sc-init) after calling this method.
```

ARGUMENTS:

- the slippery-chicken object
- the pitches: a list of pitch symbols or MIDI note numbers. Note that these will be used cyclically if there aren't enough pitches for the number of sections/subsections in the piece.
- 'set-limits-high or 'set-limits-low to set the desired curve

OPTIONAL ARGUMENTS:

- which player (symbol) to set the curve for. Default = 'all, which means the curve will apply to all players.

RETURN VALUE:

The curve created (list)

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vc (cello :midi-channel 1))))
        :set-palette '((1 ((f3 g3 a3 b3 c4))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((vc (1 1 1 1 1))))
                        (2 ((vc (1 1 1 1 1))))
                        (3 ((vc (1 1 1 1 1)))))))
  (flet ((print-em ()
                   ;;                first bars of sections 1, 2, 3
                   (loop for bar in '(1 6 11) do
                         (print (get-pitch-symbols (get-bar mini bar 'vc))))))
    (print-em)
    ;; replace existing curve. NB The y values are degrees which if default is
    ;; quarter-tone scale might be the double of what is perhaps  expected
    ;; (i.e. (note-to-degree 'c4) -> 120)
    (print (set-limits-by-section mini '(g3 a3 c4) 'set-limits-high 'vc))
    (sc-init mini)
    ;; note the new (repeating) pitches
    (print-em)))
```

```
---->>

******* section (1)
Getting notes for VC
******* section (2)
Getting notes for VC
******* section (3)
Getting notes for VC
WARNING:
   slippery-chicken::tempo-curve-to-map: No tempo-map or tempo-curve given.
Using default of crotchet/quarter = 60.
Shortening short, fast leaps...
Shortened 0 large fast leaps
(F3 G3 A3 B3 C4)
(F3 G3 A3 B3 C4)
(F3 G3 A3 B3 C4)
(1 110 5 110 6 114 10 114 11 120 15 120)
******* section (1)
Getting notes for VC
******* section (2)
Getting notes for VC
******* section (3)
Getting notes for VC
Shortening short, fast leaps...
Shortened 0 large fast leaps
(F3 F3 G3 G3 G3)
(F3 G3 G3 A3 A3)
(F3 G3 A3 B3 C4)
NIL
```

SYNOPSIS:

```
(defmethod set-limits-by-section ((sc slippery-chicken) pitches which
                                  &optional (player 'all))
```

20.89 slippery-chicken-edit/set-midi-channels

[slippery-chicken-edit] [Methods]

DATE:

```
October 25th 2018, Heidhausen
```

DESCRIPTION:

```
Change the midi-channels of events in the slippery-chicken object. All
```

events' pitch objects' midi-channel slots will be destructively overwritten when using this method.

The second argument is a list of lists. Each sublist has the player (not instrument) ID symbol as its first element, followed by one or two channel numbers. If only one channel is given (i.e. it's a two-element list e.g. (vln 1)) then it will be used for both chromatic and microtonal pitches (i.e. all the players' pitches will be written to the one given midi channel). Otherwise if a second channel is given (e.g. (vln 1 2)) then microtonal pitches will be written on the second channel.

ARGUMENTS:

- the slippery-chicken object.
- a list of player and channel info (see above and example below).

OPTIONAL ARGUMENTS:

- the start-bar in which events should begin to be processed. Default = 1
- the end bar (inclusive) in which the last events should be processed. Default = NIL = the last bar of the slippery-chicken object.

RETURN VALUE:

T

EXAMPLE:

```
(set-midi-channels +sc-obj+ '((solo 1) (computer-a 2) (computer-b 3)
                              ;; this one uses separate midi channels for
                              ;; chromatic and microtonal pitches
                              (computer-c 4 5)))
```

SYNOPSIS:

```
(defmethod set-midi-channels ((sc slippery-chicken) channel-info
                              &optional (start-bar 1) end-bar)
```

20.90 slippery-chicken-edit/set-rehearsal-letter

[slippery-chicken-edit] [Methods]

DESCRIPTION:

Add the specified rehearsal letter/number to the specified bar in one or more specified players.

NB: Since internally this method actually attaches the rehearsal letter/number to the REHEARSAL-LETTER slot of the preceding bar (bar-num - 1), no rehearsal letter/number can be attached to the first bar.

ARGUMENTS:

- A slippery-chicken object.
- An integer that is the number of the bar to which the rehearsal letter/number is to be added.
- A symbol that is the rehearsal letter/number to be added (e.g. 'A or '1)

OPTIONAL ARGUMENTS:

- The player ID or a list of player IDs to whose parts the rehearsal letter/number is to be added. If no value is given here, the rehearsal letter/number will be added to the first (top) instrument in each group of the ensemble, as specified in staff-groupings.

RETURN VALUE:

Returns T.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (va (viola :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :set-palette '((1 ((ds3 e3 fs3 af3 bf3 c4 ef4 fs4))))
        :set-map '((1 (1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                                :pitch-seq-palette ((1 2 3 4)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1))
                            (va (1 1 1 1))
                            (vc (1 1 1 1)))))))))
  (set-rehearsal-letter mini 2 'A)
  (set-rehearsal-letter mini 3 '2 '(va vc))
  (set-rehearsal-letter mini 4 'Z3))

  => T
```

SYNOPSIS:

```
(defmethod set-rehearsal-letter ((sc slippery-chicken) bar-num letter
                                 &optional players)
```

20.91 slippery-chicken-edit/set-score-order

[slippery-chicken-edit] [Methods]

DATE:

24th August 2024

DESCRIPTION:

Set the order of the players in the score. This is basically a convenience function to set the players slot of the piece object slot but also checks that your new players are in the piece already.

NB won't (yet) change order of players in CMN scores.

ARGUMENTS:

- the slippery-chicken object
- a list of players (symbols). Note that these must already be part of the piece.

RETURN VALUE:

The slippery chicken object

SYNOPSIS:

```
(defmethod set-score-order ((sc slippery-chicken) players)
```

20.92 slippery-chicken-edit/set-tempo

[slippery-chicken-edit] [Methods]

DATE:

June 14th 2025

DESCRIPTION:

Change the tempo at a given bar. If the optional argument is non nil, then the 2nd argument is interpreted as a scaler for the existing tempo markings.

ARGUMENTS:

- a slippery-chicken object
- a tempo object or BPM number; if the latter then a beat-value of quarter notes will be assumed.
- the bar number for the change (on the first event). Mid-bar changes upon request :)

OPTIONAL ARGUMENTS:

T or NIL to use the 2nd argument as a scaler instead of a BPM.

RETURN VALUE: SYNOPSIS:

```
(defmethod set-tempo ((sc slippery-chicken) tempo bar-num &optional scaler warn)
```

20.93 slippery-chicken-edit/swap-events

[slippery-chicken-edit] [Methods]

DATE:

June 10th 2025

DESCRIPTION:

Move/swap the events from one part to the other and vice-versa.

NB does not check that pitches are in range! Call force-in-range afterwards if necessary/applicable. Also, in case this method is to be called mulitple times, it does not call update-slots or update-instrument-slots

ARGUMENTS:

- the slippery-chicken obje ct
- the first player to swap from/to (symbol)
- the second player to swap from/to (symbol)
- the start bar number (inclusive)
- the end bar number (inclusive)

RETURN VALUE:

```
T
```

SYNOPSIS:

```
(defmethod swap-events ((sc slippery-chicken) player1 player2 start-bar
                        end-bar)
```

20.94 slippery-chicken-edit/swap-marks

[slippery-chicken-edit] [Methods]

DATE:

```
Thu 22 Aug 2019 17:12:27 BST
```

DESCRIPTION:

```
Replace one dynamic mark with another, or a list of marks with another
list.
```

ARGUMENTS:

```
- the slippery chicken object which contains marks to be swapped
- start-bar: the first bar to start swapping
- end-bar: the last bar to end swapping
- players: the player or players
- old-marks: a single mark or a list of marks
- new-marks: a single mark to a list. NB if a list it must be the same
length as the old-marks list. Maybe I should change this?
```

OPTIONAL ARGUMENTS:

```
nil
```

RETURN VALUE:

```
The number of swapped marks
```

EXAMPLE:

```
(let* ((mini (make-slippery-chicken
              '+mini+
              :ensemble '(((flt (flute :midi-channel 1))))
              :staff-groupings '(1)
```

```
              :tempo-map '((1 (q 60)))
              :set-palette '((set1 ((fs2 b2 d4 a4 d5 e5 a5 d6)))
                             (set2 ((b2 fs3 d4 e4 a4 d5 e5 a5 d6))))
              :set-map '((1 (set1 set1 set2 set1 set1 set2)))
              :rthm-seq-palette
              '((seq1 ((((4 4) (q) (q) q q))
                       :pitch-seq-palette (1 2)
                       :marks (pp 1)))
                (seq2 ((((4 4) (e) e q h))
                       :pitch-seq-palette (1 2 3)
                       :marks (p 1 a 1 s 1))))
              :rthm-seq-map '((1 ((flt (seq1 seq1 seq2 seq1 seq1 seq2))))))))
  (print (has-mark (get-note mini 1 1 'flt) 'fff))
  (print (swap-marks mini nil nil nil 'pp 'fff))
  (has-mark (get-note mini 1 1 'flt) 'fff))

NIL
4
=> (FFF)
```

SYNOPSIS:

```
(defmethod swap-marks ((sc slippery-chicken) start-bar end-bar
                       players old-marks new-marks)
```

20.95 slippery-chicken-edit/thin

[slippery-chicken-edit] [Methods]

DATE:

```
19th September 2018, Heidhausen
```

DESCRIPTION:

```
Thin out events in a slippery-chicken object using an activity curve (which
is deterministic, not random/stochastic). This turns existing notes into
rests. As this is an expensive method but may be called more than once, it
is up to the user to call consolidate-rests and/or update-slots when ready.
NB See note to :auto-tuplets below re. potential problems.
```

ARGUMENTS:

```
- the (fully-initialised) slippery-chicken object
```

OPTIONAL ARGUMENTS:

```
keyword arguments
- :start-bar. The bar to start at. Default = NIL = 1
- :end-bar. The bar to end at. Default = NIL = last bar in piece/object
- :players. A symbol or list of symbols for the players to be processed.
- :curve. A list of breakpoints representing the level of activity. Y-values
  should range over 0 (all events forced to rests) to 10 (no rests
  created). X-values may range over any arbitrary scale but NB that this
  will be rescaled to map over the number of bars in the whole piece,
  despite the start-bar/end-bar arguments. However this rescaling can be
  avoided via the next argument. Default = '(0 1 100 10)
- :rescale-curve. Whether to process the x-values of :curve to range over
  the number of bars in the piece. Default = T = rescale.
-:auto-tuplets. T or NIL to indicate whether auto-tuplets should be called
  when thinning a bar is finished. Bear in mind that tuplet data can become
  corrupted during this method, as individual events initially reference
  existing tuplet bracket data, but when these are forced to rests (and
  especially when they're consolidated into longer rests) these may no
  longer exist. So if e.g. generated music-xml files have extra beats in
  bars, try setting :auto-tuplets T and all may be well again. Default =
  NIL.
```

RETURN VALUE:

```
- the processed slippery-chicken object
```

SYNOPSIS:

```
(defmethod thin ((sc slippery-chicken) &key start-bar end-bar players
                                        (curve '(0 1 100 10)) auto-tuplets
                                        (rescale-curve t))
```

20.96 slippery-chicken-edit/tie

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Add a tie to a specified event object. The new tie will be placed starting
from the specified event object and spanning to the next event object. If
the next event object does not have the same pitch, its pitch will be
changed to that of the first event object.

An optional argument allows the user to adjust the steepness of the tie's
```

curvature.

NB: This method will not automatically update ties in MIDI output. To make sure that MIDI ties are also updated, use the handle-ties method.

NB: If the next event object is a rest and not a note, an error will be produced.

ARGUMENTS:

- A slippery-chicken object.
- An integer that is the number of the bar in which the tie is to be placed.
- An integer that is the number of the note to which the tie is to be attached.
- The ID of the player whose part is to be changed.

OPTIONAL ARGUMENTS:

- A positive or negative decimal number to indicate the steepness of the tie's curvature.

RETURN VALUE:

Returns T.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :set-palette '((1 ((c4 d4 e4))))
        :set-map '((1 (1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q s s (s) s))
                                :pitch-seq-palette ((1 1 2 3)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1))))))))
  (tie mini 2 1 'vn)
  (tie mini 3 2 'vn)
  (tie mini 4 2 'vn -.5))

=> T
```

SYNOPSIS:

```
(defmethod tie ((sc slippery-chicken) bar-num note-num player
                &optional curvature)
```

20.97 slippery-chicken-edit/tie-all-last-notes-over-rests

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Extend the duration of the last note of any bar that precedes a bar which
starts with a rest in the specified region, such that the rest that begins
the next measure is changed to a note and the last note of the first
measure is tied to it.

NB: This method will not automatically update ties in MIDI output. To make
    sure that MIDI ties are also updated, use the handle-ties method.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the first bar in which changes are to be made.
- An integer that is the last bar in which changes are to be made.
- A player ID or list of player IDs.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :to-next-attack. T or NIL to indicate whether ties are to extend over
  only full bars of rest or also over partial bars (until the next attacked
  note). T = until the next attacked note. Default = T.
- :tie-next-attack. T or NIL to indicate whether the new tied notes created
  should also be further extended over the next attacked note if that note
  has the same pitch as the starting note of the tie. T = also tie next
  attacked note if same pitch. Default = NIL.
- :auto-beam. T or NIL to indicate whether the new events should be
  automatically beamed after placement. T = automatically beam.
  Default = NIL.
- :last-rhythm. NIL or a rhythmic duration. If a rhythmic duration, the
  last duration of the tie will be forced to this length. Useful, for
  example, when tying into a rest bar without filling that whole
  bar. NIL = fill the bar with a tied note. Default = NIL.
- :update. T or NIL to indicate whether all slots for all events in the
  piece should be updated. This is an expensive operation so set to NIL if
  you plan on calling other similar methds, and call (update-slots sc)
  explicitly at the end. Default = T.
```

RETURN VALUE:

```
Returns NIL.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (va (viola :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 f4 g4 a4 c5 d5 f5))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e (e) e e (e) (e) e e)
                                 ((w))
                                 ((h.) q)
                                 ((w))
                                 ((w))
                                 ((e) e h.))
                                :pitch-seq-palette ((1 2 3 4 5 6 7 7)))))
        :rthm-seq-map '((1 ((vn (1 1 1))
                            (va (1 1 1))
                            (vc (1 1 1))))))))
  (tie-all-last-notes-over-rests mini 2 6 'vn)
  (tie-all-last-notes-over-rests mini 9 12 'vn :auto-beam t)
  (tie-all-last-notes-over-rests mini 3 5 '(va vc) :to-next-attack nil)
  (tie-all-last-notes-over-rests mini 9 12 'vc :tie-next-attack t)
  (tie-all-last-notes-over-rests mini 13 15 'vn :last-rhythm 'e))


=> NIL
```

SYNOPSIS:

```
(defmethod tie-all-last-notes-over-rests
    ((sc slippery-chicken)
     start-bar end-bar players
     &key
       ;; use up all rests until next attack or (if nil)
       ;; just the rest bars?
       (to-next-attack t)
       ;; if the next attack is the same note/chord as
       ;; the previous, tie to it too?
       (tie-next-attack nil)
       (last-rhythm nil)
       (update t)
       (auto-beam nil))
```

20.98 slippery-chicken-edit/tie-over-all-rests

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Extend the durations of all notes that immediately precede rests in the
specified region by changing the rests to notes and tying the previous notes
to them.

NB: This method will not automatically update ties in MIDI output. To make
    sure that MIDI ties are also updated, use the handle-ties method.
```

ARGUMENTS:

```
- A slippery-chicken object.
- The ID of the player whose part is to be changed, or a list of players or
  NIL to indicate that we should process all players
- An integer that is the number of the first bar in which notes are to be
  tied over rests.
- An integer that is the number of the last bar in which notes are to be
  tied over rests. NB: This argument does not necessarily indicate the bar
  in which the ties will stop, but rather the last bar in which a tie will
  be begun; the ties created may extend into the next bar.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :start-note. An integer that is the number of the first attacked note
  (not counting rests) in the given start-bar for which ties can be placed.
- :end-note. An integer that is the number of the last attacked note (not
  counting rests) in the given end-bar for which ties can be placed.
  NB: This argument does not necessarily indicate the note on which the
  ties will stop, but rather the last not on which a tie can begin; the
  ties created may extend to the next note.
- :auto-beam. T or NIL to indicate whether the method should automatically
  place beams for the notes of the affected measure after the ties over
  rests have been created. T = automatically beam. Default = NIL.
- :consolidate-notes. T or NIL to indicate whether the tied note are to be
  consolidated into single rhythmic units of longer durations after the
  ties over rests have been created. T = consolidate notes. Default = NIL.
- :update. T or NIL to indicate whether all slots for all events in the
  piece should be updated. This is an expensive operation so set to NIL if
  you plan on calling other similar methds, and call (update-slots sc)
  explicitly at the end. Default = T.
```

RETURN VALUE:

Returns T.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :set-palette '((1 ((c4 d4 e4))))
        :set-map '((1 (1 1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) (q) e (s) s))
                                :pitch-seq-palette ((1 2)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1 1 1)))))))
  (tie-over-all-rests mini 'vn 2 3 :start-note 2 :auto-beam t)
  (tie-over-all-rests mini 'vn 5 6 :end-note 1 :consolidate-notes t))

  => T
```

SYNOPSIS:

```
(defmethod tie-over-all-rests ((sc slippery-chicken) players
                               start-bar end-bar
                               &key
                                 (start-note 1)
                                 (end-note 9999999)
                                 (auto-beam nil)
                                 (update t)
                                 (consolidate-notes nil))
```

20.99 slippery-chicken-edit/tie-over-rest-bars

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Extend the duration of the last note in a specified bar by changing
immediately subsequent full-rest bars to notes of the same pitch and tying
them to that note.

NB: This method will not automatically update ties in MIDI output. To make
    sure that MIDI ties are also updated, use the handle-ties method.
```

ARGUMENTS:

```
- A slippery-chicken object.
```

- An integer that is the number of the bar in which the last note is to be tied.
- An ID or list of IDs of the players whose parts are to be modified.

OPTIONAL ARGUMENTS:

keyword arguments:
- :end-bar. An integer or NIL. If an integer, this is the number of the last bar of full-rests that is to be changed to a note. This can be helpful for tying into passages of multiple bars of full-rest.
- :tie-next-attack. T or NIL to indicate whether the new tied notes created should also be further extended over the next attacked note if that note has the same pitch as the starting note of the tie. T = also tie next attacked note if same pitch. Default = NIL.
- :to-next-attack. T or NIL to indicate whether ties are to extend over only full bars of rest or also over partial bars (until the next attacked note). T = until the next attacked note. Default = T.
- :auto-beam. T or NIL to indicate whether the method should automatically place beams for the notes of the affected measure after the ties over rests have been created. T = automatically beam. Default = NIL.
- :last-rhythm. NIL or a rhythmic duration. If a rhythmic duration, the last duration of the tie will be forced to this length. Useful, for example, when tying into a rest bar without filling that whole bar. NIL = fill the bar with a tied note. Default = NIL.
- :update. T or NIL to indicate whether all slots for all events in the piece should be updated. This is an expensive operation so set to NIL if you plan on calling other similar methds, and call (update-slots sc) explicitly at the end. Default = T.

RETURN VALUE:

Returns t.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (va (viola :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :set-palette '((1 ((c4 d4 e4))))
        :set-map '((1 (1 1)))
        :rthm-seq-palette '((1 ((((2 4) (q) e (s) s)
                                ((h))
```

```
                                ((s) e. e e)
                                ((h))
                                ((h))
                                ((e) q s (s)))
                               :pitch-seq-palette ((1 2 2 1 3 3 1)))))
       :rthm-seq-map '((1 ((vn (1 1))
                           (va (1 1))
                           (vc (1 1))))))))
  (tie-over-rest-bars mini 1 'vn :end-bar 2)
  (tie-over-rest-bars mini 3 'va :end-bar 5)
  (tie-over-rest-bars mini 3 '(vn vc) :end-bar 6 :tie-next-attack t)
  (tie-over-rest-bars mini 7 'vc
                      :end-bar 9
                      :to-next-attack t
                      :auto-beam t)
  (tie-over-rest-bars mini 9 'vn :end-bar 11 :last-rhythm 'e))

  => t
```

SYNOPSIS:

```
(defmethod tie-over-rest-bars ((sc slippery-chicken) bar-num players
                               &key (end-bar nil) ;; num of empty bars
                                 (tie-next-attack nil)
                                 (to-next-attack t)
                                 (last-rhythm nil)
                                 (update t)
                                 (auto-beam nil))
```

20.100 slippery-chicken-edit/tie-over-rests

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Extend the duration of a specified note that precedes a rest by changing
the rest to a note with the same pitch and adding a tie between them.

NB: This method will not automatically update ties in MIDI output. To make
    sure that MIDI ties are also updated, use the handle-ties method.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the number of the bar in which the note is located.
```

- An integer that is the number of the note within that bar which is to be extended. This number is 1-based and also counts already tied notes. If NIL, then the last note in the bar will be used.
- The ID of the player whose part is to be modified.

OPTIONAL ARGUMENTS:

keyword arguments
- :end-bar. An integer that is the number of the last bar into which the tie is to extend. This can be helpful if the user wants to tie into only the first of several consecutive full-rest bars.
- :auto-beam. T or NIL to indicate whether the method should automatically beam the beats of the modified bars after the ties have been added. T = automatically beam. Default = NIL.
- :consolidate-notes. T or NIL to indicate whether the method should consolidate tied notes into single rhythm units of longer duration. T = consolidate. Default = T.
- :update. T or NIL to indicate whether all slots for all events in the piece should be updated. This is an expensive operation so set to NIL if you plan on calling other similar methds, and call (update-slots sc) explicitly at the end. Default = T.

RETURN VALUE:

Returns T.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :set-palette '((1 ((c4 d4 e4))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) (q) e (s) s)
                                ((h))
                                ((s) e. (e) e)
                                ((h))
                                ((h))
                                ((e) q s (s)))
                               :pitch-seq-palette ((1 2 2 3 3 1)))))
        :rthm-seq-map '((1 ((vn (1 1 1))))))))
  (tie-over-rests mini 1 2 'vn)
  (tie-over-rests mini 7 1 'vn)
  (tie-over-rests mini 9 2 'vn :end-bar 10)
```

```
(tie-over-rests mini 13 1 'vn :auto-beam t :consolidate-notes nil))

=> T
```

SYNOPSIS:

```
(defmethod tie-over-rests ((sc slippery-chicken) bar-num note-num player
                           &key end-bar auto-beam (consolidate-notes t)
                             (update t))
```

20.101 slippery-chicken-edit/tie-repeated-notes

[slippery-chicken-edit] [Methods]

DATE:

```
Wed 18 Sep 2019 18:56:39 BST
```

DESCRIPTION:

```
Tie adjacent notes if they are the same pitch or chord.
NB: this method works on pitches and chords
```

ARGUMENTS:

```
- a slippery chicken object
- the first bar to start looking
- the last bar to look in
- the player or players to tie
```

OPTIONAL ARGUMENTS:

```
- :consolidate - call consolidate-all-notes at the end? Default = t
```

RETURN VALUE:

```
A list of the total number of ties between notes per instrument
```

EXAMPLE:

```
(let* ((mini (make-slippery-chicken
              '+mini+
              :ensemble '(((pno (piano :midi-channel 1))))
              :staff-groupings '(1)
```

```
                :tempo-map '((1 (q 60)))
                :set-palette '((set1 ((fs2 b2 d4 a4 d5 e5 a5 d6)))
                               (set2 ((b2 fs3 d4 e4 a4 d5 e5 a5 d6))))
                :set-map '((1 (set1 set1 set2 set1 set1 set2)))
                :rthm-seq-palette
                '((seq1 ((((4 4) q (q) q q))
                         :pitch-seq-palette (1 1 1)))
                  (seq2 ((((4 4) (e) e q e (e) e e))
                         :pitch-seq-palette (1 1 1 (1) (1)))))
                :rthm-seq-map '((1 ((pno (seq1 seq1 seq2 seq1 seq1 seq2))))))))
      (tie-repeated-notes mini nil nil nil))

=> (12)
```

SYNOPSIS:

```
(defmethod tie-repeated-notes ((sc slippery-chicken) start-bar end-bar players
                               &key (consolidate t) (check-ties? t))
```

20.102 slippery-chicken-edit/trill

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
Attach a trill mark to a specified event object by adding 'BEG-TRILL-A to
the MARKS-BEFORE slot and TRILL-NOTE with the pitch to the MARKS slot. This
method requires a specified trill pitch.

By default trills are set to span from the specified note to the next note,
though the length of the span can be specified using the optional
arguments.

NB: This is a LilyPond-only method and will not affect CMN output.
```

ARGUMENTS:

```
- A slippery-chicken object.
- The player to whose part the trill is to be added.
- An integer that is the number of the bar in which the trill is to start.
- An integer that is the number of the event object in that bar on which
  the trill is to be placed.
- A note-name symbol that is the pitch of the trill note.
```

OPTIONAL ARGUMENTS:

```
 - An integer that is the number of the event object on which the trill span
   is to stop.
 - An integer that is the number of the bar in which the trill span is to
   stop.
```

RETURN VALUE:

```
 Returns T.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :set-palette '((1 ((c4 d4 e4))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q. s s))
                                :pitch-seq-palette ((1 3 2)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1))))))))
  (trill mini 'vn 2 1 'e4)
  (trill mini 'vn 3 1 'e4 3)
  (trill mini 'vn 4 1 'e4 3 5))

=> T
```

SYNOPSIS:

```
(defmethod trill ((sc slippery-chicken) player start-bar start-event trill-note
                  &optional end-event end-bar)
```

20.103 slippery-chicken-edit/unset-cautionary-accidental

[slippery-chicken-edit] [Methods]

DESCRIPTION:

```
 A post-generation editing method: Remove the parentheses from a cautionary
 accidental (leaving the accidental itself) by setting the
 ACCIDENTAL-IN-PARENTHESES slot of the contained pitch object to NIL.

 NB: Since respell-notes is called by default within cmn-display and
     write-lp-data-for-all, that option must be explicitly set to NIL for
     this method to be effective.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the number of the bar in which the cautionary
  accidental is to be unset.
- An integer that is the number of the note in that bar for which the
  cautionary accidental is to be unset.
- The ID of the player whose part is to be changed.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether to unset the cautionary accidental for the
  written part only (for transposing instruments).
  T = written only. Default = NIL.
```

RETURN VALUE:

```
Returns NIL.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (vn (violin :midi-channel 2))))
        :set-palette '((1 ((cs4 ds4 fs4))))
        :set-map '((1 (1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 3 2 1 2 3 2)))))
        :rthm-seq-map '((1 ((cl (1 1))
                            (vn (1 1)))))))))
  (respell-notes mini)
  (unset-cautionary-accidental mini 2 5 'vn)
  (unset-cautionary-accidental mini 2 7 'cl t)
  (cmn-display mini :respell-notes nil))
```

SYNOPSIS:

```
(defmethod unset-cautionary-accidental ((sc slippery-chicken) bar-num note-num
                                        player &optional written)
```

21 sc/spectra

[Modules]

NAME:

```
spectra

File:             spectra.lsp

Class Hierarchy:  none (no classes defined)

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Definition of the spectral data for piano and other
                  instruments' notes. We'll use this data for
                  the calculation of chord dissonance values in the chord
                  class's calculate-dissonance method.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    July 27th 2015

$$ Last modified:  12:49:39 Thu Apr 18 2024 CEST

SVN ID: $Id: spectra.lsp 5359 2015-07-24 20:53:22Z medward2 $
```

21.1 spectra/get-spectra-al

[spectra] [Functions]

DATE:

```
August 6th 2015, Edinburgh
```

DESCRIPTION:

```
Create an assoc-list of spectral data by doing spectral analysis of all the
sound files contained in a given directory. The aim is to extract the
spectral data for harmonic partials, so we specifically look for ascending
partials as close as possible to integer multiples of the sample sound
file's fundamental frequency, as detected from the file name (see below).

By default we average spectra over three separate readings at 300, 400, and
500 milliseconds. We do this in order to avoid spectral anomalies at any
fixed point in any of the sound files, and of course to skip the onset
transient and any inharmonicities therein.
```

As the extensive analysis and processing here will take some computation time, in order to create our default spectral data for calculating a chord's dissonance or spectral centroid, I've performed the analysis on some of my local sample libraries. By then using the assoc-list method print-for-init I've copied the data into separate text files for reading into the +slippery-chicken-spectra+ global assoc-list and thus making these available to all slippery-chicken users. By modifying the examples at the bottom of this file, similar sample directories' spectral data could be added to this global.

ARGUMENTS:

- the path to the samples directory. This should be something suitable for passing to the (directory) function so we'll need the extension of the sound files (see "*.wav" in the example below).
- a function for processing the filenames (as a string, with no directory or extension e.g. "Stein-R(A0)-V(085)-dB(3478)-T(ML)-M(28)-P(2126-03-01)") in order to return the MIDI note number of the sample in that file. This whole approach is predicated on the assumption that sample file names will contain such pitch data. If that is not the case then get-spectra-al will be of no use. See the akoustik-piano-name and violin-ensemble-name functions below for examples.

OPTIONAL ARGUMENTS:

Keyword arguments:
- :force-harmonics: whether to look for harmonic partials or not. If this is nil then the returned spectra will be the strongest by amplitude irrespective of their harmonicity. Default = t
- :harmonic-tolerance: the maximum deviation from the exact integer partial. Default = 0.15
- :fftsize: The FFT window size for spectral analysis. Default = 4096
- :num-partials: the number of partials to identify and return. Default = 12
- :id: The default ID for the resultant assoc-list. Default = 'spectra
- :analysis-points: The times in milliseconds at which to do the spectral analysis and extract averages. If no averaging is necessary simply pass one starting value as a list. Default = '(300 400 500)

RETURN VALUE:

An assoc-list object whose keys are the MIDI note numbers associated with the spectral information of sample sound files. Each of these is in the form of a two element list: a list of fundamental scalers and a list of their amplitudes (normalised to 1.0).

EXAMPLE:

```
(get-spectra-al "/path/to/samples/*.wav"
                #'violin-ensemble-name :id 'violin-ensemble)
-->
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 21, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: VIOLIN-ENSEMBLE, tag: NIL,
data: (
NAMED-OBJECT: id: 56, tag: NIL,
data: ((0.9889743194880222d0 2.0263850081954717d0 2.9990610385449314d0
        4.0240367072839890d0 5.0325465998806150d0 6.0132921837632630d0
        7.0551329128964655d0 8.0495902746378170d0 9.0320293629542970d0
        10.017427921485291d0 11.0461040956976d0 12.028523416637086d0)
       (0.1748299156810844d0 0.6666103041159978d0 0.740572572283973d0
        0.3709016806254175 6d0 0.2529382535743205d0 0.19313279338531672d0
        0.290088638695093d0 0.20008736984066355d0 0.12280861470679348d0
        0.15534581915388804d0 0.11772253740784348d0 0.11703054477270619d0))
...
```

SYNOPSIS:

```
#+clm
(defun get-spectra-al (sample-dir name-fun
                       &key (force-harmonics t) (harmonic-tolerance 0.15)
                         (fftsize 4096) (num-partials 12) (id 'spectra)
                         ;; millisec times of fft analysis
                         (analysis-points '(300 400 500)))
```

22 sc/utilities

[Modules]

NAME:

```
 utilities

 File:             utilities.lsp

 Class Hierarchy:  none: no classes defined

 Version:          1.1.0
```

```
Project:          slippery chicken (algorithmic composition)

Purpose:          Various helper functions of a general nature.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    June 24th 2002

$$ Last modified:  17:30:39 Fri Jul 10 2026 CEST
```

22.1 utilities/a-weighting

[utilities] [Functions]

DESCRIPTION:

```
Implementation of A-weighting loudness compensation.  Formula taken from
http://en.wikipedia.org/wiki/A-weighting.  This doesn't take 1000Hz
loudness into account, rather it implements the 40-phon Fletcher-Munson
curve only.
```

ARGUMENTS:

```
The frequency in Hertz for which to find the loudness weighting.
```

OPTIONAL ARGUMENTS:

```
keyword aguments:
- :expt. A power (exponent) to raise the result to in order to
   tame/exaggerate the curve (make the db weightings less/more
   extreme). This only really makes sense if :linear t though will work
   with db values also of course.  Values < 1 result in linear values
   closer to 1 (less extreme).  Values > 1 are further from 1. Default = NIL
   i.e. no exponential function.
- :linear.  If T return amplitude values as linear scalers rather than
   logarithmic decibel values.  NB If this is NIL then returned values are
   likely to be negative (db) values.  Default = T.
- :invert.  As the weighting routine tries to tell us what relative
   loudness we'll perceive given constant amplitudes, low and high
   frequencies will return negative values as we perceive them Xdb less
   than our most sensitive frequency area.  If :invert t, just flip this
   negatives to positives so that if :linear T you get a scaler to make
   lower/higher frequences equally loud as the most sensitive frequencies.
```

RETURN VALUE:

The linear or db weighting value for the given frequency.

EXAMPLE:

```
;;; Decibels:
(a-weighting 50 :invert nil :linear nil) => -30.274979
(a-weighting 50 :invert t :linear nil) => 30.274979
;;; Linear amplitude scalers:
(a-weighting 50) => 32.639904
(a-weighting 50 :invert nil) => 0.030637344
;;; Exaggeration:
(a-weighting 50 :expt 1.1) => 46.251286
;;; Smoothing:
(a-weighting 50 :expt .5) => 5.7131343

;;; Looping through the MIDI note range by tritones returning decibel values:
(loop for midi from 0 to 127 by 6
     for freq = (midi-to-freq midi)
     collect (list (midi-to-note midi)
                   (a-weighting freq :linear nil :invert nil)))
=>
((C-1 -76.85258) (FS-1 -65.94491) (C0 -55.819363) (FS0 -46.71565)
 (C1 -38.714867) (FS1 -31.724197) (C2 -25.598646) (FS2 -20.247103)
 (C3 -15.622625) (FS3 -11.657975) (C4 -8.258142) (FS4 -5.358156)
 (C5 -2.9644737) (FS5 -1.1277018) (C6 0.13445985) (FS6 0.8842882) (C7 1.226917)
 (FS7 1.2351798) (C8 0.89729404) (FS8 0.09495151) (C9 -1.3861179)
 (FS9 -3.7814288))

;;; Similar but returning linear amplitude scalers:
(loop for midi from 0 to 127 by 6
     for freq = (midi-to-freq midi)
     collect (list (midi-to-note midi) (a-weighting freq)))
=>
((C-1 6960.316) (FS-1 1982.6475) (C0 617.9711) (FS0 216.6619) (C1 86.246864)
 (FS1 38.56647) (C2 19.051636) (FS2 10.288571) (C3 6.041312) (FS3 3.827355)
 (C4 2.5876594) (FS4 1.8531382) (C5 1.4067719) (FS5 1.1386365) (C6 0.9846389)
 (FS6 0.9032034) (C7 0.8682687) (FS7 0.86744314) (C8 0.9018521) (FS8 0.9891278)
 (C9 1.1730213) (FS9 1.5455086))
```

SYNOPSIS:

```
(defun a-weighting (f &key expt (linear t) (invert t))
```

22.2 utilities/agnostic-directory-pathname

[utilities] [Functions]

DATE:

```
March 17th 2024.
```

DESCRIPTION:

```
A replacement for directory-namestring, as that function does not return
device names when used on windows.
```

ARGUMENTS:

```
a string representing a pathname.
```

RETURN VALUE:

```
a pathname - to a directory without filename and type
```

SYNOPSIS:

```
(defun agnostic-directory-pathname (namestring)
```

22.3 utilities/all-members

[utilities] [Functions]

DESCRIPTION:

```
Find out whether the members of the list given as the second argument are
all present in the list given as the first argument.
```

ARGUMENTS:

```
- A list in which the members of the second argument will be sought.
- A list whose members will be sought in the first argument.

OPTIONAL ARGUMENT
- A comparison function.
```

RETURN VALUE:

```
T or NIL.
```

EXAMPLE:

```
(all-members '(1 2 3 4 5 6 7) '(1 2 3 7))

=> T
```

SYNOPSIS:

```
(defun all-members (list test-list &optional (test #'equal))
```

22.4 utilities/almost-flatten

[utilities] [Functions]

DATE:

```
September 4th 2013
```

DESCRIPTION:

```
Similar to flatten but allows one level of nesting
```

ARGUMENTS:

```
A list with an arbitrary level of nesting.
```

RETURN VALUE:

```
A list with a maximum of one level of nesting
```

EXAMPLE:

```
(almost-flatten '((1 (2 3 4) (5 (6 7) (8 9 10 (11) 12)) 13) 14 15 (16 17)))
=> (1 (2 3 4) 5 (6 7) 8 9 10 (11) (12) (13) 14 15 (16 17))
```

SYNOPSIS:

```
(defun almost-flatten (nested-list)
```

22.5 utilities/almost-zero

[utilities] [Functions]

DESCRIPTION:

```
Return T if a given decimal is within 0.000001 of 0.0.
```

ARGUMENTS:

```
- A number.
```

OPTIONAL ARGUMENTS:

```
- A number that is a user-specified difference for the comparison test.
```

RETURN VALUE:

```
T if the number is within the tolerance difference to zero, otherwise NIL.
```

EXAMPLE:

```
(almost-zero 0.0000007)

=> T
```

SYNOPSIS:

```
(defun almost-zero (num &optional (tolerance 0.000001))
```

22.6 utilities/amp2db

[utilities] [Methods]

DESCRIPTION:

```
Convert a standard digital amplitude value (>0.0 to 1.0) to a corresponding
decibel value.
```

ARGUMENTS:

```
- A decimal number between >0.0 and 1.0.
```

RETURN VALUE:

```
A decimal number that is a value in decibel.
```

EXAMPLE:

```
(amp2db 0.3)

=> -10.457575
```

SYNOPSIS:

```
(defmacro amp2db (amp)
```

22.7 utilities/amplitude-to-dynamic

[utilities] [Functions]

DESCRIPTION:

```
Convert a specified digital amplitude between 0.0 and 1.0 to a
corresponding dynamic between niente and ffff.
```

ARGUMENTS:

```
- A decimal number between 0.0 and 1.0.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether to print a warning if the specified
  amplitude is <0.0 or >1.0. T = warn. Default = T.
```

RETURN VALUE:

```
A symbol that is a dynamic level.
```

EXAMPLE:

```
(amplitude-to-dynamic 0.3)

=> PP
```

SYNOPSIS:

```
(defun amplitude-to-dynamic (amp &optional (warn t))
```

22.8 utilities/auto-scale-env

[utilities] [Functions]

DATE:

August 29th 2013

DESCRIPTION:

Automatically scale both the x and y values of an envelope to fit within the given ranges. Normally we'll assume that the minimum and maximum Y values are present in the original envelope and so the automatically scaled envelope will represent these with the new minimum and maximum values. However sometimes an envelope doesn't range over the possible extremes, for example (0 .3 100 .6) where the y range is from 0 to 1. If this is the case and you need a scaled envelope to take this into account, then how is the original envelopes minimum and maximum values to the keyword argument :orig-y-range.

ARGUMENTS:

- The envelope: a list of x y pairs

OPTIONAL ARGUMENTS:

keyword arguments:
- :x-min: The new minimum (starting) x value
- :x-max: The new maximum (last) x value
- :y-min: The new minimum (not necessarily starting!) y value
- :y-max: The new maximum (not necessarily starting!) y value
- :orig-y-range: a two-element list specifying the original envelope's minimum and maximum values (see above).

RETURN VALUE:

The new envelope (list).

EXAMPLE:

```
(auto-scale-env '(0 0 10 1))
=>
(0.0 0.0 100.0 10.0)

(auto-scale-env '(-1 0 .3 -3 1 1) :y-min 5 :y-max 6 :x-min 2)
=>
(2.0 5.75 65.7 5.0 100.0 6.0))

(auto-scale-env '(0 1 5 1.5 7 0 10 1) :y-min -15 :y-max -4)
=>
(0.0 -7.6666665 50.0 -4.0 70.0 -15.0 100.0 -7.6666665))
```

```
(auto-scale-env '(0 .5 100 .5) :y-min 1 :y-max 2)
=> (0.0 1.0 100.0 1.0)

(auto-scale-env '(0 .5 100 .5) :y-min 1 :y-max 2 :orig-y-range '(0 1))
=> (0.0 1.5 100.0 1.5)
```

SYNOPSIS:

```
(defun auto-scale-env (env &key
                           (x-min 0.0) (x-max 100.0)
                           (y-min 0.0) (y-max 10.0)
                           orig-y-range)
```

22.9 utilities/average

[utilities] [Functions]

DESCRIPTION:

```
Get the average value of a list of numbers.
```

ARGUMENTS:

```
- the list of numbers
```

RETURN VALUE:

```
the average of the list, as a float
```

SYNOPSIS:

```
(defun average (num-list)
```

22.10 utilities/between

[utilities] [Functions]

DESCRIPTION:

```
Return a random number between two specified numbers. If the two numbers are
integers, the random selection is inclusive. If either are floating-point
(decimal) numbers, the result will be a float between the first (inclusive)
and just less than the second (i.e. exclusive).
```

ARGUMENTS:

```
- A first, lower, number.
- A second, higher, number.

NB: The first number must always be lower than the second.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether the random seed should be fixed.
- T or NIL to indicate whether, when fixed-random is set to T, we should
  reset the random number generator (to guarantee the same random
  sequences). This would generally only be called once, perhaps at the
  start of a generation procedure.
```

RETURN VALUE:

```
An integer if both numbers are integers, or a float if one or both are
decimal numbers.
```

EXAMPLE:

```
;;; Using the defaults. This will produce a different result each time.
(loop repeat 10 collect (between 1 100))

=> (43 63 26 47 28 2 99 93 66 23)

;;; Setting fixed-random to T and using zerop to reset the random when i is 0
(loop repeat 5
   collect (loop for i from 0 to 9 collect (between 1 100 t (zerop i))))

=> ((93 2 38 81 43 19 70 18 44 26) (93 2 38 81 43 19 70 18 44 26)
    (93 2 38 81 43 19 70 18 44 26) (93 2 38 81 43 19 70 18 44 26)
    (93 2 38 81 43 19 70 18 44 26))
```

SYNOPSIS:

```
(defun between (low high &optional fixed-random restart)
```

22.11 utilities/between-extremes

[utilities] [Functions]

DATE:

April 23rd 2016, Edinburgh

DESCRIPTION:

Given a <progress> value of between 0 and 1, we'll return whatever that proportion is of the difference between <min> and <max> added to min. NB No randomness here.

ARGUMENTS:

- the minimum value (returned when <progress> is 0.0)
- the maximum value (returned when <progress> is 1.0)
- a value between

RETURN VALUE:

a number between min and max

EXAMPLE:

```
(BETWEEN-EXTREMES 0.5 1 0.5)
==> 0.75
(BETWEEN-EXTREMES 0.5 1 0.9)
==> 0.95
```

SYNOPSIS:

```
(defun between-extremes (min max progress)
```

22.12 utilities/cartesian-to-polar

[utilities] [Functions]

DESCRIPTION:

Convert a point in a 3D coordinate space from the cartesian system to polar coordinates. This differs from the normal definition of this conversion, in that the elevation is the angle from true horizontal, not vertical...

ARGUMENTS:

- The x-coordinate
- The y-coordinate
- The z-coordinate

RETURN VALUE:

A list that holds the angle (azimuth), elevation and distance of the point.

EXAMPLE:

```
(cartesian-to-polar 0 0 1)

=> (0 90 1)
```

SYNOPSIS:

```
(defun cartesian-to-polar (x y z)
```

22.13 utilities/centre-list

[utilities] [Functions]

DATE:

March 9th 2021, Heidhausen

DESCRIPTION:

Take a list of numbers and scale them so that they are symmetrical(ish) around either the mid-point (exactly) or the middle element (in terms of value).

ARGUMENTS:

- a list of numbers

OPTIONAL ARGUMENTS:

T or NIL to indicate whether the middle element should the zero point from which the other elements are offset or (if NIL) to use the calculated middle point. So if T, we won't centre around 0 but if NIL we will (i.e. if NIL we'll go equally as far in the negative direction as positive). In each case however, the returned list will by necessity range from negative to positive values.

RETURN VALUE:

A list of numbers.

EXAMPLE:

```
(centre-list '(1 2 3 4 5 6) t)
-> (-3 -2 -1 0 1 2)
(centre-list '(1 2 3 4 5 6) NIL)
-> (-2.5 -1.5 -0.5 0.5 1.5 2.5)
(centre-list '(12.2 -11 7 13 14 15 16 -2) nil)
-> '(9.7 -13.5 4.5 10.5 11.5 12.5 13.5 -4.5)
```

SYNOPSIS:

```
(defun centre-list (list &optional zero)
```

22.14 utilities/combine-into-symbol

[utilities] [Functions]

DESCRIPTION:

```
Combine a sequence of elements of any combination of type string, number,
or symbol into a symbol.
```

ARGUMENTS:

```
- A sequence of elements.
```

RETURN VALUE:

```
A symbol as the primary value, with the length of that symbol as a
secondary value.
```

EXAMPLE:

```
(combine-into-symbol "test" 1 'a)

=> TEST1A, 6
```

SYNOPSIS:

```
(defun combine-into-symbol (&rest params)
```

22.15 utilities/convert-polar-envelopes

[utilities] [Functions]

DESCRIPTION:

```
Convert a set of an angle-env and elevation-env into an envelope for the
x, y and z coordinates. Distance is assumed to be 1, if no additional
distance-env is given. The x axis represents left (-1) and right (+1).
The y axis is front (+1) to back (-1), z goes up (+1) to head-level (0).
```

ARGUMENTS:

```
- An angle-env
- An elevation-env
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
:distance-env. A distance-env...
:minimum-samples. A number - minimal amount of points between first and
last point of the envelopes at which to convert. If nil, only the original
points of the envelopes are used, this however doesn't always fully
represent the envelopes... Going from 0° to 180° is something else than
going from y = 1 to y = -1.
```

RETURN VALUE:

```
A list that holds the three envelopes for x, y and z
```

EXAMPLE:

```
(convert-polar-envelopes '(0 0  1 180) '(0 30  .5 0  1 45) :minimum-samples 5)

=> (0.0 0.0 25 0.68301266 50.0 1.0 75 0.65328145 100.0 8.6595606e-17)
=> (0.0 0.8660254 25 0.68301266 50.0 6.123234e-17 75 -0.65328145 100.0 -0.70710677)
=> (0.0 0.5 25 0.25881904 50.0 0.0 75 0.38268343 100.0 0.70710677)
```

SYNOPSIS:

```
(defun convert-polar-envelopes (angle-env elevation-env
                                &key (distance-env '(0 1 1 1))
                                  minimum-samples)
```

22.16 utilities/db2amp

[utilities] [Functions]

DESCRIPTION:

```
Convert a decibel value to a standard digital amplitude value (>0.0 to 1.0),
whereby 0dB = 1.0.
```

ARGUMENTS:

```
- A number that is a value in decibel.
```

RETURN VALUE:

```
A decimal number between >0.0 and 1.0.
```

EXAMPLE:

```
(db2amp -3)

=> 0.70794576
```

SYNOPSIS:

```
(defmacro db2amp (db)
```

22.17 utilities/decider

[utilities] [Functions]

DATE:

```
February 23rd 2023
```

DESCRIPTION:

```
Return an index, which can be used to select an element from a sequence,
when provided with a list of weights (see the example). It does that by
scaling the selector argument relative to the sum of all weights, using
rescale. Then it goes through all the weights and as soon as the selector
is smaller than the sum of the weights so far, the index of the current
weight is returned. So when given a list of weights '(2 1), the following
selectors will return:
0     => 0
1/3   => 0
19/30 => 0
2/3   => 1
1     => 1
This process is thus deterministic. By providing a random number as a
selector, you can make random choices etc.
```

ARGUMENTS:

```
 - A number between 0 and 1
 - A list of numbers, representing weights
```

RETURN VALUE:

```
 index of chosen element
```

EXAMPLE:

```
;;; simple example, choosing from a list:
(let* ((ls '(c4 d4 e4 f4 g4 a4 b4))
       (weights '(1 1 2 2 3 1 2)))
  (nth (decider 0.1 weights) ls))
=> d4

;;; making a simple melody, following a sine wave:
(let* ((ls '(c4 d4 e4 f4 g4 a4 b4))
       (weights '(1 1 2 2 3 1 2)))
  (loop for i from 0 to pi by 0.25 collect
       (nth (decider (abs (sin i)) weights) ls)))
=> (C4 E4 F4 G4 B4 B4 B4 B4 B4 A4 G4 F4 D4)

;;; make a random melody with 10 pitches:
(let* ((ls '(c4 d4 e4 f4 g4 a4 b4))
       (weights '(1 1 2 2 3 1 2)))
  (loop repeat 10 collect
       (nth (decider (random 1.0) weights) ls)))
```

SYNOPSIS:

```
(defun decider (selector weights)
```

22.18 utilities/decimal-places

[utilities] [Functions]

DATE:

```
 19-Mar-2012
```

DESCRIPTION:

```
 Round the given number to the specified number of decimal places.
```

ARGUMENTS:

- A number.
- An integer that is the number of decimal places to which to round the given number.

RETURN VALUE:

A decimal number.

EXAMPLE:

```
(decimal-places 1.1478349092347 2)

=> 1.15
```

SYNOPSIS:

```
(defun decimal-places (num places)
```

22.19 utilities/decimate-env

[utilities] [Functions]

DESCRIPTION:

Reduce the number of x,y pairs in an envelope. In every case the envelope is first stretched along the x-axis to fit the new number of points required. Then we proceed by one of three methods:

1) average: for every new output x value, interpolate 100 times from -0.5 to +0.5 around the point, then average the y value. This will catch clustering but round out spikes caused by them

2) points: also an averaging method but only using the existing points in the original envelope (unless none is present for a new x value, whereupon interpolation is used): Take an average of the (several) points nearest the new output point. This might not recreate the extremes of the original envelope but clustering is captured, albeit averaged.

3) interpolate: for each new output point, interpolate the new y value from the original envelope. This will leave out details in the case of clustering, but accurately catch peaks if there are enough output points.

In each case we create an even spread of x values, rather than clustering where clusters exist in the original.

ARGUMENTS:

```
- the original envelope (list of x,y values on any scales).
- the number of points required in the output list.
```

OPTIONAL ARGUMENTS:

```
- the method to be applied (symbol): 'points, 'average, 'interpolate.
  Default = 'points.
```

RETURN VALUE:

```
A list representing the x,y values of the new envelope
```

EXAMPLE:

```
(decimate-env '(0 0 4 4 5 5 5.1 5.1 5.3 1 5.6 5.6 6 6 10 10) 6)
=>
(0.0 0.0 1 2.0 2 4.5 3 4.425 4 8.0 5.0 10.0)
```

SYNOPSIS:

```
(defun decimate-env (env num-points &optional (method 'points))
```

22.20 utilities/degree-to-radian

[utilities] [Functions]

DESCRIPTION:

```
Convert an angle in degrees to its equivalent in radians
```

ARGUMENTS:

```
- The number in degrees
```

RETURN VALUE:

```
The number in radians
```

EXAMPLE:

```
(degree-to-radian 180)

=> 3.141592653589793d0
```

SYNOPSIS:

```
(defun degree-to-radian (degree)
```

22.21 utilities/down-up

[utilities] [Functions]

DATE:

May 8th 2016

DESCRIPTION:

This is a routine used in morphing maps but may be useful elsewhere. It interpolates between two numbers over a given number of steps before returning back to the first number.

ARGUMENTS:

- the number of steps over which the procedure should interpolate

OPTIONAL ARGUMENTS:

keyword arguments:
- :down. T or NIL: whether to first descend before ascending. Default = T
- :up. T or NIL: whether to ascend after descending. Default = T
- :start. The number to start at. Default = 1.0
- :target. The number to interpolate towards. Default = 0.0.
- :cons. Whether to return :start as the first number in the result list.
- :butlast. Whether to omit the :start when ascending. Default = T.

RETURN VALUE:

A list of numbers.

EXAMPLE:

```
(mapcar #'(lambda (x) (decimal-places x 2)) (down-up 20))
--> (0.9 0.8 0.7 0.6 0.5 0.4 0.3 0.2 0.1 0.0 0.09 0.18 0.27 0.36 0.45 0.55 0.64
     0.73 0.82 0.91)
(mapcar #'(lambda (x) (decimal-places x 2)) (down-up 20 :cons t))
--> (1.0 0.9 0.8 0.7 0.6 0.5 0.4 0.3 0.2 0.1 0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8
     0.9)
(mapcar #'(lambda (x) (decimal-places x 2)) (down-up 20 :butlast nil))
--> (0.9 0.8 0.7 0.6 0.5 0.4 0.3 0.2 0.1 0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9
     1.0)
```

SYNOPSIS:

```
(defun down-up (steps &key (down t) (up t) (start 1.0d0) (target 0.0d0)
                        (cons nil) (butlast t))
```

22.22 utilities/dynamic-to-amplitude

[utilities] [Functions]

DESCRIPTION:

```
Convert a symbol that is a dynamic level between niente and ffff to a
corresponding digital amplitude value between 0.0 and 1.0.
```

ARGUMENTS:

```
- A symbol that is a dynamic level between niente and fff.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether to print a warning when the symbol specified
  is not recognized as a dynamic. T = warn. Default = T.
```

RETURN VALUE:

```
A decimal number between 0.0 and 1.0.
```

EXAMPLE:

```
(dynamic-to-amplitude 'fff)

=> 0.9
```

SYNOPSIS:

```
(defun dynamic-to-amplitude (dynamic &optional (warn t))
```

22.23 utilities/econs

[utilities] [Functions]

DESCRIPTION:

```
Add a specified element to the end of an existing list.
```

ARGUMENTS:

```
- A list.
- An element to add to the end of the list.
```

RETURN VALUE:

```
A new list.
```

EXAMPLE:

```
(econs '(1 2 3 4) 5)

=>  '(1 2 3 4 5)
```

SYNOPSIS:

```
(defun econs (list new-back)
```

22.24 utilities/edit-file

[utilities] [Functions]

DATE:

```
April 30th 2023.
```

DESCRIPTION:

```
a simple wrapper to open a file as a string and save it into a lexical
variable. This variable is then set to the return values of the expressions
within the body of edit-file. Eventually the new value of the variable is
written into the original file. This was originally designed to be used with
the editing functions in reaper.lsp (set-track-channels etc.).
```

ARGUMENTS:

```
- A string that is the path to a file (directory and filename)
- A name for the lexical variable - this can be used within body
(without quote)
```

RETURN VALUE:

```
whatever was written into the file
```

EXAMPLE:

```
;;; set all faders of project.rpp to 0.5 and use "anything" as lexical variable
(edit-file "/E/project.rpp" anything
```

```
      (set-all-faders anything .5))
|#
#|
;;; more complex: insert a plugin on 3 tracks in two ways
(edit-file "/E/project.rpp" project
  (insert-plugin project *iem-stereo-encoder* 1)
  (insert-plugin project *iem-stereo-encoder* 2)
  (insert-plugin project *iem-stereo-encoder* 3))
;;; alternatively:
(edit-file "/E/project.rpp" project
  (loop for i from 1 to 3 with temp-var = project
     do (setf temp-var (insert-plugin temp-var *iem-stereo-encoder* i))
       finally (return temp-var)))
```

SYNOPSIS:

```
(defmacro edit-file (file var &body body)
```

22.25 utilities/env-mean-y

[utilities] [Functions]

DATE:

```
january 9th 2025
```

DESCRIPTION:

```
Get the mean value of the y values in an envelope considering the breadth of
the x axis. In other words, by sampling the envelope over many evenly spread
points, return the mean y value
```

ARGUMENTS:

```
- the envelope as a list of x y breakpoint
```

OPTIONAL ARGUMENTS:

```
- how many points to use to sample the envelope. Default = 1000.
```

RETURN VALUE:

```
the mean y value: a number
```

EXAMPLE:

```
(env-mean-y '(0 0 100 1)) = 0.5 (approx)
```

SYNOPSIS:

```
(defun env-mean-y (env &optional (granularity 1000))
```

22.26 utilities/env-plus

[utilities] [Functions]

DESCRIPTION:

```
Increase all y values of a given list of break-point pairs by a specified
amount.
```

ARGUMENTS:

```
- An envelope in the form of a list of break-point pairs.
- A number that is the amount by which all y values of the given envelope
  are to be increased.
```

RETURN VALUE:

```
A list of break-point pairs.
```

EXAMPLE:

```
(env-plus '(0 0 25 11 50 13 75 19 100 23) 7.1)

=> (0 7.1 25 18.1 50 20.1 75 26.1 100 30.1)
```

SYNOPSIS:

```
(defun env-plus (env add)
```

22.27 utilities/env-symmetrical

[utilities] [Functions]

DESCRIPTION:

```
Create a new list of break-point pairs that is symmetrical to the original
around a specified center. If no center is specified, the center value
defaults to 0.5
```

ARGUMENTS:

```
- An envelope in the form of a list of break-point pairs.
```

OPTIONAL ARGUMENTS:

```
- A number that is the center value around which the values of the
  new list are to be symmetrical.
- A number that is to be the minimum value for the y values returned.
- A number that is to be the maximum value for the y values returned.
```

RETURN VALUE:

```
An envelope in the form of a list of break-point pairs.
```

EXAMPLE:

```
;;; Default center is 0.5
(env-symmetrical '(0 0 25 11 50 13 75 19 100 23))

=> (0 1.0 25 -10.0 50 -12.0 75 -18.0 100 -22.0)

;; Specifying a center of 0
(env-symmetrical '(0 0 25 11 50 13 75 19 100 23) 0)

=> (0 0.0 25 -11.0 50 -13.0 75 -19.0 100 -23.0)

;;; Specifying minimum and maximum y values for the envelope returned
(env-symmetrical '(0 0 25 11 50 13 75 19 100 23) 0 -20 -7)

=> (0 -7 25 -11.0 50 -13.0 75 -19.0 100 -20)
```

SYNOPSIS:

```
(defun env-symmetrical (env &optional (centre .5)
                        (min most-negative-double-float)
                        (max most-positive-double-float))
```

22.28 utilities/env2gnuplot

[utilities] [Functions]

DATE:

```
24th December 2013
```

DESCRIPTION:

```
Write a data file of x,y envelope values for use with gnuplit.  Once called
start gnuplot and issue commands such as:
gnuplot> set terminal postscript default
gnuplot> set output '/tmp/env.ps'
gnuplot> plot '/tmp/env.txt' with lines.
```

ARGUMENTS:

```
- The envelope as the usual list of x y pairs
```

OPTIONAL ARGUMENTS:

```
- The pathname of the data file to write.  Default = "/tmp/env.txt".
```

RETURN VALUE:

```
Always T
```

SYNOPSIS:

```
(defun env2gnuplot (env &optional (file "/tmp/env.txt"))
```

22.29 utilities/envelope-boundaries

[utilities] [Functions]

DESCRIPTION:

```
Find sharp changes in envelope values. These are defined as when a y value
rises or falls over 30% (by default) of it's overall range within 5%
(again, by default) of its overall x axis range.
```

ARGUMENTS:

```
The envelope (a list of x y pairs).
```

OPTIONAL ARGUMENTS:

```
- jump-threshold: the minimum percentage change in y value that is deemed a
  sharp change.
- steepness-min: the maximum percentage of the overall x axis that
  constitutes a 'quick' change.
```

RETURN VALUE:

A list of x values at which boundaries are deemed to lie.

EXAMPLE:

```
(ENVELOPE-BOUNDARIES '(0 10 20 10 21 3 25 4 26 9 50 7 51 1 55 2 56 7 70 10
                     100 10))
--> (21 26 51 56)
```

SYNOPSIS:

```
(defun envelope-boundaries (envelope &optional (jump-threshold 30)
                            (steepness-min 5))
```

22.30 utilities/equal-within-tolerance

[utilities] [Functions]

DESCRIPTION:

Test whether the difference between two decimal numbers falls within a specified tolerance.

This test is designed to compensate for calculation discrepancies caused by floating-point errors (such as 2.0 vs. 1.9999997), in which the equations should yield equal numbers. It is intended to be used in place of = in such circumstances.

ARGUMENTS:

- A first number.
- A second number.

OPTIONAL ARGUMENTS:

- A decimal value that is the maximum difference allowed between the two numbers that will still return T. Default = 0.000001d0.

RETURN VALUE:

T if the two tested numbers are equal within the specified tolerance, otherwise NIL.

EXAMPLE:

```
;; An example of floating-point error
(loop for i from 0.0 below 1.1 by 0.1 collect i)

=> (0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.70000005 0.8000001 0.9000001 1.0000001)

;; Using =
(loop for i from 0.0 below 1.1 by 0.1
   for j in '(0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0)
   collect (= i j))

=> (T T T T T T T NIL NIL NIL NIL)

;; Using equal-within-tolerance
(loop for i from 0.0 below 1.1 by 0.1
   for j in '(0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0)
   collect (equal-within-tolerance i j))

=> (T T T T T T T T T T T)
```

SYNOPSIS:

```
(defun equal-within-tolerance (a b &optional (tolerance 0.000001d0))
```

22.31 utilities/exaggerate-env

[utilities] [Functions]

DESCRIPTION:

```
Makes the y values in an envelope more radically pushed towards its
extremes. Y values below the mid-point will be pushed downwards; those
above will be pushed upwards. The opposite can be accomplished by making
the exponent argument > 1 (see below).
```

ARGUMENTS:

```
- the envelope: a list of numbers representing an envelope: x y pairs
- the exponent: this determines the amount of
  exaggeration. Counterintuitively perhaps, the lower values are than 1 the
  more exaggeration takes place. Values > 1 will mean the opposite:
  understated y values, if you will.
```

OPTIONAL ARGUMENTS:

```
- easy-expt: because of the counterintuitive nature of the exponent, you
```

```
can pass values between -10 and +10 if this third argument is T. This
will be scaled to useful though not over-extreme exponents of 1.9 (-10)
to .1 (+10) with 0 equating to an exponent of 1, i.e. no change.
```

RETURN VALUE:

```
The new exaggerated envelope (a list).
```

EXAMPLE:

```
(exaggerate-env '(0 0 50 .8 100 1) 1.9)
--> (0 0.0 50 0.6894338 100 1.0)
(exaggerate-env '(0 0 50 .8 100 1) .1)
--> (0 0.0 50 0.9751001 100 1.0)
(exaggerate-env '(0 0 50 .8 100 1) -10)
--> (0 0.0 50 83.19083 100 1.0)
(exaggerate-env '(0 0 50 .8 100 1) -10 t)
--> (0 0.0 50 0.6894338 100 1.0)
(exaggerate-env '(0 0 50 .8 100 1) 0 t)
--> (0 0.0 50 0.8 100 1.0)
(exaggerate-env '(0 0 50 .8 100 1) 10 t)
--> (0 0.0 50 0.9751001 100 1.0)
```

SYNOPSIS:

```
(defun exaggerate-env (env expt &optional easy-expt)
```

22.32 utilities/factor

[utilities] [Functions]

DESCRIPTION:

```
Boolean test to check if a specified number is a multiple of a second
specified number.
```

ARGUMENTS:

```
- A number that will be tested to see if it is a multiple of the second
  number.
- A second number that is the base number for the factor test.
```

RETURN VALUE:

```
T if the first number is a multiple of the second number, otherwise NIL.
```

EXAMPLE:

```
(factor 14 7) => T
```

SYNOPSIS:

```
(defun factor (num fac)
```

22.33 utilities/filter-parameters

[utilities] [Functions]

DATE:

```
July 1st 2021, Heidhausen
```

DESCRIPTION:

```
Though generally applicable, a reaper marker specifically could look like
this:  ("MARKER" (67 709.726 "b364 F1 molto pesante" 0 0 1 B)) Allow
therefore searching of the results from (get-parameters ...) for specific
strings, returning whole parameter lists or specific elements thereof. NB
The strings are case-insensitive.
```

ARGUMENTS:

```
- a list of parameters, where each element is a 2-element list: the
parameter name and a list of associated data. I.e. this implies e.g. calling
get-parameters with the optional to-line-end set to T
- the string to search for in the data lists
```

OPTIONAL ARGUMENTS:

```
- a number to specify the nth element that will be returned from the data
lists
```

RETURN VALUE:

```
a list of matching data lists
```

EXAMPLE:

```
(filter-parameters
  (get-parameters "markers.RPP" '("MARKER") #\  t)
```

```
  "harmonic" 1)
-->
85 parameters read
(322.7273 527.37866 542.9216 686.73846 944.6802 952.6701)

;;; or without the nth option:

(filter-parameters
 (get-parameters "markers.RPP" '("MARKER") #\  t)
 "harmonic")
-->
((29 322.7273 "c harmonic resonance" 0 0 1 B
  {8F7A97DD-F0B6-4E4D-9F67-27635F109E05})
 (55 527.37866 "nice resonant harmonic pizz" 0 0 1 B
  {E380FEF5-FE44-3B47-9C43-2BA80A3D0FAC})
 (57 542.9216 "nice harmonic S trem" 0 0 1 B
  {04BB89FE-8FB8-5B4A-AB5F-EDE1F9C921CC})
 (65 686.73846 "b347 E7 nice single harmonic repeat" 0 0 1 B
  {586CDD49-BE23-F149-AABC-A98EDD6C6EEE})
 (79 944.6802 "b483: nice spe and harmonics" 0 0 1 B
  {11E361AD-22C8-0D42-AB91-ECCBCAAEA075})
 (80 952.6701 "b491 muted harmonics" 0 0 1 B
  {097D7A4D-E9F0-FE4B-8BE6-E7BC85E2969B}))
```

SYNOPSIS:

```
(defun filter-parameters (parameters string &optional get-nth)
```

22.34 utilities/flatten

[utilities] [Functions]

DESCRIPTION:

```
 Return a list of nested lists of any depth as a flat list.
```

ARGUMENTS:

```
 - A list of nested lists.
```

RETURN VALUE:

```
 A flat list.
```

EXAMPLE:

```
(flatten '((1 (2 3 4) (5 (6 7) (8 9 10 (11) 12)) 13) 14 15 (16 17)))

=> (1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17)
```

SYNOPSIS:

```
(defun flatten (nested-list)
```

22.35 utilities/force-length

[utilities] [Functions]

DATE:

```
03-FEB-2011
```

DESCRIPTION:

```
Create a new a list of a specified new length by adding or removing items
at regular intervals from the original list. If adding items and the list
contains numbers, linear interpolation will be used, but only between two
adjacent items; i.e. not with a partial increment.

NB: The function can only create new lists that have a length between 1 and
    1 less than double the length of the original list.
```

ARGUMENTS:

```
- A flat list.
- A number that is the new length of the new list to be derived from the
  original list. This number must be a value between 1 and 1 less than
  double the length of the original list.
```

RETURN VALUE: EXAMPLE:

```
;;; Shortening a list
(force-length (loop for i from 1 to 100 collect i) 17)

=> (1 7 13 20 26 32 39 45 51 57 63 70 76 82 89 95 100)

;;; Lengthening a list
(force-length (loop for i from 1 to 100 collect i) 199)

=> (1 1.5 2 2.5 3 3.5 4 4.5 5 5.5 6 6.5 7 7.5 8 8.5 9 9.5 10 10.5 11 11.5 12
    12.5 13 13.5 14 14.5 15 15.5 16 16.5 17 17.5 18 18.5 19 19.5 20 20.5 21
```

```
21.5 22 22.5 23 23.5 24 24.5 25 25.5 26 26.5 27 27.5 28 28.5 29 29.5 30
30.5 31 31.5 32 32.5 33 33.5 34 34.5 35 35.5 36 36.5 37 37.5 38 38.5 39
39.5 40 40.5 41 41.5 42 42.5 43 43.5 44 44.5 45 45.5 46 46.5 47 47.5 48
48.5 49 49.5 50 50.5 51 51.5 52 52.5 53 53.5 54 54.5 55 55.5 56 56.5 57
57.5 58 58.5 59 59.5 60 60.5 61 61.5 62 62.5 63 63.5 64 64.5 65 65.5 66
66.5 67 67.5 68 68.5 69 69.5 70 70.5 71 71.5 72 72.5 73 73.5 74 74.5 75
75.5 76 76.5 77 77.5 78 78.5 79 79.5 80 80.5 81 81.5 82 82.5 83 83.5 84
84.5 85 85.5 86 86.5 87 87.5 88 88.5 89 89.5 90 90.5 91 91.5 92 92.5 93
93.5 94 94.5 95 95.5 96 96.5 97 97.5 98 98.5 99 99.5 100)
```

SYNOPSIS:

```
(defun force-length (list new-len)
```

22.36 utilities/force-symmetrical-and-normalise

[utilities] [Functions]

DATE:

```
February 8th 2022
```

DESCRIPTION:

```
Take a list of numbers (usually samples but any of course) and, using the
average sample value, offset them to be symmetrical around 0.0
(usually). This is not the same as but is related to removing DC offset:
here we're offsetting numbers individually rather than using the
high-pass filter approach in the DSP time domain.

Note that to make the samples symmetrical we offset by the average of the
existing samples so any occasional outliers will still spike in one
direction or another, thus making a waveform view seem to be still offset.

Optionally also normalise to within -1.0 and 1.0 (or other values: see
below). If either :min or :max are nil then normalisation won't be applied.
```

ARGUMENTS:

```
a list of floats (double precision)
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
:min. The new minimum value to map to. Default = -1.0
```

```
:max. The new minimum value to map to. Default = 1.0
:verbose. Print some stats. Default = NIL.
```

RETURN VALUE:

```
the list of new samples
```

SYNOPSIS:

```
(defun force-symmetrical-and-normalise (samples &key (min -1.0)
                                                     (max 1.0) verbose)
```

22.37 utilities/get-clusters

[utilities] [Functions]

DESCRIPTION:

```
Takes a list with (ascending) numbers and creates sublists of those numbers
within <threshold> of each other.
```

ARGUMENTS:

```
A list of (ascending) numbers. NB Though the numbers don't have to be in
ascending order, the design application of the function makes most sense if
they are.
```

OPTIONAL ARGUMENTS:

```
The maximum distance between two numbers in order for them to be considered
as part of the same cluster.
```

RETURN VALUE:

```
A list with clusters in sublists.
```

EXAMPLE:

```
(get-clusters '(24 55 58 59 60 81 97 102 106 116 118 119 145 149 151 200 210
                211 214 217 226 233 235 236 237 238 239 383 411 415 419))
--> (24 (55 58 59 60) 81 (97 102 106) (116 118 119) (145 149 151) 200
        (210 211 214 217) 226 (233 235 236 237 238 239) 383 (411 415 419))

(get-clusters '(0 .1 .3 .7 1.5 1.55 2 4.3 6.3 6.4) 1)
```

```
--> ((0 0.1 0.3 0.7 1.5 1.55 2) 4.3 (6.3 6.4))

(get-clusters '(0 .1 .3 .7 1.5 1.55 2 4.3 6.3 6.4) 0.5)
--> ((0 0.1 0.3 0.7) (1.5 1.55 2) 4.3 (6.3 6.4))
```

SYNOPSIS:

```
(defun get-clusters (list &optional (threshold 5))
```

22.38 utilities/get-harmonics

[utilities] [Functions]

DESCRIPTION:

```
Return a list of the harmonic partial frequencies in Hertz from a
specified (usually fundamental) frequency.
```

ARGUMENTS:

```
- A number that is the fundamental or starting frequency in Hertz.
```

OPTIONAL ARGUMENTS:

```
keyword arguments
- :start-partial. An integer that is the number of the first harmonic
   partial to return. Default = 1.
- :min-freq. A number that is the lowest frequency in Hertz to
  return. Default = 20.
- :max-freq. A number that is the highest frequency in Hertz to
  return. Default = 20000.
- :start-freq-is-partial.  Rather than treating the first argument as the
  fundamental, treat it as the partial number indicated by this argument.
  Default = 1.
- :max-results.  The maximum number of harmonics to return.  Default =
   most-positive-fixnum
- :skip. The increment for the harmonics.  If 1, then we ascend the
   harmonics series one partial at a time; 2 would mean skipping every other
   Default = 1.
- :pitches. Return a list of pitch objects instead of frequencies. Default =
  NIL.
- :notes. Return a list of 2-element sublists: note symbols in the
  chromatics scale, with cent deviations
```

RETURN VALUE:

```
A list of numbers that are the frequencies in Hertz of harmonic partials
above the same fundamental frequency, or with the respective keyword, as
pitch objects or note symbols
```

EXAMPLE:

```
;;; Get the first 15 harmonic partials above a fundamental pitch of 64 Hertz,
;;; starting with partial 2, and specifying an upper cut-off of 1010 Hz.

(get-harmonics 63 :start-partial 2 :max-freq 1010)

=> (126 189 252 315 378 441 504 567 630 693 756 819 882 945 1008)
```

SYNOPSIS:

```
(defun get-harmonics (start-freq &key (start-partial 1) (min-freq 20)
                                      (start-freq-is-partial 1) (max-freq 20000)
                                      (skip 1)
                                      pitches notes
                                      (max-results most-positive-fixnum))
```

22.39 utilities/get-parameters

[utilities] [Functions]

DATE:

```
January 29th 2021
```

DESCRIPTION:

```
A general routine for searching text files for parameters and their
values. Here we search a file line by line, matching parameters and
returning them in a list of parameter-value pairs. This is limited, however,
to one parameter per line, and values of one word (i.e. numbers, strings,
etc. not containing space), unless optional argument to-line-end is T (see
below)
```

ARGUMENTS:

```
- the text file to search
- either a single string or list thereof to search for (case-sensitive)
- the separator character which divides the parameter name from its value
```

OPTIONAL ARGUMENTS:

```
 - the parameter-value separator (character)
 - T or NIL to indicate whether all tokens after a recognised parameter
   should be returned in a list. Default = NIL = get just one parameter.
```

RETURN VALUE:

```
 A list of parameter-value pairs
```

EXAMPLE:

```
;;; search a reaper file, where parameters are followed simply by space rather
;;; than = or : E.g. a current reaper file has lines like:
      SNAPOFFS 0
      LENGTH 0.36292517006803
      LOOP 1
      ALLTAKES 0
      FADEIN 2 0 0 2 0 1 1
      FADEOUT 2 0 0 2 0 -1 -1
      MUTE 0 0
      MIXFLAG 1
      BEAT 2
      SEL 1
      IGUID {3B79D8DF-AC08-EC4F-B93C-CAFE24FA1CBB}
      IID 3
      NAME sunni-mosque.wav
      VOLPAN 1 0 1 -1
      SOFFS 0.67933106575964
      PLAYRATE 1 1 0 -1 0 0.0025
;;; hence:
(get-parameters "~/projects/sndfilenet/reaper/sunni-mosque-split.RPP"
                '("SOFFS" "LENGTH") #\ )
-->
362 parameters read
(("LENGTH" 0.36292517) ("SOFFS" 0.67933106) ("LENGTH" 0.38848072)
 ("SOFFS" 1.0422562) ("LENGTH" 1.4923356) ("SOFFS" 1.430737)
 ("LENGTH" 1.9968253) ("SOFFS" 2.9230726) ("LENGTH" 0.5023356)
 ("SOFFS" 4.919898) ("LENGTH" 0.4907483) ("SOFFS" 5.4222336)
 ("LENGTH" 0.17068027) ("SOFFS" 5.912982) ("LENGTH" 3.6765532)
...
```

SYNOPSIS:

```
(defun get-parameters (file parameters &optional (separator #\=) to-line-end)
```

22.40 utilities/get-primes

[utilities] [Functions]

DATE:

```
July 6th 2022, Werden
```

DESCRIPTION:

```
Get all the prime numbers between minimum and maximum values
```

ARGUMENTS:

```
- the minimum (start point). Doesn't have to be a prime number.
- the maximum (end point). Doesn't have to be a prime number.
```

OPTIONAL ARGUMENTS:

```
- an integer to specify the maximum number of primes to return. If NIL
  (default) then we'll stop at max
```

RETURN VALUE:

```
a list of prime numbers
```

EXAMPLE:

```
(get-primes 5 19) -> (5 7 11 13 17 19)
(get-primes 5 19 3) -> (5 7 11)
```

SYNOPSIS:

```
(defun get-primes (min max &optional num)
```

22.41 utilities/get-sublist-indices

[utilities] [Functions]

DESCRIPTION:

```
Get the starting position of sublists within a list as though the complete
set of items were a flat list.
```

ARGUMENTS:

```
 - A list of lists.
```

RETURN VALUE:

```
 A list of integers that are the indices of the sublists.
```

EXAMPLE:

```
(get-sublist-indices '((1 2) (3 4 5 6) (7 8 9) (10 11 12 13 14) (15)))

=> (0 2 6 9 14)
```

SYNOPSIS:

```
(defun get-sublist-indices (list)
```

22.42 utilities/get-sublist-lengths

[utilities] [Functions]

DESCRIPTION:

```
 Get the lengths of all sublists in a given list.
```

ARGUMENTS:

```
 - A list of lists.
```

OPTIONAL ARGUMENTS:

```
 - T or NIL to indicate whether to first remove zeros caused by empty
   sublists from the result.
```

RETURN VALUE:

```
 A list of integers.
```

EXAMPLE:

```
;; Straightforward usage allows zeros in the result
(get-sublist-lengths '((1 2) (3 4 5 6) (7 8 9) (10 11 12 13 14) ()))

=> (2 4 3 5 0)
```

```
;; Setting the optional argument to T removes zeros from the result

(get-sublist-lengths '((1 2) (3 4 5 6) (7 8 9) (10 11 12 13 14) ()) t)

=> (2 4 3 5)
```

SYNOPSIS:

```
(defun get-sublist-lengths (list &optional (remove-zeros nil))
```

22.43 utilities/hailstone

[utilities] [Functions]

DESCRIPTION:

```
Implementation of the Collatz conjecture (see
http://en.wikipedia.org/wiki/Collatz_conjecture)

The Collatz conjecture suggests that by starting with a given number, and
if it is even dividing it by two or if it is odd multiplying it by three
and adding one, then repeating with the new result, the process will
eventually always result in one.
```

ARGUMENTS:

```
- A number to start with.
```

RETURN VALUE:

```
A list of the results collected from each iteration starting with the
specified number and ending with one.
```

EXAMPLE:

```
(hailstone 11)

=> (11 34 17 52 26 13 40 20 10 5 16 8 4 2 1)
```

SYNOPSIS:

```
(defun hailstone (n)
```

22.44 utilities/hz2ms

[utilities] [Functions]

DESCRIPTION:

```
Convert a frequency in Hertz to the equivalent number of milliseconds.
```

ARGUMENTS:

```
- A number that is a Hertz frequency.
```

RETURN VALUE:

```
A number that is the millisecond equivalent of the specified Hertz
frequency.
```

EXAMPLE:

```
(hz2ms 261.63)

=> 3.8221915
```

SYNOPSIS:

```
(defun hz2ms (hertz)
```

22.45 utilities/interleave

[utilities] [Functions]

DESCRIPTION:

```
Interleave the elements of an aribitrary number of lists. Should the lists
not be of the same length, this function will only use up as many elements
as in the shortest list.
```

ARGUMENTS:

```
As many lists as need to be interleaved.
```

RETURN VALUE:

```
A new list of interleaved elements.
```

EXAMPLE:

```
(INTERLEAVE '(1 2 3 4 5) '(a b c d) '(x y z))
--> (1 A X 2 B Y 3 C Z)

(INTERLEAVE '(1 2 3 4 5) '(a b c d e) '(v w x y z))
--> (1 A V 2 B W 3 C X 4 D Y 5 E Z)
```

SYNOPSIS:

```
(defun interleave (&rest lists)
```

22.46 utilities/interpolate

[utilities] [Functions]

DESCRIPTION:

```
Get the interpolated value at a specified point within an envelope. The
envelope must be specified in the form of a list of break-point pairs.
```

ARGUMENTS:

```
- A number that is the point within the specified envelope for which to
  return the interpolated value.
- A list of break-point pairs.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :scaler. A number that is the factor by which to scale the values of
  the break-point pairs in the given envelope before retrieving the
  interpolated value. Default = 1.
- :exp. A number that is the exponent to which the result should be
  raised. Default = 1.
- :warn. T or NIL to indicate whether the method should print a warning if
  the specified point is outside of the bounds of the x-axis specified in
  the list of break-point pairs. T = warn. Default = T.
```

RETURN VALUE: EXAMPLE:

```
;;; Using the defaults
(interpolate 50 '(0 0 100 1))
```

```
=> 0.5

;;; Specifying a different scaler
(interpolate 50 '(0 0 100 1) :scaler 2)

=> 1.0

;;; Specifying a different exponent by which the result is to be raised
(interpolate 50 '(0 0 100 1) :exp 2)

=> 0.25
```

SYNOPSIS:

```
(defun interpolate (point env &key (scaler 1) (exp 1) (warn t))
```

22.47 utilities/invert-env

[utilities] [Functions]

DATE:

```
June 15th 2017, Edinburgh
```

DESCRIPTION:

```
Invert an envelope so that its maximum value becomes its minimum,
vice-versa, and everything inbetween.
```

ARGUMENTS:

```
A list of X-Y breakpoint pairs
```

RETURN VALUE:

```
A list of X-Y breakpoint pairs exhibiting the inversion.
```

EXAMPLE:

```
(invert-env '(0 0 100 1)) -> (0 1.0 100 0.0)
(invert-env '(0 .3 40 .4 100 .9)) -> (0 0.9 40 0.79999995 100 0.3)
(invert-env '(0 -.9 40 .4 100 .9)) -> (0 0.9 40 -0.39999998 100 -0.9)
```

SYNOPSIS:

```
(defun invert-env (env)
```

22.48 utilities/list-to-string

[utilities] [Functions]

DESCRIPTION:

Convert a list to a string.

ARGUMENTS:

- A list.

OPTIONAL ARGUMENTS:

- A string that will serve as a separator between the elements.
  Default = " ".
- T or NIL to indicate whether a list value of NIL is to be returned as
  "NIL" or NIL. T = "NIL" as a string. Default = T.

RETURN VALUE: EXAMPLE:

```
;;; Using defaults
(list-to-string '(1 2 3 4 5))

=> "1 2 3 4 5"

;;; Specifying a different separator
(list-to-string '(1 2 3 4 5) "-")

=> "1-2-3-4-5"

;;; A NIL list returns "NIL" as a string by default
(list-to-string NIL)

=> "nil"

;;; Setting the second optional argument to NIL returns a NIL list as NIL
;;; rather than as "NIL" as a string
(list-to-string NIL "" nil)

=> NIL
```

SYNOPSIS:

```
(defun list-to-string (list &optional (separator " ") (nil-as-string t))
```

22.49 utilities/logarithmic-steps

[utilities] [Functions]

DESCRIPTION:

```
Create a list of numbers progressing from the first specified argument to
the second specified argument over the specified number of steps using an
exponential curve rather than linear interpolation.
```

ARGUMENTS:

```
- A number that is the starting value in the resulting list.
- A number that is the ending value in the resulting list.
- An integer that will be the length of the resulting list - 1.
```

OPTIONAL ARGUMENTS:

```
- A number that will be used as the exponent when determining the
  exponential interpolation between values. Default = 2.
```

RETURN VALUE:

```
A list of numbers.
```

EXAMPLE:

```
(logarithmic-steps 1 100 19)

=> (1.0 1.3055556 2.2222223 3.75 5.888889 8.638889 12.0 15.972222 20.555555
    25.75 31.555555 37.97222 45.0 52.63889 60.88889 69.75 79.22222 89.30556
    100.0)
```

SYNOPSIS:

```
(defun logarithmic-steps (low high num-steps &optional (exponent 2))
```

22.50 utilities/make-reaper-envelope

[utilities] [Functions]

DATE:

```
April 30th 2023.
```

DESCRIPTION:

make a reaper-envelope object.

ARGUMENTS:

- a list which represents an envelope, meaning it consists of x and y values
For example: '(0 0 50 1 100 .5)
Usually the first x value would be 0, all following x values should be ascending numbers. The length of the envelope (meaning the value of the last x value isn't important here)
Note that if you use generate-auotmation-data on the resulting reaper-envelope object, y values will range from 0 to 1. The env list can obviously have y values greater than that, but those will be reduced modulo 1 (see env-mod). An envelope like '(0 0 1 3) would then result in something like this:
'(0 0 1/3 .9999999 (+ 1/3 .001) 0 2/3 .9999999 (+ 2/3 .001) 0 1.0 .9999999)
:parameter-min and :parameter-max are applied after that operation.

OPTIONAL ARGUMENTS:

keyword arguments:
:env-type. A symbol. Wheter this envelope controls a plugin ('parmenv) or another parameter of a track. Valid options are:
volume volume-pre-fx trim-volume pan pan-pre-fx width width-pre-fx mute tempo aux-volume aux-pan aux-mute parmenv...
Any symbol that isn't recognized will create a 'parmenv.
:start-time. the time in seconds at which the first point of the envelope will be
:end-time. the time in seconds at which the last point of the envelope will be
:parameter-slot. A positive integer. When env-type is 'parmenv, this determines which parameter of the plugin the envelope controls. Open reaper and the plugin and then count the automatable parameters to find out which one you want. For example, the angle-parameter for the iem stereo encoder is the 6th.
:parameter-min. The minimm value the envelope will have in reaper. Usually it is 0 but for example for panorama it could be -1. This is less important.
:parameter-max. The maximum value the envelope will have in reaper. Usually it is 1 but for example for volume it could be 2. This is less important.
:is-visible. t or nil, wheter the envelope will be visible (opened) in the reaper-file.

RETURN VALUE:

the envelope-object

SYNOPSIS:

```
(defun make-reaper-envelope (env &key (env-type 'parmenv)
                                      start-time
                                      end-time
                                      parameter-slot
                                      parameter-min
                                      parameter-max
                                      (is-visible t))
```

22.51 utilities/make-reaper-items4

[utilities] [Functions]

DATE:

```
April 30th 2023.
```

DESCRIPTION:

```
make-reaper-items but every sndfile gets its own track and start-times are
given individually.
```

ARGUMENTS:

```
- a list of sndfiles
- a list of start-times for the sndfiles within the reaper project. If this
list is shorter than sndfiles it is looped.
```

RETURN VALUE:

```
a list of reaper-items
```

SYNOPSIS:

```
(defun make-reaper-items4 (sndfiles &optional (start-times '(0)))
```

22.52 utilities/middle

[utilities] [Functions]

DESCRIPTION:

```
Get the number value that is middle of two number values.
```

ARGUMENTS:

```
- A first number.
- A second number.
```

RETURN VALUE:

```
A number.
```

EXAMPLE:

```
(middle 7 92)

=> 49.5
```

SYNOPSIS:

```
(defun middle (lower upper)
```

22.53 utilities/mins-secs-to-secs

[utilities] [Functions]

DESCRIPTION:

```
Derive the number of seconds from a minutes-seconds value that is indicated
as a string of the form "0:00.000" or a two-item list in the form '(minutes
seconds) or three-item list in the form '(minutes seconds milliseconds)
```

ARGUMENTS:

```
- A time in minutes and seconds, as described above.
```

OPTIONAL ARGUMENTS:

```
- if a string is to be passed, then a character that denotes the separator
between minutes and seconds. Default = #\:
```

RETURN VALUE:

```
A decimal number that is a number in seconds.
```

EXAMPLE:

```
(mins-secs-to-secs '(2 1))
=> 121.0
(mins-secs-to-secs '(16 59 534)))
=> 1019.534
(mins-secs-to-secs "3:06.829"))
=> 186.829
;; using a different separator character between minutes and seconds
(mins-secs-to-secs "3-36.29" #\-) 0.0001)
=> 216.29
```

SYNOPSIS:

```
(defun mins-secs-to-secs (time &optional (post-mins #\:))
```

22.54 utilities/move-elements

[utilities] [Functions]

DATE:

```
02-Mar-2011
```

DESCRIPTION:

```
Move the specified elements from one list (if they are present in that
list) to another, deleting them from the first.
```

ARGUMENTS:

```
- A list of elements that are the elements to be moved.
- A list from which the specified elements are to be moved and deleted.
- A list to which the specified elements are to be moved.
```

OPTIONAL ARGUMENTS:

```
- A predicate by which to test that the specified elements are equal to
  elements of the source list. Default = #'eq.
```

RETURN VALUE:

```
Two values: A first list that is the source list after the items have been
moved; a second list that is the target list after the items have been
moved.
```

EXAMPLE:

```
(move-elements '(3 5 8) '(1 2 3 4 5 6 7 8 9) '(a b c d e))

=> (1 2 4 6 7 9), (8 5 3 A B C D E)
```

SYNOPSIS:

```
(defun move-elements (what from to &optional (test #'eq))
```

22.55 utilities/move-to-end

[utilities] [Functions]

DATE:

22-May-2011

DESCRIPTION:

Move a specified element of a given list to the end of the list, returning the new list.

NB: If the element exists more than once in the given list, all but one of the occurrences will be removed and only one of them will be placed at the end.

ARGUMENTS:

- An item that is an element of the list that is the second argument.
- A list.

RETURN VALUE:

A list.

EXAMPLE:

```
;;; All unique items
(move-to-end 2 '(1 2 3 4 5))

=> (1 3 4 5 2)

;;; Duplicate items
(move-to-end 2 '(1 2 3 2 4 2 5))

=> (1 3 4 5 2)
```

SYNOPSIS:

```
(defun move-to-end (what list &optional (test #'eql))
```

22.56 utilities/nconc-sublists

[utilities] [Functions]

DESCRIPTION:

```
Concatenate corresponding sublists of a given list. Each sublist in the
argument should have the same length and number of sublists etc.
```

ARGUMENTS:

```
A list of lists.
```

RETURN VALUE:

```
A list of lists.
```

EXAMPLE:

```
(nconc-sublists '(((1 2) (a b) (cat dog))
                  ((3 4) (c d) (bird fish))
                  ((5 6) (e f) (pig cow))))

=> ((1 2 3 4 5 6) (A B C D E F) (CAT DOG BIRD FISH PIG COW))
```

SYNOPSIS:

```
(defun nconc-sublists (lists)
```

22.57 utilities/nearest

[utilities] [Functions]

DATE:

```
15th May 2020, Heidhausen
```

DESCRIPTION:

```
Return the nearest number in a list to the first argument
```

ARGUMENTS:

```
- the number we're looking to get the closest to
- the list of numbers we'll search
```

RETURN VALUE:

```
the element of the list that's closest to the first argument, the list
sorted by nearest to the number, the distances to the number for the sorted
list.
```

OPTIONAL ARGUMENTS:

```
none
```

EXAMPLE:

```
(nearest 1.21 '(4 2 5 3 5 4 1.2 1.3 1.1999))
--> 1.2
```

SYNOPSIS:

```
(defun nearest (num list)
```

22.58 utilities/nearest-power-of-2

[utilities] [Functions]

DESCRIPTION:

```
Return the closest number to the specified value that is a power of two but
not greater than the specified value (unless the optional argument is T:
see below).
```

ARGUMENTS:

```
- A number.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether we can return a power of 2 greater than the
  argument, if that is nearer to the argument than the lower power or 2.
```

RETURN VALUE:

An integer that is a power of two.

EXAMPLE:

```
(nearest-power-of-2 31)

=> 16

(nearest-power-of-2 31 t)

=> 32

(nearest-power-of-2 32)

=> 32

(nearest-power-of-2 33)

=> 32
```

SYNOPSIS:

```
(defun nearest-power-of-2 (num &optional allow>)
```

22.59 utilities/nearest-prime

[utilities] [Functions]

DATE:

10th July 2026

DESCRIPTION:

Return the prime number nearest to the first argument.

ARGUMENTS:

- the integer (> 1) from which to start searching

OPTIONAL ARGUMENTS:

- whether to return prime numbers > the first argument. Default = T.
- whether to return prime numbers < the first argument. Default = T.

RETURN VALUE:

```
the prime number nearest (ingeger > 1) to the first argument. If by default
the two optional arguments are T, then the nearest > and < the first argument
will be compared to return the closest. Otherwise if one or the other is T,
then only prime numbers > or < the first argument will be returned. It is an
error to pass NIL and arguments 2 and 3
```

SYNOPSIS:

```
(defun nearest-prime (near &optional (above t) (below t))
```

22.60 utilities/now-string

[utilities] [Functions]

DATE:

```
Sat 28 Mar 2020 13:21:08 GMT - London
```

DESCRIPTION:

```
Return a string representing the current time in the format:
YEAR MONTH DAY - HOURS MINUTES SECONDS
e.g. "20200328-132227"

It is thought that this function might be useful when outputing multiple
files during the test phase of a piece. E.g.
(cmn-display +mini+
             :file (concatenate 'string "my-piece" (now-string) ".eps"))
```

ARGUMENTS:

```
None
```

OPTIONAL ARGUMENTS:

```
None
```

RETURN VALUE:

```
A string
```

EXAMPLE:

```
(now-string)

=> "20200328-132227"

(concatenate 'string "my-piece_" (now-string) ".eps")

=> "my-piece_20200328-133357.eps"
```

SYNOPSIS:

```
(defun now-string ()
```

22.61 utilities/octave-freqs

[utilities] [Functions]

DESCRIPTION:

```
A boolean test to determine whether two specified frequencies are octave
transpositions of the same pitch class.
```

ARGUMENTS:

```
- A first number that is a frequency in Hertz.
- A second number that is a frequency in Hertz.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether identical frequencies ("unison") are also
  to be considered octave transpositions of the same pitch class.
  T = unisons are also octaves. Default = T.
```

RETURN VALUE:

```
T or NIL.
```

EXAMPLE:

```
(octave-freqs 261.63 2093.04)

=> T

(octave-freqs 261.63 3000.00)
```

```
=> NIL

(octave-freqs 261.63 261.63)

=> T

(octave-freqs 261.63 261.63 nil)

=> NIL
```

SYNOPSIS:

```
(defun octave-freqs (freq1 freq2 &optional (unison-also t))
```

22.62 utilities/one-to-many

[utilities] [Functions]

DATE:

June 23rd 2020

DESCRIPTION:

Find a one-to-many relationship between the first argument and a number of equally-spaced points given in the second argument. The first argument is a number between 0.0 and 1.0 (inclusive). We calculate the proximities from this point to the number of points given and return them as a list, optionally raised to a given exponent. The list returned is scaled so that all values sum to 1.0, so this is particularly useful for, say, calculating a number of amplitude scalers for a multi-voice synthesis process.

ARGUMENTS:

- the point: a number between 0.0 and 1.0 inclusive
- the number of points to use in the calculation. This will also be the number of results returned. Alternatively this can be a list of numbers between 0.0 and 1.0. This way you can pass your own points for e.g. an unequally-spaced set. In this case though the proximity is still determined from a maximum of 1.0, not the highest number in the given list.

OPTIONAL ARGUMENTS:

- the exponent to raise proximities to. 1.0 will return a linear

```
relationship. > 1.0 will exaggerate the relationships so that those points
further away from the first argument will be pushed further away than a
linear relationship. < 1.0 will lessen the distances.
```

RETURN VALUE:

```
a list of numbers the length of which is the same as the 2nd argument and
the sum of which is 1.0
```

EXAMPLE:

```
(one-to-many .8 7) -->
(0.045112778 0.082706764 0.12030074 0.15789473 0.19548872 0.21804512 0.18045112)
(one-to-many .8 7 .7) -->
(0.06509622 0.09950039 0.1293403 0.15646003 0.18169008 0.19612299 0.17178996)
(one-to-many .8 7 1.3)  -->
(0.030845987 0.06782856 0.11039718 0.15721251 0.20752355 0.23917702 0.1870151)
;;; passing 5 points: these don't have to have min/max of 0 and 1 ...
(one-to-many .8 '(0 .1 .35 .7 .92)) -->
(0.07067137 0.10600708 0.19434628 0.3180212 0.31095406)
;;; ... and they don't have to be in ascending order either
(one-to-many .8 '(0 .1 .35 .7 .2)) -->
(0.08510638 0.12765959 0.23404254 0.38297874 0.17021276)
```

SYNOPSIS:

```
(defun one-to-many (one how-many &optional (expt 1.0))
```

22.63 utilities/os-format-path

[utilities] [Functions]

DATE:

```
March 30th 2023
```

DESCRIPTION:

```
Converts device-names ("/E/", "E:/") according to type
Windows:  "E:"
Unix:     "/E/"
If type is nil it will be set to the system this function was called on.
If a path does not contain a device-name, it is left unchanged.
A device-name is detected, when the name of the first directory consists
of only one letter and the path does not start with a ./ or ~/
```

ARGUMENTS:

```
- a string representing a path
```

OPTIONAL ARGUMENTS:

```
- a string or symbol which could be unix, linux or windows. If nil the type
of the current system will automatically be chosen.
```

RETURN VALUE:

```
- a string representing a path
```

EXAMPLE:

```
(os-format-path "/E/samples/kicks/kick.wav")
=> "/E/samples/kicks/kick.wav"
(os-format-path "E:/samples/kicks/kick.wav")
=> "/E/samples/kicks/kick.wav"
(os-format-path "/E/samples/kicks/kick.wav" 'windows)
=> "E:/samples/kicks/kick.wav"
(os-format-path "E:/samples/kicks/kick.wav" 'windows)
=> "E:/samples/kicks/kick.wav"
(os-format-path "./test.lsp" 'windows)
=> "./test.lsp"
(os-format-path "test.lsp" 'unix)
=> "test.lsp"
```

SYNOPSIS:

```
(defun os-format-path (path &optional type)
```

22.64 utilities/parse-audacity-label-file

[utilities] [Functions]

DATE:

```
August 12th 2019
```

DESCRIPTION:

```
Parse a labels file exported from Audacity and return a list of (start end
label) triplets. Times are in seconds.
```

ARGUMENTS:

```
- the path to the label file
```

RETURN VALUE:

```
a list
```

SYNOPSIS:

```
(defun parse-audacity-label-file (label-file)
```

22.65 utilities/parse-audacity-label-file-for-loops

[utilities] [Functions]

DESCRIPTION:

```
Read an audacity label file and return its loop points as groups.

NB: If this fails it's probably because there's a tab between time and
    label instead of spaces: save in emacs to detab.

NB: Beware that marker files created on different operating systems from
    the one on which this function is called might trigger errors due to
    newline character mismatches.
```

ARGUMENTS:

```
- A string that is the name of the label file to be parsed, including
  directory path and extension.
```

RETURN VALUE:

```
Returns a list of lists which are the grouped time points.

Also prints separate feedback to the listener.
```

EXAMPLE:

```
(parse-audacity-label-file-for-loops  "/path/to/24-7loops1.txt")

=>
313 markers, 50 loops read
```

```
((25.674559 25.829296 26.116327 26.649048 27.038843)
 (32.211884 32.33669 32.481815 32.618233 32.716915 32.902676 33.227757
  33.61959)
 (36.893604 37.059048 37.160633 37.27383 37.439274 37.4683 37.627937)
 (39.52907 39.81932 39.999275 40.2634 40.338867 40.605896)
 (45.612698 45.818775 46.050976 46.145306 46.275192)
 (46.4566 46.644535 46.76934 46.886894 46.971066 47.16553)
 (84.15927 84.260864 84.292786 84.355194 84.47274 84.52789 84.556915
  84.65415)
 ...
 (676.1075 676.79114 677.1503 677.57904 678.12366)
 (799.29205 799.8019 800.58984 800.96063 801.13446 801.45886)
 (804.98145 805.2016 805.5724 805.83887 806.31396))
```

SYNOPSIS:

```
(defun parse-audacity-label-file-for-loops (label-file)
```

22.66 utilities/parse-reaper-file-for-loops

[utilities] [Functions]

DATE:

```
July 1st 2021
```

DESCRIPTION:

```
Read a reaper file and return its loop points as groups.

The reaper file must contain markers with the word 'clm-loop-point-start'
and 'clm-loop-point-stop' with any number of 'clm-loop-point' markers
inbetween. This defines a new set of loop points, of which there can be any
number. Any number of markers without these names can be between the -start
and -stop markers; they will be ignored. It is assumed that the sound file
you'll process starts at time zero in the reaper file and within itself.
```

ARGUMENTS:

```
- A string that is the name of the marker file to be parsed, including
  directory path and extension.

OPTIONAL ARGUMENTS:
T or NIL to issue a warning if a marker is found beginning with
```

```
'clm-loop-point' but continuing with something other than -start or
-stop. Default = T
```

RETURN VALUE:

```
Returns a list of lists which are the grouped time points.
```

SYNOPSIS:

```
(defun parse-reaper-file-for-loops (reaper-file &optional (warn t))
```

22.67 utilities/parse-reaper-file-for-segment

[utilities] [Functions]

DATE:

```
July 5th 2021
```

DESCRIPTION:

```
Parse a reaper file and look for markers that correspond to the second
argument. What we're after is sound file segments i.e. with a start and an
end. So markers can be added with a certain symbol e.g. gran marking the
start and that symbol with -end tacked on e.g. gran-end to indicate where a
segment ends.
```

ARGUMENTS:

```
- the path to the reaper file
- the marker name we'll look for
```

RETURN VALUE:

```
a list of start-end pairs (in seconds)
```

EXAMPLE:

```
(parse-reaper-file-for-segment "barbara-markers2.RPP" 'gran)
->  '((542.9326 548.098) (598.7433 600.6894) (944.8951 945.41925)
      (947.9406 948.8009) (952.61755 959.5987) (971.784 975.9188)))))
```

SYNOPSIS:

```
(defun parse-reaper-file-for-segment (reaper-file marker)
```

22.68 utilities/parse-wavelab-marker-file-for-loops

[utilities] [Functions]

DESCRIPTION:

```
Read a wavelab marker file and return its loop points as groups.

The marker file must contain markers with the word "loop". A marker with
that name will start a new set of loop points, and nameless markers will
belong to the group until the next "loop" marker.
```

ARGUMENTS:

```
- A string that is the name of the marker file to be parsed, including
  directory path and extension.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :sampling-rate. An integer that is the sampling rate of the sound file to
  which the marker file refers. This value will affect the resulting time
  points. Default = 44100.
- :max-length. The maximum duration in seconds between two points: anything
  greater than this will result in a warning being printed.
```

RETURN VALUE:

```
Returns a list of lists which are the grouped time points.

Also prints separate feedback to the listener.
```

EXAMPLE:

```
(parse-wavelab-marker-file-for-loops "/path/to/24-7loops1.mrk")

=>
WARNING:
   utilities::parse-wavelab-marker-file-for-loops
   loop points 10:13.213 to 10:14.475 are too long (1.2620239)
WARNING:
   utilities::parse-wavelab-marker-file-for-loops
   loop points 10:33.223 to 10:34.486 are too long (1.2630615)
WARNING:
   utilities::parse-wavelab-marker-file-for-loops
```

```
   loop points 10:36.456 to 10:37.522 are too long (1.06604)


312 markers, 50 loops read

((25.674559 25.829296 26.116327 26.649048 27.038843)
 (32.211884 32.33669 32.481815 32.618233 32.716915 32.902676 33.227757
  33.61959)
 (36.893604 37.059048 37.160633 37.27383 37.439274 37.4683 37.627937)
 (39.52907 39.81932 39.999275 40.2634 40.338867 40.605896)
 (45.612698 45.818775 46.050976 46.145306 46.275192)
 (46.4566 46.644535 46.76934 46.886894 46.971066 47.16553)
 (84.15927 84.260864 84.292786 84.355194 84.47274 84.52789 84.556915
  84.65415)
 ...
 (655.91077 656.4554 656.80304 657.4519 658.04285 658.8192)
 (676.1075 676.79114 677.1503 677.57904 678.12366)
 (799.29205 799.8019 800.58984 800.96063 801.13446 801.45886)
 (804.98145 805.2016 805.5724 805.83887 806.31396))
```

SYNOPSIS:

```
(defun parse-wavelab-marker-file-for-loops
    (marker-file &key (sampling-rate 44100) (max-length 1.0))
```

22.69 utilities/partial-freqs

[utilities] [Functions]

DATE:

```
13-Dec-2011
```

DESCRIPTION:

```
A Boolean test to determine whether either of two specified frequencies
can be considered a harmonic partial of the other.
```

ARGUMENTS:

```
- A first frequency in Hertz.
- A second frequency in Hertz.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether identical frequencies ("unison") are also to
  be considered partials of each other. T = unison are partials.
  Default = T.
```

RETURN VALUE:

```
T if one of the frequencies has the ratio of a harmonic partial to the
other, otherwise NIL.
```

EXAMPLE:

```
(partial-freqs 300 900)

=> T

(partial-freqs 300 700)

=> NIL

(partial-freqs 300 300)

=> T

(partial-freqs 300 300 nil)

=> NIL
```

SYNOPSIS:

```
(defun partial-freqs (freq1 freq2 &optional (unison-also t))
```

22.70 utilities/pdivide

[utilities] [Functions]

DESCRIPTION:

```
Creates a list of proportionally related times, dividing a starting
duration into a number of smaller durations a specified number of times.
We start with a proportion as a ratio (e.g. 3/2) and divide the given
duration into two parts according to that ratio.  Then those two parts will
be divided into the same ratios.  This will iterate the number of times
indicated by the second argument.

The following are some classical proportions:
```

```
        Latin        (Greek)
(3 : 2) Sesquialtera (Diapente)
(4 : 3) Sesquitertia (Diatessaron)
(5 : 4) Sesquiquarta (Diatonus Semitonus)
(8 : 3) Duplasuperbipartiens (Diapson Diatesseron)
(9 : 8) Sesquioctava (Tonus)
```

ARGUMENTS:

```
- an integer or ratio (in Lisp terms, a rational) e.g. 3/2
- an integer >=1 specifying the number of times to iterate the process of
  dividing the duration into proportions.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :duration. The overall duration to apply the proportional divisions to.
  Units are arbitrary of course as this is just a number. Default 1.0.
- :print. If T, print each level of division as we proceed. Default NIL.
- :reverse. If T reverse the proportion (so 3/2 becomes 2/3). Default NIL.
- :alternate. If T, reverse the proportion every other division (not
  iteration) so that if we have a proportion of 3/2 on the second iteration
  we divide into 3/2 then 2/3.  Default NIL.
- :increment. If T, then each time we divide we increment both sides of the
  proportion,  so 3:2 becomes 4:3 which becomes 5:4 etc.  Default NIL.
- :halves. This will only make a difference if :increment is T: As results
  tend overall towards increasing (when numerator < denominator e.g. 2/3) or
  decreasing (numerator > denominator e.g. 3/2) numbers, we can mix things
  up by dividing the resultant list into two halves and splicing their
  elements one after the other.  Default NIL.
- :shuffle. Mix things up by shuffling the resultant list.  As this uses
  the shuffle algorithm we have fixed-seed randomness so results will be
  the same upon each call within the same Lisp implementation/version.
  Default NIL.
```

RETURN VALUE:

```
Three values: the list of ascending timings from the last generation of the
calculated proportions; the durations of each part for the last generation;
the list of ascending timings for _each_ generation of the calculated
proportions (a list of lists).
```

EXAMPLE:

```
Notice here that each generation prints the proportions along with the
```

```
durations these correspond to and the start time of each (cumulative durations).

(pdivide 3/2 4 :duration 35 :print t)

PRINTS:
Generation 1: 3 (21.00=21.00), 2 (14.00=35.00),

Generation 2: 3 (12.60=12.60), 2 (8.40=21.00), 3 (8.40=29.40), 2 (5.60=35.00),

Generation 3: 3 (7.56=7.56), 2 (5.04=12.60), 3 (5.04=17.64), 2 (3.36=21.00),
3 (5.04=26.04), 2 (3.36=29.40), 3 (3.36=32.76), 2 (2.24=35.00),

Generation 4: 3 (4.54=4.54), 2 (3.02=7.56), 3 (3.02=10.58), 2 (2.02=12.60),
3 (3.02=15.62), 2 (2.02=17.64), 3 (2.02=19.66), 2 (1.34=21.00), 3 (3.02=24.02),
2 (2.02=26.04), 3 (2.02=28.06), 2 (1.34=29.40), 3 (2.02=31.42), 2 (1.34=32.76),
3 (1.34=34.10), 2 (0.90=35.00),

RETURNS:
(0.0 4.5360003 7.5600004 10.584001 12.6 15.624001 17.640001 19.656002 21.000002
 24.024002 26.040003 28.056004 29.400003 31.416004 32.760006 34.104008
 35.000008)
(4.5360003 3.0240002 3.0240004 2.0160003 3.0240004 2.0160003 2.0160003
 1.3440002 3.0240004 2.0160003 2.0160003 1.3440002 2.0160003 1.3440001
 1.3440001 0.896)
((0.0 4.5360003 7.5600004 10.584001 12.6 15.624001 17.640001 19.656002
  21.000002 24.024002 26.040003 28.056004 29.400003 31.416004 32.760006
  34.104008 35.000008)
 (0.0 7.5600004 12.6 17.640001 21.000002 26.040003 29.400003 32.760002
  35.000004)
 (0.0 12.6 21.0 29.400002 35.0) (0.0 21.0 35.0))


(pdivide 3/2 4 :duration 35 :print t :increment t :halves t)

PRINTS:
Generation 1: 3 (21.00=21.00), 2 (14.00=35.00),

Generation 2: 4 (12.00=12.00), 3 (9.00=21.00), 5 (7.78=28.78), 4 (6.22=35.00),

Generation 3: 6 (6.55=6.55), 5 (5.45=12.00), 7 (4.85=16.85), 6 (4.15=21.00),
8 (4.15=25.15), 7 (3.63=28.78), 9 (3.29=32.07), 8 (2.93=35.00),

Generation 4: 10 (3.44=3.44), 9 (3.10=6.55), 11 (2.86=9.40), 10 (2.60=12.00),
12 (2.53=14.53), 11 (2.32=16.85), 13 (2.16=19.01), 12 (1.99=21.00),
14 (2.15=23.15), 13 (2.00=25.15), 15 (1.88=27.03), 14 (1.75=28.78),
16 (1.70=30.48), 15 (1.59=32.07), 17 (1.51=33.58), 16 (1.42=35.00),
```

```
RETURNS:
(0.0 3.4449766 5.595868 8.696347 10.6936035 13.550747 15.428142 18.025545
 19.77778 22.30621 24.0064 26.324125 27.918053 30.078053 31.58647 33.580315
 35.0)
(3.4449766 2.1508918 3.100479 1.9972568 2.8571434 1.8773947 2.5974028 1.752235
 2.5284283 1.7001898 2.317726 1.593928 2.16 1.5084175 1.9938462 1.4196872)
((0.0 3.4449766 6.5454555 9.402599 12.000002 14.52843 16.846155 19.006155
  21.000002 23.150894 25.148151 27.025547 28.777782 30.477972 32.0719 33.58032
  35.000004)
 (0.0 6.5454555 12.000002 16.846157 21.000004 25.148151 28.77778 32.0719
  35.000004)
 (0.0 12.000001 21.0 28.777779 35.0) (0.0 21.0 35.0))
```

SYNOPSIS:

```
(defun pdivide (start levels &key (duration 1.0) print reverse alternate
                                  halves shuffle increment)
```

22.71 utilities/pdivide-reaper-markers

[utilities] [Functions]

DESCRIPTION:

```
Write to standard out (terminal/interpreter) marker data for a reaper file
generated from calls to the pdivide function.
```

ARGUMENTS:

```
Takes the same arguments as the pdivide function
```

RETURN VALUE:

```
T
```

SYNOPSIS:

```
(defun pdivide-reaper-markers (&rest args)
```

22.72 utilities/periodicity

[utilities] [Functions]

DATE:

November 13th 2024

DESCRIPTION:

Think of isorhythm: you have a list of x pitches which you go through, repeating from the beginning when you get to the end, plus a similarly-handled list of y rhythms. x and y are different (lengths) so when does the cycle repeat i.e. when do we reuse the first element of each list and proceed again as at the beginning? The simple answer is x * y but it could be significantly less. This routine works that out.

So, given a list of integers > 0, representing perhaps the number of items (e.g. pitches, rhythms etc.) in an arbitrary number of lists, calculate the cycle length before we repeat, i.e. start again at the beginning of each list.

NB for convenience the argument can be a list of lists, the lengths of which will then be used as above.

ARGUMENTS:

a list of integers > 0 or a list of sublists

RETURN VALUE:

the cycle length before a repeat occurs

EXAMPLE:

```
(periodicity '(1 2 5))
10

(periodicity '(1 2 3 4 5 6))
60

(periodicity '(1 2 3 4 5 6 20))
60

(periodicity '(1 2 3 4 5 6 16))
240
```

SYNOPSIS:

```
(defun periodicity (cycle-lengths)
```

22.73 utilities/pexpand

[utilities] [Functions]

DESCRIPTION:

```
Instead of dividing an overall duration (pdivide) we start with a
proportion and expand outwards from there, keeping each newly created part
in the same proportion.  This is repeated the number of times specified in
the first argument.  Useful for generating maps (section structure).
```

ARGUMENTS:

```
The number of times to expand proportionally.
```

OPTIONAL ARGUMENTS:

```
As many integer proportions as required.  If the last argument here is t,
then instead of using letters to denote sections we use numbers instead.
```

RETURN VALUE:

```
3 values:
1) a list showing the cumulative count (e.g. bar numbers) of where major
and minor sections occur.  Topmost sections will have the labels A, B, C,
etc. with subsections such as A.A, A.B, ... C.C.C.C.  Of course, wherever a
major section starts, an arbitrary number of subsections also begin, but
only the most major section is present in the list.
2) the structure of the sections and subsections in the form of a list of
sublists for each, and containing the section labels paired with their
length.  The bottommost subsection will have a length of the sum of the
proportions, with higher subsection groupings showing multiples of this.
3) the overall length of the structure produced (also the first element of
the second returned value).
```

EXAMPLE:

```
;;; 2 generations:
(pexpand 2 3 2) =>
(1 (A) 6 (A A A B) 11 (A A A C) 16 (A A B) 21 (A A B B) 26 (A B) 31 (A B A B)
 36 (A B A C) 41 (A B B) 46 (A B B B) 51 (A C) 56 (A C A B) 61 (A C A C) 66
 (A C B) 71 (A C B B) 76 (B) 81 (B A A B) 86 (B A A C) 91 (B A B) 96 (B A B B)
 101 (B B) 106 (B B A B) 111 (B B A C) 116 (B B B) 121 (B B B B))
(125
 (((A) 75)
```

```
  (((A A) 25) (((A A A) 15) ((A A A A) 5) ((A A A B) 5) ((A A A C) 5))
   (((A A B) 10) ((A A B A) 5) ((A A B B) 5)))
  (((A B) 25) (((A B A) 15) ((A B A A) 5) ((A B A B) 5) ((A B A C) 5))
   (((A B B) 10) ((A B B A) 5) ((A B B B) 5)))
  (((A C) 25) (((A C A) 15) ((A C A A) 5) ((A C A B) 5) ((A C A C) 5))
   (((A C B) 10) ((A C B A) 5) ((A C B B) 5))))
 (((B) 50)
  (((B A) 25) (((B A A) 15) ((B A A A) 5) ((B A A B) 5) ((B A A C) 5))
   (((B A B) 10) ((B A B A) 5) ((B A B B) 5)))
  (((B B) 25) (((B B A) 15) ((B B A A) 5) ((B B A B) 5) ((B B A C) 5))
   (((B B B) 10) ((B B B A) 5) ((B B B B) 5)))))
125

;;; 3 generations:
(pexpand 3 3 2) =>
(1 (A) 6 (A A A A A B) 11 (A A A A A C) 16 (A A A A B) 21 (A A A A B B) 26
 (A A A B) 31 (A A A B A B) 36 (A A A B A C) 41 (A A A B B) 46 (A A A B B B) 51
 (A A A C) 56 (A A A C A B) 61 (A A A C A C) 66 (A A A C B) 71 (A A A C B B) 76
...
 581 (B B B A A B) 586 (B B B A A C) 591 (B B B A B) 596 (B B B A B B) 601
 (B B B B) 606 (B B B B A B) 611 (B B B B A C) 616 (B B B B B) 621
 (B B B B B B))
(625
 (((A) 375)
  (((A A) 125)
   (((A A A) 75)
    (((A A A A) 25)
     (((A A A A A) 15) ((A A A A A A) 5) ((A A A A A B) 5) ((A A A A A C) 5))
     (((A A A A B) 10) ((A A A A B A) 5) ((A A A A B B) 5)))
...
   (((B B B) 50)
    (((B B B A) 25)
     (((B B B A A) 15) ((B B B A A A) 5) ((B B B A A B) 5) ((B B B A A C) 5))
     (((B B B A B) 10) ((B B B A B A) 5) ((B B B A B B) 5)))
    (((B B B B) 25)
     (((B B B B A) 15) ((B B B B A A) 5) ((B B B B A B) 5) ((B B B B A C) 5))
     (((B B B B B) 10) ((B B B B B A) 5) ((B B B B B B) 5)))))))
625

;;; 2 generations of 3 proportional values, returning numbers for labels
(pexpand 2 3 2 4 t) =>
(1 (1) 10 (1 1 1 2) 19 (1 1 1 3) 28 (1 1 2) 37 (1 1 2 2) 46 (1 1 3) 55
 (1 1 3 2) 64 (1 1 3 3) 73 (1 1 3 4) 82 (1 2) 91 (1 2 1 2) 100 (1 2 1 3) 109
 (1 2 2) 118 (1 2 2 2) 127 (1 2 3) 136 (1 2 3 2) 145 (1 2 3 3) 154 (1 2 3 4)
... (3 4 2 2) 694 (3 4 3) 703 (3 4 3 2) 712 (3 4 3 3) 721 (3 4 3 4))
```

```
(729
 (((1) 243)
  (((1 1) 81) (((1 1 1) 27) ((1 1 1 1) 9) ((1 1 1 2) 9) ((1 1 1 3) 9))
   (((1 1 2) 18) ((1 1 2 1) 9) ((1 1 2 2) 9))
   (((1 1 3) 36) ((1 1 3 1) 9) ((1 1 3 2) 9) ((1 1 3 3) 9) ((1 1 3 4) 9)))
...
   (((3 2 2) 18) ((3 2 2 1) 9) ((3 2 2 2) 9))
   (((3 2 3) 36) ((3 2 3 1) 9) ((3 2 3 2) 9) ((3 2 3 3) 9) ((3 2 3 4) 9)))
  (((3 3) 81) (((3 3 1) 27) ((3 3 1 1) 9) ((3 3 1 2) 9) ((3 3 1 3) 9))
   (((3 3 2) 18) ((3 3 2 1) 9) ((3 3 2 2) 9))
   (((3 3 3) 36) ((3 3 3 1) 9) ((3 3 3 2) 9) ((3 3 3 3) 9) ((3 3 3 4) 9)))
  (((3 4) 81) (((3 4 1) 27) ((3 4 1 1) 9) ((3 4 1 2) 9) ((3 4 1 3) 9))
   (((3 4 2) 18) ((3 4 2 1) 9) ((3 4 2 2) 9))
   (((3 4 3) 36) ((3 4 3 1) 9) ((3 4 3 2) 9) ((3 4 3 3) 9) ((3 4 3 4) 9)))))
729
```

SYNOPSIS:

```
(defun pexpand (generations &rest proportions)
```

22.74 utilities/pexpand-find

[utilities] [Functions]

DESCRIPTION:

```
Find the cumulative number of where a label occurs in a list returned by
pexpand.
```

ARGUMENTS:

```
- the label we're looking for
- a list of the type returned by pexpand (first returned value).
```

OPTIONAL ARGUMENTS:

```
- a function to be called when the label cannot be found.  Default =
#'error but could also be #'warn or NIL.
```

RETURN VALUE:

```
An integer.
```

SYNOPSIS:

```
(defun pexpand-find (label list &optional (on-error #'error))
```

22.75 utilities/pexpand-reaper-markers

[utilities] [Functions]

DATE:

```
September 4th 2015, Edinburgh
```

DESCRIPTION:

```
Using the pexpand function, we write marker information in a format which
can be read by the Reaper (version 4/5+) DAW software. Though we can think
of the outputs of pexpand to be in beats, seconds, bars, or any arbitrary
scale, the timings of Reaper markers are in seconds, hence the need here
for an initial tempo. In other words, we treat the output of pexpand to be
beat counts; if you would prefer to interpret these as bars, simply divide
the tempo by the number of beats per bar. If there are to be tempo changes
in the mix/piece Reaper itself will update the markers' positions when the
new tempo is inserted in the project window--this is fine when you are
thinking in beats/bars but beware of changing tempo in Reaper if you are
thinking of markers with fixed timings (seconds).

Copy the output of this into the Reaper file verbatim (no enclosing < >
marks) before the <PROJBAY tag.
```

ARGUMENTS:

```
- the tempo in BPM
- the number of generations: see the pexpand function
- (&rest) the proportions: see the pexpand function
```

RETURN VALUE:

```
Always T
```

EXAMPLE:

```
(pexpand-reaper-markers 144 2 6 3 5 4)
->
  MARKER 1 7.5 "level 4" 0 0 1
  MARKER 2 15.0 "level 4" 0 0 1
  MARKER 3 22.5 "level 4" 0 0 1
  MARKER 4 30.0 "level 4" 0 0 1
  MARKER 5 37.5 "level 4" 0 0 1
...
```

```
  MARKER 108 810.0 "level 1" 0 0 1
  MARKER 109 817.5 "level 4" 0 0 1
  MARKER 110 825.0 "level 4" 0 0 1
  MARKER 111 832.5 "level 4" 0 0 1
  MARKER 112 840.0 "level 4" 0 0 1
  MARKER 113 847.5 "level 4" 0 0 1
  MARKER 114 855.0 "level 3" 0 0 1
...
  MARKER 319 2392.5 "level 4" 0 0 1
  MARKER 320 2400.0 "level 3" 0 0 1
  MARKER 321 2407.5 "level 4" 0 0 1
  MARKER 322 2415.0 "level 4" 0 0 1
  MARKER 323 2422.5 "level 4" 0 0 1
```

Here's where I pasted the data into the .RPP Reaper file:

```
  <TEMPOENVEX
    ACT 0
    VIS 1 0 1
    LANEHEIGHT 0 0
    ARM 0
    DEFSHAPE 1 -1 -1
  >
  MARKER 1 7.5 "level 4" 0 0 1
  MARKER 2 15 "level 4" 0 0 1
  MARKER 3 22.5 "level 4" 0 0 1
  MARKER 4 30 "level 4" 0 0 1
...
  MARKER 320 2400 "level 3" 0 0 1
  MARKER 321 2407.5 "level 4" 0 0 1
  MARKER 322 2415 "level 4" 0 0 1
  MARKER 323 2422.5 "level 4" 0 0 1
  <PROJBAY
  >
  <TRACK {EBF9837F-BE25-9542-B720-A1862C0DF380}
```

SYNOPSIS:

```
(defun pexpand-reaper-markers (tempo generations &rest proportions)
```

22.76 utilities/pexpand-section-length

[utilities] [Functions]

DESCRIPTION:

Return the length (integer) of any arbitrary section in the data returned by pexpand.

ARGUMENTS:

- The (rest of) the kind of list returned as the second value of a call to pexpand.
- The section ID we want the length of, either as a list or single symbol.

RETURN VALUE:

An integer or NIL if the section can't be found.

EXAMPLE:

```
(pexpand-section-length (rest (nth-value 1 (pexpand 2 3 6 4 5))) '(c a b))
=> 108

(pexpand-section-length (rest (nth-value 1 (pexpand 2 3 6 4 5))) 'c)
=> 1296
```

SYNOPSIS:

```
(defun pexpand-section-length (pexpand-list section)
```

22.77 utilities/polar-to-cartesian

[utilities] [Functions]

DESCRIPTION:

Convert a point in a 3D coordinate space from the polar system to cartesian coordinates. This differs from the normal definition of this conversion, in that the elevation is the angle from true horizontal, not vertical...

ARGUMENTS:

- The horizonal angle from the Y axis (alpha, azimuth angle) in degree.
- The vertical angle from the X axis (polar, elevation) between +-180°
- The distance from the origin (the radius), 0 <= distance <= 1

RETURN VALUE:

A list that holds the x y and z coordinates for the point.

EXAMPLE:

```
(polar-to-cartesian 0 45 1)

=> (0.0 0.70710677 0.70710677)
```

SYNOPSIS:

```
(defun polar-to-cartesian (angle elevation distance)
```

22.78 utilities/power-of-2

[utilities] [Functions]

DESCRIPTION:

```
Test whether the specified number is a power of two and return the
logarithm of the specified number to base 2.

This method returns two values: T or NIL for the test and a decimal that is
the logarithm of the specified number to base 2.
```

ARGUMENTS:

```
- A number.
```

RETURN VALUE:

```
Two values: T or NIL for the test and a decimal number that is the
logarithm of the specified number to base 2.
```

EXAMPLE:

```
(power-of-2 16)

=> T, 4.0

(power-of-2 17.3)

=> NIL, 4.1127
```

SYNOPSIS:

```
(defun power-of-2 (float)
```

22.79 utilities/prime

[utilities] [Functions]

DATE:

```
August 5th 2019, Heidhausen
```

DESCRIPTION:

```
Taken from dlocsig.lisp by Fernando Lopez Lezcano (in the CLM package):
Return T or NIL to indicated whether the argument is a prime number or not.
```

ARGUMENTS:

```
an integer (all other types, including floats, will trigger an error)
```

RETURN VALUE:

```
T or NIL
```

SYNOPSIS:

```
(defun prime (val)
```

22.80 utilities/pts2cm

[utilities] [Functions]

DESCRIPTION:

```
Convert a specified number of points to a length in centimeters at a
resolution of 72ppi.
```

ARGUMENTS:

```
- A number.
```

RETURN VALUE:

```
A number.
```

EXAMPLE:

```
(pts2cm 150)

=> 5.2916665
```

SYNOPSIS:

```
(defun pts2cm (points)
```

22.81 utilities/radian-to-degree

[utilities] [Functions]

DESCRIPTION:

```
 Convert an angle in radians to its equivalent in degrees
```

ARGUMENTS:

```
 - The number in radians
```

RETURN VALUE:

```
 The number in degrees
```

EXAMPLE:

```
(radian-to-degree pi)

=> 180
```

SYNOPSIS:

```
(defun radian-to-degree (radian)
```

22.82 utilities/random-amount

[utilities] [Functions]

DESCRIPTION:

```
 Return a random number from within a total range of <percent> of the given
 number, centering around zero. Thus, if the <number> is 100, and the
 <percent> is 5, the results will be a random number between -2.5 and +2.5.
```

ARGUMENTS:

```
A number.
```

OPTIONAL ARGUMENTS:

```
A number that will be a percent of the given number.
```

RETURN VALUE:

```
A random positive or negative number.
```

EXAMPLE:

```
;;; Using the default will return numbers within a 5% span of the given number,
;;; centering around zero. With 100 that means between -2.5 and +2.5.
(loop repeat 10 collect (random-amount 100))

=> (0.7424975 -1.4954442 -1.7126495 1.5918689 -0.43478793 -1.7916341 -1.9115914
    0.8541988 0.057197176 2.0713913)

;;; Specifying 10% of 80 will return random numbers between -4.0 and +4.0
(loop repeat 10 collect (random-amount 80 10))

=> (-0.66686153 3.0387697 3.4737322 -2.3753185 -0.8495751 -0.47580242
    -0.25743783 -1.1395472 1.3560238 -0.5958566)
```

SYNOPSIS:

```
(defun random-amount (number &optional (percent 5))
```

22.83 utilities/random-from-list

[utilities] [Functions]

DESCRIPTION:

```
Return a random element from a specified list of elements.
```

ARGUMENTS:

```
- A list.
```

OPTIONAL ARGUMENTS:

- An integer can be passed stating the length of the list, for more efficient processing. NB: There is no check to ensure this number is indeed the length of the list. If the number is less than the length of the list, only elements from the first part of the list will be returned. If it is greater than the length of the list, the method may return NIL.

RETURN VALUE:

An element from the specified list.

EXAMPLE:

```
(random-from-list '(3 5 7 11 13 17 19 23 29))

=> 13
```

SYNOPSIS:

```
(defun random-from-list (list &optional list-length) ; for efficiency
```

22.84 utilities/randomise

[utilities] [Functions]

DESCRIPTION:

Return a random decimal number close to (+ or -) the number specified (within a certain percentage of that number's value). Note that if you want the result to go from 0 to 2x the argument, then <percent> needs to be 200, not 100.

ARGUMENTS:

- A number.

OPTIONAL ARGUMENTS:

- A number that is a percentage value, such that any random number returned will be within that percentage of the original number's value. Default = 5.

RETURN VALUE:

A decimal number.

EXAMPLE:

```
(loop repeat 10 collect (randomise 100))

=> (99.413795 99.15346 98.682014 100.76199 97.74929 99.05693 100.59494 97.96452
    100.42091 100.01329)
```

SYNOPSIS:

```
(defun randomise (number &optional (percent 5))
```

22.85 utilities/read-file-as-string

[utilities] [Functions]

DATE:

```
April 30th 2023.
```

DESCRIPTION:

```
Read an entire file (not just an s-expression) a into string and return it
```

ARGUMENTS:

```
- A string that is the path to a file (directory and filename)
```

RETURN VALUE:

```
A string with the contents of the file
```

EXAMPLE:

```
(read-from-file "/path/to/lisp-lorem-ipsum.txt")
```

SYNOPSIS:

```
(defun read-file-as-string (infile)
```

22.86 utilities/read-from-file

[utilities] [Functions]

DESCRIPTION:

Read a Lisp expression from a file. This is determined by the Lisp
parenthetical syntax.

ARGUMENTS:

- A string that is a file name including directory path and extension.

RETURN VALUE:

The Lisp expression contained in the file.

EXAMPLE:

```
(read-from-file "/path/to/lisp-lorem-ipsum.txt")

=> (LOREM IPSUM DOLOR SIT AMET CONSECTETUR ADIPISCING ELIT CRAS CONSEQUAT
    CONVALLIS JUSTO VITAE CONSECTETUR MAURIS IN NIBH VEL EST TEMPUS LOBORTIS
    SUSPENDISSE POTENTI SED MAURIS MASSA ADIPISCING VITAE DIGNISSIM CONDIMENTUM
    VOLUTPAT VEL FELIS FUSCE AUGUE DUI PULVINAR ULTRICIES IMPERDIET SED
    PHARETRA EU QUAM INTEGER IN VULPUTATE VELIT ALIQUAM ERAT VOLUTPAT VIVAMUS
    SIT AMET ORCI EGET EROS CONSEQUAT TINCIDUNT NUNC ELEMENTUM ADIPISCING
    LOBORTIS MORBI AT LOREM EST EGET MATTIS ERAT DONEC AC RISUS A DUI MALESUADA
    LOBORTIS AC AT EST INTEGER AT INTERDUM TORTOR VIVAMUS HENDRERIT CONSEQUAT
    AUGUE QUISQUE ALIQUAM TELLUS NEC VESTIBULUM LOBORTIS RISUS TURPIS LUCTUS
    LIGULA IN BIBENDUM FELIS SEM PULVINAR DOLOR VIVAMUS RHONCUS NISI GRAVIDA
    PORTA VULPUTATE IPSUM LACUS PORTA RISUS A VULPUTATE MAGNA JUSTO A EST)
```

SYNOPSIS:

```
(defun read-from-file (file)
```

22.87 utilities/reflect-list

[utilities] [Functions]

DESCRIPTION:

Order a list of numbers from least to greatest, then transpose the list so
that if an element is the second lowest, it will be replaced by the second
highest etc.

ARGUMENTS:

- A list or numbers.

RETURN VALUE:

```
A list of numbers.
```

EXAMPLE:

```
(reflect-list '(1 4 3 5 9 6 2 7 8 8 9))

=> (9 6 7 5 1 4 8 3 2 2 1)
```

SYNOPSIS:

```
(defun reflect-list (list)
```

22.88 utilities/remove-all

[utilities] [Functions]

DESCRIPTION:

```
Remove all of the specified elements from a list, returning a list
containing only those elements that are not in the first argument list.
```

ARGUMENTS:

```
- A first list that is the list of items to remove.
- A second list that is the original list.
```

OPTIONAL ARGUMENTS:

```
- A predicate for testing equality between the elements of the two lists.
  Default = #'eq.
```

RETURN VALUE:

```
A list.
```

EXAMPLE:

```
(remove-all '(3 5 8 13) '(1 2 3 4 5 6 7 8 9 10 11 12 13))

=> (1 2 4 6 7 9 10 11 12)
```

SYNOPSIS:

```
(defun remove-all (rm-list list &optional (test #' eq))
```

22.89 utilities/remove-elements

[utilities] [Functions]

DESCRIPTION:

```
Remove a specified number of elements from a given list starting at a
specified position (0-based) within the list.
```

ARGUMENTS:

```
- A list.
- An integer that is the 0-based position within that list that will be the
  first element to be removed.
- An integer that is the number of elements to remove.
```

RETURN VALUE:

```
A list.
```

EXAMPLE:

```
(remove-elements '(1 2 3 4 5 6 7) 2 4)

=> (1 2 7)
```

SYNOPSIS:

```
(defun remove-elements (list start how-many)
```

22.90 utilities/remove-more

[utilities] [Functions]

DESCRIPTION:

```
Remove all instances of a list of specified elements from an original
list. The predicate used to test the presence of the specified elements in
the original list must be specified by the user (such as #'eq, #'equalp,
#'= etc.)
```

ARGUMENTS:

```
- A list.
- A predicate with which to test the presence of the specified elements.
- A sequence of elements to be removed from the given list.
```

RETURN VALUE:

```
A list.
```

EXAMPLE:

```
(remove-more '(1 2 3 4 5 5 5 6 7 7 8) #'= 5 7 2)

=> (1 3 4 6 8)
```

SYNOPSIS:

```
(defun remove-more (list test &rest remove)
```

22.91 utilities/repeat-env

[utilities] [Functions]

DATE:

```
11th December 2025.
```

DESCRIPTION:

```
Repeat an envelope any number of times. Also liften from CLM directly,
including Bill Schottstaedt's original comment here:

repeat-envelope will repeat an envelope the number of times specified by its
second argument.  Hence if you specify (repeat-envelope '(0 0 100 1) 2) the
result will be (0 0 100 1 101 0 201 1). Because the final y value was
different from the first y value, a quick ramp will be inserted between
repeats.  You can have every other repeat be a reflection of the given
envelope by setting the optional second argument to t.  In that case,
(repeat-envelope '(0 0 100 1) 2 t) the result will be (0 0 100 1 200 0).  If
you want the original x axis limits respected by the resultant envelope, set
the third argument ("x-normalized") to t.
```

ARGUMENTS:

```
- the list of x,y breakpoint pairs (the envelope)
- the number of times to repeat the envelope
```

OPTIONAL ARGUMENTS:

```
 - T or nil to reflect the envelope (see above)
 - T or nil to use the original x-axis limits (see above also)
```

RETURN VALUE:

```
 the new envelope
```

SYNOPSIS:

```
(defun repeat-env (ur-env ur-num-times &optional reflected x-normalized)
```

22.92 utilities/replace-elements

[utilities] [Functions]

DESCRIPTION:

```
 Replace the elements in list between start and end (inclusive) with the new
 list.
```

ARGUMENTS:

```
 - A list.
 - An integer that is first position of the segment of the original list to
   be replaced.
 - An integer that is the last position of the segment of the original list
   to be replaced.
 - A list that is to replace the specified segment of the original
   list. This list can be of a different length than that of the segment
   of the original specified by the start and end positions.
```

RETURN VALUE:

```
 A list.
```

EXAMPLE:

```
(replace-elements '(1 2 3 4 5 6 7 8 9) 3 7 '(dog cat goldfish))

=> (1 2 3 DOG CAT GOLDFISH 9)
```

SYNOPSIS:

```
(defun replace-elements (list start end new)
```

22.93 utilities/rescale

[utilities] [Functions]

DATE:

```
June 8th 2016, Edinburgh
```

DESCRIPTION:

```
Given a value within an original range, return its value withing a new range
```

ARGUMENTS:

```
- the value we want to rescale
- the original minimum
- the original maximum
- the new minimum
- the new maximum
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
:out-of-range. The function to call when the first argument is not within
the range of arguments two and three. This would normally be #'error,
#'warn or NIL. If #'warn or NIL, argument 1 will be hard-limited to the
original range. Default = #'error
:type-of-result. Usually this function uses float precision, but by setting
type-of-result to #'double-float or #'rationalize, it is more precise.
```

RETURN VALUE:

```
The value within the new range (a number)
```

EXAMPLE:

```
(rescale .5 0 1 0 100)
==> 50.0
```

SYNOPSIS:

```
(defun rescale (val min max new-min new-max &optional (out-of-range #'error)
                                                (type-of-result #'float))
```

22.94 utilities/reverse-env

[utilities] [Functions]

DATE:

29th November 2025

DESCRIPTION:

Reverse an envelope. This is lifted directly from CLM's envelope-reverse because we might increasingly not have CLM available to call directly.

ARGUMENTS:

A list of breakpoint pairs

RETURN VALUE:

The new breakpoint pairs list

EXAMPLE:

```
(reverse-env '(0 .1 30 .3 100 1)) --> (0 1 70 0.3 100 0.1)
```

SYNOPSIS:

```
(defun reverse-env (e)
```

22.95 utilities/round-if-close

[utilities] [Functions]

DESCRIPTION:

Round a decimal number if it is within a given tolerance to the next whole number.

ARGUMENTS:

- A decimal number.

OPTIONAL ARGUMENTS:

```
- If the given number is this amount or less than the nearest whole number,
  round the given number to the nearest whole number.
```

RETURN VALUE:

```
If the given number is within the tolerance, return the number, otherwise
return the nearest whole number.
```

EXAMPLE:

```
(round-if-close 1.999998)

=> 1.999998

(round-if-close 1.999999)

=> 2
```

SYNOPSIS:

```
(defun round-if-close (num &optional (tolerance 0.000001))
```

22.96 utilities/scale-env

[utilities] [Functions]

DESCRIPTION:

```
Scale either the x-axis values, the data values, or both of a list of
break-point pairs by specified factors.
```

ARGUMENTS:

```
- An envelope in the form of a list of break-point pairs.
- A number that is the factor by which the y values (data segment of the
  break-point pairs) are to be scaled.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :y-min. A number that is the minimum value for all y values after
  scaling.  NB The -min/-max arguments are hard-limits only; they do not
  factor into the arithmetic.
- :y-max. A number that is the maximum value for all y values after
```

```
   scaling.
 - :x-scaler. A number that is the factor by which to scale the x-axis
   values of the break-point pairs.
 - :x-min. A number that is the minimum value for all x values after
   scaling. NB: This optional argument can only be used if a value has been
   specified for the :x-scaler.
 - :x-max. A number that is the maximum value for all x values after
   scaling. NB: This optional argument can only be used if a value has been
   specified for the :x-scaler.
 - :first-x. If a number, scale the x-axis so that this is the first x-value.
   This then ignores x-scaler.
 - :last-x. If a number, scale the x-axis so that this is the last x-value.
   This then ignores x-scaler.
```

RETURN VALUE:

```
 An envelope in the form of a list of break-point pairs.
```

EXAMPLE:

```
;;; Scaling only the y values.
(scale-env '(0 53 25 189 50 7 75 200 100 3) 0.5)

=> (0 26.5 25 94.5 50 3.5 75 100.0 100 1.5)

;;; Scaling the y values and setting a min and max for those values
(scale-env '(0 53 25 189 50 7 75 200 100 3) 0.5 :y-min 20 :y-max 100)

=> (0 26.5 25 94.5 50 20 75 100 100 20)

;;; Scaling only the x-axis values
(scale-env '(0 53 25 189 50 7 75 200 100 3) 1.0 :x-scaler 2)

=> (0 53.0 50 189.0 100 7.0 150 200.0 200 3.0)

;;; Scaling the x values and setting a min and max for those values
(scale-env '(0 53 25 189 50 7 75 200 100 3) 1.0 :x-scaler 2 :x-min 9 :x-max 90)

=> (9 53.0 50 189.0 90 7.0 90 200.0 90 3.0)

;;; 'Stretching' the envelope by providing first-x and last-x values
(scale-env '(1 0 5 1 20 0) 1 :first-x 0 :last-x 100)

=> (0.0 0 21.052631 1 100.0 0)
```

SYNOPSIS:

```
(defun scale-env (env y-scaler &key x-scaler first-x last-x
                                    (x-min most-negative-double-float)
                                    (y-min most-negative-double-float)
                                    (x-max most-positive-double-float)
                                    (y-max most-positive-double-float))
```

22.97 utilities/secs-to-mins-secs

[utilities] [Functions]

DESCRIPTION:

```
Convert a number of seconds into a string of the form "24:41.723" where
seconds are always rounded to three decimal places (i.e. milliseconds).
```

ARGUMENTS:

```
- the number of seconds
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :post-mins. The string used to separate minutes and seconds. Default ":"
- :post-secs. The string used to separate seconds and milliseconds.
   Default "."
- :post-msecs. The string used to follow milliseconds. Default ""
- :same-width. Ensure minutes values are always two characters wide, like
  seconds, i.e with a leading 0.
- :round. Round to the nearest second and don't print milliseconds. Default
  NIL.
```

RETURN VALUE:

```
A string
```

EXAMPLE:

```
(secs-to-mins-secs 77.1232145)
"1:17.123"
(secs-to-mins-secs 67.1)
"1:07.100"
(secs-to-mins-secs 67.1 :same-width t)
"01:07.100"
(secs-to-mins-secs 67.1 :same-width t :post-secs "s")
"01:07s100"
```

```
(secs-to-mins-secs 67.1 :post-secs "secs" :post-mins "min" :post-msecs "msecs")
"1min07secs100msecs"
(secs-to-mins-secs 67.7 :same-width t :round t)
"01:08"
```

SYNOPSIS:

```
(defun secs-to-mins-secs (seconds &key
                                  round
                                  (post-mins ":")
                                  (post-secs ".")
                                  (post-msecs "")
                                  (same-width nil))
```

22.98 utilities/semitones

[utilities] [Functions]

DESCRIPTION:

Return the sample-rate conversion factor required for transposing an audio file by a specific number of semitones. The number of semitones can be given as a decimal number, and may be positive or negative.

ARGUMENTS:

- A number of semitones.

OPTIONAL ARGUMENTS:

- A number that is the factor required to transpose by an octave. Default = 2.0.
- A number that is the number of semitones per octave. Default = 12.

RETURN VALUE:

A number.

EXAMPLE:

```
;;; Usage with default values
(semitones 3)

=> 1.1892071
```

```
;;; Specifying a different number of semitones per octave
(semitones 3 2.0 13)

=> 1.1734605

;;; Specifying a different factor for transposing by an octave
(semitones 3 4.0)

=> 1.4142135

;;; Fractional semitones are allowed
(semitones 3.72)

=> 1.2397077

;;; Negative semitones are also allowed
(semitones -3.72)

=> 0.80664176
```

SYNOPSIS:

```
(defun semitones (st &optional (octave-size 2.0) (divisions-per-octave 12))
```

22.99 utilities/setf-last

[utilities] [Functions]

DESCRIPTION:

```
Change the last element in a given list to a specified new element.
```

ARGUMENTS:

```
- A list.
- The new last element of that list.
```

RETURN VALUE:

```
Returns the new last element.
```

EXAMPLE:

```
(let ((l '(1 2 3 4 5)))
```

```
  (setf-last l 'dog)
  l)

=> (1 2 3 4 DOG)
```

SYNOPSIS:

```
(defmacro setf-last (list new-last)
```

22.100 utilities/smooth-procession

[utilities] [Functions]

DATE:

```
23rd August 2024
```

DESCRIPTION:

```
like the procession function but ensuring that we only ever ascend or
descend values by a step of 1
```

ARGUMENTS:

```
the same as the procession function
```

OPTIONAL ARGUMENTS:

```
ditto for the keyword arguments
```

RETURN VALUE:

```
the procession list
```

SYNOPSIS:

```
(defun smooth-procession (num-results items
                          &rest keyargs &key &allow-other-keys)
```

22.101 utilities/sort-symbol-list

[utilities] [Functions]

DESCRIPTION:

Sort a list of symbols alphabetically ascending, case-insensitive.

ARGUMENTS:

A list of symbols.

RETURN VALUE:

The same list of symbols sorted alphabetically ascending, case-insensitive.

EXAMPLE:

```
(sort-symbol-list '(Lorem ipsum dolor sit amet consectetur adipiscing))

=> (ADIPISCING AMET CONSECTETUR DOLOR IPSUM LOREM SIT)
```

SYNOPSIS:

```
(defun sort-symbol-list (list)
```

22.102 utilities/splice

[utilities] [Functions]

DESCRIPTION:

Insert the elements of a first list into a second list beginning at a specified index (0-based).

ARGUMENTS:

- A list that contains the elements to be inserted into the second list.
- A list into which the elements of the first argument are to be inserted.
- An integer that is the index within the second list where the elements are to be inserted.

RETURN VALUE:

- A list.

EXAMPLE:

```
(splice '(dog cat goldfish) '(1 2 3 4 5 6 7 8 9) 3)

=> (1 2 3 DOG CAT GOLDFISH 4 5 6 7 8 9)
```

SYNOPSIS:

```
(defun splice (elements into-list where)
```

22.103 utilities/split-groups

[utilities] [Functions]

DESCRIPTION:

```
Create a list consisting of as many repetitions of a specified number as
will fit into a given greater number, with the last item in the new list
being the value of any remainder.
```

ARGUMENTS:

```
- A number that is to be split into repetitions of a specified smaller
  number (the second argument).
- The number that is to be the repeating item in the new list. This number
  must be smaller than the first number.
```

RETURN VALUE:

```
A list consisting of repetitions of the specified number, with the last
element being any possible remainder.
```

EXAMPLE:

```
(split-groups 101 17)

=> (17 17 17 17 17 16)
```

SYNOPSIS:

```
(defun split-groups (num divider)
```

22.104 utilities/split-into-sub-groups

[utilities] [Functions]

DESCRIPTION:

```
Create a new list consisting of sublists made from the elements of the
original flat list, whose lengths are determined by the second argument to
```

```
the function.

NB: The lengths given in the second argument are not required to add up to
    the length of the original list. If their sum is less than the original
    list, the resulting list of sublists will only contain a segment of the
    original elements. If their sum is greater than the length of the
    original list, the last sublist in the new list will be shorter than
    the corresponding group value.
```

ARGUMENTS:

```
- A flat list.
- A list of integers that are the lengths of the consecutive subgroups
  into which the original list is to be divided.
```

RETURN VALUE:

```
A list of lists.
```

EXAMPLE:

```
;; Used with a list of subgroup lengths whose sum is equal to the length of the
;; original list
(split-into-sub-groups '(1 2 3 4 5 6 7 8 9 10) '(2 2 3 2 1))

=> ((1 2) (3 4) (5 6 7) (8 9) (10))

;; Used with a list of subgroup lengths whose sum is less than the length of the
;; original list
(split-into-sub-groups '(1 2 3 4 5 6 7 8 9 10) '(2 1))

=> ((1 2) (3))

;; Used with a list of subgroup lengths whose sum is greater than the length of
;; the original list
(split-into-sub-groups '(1 2 3 4 5 6 7 8 9 10) '(2 3 17))

=> ((1 2) (3 4 5) (6 7 8 9 10))
```

SYNOPSIS:

```
(defun split-into-sub-groups (list groups)
```

22.105 utilities/split-into-sub-groups2

[utilities] [Functions]

DESCRIPTION:

```
Create a new list of lists by splitting the original flat list into
sublists of the specified length.

NB: The length given as the second argument is not required to be fit
    evenly into the length of the original flat list. If the original list
    is not evenly divisible by the specified length, the resulting list of
    sublists will contain a final sublist of a different length.
```

ARGUMENTS:

```
- A flat list.
- An integer that is the length of each of the sublists to be created.
```

RETURN VALUE:

```
A list of lists.
```

EXAMPLE:

```
;; The second argument fits evenly into the length of the original list.
(split-into-sub-groups2 '(1 2 3 4 5 6 7 8 9 10 11 12) 3)

=> ((1 2 3) (4 5 6) (7 8 9) (10 11 12))

;; The second argument does not fit evenly into the length of the original
;; list.

(split-into-sub-groups2 '(1 2 3 4 5 6 7 8 9 10 11 12) 5)

=> ((1 2 3 4 5) (6 7 8 9 10) (11 12))
```

SYNOPSIS:

```
(defun split-into-sub-groups2 (list length &optional shuffle)
```

22.106 utilities/split-into-sub-groups3

[utilities] [Functions]

DESCRIPTION:

```
Split a given flat list into sublists of the specified length, putting any
remaining elements, if there are any, into the last sublist.
```

ARGUMENTS:

```
- A flat list.
- An integer that is the length of the new sublists.
```

RETURN VALUE:

```
A list of lists.
```

EXAMPLE:

```
(split-into-sub-groups3 '(1 2 3 4 5 6 7 8 9 10 11 12) 3)

=> ((1 2 3) (4 5 6) (7 8 9) (10 11 12))

(split-into-sub-groups3 '(1 2 3 4 5 6 7 8 9 10 11 12) 5)

=> ((1 2 3 4 5) (6 7 8 9 10 11 12))
```

SYNOPSIS:

```
(defun split-into-sub-groups3 (list length)
```

22.107 utilities/split-into-sub-groups4

[utilities] [Functions]

DATE:

```
August 28th 2018, Heidhausen
```

DESCRIPTION:

```
Split into sub-groups with lengths defined by the integers in the second
argument list. The difference here to related functions is that the given
lengths will repeat circularly until no more elements of the first argument
remain.
```

ARGUMENTS:

```
- the list to split into sub-groups
- a list of lengths, to be repeated
```

RETURN VALUE:

```
a list of sublists
```

EXAMPLE:

```
(split-into-sub-groups4 '(1 2 3 4 5 6 7 8 9 10 11 12) '(3 4))
==> ((1 2 3) (4 5 6 7) (8 9 10) (11 12))

(split-into-sub-groups4 '(1 2 3 4 5 6 7 8 9 10 11 12 13 14) '(3 4))
==> ((1 2 3) (4 5 6 7) (8 9 10) (11 12 13 14))

(split-into-sub-groups4 '(1 2 3 4 5 6 7 8 9 10 11 12) '(1 2 3 4))
==> '((1) (2 3) (4 5 6) (7 8 9 10) (11) (12))
```

SYNOPSIS:

```
(defun split-into-sub-groups4 (list lengths)
```

22.108 utilities/srt

[utilities] [Functions]

DESCRIPTION:

```
Return the semitone transposition for a given sampling rate conversion
factor.
```

ARGUMENTS:

```
- A number that is a sample-rate conversion factor.
```

OPTIONAL ARGUMENTS:

```
- A number that is the factor required for transposing one octave.
- A number that is the number of scale degrees in an octave.
```

RETURN VALUE:

```
A number.
```

EXAMPLE:

```
;;; Using the defaults
(srt 1.73)
```

```
=> 9.4893

;;; Using a sample-rate conversion factor of 4.0 for the octave and specifying
;;; 13 divisions of the octave
(srt 1.73 4.0 13)

=> 5.14
```

SYNOPSIS:

```
(let ((last8vesize 0)
      (log8ve 0.0)) ;; so we don't have to recalculate each time
  (defun srt (srt &optional (octave-size 2.0) (divisions-per-octave 12)
              ;; MDE Tue Feb  7 16:59:45 2012 -- round so we don't get tiny
              ;; fractions of semitones due to float inaccuracies?
              (round-to 0.0001))
```

22.109 utilities/string-replace

[utilities] [Functions]

DESCRIPTION:

```
Replace specified segments of a string with a new specified string.
```

ARGUMENTS:

```
- A string that is the string segment to be replaced.
- A string that is the string with which the specified string segment is to
  be replaced.
- The string in which the specified segment is to be sought and replaced.
```

RETURN VALUE:

```
A string.
```

EXAMPLE:

```
(string-replace "flat" "\\flat" "bflat clarinet")

=> "b\\flat clarinet"
```

SYNOPSIS:

```
(defun string-replace (what with string)
```

22.110 utilities/swap-elements

[utilities] [Functions]

DESCRIPTION:

```
Swap the order of each consecutive pair of elements in a list.
```

ARGUMENTS:

```
- A list.
```

RETURN VALUE:

```
A list.
```

EXAMPLE:

```
(swap-elements '(1 2 3 4 5 6 7 8 9 10))

=> (2 1 4 3 6 5 8 7 10 9)

(swap-elements '(1 2 3 4 5 6 7 8 9))

=> (2 1 4 3 6 5 8 7 9)
```

SYNOPSIS:

```
(defun swap-elements (list)
```

22.111 utilities/update-app-src

[utilities] [Functions]

DATE:

```
June 1st 2013
```

DESCRIPTION:

```
DISABLED DUE TO SVN SERVER ACCESS RESTRICTIONS AND SERVER CHANGES.
NB This function currently works in SBCL and CCL on UNIX systems only.

For users of the slippery chicken app, this function will update the source
code of the app to the latest in the online subversion (svn) repository.
```

```
An internet connection is therefore necessary.

The first time it is run it will delete the current source code and
download all the new source code, so make sure to back up if you've
modified the source code yourself (not recommended).  When it is run from
then on, it will only update the source code that is out of date.

Once the source code is updated, you'll need to restart the app or just
Lisp for the changes to be recompiled.

**NB** The first time you call this function, you might get a "certificate
error".  In order to accept the certificate, start the terminal application
and type the following:

cd /tmp/
svn co https://svn.ecdf.ed.ac.uk/repo/user/medward2/sc-tags/sc-latest/src

That should give you a prompt in the terminal from which you can accept the
certificate.  Then the next time you try it from Lisp the certificate
should not cause a problem.

Users without the app can always download the latest source code in a
terminal by issuing the following command.
svn co https://svn.ecdf.ed.ac.uk/repo/user/medward2/sc-tags/sc-latest/src
```

ARGUMENTS:

```
The full path to the slippery-chicken application, minus the last slash.
Remember that this can't include any spaces in file/folder names
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :rm.  The path to the shell 'rm' command.  Default = "/bin/rm"
- :svn.  The path to the shell 'svn' command.  Default = "/usr/bin/svn"
```

RETURN VALUE:

```
The shell return value of the call to SVN, usually 0 on success.
```

EXAMPLE:

```
Running for the first time:
(update-app-src "/tmp/sc-app/slippery-chicken.app")
A    /tmp/sc-app/slippery-chicken.app/Contents/Resources/sc/src/sndfile.lsp
```

```
A    /tmp/sc-app/slippery-chicken.app/Contents/Resources/sc/src/osc.lsp
A    /tmp/sc-app/slippery-chicken.app/Contents/Resources/sc/src/osc-sc.lsp
[...]
Checked out revision 3608.
0

or after successfully updating a previously updated version:
...
At revision 3608.
0
```

SYNOPSIS:

```
(defun update-app-src (path-to-app &key (rm "/bin/rm") (svn "/usr/bin/svn"))
```

22.112 utilities/visualize

[utilities] [Functions]

DATE:

```
February 23rd 2023
```

DESCRIPTION:

```
Print a Visualization of an array or a list into the repl with 64 values
-> *ascii art* <-
```

ARGUMENTS:

```
- An array or a list
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
:y-range. Maximum value that the y-axis display. When nil, the graph will be
normalized.
:start. Where to start reading the sequence from. Default = 0
:abs. When t, the absolute value of all numbers is visualized
:scale. If the sequence is shorter than 64 and scale is t, the graph will
be scaled to 64 values
```

RETURN VALUE:

```
":)"
```

EXAMPLE:

```
(visualize (loop repeat 64 for i from 0 by 0.1 collect (sin i)))
(visualize (loop repeat 128 for i from 0 by 0.1 collect (sin i)) :start 64)
(visualize (loop repeat 128 for i from 0 by 0.1 collect (* (sin i) 2))
   :scale nil :start 96)
(visualize (loop repeat 55 for i from 0 by 0.1 collect (* (sin i) 2))
   :scale nil :abs t :y-range 1)
```

SYNOPSIS:

```
(defun visualize (ls &key y-range (start 0) abs (scale t))
```

22.113 utilities/wavelab-to-audacity-marker-file

[utilities] [Functions]

DESCRIPTION:

```
Write a .txt file suitable for import to audacity with the same name and in
the same directory as the file argument.
```

ARGUMENTS:

```
- A string that is the name of a wavelab marker file, including directory
  path and extension.
```

OPTIONAL ARGUMENTS:

```
- An integer that is the sampling rate of the sound file to which the
  wavelab marker file refers. This value will affect the times of the
  output.
```

RETURN VALUE:

```
Returns T and prints the number of markers read to the listener.
```

EXAMPLE:

```
(wavelab-to-audacity-marker-file "/path/to/24-7.mrk"  44100)

=> 51 markers read
```

SYNOPSIS:

```
(defun wavelab-to-audacity-marker-file (file &optional (sampling-rate 44100))
```

22.114 utilities/wide-steps

[utilities] [Functions]

DATE:

10th July 2026

DESCRIPTION:

Sample elements of a list at wide intervals, rotating as necessary.

ARGUMENTS:

- The list to sample

OPTIONAL ARGUMENTS:

- The number of steps in a rotation cycle i.e. the number of points in the list we'll sample in each rotation.

RETURN VALUE:

A new list.

EXAMPLE:

```
(wide-steps (loop for i from 1 to 32 collect i) 6) -->
(1 5 11 17 23 29 2 6 12 18 24 30 3 7 13 19 25 31 4 8 14 20 26 32 9 15 21 27 10
 16 22 28)

(wide-steps (loop for i from 1 to 7 collect i)) --> (1 3 2 4 5 6 7)
```

SYNOPSIS:

```
(defun wide-steps (list &optional (num-steps 2))
```

22.115 utilities/wrap-list

[utilities] [Functions]

DESCRIPTION:

Shift the elements of a list to start at a specified position and wrap to the beginning of the list to the list's tail.

ARGUMENTS:

```
- A list.
- An integer which is the 0-based position in the original list where the
  new list is to begin.
```

RETURN VALUE:

```
A list.
```

EXAMPLE:

```
(wrap-list '(1 2 3 4 5 6 7 8 9) 4)

=> (5 6 7 8 9 1 2 3 4)
```

SYNOPSIS:

```
(defun wrap-list (list start)
```

22.116 utilities/write-list-to-coll

[utilities] [Functions]

DATE:

```
Tue 18 Feb 2020 15:38:40 GMT
```

DESCRIPTION:

```
Turn a list of lists into a text file, formatted to be read by the MaxMSP
[coll] object. This is a bit like gen-max-coll-file (see set-palette.lsp)
but instead works with any data in a list.
```

ARGUMENTS:

```
- A list of lists in the form '((a b c) (d e f))
```

OPTIONAL ARGUMENTS:

```
keyword arguments
:file - the output file. Default = "/tmp/sc-max-coll.txt"
:base - the minimum number for coll indexing. In the resulting output file,
each list in the list of lists will be preceeded by an (increasing) integer
```

```
and a comma. This argument sets the base value of that integer. Default = 0.
:capitalize - Should any outputted text be capitalized or not?
Default = NIL.
:if-exists - what to do if the file already exists. This argument is passed
to with-open-file. More info here: http://clhs.lisp.se/Body/m_w_open.htm
Default = :supercede
:prefix - add a string prefix to the item number. Default = NIL.
:alt-label - if you do not want the items labels to be consecutive numbers,
then you can here either provide a list of lists or a function. If a list of
lists, this must be the same length as the first argument. If a function, it
must be called with the item number (which increases incrementally from
base).
```

RETURN VALUE:

```
The output file location
```

EXAMPLE:

```
(let ((l '((hello!)(how are you?)(very well thank you.)(1 2 3 4))))
      (write-list-to-coll l :base 6))

=> "/tmp/sc-max-coll.txt"

The resulting text file will looks like this when opened:

6, hello!;
7, how are you?;
8, very well thank you.;
9, 1 2 3 4;

;; DJR Tue 3 Mar 2020 13:52:34 GMT
(let ((l '((hello!)(how are you?)(very well thank you.)(1 2 3 4))))
      (write-list-to-coll l :base 15
                            :alt-label #'(lambda (count)
                                           (let ((l '(foo bar)))
                                             (nth (mod count 2) l)))
                            :prefix "yes_"))

=> "/tmp/sc-max-coll.txt"

The resulting text file will looks like this when opened:

yes_bar, hello!;
yes_foo, how are you?;
yes_bar, very well thank you.;
```

```
yes_foo, 1 2 3 4;
```

SYNOPSIS:

```
(defun write-list-to-coll (data-list &key (base 0)
                                          (file "/tmp/sc-max-coll.txt")
                                          (capitalize nil)
                                          (if-exists :supersede)
                                          ;; DJR Tue 3 Mar 2020 13:52:34 GMT
                                          (prefix "")
                                          alt-label)
```

23 clm/clm-loops

[Functions]

DESCRIPTION:

```
Generate a sound file from an existing specified sound file by shuffling
and repeating specified segments within the source sound file.

This function was first introduced in the composition "breathing Charlie"
(under the name loops): see charlie-loops.lsp in that project for examples.

The first required argument to the function is the name of the sound file,
including path and extension, looped. This must be a mono file.

The second required argument (entry-points) is a list of times, in seconds,
where attacks (or something significant) happen in the file. These are used
to create loop start/end points.

Be careful when doing shuffles as if, e.g., the transpositions list is more
than 6 elements, shuffling will take a very long time.

The entry-points are used randomly so that any segment may start at any
point and transition to any other segment (i.e. skipping intervening
segments, always forwards however). There are always two segments in use at
any time. The function randomly selects which segments are used, then a
transition (see fibonacci-transitions) from repeated segment 1 to repeated
segment 2 is made. Then the next segment is chosen and the process is
repeated (i.e. from previous segment 2 to new segment) until the
max-start-time (in seconds) is achieved.

fibonacci-transitions are first shuffled and then made into a circular
list. Then they are expanded to create the transpositions (each number
```

becomes a series of 1s and 0s--length is the number itself--with a transition from all 0s to all 1s: e.g. (fibonacci-transition 20) -> (0 0 0 0 1 0 0 1 0 1 0 1 0 1 0 1 1 1) This is then used to select one or the other of the current two segments.

The sample-rate transpositions are simply randomly permutated and selected.

ARGUMENTS:

- The name of a sound file, including path and extension.
- A list of numbers that are time in seconds. These serve as the "entry-points", i.e. loop markers within the file, and delineate the beginning and end of segments that will be shuffled and played back at random in the resulting file.

OPTIONAL ARGUMENTS:

keyword arguments.
- :max-perms. A number that is the maximum number of permutations generated for the transitions. Default = 1000.
- :fibonacci-transitions. A list of numbers that serve as the number of steps in each transition from one segment to the next. These numbers will be used as the first argument to the call to fibonacci-transition. Default = '(34 21 13 8)
- :max-start-time. A number that is the maximum time in second at which a segment can start in the resulting sound file. Default = 60.0.
- :output-dir. The directory path for the output file. Default = (get-sc-config 'default-dir).
- :srate. The sampling rate. If specified by the user, this will generally be a number. By default it takes the CLM global sample-rate, i.e. clm::*clm-srate*
- :data-format. The data format of the resulting file. This must be preceded by the clm package qualifier. See clm.html for types of data formats, such as mus-bshort, mus-l24float etc. Default is the whatever the CLM global clm::*clm-data-format* is set to.
- :header-type. The header type of the resulting file. This must be preceded by the clm package qualifier. See clm.html for possible header types, such as mus-riff, mus-aifc etc. By default it takes the CLM global clm::*clm-header-type*.
- :sndfile-extension. A string or NIL. If a string, this will be appended to the resulting sound file as a file extension. If NIL, the sound file extension will automatically be selected based on the header type. NB: This argument does not affect the header type! Default = NIL.
- :channels. An integer that is the number of channels in the resulting output. If greater than one, the segments will be automatically panned amongst the channels. Default = 1.

```
 - :transpositions. A list of number that are transpositions in
   semitones. These will be shuffled and applied randomly to each
   consecutive segment in the output. Default = '(0).
 - :num-shuffles. An integer that will indicate how many times the lists
   passed to fibonacci-transitions and entry-points will be shuffled before
   generating output. Default = - 1.
 - :suffix. A string that will be automatically appended to the end of the
   file name. Default = "".
 - :src-width. A number that represents the accuracy of the sample-rate
   conversions undertaken for transposition. The higher this number is, the
   more accurate the transposition will be, but the longer it will take to
   process the file. Default = 5.
 - :scaled-to. The normalisation target. Usually < 1.0. Default = 0.99.
 - :pan. Whether amplitude panning should be applied or simple output to
   individual channels. Default = T.
```

RETURN VALUE:

```
 Returns the name of the file generated.
```

EXAMPLE:

```
;;; A straightforward example with a number of the variables.
(clm-loops "/path/to/sndfile-3.aiff"
           '(0.180 2.164 4.371 7.575 9.4 10.864)
           :fibonacci-transitions '(1 2 3 4 5)
           :max-perms 7
           :output-dir "/tmp/"
           :channels 1
           :transpositions '(1 12 -12)
           :num-shuffles 3
           :src-width 20)

=> "/tmp/sndfile-3-loops-from-00m00.180-.wav"
```

SYNOPSIS:

```
#+clm
(defun clm-loops (sndfile entry-points
                  &key
                  (max-perms 1000)
                  (fibonacci-transitions '(34 21 13 8))
                  (max-start-time 60.0)
                  (output-dir (get-sc-config 'default-dir))
                  (srate clm::*clm-srate*)
```

```
(data-format clm::*clm-data-format*)
;; MDE Fri May 11 15:33:45 2012
(header-type clm::*clm-header-type*)
;; MDE Fri May 11 15:34:17 2012 --
(sndfile-extension nil)
(channels 1)
;; MDE Fri Jul  2 09:43:05 2021, Heidhausen -- added
(scaled-to .99)
;; semitones
(transpositions '(0))
;; added 31/7/05 to vary the order of
;; entry points, transpositions and
;; fibonacci-transitions (could be 0!)
(num-shuffles 1)
(suffix "")
(pan t)
(src-width 5))
```

24 clm/clm-loops-all

[Functions]

DESCRIPTION:

```
Similar to clm-loops, but takes a list of lists of entry points (which can
also be generated using the random-loop-points function, for example) and
produces one output sound file for each list of entry points that list
contains.
```

ARGUMENTS:

```
- A string that is the name of the source sound file including directory
  path and extension.
- A list of lists of numbers that are entry points (loop markers) in the
  specified source sound file.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :max-perms. A number that is the maximum number of permutations generated
  for the transitions. Default = 1000.
- :fibonacci-transitions. A list of numbers that serve as the number of
  steps in each transition from one segment to the next. These numbers will
  be used as the first argument to the call to fibonacci-transition.
```

```
  Default = '(34 21 13 8).
- :max-start-time. A number that is the maximum time in seconds at which a
  loop segment can start in the resulting sound file. So this is the
  approximate duration. If a list, then the durations will be used
  circularly for each output sound file. Default = 60.0.
- :output-dir. The directory path for the output file.
  Default = (get-sc-config 'default-dir).
- :srate. The sampling rate. If specified by the user, this will generally
  be a number. By default it takes the CLM global sample-rate, i.e.
  clm::*clm-srate*
- :data-format. The data format of the resulting file. This must be
  preceded by the clm package qualifier. See clm.html for types of data
  formats, such as mus-bshort, mus-l24float etc.
  Default is the whatever the CLM global clm::*clm-data-format* is set to.
- :header-type. The header type of the resulting file. This must be
  preceded by the clm package qualifier. See clm.html for possible header
  types, such as mus-riff, mus-aifc etc. By default it takes the CLM global
  clm::*clm-header-type*.
- :sndfile-extension. A string or NIL. If a string, this will be appended
  to the resulting sound file as a file extension. If NIL, the sound file
  extension will automatically be selected based on the header type.  NB:
  This argument does not affect the header type! Default = NIL.
- :channels. An integer that is the number of channels in the resulting
  output. If greater than one, the segments will be automatically panned
  amongst the channels. Default = 1.
- :do-shuffles. T or NIL to indicate whether to shuffle the lists passed to
  fibonacci-transitions and entry-points before generating output.
  T = do shuffles. Default = T.
- :start-after. A number. All loops will be excluded that start before this
  number of seconds. Default = -1.0.
- :stop-after. A number. All loops will be excluded that start after this
  number of seconds. Default =  99999999.0.
- :suffix. A string that will be automatically appended to the end of the
  file name. Default = "".
- :transpositions. A list of number that are transpositions in
  semitones. These will be shuffled and applied randomly to each
  consecutive segment in the output. Default = '(0).
- :transposition-offset. A number that is an additional number of semitones
  to be added to each transposition value before performing the
  transposition. Default = 0.0.
- :src-width. A number that represents the accuracy of the sample-rate
  conversions undertaken for transposition. The higher this number is, the
  more accurate the transposition will be, but the longer it will take to
  process the file. Default = 5.
- :scaled-to. The normalisation target. Usually < 1.0. Default = 0.99.
- :pan. Whether amplitude panning should be applied or simple output to
```

```
   individual channels. Default = T.
```

RETURN VALUE:

```
 Returns NIL.
```

EXAMPLE:

```
(clm-loops-all
 (concatenate 'string
              cl-user::+slippery-chicken-home-dir+
              "test-suite/test-sndfiles-dir-1/test-sndfile-3.aiff")
 '((0.794 0.961 1.061 1.161 1.318 1.436 1.536)
   (0.787 0.887 0.987 1.153 1.310 1.510)
   (0.749 0.889 1.056 1.213 1.413)
   (0.311 0.411 0.611 0.729)
   (0.744 0.884 1.002))
 :max-perms 6
 :fibonacci-transitions '(31 8 21 13)
 :output-dir "/tmp/"
 :channels 1
 :transpositions '(1 12 -12)
 :src-width 20)
```

SYNOPSIS:

```
#+clm
(defun clm-loops-all (sndfile entry-points-list
                      &key
                      (max-perms 1000)
                      (fibonacci-transitions '(34 21 13 8))
                      (max-start-time 60.0)
                      (output-dir (get-sc-config 'default-dir))
                      (srate clm::*clm-srate*)
                      (data-format clm::*clm-data-format*)
                      ;; MDE Fri May 11 15:33:45 2012
                      (header-type clm::*clm-header-type*)
                      ;; MDE Fri May 11 15:34:17 2012 --
                      (sndfile-extension nil)
                      (channels 1)
                      (do-shuffles t) ;; see clm-loops
                      ;; exclude all those loops who start before this
                      ;; number of seconds.
                      (start-after -1.0)
                      (stop-after 99999999.0)
```

```
(suffix "")
;; semitones
;; 6/10/06: using just one list of transpositions passed
;; onto clm-loops created the same tone structure for
;; every file generated (boring).  This list will now be
;; shuffled and 10 versions collected which will then be
;; passed (circularly) one after the other to clm-loops.
(transpositions '(0))
;; MDE Thu Aug 26 15:56:34 2021, Heidhausen -- added
(pan t)
;; MDE Fri Jul  2 09:43:05 2021, Heidhausen -- added
(scaled-to .99)
(transposition-offset 0.0)
(src-width 5))
```

25 clm/random-loop-points

[Functions]

DESCRIPTION:

```
Return a list of lists of randomly generated entry points (loop markers)
for use with clm-loops-all.

This function also produces an output text file containing the same list of
lists. This file is in Lisp syntax and can therefore be accessed using
read-from-file.
```

ARGUMENTS:

```
- A string that is the file name, including directory path and extension,
  of the output file to produce.
- A string that is the sound file for which to generate random entry
  points.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :min-points. An integer that is the least number of entry points to
  generate for each list. Default = 5.
- :max-points. An integer that is the greatest number of entry points to
  generate for each list. Default = 13.
- :min-dur. A number that is the shortest duration between two entry
  points. Default = 0.05.
```

- :num-loop-sets. An integer that is the number of lists of entry points to generate. Default = 20.
- :scalers. A list of fractions that are durations relative to the min-dur, such that, for example, a min-dur of 0.05 with a scaler of 13/8 would result in a scaled duration of 0.08125. The fractions in this list will be chosen at random when calculating the duration of the next loop segment. Default = '(1/1 2/1 3/2 5/3 8/5 13/8).

RETURN VALUE: EXAMPLE:

```
(random-loop-points
 "/tmp/outfile"
 "/path/to/test-sndfile-3.aiff"
 :min-points 3
 :max-points 7
 :min-dur 0.1
 :num-loop-sets 5
 :scalers '(1/1 2/1 3/2 5/3 7/5 11/7 13/11))

=> ((0.789 0.929 1.079) (0.028 0.228 0.368 0.487 0.687) (0.014 0.164 0.321)
    (0.256 0.406 0.524 0.681) (0.069 0.235 0.353 0.472 0.572 0.69))
```

SYNOPSIS:

```
#+clm
(defun random-loop-points (outfile sndfile
                           &key
                           ;; MDE Thu May 17 17:02:15 2012 -- could also be
                           ;; :error or anything else that with-open-file
                           ;; accepts
                           (if-outfile-exists :overwrite)
                           ;; the minimum number of time points for an output
                           ;; loop--number of looped sound segments is 1- this
                           (min-points 5)
                           ;; max number of time points--the actual number of
                           ;; points will be randomly chosen between these two
                           ;; numbers.
                           (max-points 13)
                           ;; minimum duration of a loop segment--this number
                           ;; will actually be used and scaled by scalers
                           (min-dur 0.05)
                           ;; how many sets of loops should be generated
                           (num-loop-sets 20)
                           ;; scalers for the min-dur: these are all
                           ;; proportions relative to min-dur so if we have
                           ;; 13/8 in this list and min-dur of 0.05 then the
```

```
;; duration for such a segment would be 0.08125.
;; these will be chosen at random when calculating
;; the next loop segment duration
(scalers '(1/1 2/1 3/2 5/3 8/5 13/8)))
```

26 csound/csound-p-fields-simple

[Functions]

DATE:

```
2023-02-28, Essen
```

DESCRIPTION:

```
This function is intended to be used in conjunction with
the write-csound-score method. It returns a list or -- when the given event
is a chord -- a list of lists of p-values.

The p-fields are allocated as follows:
- p4: frequency
- p5: amplitude

N.B.: The event-num and cs-instrument arguments are mandatory as they
      are required by the write-csound-score method, even though they are
      not used in this function.
```

ARGUMENTS:

```
- An event object (most likely the event currently processed by
  write-csound-score).
- A number referring as an index to the position in processing sequence.
- A reference to a Csound-instrument (number or string).
```

RETURN VALUE:

```
Returns either a list (when the event is a pitch) or a list of lists
(in case the event contains a chord) with p4- and p5-values (see
above).
```

SYNOPSIS:

```
(defun csound-p-fields-simple (event event-num cs-instrument)
```

27 emacs-utilities/validate-rthm-seq

[Functions]

DESCRIPTION:

```
Tests if a given list can be used to instantiate a rthm-seq object.
This is done by calling the make-rthm-seq function and reporting whether
the instantiation process has been successful or erraneous without actually
throwing a Common Lisp error.
```

ARGUMENTS:

```
The rthm-seq list as described in the slippery-chicken documentation.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :auto-convert. Automatically convert string input to a list? Must be a
  boolean. Default: T.
```

RETURN VALUE:

```
Either T, when the rthm-seq has been successfully validated, or the values:
- NIL
- The error condition.
```

EXAMPLE:

```
(validate-rthm-seq '((((3 4) q q q))
                     :pitch-seq-palette (1 2 3)))

;; => T

(validate-rthm-seq '((((3 4) q (q) q))
                     :pitch-seq-palette (1 2 3)))

;; => NIL, #<SIMPLE-ERROR "pitch-seq-palette::verify-and-store: ~%~
;;                  In pitch-seq ~a from palette ~a:~%~
;;                  Each pitch sequence must have ~a notes (you have ~a):
;;                  ~%~a" {7009EB3043}>
```

SYNOPSIS:

```
(defun validate-rthm-seq (rs &key (auto-convert t))
```

28 get-nearest-note/player

[Methods]

DATE:

```
Tue 11 Aug 2020 17:16:10 BST
```

DESCRIPTION:

```
Like get-note, but will return the nearest note before or after the
specified bar / event number place from, potentially, any bar.
```

ARGUMENTS:

```
- an sc object
- the bar number to start looking
- the event number within the bar to start looking
```

OPTIONAL ARGUMENTS:

```
None.
```

RETURN VALUE:

```
If found, the nearest sounding event in the stated player's part and the
number of events away (positive for forwards, negative for backwards),
otherwise NIL.
```

EXAMPLE:

```
(let ((mini
          (make-slippery-chicken
           '+mini+
           :ensemble '(((sax (alto-sax :midi-channel 1))
                        (vn (violin :midi-channel 1))))
           :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
           :set-map '((1 (1 1 1 1 1))
                      (2 (1 1 1 1 1))
                      (3 (1 1 1 1 1)))
           :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                   :pitch-seq-palette ((1 2 3 4 5))))
                               (2 ((((4 4) (w)))))
                               (3 ((((4 4) (h) q e (s) s))
                                   :pitch-seq-palette ((1 2 5)))))
```

```
           :rthm-seq-map '((1 ((sax (1 1 1 1 1))
                               (vn (2 2 2 2 2))))
                           (2 ((sax (2 2 2 2 2))
                               (vn (2 2 2 2 2))))
                           (3 ((sax (3 3 3 3 3))
                               (vn (2 2 2 2 2))))))))
    (get-nearest-note mini 11 1 'sax))
=>
EVENT: start-time: 42.000, end-time: 43.000,
       duration-in-tempo: 1.000,
       compound-duration-in-tempo: 1.000,
       amplitude: 0.700
       bar-num: 11, marks-before: NIL,
       tempo-change: NIL
       instrument-change: NIL
       display-tempo: NIL, start-time-qtrs: 42.000,
       midi-time-sig: NIL, midi-program-changes: NIL,
       midi-control-changes: NIL,
       8va: 0, player: SAX
       asco-label: NIL, asco-msgs: NIL
       set-ref: (1)
       pitch-or-chord:
PITCH: frequency: 164.814, midi-note: 52, midi-channel: 1
       pitch-bend: 0.0
       degree: 104, data-consistent: T, white-note: E3
       nearest-chromatic: E3
       src: 0.62996054, src-ref-pitch: C4, score-note: E3
       qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
       micro-tone: NIL,
       sharp: NIL, flat: NIL, natural: T,
       octave: 3, c5ths: 0, no-8ve: E, no-8ve-no-acc: E
       show-accidental: T, white-degree: 30,
       accidental: N,
       accidental-in-parentheses: NIL, marks: NIL,
       marks-before: NIL, amplitude: NIL
LINKED-NAMED-OBJECT: previous: NIL,
                     this: NIL,
                     next: NIL
NAMED-OBJECT: id: E3, tag: NIL,
data: E3
**************

       written-pitch-or-chord:
PITCH: frequency: 277.183, midi-note: 61, midi-channel: 1
       pitch-bend: 0.0
       degree: 122, data-consistent: T, white-note: C4
```

```
       nearest-chromatic: CS4
       src: 1.0594631, src-ref-pitch: C4, score-note: CS4
       qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
       micro-tone: NIL,
       sharp: T, flat: NIL, natural: NIL,
       octave: 4, c5ths: 2, no-8ve: CS, no-8ve-no-acc: C
       show-accidental: T, white-degree: 35,
       accidental: S,
       accidental-in-parentheses: NIL, marks: NIL,
       marks-before: NIL, amplitude: NIL
LINKED-NAMED-OBJECT: previous: NIL,
                     this: NIL,
                     next: NIL
NAMED-OBJECT: id: CS4, tag: NIL,
data: CS4
**************
RHYTHM: value: 4.000, duration: 1.000, rq: 1, is-rest: NIL,
        is-whole-bar-rest: NIL,
        score-rthm: 4.0, undotted-value: 4, num-flags: 0, num-dots: 0,
        is-tied-to: NIL, is-tied-from: NIL, compound-duration: 1.000,
        is-grace-note: NIL, needs-new-note: T, beam: NIL, bracket: NIL,
        rqq-note: NIL, rqq-info: NIL, marks: NIL, marks-in-part: NIL,
        letter-value: 4, tuplet-scaler: 1, bar-pos: 1,
        grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL,
                     this: NIL,
                     next: NIL
NAMED-OBJECT: id: Q, tag: NIL,
data: Q
**************

1
```

SYNOPSIS:

```
(defmethod get-nearest-note ((sc slippery-chicken) bar-num event-num player)
```

29 globals/+slippery-chicken-config-data+

[Global Parameters]

DESCRIPTION:

```
 A global to hold various user-settable configuration settings.  Use e.g.
 (set-sc-config 'default-dir "~/Desktop") or (get-sc-config 'default-dir) to
```

set or query the settings.

SYNOPSIS:

```
(defparameter +slippery-chicken-config-data+
  ;; The default directory for output of sound files, EPS files, and Lilypond
  ;; files. Don't forget the trailing slash (i.e. "/tmp/" not "/tmp").  Bear in
  ;; mind that on OSX the /tmp directory is emptied upon reboot so you shouldn't
  ;; store any files you'd like to keep in there.
  (let ((tmpdir #+(or win32 win64) "c:/tmp/" #-(or win32 win64) "/tmp/"))
    (make-instance
     'assoc-list
     :id 'slippery-chicken-config-data
     :data
     ;; MDE Sun Mar 25 10:39:07 2012 -- The following two are used in
     ;; pitch-seq::get-notes to indicate which lowest number in a pitch-seq
     ;; would indicate that we should select the highest or lowest notes
     ;; possible for the instrument/set.
     ;;
     ;; The first is used to indicate the lowest number in a pitch-seq that
     ;; would indicate that the get-notes algorithm should select the highest
     ;; notes possible for the instrument/set.
     `((pitch-seq-lowest-equals-prefers-high 5)
       ;; the lowest number in a pitch-seq that would indicate that the
       ;; get-notes algorithm should select the lowest notes possible for the
       ;; instrument/set.
       (pitch-seq-lowest-equals-prefers-low 1)
       ;; how many seconds tolerance do we have when checking bar-duration in
       ;; update-time method of rthm-seq-bar
       (rthm-seq-bar-duration-tolerance 0.003)
       ;; MDE Wed Oct 10 16:35:10 2018 -- until October 2018 an error was
       ;; signalled if the were no pitches in a set for an instrument which
       ;; should be playing. We can now have a rest-sequence generated instead
       (pitch-seq-no-pitches-error t)
       ;; MDE Thu Oct 8 16:27:19 2020, Heidhausen -- issue a warning if we scale
       ;; a rhythm/event to longer than 10xwhole?
       (rhythm-scale-warning t)
       ;; (repeatedly) warn about force-in-range turning microtones into
       ;; chromatics pitches?
       (ins-force-in-range-warning t)
       ;; MDE Sat Aug 3 16:39:40 2019 -- should shorten-large-fast-leaps issue a
       ;; warning or not?
       (shorten-large-fast-leaps-warning t)
       ;; MDE Tue Jul 16 11:17:23 2019 -- if NIL don't even warn about no
       ;; pitches being available
       (pitch-seq-no-pitches-warning t)
```

```
;; Whether to automatically open EPS files generated with CMN via
;; cmn-display.  Currently only works with SBCL and CCL on Mac OSX.
(cmn-display-auto-open #+sc-auto-open T #-sc-auto-open nil)
;; Whether to automatically open PDF files generated with via lp-display.
;; Currently only works with SBCL and CCL on Mac OSX.
(lp-display-auto-open #+sc-auto-open T #-sc-auto-open nil)
;; Whether to automatically open MIDI files generated with via midi-play.
;; Currently only works with SBCL and CCL on Mac OSX.
(midi-play-auto-open #+sc-auto-open T #-sc-auto-open nil)
;; MDE Sat Dec  7 11:56:24 2019 -- if a pitch has a mark included in this
;; list, it will be ignored
(midi-play-ignore-marks nil)
(tmp-dir ,tmpdir)
(default-dir ,tmpdir)
;; The full path to the lilypond command.  We need to set this if we'll
;; call lp-display, i.e. if we want to automatically call Lilypond and
;; open the resultant PDF directly from Lisp.
(lilypond-command nil)
;; The full path to the ffprobe cmomand.  We need to set this if we'll
;; call ffprobe (in get-sound-info, when clm is not used).
(ffprobe-command nil)
;; The default amplitude for all events that don't have amplitude/dynamic
;; set via some means such as marks.
(default-amplitude 0.7)
;; in init-instance of sc class, warn if we can't call sc-init (because
;; it will be called explicitly later)
(warn-no-sc-init t)
(warn-unused-instruments t)
;; whether to warn when there's no CMN mark for a given Lilypond mark
(warn-no-cmn-mark t)
;; sim for Lilypond
(warn-no-lp-mark t)
(warn-no-xml-mark t)
;; output harp pedal (salzedo) marks in MusicXML as a text-mark (i.e.
;; <words>) or as MusicXML <harp-pedals> (cf. xml-salzedo).
;; RP  Sun Dec 24 15:42:46 2023
(xml-salzedo-as-text t)
;; the font family used for special MusicXML marks which should
;; be explicitly set with the engraving/music font (cf. xml-salzedo for
;; an example)
;; RP  Sun Dec 24 16:33:40 2023
(xml-engraving-font "Bravura, Opus, Maestro, Leland, music")
;; warn if a pitch goes above midi note 119?
(warn-high-pitch t)
;; warn if a pitch goes below octave number -1
(warn-low-pitch t)
```

```
(warn-fingering t)
;; Bar number offsets for CMN
(cmn-bar-num-dx-for-sc -0.2)
(cmn-bar-num-dy-for-sc 1.2)
;; MDE Wed Feb  8 16:50:31 2017 -- string-chord-selection-fun (used by
;; default for selecting chords for violin, viola, cello) will call
;; best-string-diad. This tries to find a diad within a certain
;; range. These are the min/max
(best-string-diad-range (7 11))
(best-string-diad-microtones nil)
;; MDE Sat May 10 12:47:25 2014 -- whether to issue warning when we set
;; the asco-msgs slot of a rest event (because they will only be written
;; to an antescofo~ file if this happens to be a rest in the part we're
;; following and we can't know this in advance).
;; MDE Tue Dec 29 11:59:33 2020, Heidhausen -- which auxiliary routine to
;; use for auto-clefs. For years this was the best-clef-aux function but
;; we're now generally using best-clef-aux-new. Some might prefer the old
;; routine however, hence this option here.
(best-clef-aux-fun best-clef-aux-new)
(asco-msg-rest-warning t)
;; if we've added, say, an antescofo~ label to an event with a rehearal
;; letter, we'll get a warning as we can only have one antescofo label
;; per NOTE (though it's not an error to have two, the 2nd will be
;; ignored).
(asco-two-labels-warning t)
;; the default spectral data used in the chord class for things such as
;; dissonance and spectral centroid calcuation. See spectra.lsp for more
;; details about references into +slippery-chicken-spectra+ (such as
;; akoustik-piano-spectra) or see the get-spectrum function in chord.lsp
;; for a description of the format this data can take. Currently
;; slippery-chicken defines three spectra: akoustik-piano-spectra,
;; clm-piano-spectra, and violin-ensemble-spectra. Bear in mind that if
;; you change this it would be best to start slippery-chicken again (or
;; at least regenerated a piece's data) so that chord's/set's
;; dissonance/centroid values are recalculated with the new spectra.
(default-spectra akoustik-piano-spectra)
;; MDE Tue Apr 19 12:05:49 2016 -- for debugging purposes, print info
;; about the selection of pitches for an instrument given a pitch-seq and
;; set etc. (i.e. data passed to pitch-seq's get-notes method). Note that
;; chords will be printed as a sublist (i.e. in parentheses).
(verbose-pitch-selection nil)
;; MDE Thu Oct 18 11:46:46 2018 -- since Sept. 18 the slippery-chicken
;; method update-slots also calls handle-ties. This will break some older
;; pieces so allow that to be skipped if necessary.
(update-slots-handles-ties t)
;; font size for CMN bar numbers
```

```
(cmn-bar-num-size-for-sc 6)
(diapason 440)
;; MDE Thu Dec 15 11:29:25 2022, Heidhausen -- some systems need to
;; escape spaces in paths otherwise they won't open in a shell called
;; from lisp
(system-open-file-escape-spaces nil)
(autoconvert-eps-to-pdf nil)
;; LF  Thu Mar  30 14:04:33 2023
;; If true, when reaper files are written with #'write-reaper-file,
;; the pathnames to the soundfiles are converted to windows-type paths.
;; If nil, they will be unix-type paths.
(reaper-files-for-windows nil)
;; RP  Thu Mar  2 19:04:20 2023
;; The full path to the Csound command. This is required in order to be
;; to use csound-display, i.e. automatically call the Csound command to
;; render a sound file from a given orchestra and a score generated via
;; write-csound-score.
(csound-command nil)
(ignore-force-rest-bar nil)
;; LF 2025-06-18 - can this be removed?
(path-to-ppcre nil)))))
```

30 osc-sc/osc-call

[Functions]

DESCRIPTION:

```
Allow OSC (over UDP) messages to be sent for processing.  The function
waits for input and processes in an endless loop; send 'quit' to stop the
function and return to the interpreter.  Messages the function doesn't
understand will be ignored after a warning being printed.

As this function only terminates when a 'quit' message is sent via OSC, the
only way to quit from within Lisp is to send the Interrupt Command (usually
Control-C, twice).  In that case, the open sockets will remain open, and
only closed before reopening the next time this function is called.

Lisp code can be sent, e.g. in MaxMSP via a message, including
opening/closing parentheses and nested calls; symbols should be quoted as
per usual.

The first two tokens in the list must be /osc-sc and a (usually unique)
identifer e.g. ((/osc-sc 1060-osc-sc-eval (print 'dog) )) Both of these
tokens will be sent back over OSC in a list, along with the result of the
```

```
Lisp call.  If you're using MaxMSP I would recommend using
osc-sc-eval.maxpat (see below) to evaluate your Lisp code as this will
package it up with the right tokens and send the evaluated result out of
its outlet without causing conflicts with other instances of itself.

For an example MaxMSP patch, see osc-test.maxpat in the examples folder of
the documentation: http://michael-edwards.org/sc/examples/osc-test.maxpat
You'll also need http://michael-edwards.org/sc/examples/osc-sc-eval.maxpat

Note that as of November 2020, functions that return lists can also be
processed: each element will be sent back to Max et al via UDP.

NB: Currently only works in SBCL.
    Some lists (e.g. those including strings/symbols) might not be
    recognised as lists by MaxMPS's [route], so process them directly after
    [fromsymbol].
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :listen-port. The UDP port to listen to for messages. Default = 8090.
- :send-ip.  The IP address to send UDP messages back out on.
   Default = #(127 0 0 1))
- :send-port. The UDP port to send messages back out on. Default = 8091.
- :print. Print messages as they arrive. Default = NIL.
- :csound. Default = NIL, meaning incomming messages should
  be formatted similar to ("/osc-sc" "something" "(+" 1 "1"). When set to T,
  messages can also have this format: ("/osc-sc something (+ 1 1)").
```

RETURN VALUE:

```
T
```

SYNOPSIS:

```
(defun osc-call (&key
                 (listen-port 8090)
                 (send-ip #(127 0 0 1))
                 (print nil)
                 (send-port 8091)
                 csound)
```

31 reaper/insert-envelope

[Functions]

DATE:

```
April 30th 2023.
```

DESCRIPTION:

```
get the string of a reaper file and rewrite it, so that a given envelope is
inserted into the project in a given place.
```

ARGUMENTS:

```
- the string of the project file you want to edit
- the reaper-envelope object that you want to insert into the reaper project
- the number of the track, on which to insert the envelope (master is 0)
```

RETURN VALUE:

```
the edited string of the entire reaper file
```

EXAMPLE:

```
;;; insert the angle-env value of a sndfile as an envelope on track 1 of
;;; project.rpp and give it the duration of the sndfile.
(let ((snd (make-sndfile "/E/sound.wav" :angle-env '(0 0 1 1 2 .5))))
  (insert-envelope (read-file-as-string "/E/project.rpp")
                   (make-reaper-envelope (angle-env snd)
                                         :env-type 'parmenv
                                         :parameter-slot 6)
                   1 0 (duration snd)))
```

SYNOPSIS:

```
#+cl-ppcre
(defun insert-envelope (string reaper-envelope track-nr)
```

32 reaper/insert-plugin

[Functions]

DATE:

```
April 30th 2023.
```

DESCRIPTION:

```
get the string of a reaper file and rewrite it, so that a given plugin is
inserted into the project in a given place.
```

ARGUMENTS:

```
- the string of the project file you want to edit
- the binary string of the plugin you want to insert. The easiest way of
obtaining this is probably by copying it from a reaper file that you created
manually (create a reaper project file and load the plugin, the open the
file with a text editor and find and copy the string <VST ..... >.
See "slippery-chicken/src/txt/iem-stereo-encoder.txt" for an example).
To use that plugin in write-spatial-reaper-file (of possibly others), add
it to the 'plugins' slot in the definition of the reaper-file class.
- the number of the track, on which to insert the plugin (master is 0)
```

OPTIONAL ARGUMENTS:

```
- if the plugin is to be inserted before all others (t) or after all other
plugins already on the track (nil)
```

RETURN VALUE:

```
the edited string of the entire reaper file
```

EXAMPLE:

```
;;; insert the blue ripple binaural ambisonics decoder on the master track of
;;; project.rpp
(insert-plugin (read-file-as-string "/E/project.rpp") *blue-ripple-decoder* 0 t)
```

SYNOPSIS:

```
#+cl-ppcre
(defun insert-plugin (string plugin-binary-string track-nr &optional (pre nil))
```

33 reaper/reaper-layer-sounds

[Functions]

DATE:

```
October 1st 2022
```

DESCRIPTION:

Given a list of paths to sound files, create a reaper file that layers/mixes these together. We create as many mixes as possible given the number of tracks required, e.g. if there are 20 sound files and <num-tracks> is 4, then five mixes will be created, one after the other, with a gap of 10 seconds by default. So the files are mixed together in the order of the first argument, with the longest file in the mix determining the overall duration and the shorter files beginning after a wait of <indent> multiplied by the difference between the longest and shorter files' durations. E.g. with a longest file duration of 60 seconds and a shorter file of 30 seconds, given :indent 0.5 the shorter file's start time would be 15 seconds; if its duration was 20 seconds, the start time would be 20.

In case sound files are passed in alphabetical order, e.g. by (get-sndfiles...), and assuming that it is undesireable that similarly-named sound files are placed one after the other, by default the function will shuffle the order of the mixes (i.e. the groups of 4 or whatever sound files per mix). NB the 4-groups are shuffled, not the files in the groups.

ARGUMENTS:

- a list of sound file paths (strings)
- the number of tracks that should be layered/mixed

OPTIONAL ARGUMENTS:

keyword arguments:
:tempo. The BPM value that should be written to the reaper project, if desired. Default = 60.
:max-fade. The maximum length in seconds of a fade-in or fade-out. If this is too long for any given sound file, the :mid-fade multiplied by the sound file duration will be used. Default = 15 seconds.
:min-fade. The proportion of a sound file's duration that should be used if :max-fade can't be used for both fade-in and fade-out. This should generally be < 0.5 but is not enforced. Default = 0.5.
:reaper-file. The path of the reaper-file to be written. If this is NIL then no file is written and the reaper-file object is returned instead, with all mixes present. This could be used to add more tracks before writing the reaper file (see example below). Default = "/tmp/reaper-layer-sounds.rpp"
:min-channels. The minimum number of a channels a reaper track should have. Usually this will be set to be the maximum number of channels any of the files on a single track have. But if you have a bunch of stereo files you want to place on, say, a track which should become 5th order ambisonics, you can set this to 36. Note that all reaper tracks should have an even number of channels so however the channel number is achieved it will be rouned up to the nearest even number if necessary. Default = 2.
:max-channels. This can be set to override the number of channels of any

```
given sound file, should it have more channels than you want, for some
reason. Default = 4.
:gap. The gap in seconds between the end of the previous mix and the start
of the next. Default = 10 seconds.
:shuffle. Whether to shuffle the mixes before writing. See above for
details. Default = T.
:indent. How far in to start the shorter sounds, as a function of their
duration difference to the longest file (see above). 0.5 would have them
centered bang in the middle. By default the golden mean, 0.618034 (what
else?) i.e. (/ (- (sqrt 5) 1) 2).
```

RETURN VALUE:

```
If :reaper-file is given, then the path will be returned after writing,
otherwise a reaper-file object is returned.
```

EXAMPLE:

```
(let ((overlaps (reaper-overlap-sounds
                 (get-sndfiles "/Volumes/slim500/snd/samples/ambience/rain")
                 :reaper-file nil))
      (layers (reaper-layer-sounds (get-sndfiles "~/ic/projects/mete/clm") 4
                                   :reaper-file nil)))
  (add (get-first (tracks overlaps)) (tracks layers))
  (write-reaper-file layers :min-channels 36
                     :file "~/ic/projects/mete/reaper/mete-clm-with-rain.rpp"))
```

SYNOPSIS:

```
(defun reaper-layer-sounds (sndfiles num-tracks
                            &key (tempo 60)
                              (max-fade 15) ; seconds
                              (min-fade .4) ; proportion of duration
                              (reaper-file "/tmp/reaper-layer-sounds.rpp")
                              ;; e.g. if you want all tracks to be 26 channels
                              ;; despite the number of channels in the sound
                              ;; files set this here
                              (min-channels 2)
                              (max-channels 4)
                              ;; gap in seconds between groups
                              (gap 10)
                              ;; shuffle (fixed-seed) the subgroups so that the
                              ;; sound files are not simply processed by
                              ;; directory order?
                              (shuffle t)
                              (indent 0.618034))
```

34 reaper/reaper-overlap-sounds

[Functions]

DATE:

```
October 1st 2022.
```

DESCRIPTION:

```
Create a reaper file with one track that overlaps/cross-fades all the given
sound files.
```

ARGUMENTS:

```
- a list of sound files paths (strings) to overlap
```

OPTIONAL ARGUMENTS:

```
keyword arguments:

:overlap. The proportion by which to overlap the sound files. Given that
sound files can have any duration and be overlapped in any order, this
proportion is applied to the shortest of any two sound files to be
cross-faded. default = 0.5.
:reaper-file. The path of the reaper-file to be written. If this is NIL then
no file is written and the reaper-file object is returned instead, with all
mixes present. This could be used to add more tracks before writing the
reaper file (see example in reaper-layer-sounds). Default =
"/tmp/reaper-layer-sounds.rpp"
:min-channels. See reaper-layer-sounds. Default = 2.
:max-channels. See reaper-layer-sounds. Default = 4.
:track-name. The string used to name the track in reaper. Default =
"overlaps"
:tempo. The BPM value that should be written to the reaper project, if
desired. Default = 60.
```

RETURN VALUE:

```
If :reaper-file is given, then the path will be returned after writing,
otherwise a reaper-file object is returned. See reaper-layer-sounds for an
example.
```

SYNOPSIS:

```
(defun reaper-overlap-sounds (sndfiles &key (tempo 60) (min-channels 2)
                                            (max-channels 4) (overlap .5)
                                            (track-name "overlaps")
                                            (reaper-file
                                             "/tmp/reaper-layer-sounds.rpp"))
```

35 reaper/write-reaper-ambisonics-file

[Functions]

DATE:

April 30th 2023.

DESCRIPTION:

Create a reaper file that contains all the data to spatialize sndfiles using the ambisonics method. For more information on the spatialization data, see #'make-sndfile

ARGUMENTS:

- a list of sndfile objects: their angle-env and elevation-env arguments will be used for spatialization.

OPTIONAL ARGUMENTS:

keyword arguments:

:file. The path to the file that shall be generated.
:start-times. A list of start times for each soundfile on each track in seconds. If the list is shorter than the number of sndfiles, it is looped. Default = '(0) - this means all files start at 0.
:sample-rate. The sample-rate that should be written to the reaper project. Default = 48000.
:ambi-order. Should be an integer between 1 and 8, this determines the number of channel for each track.
:tempo. The BPM value that should be written to the reaper project, if desired. Default = 60.
:init-volume. Volume multipliers for all faders.
-0dB would be 1, Default = -12dB.
:encoder. The key to find the binary data for the plugin that will be inserted into all tracks except the master track in the 'plugins' slot of the reaper-file class.

Default: :iem-stereo-encoder
:decoder. The key to find the binary data for the plugin that will be inserted on the master track in the 'plugins' slot of the reaper-file class. Default: :blue-ripple-decoder
:angle-parameter-slot. The automation slot that the angle-env will controll. This should be changed when other plugins are used than the defaults. Default = 6, because the IEM stereo encoders 6th automation slot is the angle-argument. If this is a list of numbers, the first element will be used for the first angle-env of the sndfile and the second for the second env etc. (when more than one is given for each sndfile).
:elevation-parameter-slot. The automation slot that the elevation-env will controll. This should be changed when other plugins are used than the defaults. Default = 7, because the IEM stereo encoders 7th automation slot is the elevation-argument. If this is a list of numbers, the first element will be used for the first elevation-env of the sndfile and the second for the second env etc. (when more than one is given for each sndfile).
:angle-offset. An offset for the rotation of the scene, where 1 = 360°. Default is 0.5, so that the angle 0 is the front for the iem plugins.
:envs-use-start-times. If nil, all envelopes start at the minimal start-time. If t, the envelopes use the start time of the respective sndfile.
:envs-use-end-times. If nil, all envelopes end when every sndfile hast stopped. If t, end when the respective sndfile stopps.
:envs-duration. If nil, the behavior of envs-use-end-times applies. If a number, every automation envelope gets this duration - envs-use-end-times will be ignored in this case.
envs-only. If t, don't write a file or insert any plugins, just re-write the envelopes. This is usefull if you generate a file with this function and edited the project file by adding more tracks etc. (These should be added after the already existing ones). Then you could edit the envelopes of the sndfiles and set envs-only to t. This way, the file will stay the same except for the envelopes.
:envs-visible. t or nil, wheter all envelopes will be visible (opened) in the reaper-file.

RETURN VALUE:

path to the reaper file that was generated.

EXAMPLE:

```
;;; make three soundfiles into sndfile objects and spatialize them, while
;;; returning their paths in windows format.
(set-sc-config 'reaper-files-for-windows t)
(write-reaper-ambisonics-file
 `(,(make-sndfile "/E/pads.wav"
```

```
                        :angle-env '(0 0  .5 .5  .8 4  1 3.5)
                        :elevation-env '(0 0  .6 .25  2 .5))
    ,(make-sndfile "/E/synths.wav"
                        :angle-env '(0 0  .5 .5  .8 8  1 3.25)
                        :elevation-env '(0 0.5  1 .25))
    ,(make-sndfile "/E/drums.wav"
                        :angle-env '(0 .5  .5 1  .8 8.5  1 3.75)
                        :elevation-env '(0 0.5  1 .25)))
 :file "/E/spatial.rpp"
 :ambi-order 3
 :envs-use-start-times t
 :envs-use-end-times t)


;;; spatialize one sndfile with two channels, that are opposite and circling
;;; each other:
(write-reaper-ambisonics-file
 `(,(make-sndfile "/E/code/feedback/intro.wav"
                  :angle-env '((0 0  .5 .5  .8 4  1 3.5)
                               (0 .5  .5 1  .8 4.5  1 4))
                  :elevation-env '(0 0  .6 .25  2 .5)))
 :file "/E/code/test.rpp"
 :encoder :iem-multi-encoder
 :angle-parameter-slot '(7 12)
 :elevation-parameter-slot '(8 13))
```

SYNOPSIS:

```
#+cl-ppcre
(defun write-reaper-ambisonics-file (list-of-sndfiles
                                     &key file
                                       (start-times '(0))
                                       (sample-rate 48000)
                                       (ambi-order 3)
                                       (tempo 60)
                                       (init-volume .2511)
                                       (encoder :iem-stereo-encoder)
                                       (decoder :blue-ripple-decoder)
                                       (angle-parameter-slot 6)
                                       (elevation-parameter-slot 7)
                                       (angle-offset 0.5)
                                       (envs-use-start-times t)
                                       (envs-use-end-times t)
                                       envs-duration
                                       envs-only
                                       (envs-visible t))
```

36 reaper/write-reaper-file

[Methods]

DESCRIPTION:

```
Write a reaper file from the reaper-items in the data slot.
```

ARGUMENTS:

```
- a reaper-file object
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :file. The path of the reaper file to write. Default will be the id slot
  of the reaper-file object + .rpp, placed in the default-directory of
  slippery-chicken (itself /tmp by default).
- :markers. These are either a simple list of times (in seconds) or a
  mixture containing sublists such as (marker-number time-in-secs
  label-string colour), or just the first two or three of those. See the
  write-reaper-marker method for details. Default = NIL.
- :min-channels (default 2) and :max-channels (default 4). These are the
  minimum and maximum channel counts of the reaper-tracks which are usually
  dependent upon or influence the number of channels reflected in the sound
  files' playback. A sound file could have any number of channels of course
  and usually we'd use the maximum number of channels a track's sound files
  have to set the channels slot, but here we can limit these to something
  reasonable (or e.g. force 4-channel tracks even though all sound files are
  stereo).
```

RETURN VALUE:

```
The patch to the generated reaper file.
```

EXAMPLE:

```
;;; one simple way of algorithmically generating a reaper file:
(let* ((tempo 240)
       (items
         (make-reaper-items1
          (get-sndfiles
           (concatenate 'string
                        cl-user::+slippery-chicken-home-dir+
                        "tests/test-sndfiles-dir-2"))
```

```
          '(e (w) (q) q (h) (e) e. (q.) q (w) e (w) e.)
           tempo
          :input-start '(0 .1 .2)
          :play-rate '(1 1.02 1 .98 1.01 1 1.02)
          :preserve-pitch t))
       ;; NB the tempo of the reaper file is independent of the items
       (rf (make-reaper-file 'reaper-test items :tempo tempo)))
  (write-reaper-file rf))

;;; or to write a reaper file with markers only, at times given in seconds
(write-reaper-file (make-reaper-file 'test nil) :markers '(1 2 3.5 7))

;;; mixed marker data starting with simple times
(write-reaper-file (make-reaper-file 'test nil)
  :markers '(1 2 3.5 7 (49 8.021 "nice label") ; number 49, time 8.021, named
             (562 9.1 "better label" blue))) ; sim. but with a recognised colour
```

SYNOPSIS:

```
(defmethod write-reaper-file ((rf reaper-file)
                              &key file markers
                              (min-channels 2) (max-channels 4))
```

37 reaper/write-reaper-sad-file

[Functions]

DATE:

May 22nd 2023.

DESCRIPTION:

Create a reaper file that contains all the data to spatialize sndfiles using the spatial audio designer plugins. For this, the angle and elevation arguments of the sndfiles wil be converted to the cartesian coordinate system.

ARGUMENTS:

- a list of sndfile objects: their angle-env and elevation-env arguments will be used for spatialization.

OPTIONAL ARGUMENTS:

keyword arguments:

:file. The path to the file that shall be generated.
:start-times. A list of start times for each soundfile on each track in seconds. If the list is shorter than the number of sndfiles, it is looped. Default = '(0) - this means all files start at 0.
:sample-rate. The sample-rate that should be written to the reaper project. Default = 48000.
:nr-of-master-channels. = How many channels the master bus will have. The number of channels for each track is determined by the number of channels the soundfiles on each track has, but is at least 2 and not more than 8.
:tempo. The BPM value that should be written to the reaper project, if desired. Default = 60.
:init-volume. Volume multpiliers for all faders.
-0dB would be 1, Default = -12dB.
:angle-offset. An offset for the rotation of the scene, where 1 = 360°. Default is 0, so that the angle 0 is the front.
:env-conversion-srate. This number is the minimum amount of times per second that an additional point is calculated and converted from polar to cartesian coordinates. Default = 4, which means that at least every .25 seconds there will be a breakpoint. If this number is too low (< 1), the conversion from polar to cartesian won't represent the original envelope very well.
:envs-use-start-times. If nil, all envelopes start at the minimal start-time. If t, the envelopes use the start time of the respective sndfile.
:envs-use-end-times. If nil, all envelopes end when every sndfile hast stopped. If t, end when the respective sndfile stopps.
:envs-duration. If nil, the behavior of envs-use-end-times applies. If a number, every automation envelope gets this duration - envs-use-end-times will be ignored in this case.
envs-only. If t, don't write a file or insert any plugins, just re-write the envelopes. This is usefull if you generate a file with this function and edited the project file by adding more tracks etc. (These should be added after the already existing ones). Then you could edit the envelopes of the sndfiles and set envs-only to t. This way, the file will stay the same except for the envelopes.
:envs-visible. t or nil, wheter all envelopes will be visible (opened) in the reaper-file.

RETURN VALUE:

path to the reaper file that was generated.

EXAMPLE:

```
;;; make three soundfiles into sndfile objects and spatialize them, while
```

```
;;; returning their paths in windows format.
(set-sc-config 'reaper-files-for-windows t)
(write-reaper-sad-file
 `(,(make-sndfile "/E/pads.wav"
                          :angle-env '(0 0  .5 .5  .8 4  1 3.5)
                          :elevation-env '(0 0  .6 .25  2 .5))
    ,(make-sndfile "/E/synths.wav"
                          :angle-env '(0 0  .5 .5  .8 8  1 3.25)
                          :elevation-env '(0 0.5  1 .25))
    ,(make-sndfile "/E/drums.wav"
                          :angle-env '(0 .5  .5 1  .8 8.5  1 3.75)
                          :elevation-env '(0 0.5  1 .25)))
 :file "/E/spatial.rpp"
 :envs-use-start-times t
 :envs-use-end-times t)

;;; spatialize one sndfile with two channels, that are opposite and circling
;;; each other:
(write-reaper-sad-file
 `(,(make-sndfile "/E/code/feedback/intro.wav"
                  :angle-env '((0 0  .5 .5  .8 4  1 3.5)
                               (0 .5  .5 1  .8 4.5  1 4))
                  :elevation-env '(0 0  .6 .25  2 .5)))
 :file "/E/code/test.rpp")
```

SYNOPSIS:

```
#+cl-ppcre
(defun write-reaper-sad-file (list-of-sndfiles
                              &key file
                                (start-times '(0))
                                (sample-rate 48000)
                                (nr-of-master-channels 2)
                                (tempo 60)
                                (init-volume .2511)
                                (angle-offset 0)
                                (env-conversion-srate 4)
                                (envs-use-start-times t)
                                (envs-use-end-times t)
                                envs-duration
                                envs-only
                                (envs-visible t))
```

38 sc/named-object

[Classes]

NAME:

```
named-object

File:             named-object.lsp

Class Hierarchy:  None: base class of all slippery-chicken classes.

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the named-object class which is the
                  base class for all of the slippery-chicken classes.

                  The data slot of the named-object class and its
                  subclasses generally holds the original data passed when
                  creating the object. In anything but the simplest of
                  classes this may quickly become out-of-date as the object
                  is manipulated, but is nevertheless retained so that a)
                  the user can see what data was used to create an object,
                  and b) the user can derive new objects from an object's
                  original data. Data relevant to a specific subclass is
                  often stored in slots other than :data, e.g. bars,
                  rhythms, etc. so the user should not be alarmed if the
                  data slot itself does not seem to reflect changes made to
                  an object.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    4th December 2000

$$ Last modified:  19:30:44 Wed Mar 25 2026 +07

SVN ID: $Id$
```

38.1 named-object/activity-levels

[named-object] [Classes]

NAME:

```
activity-levels

File:             activity-levels.lsp

Class Hierarchy:  named-object -> activity-levels

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Class used in rthm-chain.  Used on a call-by-call basis
                  to determine (deterministically) whether a process is
                  active or not (boolean).

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    4th February 2010

$$ Last modified:  10:55:22 Fri Mar  1 2024 CET

SVN ID: $Id$
```

38.1.1 activity-levels/active

[activity-levels] [Methods]

DESCRIPTION:

```
Returns t or nil depending on whether we're active at this point.  The
object remembers where we were last time; this means if we change level
before getting to the end of a ten-list, we'll pick up where we left off
next time we return to that level.  <level> can be a floating point number:
in this case it will be rounded. But <level> must be between 0 and 10,
where 0 is always inactive, 10 is always active, and anything inbetween
will use the data lists circularly.
```

ARGUMENTS:

```
- the activity-levels object
```

OPTIONAL ARGUMENTS:

```
- the activity-level number we want to test. Although optional, it's
  expected that this argument will usually be defined.  Between 0 and 10.
  Default = 5.
```

RETURN VALUE:

```
T or NIL
```

EXAMPLE:

```
(let ((al (make-al)))
  (print (loop for i below 15 collect (active al 0)))
  (print (loop for i below 15 collect (active al 5)))
  (print (loop for i below 15 collect (active al 1)))
  (print (loop for i below 15 collect (active al 9)))
  (loop for i below 15 collect (active al 10)))

=>
(NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL)
(T T NIL NIL T NIL T T NIL NIL NIL T NIL T NIL)
(T NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL T NIL)
(T T NIL T T T T T T T T T T T NIL)
(T T T T T T T T T T T T T T T)
```

SYNOPSIS:

```
(defmethod active ((al activity-levels) &optional level)
```

38.1.2 activity-levels/flicker-round

[activity-levels] [Methods]

DATE:

```
January 19th 2021
```

DESCRIPTION:

```
Rounding is a cut-and-dry operation, usually. The part after the floating
point determines whether we round up or down: 0.5 or above goes up,
otherwise down. In some circumstances it might be preferable to have an area
in the middle that alternates between up and down. This method uses the
range between the two optional threshold arguments to select rounding up or
down: closer to the lower threshold will mean rounding down takes place more
often than up, but up will still happen occasionally. Similarly as we
approach the high threshold, rounding up will occur more often. All
deterministically of course. On the other hand, values outside the
thresholds will merely round as usual. So if you always want to
'flicker-round' then set the thresholds to 0 and 1. If you never want to
round, call round (!) or set the tresholds to 0.5 and 0.5.
```

ARGUMENTS:

```
- the activity-levels object
- the floating point number to 'flicker-round'
```

OPTIONAL ARGUMENTS:

```
the low and high threshold values: floating point numbers between 0.0 and
1.0, where the first optional argument should be less than the second, of
course.
```

RETURN VALUE:

```
An integer
```

EXAMPLE:

```
(let ((al (make-al)))
  (loop for i from 1010 to 1011 by 0.01 collect (flicker-round al i)))
-->
(1010 1010 1010 1010 1010 1010 1010 1010 1010 1010 1010 1010 1010 1010 1010
 1010 1010 1010 1010 1010 1010 1010 1010 1010 1010 1010 1010 1010 1010 1010
 1011 1010 1010 1011 1010 1010 1010 1010 1011 1010 1010 1010 1011 1011 1010
 1010 1010 1011 1011 1011 1010 1010 1011 1011 1011 1010 1011 1011 1011 1011
 1010 1011 1011 1011 1011 1010 1011 1011 1011 1011 1011 1011 1011 1011 1011
 1011 1011 1011 1011 1011 1011 1011 1011 1011 1011 1011 1011 1011 1011 1011
 1011 1011 1011 1011 1011 1011 1011 1011 1011 1011)
```

SYNOPSIS:

```
(defmethod flicker-round ((al activity-levels) float
                          &optional (threshold-low 0.3) (threshold-high 0.7))
```

38.1.3 activity-levels/make-al

[activity-levels] [Functions]

DESCRIPTION:

```
Make an activities-level object for determining (deterministically) on a
call-by-call basis whether a process is active or not (boolean).  This is
determined by nine 10-element lists (actually three versions of each) of
hand-coded 1s and 0s, each list representing an 'activity-level' (how
active the process should be).  The first three 10-element lists have only
one 1 in them, the rest being zeros.  The second three have two 1s,
etc. Activity-levels of 0 and 10 would return never active and always
active respectively.
```

ARGUMENTS:

```
None required.
```

OPTIONAL ARGUMENTS:

```
start-at (default NIL): which of the three 10-element lists to start with
(reset to).  Should be 0, 1, or 2 though if NIL will default to 0.
```

RETURN VALUE:

```
The activities-level object.
```

SYNOPSIS:

```
(defun make-al (&optional start-at id)
```

38.1.4 activity-levels/reset

[activity-levels] [Methods]

DESCRIPTION:

```
Reset the activity-levels object to restart at the first element of the 1st
(or user-specificed) 10-element list.
```

ARGUMENTS:

```
The activity-levels object.
```

OPTIONAL ARGUMENTS:

```
start-at: should be between 0 and 2; it indicates which of the 10-lists
we're going to start with.  Default = 0.
```

RETURN VALUE:

```
T
```

SYNOPSIS:

```
(defmethod reset ((al activity-levels) &optional (start-at 0) ignore)
```

38.2 named-object/activity-levels-env

[named-object] [Classes]

NAME:

```
activity-levels

File:             activity-levels-env.lsp

Class Hierarchy:  named-object -> activity-levels -> activity-levels-env

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          An extension of the activity-levels class to incorporate
                  a restarting envelope which is used to determine the
                  current 'level' argument for a call to the active method.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    30th June 2015

$$ Last modified:  17:00:41 Sat Nov 29 2025 CET

SVN ID: $Id: activity-levels.lsp 5048 2014-10-20 17:10:38Z medward2 $
```

38.2.1 activity-levels-env/active

[activity-levels-env] [Methods]

DESCRIPTION:

```
Similar to the parent class method except that the level argument in that
class is here supplied by the envelope and cycle-length.
```

ARGUMENTS:

```
- the activity-levels-env object
```

OPTIONAL ARGUMENTS:

```
- T or NIL to print debugging information.
```

RETURN VALUE:

```
 T or NIL to indicate whether we're 'active' or not.
```

EXAMPLE:

```
;;; we can see here that with a cycle-length of 10, calling active 31 times
;;; means we go through the envelope three times before arriving back at 0.
(let ((ale (make-ale '(0 0 2 10 3 1) 10 2)))
  (loop repeat 31 collect (active ale t)))
->
current: 0 level: 0.0
current: 1 level: 1.6666667
current: 2 level: 3.3333335
current: 3 level: 5.0
current: 4 level: 6.666667
current: 5 level: 8.333333
current: 6 level: 10.0
current: 7 level: 7.0
current: 8 level: 4.0
current: 9 level: 1.0
current: 0 level: 0.0
current: 1 level: 1.6666667
current: 2 level: 3.3333335
current: 3 level: 5.0
current: 4 level: 6.666667
current: 5 level: 8.333333
current: 6 level: 10.0
current: 7 level: 7.0
current: 8 level: 4.0
current: 9 level: 1.0
current: 0 level: 0.0
current: 1 level: 1.6666667
current: 2 level: 3.3333335
current: 3 level: 5.0
current: 4 level: 6.666667
current: 5 level: 8.333333
current: 6 level: 10.0
current: 7 level: 7.0
current: 8 level: 4.0
current: 9 level: 1.0
current: 0 level: 0.0
(NIL NIL NIL NIL T T T T NIL NIL NIL NIL NIL T NIL T T T T NIL NIL NIL NIL NIL
 NIL T T T NIL NIL NIL)
```

SYNOPSIS:

```
(defmethod active :around ((ale activity-levels-env) &optional print)
```

38.2.2 activity-levels-env/make-ale

[activity-levels-env] [Functions]

DESCRIPTION:

```
For an overview of what this might be used for and how it's done in its
basic form, see the description to make-al in the parent class
'activity-levels'
```

ARGUMENTS:

```
None required.
```

OPTIONAL ARGUMENTS:

```
- the envelope in x y form. y values should range from 0 to 10, as with the
  level argument to an 'active' call in the parent class. x range can be
  any arbitrary value but start with 0 to get full-range results.
- the cycle length: the envelope repeats after a specified cycle length, so
  that no matter how many times you call 'active' you always get results.
- start-at (default NIL): which of the three 10-element lists in the parent
  class to start with (reset to).  Should be 0, 1, or 2, though if NIL will
  default to 0.
```

RETURN VALUE:

```
The activities-level-env object.
```

SYNOPSIS:

```
(defun make-ale (&optional (env '(0 5 100 5)) (cycle-length 101) start-at)
```

38.3 named-object/activity-levels-pc

[named-object] [Classes]

NAME:

```
activity-levels

File:             activity-levels-pc.lsp

Class Hierarchy:  named-object -> activity-levels-pc
```

```
Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          At a minimum, this is replacement for fixed-seed weighted
                  randomness using percentages--which will produce different
                  results with different Lisp implementations--but in fact,
                  because it uses two activity-levels objects, it will
                  always produce pattern-based results. (The 'pc' stands for
                  per cent.)

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    1st March 2024

$$ Last modified:  17:09:50 Fri Mar  1 2024 CET

SVN ID: $Id$
```

38.3.1 activity-levels-pc/active

[activity-levels-pc] [Methods]

DATE:

```
1st March 2024
```

DESCRIPTION:

```
Returns T or NIL depending on whether we're active given a number between 0
and 100.
```

ARGUMENTS:

```
The activity-levels-pc object.
```

OPTIONAL ARGUMENTS:

```
the 'level', between 0 and 100 inclusive, indicating the frequency of
occurrence of the result T; comparable to a percentage chance but
deterministic and pattern-oriented (see the data slot state lists in
activity-levels.lsp). Although this is an optional argument (because of
related classes' methods) this method only really makes sense if it is
passed. Float arguments will be rounded. Default = 50.
```

RETURN VALUE:

```
 T or NIL
```

EXAMPLE:

```
(let ((alpc (make-alpc)))
  (count t (loop repeat 1000 collect (active alpc 31))))
--> 310
```

SYNOPSIS:

```
(defmethod active ((alpc activity-levels-pc) &optional level)
```

38.4 named-object/control-wave

[named-object] [Classes]

NAME:

```
 control-wave

 File:             control-wave.lsp

 Class Hierarchy:  named-object -> control-wave

 Version:          1.1.0

 Project:          slippery chicken (algorithmic composition)

 Purpose:          The creation of waveforms such as sine, cosine, sawtooth,
                   triangle, square, and pulse train, for the control of
                   musical parameters. See
                   http://michael-edwards.org/wp/?p=977 for context and
                   example code.

 Author:           Michael Edwards: m@michael-edwards.org

 Creation date:    July 6th 2016, Essen Werden, Germany

 $$ Last modified:  14:18:44 Fri Mar 22 2024 CET
```

38.5 named-object/control-wave-ins

[named-object] [Classes]

NAME:

```
control-wave

File:             control-wave-ins.lsp

Class Hierarchy:  no classes defined

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          The creation of waveforms such as sine, cosine, sawtooth,
                  triangle, square, and pulse train, for the control of
                  musical parameters. See
                  http://michael-edwards.org/wp/?p=977 for context and
                  example code.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    March 26th 2020, Heidhausen, Germany
                  (taken out of control-wav.lsp)

$$ Last modified:  09:52:01 Wed Mar 20 2024 CET
```

38.6 named-object/linked-named-object

[named-object] [Classes]

NAME:

```
linked-named-object

File:             linked-named-object.lsp

Class Hierarchy:  named-object -> linked-named-object

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Extension of named-object class to provide slots for the
                  previous and next objects in a recursive-assoc-list.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    March 10th 2002
```

```
$$ Last modified:  17:47:47 Wed Jan 31 2018 CET

SVN ID: $Id$
```

38.6.1 linked-named-object/bar-holder

[linked-named-object] [Classes]

NAME:

```
bar-holder

File:             bar-holder.lsp

Class Hierarchy:  named-object -> linked-named-object -> bar-holder

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          This class is meant to be sub-classed by piece, section
                  and sequence, all of which hold each other or, ultimately
                  a list of bars with relevant rhythms, timings, pitches
                  etc.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    16th February 2002

$$ Last modified:  08:54:15 Mon Jan 19 2026 CET

SVN ID: $Id$
```

38.6.2 bar-holder/change-pitches

[bar-holder] [Methods]

DESCRIPTION:

```
Change the pitches in the specified bars to the specified new pitches.

NB: This method requires that full bars be given, even if not all pitches
    are being changed.
```

ARGUMENTS:

- A bar-holder object (such as the PIECE slot within a slippery-chicken object).
- The ID of the player whose part is to be changed.
- An integer that is the number of the first bar in which pitches are to be changed.
- A list of lists of note-name symbols, each sublist representing a consecutive bar and containing the same number of note-name symbols as there are rhythms in that bar. A NIL in these lists means no change is to be made to the corresponding rhythm or bar (see example below). NB: This method counts tied notes rather than just attacked notes.

OPTIONAL ARGUMENTS:

keyword arguments:
- :use-last-octave. T or NIL to indicate whether the method is to require that each note-name symbols in the <new-pitches> list has an octave indicator. If this argument is set to NIL, each note-name symbol must have an octave indicator (e.g., the 4 in c4). If this argument is set to T, only the first note-name symbol in the bar is required to have an octave indicator, and all subsequent note-name symbols without octave indicators will use the last octave indicated; e.g. '((a3 b g cs4)). NB: This feature does not work with chords. Default = T.
- :written. T or NIL to indicate whether these are the written or sounding notes for a transposing instrument. Default = NIL.

RETURN VALUE:

Always returns T.

EXAMPLE:

```
;;; NIL indicates that no change is to be made; this applies to single rhythms
;;; as well as entire bars.
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((c2 d2 e2 f2 g2 a2 b2
                               c3 d3 e3 f3 g3 a3 b3
                               c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e (s) s))
                                :pitch-seq-palette ((1 (2) 3 4)))))
        :rthm-seq-map '((1 ((cl (1 1 1 1 1))
```

```
                          (vc (1 1 1 1 1))))))))
  (change-pitches (piece mini) 'cl 2 '((c4 d4 e4 f4)))
  (change-pitches (piece mini) 'vc 3 '((c3 d e f) nil (g3 nil b c4))))

=> T
```

SYNOPSIS:

```
(defmethod change-pitches ((bh bar-holder) player start-bar new-pitches
                           &key (use-last-octave t) written)
```

38.6.3 bar-holder/delete-all-marks

[bar-holder] [Methods]

DESCRIPTION:

```
Delete all marks from the MARKS slots of all events in the specified
measure range of a given bar-holder object and set the slot to NIL.

This method always applies to full bars.
```

ARGUMENTS:

```
- A bar-holder object.
- An integer that is the number of the first bar from which all marks are
  to be deleted.
- An integer that is the number of consecutive bars including the first
  bar from which all marks are to be deleted.
- The ID of the player from whose part the marks are to be deleted.
```

RETURN VALUE:

```
Always returns T.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((c2 d2 e2 f2 g2 a2 b2
                               c3 d3 e3 f3 g3 a3 b3
                               c4 d4 e4 f4 g4 a4 b4 c5))))
```

```
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e (s) s))
                                :pitch-seq-palette ((1 (2) 3 4))
                                :marks (a 1 s 2 te 3 as 4))))
        :rthm-seq-map '((1 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))))))
  (delete-all-marks (piece mini) 2 2 'vc))

=> T
```

SYNOPSIS:

```
(defmethod delete-all-marks ((bh bar-holder) start-bar num-bars player)
```

38.6.4 bar-holder/get-note

[bar-holder] [Methods]

DESCRIPTION:

Return the event object (or pitch object, if accessing a note within a chord) from a specified bar and note within a given bar-holder object.

ARGUMENTS:

- A bar-holder object (e.g. PIECE slot of a slippery-chicken object).
- An integer that is the 1-based number of the bar from which the note is to be retrieved.
- An integer or two-item list of integers that is the 1-based number of the note to retrieve within the specified bar. If an integer, the entire event object is retrieved. A two-item list of integers is used to retrieve a specific note from within a chord, in the form '(2 1), where 2 is the second note (or non-rhythm event) in the bar, and 1 is the first note in the chord counting from the bottom. NB: This argument also counts tied notes, not just attacked notes.
- The ID of the player from whose part the note is to be retrieved.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether, when accessing a pitch in a chord, to return the written or sounding pitch. T = written. Default = NIL.

RETURN VALUE:

An event object, or single pitch object if accessing a note within a chord.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((c2 d2 e2 f2 g2 a2 b2
                               c3 d3 e3 f3 g3 a3 b3
                               c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 (2) 3 4 5)))))
        :rthm-seq-map '((1 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1)))))))))
  (print (get-note (piece mini) 3 '(2 1) 'vc)) ; single pitch within a chord
  (print (get-note (piece mini) 3 2 'vc)) ; entire chord event
  (print (get-note (piece mini) 5 3 'cl)))
```

SYNOPSIS:

```
(defmethod get-note ((bh bar-holder) bar-num note-num player
                     &optional written silent)
```

38.6.5 bar-holder/piece

[bar-holder] [Classes]

NAME:

```
piece

File:             piece.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist -> assoc-list -> recursive-assoc-list ->
                  piece
                                  AND

                  named-object -> linked-named-object -> bar-holder ->
                  piece

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)
```

```
Purpose:          Implementation of the piece class which holds all the
                  note information for a whole piece in the form of
                  sections (possibly subsections), which then contain
                  player-sections, sequenzes and rthm-seq-bars.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    16th February 2002

$$ Last modified:  21:15:51 Thu Jun 12 2025 CEST

SVN ID: $Id$
```

38.6.6 piece/delete-sequenzes

[piece] [Methods]

DESCRIPTION:

```
Delete one or more consecutive sequenz objects from a given piece object by
specifying any bar number within the first sequenz object to be deleted.

This method deletes the whole sequenz object which contains the bar with a
given number.

NB: This method only deletes the sequenz object for the specified player,
    so the remaining players will have a different structure, making MIDI
    or printable output impossible. The user must be sure that each section
    has the same number of sequenzes of identical time-signature structure
    in each section.

NB: The user must call the update-slots method to ensure that changes to
    the NUM-BARS slot etc. are reflected in the given slippery-chicken
    object.
```

ARGUMENTS:

```
- A piece object.
- An integer that is the number of the bar for which the containing sequenz
  is to be deleted.
- The ID of the player from whose part the sequenz is to be deleted.
```

OPTIONAL ARGUMENTS:

```
- An integer that is the number of consecutive sequenz objects to delete,
```

```
   including the first sequenz indicated by the <bar-num> argument.
   Default = 1.
```

RETURN VALUE:

```
 Returns T.
```

EXAMPLE:

```
;;; Print the number of sequenz objects contained in section 2 of each player's
;;; part, delete two sequenz objects from each part in that section, and print
;;; the number of sequenz objects again to see the difference. Update the slots
;;; and call cmn-display for printable output.
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((hn (french-horn :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5))))
                            (2 ((((4 4) h h))
                                :pitch-seq-palette ((1 2)))))
        :rthm-seq-map '((1 ((hn (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (2 ((hn (2 2 2 2 2))
                            (vc (2 2 2 2 2))))
                        (3 ((hn (1 1 1 1 1))
                            (vc (1 1 1 1 1)))))))))
  (print (length (get-data-data 'hn (get-section mini 2))))
  (print (length (get-data-data 'vc (get-section mini 2))))
  (delete-sequenzes (piece mini) 8 'hn 2)
  (delete-sequenzes (piece mini) 8 'vc 2)
  (print (length (get-data-data 'hn (get-section mini 2))))
  (print (length (get-data-data 'vc (get-section mini 2))))
  (update-slots mini)
  (cmn-display mini))
```

SYNOPSIS:

```
(defmethod delete-sequenzes ((p piece) bar-num player &optional (how-many 1))
```

38.6.7 piece/get-nth-sequenz

[piece] [Methods]

DESCRIPTION:

```
Get the sequenz object from a specified section of a piece object by
specifying a position index and a player.

NB: This is the primary method for accessing player sequences, as it
    handles cases in which a player doesn't play in a sequence, and it is
    called automatically by slippery-chicken.

    When the specified player has no note events in the specified sequence
    and the optional argument <create-rest-seq> is set to T, this method
    creates a rest sequence (one that consists of the correct number of
    bars with the right time signatures, but in which the bars are only
    rest bars) based on a simultaneous sequence in one of the playing
    instruments.
```

ARGUMENTS:

```
- A piece object.
- The ID of the section in from which the sequenz object is to be
  returned.
- The ID of the player from whose part the sequenz object is to be
  returned.
- An integer that is the index (position) of the desired sequenz object
  within the given section. This number is 0-based.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether to convert sequenz objects that are NIL
  (i.e., the specified player has no events in the specified sequenz) to
  sequenz objects consisting of full-bar rests. T = create rest sequences.
  Default = T. NB: This argument is already called by slippery-chicken with
  a value of T, so has no effect when used as a post-generation editing
  method and can be thus considered for internal use only.
```

RETURN VALUE:

```
Returns a sequenz object.
```

EXAMPLE:

```
;;; Returns a sequenz object
```

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((hn (french-horn :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((hn (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (2 ((hn (nil nil nil nil nil))
                            (vc (1 1 1 1 1))))
                        (3 ((hn (1 1 1 1 1))
                            (vc (1 1 1 1 1)))))))))
  (get-nth-sequenz (piece mini) 3 'hn 2))

=>

SEQUENZ: pitch-curve: (1 2 3 4 5)
RTHM-SEQ: num-bars: 1
          num-rhythms: 5
          num-notes: 5
          num-score-notes: 5
          num-rests: 0
          duration: 4.0
          psp-inversions: NIL
          marks: NIL
          time-sigs-tag: NIL
          handled-first-note-tie: NIL
         (for brevity's sake, slots pitch-seq-palette and bars are not printed)
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: (1), next: NIL
BAR-HOLDER:
            start-bar: 13
            end-bar: 13
            num-bars: 1
            start-time: 48.0
            end-time: 52.0
            start-time-qtrs: 48.0
            end-time-qtrs: 52.0
            num-notes (attacked notes, not tied): 5
            num-score-notes (tied notes counted separately): 5
            num-rests: 0
```

```
            duration-qtrs: 4.0
            duration: 4.0 (4.000)
```

SYNOPSIS:

```
(defmethod get-nth-sequenz ((p piece) section player seq-num ; 0-based
                            &optional (create-rest-seq t))
```

38.6.8 piece/get-sequenz-from-bar-num

[piece] [Methods]

DESCRIPTION:

Get the specified sequenz object located at a specified bar-number location of a specified player's part in a given piece object.

ARGUMENTS:

- A piece object.
- An integer that is the number of the bar from which to return the sequenz object.
- The ID of the player from whose part the sequenz object is to be returned.

RETURN VALUE:

A sequenz object.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((hn (french-horn :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((hn (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (2 ((hn (1 1 1 1 1))
```

```
                          (vc (1 1 1 1 1))))
                     (3 ((hn (1 1 1 1 1))
                         (vc (1 1 1 1 1))))))))
  (get-sequenz-from-bar-num (piece mini) 7 'vc))

            =>
            SEQUENZ: pitch-curve: (1 2 3 4 5)
            RTHM-SEQ: num-bars: 1
            num-rhythms: 5
            num-notes: 5
            num-score-notes: 5
            num-rests: 0
            duration: 4.0
            psp-inversions: NIL
            marks: NIL
            time-sigs-tag: NIL
            handled-first-note-tie: NIL
            (for brevity's sake, slots pitch-seq-palette and bars are not printed)
            SCLIST: sclist-length: 3, bounds-alert: T, copy: T
            LINKED-NAMED-OBJECT: previous: NIL, this: (1), next: NIL
            BAR-HOLDER:
            start-bar: 7
            end-bar: 7
            num-bars: 1
            start-time: 24.0
            end-time: 28.0
            start-time-qtrs: 24.0
            end-time-qtrs: 28.0
            num-notes (attacked notes, not tied): 5
            num-score-notes (tied notes counted separately): 5
            num-rests: 0
            duration-qtrs: 4.0
            duration: 4.0 (4.000)
```

SYNOPSIS:

```
(defmethod get-sequenz-from-bar-num ((p piece) bar-num player)
```

38.6.9 piece/insert-bar

[piece] [Methods]

DESCRIPTION:

```
 Insert a rthm-seq-bar object into an existing piece object.
```

```
NB: As this is a post-generation editing method when used with this class,
    the rthm-seq-bar object must consist of event objects (with pitches),
    not just rhythm objects. If used to insert a bar into an isolated
    rthm-seq object (not a sequenz object), the inserted rthm-seq-bar could
    then consist of rhythm objects rather than events.

NB: Slippery chicken does not check to ensure that a new bar is inserted
    for each player at a given point; this is up to the user.

NB: The user must call the update-slots method to ensure that changes to
    the num-bars slot etc. are reflected in the given slippery-chicken
    object.
```

ARGUMENTS:

```
- A piece object.
- A rthm-seq-bar object.
- An integer that is the bar-number within the rthm-seq-bar object before
  which the new bar is to be inserted.
- NB: The optional arguments are actually required for use with this class.
```

OPTIONAL ARGUMENTS:

```
NB: The optional arguments are actually required for use with this class.
- An integer that is the section of in which the bar is to be inserted.
- The ID of the player into whose part the new bar is to be added.
- An integer that is the number of the sequenz object into which the bar is
  to be inserted. This is one-based.
- A list of integers that are the curve for a pitch-seq object to be
  applied to the bar that is to be inserted.
```

RETURN VALUE:

```
T
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((hn (french-horn :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1))
```

```
                  (2 (1 1 1 1 1))
                  (3 (1 1 1 1 1)))
       :rthm-seq-palette '((1 ((((4 4) h q e s s))
                               :pitch-seq-palette ((1 2 3 4 5))))
                           (2 ((((4 4) h h))
                               :pitch-seq-palette ((1 2)))))
       :rthm-seq-map '((1 ((hn (1 1 1 1 1))
                           (vc (1 1 1 1 1))))
                       (2 ((hn (1 1 1 1 1))
                           (vc (1 2 1 1 1))))
                       (3 ((hn (1 1 1 1 1))
                           (vc (1 1 1 1 1)))))))
     (new-bar (make-rthm-seq-bar '((4 4) (w)))))

  (fill-with-rhythms new-bar (loop for r in '(h q. e)
                                for p in '(c4 e4 g4)
                                collect (make-event p r)))
  ;; slippery-chicken object has 15 bars  (print (num-bars mini))
  ;; print the number of bars in the sequenz in piece=mini, section=2,
  ;; seq=2 (0=based), player='hn.
  ;; has 1 bar
  (print (num-bars (get-nth-sequenz (piece mini) 2 'hn 2)))
  ;; insert an rsb in piece=mini, section=2, seq=3 (1-based), player='hn,
  ;; before rsb=1 of the existing seq
  (insert-bar (piece mini) new-bar 1 2 'hn 3 '(1 2 3))
  ;; insert an rsb in piece=mini, section=2, seq=3 (1-based), player='vc,
  ;; before rsb=1 of the existing seq
  (insert-bar (piece mini) new-bar 1 2 'vc 3 '(1 2 3))
  ;; print the number of bars in the sequenz in piece=mini, section=2,
  ;; seq=2 (0=based), player='hn.
  ;; now has 2 bars.
  (print (num-bars (get-nth-sequenz (piece mini) 2 'hn 2)))
  ;; update slots of the sc object.
  (update-slots mini)
  ;; print the number of bars of the slippery-chicken object.
  ;; now 16.
  (print (num-bars mini)))

=>
15
1
2
16
```

SYNOPSIS:

```
(defmethod insert-bar ((p piece) (rsb rthm-seq-bar) bar-num
                       ;; these aren't actually optional but we don't
                       ;; need them in the rthm-seq method
                       &optional section player seq-num ; seq-num is 1-based!
                                 ;; this really is optional
                                 pitch-seq)
```

38.6.10 piece/rebar

[piece] [Methods]

DATE:

29-Jan-2010

DESCRIPTION:

Go through the sequences and rebar according to the first one that has the least number of bars (but following the player hierarchy).

ARGUMENTS:

- A piece object (usually provided by calling from the slippery-chicken class)

OPTIONAL ARGUMENTS:

- A list of player IDs from the given piece object, ordered in terms of importance i.e. which instrument's bar structure should take precedence.

NB: The optional arguments are actually required in this class (not in slippery-chicken) but the rebar-fun is not yet used.

RETURN VALUE:

Always T.

SYNOPSIS:

```
(defmethod rebar ((p piece) &optional instruments-hierarchy rebar-fun)
```

38.6.11 bar-holder/player-section

[bar-holder] [Classes]

NAME:

```
player-section

File:             player-section.lsp

Class Hierarchy:  named-object -> linked-named-object -> bar-holder
                  -> player-section
                  AND
                  named-object -> linked-named-object -> sclist
                  -> player-section

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of player-section class which is simply a
                  bar holder that contains a list of sequenzes for a
                  particular player.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    18th March 2002

$$ Last modified:  19:05:48 Fri Feb  2 2018 CET

SVN ID: $Id$
```

38.6.12 bar-holder/section

[bar-holder] [Classes]

NAME:

```
section

File:             section.lsp

Class Hierarchy:  named-object -> linked-named-object -> bar-holder
                  -> section

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of section class which is simply a bar
                  holder and recursive-assoc-list that contains (possibly
                  subsections which contain) player-sections.
```

```
Author:           Michael Edwards: m@michael-edwards.org

Creation date:    23rd March 2002

$$ Last modified:  15:20:01 Fri Jan 10 2020 CET

SVN ID: $Id$
```

38.6.13 section/get-all-players

[section] [Methods]

DESCRIPTION:

```
Return a list of the IDs from all players in a section object. NB: When
retrieving from a section within a slippery-chicken object, all players in
the ensemble will be returned, as the slippery-chicken object will generate
rest bars for players even when they're not active in a given section.
```

ARGUMENTS:

```
- A section object.
```

RETURN VALUE:

```
- A list of player IDs (symbols).
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (hn (french-horn :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((cl (1 1 1 1 1))
                            (hn (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
```

```
                        (2 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (3 ((hn (1 1 1 1 1))
                            (vc (1 1 1 1 1))))))))
  (print (get-all-players (get-section mini 1)))
  (print (get-all-players (get-section mini 2)))
  (print (get-all-players (get-section mini 3))))

=>
(CL HN VC)
(CL HN VC)
(CL HN VC)
```

SYNOPSIS:

```
(defmethod get-all-players ((s section))
```

38.6.14 section/get-bar

[section] [Methods]

DESCRIPTION:

```
Return the rthm-seq-bar object at the specified bar number within the given
section.

NB: The bar number is counted from the beginning of the entire piece, not
    the beginning of the section.
```

ARGUMENTS:

```
- A section object.
- An integer that is the bar number for which to return the rthm-seq-bar
  object. This number is 1-based and counts from the beginning of the
  piece, not the beginning of the section.
```

OPTIONAL ARGUMENTS:

```
NB: The <player> argument is actually required, but is listed as optional
    for reasons of class inheritance.

- The ID of the player for whose part the rthm-seq-bar object is to be
  returned.
```

RETURN VALUE:

A rthm-seq-bar object.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (hn (french-horn :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5)))
                       (2 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5)))
                       (3 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (2 2 2 2 2))
                   (3 (3 3 3 3 3)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5))))
                           (2 ((((4 4) q e s s h))
                               :pitch-seq-palette ((1 2 3 4 5))))
                           (3 ((((4 4) e s s h q))
                               :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((cl (1 1 1 1 1))
                            (hn (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (2 ((cl (2 2 2 2 2))
                            (hn (2 2 2 2 2))
                            (vc (2 2 2 2 2))))
                        (3 ((cl (3 3 3 3 3))
                            (hn (3 3 3 3 3))
                            (vc (3 3 3 3 3)))))))))
  (get-bar (get-section mini 3) 11 'hn))

=>
RTHM-SEQ-BAR: time-sig: 2 (4 4), time-sig-given: T, bar-num: 11,
              old-bar-nums: NIL, write-bar-num: NIL, start-time: 40.000,
              start-time-qtrs: 40.0, is-rest-bar: NIL, multi-bar-rest: NIL,
              show-rest: T, notes-needed: 5,
              tuplets: NIL, nudge-factor: 0.35, beams: NIL,
              current-time-sig: 2, write-time-sig: NIL, num-rests: 0,
              num-rhythms: 5, num-score-notes: 5, parent-start-end: NIL,
              missing-duration: NIL, bar-line-type: 0,
              player-section-ref: (3 HN), nth-seq: 0, nth-bar: 0,
              rehearsal-letter: NIL, all-time-sigs: (too long to print)
              sounding-duration: 4.000,
              rhythms: (
```

```
[...]
)
SCLIST: sclist-length: 6, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "3-bar1", tag: NIL,
data: ((4 4) E S S H Q)
```

SYNOPSIS:

```
(defmethod get-bar ((s section) bar-num &optional player)
```

38.6.15 section/get-sequenz

[section] [Methods]

DESCRIPTION:

```
Get the specified sequenz object from a given section object.
```

ARGUMENTS:

```
- A section object.
- The ID of the player from whose part the sequenz object is to be
  returned.
- An integer that is the number of the sequence object to be returned from
  within the given section object. This number is 1-based.
```

RETURN VALUE:

```
A sequenz object.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((f3 g3 a3 b3 c4))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5))))
                           (2 ((((4 4) q e s s h))
```

```
                              :pitch-seq-palette ((1 2 3 4 5))))
                          (3 ((((4 4) e s s h q))
                              :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                       (2 ((cl (2 2 2 2 2))
                           (vc (2 2 2 2 2))))
                       (3 ((cl (3 3 3 3 3))
                           (vc (3 3 3 3 3)))))))
  (get-sequenz (get-section mini 2) 'vc 2))

=>
SEQUENZ: pitch-curve: (1 2 3 4 5)
RTHM-SEQ: num-bars: 1
          num-rhythms: 5
          num-notes: 5
          num-score-notes: 5
          num-rests: 0
          duration: 4.0
          psp-inversions: NIL
          marks: NIL
          time-sigs-tag: NIL
          handled-first-note-tie: NIL
         (for brevity's sake, slots pitch-seq-palette and bars are not printed)
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: (1), this: (2), next: (3)
BAR-HOLDER:
            start-bar: 7
            end-bar: 7
            num-bars: 1
            start-time: 24.0
            end-time: 28.0
            start-time-qtrs: 24.0
            end-time-qtrs: 28.0
            num-notes (attacked notes, not tied): 5
            num-score-notes (tied notes counted separately): 5
            num-rests: 0
            duration-qtrs: 4.0
            duration: 4.0 (4.000)
```

SYNOPSIS:

```
(defmethod get-sequenz ((s section) player seq-num) ; 1-based
```

38.6.16 section/has-subsections

[section] [Methods]

DESCRIPTION:

Boolean test to determine whether a specified section of a slippery-chicken object has subsections.

ARGUMENTS:

- A section object.

RETURN VALUE:

T if the specified section has subsections, otherwise NIL.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :set-palette '((1 ((c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1
                    ((a (1 1 1))
                     (b (1 1 1))))
                   (2 (1 1 1 1))
                   (3
                    ((a (1 1 1))
                     (b
                      ((x (1 1 1))
                       (y (1 1 1))))))
                   (4
                    ((a (1 1 1))
                     (b (1 1 1))
                     (c (1 1 1 1)))))
        :rthm-seq-palette '((1 ((((2 4) (q) e (s) s))
                               :pitch-seq-palette ((1 2)))))
        :rthm-seq-map '((1
                         ((a ((vn (1 1 1))))
                          (b ((vn (1 1 1))))))
                        (2 ((vn (1 1 1 1))))
                        (3
                         ((a ((vn (1 1 1))))
                          (b
```

```
                          ((x ((vn (1 1 1))))
                           (y ((vn (1 1 1)))))))
                        (4
                         ((a ((vn (1 1 1))))
                          (b ((vn (1 1 1))))
                          (c ((vn (1 1 1 1)))))))))))
  (print (has-subsections (get-section mini 1)))
  (print (has-subsections (get-section mini 2))))

=>
T
NIL
```

SYNOPSIS:

```
(defmethod has-subsections ((s section))
```

38.6.17 section/num-sequenzes

[section] [Methods]

DESCRIPTION:

```
Get the number of sequenz objects in a given section object.
```

ARGUMENTS:

```
- A section object.
```

RETURN VALUE:

```
An integer that is the number of sequenz objects in the specified section
object.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vc (cello :midi-channel 1))))
        :set-palette '((1 ((f3 g3 a3 b3 c4))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
```

```
                              :pitch-seq-palette ((1 2 3 4 5)))))
       :rthm-seq-map '((1 ((vc (1 1 1 1 1))))
                      (2 ((vc (1 1 1 1 1))))
                      (3 ((vc (1 1 1 1 1))))))))
  (num-sequenzes (get-section mini 2)))

=> 5
```

SYNOPSIS:

```
(defmethod num-sequenzes ((s section))
```

38.6.18 section/re-bar

[section] [Methods]

DESCRIPTION:

```
Regroup the consecutive note and rest events of a given section object into
new bars of the specified time signature.

This method will only combine short bars into longer ones; it will not
split up longer bars and recombine them.

The method will also use the specified (or default) time signature as a
target, and may be forced to create a number of bars that are not of the
specified time signature if the number of beats in the given section object
do not correspond.

NB: The user must call the update-slots method after using this method as a
    post-generation editing method.
```

ARGUMENTS:

```
- A section object.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :start-bar. An integer that is the first bar within the specified section
  that is to be re-barred. Default = First bar of the given section.
- :end-bar. An integer that is the last bar within the specified section
  that is to be re-barred. Default = Last bar of the given section.
- :min-time-sig. The target time signature for all new bars. NB: Depending
  on the number of beats in the given section, the method may have to
```

```
  deviate from this target time signature. Default = '(2 4).
- :verbose. T or NIL to indicate whether to print feedback on the
  re-barring process to the Lisp listener. Default = NIL.
- :auto-beam.  T, NIL, or an integer. If T, the method will automatically
  attach beam indications to the corresponding events according to the beat
  unit of the time signature. If an integer, the method will beam in
  accordance with a beat unit that is equal to that integer. If NIL, the
  method will not automatically place beams. Default = T.
```

RETURN VALUE:

```
T.
```

EXAMPLE: SYNOPSIS:

```
(defmethod re-bar ((s section)
                   &key start-bar
                   end-bar
                   (min-time-sig '(2 4))
                   verbose
                   ;; could also be a beat rhythmic unit
                   (auto-beam t))
```

38.6.19 bar-holder/sequenz

[bar-holder] [Classes]

NAME:

```
sequenz

File:             sequenz.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist -> rthm-seq
                  -> sequenz
                                    AND
                  named-object -> linked-named-object -> bar-holder
                  -> sequenz

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the sequenz class which holds the
                  necessary data (pitch, rhythms etc.) for one sequenz for
```

```
                  one instrument.

 Author:          Michael Edwards: m@michael-edwards.org

 Creation date:   March 15th 2002

 $$ Last modified:  13:39:55 Wed May 21 2025 CEST

 SVN ID: $Id$
```

38.6.20 bar-holder/transpose-bars

[bar-holder] [Methods]

DESCRIPTION:

```
Transpose the pitches of the specified bars in a given player's part by a
specified number of semitones.
```

ARGUMENTS:

```
- A bar-holder object (such as the PIECE slot of a slippery-chicken
  object).
- A positive or negative integer that is the number of semitones by which
  the pitches of the specified bars are to be transposed.
- An integer that is the number of the first bar in which the pitches are
  to be transposed.
- An integer that is the number of consecutive bars, including the
  start-bar, in which the pitches are to be transposed.
- The ID of the player whose part is to be changed.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :destructively. T or NIL to indicate whether the transposed pitches are
  to replace the existing pitches of the given bar-holder object. This must
  be set to T if the pitches of the original object are to be transposed
  before, for example, generating output. If NIL, the original object will
  first be cloned, the pitches of the original object will be left
  untouched and the changes will be made to the copy. Default = NIL.
- :print-bar-nums. T or NIL to indicate whether the method should print
  feedback about which bars have been transposed to the listener.
  T = print feedback. Default = NIL.
- :chord-function. The function that is to be used for transposition of
  chords objects. Default = #'transpose (of the chord class).
```

```
 - :chord-function. The function that is to be used for transposition of
   pitch objects. Default = #'transpose (of the pitch class).
```

RETURN VALUE:

```
 Returns a list of the rthm-seq-bar objects that have been transposed.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((c2 d2 e2 f2 g2 a2 b2
                                c3 d3 e3 f3 g3 a3 b3
                                c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e (s) s))
                                :pitch-seq-palette ((1 (2) 3 4)))))
        :rthm-seq-map '((1 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))))))
  (transpose-bars (piece mini) 11 2 2 'cl
                  :destructively t
                  :print-bar-nums t))
```

SYNOPSIS:

```
(defmethod transpose-bars ((bh bar-holder) semitones start-bar num-bars player
                           &key
                           (destructively nil)
                           (print-bar-nums nil)
                           ;; the default functions are the class methods for
                           ;; pitch or chord.
                           (chord-function #'transpose)
                           (pitch-function #'transpose))
```

38.6.21 linked-named-object/instrument

[linked-named-object] [Classes]

NAME:

```
 instrument
```

```
File:             instrument.lsp

Class Hierarchy:  named-object -> linked-named-object -> instrument

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the instrument class which defines
                  musical instrument properties like range and
                  collects/stores information about what the instrument
                  plays: how many notes, in how many bars etc.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    4th September 2001

$$ Last modified:  20:25:11 Thu Jun 12 2025 CEST

SVN ID: $Id$
```

38.6.22 instrument/auto-set-subset-id

[instrument] [Methods]

DATE:

```
July 27th 2018, Heidhausen
```

DESCRIPTION:

```
Set the subset-id slot to be the same as the id slot.
```

ARGUMENTS:

```
- an instrument instance
```

RETURN VALUE:

```
- the instrument instance
```

SYNOPSIS:

```
(defmethod auto-set-subset-id ((ins instrument))
```

38.6.23 instrument/default-chord-function

[instrument] [Functions]

DESCRIPTION:

```
If an instrument is able to play chords, a function must be defined to
select pitches from a list that it can play as a chord. This function (as a
symbol) is passed as a slot to the instrument instance.

This is the default function. It returns a 2-note chord with the pitch at
index plus that below it, or that above it if there are no lower pitches
available. Or it just returns a single-pitch chord object if neither of
those cases are possible.

NB: The arguments are supplied by slippery chicken when it calls the
    function.
```

ARGUMENTS:

```
- The current number from the pitch-seq. Currently ignored by default.
- The index that the first argument was translated into by the offset and
  scaler (based on trying to get a best fit for the instrument and set).
  This can be assumed to be a legal reference into pitch-list as it was
  calculated as fitting in pitch-seq::get-notes.  (zero-based.)
- The pitch-list created from the set, taking into account the instrument's
  range and other notes already played by other instruments.
- The current pitch-seq object. Currently ignored by default.
- The current instrument object. Currently ignored by default.
- The current set object. Currently ignored by default.
```

RETURN VALUE:

```
A chord object.
```

SYNOPSIS:

```
(defun default-chord-function (curve-num index pitch-list pitch-seq
                               instrument set)
```

38.6.24 instrument/force-in-range

[instrument] [Methods]

DATE:

October 9th 2013

DESCRIPTION:

Forces a pitch to be within an instrument's range by transposing up or down the required number of octaves. NB it doesn't force a microtonal pitch to the nearest chromatic pitch so whether it's playable on an instrument is not checked here.

ARGUMENTS:

- the instrument object
- the piece object

OPTIONAL ARGUMENTS:

keyword argument:
- :sounding. Whether the pitch should be considered a sounding pitch. Default = NIL.
- see other keywords to the in-range method as these are also available

RETURN VALUE:

A pitch object within the instrument's range.

EXAMPLE:

```
(let ((cl (get-standard-ins 'b-flat-clarinet)))
  ;; needs to go down 1 octave
  (print (data (force-in-range cl (make-pitch 'e7))))
  ;; needs to go up 2 octaves
  (print (data (force-in-range cl (make-pitch 'g1))))
  ;; the t indicates we're dealing with sounding pitches so here there's no
  ;; transposition...
  (print (data (force-in-range cl (make-pitch 'd3) :sounding t)))
  ;; ... but here there is
  (print (data (force-in-range cl (make-pitch 'd3)))))
  =>
  E6
  G3
  D3
  D4
```

SYNOPSIS:

```
(defmethod force-in-range ((ins instrument) pitch
                           &key sounding try-artificial-harmonic
                             impossible-microtones (no-missing-notes t))
```

38.6.25 instrument/in-range

[instrument] [Methods]

DESCRIPTION:

Checks whether a specified pitch/chord falls within the defined range of a given instrument object or not.

ARGUMENTS:

- An instrument object.
- A pitch or chord item (either a chord object, pitch object, or note-name symbol). Chords are handled by a separate method (of the same name).

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether the pitch specified is to be compared with the given instrument object's sounding or written range. T = Sounding. Default = NIL. If T, a secondary NIL is also returned to indicate that the specified pitch is neither too high nor too low.
- T or NIL to indiciate whether an artificial harmonic should be tried if the pitch is too high for an instrument whose :harmonics slot is T. Default = NIL.
- T or NIL to indicate whether a microtonal pitch should be considered 'in-range' when the instrument can't play microtones. Default = NIL (although theoretically within range it's not playable).
- T or NIL to indicate whether pitches the instrument can't play (the missing-notes slot) should cause the method to return NIL. Default = T = 'missing notes' will not be considered in range.

RETURN VALUE:

Returns T if the specified pitch falls between the lowest-sounding/lowest-written and the highest-sounding/highest-written pitches of the given pitch object, otherwise NIL.

If the specified pitch is outside of the range, an additional value of 0 or 1 is also returned to indicate whether the specified pitch is too high (1) or too low (0). If it's out of range but could be achieved by an artificial harmonic (just the perfect 4th variety) on an instrument with :harmonics T, then the additional value will be a chord object instead (the two pitches, including diamond notehead for the second, which would achieve the desired pitch.

EXAMPLE:

```
;; Determine if a pitch provided as a note-name symbol falls within the written
;; range of a non-transposing instrument
(let ((i1 (make-instrument 'inst1 :lowest-written 'bf3 :highest-written 'a6)))
  (in-range i1 'c4))
  => T, NIL

;; Determine if a pitch provided as a note-name symbol falls within the
;; sounding range of a transposing instrument, using the optional argument T
(let ((i2 (make-instrument 'inst1 :lowest-written 'fs3 :highest-written 'c6
                           :transposition 'BF)))
  (in-range i2 'c6 T))
  => NIL, 1

;; A pitch object can be used as the specified pitch
(let ((i2 (make-instrument 'inst1 :lowest-written 'fs3 :highest-written 'c6
                           :transposition 'BF)))
  (in-range i2 (make-pitch 'd6)))
  => NIL, 1
```

SYNOPSIS:

```
(defmethod in-range ((ins instrument) pitch
                     &optional sounding try-artificial-harmonic
                       impossible-microtones (no-missing-notes t))
```

38.6.26 instrument/make-instrument

[instrument] [Functions]

DESCRIPTION:

```
Create an instrument object, specifying the values for a number of
parameters for describing characteristics of a given instrument, such as
lowest and highest pitch, transposition, clefs used by the instrument etc.

NB: The user will generally define instruments in the context of
the :instrument-palette slot of a call to make-slippery-chicken, or an
explicit call to the make-instrument-palette function.
```

ARGUMENTS:

```
- A symbol that is the instrument ID.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :staff-name. String. This is the unabbreviated instrument name that will
  be used for the first page of printed scores.
- :staff-short-name. String. This is the abbreviated instrument name that
  will be used for subsequent pages of printed scores.
- :lowest-written. Note-name symbol. This is the lowest written pitch
  available on the given instrument. Defaults to NIL. A user may only
  define either the lowest-written value or the lowest-sounding value. If a
  lowest-written value is given, the method automatically determines the
  lowest-sounding value based on the lowest-written value and the
  transposition value.
- :highest-written. Note-name symbol. This is the highest written pitch
  available on the given instrument. Defaults to NIL. A user may only
  define either the highest-written value or the highest-sounding value. If
  a highest-written value is given, the method automatically determines the
  highest-sounding value based on the highest-written value and the
  transposition value.
- :lowest-sounding. Note-name symbol. This is the lowest sounding pitch
  available on the given instrument. Defaults to NIL. A user may only
  define either the lowest-sounding value or the lowest-written value. If a
  lowest-sounding value is given, the method automatically determines the
  lowest-written value based on the lowest-sounding value and the
  transposition value.
- :highest-sounding. Note-name symbol. This is the highest sounding pitch
  available on the given instrument. Defaults to NIL. A user may only
  define either the highest-sounding value or the highest-written value. If
  a highest-sounding value is given, the method automatically determines
  the highest-written value based on the highest-sounding value and the
  transposition value.
- :transposition. Note-name symbol. This is the key of the given instrument
  (such as the "B-flat" of the "B-flat clarinet"), given as a note-name
  symbol (such as 'BF for B-flat). If a value is only given for the
  :transposition argument but not for the :transposition-semitones
  argument, and there are multiple semitone transposition options for the
  key specified, the method will choose the most common semitone
  transposition for that given key. NB: When using keyword argument
  :transposition rather than :transposition-semitones, sc will have a
  warning printed by cm with indications as to which direction the
  transposition has been undertaken.
- :transposition-semitones. Integer (positive or negative). The number of
  semitones lower that a given instrument sounds than written, e.g. -2 for
  B-flat Clarinet. If a value is only given for the
  :transposition-semitones argument but not for the :transposition
  argument, the method will automatically determine the key for the
  :transposition argument. The listener will drop into the debugger with an
  error if a key is given for the :transposition argument and the number
```

specified for the :transposition-semitones does not correspond with that key.
- :starting-clef. Symbol. This value determines the first clef that a given instrument is to use if that instrument can use different clefs. For a list of available clefs see the :clefs argument below. Default = 'treble.
- :clefs. List of symbols. All clefs that a given instrument may use in the course of a piece. Clefs available are treble, treble-8vb, alto, tenor, bass, percussion, double-treble, and double-bass. Clefs are to be given in order of preference. Defaults automatically to the value given to :starting-clef if no other clefs are specified. NB: If a separate list is indeed given here, the method will automatically add the value for :starting-clef as well, should it have been omitted. In this case, a warning will also be printed.
- :clefs-in-c. List of symbols. Similar to :clefs, but designates which clefs an instrument uses in a C-score; for example, bass clarinet may notated in bass clef for sounding pitches though it is standardly notated in treble clef for written pitches. For a list of clefs available see the :clefs argument above.
- :largest-fast-leap. Number. This value indicates the largest interval, in semitones, that a player can feasibly perform at a fast tempo on the given instrument. Default = 999. "Fast" here is determined for the whole piece by the slippery-chicken class's fast-leap-threshold slot.
- :score-write-in-c. T or NIL. Determines whether the musical material for the given instrument should be printed in C. T = print in C. Default = NIL.
- :score-write-bar-line. Integer. This argument is used for indicating system-grouping in the printed score. The given integer specifies how many instruments above this one should be grouped together with an unbroken bar-line. Default = 1.
- :midi-program. Integer. The number of the MIDI program to be used for playing back this instrument. Default = 1.
- :chords. T or NIL. Indicates whether the given instrument is capable of playing chords (starting with 2-note simultaneities, but not multiphonics).
- :subset-id. Symbol, string, number, or NIL. Indicates the ID of a specific subset of the current set to which the instrument's pitch selection is limited. No error will occur if no subset with this ID exists in a given set, i.e. some may include this subset, some may not and everything will function correctly--if the specified subset is not present in the current set the pitch selection routine will select from the whole set. In every case however, the usual set limiting according to instrument range etc. will also apply. Default = NIL.
- :microtones. T or NIL. Indicates whether the instrument can play microtones. T = can play microtones. Default = NIL. NB: If this value is set to T, a separate :microtones-midi-channel must be specified; this can

be done for the given instrument object in the :ensemble block of the make-slippery-chicken function.
- :missing-notes. A list of note-name symbols. This is a list of any notes which the given instrument can't play, for example certain quarter-tones. These are to be given by the user as written-pitch note-name symbols, but are always stored by the method as sounding pitches.
- :prefers-notes. Symbol. 'high, 'low or NIL. This value indicates whether to give preference, when choosing notes for the given instrument, to pitches from the upper or lower end of the instrument's range. When NIL, preference is given to notes from its middle register. Default = NIL.
- :chord-function. If the given instrument can play chords then it will need a reference to a function that can select chords for it. NB This should be a symbol not a function object; thus, 'my-fun not #'my-fun. Default = NIL.
- :staff-lines. How many the lines the staff should have when this instrument is playing. When we change instrument mid-piece the number of staff lines will change accordingly. Similarly, if we start with an instrument that doesn't use 5 staff lines, we'll begin a piece with the correct number. Default = 5.

RETURN VALUE:

Returns an instrument object.

EXAMPLE:

```
;; Make-instrument for the flute:       ;
  (make-instrument 'flute :staff-name "Flute" :staff-short-name "Fl."
:lowest-written 'c4 :highest-written 'd7
:starting-clef 'treble :midi-program 74 :chords nil
:microtones t :missing-notes '(cqs4 dqf4))

  =>
  INSTRUMENT: lowest-written:
  PITCH: frequency: 261.626, midi-note: 60, midi-channel: 0
  [...]
  , highest-written:
  PITCH: frequency: 2349.318, midi-note: 98, midi-channel: 0
  [...]
  lowest-sounding:
  PITCH: frequency: 261.626, midi-note: 60, midi-channel: 0
  [...]
  , highest-sounding:
  PITCH: frequency: 2349.318, midi-note: 98, midi-channel: 0
  starting-clef: TREBLE, clefs: (TREBLE), clefs-in-c: (TREBLE)
```

```
  prefers-notes: NIL, midi-program: 74
  transposition: C, transposition-semitones: 0
  score-write-in-c: NIL, score-write-bar-line: 1
  chords: NIL, chord-function: NIL,
  total-bars: 0 total-notes: 0, total-duration: 0.0
  total-degrees: 0, microtones: T
  missing-notes: (CQS4 DQF4), subset-id: NIL
  staff-name: Flute, staff-short-name : Fl.,
  largest-fast-leap: 999
  [...]
  NAMED-OBJECT: id: FLUTE, tag: NIL,
  data: NIL

;; A make-instrument for the b-flat bass clarinet ;
  (make-instrument 'bass-clarinet :staff-name "Bass Clarinet" :lowest-written 'c3
:highest-written 'g6 :staff-short-name "Bass Cl."
:chords nil :midi-program 72 :starting-clef 'treble
:microtones t :prefers-notes 'low
:missing-notes '(aqs4 bqf4 bqs4 cqs5 dqf5 gqf3 fqs3 fqf3 eqf3
dqs3 dqf3 cqs3)
:clefs '(treble) :clefs-in-c '(treble bass)
:transposition-semitones -14)

  =>
  INSTRUMENT: lowest-written:
  PITCH: frequency: 130.813, midi-note: 48, midi-channel: 0
  [...]
  , highest-written:
  PITCH: frequency: 1567.982, midi-note: 91, midi-channel: 0
  [...]
  lowest-sounding:
  PITCH: frequency: 58.270, midi-note: 34, midi-channel: 0
  [...]
  , highest-sounding:
  PITCH: frequency: 698.456, midi-note: 77, midi-channel: 0
  [...]
  NAMED-OBJECT: id: BASS-CLARINET, tag: NIL,
  data: NIL
```

SYNOPSIS:

```
(defun make-instrument (id &key
                            staff-name
                            staff-short-name
                            lowest-written
                            highest-written
```

```
lowest-sounding
highest-sounding
transposition
transposition-semitones
(starting-clef 'treble)
clefs
(largest-fast-leap 999)
score-write-in-c
(score-write-bar-line 1)
(midi-program 1)
(staff-lines 5)
chords
clefs-in-c
subset-id
microtones
missing-notes
prefers-notes
chord-function harmonics open-strings
open-string-marks
(nodes '((12 2) (7 3) (5 4) (4 5))))
```

38.6.27 instrument/natural-harmonic?

[instrument] [Methods]

DATE:

```
8th January 2019, Heidhausen
```

DESCRIPTION:

```
Determine whether a note can be played as a natural harmonic. If so, then we
return the appropriate pitch object with harmonic marks, etc. (see below).

In order for this to work, the :harmonics slot of the instrument
will need to be T and the open-strings, nodes, and open-string-marks slots
must be set appropriately. The default nodes are ((12 2) (7 3) (5 4) (4 5)
(3 6)) where the first of each pair is the nodal point in semitones above
the open string pitch and the second is the partial number produced. The
open-strings slot should be a list of descending pitches (symbols or
objects) and the open-strings-marks is something like (i ii iii iv) for
orchestral strings or (c1 c2 c3 c4 c5 c6) for guitar. Note that these two
slots must be in the same order and also that the marks must be recognised
by the output score routine if they are to be meaningful.
```

NB The note returned is the 'written' note (in the case of e.g. double-bass) but won't be transposed down an octave for e.g. harp notation.

See also the natural-harmonic function in pitch.lsp for an on-the-fly (and thus less efficient) routine that works with any arbitrary string tuning independently of instrument objects.

ARGUMENTS:

- an instrument object
- a note, either as a pitch symbol or object. This should be the written pitch in the case of transposing instruments such as the guitar or double bass.

OPTIONAL ARGUMENTS:

- frequency tolerance in cents. (A natural major third (5th partial) is 14 cents from its nearest equal-tempered neighbour.) Default = 15.

RETURN VALUE:

A pitch object with marks attached for the harmonic (circle) and string, where appropriate, plus, as a second value, and only for instruments with open-strings (e.g. orchestral strings) the note at the nodal point, with a diamond note-head mark. These could be combined into a chord for notation purposes.

EXAMPLE:

```
(natural-harmonic? (get-standard-ins 'violin) 'bf5)
-->
NIL
NIL

(natural-harmonic? (get-standard-ins 'violin) 'b5)
-->
PITCH: frequency: 987.767, midi-note: 83, midi-channel: 1
       pitch-bend: 0.0
       degree: 166, data-consistent: T, white-note: B5
       nearest-chromatic: B5
       src: 3.7754972, src-ref-pitch: C4, score-note: B5
       qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
       micro-tone: NIL,
       sharp: NIL, flat: NIL, natural: T,
       octave: 5, c5ths: 0, no-8ve: B, no-8ve-no-acc: B
```

```
       show-accidental: T, white-degree: 48,
       accidental: N,
       accidental-in-parentheses: NIL, marks: (IV HARM),
...

;;; or with scordatura:
(let ((vln (get-standard-ins 'violin)))
  (setf (open-strings vln) '(eqs5 a4 d4 g3))
  (prog1
      (natural-harmonic? vln 'eqs6)
    (setf (open-strings vln) '(e5 a4 d4 g3)))) ; reset
-->
PITCH: frequency: 1357.146, midi-note: 88, midi-channel: 1
       pitch-bend: 0.5
       degree: 177, data-consistent: T, white-note: E6
       nearest-chromatic: E6
       src: 5.1873584, src-ref-pitch: C4, score-note: ES6
       qtr-sharp: 1, qtr-flat: NIL, qtr-tone: 1,
       micro-tone: T,
       sharp: NIL, flat: NIL, natural: NIL,
       octave: 6, c5ths: 0, no-8ve: EQS, no-8ve-no-acc: E
       show-accidental: T, white-degree: 51,
       accidental: QS,
       accidental-in-parentheses: NIL, marks: (I HARM),
...
```

SYNOPSIS:

```
(defmethod natural-harmonic? ((ins instrument) note &optional (tolerance 15))
```

38.6.28 instrument/prefers-high

[instrument] [Methods]

DESCRIPTION:

```
Determine whether the PREFERS-NOTES slot of a given instrument object is
set to 'HIGH.
```

ARGUMENTS:

```
- An instrument object.
```

RETURN VALUE:

```
 Returns T if the PREFERS-NOTES slot of the given instrument object is set
 to 'HIGH, otherwise NIL.
```

EXAMPLE:

```
;; Returns T if the PREFERS-NOTES slot of the given instrument object is set to ;
;; 'HIGH                                ;
  (let ((i1 (make-instrument 'inst :prefers-notes 'high)))
(prefers-high i1))

  => T

;; Returns NIL if the PREFERS-NOTES slot of the given instrument object is not ;
;; set to 'HIGH                         ;
  (let ((i1 (make-instrument 'inst1))
(i2 (make-instrument 'inst2 :prefers-notes 'low)))
(print (prefers-high i1))
(print (prefers-high i2)))

  =>
  NIL
  NIL
```

SYNOPSIS:

```
(defmethod prefers-high ((ins instrument))
```

38.6.29 instrument/prefers-low

[instrument] [Methods]

DESCRIPTION:

```
 Determine whether the PREFERS-NOTES slot of a given instrument object is
 set to 'LOW.
```

ARGUMENTS:

```
 - An instrument object.
```

RETURN VALUE:

```
 Returns T if the PREFERS-NOTES slot of the given instrument object is set
 to 'LOW, otherwise NIL.
```

EXAMPLE:

```
;; Returns T if the PREFERS-NOTES slot of the given instrument object is set to ;
;; 'LOW                                ;
  (let ((i1 (make-instrument 'inst :prefers-notes 'low)))
(prefers-low i1))

  => T

;; Returns NIL if the PREFERS-NOTES slot of the given instrument object is not ;
;; set to 'LOW                         ;
  (let ((i1 (make-instrument 'inst1))
(i2 (make-instrument 'inst2 :prefers-notes 'high)))
(print (prefers-low i1))
(print (prefers-low i2)))

  =>
  NIL
  NIL
```

SYNOPSIS:

```
(defmethod prefers-low ((ins instrument))
```

38.6.30 instrument/set-prefers-high

[instrument] [Methods]

DATE:

```
05 Feb 2011
```

DESCRIPTION:

```
Sets the PREFERS-NOTES slot of the given instrument object to 'HIGH.
```

ARGUMENTS:

```
- An instrument object.
```

OPTIONAL ARGUMENTS:

```
(- optional ignore argument; for internal use only).
```

RETURN VALUE:

```
Returns symbol HIGH.
```

EXAMPLE:

```
;; Returns symbol HIGH by default       ;
  (let ((i1 (make-instrument 'inst)))
(set-prefers-high i1))

  => HIGH

;; Create an instrument object with only an ID, print the PREFERS-NOTES slot to ;
;; see that it is NIL by default, apply the set-prefers-high, and print the ;
;; slot again to see the changes        ;
  (let ((i1 (make-instrument 'inst)))
(print (prefers-notes i1))
(set-prefers-high i1)
(print (prefers-notes i1)))

  =>
  NIL
  HIGH

;; Reset to HIGH from LOW               ;
  (let ((i1 (make-instrument 'inst :prefers-notes 'low)))
(print (prefers-notes i1))
(set-prefers-high i1)
(print (prefers-notes i1)))

  =>
  LOW
  HIGH
```

SYNOPSIS:

```
(defmethod set-prefers-high ((ins instrument) &optional ignore)
```

38.6.31 instrument/set-prefers-low

[instrument] [Methods]

DATE:

```
05 Feb 2011
```

DESCRIPTION:

Sets the PREFERS-NOTES slot of the given instrument object to 'LOW.

ARGUMENTS:

- An instrument object.

OPTIONAL ARGUMENTS:

(- optional ignore argument; for internal use only).

RETURN VALUE:

Returns symbol LOW.

EXAMPLE:

```
;; Returns symbol LOW by default
  (let ((i1 (make-instrument 'inst)))
(set-prefers-low i1))

  => LOW

;; Create an instrument object with only an ID, print the PREFERS-NOTES slot to
;; see that it is NIL by default, apply the set-prefers-low, and print the
;; slot again to see the changes
  (let ((i1 (make-instrument 'inst)))
(print (prefers-notes i1))
(set-prefers-low i1)
(print (prefers-notes i1)))

  =>
  NIL
  LOW

;; Reset to LOW from HIGH
  (let ((i1 (make-instrument 'inst :prefers-notes 'high)))
(print (prefers-notes i1))
(set-prefers-low i1)
(print (prefers-notes i1)))

  =>
  HIGH
  LOW
```

SYNOPSIS:

```
(defmethod set-prefers-low ((ins instrument) &optional ignore)
```

38.6.32 instrument/transposing-instrument-p

[instrument] [Methods]

DESCRIPTION:

```
Determine whether a given instrument object defines a transposing
instrument.
```

ARGUMENTS:

```
- An instrument object.
```

OPTIONAL ARGUMENTS:

```
- ignore-octaves. T or NIL to indicate whether instruments that transpose
  at the octave are to be considered transposing instruments.
  T = instruments that transpose at the octave are not considered
  transposing instruments. Default = T.
```

RETURN VALUE:

```
Returns T if the given instrument object defines a transposing instrument,
otherwise NIL.
```

EXAMPLE:

```
;; Returns NIL if the instrument is not a transposing instrument ;
  (let ((i1 (make-instrument 'instrument-one)))
(transposing-instrument-p i1))

  => NIL

;; Returns T if the instrument object has been defined using a non-NIL value ;
;; for :transposition                  ;
  (let ((i2 (make-instrument 'instrument-two :transposition 'bf)))
(transposing-instrument-p i2))

  => T

;; Returns T if the instrument object has been defined using a non-0 value for ;
;; :transposition-semitones            ;
  (let ((i3 (make-instrument 'instrument-two :transposition-semitones -3)))
(transposing-instrument-p i3))
```

```
  => T

;; Setting the optional argument to NIL causes instruments that transpose at ;
;; the octave to return T.              ;
  (let ((i3 (make-instrument 'instrument-two :transposition-semitones -12)))
(transposing-instrument-p i3))

  => NIL

  (let ((i3 (make-instrument 'instrument-two :transposition-semitones -12)))
(transposing-instrument-p i3 nil))

  => T
```

SYNOPSIS:

```
(defmethod transposing-instrument-p ((ins instrument)
                                     &optional (ignore-octaves t))
```

38.6.33 linked-named-object/pitch

[linked-named-object] [Classes]

NAME:

```
 pitch

 File:             pitch.lsp

 Class Hierarchy:  named-object -> linked-named-object -> pitch

 Version:          1.1.0

 Project:          slippery chicken (algorithmic composition)

 Purpose:          Implementation of the pitch class for holding pitch
                   information: symbolic representation (eg c4), MIDI note
                   number, frequency, sampling-rate conversion etc.

 Author:           Michael Edwards: m@michael-edwards.org

 Creation date:    March 18th 2001

 $$ Last modified:  16:46:25 Thu Dec 11 2025 CET

 SVN ID: $Id$
```

38.6.34 pitch/add-mark

[pitch] [Methods]

DESCRIPTION:

```
Add a specified mark to the MARKS slot of the given pitch object.

NB: The add-mark method does not check first to see whether the mark being
    added is a legitimate mark. It does print a warning, however, when the
    specified mark is already present in the MARKS slot, though it adds it
    anyway.
```

ARGUMENTS:

```
- A pitch object.
- A symbol that is a mark.
```

RETURN VALUE:

```
A list. The method returns the entire contents of the given pitch object's
MARKS slot as a list.

Prints a warning when the specified mark is already present in the given
pitch object's MARKs slot.
```

EXAMPLE:

```
;; By default the MARKS slot of a newly created pitch object is set to NIL
(let ((p (make-pitch 'c4)))
  (marks p))

=> NIL

;; Add two marks and print the contents of the given pitch object's MARKS slot
;; to see the changes
(let ((p (make-pitch 'c4)))
  (add-mark p 'pizz)
  (add-mark p 'a)
  (print (marks p)))

=>
(A PIZZ)

;; Prints a warning when the specified mark is already present in the MARKS
;; slot, though it adds it again anyway.
```

```
(let ((p (make-pitch 'c4)))
  (add-mark p 'pizz)
  (add-mark p 'pizz)
  (marks p))

=> (PIZZ PIZZ)
WARNING:
pitch::add-mark: mark PIZZ already present but adding again!
```

SYNOPSIS:

```
(defmethod add-mark ((p pitch) mark &optional warn-rest warn-again)
```

38.6.35 pitch/cents-hertz

[pitch] [Methods]

DATE:

```
December 24th 2013
```

DESCRIPTION:

```
Convert an offset in cents into the frequency deviation of a pitch.
```

ARGUMENTS:

```
- a pitch object
- the number of cents to offset the pitch and return the frequency
  deviation for
```

RETURN VALUE:

```
The frequency deviation in Hertz of the offset pitch.
```

EXAMPLE:

```
;;; taking as a given the usual floating point round errors:
(cents-hertz (make-pitch 'a4) 1200)
=> 439.9999694824219d0 ; i.e. going up an octave from a4 would be 440 Hz higher
(cents-hertz (make-pitch 'a4) -1200)
=> -219.99998474121094d0
(cents-hertz (make-pitch 'a4) 3)
=> 0.7631253666877456d0
```

```
(cents-hertz (make-pitch 'a4) 10)
=> 2.5489090105293144d0
(cents-hertz (make-pitch 'a3) 10)
=> 1.2744545052646572d0
```

SYNOPSIS:

```
(defmethod cents-hertz ((p pitch) cents)
```

38.6.36 pitch/cmn-display-pitch-list

[pitch] [Functions]

DESCRIPTION:

```
Use CMN to display a list of pitch objects.
```

ARGUMENTS:

```
The list of pitch objects. This could also contain cmn objects if
needed, e.g. cmn::bar, cmn::interior-double-bar, cmn::line-mark, etc.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :staff. The CMN staff object to display with.  Default = cmn::treble.
- :size.  The CMN size for the staff.  Default = 20.
- :file.  The path of the file to (over)write.
   Default = "pitches.eps" in the directory (get-sc-config 'default-dir)
  (default /tmp)
- :auto-open.  Whether to open the .EPS file once written. Currently only
   available on OSX with SBCL and CCL.  Uses the default app for .EPS
   files, as if opened with 'open' in the terminal.  Default = Value of
   (get-sc-config cmn-display-auto-open).
```

RETURN VALUE:

```
A CMN score object.
```

SYNOPSIS:

```
(defun cmn-display-pitch-list
    (pitches &key (staff cmn::treble) (size 20)
     (auto-open (get-sc-config 'cmn-display-auto-open))
     (file (concatenate 'string (get-sc-config 'default-dir) "pitches.eps")))
```

38.6.37 pitch/degree-

[pitch] [Methods]

DESCRIPTION:

```
Determine the difference between the quarter-tone degree of one pitch
object and that of a second.

NB: This method does not return absolute difference; instead, it may return
    positive or negative results depending on the order in which the pitch
    objects are given. (This will aid in revealing directionality.)

NB: The DEGREE slot is measured in quarter-tones, not semitones. Thus,
    middle-C is degree 120, not 60, and the difference between two
    consecutive semitones is 2, not 1.
```

ARGUMENTS:

```
- A first pitch object.
- A second pitch object.
```

RETURN VALUE:

```
Returns a number. The number may be positive or negative.
```

EXAMPLE:

```
;; Subtracting the lower pitch object from the higher returns a positive number
(let ((p1 (make-pitch 'd4))
      (p2 (make-pitch 'c4)))
  (degree- p1 p2))

=> 4

;; Reversing the order in which the pitch objects are entered may return a
;; negative number
(let ((p1 (make-pitch 'd4))
      (p2 (make-pitch 'c4)))
  (degree- p2 p1))

=> -4
```

SYNOPSIS:

```
(defmethod degree- ((p1 pitch) (p2 pitch))
```

38.6.38 pitch/delete-marks

[pitch] [Methods]

DESCRIPTION:

```
Delete all marks stored in the MARKS slot of the given pitch object and
reset the slot to NIL.
```

ARGUMENTS:

```
- A pitch object.
```

RETURN VALUE:

```
Always returns NIL
```

EXAMPLE:

```
;; Add two marks, then delete them. The method returns NIL
(let ((p (make-pitch 'c4)))
  (add-mark p 'pizz)
  (add-mark p 'a)
  (delete-marks p))

=> NIL

;; Add two marks and print the MARKS slot to see the changes. Then apply the
;; delete-marks method and print the MARKS slot to see the changes.
(let ((p (make-pitch 'c4)))
  (add-mark p 'pizz)
  (add-mark p 'a)
  (print (marks p))
  (delete-marks p)
  (print (marks p)))

=>
(A PIZZ)
NIL
```

SYNOPSIS:

```
(defmethod delete-marks ((p pitch))
```

38.6.39 pitch/enharmonic

[pitch] [Methods]

DESCRIPTION:

```
Get the enharmonic equivalent of the given pitch object. Two chromatically
consecutive "white-note" pitches (e.g. B-sharp/C-natural) are considered
enharmonically equivalent. If there is no enharmonic equivalent, the
method just returns the same note.
```

ARGUMENTS:

```
- A pitch object.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to print a warning when no enharmonic can be found. Default = T.
```

RETURN VALUE:

```
A new pitch object i.e. this method is non-destructive.
```

EXAMPLE:

```
;; A "black-key" enharmonic equivalent
(let ((p (make-pitch 'cs4)))
  (data (enharmonic p)))

=> DF4

;; Two chromatically consecutive "white-keys" are enharmonically equivalent
(let ((p (make-pitch 'f4)))
  (data (enharmonic p)))

=> ES4

;; The method returns a pitch object with the same pitch value if there is no
;; enharmonic equivalent
(let ((p (make-pitch 'g4)))
  (data (enharmonic p)))

=> G4
```

SYNOPSIS:

```
(defmethod enharmonic ((p pitch) &key (warn t))
```

38.6.40 pitch/enharmonic-equivalents

[pitch] [Methods]

DATE:

```
25th December 2013
```

DESCRIPTION:

```
Test whether two pitches are enharmonically equivalent.
```

ARGUMENTS:

```
- pitch object 1
- pitch object 2
```

RETURN VALUE:

```
T or NIL
```

EXAMPLE:

```
(enharmonic-equivalents (make-pitch 'gs4) (make-pitch 'af4))
=> T
(enharmonic-equivalents (make-pitch 'gs4) (make-pitch 'gs4))
=> NIL
```

SYNOPSIS:

```
(defmethod enharmonic-equivalents ((p1 pitch) (p2 pitch))
```

38.6.41 pitch/find-nearest-pitch

[pitch] [Functions]

DATE:

```
April 23rd 2016, Edinburgh
```

DESCRIPTION:

```
In a given list of pitches, find the nearest pitch to the pitch given as
the 2nd argument.
```

ARGUMENTS:

- a list of pitches: either pitch objects or symbols. If symbols they will be convered to pitch objects first and optionally sorted from low to high.
- the pitch to use in the search for the nearest. Again: either a pitch object or symbol.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether the pitch list should be sorted from low to high before being searched. Default = NIL.

RETURN VALUE:

Two values: the pitch from the pitch list which is the nearest to the 2nd argument, and the index of this pitch in the list. NB if the 2nd argument is equidistant from two pitches in the list, then the first pitch in the list will be returned.

EXAMPLE:

```
(find-nearest-pitch '(b0 d1 fs1 a4) 'e1)
==>
PITCH: frequency: 36.708, midi-note: 26, midi-channel: 0
       pitch-bend: 0.0
       degree: 52, data-consistent: T, white-note: D1
       nearest-chromatic: D1
       src: 0.14030775, src-ref-pitch: C4, score-note: D1
       qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
       micro-tone: NIL,
       sharp: NIL, flat: NIL, natural: T,
       octave: 1, c5ths: 0, no-8ve: D, no-8ve-no-acc: D
       show-accidental: T, white-degree: 15,
       accidental: N,
       accidental-in-parentheses: NIL, marks: NIL,
       marks-before: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: D1, tag: NIL,
data: D1
**************

1
```

SYNOPSIS:

```
(defun find-nearest-pitch (pitch-list pitch &optional auto-sort)
```

38.6.42 pitch/in-octave

[pitch] [Functions]

DESCRIPTION:

```
Test to see if a specified pitch item falls within a specified octave. The
pitch item can be a pitch object, a numerical frequency value, or a
note-name symbol.
```

ARGUMENTS:

```
- A pitch item. This can be a pitch object, a numerical frequency value, or
  a note-name symbol.
- A number that is the specified octave designator (e.g. the "4" in "C4").
```

RETURN VALUE:

```
T if the specified pitch item falls within the specified octave, otherwise
NIL.
```

EXAMPLE:

```
;; The function returns NIL if the specified pitch item does not fall within ; ;
;; the specified octave.
(let ((p (make-pitch 'c4)))
(in-octave p 3))

=> NIL

;; The function will accept pitch objects
(let ((p (make-pitch 'c4)))
(in-octave p 4))

=> T

;; The function will accept numerical frequency values
(let ((p 261.63))
(in-octave p 4))

=> T

;; The function will accept note-name symbols
(let ((p 'c4))
(in-octave p 4))

=> T
```

SYNOPSIS:

```
(defun in-octave (pitch octave)
```

38.6.43 pitch/invert-pitch-list

[pitch] [Functions]

DESCRIPTION:

```
Using the lowest note in the list as the reference point, invert the rest
of a given list of pitch items according to their distance from
it. i.e. ascending notes return descending notes from the bottom pitch
down. This is quite different to the concept of chord inversion, where we
move from root position to first inversion, second inversion, etc.

The list of pitch items may consist of either note-name symbols, pitch
objects or frequency numbers.

NB: This function adheres to a concept of inversion more related to
    interval set theory than to the traditional inversion of a melodic
    contour. The given list of pitch items is first sorted from low to high
    before the internal semitone intervals are assessed. The resulting list
    will therefore always be in chromatic order, rather than having the
    inverted melodic contour of the original.
```

ARGUMENTS:

```
- A list of pitch items. This may consist of pitch objects, note-name
  symbols, or frequency numbers.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether the result should be a list of pitch objects
  or a list of note-name symbols. T = note-name symbols. Default = NIL.
- The package in which the process is to be performed. Default = :sc.
```

RETURN VALUE:

```
Returns list of pitch objects by default. If the first optional argument is
set to T, the function will return a list of note-name symbols instead.
```

EXAMPLE:

```
;; Simple example returning pitch symbols:
(invert-pitch-list '(c4 cs4 f4 b4) t)
=>
(C4 B3 G3 CS3)

;; The function returns a list of pitch objects by default
(let ((pl))
  (setf pl (loop for m in '(E4 G4 A4 C4) collect (make-pitch m)))
  (invert-pitch-list pl))

=>
(
PITCH: frequency: 261.626, midi-note: 60, midi-channel: 0
[...]
data: C4
[...]
PITCH: frequency: 207.652, midi-note: 56, midi-channel: 0
[...]
data: AF3
[...]
PITCH: frequency: 174.614, midi-note: 53, midi-channel: 0
[...]
data: F3
[...]
PITCH: frequency: 155.563, midi-note: 51, midi-channel: 0
[...]
data: EF3
)

;; Setting the first optional argument to T will cause the function to return a
;; list of note-name symbols instead
(let ((pl))
  (setf pl '(329.63 392.00 440.00 261.63))
  (invert-pitch-list pl t))

=> (C4 AF3 F3 EF3)
```

SYNOPSIS:

```
(defun invert-pitch-list (pitch-list &optional
                          (return-symbols nil)
                          (package :sc))
```

38.6.44 pitch/make-pitch

[pitch] [Functions]

DESCRIPTION:

```
Create a pitch object, specifying a note as either a symbol or a
number. When the note is specified as a symbol, it is treated as a
note-name; when it is specified as a number, it is treated as a frequency
in hertz.

NB If a pitch object is created from a frequency (rather than note symbol)
   then the given frequency is stored and the note/midi-note etc. nearest
   to it will be stored also.  So the frequency might not be the exact
   frequency of the reflected note.  This is by design, so that unusual
   temperaments can retain exact frequencies and show nearest notes etc.
```

ARGUMENTS:

```
- A note, either as an alphanumeric note name (e.g. cs4) or a numeric hertz
  frequency. Note that if one or more pitch objects have been initialised
  then it is not necessary to specify the octave as the last octave will be
  used.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :src-ref-pitch. A note-name symbol indicating the perceived fundamental
  pitch of a given digital audio file, to allow for later transposition of
  that audio file using note-names. Default: C4
- :midi-channel. An integer indicating which MIDI channel is to be used for
  playback of this pitch. Default: 1
```

RETURN VALUE:

```
- A pitch object.
```

EXAMPLE:

```
;; Make a pitch object using a note-name symbol
(make-pitch 'c4)

=>
PITCH: frequency: 261.626, midi-note: 60, midi-channel: 0
       pitch-bend: 0.0
```

```
       degree: 120, data-consistent: T, white-note: C4
       nearest-chromatic: C4
       src: 1.0, src-ref-pitch: C4, score-note: C4
       qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
       micro-tone: NIL,
       sharp: NIL, flat: NIL, natural: T,
       octave: 4, c5ths: 0, no-8ve: C, no-8ve-no-acc: C
       show-accidental: T, white-degree: 28,
       accidental: N,
       accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: C4, tag: NIL,
data: C4

;; Make a pitch object using a frequency in hertz and including a value for the
;; keyword argument :midi-channel, then print the DATA and MIDI-NOTE slots to
;; see the method's automatic conversion for those values.
(let ((p (make-pitch 261.63 :midi-channel 1)))
  (print (data p))
  (print (midi-note p)))

=>
C4
60

;; Make a pitch object for use with a digital audio file that includes a
;; note-name symbol for the sample-rate-conversion reference pitch; then print
;; the SRC slot of the resulting pitch object
(let ((p (make-pitch 'c4 :src-ref-pitch 'a4)))
  (src p))

=> 0.5946035487490308

;;  make two pitch objects; the 2nd uses the octave of the first
(progn (make-pitch 'cs6)
       (make-pitch 'a))
=>
PITCH: frequency: 1760.000, midi-note: 93, midi-channel: 1
       pitch-bend: 0.0
       degree: 186, data-consistent: T, white-note: A6
       nearest-chromatic: A6
...
```

SYNOPSIS:

```
(defun make-pitch (pitch &key (src-ref-pitch 'c4) (midi-channel 1))
```

38.6.45 pitch/midi-

[pitch] [Methods]

DESCRIPTION:

```
Determine the difference in number of semitones between the values of the
MIDI values of two given pitch objects.

NB: This method does not return absolute difference; instead, it may return
    positive or negative results depending on the order in which the pitch
    objects are given. (This will aid in revealing directionality.)
```

ARGUMENTS:

```
- A first pitch object.
- A second pitch object.
```

RETURN VALUE:

```
Returns a number. The number may be positive or negative.
```

EXAMPLE:

```
;; Subtracting the lower pitch object from the higher returns a positive number
(let ((p1 (make-pitch 'd4))
      (p2 (make-pitch 'c4)))
  (midi- p1 p2))

=> 2

;; Reversing the order in which the pitch objects are entered may return a
;; negative number
(let ((p1 (make-pitch 'd4))
      (p2 (make-pitch 'c4)))
  (midi- p2 p1))

=> -2
```

SYNOPSIS:

```
(defmethod midi- ((p1 pitch) (p2 pitch))
```

38.6.46 pitch/midi-play-pitch-list

[pitch] [Functions]

DATE:

```
January 2nd 2020, Heidhausen
```

DESCRIPTION:

```
Generate a MIDI file from a list of pitches.
```

ARGUMENTS:

```
the pitch list: either pitch objects or symbols
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :rhythm. The rhythm (symbol or object) to be used for all pitches. Default
  = 'e (eighth note)
- :tempo. The MIDI file tempo. Default = 120BPM.
- :file. The patch of the MIDI file to write. Default = "/tmp/tmp.mid"
```

RETURN VALUE:

```
the MIDI file path (string)
```

SYNOPSIS:

```
(defun midi-play-pitch-list (pitches
                             &key (rhythm 'e) (tempo 120) (file "/tmp/tmp.mid"))
```

38.6.47 pitch/no-accidental

[pitch] [Methods]

DESCRIPTION:

```
Set the SHOW-ACCIDENTAL and ACCIDENTAL-IN-PARENTHESES slots of a specified
pitch object to NIL, preventing any accidentals or accidentals in
parentheses from being shown for that event in the printed score.

NB: This will only be effective if the :respell-notes option for
    cmn-display and write-lp-data-for-all is set to NIL.
```

ARGUMENTS:

```
- A pitch object.
```

RETURN VALUE:

```
Always NIL.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :set-palette '((1 ((c4 cs4 fs4))))
        :set-map '((1 (1)))
        :rthm-seq-palette '((1 ((((2 4) - s s s s - - s s s s -))
                                :pitch-seq-palette ((1 2 3 2 1 2 3 2)))))
        :rthm-seq-map '((1 ((vn (1))))))))
  (no-accidental (pitch-or-chord (get-note mini 1 7 'vn)))
  (cmn-display mini :respell-notes nil)
  (write-lp-data-for-all mini :respell-notes nil))
```

SYNOPSIS:

```
(defmethod no-accidental ((p pitch))
```

38.6.48 pitch/note=

[pitch] [Methods]

DESCRIPTION:

```
Tests to see the note-name symbols (values in the DATA slots) of two given
pitch objects are equal.

NB: This method allows for the comparison of pitch objects created using
    frequency numbers and those created using note-name symbols.
```

ARGUMENTS:

```
- A first pitch object.
- A second pitch object.
```

RETURN VALUE:

```
T if the note-name symbols of the given pitch objects are equal, otherwise
NIL.
```

EXAMPLE:

```
;; Two pitch objects with equal note-name symbols return T
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'c4)))
  (note= p1 p2))

=> T
;; Two pitch objects with unequal note-name symbols return F
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'd4)))
  (note= p1 p2))

=> NIL

;; Pitch objects created using frequency numbers and those created using
;; note-name symbols can be effectively compared using this method
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 261.63)))
  (note= p1 p2))

=> T
```

SYNOPSIS:

```
(defmethod note= ((p1 pitch) (p2 pitch) &optional ignore)
```

38.6.49 pitch/partial-nodes

[pitch] [Functions]

DATE:

August 30th 2023

DESCRIPTION:

Calculate the pitches that represent the nodal/touching points to achieve a particular partial number from a given open-string. Note that despite the general rule that nodes are available from one to 1- the desired partial, not all can be used because they coincide with lower-order partials. E.g. to get the 4th partial (double octave, usually achieved by touching a perfect fourth above the open string) we have nodes at 1/4 and 3/4 but not at 2/4 because that is 1/2 i.e. the second partial.

See also https://en.wikipedia.org/wiki/String_harmonic

ARGUMENTS:

```
 - the pitch of the open string either as a symbol or pitch object
 - the desired partial number (where 1 = the open string)
```

RETURN VALUE:

```
 a list of ascending pitch objects representing the nodal/touching
 points. Note that the last pitch will always be the harmonic sounding pitch.
 last
```

EXAMPLE:

```
(partial-nodes 'd3 6)
(
PITCH: frequency: 176.199, midi-note: 53, midi-channel: 1
       pitch-bend: 0.16
...
data: F3
**************


PITCH: frequency: 880.994, midi-note: 81, midi-channel: 1
       pitch-bend: 0.02
...
data: A5
**************
)

;;; the closest pitches, rounded to a quarter-tone (the default scale) for
;;; then7th partial on the cello's D string (II) :
(mapcar #'id (partial-nodes 'd3 7))
--> (EQS3 AF3 BQS3 EQS4 BQS4 BQS5)
```

SYNOPSIS:

```
(defun partial-nodes (open-string partial)
```

38.6.50 pitch/pitch-

[pitch] [Methods]

DESCRIPTION:

```
 Get the distance in semitones between the values of two pitch objects. This
 method also takes fractional values into consideration. The
```

ARGUMENTS:

```
 - A first pitch object.
 - A second pitch object.
```

RETURN VALUE: EXAMPLE:

```
;; Get the distance between two "white-keys"
(let ((p1 (make-pitch 'd4))
      (p2 (make-pitch 'c4)))
 (pitch- p1 p2))

=> 2.0

;; Get the distance in semitones between two frequencies (rounded to the
;; nearest degree, which by default is quarter-tones)
(let ((p1 (make-pitch 293.66))
      (p2 (make-pitch 261.63)))
 (pitch- p1 p2))

=> 2.0

;; Getting the distance in semitones between pitches with fractional values can
;; return fractional results
(let ((p1 (make-pitch 'dqs4))
      (p2 (make-pitch 'c4)))
 (pitch- p1 p2))

=> 2.5
```

SYNOPSIS:

```
(defmethod pitch- ((p1 pitch) (p2 pitch))
```

38.6.51 pitch/pitch-class-eq

[pitch] [Methods]

DATE:

```
 14 Aug 2010
```

DESCRIPTION:

```
 Test whether the values of two pitch objects are of the same pitch class,
 i.e. both Cs, or F#s,irrespective of octave.
```

ARGUMENTS:

- A first pitch object.
- A second pitch object.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether or not enharmonic pitches are considered equal. T = enharmonic pitches are considered equal. Default = NIL.

RETURN VALUE:

T if the values of the two pitch objects are of the same pitch class, otherwise NIL.

EXAMPLE:

```
;; A comparison of two pitch objects with values of the same pitch class
;;; returns T
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'c5)))
  (pitch-class-eq p1 p2))

=> T

;; A comparison of two pitch objects with values of differing pitch classes
;; returns NIL
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'cs5)))
  (pitch-class-eq p1 p2))

=> NIL

;; A comparison of two pitch objects with enharmonically equivalent pitch
;; classes returns NIL by default
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'bs4)))
  (pitch-class-eq p1 p2))

=> NIL

;; Setting the optional argument to T causes the method to consider
;; enharmonically equivalent pitch classes equal
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'bs4)))
```

```
  (pitch-class-eq p1 p2 t))

=> T
```

SYNOPSIS:

```
(defmethod pitch-class-eq ((p1 pitch) (p2 pitch)
                           &optional enharmonics-are-equal)
```

38.6.52 pitch/pitch-in-range

[pitch] [Methods]

DESCRIPTION:

```
Determine whether the frequency of a given pitch object falls between the
frequencies of two other given pitch objects.
```

ARGUMENTS:

```
- A first pitch object.
- A second pitch object, which must be lower than the third.
- A third pitch object, which must be higher than the second.
```

RETURN VALUE:

```
T if the frequency value of the first specified pitch object falls between
the second and third specified pitch objects, otherwise NIL.
```

EXAMPLE:

```
;; The method returns T when the frequency value of the first pitch object
;; falls between that of the second and third pitch objects.
(let ((p (make-pitch 'c4))
      (l (make-pitch 'g3))
      (h (make-pitch 'a7)))
  (pitch-in-range p l h))

=> T

;; The method returns NIL when the frequency value of the first pitch object is
;; below the range designated by the frequency values of the other two objects.
(let ((p (make-pitch 'g3))
      (l (make-pitch 'c4))
      (h (make-pitch 'a7)))
```

```
  (pitch-in-range p l h))

=> NIL

;; The method returns NIL when the frequency value of the first pitch object is
;; above the range designated by the frequency values of the other two objects.
(let ((p (make-pitch 'a7))
      (l (make-pitch 'g3))
      (h (make-pitch 'c4)))
  (pitch-in-range p l h))

=> NIL

;; The method will also return NIL if the frequency value of the second pitch
;; object is higher than that of the third
(let ((p (make-pitch 'c4))
      (l (make-pitch 'a7))
      (h (make-pitch 'g3)))
  (pitch-in-range p l h))

=> NIL
```

SYNOPSIS:

```
(defmethod pitch-in-range ((p pitch) (lowest pitch) (highest pitch))
```

38.6.53 pitch/pitch-inc

[pitch] [Methods]

DESCRIPTION:

```
 Increment the value of a given pitch object by one degree (default) or by a
 specified number of degrees (optional argument).

 NB: The slippery-chicken package uses a quarter-tone degree system by
     default, so any function or method involving a degree argument will be
     measured in quarter-tones, not semitones. Thus, while the MIDI note
     value for 'C4 is 60 (chromatic semitones), (note-to-degree 'C4) will
     return 120. Thus, this method will increment by one quarter-tone by
     default, and any value the chooser uses for the optional argument is
     also number of quarter-tones.

 NB: This method returns a new pitch object rather than modifying the values
     of the original.
```

ARGUMENTS:

```
 - A pitch object.
```

OPTIONAL ARGUMENTS:

```
 - A number indicating the step (in degrees) by which the pitch value is to
   be incremented. Defaults = 1.
```

RETURN VALUE:

```
 Returns a pitch object.
```

EXAMPLE:

```
;; The method by default returns a pitch object and increments by one
;; quarter-tone
(let ((p (make-pitch 'c4)))
  (pitch-inc p))

=>
PITCH: frequency: 269.292, midi-note: 60, midi-channel: 0
       pitch-bend: 0.5
       degree: 121, data-consistent: T, white-note: C4
       nearest-chromatic: C4
       src: 1.0293022394180298, src-ref-pitch: C4, score-note: CS4
       qtr-sharp: 1, qtr-flat: NIL, qtr-tone: 1,
       micro-tone: T,
       sharp: NIL, flat: NIL, natural: NIL,
       octave: 4, c5ths: 0, no-8ve: CQS, no-8ve-no-acc: C
       show-accidental: T, white-degree: 28,
       accidental: QS,
       accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: CQS4, tag: NIL,
data: CQS4

;; Using the optional argument, increment steps can be changed; for example,
;; here to one semitone (= 2 quarter-tones)
(let ((p (make-pitch 'c4)))
  (data (pitch-inc p 2)))

=> CS4

;; Here the method increments by 4 quarter-tones = 1 whole-tone
```

```
(let ((p (make-pitch 'c4)))
  (data (pitch-inc p 4)))

=> D4

;; Incrementing by an additional number of quarter-tones at each pass
(let ((p (make-pitch 'c4)))
  (loop for i from 0 to 4 collect (data (pitch-inc p i))))

=> (C4 CQS4 CS4 DQF4 D4)
```

SYNOPSIS:

```
(defmethod pitch-inc ((p pitch) &optional (degrees 1))
```

38.6.54 pitch/pitch-intersection

[pitch] [Functions]

DESCRIPTION:

Return pitch objects whose values consist of pitches common to two given lists of pitch items. The given lists of pitch items can consist of pitch objects or note-name symbols, or one list of one type and the second of the other.

ARGUMENTS:

- A first list of pitch objects.
- A second list of pitch objects.

RETURN VALUE:

Returns a list of pitch objects that are common to both original lists.

EXAMPLE:

```
;; Returns a list of pitch objects
(let ((p1 '(c4 d4 e4 f4))
(p2 (loop for nn in '(d4 e4 f4 g4) collect (make-pitch nn))))
(pitch-intersection p1 p2))

(
PITCH: frequency: 293.665, midi-note: 62, midi-channel: 0
[...]
```

```
data: D4
[...]
PITCH: frequency: 329.628, midi-note: 64, midi-channel: 0
[...]
data: E4
[...]
PITCH: frequency: 349.228, midi-note: 65, midi-channel: 0
[...]
data: F4
[...]
)
```

SYNOPSIS:

```
(defun pitch-intersection (pitch-list1 pitch-list2)
```

38.6.55 pitch/pitch-list-intervals

[pitch] [Functions]

DATE:

26th August 2024

DESCRIPTION:

Get the interval structure of a list of pitches or pitch symbols.

ARGUMENTS:

the list of pitches, either as pitch objects or symbols

RETURN VALUE:

a list of intervals in semitones

EXAMPLE:

```
(pitch-list-intervals '(c2 cqs2 g2 d3 c3))
--> (0.5 6.5 7.0 -2.0)
```

SYNOPSIS:

```
(defun pitch-list-intervals (pitch-list)
```

38.6.56 pitch/pitch-list-to-symbols

[pitch] [Functions]

DESCRIPTION:

```
Return as a list the note-name values from a given list of pitch objects.
```

ARGUMENTS:

```
- A list of pitch objects.
```

OPTIONAL ARGUMENTS:

```
- The package in which to process the list of pitches. Default = :sc.
```

RETURN VALUE:

```
A list of note-name symbols
```

EXAMPLE:

```
;; Create a list of pitch objects and apply the pitch-list-to-symbols method
(let ((pl))
  (setf pl (loop for m from 0 to 127 by 13
              collect (make-pitch (midi-to-note m))))
  (pitch-list-to-symbols pl))

=> (C-1 CS0 D1 EF2 E3 F4 FS5 G6 AF7 A8)
```

SYNOPSIS:

```
(defun pitch-list-to-symbols (pitch-list &optional (package :sc))
```

38.6.57 pitch/pitch-list-to-xml

[pitch] [Functions]

DATE:

```
27th August 2024
```

DESCRIPTION:

```
Write a MusicXML file from a list of pitches or pitch symbols. The list may
also contain NILs: these will be interpreted to be rests. A slippery-chicken
object is created (via bars-to-sc) in order to accomplish this and is
returned by the function.
```

ARGUMENTS:

```
- the list of pitches
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :time-sig. The time signature (list) for the XML file. Default = '(2 4)
- :rhythm. The rhythm (symbol or object) that will be used for every pitch
  and rest. Default = 'e
- :tempo. The tempo that will be written: BPM as a number. Default = 120.
- :instrument. The instrument that the pitches will be assigned to. Must be
  part of the +slippery-chicken-standard-instrument-palette+. Default =
  'flute
- :auto-beam. Whether to automatically beam the rhythms generated. Default =
  T.
- :midi-channel. An integer that will be used to set the MIDI-CHANNEL slot
  of any event objects passed. Default = 1.
- :microtones-midi-channel. An integer that is the MIDI channel that will
  be assigned to event objects for microtonal MIDI pitches. NB: This value
  is only set when attached to event objects within a slippery-chicken
  object. Default = 2. NB: See player.lsp/make-player for details on
  microtones in MIDI output.
- :file. The path of the XML file to be written. Default = pitch-list.xml in
  the current default directory ("/tmp" if not otherwise set).
```

RETURN VALUE:

```
The slippery-chicken created in order to generated the XML file.
```

SYNOPSIS:

```
(defun pitch-list-to-xml (pitches
                          &key (time-sig '(2 4)) (rhythm 'e) (tempo 120)
                          (instrument 'flute) (auto-beam t)
                          (midi-channel 1) (microtones-midi-channel 2)
                          (file (default-dir-file "pitch-list.xml")))
```

38.6.58 pitch/pitch-max

[pitch] [Methods]

DESCRIPTION:

```
Determine which of two specified pitch objects has the greater frequency
value and return that pitch object.

NB: If the two frequency values are equal, the method returns a pitch
    object equivalent to both.
```

ARGUMENTS:

```
- A first pitch object.
- A second ptich object.
```

RETURN VALUE:

```
A pitch object.
```

EXAMPLE:

```
;; Compare two pitch objects and return the one with the greater frequency
;; value
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'd4)))
  (pitch-max p1 p2))

=>
PITCH: frequency: 293.665, midi-note: 62, midi-channel: 0
       pitch-bend: 0.0
       degree: 124, data-consistent: T, white-note: D4
       nearest-chromatic: D4
       src: 1.1224620342254639, src-ref-pitch: C4, score-note: D4
       qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
       micro-tone: NIL,
       sharp: NIL, flat: NIL, natural: T,
       octave: 4, c5ths: 0, no-8ve: D, no-8ve-no-acc: D
       show-accidental: T, white-degree: 29,
       accidental: N,
       accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: D4, tag: NIL,
data: D4

;; Comparing two pitch objects with equal frequency values returns a pitch
;; object equal to both
(let ((p1 (make-pitch 'c4))
```

```
      (p2 (make-pitch 'c4)))
  (data (pitch-max p1 p2)))

=> C4
```

SYNOPSIS:

```
(defmethod pitch-max ((p1 pitch) (p2 pitch))
```

38.6.59 pitch/pitch-member

[pitch] [Functions]

DESCRIPTION:

Test whether a specified pitch is a member of a given list of pitches.

This function can take pitch objects, note-name symbols or numerical frequency values (or lists thereof) as its arguments.

ARGUMENTS:

- A pitch item. This may be a pitch object, a note-name symbol or a numerical frequency value.
- A list of pitch items. These items may be pitch objects, note-name symbols, or numerical frequency values.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether or not the function should consider enharmonically equivalent pitches to be equal. T = enharmonics are considered equal. Default = T.
- The second optional argument allows the user to specify the test for comparison, such as note=, pitch-class-eq, or the default pitch=. If the user wants to specify his or her own, the test must take three arguments: p1, p2 and <enharmonics-are-equivalent> (which may of course be ignored).

RETURN VALUE:

Similar to Lisp's "member" function, this method returns the tail of the tested list starting with the specified pitch if the pitch is indeed a member of that list, otherwise returns NIL. NB: The list returned is a list of pitch objects.

EXAMPLE:

```
;; Returns NIL if the specified pitch item is not a member of the given list
(let ((pl '(c4 d4 e4)))
(pitch-member 'f4 pl))

=> NIL

;; Returns the tail of the given list starting from the specified pitch if that
;; pitch is indeed a member of the tested list
(let ((pl '(c4 d4 e4)))
(pitch-list-to-symbols (pitch-member 'd4 pl)))

=> (D4 E4)

;; Enharmonically equivalent pitches are considered equal by default
(let ((pl '(c4 ds4 e4)))
(pitch-list-to-symbols (pitch-member 'ef4 pl)))

=> (DS4 E4)

;; Enharmonic equivalence can be turned off by setting the first optional
;; argument to NIL
(let ((pl '(c4 ds4 e4)))
(pitch-list-to-symbols (pitch-member 'ef4 pl nil)))

=> NIL
```

SYNOPSIS:

```
(defun pitch-member (pitch pitch-list
                     &optional (enharmonics-are-equal t)
                     (test #'pitch=))
```

38.6.60 pitch/pitch-min

[pitch] [Methods]

DESCRIPTION:

```
Determine which of two specified pitch objects has the lower frequency
value and return that pitch object.

NB: If the two frequency values are equal, the method returns a pitch
    object equivalent to both.
```

ARGUMENTS:

```
 - A first pitch object.
 - A second ptich object.
```

RETURN VALUE:

```
 A pitch object.
```

EXAMPLE:

```
;; Compare two pitch objects and return the one with the lower frequency value
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'd4)))
  (pitch-min p1 p2))

=>
PITCH: frequency: 261.626, midi-note: 60, midi-channel: 0
       pitch-bend: 0.0
       degree: 120, data-consistent: T, white-note: C4
       nearest-chromatic: C4
       src: 1.0, src-ref-pitch: C4, score-note: C4
       qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
       micro-tone: NIL,
       sharp: NIL, flat: NIL, natural: T,
       octave: 4, c5ths: 0, no-8ve: C, no-8ve-no-acc: C
       show-accidental: T, white-degree: 28,
       accidental: N,
       accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: C4, tag: NIL,
data: C4

;; Comparing two pitch objects with equal frequency values returns a pitch
;; object equal to both
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'c4)))
  (data (pitch-min p1 p2)))

=> C4
```

SYNOPSIS:

```
(defmethod pitch-min ((p1 pitch) (p2 pitch))
```

38.6.61 pitch/pitch-or-chord=

[pitch] [Methods]

DATE:

```
Tue 29 Oct 2019 08:57:53 GMT - Warwick
```

DESCRIPTION:

```
Convenience method, test to see if the pitch-or-chord slots of two event
objects are the same.
```

ARGUMENTS:

```
- a pitch or chord object
- a pitch or chord object
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether or not enharmonic pitches are considered
  equal. T = enharmonic pitches are considered equal. Default = NIL.
- a number to indicate the frequency deviation allowed before returning NIL.
```

RETURN VALUE:

```
T if the values of the two specified pitch or chord objects are equal, otherwise
NIL.
```

EXAMPLE:

```
;; Comparison of equal pitch objects created using note-name symbols returns T
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'c4)))
  (pitch-or-chord= p1 p2))

=> T

(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'bs3)))
  (pitch-or-chord= p1 p2))

=> NIL

(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'bs3)))
  (pitch-or-chord= p1 p2 t))

=> T
```

```
(let ((p1 (make-pitch 'c4))
      (c1 (make-chord '(c4 e4 g4))))
  (pitch-or-chord= p1 c1))

=> NIL

(let ((c1 (make-chord '(c4 e4 g4)))
      (c2 (make-chord '(c4 e4))))
  (pitch-or-chord= c1 c2))

=> NIL

(let ((c1 (make-chord '(c4 e4 g4)))
      (c2 (make-chord '(bs4 ff4 g4))))
  (pitch-or-chord= c1 c2))

=> NIL

(let ((c1 (make-chord '(c4 e4 g4)))
      (c2 (make-chord '(bs3 ff4 g4))))
  (pitch-or-chord= c1 c2 t))

=> T

;; Chords with only one pitch are the same as pitch object with the same pitch.
(let ((c (make-chord '(c4)))
      (p (make-pitch 'c4)))
  (pitch-or-chord= c p))

=> T
```

SYNOPSIS:

```
(defmethod pitch-or-chord= ((p1 pitch) (p2 pitch)
                            &optional enharmonics-are-equal
                              (frequency-tolerance 0.01))
```

38.6.62 pitch/pitch-round

[pitch] [Methods]

DESCRIPTION:

```
Rounds the value of a specified pitch object to the nearest chromatic
semitone (non-microtonal MIDI) pitch.
```

ARGUMENTS:

```
- A pitch object.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :as-symbol. T or NIL to indicate whether the method is to return an
  entire pitch object or just a note-name symbol of the new pitch. NIL = a
  new pitch object. Default = NIL.
- :package. Used to identify a separate Lisp package in which to process
  result. This is really only applicable is combination with :as-symbol set
  to T. Default = :sc.
```

RETURN VALUE:

```
A pitch object by default.

If the :as-symbol argument is set to T, then a note-name symbol is returned
instead.
```

EXAMPLE:

```
;; Returns a pitch object by default; here an example rounding a quarter-tone
;;; note-name symbol to the nearest chromatic pitch
(let ((p (make-pitch 'CQS4)))
  (pitch-round p))

=>
PITCH: frequency: 261.626, midi-note: 60, midi-channel: 0
[...]
NAMED-OBJECT: id: C4, tag: NIL,
data: C4

;; Also rounds frequencies to the nearest chromatic pitch. This example first
;; prints the original values automatically stored with frequency 269.0
;; (rounded by default to the nearest quarter-tone), then the new value rounded
;; to the nearest chromatic semitone
(let ((p (make-pitch 269.0)))
  (print (data p))
  (print (pitch-round p :as-symbol t)))

=>
CQS4
C4
```

SYNOPSIS:

```
(defmethod pitch-round ((p pitch) &key (as-symbol nil) (package :sc))
```

38.6.63 pitch/pitch<

[pitch] [Methods]

DESCRIPTION:

```
Test to see if the frequency value of one specified pitch object is less
than that of a second.

NB: Due to the fact that a given note-name may encompass several
    fractionally different frequencies (e.g. both 261.626 and 261.627 are
    both considered to be C4), this method is not suitable for comparing
    pitch objects of which one was created using a frequency and the other
    was created using a note-name symbol.
```

ARGUMENTS:

```
- A pitch object.
- A second pitch object.
```

RETURN VALUE:

```
Returns T if the frequency value of the first pitch object is less than
that of the second, otherwise NIL.
```

EXAMPLE:

```
;; T is returned when the frequency of the first pitch is less than that of
;; the second
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'd4)))
  (pitch< p1 p2))

=> T

;; NIL is returned when the frequency of the first pitch is not less than
;; that of the second
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'd4)))
  (pitch< p2 p1))
```

```
=> NIL

;; Equivalent pitches return NIL
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'c4)))
  (pitch< p2 p1))

=> NIL

;; This method can be effectively used to compare the frequency values of two
;; pitch objects that were both created using frequency numbers
(let ((p1 (make-pitch 261.63))
      (p2 (make-pitch 293.66)))
  (pitch< p1 p2))

=> T

;; Due to sc's numerical accuracy, this method is not suitable for comparing
;; pitch objects of which one was created using a note-name symbol and the
;; other was created using a numerical frequency value. Such comparisons may
;; return misleading results.
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 261.63)))
  (pitch< p1 p2))

=> T
```

SYNOPSIS:

```
(defmethod pitch< ((p1 pitch) (p2 pitch))
```

38.6.64 pitch/pitch<=

[pitch] [Methods]

DESCRIPTION:

```
Test to see if the frequency value of one specified pitch object is less
than or equal to than that of a second.

NB: Due to the fact that a given note-name may encompass several
    fractionally different frequencies (e.g. both 261.626 and 261.627 are
    both considered to be C4), this method is not suitable for comparing
    pitch objects of which one was created using a frequency and the other
    was created using a note-name symbol.
```

ARGUMENTS:

- A pitch object.
- A second pitch object.

RETURN VALUE:

Returns T if the frequency value of the first pitch object is less than or equal to that of the second, otherwise NIL.

EXAMPLE:

```
;; T is returned when the frequency of the first pitch is less than or equal to
;; that of the second
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'd4)))
  (pitch<= p1 p2))

=> T

;; NIL is returned when the frequency of the first pitch is not less than or
;; equal to that of the second
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'd4)))
  (pitch<= p2 p1))

=> NIL

;; Equivalent pitches return T
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'c4)))
  (pitch<= p2 p1))

=> T

;; This method can be effectively used to compare the frequency values of two
;; pitch objects that were both created using frequency numbers
(let ((p1 (make-pitch 261.63))
      (p2 (make-pitch 293.66)))
  (pitch<= p1 p2))

=> T

;; Due to sc's numerical accuracy, this method is not suitable for comparing
;; pitch objects of which one was created using a note-name symbol and the
```

```
;; other was created using a numerical frequency value. Such comparisons may
;; return misleading results.
(let ((p1 (make-pitch 261.63))
      (p2 (make-pitch 'c4)))
  (pitch<= p1 p2))

=> NIL
```

SYNOPSIS:

```
(defmethod pitch<= ((p1 pitch) (p2 pitch))
```

38.6.65 pitch/pitch=

[pitch] [Methods]

DESCRIPTION:

Determines if the note-name and chromatic semitone MIDI values of two specified pitch objects are the same (or very close to each other in the case of frequency and src slot comparison).

By default, this method returns NIL when comparing enharmonic pitches. This can behavior can be changed by setting the optional argument to T, upon which enharmonic pitches are considered equal.

NB: This method may return NIL when comparing pitch objects created using frequencies with those created using note-names. The method pitch::note= may be more useful in this case.

NB: Pitch objects created using frequencies are only considered equal if their frequency values are within 0.01Hz of each other.

ARGUMENTS:

- A first pitch object.
- A second pitch object.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether or not enharmonic pitches are considered equal. T = enharmonic pitches are considered equal. Default = NIL.
- a number to indicate the frequency deviation allowed before returning NIL.

RETURN VALUE:

```
 T if the values of the two specified pitch objects are equal, otherwise
 NIL.
```

EXAMPLE:

```
;; Comparison of equal pitch objects created using note-name symbols returns T
(let ((p1 (make-pitch 'C4))
      (p2 (make-pitch 'C4)))
  (pitch= p1 p2))

=> T

;; Comparison of unequal pitch objects created using note-name symbols returns
NIL
(let ((p1 (make-pitch 'C4))
      (p2 (make-pitch 'D4)))
  (pitch= p1 p2))

=> NIL

;; Comparison of enharmonically equivalent pitch objects returns NIL by default
;; Comparison of equal pitch objects created using note-name symbols returns T
(let ((p1 (make-pitch 'CS4))
      (p2 (make-pitch 'DF4)))
  (pitch= p1 p2))

=> NIL

;; Comparison of enharmonically equivalent pitch objects return T when the
;; optional argument is set to T
;; Comparison of equal pitch objects created using note-name symbols returns T
(let ((p1 (make-pitch 'C4))
      (p2 (make-pitch 'C4)))
  (pitch= p1 p2 t))

=> T

;; Comparison of pitch objects created using frequencies with those created
;; using note-name symbols returns T
(let ((p1 (make-pitch 'C4))
      (p2 (make-pitch 261.63)))
  (pitch= p1 p2))

=> T
```

SYNOPSIS:

```
(defmethod pitch= ((p1 pitch) (p2 pitch) &optional enharmonics-are-equal
                   (frequency-tolerance 0.01)) ; (src-tolerance 0.0001))
```

38.6.66 pitch/pitch>

[pitch] [Methods]

DESCRIPTION:

```
Test to see if the frequency value of one specified pitch object is greater
than that of a second.

NB: Due to the fact that a given note-name may encompass several
fractionally different frequencies (e.g. both 261.626 and 261.627 are both
considered to be C4), this method is not suitable for comparing pitch
objects of which one was created using a frequency and the other was
created using a note-name symbol.
```

ARGUMENTS:

```
- A pitch object.
- A second pitch object.
```

RETURN VALUE:

```
Returns T if the frequency value of the first pitch object is greater than
that of the second, otherwise NIL.
```

EXAMPLE:

```
;; T is returned when the frequency of the first pitch is greater than that of
;; the second
(let ((p1 (make-pitch 'd4))
      (p2 (make-pitch 'c4)))
  (pitch> p1 p2))

=> T

;; NIL is returned when the frequency of the first pitch is not greater than
;; that of the second
(let ((p1 (make-pitch 'd4))
      (p2 (make-pitch 'c4)))
  (pitch> p2 p1))

=> NIL
```

```
;; Equivalent pitches return NIL
(let ((p1 (make-pitch 'd4))
      (p2 (make-pitch 'd4)))
  (pitch> p2 p1))

=> NIL

;; This method can be effectively used to compare the frequency values of two
;; pitch objects that were both created using frequency numbers
(let ((p1 (make-pitch 293.66))
      (p2  (make-pitch 261.63)))
  (pitch> p1 p2))

=> T

;; Due to sc's numerical accuracy, this method is not suitable for comparing
;; pitch objects of which one was created using a note-name symbol and the
;; other was created using a numerical frequency value. Such comparisons may
;; return misleading results.
(let ((p1 (make-pitch 261.63))
      (p2 (make-pitch 'c4)))
  (pitch> p1 p2))

=> T
```

SYNOPSIS:

```
(defmethod pitch> ((p1 pitch) (p2 pitch))
```

38.6.67 pitch/pitch>=

[pitch] [Methods]

DESCRIPTION:

```
Test to see if the frequency value of one specified pitch object is greater
than or equal to than that of a second.

NB: Due to the fact that a given note-name may encompass several
    fractionally different frequencies (e.g. both 261.626 and 261.627 are
    both considered to be C4), this method is not suitable for comparing
    pitch objects of which one was created using a frequency and the other
    was created using a note-name symbol.
```

ARGUMENTS:

```
 - A pitch object.
 - A second pitch object.
```

RETURN VALUE:

```
 Returns T if the frequency value of the first pitch object is greater than
 or equal to that of the second, otherwise NIL.
```

EXAMPLE:

```
;; T is returned when the frequency of the first pitch is greater than or equal
;;; to that of the second
(let ((p1 (make-pitch 'd4))
      (p2 (make-pitch 'c4)))
  (pitch>= p1 p2))

=> T

;; NIL is returned when the frequency of the first pitch is not greater than or
;; equal to that of the second
(let ((p1 (make-pitch 'd4))
      (p2 (make-pitch 'c4)))
  (pitch>= p2 p1))

=> NIL

;; Equivalent pitches return T
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'c4)))
  (pitch>= p2 p1))

=> T

;; This method can be effectively used to compare the frequency values of two
;; pitch objects that were both created using frequency numbers
(let ((p1 (make-pitch 293.66))
      (p2 (make-pitch 261.63)))
  (pitch>= p1 p2))

=> T

;; Due to sc's numerical accuracy, this method is not suitable for comparing
;; pitch objects of which one was created using a note-name symbol and the
;; other was created using a numerical frequency value. Such comparisons may
;; return misleading results.
(let ((p1 (make-pitch 'c4))
```

```
      (p2 (make-pitch 261.63)))
  (pitch>= p1 p2))

=> NIL
```

SYNOPSIS:

```
(defmethod pitch>= ((p1 pitch) (p2 pitch))
```

38.6.68 pitch/print-simple-pitch-list

[pitch] [Functions]

DESCRIPTION:

Print the data symbols of a list of pitch objects. NB If you need to print these in the context of another printing routine, you can get a string by passing NIL as the stream.

DATE:

April 10th 2012

ARGUMENTS:

- A simple list of pitch objects.

OPTIONAL ARGUMENTS:

- The stream to print to (e.g. an open file). Default: the Lisp Terminal (REPL).

RETURN VALUE:

The list of pitch data symbols.

EXAMPLE:

```
(print-simple-pitch-list (init-pitch-list '(c4 d4 e4))) ;
=>
(C4 D4 E4)
(C4 D4 E4)
```

SYNOPSIS:

```
(defun print-simple-pitch-list (pitch-list &optional (stream t))
```

38.6.69 pitch/remove-octaves

[pitch] [Functions]

DESCRIPTION:

```
Removes all but one of any pitch items in a given list that have the same
pitch-class but different octaves, keeping the lowest instance only.

The list of pitch items may be a list of pitch objects or a list of
note-name symbols.
```

ARGUMENTS:

```
- A list of pitch items. These may be pitch objects or note-name symbols.
```

OPTIONAL ARGUMENTS:

```
keyword arguments
- :as-symbol. T or NIL indicating whether the object is to return a list of
  pitch objects or a list of note-name symbols. T = return pitch
  objects. Default = NIL.
- :package. Used to identify a separate Lisp package in which to itern
  result. This is really only applicable is combination with :as-symbol set
  to T. Default = :sc.
- :allow. T or NIL to indicate whether pitch objects of certain, specified
  octave-doublings are to be kept even if they are not the lowest. This
  argument takes the form of either a single number or a list of
  numbers. NB: This number does not indicate the octave in which the pitch
  object is found, but rather pitch objects that are the specified number
  of octaves above the lowest instance of the pitch class. Thus, :allow 2
  indicates keeping the lowest pitch plus any instances of the same pitch
  class two octaves above that lowest pitch (i.e.,
  double-octaves). However, it is important to note that the function first
  removes any octave doublings that are not excepted by the :allow
  argument, which may produce confusing results. Given a list of
  consecutive octaves, such as '(C1 C2 C3 C4) and an :allow value of 2, the
  function will first remove any equal pitch classes that are are not 2
  octaves apart, resulting in C2, C3, and C4 being removed as they are one
  octave distant from C1, C2 and C3. The result of the function using these
  values would therefore be '(C1).
```

RETURN VALUE:

```
Returns a list of pitch objects by default. If the keyword argument
:as-symbol is set to T, the method returns a list of note-name symbols
```

```
instead.

If the first element of the pitch list is a number (i.e. a frequency), the
method returns a list of frequencies.
```

EXAMPLE:

```
;; The method returns a list of pitch objects by default
(remove-octaves '(c1 c2 c3 g3))

=> (
PITCH: frequency: 32.703, midi-note: 24, midi-channel: 0
[...]
data: C1
[...]
PITCH: frequency: 195.998, midi-note: 55, midi-channel: 0
[...]
data: G3
[...]
)

;; If the first element of the pitch list is a frequency, the method returns a
;; list of frequencies
(remove-octaves '(261.63 523.26 1046.52 196.00))

=> (261.63 196.0)

;; Setting keyword argument :as-symbol to T returns a list of note-name symbols
;; instead
(remove-octaves '(261.63 523.26 1046.52 196.00) :as-symbol t)

=> (C4 G3)
```

SYNOPSIS:

```
(defun remove-octaves (pitch-list &key as-symbol allow (package :sc))
```

38.6.70 pitch/remove-pitches

[pitch] [Functions]

DESCRIPTION:

```
Remove a list of specified pitch items from a given list of pitch
items. Even if only one pitch item is to be removed it should be stated as
```

a list.

The pitch items can be in the form of pitch objects, note-name symbols or numerical frequency values.

ARGUMENTS:

- A list of pitch items from which the specified list of pitches is to be removed. These can take the form of pitch objects, note-name symbols or numerical frequency values.
- A list of pitch items to remove from the given list. These can take the form of pitch objects, note-name symbols or numerical frequency values. Even if only one pitch is to be removed is must be stated as a list.

OPTIONAL ARGUMENTS:

keyword arguments:
- :enharmonics-are-equal. Set to T or NIL to indicate whether or not enharmonically equivalent pitches are to be considered the same pitch. T = enharmonically equivalent pitches are equal. Default = T.
- :return-symbols. Set to T or NIL to indicate whether the function is to return a list of pitch objects or note-name symbols. T = note-name symbols. Default = NIL.

RETURN VALUE:

Returns a list of pitch objects by default. When the keyword argument :return-symbols is set to T, the function will return a list of note-names instead.

If the specified list of pitches to be removed are not found in the given list, the entire list is returned.

EXAMPLE:

```
;; By default the function returns a list of pitch objects
(let ((pl '(c4 d4 e4)))
(remove-pitches pl '(d4 e4)))

=> (
PITCH: frequency: 261.626, midi-note: 60, midi-channel: 0
[...]
data: C4
[...]
```

```
)

;; Setting the keyword argument :return-symbols to T causes the function to
;; return a list of note-name symbols instead. Note in this example too that
;; even when only one pitch item is being removed, it must be stated as a list.
(let ((pl '(261.62 293.66 329.62)))
(remove-pitches pl '(293.66) :return-symbols t))

=> (C4 E4)

;; The function will also accept pitch objects
(let ((pl (loop for n in '(c4 d4 e4) collect (make-pitch n))))
(remove-pitches pl `(,(make-pitch 'e4)) :return-symbols t))

=> (C4 D4)

;; By default the function considers enharmonically equivalent pitches to be
;; equal
(let ((pl (loop for n in '(c4 ds4 e4) collect (make-pitch n))))
(remove-pitches pl '(ef4) :return-symbols t))

=> (C4 E4)

;; This feature can be turned off by setting the :enharmonics-are-equal keyword
;; argument to NIL. In this case here, the specified pitch is therefore not
;; found in the given list and the entire original list is returned.
(let ((pl (loop for n in '(c4 ds4 e4) collect (make-pitch n))))
(remove-pitches pl '(ef4)
:return-symbols t
:enharmonics-are-equal nil))

=> (C4 DS4 E4)
```

SYNOPSIS:

```
(defun remove-pitches (pitch-list remove
                       &key (enharmonics-are-equal t)
                       (return-symbols nil))
```

38.6.71 pitch/round-to-nearest

[pitch] [Methods]

DATE:

January 30th 2017

DESCRIPTION:

```
Round the pitch object to the nearest pitch in the current or given scale,
i.e. force frequency, midi-note, pitch bend, etc. to conform to this nearest
pitch.

NB rounding to the nearest using a microtonal :scale argument might cause
errors if the global (in-scale ...) scale is :chromatic because although
chromatic-scale is a subset of quarter-tone and twelfth-tone (so all
chromatic-pitches work) if you have microtonal pitches generated e.g. by
frequency arguments to make-pitch the note symbol in the microtonal scale
doesn't exist in :chromatic. E.g. the following breaks:

(in-scale :chromatic)
264Hz is approx. c 1/12 sharp
(round-to-nearest (make-pitch 264) :scale 'twelfth-tone)
-> fails because rounding in 1/12 tone scale gives us CTS4, which doesn't
exist in the global chromatic scale. Of course all is well if we do this
beforehand:
(in-scale :twelfth-tone)
```

ARGUMENTS:

```
the pitch object
```

OPTIONAL ARGUMENTS:

```
keyword argument:
:scale. The scale to use when rounding. (Common Music tuning object or
symbol). If a symbol, then 'chromatic-scale, 'twelfth-tone, or 'quarter-tone
only at present. Default is the current scale as set by (in-scale :...).
```

RETURN VALUE:

```
the modified pitch object
```

EXAMPLE:

```
(make-pitch 443)
-->
PITCH: frequency: 443.000, midi-note: 69, midi-channel: 1
       pitch-bend: 0.12
       degree: 138, data-consistent: T, white-note: A4
       nearest-chromatic: A4
       src: 1.6932597032572d0, src-ref-pitch: C4, score-note: A4
```

```
       qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
       micro-tone: T,
...

(round-to-nearest (make-pitch 443))
-->
PITCH: frequency: 440.000, midi-note: 69, midi-channel: 1
       pitch-bend: 0.0
       degree: 138, data-consistent: T, white-note: A4
       nearest-chromatic: A4
       src: 1.6817929, src-ref-pitch: C4, score-note: A4
       qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
       micro-tone: NIL,
...

;;; exact quarter tones round down, not up (true for quarter-sharps and flats):
(round-to-nearest (make-pitch 'dqs7) :scale 'chromatic-scale)
-->
PITCH: frequency: 2349.318, midi-note: 98, midi-channel: 1
       pitch-bend: 0.0
       degree: 196, data-consistent: T, white-note: D7
       nearest-chromatic: D7
       src: 8.979696, src-ref-pitch: C4, score-note: D7
       qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
       micro-tone: NIL,
       sharp: NIL, flat: NIL, natural: T,
       octave: 7, c5ths: 0, no-8ve: D, no-8ve-no-acc: D
       show-accidental: T, white-degree: 57,
       accidental: N,
       accidental-in-parentheses: NIL, marks: NIL,
       marks-before: NIL
LINKED-NAMED-OBJECT: previous: NIL,
                     this: NIL,
                     next: NIL
NAMED-OBJECT: id: D7, tag: NIL,
data: D7
**************
```

SYNOPSIS:

```
(defmethod round-to-nearest ((p pitch) &key (scale cm::*scale*))
```

38.6.72 pitch/set-midi-channel

[pitch] [Methods]

DESCRIPTION:

Set the MIDI-CHANNEL slot of the given pitch object.

The method takes two mandatory arguments in addition to the given pitch object, the first being the MIDI-channel used for non-microtonal pitch objects, the second that used for microtonal pitch objects.

NB: The pitch object only has one MIDI-CHANNEL slot, and determines whether that slot is set to the specified non-microtonal or microtonal midi-channel argument based on whether or not the pitch of the given pitch object is determined to be a microtone or not.

ARGUMENTS:

- A pitch object.
- A number indicating the MIDI channel which is to be used to play back non-microtonal pitches.

OPTIONAL ARGUMENTS:

- A number indicating the MIDI channel which is to be used to play back microtonal pitches. NB: See player.lsp/make-player for details on microtones in MIDI output. Default = NIL which means it will take on the value of the second argument (all pitches on the same channel, whether microtonal or not).

RETURN VALUE:

A number indicating which value has been set to the given pitch object's MIDI-CHANEL slot.

EXAMPLE:

```
;; When the pitch of the given pitch object is non-microtonal, the method sets
;; that pitch object's MIDI-CHANNEL slot to the first value specified.
(let ((p (make-pitch 'c4)))
  (set-midi-channel p 11 12)
  (midi-channel p))

=> 11

;; When the pitch of the given pitch object is microtonal, the method sets
;; that pitch object's MIDI-CHANNEL slot to the second value specified.
(let ((p (make-pitch 'cqs4)))
```

```
  (set-midi-channel p 11 12))

=> 12
```

SYNOPSIS:

```
(defmethod set-midi-channel ((p pitch) midi-channel
                             &optional microtones-midi-channel)
```

38.6.73 pitch/sort-pitch-list

[pitch] [Functions]

DESCRIPTION:

Sort a list of pitch objects from low to high based on their frequency value.

ARGUMENTS:

- A list of pitch objects.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether the method is to return a list of pitch objects or a list of note-name symbols.
- The package in which the operation is to be performed. Default = :sc.

RETURN VALUE:

Returns a list of pitch objects by default. When the first optional argument is set to T, the method returns a list of note-name symbols instead.

EXAMPLE:

```
;; Create a list of pitch objects by passing downward through a series of MIDI
;; values and print the result. Then apply the sort-pitch-list method and print
;; the result of that to see the list now ordered from low to high.
(let ((pl))
  (setf pl (loop for m from 64 downto 60
              collect (make-pitch (midi-to-note m))))
  (print (loop for p in pl collect (data p)))
  (print (sort-pitch-list pl)))
```

```
=>
(E4 EF4 D4 CS4 C4)
(
PITCH: frequency: 261.626, midi-note: 60, midi-channel: 0
[...]
data: C4
[...]
PITCH: frequency: 277.183, midi-note: 61, midi-channel: 0
[...]
data: CS4
[...]
PITCH: frequency: 293.665, midi-note: 62, midi-channel: 0
[...]
data: D4
[...]
PITCH: frequency: 311.127, midi-note: 63, midi-channel: 0
[...]
data: EF4
[...]
PITCH: frequency: 329.628, midi-note: 64, midi-channel: 0
[...]
data: E4
)

;; Setting the first optional argument to T causes the method to return a list
;; of note-name symbols instead
(let ((pl))
  (setf pl (loop for m from 64 downto 60
              collect (make-pitch (midi-to-note m))))
  (sort-pitch-list pl t))

=> (C4 CS4 D4 EF4 E4)
```

SYNOPSIS:

```
(defun sort-pitch-list (pitch-list &optional
                        (return-symbols nil)
                        (package :sc))
```

38.6.74 pitch/stretch-intervals

[pitch] [Functions]

DATE:

26th August 2024

DESCRIPTION:

Stretch a list of pitches by scaling and/or adding their intervals. Note that all pitches will be rounded to the nearest note in the current scale.

ARGUMENTS:

a list of pitches, either as pitch objects or symbols

OPTIONAL ARGUMENTS:

- the interval multiplication factor (number)
- a semitone value to add to each interval
- a symbol to skip, e.g.

RETURN VALUE:

a list of pitch objects reflecting the stretch, including the 'skip' symbol if passed and a member of the list

EXAMPLE:

```
(mapcar #'id (remove 'r (stretch-intervals '(c2 cs d3 r e4 c) 0.5 0 'r)))
--> (C2 CQS2 G2 D3 C3)

(mapcar #'id (stretch-intervals '(c2 cs d3 e4 c) 2 3))
--> (C2 F2 BF4 F7 C7)
```

SYNOPSIS:

```
(defun stretch-intervals (pitch-list &optional (multiply 1.0) (add 0.0) skip)
```

38.6.75 pitch/transpose

[pitch] [Methods]

DESCRIPTION:

Transpose the pitch information (frequency, note-name, midi-note etc.) of a given pitch object by a specified number of semitones. The number of semitones specified can be fractional; however, all fractional values will be rounded to the nearest quarter-tone frequency (or default smallest step

for the current scale).

NB: This method returns a new pitch object rather than altering the values of the current pitch object.
NB: the :destructively argument is ignored in this class

ARGUMENTS:

- A pitch object.
- A number representing the number of semitones to be transposed, and which can be fractional.

OPTIONAL ARGUMENTS:

keyword arguments:
- :as-symbol. T or NIL to indicate whether the method is to return an entire pitch object or just a note-name symbol of the new pitch. NIL = a new pitch object. Default = NIL.
- :package. Used to identify a separate Lisp package in which to process the result. This is really only applicable is combination with :as-symbol set to T. Default = :sc.

RETURN VALUE:

A pitch object by default.

If the :as-symbol argument is set to T, then a note-name symbol is returned instead.

EXAMPLE:

```
;; By default the method returns a pitch object
(let ((p (make-pitch 'c4)))
  (transpose p 2))

=>
PITCH: frequency: 293.665, midi-note: 62, midi-channel: 0
       pitch-bend: 0.0
       degree: 124, data-consistent: T, white-note: D4
       nearest-chromatic: D4
       src: 1.1224620342254639, src-ref-pitch: C4, score-note: D4
       qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
       micro-tone: NIL,
       sharp: NIL, flat: NIL, natural: T,
       octave: 4, c5ths: 0, no-8ve: D, no-8ve-no-acc: D
```

```
       show-accidental: T, white-degree: 29,
       accidental: N,
       accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: D4, tag: NIL,
data: D4

;; Setting the :as-symbol keyword argument to T returns just the note-name
;; symbol of the new pitch instead
(let ((p (make-pitch 'c4)))
  (transpose p 2 :as-symbol t))

=> D4

;; The semitones argument can be set to a decimal-point fraction, which may
;; result in quarter-tone pitch values being returned
(let ((p (make-pitch 'c4)))
  (transpose p 2.5))

=>
PITCH: frequency: 302.270, midi-note: 62, midi-channel: 0
       pitch-bend: 0.5
       degree: 125, data-consistent: T, white-note: D4
       nearest-chromatic: D4
       src: 1.1553527116775513, src-ref-pitch: C4, score-note: DS4
       qtr-sharp: 1, qtr-flat: NIL, qtr-tone: 1,
       micro-tone: T,
       sharp: NIL, flat: NIL, natural: NIL,
       octave: 4, c5ths: 0, no-8ve: DQS, no-8ve-no-acc: D
       show-accidental: T, white-degree: 29,
       accidental: QS,
       accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: DQS4, tag: NIL,
data: DQS4

;; Fractional semitone arguments are automatically rounded to the nearest
;; quarter-tone, causing x.5 and x.7, for example, to return the same result,
;; while x.3 and x.1 will return the same value as the given integer
(let ((p (make-pitch 'c4)))
  (print (transpose p 2 :as-symbol t))
  (print (loop for s from 0 to 4
            collect (transpose p (+ 2 (* s .1)) :as-symbol t)))
  (print (loop for s from 5 to 9
            collect (transpose p (+ 2 (* s .1)) :as-symbol t))))
```

```
=>
D4
(D4 D4 D4 D4 D4)
(DQS4 DQS4 DQS4 DQS4 DQS4)
```

SYNOPSIS:

```
(defmethod transpose ((p pitch) semitones &key (as-symbol nil) (package :sc)
                                          destructively)
```

38.6.76 pitch/transpose-pitch-list

[pitch] [Functions]

DESCRIPTION:

```
Transpose the values of a list of pitch objects by a specified number of
semitones.
```

ARGUMENTS:

```
- A list of pitch objects or pitch symbols. Note that any NILs in the list
  will be collected but obviously not turned into pitches.
- A number indicating the number of semitones by which the list is to be
  transposed. Or, if this is a pitch object or symbol, calculate the number
  of semitones based on the distance from this to the first pitch in the
  first argument
```

OPTIONAL ARGUMENTS:

```
keyword arguments
- :return-symbols. T or NIL indicating whether the method is to return a
  list of pitch  objects or a list of note-name symbols for those pitch
  objects. T = note-name symbols. Default = NIL.
- :package. The name of the package to perform the transpositions. Default =
  :sc.
- :lowest. Don't transpose pitches which are lower than this
  argument. Default = C-1 (midi note 0)
- :lowest. Don't transpose pitches which are higher than this
  argument. Default = B8 (midi note 119)
- :sort. Whether to sort the pitch list from highest to lowest. This will
  only work if there are no NILs in the list.
  Default = NIL
```

RETURN VALUE:

```
By default, the method returns a list of pitch objects. When the first
optional argument is set to T, a list of note-name symbols is returned
instead.
```

EXAMPLE:

```
;; Create a list of pitch objects and apply the transpose-pitch-list method
;; with the semitones argument set to 2
(let ((pl))
  (setf pl (loop for m from 60 to 71 collect (make-pitch (midi-to-note m))))
  (transpose-pitch-list pl 2))

=>
(
PITCH: frequency: 293.665, midi-note: 62, midi-channel: 0
[...]
PITCH: frequency: 311.127, midi-note: 63, midi-channel: 0
[...]
PITCH: frequency: 329.628, midi-note: 64, midi-channel: 0
[...]
PITCH: frequency: 349.228, midi-note: 65, midi-channel: 0
[...]
PITCH: frequency: 369.994, midi-note: 66, midi-channel: 0
[...]
PITCH: frequency: 391.995, midi-note: 67, midi-channel: 0
[...]
PITCH: frequency: 415.305, midi-note: 68, midi-channel: 0
[...]
PITCH: frequency: 440.000, midi-note: 69, midi-channel: 0
[...]
PITCH: frequency: 466.164, midi-note: 70, midi-channel: 0
[...]
PITCH: frequency: 493.883, midi-note: 71, midi-channel: 0
[...]
PITCH: frequency: 523.251, midi-note: 72, midi-channel: 0
[...]
PITCH: frequency: 554.365, midi-note: 73, midi-channel: 0
[...]
)

;; Perform the same action with the return-symbols optional argument set to T
(let ((pl))
  (setf pl (loop for m from 60 to 71 collect (make-pitch (midi-to-note m))))
  (print (transpose-pitch-list pl 2 t)))

=> (D4 EF4 E4 F4 FS4 G4 AF4 A4 BF4 B4 C5 CS5)
```

SYNOPSIS:

```
(defun transpose-pitch-list (pitch-list semitones
                             &key (sort nil) (return-symbols nil) (package :sc)
                               lowest highest)
```

38.6.77 pitch/transpose-pitch-list-to-octave

[pitch] [Functions]

DESCRIPTION:

Transpose the pitch values of a list of pitch objects into a specified octave. The individual initial pitch objects can have initial pitch values of different octaves.

ARGUMENTS:

- A list of pitch objects.
- A number indicating the octave in which the resulting list should be.

OPTIONAL ARGUMENTS:

keyword arguments:
- :as-symbols. Set to T or NIL to indicate whether the method is to return a list of pitch objects or a list of the note-name symbols from those pitch objects. T = return as symbols. Default = NIL.
- :package. Used to identify a separate Lisp package in which to itern result. This is really only applicable is combination with :as-symbol set to T. Default = :sc.
- :remove-duplicates. Set to T or NIL to indicate whether any duplicate pitch objects are to be removed from the resulting list. T = remove duplicates. Default = T.

RETURN VALUE:

Returns a list of pitch objects by default. When the keyword argument :as-symbols is set to T, the method returns a list of note-name symbols instead.

EXAMPLE:

```
;; Create a list of four pitch objects from random MIDI numbers and print it,
;; then apply transpose-pitch-list-to-octave, setting the octave argument to 4,
;; and print the result
```

```
(let ((pl))
  (setf pl (loop repeat 4 collect (make-pitch (midi-to-note (random 128)))))
  (print (loop for p in pl collect (data p)))
  (print (transpose-pitch-list-to-octave pl 4)))

=>
(CS7 F7 B0 D4)
(
PITCH: frequency: 493.883, midi-note: 71, midi-channel: 0
[...]
data: B4
[...]
PITCH: frequency: 293.665, midi-note: 62, midi-channel: 0
[...]
data: D4
[...]
PITCH: frequency: 277.183, midi-note: 61, midi-channel: 0
[...]
data: CS4
[...]
PITCH: frequency: 349.228, midi-note: 65, midi-channel: 0
[...]
data: F4
)

;; Setting the keyword argument :as-symbols to T return a list of note-names
;; instead
(let ((pl))
  (setf pl (loop repeat 4 collect (make-pitch (midi-to-note (random 128)))))
  (print (loop for p in pl collect (data p)))
  (print (transpose-pitch-list-to-octave pl 4 :as-symbols t)))

=>
(D5 E1 C7 AF1)
(E4 AF4 D4 C4)

;; The method removes duplicate pitch objects from the resulting list by
;; default
(let ((pl))
  (setf pl (loop repeat 4 collect (make-pitch (midi-to-note (random 128)))))
  (print (loop for p in pl collect (data p)))
  (print (transpose-pitch-list-to-octave pl 4 :as-symbols t)))

=>
(B7 AF1 AF7 G1)
(G4 AF4 B4)
```

SYNOPSIS:

```
(defun transpose-pitch-list-to-octave (pitch-list octave
                                       &key
                                       as-symbols
                                       (package :sc)
                                       (remove-duplicates t))
```

38.6.78 pitch/transpose-to-octave

[pitch] [Methods]

DESCRIPTION:

Transpose the values of a given pitch object to a specified octave.

NB: This method creates a new pitch object rather than replacing the values of the original.

ARGUMENTS:

- A pitch object.
- A number indicating the new octave.

OPTIONAL ARGUMENTS:

keyword arguments:
- :as-symbol. T or NIL to indicate whether the method is to return an entire pitch object or just a note-name symbol of the new pitch. NIL = a new pitch object. Default = NIL.
- :package. Used to identify a separate Lisp package in which to process the result. This is really only applicable is combination with :as-symbol set to T. Default = :sc.

RETURN VALUE:

A pitch object by default.

If the :as-symbol argument is set to T, then a note-name symbol is returned instead.

EXAMPLE:

```
;; Transpose the values of a pitch object containing middle-C (octave 4) to the
;;; C of the treble clef (octave 5)
```

```
(let ((p (make-pitch 'c4)))
  (transpose-to-octave p 5))

=>
PITCH: frequency: 523.251, midi-note: 72, midi-channel: 0
       pitch-bend: 0.0
       degree: 144, data-consistent: T, white-note: C5
       nearest-chromatic: C5
       src: 2.0, src-ref-pitch: C4, score-note: C5
       qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
       micro-tone: NIL,
       sharp: NIL, flat: NIL, natural: T,
       octave: 5, c5ths: 0, no-8ve: C, no-8ve-no-acc: C
       show-accidental: T, white-degree: 35,
       accidental: N,
       accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: C5, tag: NIL,
data: C5

;; Setting the :as-symbol argument to T returns a note-name symbol instead of a
;; pitch object
(let ((p (make-pitch 'c4)))
  (transpose-to-octave p 5 :as-symbol t))

=> C5
```

SYNOPSIS:

```
(defmethod transpose-to-octave ((p pitch) new-octave
                                &key
                                (as-symbol nil)
                                (package :sc))
```

38.6.79 linked-named-object/player

[linked-named-object] [Classes]

NAME:

```
 player

 File:             player.lsp

 Class Hierarchy:  named-object -> linked-named-object -> player
```

```
Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the player class which holds an
                  instrument or a assoc-list of instruments in it's data
                  slot.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    7th September 2001

$$ Last modified:  16:42:03 Sat Nov  1 2025 CET

SVN ID: $Id$
```

38.6.80 player/make-player

[player] [Functions]

DESCRIPTION:

```
Create a player object from a specified instrument-palette object and a
specified instrument or list of instruments which that player plays.

The player object is separate from the instrument object as on player in an
ensemble may perform more than one instrument ("double"), such as flute and
piccolo, clarinet and bass clarinet, or sax, flute and clarinet.
```

ARGUMENTS:

```
- A symbol which will be the ID of the resulting player object.
- An instrument-palette object. If NIL then
  +slippery-chicken-standard-instrument-palette+ will be used
- A symbol or a list of symbols that are the instruments from the
  specified instrument-palette object that the given player will play, as
  spelled and defined within the instrument-palette object. NB: If only one
  instrument is to be assigned to the given player, it should be stated as
  symbol rather than a list, to avoid errors in the DOUBLES slot.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :midi-channel. An integer that indicates the MIDI channel on which any
```

```
   non-microtonal pitch material for this player is to be played
   back. Default = 1.
 - :microtones-midi-channel. An integer that indicates the MIDI channel on
   which any microtonal pitch material for this player is to be played
   back. slippery chicken uses this channel to add MIDI pitch-bends via CM
   so that microtonal chords are possible, but due to a current glitch these
   tracks contain no pitch-bend data. A work-around for this is to simply
   open the MIDI file in a sequencer and shift the entire channel by the
   desired pitch-bend value. Default = -1 = same as :midi-channel.
 - :cmn-staff-args. A list of pairs that indicate any additional arguments
   to the call to cmn::staff for this player, such as staff size, number of
   lines etc. Instead of being real cmn function calls, as they would be in
   normal cmn, this is a simple list of pairs; e.g. '(staff-size .8
   staff-lines 3). Defaults = NIL.
 - :staff-names. A symbol, string or list of staff names, generally strings,
   for each instrument the player will play. E.g. '("violin II"). If not
   given, then the instrument's name as defined in the instrument palette
   will be used. Default = NIL.
 - :staff-short-names. A symbol, string or list of short staff
   names. E.g. '("vln2"). Default = NIL
```

RETURN VALUE:

```
 Returns a player object.
```

EXAMPLE:

```
;; Create a player object with just one instrument object
(let ((ip (make-instrument-palette
            'inst-pal
            '((picc (:transposition-semitones 12 :lowest-written d4
                     :highest-written c6))
              (flute (:lowest-written c4 :highest-written d7))
              (clar (:transposition-semitones -2 :lowest-written e3
                     :highest-written c6))
              (horn (:transposition f :transposition-semitones -7
                     :lowest-written f2 :highest-written c5))
              (vln (:lowest-written g3 :highest-written c7 :chords t))
              (vla (:lowest-written c3 :highest-written f6 :chords t))))))
  (make-player 'player-one ip 'flute))

=>
PLAYER: (id instrument-palette): INST-PAL
doubles: NIL, cmn-staff-args: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: PLAYER-ONE, tag: NIL,
```

```
data:
INSTRUMENT: lowest-written:
[...]
NAMED-OBJECT: id: FLUTE, tag: NIL,
data: NIL

;; Create a player object with two instruments, setting the midi channels using
;; the keyword arguments, then print the corresponding slots to see the changes
(let* ((ip (make-instrument-palette
            'inst-pal
            '((picc (:transposition-semitones 12 :lowest-written d4
                     :highest-written c6))
              (flute (:lowest-written c4 :highest-written d7))
              (clar (:transposition-semitones -2 :lowest-written e3
                     :highest-written c6))
              (horn (:transposition f :transposition-semitones -7
                     :lowest-written f2 :highest-written c5))
              (vln (:lowest-written g3 :highest-written c7 :chords t))
              (vla (:lowest-written c3 :highest-written f6 :chords t)))))
       (plr (make-player 'player-one ip '(flute picc)
                         :midi-channel 1
                         :microtones-midi-channel 2)))
  (print (loop for i in (data (data plr)) collect (id i)))
  (print (midi-channel plr))
  (print (microtones-midi-channel plr)))

=>
(FLUTE PICC)
1
2

;;; With specified cmn-staff-args
(let ((ip (make-instrument-palette
           'inst-pal
           '((picc (:transposition-semitones 12 :lowest-written d4
                    :highest-written c6))
             (flute (:lowest-written c4 :highest-written d7))
             (clar (:transposition-semitones -2 :lowest-written e3
                    :highest-written c6))
             (horn (:transposition f :transposition-semitones -7
                    :lowest-written f2 :highest-written c5))
             (vln (:lowest-written g3 :highest-written c7 :chords t))
             (vla (:lowest-written c3 :highest-written f6 :chords t))))))
  (make-player 'player-one ip '(flute picc)
               :midi-channel 1
               :microtones-midi-channel 2
```

```
              :cmn-staff-args '(staff-size .8 staff-lines 3)))

=>
PLAYER: (id instrument-palette): INST-PAL
doubles: T, cmn-staff-args: (#<SELF-ACTING {10097B6E73}>
                             #<SELF-ACTING {10097B6EE3}>), total-notes: 0, total-degrees: 0,
total-duration: 0.000, total-bars: 0, tessitura: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: PLAYER-ONE, tag: NIL,
data:
[...]
```

SYNOPSIS:

```
(defun make-player (id instrument-palette instruments
                    &key (cmn-staff-args nil) staff-names staff-short-names
                         (microtones-midi-channel -1) (midi-channel 1))
```

38.6.81 player/microtonal-chords-p

[player] [Methods]

DESCRIPTION:

```
Determines whether the MICROTONES-MIDI-CHANNEL slot of the given player
object is set to a value greater than 0 and is different to the midi-channel
slot, indicating that the player and its instrument are capable of
performing microtonal chords.
```

ARGUMENTS:

```
- A player object.
```

RETURN VALUE:

```
Returns T if the value stored in the MICROTONES-MIDI-CHANNEL slot of the
given player object is greater than 0, otherwise returns NIL.
```

EXAMPLE:

```
;; Returns T
(let* ((ip +slippery-chicken-standard-instrument-palette+)
       (plr (make-player 'vln ip 'violin :microtones-midi-channel 2)))
  (microtonal-chords-p plr))
```

```
=> T

;; Returns NIL
(let* ((ip +slippery-chicken-standard-instrument-palette+)
       (plr (make-player 'pno ip 'piano)))
  (microtonal-chords-p plr))

=> NIL
```

SYNOPSIS:

```
(defmethod microtonal-chords-p ((p player))
  ;; MDE Thu Dec 28 17:49:36 2017 -- updated as now make-player sets microtones
  ;; channel to midi-channel if it's not explicitly set
  (and (integer>0 (microtones-midi-channel p))
       (/= (microtones-midi-channel p)
           (midi-channel p))))

;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;

(defmethod score-write-bar-line ((p player))
  (let* ((data (data p))
         (ins (if (typep data 'assoc-list)
                  (first (data data))
                data)))
    (score-write-bar-line ins)))
```

38.6.82 player/player-get-instrument

[player] [Methods]

DESCRIPTION:

```
Get the instrument object assigned to a single-instrument player object or
get the specified instrument object assigned to a multiple-instrument
player object.

NB: This method will drop into the debugger with an error if no optional
    argument is supplied when applying the method to a multiple-instrument
    player object. It will also print a warning when supplying an optional
    argument to a player object that contains only one instrument object.
```

ARGUMENTS:

```
- A player object.
```

OPTIONAL ARGUMENTS:

```
 - Actually a required object for multiple-instrument player objects: The
   symbol that is the ID of the sought-after instrument object, as it
   appears in the instrument-palette with which the player object which
   made. If the given player object consists of only one instrument object,
   this argument is disregarded and a warning is printed. If this argument
   is simply T then the first instrument is returned (in the assoc-list, not
   the piece, of which we have no knowledge).
```

RETURN VALUE:

```
 Returns an instrument object.
```

EXAMPLE:

```
;; Returns an instrument object. Needs no optional argument when applied to a
;; player object that contains only one instrument object
(let* ((ip +slippery-chicken-standard-instrument-palette+)
       (plr (make-player 'pno ip 'piano)))
   (player-get-instrument plr))

=>
INSTRUMENT:
[...]
NAMED-OBJECT: id: PIANO, tag: NIL,
data: NIL

;; Returns the only existing instrument object and prints a warning if using
;; the optional argument when applying to a single-instrument player object
(let* ((ip +slippery-chicken-standard-instrument-palette+)
       (plr (make-player 'pno ip 'piano)))
  (id (player-get-instrument plr 'piano)))

=> PIANO
WARNING:
   player::player-get-instrument: player PNO has only 1 instrument so optional
argument PIANO is being ignored

;; Asking for a non-existent instrument obect from a single-instrument player
;; object returns the only existing instrument object instead
(let* ((ip +slippery-chicken-standard-instrument-palette+)
       (plr (make-player 'pno ip 'piano)))
  (id (player-get-instrument plr 'marimba)))
```

```
=> PIANO
WARNING:
   player::player-get-instrument: player PNO has only 1 instrument so optional
argument PIANO is being ignored

;; The ID desired instrument object must be specified when applying the method
;; to a multiple-instrument player object
(let* ((ip +slippery-chicken-standard-instrument-palette+)
       (plr (make-player 'percussion ip '(marimba vibraphone))))
  (id (player-get-instrument plr 'marimba)))

=> MARIMBA

;; Interrupts and drops into the debugger when the optional argument is omitted
;; in applying the method to a multiple-instrument player object
(let* ((ip +slippery-chicken-standard-instrument-palette+)
       (plr (make-player 'percussion ip '(marimba vibraphone))))
   (player-get-instrument plr))

=>
player::player-get-instrument: PERCUSSION doubles so you need to pass the ID of
the instrument you want.
   [Condition of type SIMPLE-ERROR]
```

SYNOPSIS:

```
(defmethod player-get-instrument ((p player) &optional ins (warn t))
```

38.6.83 player/plays-transposing-instrument

[player] [Methods]

DESCRIPTION:

```
Determine whether a given player object has one or more transposing
instrument objects assigned to it.
```

ARGUMENTS:

```
- A player object.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether instruments that transpose at the octave are
  to be considered transposing instruments.  T = instruments that transpose
  at the octave are not considered transposing instruments. Default = T.
```

RETURN VALUE:

```
Returns T if one or more of the instrument objects assigned to the given
player object has a transposition value other than C or a
transposition-semitones value other than 0.
```

EXAMPLE:

```
;; Create a player object using the 'b-flat-clarinet instrument object
;; definition from the default +slippery-chicken-standard-instrument-palette+,
;; then apply the method.
(let* ((ip +slippery-chicken-standard-instrument-palette+)
       (plr (make-player 'cl ip 'b-flat-clarinet)))
  (plays-transposing-instrument plr))

=> T

;; Create a player object using the 'flute instrument object definition from
;; the default +slippery-chicken-standard-instrument-palette+, then apply the
;; method.
(let* ((ip +slippery-chicken-standard-instrument-palette+)
       (plr (make-player 'fl ip 'flute)))
  (plays-transposing-instrument plr))

=> NIL

;; Although the intended procedure is to list single instruments as once-off
;; symbols (as in the previous example), single instruments can also be added
;; as a one-item list
(let* ((ip +slippery-chicken-standard-instrument-palette+)
       (plr (make-player 'fl ip '(flute))))
  (doubles plr))

=> NIL

;; Create a player object using a list that consists of the 'flute and
;; 'alto-sax instrument object definitions from the default
;; +slippery-chicken-standard-instrument-palette+, then apply the method to see
;; that it returns T even when only one of the instruments is transposing.
(let* ((ip +slippery-chicken-standard-instrument-palette+)
       (plr (make-player 'fl ip '(flute alto-sax))))
  (plays-transposing-instrument plr))

=> T

;; Setting the optional argument to NIL causes instruments that transpose at
```

```
;; the octave to return T.
(let* ((ip +slippery-chicken-standard-instrument-palette+)
       (plr (make-player 'db ip 'double-bass)))
  (plays-transposing-instrument plr))

=> NIL

(let* ((ip +slippery-chicken-standard-instrument-palette+)
       (plr (make-player 'db ip 'double-bass)))
  (plays-transposing-instrument plr nil))

=> T
```

SYNOPSIS:

```
(defmethod plays-transposing-instrument ((p player)
                                         &optional (ignore-octaves t) ignore)
```

38.6.84 player/reset-instrument-stats

[player] [Methods]

DATE:

23rd August 2013

DESCRIPTION:

Reset the statistics slots for each instrument the player plays.

ARGUMENTS:

- The player object

OPTIONAL ARGUMENTS:

- just-total-duration. If NIL update all statistics slots, otherwise just the total-duration slot of each instrument

RETURN VALUE:

T if the player has instruments, NIL if not.

SYNOPSIS:

```
(defmethod reset-instrument-stats ((p player) &optional just-total-duration)
```

38.6.85 player/tessitura-degree

[player] [Methods]

DESCRIPTION:

```
Return a number that represents the average pitch for a specified
instrument over the course of a piece. The number returned will be degrees
in the current scale.
```

ARGUMENTS:

```
- A player object.
```

RETURN VALUE:

```
A number that is the tessitura-degree; i.e., average pitch of the given
instrument for the entirety of the given musical data.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (va (violin :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :set-palette '((1 ((gs3 as3 b3 cs4 ds4 e4 fs4 gs4 as4 b4 cs5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q (e) s (32) 32))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1))
                            (va (1 1 1 1 1))
                            (vc (1 1 1 1 1)))))))
  (tessitura-degree (get-data 'vc (ensemble mini))))

=> 136
```

SYNOPSIS:

```
(defmethod tessitura-degree ((p player))
```

38.6.86 player/tessitura-note

[player] [Methods]

DESCRIPTION:

Return the value of the TESSITURA-DEGREE slot of a specified player object as a note-name symbol.

ARGUMENTS:

- A player object.

RETURN VALUE:

- A note-name symbol.

EXAMPLE:

```
(in-scale :chromatic)

(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (va (violin :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :set-palette '((1 ((gs3 as3 b3 cs4 ds4 e4 fs4 gs4 as4 b4 cs5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q (e) s (32) 32))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1))
                            (va (1 1 1 1 1))
                            (vc (1 1 1 1 1)))))))
  (tessitura-note (first (data (ensemble mini)))))

=> BF3
```

SYNOPSIS:

```
(defmethod tessitura-note ((p player))
```

38.6.87 player/total-bars

[player] [Methods]

DESCRIPTION:

Return the number of bars in a specified player object.

ARGUMENTS:

```
 - A player object.
```

RETURN VALUE:

```
 - An integer.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (va (violin :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :set-palette '((1 ((gs3 as3 b3 cs4 ds4 e4 fs4 gs4 as4 b4 cs5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q (e) s (32) 32))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1))
                            (va (1 1 1 1 1))
                            (vc (1 1 1 1 1))))))))
  (total-bars (first (data (ensemble mini)))))

=> 5
```

SYNOPSIS:

```
(defmethod total-bars ((p player))
```

38.6.88 player/total-degrees

[player] [Methods]

DESCRIPTION:

```
 Return a number that reflects the mean note (tessitura) of a player's
 part. This is calculated by incrementing the TOTAL-DEGREES slot of the
 corresponding instrument object for each attacked note in the player's part
 by the degree of that note (in the scale of the piece), and then dividing
 the sum by the total number of notes in the player's part.
```

ARGUMENTS:

```
 - A player object.
```

RETURN VALUE:

```
- An integer.
```

EXAMPLE:

```
(in-scale :chromatic)

(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (va (violin :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :set-palette '((1 ((gs3 as3 b3 cs4 ds4 e4 fs4 gs4 as4 b4 cs5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q (e) s (32) 32))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1))
                            (va (1 1 1 1 1))
                            (vc (1 1 1 1 1)))))))
  (total-degrees (first (data (ensemble mini)))))

=> 865
```

SYNOPSIS:

```
(defmethod total-degrees ((p player))
```

38.6.89 player/total-duration

[player] [Methods]

DESCRIPTION:

```
Get the total duration of played notes for a given player over the span of
a piece.
```

ARGUMENTS:

```
- A player object.
```

RETURN VALUE:

```
A number that is the total duration in seconds of played notes.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (va (violin :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :set-palette '((1 ((gs3 as3 b3 cs4 ds4 e4 fs4 gs4 as4 b4 cs5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q (e) s (32) 32))
                                :pitch-seq-palette ((1 2 3))))
                            (2 ((((2 4) (q) e (s) 32 32))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1))
                            (va (2 2 2 2 2))
                            (vc (1 2 1 2 1)))))))
  (print (total-duration (get-data 'vn (ensemble mini))))
  (print (total-duration (get-data 'va (ensemble mini))))
  (print (total-duration (get-data 'vc (ensemble mini)))))

=>
6.875
3.75
5.625
```

SYNOPSIS:

```
(defmethod total-duration ((p player))
```

38.6.90 player/total-notes

[player] [Methods]

DESCRIPTION:

```
Get the total number of notes (actually events) played by a specified
player (not rests or tied notes, but midi-notes) in the piece which this
instrument plays. A chord counts as 1 note/event.
```

ARGUMENTS:

```
- A player object.
```

RETURN VALUE:

```
 - An integer that is the number of notes for that player.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (va (violin :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :set-palette '((1 ((gs3 as3 b3 cs4 ds4 e4 fs4 gs4 as4 b4 cs5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q (e) s (32) 32))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1))
                            (va (1 1 1 1 1))
                            (vc (1 1 1 1 1)))))))
  (print (total-notes (get-data 'vc (ensemble mini)))))

=> 15
```

SYNOPSIS:

```
(defmethod total-notes ((p player))
```

38.6.91 linked-named-object/rhythm

[linked-named-object] [Classes]

NAME:

```
 rhythm

 File:             rhythm.lsp

 Class Hierarchy:  named-object -> linked-named-object -> rhythm

 Version:          1.1.0

 Project:          slippery chicken (algorithmic composition)

 Purpose:          Implementation of the rhythm class for parsing and
                   storing the properties of rhythms.

 Author:           Michael Edwards: m@michael-edwards.org
```

```
Creation date:    11th February 2001

$$ Last modified:  12:39:16 Wed Jun 24 2026 CEST

SVN ID: $Id$
```

38.6.92 rhythm/accented-p

[rhythm] [Methods]

DATE:

```
05 Apr 2011
```

DESCRIPTION:

```
Check the MARKS slot of a given rhythm object to determine if it possesses
an accent mark. The rhythm object may also possess other marks as well.
```

ARGUMENTS:

```
- A rhythm object.
```

RETURN VALUE:

```
If the accent mark ('a) is indeed found in the MARKS slot of the given
rhythm object, the tail of the list of marks contained in that slot is
returned; otherwise NIL is returned.
```

EXAMPLE:

```
;; Make a rhythm object, add an accent, and test for the presence of the accent
(let ((r (make-rhythm 'q)))
  (add-mark-once r 'a)
  (accented-p r))

=> (A)

;; Check if an accent mark is among all marks in the MARKS slot
(let ((r (make-rhythm 'q)))
  (add-mark-once r 's)
  (add-mark-once r 'a)
  (accented-p r))
```

```
=> (A S)

;; Add an accent and staccato, then remove the accent and test for it
(let ((r (make-rhythm 'q)))
  (add-mark-once r 'a)
  (add-mark-once r 's)
  (rm-marks r 'a)
  (accented-p r))

=> NIL
```

SYNOPSIS:

```
(defmethod accented-p ((r rhythm))
```

38.6.93 rhythm/add

[rhythm] [Methods]

DESCRIPTION:

```
Create a new rhythm object with a duration that is equal to the sum of the
duration of two other given rhythm objects.

NB: This method only returns a single rhythm rather than a list with
    ties. Thus q+s, for example, returns TQ...

If the resulting duration cannot be presented as a single, notatable
rhythm, the DATA slot of the resulting rhythm object is set to NIL, though
the VALUE and DURATION slots are still set with the corresponding numeric
values.
```

ARGUMENTS:

```
- A first rhythm object.
- A second rhythm object.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether a warning is printed when a rhythm cannot be
  made because the resulting value is 0 or a negative duration. Default =
  NIL (no warning issued).
```

RETURN VALUE:

A rhythm object. Returns NIL when the object cannot be made.

EXAMPLE:

```
;; A quarter plus an eighth makes a dotted quarter
(let ((r1 (make-rhythm 'q))
      (r2 (make-rhythm 'e)))
  (add r1 r2))

=>
RHYTHM: value: 2.6666666666666665, duration: 1.5, rq: 3/2, is-rest: NIL, score-rthm: 4.0f0.,
        undotted-value: 4, num-flags: 0, num-dots: 1, is-tied-to: NIL,
        is-tied-from: NIL, compound-duration: 1.5, is-grace-note: NIL,
        needs-new-note: T, beam: NIL, bracket: NIL, rqq-note: NIL,
        rqq-info: NIL, marks: NIL, marks-in-part: NIL, letter-value: 4,
        tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: Q., tag: NIL,
data: Q.

;; A quarter plus a triplet-eighth is presented as a triplet-half
(let ((r1 (make-rhythm 'q))
      (r2 (make-rhythm 'te)))
  (data (add r1 r2)))

=> TH

;; A quarter plus a septuplet-16th cannot be represented as a single, notatable
;; rhythm and therefore produces an object with a VALUE and DURATION but no
;; DATA
(let ((r1 (make-rhythm 4))
      (r2 (make-rhythm 28)))
  (print (value (add r1 r2)))
  (print (duration (add r1 r2)))
  (print (data (add r1 r2))))

=>
3.5
1.1428571428571428
NIL
```

SYNOPSIS:

```
(defmethod add ((r1 rhythm) (r2 rhythm) &optional warn)
```

38.6.94 rhythm/add-mark

[rhythm] [Methods]

DESCRIPTION:

```
Add an articulation, dynamic, slur or any other mark to a rhythm (also
useful in the event subclass for changing note heads etc.) Multiple marks
can be added separately and consecutively to the same rhythm object.

A warning is printed if the same mark is added to the same rhythm object
more than once.

NB: This method checks to see if the mark added is a valid mark and will
    warn if it doesn't exist (but it will still add it, in case you have
    your own processing logic for it).
```

ARGUMENTS:

```
- A rhythm object.
- A mark.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicated whether to issue a warning when trying to add marks
  to a rest. Default = NIL.
```

RETURN VALUE:

```
Always T.
```

EXAMPLE:

```
(let ((r (make-rhythm 'q)))
  (marks r))

=> NIL

(let ((r (make-rhythm 'q)))
  (add-mark r 'a))

=> T

(let ((r (make-rhythm 'q)))
  (add-mark r 's)
```

```
  (marks r))

=> (S)

(let ((r (make-rhythm 'q)))
  (add-mark r 'col-legno)
  (add-mark r 'as)
  (add-mark r 'x-head)
  (marks r))

=> (X-HEAD AS COL-LEGNO)

(let ((r (make-rhythm 'q)))
  (add-mark r 's)
  (add-mark r 's))

=> T
WARNING: rhythm::add-mark: S already present but adding again!:

(let ((r (make-rhythm 'e :is-rest t)))
  (add-mark r 'at)
  (print (is-rest r))
  (print (marks r)))

=>
T
(AT)

(let ((r (make-rhythm 'e :is-rest t)))
  (add-mark r 'at t))

=> T
WARNING:
[...]
rhythm::add-mark: add AT to rest?
```

SYNOPSIS:

```
(defmethod add-mark ((r rhythm) mark &optional warn-rest warn-again)
```

38.6.95 rhythm/add-mark-once

[rhythm] [Methods]

DATE:

26 Jul 2011 (Pula)

DESCRIPTION:

Apply the given mark to the given rhythm object, but do so only if the given rhythm object does not yet have the mark.

ARGUMENTS:

- A rhythm object.
- A mark.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether or not to print a warning when attempting to apply a mark to a rest.

RETURN VALUE:

Returns T if the mark is successfully applied (if the rhythm object did not already possess the mark), otherwise NIL if the mark was not applied because the rhythm object already had it.

EXAMPLE:

```
(let ((r (make-rhythm 'q)))
  (add-mark-once r 'a))

=> T

(let ((r (make-rhythm 'q)))
  (add-mark-once r 'a)
  (marks r))

=> (A)

(let ((r (make-rhythm 'q)))
  (add-mark-once r 'a)
  (add-mark-once r 'a))

=> NIL

(let ((r (make-rhythm 'q)))
  (add-mark-once r 'a)
  (add-mark-once r 'a)
```

```
  (marks r))

=> (A)
```

SYNOPSIS:

```
(defmethod add-mark-once ((r rhythm) mark &optional warn-rest)
```

38.6.96 rhythm/begin-slur-p

[rhythm] [Methods]

DESCRIPTION:

Check to see if the MARKS slot of a given rhythm object contains a mark for the beginning of a slur ('beg-sl). The rhythm object may also possess other marks as well.

ARGUMENTS:

- A rhythm object.

RETURN VALUE:

If the 'beg-sl mark is indeed found in the MARKS slot of the given rhythm object, the tail of the list of marks contained in that slot is returned; otherwise NIL is returned.

EXAMPLE:

```
;; Create a rhythm object, add a 'beg-sl mark and check for it
(let ((r (make-rhythm 'q)))
  (add-mark-once r 'beg-sl)
  (begin-slur-p r))

=> (BEG-SL)

;; Add several marks to a rhythm object and check for 'beg-sl
(let ((r (make-rhythm 'q)))
  (loop for m in '(a s beg-sl) do (add-mark-once r m))
  (begin-slur-p r))

=> (BEG-SL S A)

;; Add a 'beg-sl mark to a rhythm object, then delete it and check for it
```

```
(let ((r (make-rhythm 'q)))
  (add-mark-once r 'beg-sl)
  (rm-marks r 'beg-sl)
  (begin-slur-p r))

=> NIL
```

SYNOPSIS:

```
(defmethod begin-slur-p ((r rhythm))
```

38.6.97 rhythm/delete-beam

[rhythm] [Methods]

DESCRIPTION:

```
Removes indication for the start (1) or end (0) of a beam from the BEAM
slot of a given rhythm object, replacing them with NIL.
```

ARGUMENTS:

```
- A rhythm object.
```

RETURN VALUE:

```
Always returns NIL.
```

EXAMPLE:

```
;; Manually set the beam of a rhythm object and delete it to see result NIL
(let ((r (make-rhythm 'e)))
  (setf (beam r) 1)
  (delete-beam r))

=> NIL

;; Make a rthm-seq-bar object with beam indications, then check the BEAM slot
;; of each rhythm object in the rthm-seq-bar object.
(let ((rsb (make-rthm-seq-bar '((2 4) - s s e - q))))
  (loop for r in (rhythms rsb) collect (beam r)))

=> (1 NIL 0 NIL)

;; Make a rthm-seq-bar object with beam indications, delete them all, then
```

```
;; check the beam slot of each rhythm object in the rthm-seq-bar object.
(let ((rsb (make-rthm-seq-bar '((2 4) - s s e - q))))
  (loop for r in (rhythms rsb) do (delete-beam r))
  (loop for r in (rhythms rsb) collect (beam r)))

=> (NIL NIL NIL NIL)
```

SYNOPSIS:

```
(defmethod delete-beam ((r rhythm))
```

38.6.98 rhythm/delete-marks

[rhythm] [Methods]

DESCRIPTION:

```
Delete any marks in the MARKS slot of an event object created within a
rhythm object, replacing the entire list of the MARKS slot with NIL.
```

ARGUMENTS:

```
- A rhythm object.
```

RETURN VALUE:

```
Always returns NIL.
```

EXAMPLE:

```
;; The method returns NIL
(let ((r (make-rhythm (make-event 'c4 'q))))
  (loop for m in '(a s pizz) do (add-mark-once r m))
  (delete-marks r))

=> NIL

;; Create a rhythm object consisting of an event object and print the default
;; contents of the MARKS slot. Set the MARKS slot to contain three marks and
;; print the result. Apply the delete-marks method and print the result.
(let ((r (make-rhythm (make-event 'c4 'q))))
  (print (marks r))
  (loop for m in '(a s pizz) do (add-mark-once r m))
  (print (marks r))
  (delete-marks r)
```

```
  (print (marks r)))

=>
NIL
(PIZZ S A)
NI
```

SYNOPSIS:

```
(defmethod delete-marks ((r rhythm))
```

38.6.99 rhythm/duration-secs

[rhythm] [Methods]

DESCRIPTION:

```
Determine the absolute duration in seconds of a given rhythm object at a
given quarter-note tempo. If no tempo is specified, a tempo of 60 is
assumed.
```

ARGUMENTS:

```
- A rhythm object.
```

OPTIONAL ARGUMENTS:

```
- A numerical tempo value based on quarter-note beats per minute.
```

RETURN VALUE:

```
A real number (floating point) representing the absolute duration of the
given rhythm object in seconds.
```

EXAMPLE:

```
;; Determine the duration in seconds of a quarter note with a default tempo of
;;; quarter = 60
(let ((r (make-rhythm 'q)))
  (duration-secs r))

=> 1.0

;; Specifying a different tempo results in a different duration in seconds
```

```
(let ((r (make-rhythm 'q)))
  (duration-secs r 96))

=> 0.625
```

SYNOPSIS:

```
(defmethod duration-secs ((r rhythm) &optional (tempo 60))
```

38.6.100 rhythm/end-slur-p

[rhythm] [Methods]

DESCRIPTION:

Check to see if the MARKS slot of a given rhythm object contains a mark for the ending of a slur ('end-sl). The rhythm object may also possess other marks as well.

ARGUMENTS:

- A rhythm object.

RETURN VALUE:

If the 'end-sl mark is indeed found in the MARKS slot of the given rhythm object, the tail of the list of marks contained in that slot is returned; otherwise NIL is returned.

EXAMPLE:

```
;; Create a rhythm object, add a 'end-sl mark and check for it
(let ((r (make-rhythm 'q)))
  (add-mark-once r 'end-sl)
  (end-slur-p r))

=> (END-SL)

;; Add several marks to a rhythm object and check for 'end-sl
(let ((r (make-rhythm 'q)))
  (loop for m in '(a s end-sl) do (add-mark-once r m))
  (end-slur-p r))

=> (END-SL S A)
```

```
;; Add an 'end-sl mark to a rhythm object, then delete it and check for it
(let ((r (make-rhythm 'q)))
  (add-mark-once r 'end-sl)
  (rm-marks r 'end-sl)
  (end-slur-p r))

=> NIL
```

SYNOPSIS:

```
(defmethod end-slur-p ((r rhythm))
```

38.6.101 rhythm/end-tie

[rhythm] [Methods]

DATE:

```
September 14th 2018, Heidhausen
```

DESCRIPTION:

```
return T or NIL depending on whether the rhythm/event is the end of a tie
(tied-to but not tied-from)
```

ARGUMENTS:

```
the rhythm/event object
```

RETURN VALUE:

```
T or NIL
```

SYNOPSIS:

```
(defmethod end-tie ((r rhythm))
```

38.6.102 rhythm/event

[rhythm] [Classes]

NAME:

```
event

File:             event.lsp

Class Hierarchy:  named-object -> linked-named-object -> rhythm -> event

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the event class which holds data for
                  the construction of an audible event, be it a midi note,
                  a sample (with corresponding sampling-rate conversion
                  factor) or chord of these types.

                  It is generally assumed that event instances will be
                  created from (copies of) rhythm instances by promotion
                  through the sc-change-class function, hence this class is
                  derived from rhythm.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    March 19th 2001

$$ Last modified:  16:16:47 Wed Jan 21 2026 CET

SVN ID: $Id$
```

38.6.103 event/add-arrow

[event] [Methods]

DATE:

```
25 Jun 2011
```

DESCRIPTION:

```
Adds a start-arrow mark to the given event object and stores text that is
to be attached to the start and end of the given arrow for LilyPond
output. This is a little more complex than the usual mark adding process,
hence this separate method and it not being possible to add arrows to
rthm-seq objects.  Not available for CMN. See also the slippery-chicken
class method add-arrow-to-events
```

```
 NB: A separate end-arrow mark should be attached to the note where the end
     text is to appear.  Use end-arrow for this or (add-mark e 'end-arrow).
```

ARGUMENTS:

```
 - An event object.
 - A start-text string.
 - An end-text string.
```

OPTIONAL ARGUMENTS:

```
 - T or NIL to indicate whether or not to print a warning when trying to
   attach an arrow and accompanying marks to a rest. Default = NIL.
```

RETURN VALUE:

```
 Returns T.
```

EXAMPLE:

```
;; Create an event object and see that the MARKS-BEFORE and MARKS slots are set
;;; to NIL by default
(let ((e (make-event 'c4 'q)))
  (print (marks-before e))
  (print (marks e)))

=>
NIL
NIL

;; Create an event object, apply the add-arrow method, and print the
;; corresponding slots to see the changes.
(let ((e (make-event 'c4 'q)))
  (add-arrow e "start here" "end here")
  (print (marks-before e))
  (print (marks e)))

=>
((ARROW "start here" "end here"))
(START-ARROW)

;; Create an event object that is a rest and apply the add-arrow method with
;; the optional argument set to T to see the warning printed.
(let ((e (make-event nil 'q)))
  (add-arrow e "start here" "end here" t))
```

```
=> T
event::add-arrow: add arrow to rest?
```

SYNOPSIS:

```
(defmethod add-arrow ((e event) start-text end-text &optional warn-rest)
```

38.6.104 event/add-clef

[event] [Methods]

DESCRIPTION:

```
Add a clef indication to the MARKS-BEFORE slot of the given event object.

NB: This method does not check that the clef-name added is indeed a clef,
nor does it check to see if other clefs have already been attached to the
same event object.
```

ARGUMENTS:

```
- An event object.
- A clef name (symbol).
```

OPTIONAL ARGUMENTS:

```
- (Internal "ignore" arguments only; not needed by the user).
```

RETURN VALUE:

```
Returns the contents (list) of the MARKS-BEFORE slot if successful.

Returns NIL if the clef name is already present in the MARKS-BEFORE slot
and is therefore not added.
```

EXAMPLE:

```
;; Successfully adding a clef returns the contents of the MARKS-BEFORE slot
(let ((e (make-event 'c4 'q)))
  (add-clef e 'treble))

=> ((CLEF TREBLE))
```

```
;; Returns NIL if the clef name is already present
(let ((e (make-event 'c4 'q)))
  (add-clef e 'treble)
  (add-clef e 'treble))

=> NIL

;; Add a clef name to the marks-before slot and check that it's there
(let ((e (make-event 'c4 'q)))
  (add-clef e 'bass)
  (marks-before e))

=> ((CLEF BASS))
```

SYNOPSIS:

```
(defmethod add-clef ((e event) clef &optional (delete-others t) ignore1 ignore2)
```

38.6.105 event/add-mark

[event] [Methods]

DATE:

```
March 15th 2011
```

DESCRIPTION:

```
An :after method which in addition to the rhythm class's functionality will:
a) update the event's amplitude slot if we set a dynamic as a mark (note
that the :before method will similarly remove dynamics so we don't get
multiple dynamics attached to a single pitch);
b) as of 26/4/16 attach midi controller messages for pedal up/down if you
attach the marks ped, ped-up, or ped^, also una corda / tre corde
```

ARGUMENTS:

```
- the event object
- the mark (symbol)
```

OPTIONAL ARGUMENTS:

```
T or NIL to indicate whether amplitudes should be set if dynamic marks are
set. Default = T.
```

RETURN VALUE: SYNOPSIS:

```
(defmethod add-mark :after ((e event) mark
                            &optional (update-amplitude t) warn-again)
```

38.6.106 event/add-midi-program-changes

[event] [Methods]

DATE:

```
8th October 2020, Heidhausen
```

DESCRIPTION:

```
Add programme change data to an event.
```

ARGUMENTS:

```
- the event object to add to
- the MIDI channel to add the program change to. This can either by a single
  integer or a list of (usually 2) integers (for microtonal channels)
- the program change. If a number, just use this directly. If a symbol, get
  the program change for the instrument associated with this ID in the given
  instrument-palette
```

OPTIONAL ARGUMENTS:

```
- the instrument-palette object to get the program change numbers from for
  the given instrument if a symbol is passed as the 3rd argument. Default:
  +slippery-chicken-standard-instrument-palette+
```

RETURN VALUE: SYNOPSIS:

```
(proclaim '(special +slippery-chicken-standard-instrument-palette+))

(defmethod add-midi-program-changes
    ((e event) chan pc
     &optional (ins-palette +slippery-chicken-standard-instrument-palette+))
```

38.6.107 event/add-pitches

[event] [Methods]

DESCRIPTION:

Add pitches to a non-rest event. This works whether the event is a single pitch or a chord.

NB This adds to the sounding pitches, not written pitches of transposing instruments. Events attached to transposing instruments will not yet work with this method.

If pitch objects are given, then the midi-channel of the pitch is presumed to be correct before this method is called. Symbols will default to MIDI channel 1.

ARGUMENTS:

- The event object
- &rest: an arbitrary number of pitches: either pitch objects or symbols.

RETURN VALUE:

The same event but with the new pitches added.

EXAMPLE:

```
(let ((e1 (make-event 'c4 'q))
      (e2 (make-event '(c4 e4 g4) 'e)))
  (add-pitches e1 'cs3 'd5)
  (add-pitches e2 'cs2)))
```

SYNOPSIS:

```
(defmethod add-pitches ((e event) &rest pitches)
```

38.6.108 event/add-salzedo-pedal

[event] [Methods]

DESCRIPTION:

Adds a salzedo mark for harp pedalling to the given event object. The pedalling must be given as a salzedo-list (cf. salzedo-p and harp-salzedo-to-tl-set).
Not available for CMN (as of 2023-12-19).

ARGUMENTS:

- An event object.
- A salzedo list (see above).

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether or not to print a warning when trying to
  attach an arrow and accompanying marks to a rest. Default = NIL.
```

RETURN VALUE:

```
Returns T.
```

SYNOPSIS:

```
(defmethod add-salzedo-pedal ((e event) salzedo &optional warn-rest)
```

38.6.109 event/add-trill

[event] [Methods]

DATE:

```
24 Sep 2011
```

DESCRIPTION:

```
Used for adding pitched trills to printed score output. Adds the necessary
values to the MARKS and MARKS-BEFORE slots of a given event object.

NB: The main interface for adding trills by hand is
    slippery-chicken::trill, which is the class-method combination that
    should be accessed for this purpose.

NB: This method will check to see if the specified trill marks are already
    present in the MARKS and MARKS-BEFORE slots. If they are, the method
    will print a warning but will add the specified trill marks anyway.
```

ARGUMENTS:

```
- An event object.
- A pitch-symbol for the trill note.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether or not to print a warning when attaching
  trill information to a rest. Default = NIL.
```

RETURN VALUE:

```
 Always returns T.

 NB: At the moment the method will also print the reminder warning that this
 is a LilyPond-only mark.
```

EXAMPLE:

```
;; Create an event object and print the contents of the MARKS-BEFORE and MARKS
;; slots to see that they're empty by default.
(let ((e (make-event 'c4 'q)))
  (print (marks-before e))
  (print (marks e)))

=>
NIL
NIL

;; Create an event object, add a trill to the note 'D4, and print the
;; corresponding slots to see the changes
(let ((e (make-event 'c4 'q)))
  (add-trill e 'd4)
  (print (marks-before e))
  (print (marks e)))

=>
WARNING:
rhythm::validate-mark: no CMN mark for BEG-TRILL-A (but adding anyway).

(BEG-TRILL-A)
((TRILL-NOTE D4))

;; By default the method adds prints no warning when adding a mark to a rest
;; (though it still prints the warning that there is no CMN mark)
(let ((e (make-event nil 'q)))
  (add-trill e 'd4)
  (print (marks-before e))
  (print (marks e)))

=>
WARNING:
rhythm::validate-mark: no CMN mark for BEG-TRILL-A (but adding anyway).

(BEG-TRILL-A)
((TRILL-NOTE D4))

;; Set the optional argument to T to have the method print a warning when
```

```
;; attaching a mark to a rest
(let ((e (make-event nil 'q)))
  (add-trill e 'd4 t)
  (print (marks-before e))
  (print (marks e)))

=>
event::add-trill: add trill to rest?
WARNING:
rhythm::validate-mark: no CMN mark for BEG-TRILL-A (but adding anyway).

(BEG-TRILL-A)
((TRILL-NOTE D4))

;; Adding a trill that is already there will result in a warning being printed
;; but will add the mark anyway
(let ((e (make-event 'c4 'q)))
  (loop repeat 4 do (add-trill e 'd4))
  (print (marks-before e))
  (print (marks e)))

=>
WARNING:
   rhythm::add-mark: (TRILL-NOTE D4) already present but adding again!:
[...]
(BEG-TRILL-A BEG-TRILL-A BEG-TRILL-A BEG-TRILL-A)
((TRILL-NOTE D4) (TRILL-NOTE D4) (TRILL-NOTE D4) (TRILL-NOTE D4))
```

SYNOPSIS:

```
(defmethod add-trill ((e event) trill-note &optional warn-rest)
```

38.6.110 event/chord-to-single-pitch

[event] [Methods]

DATE:

```
January 21st 2023, Heidhausen
```

DESCRIPTION:

```
Any events that are chords will be convered to single-pitch events, using
the highest pitch in the chord. Rests and single-pitch events will be left
unchanged.
```

ARGUMENTS:

```
- the event object
```

OPTIONAL ARGUMENTS:

```
- the accessor function. E.g. #'lowest. If you want to access other members
of the chord object's data slot (where the pitches live) then e.g.
#'(lambda (x) (third (data x))) Default = #'highest.
```

RETURN VALUE:

```
the modified event
```

SYNOPSIS:

```
(defmethod chord-to-single-pitch ((e event) &optional (which #'highest))
```

38.6.111 event/common-notes

[event] [Methods]

DESCRIPTION:

```
Return the number of pitches common to two events. Note that the chord
method returns more information, including the common pitches and pitch
symbols.
```

ARGUMENTS:

```
- first event object
- second event object
```

OPTIONAL ARGUMENTS:

```
- T or NIL to treat enharmonic equivalents as equal
- T or NIL to treat the same pitches in different octaves as equal
```

RETURN VALUE:

```
the number of common pitches (integer). 2nd (and 3rd) returned values will
be the list of common pitch(es), also as a 3rd value: a list of symbols if
both events are chords).
```

SYNOPSIS:

```
(defmethod common-notes ((e1 event) (e2 event)
                         &optional (enharmonics-are-equal t)
                           (octaves-are-true nil))
```

38.6.112 event/delete-clefs

[event] [Methods]

DESCRIPTION:

Delete any clef names found in the MARKS-BEFORE slot of a given event object.

ARGUMENTS:

- An event object.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether or not to print a warning when there are no clef marks to delete.
- (Other internal "ignore" arguments only; not needed by the user).

RETURN VALUE:

Always NIL.

EXAMPLE:

```
;; Returns NIL when no clef marks are found to delete, and prints a warning by
;; default.
(let ((e (make-event 'c4 'q)))
  (delete-clefs e))

=> NIL
WARNING: event::delete-clefs: no clef to delete:
[...]

;; Setting the optional WARN argument to T suppresses the warning when no clefs
;; are found.
(let ((e (make-event 'c4 'q)))
  (delete-clefs e nil))

=> NIL
```

```
;; Also returns NIL when successful
(let ((e (make-event 'c4 'q)))
  (add-clef e 'treble)
  (delete-clefs e))

=> NIL

;; Create an event, add a clef, print the MARKS-BEFORE slot, delete the event,
;; print MARKS-BEFORE again to make sure it's gone
(let ((e (make-event 'c4 'q)))
  (add-clef e 'treble)
  (print (marks-before e))
  (delete-clefs e)
  (print (marks-before e)))

=>
((CLEF TREBLE))
NIL
```

SYNOPSIS:

```
(defmethod delete-clefs ((e event) &optional (warn t) ignore1 ignore2)
```

38.6.113 event/delete-written

[event] [Methods]

DESCRIPTION:

```
Delete the contents of the WRITTEN-PITCH-OR-CHORD slot of a given event
object and reset to NIL.
```

ARGUMENTS:

```
- An event object.
```

RETURN VALUE:

```
Always returns NIL.
```

EXAMPLE:

```
;; Create an event object, print the contents of the written-pitch-or-chord ;
;; slot to see it's set to NIL, set-written to -2, print the contents of the ;
;; corresponding slot to see the data of the newly created pitch object, ;
```

```
;; delete-written, print the contents of the written-pitch-or-chord slot to see ;
;; it's empty.                          ;
(let ((e (make-event 'c4 'q)))
  (print (written-pitch-or-chord e))
  (set-written e -2)
  (print (data (written-pitch-or-chord e)))
  (delete-written e)
  (print (written-pitch-or-chord e)))

=>
NIL
BF3
NIL
```

SYNOPSIS:

```
(defmethod delete-written ((e event))
```

38.6.114 event/end-arrow

[event] [Methods]

DESCRIPTION:

```
 Adds an end-arrow mark to the given event object.

 NB: This method works for LilyPond only. When used with CMN, output will
     still be generated, but no mark will be added. The method prints a
     corresponding warning when applied.
```

ARGUMENTS:

```
 - An event object.
```

RETURN VALUE:

```
 Returns T.
```

EXAMPLE:

```
;; Returns T
(let ((e (make-event 'c4 'q)))
  (end-arrow e))

=> T
```

```
WARNING:
rhythm::validate-mark: no CMN mark for END-ARROW (but adding anyway).

;; Create an event object, add end-arrow, and print the MARKS and MARKS-SLOTS
;; to see the result
(let ((e (make-event 'c4 'q)))
  (end-arrow e)
  (print (marks-before e))
  (print (marks e)))

=>
NIL
(END-ARROW)
```

SYNOPSIS:

```
(defmethod end-arrow ((e event))
```

38.6.115 event/end-trill

[event] [Methods]

DATE:

```
24 Sep 2011
```

DESCRIPTION:

```
Adds an 'end-trill-a mark to the MARKS slot of the given event object.
```

ARGUMENTS:

```
- An event object.
```

RETURN VALUE:

```
T
```

EXAMPLE:

```
;; The end-trill method returns T
(let ((e (make-event 'c4 'q)))
  (end-trill e))
```

```
=> T

;; Add an 'end-trill-a and check the MARKS slot to see that it's there
(let ((e (make-event 'c4 'q)))
  (end-trill e)
  (marks e))

=> (END-TRILL-A)
```

SYNOPSIS:

```
(defmethod end-trill ((e event))
```

38.6.116 event/enharmonic

[event] [Methods]

DESCRIPTION:

```
Change the pitch of the pitch object within the given event object to its
enharmonic equivalent.

In its default form, this method only applies to note names that already
contain an indication for an accidental (such as DF4 or BS3), while
"white-key" note names (such as B3 or C4) will not produce an enharmonic
equivalent. In order to change white-key pitches to their enharmonic
equivalents, set the :force-naturals argument to T.

NB: Doesn't work on chords.
```

ARGUMENTS:

```
- An event object.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :written. T or NIL to indicate whether the test is to handle the written
  or sounding pitch in the event. T = written. Default = NIL.
- :force-naturals. T or NIL to indicate whether to force "natural" note
  names that contain no F or S in their name to convert to their enharmonic
  equivalent (ie, B3 = CF4). NB double-flats/sharps are not implemented so
  this will only work on F/E  B/C.
```

RETURN VALUE:

An event object.

EXAMPLE:

```
;; The method alone returns an event object
(let ((e (make-event 'cs4 'q)))
  (enharmonic e))

=>
EVENT: start-time: NIL, end-time: NIL,
[...]

;; Create an event, change it's note to the enharmonic equivalent, and print
;; it.
(let ((e (make-event 'cs4 'q)))
  (enharmonic e)
  (data (pitch-or-chord e)))

=> DF4

;; Without the :force-naturals keyword, no "white-key" note names convert to
;; enharmonic equivalents
(let ((e (make-event 'b3 'q)))
  (enharmonic e)
  (data (pitch-or-chord e)))

=> B3

;; Set the :force-naturals keyword argument to T to enable switching white-key
;; note-names to enharmonic equivalents
(let ((e (make-event 'b3 'q)))
  (enharmonic e :force-naturals t)
  (data (pitch-or-chord e)))

=> CF4
```

SYNOPSIS:

```
(defmethod enharmonic ((e event) &key written force-naturals
                       ;; 1-based
                       chord-note-ref)
```

38.6.117 event/event-distance

[event] [Methods]

DESCRIPTION:

Get the distance (interval) in semitones between the pitch level of one event object and a second. Negative numbers indicate direction interval directionality. An optional argument allows distances to be always printed as absolute values (positive).

Event-distance can also be determined between chords, in which case the distance is measured between the highest pitch of one event object and the lowest of the other.

ARGUMENTS:

- A first event object.
- A second event object.

OPTIONAL ARGUMENTS:

- T or NIL for whether the value should be returned as an absolute value (i.e., always positive). Default = NIL.

RETURN VALUE:

A number.

EXAMPLE:

```
;; The semitone distance between two single pitches in ascending direction ;
                    (let ((e1 (make-event 'c4 'q))
(e2 (make-event 'e4 'q)))
(event-distance e1 e2))

                    => 4.0

;; The semitone distance between two single pitches in descending direction ;
                    (let ((e1 (make-event 'c4 'q))
(e2 (make-event 'e4 'q)))
(event-distance e2 e1))

                    => -4.0

;; Set the optional argument to T to get the absolute distance (positive ;
;; number)                              ;
                    (let ((e1 (make-event 'c4 'q))
(e2 (make-event 'e4 'q)))
```

```
(event-distance e2 e1 t))

                    => 4.0

;; The semitone distance between two chords in ascending direction ;
                    (let ((e1 (make-event '(c4 e4 g4) 'q))
(e2 (make-event '(d4 f4 a4) 'q)))
(event-distance e1 e2))

                    => 9.0
```

SYNOPSIS:

```
(defmethod event-distance ((e1 event) (e2 event) &optional absolute)
```

38.6.118 event/event-p

[event] [Functions]

DESCRIPTION:

Test to confirm that a given object is an event object.

ARGUMENTS:

- An object.

RETURN VALUE:

T if the tested object is indeed an event object, otherwise NIL.

EXAMPLE:

```
;; Create an event and then test whether it is an event object
(let ((e (make-event 'c4 'q)))
  (event-p e))
=> T

;; Create a non-event object and test whether it is an event object
(let ((e (make-rhythm 4)))
  (event-p e))
=> NIL

;; The make-rest function also creates an event
(let ((e (make-rest 4)))
```

```
  (event-p e))
=> T

;; The make-punctuation-events, make-events and make-events2 functions create
;; lists of events, not events themselves.
(let ((e (make-events '((g4 q) e s))))
  (event-p e))
=> NIL
```

SYNOPSIS:

```
(defun event-p (thing)
```

38.6.119 event/flat-p

[event] [Methods]

DESCRIPTION:

```
Determine whether the pitch of a given event object has a flat.
```

ARGUMENTS:

```
- An event object.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether the test is to handle the written or
  sounding pitch in the event. T = written. Default = NIL.
```

RETURN VALUE:

```
Returns T if the note tested has a flat, otherwise NIL (ie, is natural or
has a sharp).
```

EXAMPLE:

```
;; Returns T when the note is flat
(let ((e (make-event 'df4 'q)))
  (flat-p e))

=> T

;; Returns NIL when the note is not flat (ie, is sharp or natural)
```

```
(let ((e (make-event 'c4 'q)))
  (flat-p e))

=> NIL

(let ((e (make-event 'cs4 'q)))
  (flat-p e))

=> NIL
```

SYNOPSIS:

```
(defmethod flat-p ((e event) &optional written)
```

38.6.120 event/flat-to-sharp

[event] [Methods]

DATE:

```
Mon 10 Feb 2020 18:41:40 GMT - London
```

DESCRIPTION:

```
Change the flat (or flats) in an event to a sharp or (sharps).
```

ARGUMENTS:

```
An event object
```

OPTIONAL ARGUMENTS:

```
- clone: clone the evnet. DEFAULT = nil
- written: apply to written or sounding pitch? DEFAULT = nil (sounding)
```

RETURN VALUE:

```
An event object
```

EXAMPLE:

```
(let ((e (make-event 'bf4 'q)))
      (print-simple (flat-to-sharp e)))
```

```
=> AS4 Q,
NIL

(let ((e (make-event '(as4 bf4) 'q)))
      (print-simple (flat-to-sharp e)))

=> (AS4 AS4) Q,
NIL

;;; NB This method is non-destructive
(let ((e (make-event 'bf4 'q)))
      (print-simple (flat-to-sharp e))
      (print-simple e))

=> AS4 Q,
BF4 Q,
NIL
```

SYNOPSIS:

```
(defmethod flat-to-sharp ((e event) &optional clone written)
```

38.6.121 event/force-artificial-harmonic

[event] [Methods]

DESCRIPTION:

```
Change the pitch-or-chord content of a given event object such that the
existing pitch will be notated as an artificial harmonic.

The method creates pitch data for an artificial harmonic that will result
in the specified pitch, rather than adding an artificial harmonic to the
specified pitch. Thus, the method changes the existing pitch content by
transposing the specified pitch down two octaves and adding a new pitch one
perfect fourth above it (changing the given pitch object to a chord
object). It then also adds the mark 'flag-head to the MARKS slot of the
upper pitch for printing layout so that the upper pitch is printed as a
diamond.
```

ARGUMENTS:

```
- An event object.
```

OPTIONAL ARGUMENTS:

- an instrument object

RETURN VALUE:

The chord object which creates the harmonic.

EXAMPLE:

```
;; The method returns NIL.
(let ((e (make-event 'c7 'q)))
  (force-artificial-harmonic e))
=> NIL

;; Create an event object, apply force-artificial-harmonic, then get the new
;; pitch material
(let ((e (make-event 'c7 'q)))
  (force-artificial-harmonic e)
  (loop for p in (data (pitch-or-chord e)) collect (data p)))
=> (C5 F5)

;; Create an event object, apply force-artificial-harmonic, then get the marks
;; attached to each note in the object to see the 'flag-head
(let ((e (make-event 'c7 'q)))
  (force-artificial-harmonic e)
  (loop for p in (data (pitch-or-chord e)) collect (marks p)))
=> (NIL (FLAG-HEAD))
```

SYNOPSIS:

```
(defmethod force-artificial-harmonic ((e event)
                                      &optional instrument (warn t)
                                        (naturals t) ignore)
```

38.6.122 event/force-rest

[event] [Methods]

DESCRIPTION:

Changes a given event object to a rest by setting both the PITCH-OR-CHORD and WRITTEN-PITCH-OR-CHORD slots to NIL and the IS-REST slot to T.

ARGUMENTS:

- An event object.

RETURN VALUE:

```
- An event object
```

EXAMPLE:

```
;; The method returns an event object.  ;
                    (let ((e (make-event 'c4 'q)))
(force-rest e))

                    =>
                    EVENT: start-time: NIL, end-time: NIL,
                    [...]

;; Create an event object, apply force-rest, then print the corresponding slots ;
;; to see the effectiveness of the method ;
                    (let ((e (make-event 'c4 'q)))
(force-rest e)
(print (pitch-or-chord e))
(print (written-pitch-or-chord e))
(print (is-rest e)))

                    =>
                    NIL
                    NIL
                    T
```

SYNOPSIS:

```
(defmethod force-rest :after ((e event))
```

38.6.123 event/get-amplitude

[event] [Methods]

DESCRIPTION:

```
Return the amplitude attached to a given event object.

An optional argument allows the amplitude to be converted to and returned
as a MIDI value.
```

ARGUMENTS:

```
- An event object.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether the amplitude value is to be returned as a
  standard digital amplitude (a number between 0.0 and 1.0) or as a
  standard MIDI velocity value (a whole number between 0 and 127). T = MIDI
  value. Default = NIL.
```

RETURN VALUE:

```
If the optional argument is set to NIL, returns a real number.

If the optional argument is set to T, returns a whole number (and a
remainder).
```

EXAMPLE:

```
;; Get the amplitude as a decimal value. (Each new event object has a default
;; amplitude of 0.7).
(let ((e (make-event 'c4 'q)))
  (get-amplitude e))

=> 0.7

;; Get the amplitude as a rounded MIDI value.
(let ((e (make-event 'c4 'q)))
  (get-amplitude e t))

=> 89, -0.10000000000000853
```

SYNOPSIS:

```
(defmethod get-amplitude ((e event) &optional (midi nil))
```

38.6.124 event/get-clef

[event] [Methods]

DESCRIPTION:

```
Return the symbol associated with the key CLEF in the MARKS-BEFORE slot of
the given event object.
```

ARGUMENTS:

```
- An event object.
```

OPTIONAL ARGUMENTS:

```
 - (Internal "ignore" arguments only; not needed by the user).
```

RETURN VALUE:

```
 Returns the given clef name as a symbol if successful.

 Returns NIL if there is no clef name found in the MARKS-BEFORE slot of the
 given event object.
```

EXAMPLE:

```
;; Returns NIL when no clef is found
(let ((e (make-event 'c4 'q)))
  (get-clef e))

=> NIL

;; Returns the clef name as symbol when successful.
(let ((e (make-event 'c4 'q)))
  (add-clef e 'treble)
  (get-clef e))

=> TREBLE
```

SYNOPSIS:

```
(defmethod get-clef ((e event) &optional ignore1 ignore2 ignore3)
```

38.6.125 event/get-degree

[event] [Methods]

DESCRIPTION:

```
 Get the degree of the event, summing or averaging for chords.
```

ARGUMENTS:

```
 - an event object
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :written. T or NIL to indicate whether to use the written (in the case of
  transposing instruments) or sounding pitches. T = written. Default = NIL.
- :sum. T or NIL to indicate whether to return the sum of the degrees
  instead of a list (see below). T = degrees. Default = NIL.
- :average: sim. to sum but T or NIL to indicate whether to return the
   average instead of sum.
```

RETURN VALUE:

```
By default this returns a list (even if it's a single pitch), unless :sum T
whereupon it will return a single value: the sum of the degrees if a chord,
otherwise just the degree. :average T will return the average of the sum.
A rest would return '(0) or 0.
```

EXAMPLE:

```
;;; NB This uses the quarter-tone scale so degrees are double what they would ;
;;;    be in the chromatic-scale.       ;
(let ((event (make-event '(cs4 d4) 'e))
      (rest (make-rest 'e)))
  (print (get-degree event))
  (print (get-degree rest))
  (print (get-degree event :average t))
  (get-degree event :sum t))
(122 124)
(0)
123.0
246
```

SYNOPSIS:

```
(defmethod get-degree ((e event) &key written sum average)
```

38.6.126 event/get-dynamic

[event] [Methods]

DESCRIPTION:

```
Gets the dynamic marking attached to a given event object.

NB: This method is similar to the event::get-dynamics method, but assumes
that there is only one dynamic and returns that dynamic as a single symbol
rather than a list. If the user suspects that multiple dynamics may have
```

somehow have been added to the MARKS slot of the event class, use
get-dynamics to obtain a list of all dynamics in that slot; otherwise, this
is the method that should be generally used.

ARGUMENTS:

- An event object.

RETURN VALUE:

The symbol representing the dynamic if there is one attached to that event,
otherwise NIL.

EXAMPLE:

```
;; The method returns just the dynamic marking from the MARKS list, as a symbol
(let ((e (make-event 'c4 'q)))
  (add-mark-once e 'ppp)
  (add-mark-once e 'pizz)
  (get-dynamic e))

=> PPP

;; The method returns NIL if there is no dynamic in the MARKS list
(let ((e (make-event 'c4 'q)))
  (add-mark-once e 'pizz)
  (get-dynamic e))

=> NIL
```

SYNOPSIS:

```
(defmethod get-dynamic ((e event))
```

38.6.127 event/get-dynamics

[event] [Methods]

DESCRIPTION:

Get the list of dynamic marks from a given event object, assuming there are
multiple dynamics present. If other non-dynamic events are also contained
in the MARKS slot of the rhythm object within the given event object, these
are disregarded and only the dynamic marks are returned.

```
NB: This method is similar to the event::get-dynamic, but is intended for
    use should multiple dynamics have somehow become attached to the same
    event. The method event::get-dynamic is the method that should
    generally be used.
```

ARGUMENTS:

```
- An event object.
```

RETURN VALUE:

```
A list containing the dynamics stored in the MARKS slot of the rhythm
object within the given event object. NIL is returned if no dynamic marks
are attached to the given event object.
```

EXAMPLE:

```
;; Create an event object and get the dynamics attached to that object. These
;; are NIL by default (unless otherwise specified).
(let ((e (make-event 'c4 'q)))
  (get-dynamics e))

=> NIL

;; Create an event object, add one dynamic and one non-dynamic mark, print all
;; marks, then retrieve only the dynamics.
(let ((e (make-event 'c4 'q)))
  (add-mark-once e 'ppp)
  (add-mark-once e 'pizz)
  (print (marks e))
  (get-dynamics e))

=>
(PIZZ PPP)
(PPP)

;; Should multiple dynamics have become attached to the same event object,
;; get-dynamics will return all dynamics present in the MARKS slot
(let ((e (make-event 'c4 'q)))
  (add-mark-once e 'pizz)
  (add-mark-once e 'ppp)
  (push 'fff (marks e))
  (print (marks e))
  (get-dynamics e))

=> (FFF PPP)
```

SYNOPSIS:

```
(defmethod get-dynamics ((e event) &optional hairpins)
```

38.6.128 event/get-frequency

[event] [Methods]

DATE:

```
May 26th 2016, Edinburgh
```

DESCRIPTION:

```
Return the frequency of the event's pitch, or a list of frequencies if it's
a chord.
```

ARGUMENTS:

```
- the event object
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :written. T or NIL to indicate whether to use the written (in the case of
  transposing instruments) or sounding pitches. T = written. Default = NIL.
- :force-list. T or NIL to force a single pitch's frequency into a
  list. Default = NIL.
- :average. T or NIL to make chords average their frequency.
```

RETURN VALUE:

```
A single frequency (float) or list of frequencies in the case of a chord.
```

SYNOPSIS:

```
(defmethod get-frequency ((e event) &key written force-list average)
```

38.6.129 event/get-midi-channel

[event] [Methods]

DESCRIPTION:

Retrieve the value set for the midi-channel slot of the pitch object within a given event object. Beware that if the pitch-or-chord slots is a chord, we will merely return the channel of the first pitch in the chord.

ARGUMENTS:

- An event object.

RETURN VALUE:

An integer representing the given midi-channel value or NIL if there isn't one.

EXAMPLE:

```
;; The default midi-channel value for a newly created event-object is NIL
;;; unless otherwise specified.
(let ((e (make-event 'c4 'q)))
  (get-midi-channel e))

=> NIL

;; Create an event object, set its MIDI-channel and retrieve it
(let ((e (make-event 'c4 'q)))
  (set-midi-channel e 11 12)
  (get-midi-channel e))

=> 11
```

SYNOPSIS:

```
(defmethod get-midi-channel ((e event))
```

38.6.130 event/get-pitch-symbol

[event] [Methods]

DESCRIPTION:

Retrieve the pitch symbol (CM/CMN note-name notation) of a given event object. Returns a single symbol if the given event object consists of a single pitch; otherwise, returns a list of pitch symbols if the given event object consists of a chord.

ARGUMENTS:

```
 - An event object.
```

OPTIONAL ARGUMENTS:

```
 - T or NIL to indicate whether the test is to handle the event's
   written or sounding pitch. T = written. Default = T.
```

RETURN VALUE:

```
 A symbol, if the event object consists of only a single pitch, otherwise a
 list of pitch symbols if the event object consists of a chord.
```

EXAMPLE:

```
;; Get the pitch symbol of an event object with a single pitch
(let ((e (make-event 'c4 'q)))
  (get-pitch-symbol e))

=> C4

;; Getting the pitch symbol of an event object that consists of a chord returns
;; a list of pitch symbols
(let ((e (make-event '(c4 e4 g4) 'q)))
  (get-pitch-symbol e))

=> (C4 E4 G4)
```

SYNOPSIS:

```
(defmethod get-pitch-symbol ((e event) &optional (written t))
```

38.6.131 event/has-hairpin

[event] [Methods]

DESCRIPTION:

```
 Return T or NIL depending on whether the event has a beginning or ending
 hairpin (cresc. or dim. line).
```

ARGUMENTS:

```
 - an event object
```

RETURN VALUE:

```
The hairpin mark the event has, as a list, or NIL if it has none.
```

SYNOPSIS:

```
(defmethod has-hairpin ((e event))
```

38.6.132 event/has-mark-before

[event] [Methods]

DESCRIPTION:

```
Determine whether a specifed event object has a specified mark in its
MARKS-BEFORE slot.
```

ARGUMENTS:

```
- An event object.
- A mark.
```

RETURN VALUE:

```
Returns the specified mark if the mark exists in the MARKS-BEFORE slot,
otherwise returns NIL
```

EXAMPLE:

```
;;; Returns the specified mark if that mark is present
(let ((e (make-event 'c4 4)))
  (add-mark-before e 'ppp)
  (has-mark-before e 'ppp))

=> (PPP)

;;; Returns NIL if the specified mark is not present
(let ((e (make-event 'c4 4)))
  (add-mark-before e 'ppp)
  (has-mark-before e 'fff))

=> NIL
```

SYNOPSIS:

```
(defmethod has-mark-before ((e event) mark &optional (test #'equal))
```

38.6.133 event/highest

[event] [Methods]

DESCRIPTION:

Get the highest pitch (of a chord) in a given event object. If the given event object contains a single pitch only, that pitch is returned.

ARGUMENTS:

- An event object.

RETURN VALUE:

A pitch object.

EXAMPLE:

```
;; Returns a pitch object
(let ((e (make-event 'c4 'q)))
  (highest e))

=>
PITCH: frequency: 261.6255569458008, midi-note: 60, midi-channel: NIL
pitch-bend: 0.0
degree: 120, data-consistent: T, white-note: C4
nearest-chromatic: C4
src: 1.0, src-ref-pitch: C4, score-note: C4
qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
micro-tone: NIL,
sharp: NIL, flat: NIL, natural: T,
octave: 4, c5ths: 0, no-8ve: C, no-8ve-no-acc: C
show-accidental: T, white-degree: 28,
accidental: N,
accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: C4, tag: NIL,
data: C4

;; Returns the highest note of a chord object within an event object ;
                     (let ((e (make-event '(d4 fs4 a4) 'q)))
(data (highest e)))

=> A4
```

SYNOPSIS:

```
(defmethod highest ((e event))
```

38.6.134 event/inc-duration

[event] [Methods]

DESCRIPTION:

```
Increase the duration-in-tempo of a given event object by a specified time
in seconds. This will also result in a new value for the end-time and
compound-duration-in-tempo slots.

NB: Changing this value directly could result in incorrect timing info in a
    bar so use with caution (e.g. via make-hammer-friendly)) or use
setf/incf the duration / compound-duration slots directly.
```

ARGUMENTS:

```
- An event object.
- A value that is the increment in seconds by which the duration is to be
  extended.
```

RETURN VALUE:

```
The new duration in seconds.
```

EXAMPLE:

```
;;; Create a slippery-chicken object, assign a variable to one of the event
;;; objects it contains, print the corresponding duration slots; apply
;;; inc-duration and print the corresponding duration slots again to see the
;;; change.
(let* ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vc (cello :midi-channel 1))))
        :set-palette '((1 ((gs3 as3 b3))))
        :set-map '((1 (1)))
        :rthm-seq-palette '((1 ((((2 4) q (e) s (32) 32))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((vc (1)))))))
      (e (get-event mini 1 3 'vc)))
  (print (end-time e))
```

```
  (print (duration-in-tempo e))
  (print (compound-duration-in-tempo e))
  (inc-duration e 7.0)
  (print (end-time e))
  (print (duration-in-tempo e))
  (print (compound-duration-in-tempo e)))

=>
1.75
0.25
0.25
8.75
7.25
7.25
```

SYNOPSIS:

```
(defmethod inc-duration ((e event) inc)
```

38.6.135 event/is-chord

[event] [Methods]

DESCRIPTION:

```
Test to determine whether a given event object consists of a chord (as
opposed to a single pitch or a rest).
```

ARGUMENTS:

```
- An event object.
```

RETURN VALUE:

```
- If the given event object is a chord, the method returns a number that is
  the number of notes in the chord.
- Returns NIL if the given event object is not a chord.
```

EXAMPLE:

```
;; Returns NIL if not a chord
(let ((e (make-event 'c4 'q)))
  (is-chord e))

=> NIL
```

```
;; If a chord, returns the number of notes in the chord ;
(let ((e (make-event '(c4 e4 g4) 'q)))
  (is-chord e))

=> 3

;; A rest is not a chord
(let ((e (make-rest 'q)))
  (is-chord e))

=> NIL
```

SYNOPSIS:

```
(defmethod is-chord ((e event))
```

38.6.136 event/is-dynamic

[event] [Functions]

DESCRIPTION:

Determine whether a specified symbol belongs to the list of predefined dynamic marks.

ARGUMENTS:

- A symbol.

RETURN VALUE:

NIL if the specified mark is not found on the predefined list of possible dynamic marks, otherwise the tail of the list of possible dynamics starting with the given dynamic.

EXAMPLE:

```
(is-dynamic 'pizz)
=> NIL

(is-dynamic 'f)
=> (F FF FFF FFFF)
```

SYNOPSIS:

```
(defun is-dynamic (mark)
```

38.6.137 event/is-single-pitch

[event] [Methods]

DESCRIPTION:

```
Test to see if an event object consists of a single pitch (as opposed to a
chord or a rest).
```

ARGUMENTS:

```
- An event object.
```

RETURN VALUE:

```
Returns T if the given event object consists of a single pitch, otherwise
returns NIL.
```

EXAMPLE:

```
;; Returns T if the event object consists of a single pitch ;
(let ((e (make-event 'c4 'q)))
  (is-single-pitch e))

=> T

;; Returns NIL if the event object is a chord ;
(let ((e (make-event '(c4 e4 g4) 'q)))
  (is-single-pitch e))

=> NIL

;; Also returns NIL if the event object is a rest ;
(let ((e (make-rest 'q)))
  (is-single-pitch e))

=> NIL
```

SYNOPSIS:

```
(defmethod is-single-pitch ((e event))
```

38.6.138 event/lowest

[event] [Methods]

DESCRIPTION:

Get the lowest pitch (of a chord) in a given event object. If the given event object contains a single pitch only, that pitch is returned.

ARGUMENTS:

- An event object.

RETURN VALUE:

A pitch object.

EXAMPLE:

```
;; Returns a pitch object               ;
(let ((e (make-event 'c4 'q)))
  (lowest e))

=>
PITCH: frequency: 261.6255569458008, midi-note: 60, midi-channel: NIL
pitch-bend: 0.0
degree: 120, data-consistent: T, white-note: C4
nearest-chromatic: C4
src: 1.0, src-ref-pitch: C4, score-note: C4
qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
micro-tone: NIL,
sharp: NIL, flat: NIL, natural: T,
octave: 4, c5ths: 0, no-8ve: C, no-8ve-no-acc: C
show-accidental: T, white-degree: 28,
accidental: N,
accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: C4, tag: NIL,
data: C4

;; Returns the lowest note of a chord object within an event object ;
(let ((e (make-event '(d4 fs4 a4) 'q)))
  (data (lowest e)))

=> D4
```

SYNOPSIS:

```
(defmethod lowest ((e event))
```

38.6.139 event/make-event

[event] [Functions]

DESCRIPTION:

Create an event object for holding rhythm, pitch, and timing data.

ARGUMENTS:

- A pitch or chord. This can be one of those objects (will be added to the pitch-or-chord slot without cloning), or a pitch symbol or list of pitch symbols (for a chord).
- The event's rhythm (e.g. 'e). If this is a number, its interpretation is dependent on the value of duration (see below). NB if this is a rhythm object, it will be cloned.

OPTIONAL ARGUMENTS:

keyword arguments:
- :start-time. The start time of the event in seconds. Default = NIL.
- :is-rest. Set to T or NIL to indicate whether or not the given event is a rest. Default = NIL. NB: The make-rest method is better suited to making rests; however, if using make-event to do so, the pitch-or-chord slot must be set to NIL.
- :is-tied-to. This argument is for score output and playing purposes. Set to T or NIL to indicate whether this event is tied to the previous event (i.e. it won't sound independently). Default = NIL.
- :duration. T or NIL to indicate whether the specified duration of the event has been stated in absolute seconds, not a known rhythm like 'e. Thus (make-event 'c4 4 :duration nil) indicates a quarter note with duration 1, but (make-event '(c4 d4) 4 :duration t) indicates a whole note with an absolute duration of 4 seconds (both assuming a tempo of 60). Default = NIL.
- :amplitude sets the amplitude of the event. Possible values span from 0.0 (silent) to maximum of 1.0. Default = (get-sc-config 'default-amplitude).
- :tempo. A number to indicate the tempo of the event as a normal bpm value. Default = 60. This argument is only used when creating the rhythm from a given duration rather than from a letter value.
- :tempo-change. A number or tempo-object to attach a tempo change to the event. This is the argument you need to set in order to change playback speed.
- :midi-channel. A number from 0 to 127 indicating the MIDI channel on which the event should be played back. Default = NIL.
- :microtones-midi-channel. If the event is microtonal, this argument indicates the MIDI-channel to be used for the playback of the microtonal

notes. Default = NIL. NB: See player.lsp/make-player for details on microtones in MIDI output.
- :transposition. A number in semitones that indicates the transposition of the instrument that this event is being created for. E.g. -2 would be for a B-flat clarinet.
- :written. The given pitch or chord is the written value. In this case the sounding value will be set according to the (required) transposition argument. Default = NIL.
- :bar-num. Set the bar-num slot of the event. Usually maintained by various classes that contain events. Default = -1.

RETURN VALUE:

- An event object.

EXAMPLE:

```
;; A quarter-note (crotchet) C
(make-event 'c4 4)
=>
EVENT: start-time: NIL, end-time: NIL,
duration-in-tempo: 0.0,
compound-duration-in-tempo: 0.0,
amplitude: 0.7,
bar-num: -1, marks-before: NIL,
tempo-change: NIL
instrument-change: NIL
display-tempo: NIL, start-time-qtrs: -1,
midi-time-sig: NIL, midi-program-changes: NIL,
8va: 0
pitch-or-chord:
PITCH: frequency: 261.6255569458008, midi-note: 60, midi-channel: NIL
pitch-bend: 0.0
degree: 120, data-consistent: T, white-note: C4
nearest-chromatic: C4
src: 1.0, src-ref-pitch: C4, score-note: C4
qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
micro-tone: NIL,
sharp: NIL, flat: NIL, natural: T,
octave: 4, c5ths: 0, no-8ve: C, no-8ve-no-acc: C
show-accidental: T, white-degree: 28,
accidental: N,
accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: C4, tag: NIL,
data: C4
```

```
written-pitch-or-chord: NIL
RHYTHM: value: 4.0, duration: 1.0, rq: 1, is-rest: NIL, score-rthm: 4.0f0,
undotted-value: 4, num-flags: 0, num-dots: 0, is-tied-to: NIL,
is-tied-from: NIL, compound-duration: 1.0, is-grace-note: NIL,
needs-new-note: T, beam: NIL, bracket: NIL, rqq-note: NIL,
rqq-info: NIL, marks: NIL, marks-in-part: NIL, letter-value: 4,
tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: 4, tag: NIL,
data: 4

;; Create a whole-note (semi-breve) chord, then print its data, value, duration
;; and pitch content
(let ((e (make-event '(c4 e4 g4) 4 :duration t)))
  (print (data e))
  (print (value e))
  (print (duration e))
  (print (loop for p in (data (pitch-or-chord e)) collect (data p))))

=>
W
1.0f0
4.0
(C4 E4 G4)

;; Create a single-pitch quarter-note event which is tied to, plays back on
;; MIDI channel 1 and has an amplitude of 0.5, then print these values by
;; accessing the corresponding slots.
(let ((e (make-event 'c4 4
                     :is-tied-to t
                     :midi-channel 1
                     :amplitude 0.5)))
  (print (is-tied-to e))
  (print (midi-channel (pitch-or-chord e)))
  (print (amplitude e)))

=>
T
1
0.5

;; Create an event object that consists of a quarter-note rest and print the
;; contents of the corresponding slots
(let ((e (make-event nil 'q :is-rest t)))
  (print (pitch-or-chord e))
  (print (data e))
```

```
  (print (is-rest e)))

=>
NIL
Q
T
```

SYNOPSIS:

```
(defun make-event (pitch-or-chord rthm
                   &key
                     start-time
                     is-rest
                     is-tied-to
                     ;; MDE Fri Jan  5 10:38:22 2018 -- added
                     is-tied-from
                     duration
                     ;; MDE Wed Jun 15 20:39:05 2016 -- defaults to 1 and 2
                     (midi-channel 1)
                     (microtones-midi-channel 2)
                     ;; MDE Thu May 31 19:03:59 2012 -- allow us to auto-set the
                     ;; written-pitch-or-chord slot
                     transposition
                     ;; MDE Sat Apr 20 15:13:41 2013 -- allow us to create
                     ;; written pitch events and auto-set sounding
                     written
                     (amplitude (get-sc-config 'default-amplitude))
                     ;; MDE Thu Jun 16 13:05:21 2016
                     tempo-change
                     ;; MDE Fri Nov  2 16:37:05 2018
                     (bar-num -1)
                     (tempo 60))
```

38.6.140 event/make-events

[event] [Functions]

DESCRIPTION:

```
Make a list of events using the specified data, whereby a list indicates a
note (or chord) and its rhythm and a single datum is the rhythm of a
rest. The note/rhythm lists can include further symbols which will be
interprted as marks to be added to the event.
```

ARGUMENTS:

```
 - A list of symbols and/or sublists; see below for examples.
```

OPTIONAL ARGUMENTS:

```
 - A whole number indicating the MIDI channel on which the event is to be
   played.
 - A whole number indicating the MIDI channel on which microtonal pitches of
   the event are to be played. NB: See player.lsp/make-player for details on
   microtones in MIDI output.
```

RETURN VALUE:

```
 A list.
```

EXAMPLE:

```
;; Create a list of events including a quarter note, two rests, and a chord,
;; then print-simple its contents
(let ((e (make-events '((g4 q) e s ((d4 fs4 a4) s)))))
  (loop for i in e do (print-simple i)))
=>
G4 Q, rest E, rest S, (D4 FS4 A4) S,

;; Create a list of events to be played on MIDI-channel 3, then check the MIDI
;; channels of each sounding note. Note that in (b4 s s) the first s means
;; 1/16th note and the second s is staccato.
(let ((e (make-events '((g4 q) e s (a4 s) q e (b4 s s)) 3)))
  (loop for i in e
     when (not (is-rest i))
     collect (midi-channel (pitch-or-chord i))))
=> (3 3 3)
```

SYNOPSIS:

```
(defun make-events (data-list &optional (midi-channel 1)
                              (microtones-midi-channel 2))
```

38.6.141 event/make-events2

[event] [Functions]

DESCRIPTION:

```
 Like make-events, but rhythms and pitches are given in separate lists to
 allow for rhythms with ties using "+" etc. "Nil" or "r" given in the pitch
```

list indicates a rest; otherwise, a single note name will set a single pitch while a list of note names will set a chord. Pitches for tied notes only have to be given once.

ARGUMENTS:

- A list of rhythms.
- A list of note names (including NIL or R for rests).

OPTIONAL ARGUMENTS:

- A whole number value to indicate the MIDI channel on which to play back the event.
- A whole number value to indicate the MIDI channel on which to play back microtonal pitch material for the event. NB: See player.lsp/make-player for details on microtones in MIDI output.

RETURN VALUE:

A list.

EXAMPLE:

```
;; Create a make-events2 list and use the print-simple function to retrieve its
;; contents.
(let ((e (make-events2 '(q e e. h+s 32 q+te) '(cs4 d4 (e4 g4 b5) nil a3 r))))
  (loop for i in e do (print-simple i)))
=>
CS4 Q, D4 E, (E4 G4 B5) E., rest H, rest S, A3 32, rest Q, rest TE,

;; Create a list of events using make-events2, indicating they be played back
;; on MIDI-channel 3, then print the corresponding slots to check it
(let ((e (make-events2 '(q e. h+s 32 q+te) '(cs4 b5 nil a3 r) 3)))
  (loop for i in e
     when (not (is-rest i))
     collect (midi-channel (pitch-or-chord i))))
=>
(3 3 3)
```

SYNOPSIS:

```
(defun make-events2 (rhythms pitches
                     &optional (midi-channel 1) (microtones-midi-channel 2))
```

38.6.142 event/make-events3

[event] [Functions]

DATE:

September 12th 2020, Heidhausen

DESCRIPTION:

Make a list of events using various approaches. Either with separate lists of pitches and rhythms, make-events and make-events2 will be called according to the type of arguments given. Note that when specifying pitches, octaves don't have to be retyped if they don't change.

ARGUMENTS:

Two required arguments but their type and function vary according to requirements. See examples below.

OPTIONAL ARGUMENTS:

- A whole number value to indicate the MIDI channel on which to play back the event.
- A whole number value to indicate the MIDI channel on which to play back microtonal pitch material for the event. NB: See player.lsp/make-player for details on microtones in MIDI output.

RETURN VALUE:

a list of events

EXAMPLE:

```
;; simple list of pitch/rhythm pairs but no chords or rests (so no need for list
;; pairs)
(make-events3 '(g4 s a s b 32 c 32) nil)
;; pitch and rhythms in pairs in a single list, including chords and rests
(make-events3 '((g4 q) e s ((d4 fs4 a4) s)) nil)
;; using separate lists for rhythms and pitches, tied rhythms, single pitches, a
;; chord, and both nil and r to indicate rests
(make-events3 '(q e e. h+s 32 q+te) '(cs4 d4 (e4 g4 b5) nil a3 r))
;; just one pitch but various rhythms (but here we can't use tied rhythms)
(make-events3 '(q e e. s e s s s s) 'cs6)
;; just one rhythm but various pitches/chords and all in octave 4 without having
;; to retype (here we can't use tied rhythms either)
(make-events3 's '(c4 d e (f a) r bf)) ; r = rest here too
```

SYNOPSIS:

```
(defun make-events3 (data1 data2 &key (midi-channel 1)
                                      (microtones-midi-channel 2))
```

38.6.143 event/make-punctuation-events

[event] [Functions]

DESCRIPTION:

Given a list of numbers, a rhythm, and a note name or list of note names, create a new list of single events separated by rests.

The rhythm specified serves as the basis for the new list. The numbers specified represent groupings in the new list that are each made up of one rhythm followed by rests. Each consecutive grouping in the new list has the length of each consecutive number in the numbers list multiplied by the rhythm specified.

Notes can be a single note or a list of notes. If the latter, they'll be used one after the other, repeating the final note once reached.

ARGUMENTS:

- A list of grouping lengths.
- A rhythm.
- A note name or list of note names.

RETURN VALUE:

A list.

EXAMPLE:

```
;; Create a list of three groups that are 2, 3, and 5 16th-notes long, with the
;; first note of each grouping being a C4, then print-simple it's contents.
(let ((pe (make-punctuation-events '(2 3 5) 's 'c4)))
  (loop for e in pe do (print-simple e)))
=>
C4 S, rest S, C4 S, rest S, rest S, C4 S, rest S, rest S, rest S, rest S,

;; Create a list of "punctuated" events using a list of note names. Once the
;; final note name is reached, it is repeated for all remaining non-rest
;; rhythms.
```

```
(let ((pe (make-punctuation-events '(2 3 5 8) 'q '(c4 e4))))
  (loop for e in pe do (print-simple e)))
=>
C4 Q, rest Q, E4 Q, rest Q, rest Q, E4 Q, rest Q, rest Q, rest Q, rest Q, E4 Q,
rest Q, rest Q, rest Q, rest Q, rest Q, rest Q, rest Q,
```

SYNOPSIS:

```
(defun make-punctuation-events (distances rhythm notes)
```

38.6.144 event/make-rest

[event] [Functions]

DESCRIPTION:

```
Create an event object that consists of a rest.
```

ARGUMENTS:

```
- A rhythm (duration).
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :start-time. A number that is the start-time of the event in seconds.
- :duration. T or NIL. T indicates that the duration given is a value of
  absolute seconds rather than a known rhythm (e.g. 'e). Default = NIL.
- :tempo. Beats per minute. Note that this slot is used for duration
  calculations. If you're making events 'by hand' and you want a new tempo to
  be written to MIDI files you'll need to set the tempo-change slot with
  either a number or tempo object. If you then want the tempo to be written
  to the score, set display-tempo to T. Default = 60.
```

RETURN VALUE:

```
- An event object.
```

EXAMPLE:

```
;; Make an event object consisting of a quarter rest
(make-rest 4)
=>
EVENT: start-time: NIL, end-time: NIL,
```

```
duration-in-tempo: 0.0,
compound-duration-in-tempo: 0.0,
amplitude: 0.7,
bar-num: -1, marks-before: NIL,
tempo-change: NIL
instrument-change: NIL
display-tempo: NIL, start-time-qtrs: -1,
midi-time-sig: NIL, midi-program-changes: NIL,
8va: 0
pitch-or-chord: NIL
written-pitch-or-chord: NIL
RHYTHM: value: 4.0, duration: 1.0, rq: 1, is-rest: T, score-rthm: 4.0f0,
undotted-value: 4, num-flags: 0, num-dots: 0, is-tied-to: NIL,
is-tied-from: NIL, compound-duration: 1.0, is-grace-note: NIL,
needs-new-note: NIL, beam: NIL, bracket: NIL, rqq-note: NIL,
rqq-info: NIL, marks: NIL, marks-in-part: NIL, letter-value: 4,
tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: 4, tag: NIL,
data: 4

;; Make an event object consisting of 4 seconds of rest (rather than a quarter
;; indicated by the :duration t) starting at time-point 13.7 seconds, then
;; print the corresponding slot values.
(let ((e (make-rest 4 :start-time 13.7 :duration t)))
  (print (is-rest e))
  (print (data e))
  (print (duration e))
  (print (value e))
  (print (start-time e)))
=>
T
W
4.0
1.0f0
13.7
```

SYNOPSIS:

```
(defun make-rest (rthm &key start-time duration (tempo 60))
```

38.6.145 event/microtonal

[event] [Methods]

DATE:

January 30th 2017

DESCRIPTION:

Return numerical information about the microtonality of an event, essentially how many microtonal pitches there are.

ARGUMENTS:

- the event object

OPTIONAL ARGUMENTS:

T or NIL to determine whether numbers should always be returned (i.e. zeros when there are no microtonal pitches or any pitches at all in the case of rests)

RETURN VALUE:

NIL if the event is a rest (or there are no microtones and <force-number> is nil), otherwise two values: the number of microtonal pitches in the event (0, 1, or more if it's a chord), the number of pitches in the event (2+ for chords otherwise 0 for a rest and 1 for a non-chord).

EXAMPLE:

```
(microtonal (make-event 'cs4 'q))
NIL

(microtonal (make-event 'cs4 'q) t)
0
1

(microtonal (make-event 'cqs4 'q) t)
1
1

(microtonal (make-event 'cqs4 'q))
1
1

(microtonal (make-event '(cqs4 e4 g4) 'q))
1
3
```

```
(microtonal (make-event nil 'q))
NIL

(microtonal (make-event nil 'q) t)
0
1
```

SYNOPSIS:

```
(defmethod microtonal ((e event) &optional force-number)
```

38.6.146 event/midi-control-change

[event] [Methods]

DATE:

```
August 21st 2019, Heidhausen
```

DESCRIPTION:

```
Add a MIDI control change to the midi-control-changes list slot.
```

ARGUMENTS:

```
- the event
- the MIDI channel (integer)
- the MIDI channel (integer)
- the control change value (integer)
```

EXAMPLE:

```
;;; sustain pedal down
(midi-control-change (make-event 'c4 'q) 1 64 127)
;;; sustain pedal half down
(midi-control-change (make-event 'c4 'q) 1 64 63)
;;; sustain pedal up
(midi-control-change (make-event 'c4 'q) 1 64 0)
;;; una corda
(midi-control-change (make-event 'c4 'q) 1 67 127)
;;; tre corde
(midi-control-change (make-event 'c4 'q) 1 67 0)
;;; sost. (middle) pedal down
(midi-control-change (make-event 'c4 'q) 1 66 127)
```

RETURN VALUE:

```
The list of control-changes, including the new ones
```

SYNOPSIS:

```
(defmethod midi-control-change ((e event) channel controller-number value)
```

38.6.147 event/natural-p

[event] [Methods]

DESCRIPTION:

```
Determine whether the pitch of a given event object is a natural note (no
sharps or flats).
```

ARGUMENTS:

```
- An event object.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether the test is to handle the written or
  sounding pitch in the event. T = written. Default = NIL.
```

RETURN VALUE:

```
Returns T if the note tested is natural, otherwise NIL (ie, has a flat or
has a sharp).
```

EXAMPLE:

```
;; Returns T when the note is natural
(let ((e (make-event 'c4 'q)))
  (natural-p e))

=> T

;; Returns NIL when the note is not natural (ie, is sharp or flat)
(let ((e (make-event 'cs4 'q)))
  (natural-p e))

=> NIL
```

```
(let ((e (make-event 'df4 'q)))
  (natural-p e))

=> NIL
```

SYNOPSIS:

```
(defmethod natural-p ((e event) &optional written)
```

38.6.148 event/no-accidental

[event] [Methods]

DESCRIPTION:

```
Sets the SHOW-ACCIDENTAL and ACCIDENTAL-IN-PARENTHESES slots of the given
event object to NIL.
```

ARGUMENTS:

```
- An event object.
```

RETURN VALUE:

```
Always returns NIL.
```

EXAMPLE:

```
;; The SHOW-ACCIDENTAL slot is automatically set to T on new event objects
;; that consist of a sharp or flat note.
(let ((e (make-event 'cs4 'q)))
  (show-accidental (pitch-or-chord e)))

=> T

;; The method no-accidental sets the SHOW-ACCIDENTAL slot to NIL (and the
;; ACCIDENTAL-IN-PARENTHESES if not already).
(let ((e (make-event 'cs4 'q)))
  (no-accidental e)
  (show-accidental (pitch-or-chord e)))

=> NIL
```

SYNOPSIS:

```
(defmethod no-accidental ((e event))
```

38.6.149 event/num-notes

[event] [Methods]

DATE:

```
November 6th 2018, Heidhausen
```

DESCRIPTION:

```
Return the number of notes in the event object i.e. the number of notes in
the chord, 1 if it's a single pitch or 0 if it's a rest.
```

ARGUMENTS:

```
- the event object
```

RETURN VALUE:

```
An integer value for the number of notes in the event.
```

SYNOPSIS:

```
(defmethod num-notes ((e event))
```

38.6.150 event/output-midi

[event] [Methods]

DESCRIPTION:

```
Generate the data necessary for MIDI output for a given event object. Note
that for events that contain chords, the event's amplitude slot will be used
for all pitches unless each individual pitch object in the chord has its
amplitude slot set to a number.

NB: The given event object must contain data for start-time and
    midi-channel.
```

ARGUMENTS:

```
- An event object.
```

OPTIONAL ARGUMENTS:

```
 - A decimal number that is the number of seconds to offset the timing of
   the MIDI output.
 - A decimal number or function that is to override any other existing event
   object data for amplitude.
```

RETURN VALUE:

```
 Returns the data required for MIDI output.
```

EXAMPLE:

```
;; Simple use
(let ((e (make-event 'c4 'q
                     :start-time 0.0
                     :midi-channel 1)))
  (output-midi e))

=> (#i(midi time 0.0 keynum 60 duration 1.0 amplitude 0.7 channel 0))

;; Specifying time offset and forced amplitude value
(let ((e (make-event 'c4 'q
                     :start-time 0.0
                     :midi-channel 1)))
  (output-midi e 0.736 0.3))

=> (#i(midi time 0.736 keynum 60 duration 1.0 amplitude 0.3 channel 0))
```

SYNOPSIS:

```
(defmethod output-midi ((e event) &optional (time-offset 0.0) force-velocity)
```

38.6.151 event/pitch-

[event] [Methods]

DESCRIPTION:

```
 Determine the interval in half-steps between two pitches.

 NB: This is determined by subtracting the MIDI note value of one event from
 the other. Negative numbers may result if the greater MIDI note value is
 subtracted from the lesser.
```

ARGUMENTS:

```
 - A first event object.
 - A second event object.
```

RETURN VALUE:

```
 A number.
```

EXAMPLE:

```
(let ((e1 (make-event 'c4 'q))
      (e2 (make-event 'a3 'q)))
  (pitch- e1 e2))

=> 3.0

;; Subtracting the upper from the lower note returns a negative number
(let ((e1 (make-event 'a3 'q))
      (e2 (make-event 'c4 'q)))
  (pitch- e1 e2))

=> -3.0
```

SYNOPSIS:

```
(defmethod pitch- ((e1 event) (e2 event))
```

38.6.152 event/pitch-or-chord=

[event] [Methods]

DATE:

```
 Tue 29 Oct 2019 08:59:59 GMT - Warwick
```

DESCRIPTION:

```
 Convenience method, test to see if the pitch-or-chord slots of two event
 objects are the same.
```

ARGUMENTS:

```
 - a pitch or chord object
 - a pitch or chord object
```

OPTIONAL ARGUMENTS:

```
 - T or NIL to indicate whether or not enharmonic pitches are considered
   equal. T = enharmonic pitches are considered equal. Default = NIL.
 - a number to indicate the frequency deviation allowed before returning NIL.
```

RETURN VALUE:

```
T if the values of the two specified pitch or chord objects are equal, otherwise
NIL.
```

EXAMPLE:

```
;; Comparison of equal pitch objects created using note-name symbols returns T
(let ((p1 (make-pitch 'C4))
      (p2 (make-pitch 'C4)))
  (pitch-or-chord= p1 p2))

=> T

(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'bs3)))
  (pitch-or-chord= p1 p2))

=> NIL

(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'bs3)))
  (pitch-or-chord= p1 p2 t))

=> T

(let ((p1 (make-pitch 'c4))
      (c1 (make-chord '(c4 e4 g4))))
  (pitch-or-chord= p1 c1))

=> NIL

(let ((c1 (make-chord '(c4 e4 g4)))
      (c2 (make-chord '(c4 e4))))
  (pitch-or-chord= c1 c2))

=> NIL

(let ((c1 (make-chord '(c4 e4 g4)))
      (c2 (make-chord '(bs4 ff4 g4))))
  (pitch-or-chord= c1 c2))
```

```
=> NIL

(let ((c1 (make-chord '(c4 e4 g4)))
      (c2 (make-chord '(bs3 ff4 g4))))
  (pitch-or-chord= c1 c2 t))

=> T
```

SYNOPSIS:

```
(defmethod pitch-or-chord= ((e1 event) (e2 event)
                            &optional enharmonics-are-equal
                              (frequency-tolerance 0.01))
```

38.6.153 event/remove-dynamics

[event] [Methods]

DESCRIPTION:

Remove all dynamic symbols from the list of marks attached to a given event object.

NB: This doesn't change the amplitude.

ARGUMENTS:

- An event object.

OPTIONAL ARGUMENTS:

- T or NIL to remove hairpin marks too (e.g. cresc-beg). Default = NIL.

RETURN VALUE:

Returns the modified list of marks (which could be empty) attached to the given event object if the specified dynamic was initially present in that list and successfully removed, otherwise returns NIL.

EXAMPLE:

```
;; Create an event object, add one dynamic mark and one non-dynamic mark, print
;; all marks attached to the object, and remove just the dynamics from that
;; list of all marks.
```

```
(let ((e (make-event 'c4 'q)))
  (add-mark-once e 'ppp)
  (add-mark-once e 'pizz)
  (print (marks e))
  (remove-dynamics e))

=>
(PIZZ PPP)
(PIZZ)

;; Attempting to remove dynamics when none are present returns NIL.
(let ((e (make-event 'c4 'q)))
  (remove-dynamics e))

=> NIL
```

SYNOPSIS:

```
(defmethod remove-dynamics ((e event) &optional hairpins)
```

38.6.154 event/replace-mark

[event] [Methods]

DESCRIPTION:

```
Replace a specified mark of a given event object with a second specified
mark. If an event object contains more than one mark, individual marks can
be changed without modifying the remaining marks. NB works with strings,
symbols, lists, numbers or anything that is #'string= or #'equal
```

ARGUMENTS:

```
- An event object.
- The mark to be replaced.
- The new mark.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether the mark to be replaced is in the
  MARKS-BEFORE slot. T = it is in the MARKS-BEFORE slot. Default = NIL.
```

RETURN VALUE:

```
Returns the new value of the MARKS/MARKS-BEFORE slot of the given object.
```

EXAMPLE:

```
;; Add marks to the MARKS slot and replace 'a with 'batt ;
                     (let ((e (make-event 'c4 'q)))
(loop for m in '(a s pizz)
do (add-mark e m))
(replace-mark e 'a 'batt))

                     => (PIZZ S BATT)

;; Add marks to the MARKS-BEFORE slot and replace 'arco with 'at ;
                     (let ((e (make-event 'c4 'q)))
(loop for m in '(arco col-legno)
do (add-mark-before e m))
(replace-mark e 'arco 'at t))

                     => (COL-LEGNO AT)

                     |#
(defmethod replace-mark ((e event) what with &optional before ignore1 ignore2)
```

38.6.155 event/round-to-nearest

[event] [Methods]

DESCRIPTION:

```
Round the written and sounding pitch/chord objects to the nearest in the
current or given scale.
```

ARGUMENTS:

```
the event object
```

OPTIONAL ARGUMENTS:

```
keyword argument:
:scale. The scale to use when rounding. (Common Music tuning object or
symbol). If a symbol, then 'chromatic-scale, 'twelfth-tone, or 'quarter-tone
only at present. Default is the current scale as set by (in-scale :...).
:rm-duplicates. T or NIL to remove duplicate pitches, should the event
contain a chord and rounding created duplicate pitches.
```

RETURN VALUE:

```
the modified event object
```

SYNOPSIS:

```
(defmethod round-to-nearest ((e event) &key (scale cm::*scale*)
                                            (rm-duplicates t))
```

38.6.156 event/set-midi-channel

[event] [Methods]

DESCRIPTION:

```
Set the MIDI-channel and microtonal MIDI-channel for the pitch object
within a given event object.
```

ARGUMENTS:

```
- An event object.
- A whole number indicating the MIDI-channel to be used for playback of
  this event object.
- A whole number indicating the MIDI-channel to be used for playback of the
  microtonal pitch material of this event.
```

RETURN VALUE:

```
Returns the value of the MIDI-channel setting (a whole number) if the
MIDI-channel slot has been set, otherwise NIL.
```

EXAMPLE:

```
;; Unless specified the MIDI channel of a newly created event object defaults
;;; to NIL.
(let ((e (make-event 'c4 'q)))
  (midi-channel (pitch-or-chord e)))

=> NIL

(let ((e (make-event 'c4 'q)))
  (set-midi-channel e 7 8)
  (midi-channel (pitch-or-chord e)))

=> 7
```

SYNOPSIS:

```
(defmethod set-midi-channel ((e event) midi-channel
                             &optional microtonal-midi-channel)
```

38.6.157 event/set-midi-time-sig

[event] [Methods]

DESCRIPTION:

```
Sets a MIDI time signature for the given event object. This must be a
time-sig object, not just a time signature list.
```

ARGUMENTS:

```
- An event object.
- A time-sig object.
```

RETURN VALUE:

```
Returns a time-sig object.
```

EXAMPLE:

```
;; Creating a new event object sets the midi-time-sig slot to NIL by default
(let ((e (make-event 'c4 'q)))
  (midi-time-sig e))

=> NIL

;; The set-midi-time-sig method returns a time-sig object
(let ((e (make-event 'c4 'q)))
  (set-midi-time-sig e (make-time-sig '(3 4))))

=>
TIME-SIG: num: 3, denom: 4, duration: 3.0, compound: NIL, midi-clocks: 24,
num-beats: 3
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "0304", tag: NIL,
data: (3 4)

;; Set the midi-time-sig slot and read the data of the given time-sig object
(let ((e (make-event 'c4 'q)))
  (set-midi-time-sig e (make-time-sig '(3 4)))
  (data (midi-time-sig e)))

=> (3 4)
```

SYNOPSIS:

```
(defmethod set-midi-time-sig ((e event) time-sig)
```

38.6.158 event/set-written

[event] [Methods]

DESCRIPTION:

```
Set the written pitch (as opposed to sounding; i.e., for transposing
instruments) of a given event object. The sounding pitch remains unchanged
as a pitch object in the PITCH-OR-CHORD slot, while the written pitch is
added as a pitch object to the WRITTEN-PITCH-OR-CHORD slot.
```

ARGUMENTS:

```
- An event object.
- A number indicating the difference in semitones (positive or
  negative) between the written and sounding pitches.  E.g. to set the
  written-note for a B-flat Clarinet, this would be 2, for an E-flat
  Clarinet, it would be -3.
```

RETURN VALUE:

```
The 'written' pitch object.
```

EXAMPLE:

```
;; Returns a pitch object (here for example for a D Trumpet or Clarinet) ;
(let ((e (make-event 'c4 'q)))
  (set-written e -2))

=>
PITCH: frequency: 233.08186975464196, midi-note: 58, midi-channel: NIL
pitch-bend: 0.0
degree: 116, data-consistent: T, white-note: B3
nearest-chromatic: BF3
src: 0.8908987045288086, src-ref-pitch: C4, score-note: BF3
qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
micro-tone: NIL,
sharp: NIL, flat: T, natural: NIL,
octave: 3, c5ths: 1, no-8ve: BF, no-8ve-no-acc: B
show-accidental: T, white-degree: 27,
accidental: F,
accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: BF3, tag: NIL,
data: BF3
```

```
;; Create a single-pitch event object, set it's written pitch to two half-steps ;
;; lower, and print the corresponding data slots ;
(let ((e (make-event 'c4 'q)))
  (set-written e -2)
  (print (data (pitch-or-chord e)))
  (print (data (written-pitch-or-chord e))))

=>
C4
BF3
```

SYNOPSIS:

```
(defmethod set-written ((e event) transposition)
```

38.6.159 event/set-written-pitch-or-chord

[event] [Methods]

DATE:

```
Some time somewhere in 2013
```

DESCRIPTION:

```
Change the written (as opposed to sounding) pitch/chord data of an event. If
the second argument is NIL, do nothing.
```

ARGUMENTS:

```
- the event object
- the new pitch data (symbol, list of symbols (chord), or pitch/chord
  object)
```

OPTIONAL ARGUMENTS:

```
- an instrument for the event so that the sounding pitch can be calculated
from its transposition slot.
```

RETURN VALUE:

```
the modified event
```

SYNOPSIS:

```
(defmethod set-written-pitch-or-chord ((e event) value &optional instrument)
```

38.6.160 event/setf amplitude

[event] [Methods]

DESCRIPTION:

Change the amplitude slot of a given event object and automatically add a mark to set a corresponding dynamic.

Numbers greater than 1.0 and less than 0.0 will also be stored in the amplitude slot of the given event object without printing a warning, though corresponding dynamic marks are only available for values between 0.0 and 1.0. Any value above 1.0 or below 0.0 will result in a dynamic marking of FFFF and NIENTE respectively.

ARGUMENTS:

- An amplitude value (real number).
- An event object.

RETURN VALUE:

Returns the specified amplitude value.

EXAMPLE:

```
;; When no amplitude is specified, new event objects are created with a default
;; amplitude of 0.7.
(let ((e (make-event 'c4 'q)))
  (amplitude e))

=> 0.7

;; Setting an amplitude returns the amplitude set
(let ((e (make-event 'c4 'q)))
  (setf (amplitude e) .3))

=> 0.3

;; Create an event object, set its amplitude, then print the contents of the
;; amplitude and marks slots to see the dynamic setting.
(let ((e (make-event 'c4 'q)))
  (setf (amplitude e) .3)
  (print (amplitude e))
  (print (marks e)))
```

```
=>
0.3
(PP)

;; Setting an amplitude greater than 1.0 or less than 0.0 sets the amplitude
;; correspondingly and sets the dynamic mark to FFFF or NIENTE respectively.
(let ((e1 (make-event 'c4 'q))
      (e2 (make-event 'c4 'q)))
  (setf (amplitude e1) 1.3)
  (setf (amplitude e2) -1.3)
  (print (marks e1))
  (print (marks e2)))

=>
(FFFF)
(NIENTE)
```

SYNOPSIS:

```
(defmethod (setf amplitude) :after (value (e event))
```

38.6.161 event/setf tempo-change

[event] [Methods]

DESCRIPTION:

```
Store the tempo when a change is made.

NB: This creates a full tempo object, not just a number representing
bpm. It also sets the display-tempo object to T as a side effect.
Also note that to delete a tempo-change you should use the
delete-tempo-change method rather than passing NIL to this method.
```

ARGUMENTS:

```
- An event object.
- A number indicating the new tempo bpm.
```

RETURN VALUE:

```
Returns a tempo object.
```

EXAMPLE:

```
;; Creation of a new event object sets the tempo-change slot to NIL by default,
;; unless otherwise specified.
(let ((e (make-event 'c4 'q)))
  (tempo-change e))

=> NIL

;; The tempo-change method returns a tempo object
(let ((e (make-event 'c4 'q)))
  (setf (tempo-change e) 132))

=>
TEMPO: bpm: 132, beat: 4, beat-value: 4.0, qtr-dur: 0.45454545454545453
       qtr-bpm: 132.0, usecs: 454545, description: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: 132

;; The new tempo object is stored in the event object's tempo-change slot.
(let ((e (make-event 'c4 'q)))
  (setf (tempo-change e) 132)
  e)

=>
EVENT: start-time: NIL, end-time: NIL,
       duration-in-tempo: 0.0,
       compound-duration-in-tempo: 0.0,
       amplitude: 0.7, score-marks: NIL,
       bar-num: -1, cmn-objects-before: NIL,
       tempo-change:
TEMPO: bpm: 132, beat: 4, beat-value: 4.0, qtr-dur: 0.45454545454545453
       qtr-bpm: 132.0, usecs: 454545, description: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: 132
[...]
```

SYNOPSIS:

```
(defmethod (setf tempo-change) (value (e event))
```

38.6.162 event/sharp-p

[event] [Methods]

DESCRIPTION:

Determine whether the pitch of a given event object has a sharp.

ARGUMENTS:

- An event object.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether the test is to handle the written or sounding pitch in the event. T = written. Default = NIL.

RETURN VALUE:

Returns T if the note tested has a sharp, otherwise NIL (ie, is natural or has a flat).

EXAMPLE:

```
;; Returns T when the note is sharp
(let ((e (make-event 'cs4 'q)))
  (sharp-p e))

=> T

;; Returns NIL when the note is not sharp (ie, is flat or natural)
(let ((e (make-event 'c4 'q)))
  (sharp-p e))

=> NIL

(let ((e (make-event 'df4 'q)))
  (sharp-p e))

=> NIL
```

SYNOPSIS:

```
(defmethod sharp-p ((e event) &optional written)
```

38.6.163 event/sharp-to-flat

[event] [Methods]

DATE:

Mon 10 Feb 2020 18:31:17 GMT - London

DESCRIPTION:

Change the sharp (or sharps) in an event to a flat (or flats).

ARGUMENTS:

An event object

OPTIONAL ARGUMENTS:

None

RETURN VALUE:

An event object.

EXAMPLE:

```
(let ((e (make-event 'as4 'q)))
      (print-simple (sharp-to-flat e)))

=> BF4 Q,
NIL

(let ((e (make-event '(bf4 as4) 'q)))
      (print-simple (sharp-to-flat e)))

=> (BF4 BF4) Q,
NIL

;;; NB This method is non-destructove
(let ((e (make-event 'as4 'q)))
      (print-simple (sharp-to-flat e))
      (print-simple e))

=> BF4 Q,
AS4 Q,
NIL
```

SYNOPSIS:

```
(defmethod sharp-to-flat ((e event) &optional clone written)
```

38.6.164 event/single-pitch-chord-to-pitch

[event] [Methods]

DATE:

```
Thu 14 Nov 2019 07:44:02 GMT London
```

DESCRIPTION:

```
Turn a chord object with only one pitch in its chord slot into a pitch
object.
```

ARGUMENTS:

```
An event object
```

OPTIONAL ARGUMENTS:

```
NIL
```

RETURN VALUE:

```
An event object containing a new pitch object, if the original chord
contained a single pitch, or the original event object.
```

EXAMPLE:

```
(data (single-pitch-chord-to-pitch (make-event '(a4) 'e)))
=> A4
```

SYNOPSIS:

```
(defmethod single-pitch-chord-to-pitch ((e event))
```

38.6.165 event/sort-event-list

[event] [Functions]

DESCRIPTION:

```
Sort a list of event objects by start time from lowest to highest.
```

ARGUMENTS:

```
 - A list of event objects.
```

RETURN VALUE:

```
 A list of event objects.
```

EXAMPLE:

```
;; Create a list of event object with non-sequential start-times, sort them,
;; and return the pitches and start times of the newly ordered list.
(let ((e-list (loop repeat 8
                 for nn in '(c4 d4 e4 f4 g4 a4 b4 c5)
                 for st in '(1.0 3.0 2.0 5.0 8.0 4.0 7.0 6.0)
                 collect (make-event nn 'e :start-time st))))
  (sort-event-list e-list)
  (loop for e in e-list
     collect (get-pitch-symbol e)
     collect (start-time e)))
=>
(C4 1.0 E4 2.0 D4 3.0 A4 4.0 F4 5.0 C5 6.0 B4 7.0 G4 8.0)
```

SYNOPSIS:

```
(defun sort-event-list (event-list)
```

38.6.166 event/transpose

[event] [Methods]

DESCRIPTION:

```
 Transpose the pitch content of a given event object by a specified number
 of semitones. This method can be applied to chords or single-pitches.

 If functions are given, these will be used for the note or chord in the
 event, whereby semitones may or may not be NIL in that case (transposition
 could be dependent on the note or chord rather than being a fixed shift).

 NB: By default this method returns a modified clone of the original rather
     than changing the values of the original itself. The user can choose to
     replace the values of the original by setting the keyword argument
     :destructively to T.
```

ARGUMENTS:

```
 - An event object.
 - A number (can be positive or negative).
```

OPTIONAL ARGUMENTS:

```
 keyword arguments:
 - :destructively. T or NIL to indicate whether the method is to replace the
   pitch values of the original event object (T) or return a new event
   object with the new pitches (NIL). Default = NIL.
 - :chord-function. A function to be used for the transposition of
   chords. Default = #'transpose.
 - :pitch-function. A function to be used for the transposition of
   pitches. Default = #'transpose.
```

RETURN VALUE:

```
 An event object.
```

EXAMPLE:

```
;; Transpose returns an event object    ;
(let ((e (make-event 'c4 'q)))
  (transpose e 1))

=>
EVENT: start-time: NIL, end-time: NIL,
duration-in-tempo: 0.0,
[...]

;; By default transpose returns a modified clone, leaving the original event ;
;; object untouched.                    ;
(let ((e (make-event 'c4 'q)))
  (print (data (pitch-or-chord (transpose e 1))))
  (print (data (pitch-or-chord e))))

=>
CS4
C4

;; When the keyword argument :destructively is set to T, the data of the ;
;; original event object is replaced    ;
(let ((e (make-event 'c4 'q)))
  (transpose e 1 :destructively t)
  (data (pitch-or-chord e)))
```

```
=> CS4

;; Can transpose by 0 as well (effectively no transposition) ;
(let ((e (make-event 'c4 'q)))
  (transpose e 0 :destructively t)
  (data (pitch-or-chord e)))

=> C4

;; ...or by negative intervals          ;
(let ((e (make-event 'c4 'q)))
  (transpose e -3 :destructively t)
  (data (pitch-or-chord e)))

=> A3

;; Can transpose chords too             ;
(let ((e (make-event '(c4 e4 g4) 'q)))
  (transpose e -3 :destructively t)
  (loop for p in (data (pitch-or-chord e)) collect (data p)))

=> (A3 CS4 E4)
```

SYNOPSIS:

```
(defmethod transpose ((e event) semitones
                      &key
                      destructively
                      ;; the default functions are the class methods for pitch
                      ;; or chord.
                      (chord-function #'transpose)
                      (pitch-function #'transpose))
```

38.6.167 event/voice-chord

[event] [Methods]

DATE:

```
November 28th 2018, Heidhausen
```

DESCRIPTION:

```
Set the amplitudes of the individual pitches in a chord event. Amplitude
values should either be integers 0-127 or floating-point 0.0-1.0.
```

```
Usually, the amplitude of each pitch in a chord is set by the overall
amplitude slot of the event. However, if the pitch's amplitude slot is a
number, it will be used instead, automatically, when generating MIDI files
(and only those at present).
```

ARGUMENTS:

```
- the event object
- either a list or a function: if a list, it will be cycled through
  (returning to the beginning if necessary) to set the amplitudes of the
  pitches of the chord from bottom to top; if a function then the function
  is passed the event object and it should return a list of amplitudes which
  will then be used to set the pitch objects' amplitudes similarly.
```

OPTIONAL ARGUMENTS:

```
None
```

RETURN VALUE:

```
The (modified) event object
```

EXAMPLE:

```
(let ((e (make-event '(c4 d4 e4 f4 g5 a4 b4) 4 :start-time 0.0)))
  (print (voice-chord e '(50 60 70))) ; velocities
  (voice-chord e #'(lambda (e)
                     (loop for p in (get-pitch-symbols e) collect
                          (case p ; amplitude floats
                            (c4 .1) (d4 .2) (e4 .3) (f4 .4) (t .5))))))
```

SYNOPSIS:

```
(defmethod voice-chord ((e event) processor)
```

38.6.168 event/wrap-events-list

[event] [Functions]

DESCRIPTION:

```
Given a list of time-ascending event objects, rotate their start-times by
moving the lowest start time to a specified point in the list (determined
```

```
either by time or by nth position), assigning the subsequent start times
sequentially through the remainder of events in the list, and wrapping
around to the head of the list again to assign the final remaining start
times. If the first event doesn't start at 0, its start time will be
conserved.
```

ARGUMENTS:

```
- A flat list of event objects.
- An integer that is the number of the event object with which to start
  (nth position), or a decimal time in seconds.
```

OPTIONAL ARGUMENTS:

```
keyword argument:
- :time. T or NIL to indicate whether the second argument is a time in
  seconds or an nth index. If a time in seconds, the method skips to the
  closest event object in the list. T = time in seconds. Default = NIL.
```

RETURN VALUE:

```
Returns a flat list of event objects with adjust start-times.
```

EXAMPLE:

```
;;; Create a list of events of eighth-note durations, specifying start-times at ;
;;; 0.5-second intervals and print the pitches and start-times. Then apply the ;
;;; function and print the pitches and start-times again to see the change. ;
(let ((e-list (loop for st from 1.0 by 0.5
                 for nn in '(c4 d4 e4 f4 g4 a4 b4 c5)
                 collect (make-event nn 'e :start-time st))))
  (print
   (loop for e in e-list
      collect (get-pitch-symbol e)
      collect (start-time e)))
  (wrap-events-list e-list 3)
  (print
   (loop for e in e-list
      collect (get-pitch-symbol e)
      collect (start-time e))))
=>
(C4 1.0 D4 1.5 E4 2.0 F4 2.5 G4 3.0 A4 3.5 B4 4.0 C5 4.5)
(C4 3.5 D4 4.0 E4 4.5 F4 1.0 G4 1.5 A4 2.0 B4 2.5 C5 3.0)
```

SYNOPSIS:

```
(defun wrap-events-list (events start-at &key (time nil))
```

38.6.169 rhythm/force-rest

[rhythm] [Methods]

DESCRIPTION:

Force the given rhythm object to be a rest.

ARGUMENTS:

- A rhythm object.

RETURN VALUE:

A rhythm object.

EXAMPLE:

```
(let ((r (make-rhythm 8)))
  (force-rest r)
  (is-rest r))

=> T
```

SYNOPSIS:

```
(defmethod force-rest ((r rhythm))
```

38.6.170 rhythm/get-proportions

[rhythm] [Functions]

DATE:

15th February 2022, Heidhausen

DESCRIPTION:

Get the normalised proportions of a list of rhythms, i.e. the durational relationships of one rhythm to another. By normalised here it is meant that the returned list will sum to 1.0. Implicit there is that proportions are expressed as floating-point fractions of 1.0, not as rationals.

NB this works with the rhythm-slot only of rhythm objects, not compound-duration

ARGUMENTS:

a list of rhythm objects or symbols/numbers that represent rhythms.

RETURN VALUE:

a list of floats

EXAMPLE:

```
(get-proportions '(e e q e. s q)) --> (0.125 0.125 0.25 0.1875 0.0625 0.25)
```

SYNOPSIS:

```
(defun get-proportions (rhythms)
```

38.6.171 rhythm/has-dynamic

[rhythm] [Methods]

DATE:

December 3rd 2023

DESCRIPTION:

Determine whether a rhythm object has any dynamics in its marks slot.

ARGUMENTS:

- a rhythm object

RETURN VALUE:

The dynamic as a symbol (e.g. pp) or NIL if there are none.

SYNOPSIS:

```
(defmethod has-dynamic ((r rhythm))
```

38.6.172 rhythm/has-mark

[rhythm] [Methods]

DESCRIPTION:

Check to see if a given rhythm object possesses a specified mark.

ARGUMENTS:

- A rhythm object.
- A mark.

RETURN VALUE:

If the specified mark is indeed found in the MARKS slot of the given rhythm object, the tail of the list of marks contained in that slot is returned; otherwise NIL is returned.

EXAMPLE:

```
;; Add a specific mark and check to see if the rhythm object has it.
(let ((r (make-rhythm 'q)))
  (add-mark r 'a)
  (has-mark r 'a))

=> (A)

;; Check to see if the given rhythm object possess a mark we know it doesn't.
(let ((r (make-rhythm 'q)))
  (add-mark r 'a)
  (has-mark r 's))

=> NIL
```

SYNOPSIS:

```
(defmethod has-mark ((r rhythm) mark &optional (test #'equal))
```

38.6.173 rhythm/in-tie

[rhythm] [Methods]

DATE:

September 14th 2018, Heidhausen

DESCRIPTION:

return T or NIL depending on whether the rhythm/event is part of a tie (either tied-from or tied-to)

ARGUMENTS:

```
the rhythm/event object
```

RETURN VALUE:

```
T or NIL
```

SYNOPSIS:

```
(defmethod in-tie ((r rhythm))
```

38.6.174 rhythm/is-multiple

[rhythm] [Methods]

DESCRIPTION:

```
Determines if the value of one rhythm object is a multiple of the value of
a second rhythm object. This is established by dividing the one by the
other and checking to see if the quotient is a whole number.
```

ARGUMENTS:

```
- A first rhythm object.
- A second rhythm object.
```

RETURN VALUE:

```
Returns T if true and NIL if not. Always also returns the quotient.
```

EXAMPLE:

```
(let ((r1 (make-rhythm 'q))
      (r2 (make-rhythm 'e)))
  (is-multiple r1 r2))

=> T, 2.0

(let ((r1 (make-rhythm 'q))
      (r2 (make-rhythm 'e.)))
  (is-multiple r1 r2))

=> NIL, 1.3333333333333333
```

SYNOPSIS:

```
(defmethod is-multiple ((r1 rhythm) (r2 rhythm))
```

38.6.175 rhythm/just-attacks

[rhythm] [Functions]

DATE:

```
February 9th 2022
```

DESCRIPTION:

```
Remove any rhythms/events from a list that don't represent an attack,
i.e. remove all rests and those tied-to.
```

ARGUMENTS:

```
a list of rhythm/event objects
```

RETURN VALUE:

```
the list purged of all rests and those tied-to
```

SYNOPSIS:

```
(defun just-attacks (rthms)
```

38.6.176 rhythm/make-rhythm

[rhythm] [Functions]

DESCRIPTION:

```
Make a rhythm object.
```

ARGUMENTS:

```
- A duration either as a numeric representation of a rhythm (subdivision of
  a whole note; 2 = half note, 4 = quarter, 8 = eighth etc), a quoted
  alphabetic shorthand for a duration (ie, 'h, 'q, 'e etc.), or an absolute
  duration in seconds.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :is-rest. T or NIL to denote whether the given duration is a rest or
  not. T = rest. Default = NIL.
```

- :is-tied-to. T or NIL to denote whether the given duration is tied later to the next duration in a given rthm-seq-bar/rthm-seq object. T = tied. Default = NIL.
- :duration. Indicates whether the duration argument has been given as a duration in seconds, not a known rhythm like 'e or 8. T indicates that the duration is a duration in seconds. Default = NIL.
- :tempo. Indicates the tempo for the given rhythm and is used only when :duration is set to figure out the rhythm type (1/8, 1/4 etc.) from the two values. So this is not related to any tempi applied, rather one that is reflected in the duration-in-tempo slot of event.

RETURN VALUE:

A rhythm object.

EXAMPLE:

```
(make-rhythm 16)

=>
RHYTHM: value: 16.0, duration: 0.25, rq: 1/4, is-rest: NIL, score-rthm: 16.0,
        undotted-value: 16, num-flags: 2, num-dots: 0, is-tied-to: NIL,
        is-tied-from: NIL, compound-duration: 0.25, is-grace-note: NIL,
        needs-new-note: T, beam: NIL, bracket: NIL, rqq-note: NIL,
        rqq-info: NIL, marks: NIL, marks-in-part: NIL, letter-value: 16,
        tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: 16, tag: NIL,
data: 16

(make-rhythm 16 :is-rest t :is-tied-to t)

=>
RHYTHM: value: 16.0, duration: 0.25, rq: 1/4, is-rest: T, score-rthm: 16.0,
        undotted-value: 16, num-flags: 2, num-dots: 0, is-tied-to: T,
        is-tied-from: NIL, compound-duration: 0.25, is-grace-note: NIL,
        needs-new-note: NIL, beam: NIL, bracket: NIL, rqq-note: NIL,
        rqq-info: NIL, marks: NIL, marks-in-part: NIL, letter-value: 16,
        tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: 16, tag: NIL,
data: 16

(make-rhythm .23 :duration t)

=>
```

```
RHYTHM: value: 17.391304, duration: 0.23, rq: 23/100, is-rest: NIL, score-rthm: NIL,
        undotted-value: -1.0, num-flags: 0, num-dots: 0, is-tied-to: NIL,
        is-tied-from: NIL, compound-duration: 0.23, is-grace-note: NIL,
        needs-new-note: T, beam: NIL, bracket: NIL, rqq-note: NIL,
        rqq-info: NIL, marks: NIL, marks-in-part: NIL, letter-value: -1,
        tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: NIL
```

SYNOPSIS:

```
(defun make-rhythm (rthm &key is-rest is-tied-to is-tied-from duration
                            (tempo 60.0))
```

38.6.177 rhythm/mid-tie

[rhythm] [Methods]

DATE:

```
September 14th 2018, Heidhausen

DESCRIPTION return T or NIL depending on whether the rhythm/event is in the
middle of a tie (i.e. both tied-from or tied-to are T)
```

ARGUMENTS:

```
the rhythm/event object
```

RETURN VALUE:

```
T or NIL
```

SYNOPSIS:

```
(defmethod mid-tie ((r rhythm))
```

38.6.178 rhythm/replace-mark

[rhythm] [Methods]

DESCRIPTION:

Replace a specified mark of a given rhythm object with a second specified mark. If a rhythm object contains more than one mark, individual marks can be changed without modifying the remaining marks. NB works with strings, symbols, lists, numbers or anything that is #'string= or #'equal

ARGUMENTS:

- A rhythm object.
- The mark to be replaced.
- The new mark.

RETURN VALUE:

Returns the new value of the MARKS slot of the given object.

EXAMPLE:

```
;; Make a rhythm object, add the mark 'a, then replace 'a with 's
(let ((r (make-rhythm 'q)))
  (add-mark r 'a)
  (replace-mark r 'a 's))

=> (S)

;; Make a rhythm object, add a list of marks, replace just the 'pizz mark with
;; a 'batt mark

(let ((r (make-rhythm 'q)))
  (loop for m in '(a s pizz col-legno) do (add-mark-once r m))
  (replace-mark r 'pizz 'batt))

=> (COL-LEGNO BATT S A)
```

SYNOPSIS:

```
(defmethod replace-mark ((r rhythm) what with &optional ignore1 ignore2 ignore3)
```

38.6.179 rhythm/rhythm-equal

[rhythm] [Methods]

DESCRIPTION:

Compares the values of two rhythm objects to determine if they are equal.

NB rhythm-equal compares the values only, so rhythms with the same values will still be considered equal even if their other attributes (such as :is-rest and :is-tied-to etc.) are different.

ARGUMENTS:

- A first rhythm object.
- A second rhythm object.

RETURN VALUE:

T if the values of the given rhythm objects are equal, else NIL.

EXAMPLE:

```
(let ((r1 (make-rhythm 4))
      (r2 (make-rhythm 4)))
  (rhythm-equal r1 r2))

=> T

(let ((r1 (make-rhythm 4))
      (r2 (make-rhythm 8)))
  (rhythm-equal r1 r2))

=> NIL

(let ((r1 (make-rhythm 4 :is-rest T))
      (r2 (make-rhythm 4 :is-rest NIL)))
  (rhythm-equal r1 r2))

=> T

(let ((r1 (make-rhythm 4 :is-tied-to T))
      (r2 (make-rhythm 4 :is-tied-to NIL)))
  (rhythm-equal r1 r2))

=> T
```

SYNOPSIS:

```
(defmethod rhythm-equal ((r1 rhythm) (r2 rhythm))
```

38.6.180 rhythm/rhythm-list

[rhythm] [Functions]

DESCRIPTION:

Create a list of rhythms from symbols, possibly involving ties and not needing meters etc. (i.e. not as strict as rthm-seq).

ARGUMENTS:

- The list of rhythm symbols.

OPTIONAL ARGUMENTS:

- T or NIL indicates whether to create a circular-sclist from the result. If NIL, a simple list will be returned (default = NIL).

RETURN VALUE:

A list or circular-sclist of the rhythm objects.

EXAMPLE:

```
;; Create a list of rhythm objects
(rhythm-list '(q w+e q. h.+s e.+ts))

=>(
RHYTHM: value: 4.0f0, duration: 1.0
[...]
RHYTHM: value: 1.0f0, duration: 4.0
[...]
RHYTHM: value: 8.0f0, duration: 0.5
[...]
RHYTHM: value: 2.6666666666666665, duration: 1.5
[...]
RHYTHM: value: 1.3333333333333333, duration: 3.0
[...]
RHYTHM: value: 16.0f0, duration: 0.25
[...]
RHYTHM: value: 5.333333333333333, duration: 0.75
[...]
RHYTHM: value: 24.0f0, duration: 0.16666666666666666
)

;; Collect the data from each of the individual rhythm objects in the list.
(let ((rl (rhythm-list '(q w+e q. h.+s e.+ts))))
  (print (loop for r in rl collect (data r))))
```

```
=> (Q "W" "E" Q. "H." "S" "E." "TS")

;; Set the optional argument to T to create a circular-sclist instead
(rhythm-list '(q w+e q. h.+s e.+ts) t)

=>
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 8, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (
[...]
)

;; Create a circular-sclist and check that it's a circular-sclist using cscl-p
(let ((rl (rhythm-list '(q w+e q. h.+s e.+ts) t)))
  (cscl-p rl))

=> T
```

SYNOPSIS:

```
(defun rhythm-list (rthms &optional circular)
```

38.6.181 rhythm/rhythm/

[rhythm] [Methods]

DESCRIPTION:

```
Determines the ratio of one rhythm object's duration to that of a second
rhythm object by use of division.
```

ARGUMENTS:

```
- A rhythm object.
- A second rhythm object.
```

RETURN VALUE:

```
A number.
```

EXAMPLE:

```
(let ((r1 (make-rhythm 'q))
```

```
      (r2 (make-rhythm 'e)))
  (rhythm/ r1 r2))

=> 2.0

(let ((r1 (make-rhythm 'q))
      (r3 (make-rhythm 's.)))
  (rhythm/ r1 r3))

=> 2.6666667
```

SYNOPSIS:

```
(defmethod rhythm/ ((r1 rhythm) (r2 rhythm))
```

38.6.182 rhythm/rm-marks

[rhythm] [Methods]

DESCRIPTION:

Remove a specified mark (or a list of specified marks) from the MARKS slot of a given rhythm object. If the mark specified is not present in the given rhythm object's MARKS slot, a warning is printed. If some marks of a list of specified marks are present in the rhythm object's MARKS slot and other aren't, those that are will be removed and a warning will be printed for the rest.

ARGUMENTS:

- A rhythm object.
- A mark or list of marks.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether a warning is to be printed if the specified mark is not present in the given rhythm object's MARKS slot.

RETURN VALUE:

Always returns NIL.

EXAMPLE:

```
;; The method itself returns NIL
(let ((r (make-rhythm 'q)))
  (add-mark-once r 'a)
  (rm-marks r 'a))

=> NIL

;; Adding a list of marks to r, then removing only 's
(let ((r (make-rhythm 'q)))
  (loop for m in '(a s pizz col-legno x-head) do
       (add-mark-once r m))
  (rm-marks r 's)
  (marks r))

=> (X-HEAD COL-LEGNO PIZZ A)

;; Removing a list of marks from r
(let ((r (make-rhythm 'q)))
  (loop for m in '(a s pizz col-legno x-head) do
       (add-mark-once r m))
  (rm-marks r '(s a))
  (marks r))

=> (X-HEAD COL-LEGNO PIZZ)

;; Attempting to remove a mark that isn't present results in a warning
;; being printed by default
(let ((r (make-rhythm 'q)))
  (loop for m in '(a s pizz col-legno x-head) do
       (add-mark-once r m))
  (rm-marks r 'zippy))

=> NIL
WARNING: rhythm::rm-marks: no mark ZIPPY in (X-HEAD COL-LEGNO PIZZ S A)

;; Suppress printing the warning when the specified mark isn't present
(let ((r (make-rhythm 'q)))
  (loop for m in '(a s pizz col-legno x-head) do
       (add-mark-once r m))
  (rm-marks r 'zippy nil))

=> NIL
```

SYNOPSIS:

```
(defmethod rm-marks ((r rhythm) marks &optional (warn t))
```

38.6.183 rhythm/scale

[rhythm] [Methods]

DESCRIPTION:

Change the value of a rhythm object's duration value by a specified scaling factor. Bear in mind that after scaling, depending on the complexity of the rhythm, the data slot may no longer accurately represent the new rhythmic value (though of course the value, duration, and rq, etc., slots will).

ARGUMENTS:

- A rhythm object.
- A scaling factor.

OPTIONAL ARGUMENTS:

- <clone>. This argument determines whether a new rhythm object is made or the duration value of the old object is replaced. When set to T, a new object is made based on the duration value of the original. When set to NIL, the original duration value is replaced (see example). Default = T.

RETURN VALUE:

A rhythm object.

EXAMPLE:

```
(let ((r (make-rhythm 4)))
  (data (scale r 2)))
=> H

(let ((r (make-rhythm 4)))
  (data (scale r 3)))
=> H.

(let ((r (make-rhythm 4)))
  (data (scale r .5)))
=> E
```

SYNOPSIS:

```
(defmethod scale ((r rhythm) scaler &optional (clone t) ignore1 ignore2)
```

38.6.184 rhythm/scale-as-rests

[rhythm] [Methods]

DATE:

```
September 29th 2020, Heidhausen
```

DESCRIPTION:

```
Sometimes you don't want to actually scale a rhythm/event's duration rather
stretch it out with rests. This method will leave the rhythm object intact
(or cloned) before then cloning it a given number of times but as a rest.
```

ARGUMENTS:

```
- the rhythm object
- the scaler (integer)
```

OPTIONAL ARGUMENTS:

```
- T or NIL: whether to clone the first argument
```

RETURN VALUE:

```
a list of rhythm objects
```

SYNOPSIS:

```
(defmethod scale-as-rests ((r rhythm) scaler &optional (clone t))
```

38.6.185 rhythm/start-tie

[rhythm] [Methods]

DATE:

```
September 14th 2018, Heidhausen
```

DESCRIPTION:

```
return T or NIL depending on whether the rhythm/event is the start of a tie
(tied-from but not tied-to)
```

ARGUMENTS:

the rhythm/event object

RETURN VALUE:

T or NIL

SYNOPSIS:

```
(defmethod start-tie ((r rhythm))
```

38.6.186 rhythm/subtract

[rhythm] [Methods]

DESCRIPTION:

Create a new rhythm object with a duration that is equal to the difference between the duration of two other given rhythm objects.

NB: This method only returns a single rhythm rather than a list with ties. Thus h - e., for example, returns TQ...

If the resulting duration cannot be presented as a single rhythm, the DATA slot of the resulting rhythm object is set to NIL, though the VALUE and DURATION slots are still set with the corresponding numeric values.

If the resulting duration is equal to or less than 0, NIL is returned and an optional warning may be printed.

ARGUMENTS:

- A first rhythm object.
- A second rhythm object.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether a warning is to be printed when the resulting duration is less than or equal to 0. Default = 0.

RETURN VALUE:

A rhythm object if the resulting duration is greater than 0, else NIL and the optional warning.

EXAMPLE:

```
;; Make a new rhythm object with a duration equal to one quarter minus one
;; eighth.
(let ((r1 (make-rhythm 'q))
      (r2 (make-rhythm 'e)))
  (subtract r1 r2))

=>
RHYTHM: value: 8.0f0, duration: 0.5, rq: 1/2, is-rest: NIL, score-rthm: 8.0f0,
        undotted-value: 8, num-flags: 1, num-dots: 0, is-tied-to: NIL,
        is-tied-from: NIL, compound-duration: 0.5, is-grace-note: NIL,
        needs-new-note: T, beam: NIL, bracket: NIL, rqq-note: NIL,
        rqq-info: NIL, marks: NIL, marks-in-part: NIL, letter-value: 8,
        tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: E, tag: NIL,
data: E

;; A half minus a dotted eighth is represented as a triplet half
(let ((r1 (make-rhythm 'h))
      (r2 (make-rhythm 'e.)))
  (data (subtract r1 r2)))

=> TQ...

;; If the resulting duration is 0 or less, return NIL, with no warning by
;; default
(let ((r1 (make-rhythm 'e))
      (r2 (make-rhythm 'q)))
  (subtract r1 r2))

=> NIL

;; Setting the optional argument to t returns a warning when the resulting
;; duration is less than 0
(let ((r1 (make-rhythm 'e))
      (r2 (make-rhythm 'q)))
  (subtract r1 r2 t))

=> NIL
WARNING: rhythm::arithmetic: new duration is -0.5; can't create rhythm

;; Subtracting a septuplet-16th from a quarter results in a duration that
;; cannot be represented as a single rhythm, therefore setting the DATA to NIL
;; while VALUE and DURATION are still set.
(let ((r1 (make-rhythm 4))
      (r2 (make-rhythm 28)))
```

```
  (print (value (subtract r1 r2)))
  (print (duration (subtract r1 r2)))
  (print (data (subtract r1 r2))))

=>
4.666666666666666
0.8571428571428572
NIL
```

SYNOPSIS:

```
(defmethod subtract ((r1 rhythm) (r2 rhythm) &optional warn)
```

38.6.187 linked-named-object/sclist

[linked-named-object] [Classes]

NAME:

```
 player

 File:             sclist.lsp

 Class Hierarchy:  named-object -> linked-named-object -> sclist

 Version:          1.1.0

 Project:          slippery chicken (algorithmic composition)

 Purpose:          Implementation of a simple but self-checking (hence
                   sclist) list class.

 Author:           Michael Edwards: m@michael-edwards.org

 Creation date:    February 11th 2001

 $$ Last modified:  17:00:22 Tue Oct 14 2025 CEST

 SVN ID: $Id$
```

38.6.188 sclist/change-data

[sclist] [Classes]

NAME:

```
change-data

File:             change-data.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  change-data

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the change-data class.  Holds data
                  regarding parameter changes for a whole section
                  (e.g. tempo).  For use in change-map.  The data in the
                  <changes> slot is a three-element list: the sequence
                  number, the bar number of the sequence where the change
                  takes place (defaults to 1) and the new data (e.g. a
                  tempo value).

                  When giving this data, the sequence number and bar
                  numbers are always integers > 0, unlike sequences
                  themselves which may be given any kind of id.  Therefore
                  it's OK to sort the given data according to integer
                  precedence and perform numeric tests on them too.

                  No public interface envisaged (so no robodoc entries).

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    2nd April 2001

$$ Last modified: 20:31:51 Mon May 14 2012 BST

SVN ID: $Id$
```

38.6.189 change-data/get-change-data

[change-data] [Methods]

DESCRIPTION:

```
Get the change data (for example, from an instrument-change-map object) for
a specified sequence.
```

ARGUMENTS:

```
 - A change-data object.
 - An integer that is the number of the sequence within the given
   change-data object for which to retrieve the data.
```

OPTIONAL ARGUMENTS:

```
 - An integer that is the number of the bar within the specified sequence
   for which to return the change data.
```

RETURN VALUE:

```
 The change data of the specified sequence (and bar).
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((sax ((alto-sax tenor-sax) :midi-channel 1))
                     (db (double-bass :midi-channel 2))))
        :instrument-change-map '((1 ((sax ((1 alto-sax) (3 tenor-sax))))))
        :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s)
                                 (w))
                                :pitch-seq-palette ((1 2 3 4 5 6)))))
        :rthm-seq-map '((1 ((sax (1 1 1 1 1))
                            (db (1 1 1 1 1)))))))))
   (get-change-data
    (get-data '(1 sax) (instrument-change-map mini)) 2 2))

=> ALTO-SAX, NIL
```

SYNOPSIS:

```
(defmethod get-change-data ((cd change-data) sequence &optional (bar 1))
```

38.6.190 change-data/make-change-data

[change-data] [Functions]

DESCRIPTION:

```
 Create a change-data object, which holds data for use by a change-map
 object. The data stored in change-data object will be that of parameter
```

changes for a whole section, such as tempo values.

The data is passed to the make-change-data function as a list of three-element lists, each consisting of the number of the sequence, the number of the bar within that sequence, and the new data.

ARGUMENTS:

- An ID for the change-data object to be created.
- A list of three-item lists, each consisting of the number of the sequence in which the data is to change, the number of the bar within that sequence in which the data is to change, and the data value itself. The sequence number and bar number are always integers > 0. If no bar-number is given, it will default to 1.

RETURN VALUE:

A change-data object.

EXAMPLE:

```
(make-change-data 'cd-test '((1 1 23) (6 1 28) (18 1 35)))

=>
CHANGE-DATA:
            previous-data: NIL,
            last-data: 35
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: CD-TEST, tag: NIL,
data: ((1 1 23) (6 1 28) (18 1 35))
```

SYNOPSIS:

```
(defun make-change-data (id data)
```

38.6.191 sclist/chord

[sclist] [Classes]

NAME:

chord

File: chord.lsp

```
Class Hierarchy:  named-object -> linked-named-object -> sclist -> chord

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the chord class that is simply an
                  sclist whose data is a list of pitch instances.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    July 28th 2001

$$ Last modified:  14:23:15 Wed Jun 24 2026 CEST

SVN ID: $Id$
```

38.6.192 chord/add-harmonics

[chord] [Methods]

DESCRIPTION:

```
Adds pitches to the set which are harmonically related to the existing
pitches. The keywords are the same as for the get-harmonics function.  See
also sc-set method of the same name and get-pitch-list-harmonics. NB This
will automatically sort all pitches from high to low.
```

ARGUMENTS:

```
- a chord object
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
 see get-harmonics function
```

RETURN VALUE:

```
the same set object as the first argument but with new pitches added.
```

SYNOPSIS:

```
(defmethod add-harmonics ((c chord) &rest keywords)
```

38.6.193 chord/add-mark

[chord] [Methods]

DESCRIPTION:

```
Add the specified mark to the MARKS slot of the given chord object.
```

ARGUMENTS:

```
- A chord object.
- A mark.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether to print a warning when attempting to add a
  mark to a rest.
```

RETURN VALUE:

```
Returns the full contents of the MARKS slot of the given chord object
```

EXAMPLE:

```
;;; Returns the complete contents of the MARKS slot
(let ((chrd (make-chord '(cs4 e4 fs4 af4 bf4))))
  (add-mark chrd 'fff)
  (add-mark chrd 'pizz))

=> (PIZZ FFF)
```

SYNOPSIS:

```
(defmethod add-mark ((c chord) mark &optional warn-rest warn-again)
```

38.6.194 chord/add-pitches

[chord] [Methods]

DESCRIPTION:

```
Add the specified pitches to the given chord object. This can include
duplicate pitches for things like violin fingered and open string
unisons. If you don't want pitch duplications, call the rm-duplicates
method after calling this method.
```

ARGUMENTS:

```
- A chord object.
- The pitches to add to that object. These can be pitch objects or any data
  that can be passed to make-pitch, or indeed lists of these, as they will
  be flattened.
```

RETURN VALUE:

```
- A chord object.
```

EXAMPLE:

```
(let ((ch (make-chord '(c4 e4 g4))))
  (print (get-pitch-symbols ch))
  (add-pitches ch 'bf3)
  (print (get-pitch-symbols ch))
  (add-pitches ch 'af3 'a4 'b4)
  (print (get-pitch-symbols ch))
  (add-pitches ch '(cs5 ds5 fs5))
  (print (get-pitch-symbols ch)))

=>
(C4 E4 G4)
(BF3 C4 E4 G4)
(AF3 BF3 C4 E4 G4 A4 B4)
(AF3 BF3 C4 E4 G4 A4 B4 CS5 DS5 FS5)
```

SYNOPSIS:

```
(defmethod add-pitches ((c chord) &rest pitches)
```

38.6.195 chord/artificial-harmonic-simple?

[chord] [Methods]

DATE:

```
January 10th 2019, Heidhausen
```

DESCRIPTION:

```
Tests a chord to see if it's an artificial harmonic such as that created by
force-artificial-harmonic. Unlike artificial-harmonic? this looks for
two-note chords, the second of which has the flag-head marks and is a fourth
away i.e. this is the simplest double-octave type.
```

ARGUMENTS:

- a chord object

RETURN VALUE:

T or NIL

SYNOPSIS:

```
(defmethod artificial-harmonic-simple? ((c chord))
```

38.6.196 chord/artificial-harmonic?

[chord] [Methods]

DATE:

October 4th 2014

DESCRIPTION:

Determine whether a chord represents an artificial harmonic of the type that strings play. An artificial harmonic here is defined as a three note chord where the second note has a 'flag-head mark (i.e diamond shape) and the 1st and 3rd are related in frequency by an integer ratio.

NB What we don't do (yet) is test whether the 2nd note is the correct nodal point to produce the given pitch.

ARGUMENTS:

- a chord object

OPTIONAL ARGUMENTS:

- cents-tolerance: how many cents the top note can deviate from a pure partial frequency. E.g. the 7th harmonic is about 31 cents from the nearest tempered note.

RETURN VALUE:

If the chord is an artificial harmonic then the sounding (3rd) note is returned, otherwise NIL.

SYNOPSIS:

```
(defmethod artificial-harmonic? ((c chord) &optional (cents-tolerance 31))
```

38.6.197 chord/calculate-dissonance

[chord] [Methods]

DATE:

27th July 2015, Edinburgh

DESCRIPTION:

Calculates the dissonance of a chord as it would appear when played on the piano (by default; other instruments' spectral data can also be used: see below). This uses a roughness calculation model by Pantelis N. Vassilakis as outlined at http://www.acousticslab.org/learnmoresra/moremodel.html. The model has been perceptually verified. We use spectral data handled by spectra.lsp to sum the roughness of sine pairs up to the first 12 partials of each tone, i.e. every partial of every tone is calculated in relation to all other tones' partials, taking their amplitudes into account.

NB If notes are above/below the piano range for which we have data then we use the highest/lowest spectral data available.

NB This method will be called automatically the first time you access a chord object's dissonance slot. In that case the default keyword arguments will be used. If you want to use different arguments or recalculate dissonance after the chord has been altered, this method can be used at any time but bear in mind that it does automatically change the dissonance slot so use with caution. If you merely want to change the default spectra which are used, then you can do that at startup via (set-sc-config 'default-spectra ...). You can also call rm-diss-cen to delete centroid and dissonance slot values (this exists in the set-palette class too).

ARGUMENTS:

- the chord object

OPTIONAL ARGUMENTS:

keyword arguments:
- :num-partials. The number of partials we want to use in our calculation. Default = 12.
- :average. T or NIL to indicate whether we want to use the average spectrum for an octave with the current note of a chord in the middle. Default = T.
- :spectrum. Pass a two-element list of partial frequency scalers (list)

```
   and amplitudes (list) to use instead of the piano data. The amplitudes
   should be normalised from 0.0 to 1.0 and there should be as many of these
   as frequency scalers and :num-partials. Bear in mind that even using the
   same data for several calls, this will still give different results for
   the same chord type starting on different notes, as the perceptual
   dissonance is based on pitch height as well as interval structure.
   Default = (get-sc-config 'default-spectra) (see globals.lsp)
```

RETURN VALUE:

```
 A floating point number representing the dissonance value. Higher values
 are more dissonant.
```

EXAMPLE:

```
(loop for chord in '((c4 e4 g4)
                     (d4 fs4 a4)
                     (c4 ef4 g4)
                     (c2 e2 g2)
                     (c3 e3 g3)
                     (c4 e4 g4 b4)
                     (c4 e4 g4 b4 cs4))
   collect (calculate-dissonance (make-chord chord) :num-partials 12))
-->
(0.402377778595695d0 0.3699311905936456d0 0.4174836004632471d0
 1.8766497834038562d0 1.0522737914732592d0 0.7389679913974341d0
 1.59662137873394d0)
```

SYNOPSIS:

```
(defmethod calculate-dissonance ((c chord)
                                 &key (num-partials 12) (average t)
                                   (spectrum (get-sc-config 'default-spectra)))
```

38.6.198 chord/calculate-spectral-centroid

[chord] [Methods]

DATE:

```
 July 28th 2015
```

DESCRIPTION:

```
 Calculate the spectral centroid of a chord. This technique is of course
```

usually applied in digital signal processing to an audio signal via a Fast Fourier Transform (FFT). It's perhaps a little strange to use this on pitch data, not least of which because we will not be taking into account any phase information or interference of possible harmonic partial interactions, but nevertheless we get a good indication of the "pitch height" of a chord via this method.

NB This will be called automatically the first time you access a chord object's centroid slot. In that case the default keyword arguments will be used. If you want to use different arguments or recalculate the centroid after the chord has been altered, this method can be used at any time but bear in mind that it does automatically change the centroid slot so use with caution. You can call rm-diss-cen to delete centroid and dissonance slot values (this exists in the set-palette class too).

ARGUMENTS:

- the chord object

OPTIONAL ARGUMENTS:

keyword arguments:
- :num-partials. The number of partials we want to use in our calculation. Default = 12.
- :average. T or NIL to indicate whether we want to use the average spectrum for an octave with the current note of a chord in the middle. Default = T.
- :spectrum. Pass a two-element list of partial frequency scalers (list) and amplitudes (list) to use instead of the piano data. The amplitudes should be normalised from 0.0 to 1.0 and there should be as many of these as frequency scalers and :num-partials. Default = (get-sc-config 'default-spectra) (see globals.lsp)

RETURN VALUE:

A floating point value representing the frequency in Hertz of the spectral centroid.

EXAMPLE:

```
(calculate-spectral-centroid (make-chord '(fs3 c4 cs4 d4 ds4 e4)))
 --> 692.3215865354373d0
(calculate-spectral-centroid (make-chord '(fs3 c4 cs4 d4 ds4 e4 g6)))
 --> 772.4924013974714d0
```

SYNOPSIS:

```
(defmethod calculate-spectral-centroid ((c chord)
                                        &key (num-partials 12)
                                          (average t)
                                          (spectrum
                                           (get-sc-config 'default-spectra)))
```

38.6.199 chord/chord-equal

[chord] [Methods]

DESCRIPTION:

Test to see if two chords are equal.

NB: Two unsorted chord objects that contain the exact same pitch objects in a different order will not be considered equal and will return NIL.

NB: Equality is tested on pitch content only, not on, for example, the values of the MIDI slots of those pitch objects etc.

ARGUMENTS:

- A first chord object.
- A second chord object.

RETURN VALUE:

T or NIL. T if the pitch content of the chords is equal, otherwise NIL.

EXAMPLE:

```
;; Two chords are equal
(let ((chrd1 (make-chord '(c4 e4 gqs4 bqf4 d5 f5)
                         :midi-channel 11
                         :microtones-midi-channel 12))
      (chrd2 (make-chord '(c4 e4 gqs4 bqf4 d5 f5)
                         :midi-channel 11
                         :microtones-midi-channel 12)))
  (chord-equal chrd1 chrd2))

=> T

;; Chord objects with the same pitch objects in a different order are unequal
(let ((chrd1 (make-chord '(c4 e4 gqs4 bqf4 d5 f5)
                         :midi-channel 11
```

```
                        :microtones-midi-channel 12))
      (chrd2 (make-chord '(e4 c4 gqs4 bqf4 d5 f5)
                        :midi-channel 11
                        :microtones-midi-channel 12
                        :auto-sort nil)))
  (chord-equal chrd1 chrd2))

=> NIL

;; Only the pitch content is compared. Content of other slots is irrelevant.
(let ((chrd1 (make-chord '(e4 c4 gqs4 bqf4 d5 f5)
                         :midi-channel 11
                         :microtones-midi-channel 12))
      (chrd2 (make-chord '(e4 c4 gqs4 bqf4 d5 f5)
                         :midi-channel 7
                         :microtones-midi-channel 8)))
  (chord-equal chrd1 chrd2))

=> T
```

SYNOPSIS:

```
(defmethod chord-equal ((c1 chord) (c2 chord))
```

38.6.200 chord/chord-member

[chord] [Methods]

DESCRIPTION:

Test whether a specified pitch object is a member of a given chord object.

ARGUMENTS:

- A chord object.
- A pitch object. This must be a pitch object, not just a note-name symbol, but the pitch object can be made with either a note-name symbol or a numerical hertz frequency value.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether or not the method should consider enharmonically equivalent pitches to be equal. T = enharmonics are equal. Default = T.

RETURN VALUE:

```
Similar to Lisp's "member" function, this method returns the tail of the
data (list of pitch objects) of the tested chord object starting with the
specified pitch object if that pitch is indeed a member of that list,
otherwise returns NIL.

NB: Since the method returns the tail of the given chord (the "rest" of the
    pitches after the given pitch), the result may be different depending
    on whether that chord has been auto-sorted or not.
```

EXAMPLE:

```
;; Returns the tail of pitch objects contained starting with the tested pitch
(let ((chrd (make-chord '(c4 e4 gqs4 a4 d5 f5 bqf5)
                        :midi-channel 11
                        :microtones-midi-channel 12)))
  (pitch-list-to-symbols (chord-member chrd (make-pitch 'a4))))

=> (A4 D5 F5 BQF5)

;; The chord object's default auto-sort feature might appear to affect outcome
(let ((chrd (make-chord '(d5 c4 gqs4 a4 bqf5 f5 e4)
                        :midi-channel 11
                        :microtones-midi-channel 12)))
  (pitch-list-to-symbols (chord-member chrd (make-pitch 'a4))))

=> (A4 D5 F5 BQF5)

;; Returns NIL if the pitch is not present in the tested chord object. This
;; example uses the "pitch-list-to-symbols" function to simplify the
;; pitch-object output.
(let ((chrd (make-chord '(d5 c4 gqs4 a4 bqf5 f5 e4)
                        :midi-channel 11
                        :microtones-midi-channel 12)))
  (pitch-list-to-symbols (chord-member chrd (make-pitch 'b4))))

=> NIL

;; The optional <enharmonics-are-equal> argument is set to NIL by default
(let ((chrd (make-chord '(c4 e4 a4 d5 f5))))
  (pitch-list-to-symbols (chord-member chrd (make-pitch 'ds4))))

=> NIL

;; Setting the optional <enharmonics-are-equal> argument to T
(let ((chrd (make-chord '(c4 ef4 a4 d5 f5))))
  (pitch-list-to-symbols (chord-member chrd (make-pitch 'ds4) t)))
```

```
=> (EF4 A4 D5 F5)

;; The optional <octaves-are-true> argument is NIL by default

(let ((chrd (make-chord '(c4 ef4 a4 d5 ef5 f5))))
  (pitch-list-to-symbols (chord-member chrd (make-pitch 'c5))))

=> NIL

;; If optional <octaves-are-true> argument is set to T, any occurrence of the
;; same pitch class in a different octave will be considered part of the chord
;; and return a positive result.
(let ((chrd (make-chord '(c4 ef4 a4 d5 ef5 f5))))
  (pitch-list-to-symbols (chord-member chrd (make-pitch 'c5) nil t)))

=> (C4 EF4 A4 D5 EF5 F5)
```

SYNOPSIS:

```
(defmethod chord-member ((c chord) (p pitch)
                         &optional (enharmonics-are-equal t)
                                   (octaves-are-true nil))
```

38.6.201 chord/chord=

[chord] [Methods]

DATE:

November 11th 2013

DESCRIPTION:

Test whether the pitch objects of the two chords are pitch=. Assumes both chords are sorted by pitch height. See pitch= in the pitch class for details of comparing pitches.

RETURN VALUE:

T or NIL

SYNOPSIS:

```
(defmethod chord= ((c1 chord) (c2 chord) &optional enharmonics-are-equal
                   (frequency-tolerance 0.01)) ; (src-tolerance 0.0001))
```

38.6.202 chord/collapse

[chord] [Methods]

DATE:

```
February 1st 2018, Heidhausen
```

DESCRIPTION:

```
Collapse or shift all the pitches of a chord until a single octave.
```

ARGUMENTS:

```
- a chord object
- the octave to shift the pitches into (integer)
```

OPTIONAL ARGUMENTS:

```
none
```

RETURN VALUE:

```
a new chord object with the pitches in the given object
```

EXAMPLE:

```
(print-simple (collapse (make-chord '(d4 f4 bf4 e5 b5)) 2))
--> NIL: D2 E2 F2 BF2 B2
```

SYNOPSIS:

```
(defmethod collapse ((c chord) octave)
```

38.6.203 chord/common-notes

[chord] [Methods]

DESCRIPTION:

```
Return the integer number of pitches common to two chord objects.
```

ARGUMENTS:

```
- A first chord object.
- A second chord object.
```

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether enharmonically equivalent pitches are to be considered the same pitch. T = enharmonically equivalent pitches are considered the same pitch. Default = T.
- T or NIL to indicate whether the same pitch class in different octaves is to be considered the same pitch. T = consider the same pitch class from octaves to be the same pitch. Default = NIL.

RETURN VALUE:

Returns an integer that is the number of pitches common to the two chords objects. Two further values are returned: the list of common pitches and a list of their pitch symbols.

EXAMPLE:

```
;; The following two chord objects have 3 pitches in common
(let ((chrd-1 (make-chord '(c4 e4 g4 b4 d5 f5)))
      (chrd-2 (make-chord '(d3 f3 a3 c4 e4 g4))))
  (common-notes chrd-1 chrd-2))
=> 3

;; By default, enharmonically equivalent pitches are considered to be the same
;; pitch
(let ((chrd-1 (make-chord '(c4 e4 g4 b4 d5 f5)))
      (chrd-2 (make-chord '(d3 f3 a3 c4 ff4 g4))))
  (common-notes chrd-1 chrd-2))
=> 3

;; Setting the first optional argument to NIL causes enharmonically equivalent
;; pitches to be considered separate pitches
(let ((chrd-1 (make-chord '(c4 e4 g4 b4 d5 f5)))
      (chrd-2 (make-chord '(d3 f3 a3 c4 ff4 g4))))
  (common-notes chrd-1 chrd-2 nil))
=> 2

;; By default, the same pitch class in different octaves is considered to be a
;; separate pitch
(let ((chrd-1 (make-chord '(c4 e4 g4 b4 d5 f5)))
      (chrd-2 (make-chord '(d3 f3 a3 ff4 g4 c5))))
  (common-notes chrd-1 chrd-2 t))
=> 2

;; Setting the second optional argument to T causes all pitches of the same
```

```
;; pitch class to be considered equal regardless of their octave
(let ((chrd-1 (make-chord '(c4 e4 g4 b4 d5 f5)))
      (chrd-2 (make-chord '(d3 f3 a3 ff4 g4 c5))))
  (common-notes chrd-1 chrd-2 t t))
=> 5
```

SYNOPSIS:

```
(defmethod common-notes ((c1 chord) (c2 chord)
                         &optional (enharmonics-are-equal t)
                         (octaves-are-true nil))
```

38.6.204 chord/contains-pitches

[chord] [Methods]

DESCRIPTION:

Check to see if a given chord object contains pitch objects for all of the specified note-names. The method returns NIL if any one of the specified pitches is not found in the given chord object.

ARGUMENTS:

- An chord object.
- A list of note-name symbols or pitch objects. NB: If checking for only one pitch, that pitch must be passed as a single-item list.

RETURN VALUE:

T or NIL.

EXAMPLE:

```
;; Returns T when all specified pitches are contained in the given chord
;; object
(let ((c (make-chord '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6))))
  (contains-pitches c '(d2 e3 gf4 af5)))

=> T

;; Returns NIL if any one of the specified pitches is not contained in the
;; given chord object.
(let ((c (make-chord '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6))))
  (contains-pitches c '(d2 e3 gf4 b4 af5)))
```

```
=> NIL
```

SYNOPSIS:

```
(defmethod contains-pitches ((c chord) pitches
                             &optional enharmonics-are-equal octaves-are-true)
```

38.6.205 chord/delete-marks

[chord] [Methods]

DESCRIPTION:

Delete all marks from the MARKS slot of the given chord object.

ARGUMENTS:

- A chord object.

RETURN VALUE:

Returns NIL.

EXAMPLE:

```
;;; Make a chord object, add two marks, and print the MARKS slot to see them;
;;; apply delete-marks and print the MARKS slot again to see the change
(let ((chrd (make-chord '(cs4 e4 fs4 af4 bf4))))
  (add-mark chrd 'fff)
  (add-mark chrd 'pizz)
  (print (marks chrd))
  (delete-marks chrd)
  (print (marks chrd)))

=>
(PIZZ FFF)
NIL
```

SYNOPSIS:

```
(defmethod delete-marks ((c chord))
```

38.6.206 chord/get-highest

[chord] [Methods]

DATE:

28/10/14

DESCRIPTION:

Assuming the auto-sort slot is T (which it is by default), return the last (= highest) pitch in the chord.

ARGUMENTS:

- a chord object

RETURN VALUE:

The highest pitch (object) in the chord.

SYNOPSIS:

```
(defmethod get-highest ((c chord))
```

38.6.207 chord/get-interval-structure

[chord] [Methods]

DESCRIPTION:

Get the distances between each pitch in a given sc-set object and the lowest (or neighbouring) pitch in that object in DEGREES (which default to quarter-tones in slippery chicken). This method assumes that the given sc-set object is sorted from low to high, which is the default action for sc-set objects.

ARGUMENTS:

- An sc-set object.

OPTIONAL
- T or NIL indicating whether to return values in semitones or default of degrees. Special case: if this argument is 'frequencies, then the interval structure will be returned as frequency differences. T =

```
   semitones. Default = NIL.
 - T or NIL to indicated whether we should return intervals from pitch to
   pitch (T) or from the lowest to each pitch (NIL). Default = NIL.
```

RETURN VALUE:

```
 A list of integers.
```

EXAMPLE:

```
;;; Returns the distances in degrees (which are quarter-tones by default
;;; in slippery chicken--use (in-scale :chromatic) at the top of your code to
;;; set to the chromatic scale):

(get-interval-structure (make-chord '(c4 e4 g4)))
-> (8.0 14.0)

;;; Return semitones
(get-interval-structure (make-chord '(c4 e4 g4)) t)
-> (4.0 7.0)

;;; Interval structure not from lowest but from pitch to pitch (ascending)
(get-interval-structure (make-chord '(c4 e4 g4 b4)) t t)
->(4.0 3.0 4.0)
```

SYNOPSIS:

```
(defmethod get-interval-structure ((c chord) &optional
                                               in-semitones neighbour)
```

38.6.208 chord/get-micro-tones

[chord] [Methods]

DATE:

```
 June 3rd 2025
```

DESCRIPTION:

```
 Get the microtones from the set (i.e. those that have the micro-tone slot
 set to T, for whatever reason). As a second value get the chromatic pitches.
```

ARGUMENTS:

```
- the chord object
```

RETURN VALUE:

```
two values: a list of microtonal pitch objects and a list of chromatic
pitch objects. The union of the two will be the whole chord.
```

SYNOPSIS:

```
(defmethod get-micro-tones ((c chord))
```

38.6.209 chord/get-midi-channel

[chord] [Methods]

DESCRIPTION:

```
Get the MIDI channel of the first pitch object contained in a given chord
object.

NB: This method returns only the midi-channel of the first pitch object in
    the chord object's data list.
```

ARGUMENTS:

```
- A chord object.
```

RETURN VALUE:

```
An integer: the midi channel
```

EXAMPLE:

```
(let ((chrd (make-chord '(c4 e4 gqs4 bqf4 d5 f5)
                        :midi-channel 11
                        :microtones-midi-channel 12)))
  (get-midi-channel chrd))

=> 11
```

SYNOPSIS:

```
(defmethod get-midi-channel ((c chord))
```

38.6.210 chord/get-nearest

[chord] [Methods]

DATE:

```
June 7th 2025
```

DESCRIPTION:

```
Return the nearest pitch object in the chord to the given pitch.
```

ARGUMENTS:

```
- a chord object
- a pitch (object or symbol)
```

RETURN VALUE:

```
- a pitch object
```

SYNOPSIS:

```
(defmethod get-nearest ((c chord) pitch &optional ignore1 ignore2)
```

38.6.211 chord/get-partials-amps

[chord] [Methods]

DATE:

```
July 28th 2015, Edinburgh
```

DESCRIPTION:

```
Return a list containing for every pitch in the chord a list of two lists:
first the partial frequency scalars (which will be integers or very close
to integers) then the normalised partial amplitudes.

Note that if spectrum are given then average is ignored. If spectrum is an
assoc-list then we get the data from it otherwise if it's just a list of
freq-scalers then amps, it's fixed and we use that repeatedly.
```

ARGUMENTS:

```
- the chord object
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :average. Whether to average spectral data over one octave centring around
each of the pictures in the chord. Default = T.
- :spectrum. The spectrum data. Default = (get-sc-config 'default-spectra)
  (see globals.lsp)
```

RETURN VALUE:

```
A list of two element sublist. of numbers.
```

EXAMPLE:

```
(get-partials-amps (make-chord '(c4 e4 g4)))
-->
(((0.99666667 1.9958333 2.9941666 3.9958336 5.002501 6.015 7.0333333 8.049168
   9.086666 10.125 11.174167 12.234166)
  (0.9447856 1.0000001 0.22975834 0.13516104 0.1625952 0.05319352 0.12118169
   2.1978654e-4 0.049031425 0.010756988 0.011732774 0.0))
 ((0.9966666 1.9958333 2.9966667 4.0025005 5.0133333 6.028333 7.0575004
   8.085834 9.136668 10.193334 11.264999 12.346667)
  (0.99999994 0.73724264 0.20146085 0.10752921 0.122650035 0.03613218
   0.06638182 0.0 0.022111647 0.004390597 0.011441806 5.167861e-4))
 ((0.9966666 1.995 3.0 4.005834 5.0208335 6.042501 7.08 8.1225 9.185 10.261666
   11.45 11.405833)
  (1.0 0.48279247 0.1664997 0.066108614 0.07859015 0.027295936 0.035848983 0.0
   0.010107763 0.0027390774 0.0040924386 0.0010194636)))
```

SYNOPSIS:

```
(defmethod get-partials-amps ((c chord)
                              &key (average t)
                                (spectrum (get-sc-config 'default-spectra)))
```

38.6.212 chord/get-pitch

[chord] [Methods]

DESCRIPTION:

```
Get the pitch object located at the specified index within the given chord
object. The <ref> argument is 1-based.
```

ARGUMENTS:

```
- A chord object.
- An integer that is the index of the pitch object sought within the data
  list of the given chord object.
```

RETURN VALUE:

```
A pitch object.
```

EXAMPLE:

```
(let ((chrd (make-chord '(c4 e4 gqs4 bqf4 d5 f5)
                        :midi-channel 11
                        :microtones-midi-channel 12)))
  (get-pitch chrd 3))

=>
PITCH: frequency: 403.482, midi-note: 67, midi-channel: 12
       pitch-bend: 0.5
       degree: 135, data-consistent: T, white-note: G4
       nearest-chromatic: G4
       src: 1.5422108173370361, src-ref-pitch: C4, score-note: GS4
       qtr-sharp: 1, qtr-flat: NIL, qtr-tone: 1,
       micro-tone: T,
       sharp: NIL, flat: NIL, natural: NIL,
       octave: 4, c5ths: 0, no-8ve: GQS, no-8ve-no-acc: G
       show-accidental: T, white-degree: 32,
       accidental: QS,
       accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: GQS4, tag: NIL,
data: GQS4
```

SYNOPSIS:

```
(defmethod get-pitch ((c chord) ref)
```

38.6.213 chord/get-pitch-symbols

[chord] [Methods]

DESCRIPTION:

```
Return the data of the pitch objects from a given chord object as a list of
note-name symbols.
```

ARGUMENTS:

```
- A chord object.
```

RETURN VALUE:

```
A list of note-name symbols.
```

EXAMPLE:

```
(let ((chrd (make-chord '(c4 e4 gqs4 bqf4 d5 f5)
                        :midi-channel 11
                        :microtones-midi-channel 12)))
  (get-pitch-symbols chrd))

=> (C4 E4 GQS4 BQF4 D5 F5)
```

SYNOPSIS:

```
(defmethod get-pitch-symbols ((c chord) &optional ignore)
```

38.6.214 chord/has-notes

[chord] [Methods]

DATE:

```
16-Aug-2010
```

DESCRIPTION:

```
Tests whether a given chord object contains at least one pitch
object.

(make-chord nil) is a valid function call and creates a chord object with
no notes.
```

ARGUMENTS:

```
- A chord object.
```

RETURN VALUE:

```
Returns T if the given chord object contains at least one pitch object,
otherwise returns NIL.
```

EXAMPLE:

```
;; Returns T if the given chord object contains at least one pitch object
(let ((chrd (make-chord '(c4))))
  (has-notes chrd))

=> T

(let ((chrd (make-chord '(c4 e4 g4))))
  (has-notes chrd))

=> T

;; Otherwise returns NIL
(let ((chrd (make-chord nil)))
  (has-notes chrd))

=> NIL
```

SYNOPSIS:

```
(defmethod has-notes ((c chord))
```

38.6.215 chord/highest

[chord] [Methods]

DESCRIPTION:

```
Return the pitch object from the given chord object that has the highest
pitch data.

NB: As opposed to the "lowest" method, this method cannot handle chord
    objects whose pitches have not been auto-sorted from low to high.
```

ARGUMENTS:

```
- A chord object.
```

RETURN VALUE:

```
A pitch object
```

EXAMPLE:

```
;; Returns the last pitch object of a chord object
(let ((chrd (make-chord '(e4 c4 gqs4 bqf4 d5 f5)
                        :midi-channel 11
                        :microtones-midi-channel 12)))
  (highest chrd))

=>
PITCH: frequency: 698.456, midi-note: 77, midi-channel: 11
       pitch-bend: 0.0
       degree: 154, data-consistent: T, white-note: F5
       nearest-chromatic: F5
       src: 2.669679641723633, src-ref-pitch: C4, score-note: F5
       qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
       micro-tone: NIL,
       sharp: NIL, flat: NIL, natural: T,
       octave: 5, c5ths: 0, no-8ve: F, no-8ve-no-acc: F
       show-accidental: T, white-degree: 38,
       accidental: N,
       accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: F5, tag: NIL,
data: F5

;; Is not capable of returning the highest pitch object from chord objects that
;; have not been auto-sorted
(let ((chrd (make-chord '(e4 c4 gqs4 bqf4 f5 d5)
                        :midi-channel 11
                        :microtones-midi-channel 12
                        :auto-sort nil)))
  (data (highest chrd)))

=> D5
```

SYNOPSIS:

```
(defmethod highest ((c chord))
```

38.6.216 chord/invert

[chord] [Methods]

DATE:

```
 May 1st 2020, Heidhaisen
```

DESCRIPTION:

Invert the interval structure of a chord. By default the given bottom note becomes the new top note, as the intervals are mirrored around it, but if the optional argument is non-NIL then the result will be transposed so that the chord has the original range.

This is quite different to the concept of traditional (tonal) chord inversion, where we move from root position to first inversion, second inversion, etc. See the trad-invert method for this.

ARGUMENTS:

- a chord object

OPTIONAL ARGUMENTS:

T or NIL to indicate whether the retain the original range or not

RETURN VALUE:

a new chord object which is the interval inversion of the argument

EXAMPLE:

```
(get-pitch-symbols (invert (make-chord '(d4 f4 bf4 e5 b5))))
--> (F2 C3 FS3 B3 D4)

(get-pitch-symbols (invert (make-chord '(d4 f4 bf4 e5 b5)) t))
--> (D4 A4 EF5 AF5 B5)
```

SYNOPSIS:

```
(defmethod invert ((c chord) &optional top-to-bottom ignore)
```

38.6.217 chord/lowest

[chord] [Methods]

DESCRIPTION:

Return the pitch object from the given chord object that has the lowest pitch data. The method can handle chord objects whose pitches have not been auto-sorted from low to high.

ARGUMENTS:

```
 - A chord object.
```

RETURN VALUE:

```
 A pitch object.
```

EXAMPLE:

```
;; Returns the pitch object of the lowest pitch despite not being sorted
(let ((chrd (make-chord '(e4 c4 gqs4 bqf4 d5 f5)
                        :midi-channel 11
                        :microtones-midi-channel 12
                        :auto-sort nil)))
  (lowest chrd))

=>
PITCH: frequency: 261.626, midi-note: 60, midi-channel: 11
       pitch-bend: 0.0
       degree: 120, data-consistent: T, white-note: C4
       nearest-chromatic: C4
       src: 1.0, src-ref-pitch: C4, score-note: C4
       qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
       micro-tone: NIL,
       sharp: NIL, flat: NIL, natural: T,
       octave: 4, c5ths: 0, no-8ve: C, no-8ve-no-acc: C
       show-accidental: T, white-degree: 28,
       accidental: N,
       accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: C4, tag: NIL,
data: C4
```

SYNOPSIS:

```
(defmethod lowest ((c chord))
```

38.6.218 chord/make-chord

[chord] [Functions]

DESCRIPTION:

```
 Create a chord object from a list of note-name symbols.
```

ARGUMENTS:

- A list of note-name symbols or frequencies.

OPTIONAL ARGUMENTS:

keyword arguments:
- :id. An element of any type that is to be the ID of the chord object created.
- :auto-sort. T or NIL to indicate whether the method should first sort the individual pitch objects created from low to high before returning the new chord object. T = sort. Default = T.
- :midi-channel. An integer that is to be the MIDI channel value to which all of the chromatic pitch objects in the given chord object are to be set for playback. Default = 1.
- :microtones-midi-channel. An integer that is to be the MIDI channel value to which all of the microtonal pitch objects in the given chord object are to be set for playback. Default = 1. NB: See player.lsp/make-player for details on microtones in MIDI output.
- :force-midi-channel. T or NIL to indicate whether to force a given value to the MIDI-CHANNEL slot, even if the notes passed to the method are already pitch objects with non-zero MIDI-CHANNEL values.

RETURN VALUE:

A chord object.

EXAMPLE:

```
;; Simple usage with default values for keyword arguments
(make-chord '(c4 e4 g4 b4 d5 f5))

=>
CHORD: auto-sort: T, marks: NIL, micro-tone: NIL
SCLIST: sclist-length: 6, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (
PITCH: frequency: 261.626, midi-note: 60, midi-channel: 0
[...]
data: C4
PITCH: frequency: 329.628, midi-note: 64, midi-channel: 0
[...]
data: E4
[...]
PITCH: frequency: 391.995, midi-note: 67, midi-channel: 0
[...]
```

```
data: G4
[...]
PITCH: frequency: 493.883, midi-note: 71, midi-channel: 0
[...]
data: B4
[...]
PITCH: frequency: 587.330, midi-note: 74, midi-channel: 0
[...]
data: D5
[...]
PITCH: frequency: 698.456, midi-note: 77, midi-channel: 0
[...]
data: F5
)

;; By default the pitches are first sorted low to high
(let ((mc (make-chord '(e4 c4 g4 b4 f5 d5))))
  (loop for p in (data mc) collect (data p)))

=> (C4 E4 G4 B4 D5 F5)

;; Setting the :midi-channel and :microtones-midi-channel arguments results in
;; the MIDI-CHANNEL slot of each of the contained pitch objects being set
;; accordingly, depending on whether it is a chromatic or microtonal pitch
(let ((mc (make-chord '(cqs4 e4 gqf4 b4 dqf5 f5)
                      :midi-channel 11
                      :microtones-midi-channel 12)))
  (loop for p in (data mc) collect (midi-channel p)))

=> (12 11 12 11 12 11)
```

SYNOPSIS:

```
(defun make-chord (note-list &key (id nil) (auto-sort t) (midi-channel 1)
                   (microtones-midi-channel 1) (force-midi-channel t))
```

38.6.219 chord/make-chord-from-intervals

[chord] [Functions]

DATE:

```
May 1st 2020, Heidhausen
```

DESCRIPTION:

```
Make a chord object from a starting pitch and a list of intervals in
semitones. NB the intervalls are cumulative i.e. each number represents the
interval from the last note, not the starting pitch.
```

ARGUMENTS:

```
- the starting pitch: symbol or pitch object
- the list of intervals in semitones (may be fractional if current scale is
microtonal)
```

OPTIONAL ARGUMENTS:

```
any keyword argument accepted by make-chord
```

RETURN VALUE:

```
a chord object
```

EXAMPLE:

```
(get-pitch-symbols (make-chord-from-intervals 'c4 '(6 5 7.5) :midi-channel 7))
--> (C4 FS4 B4 GQF5)
```

SYNOPSIS:

```
(defun make-chord-from-intervals (start-pitch intervals
                                  &rest keyargs &key &allow-other-keys)
```

38.6.220 chord/morph

[chord] [Methods]

DATE:

```
May 3rd 2016, Edinburgh
```

DESCRIPTION:

```
Attempt to morph between two chords by selecting some notes from each
according to the amount given.
```

ARGUMENTS:

```
- the first chord object
- the second chord object
- the amount to morph from chord 1 to 2. This should be a number between 0
and 1 where 0 would mean all chord 1, 1 would mean all chord 2 and anything
else is an actual morph.
```

RETURN VALUE:

```
A new chord object representing a mixture of pitches from the two
chords. The length of this will depend on the relative lengths of the two
chords but the minimum will be the length of the smallest chord and the
maximum the length of the largest chord (in terms of number of pitches).

NB If you call this method on an sc-set object the subsets and related-sets
slots will be NIL (i.e. not copied over/morphed from either of the two
arguments).
```

EXAMPLE:

```
(print-simple
 (morph (make-chord '(c4 e4 g4 b4)) (make-chord '(df4 f4 af4 c5)) 0.5))
--> C4 F4 G4 C5
```

SYNOPSIS:

```
(defmethod morph ((c1 chord) (c2 chord) amount)
```

38.6.221 chord/no-accidental

[chord] [Methods]

DESCRIPTION:

```
Set the SHOW-ACCIDENTAL slot of all pitch objects within a given chord
object to NIL. This results in no accidentals for the given chord being printed
when written to a score, and also excludes the writing of any accidentals
for that chord in parentheses.
```

ARGUMENTS:

```
- A chord object.
```

RETURN VALUE:

```
Always returns NIL.
```

EXAMPLE:

```
;;; Make a chord, print the SHOW-ACCIDENTAL slots of the pitch objects it
;;; contains; then call the method and print the same slots again to see the
;;; change.

(let ((chrd (make-chord '(cs4 e4 fs4 af4 bf4))))
  (print (loop for p in (data chrd) collect (show-accidental p)))
  (no-accidental chrd)
  (print (loop for p in (data chrd) collect (show-accidental p))))

=>
(T T T T T)
(NIL NIL NIL NIL NIL)
```

SYNOPSIS:

```
(defmethod no-accidental ((c chord))
```

38.6.222 chord/num-notes

[chord] [Methods]

DATE:

```
June 22nd 2019, Heidhausen
```

DESCRIPTION:

```
A helper function to return the number of notes in a chord (which is merely
the sclist-length slot as a chord is a child class of sclist)
```

ARGUMENTS:

```
- the chord object
```

RETURN VALUE:

```
an integer
```

SYNOPSIS:

```
(defmethod num-notes ((c chord))
```

38.6.223 chord/output-midi-note

[chord] [Methods]

DESCRIPTION:

```
Generate the MIDI-related data for each pitch in a given chord object.
```

ARGUMENTS:

```
- A chord object.
- A number that is the start time in seconds of the given chord within the
  output MIDI file.
- A decimal number between 0.0 and 1.0 that is the amplitude of the given
  chord in the output MIDI file.
- A number that is the duration in seconds of the given chord in the output
  MIDI file.
```

RETURN VALUE:

```
The corresponding data in list form.
```

EXAMPLE:

```
;; Generate the MIDI-related data required for a 5-note chord that starts 100
;; seconds into the output MIDI file, with an amplitude of 0.5 and a duration
;; of 13.0 seconds.
(let ((chrd (make-chord '(cs4 e4 fs4 af4 bf4))))
  (output-midi-note chrd 100.0 0.5 13.0))

=> (#i(midi time 100.0 keynum 61 duration 13.0 amplitude 0.5 channel -1)
    #i(midi time 100.0 keynum 64 duration 13.0 amplitude 0.5 channel -1)
    #i(midi time 100.0 keynum 66 duration 13.0 amplitude 0.5 channel -1)
    #i(midi time 100.0 keynum 68 duration 13.0 amplitude 0.5 channel -1)
    #i(midi time 100.0 keynum 70 duration 13.0 amplitude 0.5 channel -1))
```

SYNOPSIS:

```
(defmethod output-midi-note ((c chord) time amplitude duration)
```

38.6.224 chord/pitch-

[chord] [Methods]

DESCRIPTION:

```
Determine the difference between the lowest pitch of two chords. This
method can be used, for example, to compare the written and sounding
versions of a chord to determine transposition.

If the lower chord is passed as the first argument, the method will return
a negative number.

NB: This method takes pitch bend into consideration when calculating.
```

ARGUMENTS:

```
- A first chord object.
- A second chord object.
```

RETURN VALUE:

```
A positive or negative decimal number.
```

EXAMPLE:

```
;; The method measures the distance between the first (lowest) pitches of the
;;; chord only.
(let ((chrd-1 (make-chord '(c4 e4 g4)))
      (chrd-2 (make-chord '(d4 e4 fs4 a4))))
  (pitch- chrd-2 chrd-1))

=> 2.0

;;; Passing the lower chord as the first argument produces a negative result
(let ((chrd-1 (make-chord '(c4 e4 g4)))
      (chrd-2 (make-chord '(d4 e4 fs4 a4))))
  (pitch- chrd-1 chrd-2))

=> -2.0
```

SYNOPSIS:

```
(defmethod pitch- ((c1 chord) (c2 chord))
```

38.6.225 chord/pitch-or-chord=

[chord] [Methods]

DATE:

```
Wed 18 Sep 2019 17:37:39 BST - London
```

DESCRIPTION:

```
Convenience method, test to see if the pitch-or-chord slots of two event
objects are the same. See other related methods in event.lsp
```

ARGUMENTS:

```
- a pitch or chord object
- a pitch or chord object
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether or not enharmonic pitches are considered
  equal. T = enharmonic pitches are considered equal. Default = NIL.
- a number to indicate the frequency deviation allowed before returning NIL.
```

RETURN VALUE:

```
T if the values of the two specified pitch or chord objects are equal,
otherwise NIL.
```

EXAMPLE:

```
;; Comparison of equal pitch objects created using note-name symbols returns T
(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'c4)))
  (pitch-or-chord= p1 p2))

=> T

(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'bs3)))
  (pitch-or-chord= p1 p2))

=> NIL

(let ((p1 (make-pitch 'c4))
      (p2 (make-pitch 'bs3)))
  (pitch-or-chord= p1 p2 t))

=> T

(let ((p1 (make-pitch 'c4))
```

```
      (c1 (make-chord '(c4 e4 g4))))
  (pitch-or-chord= p1 c1))

=> NIL

(let ((c1 (make-chord '(c4 e4 g4)))
      (c2 (make-chord '(c4 e4))))
  (pitch-or-chord= c1 c2))

=> NIL

(let ((c1 (make-chord '(c4 e4 g4)))
      (c2 (make-chord '(bs4 ff4 g4))))
  (pitch-or-chord= c1 c2))

=> NIL

(let ((c1 (make-chord '(c4 e4 g4)))
      (c2 (make-chord '(bs3 ff4 g4))))
  (pitch-or-chord= c1 c2 t))

=> T

;; Chords with only one pitch are the same as pitch object with the same pitch.
(let ((c (make-chord '(c4)))
      (p (make-pitch 'c4)))
  (pitch-or-chord= c p))

=> T
```

SYNOPSIS:

```
(defmethod pitch-or-chord= ((c1 chord) (c2 chord)
                            &optional enharmonics-are-equal
                              (frequency-tolerance 0.01))
```

38.6.226 chord/respell-chord

[chord] [Methods]

DESCRIPTION:

```
Respell the pitches of a given chord object to improve interval structure;
i.e., removing augmented intervals etc.
```

This method respells pitches from the bottom of the chord upwards. It does not process the pitches downwards again once pitches has been changed. Instead, it reattempts the whole respelling with the enharmonic of the lowest pitch to determine which spelling produces the fewest accidentals.

NB: Respelling pitches in a chord is a rather complex process and is by no means fool-proof. The process employed here is based on avoiding double accidentals; thus, since both FS4 and BS4 have single sharps, the BS4 won't be changed to C5.

ARGUMENTS:

- A chord object.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether to print feedback from the process to the listener. T = print. Default = NIL.

RETURN VALUE:

A chord object.

EXAMPLE:

```
(let ((chrd (make-chord '(a3 ds4 f4 fs5 c6))))
  (pitch-list-to-symbols (data (respell-chord chrd t))))

=> (A3 EF4 F4 GF5 C6)
```

SYNOPSIS:

```
(defmethod respell-chord ((c chord) &optional verbose)
```

38.6.227 chord/rm-pitches

[chord] [Methods]

DESCRIPTION:

Remove the specified pitches from an existing chord object.

ARGUMENTS:

```
 - A chord object.
 - The pitches to remove from that object. These can be pitch objects or any
   data that can be passed to make-pitch, or indeed lists of these, as they
   will be flattened. NB: No warning/error will be signalled if the pitches
   to be removed are not actually in the chord.
```

RETURN VALUE:

```
 - A chord object.
```

EXAMPLE:

```
(let ((ch (make-chord '(af3 bf3 c4 e4 g4 a4 b4 cs5 ds5 fs5))))
  (print (get-pitch-symbols ch))
  (rm-pitches ch 'bf3)
  (print (get-pitch-symbols ch))
  (rm-pitches ch 'af3 'a4 'b4)
  (print (get-pitch-symbols ch))
  (rm-pitches ch '(cs5 ds5 fs5))
  (print (get-pitch-symbols ch)))

=>
(AF3 BF3 C4 E4 G4 A4 B4 CS5 DS5 FS5)
(AF3 C4 E4 G4 A4 B4 CS5 DS5 FS5)
(C4 E4 G4 CS5 DS5 FS5)
(C4 E4 G4)
```

SYNOPSIS:

```
(defmethod rm-pitches ((c chord) &rest pitches)
```

38.6.228 chord/round-to-nearest

[chord] [Methods]

DATE:

```
 October 18th 2018
```

DESCRIPTION:

```
 Round the pitches in the chord to the nearest pitches in the current
 or given scale. See the pitch class for more details.
```

ARGUMENTS:

```
the chord object
```

OPTIONAL ARGUMENTS:

```
keyword argument:
:scale. The scale to use when rounding. (Common Music tuning object or
symbol). If a symbol, then 'chromatic-scale, 'twelfth-tone, or 'quarter-tone
only at present. Default is the current scale as set by (in-scale :...).
```

RETURN VALUE:

```
the modified chord object
```

SYNOPSIS:

```
(defmethod round-to-nearest ((c chord) &key (scale cm::*scale*))
```

38.6.229 chord/set-midi-channel

[chord] [Methods]

DESCRIPTION:

```
Set the MIDI channel of the pitch objects in a given chord object to the
specified values.
```

ARGUMENTS:

```
- A chord object.
- An integer that is to be the MIDI channel for chromatic pitches in the
  given chord object.
- An integer that is to be the MIDI channel for microtonal pitches in the
  given chord object. NB: See player.lsp/make-player for details on
  microtones in MIDI output.
```

RETURN VALUE:

```
Always returns NIL.
```

EXAMPLE:

```
;; Returns NIL
(let ((chrd (make-chord '(c4 e4 gqs4 bqf4 d5 f5)
                        :midi-channel 11
```

```
                          :microtones-midi-channel 12)))
  (set-midi-channel chrd 3 4))

=> NIL

;; Print the value of the MIDI slot for each of the pitch objects contained in
;; the chord object before and after setting
(let ((chrd (make-chord '(c4 e4 gqs4 bqf4 d5 f5)
                          :midi-channel 11
                          :microtones-midi-channel 12)))
  (print (loop for p in (data chrd) collect (midi-channel p)))
  (set-midi-channel chrd 3 4)
  (print (loop for p in (data chrd) collect (midi-channel p))))

=>
(11 11 12 12 11 11)
(3 3 4 4 3 3)
```

SYNOPSIS:

```
(defmethod set-midi-channel ((c chord) midi-channel
                             &optional microtones-midi-channel)
```

38.6.230 chord/similarity

[chord] [Methods]

DATE:

```
12th August 2015, Wals, Austria
```

DESCRIPTION:

```
Calculates the similarity of two chord objects on a scale of 0.0-1.0. This
weights equally the number of common notes and the interval structure
between adjacent notes.
```

ARGUMENTS:

```
- the first chord object
- the second chord object
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether enharmonic notes should be treated as
```

```
   equal. Even when NIL, enharmonic equivalents will still score high
   overall, just not perfect.
 - T or NIL to indicate whether octave equivalents such as f#1 and f#2
   should result in higher scores.
```

RETURN VALUE:

```
a number between 0.0 and 1.0, with 1.0 being the same exact chord and 0.0
being two chords with no notes and no adjacent intervals in common.
```

EXAMPLE:

```
(similarity (make-chord '(c4 e4 g4)) (make-chord '(c4 e4 g4)))
=> 1.0
(similarity (make-chord '(c4 e4 g4)) (make-chord '(df4 f4 af4)))
=> 0.5
(similarity (make-chord '(c1 cs1 f1 b1)) (make-chord '(d1 e1 fs1 gs1 as1)))
=> 0.0
(similarity (make-chord '(cs4 es4 gs4)) (make-chord '(df4 f4 af4)))
=> 1.0
(similarity (make-chord '(cs4 es4 gs4)) (make-chord '(df4 f4 af4)) nil)
=> 0.75
```

SYNOPSIS:

```
(defmethod similarity ((c1 chord) (c2 chord)
                       &optional (enharmonics-are-equal t)
                         (octaves-are-true nil))
```

38.6.231 chord/single-pitch-chord-to-pitch

[chord] [Methods]

DATE:

```
Thu 14 Nov 2019 07:44:02 GMT London
```

DESCRIPTION:

```
Turn a chord object with only one pitch in its chord slot into a pitch
object.
```

ARGUMENTS:

```
A chord object
```

OPTIONAL ARGUMENTS:

```
NIL
```

RETURN VALUE:

```
The new pitch object if the original chord contained a single pitch,
otherwise the original chord object.
```

EXAMPLE:

```
(data (single-pitch-chord-to-pitch (make-chord '(a4))))
=> A4
```

SYNOPSIS:

```
(defmethod single-pitch-chord-to-pitch ((c chord))
```

38.6.232 chord/sort-pitches

[chord] [Methods]

DESCRIPTION:

```
Sort the pitch objects contained within a given chord object and return
them as a list of pitch objects.

As an optional argument, 'ascending or 'descending can be given to indicate
whether to sort from low to high or high to low.
```

ARGUMENTS:

```
- A chord object.
```

OPTIONAL ARGUMENTS:

```
- The symbol 'ASCENDING or 'DESCENDING to indicate whether to sort the
  given pitch objects from low to high or high to low.
  Default = 'ASCENDING.
```

RETURN VALUE:

```
Returns the chord with its pitches sorted.
```

EXAMPLE:

```
;; Apply the method with no optional argument (defaults to 'ASCENDING) and
;; collect and print the data of the pitch objects in the resulting list
(let ((chrd (make-chord '(d5 c4 gqs4 bqf5 f5 e4)
                        :midi-channel 11
                        :microtones-midi-channel 12)))
  (print (loop for p in (sort-pitches chrd) collect (data p))))

=> (C4 E4 GQS4 D5 F5 BQF5)

;; Sort from high to low
(let ((chrd (make-chord '(d5 c4 gqs4 bqf5 f5 e4)
                        :midi-channel 11
                        :microtones-midi-channel 12)))
  (print (loop for p in (sort-pitches chrd 'descending) collect (data p))))

=> (BQF5 F5 D5 GQS4 E4 C4)
```

SYNOPSIS:

```
(defmethod sort-pitches ((c chord) &optional (order 'ascending))
```

38.6.233 chord/thin

[chord] [Methods]

DATE:

```
October 25th 2018, Heidhausen
```

DESCRIPTION:

```
Reduce the number of notes in a chord. See the sc-set class method for
details.
```

SYNOPSIS:

```
(defmethod thin ((c chord) &key (strength 5) remove target invert)
```

38.6.234 chord/top-to-bottom

[chord] [Methods]

DATE:

```
May 1st 2020, Heidhausen
```

DESCRIPTION:

```
Transpose a chord so that it's current top note is the new bottom note. NB
This is destructive by default
```

ARGUMENTS:

```
- a chord object
```

OPTIONAL ARGUMENTS:

```
keywords arguments:
- :destructively. Whether to modify the chord object or return a new
chord. Default = T
```

RETURN VALUE:

```
the modified chord object
```

EXAMPLE:

```
(print-simple (top-to-bottom (make-chord '(c4 e4 g4 b4))))
--> NIL: B4 EF5 FS5 BF5
```

SYNOPSIS:

```
(defmethod top-to-bottom ((c chord) &key (destructively t))
```

38.6.235 chord/transpose

[chord] [Methods]

DESCRIPTION:

```
Transpose the pitches of a given chord object by a specified number of
semitones. The specified number can be positive or negative, and may
contain a decimal segment for microtonal transposition. If passed a decimal
number, the resulting note-names will be scaled to the nearest degree of
the current tuning.
```

ARGUMENTS:

```
- A chord object.
- A positive or negative integer or decimal number indicating the number of
  semitones by which the pitches of the given chord object are to be
  transposed.
```

OPTIONAL ARGUMENTS:

```
 keyword arguments
 - :destructively. Whether to change the internal structure of the first
   argument or leave it unchanged, returning a new transposed chord object
   instead. Default = T.
 - :do-related-sets. T or NIL to indicate whether to transpose any contents
   of the RELATED-SETS slot as well. T = transpose.  Default = NIL.
 - :lowest. Don't transpose pitches in the original chord which are lower
   than  this argument. Default = NIL but eventually via the
   transpose-pitch-list function C-1 (midi note 0)
 - :highest. Don't transpose pitches in the original chord which are higher
   than this argument. Default = NIL but eventually B8 (midi note 119)
```

RETURN VALUE:

```
 the transposed chord object
```

EXAMPLE:

```
;; Returns a chord object
(let ((chrd (make-chord '(c4 e4 g4))))
  (transpose chrd 3))

=>
CHORD: auto-sort: T, marks: NIL, micro-tone: NIL
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (
[...]
)

;; Accepts positive and negative integers and decimal numbers
(let ((chrd (make-chord '(c4 e4 g4))))
  (pitch-list-to-symbols (data (transpose chrd 3))))

=> (EF4 G4 BF4)

(let ((chrd (make-chord '(c4 e4 g4))))
  (pitch-list-to-symbols (data (transpose chrd -3))))

=> (A3 CS4 E4)

(let ((chrd (make-chord '(c4 e4 g4))))
```

```
  (pitch-list-to-symbols (data (transpose chrd -3.17))))

=> (AQF3 CQS4 EQF4)
```

SYNOPSIS:

```
(defmethod transpose ((c chord) semitones
                      &key (destructively t) ; do-related-sets
                        ;; MDE Tue Aug 21 19:49:47 2018
                        lowest highest
                        ;; MDE Thu Mar  4 11:48:29 2021, Heidhausen
                        (complete-error nil))
```

38.6.236 chord/wrap

[chord] [Methods]

DATE:

```
March 5th 2016, Edinburgh
```

DESCRIPTION:

```
Take the highest pitch and wrap it around to the bottom, placing it at the
same interval to the former lowest pitch as the highest pitch is to the
next-to-highest pitch. Then, by default, we shift (transpose) the chord so
that the next-to-highest pitch is the former highest pitch.

NB If the <transpose> optional argument is T, then the lowest and highest
notes of the chord will always remain the same no matter how many times this
operation is performed.
```

ARGUMENTS:

```
- A chord object
```

OPTIONAL ARGUMENTS:

```
- the number of times to perform this operation
```

RETURN VALUE:

```
The original chord object with the new pitches (clone in advance if
necessary)
```

EXAMPLE:

```
(get-pitch-symbols (wrap (make-chord '(c4 e4 g4 b4))))
=> (C4 E4 GS4 B4)

(get-pitch-symbols (wrap (make-chord '(c4 fs4 b4 f5))))
=> (C4 GF4 C5 F5)

(get-pitch-symbols (wrap (make-chord '(df3 c4 fs4 b4 f5))))
=> (DF3 G3 FS4 C5 F5)

(get-pitch-symbols (wrap (make-chord '(df3 c4 fs4 b4 f5)) 2))
=> (DF3 GF3 C4 B4 F5)

(get-pitch-symbols (wrap (make-chord '(d4 f4 bf4 e5 b5)) 1 nil))
=> (G3 D4 F4 BF4 E5)
```

SYNOPSIS:

```
(defmethod wrap ((c chord) &optional (num-times 1) (transpose t))
```

38.6.237 sclist/circular-sclist

[sclist] [Classes]

NAME:

```
circular-sclist

File:             circular-sclist.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the circular-sclist class which offers
                  the use of a function to cycle through the values in the
                  sclist, starting at the beginning again once we've
                  reached the end.

Author:           Michael Edwards: m@michael-edwards.org
```

```
Creation date:    February 19th 2001

$$ Last modified:  11:26:03 Sat Sep 26 2020 CEST

SVN ID: $Id$
```

38.6.238 circular-sclist/afu

[circular-sclist] [Classes]

NAME:

```
afu

File:             afu.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist -> afu

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          AFU = Alternativ fuer Unentschiedener (a little play on
                  the rather silly German political party).  This translates
                  to Alternative for the Undecided. Building on the
                  circular-sclist and activity-levels class, it's meant as a
                  deterministic alternative to randomness.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    May 18th 2019

$$ Last modified:  15:11:59 Sat Mar 16 2024 CET
```

38.6.239 afu/binlist-to-proportions

[afu] [Functions]

DATE:

```
May 23rd 2019, Heidhausen
```

DESCRIPTION:

```
From a list of an arbitrary number of 1s and 0s, create a list of
```

```
corresponding proportions: Each occurrence of 1 starts a new item, with each
subsequent occurence of 0 incrementing the proportion we'll collect.

NB all 0s at the beginning of the list will be ignored until we see the
first 1.
```

ARGUMENTS:

```
a simple list of 1s and 0s
```

RETURN VALUE:

```
a list of integers
```

EXAMPLE:

```
(binlist-to-proportions '(1 0 0 1 0 1 0 1 0 0 0))
--> (3 2 2 4)

;;; the final 1 has no following 0s so its 'length' is 1
(binlist-to-proportions '(1 0 0 1 0 1 0 1 0 0 0 1))
--> (3 2 2 4 1)

;;; all leading zeros ignored
(binlist-to-proportions '(0 0 0 1 0 0 1 0 1 0 1 0 0 0 1))
--> (3 2 2 4 1)
```

SYNOPSIS:

```
(defun binlist-to-proportions (binlist)
```

38.6.240 afu/inflate-proportionally

[afu] [Functions]

DATE:

```
May 23rd 2019, Heidhausen
```

DESCRIPTION:

```
Take a list of proportions (generally integers but floating-point works too)
and create a longer list, using the proportions scaled proportionally. In
other words a list like '(1 2 3) will become '((1 2 3) (2 4 6) (3 6 9))
i.e. internally always in proportion like the original but leading of course
then to other proportions when used with permutations (see inline comments).
```

ARGUMENTS:

```
a simple list of proportions
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :invert. Whether to invert proportions (e.g. 3 becomes 1/3)  . Default =
  T.
- :rescale. Whether to scale the results to be within new
  minima/maxima. This should be a two-element list (min. max.). Default =
  '(0 1).
- :highest. The highest number to use from the original list when scaling.
  This limit is offered so that we don't end up with most of our numbers
  down at the very bottom end of our scale. Default = 4.
- :lowest. Similar to :highest but the minimum scaler we'll use.
  Default = 1.
- :reverse. Whether to alternate the original order with the reverse thereof
  when looping through the list and applying proportional scalers. Default =
  T.
- :expt. An exponent to raise our results to before the rescaling process
  (if used). Of course this completely changes proportions but there are
  times when results are far too skewed to the bottom of the range so using
  an exponent such as 0.3
```

RETURN VALUE: SYNOPSIS:

```
(defun inflate-proportionally (l &key (invert t) (rescale '(0 1))
                                   (lowest 1) (highest 4)
                                   (reverse t) (expt nil))
```

38.6.241 afu/make-afu

[afu] [Functions]

DATE:

```
May 23rd 2019, Heidhausen
```

DESCRIPTION:

```
Make an afu object. See https://michael-edwards.org/wp/?p=1227 for examples.
```

ARGUMENTS:

```
 Any keyword arguments that make-instance would accept i.e. slots such as
 start-at, level, minimum...
```

RETURN VALUE:

```
 An afu object
```

EXAMPLE:

```
(make-afu :level 3 :minimum -3 :maximum 3 :exponent .3)

AFU: start-at: 0, level: 3, period: 113
     minimum: -3, maximum: 3, num-unique: 32
     unique: (-3.0 -2.908942 -2.8265874 -2.7273216 -2.6483202 -2.5553133
              -2.4432077 -2.3418174 -2.2489507 -2.2196069 -2.1223445 -1.8698215
              -1.7599787 -1.7449956 -1.6306628 -1.5945365 -1.4747927 -1.3338206
              -1.1639004 -1.0286679 -0.86946154 -0.824985 -0.6775627 -0.5040059
              -0.29480958 -0.12831879 0.3039422 0.7751665 0.98014116 1.5123172
              2.3448153 3.0)
     binlist: (1 0 0 0 1 0 1 0 0 0 0 0 0 1 0 1 1 0 0 0 0 0 1 0 0 0 1 1 0 0 1 0
               0 0 1 0 1 0 0 0 0 0 0 1 0 1 1 0 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0
               1 0 1 0 0 0 0 0 0 1 0 1 1 0 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0 1 0
               1 0 0 0 0 0 0 1 0 1 1 0 0 0 0 0 1)
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 408, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL,
                     this: NIL,
                     next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (-2.1223445 -1.4747927 -2.5553133 -1.4747927 -0.6775627 -2.4432077
       -2.1223445 -0.6775627 -1.8698215 -2.1223445 -1.4747927 -2.5553133
       -1.4747927 -0.6775627 -2.4432077 -2.1223445 -0.6775627 -1.8698215
       -2.1223445 -1.4747927 -2.5553133 -1.4747927 -0.6775627 -2.4432077
       -2.1223445 -0.6775627 -1.8698215 -2.1223445 -1.4747927 -2.5553133
       -1.4747927 -0.6775627 -2.4432077 -0.6775627 -1.1639004 -2.7273216
       -1.1639004 -1.8698215 -2.8265874 -1.8698215 -2.4432077 -2.2196069
...
```

SYNOPSIS:

```
(defun make-afu (&rest keyargs &key &allow-other-keys)
```

38.6.242 afu/scale

[afu] [Methods]

DATE:

```
May 23rd 2019, Heidhausen
```

DESCRIPTION:

```
Scale the existing data list to be within new bounds. NB this does not check
the existing data list for its maximum and minimum values, rather it uses
the existing minimum and maximum slot values (which should be correct,
unless something went wrong).
```

ARGUMENTS:

```
- the afu object
- the new desired minimum (number). Default = NIL = use the current minimum
  slot.
```

OPTIONAL ARGUMENTS:

```
- the new desired maximum (number). Default = NIL = use the current maximum
  slot
```

RETURN VALUE:

```
the new data list
```

SYNOPSIS:

```
(defmethod scale ((a afu) new-min &optional new-max ignore1 ignore2)
```

38.6.243 circular-sclist/assoc-list

[circular-sclist] [Classes]

NAME:

```
assoc-list

File:             assoc-list.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist -> assoc-list

Version:          1.1.0
```

```
Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the assoc-list class that is somewhat
                  like the lisp association list but with more
                  error-checking.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    February 18th 2001

$$ Last modified:  16:59:19 Wed May 21 2025 CEST

SVN ID: $Id$
```

38.6.244 assoc-list/add

[assoc-list] [Methods]

DESCRIPTION:

```
Add a new element to the given assoc-list object. By default the new element
is added at the end of the data list but the optional argument can change
this.
```

ARGUMENTS:

```
- A key/data pair as a list, or a named-object.
- The assoc-list object to which it is to be added.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to add items to the front of the data list. Default = NIL = to
the end
```

RETURN VALUE:

```
Returns T if the specified named-object is successfully added to the given
assoc-list.

Returns an error if an attempt is made to add NIL to the given assoc-list
or if the given named-object is already present in the given assoc-list.
```

EXAMPLE:

```
(let ((al (make-assoc-list 'test '((jim beam)
                                   (four roses)
                                   (wild turkey)))))
  (add '(makers mark) al))

=> T

(let ((al (make-assoc-list 'test '((jim beam)
                                   (four roses)
                                   (wild turkey)))))
  (add '(makers mark) al)
  (get-data 'makers al))

=>
NAMED-OBJECT: id: MAKERS, tag: NIL,
data: MARK

(let ((al (make-assoc-list 'test '((jim beam)
                                   (four roses)
                                   (wild turkey)))))
  (add '(makers mark) al)
  (get-position 'makers al))

=> 3

(let ((al (make-assoc-list 'test '((jim beam)
                                   (four roses)
                                   (wild turkey)))))
  (add '(knob creek) al))

=> T
```

SYNOPSIS:

```
(defmethod add (named-object (al assoc-list) &optional front)
```

38.6.245 assoc-list/add-to-list-data

[assoc-list] [Methods]

DESCRIPTION:

```
Add an element of any type to the end of the data (list) associated with a
given key of a given assoc-list.

The data associated with the given key must already be a list.
```

ARGUMENTS:

```
- An item of any type.
- A given key that must be present in the given assoc-list.
- The given assoc-list.
```

RETURN VALUE:

```
Returns the whole named-object to which the new element was added.

This method will abort with an error if a key is sought which does not
exist within the given assoc-list. For such cases, use
add-to-list-data-force instead.
```

EXAMPLE:

```
(let ((al (make-assoc-list 'test '((cat felix)
                                   (dog (fido spot))
                                   (cow bessie)))))
  (add-to-list-data 'rover 'dog al))

=>
NAMED-OBJECT: id: DOG, tag: NIL,
data: (FIDO SPOT ROVER)
```

SYNOPSIS:

```
(defmethod add-to-list-data (new-element key (al assoc-list))
```

38.6.246 assoc-list/add-to-list-data-force

[assoc-list] [Methods]

DESCRIPTION:

```
Similar to add-to-list-data, but if the given key doesn't already exist in
the given assoc-list, it is first added, then the given new element is
added to that as a 1-element list.

If the given key already exists within the given assoc-list, its data must
already be in the form of a list.
```

ARGUMENTS:

```
- A (new) element of any type.
- A given key that may or may not be present in the given assoc-list.
- The the given assoc-list.
```

RETURN VALUE:

```
Returns the whole named-object to which the element was added when used
with a key that already exists within the given assoc-list.

Returns T when used with a key that does not already exist in the given
assoc-list.
```

EXAMPLE:

```
(let ((al (make-assoc-list 'test '((cat felix)
                                   (dog (fido spot))
                                   (cow bessie)))))
  (add-to-list-data-force 'rover 'dog al))

=>
NAMED-OBJECT: id: DOG, tag: NIL,
data: (FIDO SPOT ROVER)

(let ((al (make-assoc-list 'test '((cat felix)
                                   (dog (fido spot))
                                   (cow bessie)))))
  (add-to-list-data-force 'wilbur 'pig al)
  (get-keys al))

=> (CAT DOG COW PIG)
```

SYNOPSIS:

```
(defmethod add-to-list-data-force (new-element key (al assoc-list))
```

38.6.247 assoc-list/ascending-ids?

[assoc-list] [Methods]

DATE:

```
September 19th 2020, Heidhausen
```

DESCRIPTION:

```
Check whether all IDs are ascending integers (though order can be any), none
missing, optionally starting from particular integer.
```

ARGUMENTS:

```
- the assoc-list object
```

OPTIONAL ARGUMENTS:

```
- an integer representing the lowest required ID
```

RETURN VALUE:

```
T or NIL
```

EXAMPLE:

```
(ascending-ids? (make-assoc-list 'test '((1 dog) (2 cat) (3 horse)))) -> T
;;; 3 is missing
(ascending-ids? (make-assoc-list 'test '((1 dog) (2 cat) (4 horse)))) -> NIL
;; doesn't start at 2
(ascending-ids? (make-assoc-list 'test '((1 dog) (2 cat) (3 horse))) 2) -> NIL
;; missing several integers
(ascending-ids? (make-assoc-list 'test '((4 dog) (2 cat) (7 horse))) 2) -> NIL
(ascending-ids? (make-assoc-list 'test '((4 dog) (2 cat) (3 horse))) 2) -> T
```

SYNOPSIS:

```
(defmethod ascending-ids? ((al assoc-list) &optional starting-from)
```

38.6.248 assoc-list/get-data

[assoc-list] [Methods]

DESCRIPTION:

```
Return the named-object (id, tag and data) that is identified by a
specified key within a given assoc-list.

NB: This method returns the named object itself, not just the data
    associated with the key (use get-data-data for that).
```

ARGUMENTS:

```
- A symbol that is the key (id) of the named-object sought.
- The assoc-list object in which it is be sought.
```

OPTIONAL ARGUMENTS:

```
 - T or NIL to indicate whether a warning is printed if the specified key
   cannot be found within the given assoc-list. T = print. Default = T.
   Mostly we define whether we want to warn in the instance itself, but
   sometimes it would be good to warn or not on a call basis, hence the
   optional argument. If a function is passed here that will be called
   instead of warn (e.g. #'error)
```

RETURN VALUE:

```
 A named-object is returned if the specified key is found within the given
 assoc-list object.

 NIL is returned and a warning is printed if the specified key is not found
 in the given assoc-list object.
```

EXAMPLE:

```
(let ((al (make-assoc-list 'test '((jim beam)
                                   (four roses)
                                   (wild turkey)))))
  (get-data 'four al))

=>
NAMED-OBJECT: id: FOUR, tag: NIL,
data: ROSES

(let ((al (make-assoc-list 'test '((jim beam)
                                   (four roses)
                                   (wild turkey)))))
  (get-data 'jack al))

=> NIL
WARNING:
   assoc-list::get-data: Could not find data with key JACK in assoc-list with
   id TEST

(let ((al (make-assoc-list 'test '((jim beam)
                                   (four roses)
                                   (wild turkey)))))
  (get-data 'jack al t))

=> NIL
WARNING:
   assoc-list::get-data: Could not find data with key JACK in assoc-list with
   id TEST
```

```
(let ((al (make-assoc-list 'test '((jim beam)
                                   (four roses)
                                   (wild turkey)))))
  (get-data 'jack al nil))

=> NIL
```

SYNOPSIS:

```
(defmethod get-data (key (al assoc-list) &optional (warn t))
```

38.6.249 assoc-list/get-data-data

[assoc-list] [Methods]

DESCRIPTION:

```
 (Short-cut for (data (get-data ...))
 Get the data associated with the given key of the given assoc-list.
```

ARGUMENTS:

```
 - The assoc-list key symbol associated with the data list which is sought.
 - The assoc-list in which it is to be sought.
```

OPTIONAL ARGUMENTS:

```
 - T or NIL to indicate whether to print a warning if no such named-object
   can be found within the given assoc-list (default = T).
```

RETURN VALUE:

```
 If the given key is found within the given assoc-list, the data associated
 with that key is returned.
```

EXAMPLE:

```
(let ((al (make-assoc-list 'test '((jim beam)
                                   (four roses)
                                   (wild turkey)))))
  (get-data-data 'jim al))

=> BEAM
```

```
(let ((al (make-assoc-list 'test '((jim beam)
                                   (four roses)
                                   (wild turkey)))))
  (get-data-data 'jack al))

=> NIL
WARNING:
   assoc-list::get-data: Could not find data with key JACK in assoc-list with
   id TEST
```

SYNOPSIS:

```
(defmethod get-data-data (key (al assoc-list) &optional (warn t))
```

38.6.250 assoc-list/get-first

[assoc-list] [Methods]

DESCRIPTION:

```
Returns the first named-object in the DATA slot of the given assoc-list
object.
```

ARGUMENTS:

```
- An assoc-list object.
```

RETURN VALUE:

```
A named-object that is the first object in the DATA slot of the given
assoc-list object.
```

EXAMPLE:

```
(let ((al (make-assoc-list 'test '((jim beam)
                                   (four roses)
                                   (wild turkey)))))
  (get-first al))

=>
NAMED-OBJECT: id: JIM, tag: NIL,
data BEAM
```

SYNOPSIS:

```
(defmethod get-first ((al assoc-list))
```

38.6.251 assoc-list/get-keys

[assoc-list] [Methods]

DESCRIPTION:

```
Get a simple list of the keys in a given association list.
```

ARGUMENTS:

```
- An assoc-list.
```

OPTIONAL ARGUMENTS:

```
- Optional argument: T or NIL (default T) to indicate whether a warning
  should be printed when the first argument is a recursive assoc-list.
```

RETURN VALUE:

```
A list of the keys only of all top-level association list pairs in the
given assoc-list.

get-keys is a method of the assoc-list class and therefore returns only
top-level keys if accessing a recursive assoc-list.
```

EXAMPLE:

```
(let ((al (make-assoc-list 'test '((cat felix)
                                   (dog fido)
                                   (cow bessie)))))
  (get-keys al))

=> (CAT DOG COW)

(let ((al (make-assoc-list 'test '((cat felix)
                                   (dog ((scottish terrier)
                                         (german shepherd)
                                         (irish wolfhound)))
                                   (cow bessie)))))
  (get-keys al))

=> (CAT DOG COW)
```

SYNOPSIS:

```
(defmethod get-keys ((al assoc-list) &optional (warn t))
```

38.6.252 assoc-list/get-nearest

[assoc-list] [Methods]

DATE:

4/8/2015

DESCRIPTION:

In assoc-lists with numeric IDs, return the object with the closest ID to the given key.

ARGUMENTS:

- they key (number)
- the assoc-list object

RETURN VALUE:

a named-object

SYNOPSIS:

```
(defmethod get-nearest (key (al assoc-list) &optional ignore1 ignore2)
```

38.6.253 assoc-list/get-position

[assoc-list] [Methods]

DESCRIPTION:

Returns the index position (zero-based) of a named-object within a given assoc-list.

ARGUMENTS:

- The assoc-list key symbol (named-object id) of the object for which the position is sought.
- The assoc-list in which it is to be sought.

OPTIONAL ARGUMENTS:

- Optional argument: An indexing integer. In this case, get-position will search for the given object starting part-way into the list, skipping all objects located at indices lower than the given integer (default = 0).

RETURN VALUE:

```
The integer index of the named-object within the given assoc-list.

NIL is returned if the object is not present in the assoc-list starting
with the index number given as the start argument (i.e., in the entire list
if the optional start argument is omitted).
```

EXAMPLE:

```
(let ((al (make-assoc-list 'test '((jim beam)
                                   (four roses)
                                   (wild turkey)))))
  (get-position 'four al))

=> 1

(let ((al (make-assoc-list 'test '((jim beam)
                                   (four roses)
                                   (wild turkey)))))
  (get-position 'jack al))

=> NIL

(let ((al (make-assoc-list 'test '((jim beam)
                                   (four roses)
                                   (wild turkey)))))
  (get-position 'jim al 1))

=> NIL
```

SYNOPSIS:

```
(defmethod get-position (key (al assoc-list) &optional (start 0))
```

38.6.254 assoc-list/l-for-lookup

[assoc-list] [Classes]

NAME:

```
l-for-lookup

File:             l-for-lookup
```

```
Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist -> assoc-list -> l-for-lookup

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the l-for-lookup class.  The name
                  stands for L-System for Lookups (L for
                  Lindenmayer).  This provides an L-System
                  function for generating sequences of numbers
                  from rules and seeds, and then using these
                  numbers for lookups into the assoc-list.  In the
                  assoc list are stored groups of numbers, meant
                  to represent in the first place, for example,
                  rhythmic sequences.  The grouping could be as
                  follows: ((2 3 7) (11 12 16) (24 27 29) and
                  would mean that a transition should take place
                  (over the length of the number of calls
                  represented by the number of L-Sequence results)
                  from the first group to the second, then from
                  the second to the third.  When the first group
                  is in use, then we will simple cycle around the
                  given values, similar with the other groups.
                  The transition is based on a fibonacci algorithm
                  (see below).

                  The sequences are stored in the data slot. The l-sequence
                  will be a list like (3 1 1 2 1 2 2 3 1 2 2 3 2 3 3 1).
                  These are the references into the assoc-list (the 1, 2, 3
                  ids in the list below).

                  e.g. ((1 ((2 3 7) (11 16 12)))
                        (2 ((4 5 9) (13 14 17)))
                        (3 ((1 6 8) (15 18 19))))

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    15th February 2002

$$ Last modified:  12:41:03 Tue Jan 20 2026 CET

SVN ID: $Id$
```

38.6.255 l-for-lookup/count-elements

[l-for-lookup] [Functions]

DESCRIPTION:

Count the number of times each element occurs in a given list.

ARGUMENTS:

- A list of numbers or symbols (or anything which can be compared using EQL).

RETURN VALUE:

Returns a list of two-element lists, each consisting of one list element from the specified list and the number of times that element occurs in the list. If the elements are numbers, these will be sorted from low to high, otherwise symbols will be returned from most populous to least.

EXAMPLE:

```
(count-elements '(1 4 5 7 3 4 1 5 4 8 5 7 3 2 3 6 3 4 5 4 1 4 8 5 7 3 2))

=> ((1 3) (2 2) (3 5) (4 6) (5 5) (6 1) (7 3) (8 2))
```

SYNOPSIS:

```
(defun count-elements (list &optional (test #'eql))
```

38.6.256 l-for-lookup/do-lookup

[l-for-lookup] [Methods]

DESCRIPTION:

Generate an L-sequence from the rules of the specified l-for-lookup object and use it to perform the Fibonacci-based transitioning look-up of values in the specified sequences.

NB As this accesses the rules as circular lists whose states are not reset before starting, you may get (desired) different results if you call the routine more than once. In order to avoid that call (reset lflu) before do-lookup.

ARGUMENTS:

- An l-for-lookup object.
- The start seed, or axiom, that is the initial state of the L-system. This must be the key-id of one of the sequences.
- An integer that is the length of the sequence to be returned. NB: This number does not indicate the number of L-system passes, but only the number of elements in the list returned, which may be the first segment of a sequence returned by a pass that actually generates a much longer sequence.

OPTIONAL ARGUMENTS:

- A number which is the factor by which returned numerical values are to be scaled. If NIL, the method will use the value in the given l-for-lookup object's SCALER slot instead. Default = NIL. NB: The value of the given l-for-lookup object's OFFSET slot is additionally used to increase numerical values before they are returned.

RETURN VALUE:

This method returns three lists:
- The resulting sequence.
- The distribution of the values returned by the look-up.
- The L-sequence of the key-IDs.

EXAMPLE:

```
;; Create an l-for-lookup object in which the sequences are defined such that
;; the transition takes place over the 3 given lists and from x to y to z, and
;; apply the do-lookup method to see the results. Each time one of these lists
;; is accessed, it will cyclically return the next value.
(let ((lfl (make-l-for-lookup
            'lfl-test
            '((1 ((ax1 ax2 ax3) (ay1 ay2 ay3 ay4) (az1 az2 az3 az4 az5)))
              (2 ((bx1 bx2 bx3) (by1 by2 by3 by4) (bz1 bz2 bz3 bz4 bz5)))
              (3 ((cx1 cx2 cx3) (cy2 cy2 cy3 cy4) (cz1 cz2 cz3 cz4 cz5))))
            '((1 (1 2 2 2 1 1))
              (2 (2 1 2 3 2))
              (3 (2 3 2 2 2 3 3))))))
  (do-lookup lfl 1 211))

=>
(AX1 BX1 BX2 BX3 AX2 AX3 BX1 AX1 BX2 CX1 BX3 BX1 AX2 BX2 CX2 BX3 BX1 AX3 BX2
     CX3 BX3 AX1 BY1 BX1 BX2 AY1 AX2 AX3 BX3 BX1 BX2 AX1 AX2 BX3 AY2 BX1 CX1
     BY2 AX3 BX2 BX3 BX1 AX1 AY3 BX2 AX2 BY3 CY2 BX3 BX1 CX2 BY4 BX2 BX3 CX3
     CY2 BY1 AY4 BX1 CX1 BY2 BX2 AY1 BY3 CY3 BY4 AX3 BX3 BY1 BX1 AY2 AX1 BY2
```

```
     AY3 BY3 CY4 BX2 BY4 CX2 BY1 BX3 BY2 CY2 CX3 BY3 AY4 BY4 CY2 BY1 BX1 AX2
     BY2 CY3 BY3 AY1 BY4 BY1 BY2 AY2 AY3 BY3 AY4 BY4 CY4 BY1 BY2 CY2 BY3 BY4
     BY1 CY2 CY3 BY2 AY1 BY3 CY4 BY4 AY2 BY1 BY2 BY3 AY3 AY4 BY4 AY1 BY1 CZ1
     BZ1 BY2 AY2 BY3 CY2 BY4 BY1 AY3 BY2 CY2 BY3 AY4 BY4 BZ2 BY1 AZ1 AY1 AY2
     BY2 BY3 BY4 AY3 AY4 AZ2 BZ3 BY1 BY2 AY1 AY2 BZ4 AZ3 BY3 CZ2 BY4 BZ5 AY3
     BY1 CY3 BZ1 BY2 AZ4 BZ2 CZ3 BZ3 AZ5 BY3 BZ4 BY4 AY4 AZ1 AY1 BZ5 BZ1 BY1
     AZ2 AZ3 BZ2 AY2 BZ3 CY4 BY2 AZ4 BZ4 BZ5 BZ1 AZ5 AZ1 BZ2 AZ2 BY3 CZ4 BZ3
     BZ4 CY2 BZ5 BZ1 BZ2 CZ5 CZ1 BZ3 AZ3 BZ4 CZ2 BZ5),
((CX1 3) (AX3 5) (AX1 6) (BX2 11) (CX2 3) (BX3 11) (CX3 3) (BX1 12) (AX2 6)
 (AY3 7) (CY3 4) (CZ3 1) (BY4 14) (AY4 7) (AY1 8) (BY1 15) (AY2 8) (CY4 4)
 (BY2 15) (AZ4 2) (AZ5 2) (AZ1 3) (AZ2 3) (BY3 15) (CZ4 1) (CY2 9) (BZ1 5)
 (BZ25) (CZ5 1) (CZ1 2) (BZ3 5) (AZ3 3) (BZ4 5) (CZ2 2) (BZ5 5)),
(1 2 2 2 1 1 2 1 2 3 2 2 1 2 3 2 2 1 2 3 2 1 2 2 2 1 1 1 2 2 2 1 1 2 1 2 3 2 1
   2 2 2 1 1 2 1 2 3 2 2 3 2 2 2 3 3 2 1 2 3 2 2 1 2 3 2 1 2 2 2 1 1 2 1 2 3 2
   2 3 2 2 2 3 3 2 1 2 3 2 2 1 2 3 2 1 2 2 2 1 1 2 1 2 3 2 2 3 2 2 2 3 3 2 1 2
   3 2 1 2 2 2 1 1 2 1 2 3 2 2 1 2 3 2 2 1 2 3 2 1 2 2 2 1 1 1 2 2 2 1 1 1 2 2
   2 1 1 2 1 2 3 2 2 1 2 3 2 2 1 2 3 2 1 2 2 2 1 1 1 2 2 2 1 1 2 1 2 3 2 1 2 2
   2 1 1 2 1 2 3 2 2 3 2 2 2 3 3 2 1 2 3 2)
```

SYNOPSIS:

```
(defmethod do-lookup ((lflu l-for-lookup) seed stop &optional scaler)
```

38.6.257 l-for-lookup/do-lookup-linear

[l-for-lookup] [Methods]

DESCRIPTION:

```
Similar to do-lookup but here we generate a linear sequence (with
get-linear-sequence) instead of an L-System.
```

ARGUMENTS:

```
- An l-for-lookup object.
- The start seed, or axiom, that is the initial state of the linear
  system. This must be the key-id of one of the sequences.
- An integer that is the length of the sequence to be returned.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :scaler. A number which is the factor by which returned numerical values
  are to be scaled. If NIL, the method will use the value in the given
  l-for-lookup object's SCALER slot instead. Default = NIL. NB: The value
```

```
   of the given l-for-lookup object's OFFSET slot is additionally used to
   increase numerical values before they are returned.  Default = NIL.
 - :reset. T or NIL to indicate whether to reset the pointers of the given
   circular lists before proceeding. T = reset. Default = T.
```

RETURN VALUE:

```
 This method returns three lists:
 - The resulting sequence.
 - The distribution of the values returned by the look-up.
 - The L-sequence of the key-IDs.
```

EXAMPLE:

```
;; This will return the result of lookup, the number of repetitions of each
;; rule key in the result of lookup, and the linear-sequence itself.
(let* ((tune (make-l-for-lookup
              'tune
              '((1 ((2 1 8)))
                (2 ((3 4)))
                (3 ((4 5)))
                (4 ((5 1 6)))
                (5 ((6 5 7 4)))
                (6 ((4 5)))
                (7 ((4 5 1)))
                (8 ((1))))
              '((1 (1 2)) (2 (1 3 2)) (3 (1 4 3)) (4 (1 2 1)) (5 (5 3 1))
                (6 (2 5 6)) (7 (5 6 4)) (8 (3 2))))))
  (do-lookup-linear tune 1 100))
=>
(2 1 3 8 2 4 4 1 8 3 4 2 1 3 5 5 8 2 4 3 1 8 4 4 5 2 1 3 4 8 2 3 4 1 4 3 1 8 4
 5 4 2 1 3 4 8 2 3 5 6 1 8 4 3 2 1 4 4 5 8 2 3 4 1 8 3 4 5 2 1 4 3 8 2 4 5 4 1
 8 3 4 2 1 3 5 1 4 3 8 2 4 4 5 1 8 3 4 2 1 3)
((1 17) (2 15) (3 18) (4 25) (5 10) (6 1) (8 14))
(1 1 2 1 1 2 3 1 1 2 2 1 1 2 3 4 1 1 2 2 1 1 2 3 3 1 1 2 2 1 1 2 3 4 2 2 1 1 2
 3 3 1 1 2 2 1 1 2 3 4 1 1 2 2 1 1 2 3 3 1 1 2 2 1 1 2 3 4 1 1 2 2 1 1 2 3 3 1
 1 2 2 1 1 2 3 4 2 2 1 1 2 3 3 1 1 2 2 1 1 2)
```

SYNOPSIS:

```
(defmethod do-lookup-linear ((lflu l-for-lookup) seed stop
                             &key scaler (reset t))
```

38.6.258 l-for-lookup/do-simple-lookup

[l-for-lookup] [Methods]

DESCRIPTION:

```
Performs a simple look-up procedure whereby a given reference key always
returns a specific and single piece of data. This is different from
do-lookup, which performs a transitioning between lists and returns items
from those lists in a circular manner. do-simple-lookup always returns the
first element of the sequence list associated with a given key-ID.

N.B. the SCALER and OFFSET slots are ignored by this method.
```

ARGUMENTS:

```
- An l-for-lookup object.
- The start seed, or axiom, that is the initial state of the L-system. This
  must be the key-id of one of the sequences.
- An integer that is the number of elements to be returned.
```

RETURN VALUE: EXAMPLE:

```
;; Create an l-for-lookup object using three production rules and three
;; sequences of three lists. Applying do-simple-lookup returns the first
;; element of each sequence based on the L-sequence of keys created by the
;; rules of the give l-for-lookup object.
(let ((lfl (make-l-for-lookup
            'lfl-test
            '((1 ((ax1 ax2 ax3) (ay1 ay2 ay3 ay4) (az1 az2 az3 az4 az5)))
              (2 ((bx1 bx2 bx3) (by1 by2 by3 by4) (bz1 bz2 bz3 bz4 bz5)))
              (3 ((cx1 cx2 cx3) (cy2 cy2 cy3 cy4) (cz1 cz2 cz3 cz4 cz5))))
            '((1 (1 2 2 2 1 1))
              (2 (2 1 2 3 2))
              (3 (2 3 2 2 2 3 3))))))
  (do-simple-lookup lfl 1 21))

=> ((AX1 AX2 AX3) (BX1 BX2 BX3) (BX1 BX2 BX3) (BX1 BX2 BX3) (AX1 AX2 AX3)
    (AX1 AX2 AX3) (BX1 BX2 BX3) (AX1 AX2 AX3) (BX1 BX2 BX3) (CX1 CX2 CX3)
    (BX1 BX2 BX3) (BX1 BX2 BX3) (AX1 AX2 AX3) (BX1 BX2 BX3) (CX1 CX2 CX3)
    (BX1 BX2 BX3) (BX1 BX2 BX3) (AX1 AX2 AX3) (BX1 BX2 BX3) (CX1 CX2 CX3)
    (BX1 BX2 BX3))
```

SYNOPSIS:

```
(defmethod do-simple-lookup ((lflu l-for-lookup) seed stop)
```

38.6.259 l-for-lookup/fibonacci

[l-for-lookup] [Functions]

DESCRIPTION:

```
Return the longest possible list of sequential Fibonacci numbers whose
combined sum is less than or equal to the specified value. The list is
returned in descending sequential order, ending with 0.

The function also returns as a second individual value the first Fibonacci
number that is greater than the sum of the list returned.

NB: The value of the second number returned will always be equal to the sum
    of the list plus one. In most cases that number will be less than the
    number specified as the argument to the fibonacci function, and
    sometimes it will be equal to the number specified; but in cases where
    the sum of the list returned is equal to the number specified, the
    second number returned will be equal to the specified number plus one.
```

ARGUMENTS:

```
A number that is to be the test number.
```

RETURN VALUE:

```
A list of descending sequential Fibonacci numbers, of which list the last
element is 0.

Also returns as a second individual value the first Fibonacci number that
is greater than the sum of the list returned, which will always be the sum
of that list plus one.
```

EXAMPLE:

```
;; Returns a list of consecutive Fibonacci numbers from 0 whose sum is equal to
;; or less than the value specified. The second number returned is the first
;; Fibonacci number whose value is greater than the sum of the list, and will
;; always be the sum of the list plus one.
(fibonacci 5000)

 => (1597 987 610 377 233 144 89 55 34 21 13 8 5 3 2 1 1 0), 4181

;; The sum of the list
(+ 1597 987 610 377 233 144 89 55 34 21 13 8 5 3 2 1 1 0)

=> 4180
```

SYNOPSIS:

```
(defun fibonacci (max-sum)
```

38.6.260 l-for-lookup/fibonacci-divide

[l-for-lookup] [Functions]

DATE:

August 5th 2019, Heidhausen

DESCRIPTION:

Divide an integer into ever smaller proportions as determined by a Fibonacci series which sums to the largest Fibonacci number less than the given integer. The resulting list is of ascending (accumulating) numbers which are scaled to fit the given integer (optionally but by default) and (optionally but also by default) rounded. As such, this routine is useful for determining, for examples, bar numbers at which events/changes might happen.

ARGUMENTS:

- an integer

OPTIONAL ARGUMENTS:

- T or NIL: whether to scale the Fibonacci numbers to fit the given integer. Default = T.
- T or NIL: whether to round after scaling

RETURN VALUE:

A list of numbers

EXAMPLE:

```
(FIBONACCI-DIVIDE 300)
--> (116 187 230 258 274 285 291 295 297 299 300)

;;; don't scale to 300 (i.e. only increase by Fibonacci numbers but note that
;;; the Fibonacci number is added to the current number, which starts at one,
;;; hence 1+89, 90+55, 145+34 ...
(FIBONACCI-DIVIDE 300 nil)
--> (90 145 179 200 213 221 226 229 231 232 233)

;;; do scale but don't round
(FIBONACCI-DIVIDE 300 t nil)
--> (115.87983 186.69528 230.4721 257.51074 274.24893 284.54935 290.98712
     294.8498 297.4249 298.71246 300.0)
```

SYNOPSIS:

```
(defun fibonacci-divide (num &optional (scale t) (round t))
```

38.6.261 l-for-lookup/fibonacci-partial-transition

[l-for-lookup] [Functions]

DATE:

```
December 3rd 2025
```

DESCRIPTION:

```
The same as fibonacci-transition except that it doesn't complete. So we
generated more results than required, using the <progress>  argument to
determine how far along the transition we should be at the end.
```

EXAMPLE:

```
(FIBONACCI-PARTIAL-TRANSITION 100 .7) -->
(0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 0 0
 0 0 1 0 0 1 0 1 0 1 0 1 0 1 0 1 1 0 1 1 1 1)

(FIBONACCI-PARTIAL-TRANSITION 100 .5) -->
(0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0
 0 0 1 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 0 0 0)

(FIBONACCI-TRANSITION 100) -->
(0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0
 0 0 1 0 0 0 0 1 0 0 0 0 1 0 0 1 0 0 1 0 1 0 1 1 0 1 0 1 0 1 1 0 1 1 0 1 1 1 1
 0 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1)
```

SYNOPSIS:

```
(defun fibonacci-partial-transition (num-items &optional (progress 0.618)
                                                (item1 0) (item2 1) morph
                                                first-down)
```

38.6.262 l-for-lookup/fibonacci-start-at-2

[l-for-lookup] [Functions]

DESCRIPTION:

Return the longest possible list of sequential Fibonacci numbers, excluding 0 and 1, whose combined sum is less than or equal to the specified value. The list is returned in descending sequential order.

The function also returns as a second value the sum of the list.

NB: In addition to excluding 0 and 1, this function also differs from the plain fibonacci function in that the second value returned is the sum of the list rather than the first Fibonacci number greater than that sum.

ARGUMENTS:

A number that is to be the test number.

RETURN VALUE:

A list of descending sequential Fibonacci numbers, of which list the last element is 2.

Also returns as a second result the sum of the list.

EXAMPLE:

```
;; Returns a list whose sum is less than or equal to the number specified as
;; the function's only argument
(fibonacci-start-at-2 17)

=> (5 3 2), 10

(fibonacci-start-at-2 20)

=> (8 5 3 2), 18

;; Two examples showing the different results of fibonacci
;; vs. fibonacci-start-at-2

;; 1
(fibonacci 18)

=> (5 3 2 1 1 0), 13

(fibonacci-start-at-2 18)

=> (8 5 3 2), 18
```

```
;; 2
(fibonacci 20)

=> (8 5 3 2 1 1 0), 21

(fibonacci-start-at-2 20)

=> (8 5 3 2), 18
```

SYNOPSIS:

```
(defun fibonacci-start-at-2 (max-sum)
```

38.6.263 l-for-lookup/fibonacci-transition

[l-for-lookup] [Functions]

DESCRIPTION:

```
Uses Fibonacci relationships to produces a sequence that is a gradual
transition from one repeating state to a second over n repetitions. The
function gradually increases the frequency of the second repeating state
until it completely dominates.

NB: The similar but separate function fibonacci-transition-aux1 gradually
    decreases state 1 and increases state 2.
```

ARGUMENTS:

```
- An integer that is the desired number of elements in the resulting list
  (i.e., the number of repetitions over which the transition is to occur).
```

OPTIONAL ARGUMENTS:

```
- Repeating item 1 (starting state). This can be any Lisp type, including
  lists. Default = 0.
- Repeating item 2 (target state): This can also be any Lisp type.
  Default = 1.
- T or NIL to make a morph from item1 to item2. This means morphing items
  will be a morph structure. See below for an example. Default = NIL.
- T or NIL to make the morph first go down from item2 to item1 (mainly used
  by fibonacci-transitions). This can also be an envelope (see
  fibonacci-transitions for details). Default = NIL.
```

TAGS:

transitions

RETURN VALUE:

A list.

EXAMPLE:

```
;; Defaults to 0 and 1 (no optional arguments)
(fibonacci-transition 31)
=> (0 0 0 0 0 0 0 1 0 0 0 0 1 0 0 1 0 1 1 1 0 1 0 1 1 0 1 1 1 1 1)
;; Using optional arguments set to numbers
(fibonacci-transition 23 11 37)
=> (11 11 11 11 37 11 11 37 11 37 11 37 11 37 37 11 37 11 37 11 37 37 37)
;; Using lists
(fibonacci-transition 27 '(1 2 3) '(5 6 7))
=> ((1 2 3) (1 2 3) (1 2 3) (1 2 3) (5 6 7) (1 2 3) (1 2 3) (5 6 7) (1 2 3)
    (1 2 3) (5 6 7) (1 2 3) (5 6 7) (1 2 3) (5 6 7) (1 2 3) (5 6 7) (5 6 7)
    (1 2 3) (5 6 7) (5 6 7) (1 2 3) (5 6 7) (5 6 7) (5 6 7) (5 6 7) (5 6 7))
;; with morph: first going up then up/down
(fibonacci-transition 31 0 1 t) ->
(0 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.14285714285714285d0)
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.2857142857142857d0)
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.42857142857142855d0)
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.5714285714285714d0)
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.7142857142857142d0)
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.857142857142857d0) 1
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.8) #S(MORPH :I1 0 :I2 1 :PROPORTION 0.6)
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.4) #S(MORPH :I1 0 :I2 1 :PROPORTION 0.7) 1
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.7) #S(MORPH :I1 0 :I2 1 :PROPORTION 0.85) 1
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.75) 1 1 1
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.75) 1
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.75) 1 1
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.75) 1 1 1 1 1)
;; morph goes down then up
(fibonacci-transition 31 0 1 t t) -->
(0 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.666666666666667d0)
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.3333333333333334d0)
 #S(MORPH :I1 0 :I2 1 :PROPORTION 1.1102230246251565d-16)
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.25d0)
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.5d0)
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.75d0) 1
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.8) #S(MORPH :I1 0 :I2 1 :PROPORTION 0.6)
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.4) #S(MORPH :I1 0 :I2 1 :PROPORTION 0.7) 1
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.7) #S(MORPH :I1 0 :I2 1 :PROPORTION 0.85) 1
 #S(MORPH :I1 0 :I2 1 :PROPORTION 0.75) 1 1 1
```

```
#S(MORPH :I1 0 :I2 1 :PROPORTION 0.75) 1
#S(MORPH :I1 0 :I2 1 :PROPORTION 0.75) 1 1
#S(MORPH :I1 0 :I2 1 :PROPORTION 0.75) 1 1 1 1 1)
```

SYNOPSIS:

```
(defun fibonacci-transition (num-items &optional (item1 0) (item2 1) morph
                                      first-down)
```

38.6.264 l-for-lookup/fibonacci-transitions

[l-for-lookup] [Functions]

DATE:

18 Feb 2010

DESCRIPTION:

This function builds on the concept of the function fibonacci-transition by allowing multiple consecutive transitions over a specified number of repetitions. The function either produces sequences consisting of transitions from each consecutive increasing number to its upper neighbor, starting from 0 and continuing through a specified number of integers, or it can be used to produce a sequence by transitioning between each element of a user-specified list of items.

ARGUMENTS:

- An integer indicating the number of repetitions over which the transitions are to be performed.
- Either:
  - An integer indicating the number of consecutive values, starting from 0, the function is to transition (i.e. 3 will produce a sequence that transitions from 0 to 1, then from 1 to 2 and finally from 2 to 3), or
  - A list of items of any type (including lists) through which the function is to transition.

OPTIONAL ARGUMENTS:

- T or NIL to make a morph from item1 to item2, 2 to 3, etc.. This means morphing items will be a morph structure. This can also be an envelope which thus allows morphing to be switched on and off over the course of the transition. The envelope can have any x range but the y-range should be from 0 to 1. Any value < 1 means the morph object will be ignored and

either the :i1 or :i2 slot will be used instead, depending on whether the :proportion slot is > 0.5

RETURN VALUE:

A list.

EXAMPLE:

```
;; Using just an integer transitions from 0 to 1 below that integer
(fibonacci-transitions 76 4)

=> (0 0 0 0 0 0 0 1 0 0 0 0 1 0 0 1 0 1 0 1 1 0 1 0 1 1 1 1 1 1 1 2 1 1 2 1 2 1
    2 2 1 2 1 2 2 2 2 2 2 2 3 2 2 3 2 3 2 3 3 2 3 2 3 3 3 2 3 3 3 3 3 3 3 3 3 3)

;; Using a list transitions consecutively through that list
(fibonacci-transitions 152 '(1 2 3 4))

=> (1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 2 1 1 1 1 2 1 1 2 1 1 2 1 2 1 2
    2 1 2 1 2 2 1 2 2 1 2 2 2 2 2 2 2 2 2 2 2 2 3 2 2 2 2 3 2 2 3 2 2 3 2 3 2 3
    3 2 3 2 3 3 2 3 3 2 3 3 3 3 3 3 3 3 3 3 3 3 4 3 3 3 3 4 3 3 4 3 3 4 3 4 3 4
    4 3 4 3 4 4 3 4 4 3 4 4 4 4 4 3 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4)

;; A list of lists is also viable
(fibonacci-transitions 45 '((1 2 3) (4 5 4) (3 2 1)))

=> ((1 2 3) (1 2 3) (1 2 3) (1 2 3) (1 2 3) (4 5 4) (1 2 3) (1 2 3) (4 5 4)
    (1 2 3) (4 5 4) (1 2 3) (4 5 4) (1 2 3) (4 5 4) (1 2 3) (4 5 4) (1 2 3)
    (4 5 4) (4 5 4) (4 5 4) (4 5 4) (4 5 4) (3 2 1) (4 5 4) (3 2 1) (4 5 4)
    (3 2 1) (4 5 4) (3 2 1) (4 5 4) (3 2 1) (4 5 4) (3 2 1) (3 2 1) (3 2 1)
    (4 5 4) (3 2 1) (3 2 1) (3 2 1) (3 2 1) (3 2 1) (3 2 1) (3 2 1) (3 2 1))

;; using morph
(fibonacci-transitions 52 '(1 2 3) t)
=> (1 #S(MORPH :I1 1 :I2 2 :PROPORTION 0.16666666666666666d0)
      #S(MORPH :I1 1 :I2 2 :PROPORTION 0.3333333333333333d0)
      #S(MORPH :I1 1 :I2 2 :PROPORTION 0.5d0)
      #S(MORPH :I1 1 :I2 2 :PROPORTION 0.6666666666666666d0)
      #S(MORPH :I1 1 :I2 2 :PROPORTION 0.8333333333333333d0) 2
      #S(MORPH :I1 1 :I2 2 :PROPORTION 0.9)
      #S(MORPH :I1 1 :I2 2 :PROPORTION 0.5d0)
      #S(MORPH :I1 1 :I2 2 :PROPORTION 0.0d0)
      #S(MORPH :I1 1 :I2 2 :PROPORTION 0.5d0) 2
      #S(MORPH :I1 1 :I2 2 :PROPORTION 0.7)
      #S(MORPH :I1 1 :I2 2 :PROPORTION 0.85) 2
      #S(MORPH :I1 1 :I2 2 :PROPORTION 0.75) 2 2 2
```

```
      #S(MORPH :I1 1 :I2 2 :PROPORTION 0.75) 2
      #S(MORPH :I1 1 :I2 2 :PROPORTION 0.75) 2 2 2 2
      #S(MORPH :I1 2 :I2 3 :PROPORTION 0.25d0)
      #S(MORPH :I1 2 :I2 3 :PROPORTION 0.5d0)
      #S(MORPH :I1 2 :I2 3 :PROPORTION 0.75d0) 3
      #S(MORPH :I1 2 :I2 3 :PROPORTION 0.7)
      #S(MORPH :I1 2 :I2 3 :PROPORTION 0.85) 3
      #S(MORPH :I1 2 :I2 3 :PROPORTION 0.75) 3 3
      #S(MORPH :I1 2 :I2 3 :PROPORTION 0.75) 3
      #S(MORPH :I1 2 :I2 3 :PROPORTION 0.75) 3 3 3
      #S(MORPH :I1 2 :I2 3 :PROPORTION 0.75) 3 3 3 3 3 3 3 3 3)

;; using morph and a morph envelope
(fibonacci-transitions 52 '(1 2 3 3)  '(0 0 10 1 52 1))
=> (1 1 1 1 2 2 2 2 1 1 1 2 #S(MORPH :I1 1 :I2 2 :PROPORTION 0.7)
      #S(MORPH :I1 1 :I2 2 :PROPORTION 0.85) 2
      #S(MORPH :I1 1 :I2 2 :PROPORTION 0.75) 2 2 2
      #S(MORPH :I1 1 :I2 2 :PROPORTION 0.75) 2
      #S(MORPH :I1 1 :I2 2 :PROPORTION 0.75) 2 2 2 2 2 2 3 3
      #S(MORPH :I1 2 :I2 3 :PROPORTION 0.7)
      #S(MORPH :I1 2 :I2 3 :PROPORTION 0.85) 3
      #S(MORPH :I1 2 :I2 3 :PROPORTION 0.75) 3 3
      #S(MORPH :I1 2 :I2 3 :PROPORTION 0.75) 3
      #S(MORPH :I1 2 :I2 3 :PROPORTION 0.75) 3 3 3
      #S(MORPH :I1 2 :I2 3 :PROPORTION 0.75) 3 3 3 3 3 3 3 3 3)
```

SYNOPSIS:

```
(defun fibonacci-transitions (total-items levels
                              &optional morph (on-fail #'error))
```

38.6.265 l-for-lookup/get-l-sequence

[l-for-lookup] [Methods]

DESCRIPTION:

Return an L-sequence of the key-ids for the rules of a given l-for-lookup object, created using the rules of that object. This method can be called with an l-for-lookup object that contains no sequences, as it only returns a list of the key-ids for the object's rules.

Tip: It seems that systems where one rule key yields all other keys as a result makes for evenly distributed results which are different for each seed.

ARGUMENTS:

- An l-for-lookup object.
- The start seed, or axiom, that is the initial state of the L-system. This must be the key-id of one of the sequences.
- An integer that is the length of the sequence to be returned. NB: This number does not indicate the number of L-system passes, but only the number of elements in the list returned, which may be the first segment of a sequence returned by a pass that actually generates a much longer sequence.

RETURN VALUE:

A list that is the L-sequence of rule key-ids.

The second value returned is a count of each of the rule keys in the sequence created, in their given order.

EXAMPLE:

```
;; Create an l-for-lookup object with three rules and generate a new sequence
;; of 29 rule keys from those rules. The l-for-lookup object here has been
;; created with the SEQUENCES argument set to NIL, as the get-l-sequence
;; method requires no sequences. The second list returned indicates the
;; number of times each key appears in the resulting sequence (thus 1 appears 5
;; times, 2 appears 12 times etc.)
(let ((lfl (make-l-for-lookup 'lfl-test
                              NIL
                              '((1 (2))
                                (2 (1 3))
                                (3 (3 2))))))
  (get-l-sequence lfl 1 29))

=> (2 3 2 3 2 1 3 2 3 2 3 2 1 3 2 3 2 1 3 3 2 1 3 2 3 2 3 2 1), (5 12 12)

;; A similar example using symbols rather than numbers as keys and data
(let ((lfl (make-l-for-lookup 'lfl-test
                              NIL
                              '((a (b))
                                (b (a c))
                                (c (c b))))))
  (get-l-sequence lfl 'a 19))

=> (A C C B A C C B A C B C B A C C B A C), (5 5 9)
```

SYNOPSIS:

```
(defmethod get-l-sequence ((lflu l-for-lookup) seed stop-length)
```

38.6.266 l-for-lookup/get-linear-sequence

[l-for-lookup] [Methods]

DESCRIPTION:

Instead of creating L-sequences with specified rules, use them to generate a simple sequential list.

The method first returns the first element in the list whose ID matches the SEED argument, then that element is used as the ID for the next look-up. Each time a rule is accessed, the next element in the rule is returned (if there is more than one), cycling to the head of the list once its end is reached.

Seen very loosely, this method functions a bit like a first-order Markov chain, but without the randomness.

ARGUMENTS:

- An l-for-lookup object.
- The seed, which is the starting key for the resulting sequence. This must be the key-ID of one of the sequences.
- An integer that is the number of elements to be in the resulting list.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether to reset the pointers of the given circular lists before proceeding. T = reset. Default = T.
- T or NIL to indicate whether to warn if any of the given keys are not used. Default = T.

RETURN VALUE:

Two values: A list of results of user-defined length and the distribution of the results.

EXAMPLE:

```
(let ((lfl (make-l-for-lookup 'lfl-test
                              nil
                              '((1 (2 3))
                                (2 (3 1 2))
```

```
                                (3 (1))))))
  (get-linear-sequence lfl 1 23))

=> (1 2 3 1 3 1 2 1 3 1 2 2 3 1 3 1 2 1 3 1 2 2 3)
```

SYNOPSIS:

```
(defmethod get-linear-sequence ((lflu l-for-lookup) seed stop-length
                                &optional (reset t) (warn t))
```

38.6.267 l-for-lookup/make-l-for-lookup

[l-for-lookup] [Functions]

DESCRIPTION:

Create an l-for-lookup object. The l-for-lookup object uses techniques associated with Lindenmayer-systems (or L-systems) by storing a series of rules about how to produce new, self-referential sequences from the data of original, shorter sequences.

NB: This method just stores the data concerning sequences and rules. To manipulate the data and create new sequences, see do-lookup or get-l-sequence etc.

ARGUMENTS:

- A symbol that will be the object's ID.
- A sequence (list) or list of sequences, that serve(s) as the initial material, from which the new sequence is to be produced.
- A production rule or list of production rules, each consisting of a predecessor and a successor, defining how to expand and replace the individual predecessor items.

OPTIONAL ARGUMENTS:

keyword arguments:
- :auto-check-redundancy. Some rules (e.g. in simplest form, a set of keys that return only one result each) will return a result of the same length no matter how many results we ask for. This can sometimes be problematic so do a check if requested. Default = NIL.
- :scaler. Factor by which to scale the values returned by do-lookup. Default = 1. Does not modify the original data.
- :offset. Number to be added to values returned by do-lookup (after they are scaled). Default = NIL. Does not modify the original data.

RETURN VALUE:

Returns an l-for-lookup object.

EXAMPLE:

```
;; Create an l-for-lookup object based on the Lindenmayer rules (A->AB) and
;; (B->A), using the defaults for the keyword arguments
(make-l-for-lookup 'l-sys-a
                   '((1 ((a)))
                     (2 ((b))))
                   '((1 (1 2)) (2 (1))))

=>
L-FOR-LOOKUP:
[...]
              l-sequence: NIL
              l-distribution: NIL
              ll-distribution: NIL
              group-indices: NIL
              scaler: 1
              offset: 0
              auto-check-redundancy: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: L-SYS-A, tag: NIL,
data: (
[...]

;; A larger list of sequences, with keyword arguments specified
(make-l-for-lookup 'lfl-test
                             '((1 ((2 3 4) (5 6 7)))
                               (2 ((3 4 5) (6 7 8)))
                               (3 ((4 5 6) (7 8 9))))
                             '((1 (3)) (2 (3 1)) (3 (1 2)))
                             :scaler 1
                             :offset 0
                             :auto-check-redundancy nil)
```

SYNOPSIS:

```
(defun make-l-for-lookup (id sequences rules &key (auto-check-redundancy nil)
                                                  (offset 0)
                                                  (scaler 1))
```

38.6.268 l-for-lookup/remix-in

[l-for-lookup] [Functions]

DESCRIPTION:

```
Given a list (for example generated by fibonacci-transitions) where we
proceed sequentially through adjacent elements, begin occasionally mixing
earlier elements of the list back into the original list once we've reached
the third unique element in the original list.

The earlier elements are mixed back in sequentially (the list is mixed back
into itself), starting at the beginning of the original list, and inserted
at automatically selected positions within the original list.

This process results in a longer list than the original, as earlier
elements are spliced in, without removing the original elements and their
order.  If however the :replace keyword is set to T, then at the selected
positions those original elements will be replaced by the earlier
elements.  This could of course disturb the appearance of particular
results and patterns.

The :remix-in-fib-seed argument determines how often an earlier element is
re-inserted into the original list. The lower the number, the more often an
earlier element is mixed back in. A value of 1 or 2 will result in each
earlier element being inserted after every element of the original (once
the third element of the original has been reached).

NB: The affects of this method are less evident on short lists.
```

ARGUMENTS:

```
- A list.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :remix-in-fib-seed. A number that indicates how frequently an earlier
  element will be mixed back into the original list. The higher the number,
  the less often earlier elements are remixed in. Default = 13.
- :mirror. T or NIL to indicate whether the method should pass backwards
  through the original list once it has reached the end.  T = pass
  backwards. Default = NIL.
- :test. The function used to determine the third element in the list. This
  function must be able to compare whatever data type is in the
  list. Default = #'eql.
```

- :replace. If T, retain the original length of the list by replacing items rather than splicing them in (see above). Default = NIL.

RETURN VALUE:

Returns a new list.

EXAMPLE:

```
;; Straightforward usage with default values
(remix-in (fibonacci-transitions 320 '(1 2 3 4 5)))

=> (1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 2
    1 1 1 1 1 1 1 2 1 1 1 1 2 1 1 2 1 1 2 1 2 1 2 1 2 1 2 1 2 1 2 2 1 2 2 1
    2 2 2 2 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 3 2 2 1 2 2 1 2 2 1 2 1 3
    1 2 2 1 2 1 2 1 3 2 2 1 3 2 1 2 3 1 2 1 3 1 2 3 1 2 1 3 1 2 3 2 1 3 2 1 3 2
    1 3 1 3 1 2 3 1 3 1 2 1 3 3 3 2 3 2 1 3 3 1 3 1 3 1 3 3 1 3 1 3 1 3 3 3 1 3
    3 1 3 3 1 3 1 3 1 3 3 2 3 1 4 1 3 3 3 1 3 3 1 3 3 1 4 1 3 1 3 3 2 3 1 4 1 3
    3 4 1 3 3 1 4 3 2 4 1 3 1 4 3 2 4 1 3 1 4 3 4 2 3 4 1 3 4 2 4 1 3 2 4 4 1 3
    2 4 1 4 4 4 2 3 4 1 4 4 2 4 1 4 2 4 4 1 4 2 4 2 4 4 4 1 4 4 2 4 4 2 4 1 4 2
    4 4 2 5 2 4 2 4 4 4 1 4 4 2 4 5 2 4 2 4 2 4 4 2 5 2 4 2 4 5 4 2 4 5 2 4 5 2
    4 2 5 2 4 5 2 4 2 5 2 4 5 4 2 5 4 2 5 5 2 4 2 5 2 5 4 3 5 2 5 2 5 5 4 2 5 5
    2 5 5 2 5 2 5 2 5 5 3 4 2 5 2 5 5 5 2 5 5 2 5 5 3 5 2 5 2 5 5 3 5 2 5 2 5 5
    5 3 5 5 2 5 5 3 5 2 5 3 5 5 2 5 3 5 2 5 5 5 3 5 5 2 5 5 3 5 2 5 3 5 5 2)

;; A lower :remix-in-fib-seed value causes the list to be mixed back into
;; itself at more frequent intervals
(remix-in (fibonacci-transitions 320 '(1 2 3 4 5)) :remix-in-fib-seed 3)

=> (1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 2
    1 1 1 1 1 1 1 2 1 1 1 1 2 1 1 2 1 1 2 1 2 1 2 1 2 1 2 1 2 1 2 2 1 2 2 1
    2 2 2 2 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 3 2 1 2 1 2 2 1 2 1 2 2 1
    3 1 2 2 1 2 1 2 3 1 2 1 2 3 1 2 1 2 3 1 2 1 3 2 1 3 1 2 3 1 2 1 3 2 1 3 1 2
    3 1 2 1 3 3 1 2 1 3 3 2 2 1 3 3 1 3 1 3 2 1 3 1 3 3 1 3 1 3 3 1 3 1 3 3 1 3
    1 3 3 1 3 2 3 3 1 3 1 3 3 1 3 1 3 4 1 3 1 3 3 1 3 2 3 3 1 3 1 4 3 1 3 1 3 3
    2 4 1 3 3 1 4 2 3 3 1 4 1 3 4 2 3 1 4 3 2 4 1 3 4 2 3 1 4 3 2 4 1 3 4 2 4 1
    3 4 2 4 1 3 4 2 4 1 4 4 2 3 2 4 4 1 4 2 4 4 2 4 1 4 4 2 4 2 4 4 2 4 2 4 4 1
    4 2 4 4 2 4 2 4 4 2 5 2 4 4 2 4 2 4 4 2 4 2 4 5 2 4 2 4 4 2 4 2 5 4 2 4 2 5
    4 2 4 2 5 4 2 5 2 4 5 2 4 3 5 4 2 5 2 4 5 2 4 2 5 4 2 5 2 5 4 2 5 3 5 4 2 5
    2 5 5 2 5 2 4 5 3 5 2 5 5 2 5 3 5 5 2 5 2 4 5 3 5 2 5 5 3 5 2 5 5 3 5 2 5 5
    3 5 2 5 5 3 5 2 5 5 3 5 2 5 5 3 5 2 5 5 3 5 3 5 5 2 5 3 5 5 3 5 2 5 5 3 5 3
    5 5 3 5 3 5 5 2 5 3)

;; Setting the keyword argument <mirror> to T causes the method to reverse back
;; through the original list after the end has been reached
(remix-in (fibonacci-transitions 320 '(1 2 3 4 5))
```

```
          :remix-in-fib-seed 3
          :mirror t)

=> (1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 2
    1 1 1 1 1 1 1 2 1 1 1 1 2 1 1 2 1 1 2 1 2 1 2 1 2 1 2 1 2 1 2 2 1 2 2 1
    2 2 2 2 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 3 2 1 2 1 2 2 1 2 1 2 2 1
    3 1 2 2 1 2 1 2 3 1 2 1 2 3 1 2 1 2 3 1 2 1 3 2 1 3 1 2 3 1 2 1 3 2 1 3 1 2
    3 1 2 1 3 3 1 2 1 3 3 2 2 1 3 3 1 3 1 3 2 1 3 1 3 3 1 3 1 3 3 1 3 1 3 3 1 3
    1 3 3 1 3 2 3 3 1 3 1 3 3 1 3 1 3 4 1 3 1 3 3 1 3 2 3 3 1 3 1 4 3 1 3 1 3 3
    2 4 1 3 3 1 4 2 3 3 1 4 1 3 4 2 3 1 4 3 2 4 1 3 4 2 3 1 4 3 2 4 1 3 4 2 4 1
    3 4 2 4 1 3 4 2 4 1 4 4 2 3 2 4 4 1 4 2 4 4 2 4 1 4 4 2 4 2 4 4 2 4 2 4 4 1
    4 2 4 4 2 4 2 4 4 2 5 2 4 4 2 4 2 4 4 2 4 2 4 5 2 4 2 4 4 2 4 2 5 4 2 4 2 5
    4 2 4 2 5 4 2 5 2 4 5 2 4 3 5 4 2 5 2 4 5 2 4 2 5 4 2 5 2 5 4 2 5 3 5 4 2 5
    2 5 5 2 5 2 4 5 3 5 2 5 5 2 5 3 5 5 2 5 2 4 5 3 5 2 5 5 3 5 2 5 5 3 5 2 5 5
    3 5 2 5 5 3 5 2 5 5 3 5 2 5 5 3 5 2 5 5 3 5 3 5 5 2 5 3 5 5 3 5 2 5 5 3 5 3
    5 5 3 5 3 5 5 2 5 3 5 5 3 5 3 5 5 3 5 3 5 5 3 5 3 5 5 3 5 3 5 5 3 5 3 5 5 3
    5 3 5 5 3 5 3 5 5 3 5 3 5 5 3 5 3 5 5 3 5 4 5 5 3 5 3 5 5 3 5 3 5 4 3 5 3 5
    5 3 5 4 5 5 3 5 3 5 4 3 5 3 5 5 4 5 3 4 5 3 5 4 4 5 3 5 3 4 5 4 4 3 5 4 4 5
    3 4 5 4 4 3 5 4 4 5 3 4 5 4 4 3 4 5 4 4 3 4 5 4 4 3 4 4 4 4 4 5 4 3 4 4 4 4
    4 4 3 4 4 4 5 4 4 4 4 4 4 4 4 3 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
    4 4 4 3 4 4 4 4 4 4 4 4 3 4 4 4 4 3 4 4 4 4 3 4 4 4 3 4 4 3 5 4 3 4 4 4 3 4 4
    3 4 4 3 4 4 4 4 3 3 4 4 5 3 3 4 4 4 3 3 4 3 4 3 4 5 3 4 3 3 4 3 5 3 3 4 3 4 4
    3 5 3 4 3 3 5 3 4 3 3 5 3 4 3 3 5 3 4 3 3 5 3 4 3 3 5 3 4 3 3 5 3 4 2 3 5 3
    5 3 3 4 2 5 3 3 5 2 4 3 3 5 2 5 3 2 5 3 5 2 3 4 2 5 3 2 5 3 5 2 3 5 2 5 3 2
    5 2 5 3 2 5 2 4 3 2 5 2 5 2 2 5 3 5 2 2 5 2 5 2 2 5 2 5 2 3 5 2 5 2 2 5 2 5
    2 2 5 2 5 2 2 5 2 5 2 2 5 2 5 2 2 5 2 5 2 2 5 2 5 2 1 5 2 5 2 2 5 2 5 1 2 5
    2 5 1 2 5 2 5 1 2 5 1 5 2 1 5 2 5 1 2 5 1 5 2 1 5 2 5 1 2 5 1 5 1 2 5 1 5 1
    2 5 1 5 1 1 5 1 5 2 1 5 1 5 1 1 5 1 5 1 1 5 2 5 1 1 5 1 5 1 1 5 1 5 1 1 5 1
    5 1 1 5 1 5 2 1 5 1 5 1 1 5 1 5 1 1 5 1 5 1 1 5 1 5 1 1 5 1 5 1 1 5 1 5 1 1
    5 1 5 1 1 5 1 4 1)
```

SYNOPSIS:

```
(defun remix-in (list &key (remix-in-fib-seed 13) (mirror nil) (test #'eql)
                      (replace nil))
```

38.6.269 l-for-lookup/reset

[l-for-lookup] [Methods]

DESCRIPTION:

```
Sets the counters (index pointers) of all circular-sclist objects stored
within a given l-for-lookup object back to zero.
```

ARGUMENTS:

```
 - An l-for-lookup object.
```

OPTIONAL ARGUMENTS:

```
 - (an optional IGNORE argument for internal use only).
```

RETURN VALUE:

```
 Always T.
```

SYNOPSIS:

```
(defmethod reset ((lflu l-for-lookup) &optional ignore1 ignore2)
```

38.6.270 assoc-list/make-assoc-list

[assoc-list] [Functions]

DESCRIPTION:

```
 A function that provides a shortcut to creating an assoc-list, filling it
 with data, and assigning a name to it.
```

ARGUMENTS:

```
 - The name of the assoc-list to be created.
 - The data with which to fill it.
```

OPTIONAL ARGUMENTS:

```
 keyword arguments:
 - :warn-not-found. T or NIL to indicate whether a warning is printed when an
   index which doesn't exist is used for look-up.  T = warn. Default = T.
```

RETURN VALUE:

```
 Returns the assoc-list as a named-object.
```

EXAMPLE:

```
(make-assoc-list 'looney-tunes '((bugs bunny)
                                 (daffy duck)
                                 (porky pig)))
=>
```

```
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL
                     this: NIL
                     next: NIL
NAMED-OBJECT: id: LOONEY-TUNES, tag: NIL,
data: (
NAMED-OBJECT: id: BUGS, tag: NIL,
data: BUNNY

NAMED-OBJECT: id: DAFFY, tag: NIL,
data: DUCK

NAMED-OBJECT: id: PORKY, tag: NIL,
data: PIG)
```

SYNOPSIS:

```
(defun make-assoc-list (id al &key (warn-not-found t))
```

38.6.271 assoc-list/map-data

[assoc-list] [Methods]

DESCRIPTION:

```
Map a function over the data in the assoc-list, i.e. call the given
function/method on each data slot of each named-object in the assoc-list
data slot. See also recursive-assoc-list's rmap method which does pretty
much the same but acting recursively on each named-object (unless it is
itself recursive), rather than the named-object's data.
```

ARGUMENTS:

```
- The assoc-list to which the function is to be applied.
- The function to be applied.  This must take the data in the assoc-list as
  a first argument.
```

OPTIONAL ARGUMENTS:

```
- &rest: Further arguments for the function.
```

RETURN VALUE:

Returns a list of the values returned by the function call on the data.

EXAMPLE:

```
(let ((al (make-assoc-list 'al-test
                           '((1 (1 2 3 4))
                             (2 (5 6 7 8))
                             (3 (9 10 11 12))))))
  (map-data al #'(lambda (y)
                   (loop for i in y collect (* i 2)))))

=> ((2 4 6 8) (10 12 14 16) (18 20 22 24))
```

SYNOPSIS:

```
(defmethod map-data ((al assoc-list) function &rest further-arguments)
```

38.6.272 assoc-list/nmap-data

[assoc-list] [Methods]

DATE:

November 1st 2018, Heidhausen

DESCRIPTION:

A destructive version of map-data: replace the data slot with the named-object returned by the 2nd argument after it is passed one named-object after another.

ARGUMENTS:

See the map-data method in this class.

RETURN VALUE:

the assoc-list object with modified data in its elements.

EXAMPLE:

```
(let ((al (make-assoc-list 'test '((1 (2 3))
                                   (2 (4 5 6))
                                   (3 (7))))))
```

```
  (nmap-data al #'(lambda (data) (length data)))
  al)
...
data: (
NAMED-OBJECT: id: 1, tag: NIL,
data: 2
**************
NAMED-OBJECT: id: 2, tag: NIL,
data: 3
**************
NAMED-OBJECT: id: 3, tag: NIL,
data: 1
...
```

SYNOPSIS:

```
(defmethod nmap-data ((al assoc-list) function &rest further-arguments)
```

38.6.273 assoc-list/recursive-assoc-list

[assoc-list] [Classes]

NAME:

```
 recursive-assoc-list

 File:             recursive-assoc-list.lsp

 Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                   circular-sclist -> assoc-list -> recursive-assoc-list

 Version:          1.1.0

 Project:          slippery chicken (algorithmic composition)

 Purpose:          Extension of the assoc-list class to allow and
                   automatically instantiate association lists inside of
                   association lists to any level of nesting. E.g.
                   (setf x
                     '((1 one)
                       (2 two)
                       (3 ((cat "cat")
                           (dog ((mickey mouse)
                                 (donald duck)
                                 (daffy duck)
```

```
                                (uncle ((james dean)
                                        (dean martin)
                                        (fred astaire)
                                        (ginger ((wolfgang mozart)
                                                 (johann bach)
                                                 (george gershwin)))))))
                            (mouse "mouse")))
                        (4 four)))

 Author:           Michael Edwards: m@michael-edwards.org

 Creation date:    March 18th 2001

 $$ Last modified:  17:00:48 Tue Oct 14 2025 CEST

 SVN ID: $Id$
```

38.6.274 recursive-assoc-list/add

[recursive-assoc-list] [Methods]

DESCRIPTION:

```
Add a new element (key/data pair) to the given recursive-assoc-list
object.

If no value is specified for the optional argument, the new element is
added at the end of the top level. The optional argument allows for the
FULL-REF to be specified, i.e. a recursive path of keys down to the nested
level where the new element is to be placed.

N.B. All new items are added to the end of the data list, unlike in
assoc-list where that is the default but can be changed. Here it cannot be
changed.
```

ARGUMENTS:

```
- A key/data pair.
- A recursive-assoc-list object.
```

OPTIONAL ARGUMENTS:

```
- A list that is the FULL-REF, i.e. a recursive path of keys, down to the
  nested level where the new element is to be placed.
```

RETURN VALUE:

Returns T if the specified named-object is successfully added to the given recursive-assoc-list.

Returns an error if an attempt is made to add NIL to the given recursive-assoc-list or if the given named-object is already present at the same level within the given recursive-assoc-list.

EXAMPLE:

```
;; Adding an element while specifying no optional argument results in the new
;; element being placed at the end of the top level by default (evident here by
;; the fact that the ref for (MAKERS) is a single-item list)
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (add '(makers mark) ral)
  (get-all-refs ral))

=> ((JIM) (WILD) (FOUR ROSES) (FOUR VIOLETS BLUE) (FOUR VIOLETS RED DRAGON)
    (FOUR VIOLETS RED VIPER) (FOUR VIOLETS RED FOX) (FOUR VIOLETS WHITE)
    (MAKERS))

;; A list that is a path of keys (FULL-REF) to the desired recursive level must
;; be given as the optional argument in order to place the specified element
;; deeper in the given recursive-assoc-list object
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (add '(yellow sky) ral '(four violets))
  (get-all-refs ral))

=> ((JIM) (WILD) (FOUR ROSES) (FOUR VIOLETS BLUE) (FOUR VIOLETS RED DRAGON)
    (FOUR VIOLETS RED VIPER) (FOUR VIOLETS RED FOX) (FOUR VIOLETS WHITE)
    (FOUR VIOLETS YELLOW))
```

```
;; Attempting to add an element that is already present at the given level of
;; the given recursive-assoc-list object results in an error
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (add '(makers mark) ral)
  (add '(makers mark) ral))

=>
assoc-list::add: Can't add MAKERS to assoc-list with id MIXED-BAG
    because key already exists!
   [Condition of type SIMPLE-ERROR]

;; Attempting to add NIL also results in an error
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (add '() ral))

=>
assoc-list::add: named-object is NIL!
   [Condition of type SIMPLE-ERROR]
```

SYNOPSIS:

```
(defmethod add (named-object (ral recursive-assoc-list) &optional ref)
```

38.6.275 recursive-assoc-list/add-empty-parcel

[recursive-assoc-list] [Methods]

DESCRIPTION:

Add a recursive-assoc-list object with NIL data (an empty level of recursion) to the end of the top-level of a given recursive-assoc-list object.

NB: Adding an empty parcel to a given recursive-assoc-list object will cause the method get-all-refs to fail on that recursive-assoc-list object.

ARGUMENTS:

- A recursive-assoc-list object.
- A symbol that will be the ID of the new, empty recursive-assoc-list object that is to be added.

OPTIONAL ARGUMENTS:

- <new-class> The name of an existing subclass of recursive-assoc-list that the parcel should be promoted to.

RETURN VALUE:

A recursive-assoc-list object with DATA of NIL (the "empty parcel")

EXAMPLE:

```
;; Add two new empty parcels (the first a recursive-assoc-list, by default, the
;; second a rthm-seq-palette) and return the new list of REFS:
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (add-empty-parcel ral 'bricolage)
  (add-empty-parcel ral 'rsp 'rthm-seq-palette)
  (get-all-refs ral))

Mark set
=>

((JIM) (WILD) (FOUR ROSES) (FOUR VIOLETS BLUE) (FOUR VIOLETS RED DRAGON)
 (FOUR VIOLETS RED VIPER) (FOUR VIOLETS RED FOX) (FOUR VIOLETS WHITE)
 (BRICOLAGE) (RSP))
```

SYNOPSIS:

```
(defmethod add-empty-parcel ((ral recursive-assoc-list) id &optional new-class)
```

38.6.276 recursive-assoc-list/assoc-list-id-list

[recursive-assoc-list] [Functions]

DESCRIPTION:

Determine whether a given list contains only atoms which could be used as assoc-list IDs. To pass the test, a given atom must be either a symbol, a number or a string.

ARGUMENTS:

A list.

RETURN VALUE:

T or NIL indicating whether the atoms of the given list are all capable of being used as assoc-list IDs. T = all can be used as assoc-list IDs.

EXAMPLE:

```
;; All of the elements in this list are either a symbol, a number or a
;; string. The list therefore returns a T when tested.
(let ((alil '(jim beam 3 "Allegro" 5 flute)))
  (assoc-list-id-list alil))

=> T

;; This list fails, as the last element is a list (and therefore not of type
;; string, number or symbol)
(let ((alil '(jim beam 3 "Allegro" 5 (flute))))
  (assoc-list-id-list alil))

=> NIL
```

SYNOPSIS:

```
(defun assoc-list-id-list (id-list)
```

38.6.277 recursive-assoc-list/ensemble

[recursive-assoc-list] [Classes]

NAME:

```
ensemble

File:             ensemble.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist -> assoc-list -> recursive-assoc-list ->
                  ensemble

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the ensemble class.

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SVN ID: $Id$
```

38.6.278 ensemble/add-player

[ensemble] [Methods]

DESCRIPTION:

```
Add a player to an existing ensemble. It will be added at the end of the
list.
```

ARGUMENTS:

```
- The ensemble object.
- The new player, either as a player object or symbol.  If the latter this
  becomes the id of the player we'll create.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
```

- :instrument. The id of the instrument in the already existing
  instrument-palette. This is required if the player argument is a symbol.
  Default NIL.
- :instrument-palette. An instrument-palette object. Default =
  +slippery-chicken-standard-instrument-palette+.
- :midi-channel the midi-channel for the new player
- :microtones-midi-channel the microtones-midi-channel for the new player
  Default = -1 = same as :midi-channel.

RETURN VALUE:

The player object added.

SYNOPSIS:

```
(defmethod add-player ((e ensemble) player
                       &key
                         instrument
                         (instrument-palette
                          +slippery-chicken-standard-instrument-palette+)
                         ;; MDE Tue Jul 14 19:08:15 2020, Heidhausen
                         (midi-channel 1)
                         (microtones-midi-channel -1))
```

38.6.279 ensemble/auto-midi-channels

[ensemble] [Methods]

DATE:

December 28th 2017

DESCRIPTION:

This method is intended for initialisation but can also be called explicitly (but before rendering any MIDI files, if it's to make sense). It will automatically set the MIDI channels of all players in the ensemble to ascending values, if all values are currently 1 or the optional second argument is T. If using the :chromatic-scale, then the microtones-midi-channel will be the same as the main channel, otherwise it will be 1+ the main channel.

ARGUMENTS:

the ensemble object

OPTIONAL ARGUMENTS:

```
T or NIL to force the channel setting.
```

RETURN VALUE:

```
If the procedure actually runs then the last channel set (integer) otherwise
NIL.
```

SYNOPSIS:

```
(defmethod auto-midi-channels ((e ensemble) &optional force)
```

38.6.280 ensemble/balanced-load?

[ensemble] [Methods]

DATE:

```
28th August 2014
```

DESCRIPTION:

```
Determine whether the playing load is balanced across the players of the
ensemble. By default, if the least active player is playing 80% of the time
that the most active player is playing, we'll return T.
```

ARGUMENTS:

```
- an ensemble instance
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :threshold. A number between 0.0 and 1.0 which represents the lowest
  ratio between the most and least active players.
- :stats-fun. One of the player methods which tots up statistics,
  i.e. total-notes, total-degrees, total-duration, or total-bars
- :ignore. A list of players (symbols) not to count in the sorting.
```

RETURN VALUE:

```
T or NIL
```

SYNOPSIS:

```
(defmethod balanced-load? ((e ensemble) &key (threshold .8)
                                        (stats-fun #'total-duration)
                                        ignore)
```

38.6.281 ensemble/get-instrument

[ensemble] [Methods]

DATE:

November 2nd 2018, Heidhausen

DESCRIPTION:

Get the instrument object for a player in the ensemble. If the player doubles, then the optional <ins> argument is required and should be the ID of the instrument as defined in the ensemble's instrument palette (e.g. 'flute)

ARGUMENTS:

- the ensemble object
- the ID of a player in the ensemble

OPTIONAL ARGUMENTS:

- the ID of an instrument if the given player doubles
- T or NIL to issue a warning should the instrument not be found.

RETURN VALUE:

The instrument object.

SYNOPSIS:

```
(defmethod get-instrument ((e ensemble) player &optional ins (warn t))
```

38.6.282 ensemble/get-player

[ensemble] [Methods]

DESCRIPTION:

Return a player object from an ensemble, if it exists.

ARGUMENTS:

```
- An ensemble object.
- The ID of a player.
```

RETURN VALUE:

```
The player object or NIL if there's no such player.
```

SYNOPSIS:

```
(defmethod get-player ((e ensemble) player)
```

38.6.283 ensemble/get-players

[ensemble] [Methods]

DESCRIPTION:

```
Return the IDs of the players from a given ensemble object.
```

ARGUMENTS:

```
- An ensemble object.
```

RETURN VALUE:

```
- A list of symbols that are the player IDs of the given ensemble object.
```

EXAMPLE:

```
(let ((ens (make-ensemble
            'ens
            '((flt ((flute piccolo) :midi-channel 1))
              (clr ((b-flat-clarinet)))
              (tpt ((b-flat-trumpet c-trumpet) :midi-channel 2))
              (vln ((violin))))
            :instrument-palette
            +slippery-chicken-standard-instrument-palette+)))
  (get-players ens))

=> (FLT CLR TPT VLN)
```

SYNOPSIS:

```
(defmethod get-players ((e ensemble))
```

38.6.284 ensemble/lotsa-combos

[ensemble] [Methods]

DATE:

November 8th 2018, Heidhausen

DESCRIPTION:

Get various permutations of the ensemble players, ranging from a single player up to one less than the number of players in the ensemble. This uses the shuffle method so is random, but as we use fixed-seed randomness we get repeatable results.

NB This will return groups of players that are mere permutations of each others (e.g. (vln vla vc) and (vla vln vc)). As the combo-chord-possible? chord method will permutate these to try and find an instrumentation that is possible, you might think there's no point accepting permutations in this method. However, which combination (i.e. instruments in any given order) is first deemed playable for any given chord will determine the voicing selected, so different results can be expected.

ARGUMENTS:

- the ensemble object

OPTIONAL ARGUMENTS:

- an integer to determine the number of times we permutate for any given number of players. Note that depending on the number of players in the ensemble and the number we're looking to put into a combo, we might not always be able to hit this target (e.g. there are only 6 possible permutations of three values). Default = 7.

RETURN VALUE:

a list of lists of player IDs.

EXAMPLE:

```
(lotsa-combos (make-ensemble
               'ens
               '((flt ((flute piccolo)))
                 (clr ((b-flat-clarinet)))
```

```
                (bsn bassoon)
                (tpt ((b-flat-trumpet c-trumpet)))
                (trb tenor-trombone)
                (tb tuba)
                (vln ((violin))))))
-->
((CLR FLT TRB BSN TB VLN) (TRB FLT CLR BSN TB TPT) (TB TRB FLT VLN TPT BSN)
 (CLR BSN TPT TB TRB FLT) (FLT TPT VLN BSN TRB TB) (FLT TPT BSN VLN TB CLR)
 (TPT FLT VLN CLR TB TRB) (BSN CLR TB TRB FLT) (TB FLT BSN VLN TPT)
 (FLT TPT BSN CLR TB) (FLT TRB CLR VLN TB) (BSN CLR VLN TPT TRB)
 (BSN TRB FLT TB CLR) (FLT CLR TPT TB VLN) (TRB CLR TPT TB) (FLT VLN TB TRB)
 (TRB VLN TB CLR) (TB VLN TRB BSN) (TPT TRB CLR BSN) (CLR FLT TRB VLN)
 (FLT TPT TRB CLR) (TPT CLR FLT) (TB TRB FLT) (TRB VLN FLT) (TB TPT CLR)
 (FLT BSN TPT) (FLT VLN BSN) (TB TPT TRB) (TPT FLT) (CLR TRB) (FLT CLR)
 (TRB CLR) (BSN CLR) (TRB TB) (BSN TPT) (FLT CLR BSN TPT TRB TB VLN) (FLT)
 (CLR) (BSN) (TPT) (TRB) (TB) (VLN))

(lotsa-combos (make-ensemble
               'ens
               '((flt ((flute piccolo)))
                 (clr ((b-flat-clarinet)))
                 (bsn bassoon)
                 (tpt ((b-flat-trumpet c-trumpet)))
                 (trb tenor-trombone)
                 (tb tuba)
                 (vln ((violin)))))
               10) ; <<--- get more combos
-->
((CLR TPT TB FLT TRB VLN) (BSN TRB VLN TB TPT CLR) (TB CLR FLT TRB BSN TPT)
 (TPT CLR BSN FLT VLN TRB) (VLN BSN TPT CLR TB TRB) (TRB BSN VLN TPT TB CLR)
 (TB TRB CLR VLN FLT BSN) (CLR TPT VLN TB FLT TRB) (TRB BSN VLN CLR FLT TPT)
 (TPT FLT CLR TB TRB VLN) (TPT TRB VLN BSN CLR) (BSN TB VLN TPT FLT)
 (BSN FLT TRB CLR TPT) (BSN TPT FLT CLR TB) (VLN TB TRB TPT FLT)
 (FLT TRB BSN TB VLN) (FLT CLR BSN TB TPT) (TRB FLT VLN TPT BSN)
 (BSN TPT TB TRB FLT) (TPT VLN BSN TRB TB) (BSN VLN TB CLR) (VLN CLR TB TRB)
 (CLR TB TRB FLT) (FLT BSN VLN TPT) (TPT BSN CLR TB) (TRB CLR VLN TB)
 (CLR VLN TPT TRB) (TRB FLT TB CLR) (CLR TPT TB VLN) (TRB CLR TPT TB)
 (VLN TB TRB) (VLN TB CLR) (VLN TRB BSN) (TRB CLR BSN) (FLT TRB VLN)
 (TPT TRB CLR) (TPT CLR FLT) (TB TRB FLT) (TRB VLN FLT) (TB TPT CLR) (VLN BSN)
 (TPT TRB) (TPT FLT) (CLR TRB) (FLT CLR) (TRB CLR) (BSN CLR) (TRB TB) (BSN TPT)
 (FLT CLR BSN TPT TRB TB VLN) (FLT) (CLR) (BSN) (TPT) (TRB) (TB) (VLN))
```

SYNOPSIS:

```
(defmethod lotsa-combos ((e ensemble) &optional (try 7))
```

38.6.285 ensemble/make-ensemble

[ensemble] [Functions]

DESCRIPTION:

```
Make an ensemble object, specifying the players and associated
instruments.

NB: If you have an ensemble with a player doubling two instruments, be sure
    to indicate some keyword argument or other as
    (fl1 ((piccolo violin) :midi-channel 1)) works but
    (fl1 ((piccolo violin))) thinks that piccolo is a nested ensemble!!!
```

ARGUMENTS:

```
- An ID consisting of a symbol, string or number.
- A list of 2-element sublists that define the ensemble. See the above
  comment on adding a keyword argument for doubling players. An existing
  ensemble object can be passed here whereupon it will be cloned and the ID
  will be changed to the (new) given ID if it's not NIll. In this case
  however the keyword arguments will be ignored.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :instrument-palette. An instrument palette object. Default =
   +slippery-chicken-standard-instrument-palette+
- :bar-line-writers. Obsolete as no longer used.
```

RETURN VALUE:

```
An ensemble object.
```

EXAMPLE:

```
(make-ensemble 'test
               '((fl ((alto-flute piccolo) ; the flautist doubles
                      :midi-channel 1 :microtones-midi-channel 2))
                 ;; e.g. cl is the freely-named player and it plays the two
                 ;; clarinets that are defined in our instrument palette (so we
                 ;; have to use their names).
                 (cl ((b-flat-clarinet bass-clarinet)
                      :midi-channel 3 :microtones-midi-channel 4))
                 (hn (french-horn :midi-channel 5 :microtones-midi-channel 6))
```

```
                  (perc (marimba :midi-channel 7))))

(let ((ens (make-ensemble
              'ens
              '((flt ((flute piccolo) :midi-channel 1))
                (clr ((b-flat-clarinet))))
              :instrument-palette
              +slippery-chicken-standard-instrument-palette+)))
  (print ens))

=>

ENSEMBLE: bar-line-writers: NIL
          players: (FLT CLR)
          (id instrument-palette): SLIPPERY-CHICKEN-STANDARD-INSTRUMENT-PALETTE
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 2
                      linked: T
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: ENS, tag: NIL,
data: (
PLAYER: (id instrument-palette): SLIPPERY-CHICKEN-STANDARD-INSTRUMENT-PALETTE
doubles: T, cmn-staff-args: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: (FLT), next: (CLR)
NAMED-OBJECT: id: FLT, tag: NIL,
data:
[...]
data: (
INSTRUMENT: lowest-written:
[...]
NAMED-OBJECT: id: FLUTE, tag: NIL,
[...]
INSTRUMENT: lowest-written:
[...]
NAMED-OBJECT: id: PICCOLO, tag: NIL,
[...]
PLAYER: (id instrument-palette): SLIPPERY-CHICKEN-STANDARD-INSTRUMENT-PALETTE
doubles: NIL, cmn-staff-args: NIL
LINKED-NAMED-OBJECT: previous: (FLT), this: (CLR), next: NIL
NAMED-OBJECT: id: CLR, tag: NIL,
data:
INSTRUMENT: lowest-written:
```

```
[...]
NAMED-OBJECT: id: B-FLAT-CLARINET, tag: NIL,
)
```

SYNOPSIS:

```
(defun make-ensemble (id ensemble
                      &key bar-line-writers
                        (instrument-palette
                         +slippery-chicken-standard-instrument-palette+))
```

38.6.286 ensemble/num-notes

[ensemble] [Methods]

DESCRIPTION:

Get the number of attacked notes in a given slippery-chicken object. This method accesses the ensemble object within the given slippery-chicken object to perform this task.

ARGUMENTS:

- An ensemble object.

RETURN VALUE:

An integer that is the total number of attacked notes in the given slippery-chicken object.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) e e e e))
                                :pitch-seq-palette ((1 2 3 4)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1))
                            (vc (1 1 1 1 1)))))))))
  (num-notes (ensemble mini)))

=> 40
```

SYNOPSIS:

```
(defmethod num-notes ((e ensemble))
```

38.6.287 ensemble/num-players

[ensemble] [Methods]

DESCRIPTION:

```
 Get the number of players in a given ensemble object.
```

ARGUMENTS:

```
 - An ensemble object.
```

RETURN VALUE:

```
 - An integer.
```

EXAMPLE:

```
(let ((ens (make-ensemble
            'ens
            '((flt ((flute piccolo) :midi-channel 1))
              (clr ((b-flat-clarinet)))
              (tpt ((b-flat-trumpet c-trumpet) :midi-channel 2))
              (vln ((violin))))
            :instrument-palette
            +slippery-chicken-standard-instrument-palette+)))
  (num-players ens))

=> 4
```

SYNOPSIS:

```
(defmethod num-players ((e ensemble))
```

38.6.288 ensemble/players-exist

[ensemble] [Methods]

DESCRIPTION:

Produce an error message and drop into the debugger if the specified
player IDs are not found within the given ensemble object.

ARGUMENTS:

- An ensemble object.
- A list of symbols that are the IDs of the players sought.

RETURN VALUE:

T if the specified player IDs are present within the given ensemble object,
otherwise drops into the debugger with an error.

EXAMPLE:

```
;;; Returns NIL if a player with the specified ID is found in the given
;;; ensemble object.
(let ((ens (make-ensemble
            'ens
            '((flt ((flute piccolo) :midi-channel 1))
              (clr ((b-flat-clarinet)))
              (tpt ((b-flat-trumpet c-trumpet) :midi-channel 2))
              (vln ((violin))))
            :instrument-palette
            +slippery-chicken-standard-instrument-palette+)))
  (players-exist ens '(vln)))

=> T

;; Drops into the debugger with an error if no player with the specified ID is
;; found in the given ensemble object.
(let ((ens (make-ensemble
            'ens
            '((flt ((flute piccolo) :midi-channel 1))
              (clr ((b-flat-clarinet)))
              (tpt ((b-flat-trumpet c-trumpet) :midi-channel 2))
              (vln ((violin))))
            :instrument-palette
            +slippery-chicken-standard-instrument-palette+)))
  (players-exist ens '(vla)))

=>
ensemble::players-exist: VLA is not a member of the ensemble
   [Condition of type SIMPLE-ERROR]
```

SYNOPSIS:

```
(defmethod players-exist ((e ensemble) players)
```

38.6.289 ensemble/set-staff-names

[ensemble] [Methods]

DATE:

```
November 27th 2021, Heidhausen
```

DESCRIPTION:

```
Set the staff-names slot of players in the ensemble. These will be used
instead of the instruments' staff-names slot when writing the score. In fact
the new names will be passed down to the instrument objects.
```

ARGUMENTS:

```
- the ensemble object
```

OPTIONAL ARGUMENTS:

```
- names: can be nil, whereup all the player names will be passed down to the
  instrument objects as the staff-name (as a downcased string). Or can be a
  list of player (symbols) whereupon the same will happen. Or it can be a
  list of (player staff-name) pairs where the staff-name will be downcased,
  if a symbol, or used directly if a string. If the player is
  multi-instrumental the staff-name can itself be a list whereupon the
  symbol/string will be handled similarly but the order should be the same
  as the list of instruments passed when initialising the player
```

RETURN VALUE:

```
T
```

SYNOPSIS:

```
(defmethod set-staff-names ((e ensemble) &optional names short-names)
```

38.6.290 ensemble/sort-players

[ensemble] [Methods]

DATE:

27th August 2013

DESCRIPTION:

Return a hierarchical list of players sorted by e.g. how much they're playing.

ARGUMENTS:

- An ensemble object

OPTIONAL ARGUMENTS:

keyword arguments:
- :stats-fun. One of the player methods which tots up statistics, i.e. total-notes, total-degrees, total-duration, or total-bars
- :ignore. A list of players (symbols) not to count in the sorting.

RETURN VALUE:

A list of all the players in the ensemble ordered by its statistics.

SYNOPSIS:

```
(defmethod sort-players ((e ensemble) &key (stats-fun #'total-duration)
                                           ignore print)
```

38.6.291 ensemble/tessitura

[ensemble] [Methods]

DESCRIPTION:

Get the average pitch of a given slippery-chicken object. This method accesses the ensemble object within the given slippery-chicken object to perform this task.

NB: This method processes data in relationship to degrees of the current tuning system (scale), which is quarter-tone by default. It is therefore possible, when generating a piece using only chromatic pitches but within a non-chromatic tuning to get microtonal results.

ARGUMENTS:

- An ensemble object.

RETURN VALUE:

```
An integer that is the average pitch of the given slippery-chicken object
in degrees.
```

EXAMPLE:

```
;;; Change the tuning to chromatic first to get an accurate result:
(in-scale :chromatic)

=> #<tuning "chromatic-scale">

(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) e e e e))
                                :pitch-seq-palette ((1 2 3 4)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1))
                            (vc (1 1 1 1 1)))))))))
  (tessitura (ensemble mini)))

=> C4
```

SYNOPSIS:

```
(defmethod tessitura ((e ensemble))
```

38.6.292 recursive-assoc-list/get-all-refs

[recursive-assoc-list] [Methods]

DESCRIPTION:

```
Return a list of all the keys (REFS) in a given recursive-assoc-list
object. Nested keys are given in FULL-REF form, i.e. a list that is the
path of keys to the specific key.

Keys that are not part of nesting-path are also returned as lists
(single-item lists) by default. An optional argument allows these to be
returned as individual symbols rather than lists.
```

NB This will only work on the top-level object due to the creation of references when linking. If you're interested in the full refs of a sub-ral, try ths get-this-refs method.

ARGUMENTS:

- A recursive-assoc-list object.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether to return single REFS (non-nested keys) as lists or as individual symbols. T = as list. Default = T.

RETURN VALUE:

A list.

EXAMPLE:

```
;; By default all keys are returned as lists, even single (non-nested) keys
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (get-all-refs ral))

=> ((JIM) (WILD) (FOUR ROSES) (FOUR VIOLETS BLUE) (FOUR VIOLETS RED DRAGON)
    (FOUR VIOLETS RED VIPER) (FOUR VIOLETS RED FOX) (FOUR VIOLETS WHITE))

;; Setting the optional argument to NIL returns non-nested keys as symbols
;; rather than lists
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
```

```
  (get-all-refs ral nil))

=> (JIM WILD (FOUR ROSES) (FOUR VIOLETS BLUE) (FOUR VIOLETS RED DRAGON)
    (FOUR VIOLETS RED VIPER) (FOUR VIOLETS RED FOX) (FOUR VIOLETS WHITE))
```

SYNOPSIS:

```
(defmethod get-all-refs ((ral recursive-assoc-list)
                         &optional
                         (single-ref-as-list t))
```

38.6.293 recursive-assoc-list/get-data

[recursive-assoc-list] [Methods]

DESCRIPTION:

Return the named-object (or linked-named-object) that is identified by a specified key within a given recursive-assoc-list object.

NB: This method returns the named object itself, not just the data associated with the key (use assoc-list::get-data-data for that).

ARGUMENTS:

- A symbol that is the key (id) of the named-object sought, or a list of symbols that are the path to the desired named-object within the given recursive-assoc-list.
- The recursive-assoc-list object in which it is sought.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether a warning is printed if the specified key cannot be found within the given assoc-list. T = print. Default = T.

RETURN VALUE:

A named-object is returned if the specified key is found within the given recursive-assoc-list object.

NIL is returned and a warning is printed if the specified key is not found in the given recursive-assoc-list object. This applies, too, when a nested key is specified without including the other keys that are the path to that key (see example).

EXAMPLE:

```
;; Get a named-object from the top-level of the recursive-assoc-list object
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (get-data 'wild ral))

=>
NAMED-OBJECT: id: WILD, tag: NIL,
data: TURKEY

;; A list including all keys that are the path to the specified key is required
;; to get nested named-objects
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (get-data '(four violets white) ral))

=>
NAMED-OBJECT: id: WHITE, tag: NIL,
data: RIBBON

;; Searching for a key that is not present in the given recursive-assoc-list
;; object returns NIL and a warning
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
```

```
  (get-data 'johnnie ral))

=> NIL
WARNING:
   assoc-list::get-data: Could not find data with key JOHNNIE
   in assoc-list with id MIXED-BAG

;; Searching for a nested key without specifying the path to that key within a
;; list also returns a NIL and a warning
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (get-data 'fox ral))

=> NIL
WARNING:
   assoc-list::get-data: Could not find data with key FOX
   in assoc-list with id MIXED-BAG
```

SYNOPSIS:

```
(defmethod get-data :around (ids (ral recursive-assoc-list)
                                 &optional (warn t))
```

38.6.294 recursive-assoc-list/get-first

[recursive-assoc-list] [Methods]

DESCRIPTION:

```
Returns the first named-object in the DATA slot of the given
recursive-assoc-list object.
```

ARGUMENTS:

```
- A recursive-assoc-list object.
```

RETURN VALUE:

A named-object that is the first object in the DATA slot of the given recursive-assoc-list object.

EXAMPLE:

```
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (get-first ral))

=>
NAMED-OBJECT: id: JIM, tag: NIL,
data: BEAM
```

SYNOPSIS:

```
(defmethod get-first ((ral recursive-assoc-list))
```

38.6.295 recursive-assoc-list/get-first-ref

[recursive-assoc-list] [Methods]

DESCRIPTION:

Get the full reference into the given recursive-assoc-list object of the first named-object in the given recursive-assoc-list object.

NB: If the <ral> argument happens to be a recursive-assoc-list object that is contained within another recursive-assoc-list object (i.e. is a nested recursive-assoc-list object), then the result is the reference into the top-level recursive-assoc-list object, not the argument.

ARGUMENTS:

- A recursive-assoc-list object.

RETURN VALUE: EXAMPLE:

```
;; A simple call returns the first top-level named-object
```

```
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (get-first-ref ral))

=> (JIM)

;; Return the first ref of a nested recursive-assoc-list object
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (get-first-ref (get-data-data '(four violets) ral)))

=> (FOUR VIOLETS BLUE)
```

SYNOPSIS:

```
(defmethod get-first-ref ((ral recursive-assoc-list))
```

38.6.296 recursive-assoc-list/get-last

[recursive-assoc-list] [Methods]

DESCRIPTION:

```
 Get the last named-object in a given recursive-assoc-list object.

 NB: This method functions linearly, not hierarchically. The last named
     object is therefore not necessarily the deepest of a nest, but the last
     listed.
```

ARGUMENTS:

```
 - A recursive-assoc-list object.
```

RETURN VALUE:

```
A named-object (or linked-named-object).
```

EXAMPLE:

```
;; This returns '(white ribbon), not '(fox hole)
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (get-last ral))

=>
NAMED-OBJECT: id: WHITE, tag: NIL,
data: RIBBON
```

SYNOPSIS:

```
(defmethod get-last ((ral recursive-assoc-list))
```

38.6.297 recursive-assoc-list/get-last-ref

[recursive-assoc-list] [Methods]

DESCRIPTION:

```
Get the last REF (path of nested keys) of the given recursive-assoc-list
object.

NB: This method functions linearly, not hierarchically. The last-ref may
    not be the deepest nested.
```

ARGUMENTS:

```
- A recursive-assoc-list object.
```

RETURN VALUE:

```
Returns a list that is the last REF of the given recursive-assoc-list
object.
```

EXAMPLE:

```
;; Typical usage with nesting
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (get-last-ref ral))

=> (FOUR VIOLETS WHITE)

;; Returns the last-ref as a list even if the given recursive-assoc-list object
;; contains no nesting
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four roses)))))
  (get-last-ref ral))

=> (FOUR)
```

SYNOPSIS:

```
(defmethod get-last-ref ((ral recursive-assoc-list))
```

38.6.298 recursive-assoc-list/get-previous

[recursive-assoc-list] [Methods]

DESCRIPTION:

```
Get the previous named-object in the given recursive-assoc-list object by
specifying the ID of a named-object contained within that given
recursive-assoc-list object.

An optional argument allows for the retrieval of a previous named-object
that is more than one step back in the given recursive-assoc-list object
(i.e., not the named-object that immediately precedes the specified key).

If the number given for the optional <how-many> argument is greater than
```

```
the number of items in the given recursive-assoc-list object, the value
returned will be a negative number.

The method proceeds linearly, not hierarchically, when getting previous
named-objects from further down into nested assoc-lists. In other words,
the named-object immediately previous to (white ribbon) in this nested list
is (fox hole), which is at a deeper level, not (red ...) or (blue velvet),
which are at the same level:
((blue velvet)
 (red ((dragon den)
       (viper nest)
       (fox hole)))
 (white ribbon)

In order to retrieve objects that are nested more deeply, the list that is
the <keys> argument must consist of the consecutive path of keys leading to
that object. If only the key of a named object that is deeper in the list
is given, and not the path of keys to that object, a warning will be
printed that the given key cannot be found in the list.

NB: When this method is applied to keys that contain further assoc-list
    objects, the method will drop into the debugger with an error.
```

ARGUMENTS:

```
- A recursive-assoc-list object.
- A list containing one or more symbols that are either the ID of the
  specified named object or the path of keys to that object within the
  given recursive-assoc-list object.
```

OPTIONAL ARGUMENTS:

```
- An integer indicating how many steps back in the given
  recursive-assoc-list from the specified named-object to look when
  retrieving the desired object (e.g. 1 = immediately previous object, 2 =
  the one before that etc.)
```

RETURN VALUE:

```
A linked-named-object.
```

EXAMPLE:

```
;; Get the object immediately previous to that with the key WILD returns the
;; object with key JIM and data BEAM
```

```
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (get-previous ral '(wild)))

=>
LINKED-NAMED-OBJECT: previous: NIL, this: (JIM), next: (WILD)
NAMED-OBJECT: id: JIM, tag: NIL,
data: BEAM

;; Attempting to get the previous object from the key FOUR, which contains a
;; nested list, returns an error unless the first key in the nested list is
;; also included
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (get-previous ral '(four)))

=>
There is no applicable method for the generic function
  #<STANDARD-GENERIC-FUNCTION PREVIOUS (1)>
when called with arguments
  (
NAMED-OBJECT: id: FOUR, tag: NIL,

(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
```

```
  (get-previous ral '(four roses)))

=>
LINKED-NAMED-OBJECT: previous: (JIM), this: (WILD), next: (FOUR ROSES)
NAMED-OBJECT: id: WILD, tag: NIL,
data: TURKEY

;; The method defines the previous object linearly, not hierarchically; i.e.,
;; the previous object to (white ribbon) here is (fox hole) and not (red ...)
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (get-previous ral '(four violets white)))

=>
LINKED-NAMED-OBJECT: previous: (FOUR VIOLETS RED VIPER),
this: (FOUR VIOLETS RED FOX),
next: (FOUR VIOLETS WHITE)
NAMED-OBJECT: id: FOX, tag: NIL,
data: HOLE

;; Use the <how-many> argument to retrieve previous objects further back than
;; the immediate predecessor
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (get-previous ral '(four violets white) 4))

=>
LINKED-NAMED-OBJECT: previous: (FOUR ROSES),
this: (FOUR VIOLETS BLUE),
next: (FOUR VIOLETS RED DRAGON)
NAMED-OBJECT: id: BLUE, tag: NIL,
data: VELVET
```

```
;; Using a <how-many> value greater than the number of items in the given
;; recursive-assoc-list object returns a negative number
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (get-previous ral '(four violets white) 14))

=> -7
```

SYNOPSIS:

```
(defmethod get-previous ((ral recursive-assoc-list) keys
                         &optional (how-many 1))
```

38.6.299 recursive-assoc-list/get-this-refs

[recursive-assoc-list] [Methods]

DATE:

```
5th July 2019, Essen Werden
```

DESCRIPTION:

```
Get the <this> slots of each named-object (or subclass) in a
recursive-assoc-list (ral). This should work on top-level rals and sub-rals
alike, though rals with only sub-rals probably won't return anything. Unlike
get-all-refs, this doesn't traverse recursive objects.
```

ARGUMENTS:

```
- the recursive-assoc-list object
```

RETURN VALUE:

```
a list of references (most probably simple lists)
```

SYNOPSIS:

```
(defmethod get-this-refs ((ral recursive-assoc-list))
```

38.6.300 recursive-assoc-list/link-named-objects

[recursive-assoc-list] [Methods]

DESCRIPTION:

Create linked-named-objects from the named-objects in the data slots of the given recursive-assoc-list object. The linked-named-objects created hold keys that serve as pointers to the previous and next objects in the given recursive-assoc-list object, whether recursive or not.

The optional <previous> and <higher-next> arguments are only for internal recursive calls and so shouldn't be given by the user.

ARGUMENTS:

- A recursive-assoc-list object.

OPTIONAL ARGUMENTS:

- <previous>
- <higher-next>

EXAMPLE:

```
;;; The recursive-assoc-list may not be linked on creation, evident here
;;; through the value of the LINKED slot
(make-ral 'mixed-bag
                  '((jim beam)
                    (wild turkey)
                    (four ((roses red)
                           (violets ((blue velvet)
                                     (red ((dragon den)
                                           (viper nest)
                                           (fox hole)))
                                     (white ribbon)))))))

=>
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 8
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
```

```
NAMED-OBJECT: id: MIXED-BAG, tag: NIL,
data: (
NAMED-OBJECT: id: JIM, tag: NIL,
data: BEAM

NAMED-OBJECT: id: WILD, tag: NIL,
data: TURKEY
[...]

;; The recursive-assoc-list object and the named-objects it contains are linked
;; after applying the link-named-objects method
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (link-named-objects ral))

=>
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 8
                      linked: T
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: MIXED-BAG, tag: NIL,
data: (
LINKED-NAMED-OBJECT: previous: NIL, this: (JIM), next: (WILD)
NAMED-OBJECT: id: JIM, tag: NIL,
data: BEAM

LINKED-NAMED-OBJECT: previous: (JIM), this: (WILD), next: (FOUR ROSES)
NAMED-OBJECT: id: WILD, tag: NIL,
data: TURKEY
```

RETURN VALUE:

```
 the recursive-assoc-list object
```

SYNOPSIS:

```
(defmethod link-named-objects ((ral recursive-assoc-list)
                               &optional previous higher-next)
```

38.6.301 recursive-assoc-list/lisp-assoc-listp

[recursive-assoc-list] [Functions]

DESCRIPTION:

Determine whether a given list can has the structure of a lisp assoc-list. This is assessed based on each of the elements being a 2-item list, of which the first is a symbol, number or string (qualifies as a key).

The optional argument <recurse-simple-data> allows the data portion of key/data pairs to be viewed as flat lists rather than as recursive lists.

ARGUMENTS:

- A list.

OPTIONAL ARGUMENTS:

T or NIL to indicate whether to consider lists of 2-item lists in the data position of a given key/data pair to be a list or a recursive list. T = list. Default = T.

RETURN VALUE:

T or NIL. T = the tested list can be considered a Lisp assoc-list.

EXAMPLE:

```
;; A list of 2-item lists, each of whose item are all either a symbol, number,
;; or string, can be considered a Lisp assoc-list.
(let ((lal '((roses red) (3 "allegro") (5 flute))))
  (lisp-assoc-listp lal))

=> T

;; By default, lists of 2-item lists in the DATA portion of a key/data pair
;; will be considered as a simple list, rather than a recursive list, resulting
;; in the tested list passing as T.
(let ((lal '((1 2) (3 ((4 5) (6 7))) (8 9))))
  (lisp-assoc-listp lal))
```

```
=> T

;; Setting the optional argument to NIL will cause the same list to fail with
(let ((lal '((1 2) (3 ((4 5) (6 7))) (8 9))))
  (lisp-assoc-listp lal nil))

=> NIL
```

SYNOPSIS:

```
(defun lisp-assoc-listp (candidate &optional (recurse-simple-data t))
```

38.6.302 recursive-assoc-list/make-ral

[recursive-assoc-list] [Functions]

DESCRIPTION:

```
Create a recursive-assoc-list object, which allows and automatically
instantiates association lists inside of association lists to any level of
nesting.
```

ARGUMENTS:

```
- A symbol that is the object's ID.
- A list of nested lists, or a list.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :recurse-simple-data. T or NIL to indicate whether to recursively
  instantiate a recursive-assoc-list in place of data that appears to be a
  simple assoc-list (i.e. a 2-element list). If NIL, the data of 2-element
  lists whose second element is a number or a symbol will be ignored,
  therefore remaining as a list. For example, this data would normally
  result in a recursive call: (y ((2 23) (7 28) (18 2))).  T = replace
  assoc-list data with recursive-assoc-lists. Default = T.
- :full-ref. Nil or a list representing the path to a nested
  recursive-assoc-list object within the given recursive-assoc-list object,
  starting from the top level of the given object. When NIL, the given
  recursive-assoc-list object itself is the top level.  Default = NIL.
- :tag. A symbol that is another name, description etc. for the given
  recursive-assoc-list object. The tag may be used for identification but
  not for searching purposes. Default = NIL.
```

- :warn-not-found. T or NIL to indicate whether a warning is printed when
  an index which doesn't exist is used for look-up. Default = T.

RETURN VALUE:

Returns a recursive-assoc-list object.

EXAMPLE:

```
;; Create a recursive-assoc-list object with default keyword argument values
(make-ral 'mixed-bag
          '((jim beam)
            (wild turkey)
            (four ((roses red)
                   (violets ((blue velvet)
                             (red ((dragon den)
                                   (viper nest)
                                   (fox hole)))
                             (white ribbon)))))))

=>
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 8
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: MIXED-BAG, tag: NIL,
data: (
[...]

;; Use the class's get-all-refs method to show that by default, simple data is
;; recursed. The sublists in the second list in this example are processed as
;; nested lists
(let ((ral (make-ral 'ral-test
                     '((1 one)
                       (2 ((3 4) (5 6)))
                       (3 three)))))
  (get-all-refs ral))

=> ((1) (2 3) (2 5) (3))

;; Using the same data, but setting the :recurse-simple-data argument to NIL
;; will cause the method to process simple data as a unit rather than nested
```

```
;; lists
(let ((ral (make-ral 'ral-test
                     '((1 one)
                       (2 ((3 4) (5 6)))
                       (3 three))
                     :recurse-simple-data nil)))
  (get-all-refs ral))

=> ((1) (2) (3))
```

SYNOPSIS:

```
(defun make-ral (id ral &key (recurse-simple-data t) (warn-not-found t)
                 (tag nil) (full-ref nil))
```

38.6.303 recursive-assoc-list/palette

[recursive-assoc-list] [Classes]

NAME:

```
palette

File:             palette.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist -> assoc-list -> recursive-assoc-list ->
                  palette

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the palette class which adds very little
                  to its direct superclass assoc-list (as of yet) but spawns
                  a new base type for more specialised palettes.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    19th February 2001

$$ Last modified:  19:03:26 Tue Jan 13 2026 CET

SVN ID: $Id$
```

38.6.304 palette/instrument-palette

[palette] [Classes]

NAME:

```
instrument-palette

File:             instrument-palette.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist -> assoc-list -> recursive-assoc-list ->
                  palette -> instrument-palette

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the instrument-palette class which
                  intantiates instruments to be used in an ensemble
                  instance.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    6th September 2001

$$ Last modified:  12:22:46 Thu Sep 20 2018 CEST

SVN ID: $Id$
```

38.6.305 instrument-palette/auto-set-subset-id

[instrument-palette] [Methods]

DATE:

```
July 27th 2018, Heidhausen
```

DESCRIPTION:

```
Set the subset-id slot to be the same as the id slot for each instrument in
the palette.
```

ARGUMENTS:

```
- an instrument-palette instance
```

RETURN VALUE:

```
- the instrument instance
```

SYNOPSIS:

```
(defmethod auto-set-subset-id ((ip instrument-palette))
```

38.6.306 instrument-palette/make-instrument-palette

[instrument-palette] [Functions]

DESCRIPTION:

```
Create an instrument-palette object from a list of instrument descriptions
based on the keyword arguments of make-instrument.
```

ARGUMENTS:

```
- A symbol that will serve as the ID for the instrument-palette object.
- A list of instrument descriptions based on the keyword arguments of
  make-instrument.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :warn-not-found. T or NIL to indicate whether a warning is printed when
  an index which doesn't exist is used for look-up. Default = T.
```

RETURN VALUE:

```
An instrument palette.
```

EXAMPLE:

```
;; Returns an instrument-palette object
(make-instrument-palette 'inst-pal
                         '((piccolo (:transposition-semitones 12
                                     :lowest-written d4 :highest-written c6))
                           (bf-clarinet (:transposition-semitones -2
                                         :lowest-written e3
                                         :highest-written c6))
                           (horn (:transposition f :transposition-semitones -7
                                  :lowest-written f2 :highest-written c5))
```

```
                          (violin (:lowest-written g3 :highest-written c7
                                   :chords t))
                          (viola (:lowest-written c3 :highest-written f6
                                  :chords t))))

=>
INSTRUMENT-PALETTE:
PALETTE:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 5
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 5, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: INST-PAL, tag: NIL,
data: (
[...]
```

SYNOPSIS:

```
(defun make-instrument-palette (id ip &key (warn-not-found t))
```

38.6.307 instrument-palette/set-prefers-high

[instrument-palette] [Methods]

DATE:

```
05 Feb 2011
```

DESCRIPTION:

```
Set the PREFERS-NOTES slot of a specified instrument object within a given
instrument-palette object to 'HIGH. The instrument object is specified
using the ID symbol assigned to it within the instrument-palette object
definition.

NB: The optional argument is actually required, but is listed as optional
because of the attributes of the instrument class method.
```

ARGUMENTS:

```
- An instrument-palette object.
```

OPTIONAL ARGUMENTS:

```
- A symbol that is the ID of the instrument object within the
  instrument-palette object definition.
```

RETURN VALUE:

```
Returns the symbol 'HIGH.
```

EXAMPLE:

```
;; Define an instrument-palette object, then set the PREFERS-NOTES slot of the
;; instrument object 'piccolo within that instrument-palette object to 'HIGH
(let ((ip (make-instrument-palette 'inst-pal
                                   '((piccolo (:transposition-semitones 12
                                               :lowest-written d4
                                               :highest-written c6))
                                     (bf-clarinet (:transposition-semitones -2
                                                   :lowest-written e3
                                                   :highest-written c6))
                                     (horn (:transposition f
                                            :transposition-semitones -7
                                            :lowest-written f2
                                            :highest-written c5))
                                     (violin (:lowest-written g3
                                              :highest-written c7
                                              :chords t))
                                     (viola (:lowest-written c3
                                             :highest-written f6
                                             :chords t))))))
  (set-prefers-high ip 'piccolo))

=> HIGH
```

SYNOPSIS:

```
(defmethod set-prefers-high ((ip instrument-palette) &optional instrument)
```

38.6.308 instrument-palette/set-prefers-low

[instrument-palette] [Methods]

DATE:

```
05 Feb 2011
```

DESCRIPTION:

Set the PREFERS-NOTES slot of a specified instrument object within a given instrument-palette object to 'LOW. The instrument object is specified using the ID symbol assigned to it within the instrument-palette object definition.

NB: The optional argument is actually required, but is listed as optional because of the attributes of the instrument class method.

ARGUMENTS:

- An instrument-palette object.

OPTIONAL ARGUMENTS:

- A symbol that is the ID of the instrument object within the instrument-palette object definition.

RETURN VALUE:

Returns the symbol 'LOW.

EXAMPLE:

```
;; Define an instrument-palette object, then set the PREFERS-NOTES slot of the
;; instrument object 'piccolo within that instrument-palette object to 'LOW
(let ((ip (make-instrument-palette 'inst-pal
                                   '((piccolo (:transposition-semitones 12
                                               :lowest-written d4
                                               :highest-written c6))
                                     (bf-clarinet (:transposition-semitones -2
                                                   :lowest-written e3
                                                   :highest-written c6))
                                     (horn (:transposition f
                                            :transposition-semitones -7
                                            :lowest-written f2
                                            :highest-written c5))
                                     (violin (:lowest-written g3
                                              :highest-written c7
                                              :chords t))
                                     (viola (:lowest-written c3
                                             :highest-written f6
                                             :chords t))))))
  (set-prefers-low ip 'piccolo))

=> LOW
```

SYNOPSIS:

```
(defmethod set-prefers-low ((ip instrument-palette) &optional instrument)
```

38.6.309 palette/pitch-seq-palette

[palette] [Classes]

NAME:

```
pitch-seq-palette

File:             pitch-seq-palette.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist -> assoc-list -> recursive-assoc-list ->
                  palette -> pitch-seq-palette

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the pitch-seq-palette class.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    19th February 2001

$$ Last modified:  19:22:09 Tue May 20 2025 CEST

SVN ID: $Id$
```

38.6.310 pitch-seq-palette/add-inversions

[pitch-seq-palette] [Methods]

DESCRIPTION:

```
Add inversions of the existing pitch-seq objects in a given
pitch-seq-palette object to the end of that pitch-seq-palette object. (See
pitch-seq::invert for more details on slippery-chicken inversions.)
```

ARGUMENTS:

```
- A pitch-seq-palette object.
```

RETURN VALUE:

```
Always returns T.
```

EXAMPLE:

```
;; Create a pitch-seq-palette object and print the DATA of the pitch-seq
;; objects it contains; then apply the add-inversions method and print the same
;; DATA to see the changes
(let ((mpsp (make-psp 'mpsp 5 '((2 5 3 1 4)
                                (1 4 2 5 3)
                                (5 1 3 2 4)
                                (2 3 4 5 1)
                                (3 2 4 1 5)))))
  (print (loop for ps in (data mpsp)
           collect (data ps)))
  (add-inversions mpsp)
  (print (loop for ps in (data mpsp)
           collect (data ps))))

=>
((2 5 3 1 4) (1 4 2 5 3) (5 1 3 2 4) (2 3 4 5 1) (3 2 4 1 5))

((2 5 3 1 4) (1 4 2 5 3) (5 1 3 2 4) (2 3 4 5 1) (3 2 4 1 5) (4 1 3 5 2)
 (5 2 4 1 3) (1 5 3 4 2) (4 3 2 1 5) (3 4 2 5 1))
```

SYNOPSIS:

```
(defmethod add-inversions ((psp pitch-seq-palette))
```

38.6.311 pitch-seq-palette/combine

[pitch-seq-palette] [Methods]

DESCRIPTION:

```
Create a new pitch-seq-palette object by combining the pitch-seq lists from
one pitch-seq-palette object with those of another.

The method combines the contents of the two given rthm-seq-palette objects
consecutively; i.e., the first pitch-seq object of the first
pitch-seq-palette is combined with the first pitch-seq object of the other,
then the second with the second, the third with the third etc.

If one pitch-seq-palette object contains more pitch-seq objects than the
```

other, the method cycles through the shorter one until all of the members of the longer one have been handled. The new pitch-seq-palette object will therefore contain the same number of pitch-seq objects as is in the longest of the two starting pitch-seq-palette objects.

It is not necessary for the lengths of the pitch-seq objects in the two starting pitch-seq-palette objects to be the same.

NB Both pitch-seq-palettes are reset by this method.

ARGUMENTS:

- A first pitch-seq-palette object.
- A second pitch-seq-palette object.

RETURN VALUE:

A pitch-seq-palette object.

EXAMPLE:

```
;;; Combine two pitch-seq-palette objects of the same length, each of whose
;;; pitch-seqs are the same length
(let ((mpsp1 (make-psp 'mpsp1 5 '((2 5 3 1 4) (1 4 2 5 3) (5 1 3 2 4))))
      (mpsp2 (make-psp 'mpsp2 5 '((2 3 4 5 1) (3 2 4 1 5) (3 2 1 5 4)))))
  (combine mpsp1 mpsp2))

=>
PITCH-SEQ-PALETTE: num-notes: 10, instruments: NIL
PALETTE:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 3
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "MPSP1-MPSP2", tag: NIL,
data: (
PITCH-SEQ: notes: NIL
[...]
data: (2 5 3 1 4 2 3 4 5 1)
PITCH-SEQ: notes: NIL
[...]
```

```
data: (1 4 2 5 3 3 2 4 1 5)
[...]
PITCH-SEQ: notes: NIL
[...]
data: (5 1 3 2 4 3 2 1 5 4)
)

;; When combining pitch-seq-palette objects of different lengths, the method
;; cyles through the shorter of the two
(let ((mpsp1 (make-psp 'mpsp1 5 '((2 5 3 1 4) (1 4 2 5 3) (5 1 3 2 4))))
      (mpsp2 (make-psp 'mpsp2 5 '((2 3 4 5 1) (3 2 4 1 5)))))
  (loop for ps in (data (combine mpsp1 mpsp2))
        collect (data ps)))

=> ((2 5 3 1 4 2 3 4 5 1) (1 4 2 5 3 3 2 4 1 5) (5 1 3 2 4 2 3 4 5 1))

;; The two starting pitch-seq-palette objects are not required to have
;; pitch-seq objects of the same length
(let ((mpsp1 (make-psp 'mpsp1 5 '((2 5 3 1 4) (1 4 2 5 3) (5 1 3 2 4))))
      (mpsp2 (make-psp 'mpsp2 3 '((2 3 4) (3 2 4)))))
  (loop for ps in (data (combine mpsp1 mpsp2))
        collect (data ps)))

=> ((2 5 3 1 4 2 3 4) (1 4 2 5 3 3 2 4) (5 1 3 2 4 2 3 4))
```

SYNOPSIS:

```
(defmethod combine ((psp1 pitch-seq-palette) (psp2 pitch-seq-palette))
```

38.6.312 pitch-seq-palette/make-psp

[pitch-seq-palette] [Functions]

DESCRIPTION:

Create a pitch-seq-palette object from an ID, a specified number of notes, and a list of lists of numbers representing the pitch curve of the intended pitch-seq objects.

ARGUMENTS:

- A symbol that is to be the ID of the pitch-seq-palette to be created.
- An integer that is the number of notes there are to be in each pitch-seq object created.
- A list of lists, each of which contained list is a list of numbers representing the pitch curve of the intended pitch-seq object.

RETURN VALUE:

A pitch-seq-palette object.

EXAMPLE:

```
;; Returns a pitch-seq-palette object
(make-psp 'mpsp 5 '((2 5 3 1 4)
                    (1 4 2 5 3)
                    (5 1 3 2 4)
                    (2 3 4 5 1)
                    (3 2 4 1 5)))

=>
PITCH-SEQ-PALETTE: num-notes: 5, instruments: NIL
PALETTE:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 5
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 5, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: MPSP, tag: NIL,
data: (
PITCH-SEQ: notes: NIL
[...]
data: (2 5 3 1 4)
PITCH-SEQ: notes: NIL
[...]
data: (1 4 2 5 3)
PITCH-SEQ: notes: NIL
[...]
data: (5 1 3 2 4)
PITCH-SEQ: notes: NIL
[...]
data: (2 3 4 5 1)
PITCH-SEQ: notes: NIL
[...]
data: (3 2 4 1 5)
)

;; Interrupts with an error if any of the <pitch-seqs> lists is not of the
;; length specified by <num-notes>
(make-psp 'mpsp 5 '((1 2 1 1 3)
```

```
                    (1 3 2 1 5)
                    (1 3 5 6 7 8)))

=>
pitch-seq-palette::verify-and-store:
In pitch-seq MPSP-ps-3 from palette MPSP:
Each pitch sequence must have 5 notes (you have 6):
[...]
 (1 3 5 6 7 8))
   [Condition of type SIMPLE-ERROR]

;; We can also assign instruments to specific pitch-seqs by passing their names
;; (not player names, but instrument names) e.g.
(make-psp 'mpsp 5 '((1 2 1 1 3)
                    ((1 3 2 1 5) violin flute)
                    (1 3 5 6 7)))
```

SYNOPSIS:

```
(defun make-psp (id num-notes pitch-seqs)
```

38.6.313 palette/rthm-seq-palette

[palette] [Classes]

NAME:

```
 set-palette

 File:             rthm-seq-palette.lsp

 Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                   circular-sclist -> assoc-list -> recursive-assoc-list ->
                   palette -> rthm-seq-palette

 Version:          1.1.0

 Project:          slippery chicken (algorithmic composition)

 Purpose:          Implementation of the rthm-seq-palette class.

 Author:           Michael Edwards: m@michael-edwards.org

 Creation date:    19th February 2001
```

```
$$ Last modified:  19:40:01 Thu Nov 13 2025 CET

SVN ID: $Id$
```

38.6.314 rthm-seq-palette/chop

[rthm-seq-palette] [Methods]

DESCRIPTION:

```
Applies the chop method to each rthm-seq object in the given
rthm-seq-palette object (see rthm-seq-bar::chop for details). Returns a
new rthm-seq-palette object with the same structure as the argument, but
with a further level of nesting: Each rthm-seq object in the argument is
replaced by a list of rthm-seq objects that are each one "slice" of the
original rthm-seq objects.

The chop method is the basis for slippery-chicken's feature of
intra-phrasal looping.

NB: Since the chop method functions by comparing each beat of a given
    rthm-seq-bar object to the specified <chop-points> pattern for
    segmenting that beat, all rthm-seq-bar objects in the given
    rthm-seq-palette object must be evenly divisible by the beat for which
    the pattern is defined. For example, if the <chop-points> argument
    defines a quarter-note, all bars in the given rthm-seq-palette object
    must be evenly divisible by a quarter-note, and a rthm-seq-palette
    consisting of a rthm-seq object with a 2/4, a 3/4 and a 3/8 bar would
    fail at the 3/8 bar with an error.

NB: The <unit> argument must be a duplet rhythmic value (i.e. 32, 's, 'e
    etc.) and cannot be a tuplet value (i.e. 'te 'fe etc.).

NB: In order for the resulting chopped rhythms to be parsable by LilyPond
    and CMN, there can be no tuplets (triplets etc.) among the rhythms to
    be chopped. Such rhythms will result in LilyPond and CMN errors. This
    has only minimal bearing on any MIDI files produced, however, and these
    can potentially be imported into notation software.
```

ARGUMENTS:

```
- A rthm-seq-palette object.
```

OPTIONAL ARGUMENTS:

```
- <chop-points>. A list of integer pairs, each of which delineates a
```

```
   segment of the beat of the given rthm-seq-bar objects within the given
   rthm-seq-palette object, measured in the rhythmic unit specified by the
   <unit> argument. See the documentation for rthm-seq-bar::chop for more
   details.
 - <unit>. The rhythmic duration that serves as the unit of measurement for
   the chop points. Default = 's.
 - <number-bars-first>. T or NIL. This argument helps in naming (and
   therefore debugging) the newly-created bars.  If T, the bars in the
   original rthm-seq will be renumbered, starting from 1, and this will be
   reflected in the tag of the new bars.  E.g. if T, a new bar's tag may be
   new-bar-from-rs1-b3-time-range-1.750-to-2.000, if NIL this would be
   new-bar-from-rs1-time-range-1.750-to-2.000. Default = T.
```

RETURN VALUE:

```
 A rthm-seq-palette with the same top-level structure of the first argument,
 but each ID now referencing a sub-rthm-seq-palette with sequentially
 numbered rthm-seqs for each of the chopped results.
```

EXAMPLE:

```
;;; Create a rthm-seq-palette object, chop it with user-defined chop-points and
;;; a <unit> value of 'e, and print-simple the results
(let* ((rsp-orig (make-rsp
                  'sl-rsp
                  '((1
                     ((((2 4) (e) e (e) e))
                      :pitch-seq-palette (1 8)))
                    (2
                     ((((2 4) (s) e s e. (s)))
                      :pitch-seq-palette (3 5 7)))
                    (3
                     ((((3 4) q +s e. +q))
                      :pitch-seq-palette (1 7))))))
       (rsp-chopped (chop rsp-orig
                          '((1 1) (1 2) (2 2))
                          'e)))
  (print-simple rsp-chopped))

=>
rthm-seq-palette SL-RSP
rthm-seq-palette 1
rthm-seq 1
(1 8): rest 8,
rthm-seq 2
(1 4): rest E, NIL E,
```

```
rthm-seq 3
(1 8): NIL E,
rthm-seq 4
(1 8): rest 8,
rthm-seq 5
(1 4): rest E, NIL E,
rthm-seq 6
(1 8): NIL E,
rthm-seq 1
(1 8): rest S, NIL S,
rthm-seq 2
(1 4): rest S, NIL E, NIL S,
rthm-seq 3
(1 8): rest S, NIL S,
rthm-seq 4
(1 8): NIL E,
rthm-seq 5
(1 4): NIL E., rest S,
rthm-seq 6
(1 8): rest 8,
rthm-seq 1
(1 8): NIL E,
rthm-seq 2
(1 4): NIL Q,
rthm-seq 3
(1 8): rest 8,
rthm-seq 4
(1 8): rest S, NIL S,
rthm-seq 5
(1 4): rest S, NIL E.,
rthm-seq 6
(1 8): rest 8,
rthm-seq 7
(1 8): rest 8,
rthm-seq 8
(1 4): rest 4,
rthm-seq 9
(1 8): rest 8,
```

SYNOPSIS:

```
(defmethod chop ((rsp rthm-seq-palette) &optional chop-points
                                                  (unit 's)
                                                  (number-bars-first t))
```

38.6.315 rthm-seq-palette/cmn-display

[rthm-seq-palette] [Methods]

DESCRIPTION:

```
Generate printable music notation output (.EPS) of the given
rthm-seq-palette object using the Common Music Notation (CMN)
interface. The method requires at least the name of the given
rthm-seq-palette object to set, but has several additional optional
arguments for customizing output.

NB: Most of the keyword arguments are CMN attributes and share the same
    name as the CMN feature they effect.
```

ARGUMENTS:

```
- A rthm-seq-palette object.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :all-output-in-one-file. T or NIL to indicate whether to write the output
  to a multi-page file or to separate files for each page.
  T = one multi-page file. Default = T. This is a direct CMN attribute.
- :file. The file path, including the file name, of the file to be
  generated.
- :staff-separation. A number to indicate the amount of white space between
  staves belong to the same system, measured as a factor of the staff
  height. Default = 3. This is a direct CMN attribute.
- :line-separation. A number to indicate the amount of white space between
  lines of music (systems), measured as a factor of the staff
  height. Default = 5. This is a direct CMN attribute.
- :page-nums. T or NIL to indicate whether or not to print page numbers on
  the pages. T = print page numbers. Default = T.
- :no-accidentals. T or NIL to indicate whether or not to supress printing
  accidentals for each and every note (rather than once per bar).
  T = supress printing all accidentals. Default = NIL.
- :seqs-per-system. An integer indicating the number of rthm-seq objects to
  be printed in one staff system. Default = 1.
- :size. A number to indicate the font size of the CMN output.
- :auto-open. Automatically open the EPS file?.
   Default = (get-sc-config cmn-display-auto-open)
```

RETURN VALUE:

T

EXAMPLE:

```
;; A typical example with some specified keyword values for file and size
(let ((mrsp
       (make-rsp 'rsp-test
                 '((seq1 ((((2 4) q +e. s)
                           ((s) e (s) q)
                           (+e. s { 3 (te) te te } ))
                          :pitch-seq-palette ((1 2 3 4 5 6 7)
                                              (1 3 5 7 2 4 6)
                                              (1 4 2 6 3 7 5)
                                              (1 5 2 7 3 2 4))))
                   (seq2 ((((4 4) (e.) s { 3 te te te } +h)
                           ({ 3 +te (te) te } e e (h)))
                          :pitch-seq-palette (2 3 4 5 6 7 8)))
                   (seq3 ((((2 4) e e { 3 te te te })
                           ((4 4) (e) e e e s s (s) s q))
                          :pitch-seq-palette (3 4 5 6 7 8 9 10 1 2 3 7)))))))
  (cmn-display mrsp
               :file "/tmp/rmsp-output.eps"
               :size 10))
```

SYNOPSIS:

```
(defmethod cmn-display ((rsp rthm-seq-palette)
                        &key
                        (all-output-in-one-file t)
                        (file
                         (format nil "~a~a.eps"
                                 (get-sc-config 'default-dir)
                                 (string-downcase (id rsp))))
                        (staff-separation 3)
                        (line-separation 5)
                        (page-nums t)
                        (no-accidentals t)
                        (seqs-per-system 1)
                        (auto-open (get-sc-config 'cmn-display-auto-open))
                        (size 15))
```

38.6.316 rthm-seq-palette/create-psps

[rthm-seq-palette] [Methods]

DATE:

30 Mar 2006

DESCRIPTION:

Automatically create pitch-seq-palette objects for each rthm-seq object in the given rthm-seq-palette object.

The selection function given as an optional keyword argument should be able to generate a list of numbers (relative note levels) for a rthm-seq of any length; it takes two arguments only: the number of notes needed and the pitch-seq data lists (see below).

As a pitch-seq-palette usually has several options for each rthm-seq object, it's best when the selection-fun doesn't always return the same thing given the same number of notes. NB: This will silently kill the data of any pitch-seq-palette objects already supplied for any rthm-seqs in the palette.

Note that the default selection function will suffice in lots of cases. However, you may just want to use different data lists with the default function. In that case just pass these via :selection-fun-data.

ARGUMENTS:

- A rthm-seq-palette object.

OPTIONAL ARGUMENTS:

keyword arguments
- :selection-fun. This is a function that will return the pitch-seq numbers. It takes two arguments only: 1) the number of notes needed, and 2) the pitch-seq data lists. The function also needs to be able to handle being passed NIL NIL as arguments. In this case it should reset, if needs be; i.e. it's just a call to init and should return nothing. Default = #'create-psps-default.
- :pitch-seqs-per-rthm-seq. This is an integer that is the number of pitch-seqs each rthm-seq should have. NB: The method will simply cycle through the pitch-seqs given in the selection function to create the required number. Default = 3.
- :selection-fun-data. This contains the pitch-seq lists to be passed to the default selection function. There can be as many pitch-seqs in these lists as desired. The number of notes the pitch-seq will provide is the first item of the list. They need not be in ascending order. When this argument is passed a value of T, the selection function will reinitialize its default data and use that.

```
 - :overwrite. If NIL and a pitch-seq-palette was given for a particular
   rthm-seq, then a new one won't be created. Default = T.
 - :reinit. Used internally.  Do not change.

   At the moment, the default data are:
            '((1 ((3) (3) (1) (25)))
              (2 ((3 4) (5 2) (25 25) (1 25)))
              (3 ((3 4 3) (5 9 6) (1 2 4) (5 2 2) (6 2 3)))
              (4 ((3 4 3 4) (5 3 6 4) (9 4 5 11) (2 10 4 8)))
              (5 ((5 5 6 5 8) (7 7 7 4 8) (11 8 4 10 2) (7 7 4 9 9)))
              (6 ((4 5 5 3 6 6) (3 8 3 9 3 8) (9 3 9 5 10 6)))
              (7 ((8 8 8 5 9 6 9 ) (9 3 8 4 7 5 4) (3 4 3 5 3 4 3)))
              (8 ((3 3 4 3 3 1 5 4) (10 3 9 3 8 3 7 4) (3 5 8 2 8 9 4 11)))
              (9 ((3 6 4 7 4 7 3 6 7) (10 2 9 2 8 2 7 2 3)
                  (2 9 3 9 4 9 9 6 11)))
              (10 ((9 9 9 3 9 9 3 5 9 5) (8 9 8 9 5 9 9 5 6 6)))
              (12 ((1 2 5 5 5 5 5 5 5 5 4 5) (2 1 5 1 5 1 6 5 1 5 2 5)))
              (13 ((1 2 5 5 5 5 5 5 5 5 5 4 5 2) (2 1 5 1 5 1 6 5 1 5 2 5 1)))
              (14 ((1 2 5 5 5 5 5 5 5 5 4 5 2 1)
                   (2 1 5 1 5 1 6 5 1 5 2 5 1 2)))
              (15 ((1 2 5 5 5 5 5 5 5 5 4 5 2 1 2)
                   (2 1 5 1 5 1 6 5 1 5 2 5 1 2 6))))))
```

RETURN VALUE:

```
 Always returns the rthm-seq-palette argument.
```

EXAMPLE:

```
;; Create a rthm-seq-palette object that specifies pitch-seq-palettes for each
;; contained rthm-seq object and print the values of the individual
;; pitch-seq-palettes. Then apply the create-psps method using its default
;; values, and print the values of the individual pitch-seq-palettes again to
;; see the change.  NB You wouldn't normally specify pitch-seq-palettes in your
;; rthm-seq-palette as the whole point of this method is to have them created
;; algorithmically, but they are given here for purposes of comparison.
(let ((mrsp (make-rsp 'rsp-test
                      '((seq1 ((((2 4) q +e. s)
                                ((s) e (s) q)
                                (+e. s { 3 (te) te te } ))
                               :pitch-seq-palette (1 2 3 4 5 6 7)))
                        (seq2 ((((3 4) (e.) s { 3 te te te } +q)
                                ({ 3 +te (te) te } e e (q)))
                               :pitch-seq-palette (2 3 4 5 6 7 8)))
                        (seq3 ((((2 4) e e { 3 te te te })
                                ((5 8) (e) e e e s s))
```

```
                                :pitch-seq-palette (3 4 5 6 7 8 9 10 1 2)))))))
  (print
   (loop for rs in (data mrsp)
      collect
        (loop for ps in (data (pitch-seq-palette rs))
           collect (data ps))))
  (create-psps mrsp)
  (print
   (loop for rs in (data mrsp)
      collect
        (loop for ps in (data (pitch-seq-palette rs))
                collect (data ps)))))

=>
(((1 2 3 4 5 6 7)) ((2 3 4 5 6 7 8)) ((3 4 5 6 7 8 9 10 1 2)))

(((8 8 8 5 9 6 9) (9 3 8 4 7 5 4) (3 4 3 5 3 4 3))
 ((8 8 8 5 9 6 9) (9 3 8 4 7 5 4) (3 4 3 5 3 4 3))
 ((9 9 9 3 9 9 3 5 9 5) (8 9 8 9 5 9 9 5 6 6) (9 9 9 3 9 9 3 5 9 5)))

;; Use the :pitch-seqs-per-rthm-seq keyword argument to specify the number of
;; pitch-seq objects to be created for each rthm-seq. This example creates 5
;; instead of the default 3.
(let ((mrsp (make-rsp 'rsp-test
                      '((seq1 ((((2 4) q +e. s)
                                ((s) e (s) q)
                                (+e. s { 3 (te) te te } ))))
                        (seq2 ((((3 4) (e.) s { 3 te te te } +q)
                                ({ 3 +te (te) te } e e (q)))))
                        (seq3 ((((2 4) e e { 3 te te te })
                                ((5 8) (e) e e e s s))))))))
  (create-psps mrsp :pitch-seqs-per-rthm-seq 5)
  (loop for rs in (data mrsp)
            collect
              (loop for ps in (data (pitch-seq-palette rs))
                 collect (data ps))))

=>
(((8 8 8 5 9 6 9) (9 3 8 4 7 5 4) (3 4 3 5 3 4 3) (8 8 8 5 9 6 9)
  (9 3 8 4 7 5 4))
 ((3 4 3 5 3 4 3) (8 8 8 5 9 6 9) (9 3 8 4 7 5 4) (3 4 3 5 3 4 3)
  (8 8 8 5 9 6 9))
 ((9 9 9 3 9 9 3 5 9 5) (8 9 8 9 5 9 9 5 6 6) (9 9 9 3 9 9 3 5 9 5)
  (8 9 8 9 5 9 9 5 6 6) (9 9 9 3 9 9 3 5 9 5)))

;;; Now an example with our own selection-fun creating random pitch-seqs for
```

```
;;; demo purposes only:
(let ((mrsp (make-rsp 'rsp-test
                      '((seq1 ((((2 4) q +e. s)
                               ((s) e (s) q)
                               (+e. s { 3 (te) te te }))))
                        (seq2 ((((3 4) (e.) s { 3 te te te } +q)
                               ({ 3 +te (te) te } e e (q)))))
                        (seq3 ((((2 4) e e { 3 te te te })
                               ((5 8) (e) e e e s s))))))))
  (create-psps
   mrsp
   :selection-fun #'(lambda
                        (num-notes data-lists)
                      ;; NB we're not doing anything with data-lists here
                      (loop repeat num-notes collect (random 10))))
   (loop for rs in (data mrsp)
      collect
        (loop for ps in (data (pitch-seq-palette rs))
                 collect (data ps))))
=>
(((5 4 3 0 3 8 0) (1 2 8 5 3 7 8) (5 5 7 3 9 1 7))
 ((3 1 5 6 6 7 1) (7 7 8 5 5 2 4) (9 1 3 0 8 7 8))
 ((4 8 6 9 6 6 0 8 1 2) (1 5 5 7 7 2 9 3 1 2) (1 5 6 2 5 3 7 3 4 2)))
```

SYNOPSIS:

```
(defmethod create-psps ((rsp rthm-seq-palette)
                        &key
                          (selection-fun #'create-psps-default)
                          (selection-fun-data nil)
                          ;; MDE Sat Jul 14 17:20:27 2012 --
                          (reinit t)
                          ;; MDE Thu Feb  1 17:34:42 2018
                          (overwrite t)
                          ;; MDE Mon Jun 22 15:31:48 2015
                          (chords t)
                          (pitch-seqs-per-rthm-seq 3))
```

38.6.317 rthm-seq-palette/create-psps-default

[rthm-seq-palette] [Functions]

ARGUMENTS:

```
 - An integer that is the number of notes for which a pitch-seq-palette
```

object is needed.
- the pitch-seq data (see documentation for create psps method). Ideally this would only be passed the first time the function is called.

RETURN VALUE:

A list of numbers suitable for use in creating a pitch-seq object.

SYNOPSIS:

```
(defun create-psps-default (num-notes data-lists)
```

38.6.318 rthm-seq-palette/get-multipliers

[rthm-seq-palette] [Methods]

DESCRIPTION:

Get a list of factors by which a specified rhythmic unit must be multiplied in order to create the rhythms of a specified rthm-seq object within the given rthm-seq-palette object.

See also rthm-seq method for more information.

ARGUMENTS:

- A rthm-seq object.
- A rhythm unit, either as a number of a CMN shorthand symbol (i.e. 'e)
- A symbol that is the ID of the rthm-seq-object for which the multipliers is sought is also a required argument (though it is listed as an optional argument for internal reasons).

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether to round the results. T = round. Default = NIL. NB: Lisp always rounds to even numbers, meaning x.5 may sometimes round up and sometimes round down; thus (round 1.5) => 2, and (round 2.5) => 2.

RETURN VALUE:

A list of numbers.

EXAMPLE:

```
;; Returns a list of numbers, by default not rounded
(let ((mrsp
       (make-rsp 'rsp-test
                 '((seq1 ((((2 4) q +e. s)
                           ((s) e (s) q)
                           (+e. s { 3 (te) te te } ))
                          :pitch-seq-palette ((1 2 3 4 5 6 7)
                                              (1 3 5 7 2 4 6)
                                              (1 4 2 6 3 7 5)
                                              (1 5 2 7 3 2 4))))
                   (seq2 ((((4 4) (e.) s { 3 te te te } +h)
                           ({ 3 +te (te) te } e e (h)))
                          :pitch-seq-palette (2 3 4 5 6 7 8)))
                   (seq3 ((((2 4) e e { 3 te te te })
                           ((4 4) (e) e e e s s (s) s q))
                          :pitch-seq-palette (3 4 5 6 7 8 9 10 1 2 3 7)))))))
  (get-multipliers mrsp 'e 'seq1))

=> (2.0 1.0 1.5 2.0 1.1666666666666665 0.6666666666666666 0.6666666666666666)

;; Setting the option <round> argument to T returns rounded results
(let ((mrsp
       (make-rsp 'rsp-test
                 '((seq1 ((((2 4) q +e. s)
                           ((s) e (s) q)
                           (+e. s { 3 (te) te te } ))
                          :pitch-seq-palette ((1 2 3 4 5 6 7)
                                              (1 3 5 7 2 4 6)
                                              (1 4 2 6 3 7 5)
                                              (1 5 2 7 3 2 4))))
                   (seq2 ((((4 4) (e.) s { 3 te te te } +h)
                           ({ 3 +te (te) te } e e (h)))
                          :pitch-seq-palette (2 3 4 5 6 7 8)))
                   (seq3 ((((2 4) e e { 3 te te te })
                           ((4 4) (e) e e e s s (s) s q))
                          :pitch-seq-palette (3 4 5 6 7 8 9 10 1 2 3 7)))))))
  (get-multipliers mrsp 'e 'seq1 t))

=> (2 1 2 2 1 1 1)

;; The ID argument is required, even though it's listed as being optional. The
;; method interrupts with an error if no ID is supplied
(let ((mrsp
       (make-rsp 'rsp-test
                 '((seq1 ((((2 4) q +e. s)
                           ((s) e (s) q)
```

```
                          (+e. s { 3 (te) te te } ))
                         :pitch-seq-palette ((1 2 3 4 5 6 7)
                                             (1 3 5 7 2 4 6)
                                             (1 4 2 6 3 7 5)
                                             (1 5 2 7 3 2 4))))
                  (seq2 ((((4 4) (e.) s { 3 te te te } +h)
                          ({ 3 +te (te) te } e e (h)))
                         :pitch-seq-palette (2 3 4 5 6 7 8)))
                  (seq3 ((((2 4) e e { 3 te te te })
                          ((4 4) (e) e e e s s (s) s q))
                         :pitch-seq-palette (3 4 5 6 7 8 9 10 1 2 3 7)))))))
  (get-multipliers mrsp 'e))

=>
rthm-seq-palette::get-multipliers: third argument (rthm-seq ID) is required.
   [Condition of type SIMPLE-ERROR]

;;; Applying the method to the a multiple-bar rthm-seq object may return
;;; different results than applying the method to each of the bars contained
;;; within that rthm-seq object as individual one-bar rthm-seq objects, as the
;;; method measures the distances between attacked notes, regardless of ties
;;; and rests.
(let ((rs1 (make-rthm-seq '(seq1 ((((2 4) q +e. s))
                                  :pitch-seq-palette ((1 2))))))
      (rs2 (make-rthm-seq '(seq2 ((((2 4) (s) e (s) q))
                                  :pitch-seq-palette ((1 2))))))
      (rs3 (make-rthm-seq '(seq3 ((((2 4) +e. s { 3 (te) te te } ))
                                  :pitch-seq-palette ((1 2 3))))))
      (rs4 (make-rthm-seq '(seq4 ((((2 4) q +e. s)
                                   ((s) e (s) q)
                                   (+e. s { 3 (te) te te } ))
                                  :pitch-seq-palette ((1 2 3 4 5 6 7)))))))
  (print (get-multipliers rs1 'e))
  (print (get-multipliers rs2 'e))
  (print (get-multipliers rs3 'e))
  (print (get-multipliers rs4 'e)))

=>
(3.5 0.5)
(1.5 2.0)
(1.1666666666666665 0.6666666666666666 0.6666666666666666)
(3.5 1.0 1.5 3.5 1.1666666666666665 0.6666666666666666 0.6666666666666666)
```

SYNOPSIS:

```
(defmethod get-multipliers ((rsp rthm-seq-palette) rthm &optional id round)
```

38.6.319 rthm-seq-palette/invert

[rthm-seq-palette] [Methods]

DATE:

February 6th 2024

DESCRIPTION:

Turn all the rests in each rthm-seq into notes and all the notes/chords into rests. NB this method is destructive.

ARGUMENTS:

- the rthm-seq-palette object

OPTIONAL ARGUMENTS:

- T or NIL to call auto-beam on the result. Default = NIL.

RETURN VALUE:

The modified rthm-seq-palette object

SYNOPSIS:

```
(defmethod invert ((rsp rthm-seq-palette) &optional auto-beam ignore)
```

38.6.320 rthm-seq-palette/make-rsp

[rthm-seq-palette] [Functions]

DESCRIPTION:

Create a rthm-seq-palette object.

ARGUMENTS:

- A symbol that is to be the ID of the rhtm-seq-palette object created.
- A list containing rhtm-seq data to be made into rthm-seqs. Each item in this list is a list of data formatted as it would be when passed to the make-rthm-seq function.

OPTIONAL ARGUMENTS:

```
T or NIL to indicate whether to automatically generate and store inversions
of the pitch-seq-palette passed to the rthm-seq objects in the
rthm-seq-palette object created. T = generate and store. Default = NIL.
```

RETURN VALUE:

```
A rthm-seq-palette object.
```

EXAMPLE:

```
(make-rsp 'rsp-test '((seq1 ((((2 4) q +e. s)
                              ((s) e (s) q)
                              (+e. s { 3 (te) te te } ))
                             :pitch-seq-palette (1 7 3 4 5 2 6)))
                      (seq2 ((((3 4) (e.) s { 3 te te te } +q)
                              ({ 3 +te (te) te } e e (q)))
                             :pitch-seq-palette (3 1 2 5 1 7 6)))
                      (seq3 ((((2 4) e e { 3 te te te })
                              ((5 8) (e) e e e s s))
                             :pitch-seq-palette (4 4 4 5 4 4 4 5 4 3)))))

=>
RTHM-SEQ-PALETTE: psp-inversions: NIL
PALETTE:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 3
                      linked: T
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: RSP-TEST, tag: NIL,
data: (
RTHM-SEQ: num-bars: 3
[...]

;; Create two rthm-seq-palette objects, one with :psp-inversions set to NIL and
;; one with it set to T, and print the DATA of the pitch-seq-palettes of each
(let ((mrsp1 (make-rsp 'rsp-test
                       '((seq1 ((((2 4) q +e. s)
                                 ((s) e (s) q)
                                 (+e. s { 3 (te) te te } ))
                                :pitch-seq-palette (1 7 3 4 5 2 6))))
                       :psp-inversions nil))
      (mrsp2 (make-rsp 'rsp-test
```

```
                      '((seq1 ((((2 4) q +e. s)
                               ((s) e (s) q)
                               (+e. s { 3 (te) te te } ))
                              :pitch-seq-palette (1 7 3 4 5 2 6))))
                      :psp-inversions t)))
  (print
   (loop for i in (data (pitch-seq-palette (first (data mrsp1))))
      collect (data i)))
  (print
   (loop for i in (data (pitch-seq-palette (first (data mrsp2))))
      collect (data i))))

=>
((1 7 3 4 5 2 6))
((1 7 3 4 5 2 6) (7 1 5 4 3 6 2))
```

SYNOPSIS:

```
(defun make-rsp (id data &key (psp-inversions nil))
```

38.6.321 rthm-seq-palette/make-rsp-from-fragments

[rthm-seq-palette] [Functions]

DATE:

December 15th 2022

DESCRIPTION:

Define a bunch of rhythmic fragments then create a rthm-seq-palette by listing references to them in any order.

ARGUMENTS:

- a list of rhythm fragments suitable to be used in a rthm-seq (i.e. e.g. including tuplet and beaming info.), prefaced by an id
- a list of rthm-seqs using these fragments consisting of a list of the fragments to be combined into bars, each prefaced by the meter (unless it doesn't change)

OPTIONAL ARGUMENTS:

- T or NIL to indicated whether the number of rthm-seqs made should be printed

RETURN VALUE:

```
a rthm-seq-palette object where each rthm-seq will have an ID ascending from
1
```

EXAMPLE:

```
(make-rsp-from-fragments
  '((1 (- s s - (e)))
    (2 (s (s) (s) s))
    (3 ((s) - s e -))
    (4 (- s s (s) s -))
    (5 ((e) - s s -))
    (6 ((q)))
    (7 (h))
    (8 (q.)))
   ;; one rthm-seq per line: number 1 will have three bars 3/4 3/4 4/4
  '((((3 4) 1 2 3) (2 3 4) ((4 4) 5 6 1 2))
   ;; rthm-seq 2 has 5 bars: 7/8 7/8 2/4 2/4 7/8
    (((7 8) 7 8) (3 4 8) ((2 4) 3 4) (5 6) ((7 8) 5 6 8))
    (((4 4) 1 2 1 2) ((5 4) 1 2 1 2 2) (3 4 5 7))
    (((3 8) 8) ((2 4) 5 6) (7))))
```

SYNOPSIS:

```
(defun make-rsp-from-fragments (fragments references &optional verbose)
```

38.6.322 rthm-seq-palette/make-rsp-from-unit-multipliers-simp

[rthm-seq-palette] [Functions]

DESCRIPTION:

```
Make a rthm-seq-palette using lists of lists of multiples of a single unit
e.g. '(2 2 1) would result in an automatic time signature of 5/8 if passed a
unit of 'e (or 8). In that case the rhythms would be q q e. Rests may be
created by passing a number in parentheses.
```

ARGUMENTS:

```
- the ID for the rthm-seq-plaette
- the rhythm unit (symbol, number or rhythm object)
- the lists of lists multipliers, each sublist representing a rthm-seq and
  each sublist of that representing a bar.
```

RETURN VALUE:

a rthm-seq-palette object

EXAMPLE:

```
(print-simple (make-rsp-from-unit-multipliers-simp
               'test-unit-mults 's
               '(((3 4 1 2) (2 (1) 2))
                 ((2 1 2 1) (3) ((4)))
                 (((3) 4 1) (3 4 1) ((2) 3) (2 3) (2)))))
rthm-seq-palette TEST-UNIT-MULTS
rthm-seq 1
NIL: bar -1: (10 16):
note E.,
note Q,
note S,
note E,
NIL: bar -1: (5 16):
note E,
rest S,
note E,
rthm-seq 2
NIL: bar -1: (6 16):
note E,
note S,
note E,
note S,
NIL: bar -1: (3 16):
note E.,
NIL: bar -1: (4 16):
rest 4,
rthm-seq 3
NIL: bar -1: (8 16):
rest E.,
note Q,
note S,
NIL: bar -1: (8 16):
note E.,
note Q,
note S,
NIL: bar -1: (5 16):
rest E,
note E.,
NIL: bar -1: (5 16):
note E,
```

```
note E.,
NIL: bar -1: (2 16):
note E,
T
```

SYNOPSIS:

```
(defun make-rsp-from-unit-multipliers-simp (id unit seqs)
```

38.6.323 rthm-seq-palette/reset-psps

[rthm-seq-palette] [Methods]

DESCRIPTION:

```
Call the reset method (inherited from circular-sclist) for all
pitch-seq-palette objects of all rthm-seq objects in the given
rthm-seq-palette object, resetting their pointers to the head of the
sequence. This ensures that each rthm-seq starts over again at the first
note of the first given pitch-seq.

NB Since 28/10/13 there's also a reset method for the palette class which
will mean we call the reset method of all pitch-seqs here.  So this method
is now partially obsolete (but still perhaps useful in that in only resets
the psps, not the whole palette).
```

ARGUMENTS:

```
- A rthm-seq-palette object.
```

RETURN VALUE:

```
Always returns T.
```

EXAMPLE:

```
;; Create a rthm-seq-palette object whose first rthm-seq has three pitch-seq
;; objects in its pitch-seq-palette. Apply the get-next method to the
;; pitch-seq-palette object of the first rthm-seq object twice, then print the
;; data of the next pitch-seq object to show where we are. Apply the reset-psps
;; method and print the data of the next pitch-seq object to show that we've
;; returned to the beginning of the pitch-seq-palette.

(let ((mrsp
       (make-rsp 'rsp-test
```

```
                '((seq1 ((((2 4) q +e. s)
                          ((s) e (s) q)
                          (+e. s { 3 (te) te te } ))
                         :pitch-seq-palette ((1 2 3 4 5 6 7)
                                             (1 3 5 7 2 4 6)
                                             (1 4 2 6 3 7 5)
                                             (1 5 2 7 3 2 4))))
                  (seq2 ((((3 4) (e.) s { 3 te te te } +q)
                          ({ 3 +te (te) te } e e (q)))
                         :pitch-seq-palette (2 3 4 5 6 7 8)))
                  (seq3 ((((2 4) e e { 3 te te te })
                          ((5 8) (e) e e e s s))
                         :pitch-seq-palette (3 4 5 6 7 8 9 10 1 2)))))))
  (loop repeat 2
       do (get-next (pitch-seq-palette (first (data mrsp)))))
  (print (data (get-next (pitch-seq-palette (first (data mrsp))))))
  (reset-psps mrsp)
  (print (data (get-next (pitch-seq-palette (first (data mrsp)))))))

=>
(1 4 2 6 3 7 5)
(1 2 3 4 5 6 7)
```

SYNOPSIS:

```
(defmethod reset-psps ((rsp rthm-seq-palette))
```

38.6.324 rthm-seq-palette/rs-equal-durations?

[rthm-seq-palette] [Methods]

DATE:

```
June 13th 2017, Edinburgh
```

DESCRIPTION:

```
See if all the rthm-seqs in a palette (or sub-palette) have the same
duration.
```

ARGUMENTS:

```
- the rthm-seq-palette object
```

OPTIONAL ARGUMENTS:

```
- a reference (symbol or list) into the rthm-seq-palette to examine a
  sub-palette
```

RETURN VALUE:

```
If all rthm-seqs in the palette have the same duration, then their duration
in quarter notes, otherwise nil.
```

SYNOPSIS:

```
(defmethod rs-equal-durations? ((rsp rthm-seq-palette) &optional ref)
```

38.6.325 rthm-seq-palette/rsp-subseq

[rthm-seq-palette] [Methods]

DATE:

```
23rd October 2013
```

DESCRIPTION:

```
(Recursively) change all the rthm-seq objects in the palette to be a
subsequence of the existing rthm-seqs.
```

ARGUMENTS:

```
- the original rthm-seq-palette object
- the start bar (1-based)
```

OPTIONAL ARGUMENTS:

```
- the end bar (1-based and (unlike Lisp's subseq function) inclusive).  If
  NIL, we'll use the original end bar of each rthm-seq. Default = NIL.
```

RETURN VALUE:

```
The original rthm-seq-palette but with the new rthm-seqs
```

SYNOPSIS:

```
(defmethod rsp-subseq ((rsp rthm-seq-palette) start &optional end)
```

38.6.326 rthm-seq-palette/scale

[rthm-seq-palette] [Methods]

DESCRIPTION:

```
Scale the durations of the rhythm objects in a given rthm-seq-palette
object by the specified factor.

NB: As is evident in the examples below, this method does not replace the
    original data in the rthm-seq-palette object's DATA slot.
```

ARGUMENTS:

```
- A rthm-seq-palette object.
- A real number that is the scaling factor.
```

OPTIONAL ARGUMENTS:

```
(- the three IGNORE arguments are for internal purposes only).
```

RETURN VALUE:

```
Returns a rthm-seq-palette object.
```

EXAMPLE:

```
;; Returns a rthm-seq-palette object
(let ((mrsp
       (make-rsp 'rsp-test
                 '((seq1 ((((2 4) q +e. s)
                           ((s) e (s) q)
                           (+e. s { 3 (te) te te } ))
                          :pitch-seq-palette ((1 2 3 4 5 6 7)
                                              (1 3 5 7 2 4 6)
                                              (1 4 2 6 3 7 5)
                                              (1 5 2 7 3 2 4))))
                   (seq2 ((((4 4) (e.) s { 3 te te te } +h)
                           ({ 3 +te (te) te } e e (h)))
                          :pitch-seq-palette (2 3 4 5 6 7 8)))
                   (seq3 ((((2 4) e e { 3 te te te })
                           ((4 4) (e) e e e s s (s) s q))
                          :pitch-seq-palette (3 4 5 6 7 8 9 10 1 2 3 7)))))))
  (scale mrsp 2))
```

```
=>
RTHM-SEQ-PALETTE: psp-inversions: NIL
PALETTE:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 3
                      linked: T
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: RSP-TEST, tag: NIL,
data: (
RTHM-SEQ: num-bars: 3

;; Apply the method and loop through the rthm-seq objects in the
;; rthm-seq-palette object's DATA slot, using the print-simple method to see
;; the changes

(let ((mrsp
       (make-rsp 'rsp-test
                 '((seq1 ((((2 4) q +e. s)
                           ((s) e (s) q)
                           (+e. s { 3 (te) te te } ))
                          :pitch-seq-palette ((1 2 3 4 5 6 7)
                                              (1 3 5 7 2 4 6)
                                              (1 4 2 6 3 7 5)
                                              (1 5 2 7 3 2 4))))
                   (seq2 ((((4 4) (e.) s { 3 te te te } +h)
                           ({ 3 +te (te) te } e e (h)))
                          :pitch-seq-palette (2 3 4 5 6 7 8)))
                   (seq3 ((((2 4) e e { 3 te te te })
                           ((4 4) (e) e e e s s (s) s q))
                          :pitch-seq-palette (3 4 5 6 7 8 9 10 1 2 3 7)))))))
   (scale mrsp .5)
   (print-simple mrsp))

=>
rthm-seq-palette RSP-TEST
rthm-seq SEQ1
(2 8): note E, note S., note 32,
(2 8): rest 32, note S, rest 32, note E,
(2 8): note S., note 32, rest TS, note TS, note TS,
rthm-seq SEQ2
(4 8): rest S., note 32, note TS, note TS, note TS, note Q,
(4 8): note TS, rest TS, note TS, note S, note S, rest Q,
```

```
rthm-seq SEQ3
(2 8): note S, note S, note TS, note TS, note TS,
(4 8): rest S, note S, note S, note S, note 32, note 32, rest 32, note 32, note E,
```

SYNOPSIS:

```
(defmethod scale ((rsp rthm-seq-palette) scaler
                  &optional ignore1 ignore2 ignore3)
```

38.6.327 rthm-seq-palette/split-into-single-bars

[rthm-seq-palette] [Methods]

DESCRIPTION:

```
Split every rthm-seq in a palette into as many single-bar rthm-seqs as
there are bars.  This creates an extra level of recursion so that whereas a
rthm-seq that was referenced by e.g. '(long 1 a) beforehand, afterwards
each bar will be reference by '(long 1 a 1), '(long 1 a 2) etc. This is
true even for sequences that only contained one bar before processing.  The
pitch-seq-palettes of the original rthm-seqs will be used to set the
pitch-seqs of the new rthm-seqs.
```

ARGUMENTS:

```
- the rthm-seq-palette object
```

OPTIONAL ARGUMENTS:

```
- whether to clone the palette before processing (T or NIL).  Default = T.
```

RETURN VALUE:

```
A rthm-seq-palette object with extra layers of recursion.
```

SYNOPSIS:

```
(defmethod split-into-single-bars ((rsp rthm-seq-palette) &optional (clone t))
```

38.6.328 palette/set-palette

[palette] [Classes]

NAME:

set-palette

File: set-palette.lsp

Class Hierarchy: named-object -> linked-named-object -> sclist -> circular-sclist -> assoc-list -> recursive-assoc-list -> palette -> set-palette

Version: 1.1.0

Project: slippery chicken (algorithmic composition)

Purpose: Implementation of the set-palette class which extends the palette class by simply instantiating the sets given in the palette.

Note that the sets in this palette may refer to previously defined sets in order to obviate retyping note lists. Hence the reference to bcl-chord2 in the bcl-chord3 set of the example below will instantiate a set based on a transposed clone of that set previously stored as bcl-chord2.

```
(make-set-palette
 'test
 '((bcl-chord1
    ((bf1 ef2 aqf2 c3 e3 gqf3 gqs3 cs4 d4 g4 a4 cqs5
          dqf5 gs5 b5)
     :subsets
     ((tc1 (ds2 e3 a4))
      (tc2 (bf1 d4 cqs5))
      (qc1 (aqf2 e3 a4 dqf5 b5))
      (qc2 (bf1 c3 gqs3 cs4 cqs5)))
     :related-sets
     ((missing (bqs0 eqs1 f5 aqs5 eqf6 fqs6
                bqf6 dqs7 fs7)))))
   (bcl-chord2
    ((bf1 d2 fqf2 fqs2 b2 c3 f3 g3 bqf3 bqs3 fs4 gs4 a4
      cs5 gqf5)
     :subsets
     ((tc1 (d2 g3 cs5))
      (tc2 (eqs2 f3 bqf3))
      (qc1 (eqs2 c3 f3 fs4 gqf5))
      (qc2 (d2 fqs2 bqs3 gs4 a4)))
     :related-sets
     ((missing (aqs0 dqs1 ds5 gqs5 dqf6 eqf6 aqf6 cqs7
```

```
                                    e7)))))
                          (bcl-chord3
                           (bcl-chord2 :transposition 13))))

 Author:           Michael Edwards: m@michael-edwards.org

 Creation date:    August 14th 2001

 $$ Last modified:  20:03:24 Tue Jun  3 2025 CEST

 SVN ID: $Id$
```

38.6.329 set-palette/add-harmonics

[set-palette] [Methods]

DESCRIPTION:

```
 Add harmonically-related pitches to each set in the palette. See the sc-set
 class method for details and the get-harmonics function (utilities.lsp) for
 keyword arguments.
```

SYNOPSIS:

```
(defmethod add-harmonics ((sp set-palette) &rest keywords)
```

38.6.330 set-palette/auto-sequence

[set-palette] [Methods]

DATE:

```
 July 30th 2015, Edinburgh
```

DESCRIPTION:

```
 Automatically create an ordering for the sets in a set-palette based on a
 dissonance envelope and a spectral centroid envelope. For a description of
 a set or chord's dissonance or spectral centroid see the
 calculate-spectral-centroid and calculate-dissonance methods in the chord
 class.

 The envelopes describe a desired general tendency to, for example, proceed
 from less dissonant to more dissonant sets as the ordering (sequence) of
```

sets proceeds. Such an envelope would move from lower to higher values. Similarly with the spectral centroid envelope: moving from a lower to a higher value implies moving from chords with an overall lower pitch height to chords with a higher pitch height.

The envelopes should be expressed over any X axis range but with a Y axis range of 0.0 to 1.0 only. The x-axes will be stretched to fit over the number of sets in the palette. The Y axes will be stretched to fit the range of dissonance and centroid values to be found in the sets in the palette (see the quality-extremes method).

The default method is successive. It selects sets one-by-one via a sort function which compares the sets' characteristics to those of the current envelope values. Essentially, when we are comparing two sets via the sort function, we look at the deviation between the sets' dissonance values and the current desired dissonance value from the envelope, similarly with the spectral centroid. The chord with the smallest combined deviation will be chosen first. If either envelope is nil, then the sorting is based on the other envelope only. Similarly, weighting factors of any arbitrary positive number can be passed via :dissonance-weight and :centroid-weight to emphasise or deemphasise these properties when sorting. Higher values will mean that that property will take precedence over the other property.

An alternative method, indicated by :permutate T, is to get all (or at least a lot of) permutations of the sets in the palette and then score each ordering against the curves, as a form of fitness test. The advantage of this approach is that the best solution overall can be found. The successive approach, on the other hand will find the best fit at the beginning of the process but as it proceeds along the envelope and we run out of sets to select from, the fit can become worse and worse.

In any case there is no guarantee that the desired curves will be expressed exactly in the returned ordering. The function tries to find the best fit but success depends very much on the number and variety of sets in the palette.

Also taken into account when in both methods is the lowest note of the set. If :repeating-bass is NIL (the default) then the function tries to avoid repeating bass notes in two consecutive sets. Repeating highest notes are allowed.

NB Unlike the chord methods which calculate dissonance and spectral centroid, there is no way to pass spectral data here. If you want to override the default spectral data then use (set-sc-config 'default-spectra ...). See globals.lsp for more details.

ARGUMENTS:

```
 - the set-palette object
```

OPTIONAL ARGUMENTS:

```
 keyword arguments:
 - dissonance-env: The desired dissonance envelope. Y values: 0.0 to 1.0
   Default '(0 .1 62 1 100 .3)
 - centroid-env:  The desired spectral centroid envelope.Y values: 0.0 to 1.0
   Default '(0 .4 62 1 100 .2)
 - dissonance-weight: A weighting factor (scaler) applied to
   dissonance. Higher values result in dissonance playing a larger role in
   the decision process. Default 1.0
 - centroid-weight: a similar factor for spectral centroid. Default 1.0
 - verbose:  Whether to print data as the decision process proceeds.
   Default NIL
 - repeating-bass: Whether to allow bass notes to repeat between two
   consecutive chords. Default NIL
 - silent: Whether to print warnings or not. Default NIL
 - map-section: if you want a list of references suitable to be passed
   as a set-map, set this keyword to the integer ID of the section the map
   will be used for. Default = NIL.
 - permutate: use the permutation rather than the successive approach. This
   can either be T (which will limit to 2000), 'all (get all
   permutations--could take a very long time), or an integer to represent
   the maximum number of permutations we'll try. Default = NIL.
```

RETURN VALUE:

```
 If we're using the permutate method then a single list of the sets'
 set-palette references is returned. If we're using the successive method we
 return two values: A list of the full references of the sets in the set
 palette in the automatically determined order, along with a list of the
 deviations from the ideal this order represents. Either method might return
 a list suitable for passing to a set-map (see example below).
```

EXAMPLE:

```
(auto-sequence
 (recursive-set-palette-from-ring-mod '(a4 b4) 'spfrm-test
                                      :warn-no-bass nil)
 :verbose nil :centroid-weight 2 :silent t)
=>
((A4 15) (A4 13) (A4 4) (A4 7) (B4 4) (A4 14) (A4 2) (A4 20) (B4 2) (B4 17)
```

```
 (B4 14) (B4 11) (B4 9) (A4 12) (B4 21) (B4 6) (A4 6) (B4 5) (B4 7) (B4 12)
 (A4 19) (A4 5) (A4 9) (A4 3) (A4 11) (A4 10) (B4 18) (B4 10) (B4 16) (B4 3)
 (B4 19) (A4 17) (B4 13) (A4 8) (B4 15) (B4 8) (B4 20) (B4 1) (A4 21) (A4 1)
 (A4 18) (A4 16))
((0.3433064594719475d0 0.04460029910249326d0 0.38790675857444074d0)
 (0.36511199730074173d0 0.10718227303159739d0 0.4722942703323391d0)
...

(auto-sequence
 (recursive-set-palette-from-ring-mod '(a4 b4) 'spfrm-test
                                      :warn-no-bass nil)
 :verbose nil :centroid-weight 2 :silent t :permutate t :map-section 1)
=>
((1
  ((B4 1) (B4 3) (B4 15) (B4 5) (B4 10) (A4 3) (A4 17) (A4 1) (A4 14) (A4 15)
   (B4 12) (A4 4) (B4 2) (B4 18) (A4 20) (A4 5) (B4 4) (B4 6) (B4 20) (B4 17)
   (B4 16) (B4 11) (A4 6) (B4 9) (A4 7) (A4 12) (B4 21) (A4 18) (A4 19) (A4 13)
   (B4 19) (B4 7) (A4 16) (B4 13) (A4 8) (A4 2) (B4 8) (A4 9) (A4 10) (A4 11)
   (A4 21) (B4 14))))
```

SYNOPSIS:

```
(defmethod auto-sequence ((sp set-palette)
                          &key
                            (dissonance-env '(0 .1 62 1 100 .3))
                            (centroid-env '(0 .4 62 1 100 .2))
                            (dissonance-weight 1.0)
                            (centroid-weight 1.0)
                            map-section permutate verbose repeating-bass silent)
```

38.6.331 set-palette/calculate-spectral-centroid

[set-palette] [Methods]

DATE:

```
January 30th 2016
```

DESCRIPTION:

```
Calculate the spectral centroid values for each set in a set-palette. This
will set the centroid slot of each set.
```

ARGUMENTS:

```
- a set-palette object
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- sort: If T, sorts the list values from high to low, otherwise the list
  will be returned with the order of the sets in the palette. Default =
  NIL.
see this method in the chord class for a description of further keywords
```

RETURN VALUE:

```
a list of two-element lists: the reference to the set within the palette
and the centroid value.
```

SYNOPSIS:

```
(defmethod calculate-spectral-centroid ((sp set-palette)
                                        &rest keyargs &key &allow-other-keys)
```

38.6.332 set-palette/cmn-display

[set-palette] [Methods]

DESCRIPTION:

```
Generate printable music notation output (.EPS) of the given set-palette
object, including separate notation of the SUBSETS and RELATED-SETS slots,
using the Common Music Notation (CMN) interface. The method requires at
least the name of the given set-palette object, but has several additional
optional arguments for customizing output.

NB: Some of the keyword arguments are CMN attributes and share the same
    name as the CMN feature they effect.
```

ARGUMENTS:

```
- A set-palette object.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :file. The file path, including the file name, of the file to be
  generated.
- :4stave. T or NIL to indicate whether the note-heads of the output should
  be printed on 4 staves (or 2). T = 4. Default = NIL.
- :text-x-offset. Number (positive or negative) to indicate the horizontal
```

offset of any text in the output. A value of 0.0 results in all text being lined up left-flush with the note-heads below it. Units here and below are relative to CMN staff size. Default = -0.5.
- :text-y-offset. Number (positive or negative) to indicate the vertical offset of any text in the output.
- :font-size. A number indicating the size of any text font used in the output. This affects text only and not the music (see :size below for changing the size of the music). If 0 then no text will be displayed. Default = 10.0.
- :break-line-each-set. T or NIL to indicate whether each set-palette object should be printed on a separate staff or consecutively on the same staff. T = one staff per set-palette object. Default = T.
- :line-separation. A number to indicate the amount of white space between lines of music (systems), measured as a factor of the staff height. Default = 3. This is a direct CMN attribute.
- :staff-separation. A number to indicate the amount of white space between staves belong to the same system, measured as a factor of the staff height. Default = 3. This is a direct CMN attribute.
- :transposition. Nil or a number (positive or negative) to indicate the number of semitones by which the pitches of the given set-palette object should be transposed before generating the CMN output. Default = NIL (0).
- :size. A number to indicate the size of the music-font in the CMN output. This affects music only, not text.
- :use-octave-signs. T or NIL to indicate whether to automatically insert ottava spanners. Automatic placement depends on the overall pitch content. This is a slippery-chicken process and may produce different results than :automatic-octave-signs, which is a direct CMN process. T = insert octave signs. Default = NIL.
- :automatic-octave-signs. T or NIL to indicate whether to automatically insert ottava spanners. Automatic placement depends on the overall pitch content. This is a direct CMN process and may produce different results than :use-octave-signs, which is a slippery-chicken process. T = insert octave signs. Default = NIL.
- :include-missing-chromatic. T or NIL to indicate whether to also print any chromatic pitches from the complete-set that are not present in the given set-palette object. T = print. Default = T.
- :include-missing-non-chromatic. T or NIL to indicate whether to also print any microtonal pitches from the complete-set that are not present in the given set-palette object. T = print. Default = T.

RETURN VALUE:

T

EXAMPLE:

```
;; A typical example with some specified keyword values for file, font-size,
;; break-line-each-set, size, include-missing-chromatic and
;; include-missing-non-chromatic
(let ((msp (make-set-palette
            'test
         p   '((1 ((1
                    ((c3 g3 cs4 e4 fs4 a4 bf4 c5 d5 f5 gf5 af5 ef6)))
                   (2
                    ((c3 g3 cs4 e4 fs4 a4 bf4 c5 d5 f5 gf5 af5 ef6)
                     :subsets
                     ((tc1 (d2 g3 cs5))
                      (tc2 (eqs2 f3 bqf3))
                      (tc3 (b2 bqs3 gqf5)))))))
               (2 ((1 ((1 1) :transposition 5))
                   (2 ((1 2) :transposition 5))))
               (3 ((1 ((1 1) :transposition -2))
                   (2 ((1 2) :transposition -2))))))))
  (cmn-display msp
               :file "/tmp/sp-output.eps"
               :font-size 8
               :break-line-each-set nil
               :size 10
               :include-missing-chromatic nil
               :include-missing-non-chromatic nil))
```

SYNOPSIS:

```
(defmethod cmn-display ((sp set-palette) &rest keyargs &key &allow-other-keys)
```

38.6.333 set-palette/find-sets-with-pitches

[set-palette] [Methods]

DESCRIPTION:

```
Return a list of sets (as complete-set objects) from a given set-palette
object based on whether they contain specified pitches.

NB: Only sets which contain all of the specified pitches will be returned.
```

ARGUMENTS:

```
- A set-palette object.
- A list of pitches, either as pitch objects or note-name symbols.
```

```
 OPTION ARGUMENTS
 - T or NIL to indicate whether to print the notes of each successful set as
   they are being examined.
```

RETURN VALUE:

```
 A list of complete-set objects.
```

EXAMPLE:

```
;; Find sets that contain a single pitch
(let ((msp (make-set-palette
            'test
            '((1 ((1
                   ((g3 c4 e4 g4)))
                  (2
                   ((c4 d4 e4 g4)))))
              (2 ((1 ((1 1) :transposition 5))
                  (2 ((1 2) :transposition 5))))
              (3 ((1 ((1 1) :transposition -2))
                  (2 ((1 2) :transposition -2))))))))
  (find-sets-with-pitches msp '(c4)))

=>
(
COMPLETE-SET: complete: NIL
[...]
data: (BF3 C4 D4 F4)
[...]
COMPLETE-SET: complete: NIL
[...]
data: (C4 F4 A4 C5)
[...]
COMPLETE-SET: complete: NIL
[...]
data: (C4 D4 E4 G4)
[...]
COMPLETE-SET: complete: NIL
[...]
data: (G3 C4 E4 G4)
)

;; Search for a set of two pitches, printing the successfully matched sets
(let ((msp (make-set-palette
            'test
            '((1 ((1
```

```
                   ((g3 c4 e4 g4)))
                  (2
                   ((c4 d4 e4 g4)))))
              (2 ((1 ((1 1) :transposition 5))
                  (2 ((1 2) :transposition 5))))
              (3 ((1 ((1 1) :transposition -2))
                  (2 ((1 2) :transposition -2))))))))
  (print (find-sets-with-pitches msp '(c4 f4) t)))

=>
(2 1): (C4 F4 A4 C5)
(3 2): (BF3 C4 D4 F4)
(
COMPLETE-SET: complete: NIL
[...]
data: (BF3 C4 D4 F4)
COMPLETE-SET: complete: NIL
[...]
data: (C4 F4 A4 C5)
)
```

SYNOPSIS:

```
(defmethod find-sets-with-pitches ((sp set-palette) pitches &optional print)
```

38.6.334 set-palette/force-micro-tone

[set-palette] [Methods]

DESCRIPTION:

```
Change the value of the MICRO-TONE slot of all pitch objects in a given
set-palette object to the specified <value>. NB If the pitches are
microtonal and thus have associated pitch-bends and microtonal frequencies,
these will not be changed, i.e. only the micro-tone slot is changed.
```

ARGUMENTS:

```
- A set-palette object.
```

OPTIONAL ARGUMENTS:

```
- An item of any type that is to be the new value of the MICRO-TONE slot of
  all pitch objects in the given sc-set object (generally T or
  NIL). Default = NIL.
```

RETURN VALUE:

Always returns T.

EXAMPLE:

```
;; Create a set-palette object whose individual sets contain some micro-tones
;; and print the contents of all the MICRO-TONE slots to see the values. Then
;; apply the force-micro-tone method and print the slots again to see the
;; changes.

(let ((msp (make-set-palette
            'test
            '((1 ((1
                   ((bf1 ef2 aqf2 c3 e3 gqf3 gqs3 cs4 d4 g4 a4 cqs5 dqf5 gs5
                         b5)))
                  (2
                   ((bf1 d2 fqf2 fqs2 b2 c3 f3 g3 bqf3 bqs3 fs4 gs4 a4 cs5 gqf5)
                    :subsets
                    ((tc1 (d2 g3 cs5))
                     (tc2 (eqs2 f3 bqf3))
                     (tc3 (b2 bqs3 gqf5)))))))
              (2 ((1 ((1 1) :transposition 5))
                  (2 ((1 2) :transposition 5))))
              (3 ((1 ((1 1) :transposition -2))
                  (2 ((1 2) :transposition -2))))))))
  (print (loop for i in (data msp)
            collect (loop for j in (data (data i))
                          collect (loop for p in (data j)
                                        collect (micro-tone p)))))
  (force-micro-tone msp t)
  (print (loop for i in (data msp)
            collect (loop for j in (data (data i))
                          collect (loop for p in (data j)
                                        collect (micro-tone p))))))

=>
(((NIL NIL T NIL NIL T T NIL NIL NIL NIL T T NIL NIL)
  (NIL NIL T T NIL NIL NIL NIL T T NIL NIL NIL NIL T))
 ((NIL NIL T NIL NIL T T NIL NIL NIL NIL T T NIL NIL)
  (NIL NIL T T NIL NIL NIL NIL T T NIL NIL NIL NIL T))
 ((NIL NIL T NIL NIL T T NIL NIL NIL NIL T T NIL NIL)
  (NIL NIL T T NIL NIL NIL NIL T T NIL NIL NIL NIL T)))

(((T T T T T T T T T T T T T T T) (T T T T T T T T T T T T T T T))
 ((T T T T T T T T T T T T T T T) (T T T T T T T T T T T T T T T))
```

```
((T T T T T T T T T T T T T T T) (T T T T T T T T T T T T T T T)))
```

SYNOPSIS:

```
(defmethod force-micro-tone ((sp set-palette) &optional value)
```

38.6.335 set-palette/gen-max-coll-file

[set-palette] [Methods]

DATE:

26-Dec-2009

DESCRIPTION:

Write a text file from a given set-palette object suitable for reading into Max/MSP's coll object. The resulting text file has one line for each set in the palette, with the coll index being the ID of the set. The rest of the line is a list of frequency/amplitude pairs (or MIDI note number/velocity if required). 0.1 is the default amplitude for frequencies; 80 the default velocity for MIDI; if 'transp is the format argument then no amplitude or velocity data is written.

ARGUMENTS:

- A set-palette object.
- The name (and path) of the .txt file to write.

OPTIONAL ARGUMENTS:

- 'freq, 'freq-only (no amplitudes), 'midi', or 'transp to indicate whether frequencies in Hertz, MIDI note numbers, or semitone transposition factors should be generated. If the latter then the middle note will be represented by 0 transposition. Default = 'freq (frequencies).
- a list of references into the palette to define the order in which the sets are written. Default = NIL = sets will be written in the order in which they appear in the palette.
- T or NIL to indicate whether successive integers should be used as indices in the coll. Default = NIL = use combined set-palette and set IDs.

RETURN VALUE: EXAMPLE:

```
;; Generates frequencies by default
```

```
(let ((msp (make-set-palette
            'test
            '((1 ((1
                   ((g3 c4 e4 g4)))
                  (2
                   ((c4 d4 e4 g4)))))
              (2 ((1 ((1 1) :transposition 5))
                  (2 ((1 2) :transposition 5))))
              (3 ((1 ((1 1) :transposition -2))
                  (2 ((1 2) :transposition -2))))))))
  (gen-max-coll-file msp "/tmp/msp-mcf.txt"))

;; Set the optional argument to 'midi to generate MIDI note numbers instead
(let ((msp (make-set-palette
            'test
            '((1 ((1
                   ((g3 c4 e4 g4)))
                  (2
                   ((c4 d4 e4 g4)))))
              (2 ((1 ((1 1) :transposition 5))
                  (2 ((1 2) :transposition 5))))
              (3 ((1 ((1 1) :transposition -2))
                  (2 ((1 2) :transposition -2))))))))
  (gen-max-coll-file msp "/tmp/msp-mcf.txt" 'midi))
```

SYNOPSIS:

```
(defmethod gen-max-coll-file ((sp set-palette) file &optional
                                                    (format 'freq)
                                                    refs ints)
```

38.6.336 set-palette/gen-midi-chord-seq

[set-palette] [Methods]

DESCRIPTION:

```
Generate a MIDI file in which each set of the given set-palette object is
played at 1 second intervals.
```

ARGUMENTS:

```
- A set-palette object.
- The name and path for the MIDI file to be generated.
```

OPTIONAL ARGUMENTS:

- the tempo in beats per minute. Default = 60.
- subsets-id, related-sets-id. If you want to generate a MIDI file of just a named subset or related-set, pass the ID here. NB this ID will have to exist for all sets.

RETURN VALUE:

Always returns T

EXAMPLE:

```
(let ((msp (make-set-palette
            'test
            '((1 ((1
                   ((bf1 ef2 aqf2 c3 e3 gqf3 gqs3 cs4 d4 g4 a4 cqs5 dqf5 gs5
                         b5)))
                  (2
                   ((bf1 d2 fqf2 fqs2 b2 c3 f3 g3 bqf3 bqs3 fs4 gs4 a4 cs5 gqf5)
                    :subsets
                    ((tc1 (d2 g3 cs5))
                     (tc2 (eqs2 f3 bqf3))
                     (tc3 (b2 bqs3 gqf5)))))))
              (2 ((1 ((1 1) :transposition 5))
                  (2 ((1 2) :transposition 5))))
              (3 ((1 ((1 1) :transposition -2))
                  (2 ((1 2) :transposition -2))))))))
  (gen-midi-chord-seq msp "/tmp/msp-gmchs.mid"))
```

SYNOPSIS:

```
(defmethod gen-midi-chord-seq ((sp set-palette) midi-file
                               &optional (tempo 60.0)
                                 subsets-id related-sets-id)
```

38.6.337 set-palette/limit

[set-palette] [Methods]

DESCRIPTION:

Limit all the sets in a palette to a specified range. See the tl-set class limit method for details of the this method and its arguments.

RETURN VALUE:

```
The set-palette object.
```

SYNOPSIS:

```
(defmethod limit ((sp set-palette) &key upper lower do-related-sets)
```

38.6.338 set-palette/limit-shift-octave

[set-palette] [Methods]

DATE:

```
19th August 2015, Edinburgh
```

DESCRIPTION:

```
Apply the tl-set method of the same name to each set in the palette. See
that class method description for details.
```

ARGUMENTS:

```
- the set-palette object
```

OPTIONAL ARGUMENTS:

```
See descriptions in tl-set limiti-shift-octave method noting that here the
:upper and :lower arguments can be a single note symbol or pitch object, as
in tl-set, or an envelope. In the latter case, the x axis can be over any
arbitrary scale and the y values can either be note symbols or midi note
numbers.
```

RETURN VALUE:

```
The set-palette object (modified).
```

SYNOPSIS:

```
(defmethod limit-shift-octave ((sp set-palette) &key upper lower
                                                    do-related-sets)
```

38.6.339 set-palette/make-set-palette

[set-palette] [Functions]

DESCRIPTION:

Create a set-palette object.

Note that the sets in this palette may refer to previously defined sets in order to avoid retyping note lists (see example below).

ARGUMENTS:

- A symbol that is to be the ID of the resulting set-palette object.
- A recursive list of key/data pairs, of which the deepest level of data will be a list of note-name symbols.

OPTIONAL ARGUMENTS:

keyword arguments:
- :recurse-simple-data. T or NIL to indicate whether to interpret two-element data lists as recursive palettes. Default = T.
- :warn-note-found. T or NIL to indicate whether to print warnings when specified data is not found with subsequent calls to the get-data method.

RETURN VALUE:

A set-palette object.

EXAMPLE:

```
;;; Create a set-palette object
(make-set-palette
 'test
 '((1 ((1
        ((bf1 ef2 aqf2 c3 e3 gqf3 gqs3 cs4 d4 g4 a4 cqs5 dqf5 gs5 b5)
         :subsets
         ((tc1 ((ds2 e3 a4) "a-tag"))
          (tc2 (bf1 d4 cqs5))
          (tc3 (c3 cs4 gs5)))))
       (2
        ((bf1 d2 fqf2 fqs2 b2 c3 f3 g3 bqf3 bqs3 fs4 gs4 a4 cs5 gqf5)
         :subsets
         ((tc1 (d2 g3 cs5))
          (tc2 (eqs2 f3 bqf3))
          (tc3 (b2 bqs3 gqf5)))))
       (3
        ((cqs2 fs2 g2 c3 d3 fqs3 gqf3 cs4 ds4 e4 gs4 dqf5 f5 a5 bqs5)
         :subsets
         ((tc1 (cqs2 c3 f5))
          (tc2 (fs2 e4 bqs5))
```

```
          (tc3 (d3 ef4 a5)))))))
   (2 ((1 ((1 1) :transposition 5))
       (2 ((1 2) :transposition 5))
       (3 ((1 3) :transposition 5))))
   (3 ((1 ((1 1) :transposition -2))
       (2 ((1 2) :transposition -2))
       (3 ((1 3) :transposition -2))))))

=>
SET-PALETTE:
PALETTE:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 9
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: TEST, tag: NIL,
data: (
[...]

;;; NB A simple list of sets (with unique id slots) can also be passed.

;;; Create a set-palette object by referencing a set-palette-object already
;;; defined (sp1) and transposing a clone of that object.
(make-set-palette
 'test
 '((sp1
    ((bf1 ef2 aqf2 c3 e3 gqf3 gqs3 cs4 d4 g4 a4 cqs5
          dqf5 gs5 b5)
     :subsets
     ((tc1 (ds2 e3 a4))
      (tc2 (bf1 d4 cqs5))
      (qc1 (aqf2 e3 a4 dqf5 b5))
      (qc2 (bf1 c3 gqs3 cs4 cqs5)))
     :related-sets
     ((missing (bqs0 eqs1 f5 aqs5 eqf6 fqs6
                    bqf6 dqs7 fs7)))))
   (sp2
    (sp1 :transposition 13))))

=>
SET-PALETTE:
PALETTE:
```

```
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 2
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: TEST, tag: NIL,
data: (
COMPLETE-SET: complete: NIL
              num-missing-non-chromatic: 7
              num-missing-chromatic: 2
              missing-non-chromatic: (BQS BQF AQS FQS EQS EQF DQS)
              missing-chromatic: (FS F)
[...]
    subsets:
TC1: (DS2 E3 A4)
TC2: (BF1 D4 CQS5)
QC1: (AQF2 E3 A4 DQF5 B5)
QC2: (BF1 C3 GQS3 CS4 CQS5)
    related-sets:
MISSING: (BQS0 EQS1 F5 AQS5 EQF6 FQS6 BQF6 DQS7 FS7)
SCLIST: sclist-length: 15, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: SP1, tag: NIL,
data: (BF1 EF2 AQF2 C3 E3 GQF3 GQS3 CS4 D4 G4 A4 CQS5 DQF5 GS5 B5)
[...]
COMPLETE-SET: complete: NIL
              num-missing-non-chromatic: 7
              num-missing-chromatic: 2
              missing-non-chromatic: (BQS BQF AQS FQS EQS EQF DQS)
              missing-chromatic: (FS F)
TL-SET: transposition: 13
        limit-upper: NIL
        limit-lower: NIL
[...]
    subsets:
TC1: (E3 F4 BF5)
TC2: (B2 EF5 DQF6)
QC1: (AQS3 F4 BF5 DQS6 C7)
QC2: (B2 CS4 AQF4 D5 DQF6)
    related-sets:
MISSING: (BQS0 EQS1 F5 AQS5 EQF6 FQS6 BQF6 DQS7 FS7)
SCLIST: sclist-length: 15, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
```

```
NAMED-OBJECT: id: SP2, tag: NIL,
data: (B2 E3 AQS3 CS4 F4 GQS4 AQF4 D5 EF5 AF5 BF5 DQF6 DQS6 A6 C7)
**************
)
```

SYNOPSIS:

```
(defun make-set-palette (id palette
                         &key (recurse-simple-data t) (warn-not-found t))
```

38.6.340 set-palette/midi-play

[set-palette] [Methods]

DATE:

```
25th May 2017, Edinburgh
```

DESCRIPTION:

```
Write a MIDI file containing the sets in the set-palette
```

ARGUMENTS:

```
- the set-palette object
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :midi-file. The path (string) of the midi file to write. Default is to
  use the ID of the set-palette and write it into the default directory.
- :tempo. The tempo of the MIDI file in beats per minute. Default = 60.
- :auto-open. Whether to open the midi file after generating (not available
  on all operating systems). Default is (get-sc-config
  'midi-play-auto-open) which itself defaults to sc-auto-open in the
  *features* list.
- :related-sets-id :subsets-id. If you want to generate a MIDI file of just
  a named subset or related-set, pass the ID here. NB this ID will have to
  exist for all sets.
```

RETURN VALUE:

```
T
```

SYNOPSIS:

```
(defmethod midi-play ((sp set-palette)
                      &key
                        (tempo 60.0)
                        (auto-open (get-sc-config 'midi-play-auto-open))
                        ;; MDE Sat Mar 20 12:05:17 2021, Heidhausen
                        related-sets-id subsets-id
                        (midi-file
                         (format nil "~a~a.mid"
                                 (get-sc-config 'default-dir)
                                 (string-downcase (string (id sp))))))
```

38.6.341 set-palette/quality-extremes

[set-palette] [Methods]

DESCRIPTION:

```
Get the extremes (as four values) of the dissonance (min max) and spectral
centroid (min max) of a whole set palette.
```

ARGUMENTS:

```
- a set-palette object
```

RETURN VALUE:

```
Four values (i.e. to be used by multiple-value-bind and friends):
dissonance min, dissonance max, centroid min, centroid max.
```

SYNOPSIS:

```
(defmethod quality-extremes ((sp set-palette))
```

38.6.342 set-palette/recursive-set-palette-from-ring-mod

[set-palette] [Functions]

DESCRIPTION:

```
Create a set-palette object consisting of sub palette-objects whose pitch
content is generated based on ring modulation routines applied to the
specified pitches.
```

ARGUMENTS:

- A list of note-name symbols, each of which will serve as the reference pitch from which a new set-palette object is made using the set-palette-from-ring-mod method.
- A symbol that will be the ID for the top-level set-palette object. The IDs of the new set-palette objects contained in the top-level object are generated from the note-name symbols of the reference-pitches, with the IDs of the pitch sets contained with them then generated by sequential numbers.

OPTIONAL ARGUMENTS:

keyword arguments:
- :partials. A list of integers that are the partials which the method is to ring modulate, with 1 being either the reference-note or the bass note that would have the reference-note as the highest partial in the given list. Default = '(1 3 5 7).
- :warn-no-bass. T or NIL to indicate whether to issue a warning when ring-mod-bass fails to find suitable bass notes for the generated sets. T = warn. Default = T.
- :do-bass. T or NIL to indicate whether to add notes created by the ring-mod-bass function to the resulting set-palette object. T = create and add bass notes. Default = T.
- :remove-octaves. T or NIL to indicate whether to remove the upper instances of any octave-equivalent pitches from the resulting set-palette object. T = remove. Default = NIL.
- :min-bass-notes. An integer that is the minimum number of bass notes to be generated and added to the resulting set-palette object. Default = 1.
- :ring-mod-bass-octave. An integer that is the MIDI octave reference number (such as the 4 in 'C4), indicating the octave from which the bass note(s) are to be taken.
- :force-chromatic. T or NIL. If T, force all micro-tone slots of pitch objects to be NIL so that they won't be filtered out when a set is to be used by a chromatic instrument. See sc-set class force-micro-tone for more details.

RETURN VALUE:

- A set-palette object (recursive)

EXAMPLE:

```
;; Simple useage with default keyword argument values
(recursive-set-palette-from-ring-mod '(a4 b4 c4) 'rspfrm-test)

=>
```

```
SET-PALETTE:
PALETTE:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 3
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: RSPFRM-TEST, tag: NIL,
data: (
NAMED-OBJECT: id: A4, tag: NIL,
data:
SET-PALETTE:
PALETTE:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 21
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 21, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: A4, tag: NIL,
data: (
COMPLETE-SET: complete: NIL
[...]
```

SYNOPSIS:

```
(defun recursive-set-palette-from-ring-mod (reference-notes id
                                            &key
                                              (warn-no-bass t)
                                              (ring-mod-bass-octave 0)
                                              (do-bass t)
                                              remove-octaves
                                              force-chromatic
                                              (min-bass-notes 1)
                                              (partials '(1 3 5 7)))
```

38.6.343 set-palette/remove-similar

[set-palette] [Methods]

DATE:

12th August 2015, Wals, Austria

DESCRIPTION:

Remove similar sets from a palette when two sets are deemed similar. The set furthest down the data list will remain, whatever the ID. Note that this is a destructive operation.

We use the similarity chord method to remove sets from the palette, so see that method for further information and for more keyword argument descriptions.

ARGUMENTS:

- the set-palette object

OPTIONAL ARGUMENTS:

keyword arguments:
- :threshold. The lowest value that the chord class's similarity method may return in order to trigger removal. Default = 0.8
- :clone. Whether to clone the set-palette object before removing sets.
- :enharmonics-are-equal. See similarity method. Default = T.
- :octaves-are-true. See similarity method. Default = NIL.

RETURN VALUE:

the pared-down set-palette object

SYNOPSIS:

```
(defmethod remove-similar ((sp set-palette) &key (threshold 0.8)
                                                 (clone nil)
                                                 (enharmonics-are-equal t)
                                                 (octaves-are-true nil))
```

38.6.344 set-palette/ring-mod

[set-palette] [Functions]

DESCRIPTION:

Ring modulate (sum and difference tones) two pitches and return the resulting pitch and harmonic partials thereof.

ARGUMENTS:

- A first pitch, either as a numeric hertz frequencey or a note-name symbol.
- A second pitch, either as a numeric hertz frequencey or a note-name symbol. The second value needn't be higher than first.

OPTIONAL ARGUMENTS:

keyword arguments
- :return-notes. T or NIL to indicate whether to return the results as note-name symbols or frequency numbers. T = note-name symbols. Default = NIL.
- :pitch1-partials. An integer that indicates how many harmonic partials of the first pitch are to be included in the modulation. Default = 3.
- :pitch2-partials. An integer that indicates how many harmonic partials of the second pitch are to be included in the modulation. Default = 2.
- :min-freq. A number that is the the minimum frequency (hertz) that may be returned. Default = 20.
- :max-freq. A number that is the the maximum frequency (hertz) that may be returned. Default = 20000.
- :round. T or NIL to indicate whether frequency values returned are first rounded to the nearest hertz. T = round. Default = T
- :remove-duplicates. T or NIL to indicate whether any duplicate frequencies are to be removed from the resulting list before returning it. T = remove. Default = T.
- :print. T or NIL to indicate whether resulting data is to be printed as it is being generated. T = print. Default = NIL.
- :remove-octaves. T or NIL to indicate whether octave repetitions of pitches will be removed from the resulting list before returning it, keeping only the lowest instance of each pitch. This argument can also be set as a number or a list of numbers that indicates which octave repetitions will be allowed, the rest being removed. For example, :remove-octaves '(1 2) will remove all octave repetitions of a given pitch except for those that are 1 octave and 2 octaves above the given pitch; thus '(c1 c2 c3 c4 c5) would return '(c1 c2 c3), removing c4 and c5. Default = NIL.
- :scale. A variable that indicates which scale to use when converting frequencies to note-names. Default = cm::*scale* i.e. the value to which the Common Music scale is set, which in slippery chicken is *quarter-tone* by default.

RETURN VALUE:

A list of note-name symbols or frequencies.

EXAMPLE:

```
;; Apply ring modulation to 'C4 and 'D4, using 5 partials of the first pitch
;; and 3 partials of the second, removing octave repetitions, and returning the
;; results as rounded hertz-frequencies
(ring-mod 'c4 'd4
          :pitch1-partials 5
          :pitch2-partials 3
          :min-freq 60
          :max-freq 2000
          :remove-octaves t)

=> (64.0 96.0 166.0 198.0 230.0 358.0 427.0 459.0 491.0 555.0 619.0 817.0
    1079.0 1143.0 1340.0 1372.0 1404.0 1666.0 1895.0 1927.0)

;; Applying ring modulation to two frequencies, returning the results as
;; note-name symbols within the chromatic scale.
(ring-mod '261.63 '293.66
          :return-notes t
          :remove-duplicates nil
          :scale cm::*chromatic-scale*)

=> (C1 C2 G3 BF3 E4 B4 CS5 AF5 AF5 CS6 CS6 F6)
```

SYNOPSIS:

```
(defun ring-mod (pitch1 pitch2 ;; hertz or notes
                 &key (return-notes nil) (pitch1-partials 3) (pitch2-partials 2)
                 (min-freq 20) (max-freq 20000) (round t) (remove-duplicates t)
                 (print nil) remove-octaves (scale cm::*scale*))
```

38.6.345 set-palette/ring-mod-bass

[set-palette] [Functions]

DESCRIPTION:

```
Using ring-modulation techniques, invent (sensible) bass note(s) from a
list of frequencies.
```

ARGUMENTS:

```
- A list of numbers that are hertz frequencies from which the bass note(s)
  are to be generated.
```

OPTIONAL ARGUMENTS:

```
keyword arguments
- :bass-octave. An integer that is an octave indicator (e.g. the 4 in
  'C4). The method will only return any frequencies/note-names generated
  that fall in this octave. Default = 0.
- :low. A note-name symbol that is the lowest possible pitch of those
  returned. This argument further restricts the :bass-octave argument. Thus
  a :bass-octave value of 1 could be further limited to no pitches below
  :low 'DS1. Default = 'A0.
- :high. A note-name symbol that is the highest possible pitch of those
  returned. This argument further restricts the :bass-octave argument. Thus
  a :bass-octave value of 1 could be further limted to no pitches above
  :high 'FS1. Default = 'G3.
- :round. T or NIL to indicate whether the frequencies returned are rounded
  to integer values. T = round. Default = T.
- :warn. T or NIL to print a warning when no bass can be created from the
  specified frequencies/note-names. T = print warning. Default = T.
- :return-notes. T or NIL to indicate whether the resulting pitches should
  be returned as note-names instead of frequencies. T = return as
  note-names. Default = NIL.
- :scale. A variable pointing to the scale to which any translation of
  frequencies into note-names symbols should take place. By default this
  value is set to cm::*scale*, which is automatically set by slippery
  chicken to 'quarter-tone at initialisation. To return e.g. pitches rounded
  to chromatic note-names set this argument to cm::*chromatic-scale*.
```

RETURN VALUE:

```
Returns a list of frequencies by default.

Setting the :return-notes keyword argument to T will cause the method to
return note-name symbols instead.
```

EXAMPLE:

```
;; Simple usage with default keyword argument values
(ring-mod-bass '(261.63 293.66 329.63 349.23))

=> (28 29 32)

;; Return as note-names instead, in quarter-tone scale by default
(ring-mod-bass '(261.63 293.66 329.63 349.23)
               :return-notes t)

=> (A0 BF0 BQS0)

;; Set the :scale argument to cm::*chromatic-scale* to return equal-tempered
```

```
;; note-name symbols instead
(ring-mod-bass '(261.63 293.66 329.63 349.23)
               :return-notes t
               :scale cm::*chromatic-scale*)

=> (A0 BF0 C1)

;; Return pitches from bass octave 1 rather than default 0
(ring-mod-bass '(261.63 293.66 329.63 349.23 392.00)
               :return-notes t
               :scale cm::*chromatic-scale*
               :bass-octave 1)

=> (CS1 D1 F1 G1 A1 B1)

;; Further limit the notes returned by setting :low and :high values
(ring-mod-bass '(261.63 293.66 329.63 349.23 392.00)
               :return-notes t
               :scale cm::*chromatic-scale*
               :bass-octave 1
               :low 'e1
               :high 'a1)

=> (F1 G1)

;; Set the :round argument to NIL to return decimal-point frequencies
(ring-mod-bass '(261.63 293.66 329.63 349.23 392.00)
               :bass-octave 1
               :low 'e1
               :high 'a1
               :round NIL)

=> (42.76999999999998 43.45666666666667 43.80000000000001 49.16999999999999)

;; The method prints a warning by default if no bass note can be made
(ring-mod-bass '(261.63))

=>
NIL
WARNING: set-palette::ring-mod-bass: can't get bass from (261.63)!

;; This warning can be suppressed by setting the :warn argument to NIL
(ring-mod-bass '(261.63) :warn nil)

=> NIL
```

SYNOPSIS:

```
(defun ring-mod-bass (freqs &key (bass-octave 0) (low 'a0) (high 'g3) (round t)
                      (warn t) (return-notes nil) (scale cm::*scale*))
```

38.6.346 set-palette/round-to-nearest

[set-palette] [Methods]

DATE:

```
January 30th 2017
```

DESCRIPTION:

```
Rounds all pitch objects to the nearest pitch in the current or given
scale. See also sc-set, chord, slippery-chicken, event, and pitch class
methods.
```

ARGUMENTS:

```
- a set-palette object
```

OPTIONAL ARGUMENTS:

```
keyword argument:
:scale. The scale to use when rounding. (Common Music tuning object or
symbol). If a symbol, then 'chromatic-scale, 'twelfth-tone, or 'quarter-tone
only at present. Default is the current scale as set by (in-scale :...).
```

RETURN VALUE:

```
the modified set-palette
```

SYNOPSIS:

```
(defmethod round-to-nearest ((sp set-palette) &key (scale cm::*scale*))
```

38.6.347 set-palette/set-palette-from-ring-mod

[set-palette] [Functions]

DESCRIPTION:

Create a new set-palette object from the pitches returned by applying ring modulation procedures (difference and sum tones of partials).

ARGUMENTS:

- A note-name symbol that is the central pitch from which we perform the ring-modulation. See :partials below.
- A symbol that is to be the ID for the new set-palette object.

OPTIONAL ARGUMENTS:

keyword arguments
- :partials. A list of integers that are the partials which the method uses to ring modulate. We create partials ascending from the reference-note but also ascending from a fundamental calculated so that reference-note would be the highest partial in the partials list. E.g. if reference-note were 'a4 (440Hz) and :partials was '(1 2) we'd have partial frequencies of 440 and 880, as these are the ascending partials 1 and 2 from 440, but also have 220, as that is the fundamental for which 440 would be the highest partial out of (1 2). Default = '(1 3 5 7).
- :warn-no-bass. T or NIL to indicate whether to issue a warning when ring-mod-bass fails to find suitable bass notes for the generated sets. T = warn. Default = T.
- :do-bass. T or NIL to indicate whether to add notes created by the ring-mod-bass function to the resulting set-palette object. T = create and add bass notes. Default = T.
- :remove-octaves. T or NIL to indicate whether to remove the upper instances of any octave-equivalent pitches from the resulting set-palette object. T = remove. Default = NIL.
- :min-bass-notes. An integer that is the minimum number of bass notes to be generated and added to the resulting set-palette object. Default = 1.
- :ring-mod-bass-octave. An integer that is the MIDI octave reference number (such as the 4 in 'C4), indicating the octave from which the bass note(s) are to be taken.
- :force-chromatic. T or NIL. If T, force all micro-tone slots of pitch objects to be NIL so that they won't be filtered out when a set is to be used by a chromatic instrument. See sc-set class force-micro-tone for more details. If you want truly chromatic sets make sure to be (in-scale :chromatic)
- :start-id. An integer which represents the ID number for the first set. This will be incremented for each subsequent set generated.

RETURN VALUE:

A set-palette object.

EXAMPLE:

```
;; Simple usage with default keyword argument values
(set-palette-from-ring-mod 'a4 'spfrm-test)

=>
SET-PALETTE:
PALETTE:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 21
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 21, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: SPFRM-TEST, tag: NIL,
data: (
[...]

;;; Use with the :partials argument
(let ((spfrm2 (set-palette-from-ring-mod 'a4 'spfrm-test
                                         :partials '(2 4 6 8))))
  (loop for cs in (data spfrm2) collect (pitch-symbols cs)))

=> ((BQS0 CS5 E5 GQF5 B5 CS6 DQS6 FQS6 GQF6 AF6 BF6 B6 C7)
    (BQF0 B0 A2 A3 E4 A4 CS5 E5 GQF5)
    (BQS0 FQS6 GQF6 AF6 BF6 B6 C7 GQS7 AF7 AQF7 AQS7 BF7 BQF7)
    (B0 A2 A3 E4 A4 CS5 E5 GQF5 A5 B5) (BQS0 DQF7 DQS7 EQF7 EQS7 FQS7 FS7)
    (BQF0 A2 A3 E4 CS5 E5 GQF5 B5 CS6 DQS6)
    (AQS0 BQF0 B0 GQS7 AF7 AQF7 AQS7 BF7 BQF7)
    (B0 A4 E5 CS6 E6 GQF6 B6 CS7 DQS7 FQS7 GQF7) (B0 A5 A6 E7 A7)
    (B0 A3 CS5 CS6 DQS6 FQS6 GQF6 B6 C7 DQF7 DQS7 FS7 AF7) (BQS0 A5 A6 E7 A7)
    (B0 A4 A5 E6 A6 CS7 E7 GQF7 A7) (BQS0 A5 A6 E7)
    (BQS0 CS6 E6 GQF6 B6 CS7 DQS7 FQS7 GQF7 AF7 BF7 B7 C8)
    (BQF0 B0 BQS0 A2 A3 E4 A4 CS5 GQF5 A5 B5 CS6 E6)
    (BQS0 B6 CS7 DQS7 FQS7 GQF7 AF7) (B0 A3 A4 E5 A5 CS6 E6 GQF6)
    (B0 BQS0 FQS7 GQF7 AF7 BF7 B7 C8) (BQS0 CS6 FQS6 C7 DQS7 FQS7 GQS7 BQF7 C8)
    (B0 A5 A6 E7 A7) (BQS0 A5 E6 CS7 E7 GQF7 B7) (BQS0 A6 A7)
    (BQS0 AF6 B6 DQF7 FS7 AF7 AQS7)
    (BQF0 B0 BQS0 A2 A3 CS5 GQF5 CS6 DQS6 FQS6 GQF6 BF6)
    (BQS0 EQF7 FQS7 GQS7 BQF7 C8) (BQS0 A6 CS7 GQF7 A7) (B0 A5 A6)
    (BQS0 CS7 E7 GQF7 B7))

;;; Use with the :do-bass and :remove-octaves arguments
(let ((spfrm3 (set-palette-from-ring-mod 'a4 'spfrm-test
```

```
                                        :do-bass nil
                                        :remove-octaves t)))
  (loop for cs in (data spfrm3) collect (pitch-symbols cs)))
=> ((BQS1 GQF3 EF4 A4 DQF5) (DQF6 DQS6 EF6 F6 FQS6 GQF6 EQF7 EQS7)
    (BQS2 GQF3 A4 DQF5 F5 GQS5) (BQS6 C7 CQS7 DQF7 D7 DQS7)
    (BQS3 EF4 GQF4 DQF5 F5 GQS5 BF5 CQS6) (FQS7 FS7 GQF7 G7 GQS7)
    (GQF5 BF5 DQF6 AQF6 C7 DQS7 F7 GQS7) (BQS1 A4 F5 GQS5 DQS6)
    (GQS6 AQS6 BQS6 DQS7 EQF7 F7 BQF7) (BQS1 EF4 F5 GQS5 CQS6 FQS6)
    (EF7 EQS7 FQS7 GQS7 AQF7 AQS7) (F5 BQS5 GQS6 B6 D7 FS7 AF7 AQS7) (A4 GQF7)
    (A4 CS7 GQF7) (A4 CS7) (EF6 G6 BF6 F7 AQF7 BQS7)
    (BQS1 GQF3 DQF5 CQS6 DQS6 F6 AQS6) (CQS7 DQS7 F7 AQF7 BF7 BQS7)
    (E6 A6 GQF7 B7) (A4 E6 B7) (A6 CS7 E7 B7))
```

SYNOPSIS:

```
(defun set-palette-from-ring-mod (reference-note id
                                  &key
                                  (warn-no-bass t)
                                  (do-bass t)
                                  ;; the start id for the sets; will be
                                  ;; incremented
                                  (start-id 1)
                                  remove-octaves
                                  force-chromatic
                                  (min-bass-notes 1)
                                  (ring-mod-bass-octave 0)
                                  (partials '(1 3 5 7)))
```

38.6.348 set-palette/set-palette-from-spectra

[set-palette] [Functions]

DATE:

August 24th 2018

DESCRIPTION:

Create a set-palette by analysing a sound file at regular intervals and using the detected frequency components as pitches in a set.

NB CLM is needed for this method.

ARGUMENTS:

- the sndfile to analyse

OPTIONAL ARGUMENTS:

```
keyword arguments
- :id. The ID for the set-palette object to return. If NIL a suitable symbol
  will be created from the sndfile path. Default = NIL.
- :start. The time in seconds to start the analysis. Default = 0
- :stop. The time in seconds to stop the analysis. If nil then analysis will
  continue until one millisecond before the end of the sound file. Default =
  NIL.
- :interval. The time increment in seconds for the analysis start
  point. Default = 0.1
- :min-pitches. The minimum number of pitches a set may contain. Analyses
  returning less than this number of frequencies will be skipped. Default
  = 5.
- :pitch-min. The lowest pitch of any set. Pitch object or symbol. Default
  = c1.
- :pitch-max. The highest pitch of any set. Pitch object or symbol. Default
  = g7.
- :round. Round the generated pitches to the nearest in the current scale?
  Default = NIL.
- :reject-fun. A function that takes a set as an argument and returns T if
  it should be rejected, or NIL if not. Default = NIL.
- :warn-dups. T or NIL to warn if duplicate pitches are found when calling
  make-complete-set. Default = T.
- :gs-keys. Any other keyword arguments to be passed to the get-spectrum
  call (get-spectrum.lsp). In particular :fftsize will be useful here as
  larger windows will result in lower pitches being detected. In any case
  it's worth experimenting with :fftsize. Default: various but :fftsize of
  1024
```

RETURN VALUE:

```
a set-palette object. Each set in the palette has a numerically ascending
integer ID, starting from 1.
```

EXAMPLE:

```
(let ((sp (set-palette-from-spectra
           (concatenate 'string cl-user::+slippery-chicken-home-dir+
                        "test-suite/test-sndfiles-dir-1/test-sndfile-3.aiff"))))
  (print-simple sp))
=>
1: DQS2 E5 DQS6 AQF6 AQS6 B6 CQS7 DQF7
2: E5 GQS5 AQS5 B5 CQS6 AQS6
4: EQF3 FQS4 E5 AQS5 CS6 AQS6
```

```
5: EQS2 AF3 EQF4 A4 DQF5 E5 AQF5 BQF5 AQS6
6: B4 EQF5 BF5 CQS6 DQS6 EQS6 GQS6 AQS6
7: F4 A4 CQS5 EQF5 GQS5 BQS5 CS6 AQS6
23: E5 BF6 D7 EQF7 E7 FQS7
NIL
```

SYNOPSIS:

```
#+clm
(defun set-palette-from-spectra (sndfile
                                 &key
                                   id
                                   (start 0)
                                   stop
                                   (interval 0.1)
                                   (min-pitches 5)
                                   (pitch-min 'c1)
                                   (pitch-max 'g7)
                                   (round t)
                                   reject-fun
                                   (warn-dups t)
                                   (gs-keys
                                    `(:fftsize 1024
                                              :order-by clm::freq
                                              :freq-min 20
                                              :freq-max ,(note-to-freq 'b8))))
```

38.6.349 set-palette/set-palette-p

[set-palette] [Functions]

DESCRIPTION:

```
Test whether a given object is a set-palette object.
```

ARGUMENTS:

```
- A lisp object
```

EXAMPLE:

```
(let ((msp (make-set-palette
            'test
            '((1 ((1
                   ((bf1 ef2 aqf2 c3 e3 gqf3 gqs3 cs4 d4 g4 a4 cqs5 dqf5 gs5
```

```
                        b5)))
                  (2
                   ((bf1 d2 fqf2 fqs2 b2 c3 f3 g3 bqf3 bqs3 fs4 gs4 a4 cs5 gqf5)
                    :subsets
                    ((tc1 (d2 g3 cs5))
                     (tc2 (eqs2 f3 bqf3))
                     (tc3 (b2 bqs3 gqf5)))))))
              (2 ((1 ((1 1) :transposition 5))
                  (2 ((1 2) :transposition 5))))
              (3 ((1 ((1 1) :transposition -2))
                  (2 ((1 2) :transposition -2))))))))
  (set-palette-p msp))

=> T
```

RETURN VALUE:

```
t or nil
```

SYNOPSIS:

```
(defun set-palette-p (thing)
```

38.6.350 set-palette/stack

[set-palette] [Methods]

DATE:

```
30th June 2019, Corfu
```

DESCRIPTION:

```
Call the stack method for each set in the palette. See the method
description in the sc-set class for details of arguments etc.
```

RETURN VALUE:

```
- the set-palette with new stacked sets (clone before calling if necessary)
```

SYNOPSIS:

```
(defmethod stack ((sp set-palette) num-stacks &key id by-freq (up t) (down t))
```

38.6.351 set-palette/thin

[set-palette] [Methods]

DESCRIPTION:

```
Apply the thin method to each set in the palette. See the sc-set class
method for details, including keyword arguments.
```

SYNOPSIS:

```
(defmethod thin ((sp set-palette) &rest keywords &key &allow-other-keys)
```

38.6.352 set-palette/wrap

[set-palette] [Methods]

DATE:

```
March 5th 2016, Edinburgh
```

DESCRIPTION:

```
Applies the wrap method from the chord class to each set in the
palette. See the chord class method for details.
```

RETURN VALUE:

```
The set-palette object, wrapped.
```

SYNOPSIS:

```
(defmethod wrap ((sp set-palette) &optional (num-times 1) (transpose t))
```

38.6.353 palette/sndfile-palette

[palette] [Classes]

NAME:

```
sndfile-palette

File:             sndfile-palette.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
```

```
                 circular-sclist -> assoc-list -> recursive-assoc-list ->
                 palette -> sndfile-palette

 Version:        1.1.0

 Project:        slippery chicken (algorithmic composition)

 Purpose:        Implementation of the sndfile-palette class, which is a
                 simple palette that checks that all the sound files given
                 in a list for each id exist.  See comments in methods for
                 limitations and special features of this class.

 Author:         Michael Edwards: m@michael-edwards.org

 Creation date:  18th March 2001

 $$ Last modified:  17:10:00 Sat Jun 29 2024 CEST

 SVN ID: $Id$
```

38.6.354 sndfile-palette/find-sndfile

[sndfile-palette] [Methods]

DESCRIPTION:

```
Return the full directory path and file name of a specified sound file,
from within the directories given in the PATHS slot.
```

ARGUMENTS:

```
- A sndfile-palette object.
- The name of a sound file from within that object. This can be a string or
  a symbol. Unless it is given as a string, it will be handled as a symbol
  and will be converted to lowercase. Inclusion of the file extension is
  optional.
```

RETURN VALUE:

```
Returns the full directory path and file name of the specified sound file
as a string.
```

EXAMPLE:

```
(let ((msfp (make-sfp 'sfp-test
```

```
                      '((sndfile-group-1
                         (test-sndfile-1))
                        (sndfile-group-2
                         (test-sndfile-2 test-sndfile-3
                          (test-sndfile-4 :frequency 261.61)))
                        (sndfile-group-3
                         ((test-sndfile-5 :start 0.006 :end 0.182)
                          test-sndfile-6)))
                      :paths
                      '("/path/to/sndfiles-dir-1"
                        "/path/to/sndfiles-dir-2"))))
 (find-sndfile msfp 'test-sndfile-4))

=> "/path/to/sndfiles-dir-2/test-sndfile-4.aiff"
```

SYNOPSIS:

```
(defmethod find-sndfile ((sfp sndfile-palette) sndfile)
```

38.6.355 sndfile-palette/get-nearest

[sndfile-palette] [Methods]

DESCRIPTION:

```
 Get the sndfile whose frequency slot is nearest to the first argument.
```

ARGUMENTS:

```
 - the frequency we're looking for, in Hertz (a number)
 - the sndfile-palette object we'll search
```

OPTIONAL ARGUMENTS:

```
 - the IDs of the sndfile groups we'll search. Either a list of ids or a
 single id. If NIL then all groups will be searched. Default = NIL.
 - if more than one sndfile is very close in frequency, select one at random
 (see also the get-nearest-by-freq function)
```

RETURN VALUE:

```
 the nearest sndfile object
```

SYNOPSIS:

```
(defmethod get-nearest (freq (sfp sndfile-palette) &optional ids random)
```

38.6.356 sndfile-palette/get-nearest-by-freq

[sndfile-palette] [Functions]

DATE:

```
October 1st 2015, Edinburgh
```

DESCRIPTION:

```
Given a frequency and a list of sndfile objects, return the sndfile from
the list whose frequency is closest to the first argument.
```

ARGUMENTS:

```
- a frequency in Hertz (number)
- a simple list of sndfile objects, such as that contained in the data slot
 of a sndfile-palette group.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether a random file should be chosen when there are
  several sndfiles with the same frequency. (NB the function random-rep will
  be used so if you wanted repeatable results call (random-rep 1 t) in order
  to reset the random seed before calling clm-play or whatever context uses
  this function.) Default = NIL.
```

RETURN VALUE:

```
the nearest sndfile object
```

SYNOPSIS:

```
(defun get-nearest-by-freq (freq sflist &optional random)
```

38.6.357 sndfile-palette/get-sndfiles

[sndfile-palette] [Functions]

DATE:

```
5th September 2015, Edinburgh
```

DESCRIPTION:

Return a list of the full paths of sound files in the given path. Files without the extensions aif, wav, aiff, and snd are ignored.

ARGUMENTS:

The full path to the folder where the sound files are, as a string

OPTIONAL ARGUMENTS:

- a list of folders to skip i.e. just the last folder name, not the complete path
- A single pattern (string) or list of patterns that the sound file name must have--just the filename, excluding path/folders and extension. If a list then all patterns must be in the file name, not just one of them, unless (hack alert!) the first element of the list is the symbol or, in which case only one of the patterns must be present. Default = NIL.
- Sim. to :insist except this/these are patterns none of which can be in the file name. Default = NIL.

RETURN VALUE:

A list of full paths as strings.

EXAMPLE:

```
(get-sndfiles "/music/hyperboles/snd/cello/samples/"
              '("short-percussive" "weird"))
-->
("/music/hyperboles/snd/cello/samples/1/g4-III-4-001.aif"
 "/music/hyperboles/snd/cello/samples/1/g4-III-4-002.aif"
 "/music/hyperboles/snd/cello/samples/1/g4-III-4-003.aif"
 "/music/hyperboles/snd/cello/samples/1/g4-III-4-004.aif"
 "/music/hyperboles/snd/cello/samples/10/cs5-I-5-9-13-4-001.aif"
 ...
```

SYNOPSIS:

```
(defun get-sndfiles (folder &optional skip insist resist force-quotes)
```

38.6.358 sndfile-palette/kontakt-to-sfp

[sndfile-palette] [Functions]

DATE:

October 2nd 2015, Edinburgh

DESCRIPTION This function is only available on Mac OSX 64bit Intel machines and requires the nki executable which since version 1.0.6 comes with slippery-chicken (in the bin directory).

Using the public-domain C programme "nki tool" from the Linux Sampler project (http://www.linuxsampler.org/nkitool/), convert a Kontakt .nki sampler file to a sndfile-palette object. This will only include the sound file and the MIDI note number it is mapped to, but it will mean that the sndfile objects in the palette include the correct frequency. So using such a sndfile-palette makes it possible to have clm-play act as a traditional sampler, though of course much more is possible too.

The nki file could of course map more than one sample to any MIDI note (e.g. for handling different velocities). By default all of these could be included in the sndfile-palette, and the :snd-selector function passed to clm-play could select the correct file based on velocity, for instance. However, you may want to filter out some sound files. This is where the :insist and :resist keyword arguments come in handy (see below).

NB If you get the error message "zpipe: invalid or incomplete deflate data" then it's probably because the nki file is in Kontakt 4.2.2 (or higher) format so can't be converted with this tool. You could try "save as" in Kontakt 3 or 4 and rerunning this function, as those older programmes will probably save in the older format (worked for me at least once).

ARGUMENTS:

- the path to the .nki file to process
- the path to the samples i.e. the sound files which the nki file uses

OPTIONAL ARGUMENTS:

Keyword arguments:
:insist
- :insist. A single pattern (string) or list of patterns that the sound file name must have--just the filename, excluding path/folders and extension. If a list then all patterns must be in the file name, not just one of them. Default = NIL i.e. accept all.
- :resist. Sim. to :insist except this/these are patterns none of which can be in the file name. Default = NIL.
- :group. The group ID that will be assigned to the sound files. Default = NIL whereby the name of the nki file will be used.
- :converter. The path to the nki tool executable. Default = /path/to/slippery-chicken/bin/nki

RETURN VALUE: EXAMPLE: SYNOPSIS:

```
#+(and mac-osx X86-64)
(defun kontakt-to-sfp (nki samples-path
                       &key insist resist group
                         (converter
                          (concatenate 'string
                                       cl-user::+slippery-chicken-home-dir+
                                       "bin/nki")))
```

38.6.359 sndfile-palette/make-sfp

[sndfile-palette] [Functions]

DESCRIPTION:

Make a sndfile-palette object. This object is a simple palette which checks to make sure that all of the sound files in a given list exist for each given ID.

Sound files are given as as single names, without the path and without the extension. These can be given using the optional keyword arguments <paths> and <extensions>.

NB Although this class is a palette and therefore a subclass of recursive-assoc-list, the sound lists in this case cannot be nested beyond a depth of two (as in example below).

ARGUMENTS:

- An ID for the palette.
- A list of lists that contains IDs for the names of one or more groups of sound files, each paired with a list of one or more names of existing sound files. The sound file names themselves can be paired with keywords from the sndfile class, such as :start, :end, and :frequency, to define and describe segments of a given sound file.

OPTIONAL ARGUMENTS:

keyword arguments:
- :paths. A list of one or more paths to where the sound files are located.
- :extensions. A list of one or more sound file extensions for the specified sound files. The default initialization for this slot of the sndfile-palette already contains ("wav" "aiff" "aif" "snd"), so this argument can often be left unspecified.

- :auto-freq. whether to do automatic frequency detection on the sound files. T would mean using a pitch detection routine, or a function can also be passed, whereupon it will be called with the path slot as argument (the idea is that the fundamental can be extracted from the file name). Default = NIL.
- :warn-not-found. T or NIL to indicate whether a warning should be printed to the Lisp listener if the specified sound file cannot be found. T = print warning. Default = T.

RETURN VALUE:

Returns NIL.

EXAMPLE:

```
(let ((msfp (make-sfp 'sfp-test
                      '((sndfile-group-1
                         (test-sndfile-1))
                        (sndfile-group-2
                         (test-sndfile-2 test-sndfile-3
                          (test-sndfile-4 :frequency 261.61)))
                        (sndfile-group-3
                         ((test-sndfile-5 :start 0.006 :end 0.182)
                          test-sndfile-6)))
                      :paths '("/path/to/sound-files-dir-1/"
                               "/path/to/sound-files-dir-2/")))))
```

SYNOPSIS:

```
(defun make-sfp (id sfp &key paths (extensions '("wav" "aiff" "aif" "snd"))
                         auto-freq (warn-not-found t))
```

38.6.360 sndfile-palette/make-sfp-from-folder

[sndfile-palette] [Functions]

DATE:

5th September 2015, Edinburgh

DESCRIPTION:

Makes a sndfile-palette object from the sound files found in a specific folder (directory) on the file system. Allows any arbitrary levels of subfolders with the proviso that they'll be converted to a flat list, using

```
the subfolders as tags (e.g. dir/subdir1/subdir2 becomes
dir-subdir1-subdir2).

The folders can contain other files (they'll be ignored). Sound files are
those with extensions .aif .wav .aiff and .snd
```

ARGUMENTS:

```
- the folder path, as a string.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :skip. a list of folders to skip i.e. just the last folder name, not the
  complete path. Default = NIL.
- :auto-freq. whether to do automatic frequency detection on the sound
  files. Default = NIL.
- :insist. A single pattern (string) or list of patterns that the sound
  file name must have--just the filename, excluding path/folders and
  extension. If a list then all patterns must be in the file name, not
  just one of them. Default = NIL.
- :resist. Sim. to :insist except this/these are patterns none of which can
  be in the file name. Default = NIL.
```

RETURN VALUE:

```
a sndfile-palette object
```

SYNOPSIS: EXAMPLE:

```
(make-sfp-from-folder "/music/hyperboles/snd/cello/samples/"
                      :skip '("short-percussive" "weird"))
-->
SNDFILE-PALETTE: paths: NIL
                 extensions: (wav aiff aif snd)
                 num-snds: 92
PALETTE:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 16
                      linked: T
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 16, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
```

```
NAMED-OBJECT: id: AUTO, tag: NIL,
data: (
LINKED-NAMED-OBJECT: previous: NIL, this: (1), next: (10)
NAMED-OBJECT: id: 1, tag: NIL,
data: (

SNDFILE-EXT: use: T, cue-num: 2, pitch: -1, pitch-curve: -1, bandwidth: -1,
             bandwidth-curve: -1, continuity: -1, continuity-curve: -1,
             weight: -1, weight-curve: -1, energy: -1, energy-curve: -1,
             harmonicity: -1, harmonicity-curve: -1, volume: -1,
             volume-curve: -1, loop-it: NIL, bit-depth: 24, srate: 96000,
             num-frames: 1719753, bytes: 5159314, group-id: (1)
             followers: NIL

SNDFILE: path: /music/hyperboles/snd/cello/samples/1/g4-III-4-004.aif,
         snd-duration: 17.914093, channels: 1, frequency: 150.24414
         start: 0.0, end: 17.914093, amplitude: 1.0, duration 17.914093
         will-be-used: 0, has-been-used: 0
         data-consistent: T
...
|#
(defun make-sfp-from-folder (folder &key skip auto-freq insist resist
                                         (default-group 'default-group))
```

38.6.361 sndfile-palette/make-sfp-from-groups-in-wavelab-marker-file

[sndfile-palette] [Functions]

DESCRIPTION:

```
Automatically generate a sndfile-palette object using the specified sound
file from grouping defined in the specified wavelab marker file.

The <marker-file> argument can be passed as a list of marker files, in
which case these will first be concatenated.
```

ARGUMENTS:

```
- A string that is the name of the marker file, including the directory
  path and extension.
- A string that is the name of the sound file. This can either be a full
  directory path, file name, and extension, or just a base file name. If
  the latter, values for the optional arguments :paths and :extensions must
  also be specified.
```

OPTIONAL ARGUMENTS:

- :paths. NIL or a list of strings that are the directory paths to the specified sound files. If the sound file is passed with the directory path, this must be set to NIL. NB: The paths given here apply only to the sound files, not to the marker files. Default = NIL.
- :extensions. A list of strings that are the extensions to the given sound files. If the sound files are passed with their extensions, this must be set to NIL. Default = NIL.
- :warn-not-found. T or NIL to indicate whether to print a warning to the listener if the specified sound file is not found. T = print a warning. Default = NIL.
- :sampling-rate. An integer that is the sampling rate of the specified sound file. Changing this value will alter the start-times determined for each sound segment. Default = 44100.
- :print. T or NIL to indicate whether feedback about the groups found and created should be printed to the listener. T = print. Default = T.

RETURN VALUE:

Returns NIL.

EXAMPLE:

```
(make-sfp-from-groups-in-wavelab-marker-file
 "/path/to/24-7.mrk"
 "24-7"
 :paths '("/path/to/sndfile/directory/")
 :sampling-rate 44100
 :extensions '("wav"))

=>
24 markers read from /path/to/24-7.mrk
Adding tapping: 2.753 -> 4.827
Adding tapping: 5.097 -> 6.581
Adding tapping: 6.763 -> 8.538
Adding splinter: 13.878 -> 15.993
Adding tapping: 16.338 -> 18.261
Adding splinter: 19.403 -> 25.655

tapping: 4 sounds
splinter: 2 sounds
```

SYNOPSIS:

```
(defun make-sfp-from-groups-in-wavelab-marker-file (marker-file sndfile
                                                    &key
```

```
                                                    paths
                                                    extensions
                                                    warn-not-found
                                                    (sampling-rate 44100)
                                                    (print t))
```

38.6.362 sndfile-palette/make-sfp-from-reaper-markers

[sndfile-palette] [Functions]

DATE:

July 2nd 2021

DESCRIPTION:

Create a sndfile-palette object by reading a reaper-file with specially-named markers. This is specifically aimed at the situation where you want to use one longer sound file in many segments, rather than lots of different sound files. We search the reaper file for all markers whose names begin with clm-play. A marker which begins a sound file segment will then include the group name after a after a further hyphen e.g. the marker will be named clm-play-perc or clm-play-string-attack (in these cases the sndfile-palette will contain groups named perc and string-attack. A marker with the name clm-play will indicate the end of a sndfile segment, and is required. All other markers in the reaper file will be ignored.

(N.B. In all, this is a little different from the funtion which creates sndfile-palettes from wavelap-marker files but that is a different situation (marker files, not wavelab files in general, as here with reaper files) and had a slightly different goal at the time, creating groups in bundles rather than in names. So we're not reproducing functionality here with reaper files rather we're providing a slightly different functionality which sorts current needs quite a few years on.)

ARGUMENTS:

- the path to the reaper file (string)
- the path to the sound file. This must be provided as the reaper file could have marks at any point where there are several sound files playing. (string)

OPTIONAL ARGUMENTS:

- the default frequency of the sound files: either a pitch symbol, a

frequency in Hertz, 'detect for autocorrelation pitch detection, or a function to be called to e.g. extract the pitch from the file name or use another pitch detection method. Default = 'C4 (middle C)

RETURN VALUE:

a sndfile-palette object with appropriate groups as read from the reaper file.

SYNOPSIS:

```
(defun make-sfp-from-reaper-markers (reaper-file sound-file &optional
                                                              (auto-freq 'c4))
```

38.6.363 sndfile-palette/make-sfp-from-wavelab-marker-file

[sndfile-palette] [Functions]

DESCRIPTION:

Automatically create a sndfile-palette object from the specified wavelab marker file and the specified sound file (from which the marker file must have been generated).

The function will produce a sndfile-palette object with multiple groups, each of which consists of the number of sound file segments specified using the :snds-per-group argument (defaults to 8). By default the segments will be collected into the groups in chronological order. If the optional :random-every argument is given a value, every nth group will consist of random segments instead.

The sound file segments of each group will correspond to the time points stored in the marker file.

The <marker-file> argument can consist of a list of marker files, in which case these would first be concatenated.

NB: Be aware that marker files created on operating systems differing from the one on which this function is called might trigger errors due to newline character mismatches.

ARGUMENTS:

- A string that is the name of the marker file, including the directory path and extension.

- A string that is the name of the sound file. This can either be a full directory path, file name, and extension, or just a base file name. If the latter, values for the optional arguments :paths and :extensions must also be specified.

OPTIONAL ARGUMENTS:

keyword arguments:
- :snds-per-group. An integer that is the number of sound file segments to include in each group. Default = 8.
- :random-every. An integer to indicate that every nth group is to consist of random (rather than chronologically consecutive) sound file segments. Default = 999999 (i.e. essentially never)
- :paths. NIL or a list of strings that are the directory paths to the specified sound files. If the sound file is passed with the directory path, this must be set to NIL. NB: The paths given here apply only to the sound files, not to the marker files. Default = NIL.
- :sampling-rate. An integer that is the sampling rate of the specified sound file. Changing this value will alter the start-times determined for each sound segment. Default = 44100.
- :extensions. A list of strings that are the extensions to the given sound files. If the sound files are passed with their extensions, this must be set to NIL. Default = NIL.
- :warn-not-found. T or NIL to indicate whether to print a warning to the listener if the specified sound file is not found. T = print a warning. Default = NIL.
- :name. The name for the overall sndfile-palette and the base name for each group within (these will have a suffix that is an auto-incrementing number e.g. 'auto would become 'auto1 'auto2 etc.). Default = 'auto.

RETURN VALUE:

A sndfile-palette object.

EXAMPLE:

```
(make-sfp-from-wavelab-marker-file
  "/path/to/24-7.mrk"
 "24-7"
 :snds-per-group 2
 :random-every 3
 :paths '("/path/to/sound-file/directory/")
 :sampling-rate 44100
 :extensions '("wav"))
```

```
=>
SNDFILE-PALETTE: paths: (/Volumes/JIMMY/SlipperyChicken/sc/test-suite/)
                 extensions: (wav)
PALETTE:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 8
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found NIL
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 8, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: AUTO, tag: NIL,
data: (
NAMED-OBJECT: id: "auto1", tag: NIL,
data: (

SNDFILE: path: /Volumes/JIMMY/SlipperyChicken/sc/test-suite/24-7.wav,
         snd-duration: 29.652811, channels: 2, frequency: 261.62555
         start: 0.09142857, end: 1.0361905, amplitude: 1.0, duration 0.94476193
         will-be-used: 0, has-been-used: 0
         data-consistent: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "24-7", tag: NIL,
data: /Volumes/JIMMY/SlipperyChicken/sc/test-suite/24-7.wav
[...]
```

SYNOPSIS:

```
(defun make-sfp-from-wavelab-marker-file (marker-file sndfile
                                          &key
                                          (snds-per-group 8)
                                          (random-every 999999) ;; i.e. never
                                          paths
                                          (sampling-rate 44100)
                                          extensions
                                          ;; MDE Fri Oct  5 14:04:08 2012
                                          (name 'auto)
                                          warn-not-found)
```

38.6.364 sndfile-palette/set-frequency-from-filename

[sndfile-palette] [Methods]

DATE:

October 1st 2015, Edinburgh

DESCRIPTION:

Set the frequency slot of every sound file object in the palate to a value extracted from the file name. This is intended but of course not restricted to palettes created from sample libraries, where, as is often the case, the file name of each sample contains the pitch of the file. For example, from the Akoustik Piano sample library we have the file Stein-R(A1)-V(220)-dB(2446)-T(MF)-M(123)-P(404-03-01).wav. In this case the A1 in the first parentheses indicate the pitch. By writing and passing a short parsing function to read the pitch out of such a file name we can easily process each sound file in the palette. See akoustik-piano-name below for an example of how to process such file names.

See also make-sfp-from-folder.

ARGUMENTS:

- The sound file palette object

OPTIONAL ARGUMENTS:

Keyword arguments:
- :groups. A single group or list of groups (IDs) for the sound files we wish to process.
- :name-fun.The function for processing a single name. Of course the names must be consistent and each sound file must be able to be processed by this single function. NB for reasons of similar usage in get-spectra-al (see spectra.lsp) this function actually must return the MIDI note number (may be a floating point for microtonal applications), which is then converted by the sndfile class to a frequency in Hertz.
- :on-error. The function to be called when the name function cannot determine the frequency of the sound file. This could be #'error (the default), #'warn, or nil if nothing is to be done on failure.

RETURN VALUE:

The sndfile-palette object after processing.

SYNOPSIS:

```
(defmethod set-frequency-from-filename
    ((sfp sndfile-palette)
     &key groups (name-fun #'akoustik-piano-name) (on-error #'error))
```

38.6.365 palette/sndfilenet

[palette] [Classes]

NAME:

```
sndfile-palette

File:             sndfilenet.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist -> assoc-list -> recursive-assoc-list ->
                  palette -> sndfile-palette -> sndfilenet

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the sndfilenet class which extends
                  sndfile-palette to add functionality for loading and
                  playing sounds within MaxMSP.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    23rd October 2017, Essen

$$ Last modified:  17:34:58 Sat Jan 30 2021 CET

SVN ID: $Id$
```

38.6.366 sndfilenet/(setf next-sfes)

[sndfilenet] [Methods]

DATE:

```
October 21st 2017, Essen
```

DESCRIPTION:

```
A setf :after method. Set the next-sfes slot to be a sndfile-ext object. If
it is already, leave it as is. If it's an integer, assume it's a cue number
and get the right object, otherwise get the object with this ID.

If this is causing problems, use (setf (slot-value sfn 'next-sfes) ... )
```

```
If we're trying to set using a reference/id like '(group1 snd3) then it
will have to be a two-element list.
```

SYNOPSIS:

```
(defmethod (setf next-sfes) :after (value (sfn sndfilenet))
```

38.6.367 sndfilenet/analyse-followers

[sndfilenet] [Methods]

DESCRIPTION:

```
Using the followers slots of each sndfile in the palette, go through each
sndfile in the palette and generate a large number of following sounds,
i.e. emulate max-play. The results of the follow-on process are then
analysed and a warning will be issued if any sndfile seems to dominate
(defined as being present at least twice as many times as its 'fair share',
where 'fair share' would mean an even spread for all the sound files in the
palette).
```

ARGUMENTS:

```
- The sndfilenet object.
```

OPTIONAL ARGUMENTS:

```
- integer: How many times to repeat the generation process.  Default = 1000.
- T or NIL for verbose warnings: if T, the emulation list of sound files
  will be printed. Default NIL.
```

RETURN VALUE:

```
T or NIL depending on whether the analysis detects an even spread or not.
```

SYNOPSIS:

```
(defmethod analyse-followers ((sfn sndfilenet)
                              &optional (depth 1000) verbose)
```

38.6.368 sndfilenet/auto-cue-nums

[sndfilenet] [Methods]

DESCRIPTION:

Set the cue-num slot of every sndfile-ext object in the palette to be an ascending integer starting at 2. NB If a sndfile has it's :use slot set to NIL it won't be given a cue number.

ARGUMENTS:

- a sndfilenet object.

RETURN VALUE:

The cue number of the last sndfile-ext object.

SYNOPSIS:

```
(defmethod auto-cue-nums ((sfn sndfilenet))
```

38.6.369 sndfilenet/auto-followers

[sndfilenet] [Methods]

DATE:

October 20th 2017

DESCRIPTION:

Automatically generate follower sound files for every sound file in the palette. These are a mixture of the other files in the same group plus one from another group, so that we'll move through.

ARGUMENTS:

- a sndfilenet object

OPTIONAL ARGUMENTS:

- a list of references for which auto-followers should be generated. Default = NIL = all groups (which applies also if T is passed here).

RETURN VALUE:

T

SYNOPSIS:

```
(defmethod auto-followers ((sfn sndfilenet) &optional refs)
```

38.6.370 sndfilenet/get-snd-with-cue-num

[sndfilenet] [Methods]

DATE:

December 21st 2012

DESCRIPTION:

Return the (first, but generally unique) sndfile object which has the given cue-num slot.

ARGUMENTS:

- the sndfilenet object.
- the cue number (integer).

RETURN VALUE:

The sndfile/sndfile-ext object with the given cue number or NIL if it can't be found.

SYNOPSIS:

```
(defmethod get-snd-with-cue-num ((sfn sndfilenet) cue-num)
```

38.6.371 sndfilenet/make-sfn

[sndfilenet] [Functions]

DESCRIPTION:

Make a sndfilenet object. This object is a simple palette which checks to make sure that all of the sound files in a given list exist for each given ID.

Sound files are given as as single names, without the path and without the extension. These can be given using the optional keyword arguments <paths> and <extensions>.

NB Although this class is a palette and therefore a subclass of recursive-assoc-list, the sound lists in this case cannot be nested beyond a depth of two (as in example below).

ARGUMENTS:

- An ID for the palette.
- A list of lists that contains IDs for the names of one or more groups of sound files, each paired with a list of one or more names of existing sound files. The sound file names themselves can be paired with keywords from the sndfile class, such as :start, :end, and :frequency, to define and describe segments of a given sound file.

OPTIONAL ARGUMENTS:

keyword arguments:
- :paths. A list of one or more paths to where the sound files are located.
- :extensions. A list of one or more sound file extensions for the specified sound files. The default initialization for this slot of the sndfilenet already contains ("wav" "aiff" "aif" "snd"), so this argument can often be left unspecified.
- :warn-not-found. T or NIL to indicate whether a warning should be printed to the Lisp listener if the specified sound file cannot be found. T = print warning. Default = T.
- :auto-followers. T or NIL to indicate whether the auto-followers method should be called once the object is initialised. Note that if this is T, then all the followers slots of the sndfile-ext objects will be deleted as part of the process. Default = NIL.
- :analyse-followers. T or NIL to indicate whether to call the analyse-followers after initialisation (and possibly auto-followers is called). Default = NIL.

RETURN VALUE:

Returns NIL.

EXAMPLE:

```
;;; NB won't work unless you change /path/to/ below
(let ((msfn (make-sfn 'sfn-test
                      '((sndfile-group-1
                         (test-sndfile-1))
                        (sndfile-group-2
                         (test-sndfile-2 test-sndfile-3
                          (test-sndfile-4 :frequency 261.61)))
                        (sndfile-group-3
                         ((test-sndfile-5 :start 0.006 :end 0.182)
                          test-sndfile-6)))
                      :paths '("/path/to/sound-files-dir-1/"
                               "/path/to/sound-files-dir-2/")))))
```

SYNOPSIS:

```
(defun make-sfn (id sfn &key paths (extensions '("wav" "aiff" "aif" "snd"))
                          auto-freq (warn-not-found t)
                          auto-followers analyse-followers)
```

38.6.372 sndfilenet/max-play-from-id

[sndfilenet] [Methods]

DESCRIPTION:

This generates the data necessary to play the next sound in the current sound's followers list. See the sndfile-ext method for details.

ARGUMENTS:

- The sndfilenet object.
- The group ID to play the next sound from, i.e. the topmost ID to group of sounds
- The fade (in/out) duration in seconds.
- The maximum loop duration in seconds.
- The time to trigger the next file, as a percentage of the current sndfile-ext's duration.

OPTIONAL ARGUMENTS:

- whether to print data to the listener as it is generated. Default = NIL.

RETURN VALUE:

A list of values returned by the sndfile-ext method.

SYNOPSIS:

```
(defmethod max-play-from-id ((sfn sndfilenet) id fade-dur max-loop start-next
                             &optional print)
```

38.6.373 sndfilenet/preload-cues

[sndfilenet] [Methods]

DESCRIPTION:

Return a list of preload commands for each sound file in a form that a Max sflist~ can process and store (which will include the path and start/stop data: see sndfile-ext::max-cue for details).

E.g. preload 231 triplet-mphonic-001.wav 0. 3300.612305 0 1.

ARGUMENTS:

```
- the sndfilenet object.
```

RETURN VALUE:

```
The list of preload strings that osc-call can then pass on to the sflist~
```

SYNOPSIS:

```
#+(and darwin sbcl)
(defmethod preload-cues ((sfn sndfilenet))
```

38.6.374 sndfilenet/reset

[sndfilenet] [Methods]

DESCRIPTION:

```
Reset the followers' slot circular list to the beginning or to <where> and
the starting sounds for each group also.
```

ARGUMENTS:

```
- the sndfilenet object.
```

OPTIONAL ARGUMENTS:

```
- an integer to set the point at which to restart.  This can be higher than
  the number of followers as it will wrap.  Default = nil (which equates to
  0 lower down in the class hierarchy).
- whether to issue a warning if <where> is greater than the number of
  followers (i.e. that wrapping will occur).  Default = T.
```

RETURN VALUE:

```
T
```

SYNOPSIS:

```
(defmethod reset ((sfn sndfilenet) &optional (where 0) (warn t))
```

38.6.375 recursive-assoc-list/parcel-and-combine-all

[recursive-assoc-list] [Functions]

DATE:

```
December 16th 2022
```

DESCRIPTION:

```
combine a list of recursive-assoc-lists (ral) into a new ral but adding
each ral itself into a new ral with a given ID (parcelling). This allows
each of the rals in the list to have common ids but by parcelling them up we
can achieve this at the expense of a further level of recursion.
```

ARGUMENTS:

```
- a list of rals
- a list of ids to be used for parcelling. This should be as long as the
  list of rals.
```

RETURN VALUE:

```
a new ral
```

EXAMPLE:

```
;;; this is a bit convoluted but was the original motivation to write this
;;; routine: say you have a bunch of rthm-seq-palettes made from fragments and
;;; organised according to the length in quarter notes of the sequences. You can
;;; parcel them all up into recursive palettes and then into a single palette
;;; thus:
(let* ((fragments '((1 (q. q.))
                    (1a (+q. q.))
                    (2 (q h))
                    (3 (h h))
                    (4 (q h.))
                    (5 (q w))
                    (6 (q. q.+h))
                    (7 ({ 3 tq tq tq }))
                    (8 ((e.) s - e e -))))
       (rsps (mapcar #'(lambda (refs)
                         (make-rsp-from-fragments fragments refs))
                     ;; each of these lists will become an rsp, the duration of
                     ;; which is the same length (not checked)
```

```
                    '(((((3 4) 1))     ; 3/4
                       (((3 4) 2)))
                      ((((4 4) 3))     ; 4/4
                       (((4 4) 4)))
                      ((((5 4) 5))     ; 5/4
                       (((5 4) 6)))
                      ((((2 4) 7) (7) (7)) ; 6/4
                       (((2 4) 7) (7) (8)))
                      ((((4 4) 4) ((3 4) 1a)))))) ; 7/4
      ;; get the duration of the first rthm-seq in each palette and use that as
      ;; an id for the main palette below
      (durs (loop for rsp in rsps collect
                  (round (duration (get-first rsp)))))
      (rsp (parcel-and-combine-all rsps durs 'rthm-seq-palette)))
  (get-data-data '(6 2) rsp))
->  ((((2 4) { 3 TQ TQ TQ }) ({ 3 TQ TQ TQ }) ((E.) S - E E -)))
```

SYNOPSIS:

```
(defun parcel-and-combine-all (rals ids &optional new-class)
```

38.6.376 recursive-assoc-list/parcel-data

[recursive-assoc-list] [Methods]

DATE:

10 Apr 2010

DESCRIPTION:

Put all the data of a given recursive-assoc-list object into a new named-object at the top level of that recursive-assoc-list object; i.e. add a level of recursion. This is a means of making a collection of data before perhaps adding more with potentially conflicting ids.

Note that if you want to parcel some palette because of potentially conflicting IDs with another (main) palette then combine is the method to use rather than add, e.g.
(combine orig-palette (parcel-data derived-palette 'filtered))

ARGUMENTS:

- A recursive-assoc-list object.
- A symbol that is new the top-level id for the current data

RETURN VALUE:

The new recursive-assoc-list object.

EXAMPLE:

```
;; Collect all the data contained within the object 'mixed-bag and store it at
;; the top-level of 'mixed-bag within a new named-object with the id 'potpourri
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (parcel-data ral 'potpourri))

=>
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 8
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 1, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: MIXED-BAG, tag: FROM-PARCEL-DATA,
data: (
NAMED-OBJECT: id: POTPOURRI, tag: NIL,
data:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 8
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: MIXED-BAG, tag: NIL,
data: (
NAMED-OBJECT: id: JIM, tag: NIL,
data: BEAM
[...]
```

SYNOPSIS:

```
(defmethod parcel-data ((ral recursive-assoc-list) new-id)
```

38.6.377 recursive-assoc-list/r-count-elements

[recursive-assoc-list] [Methods]

DESCRIPTION:

```
Return the total number of elements recursively (across all depths) of the
given recursive-assoc-list object.
```

ARGUMENTS:

```
- A recursive-assoc-list object.
```

RETURN VALUE:

```
An integer.
```

EXAMPLE:

```
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (r-count-elements ral))

=> 8
```

SYNOPSIS:

```
(defmethod r-count-elements ((ral recursive-assoc-list))
```

38.6.378 recursive-assoc-list/ral-econs

[recursive-assoc-list] [Methods]

DESCRIPTION:

Automatically create new recursive-assoc-list objects.

This method assumes that any existing key that may be referenced will be associated with data that is already a list, to the end of which the new data will be added (an error will be signalled if this is not the case.)

ARGUMENTS:

- The data which is to be added.
- The key to which the data is to be added (see above note for cases where this key already exists).
- The recursive-assoc-list object to which this data is to be added.

RETURN VALUE:

The new data added.

EXAMPLE:

```
;;; Make an empty recursive-assoc-list object and add key/data pairs to the top
;;; level.
(let ((ral (make-ral nil nil)))
  (print (get-all-refs ral))
  (ral-econs 'beam 'jim ral)
  (ral-econs 'turkey 'wild ral)
  (ral-econs 'roses 'four ral)
  (print (get-all-refs ral))
  (print (get-data-data 'wild ral)))

=>
NIL
((JIM) (WILD) (FOUR))
(TURKEY)

;;; Add data to existing keys within a given recursive-assoc-list object
;;; Note that the data VELVET must be a list for this to succeed

(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue (velvet))
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
```

```
                                        (white ribbon)))))))))
  (print (get-all-refs ral))
  (ral-econs 'underground '(four violets blue) ral)
  (print (get-data-data '(four violets blue) ral)))

=>
((JIM) (WILD) (FOUR ROSES) (FOUR VIOLETS BLUE) (FOUR VIOLETS RED DRAGON)
 (FOUR VIOLETS RED VIPER) (FOUR VIOLETS RED FOX) (FOUR VIOLETS WHITE))
(VELVET UNDERGROUND)
```

SYNOPSIS:

```
(defmethod ral-econs (data key (ral recursive-assoc-list))
```

38.6.379 recursive-assoc-list/recursivep

[recursive-assoc-list] [Methods]

DESCRIPTION:

```
Check whether the data in a recursive-assoc-list object is really
recursive.
```

ARGUMENTS:

```
- A recursive-assoc-list object.
```

RETURN VALUE:

```
T or NIL to indicate whether or not the tested data is recursive.
T = recursive.
```

EXAMPLE:

```
;; The data in this recursive-assoc-list object is really recursive, and
;; the method therefore returns T
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
```

```
  (recursivep ral))

=> T

;; The data in this recursive-assoc-list object is not actually recursive, and
;; the method therefore returns NIL
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four roses)))))
  (recursivep ral))

=> NIL
```

SYNOPSIS:

```
(defmethod recursivep ((ral recursive-assoc-list))
```

38.6.380 recursive-assoc-list/relink-named-objects

[recursive-assoc-list] [Methods]

DESCRIPTION:

```
This method is essentially the same as the method link-named-objects, but
resets the LINKED slot to NIL and forces the link-named-objects method to
be applied again.
```

ARGUMENTS:

```
- A recursive-alloc-list object.
```

RETURN VALUE:

```
A recursive-alloc-list object.
```

EXAMPLE:

```
;; Usage as presented here; see the documentation for method link-named-objects
;; for more detail
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
```

```
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (relink-named-objects ral))

=>
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 8
                      linked: T
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: MIXED-BAG, tag: NIL,
data: (
LINKED-NAMED-OBJECT: previous: NIL, this: (JIM), next: (WILD)
NAMED-OBJECT: id: JIM, tag: NIL,
data: BEAM
[...]
```

SYNOPSIS:

```
(defmethod relink-named-objects ((ral recursive-assoc-list))
```

38.6.381 recursive-assoc-list/remove-data

[recursive-assoc-list] [Methods]

DATE:

```
February 1st 2016, Edinburgh
```

DESCRIPTION:

```
Convenience function to allow objects to be removed from a
recursive-assoc-list (and subclasses e.g. palettes and maps)
```

ARGUMENTS:

```
- the recursive assoc-list object
- as many keys as you like. If none are passed then all are removed.
```

RETURN VALUE:

The recursive-assoc-list object's new data list, i.e. with elements removed. Could of course be NIL if you've not passed any keys.

SYNOPSIS:

```
(defmethod remove-data ((ral recursive-assoc-list) &rest keys)
```

38.6.382 recursive-assoc-list/rmap

[recursive-assoc-list] [Methods]

DESCRIPTION:

Recurse over the objects in a recursive-assoc-list and call the given function for each each named-object. See also assoc-list's map-data method which does pretty much the same but acting on each named-object's data rather than the named-object itself.

ARGUMENTS:

- the recursive-assoc-list object
- the function to call (function object)

OPTIONAL ARGUMENTS:

- &rest further arguments to be passed to the function after the named-object from the recursive-assoc-list.

RETURN VALUE:

The recursive-assoc-list object

EXAMPLE:

```
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (rmap ral #'print))
```

```
=>
NAMED-OBJECT: id: JIM, tag: NIL,
data: BEAM
**************
NAMED-OBJECT: id: WILD, tag: NIL,
data: TURKEY
**************
NAMED-OBJECT: id: ROSES, tag: NIL,
data: RED
**************
NAMED-OBJECT: id: BLUE, tag: NIL,
data: VELVET
**************
NAMED-OBJECT: id: DRAGON, tag: NIL,
data: DEN
**************
NAMED-OBJECT: id: VIPER, tag: NIL,
data: NEST
**************
NAMED-OBJECT: id: FOX, tag: NIL,
data: HOLE
**************
NAMED-OBJECT: id: WHITE, tag: NIL,
data: RIBBON
**************
T
```

SYNOPSIS:

```
(defmethod rmap ((ral recursive-assoc-list) function &rest arguments)
```

38.6.383 recursive-assoc-list/sc-map

[recursive-assoc-list] [Classes]

NAME:

```
 sc-map

 File:             sc-map.lsp

 Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                   circular-sclist -> assoc-list -> recursive-assoc-list ->
                   sc-map
```

```
Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the sc-map class for mapping rhythmic
                  sequences, chords etc. to specific parts of a piece.
                  The extension to the recursive-assoc-list class is in the
                  data returned when get-data-from-palette is called: being
                  a map, the data returned by the superclass get-data
                  function is actually a reference into a palette.  Instead
                  of just returning this reference, with
                  get-data-from-palette we then use this as a lookup into
                  the palette slot.  If the reference happens to be a list,
                  then each element of the list is used as a reference into
                  the palette and the resulting objects are returned in a
                  list.

                  TODO: When in a list of references, perhaps the rthm-seq
                  references for a section, a single reference is also a
                  list this can be one of two things: the reference is to a
                  recursive palette, whereupon the data will simply be
                  returned for that reference; or, the reference is a list
                  of references that together build up an object consisting
                  of the referenced smaller objects.  This is the case
                  when, for example, 4-bar sequences in one or more
                  instruments are accompanied by groups of 4 single bar
                  sequences in others:

                  (2
                   ((bsn ((r1-1 r1-2 r1-3 r1-5) 20 1 ...))
                    (trb (2 23 3 ...))))

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    March 21st 2001

$$ Last modified:  16:08:36 Sat Feb 10 2024 CET

SVN ID: $Id$
```

38.6.384 sc-map/change-map

[sc-map] [Classes]

NAME:

```
change-map

File:             change-map.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist -> assoc-list -> recursive-assoc-list ->
                  sc-map -> change-map

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the change-map class where, unlike
                  normal sc-maps (data is given for each sequence) gives
                  data sporadically when the parameter changes, for
                  instance tempo.

                  It is assumed that maps will be typed in the order in
                  which sections occur so that previous-data slots can be
                  kept up-to-date; also, unless all data will be given,
                  that the sections (but not instrument ids) will be in
                  integer sequential order so that nearest sections can be
                  returned when given a non-existent section reference.

                  For example, the following change-map indicates tempo
                  (though tempo-maps have their own class now).
                  It has sections within sections so that the tempo for
                  section '(1 2 3) can be defined, that is, section 1 has
                  subsections 1, 2, and 3 and subsection 2 has further
                  subsections 1, 2, and 3.  This can be nested to any
                  depth.  The tempo information itself is given in sublists
                  where (3 27) means that in the third sequence of the
                  section, the tempo is 27.  (3 2 27) means the 2nd bar of
                  the third sequence has tempo 27: when only two numbers
                  are in the list, bar 1 is assumed.  The trick is, that
                  this tempo then remains, as would be expected, until the
                  next tempo change is indicated, which means that
                  requesting the tempo of section (2 2 3) with any sequence
                  and bar in the map below would return 25, because that is
                  the last tempo given in section 1 and no tempo is defined
                  for section 2.

                  (setf x
                    (make-change-map
                     'test nil
                     '((0 ((3 27) (9 3 45)))
```

```
(1
 ((1 ((1 21) (5 28) (8 35) (3 2 40) (3 1 54)))
  (2
   ((1 ((1 23) (6 28) (18 35)))
    (2 ((2 2 24) (7 28) (18 22)))
    (3 ((3 34) (7 28) (18 42)))))
  (3 ((1 22) (5 34) (10 5 25)))))
(4
 ((1 ((1 21) (5 28) (8 36) (3 2 40) (3 1 55)))
  (2 ((1 22) (5 34) (10 5 103)))))
(5 ((2 28) (6 3 45)))
(10
 ((1 ((1 21) (5 28) (8 37) (3 2 40) (3 1 56)))
  (2 ((1 22) (5 34) (10 5 27))))))))
```

You have to be careful with change-maps however as the nesting is flexible and therefore ambiguous. For instance, in the following the bcl, tape1 etc. ids are not subsections of section 1, rather these are the hint pitches assigned to the instruments in section 1 (which has no subsections). This is where the last-ref-required class slot comes in: If this slot is t (this is the second argument to make-change-map) then the last reference in a call to cm-get-data is always respected, i.e. not the last data given will be returned when the section doesn't exist, rather the last data for this reference. E.g. In the following map, if last-ref-required were nil, then the call to (cm-get-data x '(2 tape2) 1) would fail (because we can't find nearest data when references aren't numbers), but because it's t, we get the last data given for tape2 and return cs5.

```
(setf x
  (make-change-map
      'hint-pitches t
      '((1 ((bcl ((1 a4) (2 b4) (3 c5) (4 d6)))
            (tape1 ((1 a3) (2 ds2) (3 e4)))
            (tape2 ((1 a3) (2 ds2) (3 cs5)))
            (tape3 ((1 a3) (2 ds2) (3 eqf4)))))
        (2 ((bcl ((1 a4) (2 b4) (3 c5) (4 d6)))
            (tape1 ((1 a3) (2 ds2) (5 fs4)))))
        (3 ((bcl ((1 a4) (2 b4) (3 c5) (4 d6)))
            (tape1 ((1 a3) (2 ds2) (5 f4)))))))))
```

Author: Michael Edwards: m@michael-edwards.org

```
Creation date:    2nd April 2001

$$ Last modified:  10:39:42 Thu Feb  1 2024 CET

SVN ID: $Id$
```

38.6.385 change-map/cm-get-data

[change-map] [Methods]

DESCRIPTION:

```
Return the data for a specified section and player of a given change-map
object.

NB: The <section> argument may require the ID of the player as well if
    used, for example, with the instrument-change-map subclass of this
    class .
```

ARGUMENTS:

```
- A change-map object.
- A simple key reference into the given change-map object or a list of
  references. NB: This reference may require a player ID if, for example,
  used with an instrument-change-map subclass of this class.
```

OPTIONAL ARGUMENTS:

```
- The ID of the sequence from which to return the change-map data.
- An integer that is the bar number for which to return the change-map
  data.
```

RETURN VALUE:

```
The change-map data stored at the specified location within the given
change-map.
```

EXAMPLE:

```
;;; An example using the instrument-change-map subclass of change-map

(let ((mini
       (make-slippery-chicken
        '+mini+
```

```
        :ensemble '(((sax ((alto-sax tenor-sax) :midi-channel 1))))
        :instrument-change-map '((1 ((sax ((1 alto-sax) (3 tenor-sax)))))
                                 (2 ((sax ((2 alto-sax) (5 tenor-sax)))))
                                 (3 ((sax ((3 alto-sax) (4 tenor-sax))))))
        :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((sax (1 1 1 1 1))))
                        (2 ((sax (1 1 1 1 1))))
                        (3 ((sax (1 1 1 1 1)))))))
  (cm-get-data (instrument-change-map mini) '(2 sax) 4))

=> ALTO-SAX
```

SYNOPSIS:

```
(defmethod cm-get-data ((cm change-map) section
                        &optional (sequence 1) (bar 1))
```

38.6.386 change-map/find-nearest

[change-map] [Methods]

DESCRIPTION:

```
Return the nearest change-data object to the specified section within the
given change-map. NB: The section may require a player ID if used, for
example, with an instrument-change-map subclass of change-map.
```

ARGUMENTS:

```
- A section indication, either as a single reference ID or a list of
  reference IDs into the given change-map.
- A change-map object.
```

RETURN VALUE:

```
Returns a change-data object.
```

EXAMPLE:

```
(let ((mini
```

```
       (make-slippery-chicken
        '+mini+
        :ensemble '(((sax ((alto-sax tenor-sax) :midi-channel 1))))
        :instrument-change-map '((1 ((sax ((1 alto-sax) (3 tenor-sax)))))
                                 (2 ((sax ((2 alto-sax) (5 tenor-sax)))))
                                 (3 ((sax ((3 alto-sax) (4 tenor-sax))))))
        :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((sax (1 1 1 1 1))))
                        (2 ((sax (1 1 1 1 1))))
                        (3 ((sax (1 1 1 1 1)))))))
  (find-nearest '(4 sax) (instrument-change-map mini)))

=>
CHANGE-DATA:
            previous-data: TENOR-SAX,
            last-data: TENOR-SAX
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: (2 SAX), this: (3 SAX), next: NIL
NAMED-OBJECT: id: SAX, tag: NIL,
data: ((3 1 ALTO-SAX) (4 1 TENOR-SAX))
```

SYNOPSIS:

```
(defmethod find-nearest (section (cm change-map))
```

38.6.387 change-map/instrument-change-map

[change-map] [Classes]

NAME:

```
 instrument-change-map

 File:             instrument-change-map

 Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                   circular-sclist -> assoc-list -> recursive-assoc-list ->
                   sc-map -> change-map -> instrument-change-map

 Version:          1.1.0
```

```
Project:          slippery chicken (algorithmic composition)

Purpose:          Extends change-map to check that instruments defined in
                  the map have data for the first bar of the first section.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    12th April 2002

$$ Last modified:  10:29:44 Thu Feb  1 2024 CET

SVN ID: $Id$
```

38.6.388 instrument-change-map/get-first-for-player

[instrument-change-map] [Methods]

DESCRIPTION:

```
Return the first instrument object assigned to a given player in cases
where a player has been assigned more than one instrument.
```

ARGUMENTS:

```
- An instrument-change-map
- The ID of the player for whom to return the first instrument.
```

RETURN VALUE:

```
The ID of the first instrument object assigned to the specified player.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((sax ((alto-sax tenor-sax) :midi-channel 1))
                     (db (double-bass :midi-channel 2))))
        :instrument-change-map '((1 ((sax ((1 alto-sax) (3 tenor-sax))))))
        :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((sax (1 1 1 1 1))
```

```
                            (db (1 1 1 1 1))))))))
  (get-first-for-player (instrument-change-map mini) 'sax))

=> ALTO-SAX, T
```

SYNOPSIS:

```
(defmethod get-first-for-player ((icm instrument-change-map)
                                 player)
```

38.6.389 instrument-change-map/make-instrument-change-map

[instrument-change-map] [Functions]

DESCRIPTION:

Create an instrument-change-map object.

ARGUMENTS:

- An ID for the instrument-change-map to be created.
- A list of lists. The top level of these lists consists of the section IDs for the given slippery-chicken object paired with lists of data for the specified players for each section. Each player list then consists of the ID of the player paired with a list of two-item lists pairing measure numbers with the instrument to which that player is to change, e.g.:

```
'((1 ((fl ((1 flute) (3 piccolo) (5 flute)))
      (cl ((1 b-flat-clarinet) (2 bass-clarinet) (6 b-flat-clarinet)))))
  (2 ((fl ((2 piccolo) (4 flute)))
      (cl ((2 bass-clarinet) (3 b-flat-clarinet))))))
```

OPTIONAL ARGUMENTS:

- :warn-not-found. T or NIL to indicate whether a warning is printed when an index which doesn't exist is used for look-up. T = warn. Default = T.

RETURN VALUE: EXAMPLE:

```
(make-instrument-change-map
 'icm-test
 '((1 ((fl ((1 flute) (3 piccolo) (5 flute)))
       (cl ((1 b-flat-clarinet) (2 bass-clarinet) (6 b-flat-clarinet)))))
   (2 ((fl ((2 piccolo) (4 flute)))
       (cl ((2 bass-clarinet) (3 b-flat-clarinet)))))))
```

```
=>

INSTRUMENT-CHANGE-MAP:
CHANGE-MAP: last-ref-required: T
SC-MAP: palette id: NIL
RECURSIVE-ASSOC-LIST: recurse-simple-data: NIL
                      num-data: 4
                      linked: T
                      full-ref: NIL
ASSOC-LIST: warn-not-found NIL
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: ICM-TEST, tag: NIL,
data: (
NAMED-OBJECT: id: 1, tag: NIL,
data:
CHANGE-MAP: last-ref-required: T
SC-MAP: palette id: NIL
RECURSIVE-ASSOC-LIST: recurse-simple-data: NIL
                      num-data: 2
                      linked: T
                      full-ref: (1)
ASSOC-LIST: warn-not-found NIL
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "sub-ral-of-ICM-TEST", tag: NIL,
data: (
CHANGE-DATA:
            previous-data: NIL,
            last-data: FLUTE
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: (1 FL), next: (1 CL)
NAMED-OBJECT: id: FL, tag: NIL,
data: ((1 1 FLUTE) (3 1 PICCOLO) (5 1 FLUTE))
**************


CHANGE-DATA:
            previous-data: NIL,
            last-data: B-FLAT-CLARINET
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: (1 FL), this: (1 CL), next: (2 FL)
NAMED-OBJECT: id: CL, tag: NIL,
```

```
data: ((1 1 B-FLAT-CLARINET) (2 1 BASS-CLARINET) (6 1 B-FLAT-CLARINET))
**************
)
**************

**************


NAMED-OBJECT: id: 2, tag: NIL,
data:
CHANGE-MAP: last-ref-required: T
SC-MAP: palette id: NIL
RECURSIVE-ASSOC-LIST: recurse-simple-data: NIL
                      num-data: 2
                      linked: T
                      full-ref: (2)
ASSOC-LIST: warn-not-found NIL
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "sub-ral-of-ICM-TEST", tag: NIL,
data: (
CHANGE-DATA:
            previous-data: FLUTE,
            last-data: FLUTE
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: (1 CL), this: (2 FL), next: (2 CL)
NAMED-OBJECT: id: FL, tag: NIL,
data: ((2 1 PICCOLO) (4 1 FLUTE))
**************


CHANGE-DATA:
            previous-data: B-FLAT-CLARINET,
            last-data: B-FLAT-CLARINET
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: (2 FL), this: (2 CL), next: NIL
NAMED-OBJECT: id: CL, tag: NIL,
data: ((2 1 BASS-CLARINET) (3 1 B-FLAT-CLARINET))
**************
)
**************

**************
)
```

SYNOPSIS:

```
(defun make-instrument-change-map (id icm &key (warn-not-found nil))
```

38.6.390 change-map/simple-change-map

[change-map] [Classes]

NAME:

```
simple-change-map

File:             simple-change-map.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist -> assoc-list -> recursive-assoc-list ->
                  sc-map -> change-map -> simple-change-map

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the simple-change-map class which gives
                  data associated with a non-recursive list of number ids.
                  For example, good for specifying data which changes at
                  specific bar numbers.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    March 31st 2002

$$ Last modified: 20:16:16 Mon May 14 2012 BST

SVN ID: $Id$
```

38.6.391 simple-change-map/make-simple-change-map

[simple-change-map] [Functions]

DESCRIPTION:

```
Create a simple-change-map object, which stores data associated with a
non-recursive list of number IDs. This object could be good, for example,
for specifying data which changes at specific bar numbers.
```

ARGUMENTS:

```
 - An ID for the simple-change-map object to be created.
 - A list of non-recursive lists consisting of ID/data pairs, of which the
   first item is a numerical ID.
```

RETURN VALUE:

```
 A simple-change-map object.
```

EXAMPLE:

```
(make-simple-change-map 'bar-map '((1 3) (34 3) (38 4)))

=>
SIMPLE-CHANGE-MAP:
CHANGE-MAP: last-ref-required: NIL
SC-MAP: palette id: NIL
RECURSIVE-ASSOC-LIST: recurse-simple-data: NIL
                      num-data: 3
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found NIL
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: BAR-MAP, tag: NIL,
data: (
NAMED-OBJECT: id: 1, tag: NIL,
data: 3
**************


NAMED-OBJECT: id: 34, tag: NIL,
data: 3
**************


NAMED-OBJECT: id: 38, tag: NIL,
data: 4
**************
)
**************
```

SYNOPSIS:

```
(defun make-simple-change-map (id scm)
```

38.6.392 simple-change-map/scm-get-data

[simple-change-map] [Methods]

DESCRIPTION:

```
Get the data associated with the specified key within a given
simple-change-map object.
```

ARGUMENTS:

```
- An integer that is an existing key ID within the given simple-change-map
  object.
- A simple-change-map-object.
```

RETURN VALUE:

```
The data associated with the specified key ID, as a named object.
```

EXAMPLE:

```
(let ((scm (make-simple-change-map 'bar-map '((1 3) (34 3) (38 4)))))
  (scm-get-data 34 scm))

=>
NAMED-OBJECT: id: 34, tag: NIL,
data: 3
```

SYNOPSIS:

```
(defmethod scm-get-data (ref (scm simple-change-map))
```

38.6.393 simple-change-map/tempo-map

[simple-change-map] [Classes]

NAME:

```
tempo-map

File:             tempo-map.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist -> assoc-list -> recursive-assoc-list ->
                  sc-map -> change-map -> simple-change-map -> tempo-map
```

```
Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Allow the specification of tempi by just a number
                  (defaulting to crotchet number) or a list where the first
                  element would be the beat, the second the speed.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    6th May 2006

$$ Last modified:  14:44:23 Tue Apr  1 2025 CEST

SVN ID: $Id$
```

38.6.394 sc-map/count-ref

[sc-map] [Methods]

DATE:

```
February 13th 2018, Heidhausen
```

DESCRIPTION:

```
Count the number of times a map reference occurs, e.g. how often a
particular rthm-seq or set is used in a piece.
```

ARGUMENTS:

```
- the sc-map reference
- the reference we're counting
```

RETURN VALUE:

```
an integer reflecting the number of occurences of <ref>
```

SYNOPSIS:

```
(defmethod count-ref ((scm sc-map) ref)
```

38.6.395 sc-map/delete-nth-in-map

[sc-map] [Methods]

DATE:

05 Feb 2011

DESCRIPTION:

Replace the element at the specified location within the specified list of a given sc-map object with NIL.

ARGUMENTS:

- A list that is the map-ref; i.e., the path of IDs into the list to be searched.
- An integer that is the zero-based index of the element to be returned from the specified list.
- An sc-map object.

RETURN VALUE:

Always returns NIL

EXAMPLE:

```
(let ((mscm (make-sc-map 'scm-test
                         '((1
                            ((vn (1 2 3 4 5))
                             (va (2 3 4 5 1))
                             (vc (3 4 5 1 2))))
                           (2
                            ((vn (6 7 8))
                             (va (7 8 6))
                             (vc (8 6 7))))
                           (3
                            ((vn (9))
                             (va (9))
                             (vc (9)))))))))
  (delete-nth-in-map '(1 vn) 1 mscm)
  (get-data-from-palette '(1 vn) mscm))

=>
NAMED-OBJECT: id: VN, tag: NIL,
data: (1 NIL 3 4 5)
```

SYNOPSIS:

```
(defmethod delete-nth-in-map (map-ref nth (scm sc-map))
```

38.6.396 sc-map/double

[sc-map] [Methods]

DATE:

13-Feb-2011

DESCRIPTION:

Change the specified sequences of one or more specified players within an existing sc-map object to double the rhythms of the corresponding sequences of another specified player.

This allows an existing map, for example, to have several players playing in rhythmic unison.

ARGUMENTS:

- An sc-map object.
- A section reference (i.e. section ID or list of section-subsection IDs).
- An integer that is the 1-based number of the first sequence within the given section to be changed.
- An integer that is the 1-based number of the last sequence within the given section to be changed.
- The ID of the player whose part is to serve as the source for the doubling.
- An ID or list of IDs of the player(s) whose parts are to be changed.

RETURN VALUE:

Returns NIL.

EXAMPLE:

```
;;; Create an sc-map with parts for players 'fl and 'cl containing only NILs
;;; and print the corresponding data. Double the second and third sequence of
;;; the 'vn part of that section into the 'fl and 'cl parts and print the same
;;; data again to see the change.
(let ((scm (make-sc-map 'sc-m
                        '((1
```

```
                          ((fl (nil nil nil))
                           (cl (nil nil nil))
                           (vn (set1 set3 set2))
                           (va (set2 set3 set1))
                           (vc (set3 set1 set2))))
                         (2
                          ((vn (set1 set2 set1))
                           (va (set2 set1 set3))
                           (vc (set1 set3 set3))))
                         (3
                          ((vn (set1 set1 set3))
                           (va (set1 set3 set2))
                           (vc (set3 set2 set3))))))))
  (print (get-data-data '(1 fl) scm))
  (print (get-data-data '(1 cl) scm))
  (double scm 1 2 3 'vn '(fl cl))
  (print (get-data-data '(1 fl) scm))
  (print (get-data-data '(1 cl) scm)))

=>
(NIL NIL NIL)
(NIL NIL NIL)
(NIL SET3 SET2)
(NIL SET3 SET2)
```

SYNOPSIS:

```
(defmethod double ((scm sc-map) section-ref start-seq end-seq master-player
                   doubling-players)
```

38.6.397 sc-map/get-all-data-from-palette

[sc-map] [Methods]

DESCRIPTION:

```
Given an sc-map object that has been bound to a palette object of any type,
return all of the palette data contained in the given sc-map object as it
has been allocated to the map, in the order in which it appears in the map.

The given sc-map object must be bound to a palette object for this method
to work. If no palette object has been bound to the given sc-map object,
the method returns NIL and prints a warning.
```

ARGUMENTS:

- An sc-map object.

RETURN VALUE:

- A list of objects, the type depending on the given palette.

EXAMPLE:

```
;; Create a set-palette object and an sc-map object, bind them using the
;; <palette> argument of the make-sc-map function, and print the results of
;; applying the get-all-data-from-palette method by printing the data of each
;; of the objects in the list it returns as note-name symbols.
(let* ((sp (make-set-palette 'set-pal '((set1 ((c2 b2 a3 g4 f5 e6)))
                                       (set2 ((d2 c3 b3 a4 g5 f6)))
                                       (set3 ((e2 d3 c4 b4 a5 g6))))))
       (scm (make-sc-map 'sc-m '((sec1
                                  ((vn (set1 set3 set2))
                                   (va (set2 set3 set1))
                                   (vc (set3 set1 set2))))
                                 (sec2
                                  ((vn (set1 set2 set1))
                                   (va (set2 set1 set3))
                                   (vc (set1 set3 set3))))
                                 (sec3
                                  ((vn (set1 set1 set3))
                                   (va (set1 set3 set2))
                                   (vc (set3 set2 set3)))))
                           :palette sp)))
  (loop for cs in (get-all-data-from-palette scm)
     collect (pitch-list-to-symbols (data cs))))

=>
((C2 B2 A3 G4 F5 E6) (E2 D3 C4 B4 A5 G6) (D2 C3 B3 A4 G5 F6)
 (D2 C3 B3 A4 G5 F6) (E2 D3 C4 B4 A5 G6) (C2 B2 A3 G4 F5 E6)
 (E2 D3 C4 B4 A5 G6) (C2 B2 A3 G4 F5 E6) (D2 C3 B3 A4 G5 F6)
 (C2 B2 A3 G4 F5 E6) (D2 C3 B3 A4 G5 F6) (C2 B2 A3 G4 F5 E6)
 (D2 C3 B3 A4 G5 F6) (C2 B2 A3 G4 F5 E6) (E2 D3 C4 B4 A5 G6)
 (C2 B2 A3 G4 F5 E6) (E2 D3 C4 B4 A5 G6) (E2 D3 C4 B4 A5 G6)
 (C2 B2 A3 G4 F5 E6) (C2 B2 A3 G4 F5 E6) (E2 D3 C4 B4 A5 G6)
 (C2 B2 A3 G4 F5 E6) (E2 D3 C4 B4 A5 G6) (D2 C3 B3 A4 G5 F6)
 (E2 D3 C4 B4 A5 G6) (D2 C3 B3 A4 G5 F6) (E2 D3 C4 B4 A5 G6))

;; Applying the method to an sc-map object that is not bound to a palette
;; object returns NIL
(let ((scm (make-sc-map 'sc-m '((sec1
                                 ((vn (set1 set3 set2))
```

```
                                  (va (set2 set3 set1))
                                  (vc (set3 set1 set2))))
                                (sec2
                                 ((vn (set1 set2 set1))
                                  (va (set2 set1 set3))
                                  (vc (set1 set3 set3))))
                                (sec3
                                 ((vn (set1 set1 set3))
                                  (va (set1 set3 set2))
                                  (vc (set3 set2 set3))))))))
  (get-all-data-from-palette scm))

=>
NIL
WARNING:
   sc-map::get-all-data-from-palette:
   palette slot is nil so can't return data from it.
```

SYNOPSIS:

```
(defmethod get-all-data-from-palette ((scm sc-map))
```

38.6.398 sc-map/get-data-from-palette

[sc-map] [Methods]

DESCRIPTION:

Given an sc-map object that has been bound to a palette object of any type, return the palette data contained at the location within the given sc-map object as specified by the <IDs> argument.

Deeper levels of the map can be accessed by specifying a path of IDs into the given sc-map object.

If no palette object has been bound to the given sc-map object, the method returns the contents of the sc-map object at the specified location instead.

ARGUMENTS:

- A symbol or list of symbols that is/are the ID or path of nested IDs within the given sc-map object for which the data is sought.
- The sc-map object in which the data is sought.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether to print a warning if the specified ID is
  not found in the given sc-map object. T = print warning. Default = T.

RETURN VALUE:

The named object or list of named objects associated with the specified ID
or path of IDs.

If the specified ID is not found within the given sc-map object, the method
returns NIL. If the optional <warn> argument is set to T, a warning is also
printed in this case.

EXAMPLE:

```
;;; Create a palette object and an sc-map object and bind them using the
;;; <palette> keyword argument of the make-sc-map function. Then apply the
;;; get-data-from-palette object to a nested ID in the sc-map object. Loop
;;; through the data of the named objects in the list returned and return them
;;; as note-name symbols.
(let* ((sp (make-set-palette 'set-pal '((set1 ((c2 b2 a3 g4 f5 e6)))
                                       (set2 ((d2 c3 b3 a4 g5 f6)))
                                       (set3 ((e2 d3 c4 b4 a5 g6))))))
       (scm (make-sc-map 'sc-m '((sec1
                                  ((vn (set1 set3 set2))
                                   (va (set2 set3 set1))
                                   (vc (set3 set1 set2))))
                                 (sec2
                                  ((vn (set1 set2 set1))
                                   (va (set2 set1 set3))
                                   (vc (set1 set3 set3))))
                                 (sec3
                                  ((vn (set1 set1 set3))
                                   (va (set1 set3 set2))
                                   (vc (set3 set2 set3)))))
                           :palette sp)))
  (loop for cs in (get-data-from-palette '(sec1 vn) scm)
     collect (pitch-list-to-symbols (data cs))))

=> ((C2 B2 A3 G4 F5 E6) (E2 D3 C4 B4 A5 G6) (D2 C3 B3 A4 G5 F6))

;; If applied to an sc-map object that is not bound to a palette, the contents
;; of the sc-map object at the specified location are returned and a warning is
;; printed by default
(let ((scm (make-sc-map 'sc-m '((sec1
                                 ((vn (set1 set3 set2))
                                  (va (set2 set3 set1))
```

```
                                  (vc (set3 set1 set2))))
                                (sec2
                                 ((vn (set1 set2 set1))
                                  (va (set2 set1 set3))
                                  (vc (set1 set3 set3))))
                                (sec3
                                 ((vn (set1 set1 set3))
                                  (va (set1 set3 set2))
                                  (vc (set3 set2 set3))))))))
   (get-data-from-palette '(sec1 vn) scm))

=>
NAMED-OBJECT: id: VN, tag: NIL,
data: (SET1 SET3 SET2)
**************
, NO-PALETTE
```

SYNOPSIS:

```
(defmethod get-data-from-palette (ids (scm sc-map) &optional (warn t))
```

38.6.399 sc-map/get-nth-from-map

[sc-map] [Methods]

DESCRIPTION:

```
Get the element located at the nth position within a given sc-map
object. Both the map-ref (the path of IDs into the list to be searched) and
the nth must be specified.
```

ARGUMENTS:

```
- A list that is the map-ref; i.e., the path of IDs into the list to be
  searched.
- An integer that is the zero-based index of the element to be returned
  from the specified list.
- An sc-map object.
```

RETURN VALUE:

```
Returns the element located at the given index.

Returns NIL if the index does not exist.
```

EXAMPLE:

```
;; Specify the path of IDs into the desired list ("map-ref") as a list, then
;; the position to be read from within the list located there.
(let ((mscm (make-sc-map 'scm-test
                         '((1
                            ((vn (1 2 3 4 5))
                             (va (2 3 4 5 1))
                             (vc (3 4 5 1 2))))
                           (2
                            ((vn (6 7 8))
                             (va (7 8 6))
                             (vc (8 6 7))))
                           (3
                            ((vn (9))
                             (va (9))
                             (vc (9))))))))
  (get-nth-from-map '(1 vn) 1 mscm))

=> 2

;; Returns NIL if the specified index does not exist
(let ((mscm (make-sc-map 'scm-test
                         '((1
                            ((vn (1 2 3 4 5))
                             (va (2 3 4 5 1))
                             (vc (3 4 5 1 2))))
                           (2
                            ((vn (6 7 8))
                             (va (7 8 6))
                             (vc (8 6 7))))
                           (3
                            ((vn (9))
                             (va (9))
                             (vc (9))))))))
  (get-nth-from-map '(3 vn) 1 mscm))

=> NIL
```

SYNOPSIS:

```
(defmethod get-nth-from-map (map-ref nth (scm sc-map))
```

38.6.400 sc-map/get-nth-from-palette

[sc-map] [Methods]

DESCRIPTION:

Given an sc-map object that is bound to a palette object of any type, return the data of the palette object located at the nth position of the list found at the specified ID or path of nested IDs.

If the given sc-map object is not bound to a palette object, NIL is returned instead.

ARGUMENTS:

- An ID or list of IDs that are the path to the list within the given sc-map object from which the specified nth position is to be returned.
- A zero-based integer that is the position within the list found at the path specified from which the given element is to be returned.
- An sc-map object.

RETURN VALUE:

- An element/object of the type contained within the given palette object of the given sc-map object.

EXAMPLE:

```
;;; Create a set-palette object and an sc-map object, bind them using the
;;; <palette> object of the make-sc-map function, and apply the
;;; get-nth-from-palette method
(let* ((sp (make-set-palette 'set-pal '((set1 ((c2 b2 a3 g4 f5 e6)))
                                       (set2 ((d2 c3 b3 a4 g5 f6)))
                                       (set3 ((e2 d3 c4 b4 a5 g6))))))
      (scm (make-sc-map 'sc-m '((sec1
                                 ((vn (set1 set3 set2))
                                  (va (set2 set3 set1))
                                  (vc (set3 set1 set2))))
                                (sec2
                                 ((vn (set1 set2 set1))
                                  (va (set2 set1 set3))
                                  (vc (set1 set3 set3))))
                                (sec3
                                 ((vn (set1 set1 set3))
                                  (va (set1 set3 set2))
                                  (vc (set3 set2 set3)))))
                          :palette sp)))
  (get-nth-from-palette '(sec1 vn) 0 scm))

=>
COMPLETE-SET: complete: NIL
```

```
              num-missing-non-chromatic: 12
              num-missing-chromatic: 6
              missing-non-chromatic: (BQS BQF AQS AQF GQS GQF FQS EQS EQF DQS
                                      DQF CQS)
              missing-chromatic: (BF AF FS EF D CS)
TL-SET: transposition: 0
        limit-upper: NIL
        limit-lower: NIL
SC-SET: auto-sort: T, used-notes:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 0
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 0, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: USED-NOTES, tag: NIL,
data: NIL
**************


**** N.B. All pitches printed as symbols only, internally they are all
pitch-objects.


    subsets:
    related-sets:
SCLIST: sclist-length: 6, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: SET1, tag: NIL,
data: (C2 B2 A3 G4 F5 E6)
**************

;;; Applying the method to an sc-map object that is not bound to a palette
;;; object returns NIL
(let ((scm (make-sc-map 'sc-m '((sec1
                                 ((vn (set1 set3 set2))
                                  (va (set2 set3 set1))
                                  (vc (set3 set1 set2))))
                                (sec2
                                 ((vn (set1 set2 set1))
                                  (va (set2 set1 set3))
                                  (vc (set1 set3 set3))))
                                (sec3
                                 ((vn (set1 set1 set3))
```

```
                                  (va (set1 set3 set2))
                                  (vc (set3 set2 set3))))))))
  (get-nth-from-palette '(sec1 vn) 0 scm))

=> NIL
```

SYNOPSIS:

```
(defmethod get-nth-from-palette (sc-map-ref nth (scm sc-map))
```

38.6.401 sc-map/make-sc-map

[sc-map] [Functions]

DESCRIPTION:

```
Create an sc-map object, which will be used for mapping rhythmic sequences,
chords etc. to specific parts of a piece.
```

ARGUMENTS:

```
- The ID of the resulting sc-map object.
- A list of data, most likely recursive.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :warn-not-found. T or NIL to indicate whether a warning is printed when
  an index which doesn't exist is used for look-up.  T = warn. Default = T.
- :recurse-simple-data. T or NIL to indicate whether to recursively
  instantiate a recursive-assoc-list in place of data that appears to be a
  simple assoc-list (i.e. a 2-element list). If NIL, the data of 2-element
  lists whose second element is a number or a symbol will be ignored,
  therefore remaining as a list. For example, this data would normally
  result in a recursive call: (y ((2 23) (7 28) (18 2))).
  T = recurse. Default = T.
- :replacements. A list of lists in the format '(((1 2 vla) 3 20b) ((2 3
  vln) 4 16a)) that indicate changes to individual elements of lists within
  the given sc-map object. (Often sc-map data is generated algorithmically,
  but individual elements of the lists need to be changed.) Each such list
  indicates a change, the first element of the list being the reference
  into the sc-map (the viola voice of section 1 subsection 2 in the first
  element here, for example), the second element being the nth of the data
  list to change for this key, and the third being the new data.
- :palette. A palette object or NIL. If a palette object is specified or
```

```
   defined here, it will be automatically bound to the given sc-map
   object. Default = NIL
```

RETURN VALUE:

```
 An sc-map object.
```

EXAMPLE:

```
;; Create an sc-map object with contents that could be used as a rthm-seq-map
(make-sc-map 'scm-test
             '((1
                ((vn (1 2 3 4 5))
                 (va (2 3 4 5 1))
                 (vc (3 4 5 1 2))))
               (2
                ((vn (6 7 8))
                 (va (7 8 6))
                 (vc (8 6 7))))
               (3
                ((vn (9))
                 (va (9))
                 (vc (9))))))

=>
SC-MAP: palette id: NIL
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 9
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: SCM-TEST, tag: NIL,
data: (
NAMED-OBJECT: id: 1, tag: NIL,
data:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 3
                      linked: NIL
                      full-ref: (1)
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
```

```
NAMED-OBJECT: id: "sub-ral-of-SCM-TEST", tag: NIL,
data: (
NAMED-OBJECT: id: VN, tag: NIL,
data: (1 2 3 4 5)
[...]

;;; Create an sc-map object and automatically bind it to a set-palette object
;;; using the <palette> keyword argument. Then read the PALETTE slot of the
;;; sc-map created to see its contents.
(let ((scm
       (make-sc-map
        'scm-test
        '((1
           ((vn (1 2 3 4 5))
            (va (2 3 4 5 1))
            (vc (3 4 5 1 2))))
          (2
           ((vn (6 7 8))
            (va (7 8 6))
            (vc (8 6 7))))
          (3
           ((vn (9))
            (va (9))
            (vc (9)))))
        :palette (make-set-palette 'set-pal
                                   '((set1 ((c2 b2 a3 g4 f5 e6)))
                                     (set2 ((d2 c3 b3 a4 g5 f6)))
                                     (set3 ((e2 d3 c4 b4 a5 g6))))))))
  (palette scm))

=>

SET-PALETTE:
PALETTE:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 3
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: SET-PAL, tag: NIL,
data: (
COMPLETE-SET: complete: NIL
[...]
```

```
data: (C2 B2 A3 G4 F5 E6)
[...]
COMPLETE-SET: complete: NIL
[...]
data: (D2 C3 B3 A4 G5 F6)
[...]
COMPLETE-SET: complete: NIL
[...]
data: (E2 D3 C4 B4 A5 G6)
)

;;; An example using replacements
(make-sc-map 'sc-m
             '((1
                ((vn (set1 set3 set2))
                 (va (set2 set3 set1))
                 (vc (set3 set1 set2))))
               (2
                ((vn (set1 set2 set1))
                 (va (set2 set1 set3))
                 (vc (set1 set3 set3))))
               (3
                ((vn (set1 set1 set3))
                 (va (set1 set3 set2))
                 (vc (set3 set2 set3)))))
             :replacements '(((1 va) 2 set2)))

=>

SC-MAP: palette id: NIL
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 9
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: SC-M, tag: NIL,
data: (
NAMED-OBJECT: id: 1, tag: NIL,
data:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 3
                      linked: NIL
                      full-ref: (1)
```

```
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "sub-ral-of-SC-M", tag: NIL,
data: (
NAMED-OBJECT: id: VN, tag: NIL,
data: (SET1 SET3 SET2)
**************


NAMED-OBJECT: id: VA, tag: NIL,
data: (SET2 SET2 SET1)
**************


NAMED-OBJECT: id: VC, tag: NIL,
data: (SET3 SET1 SET2)
**************
)

[...]
```

SYNOPSIS:

```
(defun make-sc-map (id scm &key (palette nil) (warn-not-found t)
                    (recurse-simple-data t) (replacements nil))
```

38.6.402 sc-map/num-sequences

[sc-map] [Methods]

DATE:

January 27th 2018

DESCRIPTION:

Return the number of sequences in the map as a whole (i.e. counting all sections and subsections). This method also works on subsections. Note that the first time this method is used it sets the :num-sequences slot which will then be used in subsequent queries until the data is changed. However, if you've not specifically called this method on a subsection, its slot will not yet be set.

ARGUMENTS:

- an sc-map object

RETURN VALUE:

An integer

SYNOPSIS:

```
(defmethod num-sequences ((scm sc-map))
```

38.6.403 sc-map/remove-ref

[sc-map] [Methods]

DATE:

August 24th 2018

DESCRIPTION:

Remove or replace a reference from a map. If there's no replacement then the data list will be shorter (if the reference is present) thereafter (which could cause problems .

ARGUMENTS:

- an sc-map object
- the section reference (symbol, list, etc.)
- the reference in the data list that should be removed

OPTIONAL ARGUMENTS:

- the replacement reference, if any. If, for example in a rthm-seq-map, you want to replace a rthm-seq reference with nil, so that the player sits out a sequence, then pass '(nil) as the replacement.

RETURN VALUE:

The sc-map object.

SYNOPSIS:

```
(defmethod remove-ref ((scm sc-map) section ref &optional replacement)
```

38.6.404 sc-map/rthm-seq-map

[sc-map] [Classes]

NAME:

```
rthm-seq-map

File:             rthm-seq-map.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist -> assoc-list -> recursive-assoc-list ->
                  sc-map -> rthm-seq-map

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the rthm-seq-map class which maps
                  references to rthm-seq objects for the players in the
                  piece.  Extensions to the sc-map superclass are the
                  collection of all the players in the piece and a
                  check to make sure that each list each instrument has the
                  same number of rthm-seq references for each section.

                  Instances of this class must declare sections and
                  players so if the piece is in one section, give it
                  the label 1 or whatever, e.g.

                  '((1
                     ((vln (2 20 1 9 10 22 16 25 6 14 21 17 4 9 13 2))
                      (vla (2 23 3 7 13 22 19 3 8 12 23 14 2 10 15 4))
                      (vc (2 21 3 12 11 22 16 1 8 17 23 20 24 9 12 2)))))


Author:           Michael Edwards: m@michael-edwards.org

Creation date:    July 28th 2001

$$ Last modified:  15:06:07 Mon Apr 14 2025 CEST

SVN ID: $Id$
```

38.6.405 rthm-seq-map/add-player

[rthm-seq-map] [Methods]

DATE:

January 18th 2018, Heidhausen

DESCRIPTION:

Add a player to a rthm-seq-map. The data (3rd) argument defines the rthm-seq references (IDs) to be used for the new player. This is a very flexible argument:

If not given, then NIL will be added to the map, resulting in the new player not playing at all.

If a single reference ID is given, then that will be used throughout each section of the map.

If a list of references is given, then items will be popped out of the list sequentially and used for the new player. In this case the length of the list does not matter as if it's too long the last elements will simply be unused, and if it's too short then NIL will be returned when the list is exhausted, resulting in the new player not playing from that point onwards. The exception here is if the cycle (4th) argument is T, whereupon the list of references will be used circularly, restarting the list when the end is reached.

Otherwise a function can be passed: this function should take four arguments: the rthm-seq-map object (which it can then analyse/query to make decisions); the full reference to the current (sub)section for which references should be provided; the number of references which are needed for this (sub)section; and the player being added. It must return a list of references for each given section.

ARGUMENTS:

- the rthm-seq-map object
- the new player to add (ID only: the same which will be referenced in the ensemble slot of a slippery-chicken object)

OPTIONAL ARGUMENTS:

keyword arguments:
- :data. the rthm-seq-palette references to use for the new player or a function. See above. Default NIL.
- :cycle. T or NIL to cycle the references. See above. Default NIL.

RETURN VALUE:

the same rthm-seq-map object but with the new player now added.

EXAMPLE:

```
;; will use rthm-seq 1 throughout the piece
(add-player m 'cl 1)
;; will use 1 2 3 4 5 then nil thereafter
(add-player m 'ob '(1 2 3 4 5))
;; forcing a reference into a recursive rthm-seq-palette to be used just once
(add-player m 'db '((bass dense 1)))
;; forcing it to be used repeatedly
(add-player m 'db '((bass dense 1)) t)
;; a simple example with a function: rs1 for section 1, rs3 for section three
;;; otherwise rs1a
(add-player m 'bsn #'(lambda (rsm section-ref num-refs player)
                            (ml (case section-ref
                                  (sec1 'rs1)
                                  (sec3 'rs3)
                                  (t 'rs1a))
                                num-refs)))
;; db will be added but it won't play anything (unless references are added
;;; later but before make-slippery-chicken is called)
(add-player m 'db)
```

SYNOPSIS:

```
(defmethod add-player ((rsm rthm-seq-map) player &key data cycle)
```

38.6.406 rthm-seq-map/add-repeats

[rthm-seq-map] [Methods]

DATE:

30-Dec-2010

DESCRIPTION:

Generate repeating sequences at given cycle points using recurring-event data. This extends the number of items in the map.

ARGUMENTS:

- A rthm-seq-map object.
- A list of two-item lists of integers that determine the cycle

pattern. This list will have the format of the recurring-event class's DATA slot (see recurring-event.lsp).
- A list of two-item lists of integers that determine the number of repeats made (or references into the :repeats list). This list is also processed cyclically (i.e. the recurring-event class's RETURN-DATA-CYCLE slot).

OPTIONAL ARGUMENTS:

keyword arguments:
- :section. The section map reference. Default = 1.
- :repeats-indices. A list of the number of repeat bars returned by the cycle data (i.e. recurring-event class's RETURN-DATA slot). Generally this will remain NIL and the number of repeats will be expressed directly in the third argument, but it could be useful to use references into this list there instead, since the recurring-event class already makes this possible. Default = NIL.
- :start. An integer that is the number of the bar/rthm-seq where the process is to begin. Default = 1.
- :end. An integer that is the number of the bar/rthm-seq where the process is to end. NIL = process all bars/rthm-seqs. Default = NIL.
- :print. T or NIL to indicate whether to print the rthm-seq ID and the number repetitions to the listener. T = print. Default = NIL.
- :repeat-rest-seqs. T or NIL to indicate whether sequences consisting of just rests should be repeated also. This will need the :palette in order to work. Default = T.
- :palette. The rthm-seq-palette to check that sequences are not rest sequences, if about to repeat them and repeat-rest-seqs is NIL.

RETURN VALUE:

An integer that is the number of sequences added.

EXAMPLE:

```
;;; Straightforward usage, additionally printing the DATA slot before and after
;;; applying the method
(let ((mrsm
       (make-rthm-seq-map
        'rsm-test
        '((1 ((vn (1 2 3 2 1 3 1 3 2 3 1 2 1 3 1 3 2 1)))))
        :palette (make-rsp
                  'rs-pal
                  '((1 ((((2 4) q e s s))))
                    (2 ((((2 4) e s s q))))
                    (3 ((((2 4) s s q e)))))))))
```

```
  (print (get-data-data '(1 vn) mrsm))
  ;; so there'll be a repeat after two events three times in a row, then after
  ;; three events twice in a row. The number of repeats will be 5 three times
  ;; in a row, then 8 twice in a row.
  (add-repeats mrsm '((2 3) (3 2)) '((5 3) (8 2)))
  (print (get-data-data '(1 vn) mrsm)))

=>
(1 2 3 2 1 3 1 3 2 3 1 2 1 3 1 3 2 1)
(1 2 3 3 3 3 3 2 1 1 1 1 1 3 1 1 1 1 1 3 2 3 3 3 3 3 3 3 3 1 2 1 1 1 1 1 1
 1 1 3 1 1 1 1 1 3 2 2 2 2 2 1)

;;; Using the :start, :end, and :print arguments
(let ((mrsm
       (make-rthm-seq-map
        'rsm-test
        '((1 ((vn (1 2 3 2 1 3 1 3 2 3 1 2 1 3 1 3 2 1)))))
        :palette (make-rsp
                  'rs-pal
                  '((1 ((((2 4) q e s s))))
                    (2 ((((2 4) e s s q))))
                    (3 ((((2 4) s s q e)))))))))
  (print (get-data-data '(1 vn) mrsm))
  (add-repeats mrsm '((1 6) (2 6)) '((11 6) (23 3))
               :start 3
               :end 11
               :print t)
  (print (get-data-data '(1 vn) mrsm)))

=>
(1 2 3 2 1 3 1 3 2 3 1 2 1 3 1 3 2 1)
2 x 11
1 x 11
3 x 11
1 x 11
3 x 11
2 x 11
1 x 23
(1 2 3 2 2 2 2 2 2 2 2 2 2 1 1 1 1 1 1 1 1 1 1 3 3 3 3 3 3 3 3 3 3 1
 1 1 1 1 1 1 1 1 1 3 3 3 3 3 3 3 3 3 3 2 2 2 2 2 2 2 2 2 2 3 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 3 1 3 2 1)
```

SYNOPSIS:

```
(defmethod add-repeats ((rsm rthm-seq-map) repeat-every repeats &key
                        (repeat-rest-seqs t) palette
```

```
                        (section 1) repeats-indices (start 1) end print)
```

38.6.407 rthm-seq-map/add-repeats-simple

[rthm-seq-map] [Methods]

DESCRIPTION:

```
Add repeats of a specified rthm-seq within the given rthm-seq-map object a
specified number of times.
```

ARGUMENTS:

```
- A rthm-seq-map object.
- An integer that is the number of the rthm-seq (position within the
  rthm-seq-map) to be repeated.
- An integer that is the number of times that rthm-seq is to be repeated.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :section. An integer that is the ID of the section in which the repeat
  operation is to be performed.
- :print. T or NIL to indicate whether to print the rthm-seq ID and the
  number repetitions to the listener. T = print. Default = NIL.
```

RETURN VALUE:

```
Always returns T.
```

EXAMPLE:

```
;;; Print the DATA of the given rthm-seq-map, apply the method, and print again
;;; to see the difference.
(let ((mrsm
       (make-rthm-seq-map
        'rsm-test
        '((1 ((vn (1 2 3 2 1 3 1 3 2 3 1 2 1 3 1 3 2 1)))))
        :palette (make-rsp
                  'rs-pal
                  '((1 ((((2 4) q e s s))))
                    (2 ((((2 4) e s s q))))
                    (3 ((((2 4) s s q e)))))))))
  (print (get-data-data '(1 vn) mrsm))
  (add-repeats-simple mrsm 3 13)
```

```
  (print (get-data-data '(1 vn) mrsm)))

=>
(1 2 3 2 1 3 1 3 2 3 1 2 1 3 1 3 2 1)
(1 2 3 3 3 3 3 3 3 3 3 3 3 3 3 2 1 3 1 3 2 3 1 2 1 3 1 3 2 1)
```

SYNOPSIS:

```
(defmethod add-repeats-simple ((rsm rthm-seq-map) start-seq repeats &key
                               (section 1) print)
```

38.6.408 rthm-seq-map/check-num-sequences

[rthm-seq-map] [Functions]

DESCRIPTION:

```
Check to ensure that each player in each section of the given rthm-seq-map
object has the same number of references as every other instrument. If not,
drop into the debugger with an error.

NB: This function is called automatically every time make-rthm-seq-map is
    called so it shouldn't generally be necessary for the user to call this
    method.  However, if the rthm-seq-map is changed somehow, it might be a
    good idea to recheck.
```

ARGUMENTS:

```
- A rthm-seq-map object.
```

RETURN VALUE:

```
Returns T if all players have the same number of references in each
section, otherwise drops into the debugger with an error.
```

EXAMPLE:

```
;;; Passes the test:
(let ((rsmt (make-rthm-seq-map
             'rsm-test
             '((sec1 ((vn (rs1a rs3a rs2a))
                      (va (rs1b rs3b rs2b))
                      (vc (rs1a rs3b rs2a))))
               (sec2 ((vn (rs1a rs2a rs1a))
                      (va (rs1a rs2a rs1b))
```

```
                      (vc (rs1a rs2b rs1a))))
               (sec3 ((vn (rs1a rs1a rs3a))
                      (va (rs1a rs1a rs3b))
                      (vc (rs1a rs1b rs3a))))
               (sec4 ((vn (rs1a rs1a rs1a))
                      (va (rs1a rs1a rs1b))
                      (vc (rs1a rs1b rs1a))))))))
  (check-num-sequences rsmt))

=> T

;;; Doesn't pass the test; drops into debugger with an error.
(let ((rsmt (make-rthm-seq-map
             'rsm-test
             '((sec1 ((vn (rs1a rs3a rs2a))
                      (va (rs1b rs3b))
                      (vc (rs1a rs3b rs2a))))
               (sec2 ((vn (rs1a))
                      (va (rs1a rs2a rs1b))
                      (vc (rs1a rs2b rs1a))))
               (sec3 ((vn (rs1a rs3a))
                      (va (rs1a))
                      (vc (rs1a rs1b rs3a))))
               (sec4 ((vn (rs1a))
                      (va (rs1a rs1a rs1b))
                      (vc (rs1a rs1a)))))))
  (check-num-sequences rsmt))

=>
rthm-seq-map::check-num-sequences: In rthm-seq-map RSM-TEST-5, instrument VA:
Each instrument must have the same number of sequences for any given section:
(RS1B RS3B)
   [Condition of type SIMPLE-ERROR]
```

SYNOPSIS:

```
(defun check-num-sequences (rsm)
```

38.6.409 rthm-seq-map/get-map-refs

[rthm-seq-map] [Methods]

DATE:

```
 29-Dec-2010
```

DESCRIPTION:

```
Return the list of rthm-seq-palette references for the given player and
section.
```

ARGUMENTS:

```
- A rthm-seq-map object.
- The ID of the section in which the references are sought.
- The ID of the player for whom the references are sought.
```

RETURN VALUE:

```
A list of references (each of which might also  be a list).
```

EXAMPLE:

```
(let ((rsmt (make-rthm-seq-map
             'rsm-test-5
             '((sec1 ((vn (rs1 rs3 rs2))
                      (va (rs2 rs3 rs1))
                      (vc (rs3 rs1 rs2))))
               (sec2 ((vn (rs1 rs2 rs1))
                      (va (rs2 rs1 rs3))
                      (vc (rs1 rs3 rs3))))
               (sec3 ((vn (rs1 rs1 rs3))
                      (va (rs1 rs3 rs2))
                      (vc (rs3 rs2 rs3)))))
             :palette (make-rsp
                       'rs-pal
                       '((rs1 ((((2 4) q e s s))))
                         (rs2 ((((2 4) e s s q))))
                         (rs3 ((((2 4) s s q e)))))))))
  (get-map-refs rsmt 'sec3 'vc))

=> (RS3 RS2 RS3)
```

SYNOPSIS:

```
(defmethod get-map-refs ((rsm rthm-seq-map) section player)
```

38.6.410 rthm-seq-map/get-time-sig-ral

[rthm-seq-map] [Methods]

DESCRIPTION:

Collate the IDs of all rthm-seq objects in a given rthm-seq-map into groups based on identical time signatures and return this as a recursive-assoc-list object that has the same structure of the map.

Instead of having the list of rthm-seq references for each section/instrument, the method returns an assoc-list whose keys are the time-sig tags, with the corresponding data being circular-sclists of rthm-seq IDs.

The result is a recursive-assoc-list for each instrument/section that can be queried to find the rthm-seq refs of all rthm-seqs that share the same bar/time-sig structure; e.g., all those that have a 2/4 bar followed by a 3/4 bar etc.

ARGUMENTS:

- A rthm-seq-map object.
- A rthm-seq-palette object.

RETURN VALUE:

A recursive-assoc-list object.

EXAMPLE:

```
(let* ((mini
        (make-slippery-chicken
         '+mini+
         :ensemble '(((sax (alto-sax :midi-channel 1))))
         :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
         :set-map '((1 (1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1)))
         :rthm-seq-palette '((1 ((((4 4) h q e s s) ((2 4) h))))
                             (2 ((((4 4) h q e s s))))
                             (3 ((((4 4) h q e s s))))
                             (4 ((((4 4) h q q) ((2 4) q q))))
                             (5 ((((4 4) h q e s s))))
                             (6 ((((4 4) h q q) ((2 4) q q)))))
         :rthm-seq-map '((1 ((sax (1 2 3 5 2 4 6 2 3 1 3 2 3 2 1 3 2)))))))
       (tsral (get-time-sig-ral (rthm-seq-map mini) (rthm-seq-palette mini))))
  (get-data-data 1 tsral))

=>
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 1
                      linked: T
```

```
                    full-ref: (1)
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 1, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "sub-ral-of-+MINI+-RTHM-SEQ-MAP", tag: NIL,
data: (
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: (1 SAX), next: NIL
NAMED-OBJECT: id: SAX, tag: NIL,
data: (
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "0404-0204", tag: NIL,
data: (4 6 1)
**************


CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "0404", tag: NIL,
data: (5 3 2)
**************
)
**************
)
**************
```

SYNOPSIS:

```
(defmethod get-time-sig-ral ((rsm rthm-seq-map) (rsp rthm-seq-palette))
```

38.6.411 rthm-seq-map/make-rthm-seq-map

[rthm-seq-map] [Functions]

DESCRIPTION:

```
Make a rthm-seq-map object.
```

ARGUMENTS:

```
- The ID of the rthm-seq-map object to be made.
- A list of nested lists, generally taking the form
  '((section1 ((player1 (rthm-seq ids))
               (player2 (rthm-seq ids))
               (etc... (etc...))))
    (section2 ((player1 (rthm-seq ids))
               (player2 (rthm-seq ids))
               (etc...)))
    (etc...))
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :palette. A palette object or NIL. If a palette object is specified or
  defined here, it will be automatically bound to the given rthm-seq-map
  object. Default = NIL.
- :warn-not-found. T or NIL to indicate whether a warning is printed when
  an index which doesn't exist is used for look-up.
  T = warn. Default = NIL.
- :replacements. A list of lists in the format
  '(((1 2 vla) 3 20b) ((2 3 vln) 4 16a)) that indicate changes to
  individual elements of lists within the given rthm-seq-map object. (Often
  rthm-seq-map data is generated algorithmically but individual elements of
  the lists need to be changed.)  Each such list indicates a change, the
  first element of the list being the reference into the rthm-seq-map (the
  vla player of section 1, subsection 2 in the first example here), the
  second element is the nth of the data list for this key to change, and
  the third is the new data. Default = NIL.
- :recurse-simple-data. T or NIL to indicate whether to recursively
  instantiate a recursive-assoc-list in place of data that appears to be a
  simple assoc-list (i.e. a 2-element list). If NIL, the data of 2-element
  lists whose second element is a number or a symbol will be ignored,
  therefore remaining as a list. For example, this data would normally
  result in a recursive call: (y ((2 23) (7 28) (18 2))).
  T = recurse. Default = T.
```

RETURN VALUE:

```
A rthm-seq-map object.
```

EXAMPLE:

```
;;; Straightforward usage
(make-rthm-seq-map 'rsm-test
                   '((1 ((vn (1 2 3 4))
```

```
                        (va (2 3 4 1))
                        (vc (3 4 1 2))))
                    (2 ((vn (4 5 6))
                        (va (5 6 4))
                        (vc (6 4 5))))
                    (3 ((vn (7 8 9 1 2))
                        (va (8 9 1 2 7))
                        (vc (9 1 2 7 8))))))

=>

RTHM-SEQ-MAP: num-players: 3
              players: (VA VC VN)
SC-MAP: palette id: NIL
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 9
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found NIL
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: RSM-TEST, tag: NIL,
data: (
[...]

;;; An example using the :replacements argument and binding directly to a
;;; specified rthm-seq-palette object.
(make-rthm-seq-map 'rsm-test
                   '((1 ((vn (1 2 3 4))
                         (va (2 3 4 1))
                         (vc (3 4 1 2))))
                     (2 ((vn (4 5 6))
                         (va (5 6 4))
                         (vc (6 4 5))))
                     (3 ((vn (7 8 9 1 2))
                         (va (8 9 1 2 7))
                         (vc (9 1 2 7 8)))))
                   :palette (make-rsp
                             'rs-pal
                             '((rs1 ((((2 4) q e s s))))
                               (rs2 ((((2 4) e s s q))))
                               (rs3 ((((2 4) s s q e))))))
                   :replacements '(((1 vn) 2 7)
                                   ((2 va) 1 1)
                                   ((3 vc) 1 0)))
```

```
=>
RTHM-SEQ-MAP: num-players: 3
              players: (VA VC VN)
SC-MAP: palette id: RS-PAL
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 9
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found NIL
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: RSM-TEST, tag: NIL,
data: (
[...]
```

SYNOPSIS:

```
(defun make-rthm-seq-map (id rsm &key (palette nil) (warn-not-found nil)
                                      (replacements nil)
                                      (recurse-simple-data t))
```

38.6.412 rthm-seq-map/rsm-count-notes

[rthm-seq-map] [Functions]

DESCRIPTION:

Returns the number of notes in the given rthm-seq-map object for the specified player and rthm-seq-palette.

ARGUMENTS:

- A rthm-seq-map object.
- The ID of the player whose notes are to be counted.
- The rthm-seq-palette object whose rthm-seq object IDs are referred to by the given rthm-seq-map object.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether to count just the number of notes that need new events (i.e., not counting tied notes; also not counting chords, since chords need only one event) or the total number of notes in that player's part in the score. T = count just attacked notes. Default = T.

RETURN VALUE:

```
Returns an integer that is the number of notes counted.
```

EXAMPLE:

```
(let ((rsmt (make-rthm-seq-map
             'rsm-test
             '((sec1 ((vn (rs1 rs3 rs2))
                      (va (rs2 rs3 rs1))
                      (vc (rs3 rs1 rs2))))
               (sec2 ((vn (rs1 rs2 rs1))
                      (va (rs2 rs1 rs3))
                      (vc (rs1 rs3 rs3))))
               (sec3 ((vn (rs1 rs1 rs3))
                      (va (rs1 rs3 rs2))
                      (vc (rs3 rs2 rs3)))))))
      (rspt (make-rsp
             'rs-pal
             '((rs1 ((((2 4) q (e) s s))))
               (rs2 ((((2 4) e +s (s) q))))
               (rs3 ((((2 4) (s) s +q e)))))))))
  (print (rsm-count-notes rsmt 'vn rspt))
  (print (rsm-count-notes rsmt 'va rspt nil)))

=>
23
27
```

SYNOPSIS:

```
(defun rsm-count-notes (rthm-seq-map player palette &optional (just-attacks t))
```

38.6.413 rthm-seq-map/rthm-chain

[rthm-seq-map] [Classes]

NAME:

```
rthm-chain

File:             rthm-chain.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist -> assoc-list -> recursive-assoc-list ->
```

sc-map -> rthm-seq-map -> rthm-chain

Version: 1.1.0

Project: slippery chicken (algorithmic composition)

Purpose: Algorithmic generation of rthm-seqs that include slower-moving counterpoint and a means to control activity development through curves. Here we generate a rthm-seq-map and its associated palette algorithmically.

Say we have 9 irregular 1 beat duration patterns; these would enter in the sequence defined by (procession x 9) where x would be the number of patterns to generate.

Rests are inserted at regular but changing intervals e.g

3x every 2 beats (6)
2x every 3 beats (6)
3x every 5 beats (15)
2x every 8 beats (16)

e, q, and q. rests are used by default, in a sequence determined by a recurring-event instance.

In order to make music that 'progresses' we have curves with y values from 1-10 indicating how much activity there should be: 1 would mean only 1 in 10 beats would have notes in/on them, 10 would indicate that all do. We use the patterns given in activity-levels::initialize-instance, where 1 means 'play', 0 means 'rest'. There are three examples of each level so that if we stick on one level of activity for some time we won't always get the same pattern: these will instead be cycled through.

A slower moving (bass) line is also added that is made up of 2 or 3 beat groups---if the activity curve indicates a rest, then the whole 2-3 beat group is omitted.

There are also 'sticking points' where a rhythm will be repeated a certain number of times (either s, e, e., or q by default). Sticking happens after rests. This can be controlled with an activity envelope too, also indicating one of the 10 patterns above (but also including 0). A 0 or 1 unit here would refer to a certain number of repeats

```
                (1) or none (0).  How many repeats could be determined by
                something like: (procession 34 '(2 3 5 8 13) :peak 1
                :expt 3) There's always a slower group to accompany the
                sticking points: simply the next in the sequence,
                repeated for as long as we stick

                The harmonic-rthm curve specifies how many slower-rthms
                will be combined into a rthm-seq (each rthm-seq has a
                single harmony).  The default is 2 bars (slower-rthms)
                per rthm-seq.


 Author:            Michael Edwards: m@michael-edwards.org

 Creation date:     4th February 2010

 $$ Last modified:  19:32:33 Wed Oct 29 2025 CET

 SVN ID: $Id$
```

38.6.414 rthm-chain/add-voice

[rthm-chain] [Methods]

DESCRIPTION:

```
Add a new voice to an existing rthm-chain object based on the rhythmic
material and slot values already contained in that object.

The main rthm-chain algorithm generates only two voices. Rather than
generate further voices in the same fashion by which the first two were
created, this method uses the already created rthm-seqs in the given
rthm-chain object to create a new voice.

The challenge here is that each rthm-seq potentially has its own time
signature structure: There could be a 2/4 bar followed by 5/4 then 3/16,
for example, or any other combination of any meter. So the method first
analyses the time-signature structure of the existing rthm-seqs and saves
those with the same bar/meter structure together in the order in which they
occur. When creating the extra voice then, the method actually starts ahead
of the main voice by choosing <offset> number of similar rthm-seqs in
advance.  NB Your data might well produce only one rthm-seq with a
particular metric structure, which might mean that calling this method
produces rhythmic doubling instead of an independent part.  In that case
you might try adding more rhythmic fragments and regenerating your piece.
```

ARGUMENTS:

- A rthm-chain object.
- The reference (key path) of the player within the given rthm-chain object whose rthm-seq-map is to serve as the 'parent voice', e.g. '(1 cl).
- A symbol that will be the ID of the new player.

OPTIONAL ARGUMENTS:

- An integer that indicates an offset into the group of similar rthm-seq objects from which the new voice is to begin. (The generated voice will thus be ahead of the main voice). Default = 1.

RETURN VALUE:

Returns T.

EXAMPLE:

```
(let ((rch
       (make-rthm-chain
        'test-rch 150
        '((((e) e) ; 4 in total
           (- s (s) (s) s -)
           ({ 3 (te) - te te - })
           ((e.) s))
          (({ 3 (te) te (te) }) ; what we transition to
           ({ 3 - te (te) te - })
           ({ 3 (te) - te te - })
           ({ 3 (te) (te) te })))
        '((((q q) ; the 2/4 bars: 5 total
            ((q) q)
            ((q) q)
            ((q) (s) e.)
            (- e e - (e) e))
           (({ 3 te+te te+te te+te }) ; what we transition to
            (q - s e. -)
            (q (s) e.)
            (q (s) - s e -)
            ({ 3 te+te te+te - te te - })))
          ((((e.) s (e) e (s) e.) ; the 3/4 bars: 4 total
            (- e e - (e) e (q))
            (- e. s - - +e e - (q))
            (q (e.) s (q)))
           (({ 3 (te) (te) te+te te+te } (q)) ; what we transition to
```

```
           (- e. s - (q) (s) - s e -)
           ({ 3 te+te te } (q) q)
           ({ 3 - te te te - } (e) e { 3 (te) (te) te }))))
       :players '(fl cl))))
  (add-voice rch '(1 cl) 'ob))
```

SYNOPSIS:

```
(defmethod add-voice ((rc rthm-chain) parent new-player &optional (offset 1))
```

38.6.415 rthm-chain/hash-least-used

[rthm-chain] [Functions]

DESCRIPTION:

Return the least used key in a hash table. This may be used to retrieve the number of times the keys have been used as indices, for example.

ARGUMENTS:

- A hash table. This must be a lisp hash table object whose keys and values are all numbers.

OPTIONAL ARGUMENTS:

keyword arguments:
- :start. The lowest key value we'll test. Default = 0.
- :end. The highest key value we'll test. Default = number of items in the hash table.
- :ignore. A list of keys to ignore when processing. NIL = process all keys. Default = NIL.
- :auto-inc. T or NIL to determine whether the function will automatically increment the count of the returned key. T = automatically increment. Default = T.
- :invert. T or NIL to invert the functionality i.e. to return the most used rather than the least. Default = NIL.

RETURN VALUE:

The key of the least used element in the hash table.

EXAMPLE:

```
(let ((h (make-hash-table)))
  (loop for i below 100 do
       (setf (gethash i h) 10000))
  (setf (gethash 10 h) 5
        (gethash 11 h) 4
        (gethash 12 h) 3
        (gethash 13 h) 2)
  (print (hash-least-used h :auto-inc nil))
  (print (hash-least-used h :auto-inc t))
  (print (hash-least-used h :auto-inc t))
  (print (hash-least-used h :auto-inc nil :start 12))
  (setf (gethash 2 h) 0)
  (print (hash-least-used h :auto-inc nil :start 3 :end 11))
  (print (hash-least-used h :auto-inc nil :end 11))
  (print (hash-least-used h :auto-inc nil :ignore '(2))))

=>
13
13
12
13
11
2
13
```

SYNOPSIS:

```
(defun hash-least-used (hash &key (start 0) end ignore (auto-inc t)
                                invert)
```

38.6.416 rthm-chain/hash-least-useds

[rthm-chain] [Functions]

DATE:

Mary 26th 2016, Edinburgh

DESCRIPTION:

Same as hash-least-used except it will return the <num> least used items (3 by default).

SYNOPSIS:

```
(defun hash-least-useds (hash &key (start 0) end ignore (auto-inc t)
                                 invert (num 3))
```

38.6.417 rthm-chain/make-rthm-chain

[rthm-chain] [Functions]

DESCRIPTION:

```
Create an instance of a rthm-chain object. The rthm-chain class enables the
algorithmic generation of a rthm-seq-map (with just one section) and its
associated rthm-seq-palette, which consists in turn of algorithmically
generated rthm-seq objects.

The rhythm-seq objects are made up of both faster material based on 1-beat
groups and slower-moving counterpoint based on 2- or 3-beat groups.

The rthm-chain class also allows for control of the degree of activity in
the parts over time through user-specified envelopes.

Rests are automatically inserted at regular but changing intervals.

Specified 'sticking points' cause individual rhythms to be repeated a
certain number of times. Sticking happens after rests and can also be
controlled with an activity envelope.

NB: Because this method uses the procession method internally, each
    collection of 1-beat-rthms and slower-rthms defined must contain at
    least four items.

NB: Since this method automatically inserts rests into the chains, the user
    may like to implement the various tie-over-rests post-generation
    editing methods. If this is done, the handle-ties method may also be
    recommended, as the tie-over-rests methods only affect printed output
    and not MIDI output.
```

ARGUMENTS:

```
- A number, symbol, or string that is to be the ID of the new rthm-chain
  object.
- An integer that is the number of beats to be generated prior to adding
  additional material created from sticking points and the automatic
  addition of rests.  For generating a whole piece this will generally be
  in the hundreds, not dozens.  Lower numbers might create very limited
  results or make the use of the add-voice method next to useless.
- A list with sublists of rhythms that are to be the 1-beat rhythms used to
  construct the faster-moving material of the rthm-seq-palette. Each
  sublist represents the repertoire of rhythms that will be used by the
  procession method. Each sublist must contain the same number of rthms but
```

their number and the number of sublists is open. A transition will be made from one group of rhythms to the next over the whole output (i.e. not one unit to another within e.g. the 1-beat rhythms) according to a fibonacci-transition method. NB Here and below, at least two lists are required: what we start with, and what we transition to. If no transition is required you could of course just duplicate the rhythms via a let variable: (list rhythms rhythms).
- A list with sublists of 2-beat and 3-beat full bars of rhythms used to construct the slower-moving counterpoint material of the rthm-seq-palette. This will be turned into a rthm-chain-slow object, and will therefore remain as lists of unparsed rhythms. Each sublist must contain the same number of rthms but their number and the number of sublists is open. A transition will be made from one group of rhythms to the next over the whole output (i.e. not one unit to another within e.g. the 1-beat rhythms) according to a fibonacci-transition method. NB: The rhythm units of slower-rthms must be expressed in single beats; e.g., a 2/4 bar must consist of q+q rather than h. The consolidate-notes method can be called afterwards if desired.

OPTIONAL ARGUMENTS:

keyword arguments:
- :players. A list of two player IDs. When used in conjunction with a slippery-chicken object (which is the standard usage), these must be IDs as they are defined in that object's ENSEMBLE slot. The first player will play the 1-beat rhythms, the second the slower rhythms. Default = '(player1 player2).
- :section-id. An integer that will be used as the ID of the rthm-seq-map created. NB: rthm-chain only creates rthm-seq-maps with one section, making it possible to create several different rthm-seq-map objects for different sections in the given piece, and requiring that these be manually assigned IDs. Additionally, any ID given here must match an existing ID within the other maps. Default = 1.
- :activity-curve. A list of break-point pairs with y values from 1 to 10 indicating the amount of activity there should be over the course of the piece. A value of 1 indicates that only 1 in 10 beats will have notes in/on them, and a value of 10 indicates that all beats will have notes. This process uses the patterns given in activity-levels::initialize-instance, where 1 means 'play' and 0 means 'rest'. There are three templates for each level, so that if the curve remains on one level of activity for some time it won't always return the same pattern; these will be rotated instead. If the activity curve indicates a rest for one of the slower-rhythms groups, the whole 2-3 beat group is omitted. Default = '(0 10 100 10).
- :do-rests. T or NIL to indicate whether to apply the automatic rest-insertion algorithm. T = use. Default = T.

- :rests. A list of rhythmic duration units from which the durations will be drawn when using the automatic rest-insertion algorithm. The specified rests are used in a sequence determined by a recurring-event object. Default = '(e q q. w). NB: Each of these values must not resolve to less than one-quarter of the beat basis, either alone or in combination, as this could result in an attempt to create meters from fractional beats (e.g. 3.25). An error message will be printed in such cases.
- :rest-cycle. A list of 2-item lists that indicate the pattern by which rests of specific rhythmic durations will be selected from the RESTS slot for automatic insertion. The first number of each pair is a 0-based position referring to the list of rests in the RESTS slot, and the second number is the number of times the rest at that particular position should be inserted. (This number does not mean that the selected rest will be inserted that many times at once, but rather that each consecutive time the rest algorithm selects one rest to be inserted, it will insert that specific rest, for the specified number of consecutive times.) For example, (0 3) indicates that for the next three times that the rest algorithm selects one rest to insert, it will select the rest located at position 0 in the list of rests in the RESTS slot (e by default). Default ='((0 3) (1 1) (0 2) (2 1) (1 1) (3 1)).
- :rest-re. A list of 2-item lists that indicate the pattern by which rests will be automatically inserted. The first number of each pair determines how many events occur before inserting a rest, and the second number of each pair determines how many times that period will be repeated. For example, (2 3) indicates that a rest will be inserted every two events, three times in a row. The list passed here will be treated as data for a recurring-event object that will be repeatedly cycled through. Default = '((2 3) (3 2) (2 2) (5 1) (3 3) (8 1)).
- :do-rests-curve. A list of break-point pairs with y values of either 0 or 1 indicating whether the do-rests algorithm is active or disabled. These values are interpolated between each pair, with all values 0.5 and higher being rounded up to 1 and all below 0.5 rounded to 0. Default = NIL.
- :do-sticking. T or NIL to indicate whether the method should apply the sticking algorithm. T = apply. Default = T.
- :sticking-rthms. A list of rhythmic units that will serve as the rhythms employed by the sticking algorithm. These are generated at initialization if not specified here. NB: This list is used to create a list using the procession algorithm at initialization, so it is best to apply something similar to the default if not accepting the default. If a circular-sclist object is provided here, it will be used instead of the default procession. Default = '(e e e. q e s).
- :sticking-repeats. A list of integers to indicate the number of repetitions applied in sticking segments. When the values of this list have been exhausted, the method cycles to the beginning and continues drawing from the head of the list again. NB: This list is made into a

circular-sclist object when the given rthm-chain object is initialized unless a circular-sclist object is explicitly provided. Default = '(3 5 3 5 8 13 21).

- :sticking-curve. A list of break-point pairs that acts as an activity envelope to control the sticking, which always occurs after rests. As with the activity curve, this curve can take y values up to 10, but also allows 0. A y value of 0 or 1 here refers to either a specific number of repeats (1) or none (0). The number of repeats may be determined, for example, by use of the procession method, such as (procession 34 '(2 3 5 8 13) :peak 1 :expt 3). Every sticking point is accompanied by a slower group, which is simply chosen in sequence and repeated for the duration of the sticking period. Default = '(0 2 100 2).
- :do-sticking-curve. A list of break-point pairs that can be used, alternatively, to control whether the sticking algorithm is being applied or not at any given point over the course of the piece. The y values for this curve should be between 0 and 1, and the decimal fractions achieved from interpolation will be rounded. The 1 values resulting from this curve will only be actively applied to if do-sticking is set to T. Default = NIL.
- :harmonic-rthm-curve. A list of break-point pairs that indicates how many slower-rthms will be combined into one rthm-seq (each rthm-seq has a single harmony). The default is 2 bars (slower-rthms) per rthm-seq, i.e. '(0 2 100 2).
- :split-data. NIL or a two-item list of integers that are the minimum and maximum beat duration of bars generated. If NIL, the bars will not be split. These values are targets only; the method may create bars of different lengths if the data generated cannot be otherwise split. NB: The values given here will apply to a different beat basis depending on time signature of each individual bar, rather than on a consistent beat basis, such as quarters or eighths. Since this method produces bars of different lengths with time signatures of differing beat bases (e.g. 16, 8, 4 etc.) before it applies the split algorithm, a minimum value of 4, for example, can result in bars of 4/16, 4/8, 4/4 etc. Default = '(2 5)
- :1-beat-fibonacci. T or NIL to indicate whether the sequence of 1-beat rhythms is to be generated using the fibonacci-transitions method or the processions method. T = use fibonacci-transitions method. Default = NIL.
- :slow-fibonacci. T or NIL to indicate whether the sequence of the slow rhythms will be generated using the fibonacci-transitions method or the processions method. This affects the order in which each 2- or 3-beat unit is used when necessary, not the order in which each 2- or 3-beat unit is selected; the latter is decided by the next element in the DATA slot of the rthm-chain-slow object, which simply cycles through '(2 3 2 2 3 2 2 3 3 3). T = use fibonacci-transitions method. Default = NIL.

RETURN VALUE:

A rthm-chain object.

EXAMPLE:

```
;; An example using a number of the keyword arguments.
(make-rthm-chain
 'test-rch 23
 '((((e) e) ; 4 in total
    (- s (s) (s) s -)
    ({ 3 (te) - te te - })
    ((e.) s))
   (({ 3 (te) te (te) }) ; what we transition to
    ({ 3 - te (te) te - })
    ({ 3 (te) - te te - })
    ({ 3 (te) (te) te })))
 '((((q q) ; the 2/4 bars: 5 total
     ((q) q)
     ((q) q)
     ((q) (s) e.)
     (- e e - (e) e))
    (({ 3 te+te te+te te+te }) ; what we transition to
     (q - s e. -)
     (q (s) e.)
     (q (s) - s e -)
     ({ 3 te+te te+te - te te - })))
   ((((e.) s (e) e (s) e.) ; the 3/4 bars: 4 total
     (- e e - (e) e (q))
     (- e. s - - +e e - (q))
     (q (e.) s (q)))
    (({ 3 (te) (te) te+te te+te } (q)) ; what we transition to
     (- e. s - (q) (s) - s e -)
     ({ 3 te+te te } (q) q)
     ({ 3 - te te te - } (e) e { 3 (te) (te) te }))))
 :players '(fl cl)
 :slow-fibonacci t
 :activity-curve '(0 1 100 10)
 :harmonic-rthm-curve '(0 1 100 3)
 :do-sticking t
 :do-sticking-curve '(0 1 25 0 50 1 75 0 100 1)
 :sticking-curve '(0 0 100 10)
 :sticking-repeats '(3 5 7 11 2 7 5 3 13)
 :sticking-rthms '(e s. 32 e.)
 :split-data '(4 7))
```

```
=>
RTHM-CHAIN: 1-beat-rthms: (((E E) (S S S S) (TE TE TE) (E. S))
                           ((TE TE TE) (TE TE TE) (TE TE TE) (TE TE TE)))
            slower-rthms: ((((Q Q) ((Q) Q) ((Q) Q) ((Q) (S) E.)
                            (- E E - (E) E))
                           (({ 3 TE+TE TE+TE TE+TE }) (Q - S E. -) (Q (S) E.)
                            (Q (S) - S E -) ({ 3 TE+TE TE+TE - TE TE - })))
                          ((((E.) S (E) E (S) E.) (- E E - (E) E (Q))
                            (- E. S - - +E E - (Q)) (Q (E.) S (Q)))
                           (({ 3 (TE) (TE) TE+TE TE+TE } (Q))
                            (- E. S - (Q) (S) - S E -) ({ 3 TE+TE TE } (Q) Q)
                            ({ 3 - TE TE TE - } (E) E { 3 (TE) (TE) TE }))))
            1-beat-fibonacci: NIL
            num-beats: 23
            slow-fibonacci: T
            num-1-beat-rthms: 4
            num-1-beat-groups: 2
            sticking-curve: (0.0 0 22 10)
            harmonic-rthm-curve: (0.0 1 22 3)
            beat: 4
            do-sticking: T
            do-rests: T
            do-sticking-curve: (0.0 1 5.5 0 11.0 1 16.5 0 22 1)
            do-rests-curve: NIL
            sticking-al: (not printed for brevity's sake)
            sticking-rthms: (E S. E S. 32 E 32 E E E. S. 32 S. E. S. 32 S. 32
                             E. E)
            sticking-repeats: (3 5 3 5 7 3 7 3 3 11 5 7 5 11 5 7 5 7 11 3 7 3 3
                               11 5 7 5 11 3 7 3 7 11 5 7 5 5 11 3 11 3 2 11 2
                               11 2 7 2 7 2 2 5 5 3 5)
            activity-curve: (0.0 1 22 10)
            main-al: (not printed for brevity's sake)
            slower-al: (not printed for brevity's sake)
            num-slower-bars: 22
            rcs: (not printed for brevity's sake)
            rests: (E Q Q. W)
            rest-re: (not printed for brevity's sake)
            rest-cycle: ((0 3) (1 1) (0 2) (2 1) (1 1) (3 1))
            num-rthm-seqs: 19
            section-id: 1
            split-data: (4 7)
RTHM-SEQ-MAP: num-players: 2
              players: (FL CL)
SC-MAP: palette id: RTHM-CHAIN-RSP
[...]
```

SYNOPSIS:

```
(defun make-rthm-chain (id num-beats 1-beat-rthms slower-rthms &key
                        (1-beat-fibonacci nil)
                        (slow-fibonacci nil)
                        (players '(player1 player2))
                        (section-id 1)
                        (rests '(e q q. w))
                        (do-rests t)
                        (do-rests-curve nil)
                        (rest-re '((2 3) (3 2) (2 2) (5 1) (3 3) (8 1)))
                        (rest-cycle '((0 3) (1 1) (0 2) (2 1) (1 1) (3 1)))
                        (activity-curve '(0 10 100 10))
                        (sticking-curve '(0 2 100 2))
                        (harmonic-rthm-curve '(0 2 100 2))
                        (do-sticking t)
                        (do-sticking-curve nil)
                        (sticking-repeats '(3 5 3 5 8 13 21))
                        (sticking-rthms '(e e e. q e s))
                        (split-data '(2 5)))
```

38.6.418 rthm-chain/procession

[rthm-chain] [Functions]

DATE:

26-Jan-2010

DESCRIPTION:

Generate a list of a specified length consisting of items extrapolated from a specified starting list. All elements of the resulting list will be members of the original list.

The method generates the new list by starting with the first 3 elements of the initial list and successively adding consecutive elements from the initial list until all elements have been added.

ARGUMENTS:

- An integer that is the number of items in the list to be generated.
- A list of at least 4 starting items or an integer >=4. If an integer is given rather than a list, the method will process a list of <integer> consecutive numbers from 1 (by default).

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :peak. A decimal number >0.0 and <=1.0. This number indicates the target
  location in the new list at which the last element is to finally occur,
  whereby e.g. 0.7 = ~70% of the way through the resulting list. This is an
  approximate value only. The last element may occur earlier or later
  depending on the values of the other arguments. In particular, initial
  lists with a low number of items are likely to result in new lists in
  which the final element occurs quite early on, perhaps even nowhere near
  the specified peak value. Default = 0.7.
- :expt. An exponent (floating point number) to indicate the "curve" that
  determines the intervals at which each successive element of the initial
  list is introduced to the new list. A higher number indicates a steeper
  exponential curve. Default = 1.3.
- :start-at. The number to start counting from if the 2nd argument is an
  integer instead of a list. Default = 1.
- :orders. The patterns by which the elements are added. The method
  cyclically applies these orders, the numbers 1, 2, and 3 representing the
  three least used elements at each pass. These orders must therefore
  contain all of the numbers 1, 2, and 3, and those numbers only.
  Default = '((1 2 1 2 3) (1 2 1 1 3) (1 2 1 3)).
```

RETURN VALUE:

```
Returns two values, the first being the new list, with a secondary value
that is a list of 2-item lists that show the distribution of each element
in the new list.
```

EXAMPLE:

```
(procession 300 30 :peak 0.1)

=>
(1 2 1 2 3 4 5 4 4 6 7 8 7 9 10 11 10 11 12 13 14 13 13 15 16 17 16 18 19 20 19
   20 21 22 23 22 22 24 25 26 25 27 28 29 28 29 30 3 5 3 3 6 8 9 8 12 14 15 14
   15 17 18 21 18 18 23 24 26 24 27 1 2 1 2 30 5 6 5 5 7 9 10 9 11 12 16 12 16
   17 19 20 19 19 21 23 25 23 26 27 28 27 28 29 4 6 4 4 30 7 8 7 10 11 13 11 13
   14 15 17 15 15 20 21 22 21 24 25 26 25 26 29 1 2 1 1 30 3 6 3 8 9 10 9 10 12
   14 16 14 14 17 18 20 18 22 23 24 23 24 27 28 29 28 28 30 2 5 2 6 7 8 7 8 11
   12 13 12 12 16 17 19 17 20 21 22 21 22 25 26 27 26 26 29 3 4 3 30 5 6 5 6 9
   10 11 10 10 13 15 16 15 18 19 20 19 20 23 24 25 24 24 27 1 29 1 30 2 4 2 4 7
   8 9 8 8 11 13 14 13 16 17 18 17 18 21 22 23 22 22 25 27 28 27 29 3 5 3 5 30
   6 7 6 6 9 11 12 11 14 15 16 15 16 19 20 21 20 20 23 25 26 25 28 1 29 1 29 30
   2 4 2 2 7 9 10 9 12 13 14 13 14 17 18), ((2 12) (20 11) (14 11) (13 11)
```

```
   (9 11) (6 11) (1 11) (29 10) (25 10) (22 10) (18 10) (17 10) (16 10) (15 10)
   (12 10) (11 10) (10 10) (8 10) (7 10) (5 10) (4 10) (3 10) (30 9) (28 9)
   (27 9) (26 9) (24 9) (23 9) (21 9) (19 9))

(procession 300 30 :peak 0.9)

=>
(1 2 1 2 3 1 3 1 1 4 2 3 2 4 3 4 3 4 5 2 4 2 2 5 1 3 1 5 3 4 3 4 5 1 5 1 1 6 2
   5 2 6 4 5 4 5 6 3 6 3 3 7 5 6 5 7 2 6 2 6 7 6 7 6 6 8 4 7 4 8 7 8 7 8 9 7 8
   7 7 9 8 9 8 10 8 9 8 9 10 8 9 8 8 10 9 10 9 11 9 10 9 10 11 10 11 10 10 12
   10 11 10 12 11 12 11 12 13 11 12 11 11 13 12 13 12 14 12 13 12 13 14 13 14
   13 13 15 13 14 13 15 11 14 11 14 15 14 15 14 14 16 15 16 15 17 15 16 15 16
   17 16 17 16 16 18 16 17 16 18 17 18 17 18 19 17 18 17 17 19 18 19 18 20 18
   19 18 19 20 19 20 19 19 21 15 20 15 21 20 21 20 21 22 20 21 20 20 22 21 22
   21 23 21 22 21 22 23 22 23 22 22 24 23 24 23 25 23 24 23 24 25 24 25 24 24
   26 23 25 23 26 25 26 25 26 27 26 27 26 26 28 25 27 25 28 27 28 27 28 29 27
   28 27 27 29 28 29 28 30 24 29 24 29 30 26 29 26 26 30 28 29 28 30 19 29 19
   29 30 22 25 22 22 30 12 27 12 30 14 16 14 16 30 17), ((8 12) (22 11)
   (16 11) (14 11) (12 11) (11 11) (10 11) (4 11) (3 11) (2 11) (26 10)
   (19 10) (17 10) (15 10) (13 10) (9 10) (7 10) (6 10) (5 10) (1 10)
   (29 9) (28 9) (27 9) (25 9) (24 9) (23 9) (21 9) (20 9) (18 9) (30 8))
```

SYNOPSIS:

```
(defun procession (num-results items
                   &key
                     ;; what proportion of the way through should we aim to
                     ;; reach the max number of items?  NB This is approximate
                     ;; only: you may find the first occurrence of the highest
                     ;; element earlier or later depending on the values of the
                     ;; other arguments.  In particular, with a low number of
                     ;; items the highest element will be hit very early on,
                     ;; perhaps nowhere near the peak argument.
                     (peak 0.7)
                     ;; for an exponential curve going from 3 to num <items>
                     (expt 1.3)
                     ;; MDE Thu Sep 17 16:11:03 2020, Heidhausen -- allow
                     ;; counting from another number if items is an integer
                     (start-at 1)
                     ;; these are the orders we'll use at the beginning
                     ;; (cyclically). They will then be used when we've gone
                     ;; beyond 3 items by always using the 3 least used items.
                     ;; NB This must contain the numbers 1, 2, and 3 only but
                     ;; there can be 1 or any number of sublists.
                     (orders '((1 2 1 2 3) (1 2 1 1 3) (1 2 1 3))))
```

38.6.419 rthm-chain/procession-mirror

[rthm-chain] [Functions]

DESCRIPTION:

```
Perform the same operation as the procession function, but instead of just
progressing upwards, go backwards to return to the beginning also.
```

ARGUMENTS:

```
The same as for procession.
```

RETURN VALUE:

```
A list of the procession. Note that at the mirror point there is no
repetition, and we don't include the first element at the end. Note also
that we don't return statistics as we do with procession.
```

EXAMPLE:

```
(procession-mirror 33 '(1 2 3 4 5 6 7))
--> (1 2 1 2 3 3 4 3 3 5 4 6 4 7 5 6 5 6 6 5 7 4 6 4 5 3 3 4 3 3 2 1 2)
```

SYNOPSIS:

```
(defun procession-mirror (&rest args)
```

38.6.420 rthm-chain/reset

[rthm-chain] [Methods]

DESCRIPTION:

```
Reset the various circular-sclist objects within the given rthm-chain
object to their initial state.
```

ARGUMENTS:

```
- A rthm-chain object.
```

OPTIONAL ARGUMENTS:

```
(- :where. This argument is ignored by the method as it is only present due
   to inheritance.)
```

RETURN VALUE:

Returns T.

EXAMPLE:

```
;;; Print the results of applying get-next to the STICKING-RTHMS slot of the
;;; given rthm-chain object, repeat, reset, and print again to see that the
;;; get-next now begins at the beginning of the slot again.

(let ((rch
       (make-rthm-chain
        'test-rch 150
        '((((e) e) ; 4 in total
           (- s (s) (s) s -)
           ({ 3 (te) - te te - })
           ((e.) s))
          (({ 3 (te) te (te) }) ; what we transition to
           ({ 3 - te (te) te - })
           ({ 3 (te) - te te - })
           ({ 3 (te) (te) te })))
        '((((q q) ; the 2/4 bars: 5 total
            ((q) q)
            ((q) q)
            ((q) (s) e.)
            (- e e - (e) e))
           (({ 3 te+te te+te te+te }) ; what we transition to
            (q - s e. -)
            (q (s) e.)
            (q (s) - s e -)
            ({ 3 te+te te+te - te te - })))
          ((((e.) s (e) e (s) e.) ; the 3/4 bars: 4 total
            (- e e - (e) e (q))
            (- e. s - - +e e - (q))
            (q (e.) s (q)))
           (({ 3 (te) (te) te+te te+te } (q)) ; what we transition to
            (- e. s - (q) (s) - s e -)
            ({ 3 te+te te } (q) q)
            ({ 3 - te te te - } (e) e { 3 (te) (te) te }))))))))
  (print
   (loop repeat 19
      collect (data (get-next (sticking-rthms rch)))))
  (print
   (loop repeat 19
      collect (data (get-next (sticking-rthms rch)))))
  (reset rch)
```

```
  (print
   (loop repeat 19
      collect (data (get-next (sticking-rthms rch))))))

=>
(E E E E E. E E. E E Q E E. E Q E. Q E. Q E)
(E E E E E E. E E. E E Q E E. E Q E. Q E. Q)
(E E E E E. E E. E E Q E E. E Q E. Q E. Q E)
```

SYNOPSIS:

```
(defmethod reset ((rc rthm-chain) &optional ignore1 ignore2)
```

38.6.421 rthm-chain/rthm-chain-gen

[rthm-chain] [Methods]

DESCRIPTION:

Generate a chain of rhythms using the procession function (internally).

The basic algorithm for generating a rthm-chain object of two parts is as follows: The user provides an arbitrary number of 1-beat rthms (e.g. s s (e)) and 2-3 beat slower-moving counterpoints. The method generates a sequence from these using the procession function. Next the activity curve is applied to this, and after that the insertion of rests. Then the 'sticking points' are generated: These come after the rests, and the activity curves applied to these count inserted rests not seqs or beats.

NB: Rests are put into the given rthm-seq object mid-sequence, so sticking points won't come directly after the rests, rather, at the end of the seq.

The activity curves that turn notes into rests will be queried every beat, so if an activity level is changed, the method won't wait until the end of the previous level's ten beats.

NB: This method is not generally called by the user (though it can be of course); rather, it's called by the init function.

ARGUMENTS:

- A rthm-chain object.

OPTIONAL ARGUMENTS:

keyword arguments:
- :rests. T or NIL to indicate whether rests are to be automatically inserted. T = automatically insert. Default = T.
- :stick. T or NIL to indicate whether to generate the sticking points. T = generate sticking points. Default = T.
- :num-beats. NIL or an integer to indicate how many beats are to be used for the algorithm. NB: The method will generate considerably more beats if also generating sticking points and inserting rests; this number merely refers to the number of standard 1-beat rhythms to be generated. If NIL, the method will obtain the number of beats from the NUM-BEATS slot of the rthm-chain instance. Default = NIL.
- :use-fibonacci. T or NIL to indicate whether to use the fibonacci-transitions method when generating the sequence from the 1-beat rhythms (in which case these will be repeated) or the procession algorithm (in which case they'll be alternated). T = use the fibonacci-transitions method. Default = T.
- :section-id. An integer that is the section ID of the rthm-chain object to be generated. This will determine the section of the rthm-seq-map into which the references will be placed. The rthm-seq objects themselves will also be parcelled up into an object with this ID, so ID conflicts can be avoided if combining two or more sections generated by separate rthm-chain objects. Default = 1.
- :wrap. An integer or NIL to determine the position within the list of 1-beat rhythms and slow rhythms from which the generated rhythm chain will begin. NIL = begin at the beginning. Default = NIL.
- :split. T or NIL to indicate whether to split up longer generated bars (e.g. 7/4) into smaller bars. If this is a two-element list it represents the min/max number of beats in a bar (where a 6/8 bar is two compound beats). Default = '(2 5).

RETURN VALUE:

the number of rthm-seqs we've generated

SYNOPSIS:

```
(defmethod rthm-chain-gen ((rc rthm-chain)
                           &key
                           (use-fibonacci t)
                           (rests t)
                           (stick t)
                           (section-id 1)
                           num-beats
                           wrap)
```

38.6.422 rthm-chain/split

[rthm-chain] [Methods]

DATE:

29-Jan-2011

DESCRIPTION:

Split the longer generated bars into smaller ones where possible.

ARGUMENTS:

- A rthm-chain object.

OPTIONAL ARGUMENTS:

keyword arguments:
- :min-beats. An integer that is the minimum number of beats in the resulting bars. This is a target-length only, and may not be adhered to strictly if durations do not allow. Default = 2.
- :max-beats. An integer that is the maximum number of beats in the resulting bars. This is a target-length only, and may not be adhered to strictly if durations do not allow. Default = 5.
- :warn. T or NIL to indicate whether to print a warning to the listener if the current bar cannot be split. T = print. Default = NIL.
- :clone. T or NIL to indicate whether the rthm-seq of the given rthm-chain object should be changed in place or changes should be made to a copy of that object. T = create a copy to be changed. Default = T.

RETURN VALUE:

Returns T.

EXAMPLE:

```
;;; Make a rthm-chain object using make-rthm-chain with the :split-data
;;; argument set to NIL and print the number of bars in each resulting rthm-seq
;;; object. Apply the split method and print the number of bars again to see
;;; the change.

(let* ((rch
        (make-rthm-chain
         'test-rch 150
```

```
         '((((e) e) ; 4 in total
            (- s (s) (s) s -)
            ({ 3 (te) - te te - })
            ((e.) s))
           (({ 3 (te) te (te) }) ; what we transition to
            ({ 3 - te (te) te - })
            ({ 3 (te) - te te - })
            ({ 3 (te) (te) te })))
         '((((q q) ; the 2/4 bars: 5 total
             ((q) q)
             ((q) q)
             ((q) (s) e.)
             (- e e - (e) e))
            (({ 3 te+te te+te te+te }) ; what we transition to
             (q - s e. -)
             (q (s) e.)
             (q (s) - s e -)
             ({ 3 te+te te+te - te te - })))
           ((((e.) s (e) e (s) e.) ; the 3/4 bars: 4 total
             (- e e - (e) e (q))
             (- e. s - - +e e - (q))
             (q (e.) s (q)))
            (({ 3 (te) (te) te+te te+te } (q)) ; what we transition to
             (- e. s - (q) (s) - s e -)
             ({ 3 te+te te } (q) q)
             ({ 3 - te te te - } (e) e { 3 (te) (te) te }))))
         :split-data nil)))
  (print
   (loop for rs in (data (get-data-data 1 (palette rch)))
      collect (num-bars rs)))
  (split rch :min-beats 1 :max-beats 3 :clone nil)
  (print
   (loop for rs in (data (get-data-data 1 (palette rch)))
      collect (num-bars rs))))

=>
(1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 1 1 2 2 2 2 2 2 2 2 2 2 2 2
 2 2 2 2 2 2 2 1 1 2 2 1 1 2 2 2 2 1 1 2 2 2 2 2 2 2 2 1 1 2 2 2 2 2 2 2
 2 2)
(1 1 4 4 2 7 4 4 3 3 4 4 2 9 7 7 2 5 5 5 2 6 1 1 2 13 3 3 4 4 5 5 2 7 5 5
 7 7 9 9 2 7 2 9 3 3 2 9 1 1 5 5 9 9 3 3 2 7 5 5 4 4 2 11 1 1 2 10 2 9 2 6
 7 7 7 7)
```

SYNOPSIS:

```
(defmethod split ((rc rthm-chain) &key
```

```
                                    (min-beats 2) (max-beats 5) warn (clone t))
```

38.6.423 rthm-seq-map/set-map-refs

[rthm-seq-map] [Methods]

DATE:

30-Dec-2010

DESCRIPTION:

Change the reference IDs of the specified rthm-seq objects in the given rthm-seq-map object.

ARGUMENTS:

- A rthm-seq-map object.
- The ID of the section in which references are to be set.
- The ID of the player for whom the references are to be set.
- A list of the new rthm-seq IDs (references)

RETURN VALUE:

Returns the modified named object whose ID is the specified player.

EXAMPLE:

```
(let ((rsmt (make-rthm-seq-map
             'rsm-test-5
             '((sec1 ((vn (rs1 rs3 rs2))
                      (va (rs2 rs3 rs1))
                      (vc (rs3 rs1 rs2))))
               (sec2 ((vn (rs1 rs2 rs1))
                      (va (rs2 rs1 rs3))
                      (vc (rs1 rs3 rs3))))
               (sec3 ((vn (rs1 rs1 rs3))
                      (va (rs1 rs3 rs2))
                      (vc (rs3 rs2 rs3)))))
             :palette (make-rsp
                       'rs-pal
                       '((rs1 ((((2 4) q e s s))))
                         (rs2 ((((2 4) e s s q))))
                         (rs3 ((((2 4) s s q e)))))))))
  (set-map-refs rsmt 'sec2 'vc '(rs2 rs3 rs2)))
```

```
=>
NAMED-OBJECT: id: VC, tag: NIL,
data: (RS2 RS3 RS2)
```

SYNOPSIS:

```
(defmethod set-map-refs ((rsm rthm-seq-map) section player new-refs)
```

38.6.424 sc-map/set-map

[sc-map] [Classes]

NAME:

```
 set-map

 File:             set-map.lsp

 Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                   circular-sclist -> assoc-list -> recursive-assoc-list ->
                   sc-map -> set-map

 Version:          1.1.0

 Project:          slippery chicken (algorithmic composition)

 Purpose:          Implementation of the set-map class for mapping sets for
                   a piece.

 Author:           Michael Edwards: m@michael-edwards.org

 Creation date:    March 11th 2010

 $$ Last modified:  19:47:46 Thu Feb  8 2024 CET

 SVN ID: $Id$
```

38.6.425 set-map/cmn-display

[set-map] [Methods]

DATE:

```
 January 30th 2016
```

DESCRIPTION:

Generate musical notation of the set-map object using CMN.

ARGUMENTS:

a set-map object

OPTIONAL ARGUMENTS:

see the set-palette class for a description of the keyword arguments.

RETURN VALUE:

T

SYNOPSIS:

```
(defmethod cmn-display ((sm set-map) &rest keyargs &key &allow-other-keys)
```

38.6.426 set-map/gen-midi-chord-seq

[set-map] [Methods]

DESCRIPTION:

Write a midi file with each set in the set-map played as a chord at 1 second intervals.

ARGUMENTS:

- A set-map object.
- The path+file-name for the midi file to be written.

OPTIONAL ARGUMENTS:

- the tempo in beats per minute. Default = 60.
- subsets-id, related-sets-id. If you want to generate a MIDI file of just a named subset or related-set, pass the ID here. NB this ID will have to exist for all sets.

RETURN VALUE:

Returns T.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :set-palette '((1 ((c3 e3 g3 a3 c4 d4 g4 a4 b4 e5))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) q e s s))
                                :pitch-seq-palette ((1 2 3 4)))))
        :rthm-seq-map '((1 ((vn (1 1 1))))))))
  (gen-midi-chord-seq (set-map mini) "/tmp/mchsq.mid"))

=> T
```

SYNOPSIS:

```
(defmethod gen-midi-chord-seq ((sm set-map) midi-file
                               &optional (tempo 60.0) subsets-id
                                 related-sets-id)
```

38.6.427 set-map/midi-play

[set-map] [Methods]

DESCRIPTION:

```
Write a MIDI file containing the sets in order of the set-map.
```

ARGUMENTS:

```
- the set-map object
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :midi-file. The path (string) of the midi file to write. Default is to
  use the ID of the set-palette and write it into the default directory.
- :tempo. The tempo of the MIDI file in beats per minute. Default = 60.
- :auto-open. Whether to open the midi file after generating (not available
  on all operating systems). Default is (get-sc-config
  'midi-play-auto-open) which itself defaults to sc-auto-open in the
  *features* list.
- :related-sets-id :subsets-id. If you want to generate a MIDI file of just
  a named subset or related-set, pass the ID here. NB this ID will have to
  exist for all sets.
```

RETURN VALUE:

```
T
```

SYNOPSIS:

```
(defmethod midi-play ((sm set-map)
                      &key
                        ;; MDE Sat Mar 20 12:05:17 2021, Heidhausen
                        related-sets-id subsets-id
                        (tempo 60.0)
                        (auto-open (get-sc-config 'midi-play-auto-open))
                        (midi-file
                         (format nil "~a~a.mid"
                                 (get-sc-config 'default-dir)
                                 (string-downcase (string (id sm))))))
```

38.6.428 sc-map/shorten

[sc-map] [Methods]

DATE:

```
July 10th 2020, Heidhausen
```

DESCRIPTION:

```
shorten the length of the map for a given section
```

ARGUMENTS:

```
- the section reference
- the new length (integer)
- the sc-map object
```

RETURN VALUE:

```
the shortened sc-map object
```

SYNOPSIS:

```
(defmethod shorten (section new-length (scm sc-map))
```

38.6.429 recursive-assoc-list/set-data

[recursive-assoc-list] [Methods]

DESCRIPTION:

Replace the named-object associated with a specified key within a given recursive-assoc-list object. This method replaces the whole named-object, not just the data of that object.

ARGUMENTS:

- A key present within the given recursive-assoc-list object. This must be a list that is the FULL-REF (path of keys) if replacing a nested named-object. If replacing a named-object at the top level, the key can be given either as a single-item list or an individual symbol.
- A key/data pair as a list.
- The recursive-assoc-list object in which to find and replace the named-object associated with the specified key.

RETURN VALUE:

Returns the new named-object.

Returns NIL when the specified key is not found within the given recursive-assoc-list object.

EXAMPLE:

```
;;; Replace a named-object at the top level using a single symbol
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (set-data 'wild '(makers mark) ral))

=>
NAMED-OBJECT: id: MAKERS, tag: NIL,
data: MARK

;; The same can be done stating the top-level key as a single-item list. Apply
```

```
;; the get-all-refs method in this example to see the change
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (set-data '(wild) '(makers mark) ral)
  (get-all-refs ral))

=> ((JIM) (MAKERS) (FOUR ROSES) (FOUR VIOLETS BLUE) (FOUR VIOLETS RED DRAGON)
    (FOUR VIOLETS RED VIPER) (FOUR VIOLETS RED FOX) (FOUR VIOLETS WHITE))

;; Replace a nested named-object using a list that is the FULL-REF to that
;; object. Print the application of the method as well as the results from
;; applying the get-all-refs method in this example to see the effects
(let ((ral (make-ral 'mixed-bag
                     '((jim beam)
                       (wild turkey)
                       (four ((roses red)
                              (violets ((blue velvet)
                                        (red ((dragon den)
                                              (viper nest)
                                              (fox hole)))
                                        (white ribbon)))))))))
  (print (set-data '(four violets red fox) '(bee hive) ral))
  (print (get-all-refs ral)))

=>
NAMED-OBJECT: id: BEE, tag: NIL,
data: HIVE
**************

((JIM) (WILD) (FOUR ROSES) (FOUR VIOLETS BLUE) (FOUR VIOLETS RED DRAGON)
 (FOUR VIOLETS RED VIPER) (FOUR VIOLETS RED BEE) (FOUR VIOLETS WHITE))
```

SYNOPSIS:

```
(defmethod set-data (key new-value (ral recursive-assoc-list))
```

38.6.430 recursive-assoc-list/set-slot

[recursive-assoc-list] [Methods]

DESCRIPTION:

```
Set the specified slot of an object with a recursive-assoc-list structure
to the specified value. This is particularly useful for changing the
parameters of instrument objects within an instrument palette, for example.

NB Setting, for instance, the high-sounding slot of an instrument will do
nothing to the highest-written slot so do take care to set all the slots
you need to in order to effect pitch selection etc. Or better still use the
set-standard-range function.
```

ARGUMENTS:

```
- The name of the slot whose value is to be set.
- The value to which that slot is to be set.
- The key within the given recursive-assoc-list object for which the slot
  is to be set.
- The recursive-assoc-list object in which the slot is to be changed.
```

RETURN VALUE:

```
The value to which the slot has been set.
```

EXAMPLE:

```
(set-slot 'largest-fast-leap 10 'oboe
          +slippery-chicken-standard-instrument-palette+)

=> 10
```

SYNOPSIS:

```
(defmethod set-slot (slot value id (ral recursive-assoc-list))
```

38.6.431 recursive-assoc-list/set-standard-instrument-slot

[recursive-assoc-list] [Methods]

DESCRIPTION:

```
A convenience method to set slots of instruments in the
+slippery-chicken-standard-instrument-palette+.
```

DATE:

```
July 20th 2015
```

EXAMPLE:

```
(set-standard-instrument-slot 'highest-written 'f4 'double-bass)
```

SYNOPSIS:

```
(defmethod set-standard-instrument-slot (slot value id)
```

38.6.432 assoc-list/reindex

[assoc-list] [Methods]

DATE:

```
14th June 2017, Edinburgh
```

DESCRIPTION:

```
Re-index the named-objects to consecutive integers starting from a given
number.
```

ARGUMENTS:

```
- the assoc-list object
```

OPTIONAL ARGUMENTS:

```
- the integer to start reindexing at
```

RETURN VALUE:

```
the assoc-list object
```

SYNOPSIS:

```
(defmethod reindex ((al assoc-list) &optional (start 1))
```

38.6.433 assoc-list/remove-data

[assoc-list] [Methods]

DATE:

February 1st 2016

DESCRIPTION:

Convenience function to allow objects to be removed from an assoc-list (and subclasses e.g. palettes and maps)

ARGUMENTS:

- the assoc-list object
- as many keys as you like for the members of the assoc-list. If none are passed then all are removed.

RETURN VALUE:

The assoc-list object's new data list, i.e. with elements removed. Could of course be NIL if you've not passed any keys.

SYNOPSIS:

```
(defmethod remove-data ((al assoc-list) &rest keys)
```

38.6.434 assoc-list/remove-when

[assoc-list] [Methods]

DATE:

June 28th 2017, Edinburgh

DESCRIPTION:

Use a function to test each object in an assoc-list and if the test returns T, then remove the object. Works in recursive-assoc-list class too.

ARGUMENTS:

- an assoc-list object
- a function that returns T or NIL after interrogating a named-object or subclass's slots

RETURN VALUE:

```
a list of the IDs of the objects removed.
```

EXAMPLE:

```
(let ((al (make-assoc-list 'mixed-bag
                           '((jim beam)
                             (wild turkey)
                             (four roses)))))
  (remove-when al #'(lambda (x) (eq (data x) 'roses))))
```

SYNOPSIS:

```
(defmethod remove-when ((al assoc-list) test)
```

38.6.435 assoc-list/replace-data

[assoc-list] [Methods]

DESCRIPTION:

```
A convenience method that's essentially the same as set-data but without
having to pass the new ID in a list along with the new value.
```

ARGUMENTS:

```
- The key (number, symbol, string) to the existing data.
- The new value to be associated with this key (any data).
- The assoc-list object
```

RETURN VALUE:

```
The named-object for the datum.
```

SYNOPSIS:

```
(defmethod replace-data (key new-value (al assoc-list))
```

38.6.436 assoc-list/set-data

[assoc-list] [Methods]

DESCRIPTION:

Replace the named-object associated with a specified key within a given assoc-list object. This method replaces the whole named-object, not just the data of that object.

ARGUMENTS:

- A key present within the given assoc-list object.
- A key/data pair as a list.
- The assoc-list object in which to find and replace the named-object associated with the specified key.

RETURN VALUE:

Returns the new named-object.

Returns NIL when the given key is not present within the given assoc-list.

EXAMPLE:

```
(let ((al (make-assoc-list 'test '((cat felix)
                                   (dog fido)
                                   (cow bessie)))))
  (set-data 'dog '(dog spot) al))

=>
NAMED-OBJECT: id: DOG, tag: NIL,
data: SPOT

(let ((al (make-assoc-list 'test '((cat felix)
                                   (dog fido)
                                   (cow bessie)))))
  (set-data 'pig '(pig wilbur) al))

=> NIL
WARNING:
   assoc-list::set-data: Could not find data with key PIG in assoc-list with id
   TEST

(let ((al (make-assoc-list 'test '((cat felix)
                                   (dog fido)
                                   (cow bessie)))))
  (set-data 'dog '(pig wilbur) al))

=>
NAMED-OBJECT: id: PIG, tag: NIL,
data: WILBUR
```

SYNOPSIS:

```
(defmethod set-data (key new-value (al assoc-list))
```

38.6.437 assoc-list/set-nth-of-data

[assoc-list] [Methods]

DESCRIPTION:

```
 Replace a given member of a given data list within a given assoc-list.
```

ARGUMENTS:

```
 - The key (named-object id) associated with the data to be changed.
 - The zero-based integer index of the member of the list to be changed.
 - The new value.
 - The assoc-list in which the change is to be made.

 The data to be modified must already be in the form of a list.

 The index integer given must be less than the length of the data list to be
 modified.
```

RETURN VALUE:

```
 Returns the new value only.
```

EXAMPLE:

```
(let ((al (make-assoc-list 'test '((cat felix)
                                   (dog (fido spot rover))
                                   (cow bessie)))))
  (set-nth-of-data 'dog 0 'snoopy al))

=> SNOOPY

(let ((al (make-assoc-list 'test '((cat felix)
                                   (dog (fido spot rover))
                                   (cow bessie)))))
  (set-nth-of-data 'dog 0 'snoopy al)
  (get-data 'dog al))

=>
NAMED-OBJECT: id: DOG, tag: NIL,
data: (SNOOPY SPOT ROVER)
```

SYNOPSIS:

```
(defmethod set-nth-of-data (key nth new-value (al assoc-list))
```

38.6.438 assoc-list/wolfram

[assoc-list] [Classes]

NAME:

```
rhythm

File:             wolfram.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist -> assoc-list -> wolfram

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the wolfram class for one-dimensional
                  binary cellular automata introduced by Stephen Wolfram in
                  1983. The rule number is the decimal representation of
                  the binary number indicated by the eight transition rules
                  (i.e. the results given the triplet test cases; see the
                  rules slot below).

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    7th June 2017, Edinburgh

$$ Last modified:  10:56:40 Sat Aug 24 2024 CEST

SVN ID: $Id: wolfram.lsp 6210 2017-04-07 11:42:29Z medward2 $
```

38.6.439 wolfram/defwolfram

[wolfram] [Methods]

DATE:

```
June 6th 2017, Edinburgh
```

DESCRIPTION:

Define your own Wolfram function to create an object with the flavour in the rules passed. See below for several examples.

In addition, the name of the newly created function will be added to an internal list of all functions defined by this macro, so that (try-all) can be called (see below).

ARGUMENTS:

- the name of the function you're defining (symbol)
- the list of rules

OPTIONAL ARGUMENTS:

- the default width (integer>0) for the Wolfram object that will be created
- the default initial-state: 1 or 0.

RETURN VALUE:

the name of the function (symbol), just as with defun.

SYNOPSIS:

```
(defmacro defwolfram (fun rules &optional (w 100) (is 0))
```

38.6.440 wolfram/generate

[wolfram] [Methods]

DATE:

June 6th 2017, Edinburgh

DESCRIPTION:

Generate the cellular automata rows using the given rules. N.B. Each time this method is called, the results (data slot) of the previous calls are deleted.

At the beginning, we hava a row. This is a list of length <width>, with all elements being <initial-state>. However, we set the middle element to the <start> argument given to this method. We then loop through the elements of the list and make a triplet out of the element to the left of the current element (or <initial-state> when we begin the loop), the current element,

```
and the element to the right of current (or, again, <initial-state> if we're
at the last element of the row). The we loop through the <rules> and when
the current triplet matches the rule we return the 0 or 1 associated with
the triplet in the rule (i.e. the second element of the matching rule).

so e.g. we start with:
(0 0 0 0 0 0 0 0 0 0 0 0 0 0 0)

set the middle element to default of 1 (first generation)
(0 0 0 0 0 0 0 1 0 0 0 0 0 0 0)

then proceed with the loop. Using e.g. the r30 rules:
(((1 1 1) 0) ((1 1 0) 0) ((1 0 1) 0) ((1 0 0) 1) ((0 1 1) 1)
             ((0 1 0) 1) ((0 0 1) 1) ((0 0 0) 0))

all those triplet 0s return 0 (because of the last rule) but when we get to
the 1 we first of all have (0 0 1) which returns 1, then (0 1 0) which also
returns 1, and finally--before we only have 0 triplets again--(1 0 0), also
returning 1. So we end up with this as the second generation:

(0 0 0 0 0 0 1 1 1 0 0 0 0 0 0)

(0 0 0 0 0 1 1 0 0 1 0 0 0 0 0)
```

ARGUMENTS:

```
- the wolfram object
- the number of generations (rows) to generate
```

OPTIONAL ARGUMENTS:

```
<start> is the value of the middle cell of the first row (or just to the
left of middle, if width is an even number). All cells to the left and
right of it will be in <initial-state>. This should be 0 or 1 only.
```

RETURN VALUE:

```
The data slot, which will be the rows in the form of a list of
named-objects containing the ID (starting from 1) and the list of cell
values.
```

SYNOPSIS:

```
(defmethod generate ((w wolfram) generations &optional (start 1))
```

38.6.441 wolfram/make-wolfram-r110

[wolfram] [Functions]

DATE:

```
June 7th 2017, Edinburgh.
```

DESCRIPTION:

```
#b01101110 = 110
This is the one Wolfram suggests might be Turing complete. This was proved
in 2000. "Rule 110, like the Game of Life, exhibits what Wolfram calls
"Class 4 behavior", which is neither completely stable nor completely
chaotic. Localized structures appear and interact in various
complicated-looking ways" (https://en.wikipedia.org/wiki/Rule_110)
```

ARGUMENTS:

```
none required
```

OPTIONAL ARGUMENTS:

```
- the width of a row. Default = 100.
- the initial state of the object. Default = 0.
```

RETURN VALUE:

```
the function name as a symbol but when called a wolfram object with the
given rules.
```

EXAMPLE:

```
so this one only goes up to column 7 when called with width of 12:
      .
     ..
    ...
   .. .
  .....
 ..   .
...  ..
. . ...
..... .
.   ...
.  .. .
```

```
. .....
...   .
. .  ..
... ...
. ... .
then the last 2 lines repeat

but called with width of 60:
                                .
                               ..
                              ...
                             .. .
                            .....
                           ..   .
                          ...  ..
                         .. . ...
                        ....... .
                       ..     ...
                      ...    .. .
                     .. .   .....
                    .....  ..   .
                   ..   . ...  ..
                  ...  .... . ...
                .. . ..  ..... .
               ........ ..   ...
             ..      ....  .. .
            ...     ..  . .....
          .. .    ... ....   .
         .....   .. ...  .  ..
        ..   .  ..... . .. ...
       ...  .. ..   ........ .
      .. . ......  ..      ...
     ........    . ...     .. .
   ..     .   .... .    .....
  ...    ..  ..  ...   ..   .
 .. .   ... ... .. .  ...  ..
.....  .. ... ...... .. . ...
..   . ..... ...    ........ .
...  ....   ... .   ..      ...
. . ..  .  .. ...  ...     .. .
...... .. ..... . .. .    .....
.    ......   ........   ..   .
.   ..    .  ..      .  ...  ..
.  ...   .. ...     .. .. . ...
. .. .  ..... .    .......... .
...... ..   ...   ..        ...
```

```
.    ....  .. .  ...       .. .
.   ..  . ..... .. .      .....
.  ... ....   ......     ..   .
. .. ...  .  ..    .    ...  ..
....... . .. ...   ..   .. . ...
.    ........ .  ...  ....... .
.   ..      ... .. . ..     ...
.  ...     .. .........    .. .
. .. .    .....       .   .....
......   ..   .      ..  ..   .
.    .  ...  ..     ... ...  ..
.   .. .. . ...    .. ... . ...
.  .......... .   ..... ..... .
. ..        ...  ..   ...   ...
....       .. . ...  .. .  .. .
.  .      ....... . ..... .....
. ..     ..     .....   ...   .
|#
(defwolfram make-wolfram-r110
    '(((1 1 1) 0) ((1 1 0) 1) ((1 0 1) 1) ((1 0 0) 0) ((0 1 1) 1)
      ((0 1 0) 1) ((0 0 1) 1) ((0 0 0) 0)))
```

38.6.442 wolfram/make-wolfram-r126

[wolfram] [Functions]

DATE:

```
June 6th 2017, Edinburgh.
```

DESCRIPTION:

```
#b01111110 = 126
```

ARGUMENTS:

```
none required
```

OPTIONAL ARGUMENTS:

```
- the width of a row. Default = 100.
- the initial state of the object. Default = 0.
```

RETURN VALUE:

```
 the function name as a symbol but when called a wolfram object with the
 given rules.
```

EXAMPLE:

```
width: 12, initial-state: 0
      .
     ...
    .. ..
   .......
  ..     ..
 ....   ....
..  .. ..  .
............
.           .
..         ..
...       ...
. ..     .. .
.....   .....
.    ....   .
.. ..  .. ..
............
using 1, the following repeats but not forever
............

.           .
..         ..
 ..       ..
....     ....
   ..   ..
. ........ .
...       ...
  ..     ..
.....   .....
    ....
.  ..  ..  .
```

SYNOPSIS:

```
(defwolfram make-wolfram-r126
    '(((1 1 1) 0) ((1 1 0) 1) ((1 0 1) 1) ((1 0 0) 1) ((0 1 1) 1)
      ((0 1 0) 1) ((0 0 1) 1) ((0 0 0) 0)))
```

38.6.443 wolfram/make-wolfram-r150

[wolfram] [Functions]

DATE:

June 7th 2017, Edinburgh.

DESCRIPTION:

#b10010110 = 150

ARGUMENTS:

none required

OPTIONAL ARGUMENTS:

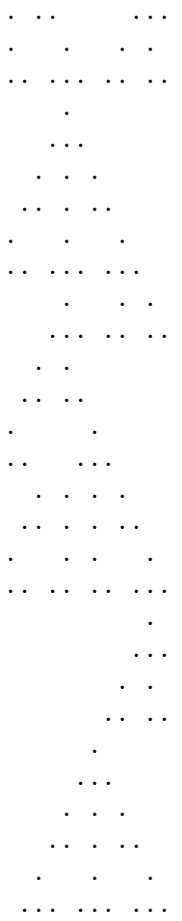
- the width of a row. Default = 100.
- the initial state of the object. Default = 0.

RETURN VALUE:

the function name as a symbol but when called a wolfram object with the given rules.

EXAMPLE:

```
width: 12, initial-state: 0
non-repeating  but lovely patterns
      .
     ...
    . . .
   .. . ..
  .   .   .
 ... ... ...
. .   .   .
. .. ... ...
.     .   .
..   ... ...
  . . .   .
 .. . .. ...
.   .     .
.. ...   ...
    . . . .
   .. . . ..
  .   . .
 ... .. ..
. .       .
```



SYNOPSIS:

```
(defwolfram make-wolfram-r150
    '(((1 1 1) 1) ((1 1 0) 0) ((1 0 1) 0) ((1 0 0) 1) ((0 1 1) 0)
      ((0 1 0) 1) ((0 0 1) 1) ((0 0 0) 0)))
```

38.6.444 wolfram/make-wolfram-r151

[wolfram] [Functions]

DATE:

```
June 7th 2017, Edinburgh.
```

DESCRIPTION:

```
#b10010111 = 151
```

ARGUMENTS:

```
none required
```

OPTIONAL ARGUMENTS:

```
- the width of a row. Default = 100.
- the initial state of the object. Default = 0.
```

RETURN VALUE:

```
the function name as a symbol but when called a wolfram object with the
given rules.
```

EXAMPLE:

```
      .
.............
 ...........
. ......... .
.  .......  .
... ..... ...
 .   ...   .
.....   .....
 ... ... ...
. .       . .
. ......... .
.  .......  .
then the last 7 lines repeat
|#
(defwolfram make-wolfram-r151
    '(((1 1 1) 1) ((1 1 0) 0) ((1 0 1) 0) ((1 0 0) 1) ((0 1 1) 0)
      ((0 1 0) 1) ((0 0 1) 1) ((0 0 0) 1)))
```

38.6.445 wolfram/make-wolfram-r159

[wolfram] [Functions]

DATE:

```
June 7th 2017, Edinburgh.
```

DESCRIPTION:

```
#b10011111 = 159
```

ARGUMENTS:

```
none required
```

OPTIONAL ARGUMENTS:

```
- the width of a row. Default = 100.
- the initial state of the object. Default = 0.
```

RETURN VALUE:

```
the function name as a symbol but when called a wolfram object with the
given rules.
```

EXAMPLE:

```
      .
.............
...........
.......... .
.........  .
........ ...
.......  ..
...... ... .
.....  ..  .
.... ... ...
...  ..  ..
.. ... ... .
.  ..  ..  .
then the last 4 lines repeat
|#
(defwolfram make-wolfram-r159
    '(((1 1 1) 1) ((1 1 0) 0) ((1 0 1) 0) ((1 0 0) 1) ((0 1 1) 1)
      ((0 1 0) 1) ((0 0 1) 1) ((0 0 0) 1)))
```

38.6.446 wolfram/make-wolfram-r190

[wolfram] [Functions]

DATE:

```
June 6th 2017, Edinburgh.
```

DESCRIPTION:

```
#b10111110 = 190
```

ARGUMENTS:

```
none required
```

OPTIONAL ARGUMENTS:

```
- the width of a row. Default = 100.
- the initial state of the object. Default = 0.
```

RETURN VALUE:

```
the function name as a symbol but when called a wolfram object with the
given rules.
```

EXAMPLE:

```
width: 12, initial-state: 0
      .
     ...
    ... .
   ... ...
  ... ... .
 ... ... ...
... ... ...
.. ... ... .
. ... ... ..
.... ... ..
... ... .. .
.. ... .. ..
. ... .. ..
.... .. .. .
... .. .. ..
.. .. .. ..
. .. .. .. .
then last three rows repeat
```

SYNOPSIS:

```
(defwolfram make-wolfram-r190
    '(((1 1 1) 1) ((1 1 0) 0) ((1 0 1) 1) ((1 0 0) 1) ((0 1 1) 1)
      ((0 1 0) 1) ((0 0 1) 1) ((0 0 0) 0)))
```

38.6.447 wolfram/make-wolfram-r25

[wolfram] [Functions]

DATE:

June 6th 2017, Edinburgh.

DESCRIPTION:

#b00011001 = 25

ARGUMENTS:

none required

OPTIONAL ARGUMENTS:

- the width of a row. Default = 100.
- the initial state of the object. Default = 0.

RETURN VALUE:

the function name as a symbol but when called a wolfram object with the given rules.

EXAMPLE:

```
width: 12, initial-state: 0
      .
.....  .....
.    . .
 ...    ....
 .  ... .
  . .    ...
.    ... .
 ... .    ..
 .    ... .
  ... .    .
. . .    ...
   ... .  ..
.. .    . .
.   ...    .
 .. .  ...
 .   . .  ..
```

```
  ..    . .
. . ...    .
    .  ...
...  . .  ..
.  .    . .
 .  ...    .
  . .  ...
.    . .  ..
 ...    . .
 .  ...    .
  . .  ...
.    . .  ..
then last four rows repeat

also very nice starting with 0 instead of 1 (no obvious repeats):
(generate w 100 0)
............
.
 ...........
 .
  ..........
. .
   .........
.. .
.   ........
 .. .
 .   .......
  .. .
. .   ......
   .. .
.. .   .....
.   .. .
 .. .   ....
 .   .. .
  .. .   ...
. .   .. .
   .. .   ..
.. .   .. .
.   .. .   .
 .. .   ..
 .   .. . ..
  .. .    .
. .   ...  .
   .. .  .
.. .   .  ..
.   ..  . .
```

```
 .. . .    .
 .     ...
  .... .  ..
. .     . .
   ....    .
.. .   ...
```

SYNOPSIS:

```
(defwolfram make-wolfram-r25
    '(((1 1 1) 0) ((1 1 0) 0) ((1 0 1) 0) ((1 0 0) 1)
      ((0 1 1) 1) ((0 1 0) 0) ((0 0 1) 0) ((0 0 0) 1)))
```

38.6.448 wolfram/make-wolfram-r30

[wolfram] [Functions]

DATE:

```
June 6th 2017, Edinburgh.
```

DESCRIPTION:

```
This is the default Wolfgram object: #b00011110 = 30
```

ARGUMENTS:

```
none required
```

OPTIONAL ARGUMENTS:

```
- the width of a row. Default = 100.
- the initial state of the object. Default = 0.
```

RETURN VALUE:

```
the function name as a symbol but when called a wolfram object with the
given rules.
```

EXAMPLE:

```
;; so width: 12, initial-state: 0
(let ((w (make-wolfram-r30 12 0)))
  (generate w 100)
```

```
  (print-matrix w :row-number nil :off "." :on 'X)
  (print-matrix w :row-number nil))
      .
     ...
    ..  .
   .. ....
  ..  .   .
 .. .... ...
..  .    .
. ....  ...
. .   ...  .
. .. ..  ...
. .  . ...
. .... .  .
. .    .....
. ..  ..
. . ... .
. . .   ..
. . .. .. .
. . .  .  ..
. . .......
. . .      .
. . ..    ..
. . . .  ..
. . . .... .
. . . .    .
. . . ..  ..
. . . . ...
. . . . .  .
. . . . ....
. . . . .
. . . . ..
. . . . . .
. . . . . ..
then last two rows repeat
```

SYNOPSIS:

```
(defwolfram make-wolfram-r30
    '(((1 1 1) 0) ((1 1 0) 0) ((1 0 1) 0) ((1 0 0) 1) ((0 1 1) 1)
      ((0 1 0) 1) ((0 0 1) 1) ((0 0 0) 0)))
```

38.6.449 wolfram/make-wolfram-r62

[wolfram] [Functions]

DATE:

June 7th 2017, Edinburgh.

DESCRIPTION:

#b00111110 = 62

ARGUMENTS:

none required

OPTIONAL ARGUMENTS:

- the width of a row. Default = 100.
- the initial state of the object. Default = 0.

RETURN VALUE:

the function name as a symbol but when called a wolfram object with the given rules.

EXAMPLE:

```
      .
     ...
    ..  .
   .. ....
  .. ..   .
 .. .. . ...
.. .. ....
. .. ..   .
... .. . ...
.  .. ....
.... ..   .
.   .. . ...
.. .. ....
. .. ..   .
... .. . ...
.  .. ....
then the last 6 lines repeat
|#
(defwolfram make-wolfram-r62
    '(((1 1 1) 0) ((1 1 0) 0) ((1 0 1) 1) ((1 0 0) 1) ((0 1 1) 1)
      ((0 1 0) 1) ((0 0 1) 1) ((0 0 0) 0)))
```

38.6.450 wolfram/make-wolfram-r73

[wolfram] [Functions]

DATE:

```
June 8th 2017, Edinburgh.
```

DESCRIPTION:

```
#b01001001 = 73
Suggested by Orestis Papadopoulos
```

ARGUMENTS:

```
none required
```

OPTIONAL ARGUMENTS:

```
- the width of a row. Default = 100.
- the initial state of the object. Default = 0.
```

RETURN VALUE:

```
the function name as a symbol but when called a wolfram object with the
given rules.
```

EXAMPLE:

```
      .
.....   ....
.   . . .  .
  .
.   ........
  . .      .
.     ....
  ... .  . .
. . .
      ......
..... .    .
.   .   ..
  .   . .. .
then the last 2 lines repeat
|#
(defwolfram make-wolfram-r73
    '(((1 1 1) 0) ((1 1 0) 1) ((1 0 1) 0) ((1 0 0) 0) ((0 1 1) 1)
      ((0 1 0) 0) ((0 0 1) 0) ((0 0 0) 1)))
```

38.6.451 wolfram/make-wolfram-r82

[wolfram] [Functions]

DATE:

```
June 6th 2017, Edinburgh.
```

DESCRIPTION:

```
#b01010010 = 82
```

ARGUMENTS:

```
none required
```

OPTIONAL ARGUMENTS:

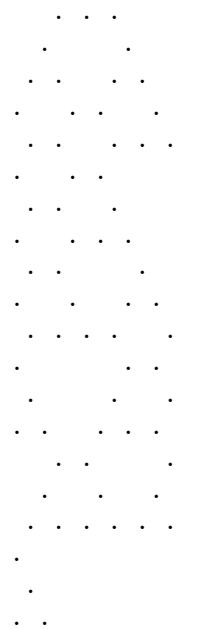
```
- the width of a row. Default = 100.
- the initial state of the object. Default = 0.
```

RETURN VALUE:

```
the function name as a symbol but when called a wolfram object with the
given rules.
```

EXAMPLE:

```
width: 12, initial-state: 0
non-repeating  but lovely patterns
      .
     . .
    .   .
   . . . .
  .       .
 . .     . .
.   .   .
 . . . . .
.         .
 .       . .
. .     .
   .   . .
  . . .   .
 .     . . .
. .   .
```



SYNOPSIS:

```
(defwolfram make-wolfram-r82
    '(((1 1 1) 0) ((1 1 0) 1) ((1 0 1) 0) ((1 0 0) 1) ((0 1 1) 0)
      ((0 1 0) 0) ((0 0 1) 1) ((0 0 0) 0)))
```

38.6.452 wolfram/make-wolfram-r93

[wolfram] [Functions]

DATE:

```
June 7th 2017, Edinburgh.
```

DESCRIPTION:

```
#b01011101 = 93
```

ARGUMENTS:

```
none required
```

OPTIONAL ARGUMENTS:

```
 - the width of a row. Default = 100.
 - the initial state of the object. Default = 0.
```

RETURN VALUE:

```
 the function name as a symbol but when called a wolfram object with the
 given rules.
```

EXAMPLE:

```
      .
..... ......
.   ...    .
... . .... .
. .....  ...
...   .. . .
. ... ......
... ...    .
. ... .... .
... ...  ...
. ... .. . .
... ........
. ...      .
... ...... .
. ...    ...
... .... . .
. ...  .....
... .. .   .
. ........ .
...      ...
. ...... . .
...    .....
. .... .   .
...  ..... .
. .. .   ...
........ . .
.      .....
...... .   .
.    ..... .
.... .   ...
.  ..... . .
.. .   .....
...... .   .
.    ..... .
then the last 5 lines repeat
|#
```

```
(defwolfram make-wolfram-r93
    '(((1 1 1) 0) ((1 1 0) 0) ((1 0 1) 0) ((1 0 0) 1) ((0 1 1) 1)
      ((0 1 0) 1) ((0 0 1) 0) ((0 0 0) 1)))
```

38.6.453 wolfram/print-matrix

[wolfram] [Methods]

DATE:

```
June 6th 2017, Edinburgh
```

DESCRIPTION:

```
Print the state of the cells to the terminal or another open stream . On
and off could theoretically be any object but of course single characters
are clearest. See the defwolfram definitions below for examples.
```

ARGUMENTS:

```
- the wolfram object
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :stream. The stream to print to. Default t which means it prints to the
  terminal.
- :on. what to print when a cell has the value 1. Default: the . character
- :off. what to print when a cell has the value 0. Default: space
- :row-number: whether to print row numbers. Default t.
```

RETURN VALUE:

```
T
```

SYNOPSIS:

```
(defmethod print-matrix ((w wolfram) &key (stream t) (on #\.) (off #\ )
                                          (row-number nil))
```

38.6.454 wolfram/try-all

[wolfram] [Functions]

DATE:

June 10th 2017, Edinburgh

DESCRIPTION:

Try all of the Wolfram rules defined via defwolfram. An instance of each rule set defined will be created, the rows will be generated, and print-matrix will be called. This allows you to see and compare the results of each rule set using the same parameters.

ARGUMENTS:

none required

OPTIONAL ARGUMENTS:

keyword arguments:
:width. The width slot (number of columns) for the Wolfram objects. Any positive ingeger. Default = 100.
:initial-state. The initial-state (i.e. default) slot for the Wolfram objects. 0 or 1. Default = 0.
:start. The value of the middle cell of the first row. See generate method for more details. 0 or 1. Default = 1.
:generations. The number of rows to generate. Any positive integer. Default = 30.
:stream, :on, :off, :row-number: see print-matrix method.

RETURN VALUE:

T

SYNOPSIS:

```
(defun try-all (&key (width 100) (initial-state 0) (start 1) (generations 30)
                  (stream t) (on #\.) (off #\ ) (row-number nil))
```

38.6.455 circular-sclist/at-start

[circular-sclist] [Methods]

DESCRIPTION:

Determines whether the pointer for the given circular-sclist object is at the head of its list.

ARGUMENTS:

```
 - A circular-sclist object.
```

RETURN VALUE: EXAMPLE:

```
;; At creation the pointer is located at the start of the list
(let ((cscl (make-cscl '(0 1 2 3 4))))
  (at-start cscl))
=> T

;; Retrieve a number of the items using get-next, then determine whether the
;; pointer is located at the start of the list
(let ((cscl (make-cscl '(0 1 2 3 4))))
  (loop repeat 7 do (get-next cscl))
  (at-start cscl))
=> NIL
```

SYNOPSIS:

```
(defmethod at-start ((cscl circular-sclist))
```

38.6.456 circular-sclist/cycle-repeats

[circular-sclist] [Classes]

NAME:

```
 rthm-chain

 File:             cycle-repeats.lsp

 Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                   circular-sclist -> cycle-repeats

 Version:          1.1.0

 Project:          slippery chicken (algorithmic composition)

 Purpose:          class used originally in rthm-chain to determine cyclic
                   repeats of data. E.g. passing the data list ((0 5) (1 1)
                   (0 3) (1 1) (0 2) (2 1)) means we'll have 0 five times,
                   then 1 once, then 0 three times etc.  This repeats
                   indefinitely when the get-next method is called.

 Author:           Michael Edwards: m@michael-edwards.org
```

```
Creation date:    4th February 2010

$$ Last modified:  11:24:36 Sat Jan 17 2026 CET

SVN ID: $Id$
```

38.6.457 circular-sclist/get-current

[circular-sclist] [Methods]

DESCRIPTION:

```
Return the current list element without incrementing the current slot.
```

ARGUMENTS:

```
- The circular-sclist object
```

RETURN VALUE:

```
The current list element, which can be of any type.
```

EXAMPLE:

```
(let ((cscl (make-cscl '(0 1 2 3 4))))
  (loop for i below 10 do
     (print (if (oddp i) (get-next cscl) (get-current cscl)))))
(let ((cscl (make-cscl '(0 1 2 3 4))))
  (loop for i below 10
     collect (if (oddp i) (get-next cscl) (get-current cscl))))
=>
(0 0 1 1 2 2 3 3 4 4)
```

SYNOPSIS:

```
(defmethod get-current ((cscl circular-sclist))
```

38.6.458 circular-sclist/get-last

[circular-sclist] [Methods]

DESCRIPTION:

```
Return the the most recent item retrieved in a circular-sclist object.
```

ARGUMENTS:

```
- A circular-sclist object.
```

RETURN VALUE:

```
An item from the given circular-sclist object.
```

EXAMPLE:

```
;; Retrieves the final item in the list at creation
(let ((cscl (make-cscl '(0 1 2 3 4))))
  (get-last cscl))
=> 4

;; Get and print a number of items from the list using get-next, then return
;; the most recent item retrieved using get-last
(let ((cscl (make-cscl '(0 1 2 3 4))))
  (loop repeat 7 do (print (get-next cscl)))
  (get-last cscl))
=> 1
```

SYNOPSIS:

```
(defmethod get-last ((cscl circular-sclist))
```

38.6.459 circular-sclist/get-next

[circular-sclist] [Methods]

DESCRIPTION:

```
Get the next item in a given circular-sclist object. The class
automatically keeps track of the last item retrieved. If the final item of
the given circular-sclist object was the last item retrieved, the method
begins again at the beginning of the list.
```

ARGUMENTS:

```
- A circular-sclist object.
```

RETURN VALUE:

```
An item from the given circular-sclist object.
```

EXAMPLE:

```
;; Repeatedly calling get-next retrieves each subsequent item from the
;; given circular-sclist object. When the list has been exhausted, retrieval
;; begins again from the head of the list.
(let ((cscl (make-cscl '(0 1 2 3 4))))
  (loop repeat 10 collect (get-next cscl)))
=>
(0 1 2 3 4 0 1 2 3 4)
```

SYNOPSIS:

```
(defmethod get-next ((cscl circular-sclist))
```

38.6.460 circular-sclist/make-cscl

[circular-sclist] [Functions]

DESCRIPTION:

```
Create a circular-sclist object from a specified list of items. The items
themselves may also be lists.
```

ARGUMENTS:

```
- A list.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :id. A symbol that will be used as the ID for the created circular-sclist
  object. Default = NIL.
- :bounds-alert. T or NIL to indicate whether or not to print a warning if
  when an attempt is made to access the object using an out-of-bounds index
  number (i.e., not enough elements in the list).  T = print a
  warning. Default = T.
- :copy. T or NIL to indicate whether the given data list should be copied
  (any slippery-chicken class instances will be cloned), with subsequent
  modifications being applied to the copy. T = copy. Default = T.
```

RETURN VALUE:

```
A circular-sclist object.
```

EXAMPLE:

```
;; Returns a circular-sclist object with ID of NIL, bounds-alert=T and copy=T
;; by default
(make-cscl '(1 2 3 4))

=>
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 4, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (1 2 3 4)

;; Can be created using nested lists
(let ((cscl (make-cscl '((1 (4 5 6))
                         (2 (7 8 9))
                         (3 (10 11 12))))))
  (data cscl))

=> ((1 (4 5 6)) (2 (7 8 9)) (3 (10 11 12)))

;; Setting the ID
(make-cscl '(1 2 3 4) :id 'test-cscl)

=>
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 4, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: TEST-CSCL, tag: NIL,
data: (1 2 3 4)

;; By default, attempts to access the object with an out-of-bounds index result
;; in a warning being printed
(let ((cscl (make-cscl '(1 2 3 4))))
  (get-nth 11 cscl))

=>
NIL
WARNING: sclist::sclist-check-bounds: Illegal list reference: 11
(length = 4) (sclist id = NIL)

;; This can be suppressed by creating the object with :bounds-alert set to NIL
(let ((cscl (make-cscl '(1 2 3 4) :bounds-alert nil)))
  (get-nth 11 cscl))

=> NIL
```

SYNOPSIS:

```
(defun make-cscl (list &key (id nil) (bounds-alert t) (copy t))
```

38.6.461 circular-sclist/popcorn

[circular-sclist] [Classes]

NAME:

```
assoc-list

File:             popcorn.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist -> popcorn

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Inspired by popping popcorn, generate a series of values
                  ranging between > 0.0 and <= 1.0 by (optionally fixed)
                  random selection. Given 1 or more starting values (not
                  zero) we generate tendentially increasing new values
                  until we reach 1.0. This is not a linear process,
                  rather, we get spike values that increase the average
                  value and thus increase the chance of further spikes.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    3rd February 2011 (Ko Lanta, Thailand)

$$ Last modified:  20:44:58 Wed Aug 28 2024 CEST

SVN ID: $Id$
```

38.6.462 popcorn/fit-to-length

[popcorn] [Methods]

DESCRIPTION:

```
Change the length of the list of kernels contained in a given popcorn
object by adding or removing items at regular intervals. If adding items,
linear interpolation will be used.
```

```
NB: The new length must be between 1 and 1 less than double the original
    length.
```

ARGUMENTS:

```
- A popcorn object.
- An integer that is the new length of the list of the KERNELS slot of the
  given popcorn object.
```

RETURN VALUE:

```
Returns the integer that is the new length of the KERNELS slot.
```

EXAMPLE:

```
(let ((ppcn (make-popcorn '(0.01 0.02) :min-spike 3.0 :max-spike 5.0)))
  (fit-to-length ppcn 100))

=> 100
```

SYNOPSIS:

```
(defmethod fit-to-length ((pc popcorn) length)
```

38.6.463 popcorn/get-kernel

[popcorn] [Methods]

DESCRIPTION:

```
Generate the next value for the KERNELS slot of a given popcorn object and
change the internal state, with the help of the get-kernel-aux method.

This method is called automatically from within the heat method.
```

ARGUMENTS:

```
- A popcorn object.
```

RETURN VALUE:

```
The next value for the given popcorn object's KERNEL slot.
Returns NIL when the kernel value is > 1.0.
```

SYNOPSIS:

```
(defmethod get-kernel ((pc popcorn))
```

38.6.464 popcorn/heat

[popcorn] [Methods]

DESCRIPTION:

```
Generate a series of values for the KERNELS slot of a popcorn object,
ranging between >0.0 and <= 1.0, by (optionally fixed) random selection.
If calling heat explicitly on a previously heated object, all kernels and
associated data will be deleted before being regenerated.

Taking the one or more starting values of the popcorn object, the method
generates tendentially increasing new values until it reaches 1.0. This is
not a linear process; rather, the method produces spike values based on the
min-spike and max-spike values of the given popcorn object that increase
the average value and thus increase the chance of further spikes.

NB: This method is called within the initialize-instance for the popcorn
    object, and as such is not necessarily needed to be accessed directly
    by the user.
```

ARGUMENTS:

```
- An popcorn object.
```

RETURN VALUE:

```
Returns the popcorn object with a newly generated list of 'kernel' values.
```

EXAMPLE:

```
(let ((ppcn (make-popcorn '(0.01 0.02) :min-spike 3.0 :max-spike 5.0)))
  (print ppcn)
  (setf (min-spike ppcn) 4.0)
  (heat ppcn))
=>
POPCORN: kernels: (0.01 0.02 0.016648924 0.018915312 0.016573396
                   0.017766343 0.018711153 0.017729789 0.017080924
                   0.018266398 0.018132625 0.019022772 0.017662765
[...]
POPCORN: kernels: (0.01 0.02 0.015828498 0.015408514 0.015781755 0.01670348
                   0.019892192 0.017849509 0.016623463 0.019682804 0.017869182
                   0.019521425 0.017451862 0.017689057 0.01758664 0.01863435
[...]
```

SYNOPSIS:

```
(defmethod heat ((pc popcorn))
```

38.6.465 popcorn/make-popcorn

[popcorn] [Functions]

DESCRIPTION:

```
Make a popcorn object. This method uses the heat method internally to
generate a series of decimal values ('kernels'), ranging between > 0.0 and
<= 1.0, by (optionally fixed) random selection.

Taking the one or more starting values, the method generates tendentially
increasing new values until it reaches 1.0. This is not a linear process;
rather, the method produces spike values based on the min-spike and
max-spike values specified, which increase the average value of the kernels
generated so far and thus increase the chance of further spikes.
```

ARGUMENTS:

```
- A list of at least two decimal numbers from which the 'kernel' values
  will be generated. These values must be >0.0 and <1.0.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :id. An optional ID for the popcorn object to be created. Default = NIL.
- :fixed-random. T or NIL to indicate whether the 'kernel' values generated
  by the subsequent heat method are to be based on a fixed random seed.
  T = fixed random. Default = T.
- :max-spike. A decimal number that is the highest possible 'spike' value
  that the heat method may produce when generating the 'kernel'
  values. This is a sudden high value that will itself not be present in
  the final data, but will go towards skewing the mean, thus increasing the
  kernel values more rapidly and increasing the chance of more spikes
  occurring. Default = 4.0.
- :min-spike. A decimal number that is the lowest possible 'spike' value
  that the heat method may produce when generating the 'kernel'
  values. This is a sudden high value that will itself not be present in
  the final data, but will go towards skewing the mean, thus increasing the
  kernel values more rapidly and increasing the chance of more spikes
  occurring. Default = 2.0.
```

RETURN VALUE:

```
- A popcorn object.
```

EXAMPLE:

```
(make-popcorn '(0.02 0.03) :max-spike 4.2 :min-spike 3.7)

=>
POPCORN: kernels: (0.02 0.03 0.025828497 0.02540851 0.02578175 0.026703479
                   0.029892191 0.027849507 0.026623461 0.029682804 0.02786918
                   0.029521424 0.02745186 0.027689056 0.02758664 0.028634349
                   0.028176062 0.028434621 0.028410202 0.02834666 0.027676953
                   0.027972711 0.027877634 0.028453272 0.027664827 0.029336458
                   0.028315568 0.029327389 0.10877271 0.032779325 0.095442966
                   0.10383448 0.03631042 0.054371007 0.0775562 0.057371408
                   0.05496178 0.10499479 0.048501145 0.09311144 0.07531821
                   0.08538791 0.05866453 0.06692247 0.052130517 0.09605096
                   0.102914646 0.061326876 0.09510137 0.0927515 0.08405721
                   0.09921508 0.1054862 0.09474778 0.07701611 0.069283865
                   0.082345024 0.090727165 0.081423506 0.0918279 0.06942183
                   0.09431985 0.0790893 0.07795428 0.061114937 0.21615848
                   0.17666964 0.09314137 0.11025161 0.1909036 0.23906681
                   0.17467138 0.22562174 0.1757016 0.16630511 0.23570478
                   0.18461326 0.2358803 0.14396386 0.121555254 0.082086496
                   0.094552115 0.08456006 0.10379071 0.113467366 0.12590313
                   0.2211197 0.2096048 0.19645368 0.17204309 0.18469864
                   0.14422922 0.20209482 0.11207011 0.1176545 0.22522071
                   0.23593009 0.13767788 0.1589861 0.23501754 0.14337942
                   0.14403008 0.3852736 0.19077776 0.15493082 0.15311162
                   0.31107113 0.10612649 0.36018372 0.31991273 0.17881061
                   0.2653634 0.26506728 0.31478146 0.31331018 0.33569553
                   0.3001081 0.1574295 0.4698523 0.12513468 0.2010088
                   0.17438973 0.24960503 0.27139995 0.31985858 0.14607468
                   0.34586 0.52092844 0.5461051 0.33965456 0.24476483
                   0.45786726 0.23932996 0.18096672 0.5287333 0.45701692
                   0.58791053 0.5219719 0.39459002 0.56624746 0.37368405
                   0.21688993 0.3374743 0.6648663 0.44353223 0.16596928
                   0.3590309 0.17943183 0.673855 0.6455428 0.21892962
                   0.31195784 0.37920266 0.73120433 0.713979 0.5987564
                   0.29621923 0.5414667 0.64287895 0.56254905 0.514681
                   0.3153673 0.52838445 0.71745664 0.8074915 0.47637874
                   0.409207 0.49155992 0.777411 0.6339724 0.3673042 0.5411029
                   0.6993387 0.3566729 0.49429625 0.89963627 0.36773333
                   0.575006 0.74177176 0.53539884 0.4392826 0.45671058
                   0.2824728 0.60876155 0.2798523 0.47930354)
total: 44.67911, numk: 186, mink: 0.02, maxk: 0.89963627
min-spike: 3.7, max-spike: 4.2, fixed-random: T, mean: 0.24021028
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
```

```
data: (0.02 0.03)
```

SYNOPSIS:

```
(defun make-popcorn (starting-values &key (id nil) (fixed-random t)
                     (max-spike 4.0) (min-spike 2.0))
```

38.6.466 popcorn/scale

[popcorn] [Methods]

DESCRIPTION:

```
Scale the list of number values in the KERNEL slot of a given popcorn
object to a new range using specified maximum value and optional minimum
value.

NB: This method does not change the internal state of the given popcorn
    object except for the KERNELS slot.
```

ARGUMENTS:

```
- A popcorn object.
- A number that is the new maximum value for the scaled list.
```

OPTIONAL ARGUMENTS:

```
- A number that is the new minimum value for the scaled list.
```

RETURN VALUE:

```
The new contents of the given popcorn object's KERNELS slot after scaling.
```

EXAMPLE:

```
;; Specifying a new maximum value only
(let ((ppcn (make-popcorn '(0.01 0.02) :min-spike 3.0 :max-spike 5.0)))
  (scale ppcn 10.0))

=> (0.0 0.10578585 0.061657257 0.057214428 0.061162785 0.070913345 0.10464539
    0.08303669 0.07006687 0.102430366 0.08324481 0.1007232 0.07883015
    0.08133934 0.0802559 0.09133919 0.08649117 0.08922634 0.08896804 0.08829583
    0.081211284 0.084339984 0.08333421 0.089423634 0.08108301 0.09876652
    0.08796693 0.09867058 0.6236897 0.1155461 0.5345579 0.59066933 0.13915743
```

```
    0.2599228 0.4149548 0.27998555 0.2638731 0.59842795 0.22067288 0.51896775
    0.39999005 0.467323 0.28863218 0.3438505 0.24494135 0.5386234 0.58451873
    0.30643454 0.53227377 0.516561 0.458425 0.55978096 0.60171384 0.52990943
    0.41134343 0.35964036 0.44697618 0.5030249 0.4408143 0.51038516 0.36056283
    0.527048 0.42520615 0.41761667 0.30501738 1.2049319 0.9747751 0.48793882
    0.5876642 1.0577364 0.3861802 0.8744062 1.1359217 0.87969404 0.8314643
    1.1876756 0.92543525 1.1885763 0.7167921 0.60177433 0.39919102 0.46317393
    0.4118872 0.5105933 0.56026125 0.62409085 1.112814 1.0537107 0.98620933
    0.86091584 0.92587364 0.71815413 1.0151639 0.5530894 0.5817527 1.1338633
    1.1888319 0.68452775 0.79389757 1.1841481 0.7137923 0.717132 2.3223429
    1.0770427 0.8475252 0.83587736 1.847246 0.53504527 2.1616995 1.9038564
    1.0004206 1.5545927 1.552697 1.8710022 1.861582 2.004909 1.7770531
    0.86352354 1.5642304 0.63962364 1.0099988 0.8800593 1.2472185 1.353609
    1.5901572 0.7418409 1.7170814 3.3055406 3.700319 2.1952631 1.5035022
    3.057052 1.4638814 1.0384043 1.1837897 3.0469408 2.803617 3.2614107
    2.4135168 3.556123 2.27436 1.2306889 2.0333362 4.5721135 2.941492 0.8966192
    2.3189502 0.9958008 4.344239 4.3671365 1.2785082 1.952021 2.438866
    4.3924103 4.46023 3.7118216 1.746746 3.3397071 3.9984121 3.4766433 3.165725
    1.8711188 3.2547336 4.482818 3.3188024 2.6877654 2.3054938 2.7741609
    3.796511 3.5835814 2.064381 3.0545063 3.9559705 2.0038147 2.7878509
    3.4762452 1.9495757 2.9715302 3.7937658 2.7762475 2.3023481 2.3882763
    1.5292001 3.1379611 1.5162798 2.4996707 4.362247 3.1643825 2.902113
    1.5552855 3.569274 3.6554635 3.8193665 3.2386634 5.418084 1.488541
    4.5816646 4.1958213 2.411787 2.6187074 3.1729605 2.959683 2.3334894
    5.325289 3.2408857 4.67207 3.0460484 6.0358443 6.879726 3.3280933 5.5901675
    1.8741251 3.5842674 4.855096 6.005389 1.7205821 3.8116035 3.439082 5.024595
    2.205073 4.140361 1.8645307 2.511795 5.744685 2.0451677 2.311025 6.787981
    6.533982 3.840785 2.2128632 6.444055 2.7525501 8.19589 7.3742037 2.5753407
    8.9812355 3.0030684 5.501138 6.7223954 4.8878922 3.2250557 2.3134975
    8.762646 3.072827 7.0158014 7.426256 5.388799 10.0 7.367759 7.078608
    8.373905 9.210589 7.072851 2.7709346 7.233898)


;; Using both a new maximum and new minimum value
(let ((ppcn (make-popcorn '(0.01 0.02) :min-spike 3.0 :max-spike 5.0)))
  (scale ppcn 8.0 5.0))

=> (5.0 5.031736 5.018497 5.017164 5.0183487 5.021274 5.0313935 5.024911
    5.02102 5.0307293 5.0249734 5.030217 5.023649 5.0244017 5.024077 5.027402
    5.0259476 5.0267677 5.0266905 5.026489 5.0243635 5.025302 5.025 5.026827
    5.024325 5.02963 5.02639 5.029601 5.187107 5.0346637 5.1603675 5.177201
    5.041747 5.0779767 5.1244864 5.083996 5.079162 5.179528 5.0662017 5.15569
    5.119997 5.140197 5.08659 5.103155 5.0734825 5.161587 5.1753554 5.0919304
    5.1596823 5.1549683 5.1375275 5.1679344 5.1805143 5.1589727 5.123403
    5.107892 5.134093 5.1509075 5.132244 5.1531157 5.108169 5.1581144 5.127562
    5.125285 5.091505 5.3614798 5.292433 5.146382 5.176299 5.317321 5.1158543
```

```
5.262322 5.3407764 5.2639084 5.2494392 5.3563027 5.277631 5.356573 5.215038
5.1805325 5.119757 5.1389523 5.123566 5.153178 5.1680784 5.1872272 5.333844
5.3161135 5.2958627 5.2582746 5.277762 5.2154465 5.304549 5.165927
5.1745257 5.340159 5.3566494 5.2053585 5.238169 5.3552446 5.2141376
5.2151394 5.696703 5.323113 5.2542577 5.2507634 5.554174 5.1605134 5.64851
5.571157 5.300126 5.4663777 5.465809 5.5613008 5.5584745 5.601473 5.533116
5.259057 5.4692693 5.191887 5.3029995 5.2640176 5.3741655 5.4060826
5.477047 5.2225523 5.5151243 5.991662 6.110096 5.658579 5.4510508 5.9171157
5.4391646 5.3115215 5.355137 5.9140825 5.8410854 5.978423 5.7240553
6.066837 5.682308 5.369207 5.610001 6.3716345 5.8824477 5.2689857 5.6956854
5.2987404 6.303272 6.310141 5.3835526 5.5856066 5.73166 6.3177233 6.338069
6.1135464 5.524024 6.001912 6.199524 6.042993 5.9497175 5.5613356 5.9764204
6.3448453 5.9956408 5.8063297 5.691648 5.832248 6.138953 6.075074 5.619314
5.916352 6.1867914 5.6011443 5.836355 6.0428734 5.5848727 5.891459
6.1381297 5.8328743 5.6907043 5.716483 5.4587603 5.9413886 5.454884
5.7499013 6.3086743 5.9493146 5.870634 5.4665856 6.070782 6.096639 6.14581
5.971599 6.6254253 5.4465623 6.3744993 6.258746 5.723536 5.785612 5.951888
5.887905 5.700047 6.5975866 5.9722657 6.401621 5.9138145 6.8107533 7.063918
5.998428 6.6770506 5.5622377 6.07528 6.4565287 6.8016167 5.516175 6.1434813
6.0317245 6.5073786 5.661522 6.2421083 5.559359 5.7535386 6.723406 5.61355
5.6933074 7.0363946 6.9601946 6.1522355 5.663859 6.9332166 5.825765
7.4587674 7.212261 5.772602 7.694371 5.9009204 6.6503415 7.016719 6.4663677
5.967517 5.6940494 7.6287937 5.9218483 7.1047406 7.227877 6.61664 8.0
7.210328 7.123583 7.5121717 7.763177 7.1218557 5.8312807 7.17017)
```

SYNOPSIS:

```
(defmethod scale ((pc popcorn) max &optional (min 0.0) ignore1 ignore2)
```

38.6.467 circular-sclist/recurring-event

[circular-sclist] [Classes]

NAME:

```
rthm-chain

File:             recurring-event.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist -> recurring-event

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)
```

Purpose: class used in rthm-chain
This class allows for the periodic/cyclic return of given data. It is intended for situations where you want to do/collect something every several events, but the cycle period changes. E.g. (the data slot is) something like '((2 3) (3 2) (5 3) (8 2)) which means every two events three times, then every 3 events twice, every 5 events thrice etc.

If you want to return specific data on these cycle points, provide it in the return-data slot, with the indices into this data in the return-data-cycle slot.

```
simple example, without return-data
(let* ((re (make-re '((2 3) (3 2) (5 3) (8 2))
                    :return-data nil
                    :return-data-cycle nil)))
  (loop repeat 100 collect (on-it re)))

=> (NIL NIL T NIL T NIL T NIL NIL T NIL NIL T NIL NIL NIL
    NIL T NIL NIL NIL NIL T NIL NIL NIL NIL T NIL NIL NIL
    NIL NIL NIL NIL T NIL NIL NIL NIL NIL NIL NIL T NIL T
    NIL T NIL T NIL NIL T NIL NIL T NIL NIL NIL NIL T NIL
    NIL NIL NIL T NIL NIL NIL NIL T NIL NIL NIL NIL NIL
    NIL NIL T NIL NIL NIL NIL NIL NIL NIL T NIL T NIL T
    NIL T NIL NIL T NIL NIL T NIL)

(let* ((re (make-re '((2 3) (3 2) (5 3) (8 2))
                    ;; the data about to be collected
                    :return-data '(a b c d)
                    ;; the indices into the data; this
                    ;; means we'll return A (nth 0)
                    ;; thrice, D (nth 3) twice, C once,
                    ;; and B 5x
                    :return-data-cycle
                    '((0 3) (3 2) (2 1) (1 5)))))
  (loop repeat 100 collect (get-it re)))

=> (NIL NIL A NIL A NIL A NIL NIL D NIL NIL D NIL NIL NIL
    NIL C NIL NIL NIL NIL B NIL NIL NIL NIL B NIL NIL NIL
    NIL NIL NIL NIL B NIL NIL NIL NIL NIL NIL NIL B NIL B
    NIL A NIL A NIL NIL A NIL NIL D NIL NIL NIL NIL D NIL
    NIL NIL NIL C NIL NIL NIL NIL B NIL NIL NIL NIL NIL
    NIL NIL B NIL NIL NIL NIL NIL NIL NIL B NIL B NIL B
    NIL A NIL NIL A NIL NIL A NIL)
```

```
Author:           Michael Edwards: m@michael-edwards.org

Creation date:    4th February 2010

$$ Last modified:  16:18:50 Sat Jun 29 2024 CEST

SVN ID: $Id$
```

38.6.468 recurring-event/get-it

[recurring-event] [Methods]

DESCRIPTION:

```
Get the next element from the return-data. This method is most effective
when called repeatedly (e.g. within a loop) when the return-data and
return-data-cycle slots have been set. In those cases the return-data-cycle
element will be used as look-up into return-data. If no return-data has
been specified, then the element itself will be returned.
```

ARGUMENTS:

```
- A recurring-event object.
```

RETURN VALUE:

```
Data from the return-data slot (or the return-data-cycle element) when
we're on a boundary, otherwise NIL.
```

EXAMPLE:

```
;;; Used together with return-data
(let ((re (make-re '((2 3) (3 2) (5 3) (8 2))
                   :return-data '(a b c d)
                   :return-data-cycle '((0 3) (3 2) (2 1) (1 5)))))
  (loop repeat 50 collect (get-it re)))

=> (NIL NIL A NIL A NIL A NIL NIL D NIL NIL D NIL NIL NIL NIL C NIL NIL NIL NIL
    B NIL NIL NIL NIL B NIL NIL NIL NIL NIL NIL NIL B NIL NIL NIL NIL NIL NIL
    NIL B NIL B NIL A NIL A)

;;; Used without return-data
(let ((re (make-re '((2 3) (3 2) (5 3) (8 2))
                   :return-data-cycle '((0 3) (3 2) (2 1) (1 5)))))
```

```
  (loop repeat 50 collect (get-it re)))

=> (NIL NIL 0 NIL 0 NIL 0 NIL NIL 3 NIL NIL 3 NIL NIL NIL NIL 2 NIL NIL NIL NIL
    1 NIL NIL NIL NIL 1 NIL NIL NIL NIL NIL NIL NIL 1 NIL NIL NIL NIL NIL NIL
    NIL 1 NIL 1 NIL 0 NIL 0)
```

SYNOPSIS:

```
(defmethod get-it ((re recurring-event))
```

38.6.469 recurring-event/make-re

[recurring-event] [Functions]

DESCRIPTION:

```
Make an instance of a recurring-event object, which allows for the
periodic/cyclic return of given data. The recurring-event object is
intended for situations in which the user would like to perform an action
or collect data every several events, but with a varying cycle period.
```

ARGUMENTS:

```
- A list of two-item lists that indicate the period pattern by which the
  action or data collection is to be performed. For example, a value such
  as '((2 3) (3 2) (5 3) (8 2)) will result in the action being performed
  every 2 events three times, then every 3 events twice, every 5 events
  thrice etc.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :return-data. If the recurring-event object is to be used to collect
  data, that data can be specified in this slot, with the indices into this
  data in the return-data-cycle slot. The return-data and return-data-cycle
  slots must be used together.
- :return-data-cycle. If data is specified using :return-data, the indices
  into that data must be specified here. For example, the value
  '((0 3) (3 2) (2 1) (1 5)) will the data item at (nth 0) thrice, that at
  (nth 3) twice, that at (nth 2) once, and that at (nth 1) five times.
- :id. An optional ID can also be specified for the recurring-event object
  created.
```

RETURN VALUE:

```
 A recurring-event object.
```

EXAMPLE:

```
;;; Simple usage with no specified data
(make-re '((2 3) (3 2) (5 3) (8 2)))
=>
RECURRING-EVENT: current-period: 2, current-repeats: 3
                 pcount: -1, rcount: 0
                 return-data: NIL, return-data-cycle: NIL
CIRCULAR-SCLIST: current 1
SCLIST: sclist-length: 4, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: ((2 3) (3 2) (5 3) (8 2))

;;; Usage with specified :return-data and :return-data-cycle
(make-re '((2 3) (3 2) (5 3) (8 2))
         :return-data '(a b c d)
         :return-data-cycle '((0 3) (3 2) (2 1) (1 5)))

=>
RECURRING-EVENT: current-period: 2, current-repeats: 3
                 pcount: -1, rcount: 0
                 return-data: (A B C D), return-data-cycle:
CYCLE-REPEATS: folded: ((0 3) (3 2) (2 1) (1 5))
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 11, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (0 0 0 3 3 2 1 1 1 1 1)
**************

CIRCULAR-SCLIST: current 1
SCLIST: sclist-length: 4, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: ((2 3) (3 2) (5 3) (8 2))
```

SYNOPSIS:

```
(defun make-re (data &key return-data return-data-cycle id)
```

38.6.470 recurring-event/on-it

[recurring-event] [Methods]

DESCRIPTION:

```
Test to determine whether the method is currently at a period boundary. The
object keeps track of its own internal state and position counter. This
method is most effective when called repeatedly in a loop.

Note that NIL is returned for the full first cycle rather than T then nil
as it is assumed that the beginning is a special case where state would
not want to be changed.
```

ARGUMENTS:

```
- A recurring-event object.
```

RETURN VALUE:

```
T or NIL.
```

EXAMPLE:

```
;;; Straightforward usage
(let ((re (make-re '((2 3) (3 2) (5 3) (8 2))
                   :return-data '(a b c d)
                   :return-data-cycle '((0 3) (3 2) (2 1) (1 5)))))
  (loop repeat 50 collect (on-it re)))

=> (NIL NIL T NIL T NIL T NIL NIL T NIL NIL T NIL NIL NIL NIL T NIL NIL NIL NIL
    T NIL NIL NIL NIL T NIL NIL NIL NIL NIL NIL NIL T NIL NIL NIL NIL NIL NIL
    NIL T NIL T NIL T NIL T)
```

SYNOPSIS:

```
(defmethod on-it ((re recurring-event))
```

38.6.471 circular-sclist/reset

[circular-sclist] [Methods]

DESCRIPTION:

```
Reset the pointer of a given circular-sclist object. The pointer is reset
to 0 by default, but the desired index may be specified using the optional
argument.

NB: An immediately subsequent get-next call will retrieve the item at the
```

index to which the pointer is reset. An immediately subsequent
get-last call will retrieve the item at the index one-less
than the value to which the pointer is set.

ARGUMENTS:

- A circular-sclist object.

OPTIONAL ARGUMENTS:

- An index integer to which the pointer for the given circular-sclist
  object should be reset.

RETURN VALUE:

Returns T.

EXAMPLE:

```
;; Resets to 0 by default. Here: Get a number of items using get-next, reset
;; the pointer, and apply get-next again.
(let ((cscl (make-cscl '(0 1 2 3 4))))
  (loop repeat 8 do (print (get-next cscl)))
  (reset cscl)
  (get-next cscl))
=> 0

;; Reset to a specified index
(let ((cscl (make-cscl '(0 1 2 3 4))))
  (loop repeat 8 do (print (get-next cscl)))
  (reset cscl 3)
  (get-next cscl))
=> 3

;; By default, get-last will then retrieve the item at index one less than the
;; reset value
(let ((cscl (make-cscl '(0 1 2 3 4))))
  (loop repeat 8 do (print (get-next cscl)))
  (reset cscl 3)
  (get-last cscl))
=> 2
```

SYNOPSIS:

```
(defmethod reset ((cscl circular-sclist) &optional where (warn t))
```

38.6.472 circular-sclist/rthm-chain-slow

[circular-sclist] [Classes]

NAME:

```
rthm-chain

File:             rthm-chain-slow.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  circular-sclist -> rthm-chain-slow

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          class used in rthm-chain

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    4th February 2010

$$ Last modified: 11:20:15 Mon Dec 17 2012 ICT

SVN ID: $Id$
```

38.6.473 sclist/combine

[sclist] [Methods]

DESCRIPTION:

```
Combine the contents of two given sclist objects into one list.  NB This
changes the data list of a clone of the first argument by appending a
copy of the data list of the second argument i.e. it creates a wholly new
sclist object which it then returns.

NB see also the combine-all function (utilities.lsp)
```

ARGUMENTS:

```
- A first sclist object.
- A second sclist object.
```

RETURN VALUE:

Returns an sclist object.

EXAMPLE:

```
;; Combine the contents of two sclist objects to make a new one
(let ((scl1 (make-sclist '(0 1 2 3 4)))
      (scl2 (make-sclist '(5 6 7 8 9))))
  (combine scl1 scl2))

=>
SCLIST: sclist-length: 10, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (0 1 2 3 4 5 6 7 8 9)
```

SYNOPSIS:

```
(defmethod combine ((scl1 sclist) (scl2 sclist))
```

38.6.474 sclist/get-last

[sclist] [Methods]

DESCRIPTION:

Returns the last element in the data list of a given sclist.

ARGUMENTS:

- An sclist.

RETURN VALUE:

The last object in the data list.

EXAMPLE:

```
(let ((al (make-assoc-list 'test '((jim beam)
                                   (four roses)
                                   (wild turkey)))))
  (get-last al))

=>
NAMED-OBJECT: id: WILD, tag: NIL,
data TURKEY
```

SYNOPSIS:

```
(defmethod get-last ((scl sclist))
```

38.6.475 sclist/get-nth

[sclist] [Methods]

DESCRIPTION:

```
 Get the nth element (zero-based) of data in a given sclist object.
```

ARGUMENTS:

```
 - An index integer.
 - An sclist object.
```

RETURN VALUE:

```
 Returns the item at index n within the given sclist object.

 Returns NIL and prints a warning if the specified index is greater than the
 number of items in the given list (minus 1).
```

EXAMPLE:

```
;; Get the 3th item from the given sclist object
(let ((scl (make-sclist '(cat dog cow pig sheep))))
  (get-nth 3 scl))

=> PIG

;; Returns NIL and prints a warning when the specified index is beyond the
;; bounds of the given list
(let ((scl (make-sclist '(cat dog cow pig sheep))))
  (get-nth 31 scl))

=>
NIL
WARNING: sclist::sclist-check-bounds: Illegal list reference: 31
(length = 5) (sclist id = NIL)
```

SYNOPSIS:

```
(defmethod get-nth (index (i sclist))
```

38.6.476 sclist/intervals-mapper

[sclist] [Classes]

NAME:

```
intervals-mapper

File:             intervals-mapper.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  intervals-mapper

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of a scale object that can initialize its
                  complete pitch list from the interval structure of a
                  given list of notes.  Given the scale, it's a cinch to
                  generate note sequences based on note offset patterns
                  e.g.
                  (let ((s (make-intervals-mapper 'c0 '(d e gs as d ef g a
                                                        bf cs d ef gf)))
                        (pat '(-1 2 4  3 6 -2 -1 2 6  7 3 6 2)))
                    (loop for p in pat collect
                         (data (intervals-mapper-note s p 4))))
                  -> (A3 EF4 F4 E4 BF4 F3 A3 EF4 BF4 D5 E4 BF4 EF4)

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    August 3rd 2010 Edinburgh

$$ Last modified:  16:31:36 Sat Jan 10 2026 CET


SVN ID: $Id$
```

38.6.477 intervals-mapper/get-pitch-symbols

[intervals-mapper] [Methods]

DESCRIPTION:

```
Get the pitches contained in a given intervals-mapper object returned as a
list of note-name symbols.
```

ARGUMENTS:

- An intervals-mapper object.

RETURN VALUE:

A list of note-name pitch symbols.

EXAMPLE:

```
(let ((im (make-intervals-mapper 'c0 '(d e gs as d ef g a bf cs d ef gf))))
  (get-pitch-symbols im))

=> (C0 D0 FS0 AF0 C1 CS1 F1 G1 AF1 B1 C2 CS2 E2 FS2 BF2 C3 E3 F3 A3 B3 C4 EF4
    E4 F4 AF4 BF4 D5 E5 AF5 A5 CS6 EF6 E6 G6 AF6 A6 C7 D7 FS7 AF7 C8 CS8 F8 G8
    AF8 B8 C9 CS9 E9 FS9)
```

SYNOPSIS:

```
(defmethod get-pitch-symbols ((im intervals-mapper) &optional ignore)
```

38.6.478 intervals-mapper/get-scale

[intervals-mapper] [Methods]

DESCRIPTION:

Create a scale (sequence of pitches) beginning with the specified starting note and extending up to MIDI note 119 (by default: settable in the highest slot) by cycling through the interval structure of the STEPS slot.

The scale will only repeat at octaves if the interval structure of the list of pitches passed at initialisation creates that result.

NB: This method is usually only called automatically at initialisation.

ARGUMENTS:

- An intervals mapper object.
- A note-name pitch symbol (e.g. 'c0) that is the pitch on which to begin the scale, and which is to be stored in the TONIC slot.

RETURN VALUE:

A list of the pitch objects in the new scale. These are also stored in the SCALE-PITCHES slot.

EXAMPLE:

```
(let ((im (make-intervals-mapper 'c0 '(d e gs as d ef g a bf cs d ef gf))))
  (pitch-list-to-symbols (get-scale im 'd4)))

=> (D4 E4 AF4 BF4 D5 EF5 G5 A5 BF5 CS6 D6 EF6 FS6 AF6 C7 D7 FS7 G7 B7 CS8 D8 F8
    FS8 G8 BF8 C9 E9 FS9)
```

SYNOPSIS:

```
(defmethod get-scale ((im intervals-mapper) start-note)
```

38.6.479 intervals-mapper/get-steps

[intervals-mapper] [Methods]

DESCRIPTION:

```
Extract the interval structure of the list of note-name pitch symbols
passed as the data to the instance of the intervals-mapper object upon
initialization. The interval structure is returned as a list of semitone
values.
```

ARGUMENTS:

```
- An intervals-mapper object.
```

RETURN VALUE:

```
A list of integers that are the numbers of semitones between each
consecutive pitch in the original data list.
```

EXAMPLE:

```
(let ((im (make-intervals-mapper 'c0 '(d e gs as d ef g a bf cs d ef gf))))
  (get-steps im))

=> (2 4 2 4 1 4 2 1 3 1 1 3)
```

SYNOPSIS:

```
(defmethod get-steps ((im intervals-mapper))
```

38.6.480 intervals-mapper/intervals-mapper-degree

[intervals-mapper] [Methods]

DATE:

14-Aug-2010

DESCRIPTION:

Return the scale degree number of a specified pitch class in relation to a specified octave within the given intervals-mapper object.

To determine the scale degree number, this method begins at the first pitch >= C in the specified octave and passes consecutively through each subsequent pitch in the interval-mapper object's full scale, counting each step until it first encounters the pitch class of the specified pitch.

If there are no more instances of the specified pitch class, the method returns NIL.

The method takes as its pitch class a pitch object, which includes an octave indicator. For the purposes of this method, solely the pitch-class name is extracted from the pitch object.

ARGUMENTS:

- An intervals-mapper object.
- An instance of a pitch object whose pitch class is being sought.
- An integer that is the octave in relationship to which the scale degree is sought.

OPTIONAL ARGUMENTS:

- T or NIL to indication whether to return the position of the found pitch object within the complete scale list of the given intervals-mapper object rather than the scale degree. T = return the position. Default = NIL.

RETURN VALUE:

Returns an integer that is either the scale degree of the specified pitch class in relation to the specified octave, counting from 1, or the position of the found pitch object within the interval-mapper object's complete scale.

Returns NIL if no instances of the specified pitch class are found.

EXAMPLE:

```
;;; The desired pitch class BF is found within the specified octave
(let ((im (make-intervals-mapper 'c0 '(d e gs as d ef g a bf cs d ef gf))))
  (intervals-mapper-degree im (make-pitch 'bf4) 4))

=> 6

;;; The desired pitch class B is found outside of the specified octave
(let ((im (make-intervals-mapper 'c0 '(d e gs as d ef g a bf cs d ef gf))))
  (intervals-mapper-degree im (make-pitch 'b4) 5))

=> 20

;;; Return the position instead
(let ((im (make-intervals-mapper 'c0 '(d e gs as d ef g a bf cs d ef gf))))
  (intervals-mapper-degree im (make-pitch 'b4) 5 t))

=> 45

;;; The desired pitch class BF is not present in any of the octaves beginning
;;; including and above the specified octave

(let ((im (make-intervals-mapper 'c0 '(d e gs as d ef g a bf cs d ef gf))))
  (intervals-mapper-degree im (make-pitch 'bf4) 5))

=> NIL
```

SYNOPSIS:

```
(defmethod intervals-mapper-degree ((im intervals-mapper) pitch octave
                                    &optional return-nth)
```

38.6.481 intervals-mapper/intervals-mapper-note

[intervals-mapper] [Methods]

DESCRIPTION:

```
Get the pitch object that constitutes the specified scale degree of the
specified octave within an intervals-mapper object; or, if keyword argument
<nth> is set to T, get the position of this pitch object within the full
range of the intervals-mapper object's complete scale.

As there is no concept of a tonic that repeats at octaves, degree 1 in any
```

given octave is simply the first note >= the C in that octave.

If a note-name pitch symbol is given for the keyword argument <tonic>, the given intervals-mapper object's TONIC slot will be changed, and its scale-pitches will be recalculated accordingly before getting the pitch.

ARGUMENTS:

- An intervals-mapper object.
- An integer that is the scale degree (1-based) of the desired pitch within the specified octave, counting from the first note of the scale above or on the C in that octave. If this number is higher than the number of pitches in the span of an octave, a pitch from a higher octave will accordingly be returned. Similarly, a negative number can also be given here to indicate that the pitch is to be collected from that many degrees below the specified octave.
- An integer that indicates the octave from which the pitch is to be returned (e.g. 4 for the octave starting on middle C, 5 for the octave starting on the C above that etc.)

OPTIONAL ARGUMENTS:

keyword arguments
- :tonic. NIL or a note-name pitch symbol that is the new starting note for above which the intervals-mapper scale is to be re-mapped before the pitch is returned. This will usually be the lowest pitch on which the scale should start. If NIL, no changes will be made to the object's existing scale pitches before returning the desired pitch. Default = NIL.
- :nth. T or NIL to indicate whether instead of returning the pitch object itself, the method should return an integer that is the position of that pitch within the full span of the interval-mapper object's complete scale. T = return the position. Default = NIL.

RETURN VALUE:

A pitch object by default, or its position in the scale list if the :nth argument is set to T.

EXAMPLE:

```
;;; Returns a pitch object by default
(let ((im (make-intervals-mapper 'c0 '(d e gs as d ef g a bf cs d ef gf))))
  (intervals-mapper-note im 3 4))

=>
```

```
PITCH: frequency: 329.628, midi-note: 64, midi-channel: 0
       pitch-bend: 0.0
       degree: 128, data-consistent: T, white-note: E4
       nearest-chromatic: E4
       src: 1.2599211, src-ref-pitch: C4, score-note: E4
       qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
       micro-tone: NIL,
       sharp: NIL, flat: NIL, natural: T,
       octave: 4, c5ths: 0, no-8ve: E, no-8ve-no-acc: E
       show-accidental: T, white-degree: 30,
       accidental: N,
       accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: E4, tag: NIL,
data: E4

;;; Used with negative degree numbers
(let ((im (make-intervals-mapper 'c0 '(d e gs as d ef g a bf cs d ef gf))))
  (data (intervals-mapper-note im -3 2)))

=> F1

;;; Use with a new tonic and setting nth to T to return the position
(let ((im (make-intervals-mapper 'c0 '(d e gs as d ef g a bf cs d ef gf))))
  (intervals-mapper-note im 11 5 :tonic 'd1 :nth t))

=> 29
```

SYNOPSIS:

```
(defmethod intervals-mapper-note ((im intervals-mapper) degree octave
                                  &key tonic nth)
```

38.6.482 intervals-mapper/make-intervals-mapper

[intervals-mapper] [Functions]

DESCRIPTION:

```
Returns an intervals-mapper object starting with the specified pitch
('tonic') and using the interval structure of the specified list of
pitches, creating a complete pitch list from the interval structure of list
of pitches.

NB The pitches specified aren't necessarily used in the resulting complete
```

```
    pitch list; rather, their interval structure is mapped above the
    specified starting pitch and repeated upwards until the method reaches
    MIDI note 127.

 NB The scale will only repeat at octaves if that is the interval structure
    of the list of pitches it is passed.
```

ARGUMENTS:

```
 - A note-name pitch symbol that is the starting pitch, (e.g. 'c0).
 - A list of note-name pitch symbols that provides the interval structure
   for the resulting scale.
```

RETURN VALUE:

```
 An intervals-mapper object.
```

EXAMPLE:

```
;;; A scale without repeating octaves:
(make-intervals-mapper 'c0 '(d e gs as d ef g a bf cs d ef gf))

=>
INTERVALS-MAPPER: steps: (2 4 2 4 1 4 2 1 3 1 1 3),
scale-pitches (pitch objects): (C0 D0 FS0 AF0 C1 CS1 F1 G1 AF1 B1 C2 CS2 E2 FS2
                                BF2 C3 E3 F3 A3 B3 C4 EF4 E4 F4 AF4 BF4 D5 E5
                                AF5 A5 CS6 EF6 E6 G6 AF6 A6 C7 D7 FS7 AF7 C8
                                CS8 F8 G8 AF8 B8 C9 CS9 E9 FS9)
tonic: C0
SCLIST: sclist-length: -1, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (D E GS AS D EF G A BF CS D EF GF)
```

SYNOPSIS:

```
(defun make-intervals-mapper (tonic notes &optional (highest 119))
```

38.6.483 sclist/limits

[sclist] [Methods]

DATE:

```
 January 12th 2019, Heidhausen
```

DESCRIPTION:

```
Get the lowest and highest values of the list
```

ARGUMENTS:

```
- an sclist object
```

RETURN VALUE:

```
two values: the minimum and maximum values in the list
```

SYNOPSIS:

```
(defmethod limits ((scl sclist))
```

38.6.484 sclist/make-sclist

[sclist] [Functions]

DESCRIPTION:

```
Create an sclist object with the specified list.
```

ARGUMENTS:

```
- A list of numbers or symbols.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :id. A symbol that will be the ID of the given sclist object.
  Default = NIL.
- :bounds-alert. T or NIL to indicate whether a warning should be issued
  when a request is given to set or get an out-of-bounds element (i.e. not
  enough elements in list). T = print warning. Default = NIL.
- :copy. T or NIL to indicate whether the data in the list should be copied
  (any slippery-chicken class instances will be cloned), with subsequent
  modifications being applied to the copy. T = copy. Default = T.
```

RETURN VALUE:

```
Returns an sclist object.
```

EXAMPLE:

```
;; Create a simple object with just a list of numbers
(make-sclist '(1 2 3 4 5 6 7))

=>
SCLIST: sclist-length: 7, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (1 2 3 4 5 6 7)

;; Create the same object and assign an ID to it
(make-sclist '(1 2 3 4 5 6 7) :id 'number-list)

=>
SCLIST: sclist-length: 7, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NUMBER-LIST, tag: NIL,
data: (1 2 3 4 5 6 7)
```

SYNOPSIS:

```
(defun make-sclist (list &key (id nil) (bounds-alert t) (copy t))
```

38.6.485 sclist/max-items

[sclist] [Methods]

DATE:

```
September 21st 2018, Heidhausen
```

DESCRIPTION:

```
A destructive method to reduce the number of items in the list to a maximum
of the 2nd argument. The remaining elements can be from either the beginning
of the list, the middle, or the end. If the list has fewer then or the same
number of elements as the 2nd argument, the list will remain unchanged.
```

ARGUMENTS:

```
- the sclist object
- the number of elements the list should be reduced to (integer, >= 0)
```

OPTIONAL ARGUMENTS:

```
- a symbol ('start 'middle or 'end) to indicate which elements remain. See
  examples below. Default = 'start
```

RETURN VALUE:

```
 the (potentially) modified sclist object (NB this method is destructive!)
```

EXAMPLE:

```
(max-items (make-sclist '(1 2 3 4 5 6 7)) 3)
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL,
                     this: NIL,
                     next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (1 2 3)
**************

(max-items (make-sclist '(1 2 3 4 5 6 7)) 3 'end)


SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL,
                     this: NIL,
                     next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (5 6 7)
**************

(max-items (make-sclist '(1 2 3 4 5 6 7)) 3 'middle)
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL,
                     this: NIL,
                     next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (3 4 5)
**************
```

SYNOPSIS:

```
(defmethod max-items ((scl sclist) max &optional (from 'start))
```

38.6.486 sclist/new-limits

[sclist] [Methods]

DATE:

```
 January 12th 2019, Heidhausen
```

DESCRIPTION:

```
Rescale the numbers in the list to be within a new range.
```

ARGUMENTS:

```
- an sclist object
```

OPTIONAL ARGUMENTS:

```
- the new minimum value. Default = 0.0
- the new maximum value. Default = 100.0
```

RETURN VALUE:

```
the new data list
```

EXAMPLE:

```
(new-limits (make-sclist '(-1.4 5 6 2.3 7.3 -2.1)))
-->
(7.4468083 75.53192 86.17022 46.80851 100.0 0.0)
```

SYNOPSIS:

```
(defmethod new-limits ((scl sclist) &optional (new-min 0.0) (new-max 100.0))
```

38.6.487 sclist/pitch-seq

[sclist] [Classes]

NAME:

```
pitch-seq

File:             pitch-seq.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  pitch-seq

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)
```

Purpose: Implementation of the pitch-seq class. This describes the pitch curves for a given rhythmic sequence. These are normally simple lists of notes indicating pitch height (and later mapped onto sets); chords are indicated by placing a number in parentheses.

Author: Michael Edwards: m@michael-edwards.org

Creation date: 19th February 2001

$$ Last modified: 19:34:19 Wed Apr 9 2025 CEST

SVN ID: Id

38.6.488 pitch-seq/get-notes

[pitch-seq] [Methods]

DESCRIPTION:

This gets notes from the sets, limiting the notes used to the range of the instrument and any other ranges defined in the slippery-chicken class. If either the instrument or set are missing it just gets the relative pitches we use to display a pitch sequence.

limit-high and limit-low are pitch objects. They are extra range definers that are given to the slippery-chicken object to control the pitch curve of an instrument over the duration of the whole piece. They always refer to sounding pitches.

The order of operations for selecting pitches are as follows:

1) Limit the set object to the instrument's range.

2) Remove the notes that have already been selected for other instruments. This is where the slippery-chicken slot :instrument-hierarchy plays an important role. This can be skipped if the <avoid-used-notes> argument is nil.

3) If there is a subset with the same ID as the ID slot for the player, use only those pitches common to that subset and those in step 2. If not, try again using the subset-id of the instrument.

4) If the ratio between the number of pitches now available and the number of different numbers in the pitch-seq is less than the slippery-chicken slot pitch-seq-index-scaler-min*, add notes from those used by other

instruments until there are enough; the lowest number in the pitch-seq will now select the lowest pitch in the set that is in the instrument's range.

If however there are enough pitches without adding pitches already used by other instruments, then where in the available pitches the lowest number of the pitch-seq will be placed depends on whether the prefers-notes slot of the instrument has been set to be high or low. If high, then the highest number in the pitch-seq will result in the highest pitch in the available pitches that is in the instrument's range. If low, then the lowest number in the pitch-seq will result in the lowest pitch in the available pitches that is in the instrument's range. If the user hasn't set this slot, then the range of the pitch-seq will correspond to the middle of the available pitches.

There are two caveats here if the instrument's prefers-notes slot is NIL: 1) If the lowest number in the pitch-seq is 5 or higher, this will have the same effect as the prefers-notes slot being high. Similarly, if the lowest number is 1, it will have the same effect as the prefers-notes slot being low. These two numbers (5 and 1) are actually global slippery chicken configuration data: (get-sc-config pitch-seq-lowest-equals-prefers-high) and (get-sc-config pitch-seq-lowest-equals-prefers-low) so can be set using the set-sc-config function.

* The question as to how many pitches are enough pitches before adding used notes is determined by the pitch-seq-index-scaler-min argument, which is by default 0.5 (in the slippery-chicken slot that's usually used and passed to this method). As the pitch-seq notes must be offset and scaled before they can be used as indices, there's a minimum scaler that's considered acceptable; anything below this would result in more notes being added.

5) If at this point there are no available pitches, the function will by default trigger an error and exit (see however pitch-seq-no-pitches-error in globals.lsp). This could happen if the value of set-limits, both high and low, took the available pitches outside of the instrument's range, for instance.

6) The pitch-seq numbers are now offset and scaled, then rounded in order to use them as indices into the pitch list. If a number is in parentheses then this is where the instrument's chord function would be called. As notes are selected, the set marks them as used for the next time around. Also, there's an attempt to avoid melodic octaves on adjacent notes; however, if the set is full of octaves this won't be possible; in that case a warning will be issued and the octave will be

used.

ARGUMENTS:

- A pitch-seq object.
- An instrument object.
- A player object.
- An sc-set object.
- A hint pitch (ignored for now).
- A pitch-object defining the highest possible note.
- A pitch-object defining the lowest possible note.
- The sequence number (for diagnostics).
- The last note of the previous sequence, as a pitch object.
- The lowest scaler that will be accepted before adding notes from those used; i.e., if the pitch-seq needs 6 notes and only 3 are available, there would be note repetition, but as this would create a scaler of 0.5, that would be acceptable
- Whether to avoid lines jumping an octave in either direction (passed by the slippery chicken slot).
- Whether to remove notes already chosen for other instruments before selecting notes for this one.

RETURN VALUE:

Returns the list of pitch objects that forms the notes slot of the given pitch-seq

SYNOPSIS:

```
(defmethod get-notes ((ps pitch-seq) instrument player set hint-pitch limit-high
                      limit-low seq-num last-note-previous-seq
                      pitch-seq-index-scaler-min avoid-melodic-octaves
                      avoid-used-notes)
```

38.6.489 pitch-seq/invert

[pitch-seq] [Methods]

DESCRIPTION:

Invert the pitch sequence contour attached to a given pitch-seq object. The inversion uses only the same numbers from the original pitch contour list.

ARGUMENTS:

- A pitch-seq object.

RETURN VALUE:

A pitch-seq object.

EXAMPLE:

```
(let ((ps (make-pitch-seq '(pseq1 (1 2 1 3 4 7)))))
  (data (invert ps)))

=> (7 4 7 3 2 1)
```

SYNOPSIS:

```
(defmethod invert ((ps pitch-seq) &optional ignore1 ignore2)
```

38.6.490 pitch-seq/make-pitch-seq

[pitch-seq] [Functions]

DESCRIPTION:

Create a pitch-seq object.

This function can be either called with one argument, consisting of a two-item list, in which the first item is the pitch-seq ID and the second is a list of numbers representing the pitch curve of the intended pitch sequence; or it can be created with two arguments, the first of which being the list of numbers representing the pitch curve and the second being the pitch-seq's ID.

NB We can assign a pitch-seq exclusively to particular instruments in the ensemble simply by passing their names as symbols along with the curve data. See below for an example.

ARGUMENTS:

- A two-item list, of which the first item is a symbol to be used as the object's ID, and the second is a list of integers representing the general contour of the pitch sequence.

OPTIONAL ARGUMENTS:

- If the optional argument format is used, the first argument is to be
  a list of numbers representing the general contour of the pitch sequence,
  and the second is to be a symbol for the pitch-seq object's ID.

RETURN VALUE:

- A pitch-seq object.

EXAMPLE:

```
;; The first creation option is using one argument that is a two-item list,
;; whereby the first item is a symbol to be used as the pitch-seq object's ID
;; and the second is a list of numbers representing the general contour of the
;; pitch sequence.
(make-pitch-seq '(pseq1 (1 2 1 1 3)))

=>
PITCH-SEQ: notes: NIL
highest: 3
lowest: 1
original-data: (1 2 1 1 3)
user-id: T
instruments: NIL
relative-notes: (not printed for sake of brevity)
relative-notes-length: 25
SCLIST: sclist-length: 5, bounds-alert: T, copy: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: PSEQ1, tag: NIL,
data: (1 2 1 1 3)

;; The second creation option uses two arguments, the first of which is a list
;; of numbers representing the general contour of the pitch sequence, the
;; second of which is a symbol which will be used as the pith-seq object's ID.
(make-pitch-seq '(2 1 1 3 1) 'pseq2)

=>
PITCH-SEQ: notes: NIL
highest: 3
lowest: 1
original-data: (2 1 1 3 1)
user-id: NIL
instruments: NIL
relative-notes: (not printed for sake of brevity)
relative-notes-length: 25
SCLIST: sclist-length: 5, bounds-alert: T, copy: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
```

```
NAMED-OBJECT: id: PSEQ2, tag: NIL,
data: (2 1 1 3 1)

;; An example assigning a pitch-seq only to specific instruments: ; ;
(make-pitch-seq '((1 2 1 1 3) violin flute) 'ps1))
```

SYNOPSIS:

```
(defun make-pitch-seq (id-data &optional (id nil))
```

38.6.491 sclist/plot

[sclist] [Methods]

DESCRIPTION:

Create text and data files suitable for plotting the numbers in the data (or other) slot with gnuplot. The file name should be given without extension, as the method will create a .txt and a .data file, for the command and data files respectively. It is assumed that the data to be plotted is a list of numbers. An error will be triggered if not.

NB gnuplot's 'postscript terminal' is used and to my knowledge there is no pdf terminal, so if you want a pdf you'll have to use something like ps2pdf

The user must then call gnuplot in a terminal, in a manner such as "gnuplot sclist.txt; open sclist.ps".

The method will create files that draw data points connected by lines by default.

ARGUMENTS:

- An sclist object.
- A string that is the directory path and base file name (without extension) of the files to create.

OPTIONAL ARGUMENTS:

- The slot to plot. By default this will be the data slot, but derived classes may want to plot a different list slot, e.g. the kernels slot of the popcorn class.
- T or NIL to indicate whether to connect points by lines. T = draw lines. Default = T.

RETURN VALUE:

```
 Returns the number of points plotted.
```

EXAMPLE:

```
(let ((ppcn (make-popcorn '(0.01 0.02) :min-spike 3.0 :max-spike 5.0)))
  (fit-to-length ppcn 100)
  (plot ppcn "/tmp/ppcn" 'kernels))

then in a terminal:
gnuplot ppcn.txt; ps2pdf ppcn.ps

this will create the postscript file ppcn.ps
```

SYNOPSIS:

```
(defmethod plot ((scl sclist) file &optional (slot 'data) (lines t))
```

38.6.492 sclist/reaper-file

[sclist] [Classes]

38.6.493 sclist/reaper-track

[sclist] [Classes]

38.6.494 sclist/rthm-seq

[sclist] [Classes]

NAME:

```
 rthm-seq

 File:             rthm-seq.lsp

 Class Hierarchy:  named-object -> linked-named-object -> sclist -> rthm-seq

 Version:          1.1.0

 Project:          slippery chicken (algorithmic composition)

 Purpose:          Implementation of the rthm-seq class which holds the bars
                   and rhythms of a sequence (multiple bars).  This will
                   generally be stored in a rthm-seq-palette and referenced
```

```
later in the rthm-seq-map.

The data used to create such an object will look
something like:

(rthm1 ((((2 4) q (q))
          (s x 4 (e) e)
          ((3 8) (e) e (e)))
        :pitch-seq-palette '((psp1 (1 2 1 2 3 2 1))
                             (psp2 (3 2 4 6 1 5 7))
                             (psp3 (2 3 4 1 3 4 5)))))

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    14th February 2001

$$ Last modified:  17:02:05 Fri Oct 10 2025 CEST

SVN ID: $Id$
```

38.6.495 rthm-seq/add-bar

[rthm-seq] [Methods]

DESCRIPTION:

```
Add a rthm-seq-bar object to the end of a given rthm-seq object.

NB: If the rthm-seq-bar object is added without specifying a
    pitch-seq-palette, the method automatically adds data to the existing
    pitch-seq-palette.
```

ARGUMENTS:

```
- A rhtm-seq object.
- A rthm-seq-bar object.
```

OPTIONAL ARGUMENTS:

```
- A pitch-seq-palette.
```

RETURN VALUE:

```
Returns the new value of the DURATION slot of the given rthm-seq object.
```

EXAMPLE:

```
;; Returns the new value of the DURATION slot
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                           ((e) q (e))
                           ((3 8) s s e. s))
                          :pitch-seq-palette ((1 2 3 1 1 2 3 4))))))
  (add-bar rs (make-rthm-seq-bar '((5 8) e e+32 s. +q))))

=> 10.5

;; Apply the method and print the rhythms objects of the given rthm-seq object
;; to see the changes
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                           ((e) q (e))
                           ((3 8) s s e. s))
                          :pitch-seq-palette ((1 2 3 1 1 2 3 4))))))
  (add-bar rs (make-rthm-seq-bar '((5 8) e e+32 s. +q)))
  (print-simple rs))

=>
rthm-seq NIL
(2 4): note Q, note E, note S, note S,
(2 4): rest E, note Q, rest E,
(3 8): note S, note S, note E., note S,
(5 8): note E, note E, note 32, note S., note Q,

;; Apply the method and print the DATA slot of the updated PITCH-SEQ-PALETTE
;; slot to see the new notes that have been automatically added
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                           ((e) q (e))
                           ((3 8) s s e. s))
                          :pitch-seq-palette ((1 2 3 1 1 2 3 4))))))
  (add-bar rs (make-rthm-seq-bar '((5 8) e e+32 s. +q)))
  (data (first (data (pitch-seq-palette rs)))))

=> (1 2 3 1 1 2 3 4 3 4 3)
```

SYNOPSIS:

```
(defmethod add-bar ((rs rthm-seq) (rsb rthm-seq-bar) &optional psp)
```

38.6.496 rthm-seq/change-pitches

[rthm-seq] [Methods]

DATE:

February 3rd 2018, Heidhausen

DESCRIPTION:

Change the pitches in a rthm-seq. The replacement pitches list can contain fewer pitches than in the seq: the method stops when it runs out. If there are more pitches than necessary, a list of unused pitches will be returned.

ARGUMENTS:

- the rthm-seq object
- a list of pitch objects or symbols as replacements
- the bar within the sequence to start at (counting from 1)
- the event within the start bar to start at (counting from 1)

OPTIONAL ARGUMENTS:

keyword arguments:
:written: whether to change the written or sounding pitches. Default = NIL = sounding pitches.

RETURN VALUE:

if all pitches given are used, then NIL, otherwise the unused pitches as a list.

SYNOPSIS:

```
(defmethod change-pitches ((rs rthm-seq)
                           pitch-list
                           (start-bar integer) ; don't use bar-holder method
                           start-event
                           &key written)
```

38.6.497 rthm-seq/chop

[rthm-seq] [Methods]

DESCRIPTION:

Applies the chop method to each rthm-seq-bar object contained in the given rthm-seq object (see rthm-seq-bar::chop for details), returning a list of rthm-seq objects, each of which contains just one of the rthm-seq-bar objects created with chop.

The chop method is the basis for slippery-chicken's feature of intra-phrasal looping.

NB: Since the chop method functions by comparing each beat of a given rthm-seq-bar object to the specified <chop-points> pattern for segmenting that beat, all rthm-seq-bar objects in the given rthm-seq object must be evenly divisible by the beat for which the pattern is defined. For example, if the <chop-points> argument defines a quarter-note, all bars in the given rthm-seq object must be evenly divisible by a quarter-note, and a rthm-seq object consisting of a 2/4, a 3/4 and a 3/8 bar would fail at the 3/8 bar with an error.

NB: The <unit> argument must be a duplet rhythmic value (i.e. 32, 's, 'e etc.) and cannot be a tuplet value (i.e. 'te 'fe etc.).

NB: In order for the resulting chopped rhythms to be parsable by LilyPond and CMN, there can be no tuplets (triplets etc.) among the rhythms to be chopped. Such rhythms will result in LilyPond and CMN errors. This has only minimal bearing on any MIDI files produced, however, and these can potentially be imported into notation software.

ARGUMENTS:

- A rthm-seq object.

OPTIONAL ARGUMENTS:

- <chop-points>. A list of integer pairs, each of which delineates a segment of the beat of the given rthm-seq-bar object measured in the rhythmic unit specified by the <unit> argument. See the documentation for rthm-seq-bar::chop for more details.
- <unit>. The rhythmic duration that serves as the unit of measurement for the chop points. Default = 's.
- <number-bars-first>. T or NIL. This argument helps in naming (and therefore debugging) the newly-created bars. If T, the bars in the original rthm-seq will be renumbered, starting from 1, and this will be reflected in the tag of the new bars. E.g. if T, a new bar's tag may be new-bar-from-rs1-b3-time-range-1.750-to-2.000, if NIL this would be new-bar-from-rs1-time-range-1.750-to-2.000. Default = T.

RETURN VALUE:

A list of rthm-seq objects.

EXAMPLE:

```
;; Create a rthm-seq with three bars, all having a quarter-note beat basis,
;; apply chop, and print-simple the resulting list of new rthm-seq-bar
;; objects. The rthm-seq numbers printed with this are the IDs of the rthm-seq
;; objects, not the bar-nums of the individual rthm-seq-bar objects.
(let* ((rs (make-rthm-seq '(seq1 ((((2 4) q e s s)
                                   ((e) q (e))
                                   (s s (e) e. s))
                                  :pitch-seq-palette ((1 2 3 4 5 6 7 8 9)
                                                      (9 8 7 6 5 4 3 2 1))))))
       (ch (chop rs
                 '((1 1) (1 2) (1 3) (1 4) (2 2) (2 3) (2 4) (3 3) (3 4) (4 4))
                 's)))
  (loop for rs-obj in ch do (print-simple rs-obj)))

=>
rthm-seq 1
(1 16): NIL S,
rthm-seq 2
(1 8): NIL E,
rthm-seq 3
(3 16): NIL E.,
rthm-seq 4
(1 4): NIL Q,
rthm-seq 5
(1 16): rest 16,
rthm-seq 6
(1 8): rest 8,
rthm-seq 7
(3 16): rest 16/3,
rthm-seq 8
(1 16): rest 16,
rthm-seq 9
(1 8): rest 8,
rthm-seq 10
(1 16): rest 16,
rthm-seq 11
(1 16): NIL S,
rthm-seq 12
(1 8): NIL E,
rthm-seq 13
(3 16): NIL E, NIL S,
rthm-seq 14
(1 4): NIL E, NIL S, NIL S,
rthm-seq 15
(1 16): rest 16,
rthm-seq 16
```

```
(1 8): rest S, NIL S,
rthm-seq 17
(3 16): rest S, NIL S, NIL S,
rthm-seq 18
(1 16): NIL S,
rthm-seq 19
(1 8): NIL S, NIL S,
rthm-seq 20
(1 16): NIL S,
rthm-seq 21
(1 16): rest 16,
rthm-seq 22
(1 8): rest 8,
rthm-seq 23
(3 16): rest E, NIL S,
rthm-seq 24
(1 4): rest E, NIL E,
rthm-seq 25
(1 16): rest 16,
rthm-seq 26
(1 8): rest S, NIL S,
rthm-seq 27
(3 16): rest S, NIL E,
rthm-seq 28
(1 16): NIL S,
rthm-seq 29
(1 8): NIL E,
rthm-seq 30
(1 16): rest 16,
rthm-seq 31
(1 16): rest 16,
rthm-seq 32
(1 8): rest 8,
rthm-seq 33
(3 16): rest 16/3,
rthm-seq 34
(1 4): rest 4,
rthm-seq 35
(1 16): rest 16,
rthm-seq 36
(1 8): rest 8,
rthm-seq 37
(3 16): rest 16/3,
rthm-seq 38
(1 16): rest 16,
rthm-seq 39
```

```
(1 8): rest 8,
rthm-seq 40
(1 16): rest 16,
rthm-seq 41
(1 16): NIL S,
rthm-seq 42
(1 8): NIL S, NIL S,
rthm-seq 43
(3 16): NIL S, NIL S, rest S,
rthm-seq 44
(1 4): NIL S, NIL S, rest E,
rthm-seq 45
(1 16): NIL S,
rthm-seq 46
(1 8): NIL S, rest S,
rthm-seq 47
(3 16): NIL S, rest E,
rthm-seq 48
(1 16): rest 16,
rthm-seq 49
(1 8): rest 8,
rthm-seq 50
(1 16): rest 16,
rthm-seq 51
(1 16): NIL S,
rthm-seq 52
(1 8): NIL E,
rthm-seq 53
(3 16): NIL E.,
rthm-seq 54
(1 4): NIL E., NIL S,
rthm-seq 55
(1 16): rest 16,
rthm-seq 56
(1 8): rest 8,
rthm-seq 57
(3 16): rest E, NIL S,
rthm-seq 58
(1 16): rest 16,
rthm-seq 59
(1 8): rest S, NIL S,
rthm-seq 60
(1 16): NIL S,

;; Attempting to apply the method to a rthm-seq object in which not all bars
;; have time-signatures that are divisible by the beat defined in the
```

```
;; <chop-points> argument will result in dropping into the debugger with an
;; error
(let* ((rs (make-rthm-seq '(seq1 ((((2 4) q e s s)
                                   ((e) q (e))
                                   ((3 8) (e) e. s))
                                  :pitch-seq-palette ((1 2 3 4 5 6 7)
                                                      (9 8 7 6 5 4 3))))))
       (ch (chop rs
                 '((1 1) (1 2) (1 3) (1 4) (2 2) (2 3) (2 4) (3 3) (3 4) (4 4))
                 's)))
  (loop for rs-obj in ch do (print-simple rs-obj)))

=>
rthm-seq-bar::get-beats: Can't find an exact beat of rhythms
   (dur: 0.75 beat-dur: 0.5)!
   [Condition of type SIMPLE-ERROR]
```

SYNOPSIS:

```
(defmethod chop ((rs rthm-seq) &optional chop-points (unit 's)
                 (number-bars-first t))
```

38.6.498 rthm-seq/clear-ties-beg-end

[rthm-seq] [Methods]

DESCRIPTION:

```
Deletes ties to the first rhythm(s) of a rthm-seq, making them rests
instead of tied notes.  Also makes sure the last rhythm is not tied from
(into another sequence). Useful if you're making rthm-seqs from other
(larger) rthm-seqs or algorithmically.
```

ARGUMENTS:

```
- the rthm-seq object
```

RETURN VALUE:

```
T if any ties were cleared, NIL otherwise.
```

SYNOPSIS:

```
(defmethod clear-ties-beg-end ((rs rthm-seq))
```

38.6.499 rthm-seq/combine

[rthm-seq] [Methods]

DESCRIPTION:

```
Combine two rthm-seqs into one, updating slots for the new object, which is
a clone.

NB: The MARKS slot is ignored for now (it is as of yet

NB: This method sets the values of the individual slots but leaves the DATA
    slot untouched (for cases in which the user might want to see where the
    new data originated from, or otherwise use the old data somehow, such
    as in a new rthm-seq object).
```

ARGUMENTS:

```
- A first rthm-seq object.
- A second rthm-seq object.
```

RETURN VALUE:

```
- A rthm-seq object.
```

EXAMPLE:

```
;; The method returns a rthm-seq object
(let ((rs1 (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 1 1 2 3 4)))))
      (rs2 (make-rthm-seq '((((4 4) h+e (e) { 3 te te te })
                            ((5 8) e e+32 s. +q)
                            ((3 4) (q) q q))
                           :pitch-seq-palette ((1 2 3 4 1 2 3 1 2))))))
  (combine rs1 rs2))

=>
RTHM-SEQ: num-bars: 6
          num-rhythms: 25
          num-notes: 17
          num-score-notes: 21
          num-rests: 4
          duration: 15.0
          psp-inversions: NIL
```

```
          marks: NIL
          time-sigs-tag: NIL
          handled-first-note-tie: NIL
         (for brevity's sake, slots pitch-seq-palette and bars are not printed)
SCLIST: sclist-length: 6, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "NIL-NIL", tag: NIL,
data: ((((2 4) Q+E S S) ((E) Q (E)) ((3 8) S S E. S)) PITCH-SEQ-PALETTE
       ((1 2 3 1 1 2 3 4))
       (((4 4) H+E (E) { 3 TE TE TE }) ((5 8) E E+32 S. +Q) ((3 4) (Q) Q Q))
       PITCH-SEQ-PALETTE ((1 2 3 4 1 2 3 1 2)))

;; With the same combine call, print the collected contents of the BARS slot
;; and the PITCH-SEQ-PALETTE slot of the new rthm-seq object
(let* ((rs1 (make-rthm-seq '((((2 4) q+e s s)
                             ((e) q (e))
                             ((3 8) s s e. s))
                            :pitch-seq-palette ((1 2 3 1 1 2 3 4)))))
       (rs2 (make-rthm-seq '((((4 4) h+e (e) { 3 te te te })
                             ((5 8) e e+32 s. +q)
                             ((3 4) (q) q q))
                            :pitch-seq-palette ((1 2 3 4 1 2 3 1 2)))))
       (crs (combine rs1 rs2)))
  (print-simple crs)
  (print (data (get-first (pitch-seq-palette crs)))))

=>
rthm-seq NIL-NIL
(2 4): note Q, note E, note S, note S,
(2 4): rest E, note Q, rest E,
(3 8): note S, note S, note E., note S,
(4 4): note H, note E, rest E, note TE, note TE, note TE,
(5 8): note E, note E, note 32, note S., note Q,
(3 4): rest 4, note Q, note Q,
(1 2 3 1 1 2 3 4 1 2 3 4 1 2 3 1 2)
```

SYNOPSIS:

```
(defmethod combine ((rs1 rthm-seq) (rs2 rthm-seq))
```

38.6.500 rthm-seq/delete-marks

[rthm-seq] [Methods]

DESCRIPTION:

Delete all marks from the MARKS slot of the specified rthm-seq object and replace them with NIL.

ARGUMENTS:

- A rthm-seq object

RETURN VALUE:

NIL

EXAMPLE:

```
(let ((mrs (make-rthm-seq '(seq1 ((((2 4) q e (s) s))
                                  :pitch-seq-palette ((1 2 3))
                                  :marks (ff 1 a 1 pizz 1 ppp 2 s 2))))))
  (print (marks mrs))
  (delete-marks mrs)
  (print (marks mrs)))

=>
((FF 1) (A 1) (PIZZ 1) (PPP 2) (S 2))
NIL
```

SYNOPSIS:

```
(defmethod delete-marks ((rs rthm-seq))
```

38.6.501 rthm-seq/filter

[rthm-seq] [Methods]

DATE:

February 6th 2024

DESCRIPTION:

filter a rthm-seq for rhythm objects that have certain characteristics, turning those which don't into rests. E.g. only accented notes could be retained.

NB The marks slot as given when the rthm-seq was initialised will be out of sync with the remaining notes and rests. Also the pitch-seq-palette will be deleted and re-initialised to contain a list of only 3s. Call create-psps perhaps on the rthm-seq-palette containing the rthm-seq to re-initialise to something useful.

ARGUMENTS:

```
- a rthm-seq object
- a function which takes a rhythm object and returns T if it should be
retained, otherwise it will be turned into a rest.
```

RETURN VALUE:

```
The modified rthm-seq.
```

EXAMPLE:

```
(let ((mrs (make-rthm-seq '(seq1 ((((4 4) q e (s) s - s x 4 - - e s s -))
                                  :marks (ff 1 a 1 3 6 8 10 pizz 1 ppp
                                          2 s 2))))))
  (filter mrs #'(lambda (r) (has-mark r 'a)))
  (print (= 5 (num-notes mrs)))
  (print (= 5 (num-rests mrs)))
  (print-simple mrs nil))

T
T
rthm-seq SEQ1
no player: bar -1: (4 4): note Q, rest E, rest S, note S, rest E, note S, rest|#
#|S, note E, rest S, note S,
```

SYNOPSIS:

```
(defmethod filter ((rs rthm-seq) test)
```

38.6.502 rthm-seq/get-bar

[rthm-seq] [Methods]

DESCRIPTION:

```
Get a specified rthm-seq-bar object from within a rthm-seq object.
```

ARGUMENTS:

```
- A rthm-seq object.
- An integer that is the 1-based number of the desired bar to return from
  within the given rthm-seq object.
```

RETURN VALUE:

Returns a rthm-seq-bar object.

EXAMPLE:

```
;;; Returns a rthm-seq-bar object
(let ((rs (make-rthm-seq '(seq1 ((((2 4) q e s s)
                                  ((e) q (e))
                                  ((3 8) s s e. s)))))))
  (get-bar rs 2))

=>
RTHM-SEQ-BAR: time-sig: 0 (2 4), time-sig-given: NIL, bar-num: -1,
              old-bar-nums: NIL, write-bar-num: NIL, start-time: -1.000,
              start-time-qtrs: -1.0, is-rest-bar: NIL, multi-bar-rest: NIL,
              show-rest: T, notes-needed: 1,
              tuplets: NIL, nudge-factor: 0.35, beams: NIL,
              current-time-sig: 6, write-time-sig: NIL, num-rests: 2,
              num-rhythms: 3, num-score-notes: 1, parent-start-end: NIL,
              missing-duration: NIL, bar-line-type: 0,
              player-section-ref: NIL, nth-seq: NIL, nth-bar: NIL,
              rehearsal-letter: NIL, all-time-sigs: (too long to print)
              sounding-duration: NIL,
              rhythms: (
[...]

(let ((rs (make-rthm-seq '(seq1 ((((2 4) q e s s)
                                  ((e) q (e))
                                  ((3 8) s s e. s)))))))
  (print-simple (get-bar rs 2)))

=> (2 4): rest E, note Q, rest E,
```

SYNOPSIS:

```
(defmethod get-bar ((rs rthm-seq) bar-num &optional ignore)
```

38.6.503 rthm-seq/get-first

[rthm-seq] [Methods]

DESCRIPTION:

Return the first rhythm/event object in the specified rthm-seq object.

ARGUMENTS:

```
 - A rthm-seq object.
```

RETURN VALUE:

```
 A rhythm/event object.
```

EXAMPLE:

```
(let ((mrs (make-rthm-seq '(seq1 ((((2 4) q e (s) s))
                                  :pitch-seq-palette ((1 2 3))
                                  :marks (ff 1 a 1 pizz 1 ppp 2 s 2))))))
  (get-first mrs))

=>
RHYTHM: value: 4.000, duration: 1.000, rq: 1, is-rest: NIL,
        is-whole-bar-rest: NIL,
        score-rthm: 4.0, undotted-value: 4, num-flags: 0, num-dots: 0,
        is-tied-to: NIL, is-tied-from: NIL, compound-duration: 1.000,
        is-grace-note: NIL, needs-new-note: T, beam: NIL, bracket: NIL,
        rqq-note: NIL, rqq-info: NIL, marks: (PIZZ A FF), marks-in-part: NIL,
        letter-value: 4, tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: Q, tag: NIL,
data: Q
```

SYNOPSIS:

```
(defmethod get-first ((rs rthm-seq))
```

38.6.504 rthm-seq/get-last

[rthm-seq] [Methods]

DESCRIPTION:

```
 Return the last rhythm/event object in the specified rthm-seq object.
```

ARGUMENTS:

```
 - A rthm-seq object.
```

RETURN VALUE:

```
 A rhythm/event object.
```

EXAMPLE:

```
(let ((mrs (make-rthm-seq '(seq1 ((((2 4) q e (s) s))
                                  :pitch-seq-palette ((1 2 3))
                                  :marks (ff 1 a 1 pizz 1 ppp 2 s 2))))))
  (get-last mrs))

=>
RHYTHM: value: 16.000, duration: 0.250, rq: 1/4, is-rest: NIL,
        is-whole-bar-rest: NIL,
        score-rthm: 16.0, undotted-value: 16, num-flags: 2, num-dots: 0,
        is-tied-to: NIL, is-tied-from: NIL, compound-duration: 0.250,
        is-grace-note: NIL, needs-new-note: T, beam: NIL, bracket: NIL,
        rqq-note: NIL, rqq-info: NIL, marks: NIL, marks-in-part: NIL,
        letter-value: 16, tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: S, tag: NIL,
data: S
```

SYNOPSIS:

```
(defmethod get-last ((rs rthm-seq))
```

38.6.505 rthm-seq/get-last-attack

[rthm-seq] [Methods]

DESCRIPTION:

```
Gets the rhythm object for the last note that needs an attack (i.e. not a
rest and not a tied note) in a given rthm-seq object.
```

ARGUMENTS:

```
- A rthm-seq object.
```

OPTIONAL ARGUMENTS:

```
- T or NIL indicating whether to print a warning message if the given index
  (minus 1) is greater than the number of attacks in the rthm-seq object
  (default = T). This is a carry-over argument from the get-nth-attack
  method called within the get-last-attack method and not likely to be
  needed for use with get-last-attack.
```

RETURN VALUE:

A rhythm object.

EXAMPLE:

```
;; Returns a rhythm object
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (get-last-attack rs))

=>
RHYTHM: value: 16.000, duration: 0.250, rq: 1/4, is-rest: NIL,
        score-rthm: 16.0f0, undotted-value: 16, num-flags: 2, num-dots: 0,
        is-tied-to: NIL, is-tied-from: NIL, compound-duration: 0.250,
        is-grace-note: NIL, needs-new-note: T, beam: NIL, bracket: NIL,
        rqq-note: NIL, rqq-info: NIL, marks: NIL, marks-in-part: NIL,
        letter-value: 16, tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: S, tag: NIL,
data: S
```

SYNOPSIS:

```
(defmethod get-last-attack ((rs rthm-seq) &optional (warn t))
```

38.6.506 rthm-seq/get-last-bar

[rthm-seq] [Methods]

DESCRIPTION:

Get the last rthm-seq-bar object of a given rthm-seq object.

ARGUMENTS:

- A rthm-seq object.

RETURN VALUE:

A rthm-seq-bar object.

EXAMPLE:

```
;;; The method returns a rthm-seq-bar object
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                           ((e) q (e))
                           ((3 8) s s e. s))
                          :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (get-last-bar rs))

=>
RTHM-SEQ-BAR: time-sig: 6 (3 8), time-sig-given: T, bar-num: -1,
[...]
data: ((3 8) S S E. S)
```

SYNOPSIS:

```
(defmethod get-last-bar ((rs rthm-seq))
```

38.6.507 rthm-seq/get-last-event

[rthm-seq] [Methods]

DESCRIPTION:

```
Get the last event object (or rhythm object) of a given rthm-seq-bar
object.
```

ARGUMENTS:

```
- A rthm-seq object.
```

RETURN VALUE:

```
Returns an event (or rhythm) object.
```

EXAMPLE:

```
;; The last event is a rhythm object
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                           ((e) q (e))
                           ((3 8) s s e. s))
                          :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (get-last-event rs))

=>
RHYTHM: value: 16.000, duration: 0.250, rq: 1/4, is-rest: NIL,
        score-rthm: 16.0f0, undotted-value: 16, num-flags: 2, num-dots: 0,
```

```
        is-tied-to: NIL, is-tied-from: NIL, compound-duration: 0.250,
        is-grace-note: NIL, needs-new-note: T, beam: NIL, bracket: NIL,
        rqq-note: NIL, rqq-info: NIL, marks: NIL, marks-in-part: NIL,
        letter-value: 16, tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: S, tag: NIL,
data: S

;; The last event is an event object
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                           ((e) q (e))
                           ((3 8) s s e. ,(make-event 'c4 's)))
                          :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (get-last-event rs))

=>
EVENT: start-time: NIL, end-time: NIL,
[...]
PITCH: frequency: 261.6255569458008, midi-note: 60, midi-channel: NIL
[...]
RHYTHM: value: 16.000, duration: 0.250, rq: 1/4, is-rest: NIL,
[...]
NAMED-OBJECT: id: S, tag: NIL,
data: S
```

SYNOPSIS:

```
(defmethod get-last-event ((rs rthm-seq))
```

38.6.508 rthm-seq/get-multipliers

[rthm-seq] [Methods]

DESCRIPTION:

```
Get a list of factors by which a specified rhythmic unit must be multiplied
in order to create the rhythms of a given rthm-seq object.

NB: The get-multipliers method determines durations in the source rhythmic
    material based on attacked notes only, so beginning ties will be
    ignored and rests following an attack will count the same as if the
    attacked note were tied to another note with the same duration as the
    rest. For this reason, the results returned by the method when applied
    to a rthm-seq object that contains multiple bars may differ from
    applying the method to multiple rthm-seqs with single bars, albeit with
    the same rhythms when seen as a group (see example below).
```

ARGUMENTS:

```
- A rthm-seq object.
- A rhythm unit, either as a number or a shorthand symbol (i.e. 's)
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether to round the results. T = round.
  Default = NIL. NB: Lisp always rounds to even numbers, meaning x.5 may
  sometimes round up and sometimes round down; thus (round 1.5) => 2, and
  (round 2.5) => 2.
```

RETURN VALUE:

```
A list of numbers.
```

EXAMPLE:

```
;;; By default the method returns the list of multipliers un-rounded
(let ((rs (make-rthm-seq '(seq1 ((((2 4) q e s s))
                                 :pitch-seq-palette ((1 2 3 4)))))))
  (get-multipliers rs 'e))

=> (2.0 1.0 0.5 0.5)

;; Setting the optional argument to T rounds the results before returning
(let ((rs (make-rthm-seq '(seq1 ((((2 4) q e s s))
                                 :pitch-seq-palette ((1 2 3 4)))))))
  (get-multipliers rs 'e t))

=> (2 1 0 0)

;;; Applying the method to the a multiple-bar rthm-seq object may return
;;; different results than applying the method to each of the bars contained
;;; within that rthm-seq object as individual one-bar rthm-seq objects, as the
;;; method measures the distances between attacked notes, regardless of ties
;;; and rests.
(let ((rs1 (make-rthm-seq '(seq1 ((((2 4) q +e. s))
                                  :pitch-seq-palette ((1 2))))))
      (rs2 (make-rthm-seq '(seq2 ((((2 4) (s) e (s) q))
                                  :pitch-seq-palette ((1 2))))))
      (rs3 (make-rthm-seq '(seq3 ((((2 4) +e. s { 3 (te) te te } ))
                                  :pitch-seq-palette ((1 2 3))))))
      (rs4 (make-rthm-seq '(seq4 ((((2 4) q +e. s)
                                   ((s) e (s) q)
```

```
                                  (+e. s { 3 (te) te te } ))
                                 :pitch-seq-palette ((1 2 3 4 5 6 7)))))))
  (print (get-multipliers rs1 'e))
  (print (get-multipliers rs2 'e))
  (print (get-multipliers rs3 'e))
  (print (get-multipliers rs4 'e)))

=>
(3.5 0.5)
(1.5 2.0)
(1.1666666666666665 0.6666666666666666 0.6666666666666666)
(3.5 1.0 1.5 3.5 1.1666666666666665 0.6666666666666666 0.6666666666666666)
```

SYNOPSIS:

```
(defmethod get-multipliers ((rs rthm-seq) rthm &optional round ignore)
```

38.6.509 rthm-seq/get-nth-attack

[rthm-seq] [Methods]

DESCRIPTION:

```
Gets the rhythm object for the nth note in a given rthm-seq object that
needs an attack, i.e. not a rest and not tied.
```

ARGUMENTS:

```
- The zero-based index number indicating which attack is sought.
- The given rthm-seq object in which to search.
```

OPTIONAL ARGUMENTS:

```
- T or NIL indicating whether to print an error message if the given index
  is greater than the number of attacks (minus 1) in the rthm-seq object
  (default = T).
```

RETURN VALUE:

```
A rhythm object.
```

EXAMPLE:

```
;; The method returns a rhythm object when successful
```

```
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (get-nth-attack 4 rs))

=>
RHYTHM: value: 16.000, duration: 0.250, rq: 1/4, is-rest: NIL,
        score-rthm: 16.0f0, undotted-value: 16, num-flags: 2, num-dots: 0,
        is-tied-to: NIL, is-tied-from: NIL, compound-duration: 0.250,
        is-grace-note: NIL, needs-new-note: T, beam: NIL, bracket: NIL,
        rqq-note: NIL, rqq-info: NIL, marks: NIL, marks-in-part: NIL,
        letter-value: 16, tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: S, tag: NIL,
data: S

;; By default, the method drops into the debugger with an error when the
;; specified index is greater than the number of items in the given rthm-seq
;; object.
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (get-nth-attack 11 rs))

=>
rthm-seq::get-nth-attack: Couldn't get attack with index 11
   [Condition of type SIMPLE-ERROR]

;; This error can be suppressed, simply returning NIL, by setting the optional
;; argument to NIL.
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (get-nth-attack 11 rs nil))

=> NIL, 0, NIL
```

SYNOPSIS:

```
(defmethod get-nth-attack (index (rs rthm-seq)
                           &optional (error t))
```

38.6.510 rthm-seq/get-nth-bar

[rthm-seq] [Methods]

DESCRIPTION:

```
Get the nth rthm-seq-bar object from a given rthm-seq object.
```

ARGUMENTS:

```
- A rthm-seq object.
- An index number (zero-based).
```

RETURN VALUE:

```
Returns a rthm-seq-bar object if successful.

Returns NIL and prints a warning if the specified index number is greater
than the number of rthm-seq-bar objects (minus one) in the given rthm-seq
object.
```

EXAMPLE:

```
;;; The method returns a rhtm-seq-bar object when successful
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                           ((e) q (e))
                           ((3 8) s s e. s))
                          :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (get-nth-bar 1 rs))

=>
RTHM-SEQ-BAR: time-sig: 0 (2 4), time-sig-given: NIL, bar-num: -1,
[...]
NAMED-OBJECT: id: "NIL-bar2", tag: NIL,
data: ((E) Q (E))

;; Returns a warning and prints NIL when the specified index number is greater
;; than the number of rthm-seq-bar objects in the given rthm-seq object
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                           ((e) q (e))
                           ((3 8) s s e. s))
                          :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (get-nth-bar 11 rs))

=> NIL
WARNING: rthm-seq::rthm-seq-check-bounds: Illegal list reference: 11
```

SYNOPSIS:

```
(defmethod get-nth-bar (nth (rs rthm-seq))
```

38.6.511 rthm-seq/get-nth-non-rest-rhythm

[rthm-seq] [Methods]

DESCRIPTION:

```
Get the nth non-rest rhythm object stored in the given rthm-seq object.
```

ARGUMENTS:

```
- The zero-based index number indicating which non-rest-rhythm is sought.
- The given rthm-seq object in which to search.
```

OPTIONAL ARGUMENTS:

```
- T or NIL indicating whether to print an error message if the given index
  is greater than the number of non-rest rhythms (minus 1) in given
  rthm-seq object.  (Default = T.)
```

RETURN VALUE:

```
A rhythm object.

Returns NIL if the given index is higher than the highest possible index of
non-rest rhythms in the given rthm-seq-bar object.
```

EXAMPLE:

```
;; The method returns a rhythm object when successful
(let ((rs (make-rthm-seq '((((2 4) q e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 4 1 1 2 3 4))))))
  (get-nth-non-rest-rhythm 4 rs))

=>
RHYTHM: value: 4.000, duration: 1.000, rq: 1, is-rest: NIL,
        score-rthm: 4.0f0, undotted-value: 4, num-flags: 0, num-dots: 0,
        is-tied-to: NIL, is-tied-from: NIL, compound-duration: 1.000,
        is-grace-note: NIL, needs-new-note: T, beam: NIL, bracket: NIL,
        rqq-note: NIL, rqq-info: NIL, marks: NIL, marks-in-part: NIL,
```

```
       letter-value: 4, tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: Q, tag: NIL,
data: Q

;; By default, the method drops into the debugger with an error when the
;; specified index is greater than the number of items in the given rthm-seq
;; object.
(let ((rs (make-rthm-seq '((((2 4) q e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 4 1 1 2 3 4))))))
  (get-nth-non-rest-rhythm 11 rs))

=>
rthm-seq::get-nth-non-rest-rhythm: Couldn't get non-rest rhythm with index 11
   [Condition of type SIMPLE-ERROR]

;; This error can be suppressed, simply returning NIL, by setting the optional
;; argument to NIL.
(let ((rs (make-rthm-seq '((((2 4) q e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 4 1 1 2 3 4))))))
  (get-nth-non-rest-rhythm 11 rs nil))

=> NIL
```

SYNOPSIS:

```
(defmethod get-nth-non-rest-rhythm (index (rs rthm-seq)
                                    &optional (error t))
```

38.6.512 rthm-seq/get-nth-rhythm

[rthm-seq] [Methods]

DESCRIPTION:

```
Gets the rhythm (or event) object for the nth note in a given rthm-seq
object.
```

ARGUMENTS:

```
- The zero-based index number indicating which attack is sought.
- The given rthm-seq object in which to search.
```

OPTIONAL ARGUMENTS:

```
 - T or NIL indicating whether to print an error message if the given index
   is greater than the number of attacks (minus 1) in the rthm-seq object
   (default = T).
```

RETURN VALUE:

```
 A rhythm or event object.
```

EXAMPLE:

```
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (get-nth-rhythm 4 rs))

=>

RHYTHM: value: 8.000, duration: 0.500, rq: 1/2, is-rest: T,
        score-rthm: 8.0f0, undotted-value: 8, num-flags: 1, num-dots: 0,
        is-tied-to: NIL, is-tied-from: NIL, compound-duration: 0.500,
        is-grace-note: NIL, needs-new-note: NIL, beam: NIL, bracket: NIL,
        rqq-note: NIL, rqq-info: NIL, marks: NIL, marks-in-part: NIL,
        letter-value: 8, tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: E, tag: NIL,
data: E
**************
```

SYNOPSIS:

```
(defmethod get-nth-rhythm (index (rs rthm-seq) &optional (error t))
```

38.6.513 rthm-seq/get-rhythm-list

[rthm-seq] [Methods]

DATE:

```
 February 10th 2022, Heidhausen
```

DESCRIPTION:

Sometimes (e.g. in reaper.lsp) you just want a rthm-seq's original rhythm data but not listed in bars, or with meter data, so that you can pass it to e.g. rhythm-list, i.e. take everything outside of rthm-seqs for handling outside of make-slippery-chicken

ARGUMENTS:

- a rthm-seq object

RETURN VALUE:

a list of rhythm symbols

EXAMPLE:

```
(let ((rs (make-rthm-seq
           '((((4 4) w) (+s (s) (e) (q) (h)) ((h) (s) e.+q) (+w)
               (+s (s) (e) (q) (h)) ((e) h..) (+w) (+h.. (e)) ()
               ((h) (q) (e.) s) (+e (e) (q) (h)) (s (s) (e) (q) (h))
              ((h) (q) (s) e (s)) ((q) h.) (+q. (e) (h)))))))
  (get-rhythm-list rs))
--->
(W +S (S) (E) (Q) (H) (H) (S) E.+Q +W +S (S) (E) (Q) (H) (E) H.. +W +H.. (E)
 (H) (Q) (E.) S +E (E) (Q) (H) S (S) (E) (Q) (H) (H) (Q) (S) E (S) (Q) H. +Q.
 (E) (H))
```

SYNOPSIS:

```
(defmethod get-rhythm-list ((rs rthm-seq))
```

38.6.514 rthm-seq/get-rhythms

[rthm-seq] [Methods]

DESCRIPTION:

Get the rhythm objects in a given rthm-seq object, contained in a list.

ARGUMENTS:

- A rthm-seq object.

RETURN VALUE:

A list.

EXAMPLE:

```
;; Returns a list of rhythm objects
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 1 1 2 3 4))))))
  (get-rhythms rs))

=>
(
RHYTHM: value: 4.000, duration: 1.000, rq: 1, is-rest: NIL,
[...]
RHYTHM: value: 8.000, duration: 0.500, rq: 1/2, is-rest: NIL,
[...]
RHYTHM: value: 16.000, duration: 0.250, rq: 1/4, is-rest: NIL,
[...]
RHYTHM: value: 16.000, duration: 0.250, rq: 1/4, is-rest: NIL,
[...]
RHYTHM: value: 8.000, duration: 0.500, rq: 1/2, is-rest: T,
[...]
RHYTHM: value: 4.000, duration: 1.000, rq: 1, is-rest: NIL,
[...]
RHYTHM: value: 8.000, duration: 0.500, rq: 1/2, is-rest: T,
[...]
RHYTHM: value: 16.000, duration: 0.250, rq: 1/4, is-rest: NIL,
[...]
RHYTHM: value: 16.000, duration: 0.250, rq: 1/4, is-rest: NIL,
[...]
RHYTHM: value: 5.333, duration: 0.750, rq: 3/4, is-rest: NIL,
[...]
RHYTHM: value: 16.000, duration: 0.250, rq: 1/4, is-rest: NIL,
[...]
)

;; Get just the rhythm labels from the same rthm-seq object
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 1 1 2 3 4))))))
  (loop for r in (get-rhythms rs) collect (data r)))

=> ("Q" "E" S S E Q E S S E. S)
```

SYNOPSIS:

```
(defmethod get-rhythms ((rs rthm-seq))
```

38.6.515 rthm-seq/get-time-sigs

[rthm-seq] [Methods]

DESCRIPTION:

```
Return a list of time-sig objects for each of the rthm-seq-bar objects in a
given rthm-seq object.

One time signature is returned for each rthm-seq-bar object, even if two or
more consecutive objects have the same time signature.

Optionally, this method can return a list of time signatures in list form
(e.g. ((2 4) (3 4)) etc.) rather than a list of time-sig objects.
```

ARGUMENTS:

```
- A rthm-seq object.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether to return the time signatures as time-sig
  objects or a list of two-item lists. T = time-sig objects. Default = T.
```

RETURN VALUE:

```
Returns a list of time-sig objects by default. Optionally a list of time
signatures as two-item lists can be returned instead.
```

EXAMPLE:

```
;; Return a list of time-sig objects, one for each rthm-seq-bar object even if
;; consecutive rthm-seq-bar objects have the same time signature
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 1 1 2 3 4))))))
  (get-time-sigs rs))

=> (
TIME-SIG: num: 2, denom: 4, duration: 2.0, compound: NIL, midi-clocks: 24, num-beats: 2
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
```

```
NAMED-OBJECT: id: "0204", tag: NIL,
data: (2 4)
**************


TIME-SIG: num: 2, denom: 4, duration: 2.0, compound: NIL, midi-clocks: 24, num-beats: 2
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "0204", tag: NIL,
data: (2 4)
**************


TIME-SIG: num: 3, denom: 8, duration: 1.5, compound: T, midi-clocks: 24, num-beats: 1
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "0308", tag: NIL,
data: (3 8)
**************
)

;; Return the same as a list of two-item lists instead
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 1 1 2 3 4))))))
  (get-time-sigs rs t))

=> ((2 4) (2 4) (3 8))
```

SYNOPSIS:

```
(defmethod get-time-sigs ((rs rthm-seq) &optional as-list)
```

38.6.516 rthm-seq/insert-bar

[rthm-seq] [Methods]

DESCRIPTION:

```
Insert a rthm-seq-bar object into the given rthm-seq object and
re-initialize it. If there's a pitch-seq/pitch-seq-palette given (list of
numbers, or list of lists), splice this in at the appropriate location.

NB: This method sets the values of the individual slots but leaves the DATA
```

slot untouched (for cases in which the user might want to see where the new data originated from, or otherwise use the old data somehow, such as in a new rthm-seq object).

ARGUMENTS:

- A rthm-seq object.
- A rthm-seq-bar object.
- A bar number (integer). This argument is the bar number of the bar to be inserted, relative to the rthm-seq and 1-based; e.g., if 3, then it will come before the present third bar.

OPTIONAL ARGUMENTS:

- A pitch-seq object.
- (three ignore arguments for sc-internal use only)

RETURN VALUE:

Returns T if successful.

Drops into the debugger with an error if the specified bar-number argument is greater than the number of rthm-seq-bar objects in the given rthm-seq.

EXAMPLE:

```
;; The method returns T when successful
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (insert-bar rs (make-rthm-seq-bar '((3 4) q. e e s s)) 3))

=> T

;; Create a rthm-seq object with three rthm-seq-bars and print the contents of
;; the NUM-BARS slot to confirm that it contains 3 objects. Insert a bar before
;; the third item and print the value of the NUM-BARS slot again to confirm
;; that there are now 4 objects. Use print-simple and get-nth-bar to confirm
;; that the 3rd object (with a zero-based index of 2) is indeed the one
;; inserted.

(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
```

```
                          :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (print (num-bars rs))
  (insert-bar rs (make-rthm-seq-bar '((3 4) q. e e s s)) 3)
  (print (num-bars rs))
  (print-simple (get-nth-bar 2 rs)))

=>
3
4
(3 4): note Q., note E, note E, note S, note S,

;; Attempting to insert a bar with an index number greater than the number of
;; objects currently in the rthm-seq object drops into the debugger with an
;; error
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (insert-bar rs (make-rthm-seq-bar '((3 4) q. e e s s)) 11))

=>
rthm-seq::insert-bar: only 3 bars in rthm-seq!
   [Condition of type SIMPLE-ERROR]

;; Inserting a rthm-seq-bar using the optional pitch-seq argument splices the
;; specified value of that argument into the existing pitch-seq-palette
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 1 1 2 3 4))))))
  (insert-bar rs (make-rthm-seq-bar '((3 4) q. e e s s)) 3 '((1 2 3 4 5)))
  (data (get-first (pitch-seq-palette rs))))

=> (1 2 3 1 1 2 3 4 5 1 2 3 4)
```

SYNOPSIS:

```
(defmethod insert-bar ((rs rthm-seq) (rsb rthm-seq-bar) bar-num
                       &optional pitch-seq ignore1 ignore2 ignore3)
```

38.6.517 rthm-seq/invert

[rthm-seq] [Methods]

DATE:

February 6th 2024

DESCRIPTION:

Turn all the rests in each rthm-seq-bar into notes and all the notes/chords into rests. NB this method is destructive.

ARGUMENTS:

- the rthm-seq object

OPTIONAL ARGUMENTS:

- T or NIL to call auto-beam on the result. Default = NIL.

RETURN VALUE:

The modified rthm-seq object

SYNOPSIS:

```
(defmethod invert ((rs rthm-seq) &optional auto-beam ignore)
```

38.6.518 rthm-seq/make-rhythms

[rthm-seq] [Functions]

DATE:

11 Feb 2010

DESCRIPTION:

Initialize a group of rhythms, taking advantage of rthm-seq's ability to add tuplet and beaming info.

ARGUMENTS:

- A list of rhythms equalling one full bar
- The time signature of that bar as a list (e.g (2 4))

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether to divide the resulting list into sublists, each of which are the equivalent of one beat long. Default = NIL.

RETURN VALUE:

- A list

EXAMPLE:

```
;; Apply the function and test that the result is a list
(let ((rs (make-rhythms '(q e s s) '(2 4))))
  (listp rs))

=> T

;; Apply the function and see that we've created a list with 4 elements
(let ((rs (make-rhythms '(q e s s) '(2 4))))
  (length rs))

=> 4

;; Apply the function with the optional split-into-beats argument set to T and
;; see that we now have two lists, each equalling one beat in combined
;; length. Print the data of the contents.
(let ((rs (make-rhythms '(q e s s) '(2 4) t)))
  (print (length rs))
  (print (loop for b in rs collect (length b)))
  (print (loop for b in rs
           collect (loop for r in b
                     collect (data r)))))

=>
2
(1 3)
((Q) (E S S))

;; Apply the function using beam indications then print the BEAM slots of the
;; individual rhythm objects contained in the result
(let ((rs (make-rhythms '(q - e s s -) '(2 4))))
  (loop for r in rs collect (beam r)))

=> (NIL 1 NIL 0)

;; Apply the function using tuplet indications then print the BRACKET slots of
;; the individual rhythms objects contained in the result
(let ((rs (make-rhythms '( { 3 te te te } - e s s -) '(2 4))))
  (loop for r in rs collect (bracket r)))

=> (((1 3)) (-1) (1) NIL NIL NIL)
```

SYNOPSIS:

```
(defun make-rhythms (bar time-sig &optional split-into-beats)
```

38.6.519 rthm-seq/make-rthm-seq

[rthm-seq] [Functions]

DESCRIPTION:

```
Creates a rthm-seq object from a list of at least bars and generally also a
list of pitch sequences.
```

ARGUMENTS:

```
- A list with the following items:
  - A symbol that will be used as the ID of the seq
  - Another list, containing two items:
    - A list of rthm-seq-bars and
    - A list of pitch-seqs attached to the :pitch-seq-palette accessor
```

OPTIONAL ARGUMENTS:

```
keyword argument
- :psp-inversions. T or NIL to indicate whether to also automatically
  generate and add inverted forms of the specified pitch-seq objects.
  T = generate and add. Default = NIL.
- :warn-time-sigs. T or NIL to indicate whether to issue a warning when a
  time-signature is given where it's not needed (i.e. two bars of 4/4)
```

RETURN VALUE:

```
Returns a rthm-seq object.
```

EXAMPLE:

```
;; Make a rthm-seq object with the ID seq1 that contains one 2/4 bar of
;; rhythms and one pitch sequence in the pitch-seq-palette
(make-rthm-seq '(seq1 ((((2 4) q e s s))
                       :pitch-seq-palette ((1 2 3 4)))))

=>
RTHM-SEQ: num-bars: 1
          num-rhythms: 4
          num-notes: 4
```

```
          num-score-notes: 4
          num-rests: 0
          duration: 2.0
          psp-inversions: NIL
          marks: NIL
          time-sigs-tag: NIL
          handled-first-note-tie: NIL
         (for brevity's sake, slots pitch-seq-palette and bars are not printed)
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: SEQ1, tag: NIL,
data: ((((2 4) Q E S S)) PITCH-SEQ-PALETTE (1 2 3 4))

;; A rthm-seq object with two bars of rhythms and two pitch-seqs in the
;; pitch-seq-palette. There must be as many items in each pitch-seq list as
;; there are rhythms in each rthm-seq-bar.
(make-rthm-seq '(seq1 ((((2 4) q e s s)
                        ((e) q (e)))
                       :pitch-seq-palette ((1 2 3 4 5)
                                           (2 4 6 8 10)))))

=>
RTHM-SEQ: num-bars: 2
          num-rhythms: 7
          num-notes: 5
          num-score-notes: 5
          num-rests: 2
          duration: 4.0
          psp-inversions: NIL
          marks: NIL
          time-sigs-tag: NIL
          handled-first-note-tie: NIL
         (for brevity's sake, slots pitch-seq-palette and bars are not printed)
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: SEQ1, tag: NIL,
data: ((((2 4) Q E S S) ((E) Q (E))) PITCH-SEQ-PALETTE
       ((1 2 3 4 5) (2 4 6 8 10)))

;; The pitch-seq-palette may be omitted, and time signatures may be changed
(make-rthm-seq '(seq1 ((((2 4) q e s s)
                        ((e) q (e))
                        ((3 8) s s e. s)))))

=>
RTHM-SEQ: num-bars: 3
```

```
          num-rhythms: 11
          num-notes: 9
          num-score-notes: 9
          num-rests: 2
          duration: 5.5
          psp-inversions: NIL
          marks: NIL
          time-sigs-tag: NIL
          handled-first-note-tie: NIL
         (for brevity's sake, slots pitch-seq-palette and bars are not printed)
SCLIST: sclist-length: 1, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: SEQ1, tag: NIL,
data: ((((2 4) Q E S S) ((E) Q (E)) ((3 8) S S E. S)))

;;; With :psp-inversions set to T, the inverted forms of the specified
;;; pitch-seq are automatically generated and added
(let ((mrs
       (make-rthm-seq '(seq1 ((((2 4) q e s s))
                              :pitch-seq-palette ((1 2 3 4))))
                      :psp-inversions t)))
  (data (pitch-seq-palette mrs)))

=> (
PITCH-SEQ: notes: NIL
           highest: 4
           lowest: 1
           original-data: (1 2 3 4)
           user-id: NIL
           instruments: NIL
           relative-notes: (not printed for sake of brevity)
           relative-notes-length: 25
SCLIST: sclist-length: 4, bounds-alert: T, copy: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "rthm-seq-SEQ1-pitch-seq-palette-ps-1", tag: NIL,
data: (1 2 3 4)
**************


PITCH-SEQ: notes: NIL
           highest: 4
           lowest: 1
           original-data: (4 3 2 1)
           user-id: NIL
           instruments: NIL
           relative-notes: (not printed for sake of brevity)
```

```
          relative-notes-length: 25
SCLIST: sclist-length: 4, bounds-alert: T, copy: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "rthm-seq-SEQ1-pitch-seq-palette-ps-1-inverted", tag: NIL,
data: (4 3 2 1)
**************
)
```

SYNOPSIS:

```
(defun make-rthm-seq (rs &key (psp-inversions nil) (warn-time-sigs t))
```

38.6.520 rthm-seq/make-rthm-seq-from-fragments

[rthm-seq] [Functions]

DATE:

January 2010

DESCRIPTION:

Make a rthm-seq object from a predefined list of rhythm fragments.

NB: No pitch-seqs can be passed as yet.

ARGUMENTS:

- The ID of the rthm-seq object to be made.
- A list of rhythm lists (fragments) paired with key IDs. The rhythm lists take the form of rthm-seq-bar definitions without the time signatures.
- A list of lists containing any combination of the key IDs from the list of fragments. These will be collated to create the resulting rthm-seq object. Each element will make up one whole bar.
- A list of meters. These can be given either as single numerators, whereby the optional <default-beat> argument will then be the denominator) or two-item lists consisting of (num denom). There must be one meter for each item in the list of references, and the meters must correspond to the number of beats in the corresponding item from the list of references.

RETURN VALUE:

A rthm-seq object.

EXAMPLE:

```
(let ((frags '((1 (- s s - (e)))
               (2 (s (s) (s) s))
               (3 ((s) - s e -))
               (4 (- s s (s) s -))
               (5 ((e) - s s -))
               (6 ((q))))))
  (make-rthm-seq-from-fragments
   'test-rs frags
   '((1 2 3) (1 4) (6 1) (5 6))
   '((3 4) (2 4) (2 4) (2 4))))

=>
RTHM-SEQ: num-bars: 4
          num-rhythms: 25
          num-notes: 15
          num-score-notes: 15
          num-rests: 10
          duration: 9.0
          psp-inversions: NIL
          marks: NIL
          time-sigs-tag: NIL
          handled-first-note-tie: NIL
         (for brevity's sake, slots pitch-seq-palette and bars are not printed)
SCLIST: sclist-length: 1, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: TEST-RS, tag: NIL,
data: ((((3 4) - S S - (E) S (S) (S) S (S) - S E -)
        ((2 4) - S S - (E) - S S (S) S -) ((Q) - S S - (E)) ((E) - S S - (Q))))
```

SYNOPSIS:

```
(defun make-rthm-seq-from-fragments (id fragments references meters
                                     &optional (default-beat 4))
```

38.6.521 rthm-seq/make-rthm-seq-from-unit-multipliers

[rthm-seq] [Functions]

DESCRIPTION:

```
Given a rhythmic unit, e.g. 32, a list of multipliers (e.g. '(7 9 16)),
and a time signature (e.g. '(4 4)), return a rthm-seq object made up of
bars whose rhythms are multiples of the specified unit by the numbers in
the multipliers list.
```

```
At this point the unit should be a whole number divisor of the beat in the
time signature, i.e. quintuple eighths won't work in 4/4.

NB: Setting the auto-beam keyword argument to T can result in errors if
    creating durations longer than 1 beat, as auto-beam will call
    get-beats. :auto-beam is therefore set to NIL by default.
```

ARGUMENTS:

```
- A rhythmic duration unit.
- A list of multipliers.
- A time signature.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :tag. A symbol that is another name, description etc. for the given
  object. The tag may be used for identification but not for searching
  purposes. Default = NIL.
- :auto-beam.  T or NIL. When T, the function will attempt to automatically
  set beaming indicators among the resulting rthm-seq-bar objects. This can
  result in errors if the resulting rhythms have a duration of more than 1
  beat. Default = NIL.
- :id. A symbol that will be the ID of the given object.
  Default = "from-multipliers".
- :tuplet. An integer or NIL. If an integer, the function will
  automatically place tuplet brackets of that value above beats consisting
  of tuplet rhythms. NB: This function will only place the same value over
  all tuplets.  Default = NIL.
```

RETURN VALUE:

```
Returns a rthm-seq object.
```

EXAMPLE:

```
;; Make a rthm-seq object using the rhythmic unit of a 16th-note, rhythms that
;; are 4, 2, 2, 4 and 4 16th-notes long, and a time signature of 2/4; then
;; print-simple the object returned to see the results.
(let ((rs (make-rthm-seq-from-unit-multipliers 's '(4 2 2 4 4) '(2 4))))
  (print-simple rs))

=>
rthm-seq from-multipliers
(2 4): note Q, note E, note E,
```

```
(2 4): note Q, note Q,

;; Make a rthm-seq object using the rhythmic unit of a 32nd note, combinations
;; of irregular duration, and a time signature of 4/4; then print-simple the
;; returned object to see the results.
(let ((rs (make-rthm-seq-from-unit-multipliers 32 '(7 9 16) '(4 4))))
  (print-simple rs))

=>
rthm-seq from-multipliers
(4 4): note E.., note 32, note Q, note H

;; The print-simple output of the above example disregards the ties. We can
;; check to make sure that there are only three attacked rhythms in the result
;; by reading the values of the IS-TIED-FROM and IS-TIED-TO slots, which show
;; that the 32 is tied to the Q
(let ((rs (make-rthm-seq-from-unit-multipliers 32 '(7 9 16) '(4 4))))
  (loop for b in (bars rs)
       collect (loop for r in (rhythms b) collect (is-tied-from r))
       collect (loop for r in (rhythms b) collect (is-tied-to r))))

=> ((NIL T NIL NIL) (NIL NIL T NIL))

;;; Using with a tuplet rhythm ('te) and setting the :tuplet option to 3 so
;;; that triplet brackets are automatically placed.
(let ((rs (make-rthm-seq-from-unit-multipliers 'te '(7 9 16) '(4 4)
                                               :tuplet 3)))
  (loop for b in (bars rs)
     collect (loop for r in (rhythms b) collect (bracket r))))

=> ((NIL NIL ((1 3)) (1) NIL) (NIL ((1 3)) (1) NIL NIL)
    (NIL NIL ((1 3)) (1) NIL))
```

SYNOPSIS:

```
(defun make-rthm-seq-from-unit-multipliers (unit multipliers time-sig
                                            &key
                                              ;; a number for brackets over
                                              ;; each beat.
                                              (tuplet nil)
                                              (tag nil)
                                              (auto-beam nil) ; see above
                                              (id "from-multipliers"))
```

38.6.522 rthm-seq/make-rthm-seq-from-unit-multipliers-simp

[rthm-seq] [Functions]

DATE:

June 6th 2020

DESCRIPTION:

Make a rthm-seq using lists of multiples of a single unit e.g. '(2 2 1) would result in an automatic time signature of 5/8 if passed a unit of 'e (or 8). In that case the rhythms would be q q e. Rests may be created by passing a number in parentheses.

See also make-rthm-seq-from-unit-multipliers for a related method.

ARGUMENTS:

- the ID for the rthm-seq
- the rhythm unit (symbol, number or rhythm object)
- the lists of multipliers, each sublist representing a bar

RETURN VALUE:

a rthm-seq object

EXAMPLE:

```
(print-simple (make-rthm-seq-from-unit-multipliers-simp 1 'e
                  '((3 4 1 2) (2 1 2))))
-->
rthm-seq 1
NIL: bar -1: (10 8):
note Q.,
note H,
note E,
note Q,
NIL: bar -1: (5 8):
note Q,
note E,
note Q,
```

SYNOPSIS:

```
(defun make-rthm-seq-from-unit-multipliers-simp (id unit multipliers)
```

38.6.523 rthm-seq/rs-subseq

[rthm-seq] [Methods]

DATE:

30th September 2013

DESCRIPTION:

Extract a new rthm-seq object from the bars of an existing rthm-seq.

NB other -subseq methods are more like Lisp's subseq but as this is for the end user it's a little different in the use of its indices.

ARGUMENTS:

- the original rthm-seq object
- the start bar (1-based)

OPTIONAL ARGUMENTS:

- the end bar (1-based and (unlike Lisp's subseq function) inclusive). If NIL, we'll use the original end bar. Default = NIL.

RETURN VALUE:

A new rthm-seq object.

EXAMPLE: SYNOPSIS:

```
(defmethod rs-subseq ((rs rthm-seq) start &optional end)
```

38.6.524 rthm-seq/scale

[rthm-seq] [Methods]

DESCRIPTION:

Scale the durations of the rhythm objects in a given rthm-seq object by the specified factor.

NB: As is evident in the examples below, this method does not replace the original data in the rthm-seq object's DATA slot.

ARGUMENTS:

- A rthm-seq object.
- A real number that is the scaling factor.

RETURN VALUE:

Returns a rthm-seq object.

EXAMPLE:

```
;; The method returns a rthm-seq object.
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 1 1 2 3 4))))))
  (scale rs 3))

=>
RTHM-SEQ: num-bars: 3
          num-rhythms: 11
          num-notes: 8
          num-score-notes: 9
          num-rests: 2
          duration: 16.5
          psp-inversions: NIL
          marks: NIL
          time-sigs-tag: NIL
          handled-first-note-tie: NIL
         (for brevity's sake, slots pitch-seq-palette and bars are not printed)
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: ((((2 4) Q+E S S) ((E) Q (E)) ((3 8) S S E. S)) PITCH-SEQ-PALETTE
       ((1 2 3 1 1 2 3 4)))

;; Create a rthm-seq object, scale the durations by 3 times using the scale
;; method, and print-simple the corresponding slots to see the results
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 1 1 2 3 4))))))
  (print-simple (scale rs 3)))

=>
```

```
rthm-seq NIL
(6 4): note H., note Q., note E., note E.,
(6 4): rest Q., note H., rest Q.,
(9 8): note E., note E., note E., note E.,
```

SYNOPSIS:

```
(defmethod scale ((rs rthm-seq) scaler
                  &optional ignore1 ignore2 ignore3)
```

38.6.525 rthm-seq/set-nth-attack

[rthm-seq] [Methods]

DESCRIPTION:

Sets the value of the nth rhythm object of a given rthm-seq object that needs an attack; i.e., not a rest and not a tied note.

NB: This method does not check to ensure that the resulting rthm-seq bars contain the right number of beats.

ARGUMENTS:

- A zero-based index number for the attacked note to change.
- An event.
- A rthm-seq object.

OPTIONAL ARGUMENTS:

- T or NIL indicating whether to print an error message if the given index is greater than the number of attacks (minus 1) in the rthm-seq object (default = T).

RETURN VALUE:

- An event object.

EXAMPLE:

```
;; The method returns an event object
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
```

```
                          :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (set-nth-attack 2 (make-event 'c4 'q) rs))

=>
EVENT: start-time: NIL, end-time: NIL,
[...]
       pitch-or-chord:
PITCH: frequency: 261.6255569458008, midi-note: 60, midi-channel: NIL
[...]
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: C4, tag: NIL,
data: C4
[...]
       written-pitch-or-chord: NIL
RHYTHM: value: 4.000, duration: 1.000, rq: 1, is-rest: NIL,
[...]
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: Q, tag: NIL,
data: Q

;; Create a rthm-seq object, apply set-nth-attack, print the corresponding
;; slots to see the change
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (set-nth-attack 2 (make-event 'c4 'q) rs)
  (print-simple rs))

=>
rthm-seq NIL
(2 4): note Q, note E, note S, C4 Q,
(2 4): rest E, note Q, rest E,
(3 8): note S, note S, note E., note S,

;; By default, the method drops into the debugger with an error when the
;; specified index is greater than the number of items in the given rthm-seq
;; object.
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (set-nth-attack 11 (make-event 'c4 'q) rs))

=>
rthm-seq::set-nth-attack: Can't set attack 11 as only 8 notes in the rthm-seq
```

```
   [Condition of type SIMPLE-ERROR]

;; This error can be suppressed, simply returning NIL, by setting the optional
;; argument to NIL.
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (set-nth-attack 11 (make-event 'c4 'q) rs nil))

=> NIL
```

SYNOPSIS:

```
(defmethod set-nth-attack (index (e event) (rs rthm-seq)
                           &optional (error t))
```

38.6.526 rthm-seq/set-nth-bar

[rthm-seq] [Methods]

DESCRIPTION:

```
 Change the contents of the nth rthm-seq-bar object in the given rthm-seq.
```

ARGUMENTS:

```
 - A zero-based index number for the bar to change.
 - A rthm-seq-bar object containing the new bar.
 - A rthm-seq object.
```

RETURN VALUE:

```
 A rthm-seq-bar object.
```

EXAMPLE:

```
;; The method returns what is passed to it as the new-bar argument (generally a
;; rthm-seq-bar object.
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (set-nth-bar 1 (make-rthm-seq-bar '((2 4) (s) e (s) q)) rs))
```

```
=>
RTHM-SEQ-BAR: time-sig: 0 (2 4), time-sig-given: T, bar-num: -1,
[...]
data: ((2 4) (S) E (S) Q)

;; Create a rthm-seq object, change the second bar (index 1) using the
;; set-nth-bar method, and print the contents of the rhythms data to see the
;; changes.
(let ((rs (make-rthm-seq '((((2 4) q+e s s)
                            ((e) q (e))
                            ((3 8) s s e. s))
                           :pitch-seq-palette ((1 2 3 4 1 1 2 3))))))
  (set-nth-bar 1 (make-rthm-seq-bar '((2 4) (s) e (s) q)) rs)
  (print-simple rs))

=>
rthm-seq NIL
(2 4): note Q, note E, note S, note S,
(2 4): rest S, note E, rest S, note Q,
(3 8): note S, note S, note E., note S,
```

SYNOPSIS:

```
(defmethod set-nth-bar (index new-bar (rs rthm-seq))
```

38.6.527 rthm-seq/split

[rthm-seq] [Methods]

DATE:

```
27 Jan 2011
```

DESCRIPTION:

```
Splits the rthm-seq-bar objects of a given rthm-seq object into multiple
smaller rthm-seq-bar objects, creating a new rthm-seq object with a greater
number of bars than the original. This will only work if the given
rthm-seq-bar objects can be split into whole beats; e.g., a 4/4 bar will
not be split into 5/8 + 3/8.

The keyword arguments :min-beats and :max-beats serve as guidelines rather
than strict cut-offs. In some cases, the method may only be able to
effectively split the given rthm-seq-bar by dividing it into segments that
slightly exceed the length stipulated by these arguments (see example
```

below).

Depending on the min-beats/max-beats arguments stipulated by the user or the rhythmic structure of the given rthm-seq-bar objects, the given rthm-seq-bar or rthm-seq objects may not be splittable, in which case NIL is returned. If the keyword argument :warn is set to T, a warning will be also printed in such cases.

NB: This method sets the values of the individual slots but leaves the DATA slot untouched (for cases in which the user might want to see where the new data originated from, or otherwise use the old data somehow, such as in a new rthm-seq object).

ARGUMENTS:

- A rthm-seq object.

OPTIONAL ARGUMENTS:

keyword arguments
- :min-beats. This argument takes an integer value to indicate the minimum number of beats in any of the new rthm-seq-bar objects created. This serves as a guideline only and may occasionally be exceeded in value by the method. Default value = 2.
- :max-beats. This argument takes an integer value to indicate the maximum number of beats in any of the new rthm-seq-bar objects created. This serves as a guideline only and may occasionally be exceeded in value by the method. Default value = 5.
- :warn. Indicates whether to print a warning if the rthm-seq-bar object is unsplittable. Value T = print a warning. Defaults to NIL.

RETURN VALUE:

A rthm-seq object.

EXAMPLE:

```
;; The method returns a new rthm-seq object
(let ((rs (make-rthm-seq '((((4 4) q e s s (e) e e (e))
                           ((3 4) s s e s e s e. s)
                           ((5 4) h q. e e s s))
                          :pitch-seq-palette ((1 2 3 4 5 6 1 2 3 4 5 6 7 8 1 2
                                               3 4 5 6))))))
  (split rs))
```

```
=>

RTHM-SEQ: num-bars: 5
          num-rhythms: 22
          num-notes: 20
          num-score-notes: 20
          num-rests: 2
          duration: 12.0
          psp-inversions: NIL
          marks: NIL
          time-sigs-tag: NIL
          handled-first-note-tie: NIL
         (for brevity's sake, slots pitch-seq-palette and bars are not printed)
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: ((((4 4) Q E S S (E) E E (E)) ((3 4) S S E S E S E. S)
        ((5 4) H Q. E E S S))
       PITCH-SEQ-PALETTE ((1 2 3 4 5 6 1 2 3 4 5 6 7 8 1 2 3 4 5 6)))

;; Without setting the :min-beats and :max-beats arguments, the following
;; rthm-seq object is broken down from 3 to 5 rthm-seq-bar objects
(let* ((rs (make-rthm-seq '((((4 4) q e s s (e) e e (e))
                            ((3 4) s s e s e s e. s)
                            ((5 4) h q. e e s s))
                           :pitch-seq-palette ((1 2 3 4 5 6 1 2 3 4 5 6 7 8 1 2
                                                3 4 5 6)))))
       (rssplt (split rs)))
  (print-simple rssplt))

=>
rthm-seq NIL
(2 4): note Q, note E, note S, note S,
(2 4): rest E, note E, note E, rest E,
(3 4): note S, note S, note E, note S, note E, note S, note E., note S,
(2 4): note H,
(3 4): note Q., note E, note E, note S, note S,

;; Setting :min-beats to 4 affects the resulting subdivisions to larger bars
(let* ((rs (make-rthm-seq '((((4 4) q e s s (e) e e (e))
                            ((3 4) s s e s e s e. s)
                            ((5 4) h q. e e s s))
                           :pitch-seq-palette ((1 2 3 4 5 6 1 2 3 4 5 6 7 8 1 2
                                                3 4 5 6)))))
       (rssplt (split rs :min-beats 4)))
  (print-simple rssplt))
```

```
=>
rthm-seq NIL
(4 4): note Q, note E, note S, note S, rest E, note E, note E, rest E,
(3 4): note S, note S, note E, note S, note E, note S, note E., note S,
(5 4): note H, note Q., note E, note E, note S, note S,

;; Even though :max-beats is set to 2, an occasional 3/4 bar is constructed
(let* ((rs (make-rthm-seq '((((4 4) q e s s (e) e e (e))
                            ((3 4) s s e s e s e. s)
                            ((5 4) h q. e e s s))
                           :pitch-seq-palette ((1 2 3 4 5 6 1 2 3 4 5 6 7 8 1 2
                                                3 4 5 6)))))
       (rssplt (split rs :max-beats 2)))
  (print-simple rssplt))

=>
rthm-seq NIL
(2 4): note Q, note E, note S, note S,
(2 4): rest E, note E, note E, rest E,
(3 4): note S, note S, note E, note S, note E, note S, note E., note S,
(2 4): note H,
(3 4): note Q., note E, note E, note S, note S,
```

SYNOPSIS:

```
(defmethod split ((rs rthm-seq)
                  &key (min-beats 2) (max-beats 5) warn (clone t))
```

38.6.528 rthm-seq/split-into-single-bars

[rthm-seq] [Methods]

DESCRIPTION:

```
Split a rthm-seq into single bar rthm-seqs.  The pitch-seq-palette will be
used to set the pitch-seqs of the new rthm-seqs.
```

ARGUMENTS:

```
- a rthm-seq object
```

OPTIONAL ARGUMENTS:

```
- whether to clone each bar or just the original.  Default = T = clone.
```

RETURN VALUE:

```
a list of rthm-seq objects
```

SYNOPSIS:

```
(defmethod split-into-single-bars ((rs rthm-seq) &optional (clone t))
```

38.6.529 sclist/rthm-seq-bar

[sclist] [Classes]

NAME:

```
rthm-seq-bar

File:             rthm-seq-bar.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                  rthm-seq-bar

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the rthm-seq-bar class, objects of
                  which make up the individual bars that reside in a
                  rhythmic sequence.  This class is responsible for parsing
                  lists containing rhythms and time signatures (but not
                  parsing these things themselves--that is done by separate
                  classes).

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    13th February 2001

$$ Last modified:  12:42:01 Wed Jun 24 2026 CEST

SVN ID: $Id$
```

38.6.530 rthm-seq-bar/add-half-beat-rest

[rthm-seq-bar] [Methods]

DATE:

August 25th 2016, Edinburgh

DESCRIPTION:

Adds a half beat rest at the end of the bar. This is a destructive method. It will change the time signature, so a 2/4 bar becomes a 5/8 bar, with an 1/8th rest at the end.

ARGUMENTS:

- rthm-seq-bar object

OPTIONAL ARGUMENTS:

- ignore: don't pass this; it's only there so we can have a bar number passed in the slippery-chicken class.

RETURN VALUE:

The modified rthm-seq-bar object passed as an argument

SYNOPSIS:

```
(defmethod add-half-beat-rest ((rsb rthm-seq-bar) &optional ignore)
```

38.6.531 rthm-seq-bar/all-rests?

[rthm-seq-bar] [Methods]

DESCRIPTION:

Test whether all rhythms in a rthm-seq-bar object are rests.

ARGUMENTS:

- A rthm-seq-bar object.

RETURN VALUE:

T if all rhythms are rests, otherwise NIL

EXAMPLE:

```
(let ((rsb (make-rthm-seq-bar '((2 4) (q) (e) (s) (s)))))
  (all-rests? rsb))

=> T

(let ((rsb (make-rthm-seq-bar '((2 4) q e s s))))
  (all-rests? rsb))

=> NIL
```

SYNOPSIS:

```
(defmethod all-rests? ((rsb rthm-seq-bar))
```

38.6.532 rthm-seq-bar/auto-beam

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Automatically add beaming indications to the rhythm objects of the given
rthm-seq-bar object.  This will only set one beam group per beat.

NB: This method does not modify the data slot of the rthm-seq-bar object
    itself. Instead, it modifies the beam value for the individual rhythms.
```

ARGUMENTS:

```
- A rthm-seq-bar object.
```

OPTIONAL ARGUMENTS:

```
- The beat basis for the given rthm-seq-bar. This will affect which notes
  get beamed together. This value can be either numeric (4, 8 16 etc.) or
  alphabetic (q, e, s etc). If no beat is given, the method defaults this
  value to NIL and takes the beat from the current time signature.
- Check-dur. This can be t, nil, #'warn or #'error, where t is the same as
  #'error. If T, the method will make sure there is a complete beat of
  rhythms for each beat of the bar and issue an error if a full beat's
  worth cannot be returned--this may not mean that your bar is
  malformed. Or if you pass a symbol like 'silent the
  duration will be checked and NIL returned if we can't get an exact beat's
  worth of rthms. Default = T.
```

RETURN VALUE:

```
 Returns the rthm-seq-bar-object
```

EXAMPLE:

```
(let ((rsb (make-rthm-seq-bar '((2 4) e e s s s s))))
  (auto-beam rsb))

=> NIL

(let ((rsb (make-rthm-seq-bar '((2 4) e e s s s s))))
  (auto-beam rsb)
  (loop for r in (rhythms rsb) collect (beam r)))

=> (1 0 1 NIL NIL 0)

(let ((rsb (make-rthm-seq-bar '((2 4) e e s s s s))))
  (auto-beam rsb 8)
  (loop for r in (rhythms rsb) collect (beam r)))

=> (NIL NIL 1 0 1 0)

(let ((rsb (make-rthm-seq-bar '((2 4) e e s s s s))))
  (auto-beam rsb 8 t)
  (loop for r in (rhythms rsb) collect (beam r)))

=> (NIL NIL 1 0 1 0)

(let ((rsb (make-rthm-seq-bar '((2 4) e e s s s s))))
  (auto-beam rsb 8 nil)
  (loop for r in (rhythms rsb) collect (beam r)))

=> (NIL NIL 1 0 1 0)
```

SYNOPSIS:

```
(defmethod auto-beam ((rsb rthm-seq-bar) &optional (beat nil) (check-dur t)
                                                   ignore)
```

38.6.533 rthm-seq-bar/auto-put-tuplet-bracket-on-beats

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
 Given a rthm-seq-bar object with tuplet rhythms and an indication of which
```

tuplet value to place, this method will automatically add the appropriate tuplet bracket to the beats of the bar in the printed score output. If the TUPLET argument is set to NIL, the method will proceed on the basis of best-guessing rules.

NB: This method may produce results that encapsulate an entire beat when applying brackets to a portion of that beat. Thus bracketing the rhythm (e ts ts ts) will return
{ 3 e. ts ts ts } rather than
(e { 3 ts ts ts })

ARGUMENTS:

- A rthm-seq-bar object
- An integer indicating the tuplet value (e.g. 3 for triplets, 5 for quintuplets etc.)

RETURN VALUE:

Returns T.

OPTIONAL ARGUMENTS:

- An integer indicating beat basis for the bar, or NIL. If NIL (default), the beat is taken from the time signature.
- An integer indicating the beat number within the bar to look for tuplets, or T. If T (default), all beats in the bar will be examined for possible tuplets.
- T or NIL to indicate whether to delete the tuplet bracket indicators already present in the given rthm-seq-bar object. T = delete. Default = T.

EXAMPLE:

```
(let ((rsb (make-rthm-seq-bar '((2 4) te te te q))))
  (tuplets rsb))

=> NIL

(let ((rsb (make-rthm-seq-bar '((2 4) te te te q))))
  (loop for r in (rhythms rsb) collect (bracket r))

=> (NIL NIL NIL NIL)

(let ((rsb (make-rthm-seq-bar '((2 4) te te te q))))
```

```
  (auto-put-tuplet-bracket-on-beats rsb 3))

=> T

(let ((rsb (make-rthm-seq-bar '((2 4) te te te q))))
  (auto-put-tuplet-bracket-on-beats rsb 3)
  (print (tuplets rsb))
  (print (loop for r in (rhythms rsb) collect (bracket r))))

=>
((3 0 2))
(((1 3)) (-1) (1) NIL)

(let ((rsb (make-rthm-seq-bar '((2 4) te te te q))))
  (auto-put-tuplet-bracket-on-beats rsb nil)
  (tuplets rsb))

=> ((3 0 2))

;;; The method may bracket the entire beat, returning ((3 1 4)) rather than
;;; ((3 2 4))
(let ((rsb (make-rthm-seq-bar '((2 4) q e ts ts ts))))
  (auto-put-tuplet-bracket-on-beats rsb 3)
  (tuplets rsb))

=> ((3 1 4))
```

SYNOPSIS:

```
(defmethod auto-put-tuplet-bracket-on-beats ((rsb rthm-seq-bar) tuplet
                                             &optional
                                             (beat nil)
                                             ;; can be a beat number or t for
                                             ;; all
                                             (beat-number t)
                                             ;; delete the tuplets already
                                             ;; there?
                                             (delete t))
```

38.6.534 rthm-seq-bar/auto-tuplets

[rthm-seq-bar] [Methods]

DESCRIPTION:

Automatically place the data necessary for tuplet brackets in rthm-seq-bar

```
objects that contain tuplet rhythms.
```

ARGUMENTS:

```
- A rthm-seq-bar object.
```

OPTIONAL ARGUMENTS:

```
- A function to be performed on fail. Default = #'error.
```

RETURN VALUE:

```
Returns T if successful, and the rsb as a second value in any case
```

EXAMPLE:

```
;;; Make a rthm-seq-bar object and print the values of the BRACKET slots for
;;; the rhythm objects it contains. Then apply auto-brackets and print the same
;;; again to see the change.
(let ((rsb (make-rthm-seq-bar '((4 4) tq tq tq +q fs fs fs fs fs))))
  (print (loop for r in (rhythms rsb) collect (bracket r)))
  (auto-tuplets rsb)
  (print (loop for r in (rhythms rsb) collect (bracket r))))

=>
(NIL NIL NIL NIL NIL NIL NIL NIL NIL)
(((1 3)) (-1) (1) NIL ((2 5)) (-2) (-2) (-2) (2))
```

SYNOPSIS:

```
(defmethod auto-tuplets ((rsb rthm-seq-bar) &optional (on-fail #'error))
```

38.6.535 rthm-seq-bar/check-beams

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Check the BEAM slots of the event objects within a specified rthm-seq-bar
object to ensure that every beginning beam indication (slot value of 1) is
coupled with a corresponding closing beam indication (slot value of 0), and
print a warning and return NIL if this is not the case.
```

ARGUMENTS:

- A rthm-seq-bar object.

OPTIONAL ARGUMENTS:

keyword arguments:
- :auto-beam. T or NIL to indicate the method should apply the auto-beam algorithm to the given bar after the check has determined that the beaming is wrong (and :on-fail is not NIL). T = auto-beam. Default = NIL.
- :print. T or NIL to indicate whether the method should print feedback of the checking process to the Lisp listener. T = print feedback. Default = NIL.
- :fix. T or NIL to indicate that when a beam has been placed over a rhythm with no flags (e.g. a 1/4 note), then we delete beams over that note and try again. Default = T.
- :on-fail. The function that should be applied when the check does not pass. May be NIL for no warning/error or #'error if processing should stop. Default = #'warn.

RETURN VALUE:

T if the check passes, otherwise NIL.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((sax ((alto-sax tenor-sax) :midi-channel 1))))
        :instrument-change-map '((1 ((sax ((1 alto-sax) (3 tenor-sax)))))
                                 (2 ((sax ((2 alto-sax) (5 tenor-sax)))))
                                 (3 ((sax ((3 alto-sax) (4 tenor-sax))))))
        :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h e (s) (s) e+s+s))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((sax (1 1 1 1 1))))
                        (2 ((sax (1 1 1 1 1))))
                        (3 ((sax (1 1 1 1 1))))))))
  (check-beams (get-bar mini 1 'sax)))

=> T

(let ((mini
```

```
       (make-slippery-chicken
        '+mini+
        :ensemble '(((sax ((alto-sax tenor-sax) :midi-channel 1))))
        :instrument-change-map '((1 ((sax ((1 alto-sax) (3 tenor-sax)))))
                                 (2 ((sax ((2 alto-sax) (5 tenor-sax)))))
                                 (3 ((sax ((3 alto-sax) (4 tenor-sax))))))
        :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h e (s) (s) e+s+s))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((sax (1 1 1 1 1))))
                        (2 ((sax (1 1 1 1 1))))
                        (3 ((sax (1 1 1 1 1)))))))
  (setf (beam (nth 1 (rhythms (get-bar mini 1 'sax)))) 1)
  (check-beams (get-bar mini 1 'sax)))

=> NIL
```

SYNOPSIS:

```
(defmethod check-beams ((rsb rthm-seq-bar) &key auto-beam print (fix t)
                                             (on-fail #'warn))
```

38.6.536 rthm-seq-bar/check-tuplets

[rthm-seq-bar] [Methods]

DESCRIPTION:

Check the qualities of the tuplets brackets in a given rthm-seq-bar object to make sure they are all formatted properly (i.e. each starting tuplet bracket has a closing tuplet bracket etc.). If an error is found, the method will try to fix it, then re-check, and only issue an error then if another is found.

NB this won't check whether tuplet brackets start and stop on notes spanning a whole beat (as that wouldn't make sense with nested tuplets) so it won't guarantee that a bar can be notated without errors.

ARGUMENTS:

- A rthm-seq-bar object.

OPTIONAL ARGUMENTS:

```
 - The function to use if something is not ok with the tuplets. This
   defaults to #'error, but could also be #'warn for example
```

RETURN VALUE:

```
 T if all tuplets brackets are ok, otherwise performs the on-fail function
 and returns NIL.
```

EXAMPLE:

```
(let ((rsb (make-rthm-seq-bar '((4 4) { 3 te te te } q q q))))
  (setf (bracket (get-nth-event 2 rsb)) nil)
  (check-tuplets rsb #'warn))

=> rthm-seq-bar::check-tuplets: got a nil bracket when brackets still open.
```

SYNOPSIS:

```
(defmethod check-tuplets ((rsb rthm-seq-bar) &optional (on-fail #'error))
```

38.6.537 rthm-seq-bar/chop

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
 Creates a list of new rthm-seq-bar objects, with new time signatures, which
 are formed by systematically chopping the bar represented by the current
 rthm-seq-bar into segments.

 The method creates these segments based on chop-point pairs specified in
 the <chop-points> argument, which is a list of 2-element lists, each of
 which specifies the start and end points of a rhythmic span within the
 bounds of a given beat, measured in the unit specified by the <unit>
 argument.

 The chop points specified are used to individually process each beat in the
 given rthm-seq-bar object; thus, chop-points specified for the subdivisions
 of a quarter-note will not work if applied to a 5/8 bar.

 The method fills each newly created rthm-seq-bar object with one rhythmic
 duration that is equal to the length of the bar. If the beginning of the
 given chop segment coincides with an attack in the original bar, the result
 is a sounding note; if not, the result is a rest. NB: In this abstraction
 of the class for the sake of this documentation, sounding notes will appear
```

as NIL.

The chop method is the basis for slippery-chicken's feature of intra-phrasal looping.

NB: The <unit> argument must be a duplet rhythmic value (i.e. 32, 's, 'e etc.) and cannot be a tuplet value (i.e. 'te 'fe etc.).

NB: In order for the resulting chopped rhythms to be parsable by LilyPond and CMN, there can be no tuplets (triplets etc.) among the rhythms to be chopped. Such rhythms will result in LilyPond and CMN errors. This has only minimal bearing on any MIDI files produced, however, and these can potentially be imported into notation software.

ARGUMENTS:

- A rthm-seq-bar object.

OPTIONAL ARGUMENTS:

- <chop-points> A list of integer pairs, each of which delineates a segment of the beat of the given rthm-seq-bar object measured in the rhythmic unit specified by the <unit> argument. Thus, if all possible spans of sixteenth-notes within a quarter-note, starting from the first sixteenth, were delineated, they would span from 1 to 4 (the full quarter), 1 to 3 (the first dotted 8th of the quarter), 1 to 2 (the first 8th) and 1 to 1 (the first 16th of the quarter); the process could continue then with all rhythmic durations contained within the bounds of the same quarter starting on the second 16th, etc. The default chop-points for a quarter are '((1 4) (1 3) (1 2) (2 4) (2 3) (3 4) (1 1) (2 2) (3 3) (4 4)).
- <unit>. The rhythmic duration that serves as the unit of measurement for the chop points. Default = 's.
- <rthm-seq-id>. A symbol that will be the ID for the list created.

RETURN VALUE:

A list of rthm-seq-bar objects.

EXAMPLE:

```
;; Systematically subdivide each quarter-note of a 2/4 bar containing two ; ;
;; quarter-notes into all possible segments whose durations are multiples of a ; ;
;; sixteenth-note unit, and print-simple the resulting list. The quarter-note ; ;
;; subdivision is re-specified here sightly differently to the default for the ; ;
;; sake of systematic clarity. Only those segments whose start point coincide ; ;
```

```
;; with an attack in the original bar, i.e. those that begin on the first ; ;
;; sixteenth of each  beat, will be assigned a NIL (which will later become ; ;
;; a sounding note); all others are assigned a rest. ; ;

(let* ((rsb (make-rthm-seq-bar '((2 4) q q)))
(ch (chop rsb
'((1 4) (1 3) (1 2) (1 1)
(2 4) (2 3) (2 2)
(3 4) (3 3)
(4 4))
's)))
(loop for b in ch do (print-simple b)))

=>
(1 4): NIL Q,
(3 16): NIL E.,
(1 8): NIL E,
(1 16): NIL S,
(3 16): rest 16/3,
(1 8): rest 8,
(1 16): rest 16,
(1 8): rest 8,
(1 16): rest 16,
(1 16): rest 16,
(1 4): NIL Q,
(3 16): NIL E.,
(1 8): NIL E,
(1 16): NIL S,
(3 16): rest 16/3,
(1 8): rest 8,
(1 16): rest 16,
(1 8): rest 8,
(1 16): rest 16,
(1 16): rest 16,

;; The same thing, but returning all possible segments within the bounds of a ; ;
;; quarter-note whose durations that are multiple of an 8th-note unit ; ;
(let* ((rsb (make-rthm-seq-bar '((2 4) q q)))
(choprsb (chop rsb
'((1 2) (1 1) (2 2))
'e)))
(loop for b in choprsb do (print-simple b)))

=>
(1 4): NIL Q,
(1 8): NIL E,
```

```
(1 8): rest 8,
(1 4): NIL Q,
(1 8): NIL E,
(1 8): rest 8,

;; Adapt the 16th-note example above to a starting rthm-seq-bar object with ; ;
;; more complex rhythmic content. Note here, too, that the rthm-seq-bar object ; ;
;; being segmented contains rhythmic durations smaller than the <unit> ; ;
;; argument.                              ; ;
(let* ((rsb (make-rthm-seq-bar '((4 4) - (s) (32) 32 (s) s - - +s+32 (32) (e) -
(q) (s) s (e))))
(choprsb (chop rsb
'((1 4) (1 3) (1 2) (1 1)
(2 4) (2 3) (2 2)
(3 4) (3 3)
(4 4))
's)))
(loop for b in choprsb do (print-simple b)))

=>
(1 4): rest S, rest 32, NIL 32, rest S, NIL S,
(3 16): rest S, rest 32, NIL 32, rest S,
(1 8): rest S, rest 32, NIL 32,
(1 16): rest 16,
(3 16): rest 32, NIL 32, rest S, NIL S,
(1 8): rest 32, NIL 32, rest S,
(1 16): rest 32, NIL 32,
(1 8): rest S, NIL S,
(1 16): rest 16,
(1 16): NIL S,
(1 4): rest 4,
(3 16): rest 16/3,
(1 8): rest 8,
(1 16): rest 16,
(3 16): rest 16/3,
(1 8): rest 8,
(1 16): rest 16,
(1 8): rest 8,
(1 16): rest 16,
(1 16): rest 16,
(1 4): rest 4,
(3 16): rest 16/3,
(1 8): rest 8,
(1 16): rest 16,
(3 16): rest 16/3,
(1 8): rest 8,
```

```
(1 16): rest 16,
(1 8): rest 8,
(1 16): rest 16,
(1 16): rest 16,
(1 4): rest S, NIL S, rest E,
(3 16): rest S, NIL S, rest S,
(1 8): rest S, NIL S,
(1 16): rest 16,
(3 16): NIL S, rest E,
(1 8): NIL S, rest S,
(1 16): NIL S,
(1 8): rest 8,
(1 16): rest 16,
(1 16): rest 16,

;; The same again with a <unit> of eighths ; ;
(let* ((rsb (make-rthm-seq-bar '((4 4) - (s) (32) 32 (s) s - - +s+32 (32) (e) -
(q) (s) s (e))))
(choprsb (chop rsb
'((1 2) (1 1) (2 2))
'e)))
(loop for b in choprsb do (print-simple b)))

=>
(1 4): rest S, rest 32, NIL 32, rest S, NIL S,
(1 8): rest S, rest 32, NIL 32,
(1 8): rest S, NIL S,
(1 4): rest 4,
(1 8): rest 8,
(1 8): rest 8,
(1 4): rest 4,
(1 8): rest 8,
(1 8): rest 8,
(1 4): rest S, NIL S, rest E,
(1 8): rest S, NIL S,
(1 8): rest 8,
```

SYNOPSIS:

```
(defmethod chop ((rsb rthm-seq-bar)
                 &optional chop-points (unit 's) rthm-seq-id)
```

38.6.538 rthm-seq-bar/consolidate-notes

[rthm-seq-bar] [Methods]

DESCRIPTION:

Combine consecutive tied notes into one (or a few) notes of a longer rhythmic duration.

NB: This method is the core method that is called for rthm-seq objects or slippery-chicken objects, at which point it takes ties (and perhaps another couple of things) into consideration, after the tie slots etc. have been updated. As such, though it will work to a certain degree when called directly on a rthm-seq-bar object, it should primarily be used when getting a rthm-seq-bar from within a rthm-seq object or slippery-chicken object.

ARGUMENTS:

- A rthm-seq-bar object.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether the method should make sure that an exact beat's worth of rhythms is handled. T = check durations. Default = NIL.

RETURN VALUE:

The rthm-seq-bar object.

EXAMPLE:

```
;;; Create a slippery-chicken object, print-simple a bar from that object,
;;; apply the consolidate-notes method to that bar, and print-simple that bar
;;; again to see the changes.

(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :set-palette '((1 ((gs4 af4 bf4))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e +e +e +e e +s +s +s e.))
                               :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((vn (1 1 1))))))))
  (print-simple (get-bar mini 2 'vn))
  (consolidate-notes (get-bar mini 2 'vn))
  (print-simple (get-bar mini 2 'vn)))
```

```
=>
(4 4): GS4 E+, +GS4 E+, +GS4 E+, +GS4 E, AF4 E+, +AF4 S+, +AF4 S+, +AF4 S,
BF4 E.,
(4 4): GS4 H, AF4 Q+, +AF4 S, BF4 E.,
```

SYNOPSIS:

```
(defmethod consolidate-notes ((rsb rthm-seq-bar)
                              ;; MDE Wed Nov 28 11:40:59 2012 -- added auto-beam
                              &optional check-dur beat (auto-beam t))
```

38.6.539 rthm-seq-bar/consolidate-rests

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Consolidate two or more consecutive rests into one longer rhythmic
unit. This method works on the basis of beats, striving to consolidate into
beats first.

NB: The user may find it helpful to adjust the :beat and :min values, and
    even to call the method more than once consecutively. For multiple
    calls, the method consolidate-rests-max may also be useful.
```

ARGUMENTS:

```
- A rthm-seq-bar object.
```

OPTIONAL ARGUMENTS:

```
keyword arguments
- :beat. The beat basis into which rests are to be consolidated. If no
  value is given for this option, the method will take the beat from the
  time signature.
- :min. A seldom-used argument that will only make a difference when there
  are a number of rests of the same duration followed by a note.  This is
  then the minimum duration that such rests may have if they are to be
  consolidated. Default = NIL.
- :warn. T or NIL to indicate whether the method should print a warning to
  the Lisp listener if it is mathematically unable to consolidate the
  rests. T = print warning. Default = NIL.
- :auto-tuplets. Whether to force a call to the auto-tuplets method before
  returning. This will happen by default anyway if check-tuplets fails, but
  sometimes that doesn't fail, exactly, but auto-tuplets still makes
```

```
   sense. If, for example, Lilypond fails with strange error messages,
   setting this to T might help. Default = NIL.
```

RETURN VALUE:

```
 The rthm-seq-bar object
```

EXAMPLE:

```
;;; Returns a list of rhythm/event objects
(let ((rsb (make-rthm-seq-bar '((4 4) (e) (e) (e) (e) (e) (s) (s) (s) e.))))
  (consolidate-rests rsb))

=>
(
EVENT: start-time: NIL, end-time: NIL,
[...]
data: 4
[...]
EVENT: start-time: NIL, end-time: NIL,
[...]
data: 4
[...]
EVENT: start-time: NIL, end-time: NIL,
[...]
data: 4
[...]
RHYTHM: value: 16.000, duration: 0.250, rq: 1/4, is-rest: T,
[...]
data: S
[...]
RHYTHM: value: 5.333, duration: 0.750, rq: 3/4, is-rest: NIL,
[...]
data: E.
)

;;; Consolidating on the basis of the time-signature's beat by default
(let ((rsb (make-rthm-seq-bar '((4 4) (e) (e) (e) (e) (e) (s) (s) (s) e.))))
  (consolidate-rests rsb)
  (loop for r in (rhythms rsb) collect (data r)))

=> (4 4 4 S E.)

;; Changing the :beat may effect the outcome
(let ((rsb (make-rthm-seq-bar '((4 4) (e) (e) (e) (e) (e) (s) (s) (s) e.))))
  (consolidate-rests rsb :beat 2)
```

```
  (loop for r in (rhythms rsb) collect (data r)))

=> (2 E E S E.)

;; Calling multiple times may further consolidate the results
(let ((rsb (make-rthm-seq-bar '((2 2) (e) (e) (e) (e) (e) (s) (s) (s) e.))))
  (consolidate-rests rsb)
  (print (loop for r in (rhythms rsb) collect (data r)))
  (consolidate-rests rsb)
  (print (loop for r in (rhythms rsb) collect (data r))))

=>
(2 E E S E.)
(2 Q S E.)
```

SYNOPSIS:

```
(defmethod consolidate-rests ((rsb rthm-seq-bar)
                              &key beat min warn auto-tuplets)
```

38.6.540 rthm-seq-bar/consolidate-rests-max

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Similar to consolidate-rests, but calls that method repeatedly until no
more changes can be made to the given rthm-seq-bar object.

NB This will still only reduce rests down to a maximum of a beat.  If you
   need e.g. two quarter rests reduced to a single half rest in a 4/4 bar,
   specify :beat 2
```

ARGUMENTS:

```
- A rthm-seq-bar object.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :beat. The beat basis into which rests are to be consolidated. If no
  value is given for this option, the method will take the beat from the
  time signature.
- :min. A seldom-used argument that will only make a difference when there
  are a number of rests of the same duration followed by a note.  This is
```

```
   then the minimum duration that such rests may have if they are to be
   consolidated. Default = NIL.
 - :warn. T or NIL to indicate whether the method should print a warning to
   the Lisp listener if it is mathematically unable to consolidate the
   rests. T = print warning. Default = NIL.
```

RETURN VALUE:

```
The rthm-seq-bar-object
```

EXAMPLE:

```
;;; Two examples with the same result; the first calling consolidate-rests
;;; twice, the second calling consolidate-rests-max
(let ((rsb (make-rthm-seq-bar '((2 2) (e) (e) (e) (e) (e) (s) (s) (s) e.))))
  (consolidate-rests rsb)
  (consolidate-rests rsb)
  (loop for r in (rhythms rsb) collect (data r)))

=> (2 Q S E.)

(let ((rsb (make-rthm-seq-bar '((2 2) (e) (e) (e) (e) (e) (s) (s) (s) e.))))
  (consolidate-rests-max rsb)
  (loop for r in (rhythms rsb) collect (data r)))

=> (2 Q S E.)
```

SYNOPSIS:

```
(defmethod consolidate-rests-max ((rsb rthm-seq-bar) &key beat min warn)
```

38.6.541 rthm-seq-bar/delete-beams

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Remove any beaming indications from the rthm-seq-bar object.

NB: This method changes the data for the rthm-seq-bar object's BEAMS slot
    and the individual BEAM slots of the RHYTHMs contained within the
    rthm-seq-bar's RHYTHMS slot. It does not change the value of the
    rthm-seq-bar's DATA slot.

NB: Neither the presence nor absence of beams are not reflected in the
    output of the print-simple method.
```

ARGUMENTS:

```
 - A rthm-seq-bar object.
```

RETURN VALUE:

```
 Returns T.
```

EXAMPLE:

```
(let ((rsb (make-rthm-seq-bar '((2 4) - s s - s - s s s - s s))))
  (delete-beams rsb))

=> T

(let ((rbs (make-rthm-seq-bar '((2 4) - s s - s - s s s - s s))))
  (delete-beams rsb)
  (beams rsb))

=> NIL

(let ((rsb (make-rthm-seq-bar '((2 4) - s s - s - s s s - s s))))
  (delete-beams rsb)
  (loop for r in (rhythms rbs) collect (beam r)))

=> (NIL NIL NIL NIL NIL NIL NIL NIL)

(let ((rbs (make-rthm-seq-bar '((2 4) - s s - s - s s s - s s))))
  (delete-beams rsb)
  (print rsb))

=>
RTHM-SEQ-BAR: time-sig: 1 (2 4)
              time-sig-given: T
              bar-num: -1
              old-bar-nums: NIL
              write-bar-num: NIL
              start-time: -1.0
              start-time-qtrs: -1.0
              is-rest-bar: NIL
              multi-bar-rest: NIL
              show-rest: T
              notes-needed: 8
              tuplets: NIL
              nudge-factor: 0.35
```

```
              beams: NIL
[...]
NAMED-OBJECT: id: NIL, tag: NIL,
data: ((2 4) - S S - S - S S S - S S)
```

SYNOPSIS:

```
(defmethod delete-beams ((rsb rthm-seq-bar))
```

38.6.542 rthm-seq-bar/delete-marks

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Delete all marks from the rhythm (or event) objects contained within a given
rthm-seq-bar object.

ARGUMENT
- A rthm-seq-bar object.
```

RETURN VALUE:

```
Always returns NIL.
```

EXAMPLE:

```
;; Create a rthm-seq-bar object and print the contents of the MARKS slots of
;; the contained event objects to see they're set to NIL by default. Fill them
;; each with a 's (staccato) mark and print the results. Apply the delete-marks
;; method and print the results again to see that the values have been reset to
;; NIL.
(let ((rsb (make-rthm-seq-bar
            (list
             '(3 8)
             (make-event 'cs4 'e)
             (make-event 'cs4 'e)
             (make-event 'cs4 'e)))))
  (print (loop for e in (rhythms rsb) collect (marks e)))
  (loop for e in (rhythms rsb) do (add-mark-once e 's))
  (print (loop for e in (rhythms rsb) collect (marks e)))
  (delete-marks rsb)
  (print (loop for e in (rhythms rsb) collect (marks e))))

=>
```

```
(NIL NIL NIL)
((S) (S) (S))
(NIL NIL NIL)
```

SYNOPSIS:

```
(defmethod delete-marks ((rsb rthm-seq-bar))
```

38.6.543 rthm-seq-bar/delete-tuplets

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Removes all indications for tuplet brackets from a given rthm-seq-bar
object.

NB: This method does not alter the tuplet rhythmic durations; it only
removes the tuplet bracket from the score.
```

ARGUMENTS:

```
- A rthm-seq-bar.
```

RETURN VALUE:

```
The rthm-seq-bar object
```

EXAMPLE:

```
(let ((rsb (make-rthm-seq-bar '((2 4) { 3 te te te } q))))
  (tuplets rsb))

=> ((3 0 2))

(let ((rsb (make-rthm-seq-bar '((2 4) { 3 te te te } q))))
  (delete-tuplets rsb))

=> NIL

(let ((rsb (make-rthm-seq-bar '((2 4) { 3 te te te } q))))
  (delete-tuplets rsb)
  (tuplets rsb))

=> NIL
```

```
(let ((rsb (make-rthm-seq-bar '((2 4) { 3 te te te } q))))
  (loop for r in (rhythms rsb) collect (bracket r)))

=> (((1 3)) (-1) (1) NIL)

(let ((rsb (make-rthm-seq-bar '((2 4) { 3 te te te } q))))
  (delete-tuplets rsb)
  (loop for r in (rhythms rsb) collect (bracket r)))

=> (NIL NIL NIL NIL)
```

SYNOPSIS:

```
(defmethod delete-tuplets ((rsb rthm-seq-bar))
```

38.6.544 rthm-seq-bar/delete-written

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Delete the contents of the WRITTEN-PITCH-OR-CHORD slot of a pitch object
within a given event object and reset to NIL.
```

ARGUMENTS:

```
- A rthm-seq-bar object.
```

RETURN VALUE:

```
Always returns NIL.
```

EXAMPLE:

```
;; Create a rthm-seq-bar object consisting of events and print the contents of
;; the WRITTEN-PITCH-OR-CHORD slots to see they're set to NIL. Apply the
;; set-written method with a value of -2 and print the contents of the
;; WRITTEN-PITCH-OR-CHORD slots to see the data of the newly created pitch
;; objects. Apply the delete-written method and print the contents of the
;; WRITTEN-PITCH-OR-CHORD slots to see they're empty.
(let ((rsb (make-rthm-seq-bar
            (list
             '(3 8)
             (make-event 'cs4 'e)
```

```
            (make-event 'cs4 'e)
            (make-event 'cs4 'e)))))
  (print (loop for p in (rhythms rsb)
               collect (written-pitch-or-chord p)))
  (set-written rsb -2)
  (print (loop for p in (rhythms rsb)
               collect (get-pitch-symbol p)))
  (delete-written rsb)
  (print (loop for p in (rhythms rsb)
               collect (written-pitch-or-chord p))))

=>
(NIL NIL NIL)
(B3 B3 B3)
(NIL NIL NIL)
```

SYNOPSIS:

```
(defmethod delete-written ((rsb rthm-seq-bar))
```

38.6.545 rthm-seq-bar/enharmonic

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Change the pitches of the events within a given rthm-seq-bar object to
their enharmonic equivalents.

In its default form, this method only applies to note names that already
contain an indication for an accidental (such as DF4 or BS3), while
"white-key" note names (such as B3 or C4) will not produce an enharmonic
equivalent. In order to change white-key pitches to their enharmonic
equivalents, set the :force-naturals argument to T.
```

ARGUMENTS:

```
- A rthm-seq-bar object.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :written. T or NIL to indicate whether the test is to handle the
  written or sounding pitch in the event. T = written.
  Default = NIL.
```

```
 - :force-naturals. T or NIL to indicate whether to force "natural"
   note names that contain no F or S in their name to convert to
   their enharmonic equivalent (e.g. B3 = CF4). Default = NIL.
 - :pitches. All sharp/flat pitches are changed by default but if a
   list of pitch objects or symbols is given, then only those
   pitches will be changed.  Note that if written is T, then this
   pitch list should be the written not sounding pitches.
   Default = NIL.
```

RETURN VALUE:

```
 Always returns T.
```

EXAMPLE:

```
;; The method returns T.                ; ; ; ; ;
  (let ((rsb (make-rthm-seq-bar
(list '(3 8)
(make-event 'cs4 'e)
(make-event 'cs4 'e)
(make-event 'cs4 'e)))))
(enharmonic rsb))

  => T

;; Create a rthm-seq-bar object with events, apply the enharmonic method, and ; ; ; ; ;
;; print the corresponding slots to see the changes ; ; ; ; ; ;
  (let ((rsb (make-rthm-seq-bar
(list
'(3 8)
(make-event 'cs4 'e)
(make-event 'cs4 'e)
(make-event 'cs4 'e)))))
(enharmonic rsb)
(loop for p in (rhythms rsb)
collect (get-pitch-symbol p)))

  => (DF4 DF4 DF4)

;; By default, the method will not change white-key pitches ; ; ; ; ;
  (let ((rsb (make-rthm-seq-bar
(list
'(3 8)
(make-event 'c4 'e)
(make-event 'c4 'e)
(make-event 'c4 'e)))))
```

```
(enharmonic rsb)
(loop for p in (rhythms rsb)
collect (get-pitch-symbol p)))

  => (C4 C4 C4)

;; This can be forced by setting the :force-naturals argument to T ; ; ; ; ;
  (let ((rsb (make-rthm-seq-bar
(list
'(3 8)
(make-event 'c4 'e)
(make-event 'c4 'e)
(make-event 'c4 'e)))))
(enharmonic rsb :force-naturals t)
(loop for p in (rhythms rsb)
collect (get-pitch-symbol p)))

  => (BS3 BS3 BS3)

;; Apply the set-written method to fill the WRITTEN-PITCH-OR-CHORD slot, print ; ; ; ; ;
;; its contents, apply the enharmonic method with the :written keyword argument ; ; ; ; ;
;; set to T, then print the pitch data of the same slot again to see the ; ; ; ; ;
;; change.                              ; ; ; ; ;
  (let ((rsb (make-rthm-seq-bar
(list
'(3 8)
(make-event 'cs4 'e)
(make-event 'cs4 'e)
(make-event 'cs4 'e)))))
(set-written rsb -3)
(print (loop for p in (rhythms rsb)
collect (get-pitch-symbol p)))
(enharmonic rsb :written t)
(print (loop for p in (rhythms rsb)
collect (get-pitch-symbol p))))

  =>
  (BF3 BF3 BF3)
  (AS3 AS3 AS3)
```

SYNOPSIS:

```
(defmethod enharmonic ((rsb rthm-seq-bar) &key written force-naturals
                       ;; MDE Wed Apr 18 11:34:01 2012
                       pitches)
```

38.6.546 rthm-seq-bar/events-update-time

[rthm-seq-bar] [Functions]

DATE:

```
March 16th 2016, Edinburgh
```

DESCRIPTION:

```
Update the start-time and related slots of any events for which timing
information is missing or out-of-sync.
```

ARGUMENTS:

```
- a list of event objects
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :start-time. The start time in seconds of the first event. Default = 0.0
- :start-time-qtrs. The start time in quarter notes of the first event (used
  in MIDI files). If NIL, then this will be calculated as a function of
  :start-time and :tempo. If not NIL, this will mean start-time is ignored
  when writing MIDI files. Default = nil.
- :tempo. The tempo in beats per minute (or a tempo object)
- :max-start-time. A maximum time in seconds above which we stop processing
  (and returning) events.
```

RETURN VALUE:

```
two values: the list of events with their times updated, and the time the
last event finishes.
```

EXAMPLE:

```
(event-list-to-midi-file
  (events-update-time (make-events2 '(q q q) '(g4 g4 g4) 1))
  :midi-file "/tmp/test.mid" :start-tempo 120)
```

SYNOPSIS:

```
(defun events-update-time (events &key (start-time 0.0) start-time-qtrs
                                  max-start-time (tempo 60.0))
```

38.6.547 rthm-seq-bar/fill-with-rhythms

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Any rhythms (or event objects) in the existing rthm-seq-bar object will be
deleted, and then rhythm (or event) objects will be taken one by one from
the <rhythms> argument until the bar is full.

If too few rhythm or event objects are given, a warning will be printed
that there are too few beats in the bar.

If there are too many and the last rhythm or event object to be placed in
the bar fills out the bar evenly, no warning is printed and the remaining
rhythm or event objects are ignored. If :is-full-error is T (default) and
the last rhythm or event object that the method attempts to place in the
bar is too long to fit evenly into the bar, the method will drop into the
debugger with an error. If in this case :is-full-error is NIL the bar will
remain underfull and the index of the last rhythm added before over-filling
will be returned.

The :transposition, :midi-channel, and :microtones-midi-channel arguments
can only be used in conjunction with event objects.

The number of rhythms (or event objects) used is returned.

NB: This method does not change the DATA slot of the rthm-seq-bar object
    itself to reflect the new rhythms. Instead, it changes the contents of
    the RHYTHMS slot within that object and changes the DATA of the
    rthm-seq-bar object to NIL. It also assigns the ID of the named-object
    to "rhythms-inserted-by-fill-with-rhythms".
```

ARGUMENTS:

```
- A rthm-seq-bar object.
- A list of rhythm objects or event objects.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :transposition. An integer or NIL to indicate the transposition in
  semitones for written pitches of any event objects passed. If NIL, no
  written-pitches will be created. Default = NIL.
- :midi-channel. An integer that will be used to set the MIDI-CHANNEL slot
  of any event objects passed. Default = 0.
```

- :microtones-midi-channel. An integer that is the MIDI channel that will be assigned to event objects for microtonal MIDI pitches. NB: This value is only set when attached to event objects within a slippery-chicken object. Default = 0. NB: See player.lsp/make-player for details on microtones in MIDI output.
- :new-id. An optional ID for new rhythm or event objects added. Default = "rhythms-inserted-by-fill-with-rhythms".
- :warn. T or NIL to indicate whether a warning should be printed if there are not enough rhythms to create a full bar. T = warn. Default = T.
- :is-full-error. T or NIL to indicate whether the last rhythm or event object that the method attempts to add to the bar is too long to fit evenly into the bar. T = drop into the debugger with an error if this is the case. Default = T.

RETURN VALUE:

The number of rhythm or event objects used, which could be 0. NIL will be returned if, for example, an empty list of rhythms were passed, so do check for a NIL result.

EXAMPLE:

```
(let ((rsb (make-rthm-seq-bar '((3 4) q q q))))
  (fill-with-rhythms rsb (loop for r in '(e e e e e e)
                           collect (make-rhythm r))))

=> 6

(let ((rsb (make-rthm-seq-bar '((3 4) q q q))))
  (fill-with-rhythms rsb (loop for r in '(e e e e e e)
                           collect (make-rhythm r)))
  (print-simple rsb))

=> NIL
(3 4): note E, note E, note E, note E, note E, note E,

(let ((rsb (make-rthm-seq-bar '((3 4) q q q))))
  (fill-with-rhythms rsb (loop for r in '(e e e e e e)
                           collect (make-rhythm r)))
  (print rsb))

=>
RTHM-SEQ-BAR: time-sig: 0 (3 4)
              time-sig-given: T
              bar-num: -1
              old-bar-nums: NIL
```

```
              write-bar-num: NIL
              start-time: -1.0
              start-time-qtrs: -1.0
              is-rest-bar: NIL
              multi-bar-rest: NIL
              show-rest: T
              notes-needed: 6
              tuplets: NIL
              nudge-factor: 0.35
              beams: NIL
              current-time-sig: 0
              write-time-sig: T
              num-rests: 0
              num-rhythms: 6
              num-score-notes: 6
              rhythms: (
RHYTHM: value: 8.0, duration: 0.5, rq: 1/2, is-rest: NIL, score-rthm: 8.0,
        undotted-value: 8, num-flags: 1, num-dots: 0, is-tied-to: NIL,
        is-tied-from: NIL, compound-duration: 0.5, is-grace-note: NIL,
        needs-new-note: T, beam: NIL, bracket: NIL, rqq-note: NIL,
        rqq-info: NIL, marks: NIL, marks-in-part: NIL, letter-value: 8,
        tuplet-scaler: 1, grace-note-duration: 0.05,
LINKED-NAMED-OBJECT: previous: NIL
                     this: NIL
                     next: NIL
NAMED-OBJECT: id: E, tag: NIL,
data: E
[...]
NAMED-OBJECT: id: "rhythms-inserted-by-fill-with-rhythms", tag: NIL,
data: NIL

;;; Using the :transpositions and :midi-channel arguments
(let ((rsb (make-rthm-seq-bar '((4 4) q q q q))))
  (fill-with-rhythms rsb (loop for r in '(h q e s s)
                               for p in '(c4 dqs4 e4 gqf4 a4)
                             collect (make-event p r))
                     :microtones-midi-channel 12
                     :transposition -14
                     :midi-channel 11)
  (print
   (loop for e in (rhythms rsb)
      collect (data (pitch-or-chord e))))
  (print
   (loop for e in (rhythms rsb)
      collect (data (written-pitch-or-chord e))))
  (print
```

```
   (loop for e in (rhythms rsb)
      collect (midi-channel (pitch-or-chord e)))))

=>
(C4 DQS4 E4 GQF4 A4)
(D5 EQS5 FS5 AQF5 B5)
(11 12 11 12 11)
```

SYNOPSIS:

```
(defmethod fill-with-rhythms ((rsb rthm-seq-bar) rhythms
                              &key
                                ;; 24.3.11 add this too to make sure written
                                ;; pitch is set--this is the instrument
                                ;; transposition e.g. -14 for bass clarinet
                                transposition
                                (midi-channel 0)
                                (microtones-midi-channel 0)
                                (new-id "rhythms-inserted-by-fill-with-rhythms")
                                (warn t)
                                (is-full-error t))
```

38.6.548 rthm-seq-bar/find-repeated-pitches

[rthm-seq-bar] [Methods]

DATE:

```
January 15th 2026
```

DESCRIPTION:

```
find the indices of repeated notes within a bar
```

ARGUMENTS:

```
- the rthm-seq-bar object
```

RETURN VALUE:

```
a list of indices (integers) to the repeated notes
```

EXAMPLE:

```
(let ((rsb (make-rest-bar '(5 8))))
  (setf (rhythms rsb)
        (make-events3 's '(c4 d c c f (e f) (e f) fs fs fs)))
  (find-repeated-pitches rsb))
--> (3 6 8 9)
```

SYNOPSIS:

```
(defmethod find-repeated-pitches ((rsb rthm-seq-bar))
```

38.6.549 rthm-seq-bar/force-all-rests

[rthm-seq-bar] [Methods]

DATE:

```
November 7th 2018, Heidhausen
```

DESCRIPTION:

```
Makes all events in the bar rests. This differs from force-rest-bar in that
it doesn't replace the events with a single full-bar rest
```

ARGUMENTS:

```
- a rthm-seq-bar object
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate that any marks (e.g. dynamics) should be retained,
  whether or not these make sense on rests. The exception is that slurs
  won't be retained.  Default = NIL.
```

RETURN VALUE:

```
the modified rthm-seq-bar object
```

SYNOPSIS:

```
(defmethod force-all-rests ((rsb rthm-seq-bar) &optional retain-marks)
```

38.6.550 rthm-seq-bar/force-rest-bar

[rthm-seq-bar] [Methods]

DESCRIPTION:

Force all rhythms of a rthm-seq-bar object to be replaced by a whole-bar rest.

NB: This method changes the value of the RHYTHMS slot of the rthm-seq-bar but not the value of the rthm-seq-bar DATA slot.

ARGUMENTS:

- A rthm-seq-bar object.

RETURN VALUE:

Returns a rthm-seq-bar object.

EXAMPLE:

```
(let ((rsb (make-rthm-seq-bar '((2 4) q e s s))))
  (force-rest-bar rsb))

=>
RTHM-SEQ-BAR: time-sig: 1 (2 4)
              time-sig-given: T
              bar-num: -1
              old-bar-nums: NIL
              write-bar-num: NIL
              start-time: -1.0
              start-time-qtrs: -1.0
              is-rest-bar: T
[...]
RHYTHM: value: 2.0, duration: 2.0, rq: 2, is-rest: T,
[...]
NAMED-OBJECT: id: NIL, tag: NIL,
data: ((2 4) Q E S S)

(let ((rsb (make-rthm-seq-bar '((2 4) q e s s))))
  (force-rest-bar rsb)
  (print-simple rsb))

=>
(2 4): rest 2,
```

SYNOPSIS:

```
(defmethod force-rest-bar ((rsb rthm-seq-bar))
```

38.6.551 rthm-seq-bar/get-beats

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Try to organise the events of a bar into sublists of a beat's worth of
events.

If beat is nil, we'll get the beat from the time-sig. If check-dur is T
we'll make sure we get a complete beat of rhythms for each beat of the bar
Since May 1 2012, check-dur can be T, NIL, #'warn or #'error, where t is
the same as #'error. If you pass a symbol like 'silent the duration will be
checked and NIL returned if we can't get an exact beat's worth of rthms.
```

ARGUMENTS:

```
- a rthm-seq-bar object
```

OPTIONAL ARGUMENTS:

```
- the beat, as a symbol or rhythm object (see above)
- whether to check the duration of the beats to see if they're full (see
above)
```

RETURN VALUE:

```
A list of lists of event/rhythm objects
```

EXAMPLE:

```
NB the elements returned will actually be rhythm objects but I'll just show the
rhythm symbol for clarity:
(get-beats (make-rthm-seq-bar '((4 4) s x 4 q e e s e.)))
-->
((s s s s) (q) (e e) (s e.))
```

SYNOPSIS:

```
(defmethod get-beats ((rsb rthm-seq-bar) &optional beat check-dur)
```

38.6.552 rthm-seq-bar/get-last-attack

[rthm-seq-bar] [Methods]

DESCRIPTION:

Gets the rhythm object for the last note that needs an attack (i.e. not a rest and not a tied note) in a given rthm-seq-bar object.

ARGUMENTS:

- The given rthm-seq-bar object.

OPTIONAL ARGUMENTS:

- T or NIL indicating whether to print a warning message if the given index (minus one) is greater than the number of attacks in the RHYTHMS list (default = T). This is a carry-over argument from the get-nth-attack method called within the get-last-attack method and not likely to be needed for use with get-last-attack.

RETURN VALUE:

A rhythm object.

Returns NIL if the given index is higher than the highest possible index of attacks in the given rthm-seq-bar object.
Get the rhythm object of the last

EXAMPLE:

```
(let ((rsb (make-rthm-seq-bar '((3 4) q+e (e) s (s) e))))
(get-last-attack rsb))

=>
RHYTHM: value: 8.0, duration: 0.5, rq: 1/2, is-rest: NIL, score-rthm: 8.0,
undotted-value: 8, num-flags: 1, num-dots: 0, is-tied-to: NIL,
is-tied-from: NIL, compound-duration: 0.5, is-grace-note: NIL,
needs-new-note: T, beam: NIL, bracket: NIL, rqq-note: NIL,
rqq-info: NIL, marks: NIL, marks-in-part: NIL, letter-value: 8,
tuplet-scaler: 1, grace-note-duration: 0.05,
LINKED-NAMED-OBJECT: previous: NIL
this: NIL
next: NIL
NAMED-OBJECT: id: E, tag: NIL,
data: E
```

SYNOPSIS:

```
(defmethod get-last-attack ((rsb rthm-seq-bar) &optional (warn t))
```

38.6.553 rthm-seq-bar/get-last-attacks

[rthm-seq-bar] [Methods]

DATE:

```
July 24th 2015, Glenferness
```

DESCRIPTION:

```
Return the last nth attacks in a bar, with any notes tied from them.
```

ARGUMENTS:

```
- a rthm-seq-bar object
- an integer indicating how many attacks to return
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether a warning should be issued if the required
  number of attacks cannot be returned.
```

RETURN VALUE:

```
A list of lists of rhythm or event objects. The length of the outer list
will be the same as the second argument. The sublists will contain the
attack event with any subsequent events tied from it. NB if a request is
made for more attacks than the bar contains the method returns NIL (rather
than all the attacks in the bar).
```

SYNOPSIS:

```
(defmethod get-last-attacks ((rsb rthm-seq-bar) how-many &optional (warn t))
```

38.6.554 rthm-seq-bar/get-last-event

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Get the last event object (or rhythm object) of a given rthm-seq-bar
object.
```

ARGUMENTS:

```
 - A rthm-seq-bar object.
```

RETURN VALUE:

```
 Returns a rhythm object.
```

EXAMPLE:

```
;; Returns a rhythm object.
(let ((rsb (make-rthm-seq-bar '((2 4) s s e q))))
(get-last-event rsb))

=>
RHYTHM: value: 4.000, duration: 1.000, rq: 1, is-rest: NIL,
score-rthm: 4.0f0, undotted-value: 4, num-flags: 0, num-dots: 0,
is-tied-to: NIL, is-tied-from: NIL, compound-duration: 1.000,
is-grace-note: NIL, needs-new-note: T, beam: NIL, bracket: NIL,
rqq-note: NIL, rqq-info: NIL, marks: NIL, marks-in-part: NIL,
letter-value: 4, tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: Q, tag: NIL,
data: Q
```

SYNOPSIS:

```
(defmethod get-last-event ((rsb rthm-seq-bar))
```

38.6.555 rthm-seq-bar/get-nearest-by-start-time

[rthm-seq-bar] [Methods]

DATE:

```
 December 16th 2019, Essen Werden
```

DESCRIPTION:

```
 Find the event with the closest start-time to a given target.
```

ARGUMENTS:

```
 - a rthm-seq-bar object, with times appropriately initialised (e.g. by
   make-slippery-chicken)
 - the target: either start time in seconds (float) or an event object
```

RETURN VALUE:

```
two values: the closest event and it's position in the bar (0-based)
```

SYNOPSIS:

```
(defmethod get-nearest-by-start-time ((rsb rthm-seq-bar) target)
```

38.6.556 rthm-seq-bar/get-nth-attack

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Gets the rhythm object for the nth note in a given rthm-seq-bar that needs
an attack, i.e. not a rest and not tied.
```

ARGUMENTS:

```
- The zero-based index number indicating which attack is sought.
- The given rthm-seq-bar object in which to search.
```

OPTIONAL ARGUMENTS:

```
- T or NIL indicating whether to print a warning message if the given index
  is greater than the number of attacks in the RHYTHMS list (minus one to
  compensate for the zero-based indexing) (default = T).
```

RETURN VALUE:

```
A rhythm object.

Returns NIL if the given index is higher than the highest possible index of
attacks in the given rthm-seq-bar object.
```

EXAMPLE:

```
;; The method returns a rhythm object when successful
(let ((rsb (make-rthm-seq-bar '((3 4) q+e (e) s (s) e))))
(get-nth-attack 0 rsb))

=>
RHYTHM: value: 4.0, duration: 1.0, rq: 1, is-rest: NIL, score-rthm: 4.0,
undotted-value: 4, num-flags: 0, num-dots: 0, is-tied-to: NIL,
```

```
is-tied-from: NIL, compound-duration: 1.0, is-grace-note: NIL,
needs-new-note: T, beam: NIL, bracket: NIL, rqq-note: NIL,
rqq-info: NIL, marks: NIL, marks-in-part: NIL, letter-value: 4,
tuplet-scaler: 1, grace-note-duration: 0.05,
LINKED-NAMED-OBJECT: previous: NIL
this: NIL
next: NIL
NAMED-OBJECT: id: "Q", tag: NIL,
data: Q

(let ((rsb (make-rthm-seq-bar '((3 4) q+e (e) s (s) e))))
(data (get-nth-attack 1 rsb)))

=> S

(Let ((rsb (make-rthm-seq-bar '((3 4) q+e (e) s (s) e))))
(get-nth-attack 3 rsb))

=> NIL
WARNING: rthm-seq-bar::get-nth-attack:  index (3) < 0 or >= notes-needed (3)

(Let ((rsb (make-rthm-seq-bar '((3 4) q+e (e) s (s) e))))
(get-nth-attack 3 rsb nil))

=> NIL
```

SYNOPSIS:

```
(defmethod get-nth-attack (index (rsb rthm-seq-bar) &optional (warn t))
```

38.6.557 rthm-seq-bar/get-nth-attack-with-tied

[rthm-seq-bar] [Methods]

DATE:

```
July 24th 2015, Glenferness
```

DESCRIPTION:

```
Same as get-nth-attack method but will always return a list of the attacked
event plus any following events this is tied to.
```

ARGUMENTS:

- The zero-based index number indicating which attack is sought.
- The given rthm-seq-bar object in which to search.

OPTIONAL ARGUMENTS:

- T or NIL indicating whether to print a warning message if the given index is greater than the number of attacks in the RHYTHMS list (minus one to compensate for the zero-based indexing) (default = T).

RETURN VALUE:

A rhythm object.

Returns NIL if the given index is higher than the highest possible index of attacks in the given rthm-seq-bar object.

SYNOPSIS:

```
(defmethod get-nth-attack-with-tied (index (rsb rthm-seq-bar)
                                     &optional (warn t))
```

38.6.558 rthm-seq-bar/get-nth-event

[rthm-seq-bar] [Methods]

DESCRIPTION:

Get the nth event (rhythm) in the given rthm-seq-bar object. This is a zero-based index.

The method defaults to interrupting with an error if the n-value is greater than the number of items in the rthm-seq-bar. This can be disabled using the optional argument.

ARGUMENTS:

- A rthm-seq-bar object.
- An index number.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether to interrupt and drop into the debugger with an error. Default = T.

RETURN VALUE:

A rhythm object when successful.

Returns NIL when the specified index number is greater than the number of events in the rthm-seq-bar object. Also prints an error in this case by default, which can be suppressed by setting the optional argument to NIL.

EXAMPLE:

```
;; Zero-based indexing. Returns a rhythm object when successful.
(let ((rsb (make-rthm-seq-bar '((2 4) q e s s))))
(get-nth-event 0 rsb))

=>
RHYTHM: value: 4.000, duration: 1.000, rq: 1, is-rest: NIL,
score-rthm: 4.0f0, undotted-value: 4, num-flags: 0, num-dots: 0,
is-tied-to: NIL, is-tied-from: NIL, compound-duration: 1.000,
is-grace-note: NIL, needs-new-note: T, beam: NIL, bracket: NIL,
rqq-note: NIL, rqq-info: NIL, marks: NIL, marks-in-part: NIL,
letter-value: 4, tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: Q, tag: NIL,
data: Q

;; Interrupts with an error and drops into the debugger by default if the
;; specified index number is greater than the number of events in the
;; rthm-seq-bar.
(let ((rsb (make-rthm-seq-bar '((2 4) q e s s))))
(get-nth-event 4 rsb))

=>
rthm-seq-bar::get-nth-event: Couldn't get event with index 4
[Condition of type SIMPLE-ERROR]

;; The error can be suppressed by setting the optional argument to NIL
(let ((rsb (make-rthm-seq-bar '((2 4) q e s s))))
(get-nth-event 4 rsb nil))

=> NIL
```

SYNOPSIS:

```
(defmethod get-nth-event (index (rsb rthm-seq-bar)
                          &optional (error t))
```

38.6.559 rthm-seq-bar/get-nth-non-rest-rhythm

[rthm-seq-bar] [Methods]

DESCRIPTION:

Get the nth non-rest rhythm object stored in the given rthm-seq-bar.

ARGUMENTS:

- The zero-based index number indicating which non-rest-rhythm is sought.
- The given rthm-seq-bar object in which to search.

OPTIONAL ARGUMENTS:

- T or NIL indicating whether to print an error message if the given index is greater than the number of non-rest rhythms in the RHYTHMS list (minus one to compensate for the zero-based indexing). (Default = T).

RETURN VALUE:

A rhythm object.

Returns NIL if the given index is higher than the highest possible index of non-rest rhythms in the given rthm-seq-bar object.

EXAMPLE:

```
;; The method returns a rhythm object when successful ; ;
(let ((rsb (make-rthm-seq-bar '((2 4) e (e) s s (s) s))))
(get-nth-non-rest-rhythm 0 rsb))

=>
RHYTHM: value: 8.0, duration: 0.5, rq: 1/2, is-rest: NIL, score-rthm: 8.0,
undotted-value: 8, num-flags: 1, num-dots: 0, is-tied-to: NIL,
is-tied-from: NIL, compound-duration: 0.5, is-grace-note: NIL,
needs-new-note: T, beam: NIL, bracket: NIL, rqq-note: NIL,
rqq-info: NIL, marks: NIL, marks-in-part: NIL, letter-value: 8,
tuplet-scaler: 1, grace-note-duration: 0.05,
LINKED-NAMED-OBJECT: previous: NIL
this: NIL
next: NIL
NAMED-OBJECT: id: E, tag: NIL,
data: E
```

```
(let ((rsb (make-rthm-seq-bar '((2 4) e (e) s s (s) s))))
(data (get-nth-non-rest-rhythm 1 rsb)))

=> S

(let ((rsb (make-rthm-seq-bar '((2 4) e (e) s s (s) s))))
(data (get-nth-non-rest-rhythm 4 rsb)))

=>
Evaluation aborted on #<SIMPLE-ERROR>
rthm-seq-bar::get-nth-non-rest-rhythm: Couldn't get non-rest rhythm with index
4 for bar number -1
[Condition of type SIMPLE-ERROR]

(let ((rsb (make-rthm-seq-bar '((2 4) e (e) s s (s) s))))
(get-nth-non-rest-rhythm 4 rsb nil))

=> NIL
```

SYNOPSIS:

```
(defmethod get-nth-non-rest-rhythm (index (rsb rthm-seq-bar)
                                    &optional (error t))
```

38.6.560 rthm-seq-bar/get-nth-rest

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Gets the rhythm object of the nth rest in a given rthm-seq-bar.
```

ARGUMENTS:

```
- The zero-based index number indicating which rest is sought.
- The given rthm-seq-bar object in which to search.
```

OPTIONAL ARGUMENTS:

```
- T or NIL indicating whether to print an error message if the given index
  is greater than the number of rests in the RHYTHMS list (minus one to
  compensate for the zero-based indexing) (default = T).
```

RETURN VALUE:

```
A rhythm object.

Returns NIL if the given index is higher than the highest possible index of
rests in the given rthm-seq-bar object.
```

EXAMPLE:

```
(let ((rsb (make-rthm-seq-bar '((3 4) e (e) s s (s) s (q)))))
(get-nth-rest 0 rsb))

=>
RHYTHM: value: 8.0, duration: 0.5, rq: 1/2, is-rest: T, score-rthm: 8.0,
undotted-value: 8, num-flags: 1, num-dots: 0, is-tied-to: NIL,
is-tied-from: NIL, compound-duration: 0.5, is-grace-note: NIL,
needs-new-note: NIL, beam: NIL, bracket: NIL, rqq-note: NIL,
rqq-info: NIL, marks: NIL, marks-in-part: NIL, letter-value: 8,
tuplet-scaler: 1, grace-note-duration: 0.05,
LINKED-NAMED-OBJECT: previous: NIL
this: NIL
next: NIL
NAMED-OBJECT: id: E, tag: NIL,
data: E

(let ((rsb (make-rthm-seq-bar '((3 4) e (e) s s (s) s (q)))))
(data (get-nth-rest 2 rsb)))

=> Q

(let ((rsb (make-rthm-seq-bar '((3 4) e (e) s s (s) s (q)))))
(get-nth-rest 3 rsb t))

Evaluation aborted on #<SIMPLE-ERROR>
rthm-seq-bar::get-nth-rest: Couldn't get rest with index 3
[Condition of type SIMPLE-ERROR]

(let ((rsb (make-rthm-seq-bar '((3 4) e (e) s s (s) s (q)))))
(get-nth-rest 3 rsb nil))

=> NIL
```

SYNOPSIS:

```
(defmethod get-nth-rest (index (rsb rthm-seq-bar)
                         &optional (error t))
```

38.6.561 rthm-seq-bar/get-pitch-symbols

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Return a list of the pitch symbols for the events in the bar. Note that
chord pitches will be in a sublist such as ((b4 ds5) eqf5 (e5 g5)).
```

ARGUMENTS:

```
- a rthm-seq-bar object
```

RETURN VALUE:

```
A list of pitch symbols
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :set-palette '((1 ((gs4 bf4 c4))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                :pitch-seq-palette ((1 2 1 2 1 1 3 1)))))
        :rthm-seq-map '((1 ((vn (1 1 1))))))))
  (get-pitch-symbols (get-bar mini 1 'vn)))
  =>
  (C4 GS4 C4 GS4 C4 C4 BF4 C4)
```

SYNOPSIS:

```
(defmethod get-pitch-symbols ((rsb rthm-seq-bar) &optional written)
```

38.6.562 rthm-seq-bar/get-rhythm-symbols

[rthm-seq-bar] [Methods]

DATE:

```
01-May-2012
```

DESCRIPTION:

Return the rhythms of a given rthm-seq-bar object as a list of rhythm symbols.

ARGUMENTS:

- A rthm-seq-bar object.

RETURN VALUE:

- A list of rhythm symbols.

EXAMPLE:

```
  (let ((rsb (make-rthm-seq-bar '((4 4) q e s s q. e))))
(get-rhythm-symbols rsb))

  => (Q E S S Q. E)
```

SYNOPSIS:

```
(defmethod get-rhythm-symbols ((rsb rthm-seq-bar))
```

38.6.563 rthm-seq-bar/get-time-sig

[rthm-seq-bar] [Methods]

DESCRIPTION:

Return the time-sig object for the given rthm-seq-bar object.

ARGUMENTS:

- A rthm-seq-bar object.

RETURN VALUE:

A time-sig object.

EXAMPLE:

```
(let ((rsb (make-rthm-seq-bar '((2 4) q e s s))))
  (get-time-sig rsb))
=>
TIME-SIG: num: 2, denom: 4, duration: 2.0, compound: NIL, midi-clocks: 24,
```

```
num-beats: 2
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "0204", tag: NIL,
data: (2 4)
```

SYNOPSIS:

```
(defmethod get-time-sig ((rsb rthm-seq-bar) &optional ignore)
```

38.6.564 rthm-seq-bar/get-time-sig-as-list

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Get the time signature for a given rthm-seq-bar object in list form.
```

ARGUMENTS:

```
- A rthm-seq-bar object.
```

RETURN VALUE:

```
A list.
```

EXAMPLE:

```
(let ((rsb (make-rthm-seq-bar '((2 4) q e s s))))
  (get-time-sig-as-list rsb))
=> (2 4)
```

SYNOPSIS:

```
(defmethod get-time-sig-as-list ((rsb rthm-seq-bar))
```

38.6.565 rthm-seq-bar/invert

[rthm-seq-bar] [Methods]

DATE:

```
11th July 2020, Heidhausen
```

DESCRIPTION:

Turn all the rests in a rthm-seq-bar into notes and all the notes/chords into rests. NB this method is destructive.

ARGUMENTS:

- the rthm-seq-bar object

OPTIONAL ARGUMENTS:

- a list of pitches/chords (either objects or symbols/lists of symbols) to replace the rests with. This will be used cyclically if there aren't enough for the number of generated rests. They will not all be used up if there aren't enough rests. If NIL then all rests will be replaced with middle Cs.
- T or NIL to tie notes rather than use new ones when more than one consecutive rest is created.

RETURN VALUE:

the modified rthm-seq-bar object

EXAMPLE:

```
(let ((r (make-rest 'e))) (setf (pitch-or-chord r) '(c4 e4)) r)
--->
EVENT: start-time: NIL, end-time: NIL,
       duration-in-tempo: 0.000,
       compound-duration-in-tempo: 0.000,
       amplitude: 0.300
       bar-num: -1, marks-before: NIL,
       tempo-change: NIL
       instrument-change: NIL
       display-tempo: NIL, start-time-qtrs: -1.000,
       midi-time-sig: NIL, midi-program-changes: NIL,
       midi-control-changes: NIL,
       8va: 0, player: NIL
       asco-label: NIL, asco-msgs: NIL
       set-ref: NIL
       pitch-or-chord:
CHORD: auto-sort: T, marks: NIL, micro-tone: NIL, micro-tonality: 0.0
centroid: NIL, dissonance: NIL
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL,
                     this: NIL,
                     next: NIL
```

```
NAMED-OBJECT: id: NIL, tag: NIL,
data: (
PITCH: frequency: 261.626, midi-note: 60, midi-channel: NIL
...
```

SYNOPSIS:

```
(defmethod invert ((rsb rthm-seq-bar) &optional pitch-list tie)
```

38.6.566 rthm-seq-bar/make-rest-bar

[rthm-seq-bar] [Functions]

DESCRIPTION:

```
Make a rthm-seq-bar object that consists of a bar of rest.
```

ARGUMENTS:

```
- The time signature of the rthm-seq-bar object to be made, as a quoted
  list.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether to print the time signature in score output.
- show-rest. This argument indicates whether or not to print the rest in
  the printed score output (CMN/LilyPond). Default = T.

The remaining optional arguments are set internally by the
slippery-chicken class, but can be read by the user for debugging.
- missing-duration: Indicates whether the bar is missing a duration.
- player-section-ref: The current player and section of the given
  rthm-seq-bar object.
- nth-seq: The current sequenz (with a "z") of the given rthm-seq-bar
  object.
- nth-bar: The current bar number of the given rthm-seq-bar object.
```

RETURN VALUE:

```
A rthm-seq-bar object.
```

EXAMPLE:

```
  (let ((rsb-rb (make-rest-bar '(2 4) nil t)))
```

```
(format t "~%time-sig: ~a~%is-rest-bar: ~a~%write-time-sig: ~
            ~a~%show-rest: ~a~%"
(data (get-time-sig rsb-rb))
(is-rest-bar rsb-rb)
(write-time-sig rsb-rb)
(show-rest rsb-rb))
(print-simple rsb-rb)
rsb-rb)

  =>
  RTHM-SEQ-BAR: time-sig: 0 (2 4), time-sig-given: T, bar-num: -1,
  [...]

  time-sig: (2 4)
  is-rest-bar: T
  write-time-sig: NIL
  show-rest: T
  (2 4): rest 2,
```

SYNOPSIS:

```
(defun make-rest-bar (time-sig &optional
                                write-time-sig
                                (show-rest t)
                                missing-duration
                                player-section-ref nth-seq
                                nth-bar)
```

38.6.567 rthm-seq-bar/make-rsb-from-unit-multipliers

[rthm-seq-bar] [Functions]

DATE:

```
Jun 6th 2020, Heidhausen
```

DESCRIPTION:

```
Make a rthm-seq-bar using multiples of a single unit e.g. '(2 2 1) would
result in an automatic time signature of 5/8 if passed a unit of 'e (or
8). In that case the rhythms would be q q e. Rests may be created by passing
a number in parentheses.
```

ARGUMENTS:

```
 - the rhythm unit (symbol, number or rhythm object)
 - the list of multipliers
```

RETURN VALUE:

```
 a rthm-seq-bar object
```

EXAMPLE:

```
(print-simple (make-rsb-from-unit-multipliers 'e '(3 4 (1) 2)))
-->
NIL: bar -1: (10 8):
note Q.,
note H,
rest E,
note Q,
```

SYNOPSIS:

```
(defun make-rsb-from-unit-multipliers (unit multipliers)
```

38.6.568 rthm-seq-bar/make-rthm-seq-bar

[rthm-seq-bar] [Functions]

DESCRIPTION:

```
 Public interface for creating a rthm-seq-bar object, each instance of which
 which holds one of the individual bars that reside in a rhythmic
 sequence.

 This class is responsible for parsing lists containing rhythms and time
 signatures, but not for parsing these things themselves--that is done by
 separate classes.

 A { followed by a number means that all the notes from now to the } will be
 enclosed in a bracket with the number inside.  This may be nested.  A -
 indicates beaming: the first - indicates the start of a beam, the second
 the end of that beam.
```

ARGUMENTS:

```
 - A list of rhythmic durations, which may include ties and dots. Durations
   may be written as numeric (integer) values or may use the CM/CMN/SCORE
   alphabetic shorthand s=16, e=8, q=4, h=2, w=1. NB: Repeating rhythms can
```

be indicated using a shorthand notation consisting of a multiplication symbol ('x'), e.g.: (make-rthm-seq-bar '((4 4) s x 16)). This may also be a list of rhythm or event objects.

make-rthm-seq-bar requires a time signature. If no time signature is provided, the most recently defined time signature will be used. If one is provided, it must be included as the first element of the data list. The time signature is formulated as a list containing two integers, the first being the number of beats in the bar and the second being the beat unit for the bar.

OPTIONAL ARGUMENTS:

- A name (symbol) for the object ID.

RETURN VALUE:

Returns a rthm-seq-bar.

EXAMPLE:

```
(make-rthm-seq-bar '((2 4) q e s s))

=>
RTHM-SEQ-BAR:
[...]
NAMED-OBJECT: id: NIL, tag: NIL,
data: ((2 4) Q E S S)

(make-rthm-seq-bar '((2 4) q e s s) 'test)
=>
RTHM-SEQ-BAR:
[...]
NAMED-OBJECT: id: TEST, tag: NIL,
data: ((2 4) Q E S S)

(make-rthm-seq-bar '((2 4) q \+16\.+32 e))
=>
RTHM-SEQ-BAR:
[...]
NAMED-OBJECT: id: NIL, tag: NIL,
data: ((2 4) Q +16.+32 E)

(make-rthm-seq-bar '((2 4) { 3 te te te } q))
=>
```

```
RTHM-SEQ-BAR:
[...]
NAMED-OBJECT: id: NIL, tag: NIL,
data: ((2 4) { 3 TE TE TE } Q)
```

SYNOPSIS:

```
(defun make-rthm-seq-bar (rhythms &optional name)
```

38.6.569 rthm-seq-bar/pad-right

[rthm-seq-bar] [Methods]

DATE:

May 13th 2025

DESCRIPTION:

If a bar is underfull, try and fill it with one or more events: either rests or tied pitches.

ARGUMENTS:

- the rthm-seq-bar object

OPTIONAL ARGUMENTS:

keyword arguments:
- :pitch. a pitch (object, symbol) or chord (object, list of pitch symbols) that will be used to create the events instead of rests. Note that if the duration requires more than one event (e.g. a 1/4 and a 16th note) then the pitch will be used (and tied) for all. Note also that the pitch-or-chord slot will be set in any case, not the written-pitch-or-chord slot.
- :dots. T or nil to indicated whether tied notes should be turned into dotted notes where possible.

RETURN VALUE:

T or NIL: whether it was possible to fill the bar or not. Note that the rthm-seq-bar object passed here is altered.

SYNOPSIS:

```
(defmethod pad-right ((rsb rthm-seq-bar) &key pitch dots)
```

38.6.570 rthm-seq-bar/remove-dynamics

[rthm-seq-bar] [Methods]

DATE:

```
June 13th
```

DESCRIPTION:

```
remove all dynamics attached to events in the bar
```

ARGUMENTS:

```
- a rthm-seq-bar object
```

OPTIONAL ARGUMENTS:

```
- T or NIL to remove hairpin marks too (e.g. cresc-beg). Default = NIL.
```

RETURN VALUE:

```
T
```

SYNOPSIS:

```
(defmethod remove-dynamics ((rsb rthm-seq-bar) &optional hairpins)
```

38.6.571 rthm-seq-bar/remove-rhythms

[rthm-seq-bar] [Methods]

DATE:

```
February 8th 2017
```

DESCRIPTION:

```
remove a specified number of rhythms/events from a rthm-seq-bar object.
```

ARGUMENTS:

```
- the rthm-seq-bar object
- the rhythm/event to start removing at (1-based)
- how many rhythms/events to remove
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether the tuplets slot of the rthm-seq-bar should
  be updated to reflect changes. Default = T.
- T or NIL to indicated whether the bracket slot of the starting
  rhythm/event should take the value of the last rhythm/event removed
```

RETURN VALUE:

```
the list of rhythm/event objects from the rhythms slot of the rthm-seq-bar
```

SYNOPSIS:

```
(defmethod remove-rhythms ((rsb rthm-seq-bar) start how-many
                           &optional (update-tuplets t)
                             (inherit-last-tuplet t))
```

38.6.572 rthm-seq-bar/replace-rhythms

[rthm-seq-bar] [Methods]

DATE:

```
February 16th 2017, Edinburgh
```

DESCRIPTION:

```
Replace the rhythms/events in a rthm-seq-bar object, with an option to
automatically beam the result. The bar is checked to make sure it is full
after replacing and an error will be issued if it is not.
```

ARGUMENTS:

```
- a rthm-seq-bar object
- the index (1-based) of the rhythm/events to start at (integer)
- the number of rhythms/events to replace (integer). If NIL this will be set
  to the length of the new rhythms (next argument)
- a list of the new rhythm/event objects
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether the automatic beaming routine should be
  called after the replacement
- T or nil to indicate whether pitches should be retain if the bar contains
```

```
events as opposed to just rhythm objects. If the bar will have more
rhythms after this call (i.e. shorter rhythms passed) then the last pitch
in the original will be used to set the pitch of the remaining new events.
Or this can be a list of 1-based indices of the new rhythms: each of these
will be set to the pitches of the original bar, starting at its first
pitch and moving through from there
```

RETURN VALUE:

```
What the is-full method returns: two values: T or NIL to indicate whether
the bar is full, and the difference between the duration of the time
signature and the rhythms (usually a small floating-point error difference)
```

SYNOPSIS:

```
(defmethod replace-rhythms ((rsb rthm-seq-bar) start-rhythm replace-num-rhythms
                            new-rhythms &optional auto-beam keep-pitches)
```

38.6.573 rthm-seq-bar/reset-8va

[rthm-seq-bar] [Methods]

DATE:

```
22 Sep 2011
```

DESCRIPTION:

```
Reset the 8VA slots of all event objects within a given rthm-seq-object to
0 (no ottava/ottava bassa transposition).
```

ARGUMENTS:

```
- A rthm-seq-bar object
```

RETURN VALUE:

```
Always returns NIL.
```

EXAMPLE:

```
;; Create a rthm-seq-bar object consisting of event objects, print the default ; ; ; ; ; ;
;; value of the 8VA slots for those events. Set the 8VA slots to 1 and print ; ; ; ; ; ;
;; the value of those slots to see the change. Apply the reset-8va method to ; ; ; ; ; ;
```

```
;; remove any values and reset the slots to NIL, and print the results. ; ; ; ; ; ;

  (let ((rsb (make-rthm-seq-bar
(list
'(3 8)
(make-event 'cs4 'e)
(make-event 'cs4 'e)
(make-event 'cs4 'e)))))
(print (loop for e in (rhythms rsb) collect (8va e)))
(set-8va rsb 1)
(print (loop for e in (rhythms rsb) collect (8va e)))
(reset-8va rsb)
(print (loop for e in (rhythms rsb) collect (8va e))))

  =>
  (0 0 0)
  (1 1 1)
  (0 0 0)
```

SYNOPSIS:

```
(defmethod reset-8va ((rsb rthm-seq-bar))
```

38.6.574 rthm-seq-bar/respell-bar

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Scan the specified rthm-seq-bar object for enharmonically equivalent
pitches and unify their spelling.  This method won't generally be called by
the user directly, rather, it's called by the respell-bars method in the
slippery-chicken class.

Clearly, this method will not work if the rthm-seq-bar only contains rhythm
objects: it's made to be called when these have been promoted to event
objects during the initialization of a slippery-chicken object.

NB: Although this method focuses on just one rthm-seq-bar object, the
    parent slippery-chicken object and player ID are needed in order to
    determine ties that may exist into the next bar from the present bar.

NB: The slippery-chicken class version of the method of this name uses
    pitches from previous bars as well when respelling a given rthm-seq-bar
    object, so different results are normal. Users should generally call
```

```
     the slippery-chicken method rather than calling this one directly on
     individual bars.
```

ARGUMENTS:

```
 - A rthm-seq-bar object.
 - A slippery-chicken object.
 - A player ID (symbol).
```

OPTIONAL ARGUMENTS:

```
 - T or NIL to indicate whether to process only written or only sounding
   pitches. T = only written. Default = NIL.
 - The last attack (event object) of the previous bar. This is usually
   supplied by the calling method. Default = NIL.
```

RETURN VALUE:

```
 Returns the rthm-seq-bar object it was passed.
```

EXAMPLE:

```
;;; Create a slippery-chicken object using pitches GS4 and AF4, print the ; ; ; ; ; ;
;;; pitches of a specified bar within that object. Apply respell-bar and print ; ; ; ; ; ;
;;; the same pitches again to see the difference. ; ; ; ; ; ;

  (let ((mini
(make-slippery-chicken
'+mini+
:ensemble '(((vn (violin :midi-channel 1))))
:set-palette '((1 ((gs4 af4 bf4))))
:set-map '((1 (1 1 1)))
:rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
:pitch-seq-palette ((1 2 1 1 1 1 1 1)))))
:rthm-seq-map '((1 ((vn (1 1 1)))))))))
(print (loop for r in (rhythms (get-bar mini 2 'vn))
collect (get-pitch-symbol r)))
(respell-bar (get-bar mini 2 'vn) mini 'vn)
(print (loop for r in (rhythms (get-bar mini 2 'vn))
collect (get-pitch-symbol r))))
  =>
  (GS4 AF4 GS4 GS4 GS4 GS4 GS4 GS4)
  (GS4 GS4 GS4 GS4 GS4 GS4 GS4 GS4)
```

SYNOPSIS:

```
(defmethod respell-bar ((rsb rthm-seq-bar) sc player
                        &optional written last-attack-previous-bar)
```

38.6.575 rthm-seq-bar/scale

[rthm-seq-bar] [Methods]

DESCRIPTION:

Change the values of a rthm-seq-bar objects rhythm durations by a specified scaling factor.

This method always returns a new rthm-seq-bar object, recreating scaled rhythms with beams etc. where appropriate. See time-sig::scale for details on how the new meter is created.

ARGUMENTS:

- A rthm-seq-bar object.
- A number that is the scaling factor.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether to preserve the original meter (duple, triple, quadruple etc.)
- (two ignore arguments for internal use only)

RETURN VALUE:

Returns a rthm-seq-bar object

EXAMPLE:

```
;;; Create a rthm-seq-bar object and scale its durations by a fact of
;;; 2. Returns a rthm-seq-bar object.
(let ((rsb (make-rthm-seq-bar '((2 4) q e s s))))
(scale rsb 2))

=>
RTHM-SEQ-BAR: time-sig: 19 (2 2), time-sig-given: T, bar-num: -1,
[...]
RHYTHM: value: 2.000, duration: 2.000, rq: 2, is-rest: NIL,
[...]
data: H
[...]
```

```
RHYTHM: value: 4.000, duration: 1.000, rq: 1, is-rest: NIL,
[...]
data: Q
[...]
RHYTHM: value: 8.000, duration: 0.500, rq: 1/2, is-rest: NIL,
[...]
data: E
[...]
RHYTHM: value: 8.000, duration: 0.500, rq: 1/2, is-rest: NIL,
[...]
data: E
[...]

;;; Use the print-simple method to see formatted results
(let ((rsb (make-rthm-seq-bar '((2 4) q e s s))))
(print-simple (scale rsb .5)))

=>
(2 8): note E, note S, note 32, note 32,

;;; Set the optional <preserve-meter> argument to NIL to allow the method to
;;; return results in a different metric quality (this returns a quadruple
;;; meter rather than a duple)
(let ((rsb (make-rthm-seq-bar '((6 8) q e q s s))))
(print-simple (scale rsb 2 nil)))

=>
(12 8): note H, note Q, note H, note E, note E,
```

SYNOPSIS:

```
(defmethod scale ((rsb rthm-seq-bar) scaler
                  &optional (preserve-meter t) ignore1 ignore2)
```

38.6.576 rthm-seq-bar/set-8va

[rthm-seq-bar] [Methods]

DATE:

23-Sep-2011

DESCRIPTION:

Set the 8VA (ottava) slots of the event objects within a given rthm-seq-bar

object. This number can be positive or negative. Only the values 1, 0 and -1 are valid for the number of octaves to be transposed.

ARGUMENTS:

- A rthm-seq-bar object.
- A number indicating the number of octaves to be transposed in either direction (ottava/ottava bassa).

RETURN VALUE:

Always returns NIL.

EXAMPLE:

```
;; The method returns NIL               ; ; ; ; ;

  (let ((rsb (make-rthm-seq-bar
(list
'(3 8)
(make-event 'cs4 'e)
(make-event 'cs4 'e)
(make-event 'cs4 'e)))))
(set-8va rsb 1))

  => NIL

;; Create a rthm-seq-bar object with event objects, set the 8va slot to 1, and ; ; ; ; ;
;; access and print it to see it's new value. ; ; ; ; ;

  (let ((rsb (make-rthm-seq-bar
(list
'(3 8)
(make-event 'cs4 'e)
(make-event 'cs4 'e)
(make-event 'cs4 'e)))))
(set-8va rsb 1)
(loop for e in (rhythms rsb) collect (8va e)))

  => (1 1 1)
```

SYNOPSIS:

```
(defmethod set-8va ((rsb rthm-seq-bar) 8va)
```

38.6.577 rthm-seq-bar/set-amplitudes

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Add a specified amplitude (between 0.0 and 1.0) to all non-rest event
objects in a specified rthm-seq-bar object.
```

ARGUMENTS:

```
- A rthm-seq-bar object.
- A number that is an amplitude value between 0.0 and 1.0.
```

RETURN VALUE:

```
Returns the amplitude value set.
```

EXAMPLE:

```
  (let* ((mini
(make-slippery-chicken
'+sc-object+
:ensemble '(((va (viola :midi-channel 2))))
:set-palette '((1 ((c3 d3 e3 f3 g3 a3 b3 c4))))
:set-map '((1 (1 1 1)))
:rthm-seq-palette '((1 ((((4 4) e (e) e (e) (e) e e e))
:pitch-seq-palette ((1 2 3 4 5)))))
:rthm-seq-map '((1 ((va (1 1 1))))))))
(set-amplitudes (get-bar mini 2 'va) 0.9)
(cmn-display mini))
```

SYNOPSIS:

```
(defmethod set-amplitudes ((rsb rthm-seq-bar) amp)
```

38.6.578 rthm-seq-bar/set-dynamics

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Add a specified dynamic mark to all attacked event objects (i.e. not rests
and not notes that are tied to) in a specified rthm-seq-bar-object. This
method was created mainly to make it easy to set amplitudes for a range of
```

```
notes (e.g. with map-over-bars), so that they are, for example, reflected
in MIDI velocities.  If used over many notes the score will probably then be
littered with extraneous dynamic marks.  These can then be removed, if so
desired, with the slippery-chicken class remove-extraneous-dynamics method.
```

ARGUMENTS:

```
- A rthm-seq-bar object.
- A dynamic mark.
```

RETURN VALUE:

```
The specified dynamic mark
```

EXAMPLE:

```
  (let* ((mini
(make-slippery-chicken
'+sc-object+
:ensemble '(((va (viola :midi-channel 2))))
:set-palette '((1 ((c3 d3 e3 f3 g3 a3 b3 c4))))
:set-map '((1 (1 1 1)))
:rthm-seq-palette '((1 ((((4 4) e (e) e (e) (e) e e e))
:pitch-seq-palette ((1 2 3 4 5)))))
:rthm-seq-map '((1 ((va (1 1 1)))))))
(set-dynamics (get-bar mini 2 'va) 'ppp))

  => PPP
```

SYNOPSIS:

```
(defmethod set-dynamics ((rsb rthm-seq-bar) dynamic)
```

38.6.579 rthm-seq-bar/set-midi-channel

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Set the MIDI-channel and microtonal MIDI-channel for the pitch object
of an event object within a given rthm-seq-bar object. Sets the
MIDI-CHANNEL slot of all event objects contained in the rthm-seq-bar object
to the same channel.
```

ARGUMENTS:

```
 - A rthm-seq-bar object.
 - A whole number indicating the MIDI channel to be used for the
   equal-tempered pitch material of the given rthm-seq-bar object.
 - A whole number indicating the MIDI channel to be used for microtonal
   pitch material of the given rthm-seq-bar object.
```

RETURN VALUE:

```
 Always returns NIL.
```

EXAMPLE:

```
;; Create a rthm-seq-bar using event objects and check the MIDI-CHANNEL slots ; ; ; ; ; ;
;; of those event objects to see that they are NIL by default. ; ; ; ; ; ;
  (let ((rsb (make-rthm-seq-bar
(list
'(3 8)
(make-event 'cs4 'e)
(make-event 'cs4 'e)
(make-event 'cs4 'e)))))
(loop for p in (rhythms rsb)
collect (midi-channel (pitch-or-chord p))))

  => (NIL NIL NIL)

;; Apply the set-midi-channel method to the rthm-seq-bar object and read and ; ; ; ; ; ;
;; print the MIDI-CHANNEL slots of each of the individual events to see that ; ; ; ; ; ;
;; they've been set.                    ; ; ; ; ; ;
  (let ((rsb (make-rthm-seq-bar
(list
'(3 8)
(make-event 'cs4 'e)
(make-event 'cs4 'e)
(make-event 'cs4 'e)))))
(set-midi-channel rsb 13 14)
(loop for p in (rhythms rsb)
collect (midi-channel (pitch-or-chord p))))

  => (13 13 13)
```

SYNOPSIS:

```
(defmethod set-midi-channel ((rsb rthm-seq-bar) midi-channel
                             &optional microtonal-midi-channel)
```

38.6.580 rthm-seq-bar/set-nth-attack

[rthm-seq-bar] [Methods]

DESCRIPTION:

Sets the value of the nth rhythm object of a given rthm-seq-bar that needs an attack; i.e., not a rest and not a tied note.

NB: This method does not check to ensure that the resulting rthm-seq-bar contains the right number of beats.

ARGUMENTS:

- A zero-based index number for the attacked note to change.
- An event.
- A rthm-seq-bar object.

OPTIONAL ARGUMENTS:

- T or NIL indicating whether to print a warning message if the given index (minus one) is greater than the number of attacks in the RHYTHMS list. Default = T.

RETURN VALUE:

An event object.

EXAMPLE:

```
(let ((rsb (make-rthm-seq-bar '((2 4) q+e s s))))
(set-nth-attack 1 (make-event 'e4 'q) rsb))

=>
EVENT: start-time: NIL, end-time: NIL,
[...]
PITCH: frequency: 329.6275526703903d0, midi-note: 64, midi-channel: NIL
[...]
NAMED-OBJECT: id: E4, tag: NIL,
data: E4
[...]
RHYTHM: value: 4.0, duration: 1.0, rq: 1, is-rest: NIL, score-rthm: 4.0,
[...]
NAMED-OBJECT: id: Q, tag: NIL,
data: Q
```

```
(let ((rsb (make-rthm-seq-bar '((2 4) q+e s s))))
(set-nth-attack 2 (make-event 'e4 'q) rsb)
(loop for r in (rhythms rsb) collect (data r)))

=> ("Q" "E" S Q)

(let ((rsb (make-rthm-seq-bar '((2 4) q+e s s))))
(set-nth-attack 3 (make-event 'e4 'q) rsb))

=> NIL
rthm-seq-bar::set-nth-attack: index (3) < 0 or >= notes-needed (3)

(let ((rsb (make-rthm-seq-bar '((2 4) q+e s s))))
(set-nth-attack 3 (make-event 'e4 'q) rsb nil))

=> NIL
```

SYNOPSIS:

```
(defmethod set-nth-attack (index (e event) (rsb rthm-seq-bar)
                           &optional (warn t))
```

38.6.581 rthm-seq-bar/set-written

[rthm-seq-bar] [Methods]

DATE:

20 Jul 2011 (Pula)

DESCRIPTION:

Set the written pitch (as opposed to sounding; i.e., for transposing instruments) of an event object within a given rthm-seq-bar object. The sounding pitch remains unchanged as a pitch object in the PITCH-OR-CHORD slot, while the written pitch is added as a pitch object to the WRITTEN-PITCH-OR-CHORD slot.

ARGUMENTS:

- A rthm-seq-bar-object
- A number (positive or negative) indicating the transposition by semitones. See this method in the event class for more information and examples.

RETURN VALUE:

Always returns T.

EXAMPLE:

```
;; The method returns NIL               ; ; ; ; ;
  (let ((rsb (make-rthm-seq-bar
(list
'(3 8)
(make-event 'cs4 'e)
(make-event 'cs4 'e)
(make-event 'cs4 'e)))))
(set-written rsb -2))

  => T

;; Set the written pitch transposition to 2 semitones lower, then check the ; ; ; ; ;
;; data of the WRITTEN-PITCH-OR-CHORD slot of each event to see the ; ; ; ; ;
;; corresponding pitches                ; ; ; ; ;
  (let ((rsb (make-rthm-seq-bar
(list
'(3 8)
(make-event 'cs4 'e)
(make-event 'cs4 'e)
(make-event 'cs4 'e)))))
(set-written rsb -2)
(loop for p in (rhythms rsb)
collect (get-pitch-symbol p)))

  => (B3 B3 B3)
```

SYNOPSIS:

```
(defmethod set-written ((rsb rthm-seq-bar) transposition)
```

38.6.582 rthm-seq-bar/split

[rthm-seq-bar] [Methods]

DATE:

27 Jan 2011

DESCRIPTION:

Splits a given rthm-seq-bar into multiple smaller rthm-seq-bar objects. This will only work if the given rthm-seq-bar object can be split into whole beats; e.g. a 4/4 bar will not be split into 5/8 + 3/8.

The keyword arguments :min-beats and :max-beats serve as guidelines rather than strict cut-offs. In some cases, the method may only be able to effectively split the given rthm-seq-bar by dividing it into segments that exceed the length stipulated by these arguments (see example below).

Depending on the min-beats/max-beats arguments stipulated by the user or the rhythmic structure of the given rthm-seq-bar object, the given rthm-seq-bar may not be splittable, in which case NIL is returned. If the keyword argument :warn is set to T, a warning will be also be printed in such cases.

NB The method does not copy over and update bar start-times (this is meant to be done at the rthm-seq stage, not once the whole piece has been generated).

ARGUMENTS:

- A rthm-seq-bar object.

OPTIONAL ARGUMENTS:

keyword arguments:
- :min-beats. This argument takes an integer value to indicate the minimum number of beats in any of the new rthm-seq-bar objects created. This serves as a guideline only and may occasionally be exceeded in value by the method. Default value = 2.
- :max-beats. This argument takes an integer value to indicate the maximum number of beats in any of the new rthm-seq-bar objects created. This serves as a guideline only and may occasionally be exceeded in value by the method. Default value = 5.
- :warn. Indicates whether to print a warning if the rthm-seq-bar object is unsplittable. Value T = print a warning. Defaults to NIL.

RETURN VALUE:

Returns a list of rthm-seq-bar objects if successful, NIL if not.

EXAMPLE:

```
  (let* ((rsb (make-rthm-seq-bar '((7 4) h. e e +e. e. e q)))
(rsb-splt (split rsb)))
```

```
(loop for i in rsb-splt collect
(loop for r in (rhythms i) collect (data r))))

  => ((H.) (E E "E." E. E Q))

  (let* ((rsb (make-rthm-seq-bar '((7 4) h. e e +e. e. e q)))
(rsb-splt (split rsb)))
(loop for i in rsb-splt do (print-simple i)))

  =>
  (3 4): note H.,
  (4 4): note E, note E, note E., note E., note E, note Q,

  (let* ((rsb (make-rthm-seq-bar '((7 4) h. e e +e. e. e q)))
(rsb-splt (split rsb :min-beats 1 :max-beats 3)))
(loop for i in rsb-splt do (print-simple i)))

  =>
  (3 4): note H.,
  (1 4): note E, note E,
  (2 4): note E., note E., note E,
  (1 4): note Q,

  (let ((rsb (make-rthm-seq-bar '((7 4) h. e e +e. e. e q))))
(split rsb :max-beats 1 :warn t))

  => NIL
  WARNING: rthm-seq-bar::split: couldn't split bar:
```

SYNOPSIS:

```
(defmethod split ((rsb rthm-seq-bar) &key
                                      (min-beats 2) (max-beats 5) warn ignore)
```

38.6.583 rthm-seq-bar/time-sig-equal

[rthm-seq-bar] [Methods]

DESCRIPTION:

Check to see if two given rthm-seq-bar objects have the same time signature.

ARGUMENTS:

- Two rthm-seq-bar objects.

RETURN VALUE:

```
T if the given rthm-seq-bar objects have the same time signature.
NIL if the given rthm-seq-bar objects have different times signatures.
```

EXAMPLE:

```
(let ((rsb1 (make-rthm-seq-bar '((2 4) q e s s)))
      (rsb2 (make-rthm-seq-bar '((2 4) s s e q))))
  (time-sig-equal rsb1 rsb2))

=> T

(let ((rsb1 (make-rthm-seq-bar '((2 4) q e s s)))
      (rsb2 (make-rthm-seq-bar '((3 4) q+e e s s s s))))
  (time-sig-equal rsb1 rsb2))
=> NIL
```

SYNOPSIS:

```
(defmethod time-sig-equal ((rsb1 rthm-seq-bar) (rsb2 rthm-seq-bar))
```

38.6.584 rthm-seq-bar/transpose

[rthm-seq-bar] [Methods]

DESCRIPTION:

```
Transpose the pitches of event objects stored in a rthm-seq-bar object by a
specified number of semitones (positive for up, negative for down).
```

ARGUMENTS:

```
- A rthm-seq-bar object.
- A whole number (positive or negative).
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :destructively. Set to T or NIL to indicate whether the slot values of
  the original rthm-seq-bar object should be changed or not (even though
  the method always returns a clone). T = change the originals.
  Default = NIL.
- :chord-function. A function to be used for the transposition of
  chords. Default = #'transpose.
- :pitch-function. A function to be used for the transposition of
  pitches. Default = #'transpose.
```

RETURN VALUE:

```
This method returns a clone of the rthm-seq-bar object whether the keyword
argument :destructively is set to T or NIL. It does change the
corresponding slot values of the original when set to T even though it
returns the clone.
```

EXAMPLE:

```
;; Create a rthm-seq-bar object using make-event, transpose the contained
;; pitches destructively, and read the values of the corresponding slots to see
;; the change.
(let ((rsb (make-rthm-seq-bar
            (list
             '(3 8)
             (make-event 'cs4 'e)
             (make-event 'cs4 'e)
             (make-event 'cs4 'e)))))
  (transpose rsb 3 :destructively t)
  (loop for p in (rhythms rsb)
     collect (data (pitch-or-chord p))))

=> (E4 E4 E4)

;; Do the same thing without the :destructively keyword being set to T
(let ((rsb (make-rthm-seq-bar
            (list
             '(3 8)
             (make-event 'cs4 'e)
             (make-event 'cs4 'e)
             (make-event 'cs4 'e)))))
  (transpose rsb 3)
  (loop for p in (rhythms rsb)
     collect (data (pitch-or-chord p))))

=> (CS4 CS4 CS4)
```

SYNOPSIS:

```
(defmethod transpose ((rsb rthm-seq-bar) semitones
                      &key
                      ;; when t, then the events will be replaced by the
                      ;; transposition.
                      (destructively nil)
                      ;; the default functions are the class methods for pitch
```

```
;; or chord.
(chord-function #'transpose)
(pitch-function #'transpose))
```

38.6.585 sclist/sc-nthcdr

[sclist] [Methods]

DESCRIPTION:

Get the tail (rest) of a given sclist object beginning with and including the specified zero-based index number.

NB: This method is destructive and replaces the contents of the given list with the sublist returned by the method.

ARGUMENTS:

- An index number.
- An sclist object

RETURN VALUE:

Returns a list.

Returns NIL if the specified index is greater (minus 1) than the number of items in the given list.

EXAMPLE:

```
;; Create an sclist object and get the tail of the list starting at place
;; 4. The subset returned replaces the data of the original.
(let ((scl (make-sclist '(0 1 2 3 4 5 6 7 8 9))))
  (sc-nthcdr 4 scl)
  (data scl))

=> (4 5 6 7 8 9)

(let ((scl (make-sclist '(0 1 2 3 4 5 6 7 8 9))))
  (sc-nthcdr 14 scl))

=> NIL
```

SYNOPSIS:

```
(defmethod sc-nthcdr (nth (scl sclist))
```

38.6.586 sclist/sc-set

[sclist] [Classes]

NAME:

```
player

File:             sc-set.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist -> chord ->
                  sc-set

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the sc-set class which holds pitch set
                  information for harmonic and pitch manipulation.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    August 10th 2001

$$ Last modified:  13:32:57 Wed Jun 24 2026 CEST

SVN ID: $Id$
```

38.6.587 sc-set/add

[sc-set] [Methods]

DESCRIPTION:

```
Create a new sc-set object from the data of two other specified sc-set
objects.

NB: Any subsets contained in the original sc-set objects are lost in the
    process.
```

ARGUMENTS:

```
- A first sc-set object.
- A second sc-set object.
```

OPTIONAL ARGUMENTS:

```
 (- optional argument <ignore> is internal only)
```

RETURN VALUE: EXAMPLE:

```
(let ((mscs1 (make-sc-set '(d2 a2 e3 b3 gf4 df5 af5)))
      (mscs2 (make-sc-set '(f2 c3 g3 d4 bf4 f5 c6))))
  (add mscs1 mscs2))

=>
SC-SET: auto-sort: T, used-notes:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 0
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 0, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: USED-NOTES, tag: NIL,
data: NIL

**** N.B. All pitches printed as symbols only, internally they are all
pitch-objects.

    subsets:
    related-sets:
SCLIST: sclist-length: 14, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (D2 F2 A2 C3 E3 G3 B3 D4 GF4 BF4 DF5 F5 AF5 C6)
```

SYNOPSIS:

```
(defmethod add ((s1 sc-set) (s2 sc-set) &optional ignore)
```

38.6.588 sc-set/add-harmonics

[sc-set] [Methods]

DESCRIPTION:

```
 Adds pitches to the set which are harmonically related to the existing
 pitches.  The keywords are the same as for the get-harmonics function.  NB
 This will automatically sort all pitches from high to low (irrespective of
 the auto-sort slot).
```

ARGUMENTS:

```
 - an sc-set object
```

OPTIONAL ARGUMENTS:

```
 keyword arguments:
  see get-harmonics function
```

RETURN VALUE:

```
 the same set object as the first argument but with new pitches added.
```

EXAMPLE:

```
;;; treat the existing pitches as fundamentals. Note that the existence of E6
;;; twice in the result is not a mistake: though E6 is the nearest pitch, the
;;; frequencies are different.
(let ((s (make-sc-set '(c4 e4) :id 'test)))
  (add-harmonics s :start-partial 3 :max-results 3))
=>
SC-SET: auto-sort: T, rm-dups: T, warn-dups: T used-notes:
[...]
data: (C4 E4 G5 B5 C6 E6 E6 AF6)

;;; treat the existing pitches as partials and add the fundamentals and
;;; harmonics
(let ((s (make-sc-set '(c4 e4) :id 'test)))
  (add-harmonics s :start-freq-is-partial 3 :max-results 3))
SC-SET: auto-sort: T, rm-dups: T, warn-dups: T used-notes:
[...]
data: (F2 A2 F3 A3 C4 E4)
```

SYNOPSIS:

```
(defmethod add-harmonics ((s sc-set) &rest keywords)
```

38.6.589 sc-set/average-pitch

[sc-set] [Methods]

DATE:

```
 July 24th 2015, Glenferness
```

DESCRIPTION:

```
Calculate the average frequency of all the pitches in the chord and return
this as a new pitch-object.
```

ARGUMENTS:

```
- the sc-set object
```

RETURN VALUE:

```
a pitch-object
```

EXAMPLE:

```
(average-pitch (make-complete-set '(c4 cs4  d4 e4 ef4 g4 b4)))
=>
PITCH: frequency: 337.015, midi-note: 64, midi-channel: 0
       pitch-bend: 0.38
       degree: 64, data-consistent: T, white-note: E4
       nearest-chromatic: E4
       src: 1.2881585030063512d0, src-ref-pitch: C4, score-note: E4
       qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
       micro-tone: T,
       sharp: NIL, flat: NIL, natural: T,
       octave: 4, c5ths: 0, no-8ve: E, no-8ve-no-acc: E
       show-accidental: T, white-degree: 37,
       accidental: N,
       accidental-in-parentheses: NIL, marks: NIL,
       marks-before: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: E4, tag: NIL,
data: E4
```

SYNOPSIS:

```
(defmethod average-pitch ((s sc-set))
```

38.6.590 sc-set/create-chord

[sc-set] [Methods]

DESCRIPTION:

```
Create a chord object from the pitches of the given sc-set object. As of
1.0.6 sc-set is a subset of chord instead of sc-list so this shouldn't be
necessary, but it's still here for legacy code purposes.
```

ARGUMENTS:

```
- An sc-set object.
```

OPTIONAL ARGUMENTS:

```
- either a subsets-id or related-sets-id in order to use the respective
  notes instead of all the notes of the set.
```

RETURN VALUE:

```
A chord object.
```

EXAMPLE:

```
(let ((mscs (make-sc-set '(d2 c3 d4 df5 c6))))
  (create-chord mscs))

=>
CHORD: auto-sort: T, marks: NIL, micro-tone: NIL
SCLIST: sclist-length: 5, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (
PITCH: frequency: 73.416, midi-note: 38, midi-channel: 0
[...]
)
```

SYNOPSIS:

```
(defmethod create-chord ((s sc-set) &optional subsets-id related-sets-id)
```

38.6.591 sc-set/create-event

[sc-set] [Methods]

DESCRIPTION:

```
Create an event object (that is a chord) from a given sc-set object,
specifying a rhythmic value and a start-time (in seconds).
```

ARGUMENTS:

```
- An sc-set object.
- A rhythmic unit, either as a numerical value (32, 16 etc) or a symbol
  that is an alphabetic shorthand ('e, 's etc).
- A number that is the start time in seconds.
```

OPTIONAL ARGUMENTS:

- A number that is the start-time in quarter-notes rather than seconds (see event class documentation for more details)
- subsets-id, related-sets-id: pass an ID to use just the notes of the sub/related set, rather than all the notes of the set.

RETURN VALUE:

An event object.

EXAMPLE:

```
;; Create an event from the specified sc-set object that is a quarter-note
;; chord starting at 0.0 seconds
(let ((mscs (make-sc-set '(d2 c3 d4 df5 c6))))
  (create-event mscs 'q 0.0))

=>
EVENT: start-time: 0.000, end-time: NIL,
       duration-in-tempo: 0.000,
       compound-duration-in-tempo: 0.000,
       amplitude: 0.700
       bar-num: -1, marks-before: NIL,
       tempo-change: NIL
       instrument-change: NIL
       display-tempo: NIL, start-time-qtrs: 0.000,
       midi-time-sig: NIL, midi-program-changes: NIL,
       8va: 0
       pitch-or-chord:
CHORD: auto-sort: T, marks: NIL, micro-tone: NIL
SCLIST: sclist-length: 5, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (
PITCH: frequency: 73.416, midi-note: 38, midi-channel: 0
[...]
RHYTHM: value: 4.000, duration: 1.000, rq: 1, is-rest: NIL,
        score-rthm: 4.0f0, undotted-value: 4, num-flags: 0, num-dots: 0,
        is-tied-to: NIL, is-tied-from: NIL, compound-duration: 1.000,
        is-grace-note: NIL, needs-new-note: T, beam: NIL, bracket: NIL,
        rqq-note: NIL, rqq-info: NIL, marks: NIL, marks-in-part: NIL,
        letter-value: 4, tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: Q, tag: NIL,
data: Q
```

SYNOPSIS:

```
(defmethod create-event ((s sc-set) rhythm start-time
                         &optional start-time-qtrs subsets-id related-sets-id)
```

38.6.592 sc-set/force-micro-tone

[sc-set] [Methods]

DESCRIPTION:

Change the value of the MICRO-TONE slot of all pitch objects in a given sc-set object to the specified <value>.

NB If the pitches are microtonal and thus have associated pitch-bends and microtonal frequencies, these will not be changed, i.e. only the micro-tone slot is changed.

NB: Although the MICRO-TONE slot is generally used as a boolean, this method allows the user to force-set it to any value.

ARGUMENTS:

- An sc-set object.

OPTIONAL ARGUMENTS:

- An item of any type that is to be the new value of the MICRO-TONE slot of all pitch objects in the given sc-set object (generally T or NIL). Default = NIL.

RETURN VALUE:

Always returns NIL.

EXAMPLE:

```
;; Create an sc-set object that contains micro-tones and print the MICRO-TONE
;; slot of all of the contained pitch objects to see their values:
(let ((mscs (make-sc-set '(d2 cqs3 fs3 cs4 e4 c5 aqf5 ef6))))
  (loop for p in (data mscs) do (print (micro-tone p))))

=>
NIL
T
```

```
NIL
NIL
NIL
NIL
T
NIL

;; Now apply the force-micro-tone method to the same set using the default
;; value of NIL and print the results

(let ((mscs (make-sc-set '(d2 cqs3 fs3 cs4 e4 c5 aqf5 ef6))))
  (force-micro-tone mscs)
    (loop for p in (data mscs) do (print (micro-tone p))))

=>
NIL
NIL
NIL
NIL
NIL
NIL
NIL
NIL

;; Using the same sc-set, force all the values to T
(let ((mscs (make-sc-set '(d2 cqs3 fs3 cs4 e4 c5 aqf5 ef6))))
  (force-micro-tone mscs 't)
    (loop for p in (data mscs) do (print (micro-tone p))))

=>
T
T
T
T
T
T
T
T
```

SYNOPSIS:

```
(defmethod force-micro-tone ((s sc-set) &optional value)
```

38.6.593 sc-set/get-chromatic

[sc-set] [Methods]

DESCRIPTION:

```
Return those notes of a given sc-set object that are normal chromatic notes
(i.e. no microtones).

If a number is given for the <octave> argument, the method will transpose
all returned pitches into the specified octave, in which case any duplicate
pitches are removed.
```

ARGUMENTS:

```
- An sc-set object.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :octave. NIL or an integer that is the octave designator to which all
  resulting pitches are to be transposed (i.e. the "4" in "C4" etc.)
  Default = NIL.
- :remove-duplicates. T or NIL to indicate whether any duplicate pitches
  within an octave that are created by use of the :octave keyword argument
  are to be removed. T = remove duplicates. Default = NIL.
- :as-symbols. T or NIL to indicate whether to return the results of the
  method as a list of note-name symbols rather than a list of pitch
  objects. T = return as note-name symbols. Default = NIL.
- :package. The package in which the pitches are to be handled.
  Default = :sc.
- :invert. Get the micro-tone pitches instead.
```

RETURN VALUE:

```
Returns a list of pitch objects by default.

When the :as-symbols argument is set to T, a list of note-name symbols is
returned instead.
```

EXAMPLE:

```
;;; Returns a list of pitch objects by default
(let ((mscs (make-sc-set '(d2 cqs3 fs3 gqf3 cs4 e4 fqs4 c5 af5 bqf5 d6))))
  (get-chromatic mscs))
```

```
=>
(
PITCH: frequency: 73.416, midi-note: 38, midi-channel: 0
       pitch-bend: 0.0
       degree: 76, data-consistent: T, white-note: D2
       nearest-chromatic: D2
       src: 0.28061550855636597, src-ref-pitch: C4, score-note: D2
       qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
       micro-tone: NIL,
       sharp: NIL, flat: NIL, natural: T,
       octave: 2, c5ths: 0, no-8ve: D, no-8ve-no-acc: D
       show-accidental: T, white-degree: 15,
       accidental: N,
       accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: D2, tag: NIL,
data: D2

PITCH: frequency: 184.997, midi-note: 54, midi-channel: 0
[...]
)

;; Setting the :as-symbols argument to T returns a list of note-name symbols
;; instead
(let ((mscs (make-sc-set '(d2 cqs3 fs3 gqf3 cs4 e4 fqs4 c5 af5 bqf5 d6))))
  (get-chromatic mscs
                 :as-symbols t))

=> (D2 FS3 CS4 E4 C5 AF5 D6)

;; Giving an integer as the :octave argument transposes all returned  pitches
;; to the specified octave, removing any duplicates by default.
(let ((mscs (make-sc-set '(d2 cqs3 fs3 gqf3 cs4 e4 fqs4 c5 af5 bqf5 d6))))
  (get-chromatic mscs
                 :as-symbols t
                 :octave 4))

=> (FS4 CS4 E4 C4 AF4 D4)

;; Setting the :invert argument to T returns the non-chromatic elements of the
;; given sc-set object instead
(let ((mscs (make-sc-set '(d2 cqs3 fs3 gqf3 cs4 e4 fqs4 c5 af5 bqf5 d6))))
  (get-chromatic mscs
                 :as-symbols t
                 :invert t))
```

```
=> (CQS3 GQF3 FQS4 BQF5)
```

SYNOPSIS:

```
(defmethod get-chromatic ((s sc-set)
                          &key
                          (octave nil)
                          (remove-duplicates t) ;; only if octave!
                          (as-symbols nil)
                          (package :sc)
                          (invert nil))
```

38.6.594 sc-set/get-degrees

[sc-set] [Methods]

DESCRIPTION:

```
Return the pitches contained in the given sc-set object as a list of
DEGREES (which default to quarter-tones in slippery chicken).
```

ARGUMENTS:

```
- An sc-set object.
```

RETURN VALUE:

```
A list of integers.
```

EXAMPLE:

```
(let ((mscs (make-sc-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6))))
  (get-degrees mscs))

=> (76 82 90 96 104 110 118 124 132 140 146 154 160 168)
```

SYNOPSIS:

```
(defmethod get-degrees ((s sc-set))
```

38.6.595 sc-set/get-freqs

[sc-set] [Methods]

DESCRIPTION:

```
Return the pitches of a given sc-set object as a list of Hz frequencies
```

ARGUMENTS:

```
- An sc-set object.
```

RETURN VALUE:

```
A list of numbers
```

EXAMPLE:

```
(let ((mscs (make-sc-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6))))
  (get-freqs mscs))

=> (73.41618871368837 87.30705289160142 109.99999810639679 130.8127784729004
    164.81377633519514 195.99771591817216 246.94163930037348 293.6647548547535
    369.99440456398133 466.1637395092839 554.3652698843016 698.4564231328113
    830.6093584209975 1046.5022277832031)
```

SYNOPSIS:

```
(defmethod get-freqs ((s sc-set))
```

38.6.596 sc-set/get-midi

[sc-set] [Methods]

DESCRIPTION:

```
Return the pitches of a given sc-set object as a list of their equivalent
MIDI note numbers.
```

ARGUMENTS:

```
- An sc-set object.
```

RETURN VALUE:

```
A list of numbers
```

EXAMPLE:

```
(let ((mscs (make-sc-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6))))
  (get-midi mscs))

=> (38 41 45 48 52 55 59 62 66 70 73 77 80 84)
```

SYNOPSIS:

```
(defmethod get-midi  ((s sc-set))
```

38.6.597 sc-set/get-non-chromatic

[sc-set] [Methods]

DESCRIPTION:

Return those notes of a given sc-set object that are microtones (i.e. no "normal" chromatic notes).

If a number is given for the <octave> argument, the method will transpose all returned pitches into the specified octave, in which case any duplicate pitches are removed.

ARGUMENTS:

- An sc-set object.

OPTIONAL ARGUMENTS:

keyword arguments:
- :octave. NIL or an integer that is the octave designator to which all resulting pitches are to be transposed (i.e. the "4" in "C4" etc.) Default = NIL.
- :as-symbols. T or NIL to indicate whether to return the results of the method as a list of note-name symbols rather than a list of pitch objects. T = return as note-name symbols. Default = NIL.
- :package. The package in which the pitches are to be handled. Default = :sc.

RETURN VALUE:

Returns a list of pitch objects by default.

When the :as-symbols argument is set to T, a list of note-name symbols is returned instead.

EXAMPLE:

```
;; Returns a list of pitch objects by default
(let ((mscs (make-sc-set '(d2 cqs3 fs3 gqf3 cs4 e4 fqs4 c5 af5 bqf5 d6))))
  (get-non-chromatic mscs))

=>
=> (
PITCH: frequency: 134.646, midi-note: 48, midi-channel: 0
       pitch-bend: 0.5
       degree: 97, data-consistent: T, white-note: C3
       nearest-chromatic: C3
       src: 0.5146511197090149, src-ref-pitch: C4, score-note: CS3
       qtr-sharp: 1, qtr-flat: NIL, qtr-tone: 1,
       micro-tone: T,
       sharp: NIL, flat: NIL, natural: NIL,
       octave: 3, c5ths: 0, no-8ve: CQS, no-8ve-no-acc: C
       show-accidental: T, white-degree: 21,
       accidental: QS,
       accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: CQS3, tag: NIL,
data: CQS3

PITCH: frequency: 190.418, midi-note: 54, midi-channel: 0
[...]
)

;; Setting :as-symbols to T returns a list of note-names instead
(let ((mscs (make-sc-set '(d2 cqs3 fs3 gqf3 cs4 e4 fqs4 c5 af5 bqf5 d6))))
  (get-non-chromatic mscs
                     :as-symbols t))

=> (CQS3 GQF3 FQS4 BQF5)

;; Giving an integer as the :octave argument transposes all returned  pitches
;; to the specified octave, removing any duplicates
(let ((mscs (make-sc-set '(d2 cqs3 fs3 gqf3 cs4 e4 fqs4 c5 af5 bqf5 cqs6 d6))))
  (get-non-chromatic mscs
                     :as-symbols t
                     :octave 4))

=> (GQF4 FQS4 BQF4 CQS4)
```

SYNOPSIS:

```
(defmethod get-non-chromatic ((s sc-set)
                              &key
                              (octave nil)
                              (as-symbols nil)
                              (package :sc))
```

38.6.598 sc-set/get-semitones

[sc-set] [Methods]

DESCRIPTION:

Get the distances in semitones of each pitch in a given sc-set object to a static reference pitch.

Though this method can be used in other contexts, it was devised as an aid for transposing audio samples (sound files), and the reference pitch is therefore generally the perceived fundamental pitch of the audio sample to be transposed.

ARGUMENTS:

- An sc-set object.

OPTIONAL ARGUMENTS:

- An optional note-name symbol sets the value of the <reference-pitch>, which is the basis pitch to which the resulting number of semitones refer. This will generally be the perceived fundamental pitch of the sample (sound file) being modified ("transposed").
- The optional <offset> argument takes a number that is the number of semitones to add to the pitch of the given set prior to determining its distance in semitones from the reference pitch.

RETURN VALUE:

A list of positive and negative numbers.

EXAMPLE:

```
;; Chromatic example
(let ((mscs (make-sc-set '(d2 fs3 cs4 c5 af5 d6))))
  (get-semitones mscs))

=> (-22.0 -6.0 1.0 12.0 20.0 26.0)
```

```
;; Quarter-tone example; results can be decimal fractions of semitone
(let ((mscs (make-sc-set '(d2 cqs3 fs3 gqf3 cs4 fqs4 c5 af5 bqf5 cqs6 d6))))
  (get-semitones mscs))

=> (-22.0 -11.5 -6.0 -5.5 1.0 5.5 12.0 20.0 22.5 24.5 26.0)
```

SYNOPSIS:

```
(defmethod get-semitones ((s sc-set) &optional
                          (reference-pitch 'c4)
                          (offset 0))
```

38.6.599 sc-set/get-semitones-from-middle-note

[sc-set] [Methods]

DESCRIPTION:

```
Return a list of numbers that are the distances in semitones of each pitch
in a given sc-set object from the middle note of that object.

NB: If the given sc-object contains an even number of pitch objects, the
    middle note is determined to be the first note of the second half of
    the set.
```

ARGUMENTS:

```
- An sc-set object.
```

OPTIONAL ARGUMENTS:

```
- A symbol that is the key of one of the key/data pairs contained in the
  SUBSETS slot of the given sc-set object.
```

RETURN VALUE:

```
A list of positive and negative numbers.
```

EXAMPLE:

```
;; With an odd number of items in the sc-set object, the method returns the
;; same number of positive and negative numbers (non-zero)
(let ((mscs (make-sc-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5))))
```

```
  (get-semitones-from-middle-note mscs))

=> (-21.0 -18.0 -14.0 -11.0 -7.0 -4.0 0.0 3.0 7.0 11.0 14.0 18.0 21.0)

;; With an even number of items in the sc-set object, the middle note is
;; considered to be the first note of the second half of the set
(let ((mscs (make-sc-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6))))
  (get-semitones-from-middle-note mscs))

=> (-24.0 -21.0 -17.0 -14.0 -10.0 -7.0 -3.0 0.0 4.0 8.0 11.0 15.0 18.0 22.0)

;; Setting the optional <subset> argument to a symbol that is the key of a
;; given key/data pair in the sc-object's SUBSETS slot applies the method to
;; that subset only
(let ((mscs (make-sc-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6)
                         :subsets '((fl (df5 f5 af5 c6))
                                    (va (c3 e3 g3 b3 d4 gf4))))))
  (get-semitones-from-middle-note mscs 'fl))

=> (-7.0 -3.0 0.0 4.0)
```

SYNOPSIS:

```
(defmethod get-semitones-from-middle-note ((s sc-set) &optional subset)
```

38.6.600 sc-set/get-srts

[sc-set] [Methods]

DESCRIPTION:

```
Get the sampling-rate conversion factors for the given sc-set object,
whereby 1.0 = unison, 2.0 = one octave higher and 0.5 = one octave lower
etc.
```

ARGUMENTS:

```
- An sc-set object.
```

OPTIONAL ARGUMENTS:

```
- An optional note-name symbol sets the value of the <reference-pitch>,
  which is the basis pitch to which the resulting factors refer. This will
  generally be the perceived fundamental pitch of the sample (sound file)
  being modified ("transposed").
```

- The optional <offset> argument takes a number that is the number of semitones to add to the pitch of the given set prior to determining the sampling-rate conversion factors.

RETURN VALUE:

Returns a list of numbers.

EXAMPLE:

```
;; Returns a list of factors that are the sampling-rate conversion factor
;; compared to a 'C4 by default:
(let ((mscs (make-sc-set '(d2 fs3 cs4 c5 af5 d6))))
  (get-srts mscs))

=> (0.28061550855636597 0.7071067690849304 1.0594631433486938 2.0
    3.17480206489563 4.4898481369018555)

;; Comparing the same set against a higher reference-pitch will return lower
;; values
(let ((mscs (make-sc-set '(d2 fs3 cs4 c5 af5 d6))))
  (get-srts mscs 'd4))

=> (0.25 0.6299605220704482 0.9438743681693953 1.781797458637491
    2.8284271254540463 4.0)

;; Conversely, comparing the same set against the default reference-pitch but
;; with a positive offset will return higher values

(let ((mscs (make-sc-set '(d2 fs3 cs4 c5 af5 d6))))
  (get-srts mscs 'c4 2))

=> (0.3149802585215549 0.7937005124004939 1.1892071699914617 2.244924096618746
    3.563594828739576 5.039684136344879)
```

SYNOPSIS:

```
(defmethod get-srts ((s sc-set) &optional
                               (reference-pitch 'c4)
                               (offset 0))
```

38.6.601 sc-set/has-pitches-in-range

[sc-set] [Methods]

DATE:

```
June 28th 2017, Edinburgh
```

DESCRIPTION:

```
Tests whether a set has pitches within a certain range.
```

ARGUMENTS:

```
- the sc-set object
- the lower pitch of the range (pitch object or symbol)
- the upper pitch of the range (pitch object or symbol)
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate that only chromatic pitches count, i.e. not
  microtones
```

RETURN VALUE:

```
T or NIL
```

SYNOPSIS:

```
(defmethod has-pitches-in-range ((s sc-set) lower upper &optional chromatic)
```

38.6.602 sc-set/least-used-octave

[sc-set] [Methods]

DATE:

```
18th August 2015, Edinburgh
```

DESCRIPTION:

```
Return the octave with the least notes in it. This includes octaves between
the lowest and highest notes inclusive, and may include an octave within
those ranges with no notes at all.
```

ARGUMENTS:

```
- the sc-set object
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :highest-wins. If two octaves share the least number of notes, return the
   highest when T. Default = T.
- :avoiding. An integer or list of integers. Don't return any octave in
   this list or don't return this single integer argument. Default = NIL.
```

RETURN VALUE:

```
An integer representing the octave with the least notes.
```

EXAMPLE:

```
(least-used-octave (make-sc-set '(cs2 d4 e4 g5)))
--> 3
(least-used-octave (make-sc-set '(cs2 d3 ds4 e4 g5)) :highest-wins nil)
--> 2
(least-used-octave (make-sc-set '(cs2 d3 ds4 e4 g5)))
--> 5
```

SYNOPSIS:

```
(defmethod least-used-octave ((s sc-set) &key (highest-wins t) avoiding)
```

38.6.603 sc-set/make-sc-set

[sc-set] [Functions]

DESCRIPTION:

```
Create an sc-set object, which holds pitch-set information for harmonic and
pitch manipulation.
```

ARGUMENTS:

```
- A list of note-name symbols or frequencies that is to be the set
  (pitch-set) for the given sc-set object.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :id. A symbol that is to be the ID of the given sc-set object.
- :subsets. An assoc-list of key/data pairs, in which the data is a list of
  note-name symbols that are a subset of the main set. One use for this
  keyword argument might be to create subsets that particular instruments
```

```
   can play; these could for instance be selected in the chord-function
   passed to the instrument object.  In any case, if the instrument has a
   subset-id slot, and the current set contains a subset with that ID, the
   pitches the instrument may play are limited to that subset.
 - :related-sets. An assoc-list of key/data pairs, similar to :subsets, only
   that the pitches given here do not have to be part of the main set. This
   can be used, for example, for pitches missing from the main set.
 - :auto-sort. T or NIL to indicate whether the specified pitches (note-name
   symbols) are to be automatically sorted from lowest to highest.
   T = sort. Default = T.
```

RETURN VALUE:

```
 An sc-set object.
```

EXAMPLE:

```
;; Simplest usage, with no keyword arguments; returns an sc-set object
(make-sc-set '(d2 cs3 fs3 cs4 e4 c5 af5 ef6))

=>
SC-SET: auto-sort: T, used-notes:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 0
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 0, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: USED-NOTES, tag: NIL,
data: NIL

N.B. All pitches printed as symbols only, internally they are all
pitch-objects.

    subsets:
    related-sets:
SCLIST: sclist-length: 8, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (D2 CS3 FS3 CS4 E4 C5 AF5 EF6)

;; With keyword arguments
(make-sc-set '(d2 cs3 fs3 cs4 e4 c5 af5 ef6)
             :id 'scs1
```

```
            :subsets '((violin (e4 c5 af5 ef6))
                       (viola (cs4 e4)))
            :related-sets '((missing (ds2 e2 b3 cs6 d6))))

=>
SC-SET: auto-sort: T, used-notes:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 0
                      linked: NIL
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 0, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: USED-NOTES, tag: NIL,
data: NIL

N.B. All pitches printed as symbols only, internally they are all
pitch-objects.

    subsets:
VIOLIN: (E4 C5 AF5 EF6)
VIOLA: (CS4 E4)
    related-sets:
MISSING: (DS2 E2 B3 CS6 D6)
SCLIST: sclist-length: 8, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: SCS1, tag: NIL,
data: (D2 CS3 FS3 CS4 E4 C5 AF5 EF6)
```

SYNOPSIS:

```
(defun make-sc-set (sc-set &key id subsets related-sets (auto-sort t) tag
                            (warn-dups t) (rm-dups t))
```

38.6.604 sc-set/most-used-octave

[sc-set] [Methods]

DATE:

```
18th August 2015, Edinburgh
```

DESCRIPTION:

```
Return the octave with the least notes in it. See least-used-octave for
```

```
more details and argument descriptions.
```

SYNOPSIS:

```
(defmethod most-used-octave ((s sc-set) &key (highest-wins t) avoiding)
```

38.6.605 sc-set/pitch-symbols

[sc-set] [Methods]

DESCRIPTION:

```
Return the pitches of a given sc-set object as a list of note-name
symbols.
```

ARGUMENTS:

```
- An sc-set object.
```

RETURN VALUE:

```
A list of note-name symbols.
```

EXAMPLE:

```
(let ((mscs (make-sc-set '(d2 c3 d4 df5 c6))))
  (pitch-symbols mscs))

=> (D2 C3 D4 DF5 C6)
```

SYNOPSIS:

```
(defmethod pitch-symbols ((s sc-set))
```

38.6.606 sc-set/round-inflections

[sc-set] [Methods]

DESCRIPTION:

```
Get the microtones of a given sc-set object, rounded to the nearest
chromatic note.

This method returns only the rounded microtones, and not any of the pitches
```

of the original sc-set that are already chromatic.

By default, this method only gets those microtones that are less than a quarter-tone. This behavior can be changed by setting the :qtr-tones-also argument to T.

An optional argument allows for all pitches to be moved to a specified octave, in which case any duplicate pitches are removed.

ARGUMENTS:

- An sc-set object.

OPTIONAL ARGUMENTS:

keyword arguments:
- :qtr-tones-also. T or NIL to indicate whether quarter-tones are also to be rounded to the nearest chromatic pitch and returned. T = round and return. Default = NIL.
- :octave. NIL or an integer that is the octave designator to which all resulting pitches are to be transposed (i.e. the "4" in "C4" etc.) Default = NIL.
- :remove-duplicates. T or NIL to indicate whether any duplicate pitches within an octave that are created by use of the :octave keyword argument are to be removed. T = remove duplicates. Default = NIL.
- :as-symbols. T or NIL to indicate whether to return the results of the method as a list of note-name symbols rather than a list of pitch objects. T = return as note-name symbols. Default = NIL.
- :package. The package in which the pitches are to be handled. Default = :sc.

RETURN VALUE:

A list of pitch objects.

EXAMPLE:

```
;; First set the *scale* environment of CM (which is used by slippery chicken)
;; to twelfth-tones
(setf cm::*scale* (cm::find-object 'twelfth-tone))

=> #<tuning "twelfth-tone">

;; By default the method returns a list of pitch objects.
(let ((mscs (make-sc-set '(c4 cts4 css4 cqs4 cssf4 cstf4 cs4))))
```

```
  (round-inflections mscs))

=>
(
PITCH: frequency: 261.626, midi-note: 60, midi-channel: 0
[...]
data: C4
PITCH: frequency: 261.626, midi-note: 60, midi-channel: 0
[...]
data: C4
[...]
PITCH: frequency: 277.183, midi-note: 61, midi-channel: 0
[...]
data: CS4
[...]
PITCH: frequency: 277.183, midi-note: 61, midi-channel: 0
[...]
data: CS4
)

;; Setting the :as-symbols argument to T returns a list of note-name symbols
;; instead
(let ((mscs (make-sc-set '(c4 cts4 css4 cqs4 cssf4 cstf4 cs4))))
  (round-inflections mscs :as-symbols t))

=> (C4 C4 CS4 CS4)

;; Setting the :qtr-tones-also argument to T returns causes quarter-tones to be
;; rounded and returned as well.
(let ((mscs (make-sc-set '(c4 cts4 css4 cqs4 cssf4 cstf4 cs4))))
  (round-inflections mscs
                     :qtr-tones-also T
                     :as-symbols t))

=> (C4 C4 C4 CS4 CS4)

;; Specifying an octave transposes all returned pitches to that octave,
;; removing any duplicates by default
(let ((mscs (make-sc-set '(c2 cts3 css4 cqs5 cssf6 cstf7 cs8))))
  (round-inflections mscs
                     :qtr-tones-also T
                     :octave 4
                     :as-symbols t))

=> (C4 CS4)
```

```
;; The removal of the duplicates can be turned off by setting the
;; :remove-duplicates argument to NIL
(let ((mscs (make-sc-set '(c2 cts3 css4 cqs5 cssf6 cstf7 cs8))))
  (round-inflections mscs
                     :qtr-tones-also T
                     :octave 4
                     :remove-duplicates NIL
                     :as-symbols t))

=> (C4 C4 C4 CS4 CS4)
```

SYNOPSIS:

```
(defmethod round-inflections ((s sc-set)
                              &key
                              qtr-tones-also
                              octave
                              (remove-duplicates t) ;; only if octave!
                              (as-symbols nil)
                              (package :sc))
```

38.6.607 sc-set/round-to-nearest

[sc-set] [Methods]

DATE:

January 30th 2017

DESCRIPTION:

Round the (potentially very microtonal) pitches in the set to the nearest pitch of the current or given scale. Note that this is a destructive operation.

ARGUMENTS:

- the sc-set object

OPTIONAL ARGUMENTS:

keyword argument:
:scale. The scale to use when rounding. (Common Music tuning object or symbol). If a symbol, then 'chromatic-scale, 'twelfth-tone, or 'quarter-tone only at present. Default is the current scale as set by (in-scale :...).

RETURN VALUE:

```
the modified sc-set object
```

SYNOPSIS:

```
(defmethod round-to-nearest ((s sc-set) &key (scale cm::*scale*))
```

38.6.608 sc-set/set-position

[sc-set] [Methods]

DESCRIPTION:

```
Get the position (zero-index) of a specified pitch object within a given
sc-set object.
```

ARGUMENTS:

```
- A pitch object.
- An sc-set object.
```

RETURN VALUE:

```
An integer.
```

EXAMPLE:

```
(let ((mscs (make-sc-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6))))
  (set-position (make-pitch 'e3) mscs))

=> 4
```

SYNOPSIS:

```
(defmethod set-position ((p pitch) (s sc-set))
```

38.6.609 sc-set/stack

[sc-set] [Methods]

DESCRIPTION:

Extend the pitch content of a given sc-set object by adding new pitch objects which have the same interval structure as the original set.

The method analyzes the interval structure of the original set from the bottom note to the top and adds new sets to the top and bottom of the original set symmetrically; i.e., with the identical interval structure above the original set and inverted interval structure below.

The second argument indicates how many times this procedure is carried out.

NB: The method assumes that the pitch content of the original sc-set object is sorted from low to high.

See also: the make-stack method in the complete-set class to make a stack from a simple list of note-name symbols.

ARGUMENTS:

- An sc-set object.
- An integer that is the number of new sets to be added to each end of the original set.

OPTIONAL ARGUMENTS:

keyword arguments:
- :id. A symbol that will be the ID of the new sc-set object. Default NIL.
- :by-freq. If T then use frequencies when calculating the interval structure instead of degrees (semitones if default scale is chromatic). In this case the frequencies of pitches will be retained but their symbolic value will be rounded to the nearest note in the current scale (and pitch bends will be set accordingly). Default NIL.
- :up. Apply the process upwards in pitch space. Default = T.
- :down. Apply the process downwards in pitch space. Default = T.
- :respell. Respell the chord after the stacking process to (hopefully get better spellings. Default = T

RETURN VALUE:

A new sc-set object. This will have the same tag and id (unless given as a keyword argument here) slots as the original sc-set but not subsets, related-sets, used-notes, etc.

EXAMPLE:

```
;; Extends the original set with new sets that have the identical interval
;; structure upwards and inverted interval structure downwards.
(let ((set (make-sc-set '(c4 e4 g4))))
  (stack set 3))
=>
SC-SET: auto-sort: T, used-notes:
[...]
data: (EF2 GF2 BF2 DF3 F3 AF3 C4 E4 G4 B4 D5 GF5 A5 DF6 E6)

;;; or by calling the make-stack function, which returns a complete-set object
;;; (subclass of tl-set and sc-set).  Called with (in-scale :chromatic):
(make-stack 'test '(430 441 889 270) 1 :by-freq t)
=>
COMPLETE-SET: complete: NIL
[...]
data: (G2 A2 CS4 A4 A4 A5 C6 C6 FS6)

Note the difference of doing :by-freq and by interval:

(let ((set (make-sc-set '(c4 e4 g4))))
        (print (get-pitch-symbols (stack set 1 :by-freq t)))
        (get-pitch-symbols (stack set 1 :by-freq nil)))
-->
(C3 G3 C4 E4 G4 BF4 C5)
(F3 AF3 C4 E4 G4 B4 D5)
```

SYNOPSIS:

```
(defmethod stack ((s sc-set) num-stacks &key id by-freq (up t) (down t)
                                            (respell t))
```

38.6.610 sc-set/subset-from-harp-salzedo

[sc-set] [Methods]

DESCRIPTION:

```
This method creates a subset from a given sc-set according to a specific
harp-pedaling indicated by a salzedo-list (cf. harp-salzedo-to-tl-set) and
returns either a new sc-set or a list of symbols (when :as-symbols = T)
in order to be further used as either an independent/new sc-set or a
pitch-list to be used in the subset-slot of e.g. a sc-set.
```

ARGUMENTS:

```
- A sc-set object which is the actual set to derive the subset from.
```

- A list, which is a salzedo list with the harp pedalling
  (cf. salzedo-to-set).

OPTIONAL ARGUMENTS:

- :id. The id of the subset to be generated. When :as-symbols = T, this
  argument will be ignored.
- :lowest. The lowest pitch to be included in the subset. Default: 'b0
- :highest. The highest pitch to be included in the subset. Default: 'gs7
- :as-symbols. A boolean indicating whether a new sc-list object
  should (NIL) or a list of pitch symbols (NIL) should be returned.
  Default = NIL.
- :subsets. Inherited from make-sc-set. This will be ignored when
  :as-symbols = T.
- :related-sets. Inherited from make-sc-set. This will be ignored when
  :as-symbols = T.
- :auto-sort. Inherited from make-sc-set. This will be ignored when
  :as-symbols = T. Default = T.

RETURN VALUE:

Either a sc-set or a list of pitch symbols.

EXAMPLE:

```
(subset-from-harp-salzedo (make-sc-set '(C1 D1 E1 F1 GS1 AF1 AF2 BF2 C3 D3
                                         E3 F3 GS3
                                         AF3 BF3 C4 D4 E4 F4 GS4 AF4 BF4 C5
                                         D5 E5 F5
                                         C6 D6 E6 F6 GS6 AF6 BF6 C7 D7 E7))
                          '(0 1 -1 0 1 0 -1)
                          :as-symbols t)

=>
(D1 E1 AF1 AF2 BF2 D3 E3 AF3 BF3 D4 E4 AF4 BF4 D5 E5 D6 E6 AF6 BF6 D7 E7)
```

SYNOPSIS:

```
(defmethod subset-from-harp-salzedo ((s sc-set) salzedo
                                     &key
                                     id
                                     (lowest 'b0)
                                     (highest 'gs7)
                                     as-symbols
                                     subsets
                                     related-sets
                                     (auto-sort t))
```

38.6.611 sc-set/subset-get-srts

[sc-set] [Methods]

DESCRIPTION:

```
Get the sampling-rate conversion factors for the specified subset of a
given sc-set object, whereby 1.0 = unison, 2.0 = one octave higher and 0.5
= one octave lower etc.
```

ARGUMENTS:

```
- An sc-set object.
- A symbol that is the key of one of the key/data pairs stored in the
  SUBSETS slot of the given sc-set object.
```

OPTIONAL ARGUMENTS:

```
- The optional <reference-pitch> is the basis pitch to which the resulting
  factors refer. This will generally be the perceived fundamental pitch of
  the sample (sound file) being modified ("transposed").
- The optional <offset> argument is the number of semitones to add to the
  pitch of the given set prior to determining the sampling-rate conversion
  factors.
```

RETURN VALUE: EXAMPLE:

```
;;; Create an sc-set object with two subsets named 'FL and 'VA, then get the
;;; sampling-rate conversion factors for the 'FL subset
(let ((mscs (make-sc-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6)
                         :subsets '((fl (df5 f5 af5 c6))
                                    (va (c3 e3 g3 b3 d4 gf4))))))
  (subset-get-srts mscs 'fl))

=> (2.1189262866973877 2.669679641723633 3.17480206489563 4.0)
```

SYNOPSIS:

```
(defmethod subset-get-srts  ((s sc-set) subset &optional
                             (reference-pitch 'c4)
                             (offset 0))
```

38.6.612 sc-set/subsets-remove

[sc-set] [Methods]

DATE:

June 2nd 2017, Edinburgh

DESCRIPTION:

Remove pitches from all subsets of an sc-set object

ARGUMENTS:

- the sc-set object
- &rest the pitches to remove, as pitch objects or symbols

RETURN VALUE:

the sc-set

SYNOPSIS:

```
(defmethod subsets-remove ((s sc-set) &rest pitches)
```

38.6.613 sc-set/thin

[sc-set] [Methods]

DATE:

June 1st 2017, Edinburgh

DESCRIPTION:

Remove pitches from a set using a deterministic algorithm. From bottom to top (or vice-versa) pitches will be selected or rejected using the activity-levels class, to which the strength (1-10) is passed. The pitches removed are also removed from any subsets.

By default the set will be reduced by 1/3 of its pitches but it is expected that either the :remove or :target keyword will be used (see below).

ARGUMENTS:

- an sc-set object

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :strength. An integer from 1-10 indicating the likelihood of removing
  individual pitches. 1 would be 10%, 10, 100%. Default = 5.
- :remove. An integer indicating how many pitches to remove. Default = NIL
- :target. An integer indicating how many pitches the set should be left
  with. Default = NIL
- :invert. T or NIL to remove pitches starting at the top or bottom. T =
  from top, NIL = from bottom. Default = NIL
```

RETURN VALUE:

```
The (thinned) sc-set object.
```

EXAMPLE: SYNOPSIS:

```
(defmethod thin ((s sc-set) &key (strength 5) remove target invert)
```

38.6.614 sc-set/tl-set

[sc-set] [Classes]

NAME:

```
tl-set

File:             tl-set.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist -> chord ->
                  sc-set -> tl-set

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the tl-set class that extends
                  set to incorporate transposition and limiting to
                  certain pitch ranges.  NB As of yet, once a set is
                  transposed or limited, it can't be retransposed from its
                  original pitches, only from the current set, i.e these
                  methods are destructive!

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    13th August 2001
```

```
$$ Last modified:  19:07:51 Wed Jun  4 2025 CEST

SVN ID: $Id$
```

38.6.615 tl-set/complete-set

[tl-set] [Classes]

NAME:

```
complete-set

File:             complete-set.lsp

Class Hierarchy:  named-object -> linked-named-object -> sclist -> chord ->
                  sc-set -> tl-set -> complete-set

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the complete-set class which as an
                  extension of the tl-set class allows checking for full
                  sets: ones in which every note of the current scale is
                  present.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    10th August 2001

$$ Last modified:  17:38:48 Wed Feb  7 2024 CET

SVN ID: $Id$
```

38.6.616 complete-set/make-complete-set

[complete-set] [Functions]

DESCRIPTION:

```
Create a complete-set object, which as an extension of the tl-set class
allows checking for full sets: ones in which every note of *standard-scale*
is present.
```

ARGUMENTS:

- A set of pitches. This can either take the form of a list of note-name symbols or a complete-set, tl-set or sc-set object.

OPTIONAL ARGUMENTS:

keyword arguments:
- :id. An number, symbol or string that is to be the ID of the given complete-set object (see doc for sc-set).
- :tag. A number, symbol or string that is secondary name, description, tag etc. for the given complete-set object. The :tag serves for identification but not searching purposes (see doc for named-object).
- :subsets. An assoc-list of key/data pairs, in which the data is a list of note-name symbols that are a subset of the main set. One use for this keyword argument might be to create subsets that particular instruments can play; these would then be selected in the chord-function passed to the instrument object (see doc for sc-set).
- :related-sets. An assoc-list of key/data pairs, similar to :subsets, only that the pitches given here do not have to be part of the main set. This can be used, for example, for pitches missing from the main set (see doc for sc-set).
- :auto-sort. T or NIL to indicate whether the specified pitches (note-name symbols) are to be automatically sorted from lowest to highest.
 T = sort. Default = T. (see doc for sc-set)
- :transposition. A number that is the number of semitones by which the pitches of the new complete-set are to be transposed when the object is created. Default = 0. (see doc for tl-set)
- :limit-upper. A note-name symbol or a pitch object to indicate the highest possible pitch in the given complete-set object to be created. (see doc for tl-set)
- :limit-lower. A note-name symbol or a pitch object to indicate the lowest possible pitch in the complete-set object to be created. (see doc for tl-set)
- :complete. T, NIL, or 'CHROMATIC. This argument can be given at init, and if the set is not complete in the sense of T or 'CHROMATIC (all chromatic, equally-tempered notes are present in the set), a warning is printed. If the set is neither T nor 'CHROMATIC at init, then no warning will be issued. In both cases the COMPLETE slot of the given complete-set object will be set after checking the set.
- :midi-channel. The channel to set chromatic pitches to. Default = 1.
- :microtones-midi-channel The channel to set microtonal pitches to. Default = 2.

RETURN VALUE:

A complete-set object.

EXAMPLE:

```
;; Create a complete set using a list of note-name symbols and the default
;; values for the keyword arguments
(make-complete-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6))

=>
COMPLETE-SET: complete: NIL
...
data: (D2 F2 A2 C3 E3 G3 B3 D4 GF4 BF4 DF5 F5 AF5 C6)

;; A new complete-set object can be created from tl-set and sc-set objects
(let ((mcs (make-complete-set
            (make-tl-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5)))))
  (pitch-symbols mcs))

=> (D2 F2 A2 C3 E3 G3 B3 D4 GF4 BF4 DF5 F5 AF5)

(let ((mcs (make-complete-set
            (make-sc-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5)))))
  (pitch-symbols mcs))

=> (D2 F2 A2 C3 E3 G3 B3 D4 GF4 BF4 DF5 F5 AF5)

;; Using the other keyword arguments
(make-complete-set '(d2 f2 a2 e3 g3 b3 d4 gf4 bf4 df5 f5 af5)
             :id 'csset
             :subsets '((low (d2 f2 a2))
                        (mid (b3 d4)))
             :related-sets '((not-playable (dqs2 eqf3)))
             :transposition 3
             :limit-upper 'g5
             :limit-lower 'e2)

=>
COMPLETE-SET: complete: NIL
...
data: (F2 AF2 C3 G3 BF3 D4 F4 A4 CS5 E5)
```

SYNOPSIS:

```
(defun make-complete-set (set &rest keys
                          &key id tag subsets related-sets
                            (transposition 0) (auto-sort t) (warn-dups t)
                            (rm-dups t) limit-upper limit-lower complete
                            ;; MDE Fri Aug 24 14:56:50 2018
```

```
                          (midi-channel 1) (microtones-midi-channel 2))
```

38.6.617 complete-set/make-stack

[complete-set] [Functions]

DESCRIPTION:

```
Make a complete-set containing stacks based on the given notes.  See
documentation for the stack method in the sc-set class for details of
what this algorithm does.
```

ARGUMENTS:

```
- an ID for the set to be created (symbol, string, number)
- a list of notes as either symbols or pitch objects
- the number of stacks to create
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :by-freq. Use the frequencies of the pitches to create the stack instead
  of the interval structure. Default = NIL.
- :up. Apply the process upwards in pitch space. Default = T.
- :down. Apply the process downwards in pitch space. Default = T.
- :respell. Respell the chord after the stacking process to (hopefully get
  better spellings. Default = T.
```

RETURN VALUE:

```
a complete-set object
```

SYNOPSIS:

```
(defun make-stack (id notes num-stacks &key by-freq (up t) (down t) (respell t))
```

38.6.618 tl-set/harp-salzedo-to-tl-set

[tl-set] [Functions]

DESCRIPTION:

```
This function creates a tl-set from a list of harp pedalling indications
(similar to those introduced by Salzedo). Thus, the resulting tl-set
represents all producible pitches on a concert harp within a given range
(defined by the :lowest and :highest slots).
```

ARGUMENTS:

```
A 7-item list with the position of the pedals, where -1 indicates lowering
the respective note a half step, 0 means no alteration, 1 means raising the
note a half. The order of notes reflects the pedal structure of the concert
harp (i.e. D C B E F G A). Thus, '(0 1 0 0 0 -1 0) leads to the following
alteration: C# D E F Gb A B
```

OPTIONAL ARGUMENTS:

```
keyword-arguments:
- :id. The id of the sc-st to be generated.
- :lowest. This is the lowest note to be contained in the set.
  Default = 'b0
  +slippery-chicken-standard-instrument-palette+.
- :highest. This is the highest note to be contained in the set.
  Default = 'gs7
- :limit-lower. A note-name symbol or a pitch object to indicate the
  lowest possible pitch in the tl-set object to be created (cf.
  make-tl-set). Default = The value of :lowest-written 'harp in
  +slippery-chicken-standard-instrument-palette+.
- :limit-upper. A note-name symbol or a pitch object to indicate the
  highest possible pitch in the tl-set object to be created (cf.
  make-tl-set). Default = The value of :highest-written 'harp in
  +slippery-chicken-standard-instrument-palette+.
- :subsets. Inherited from make-tl-set.
- :related-sets. Inherited from make-tl-set.
- :auto-sort. Inherited from make-tl-set. Default = T.
```

RETURN VALUE:

```
An tl-set generated from the given data.
```

EXAMPLE:

```
(harp-salzedo-to-tl-set '(0 0 -1 0 0 1 -1))

=>
TL-SET: transposition: 0
        limit-upper: NIL
        limit-lower: NIL
SC-SET: auto-sort: T, rm-dups: T, warn-dups: T used-notes:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
                      num-data: 0
                      linked: NIL
```

```
                      full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 0, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL,
                     this: NIL,
                     next: NIL
NAMED-OBJECT: id: USED-NOTES, tag: NIL,
data: NIL
**************
,

**** N.B. All pitches printed as symbols only, internally they are all
pitch-objects.


    subsets:
    related-sets:
CHORD: auto-sort: T, marks: NIL, micro-tone: NIL, micro-tonality: 0.0
centroid: NIL, dissonance: NIL
SCLIST: sclist-length: 48, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL,
                     this: NIL,
                     next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (C1 D1 E1 F1 Gs1 Af1 Bf1 C2 D2 E2 F2 Gs2 Af2 Bf2 C3 D3 E3 F3 Gs3 Af3 Bf3
       C4 D4 E4 F4 Gs4 Af4 Bf4 C5 D5 E5 F5 Gs5 Af5 Bf5 C6 D6 E6 F6 Gs6 Af6 Bf6
       C7 D7 E7 F7 Gs7 Af7)
**************
```

SYNOPSIS:

```
(defun harp-salzedo-to-tl-set (salzedo
                               &key
                                 id
                                 (lowest 'b0)
                                 (highest 'gs7)
                                 (limit-lower (lowest-written
                                               (get-standard-ins 'harp)))
                                 (limit-upper (highest-written
                                               (get-standard-ins 'harp)))
                                 subsets
                                 related-sets
                                 (auto-sort t))
```

38.6.619 tl-set/limit

[tl-set] [Methods]

DESCRIPTION:

```
Remove pitch objects from a given tl-set whose pitch content is higher or
lower than the pitches specified. Any pitch objects whose pitch content is
equal to the limit pitches specified will be retained. This method is
therefore destructive.

NB: C0 and B10 are the highest and lowest possible pitches of the
    quarter-tone scale defined in scale.lsp (16.35 and 31608.55 Hz
    respectively).

NB: The keyword arguments for which the lower and upper limits are to be
    specified are optional arguments, but are required in order for any
    effect to be had.

NB: Whether a pitch is higher or lower than the specified pitches is
    determined by their frequencies. If a pitch happens to be microtonal it
    could be that the nearest pitch symbol is an approximation, i.e. not
    quite the same as the frequency, so limiting might appear to be working
    incorrectly.
```

ARGUMENTS:

```
- The tl-set object.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :upper. A note-name symbol that is the upper limit for the limiting
  process.
- :lower. A note-name symbol that is the lower limit for the limiting
  process.
- :do-related-sets. T or NIL to indicate whether the RELATED-SETS slot of
  the given tl-set object is to be transposed as well or left unhandled. T
  = transpose. Default = NIL.
```

RETURN VALUE:

```
the modified tl-set object.
```

EXAMPLE:

```
;;; By default the method does not transpose the pitches of the RELATED-SETS
;;; slot
(let ((mtls (make-tl-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6)
                         :subsets '((fl (df5 f5 af5))
                                    (vla (e3 g3 b3)))
                         :related-sets '((missing (fs2 b5))))))
  (limit mtls :upper 'df5 :lower 'c3))

=>
TL-SET: transposition: 0
        limit-upper:
PITCH: frequency: 554.365, midi-note: 73, midi-channel: 0
[...]
    subsets:
FL: (DF5)
VLA: (E3 G3 B3)
    related-sets:
MISSING: (FS2 B5)
SCLIST: sclist-length: 14, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (C3 E3 G3 B3 D4 GF4 BF4 DF5)

;; Setting the :do-related-sets argument to T results in any RELATED-SETS pitch
;; content being transposed as well
(let ((mtls (make-tl-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6)
                         :subsets '((fl (df5 f5 af5))
                                    (vla (e3 g3 b3)))
                         :related-sets '((missing (fs2 b5))))))
  (limit mtls :upper 'c6 :lower 'c3 :do-related-sets t))

=>
[...]
    subsets:
FL: (DF5 F5 AF5)
VLA: (E3 G3 B3)
    related-sets:
MISSING: (B5)
SCLIST: sclist-length: 14, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (C3 E3 G3 B3 D4 GF4 BF4 DF5 F5 AF5 C6)
```

SYNOPSIS:

```
(defmethod limit ((tls tl-set) &key upper lower do-related-sets)
```

38.6.620 tl-set/limit-for-instrument

[tl-set] [Methods]

DESCRIPTION:

Remove any pitch objects from the given tl-set object which are outside of the range of the specified instrument object.

The pitch objects returned after that operation can then be reduced again by applying further limits specified by the :upper and :lower keyword arguments.

NB: This method returns a list of pitch objects, not a tl-set object, though it does destructively alter the data of the given tl-set object accordingly.

NB: This method will return NIL if the pitch objects of the given tl-set object are microtonal while the given instrument object is set to be a non-microtonal instrument (see example).

NB: Whether a pitch is higher or lower than the specified pitches is determined by their frequencies. If a pitch happens to be microtonal it could be that the nearest pitch symbol is an approximation, i.e. not quite the same as the frequency, so limiting might appear to be working incorrectly.

ARGUMENTS:

- A tl-set object.
- An instrument object.

OPTIONAL ARGUMENTS:

keyword arguments:
- :upper. A pitch object or note-name symbol that is the uppermost possible pitch (inclusive) of the pitch objects returned, as a further limitation after the range of the instrument object has been applied.
- :lower. A pitch object or note-name symbol that is the lowermost possible pitch (inclusive) of the pitch objects returned, as a further limitation after the range of the instrument object has been applied.
- :do-related-sets. T or NIL to indicate whether to apply the specified range restrictions to the RELATED-SETS slot of the given tl-set object as well. NB: These will be modified within the original tl-set object but not returned as part of the resulting list. T = apply to RELATED-SETS as well. Default = NIL.

RETURN VALUE:

A list of pitch objects.

EXAMPLE:

```
;;; Returns a list of pitch objects, limited only by the range of the given
;;; instrument object by default
(let ((mtls (make-tl-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6)
                         :related-sets '((other-notes (b4 e5 fs5 c6)))))
      (mi (make-instrument 'flute
                           :staff-name "Flute" :staff-short-name "Fl."
                           :lowest-written 'c4 :highest-written 'd7
                           :starting-clef 'treble :midi-program 74 :chords nil
                           :microtones t :missing-notes '(cqs4 dqf4))))
  (limit-for-instrument mtls mi))

=>
(
PITCH: frequency: 293.665, midi-note: 62, midi-channel: 0
[...]
data: D4
[...]
PITCH: frequency: 369.994, midi-note: 66, midi-channel: 0
[...]
data: GF4
[...]
PITCH: frequency: 466.164, midi-note: 70, midi-channel: 0
[...]
data: BF4
[...]
PITCH: frequency: 554.365, midi-note: 73, midi-channel: 0
[...]
data: DF5
[...]
PITCH: frequency: 698.456, midi-note: 77, midi-channel: 0
[...]
data: F5
[...]
PITCH: frequency: 830.609, midi-note: 80, midi-channel: 0
[...]
data: AF5
[...]
PITCH: frequency: 1046.502, midi-note: 84, midi-channel: 0
[...]
data: C6
```

```
)

;;; Further restrict the pitches returned by setting values for the :upper and
;;; :lower keyword arguments and print the new pitch content of the given
;;; tl-set object to see the destructive modification
(let ((mtls (make-tl-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6)
                         :related-sets '((other-notes (b4 e5 fs5 c6)))))
      (mi (make-instrument 'flute
                           :staff-name "Flute" :staff-short-name "Fl."
                           :lowest-written 'c4 :highest-written 'd7
                           :starting-clef 'treble :midi-program 74 :chords nil
                           :microtones t :missing-notes '(cqs4 dqf4))))
  (limit-for-instrument mtls mi :upper 'b5 :lower 'c5)
  (pitch-symbols mtls))

=> (DF5 F5 AF5)

;;; By default the RELATED-SETS slot of the given tl-set object is not affected
(let ((mtls (make-tl-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6)
                         :related-sets '((other-notes (b4 e5 fs5 c6)))))
      (mi (make-instrument 'flute
                           :staff-name "Flute" :staff-short-name "Fl."
                           :lowest-written 'c4 :highest-written 'd7
                           :starting-clef 'treble :midi-program 74 :chords nil
                           :microtones t :missing-notes '(cqs4 dqf4))))
  (limit-for-instrument mtls mi :upper 'b5 :lower 'c5)
  (loop for nobj in (data (related-sets mtls))
     collect (loop for p in (data nobj)
                collect (data p))))

=> ((B4 E5 FS5 C6))


;;; Setting the :do-related-sets argument to T will cause the method to be
;;; applied to the RELATED-SETS slot as well.
(let ((mtls (make-tl-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6)
                         :related-sets '((other-notes (b4 e5 fs5 c6)))))
      (mi (make-instrument 'flute
                           :staff-name "Flute" :staff-short-name "Fl."
                           :lowest-written 'c4 :highest-written 'd7
                           :starting-clef 'treble :midi-program 74 :chords nil
                           :microtones t :missing-notes '(cqs4 dqf4))))
  (limit-for-instrument mtls mi :upper 'b5 :lower 'c5 :do-related-sets t)
  (print (pitch-symbols mtls))
  (print (loop for nobj in (data (related-sets mtls))
            collect (loop for p in (data nobj)
```

```
                      collect (data p)))))

=>
(DF5 F5 AF5)
((E5 FS5))

;;; The method will return NIL if a set of only microtonal pitches (which
;;; e.g. ring-mod might return) is given in combination with an instrument
;;; object which is not microtone-capable (such as the 'piano object of the
;;; +slippery-chicken-standard-instrument-palette+
(let ((mtls (make-tl-set '(dqs2 fqs2 aqf2 gqs3 bqf3 gqf4 bqf4 dqf5 fqs5)))
      (pno (get-data 'piano
                     +slippery-chicken-standard-instrument-palette+)))
  (limit-for-instrument mtls pno :lower 'e5 :upper 'd6))

=> NIL
```

SYNOPSIS:

```
(defmethod limit-for-instrument ((tls tl-set) (ins instrument)
                                 &key upper lower do-related-sets)
```

38.6.621 tl-set/limit-shift-octave

[tl-set] [Methods]

DATE:

August 18th 2015, Edinburgh

DESCRIPTION:

Like the limit method, we restrict pitches in the set to be within specified limits, but instead of removing pitches we shift them to octaves within the limits. Rather than just putting the outlying pitches up or down and octave or two, we put them into the least-used octave of the set so as to balance pitch spread.

ARGUMENTS:

- the tl-set object

OPTIONAL ARGUMENTS:

keyword arguments:

- :upper. A note-name symbol that is the upper limit for the limiting process.
- :lower. A note-name symbol that is the lower limit for the limiting process.
- :do-related-sets. T or NIL to indicate whether the RELATED-SETS slot of the given tl-set object is to be transposed as well or left unhandled. T = transpose. Default = NIL.
- :highest-wins. T or NIL to indicate whether in the case of a tie when searching for the least used octave in a set, the highest octave will be used to move notes into. Default = T.

RETURN VALUE:

The tl-set object.

EXAMPLE:

```
;;; the C2 will be shifted to C5 as the 5th octave is the least used in the
;;; original set. The DS5 will be shifted down to DS2 as once we've shifted the
;;; C2, the second octave is now the least used, plus it's within range.
(get-pitch-symbols
 (limit-shift-octave (make-tl-set '(c2 e2 cs3 f3 g3 d4 a4 ds5))
                     :upper 'c5 :lower 'd2))
--> (DS2 E2 CS3 F3 G3 D4 A4 C5)
```

SYNOPSIS:

```
(defmethod limit-shift-octave ((tls tl-set)
                               &key upper lower (highest-wins t)
                                 do-related-sets)
```

38.6.622 tl-set/make-tl-set

[tl-set] [Functions]

DESCRIPTION:

Create a tl-set object, which extends the sc-set class by incorporating transposition and limiting to certain pitch ranges.

NB: As of yet, once a set is transposed or limited, it can't be re-transposed from its original pitches, only from the current set; i.e these methods are destructive!

ARGUMENTS:

- A list of note-name symbols that is to be the set (pitch-set) for the given tl-set object.

OPTIONAL ARGUMENTS:

keyword arguments:
- :id. A symbol that is to be the ID of the given tl-set object.
- :subsets. An assoc-list of key/data pairs, in which the data is a list of note-name symbols that are a subset of the main set. One use for this keyword argument might be to create subsets that particular instruments can play; these would then be selected in the chord-function passed to the instrument object.
- :related-sets. An assoc-list of key/data pairs, similar to :subsets, only that the pitches given here do not have to be part of the main set. This can be used, for example, for pitches missing from the main set.
- :limit-upper. A note-name symbol or a pitch object to indicate the highest possible pitch in the tl-set object to be created.
- :limit-lower. A note-name symbol or a pitch object to indicate the lowest possible pitch in the tl-set object to be created.
- :transposition. A number that is the number of semitones by which the pitches of the new tl-set are to be transposed when the object is created. Default = 0.
- :auto-sort. T or NIL to indicate whether the specified pitches (note-name symbols) are to be automatically sorted from lowest to highest. T = sort. Default = T.
- :tag. The tag (symbol, string) to be attached to the object; from the named-object base class. Default = NIL.
- :warn-dups. To or NIL to indicate whether a warning should be issued if duplicate pitches are found. Default = T.

RETURN VALUE:

A tl-set object.

EXAMPLE:

```
;; Simple usage with default values for keyword arguments
(make-tl-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6))

=>
TL-SET: transposition: 0
        limit-upper: NIL
        limit-lower: NIL
SC-SET: auto-sort: T, used-notes:
RECURSIVE-ASSOC-LIST: recurse-simple-data: T
```

```
                    num-data: 0
                    linked: NIL
                    full-ref: NIL
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 0, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: USED-NOTES, tag: NIL,
data: NIL

N.B. All pitches printed as symbols only, internally they are all
pitch-objects.

    subsets:
    related-sets:
SCLIST: sclist-length: 14, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (D2 F2 A2 C3 E3 G3 B3 D4 GF4 BF4 DF5 F5 AF5 C6)

;; Adding subsets and related-sets
(make-tl-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6)
             :subsets '((fl (df5 f5 af5))
                        (vla (e3 g3 b3)))
             :related-sets '((missing (fs2 b5))))
=>
TL-SET: transposition: 0
[...]
    subsets:
FL: (DF5 F5 AF5)
VLA: (E3 G3 B3)
    related-sets:
MISSING: (FS2 B5)
SCLIST: sclist-length: 14, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (D2 F2 A2 C3 E3 G3 B3 D4 GF4 BF4 DF5 F5 AF5 C6)

;; Limiting the pitch range of the tl-set object, once using a note-name
;; symbol and once using a pitch object
(make-tl-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6)
             :limit-upper 'g5
             :limit-lower (make-pitch 'd3))

=>
TL-SET: transposition: 0
```

```
        limit-upper:
PITCH: frequency: 783.991, midi-note: 79, midi-channel: 0
[...]
        limit-lower:
PITCH: frequency: 146.832, midi-note: 50, midi-channel: 0
[...]
data: (E3 G3 B3 D4 GF4 BF4 DF5 F5)

;; Applying a transposition by semitones
(make-tl-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6)
             :transposition 3)

=>
TL-SET: transposition: 3
[...]
data: (F2 AF2 C3 EF3 G3 BF3 D4 F4 A4 CS5 E5 AF5 B5 EF6)
```

SYNOPSIS:

```
(defun make-tl-set (set &key id tag subsets related-sets
                            limit-upper limit-lower
                            (rm-dups t) (warn-dups t)
                            (transposition 0)
                            (auto-sort t))
```

38.6.623 tl-set/stack

[tl-set] [Methods]

DESCRIPTION:

```
Extend the pitch content of a given sc-set object by adding new pitch
objects which have the same interval structure as the original set.

NB: See documentation for the stack method in the sc-set class for usage.
```

SYNOPSIS:

```
(defmethod stack ((tls tl-set) num-stacks &key id by-freq respell)
```

38.6.624 tl-set/transpose

[tl-set] [Methods]

DESCRIPTION:

Transpose the pitches of a given tl-set by a specified number of semitones.

The contents of the SUBSETS slot are automatically transposed as well, but the RELATED-SETS slot is left untransposed by default. An optional argument allows for RELATED-SETS slot to be transposed as well.

ARGUMENTS:

- A tl-set object.
- A positive or negative integer that is the number of semitones by which the pitch content of the given tl-set object is to be transposed.

OPTIONAL ARGUMENTS:

keyword arguments:
- :do-related-sets. T or NIL to indicate whether to transpose any contents of the RELATED-SETS slot as well. T = transpose. Default = NIL.

(- additional <ignore> arguments are for internal use only)

RETURN VALUE:

The tl-set object.

EXAMPLE:

```
;; By default the RELATED-SETS are left untransposed
(let ((mtls (make-tl-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6)
                         :subsets '((fl (df5 f5 af5))
                                    (vla (e3 g3 b3)))
                         :related-sets '((missing (fs2 b5))))))
  (transpose mtls 3))

=>
TL-SET: transposition: 3
        limit-upper: NIL
        limit-lower: NIL
SC-SET: auto-sort: T, used-notes:
[...]
    subsets:
FL: (E5 AF5 B5)
VLA: (G3 BF3 D4)
    related-sets:
MISSING: (FS2 B5)
SCLIST: sclist-length: 14, bounds-alert: T, copy: T
```

```
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
pNAMED-OBJECT: id: NIL, tag: NIL,
data: (F2 AF2 C3 EF3 G3 BF3 D4 F4 A4 CS5 E5 AF5 B5 EF6)

;; Set the <do-related-sets> argument to T for the RELATED-SETS contents to be
;; transposed as well
(let ((mtls (make-tl-set '(d2 f2 a2 c3 e3 g3 b3 d4 gf4 bf4 df5 f5 af5 c6)
                         :subsets '((fl (df5 f5 af5))
                                    (vla (e3 g3 b3)))
                         :related-sets '((missing (fs2 b5))))))
  (transpose mtls 3 :do-related-sets t))

=>
TL-SET: transposition: 3
        limit-upper: NIL
        limit-lower: NIL
SC-SET: auto-sort: T, used-notes:
[...]
    subsets:
FL: (E5 AF5 B5)
VLA: (G3 BF3 D4)
    related-sets:
MISSING: (A2 D6)
SCLIST: sclist-length: 14, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (F2 AF2 C3 EF3 G3 BF3 D4 F4 A4 CS5 E5 AF5 B5 EF6)
```

SYNOPSIS:

```
(defmethod transpose :before ((tls tl-set) semitones
                              &key destructively do-related-sets
                                ;; MDE Wed Aug 22 09:43:29 2018
                                lowest highest)
```

38.6.625 sclist/sc-subseq

[sclist] [Methods]

DESCRIPTION:

```
Return a subsequence from a given sclist based on starting and finishing
indices.

NB: This method uses Common Lisp's subseq function and thus inherits its
```

attributes, whereby the START argument indicates the zero-based index of the first list item to be returned and the FINISH argument indicates the zero-based index of the first list item after that NOT to be returned.

ARGUMENTS:

- An sclist object.
- An integer indicating the zero-based index of the first list item to be returned.
- An integer indicating the zero-based index of the first list item after the START item to not be returned.

OPTIONAL ARGUMENTS:

- (fun #'error). By default an error will be signalled if the requested subseq is out of bounds. If you prefer, this could be a warning instead by passing #'warn, or nothing at all if NIL.
- reverse. T or NIL to indicate if the data list should be reversed before calling subseq

RETURN VALUE:

A list.

An error is returned if the user attempts to apply the method with START and FINISH arguments that are beyond the bounds of the given sclist object.

EXAMPLE:

```
;; Returns a sublist from the given list. The START argument indicates the
;; zero-based index of the first item in the given list to be returned and the
;; FINISH argument indicates the zero-based index of the first item after that
;; to NOT be returned.
(let ((scl (make-sclist '(1 2 3 4 5 6 7 8 9))))
  (sc-subseq scl 2 7))

=> (3 4 5 6 7)

;; Drops into the debugger with an error if one of the indexing arguments is
;; beyond the bounds of the given sclist object
(let ((scl (make-sclist '(1 2 3 4 5 6 7 8 9))))
  (sc-subseq scl 0 15))

=>
```

```
sclist::sc-subseq: Illegal indices for above list: 0 15 (length = 9)
   [Condition of type SIMPLE-ERROR]

(let ((scl (make-sclist '(1 2 3 4 5 6 7 8 9))))
  (sc-subseq scl 0 15 NIL))
=>
NIL
```

SYNOPSIS:

```
(defmethod sc-subseq ((scl sclist) start finish &optional (fun #'error) reverse)
```

38.6.626 sclist/sclist-econs

[sclist] [Methods]

DESCRIPTION:

```
 Add a single item to the end of a given sclist object.

 NB: This method destructively modifies the list.

 NB: This method adds any element specified as a single item. For combining
     two lists into one see sclist/combine.

 NB: Though related to Lisp's cons function, remember that the order of
    arguments here is the other way round i.e. element after list, not
    before.
```

ARGUMENTS:

```
 - An sclist object.
 - An item to add to the end of the given sclist object.
```

RETURN VALUE:

```
 - The new value (list) of the given sclist object.
```

EXAMPLE:

```
;; Add a single integer to the end of a list of integers
(let ((scl (make-sclist '(0 1 2 3 4))))
  (sclist-econs scl 5))

=> (0 1 2 3 4 5)
```

SYNOPSIS:

```
(defmethod sclist-econs ((scl sclist) element)
```

38.6.627 sclist/sclist-remove-elements

[sclist] [Methods]

DESCRIPTION:

```
Remove a specified number of consecutive items from a given sclist object.

NB: This is a destructive method and replaces the data of the given sclist
object with the newly created sublist.
```

ARGUMENTS:

```
- An sclist object.
- The index integer within the given list with which to start (inclusive
  and zero-based).
- An integer that is the number of items to remove.
```

RETURN VALUE:

```
Returns
```

EXAMPLE:

```
;;; Returns an sclist object.
(let ((scl (make-sclist '(0 1 2 3 4 5 6 7 8 9))))
  (sclist-remove-elements scl 3 4))

=>
SCLIST: sclist-length: 6, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: (0 1 2 7 8 9)

;; Drops into the debugger with an error if the given sclist object has fewer
;; items than specified for the START or HOW-MANY arguments
(let ((scl (make-sclist '(0 1 2 3 4 5 6 7 8 9))))
  (data (sclist-remove-elements scl 3 41)))

=>
remove-elements: arguments 2 and 3 must be integers < the length of argument 1:
```

```
3 41 10
   [Condition of type SIMPLE-ERROR]
```

SYNOPSIS:

```
(defmethod sclist-remove-elements ((scl sclist) start how-many)
```

38.6.628 sclist/sclist-remove-if

[sclist] [Methods]

DATE:

```
4.6.2025
```

DESCRIPTION:

```
apply the common-lisp remove-if function to the data slot of an sclist,
updating other slots as needed.
```

ARGUMENTS:

```
- the sclist object
- the test function that will lead to an element being removed
```

OPTIONAL ARGUMENTS:

```
any keyword arguments that can be passed to remove-if, i.e. &key :from-end
:start :end :count :key
```

RETURN VALUE:

```
the modified sclist object
```

SYNOPSIS:

```
(defmethod sclist-remove-if ((scl sclist) test
                             &rest keyargs &key &allow-other-keys)
```

38.6.629 sclist/set-nth

[sclist] [Methods]

DESCRIPTION:

```
 Set the nth element (zero-based) of a given sclist object.

 NB: This doesn't auto-grow the list.
```

ARGUMENTS:

```
 - An index integer.
 - An sclist object.
```

RETURN VALUE:

```
 Returns the item added if successful.

 Returns NIL and prints a warning if the specified index number is greater
 than the number of items in the list (minus 1)
```

EXAMPLE:

```
;; Returns the item added
(let ((scl (make-sclist '(cat dog cow pig sheep))))
  (set-nth 3 'horse scl))

=> HORSE

;; Access the DATA slot to see the change
(let ((scl (make-sclist '(cat dog cow pig sheep))))
  (set-nth 3 'horse scl)
  (data scl))

=> (CAT DOG COW HORSE SHEEP)

;; Returns NIL and prints a warning if the index number is beyond the bounds of
;; the list
(let ((scl (make-sclist '(cat dog cow pig sheep))))
  (set-nth 31 'horse scl))

=> NIL
WARNING: sclist::sclist-check-bounds: Illegal list reference: 31
(length = 5) (sclist id = NIL)
```

SYNOPSIS:

```
(defmethod set-nth (index new-element (i sclist))
```

38.6.630 linked-named-object/sndfile

[linked-named-object] [Classes]

NAME:

```
sndfile

File:             sndfile.lsp

Class Hierarchy:  named-object -> linked-named-object -> sndfile

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the sndfile class that holds
                  information about a sound file as well as specifying
                  desired parameters

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    March 21st 2001

$$ Last modified:  12:16:23 Sat Feb 15 2025 CET
```

38.6.631 sndfile/centroid

[sndfile] [Methods]

DATE:

```
February 7th 2023, Heidhausen
```

DESCRIPTION:

```
Calculate the average spectral centroid of a sound file, between the start
and end points given. This requires CLM's scentroid.ins to be compiled
and loaded in advance. Note that the averaging is done by FFT analysis every
10 milliseconds and with an FFT size of 4096 samples. FFT frames that have
an RMS amplitude of < -40dbFS will be ignored.
```

ARGUMENTS:

```
the sndfile object
```

RETURN VALUE:

```
The frequency of the centroid in Hertz.
```

SYNOPSIS:

```
(defmethod centroid ((sf sndfile))
```

38.6.632 sndfile/get-sound-info

[sndfile] [Functions]

DATE:

```
March 17th 2024.
```

DESCRIPTION:

```
Get duration, srate, number of channels and bits per sample for a
sound (or video) file. This function uses clm, if installed, or ffprobe
as an alternative and fallback method, when clm is not available.
Use (set-sc-config 'ffprobe-command "new/path") to change the default path
for ffprobe.

NB: CLM is not recommended for use with video files.
```

ARGUMENTS:

```
- filename. The path to the sound or video file.
```

OPTIONAL ARGUMENTS:

```
- ffprobe. Default=nil. When nil, use clm if possible, else ffprobe.
  When t, use ffprobe with the default path to ffprobe. When a string,
  interpret that string as path to the ffprobe executable.
```

RETURN VALUE:

```
a list containing srate, number of channels, bits per sample, duration,
the file size in byte, the number of framples, the video fps, width and
heigth as numbers.
```

SYNOPSIS:

```
(defun get-sound-info (filename &optional ffprobe)
```

38.6.633 sndfile/make-sndfile

[sndfile] [Functions]

DESCRIPTION:

```
Create a sndfile object to hold data about an existing sound file,
specifying at least the path and file name of that sound file. Optional
arguments allow for the specification of segments within the given sound
file and its perceived fundamental frequency (for src-based
transposition).

If the first argument ("path") is a list, then this sndfile object has been
created from within a sndfile-palette object. In this case, the first item
in the list will be the full path to the sound file, as defined in the
sndfile-palette object; the second item in the list is the list given by
the user containing the data slots to be used, whereby the first item of
that list must be the ID of the object.

NB: This function creates an object of the class sndfile, which is contains
    data concerning an existing sound file. It does not create an actual
    sound file.
```

ARGUMENTS:

```
- A path and file name of an existing sound file; or a list as explained
  above.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :id. An ID for the sndfile. Will be set automatically if created from
  within a sndfile-palette.  Default nil.
- :data. The given file name, including path and extension, usually set
  automatically to be the given path if nil.  Default nil.
- :duration. A number in seconds which is the duration of the segment of
  the specified sound file which the user would like to use. This should
  not be specified if :end has been specified. Default nil.
- :end. A number in seconds which is the end time within the source sound
  file for the segment of the file which the user would like to use. This
  should not be specified if :duration has been specified. Default nil.
- :start. A number in seconds which is the start time within the source
  sound file for the segment of the file which the user would like to
  use. Defaults to 0.0.
- :frequency. A number or note-name symbol. This frequency will serve as
  the reference pitch for any src transpositions of this file. This can be
```

any value, but will most likely be specified if the source sound file has a perceptible fundamental pitch. If given as a number, this number will be handled as a frequency in Hertz. Default = 'C4.
- :amplitude. An number that is the amplitude which the user would like to designate for this sound file. This number may be of any value, as slippery chicken normalizes all sound file events; however, standard practice would suggest that this should fall between 0.0 and 1.0. Default = 1.0
- :angle-env. used for spatialization, fore example with #'write-reaper-ambisonics-file. Can be an env (list of breakpoints) or a list of envs. A list of envs can be used to spatialize different channels of the soundfile differently. See the examples in write-reaper-ambisonics-file. This then represents the azimuth angle in a polar coordinate system. 0 and 1 represend 0° and 360° and are assumed to be in the front. 0.5 would thus be 180° and located behind the listener.
- :elevation-env. same as angle-env but represents the elevation angle. (the horizontal angle from the x-axis). -.5 represents the lowest point at -90°, 0 the level of the listening position and .5 the top (90°). This is, so that the span from 0 to 1 can represent the entire 360°.
- :distance-env. same as angle-env but represents the distance from the listening position. This is not really relevant for use in ambisonics. There the distance is usually 1.

RETURN VALUE:

A sndfile object.

EXAMPLE:

```
;; Example specifying the full path, a start and end time, and a base frequency
(make-sndfile "/path/to/sndfile-1.aiff"
              :start 0.3
              :end 1.1
              :frequency 654)

=>

SNDFILE: path: /path/to/sndfile-1.aiff,
         snd-duration: 3.011043, channels: 1, frequency: 654
         start: 0.3, end: 1.1, amplitude: 1.0, duration 0.8
         will-be-used: 0, has-been-used: 0
         data-consistent: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: /path/to/sndfile-1.aiff
```

```
;; Example using the sndfile-palette list as the first argument
(make-sndfile '("/path/to/sndfile-1.aiff"
                (nil :start 0.3 :end 1.1)))

=>

SNDFILE: path: /path/to/sndfile-1.aiff,
         snd-duration: 3.011043, channels: 1, frequency: 261.62555
         start: 0.3, end: 1.1, amplitude: 1.0, duration 0.8
         will-be-used: 0, has-been-used: 0
         data-consistent: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "/Volumes/JIMMY/SlipperyChicken/sndfile-2.aiff", tag: NIL,
data: /path/to/sndfile-1.aiff
```

SYNOPSIS:

```
(defun make-sndfile (path &key id data duration end (start 0.0)
                          (force-ffprobe nil)
                          (frequency nil)
                          (amplitude 1.0)
                          (angle-env '(0 0  100 0))
                          (elevation-env '(0 0  100 0))
                          (distance-env '(0 1  100 1)))
```

38.6.634 sndfile/reaper-item

[sndfile] [Classes]

NAME:

```
 reaper

 File:             reaper.lsp

 Class Hierarchy:  named-object -> linked-named-object -> sndfile ->
                   reaper-item
 Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                   reaper-track
 Class Hierarchy:  named-object -> linked-named-object -> sclist ->
                   reaper-file

 Version:          1.1.0

 Project:          slippery chicken (algorithmic composition)
```

```
Purpose:          Class and routine definition for the generation of partial
                  and/or complete reaper files from rhythms, soundfiles.
                  NB Because this class can write reaper files that include
                  video as well as sound files (and we need to distinguish
                  between the two), the ffprobe command is needed here.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    January 21st 2021

$$ Last modified:  12:37:33 Mon Feb 24 2025 CET

SVN ID: $Id: sclist.lsp 963 2010-04-08 20:58:32Z medward2 $
```

38.6.635 sndfile/reset-usage

[sndfile] [Methods]

DESCRIPTION:

```
Reset the WILL-BE-USED and HAS-BEEN-USED slots of the given sndfile object
to 0. These slots keep track of how many times a sound will be used and has
been used, which is useful for purposes such as incrementing
start-time. These slots are set internally and are not intended to be set by
the user.
```

ARGUMENTS:

```
- A sndfile object.
```

RETURN VALUE:

```
Returns 0.
```

EXAMPLE:

```
;; First set the values of the WILL-BE-USED and HAS-BEEN-USED slots, as these
;; are 0 when a new sndfile object is created using make-sndfile. Set the
;; values, print them; reset both using reset-usage, and print again to see
;; the change.
(let ((sf-1 (make-sndfile "/path/to/sndfile-1.aiff"))))
  (setf (will-be-used sf-1) 11)
  (setf (has-been-used sf-1) 13)
  (print (will-be-used sf-1))
```

```
  (print (has-been-used sf-1))
  (reset-usage sf-1)
  (print (will-be-used sf-1))
  (print (has-been-used sf-1)))

=>
11
13
0
0
```

SYNOPSIS:

```
(defmethod reset-usage ((sf sndfile))
```

38.6.636 sndfile/sndfile-ext

[sndfile] [Classes]

NAME:

```
 sndfile-ext

 File:             sndfile-ext.lsp

 Class Hierarchy:  named-object -> linked-named-object -> sndfile ->
                   sndfile-ext

 Version:          1.1.0

 Project:          slippery chicken (algorithmic composition)

 Purpose:          Extension of the sndfile class to hold further properties
                   which quantify the character of the sound file, and
                   specifies sound files which can follow the current one.
                   Specifically created to interface with the sndfilenet
                   project in MaxMSP via OSC (see osc.lsp and osc-sc.lsp).

 Author:           Michael Edwards: m@michael-edwards.org

 Creation date:    16th December 2012, Koh Mak, Thailand

 $$ Last modified:  15:37:48 Sat Jun 22 2024 CEST

 SVN ID: $Id$
```

38.6.637 sndfile-ext/get-next

[sndfile-ext] [Methods]

DESCRIPTION:

```
Get the next sound file from the <followers> slot.
```

ARGUMENTS:

```
- A sndfile-ext object.
```

RETURN VALUE:

```
A sndfile-ext object.
```

SYNOPSIS:

```
(defmethod get-next ((sfe sndfile-ext))
```

38.6.638 sndfile-ext/make-sndfile-ext

[sndfile-ext] [Functions]

DESCRIPTION:

```
Make a sndfile-ext (extension of sndfile) object which holds the usual
sndfile data as well as a host of others to do with the characteristics of
the sound file.  In addition, the followers slot specifies sound files
which can follow the current one.  The bit-depth, srate, num-frames, and
bytes slots will be filled automatically if the path to an existing sound
file is given.
```

ARGUMENTS:

```
See defclass slot descriptions
```

RETURN VALUE:

```
A sndfile-ext object or NIL if path is NIL
```

SYNOPSIS:

```
(defun make-sndfile-ext (path &key id data duration end (start 0.0)
                              (frequency nil) (amplitude 1.0) (cue-num -1)
```

```
                                (use t) (pitch -1) (pitch-curve -1)
                                (bandwidth -1)
                                (force-ffprobe nil)
                                (bandwidth-curve -1) (continuity -1)
                                (continuity-curve -1) (weight -1)
                                (weight-curve -1)
                                (energy -1) (energy-curve -1) (harmonicity -1)
                                (harmonicity-curve -1) (volume -1)
                                (volume-curve -1) (loop-it nil) (bit-depth -1)
                                (srate -1) (num-frames -1)
                                (bytes -1) followers group-id)
```

38.6.639 sndfile-ext/max-cue

[sndfile-ext] [Methods]

DESCRIPTION:

```
Generate the data necessary to preload the sound file in a MaxMSP sflist~
object.
```

ARGUMENTS:

```
- The sndfile-ext object.
```

OPTIONAL ARGUMENTS:

```
- Whether to issue an error if the cue-num slot has not been set to a value
  above 1.  Default = #'error.  Could also be #'warn and NIL.
```

RETURN VALUE:

```
A list of data suitable to be passed via OSC to the sflist~ object.
```

EXAMPLE:

```
(let* ((sf1 (make-sndfile-ext
             (concatenate 'string
                          cl-user::+slippery-chicken-home-dir+
                          "test-suite/sndfile-1.aiff")
             :cue-num 2 :start 0.3 :end 1.1 :frequency 653)))
  (max-cue sf1))

=>
("preload" 2 "/Users/medward2/lisp/sc/test-suite/sndfile-1.aiff" 300.0 1100.0)
```

SYNOPSIS:

```
(defmethod max-cue ((sfe sndfile-ext) &optional (on-fail #'error))
```

38.6.640 sndfile-ext/max-play

[sndfile-ext] [Methods]

DESCRIPTION:

```
Generate the data necessary for MaxMSP to play the sndfile using the
sflist~ and sfplay~.

NB fade-dur could be 0 (= no fade)
```

ARGUMENTS:

```
- The sndfile-ext object.
- The fade (in/out) duration in seconds.
- The maximum loop duration in seconds.
- The time to trigger the next file, as a percentage of the current
  sndfile-ext's duration.
```

RETURN VALUE:

```
A list of values to be passed via OSC to sndfilenet-aux.maxpat:
cue-number number-of-channels loop speed fade-dururation
fade-out-start-time delay-to-next-snfile-start amplitude
```

EXAMPLE:

```
(let* ((sf1 (make-sndfile-ext
             (concatenate 'string
                          cl-user::+slippery-chicken-home-dir+
                          "test-suite/sndfile-1.aiff")
             :start 0.3 :end 1.1 :frequency 653)))
  (max-play sf1 20))

=>
(-1 1 0 1.0 0.32000002 0.48)
```

SYNOPSIS:

```
(defmethod max-play ((sfe sndfile-ext) fade-dur max-loop start-next
                     &optional print)
```

38.6.641 sndfile-ext/proximity

[sndfile-ext] [Methods]

DESCRIPTION:

```
In terms of the characteristics expressed in the various class slots,
evaluate the proximity of one sndfile-ext object to another.  The closer
they are in character, the closer to 0 the returned value will be.  The
order of the two sndfile-ext objects passed to the method is unimportant as
the return value is always >= 0.0.  The more slots match, the lower
(closer) the result will be, i.e., slots are only compared if they have a
value >= 0 (they default to -1), so it could be that in one comparison 5/6
slots match exactly, and in another 2/3 match; in both cases the
non-matching slots is off by 1; in that case the first comparison will
return a lower value: 0.067 vs 0.233.
```

ARGUMENTS:

```
- first sndile-ext object
- second sndile-ext object
```

RETURN VALUE:

```
A floating point number >= 0.0 where 0.0 indicates an exact match.
```

EXAMPLE:

```
(let ((sf3 (make-sndfile-ext
            nil :pitch 3 :pitch-curve 4 :bandwidth 10 :energy 2
            :harmonicity-curve 0))
      (sf4 (make-sndfile-ext
            nil :pitch 3 :pitch-curve 4 :bandwidth 10 :energy 2
            :harmonicity-curve 1)))
  ;; harmonicity-curve is slightly different so we get a result > 0
  (print (proximity sf3 sf4))
  (set-characteristic sf4 'harmonicity-curve 0)
  ;; now they're the same so we get 0.0
  (proximity sf3 sf4))

=>
0.12857144
0.0
```

SYNOPSIS:

```
(defmethod proximity ((sfe1 sndfile-ext) (sfe2 sndfile-ext))
```

38.6.642 sndfile-ext/reset

[sndfile-ext] [Methods]

DESCRIPTION:

Reset the <followers> circular list to the first or any other following sound file.

ARGUMENTS:

A sndfile-ext object.

OPTIONAL ARGUMENTS:

The position (index) to reset to (will default to 0 i.e. the beginning of the list). NB This position may be higher than the number of followers attached to any given sndfile-ext object as it will wrap around.

RETURN VALUE:

T

SYNOPSIS:

```
(defmethod reset ((sfe sndfile-ext) &optional where (warn t))
```

38.6.643 sndfile-ext/set-characteristic

[sndfile-ext] [Methods]

DESCRIPTION:

Set the chracteristic of a sndfile-ext object to a given value. The value for the slot is first checked to correspond to an accepted range; if not an error (default) or warning (or nothing) will be issued.

ARGUMENTS:

- a sndfile-ext object.
- the characteristic, i.e. one of the class slot names, as a symbol.
- the new value, as an accepted integer (see characteristics slot for accepted range).

OPTIONAL ARGUMENTS:

```
 - the function to call if the given value is out of range. Default =
 #'error but could also be #'warn, or NIL (if no error message should be
 issued).
```

RETURN VALUE:

```
 NIL on fail otherwise <value>
```

EXAMPLE:

```
(let ((sf4 (make-sndfile-ext
            nil :pitch 3 :pitch-curve 4 :bandwidth 10 :energy 2
            :harmonicity-curve 1)))
  ;; out of range but no error/warning
  (print (set-characteristic sf4 'harmonicity-curve 15 nil))
  ;; out of range and warn
  (print (set-characteristic sf4 'harmonicity-curve -1 #'warn))
  ;; in range
  (set-characteristic sf4 'harmonicity-curve 0))

=>
NIL
WARNING:
   sndfile-ext::set-characteristic: No such characteristic: HARMONICITY-CURVE -1
NIL
0
```

SYNOPSIS:

```
(defmethod set-characteristic ((sfe sndfile-ext) characteristic value
                               &optional (on-fail #'error))
```

38.6.644 sndfile-ext/vidfile

[sndfile-ext] [Classes]

NAME:

```
 sndfile-ext

 File:             vidfile.lsp

 Class Hierarchy:  named-object -> linked-named-object -> sndfile ->
                   sndfile-ext -> vidfile
```

```
Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Extension of the sndfile-ext class to hold further
                  properties of video files. Essentially, for now--and for
                  the main purpose of allowing video files to be treated
                  like sound files and placed into reaper files via
                  reaper-play--we'll just make a cursory distinction between
                  sound files and video files. This will allow for further
                  functionality in the future, where necessary and relevant
                  for slippery-chicken. See the sndfile-palette class which
                  separates true sound files from video files and allows
                  handling of both. NB clm-play and others won't be able to
                  process the (usually compressed) audio in these files.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    19th March 2024, Heidhausen, Germany

$$ Last modified:  15:50:21 Sat Jun 22 2024 CEST
```

38.6.645 sndfile/spectrum

[sndfile] [Methods]

DATE:

```
18th April 2024
```

DESCRIPTION:

```
This is a simple convenience method to access the get-spectrum function
defined in get-spectrum.lsp
```

ARGUMENTS:

```
- the sndfile object
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
any keyword arguments that the get-spectrum function (get-spectrum.lsp) can
handle. E.g. :start-analysis 0.1 (seconds) or :order-by 'freq or 'amp
```

RETURN VALUE:

```
two values: a list of the frequencies detected and a list of their
amplitudes. The ordering of the lists depends on the :order-by keyword
argment (see above).
```

SYNOPSIS:

```
#+clm
(defmethod spectrum ((sf sndfile) &rest keyargs &key &allow-other-keys)
```

38.6.646 sndfile/stereo

[sndfile] [Methods]

DESCRIPTION:

```
Test whether the CHANNELS slot of a given sndfile object is set to 2.
```

ARGUMENTS:

```
- A sndfile object.
```

RETURN VALUE:

```
Returns T if the CHANNELS slot is set to 2, otherwise returns NIL.
```

EXAMPLE:

```
;; The method make-sndfile creates a sndfile object with the CHANNELS slot set
;; to NIL. Make a sndfile object, test to see whether the value of the CHANNELS
;; slot is 2; set the CHANNELS slot to 2 and test again.
(let ((sf-1 (make-sndfile "/path/to/sndfile-1.aiff")))
  (print (stereo sf-1))
  (setf (channels sf-1) 2)
  (print (stereo sf-1)))

=>
NIL
T
```

SYNOPSIS:

```
(defmethod stereo ((sf sndfile))
```

38.6.647 linked-named-object/tempo

[linked-named-object] [Classes]

NAME:

```
tempo

File:             tempo.lsp

Class Hierarchy:  named-object -> linked-named-object -> tempo

Version:          1.1.0

Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the tempo class which holds very simple
                  tempo information, simply the type of beat and the number
                  of beats per minute etc.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    March 11th 2001

$$ Last modified:  15:51:49 Sat May 24 2025 CEST

SVN ID: $Id$
```

38.6.648 tempo/make-tempo

[tempo] [Functions]

DESCRIPTION:

```
Make a tempo object.
```

ARGUMENTS:

```
- A number indicating beats per minute. If this is >= 10000 we'll treat the
argument as a usecs slot (number of microseconds quarter note) and calcuate
the BPM from that. That gives us a maximum BPM of 5999 before we start
thinking these are usecs (should be fine).
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
```

- :beat. Sets the "beat" value of the beats per minute; i.e., 'q (or 4) for "quarter = xx bpm" etc. Default = 4.
- :id. Sets the ID of the tempo object.
- :description. A text description (string) of the tempo, such as "Allegro con brio" etc.

RETURN VALUE:

A tempo object.

EXAMPLE:

```
;; Default beat is a quarter, thus the following makes a tempo object of
;; quarter=60.
(make-tempo 60)

=>
TEMPO: bpm: 60, beat: 4, beat-value: 4.0, qtr-dur: 1.0
       qtr-bpm: 60.0, usecs: 1000000, description: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: 60

;; Set the beat using the :beat keyword argument. Thus, the following makes a
;; tempo object of dotted-quarter = 96.
(make-tempo 96 :beat 'q.)

;; Add a text description, which is stored in the tempo object's DESCRIPTION
;; slot.
(let ((tt (make-tempo 76 :beat 2 :description "Allegretto")))
  (description tt))

=> "Allegretto"
```

SYNOPSIS:

```
(defun make-tempo (bpm &key (beat 4) id description)
```

38.6.649 tempo/tempo-equal

[tempo] [Methods]

DESCRIPTION:

Test to determine whether the values of two tempo objects are equal.

ARGUMENTS:

```
 - A first tempo object.
 - A second tempo object.
```

RETURN VALUE:

```
 Returns T if the values of the two tempo objects are equal, otherwise NIL.
```

EXAMPLE:

```
;; Equal
(let ((tt1 (make-tempo 60))
      (tt2 (make-tempo 60)))
  (tempo-equal tt1 tt2))

=> T

;; Not equal
(let ((tt1 (make-tempo 60))
      (tt2 (make-tempo 96)))
  (tempo-equal tt1 tt2))

=> NIL
```

SYNOPSIS:

```
(defmethod tempo-equal ((t1 tempo) (t2 tempo))
```

38.6.650 linked-named-object/time-sig

[linked-named-object] [Classes]

NAME:

```
 time-sig

 File:             time-sig.lsp

 Class Hierarchy:  named-object -> linked-named-object -> sclist -> time-sig

 Version:          1.1.0

 Project:          slippery chicken (algorithmic composition)
```

```
Purpose:          Implementation of a time-sig class that stores
                  information about time signatures, allows comparison of
                  time signatures etc.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    12th February 2001

$$ Last modified:  11:04:26 Fri Apr 29 2022 CEST

SVN ID: $Id$
```

38.6.651 time-sig/beat-duration

[time-sig] [Methods]

DESCRIPTION:

```
Get the duration in seconds of one beat of the given time-signature at a
tempo of quarter=60.
```

ARGUMENTS:

```
- A time-sig object.
```

RETURN VALUE:

```
A number.
```

EXAMPLE:

```
;; Beat duration in seconds for time-signature 2/4 at quarter=60
(let ((ts (make-time-sig '(2 4))))
  (beat-duration ts))

=> 1.0

;; Beat duration in seconds for 6/8 at quarter=60
(let ((ts (make-time-sig '(6 8))))
  (beat-duration ts))

=> 1.5
```

SYNOPSIS:

```
(defmethod beat-duration ((ts time-sig))
```

38.6.652 time-sig/get-beat-as-rhythm

[time-sig] [Methods]

DESCRIPTION:

Get the beat unit of a given time-sig object and return it as a rhythm.

ARGUMENTS:

- A time-sig object.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether to consider the beat of a compound meter to be the denominator of the time signature (such as 8 for 6/8) or the beat duration derived from the traditionally understood beat of that meter (such as Q. for 6/8). NIL = denominator. Default = NIL.

RETURN VALUE:

A rhythm object.

EXAMPLE:

```
;; Returns a rhythm object
(let ((ts (make-time-sig '(2 4))))
  (get-beat-as-rhythm ts))

=>
RHYTHM: value: 4.000, duration: 1.000, rq: 1, is-rest: NIL,
        score-rthm: 4.0f0, undotted-value: 4, num-flags: 0, num-dots: 0,
        is-tied-to: NIL, is-tied-from: NIL, compound-duration: 1.000,
        is-grace-note: NIL, needs-new-note: T, beam: NIL, bracket: NIL,
        rqq-note: NIL, rqq-info: NIL, marks: NIL, marks-in-part: NIL,
        letter-value: 4, tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: 4, tag: NIL,
data: 4

;; Default for compound meters is to return the denominator of the time
;; signature
(let ((ts (make-time-sig '(6 8))))
  (data (get-beat-as-rhythm ts)))
```

```
=> 8

;; Setting the optional argument to T returns the compound beat of a compound
;; meter rather than the denominator of the time signature
(let ((ts (make-time-sig '(6 8))))
  (data (get-beat-as-rhythm ts t)))

=> Q.
```

SYNOPSIS:

```
(defmethod get-beat-as-rhythm ((ts time-sig) &optional handle-compound)
```

38.6.653 time-sig/get-whole-bar-rest

[time-sig] [Methods]

DESCRIPTION:

```
Create an event object consisting of a rest equal in duration to one full
bar of the given time-sig object.
```

ARGUMENTS:

```
- A time-sig object.
```

RETURN VALUE:

```
Returns an event object.
```

EXAMPLE:

```
;; Returns an event object
(let ((ts (make-time-sig '(2 4))))
  (get-whole-bar-rest ts))

=>
EVENT: start-time: NIL, end-time: NIL,
       duration-in-tempo: 0.0,
       compound-duration-in-tempo: 0.0,
       amplitude: 0.7
       bar-num: -1, marks-before: NIL,
       tempo-change: NIL
       instrument-change: NIL
       display-tempo: NIL, start-time-qtrs: -1,
```

```
       midi-time-sig: NIL, midi-program-changes: NIL,
       8va: 0
       pitch-or-chord: NIL
       written-pitch-or-chord: NIL
RHYTHM: value: 2.000, duration: 2.000, rq: 2, is-rest: T,
        score-rthm: 2.0f0, undotted-value: 2, num-flags: 0, num-dots: 0,
        is-tied-to: NIL, is-tied-from: NIL, compound-duration: 2.000,
        is-grace-note: NIL, needs-new-note: NIL, beam: NIL, bracket: NIL,
        rqq-note: NIL, rqq-info: NIL, marks: NIL, marks-in-part: NIL,
        letter-value: 2, tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: 2, tag: NIL,
data: 2

;; The rhythmic value of the event object returned is equal to the rhythmic
;; duration of a full bar in the given time signature, the PITCH-OR-CHORD slot
;; is set to NIL, and the IS-REST slot is set to T.
(let* ((ts (make-time-sig '(2 4)))
       (tswbr (get-whole-bar-rest ts)))
  (print (value tswbr))
  (print (pitch-or-chord tswbr))
  (print (is-rest tswbr)))

=>
2.0
NIL
T
```

SYNOPSIS:

```
(defmethod get-whole-bar-rest ((ts time-sig))
```

38.6.654 time-sig/is-compound

[time-sig] [Methods]

DESCRIPTION:

```
Determine whether the value of a given time-sig object is a compound time
signature.
```

ARGUMENTS:

```
- A time-sig object.
```

RETURN VALUE:

```
T if the value of the given time-sig object is a compound time signature,
otherwise NIL.
```

EXAMPLE:

```
;; Testing a time-sig object with a 2/4 time signature returns NIL
(let ((ts (make-time-sig '(2 4))))
  (is-compound ts))

=> NIL

;; Testing a time-sig object with a 6/8 time signature returns T
(let ((ts (make-time-sig '(6 8))))
  (is-compound ts))

=> T
```

SYNOPSIS:

```
(defmethod is-compound ((ts time-sig))
```

38.6.655 time-sig/make-time-sig

[time-sig] [Functions]

DESCRIPTION:

```
Create a time-sig object. In addition to the numerator and denominator
values, the object also stores other automatically calculated information,
such as whether the signature is simple or compound, the duration of one
bar of the given time signature in seconds, the number of midi-clocks, etc.
```

ARGUMENTS:

```
- A two-item list of numbers, the first being the numerator (number of
  beats per measure), the second being the denominator (beat type).
```

RETURN VALUE:

```
- A time-sig object.
```

EXAMPLE:

```
(make-time-sig '(2 4))
```

```
=>
TIME-SIG: num: 2, denom: 4, duration: 2.0, compound: NIL, midi-clocks: 24, num-beats: 2
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "0204", tag: NIL,
data: (2 4)
```

SYNOPSIS:

```
(defun make-time-sig (ts)
```

38.6.656 time-sig/make-time-sig-from-duration

[time-sig] [Functions]

DESCRIPTION:

```
This tries to generate a reasonable time signature from a given duration in
seconds. Will not do anything sophisticated like decide between simple and
compound times.

This will usually result in a numerator >= 2, but tt's not always easy to
decide on the best time-sig, e.g. 2/8 is usually preferable to 1/4, but
1/16 is better than 2/32. Hence there's a call to a routine that
substitutes less-preferable for more-preferable time-sigs.
```

ARGUMENTS:

```
- a duration in seconds
```

OPTIONAL ARGUMENTS:

```
- the tempo. This will be used to convert the first argument to quarter
  notes
- overrides. Pairwise list of preferred time signatures, where equivalents
  can be found (e.g. 3/8 and 6/16). See example below and
  get-preferred-time-sig
```

RETURN VALUE:

```
a time-sig object
```

EXAMPLE:

```
(make-time-sig-from-duration 2) -> 2/4
(make-time-sig-from-duration 1.75) -> 7/16
(make-time-sig-from-duration 1.5) -> 3/8
(make-time-sig-from-duration 1.5 60 '(((6 16) (3 8)))) -> 6/16
```

SYNOPSIS:

```
(defun make-time-sig-from-duration (dur-secs
                                    &optional
                                      (tempo 60.0)
                                      overrides)
```

38.6.657 time-sig/scale

[time-sig] [Methods]

DESCRIPTION:

```
Scale the value of the given time-sig object by a specified factor.
```

ARGUMENTS:

```
- A time-sig object.
- A number (scaling factor).
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether or not to preserve the meter by maintaining
  the same number of beats as the numerator of the time signature. T =
  preserve the meter. Default = T.
```

RETURN VALUE:

```
A time-sig object.
```

EXAMPLE:

```
;; Scaling a (2 4) time-sig object by 3 creates a new time-sig object with a
;; value of 6/4
(let ((ts (make-time-sig '(2 4))))
  (scale ts 3))

=>
TIME-SIG: num: 6, denom: 4, duration: 6.0, compound: NIL, midi-clocks: 24, num-beats: 6
```

```
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "0604", tag: NIL,
data: (6 4)

;; Scaling a (2 4) time-sig object by 2 by default preserves the meter
(let ((ts (make-time-sig '(2 4))))
  (data (scale ts 2)))

=> (2 2)

;; Scaling a (2 4) time-sig object by 2 with the optional argument set to NIL
;; changes the meter and results in a 4/4
(let ((ts (make-time-sig '(2 4))))
  (data (scale ts 2 nil)))

=> (4 4)

;; Halving the value of a time-sig object is achieved using a factor of .5
(let ((ts (make-time-sig '(2 4))))
  (data (scale ts .5)))

=> (2 8)
```

SYNOPSIS:

```
(defmethod scale ((ts time-sig) scaler
                  &optional (preserve-meter t) ignore1 ignore2)
```

38.6.658 time-sig/time-sig-equal

[time-sig] [Methods]

DESCRIPTION:

```
Determine whether the values of two given time-sig objects are the same. If
they are identical in signature, return T; if they are different signatures
but have the same duration (e.g. 2/4, 4/8, 8/16 etc.) return
TIME-SIG-EQUAL-DURATION; otherwise return NIL.
```

ARGUMENTS:

```
- A first time-sig object.
- A second time-sig object.
```

RETURN VALUE:

```
 Returns T if the time signatures are identical; returns
 TIME-SIG-EQUAL-DURATION if they are different signatures with the same
 duration; otherwise NIL.
```

EXAMPLE:

```
;; Two identical signatures return T
(let ((ts1 (make-time-sig '(2 4)))
      (ts2 (make-time-sig '(2 4))))
  (time-sig-equal ts1 ts2))

=> T

;; Two different signatures of the same duration return TIME-SIG-EQUAL-DURATION
(let ((ts1 (make-time-sig '(2 4)))
      (ts2 (make-time-sig '(4 8))))
  (time-sig-equal ts1 ts2))

=> TIME-SIG-EQUAL-DURATION

;; Two completely different signatures return NIL
(let ((ts1 (make-time-sig '(2 4)))
      (ts2 (make-time-sig '(3 4))))
  (time-sig-equal ts1 ts2))

=> NIL
```

SYNOPSIS:

```
(defmethod time-sig-equal ((ts1 time-sig) (ts2 time-sig))
```

39 sc/slippery-chicken

[Classes]

NAME:

```
 slippery-chicken

 File:             slippery-chicken.lsp

 Class Hierarchy:  named-object -> slippery-chicken

 Version:          1.1.0
```

```
Project:          slippery chicken (algorithmic composition)

Purpose:          Implementation of the slippery-chicken class.

Author:           Michael Edwards: m@michael-edwards.org

Creation date:    March 19th 2001

$$ Last modified:  19:53:04 Wed Feb  4 2026 CET
```

39.1 slippery-chicken/auto-set-written

[slippery-chicken] [Methods]

DESCRIPTION:

```
Automatically set the WRITTEN-PITCH-OR-CHORD slot for all events where the
player plays a transposing instrument (which can change of course).
```

ARGUMENTS:

```
- A slippery-chicken object.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :start-bar. NIL or an integer that is the first bar for which the written
  pitch/chord data is to be set. If NIL, start from bar 1. Default = NIL.
- :end-bar. NIL or an integer that is the last bar for which the written
  pitch/chord data is to be set. If NIL, do all bars. Default = NIL.
- :players. NIL, the ID or a list of IDs for the player(s) whose part(s)
  are to be affected. If NIL, all players' parts will be affected.
  Default = NIL.
```

RETURN VALUE:

```
T
```

EXAMPLE:

```
;;; Create a slippery-chicken object, set all the written-pitch-or-chord ;
;;; slots to NIL and print the results. Apply the method and print the results ;
;;; again to see the difference.        ;
(let ((mini
```

```
       (make-slippery-chicken
        '+mini+
        :ensemble '(((hn (french-horn :midi-channel 1))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((hn (1 1 1 1 1))))))))
  (next-event mini 'hn nil 1)
  (loop for ne = (next-event mini 'hn)
     while ne
do (setf (written-pitch-or-chord ne) nil))
  (next-event mini 'hn nil 1)
  (print
   (loop for ne = (next-event mini 'hn)
      while ne
collect (written-pitch-or-chord ne)))
  (auto-set-written mini)
  (next-event mini 'hn nil 1)
  (print
   (loop for ne = (next-event mini 'hn)
      while ne
collect (data (written-pitch-or-chord ne)))))

=>
(NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL NIL
NIL NIL NIL NIL NIL NIL NIL)
(C4 D4 E4 FS4 G4 C4 D4 E4 FS4 G4 C4 D4 E4 FS4 G4 C4 D4 E4 FS4 G4 C4 D4 E4
FS4 G4)
```

SYNOPSIS:

```
(defmethod auto-set-written ((sc slippery-chicken) &key start-bar end-bar
                             players)
```

39.2 slippery-chicken/change-bar-line-type

[slippery-chicken] [Methods]

DESCRIPTION:

```
Change single to double or repeat bar lines and vice-versa.  NB This is a
score function only, i.e., if you add repeat bar lines these will not (yet)
be reflected in playback with MIDI or CLM.
```

ARGUMENTS:

```
 - the slippery-chicken object
 - the bar number at the end of which you want the bar line to change
 - bar line type: 0 = normal, 1 = double bar, 2 = final double bar, 3 =
   begin repeat, 4 = begin and end repeat, 5 = end repeat. As remembering
   these numbers is a bit arbitrary, you can also just pass symbols, if you
   prefer: 'normal, 'double-bar, 'final-double, 'begin-repeat,
   'begin-end-repeat, 'end-repeat
```

RETURN VALUE:

```
 returns the bar-line type
```

EXAMPLE:

```
(let ((min
       (make-slippery-chicken
        '+minimum+
        :instrument-palette +slippery-chicken-standard-instrument-palette+
        :ensemble '(((fl (flute :midi-channel 1))))
        :set-palette '((1 ((c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1)))
        :rthm-seq-palette '((1 ((((4 4) - e e e e - - e e e e -)))))
        :rthm-seq-map '((1 ((fl (1))))))))
;; this piece only has one bar so the bar line will be 2 by default ;
  (print (bar-line-type (get-bar min 1 'fl)))
  (change-bar-line-type min 1 'double-bar)
  (bar-line-type (get-bar min 1 'fl)))
=>
...
2
1
```

SYNOPSIS:

```
(defmethod change-bar-line-type ((sc slippery-chicken) bar-num type)
```

39.3 slippery-chicken/check-beams

[slippery-chicken] [Methods]

DESCRIPTION:

```
 See the description, with example, for check-beams in rthm-seq-bar.lsp.
 Players can either be a single symbol, a list of symbols, or nil (whereby
 all players will be checked).
```

SYNOPSIS:

```
(defmethod check-beams ((sc slippery-chicken) &key start-bar end-bar players
                        auto-beam print (on-fail #'warn))
```

39.4 slippery-chicken/check-ties

[slippery-chicken] [Methods]

DESCRIPTION:

Check that all ties are started and ended properly. If the optional argument <same-spellings> is set to T, all tied pitches will be forced to have the same enharmonic spellings.

ARGUMENTS:

- A slippery-chicken object.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether to force all tied pitches to have the same enharmonic spellings.
- a function to call on error, or NIL for no action.

RETURN VALUE:

T if all tie data is ok, otherwise performs the on-fail function and returns NIL.

EXAMPLE:

```
;;; Create a slippery-chicken object, manually create a problem with the ties,
;;; and call check-ties with a #'warn as the on-fail function.
(let* ((mini
        (make-slippery-chicken
         '+mini+
         :ensemble '(((cl (b-flat-clarinet :midi-channel 1))))
         :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
         :set-map '((1 (1)))
         :rthm-seq-palette '((1 ((((4 4) { 3 tq tq tq } +q e (s) s)))))
         :rthm-seq-map '((1 ((cl (1)))))))
       (e4 (get-event mini 1 4 'cl)))
  (setf (is-tied-to e4) nil)
```

```
  (check-ties mini nil #'warn))

=> WARNING: slippery-chicken::check-ties: bad tie, CL bar 1
```

SYNOPSIS:

```
(defmethod check-ties ((sc slippery-chicken)
                       &optional same-spellings (on-fail #'error))
```

39.5 slippery-chicken/check-time-sigs

[slippery-chicken] [Methods]

DATE:

28-Jan-2011

DESCRIPTION:

Check every bar in the given slippery-chicken object to see if all players have the same time signature. Drops into the debugger with an error if not.

ARGUMENTS:

- A slippery-chicken object.

RETURN VALUE:

T if all players have the same time signature at the same time, otherwise drops into the debugger with an error.

EXAMPLE:

```
;; A successful test                    ;
(let* ((mini
        (make-slippery-chicken
         '+mini+
         :ensemble '(((vn (violin :midi-channel 1))
                      (va (viola :midi-channel 2))
                      (vc (cello :midi-channel 3))))
         :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
         :set-map '((1 (1 1 1)))
         :rthm-seq-palette '((1 ((((4 4) { 3 tq tq tq } +q e (s) s)))))
         :rthm-seq-map '((1 ((vn (1 1 1))
```

```
                            (va (1 1 1))
                            (vc (1 1 1))))))))
  (check-time-sigs mini))

=> T

;; A failing test                       ;
(let* ((mini
        (make-slippery-chicken
         '+mini+
         :ensemble '(((vn (violin :midi-channel 1))
                      (va (viola :midi-channel 2))
                      (vc (cello :midi-channel 3))))
         :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
         :set-map '((1 (1 1 1)))
         :rthm-seq-palette '((1 ((((4 4) { 3 tq tq tq } +q e (s) s)))))
         :rthm-seq-map '((1 ((vn (1 1 1))
                             (va (1 1 1))
                             (vc (1 1 1))))))))
  (setf (time-sig (get-bar mini 1 'vn)) '(3 4))
  (check-time-sigs mini))

=>
slippery-chicken::check-time-sigs: time signatures are not the same at bar 1
[Condition of type SIMPLE-ERROR]
```

SYNOPSIS:

```
(defmethod check-time-sigs ((sc slippery-chicken))
```

39.6 slippery-chicken/check-tuplets

[slippery-chicken] [Methods]

DESCRIPTION:

```
Check the qualities of the tuplet brackets in a given slippery-chicken
object to make sure they are all formatted properly (i.e. each starting
tuplet bracket has a closing tuplet bracket etc.). If an error is found,
the method will try to fix it, then re-check, and only issue an error then
if another is found.
```

ARGUMENTS:

```
- A slippery-chicken object.
```

OPTIONAL ARGUMENTS:

```
- The function to use if something is not ok with the tuplets. This
  defaults to #'error, but could also be #'warn for example
```

RETURN VALUE:

```
T if all tuplets brackets are ok, otherwise performs the on-fail function
and returns NIL.
```

EXAMPLE:

```
;;; Create a slippery-chicken object, manually add an error to the tuplet data ;
;;; and call check-tuplets with #'warn as the on-fail function. ;
(let* ((mini
        (make-slippery-chicken
         '+mini+
         :ensemble '(((cl (b-flat-clarinet :midi-channel 1))))
         :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
         :set-map '((1 (1)))
         :rthm-seq-palette '((1 ((((4 4) { 3 tq tq tq } +q e (s) s)))))
         :rthm-seq-map '((1 ((cl (1)))))))
       (e1 (get-event mini 1 1 'cl)))
  (setf (bracket e1) nil)
  (check-tuplets mini #'warn))

=> rthm-seq-bar::check-tuplets: Can't close non-existent bracket.
```

SYNOPSIS:

```
(defmethod check-tuplets ((sc slippery-chicken) &optional (on-fail #'error))
```

39.7 slippery-chicken/clm-play

[slippery-chicken] [Methods]

DESCRIPTION:

```
Using the sound files (samples) defined for the given reference (group ID)
in the sndfile-palette slot of the slippery-chicken object, use CLM to
generate a new sound file using the pitch and timing information of one or
more players' parts from the slippery-chicken object.

NB: The sound file will begin with the first sounding event in the section
    at 0.0 seconds. If there are any leading rests in the player's part,
```

these will be omitted in the output file.

By grouping sound files in the sndfile-palette slot, the user can generate a CLM sound file version of the piece in various 'flavours'; perhaps, for example, using exclusively source sound files consisting of string samples, or percussion sounds, or a variety of sounds, as desired. See below for an example of a sndfile-palette.

By default, as :pitch-synchronous defaults to NIL (see below) this method does not attempt to match the pitches of the output sound file to those generated for the slippery-chicken object, rather, it generates its own sequence of pitches based on pitches from the current set. Instead of using the pitches of the specified players' parts--which might produce extreme sound file transpositions both upwards and downwards--it accesses each note of the current set (assigned by the set-map to each rthm-seq) from the bottom up, one voice after another. If do-src is T, transposition will then be calculated such that the frequency of the sound file (defaults to C4 in the sndfile object) will be shifted to the pitch of the given pitch of the set. When the set changes it will play the same respective note number of that set, etc. This means that, in effect, any given voice will retain the same pitch for the duration of the rthm-seq (set) but as the sound files cycle around, and these often have quite different pitch/spectral content, there's not necessarily a sense of pitch repetition. If you find that this transposition process still yields some extreme transpositions, the :note-number keyword can be specified to indicate an index into the current set of pitches to serve as the lowest voice. If you increase :note-number the transpositions will get higher. If the number of voices plus this index exceeds the number of pitches in the set, the method will wrap around to the lowest pitch of the set. Bear in mind then that when :pitch-synchronous is NIL, the pitch-seq curves given in the rthm-seq-palette--and therefore all the carefully chosen pitches in the piece--will be ignored in favour of representing the transpositions and therefore intervals present in the set. That might seem like a strange thing to (want to) do, but it's served me (Michael Edwards) very well on many occasions, so it's still the default.

If instead of the above method the user would like the pitches in the resulting sound file to be transposed to match the pitches of the players' notes, the keyword argument :pitch-synchronous can be set to T (and do-src should be left set to T as well). This will also work with chords.

See also make-sfp-from-wavelab-marker-file in sndfile-palette.lsp for automatically creating a sndfile-palette from markers in a Steinberg Wavelab marker file.

Event amplitudes are as yet unused by this method.

NB: CLM's nrev instrument must be loaded before calling this method. If you're using the slippery-chicken app, then this is done for you at startup.

ARGUMENTS:

- A slippery chicken object.
- The ID of the starting section.
- The IDs of the player(s) whose events are to be used to obtain the rhythmic structure (and optionally, pitch content) of the resulting sound file. This can be a single symbol for an individual player, a list of player IDs, or NIL. If NIL, the event from all players' parts will be reflected in the output file. Default = NIL.
- The ID of the sound file group in the sndfile-palette slot of the slippery-chicken object that contains the source sound files from which the new sound file is to be generated.

OPTIONAL ARGUMENTS:

keyword arguments:
- :num-sections. An integer or NIL to indicate how many sections should be generated, including the starting section. If NIL, sound file data will be generated for all sections of the piece. NB If there are sub-sections this will include them in the total, i.e., if section 1 has 3 subsections and :num-sections is 2, we'll generate data from the first two subsections of section 1, not all subsections of main sections 1 and 2. Default = NIL.
- :from-sequence. An integer that is the number of the first sequence within the specified starting section to be used to generate the output file. This argument can only be used when num-sections = 1. Default = 1.
- :num-sequences. NIL or an integer that indicates how many sequences are to be generated, including that specified by :from-sequence. If NIL, all sequences will be played. This argument can only be used when num-sections = 1 and the section has no subsections. Default = NIL.
- :srate. A number that is the sampling rate of the output file (independent of the input file). This and the following two arguments default to the CLM package globals. See clm.html for more options. Default = clm::*clm-srate*.
- :header-type: A CLM package header-type specification to designate the output sound file format. For example, clm::mus-riff will produce .wav files, clm::mus-aiff will produce .aiff files. The value of this argument defaults to the CLM package globals. See clm.html for more options. Default = clm::*clm-header-type*.
- :data-format. A CLM package data-format specification to designate the

output sound file sample data format. For example, clm::mus-float will produce a 32-bit little-endian floating-point format; clm::mus-l24int will produce little-endian 24-bit integer; mus-bshort will produce 16-bit big-endian files, and mus-bfloat will produce 32-bit floating-point big-endian files. NB: AIFF and AIFC files are not compatible with little endian formats. The value of this argument defaults to the CLM package globals. See clm.html for more options.
Default = clm::*clm-data-format*.
- :sndfile-extension. NIL or a string that will be the extension of the output sound file (e.g. ".wav", ".aif"). If NIL, the method will determine the extension auto matically based on the header-type. NB: The extension does not determine the output sound file format; that is determined by :header-type. Default = NIL.
- :channels. An integer that is the number of channels in the output sound file, limited only by the sound file format specified. Note that both stereo and mono sounds from the palette will be randomly panned between any two adjacent channels. Default = 2.
- :rev-amt. A number that determines the amount of reverberation for the resulting sound file, passed to CLM's nrev.
NB: 0.1 is a lot. Default = 0.0.
- time-offset. A number that is an offset time in seconds. This produces a lead time of a specified number of seconds of silence prior to the sound output.
- :play. T or NIL to indicate whether CLM should play the output file automatically immediately after it has been written.
T = play. Default = NIL.
- :inc-start. T or NIL to indicate whether playback of the source sound files is to begin at incrementally increasing positions in those files or at their respective 0.0 positions every time. If T, the method will increment the position in the source sound file from which playback is begun such that it reaches the end of the source sound file the last time it is 'played'. T = increment start times. Default = NIL.
- :ignore-rests. T or NIL to indicate whether silence should be incorporated into the resulting sound file to correspond with rests in the player's parts. If T, the sound files will play over the duration of rests. However, this is only true on a bar-by-bar basis; i.e., notes at the end of one bar will not be continued over into a rest in the next bar. This implies that rests at the start of a bar will not be turned into sounding notes. T = ignore resets. Default = T.
- :sound-file-palette-ref2. The ID of a sound file group in the given slippery-chicken object's sndfile-palette slot. If this reference is given, the method will invoke fibonacci-transitions to transition from the first specified group of source sound files to this one. If NIL, only one group of source sound files will be used. Default = NIL.
- :do-src. T, NIL, a number, or a note-name pitch symbol to indicate whether transposition of the source sound files for playback will be calculated

such that the perceived fundamental frequencies of those sound files are shifted to match the pitches of the current set. If do-src is a number (frequency in Hertz) or a note-name pitch symbol, the method will use only that pitch instead of the sound files' frequencies when transposing to the events' pitches. NB Whichever is used, after being converted to a sample rate conversion factor, this is always multiplied by the src-scaler (see below). T = match sound files' frequencies to set pitches. NIL means no transposition will be performed. Default = T.

- :pitch-synchronous: T or NIL to indicate whether the source sound files are to be transposed to match the pitches of the events in the given players' part. This will only be effective if the given source sound file has a perceptible frequency that has been specified using the sndfile object's :frequency slot in the sndfile-palette. :do-src must also be T for this to work. T = match pitches. Default = NIL.
- :reset-snds-each-rs. T or NIL to indicate whether to begin with the first source sound file of the specified group at the beginning of each rthm-seq. T = begin with the first sound file. Default = T.
- :reset-snds-each-player. T or NIL to indicate whether to begin with the first source sound file of the specified group for the beginning of each player's part. T = begin with the first sound file. Default = T.
- :play-chance-env. A list of break-point pairs that determines the chance that a given event from the source player's part will be reflected in the new sound file. It is determined by random selection but uses a fixed seed that is re-initialized each time clm-play is called. The following default ensures every note will play: '(0 100 100 100).
- :play-chance-env-exp. A number that will be applied as the exponent to the play-chance-env's y values to create an exponential interpolation between break-point pairs. Default = 0.5.
- :max-start-time. A number that is the last time-point in seconds for which events will be processed for the output file. If a maximum start time is specified here (in seconds), events after this will be skipped. The default value of 99999999 seconds (27778 hours) will result in all events being reflected in the sound file.
- :time-scaler. A number that will be the factor by which all start times are scaled for the output file (in effect a tempo scaler). If :ignore-rests is T, this will also have an indirect effect on durations. This argument should not be confused with :duration-scaler. Default = 1.0.
- :duration-scaler. A number that is the factor by which the duration of all events in the output sound file will be scaled. This does not alter start times, and will therefore result in overlapping sounds if greater than 1.0. This is not to be confused with :time-scaler.
  Instead of a simple number, :duration-scaler can also be an envelope (a list of break-point pairs). This means you could smoothly transition from staccato-like to overlapping sounds in a single clm-play call.
  Default = 1.0.

- :normalise. A decimal number that will be the maximum amplitude of the resulting output file; i.e., to which the samples will be scaled. Can also be NIL, whereupon no normalisation will be performed. Default = 0.99
- :amp-env. A list of break-point pairs that will govern the amplitude envelope applied to all source-sound files as it is being written to the new output file. NB: If the user wants to maintain the original attack of the source sound file and is not employing the :inc-start option, this should be set to '(0 1 ...). If :inc-start is T, the resulting sound file will probably contain clicks from non-zero crossings. Default = '(0 0 5 1 60 1 100 0).
- :src-width. An integer that reflects the accuracy of the sample-rate conversion. The higher the value, the more accurate the transposition, but the slower the processing. Values of 100 might be useful for very low transpositions. Default = 20.
- :src-scaler: A number that is the factor by which all sample-rate conversion values will be scaled (for increasing or decreasing the transposition of the overall resulting sound file). Instead of a simple number, :src-scaler can also be an envelope (a list of break-point pairs), similar to :duration-scaler. Default = 1.0.
- :note-number. A number that is an index, representing the the nth pitch of the current set or chord (from the bottom) to be used for the lowest player. Default = 0.
- :duration-run-over. T or NIL to indicate whether the method will allow a sound file event to extend past the end of specified segment boundaries of a sound file in the sndfile-palette. T = allow. Default = NIL.
- :short-file-names. T or NIL to indicate whether abbreviated output file names will be automatically created instead of the usually rather long names. T = short. Default = NIL.
- :output-name-uniquifier. A user-specified string that will be incorporated into the file name, either at the end or the beginning depending on whether short-file-names is T or NIL. Default = "".
- :check-overwrite. T or NIL to indicate whether to query the user before overwriting existing sound files. T = query. Default = T.
- :print-secs. T or NIL to indicate whether CLM should print the seconds computed as it works. T = print. Default = NIL.
- :simulate. T or NIL to indicate whether only the sound file sequencing information should be calculated and printed for testing purposes, without generating a sound file. T = simulate. Default = NIL.
- :sndfile-palette. NIL or a file name including path and extension that contains an external definition of a sndfile-palette. This will replace any sndfile-palette defined in the slippery-chicken object. If NIL, the one in the slippery-chicken object will be used. Default = NIL.
- :chords. NIL or a list of lists consisting of note-name symbols to be used as the pitches for the resulting sound file in place of the pitches from the set-map. There must be one chord specified for each sequence. If

NIL, the pitches from the set-map will be used. Default = NIL.
- :chord-accessor. Sometimes the chord stored in the palette is not a simple list of data so we need to access the nth of the chord list. Default = NIL.
- :clm-ins. The CLM instrument that should be called to generate sound file output. This must accept required and keyword arguments like samp5 (see src/samp5.lsp or more simply src/sine.lsp) but can of course choose to ignore them. Remember that your CLM instruments will (most probably) be in the CLM package so you'll need the clm:: qualifer. Default = #'clm::samp5.
- :clm-ins-args. This should be a list of keyword arguments and values to pass to each CLM instrument call (e.g. '(:cutoff-freq 2000 :q .6)) or a function which takes two arguments (the current event and the event number) and returns a list of keyword arguments perhaps based on those. Default = NIL.
- :pan-min-max. Each event is panned randomly between any two adjacent channels in the output sound file. The values used are chosen from the following list of degrees: (15 25 35 45 55 65 75). However, these can be scaled to be within new limits by passing a two-element list to :pan-mix-max. E.g. '(40 50) would limit most values to the middle area of stereo space. NB if stereo soundfiles are used as input the original panning will generally be retained. See samp5.lsp for details of how such files are handled. Default = NIL (i.e. use the list given above).
- :pan-fun. If you want to take charge of selecting pan positions yourself, pass a function via this keyword. The function must take one argument: the current event, though of course, it can ignore it completely if preferred. The function should return a degree value: a number between 0 and 90. Bear in mind if using samp5 with more than 2 output channels that this will still result in the sound moving around multichannel space as all you would be setting is the pan value between any two adjacent channels. If this isn't your cup of tea, pass a new instrument entirely via :clm-ins. See example below. Default = NIL.
- :snd-selector. By default the sound files in the given group are cycled through, one after the other, returning to the beginning when the end is reached. This can be changed by passing a function to :snd-selector. This function must take three arguments: the circular-sclist object that contains the group's sndfile objects (the sndfiles are in a simple list in the data slot); the pitch of the current event (if the event happens to be a chord, each pitch of the chord will be handled separately i.e. the :snd-selector function will be called for each pitch); and the event object. Of course any of these arguments may be ignored by the :snd-selector function; indeed it is imagined that the event object will be mostly ignored and the pitch object used instead, but it is nevertheless passed for completeness. For example, the following function, declared on-the-fly using lambda rather than separately with defun, makes clm-play act as a traditional sampler would: Assuming our

sndfile-palette group has a list of sound files whose frequencies really correspond to their perceived fundamentals (e.g. see the set-frequency-from-filename method, and the kontakt-to-sfp function) we choose the nearest sound file in the group to the frequency of the current event's pitch:

```
:snd-selector #'(lambda (sflist pitch event)
                  (declare (ignore event))
                  (get-nearest-by-freq
                   (frequency pitch) (data sflist)))
```

NB If :pitch-synchronous is NIL, then the pitch-or-chord slot of the event will be the pitch selected from the current set, rather than the actual pitch of the instrument in question.
Default = NIL i.e. use the circular selection method.

- :decay-time. Reverb decay time. Default = 3 seconds.
- :snd-transitions. Set the transition from sound-file-ref to sound-file-ref over the course of the call to clm-play using a custom envelope instead of the default fibonacci-transitions method, e.g. '(0 0 100 1). x-axis is arbitrary, y-axis should range from 0 to 1. When reading, y-values will be rounded.

RETURN VALUE:

Total events generated (integer).

EXAMPLE:

```
;;; An example using some of the more frequent arguments
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (hn (french-horn :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1 1 1))
                   (2 (1 1 1 1 1 1 1))
                   (3 (1 1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h (q) e (s) s))
                                :pitch-seq-palette ((1 (2) 3))))
                            (2 ((((4 4) (q) e (s) s h))
                                :pitch-seq-palette ((1 2 (3)))))
                            (3 ((((4 4) e (s) s h (q)))
                                :pitch-seq-palette ((2 3 3))))
                            (4 ((((4 4) (s) s h (q) e))
```

```
                              :pitch-seq-palette ((3 (1) 2)))))
        :rthm-seq-map '((1 ((cl (2 3 2 4 1 3 1))
                            (hn (2 4 1 2 3 1 3))
                            (vc (1 2 2 3 4 1 3))))
                       (2 ((cl (4 2 1 3 3 1 2))
                            (hn (2 1 4 3 2 1 3))
                            (vc (2 3 4 3 1 2 1))))
                       (3 ((cl (3 1 2 4 3 1 2))
                            (hn (3 4 2 1 3 2 1))
                            (vc (3 2 3 1 4 2 1)))))
        :snd-output-dir (get-sc-config 'default-dir)
        :sndfile-palette '(((sndfile-grp-1
                             ((test-sndfile-1.aiff)
                              (test-sndfile-2.aiff)
                              (test-sndfile-3.aiff)))
                            (sndfile-grp-2
                             ((test-sndfile-4.aiff :frequency 834)
                              (test-sndfile-5.aiff)
                              (test-sndfile-6.aiff))))
                           ("/path/to/sndfiles-dir-1"
                            "/path/to/sndfiles-dir-2")))))
  (clm-play mini 2 '(cl vc) 'sndfile-grp-1
            :num-sections 1
            :srate 48000
            :header-type clm::mus-aiff
            :data-format clm::mus-b24int
            :rev-amt 0.05
            :inc-start t
            :ignore-rests nil
            :sound-file-palette-ref2 'sndfile-grp-2
            :pitch-synchronous t
            :reset-snds-each-rs nil
            :reset-snds-each-player nil))

;;; A minimal working example demonstrating envelope-arguments
;;; to duration-scaler and src-scaler
(let* ((mini
        (make-slippery-chicken
         '+mini+
         :ensemble '(((vn (violin :midi-channel 1))))
         :set-palette '((1 ((c4 d4 e4 f4 g4 a4 b4 c5))))
         :set-map '((1 (1 1 1)))
         :rthm-seq-palette '((1 ((((2 4) (s) (s) e e e))
                                 :pitch-seq-palette ((1 2 3)))))
         :rthm-seq-map '((1 ((vn (1 1 1)))))
         :snd-output-dir "/tmp/"
```

```
        :sndfile-palette '(((sndfile-grp-1
                             ((test-sndfile-1.aiff)
                              (test-sndfile-2.aiff)
                              (test-sndfile-3.aiff))))
                           ("/path/to/sndfiles-dir-1")))))
  (clm-play mini 1 'vn 'sndfile-grp-1
            :src-scaler '(0 .5  80 1  100 2)
            :do-src 500
            :src-width 5
            :duration-scaler '(0 .1  80 1  100 2)
            :check-overwrite nil
            :play nil :header-type clm::mus-riff))

;;; if you don't want amplitude panning, rather a random placement of each sound
;;; in one channel only, then this should work (no matter how many channels
;;; there are in your output file):

    :pan-fun #'(lambda (event)
                 (declare (ignore event))
                 (nth (random-rep 2) '(0 90)))
```

SYNOPSIS:

```
#+clm
(defmethod clm-play ((sc slippery-chicken) section players
                     sound-file-palette-ref
                     &key
                     sound-file-palette-ref2
                     (play-chance-env '(0 100 100 100))
                     (max-start-time 99999999)
                     (play-chance-env-exp 0.5)
                     (time-scaler 1.0)
                     (normalise .99)
                     scaled-by
                     (simulate nil)
                     (from-sequence 1)
                     (num-sequences nil)
                     (num-sections nil)
                     (ignore-rests t)
                     (time-offset 0.0)
                     (chords nil)
                     (chord-accessor nil)
                     (note-number 0)
                     (play nil)
                     (amp-env '(0 0 5 1 60 1 100 0))
                     (inc-start nil)
```

```
(src-width 20)
;; either a number or an envelope
(src-scaler 1.0)
(do-src t)
(pitch-synchronous nil)
(rev-amt 0.0)
;; either a number or an envelope
(duration-scaler 1.0)
(short-file-names nil)
(check-overwrite t)
(reset-snds-each-rs t)
(reset-snds-each-player t)
(duration-run-over nil)
(channels 2)
(srate clm::*clm-srate*)
(header-type clm::*clm-header-type*)
(data-format clm::*clm-data-format*)
(print-secs nil)
(output-name-uniquifier "")
(sndfile-extension nil)
(sndfile-palette nil)
;; MDE Sat Oct  3 18:45:28 2015 -- for Cameron!
pan-fun
;; MDE Thu Oct  1 21:03:59 2015
(pan-min-max nil)  ; actually '(15 75) by default below
;; MDE Thu Oct  1 19:13:49 2015
snd-selector
;; MDE Mon Nov 4 10:10:35 2013 -- the following were
;; added so we could use instruments other than samp5 MDE
;; Thu Mar 21 15:42:53 2024, Heidhausen -- now defaults to
;; nil because samp5 fun is not available until it's
;; compiled and this happens within clm-play now rather
;; than in advance
(clm-ins nil)    ;#'clm::samp5)
;; either a list or a function (see above)
clm-ins-args
;; DJR Thu 22 Aug 2019 15:08:39 BST
;; clm::with-sound let's us set this, so why doesn't
;; clm-play? Voila!
(decay-time 3)
;; DJR Mon 16 Sep 2019 01:26:11 BST
;; We can now set snd-transitions with a custom envelope,
;; e.g. '(0 0 100 1). x-values are arbitrary, y-values
;; are from 0 (for sound-file-palette-ref) and 1 (for
;; sound-file-palette-ref2).
snd-transitions)
```

39.8 slippery-chicken/clone

[slippery-chicken] [Methods]

DESCRIPTION:

```
Copy (clone) the specified instance and all data associated with the
slippery-chicken object.
```

ARGUMENTS:

```
- A slippery-chicken object.
```

RETURN VALUE:

```
A slippery-chicken object.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :instrument-palette +slippery-chicken-standard-instrument-palette+
        :ensemble '(((fl (flute :midi-channel 1))))
        :set-palette '((1 ((c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1)))
        :rthm-seq-palette '((1 ((((4 4) - e e e e - - e e e e -)))))
        :rthm-seq-map '((1 ((fl (1))))))))
  (clone mini))
```

SYNOPSIS:

```
(defmethod clone ((sc slippery-chicken))
```

39.9 slippery-chicken/cmn-display

[slippery-chicken] [Methods]

DESCRIPTION:

```
Write the data stored in a given slippery-chicken object to disk as an EPS
(Encapsulated Postscript) file using CMN.

Several of the keyword arguments for this method are passed directly to CMN
and therefore have identical names to CMN functions.
```

```
NB: This might fail if LilyPond files are generated first. If this happens,
    re-evaluate the slippery-chicken object and call cmn-display again.
```

ARGUMENTS:

```
- A slippery-chicken object.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :file. A string that is the directory path with file name and extension
  for the .eps file to be created. Default = "cmn.eps" in the directory
  (get-sc-config 'default-dir) (default "/tmp/")
- :players. NIL or a list of player IDs to indicate whether all players'
  parts should be printed to the score. If NIL, all players' parts will be
  written to the score. If a single symbol or a list of player IDs, only
  the parts of those players will be written to the score. Default = NIL.
- :in-c. T or NIL to indicate whether the output should be printed at
  sounding pitch (in C) or at written pitch. NB: If in C, piccolo and
  double bass maintain their usual octave transpositions.
  T = print at sounding pitch. Default = NIL.
- :respell-notes. T, a list of player IDs paired with a sequence of bar and
  note numbers, or NIL to indicate whether to the cmn-display method should
  call the respell-notes method to the pitches contained in the
  slippery-chicken object according to slippery chicken's enharmonics
  algorithm. If T, the all of the pitches in the object will be considered
  and slippery chicken will convert a number of the pitches to their
  enharmonic equivalents to create more sensible linear pitch progression
  within bars. If a list of player IDs paired with a sequence of bar and
  event numbers is passed, in the form
  '((vln (13 2) (14 3)) (cl (14 3 t))), only the specified pitches are
  changed; e.g., (13 2) = bar 13 note 2 (1-based and counting tied notes
  but not rests). If an additional T is included after the bar number and
  event number (as in the cl example above), only the spelling of the
  written pitch for that event will be changed (the default is to change
  the sounding note spelling only). Chords are not respelled by the default
  algorithm, so if these need to be respelled, this should be indicated by
  sub-grouping the note number portion of the given bar/note pair into a
  2-item sublist, in which the first number is the position of the chord in
  among the attacked notes of that bar and the second number is the
  position of the desired pitch within the chord, counted from the bottom
  up, e.g. (vln (13 (2 1))). If NIL, no changes will be made. Default = T.
- :auto-clefs. T or NIL to indicate whether the cmn-display method should
  call the auto-clefs method, which automatically insert clef changes into
  the parts of those instruments that use more than one clef.
```

```
  T = automatically place clef changes. Default = T.
- :start-bar. An integer that indicates the first bar of the object to be
  written to the resulting .eps file. NIL = the first bar. Default = NIL.
- :end-bar. The last bar to be written to the resulting .eps file. NIL =
  the last bar of the slippery-chicken object. Default = NIL.
- :title. T, a string, or NIL to indicate whether to write the title of the
  given slippery-chicken object to the resulting .eps file. If T, the TITLE
  slot of the slippery-chicken object will be used. If a string, the
  specified string will be used instead. If NIL, no title will be included
  in the output. Default = T.
- :size. A number to indicate the overall size of the symbols in the CMN
  output. Default = 15.
- :page-nums. T or NIL to indicate whether page numbers are to be written.
  T = write page numbers. Default = T.
- :empty-staves. T or NIL to indicate whether an empty staff should be
  displayed under each instrument. This can be useful for making editing
  notes by hand. T = print empty staff. Default = NIL.
- :display-sets. T or NIL to indicate whether to print the set of pitches
  used for each rthm-seq on a separate treble-bass grand staff at the
  bottom of each system in the score. T = print. Default = NIL.
- :write-section-info. T or NIL to indicate whether to write the section ID
  into the score. NB: This might not work without first regenerating the
  slippery-chicken object. T = write section IDs. Default = NIL.
- :display-time. T or NIL to indicate whether the elapsed time in
  (mins:secs) should printed above each measure in the resulting score.
  T = print time. Default = NIL.
- :staff-separation. A number that governs the amount of white space to be
  placed between staves, measured in CMN's units. Default = 3.
- :line-separation. A number that governs the amount of white space to be
  placed between systems (i.e. not groups, but a line of music for the
  whole ensemble), measured in CMN's units. Default = 5.
- :group-separation. A number that governs the amount of white space placed
  between groups in a system, measured in CMN's units. Default = 2.
- :system-separation. An indication for how CMN determines the amount of
  white space between systems. If cmn::page-mark, only one system will be
  written per page. Default cmn::line-mark.
- :page-height. A number to indicate the height of the page in centimeters.
  Default = 29.7.
- :page-width. A number to indicate the width of the page in
  centimeters. Default = 21.0.
- :all-output-in-one-file. T or NIL to indicate whether to write a separate
  file for each page of the resulting score. T = write all pages to the
  same multi-page file. Default = T.
- :one-line-per-page. T or NIL to indicate whether to write just one line
  (system) to each page. T = one line per page. Default = NIL.
- :start-bar-numbering. An integer that indicates the number to be given as
```

the first bar number in the resulting EPS file. The bars will be numbered every five bars starting from this number. NB: The value of this argument is passed directly to a CMN function. If a value is given for this argument, slippery chicken's own bar-number writing function will be disabled. NB: It is recommended that a value not be passed for this argument if a value is given for :auto-bar-nums. NIL = bar 1. Default = NIL.

- :auto-bar-nums. An integer or NIL to indicate a secondary bar numbering interval. This is separate from and in addition to the bar-number written in every part every 5 bars. It corresponds to CMN's automatic-measure-numbers. If set to e.g. 1, a bar number will be printed for every measure at the top of each system, or if :by-line, a bar number will be printed at the start of each line. NB: The value of this argument is passed directly to a CMN function. If a value is given for this argument, slippery chicken's own bar-number writing function will be disabled. NB: It is recommended that a value not be passed for this argument if a value is given for :start-bar-numbering. NIL = no secondary bar numbering. Default = T.
- :rehearsal-letters-all-players. T or NIL to indicate whether rehearsal letters should be placed above the staves of all instruments in a score (this can be useful when generating parts). If NIL, rehearsal letters are only placed above the staves of the instruments at the top of each group. T = place rehearsal letters above all instruments. Default = NIL.
- :tempi-all-players. T or NIL to indicate whether to print the tempo above all players' parts in the score. T = print above all players' parts. Default = NIL.
- :process-event-fun. A user-defined function that takes one argument, namely an event object. The specified function will then be called for each event in the piece. This could be used, for example, to algorithmically add accents, dynamics, or change the colour of notes, etc. If NIL, no function will be applied. Default = NIL.
- :automatic-octave-signs. T or NIL to indicate whether ottava signs should be inserted automatically when notes would otherwise need many ledger lines. T = automatically insert. Default = NIL.
- :multi-bar-rests. T or NIL to indicate whether multiple bars of wrests should be consolidated when writing parts. T = consolidate. NIL = write each consecutive rest bar separately. Default = NIL.
- :display-marks-in-part. T or NIL to indicate whether to print the marks stored in the MARKS-IN-PART slot of each rhythm object in the score. If NIL, the indications stored in the MARKS-IN-PART slot are added to parts only. T = also print to score. Default = NIL.
- :add-postscript. NIL or postscript code to be added to the .eps file after it has been generated. See the add-ps-to-file function for details. Default = NIL.
- :auto-open. Whether to open the .EPS file once written. Currently only available on OSX with SBCL and CCL. Uses the default app for .EPS

files, as if opened with 'open' in the terminal. Default = Value of (get-sc-config cmn-display-auto-open).

RETURN VALUE:

Always T.

EXAMPLE:

```
;;; The simplest usage
(let ((mini
       (make-slippery-chicken
        '+mini+
        :title "mini"
        :ensemble '(((vn (violin :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1)))
        :rthm-seq-palette '((1 ((((2 4) (s) (s) e e e))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((vn (1))))))))
  (cmn-display mini :file "/tmp/mini.eps"))

;;; Used with some of the more frequently implemented keyword arguments
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (hn (french-horn :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5)))
                       (2 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5)))
                       (3 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (2 2 2 2 2))
                   (3 (3 3 3 3 3)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5))))
                            (2 ((((4 4) q e s s h))
                                :pitch-seq-palette ((1 2 3 4 5))))
                            (3 ((((4 4) e s s h q))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((cl (1 3 2 1 2))
                            (hn (3 1 1 2 2))
                            (vc (1 1 3 2 2))))))
```

```
                        (2 ((cl (3 1 1 2 2))
                            (hn (1 3 1 2 2))
                            (vc (3 2 2 1 1))))
                        (3 ((cl (1 1 3 2 2))
                            (hn (2 1 1 2 3))
                            (vc (3 1 1 2 2))))))))
  (cmn-display mini
               :file "/tmp/cmn.eps"
               :players '(cl vc)
               :in-c nil
               :respell-notes nil
               :auto-clefs nil
               :start-bar 8
               :end-bar 13
               :title "CMN Fragment"
               :size 13
               :page-nums nil
               :empty-staves t
               :display-sets t
               :write-section-info t
               :display-time t
               :staff-separation 2
               :line-separation 3))

=> T
```

SYNOPSIS:

```
(defmethod cmn-display ((sc slippery-chicken)
                        &key
                        (respell-notes t)
                        (start-bar nil)
                        (start-bar-numbering nil)
                        (end-bar nil)
                        ;; MDE Fri Apr  6 13:27:08 2012
                        (title t)
                        (file (format nil "~a~a.eps"
                                      (get-sc-config 'default-dir)
                                      (filename-from-title (title sc))))
                        (all-output-in-one-file t)
                        (one-line-per-page nil)
                        (staff-separation 3)
                        (line-separation 5)
                        (empty-staves nil)
                        (write-section-info nil)
                        (group-separation 2)
```

```
(system-separation #+cmn cmn::line-mark)
(process-event-fun nil)
(display-sets nil)
(rehearsal-letters-all-players nil)
(display-marks-in-part nil)
(tempi-all-players nil)
(players nil)
(page-height 29.7)
(page-width 21.0)
(size 15)
(auto-bar-nums t)
(page-nums t)
(in-c nil)
(auto-clefs t)
(multi-bar-rests nil)
(automatic-octave-signs nil)
(display-time nil)
(auto-open (get-sc-config 'cmn-display-auto-open))
(add-postscript nil))
```

39.10 slippery-chicken/copy-bars

[slippery-chicken] [Methods]

DESCRIPTION:

```
Copy the rhythmic contents (rthm-seq-bar objects) from the specified bars
of one specified player's part to another.  NB No check is performed to
ensure that the copied notes are within the new instrument's range.
```

ARGUMENTS:

```
- A slippery-chicken object.
- A 1-based integer or assoc-list reference (section seq-num bar-num) that
  is the number of the first bar in the source player's part whose rhythmic
  contents are to be copied.
- A 1-based integer or assoc-list reference (section seq-num bar-num) that
  is the number of the first bar in the target player's part to which the
  rhythmic contents are to be copied.
- The ID of the source player's part.
- The ID of the target player's part.
- NIL or an integer that is the number of bars to copy, including the
  start-bar. When NIL, all bars in the piece starting from <to-start-bar>
  will be copied.
```

OPTIONAL ARGUMENTS:

```
 - T or NIL to indicate whether to print feedback to the listener about the
   copying process. T = print. Default = NIL.
```

RETURN VALUE:

```
T
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((hn (french-horn :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5))))
                           (2 ((((4 4) h h))
                                :pitch-seq-palette ((1 2)))))
        :rthm-seq-map '((1 ((hn (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (2 ((hn (2 2 2 2 2))
                            (vc (2 2 2 2 2))))
                        (3 ((hn (1 1 1 1 1))
                            (vc (1 1 1 1 1))))))))
  (copy-bars mini 7 2 'vc 'hn 2 t))

=> T
```

SYNOPSIS:

```
(defmethod copy-bars ((sc slippery-chicken) from-start-bar to-start-bar
                      from-player to-player num-bars
                      &optional (print-bar-nums nil))
```

39.11 slippery-chicken/count-microtones

[slippery-chicken] [Methods]

DATE:

```
January 30th 2017
```

DESCRIPTION:

```
Count the number of microtonal pitches in the slippery-chicken object.
```

ARGUMENTS:

```
- the slippery-chicken object
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :players. the players for whom the count should be executed; either a
  single player symbol or a list of symbols
- :start-bar. The bar number in which the counting should begin. Default =
  1
- :end-bar. The bar number in which counting should end. Default = NIL
  which means count through the final bar.
```

RETURN VALUE:

```
Four values: the number of microtones, the number of non-microtones, the
total number of pitches, the percentage of microtonal pitches.

Note that the total number of pitches will almost certainly be different
from the number of events as some events will be rests, others single pitch
notes, yet others chords.
```

EXAMPLE:

```
(count-microtones +jitterbug+)
1420
2312
3732
38.049305
```

SYNOPSIS:

```
(defmethod count-microtones ((sc slippery-chicken) &key players (start-bar 1)
                                                        end-bar)
```

39.12 slippery-chicken/count-notes

[slippery-chicken] [Methods]

DESCRIPTION:

Returns the number of notes between start-bar and end-bar (both inclusive).

ARGUMENTS:

- A slippery-chicken object.
- An integer that is the first bar in which notes will be counted. If NIL, then 1
- An integer that is the last bar in which notes will be counted. If NIL, then the last bar of the piece.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether to count just the number of attacked notes (not including ties) or the number of note events (including ties). T = just attacked notes. Default = NIL.
  NB: A chord counts as one note only.
- NIL or a list of one or more IDs of the players whose notes should be counted. This can be a single symbol or a list of players. If NIL, the notes in all players' parts will be counted. Default = NIL.

RETURN VALUE:

An integer that is the number of notes.

EXAMPLE:

```
;;; Using defaults
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h (q) e (s) s)
                                 (q (e) s +s h)
                                 ((e) s (s) (q) h))
                                :pitch-seq-palette ((1 2 3 4 5 1 3 2)))))
        :rthm-seq-map '((1 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (2 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
```

```
                      (3 ((cl (1 1 1 1 1))
                          (vc (1 1 1 1 1))))))))
  (count-notes mini 2 11))

=> 62

;;; Counting all notes just for player 'vc ;
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h (q) e (s) s)
                                 (q (e) s +s h)
                                 ((e) s (s) (q) h))
                                :pitch-seq-palette ((1 2 3 4 5 1 3 2)))))
        :rthm-seq-map '((1 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (2 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (3 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))))))
  (count-notes mini 2 11 nil 'vc))

=> 31

;;; Counting just the attacked notes for player 'vc ;
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h (q) e (s) s)
                                 (q (e) s +s h)
                                 ((e) s (s) (q) h))
                                :pitch-seq-palette ((1 2 3 4 5 1 3 2)))))
        :rthm-seq-map '((1 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
```

```
                        (2 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (3 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))))))
  (count-notes mini 2 11 t 'vc))

=> 27
```

SYNOPSIS:

```
(defmethod count-notes ((sc slippery-chicken) start-bar end-bar
                        &optional just-attacks players)
```

39.13 slippery-chicken/csound-play

[slippery-chicken] [Methods]

DATE:

2023-03-02, Essen

DESCRIPTION:

This method outputs a soundfile from a slippery-chicken object by generating a Csound-score via write-csound-score and calling the Csound command with a given Csound-Orchestra as an additional argument. Hence, this method's arguments resemble those of write-csound-score, except for the argument indicating the orchestra file and a few keyword-arguments which primarily deal with the soundfile output.

In order to work properly, please make sure to have Csound properly installed on your system and set the value of 'csound-command via (set-sc-config ...) according to your system's configuration.

ARGUMENTS:

- A slippery chicken object.
- The ID(s) of the player(s) whose events are used to obtain the rhythmic structure of the score events. This must be a list with one or more symbols. See write-csound-score for more detail.
- The ID(s) (number) or name(s) (string) of the Csound instruments according to the resp. instrument definition in the Csound orchestra. This must be a list with one or more values. See write-csound-score for more information.
- The path to the Csound-orchestra file (.orc) to be used for rendering

```
   the output soundfile. Make sure that it contains instrument definitions
   for all Csound-instruments set in the preceding argument. This must be
   a string.
```

OPTIONAL ARGUMENTS:

```
 keyword arguments:
 - :csound-flags. A list of strings containing the flags, for the Csound
   command. Altering these might be useful e.g. to change the output format
   of the soundfile. N.B.: The "-o" argument, usually preceding the output
   soundfile name, is already set per default and should not be included in
   this list. For a full list of available arguments, see the Csound-manual.
   Default = '("-d" "-W" "-f")
 - :sndfile-suffix. A string to be added just before the file-extension of
   the generated soundfile. Will be overridden when manually setting the
   sndfile-path via :sndfile. Default = "".
 - :sndfile-extension. A string specifying the file extension of the
   soundfile to be generated by the Csound command. This might be altered
   in case the output format is changed e.g. via :csound-flags. Will be
   overridden when manually setting the sndfile-path via :sndfile.
   Default = ".wav".
 - :sndfile. A string indicating the path and filename to the soundfile to be
   generated with this method. Please bear in mind to alter the suffix if you
   change the format via :csound-flags. Default is a filename extracted from
   the title of the sc piece, placed in the (get-sc-config 'default-dir)
   directory (per default /tmp).
 - The other keyword arguments are exactly the same as write-csound-score.
   Please see write-csound-score for detailed documentation.
```

RETURN VALUE:

```
 Returns the path of the soundfile written, as a string.
```

EXAMPLE:

```
;;; An example using the csound-p-fields-simple function.
;;; The Csound orchestra might consist of two instruments producing
;;; tones based on simple waveforms (sines, square waves etc.)
(let ((mini
        (make-slippery-chicken
         '+mini+
         :ensemble '(((pno (piano :midi-channel 1))
                      (vln (violin :midi-channel 2))))
         :set-palette '((1 ((f3 g3 as3 a3 bf3 b3 c4 d4 e4 f4 g4 a4 bf4 cs5))))
         :set-map '((1 (1 1 1 1 1 1 1))
```

```
                    (2 (1 1 1 1 1 1 1))
                    (3 (1 1 1 1 1 1 1)))
         :tempo-map '((1 (q 60)))
         :rthm-seq-palette '((1 ((((4 4) h (q) e (s) s))
                                 :pitch-seq-palette ((1 (2) 3))))
                             (2 ((((4 4) (q) e (s) s h))
                                 :pitch-seq-palette ((1 2 3))))
                             (3 ((((4 4) e (s) s h (q)))
                                 :pitch-seq-palette ((2 3 3))))
                             (4 ((((4 4) (s) s h (q) e))
                                 :pitch-seq-palette ((3 1 (2))))))
         :rthm-seq-map '((1 ((pno (1 2 1 2 1 2 1))
                             (vln (1 2 1 2 1 2 1))))
                         (2 ((pno (3 4 3 4 3 4 3))
                             (vln (3 4 3 4 3 4 3))))
                         (3 ((pno (1 2 1 2 1 2 1))
                             (vln (1 2 1 2 1 2 1)))))))
  (csound-play mini
               '(pno vln)
               '(1 2)
               "mini-orchestra.orc"
               :comments nil))
```

SYNOPSIS:

```
(defmethod csound-play (sc
                        players
                        csound-instruments
                        ;; RP  Thu Mar  2 21:17:59 2023
                        ;;; unique to csound-play
                        orchestra-file
                        &key
                          ;; RP  Thu Mar  2 19:10:21 2023
                          ;; specific to csound-play
                          (csound-flags '("-d" "-W" "-f"))
                          ;; add something just before the
                          ;; sndfile-extension?
                          (sndfile-suffix "")
                          ;; choose the right file extension
                          ;; according to your format!
                          (sndfile-extension ".wav")
                          (sndfile (format nil "~a~a~a~a"
                                           (get-sc-config 'default-dir)
                                           (filename-from-title (title sc))
                                           sndfile-suffix
                                           sndfile-extension))
```

```
;; RP  Thu Mar  2 19:32:09 2023
;; analogous to write-csound-score
  (start-section 1)
  (offset 0)
  (suffix "")
  (csound-file
   (format nil "~a~a~a.sco"
           (get-sc-config 'default-dir)
           (filename-from-title (title sc))
           suffix))
  (comments t)
  (delimiter #\space)
  (from-sequence 1)
  (num-sequences nil)
  (chords t)
  (p-fields #'csound-p-fields-simple)
  (num-sections nil))
```

39.14 slippery-chicken/empty-bars?

[slippery-chicken] [Methods]

DESCRIPTION:

Test whether players have rest/empty bars within a given range.

ARGUMENTS:

- the slippery-chicken object
- the start bar (integer, inclusive)
- the end bar (integer, inclusive)

OPTIONAL ARGUMENTS:

- the players we want to test. Either a list of player IDs, a single ID, or nil if all players should be tested. Default = NIL = all players.
- T or NIL to indicate whether a bar of only rests (instead of a full bar rest) should be considered empty. Default = NIL.

RETURN VALUE:

T if all players have rest bars within the given range, NIL if no.

SYNOPSIS:

```
(defmethod empty-bars? ((sc slippery-chicken) start-bar end-bar
                        &optional players all-rests)
```

39.15 slippery-chicken/event-list-to-csound-score

[slippery-chicken] [Functions]

DESCRIPTION:

```
This function generates a Csound-score file (.sco) from an event-list.
Its structure is very similar to write-csound-score except that the
allocation process between players and Csound-instruments ist structured
differently. This function uses the event-list's MIDI-channels for
allocating the data to Csound-instruments (if desired).
NB: The function events-update-time might be useful in this context (also
refer to event-list-to-midi-file).
```

ARGUMENTS:

```
- A list of event objects.
- The MIDI-channel(s) to take into account for generating the score.
  This must be a list with one or more symbols or NIL. When NIL, all
  MIDI-channels present in the event-list will be used. When a list,
  only the MIDI-channels contained in this list will be used as a basis
  for the rhythmic structure of the Csound-score.
- The ID(s) (as a number) or name(s) (as a string) of the Csound instruments
  according to the resp. instrument definition in the Csound orchestra as
  list of one or more items.
  Each MIDI-channel of the second argument of this method needs to be
  assigned to one Csound-instrument. Thus, the length of this list must
  be exactly the same as the previous list, except when using NIL in the
  MIDI-channels list. In the latter case, the Csound instruments defined
  in this list will be allocated cyclically to the available MIDI-channels
  in ascending order of the MIDI-channels.
  E.g.: When the MIDI-channels 1, 2, 3, and 10 are present in the
  event-list, and this argument is set to '("ins1" "ins2"), the value of
  this slot will internally be changed to '("ins1" "ins2" "ins1" "ins2").
```

OPTIONAL ARGUMENTS:

```
keyword-arguments:
- :suffix. A string to be added just before the .sco of the generated
  csound score-file. Will be overridden when manually setting the path
  via :csound-file. Default = "".
- :csound-file. A string indicating the path and filename to the .sco-file
  to be generated with this method. Default is "tmp.suffix.sco", placed
  in the (get-sc-config 'default-dir) directory (per default /tmp).
- :offset. An integer or floating point value to indicate the global
  offset of all score events (in seconds). Default = 0.
```

- :comments. A boolean indicating whether additional comments should be included into the generated score. Default = T.
- :delimiter. A character which separates each score statement. Csound recommends using spaces, though #\tab does also work an might, in some cases, improve legibility. Default = #\space.
- :chords. Either a boolean or an integer which determines how to deal with chords in events. When T, all notes of a chord are parsed as i-statements in the Csound-score. When NIL, chords will be ignored and treated like rests (i.e. nothing will be generated). When an integer is given, the number determines the max. amount of pitches being parsed as i-statements. They are chosen chronologically from the given chord. Default = T.
- :p-fields. This should be either NIL, a list of lists or a function used to generate the p-fields starting from p4. Cf. the documentation of write-csound-score for more detail. Default: #'csound-p-fields-simple.

RETURN VALUE:

Returns the path of the file written, as a string.

EXAMPLE:

```
(let* ((notes '(c4 d4 e4 f4 g4))
       (notes-len (length notes))
       (rthms '(q e s))
       (rthms-len (length rthms))
       (events (loop repeat 20
                     collect
                     (make-event (nth (random notes-len) notes)
                                 (nth (random rthms-len) rthms)
                                 :midi-channel (1+ (random 3))))))
  (events-update-time events)
  (event-list-to-csound-score events
                              nil ;; use all midi-channels
                              '(1 2)
                              :csound-file "/tmp/example.sco"))

;; => /tmp/example.sco
```

SYNOPSIS:

```
(defun event-list-to-csound-score (event-list
                                   ;; when NIL, use all available
                                   ;; midi-channels and allocate them
```

```
;; to the given csound instruments
;; in a cyclical order
midi-channels
csound-instruments
&key
  ;; add something just before .sco?
  (suffix "")
  (csound-file
   (format nil "~atmp~a.sco"
           (get-sc-config 'default-dir)
           suffix))
  ;; offset in seconds
  (offset 0)
  ;; add a comments section?
  (comments t)
  (delimiter #\space)
  (chords t)
  (p-fields #'csound-p-fields-simple))
```

39.16 slippery-chicken/find-boundaries

[slippery-chicken] [Methods]

DESCRIPTION:

```
This methods tries to find structural boundaries (i.e section starts) in a
complete piece. It does so by looking purely at event densities across bars
and groups of bars; when these change sharply, a section change is noted.
```

ARGUMENTS:

```
- the slippery-chicken object to analyse
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :sum-bars: when looking at groups of bars, how many bars at a time?
  Default = 3
- :jump-threshold: when looking at the sharpness of the envelope, what
  percentage change in number of events per bar (based in the min/max in the
  overall envelope) is considered a steep enough change. Default = 50 (%)
- :min-length: the minimum length in bars between boundaries. Default = NIL
  = no minimum.
```

RETURN VALUE:

```
A list of bar numbers
```

SYNOPSIS:

```
(defmethod find-boundaries ((sc slippery-chicken)
                            &key (sum-bars 3) (jump-threshold 50) min-length
                            skip-rest-bars)
```

39.17 slippery-chicken/find-end-tie

[slippery-chicken] [Methods]

DATE:

```
November 8th 2018, Heidhausen
```

DESCRIPTION:

```
Find the event which is the last in a group of tied notes.
```

ARGUMENTS:

```
- the slippery-chicken object
- the event object at the start or in the middle of a the group of tied
  notes
```

OPTIONAL ARGUMENTS:

```
- the function, generally some kind of error, to call if the given event is
  not tied from. Default = #'error.
```

RETURN VALUE:

```
the last event object which is tied to or NIL if the given event is not tied
from.
```

SYNOPSIS:

```
(defmethod find-end-tie ((sc slippery-chicken) (e event)
                         &optional (on-fail #'error))
```

39.18 slippery-chicken/find-gps

[slippery-chicken] [Methods]

DATE:

June 9th 2025

DESCRIPTION:

Find all the bar-numbers of the general pauses (hence find-gps: nothing to do with map coordinates) across a given bar range. A general pause here is defined as is when all the players have a complete rest bar. NB Consecutive rest bars count as part of the same general pause.

ARGUMENTS:

- the slippery- chicken object

OPTIONAL ARGUMENTS:

keyword arguments
- the start bar. Default = 1.
- the end bar. Default = NIL = end of piece.

RETURN VALUE:

a list of bar numbers

SYNOPSIS:

```
(defmethod find-gps ((sc slippery-chicken) &key (start-bar 1) end-bar)
```

39.19 slippery-chicken/find-note

[slippery-chicken] [Methods]

DATE:

09-Apr-2011

DESCRIPTION:

Print to the Listener the numbers of all bars in a specified player's part of a given slippery-chicken object in which the specified pitch is found. Note that pitch symbols rather than objects are searched for so that small microtonal variations are be ignored.

ARGUMENTS:

- A slippery-chicken object.
- A player ID.
- A note-name pitch symbol or a list of note-name pitch symbols for the pitch to be sought. If a list, this will be handled as a chord.

OPTIONAL ARGUMENTS:

keyword arguments:
- :written. T or NIL to indicate whether to look for the specified pitch as as a written note only. T = as written only. Default = NIL.
- :start-bar. An integer that is the first bar in which to search for the given pitch. This number is inclusive. Default = 1.
- :end-bar. An integer that is the last bar in which to search for the given pitch. This number is inclusive. Default = number of bars in the given slippery-chicken object.

RETURN VALUE:

Returns a list of all corresponding event objects found. Prints the bar numbers of the results directly to the Lisp listener.

EXAMPLE:

```
;;; Prints the bar number for all occurrences in the entire piece by default ;
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h (q) e (s) s)
                                 (q (e) s +s h)
                                 ((e) s (s) (q) h))
                                :pitch-seq-palette ((1 2 3 4 5 1 3 2)))))
        :rthm-seq-map '((1 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (2 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (3 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1)))))))))
  (find-note mini 'vc 'f4))
```

```
=>
bar 1
bar 3
bar 4
bar 6
bar 7
bar 9
bar 10
bar 12
bar 13
bar 15
bar 16
bar 18
bar 19
bar 21
bar 22
bar 24
bar 25
bar 27
bar 28
bar 30
bar 31
bar 33
bar 34
bar 36
bar 37
bar 39
bar 40
bar 42
bar 43
bar 45

;;; Examples of use specifying the optional arguments ;
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h (q) e (s) s)
                                 (q (e) s +s h)
                                 ((e) s (s) (q) h))
                                :pitch-seq-palette ((1 2 3 4 5 1 3 2)))))
```

```
        :rthm-seq-map '((1 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (2 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (3 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1)))))))
  (find-note mini 'cl 'f3)
  (find-note mini 'cl 'f3 :written t)
  (find-note mini 'vc 'f4 :start-bar 3 :end-bar 17))
```

SYNOPSIS:

```
(defmethod find-note ((sc slippery-chicken) player note &key (written nil)
                      start-bar end-bar)
```

39.20 slippery-chicken/find-rehearsal-letters

[slippery-chicken] [Methods]

DESCRIPTION:

```
Return in list form the numbers of bars in the given slippery-chicken
object that have rehearsal letters.
```

ARGUMENTS:

```
- A slippery-chicken object.
```

RETURN VALUE:

```
A list of numbers.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :rehearsal-letters '(2 5 7)
        :set-palette '((1 ((c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) (s) (s) e e e))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1 1 1)))))))
```

```
  (find-rehearsal-letters mini))

=> (2 5 7)
```

SYNOPSIS:

```
(defmethod find-rehearsal-letters ((sc slippery-chicken))
```

39.21 slippery-chicken/get-all-events

[slippery-chicken] [Methods]

DATE:

```
August 22nd 2013 (Edinburgh)
```

DESCRIPTION:

```
Return a flat list containing all the events of a slippery-chicken object
fo the given player.
```

ARGUMENTS:

```
- The slippery-chicken object
- The player (symbol)
```

RETURN VALUE:

```
A list of event objects
```

SYNOPSIS:

```
(defmethod get-all-events ((sc slippery-chicken) player)
```

39.22 slippery-chicken/get-all-section-refs

[slippery-chicken] [Methods]

DESCRIPTION:

```
Return all section IDs as a list of lists. Subsection IDs will be contained
in the same sublists as their enclosing sections.
```

ARGUMENTS:

- A slippery-chicken object.

RETURN VALUE:

A list of lists.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((sax (alto-sax :midi-channel 1))))
        :set-palette '((1 ((e3 fs3 b3 cs4 fs4 gs4 ds5 f5))))
        :set-map '((1 (1 1 1))
                   (2 (1 1 1))
                   (3 ((a (1 1 1))
                       (b ((x (1 1 1))
                           (y (1 1 1))))))
                   (4 ((a (1 1 1))
                       (b (1 1 1))
                       (c (1 1 1 1))))
                   (5 (1 1 1))
                   (6 (1 1 1))
                   (7 (1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((sax (1 1 1))))
                        (2 ((sax (1 1 1))))
                        (3 ((a ((sax (1 1 1))))
                            (b ((x ((sax (1 1 1))))
                                (y ((sax (1 1 1)))))))))
                        (4 ((a ((sax (1 1 1))))
                            (b ((sax (1 1 1))))
                            (c ((sax (1 1 1 1))))))
                        (5 ((sax (1 1 1))))
                        (6 ((sax (1 1 1))))
                        (7 ((sax (1 1 1)))))))
  (get-all-section-refs mini))

=> ((1) (2) (3 A) (3 B X) (3 B Y) (4 A) (4 B) (4 C) (5) (6) (7))
```

SYNOPSIS:

```
(defmethod get-all-section-refs ((sc slippery-chicken))
```

39.23 slippery-chicken/get-bar

[slippery-chicken] [Methods]

DESCRIPTION:

Get the rthm-seq-bar object located at a specified bar number within a given player's part.

ARGUMENTS:

- A slippery-chicken object.
- An integer that is the number of the bar within the overall piece for which the rthm-seq-bar object is sought.
- The ID of the player from whose part the rthm-seq-bar object is sought. If this is passed as NIL, the method will return the rthm-seq-bar objects for all players in the ensemble at the specified bar number. NB: Although listed as an optional argument, the player ID is actually required. It is listed as optional due to method inheritance.

OPTIONAL ARGUMENTS:

- (see the comment on the <player> argument above.

RETURN VALUE:

A rthm-seq-bar object (or objects).

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s)
                                 (q e s s h)
                                 (e s s q h))
                                :pitch-seq-palette ((1 2 3 4 5
                                                     1 3 2 4 5
                                                     3 5 2 4 1)))))
```

```
        :rthm-seq-map '((1 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (2 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (3 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1)))))))
  (get-bar mini 17 'cl))

=>
RTHM-SEQ-BAR: time-sig: 2 (4 4), time-sig-given: NIL, bar-num: 17,
old-bar-nums: NIL, write-bar-num: NIL, start-time: 64.000,
start-time-qtrs: 64.0, is-rest-bar: NIL, multi-bar-rest: NIL,
show-rest: T, notes-needed: 5,
tuplets: NIL, nudge-factor: 0.35, beams: NIL,
current-time-sig: 2, write-time-sig: NIL, num-rests: 0,
num-rhythms: 5, num-score-notes: 5, parent-start-end: NIL,
missing-duration: NIL, bar-line-type: 0,
player-section-ref: (2 CL), nth-seq: 0, nth-bar: 1,
rehearsal-letter: NIL, all-time-sigs: (too long to print)
sounding-duration: 4.000,
rhythms: (
[...]
```

SYNOPSIS:

```
(defmethod get-bar ((sc slippery-chicken) bar-num &optional player)
```

39.24 slippery-chicken/get-bar-from-ref

[slippery-chicken] [Methods]

DESCRIPTION:

```
Return a rthm-seq-bar object from the piece by specifying its section,
sequence number, bar number, and the player. Sequenz-num and bar-num are
1-based.
```

ARGUMENTS:

```
- A slippery-chicken object.
- A section ID (number or list).
- A player ID.
- An integer that is the number of the sequence in the section from which
  the bar is to be returned (1-based).
- An integer that is the number of the bar within the given sequence
  (1-based).
```

RETURN VALUE:

```
A rthm-seq-bar object.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s)
                                 (q e s s h)
                                 (e s s q h))
                                :pitch-seq-palette ((1 2 3 4 5
                                                     1 3 2 4 5
                                                     3 5 2 4 1)))))
        :rthm-seq-map '((1 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (2 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1))))
                        (3 ((cl (1 1 1 1 1))
                            (vc (1 1 1 1 1)))))))))
  (get-bar-from-ref mini 2 'vc 3 2))
=>
RTHM-SEQ-BAR: time-sig: 2 (4 4), time-sig-given: NIL, bar-num: 23,
old-bar-nums: NIL, write-bar-num: NIL, start-time: 88.000,
start-time-qtrs: 88.0, is-rest-bar: NIL, multi-bar-rest: NIL,
show-rest: T, notes-needed: 5,
tuplets: NIL, nudge-factor: 0.35, beams: NIL,
current-time-sig: 2, write-time-sig: NIL, num-rests: 0,
num-rhythms: 5, num-score-notes: 5, parent-start-end: NIL,
missing-duration: NIL, bar-line-type: 0,
player-section-ref: (2 VC), nth-seq: 2, nth-bar: 1,
rehearsal-letter: NIL, all-time-sigs: (too long to print)
sounding-duration: 4.000,
rhythms: (
[...]
```

SYNOPSIS:

```
(defmethod get-bar-from-ref ((sc slippery-chicken) section player
```

```
sequenz-num bar-num)
```

39.25 slippery-chicken/get-bar-num-from-ref

[slippery-chicken] [Methods]

DESCRIPTION:

```
Get the bar number of a given rthm-seq-bar object by specifying the
section, sequenz, and number of the bar within that sequenz.
```

ARGUMENTS:

```
- A slippery-chicken object.
- The ID of the section in which the given rthm-seq-bar object is located.
- An integer that is the number of the sequence (starting from 1 not 0)
  within that section in which the rthm-seq-bar object is located.
- The number of the bar within the given rthm-seq-bar object for which the
  overall bar number (within the entire piece) is sought.
```

RETURN VALUE:

```
An integer.
```

EXAMPLE:

```
(let ((mini
(make-slippery-chicken
'+mini+
:ensemble '(((cl (b-flat-clarinet :midi-channel 1))
(vc (cello :midi-channel 2))))
:set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
:set-map '((1 (1 1 1 1 1))
(2 (1 1 1 1 1))
(3 (1 1 1 1 1)))
:rthm-seq-palette '((1 ((((4 4) h q e s s)
(q e s s h)
(e s s q h))
:pitch-seq-palette ((1 2 3 4 5
1 3 2 4 5
3 5 2 4 1)))))
:rthm-seq-map '((1 ((cl (1 1 1 1 1))
(vc (1 1 1 1 1))))
(2 ((cl (1 1 1 1 1))
(vc (1 1 1 1 1))))
```

```
(3 ((cl (1 1 1 1 1))
(vc (1 1 1 1 1))))))))
(get-bar-num-from-ref mini 2 4 3))

=> 27
```

SYNOPSIS:

```
(defmethod get-bar-num-from-ref ((sc slippery-chicken) section
                                 sequenz-num bar-num)
```

39.26 slippery-chicken/get-clef

[slippery-chicken] [Methods]

DATE:

```
11-Apr-2011
```

DESCRIPTION:

```
Get the clef symbol attached to a specified event.

NB: The very first clef symbol in the very first measure of a given
    player's part is determined by the corresponding instrument object and
    attached to differently; as such, it cannot be retrieved using this
    method.
NB: All clef symbols after the starting clef are added using the auto-clefs
    method, either directly or by default in the cmn-display or
    write-lp-data-for-all methods.
```

ARGUMENTS:

```
- A slippery-chicken object.
- (NB: The optional arguments are actually required.)
```

OPTIONAL ARGUMENTS:

```
NB: The optional arguments are actually required.
- An integer that is the number of the bar from which to return the clef
  symbol.
- An integer that is the number of the event object within that bar from
  which to retrieve the clef symbol.
- The ID of the player from whose part the clef symbol is to be returned.
```

RETURN VALUE:

```
A clef symbol.
```

EXAMPLE:

```
(let ((mini
(make-slippery-chicken
'+mini+
:ensemble '(((vc (cello :midi-channel 1))))
:tempo-map '((1 (q 96)))
:set-palette '((1 ((g2 f4 e5))))
:set-map '((1 (1 1 1)))
:rthm-seq-palette '((1 ((((5 4) e e e e e e e e e))
:pitch-seq-palette ((1 1 2 2 2 2 3 3 3 1)))))
:rthm-seq-map '((1 ((vc (1 1 1))))))))
(auto-clefs mini)
(get-clef mini 1 3 'vc))

=> TENOR
```

SYNOPSIS:

```
(defmethod get-clef ((sc slippery-chicken) &optional bar-num event-num player)
```

39.27 slippery-chicken/get-current-instrument-for-player

[slippery-chicken] [Methods]

DESCRIPTION:

```
Get the currently active instrument for a given player in a specified
sequence of a slippery-chicken object, as defined in the
instrument-change-map.
```

ARGUMENTS:

```
- The ID of the section from which to retrieve the current instrument for
  the specified player. This can also be a reference, e.g. in the form
  '(2 1).
- The ID of the player for whom the current instrument is sought.
- The number of the sequence from which to retrieve the current
  instrument. This is a 1-based number. A slippery-chicken object.
```

RETURN VALUE:

An instrument object.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((sax ((alto-sax tenor-sax) :midi-channel 1))
                     (db (double-bass :midi-channel 2))))
        :instrument-change-map '((1 ((sax ((1 alto-sax) (3 tenor-sax)))))
                                 (2 ((sax ((2 alto-sax) (5 tenor-sax))))))
        :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((sax (1 1 1 1 1))
                            (db (1 1 1 1 1))))
                        (2 ((sax (1 1 1 1 1))
                            (db (1 1 1 1 1)))))))))
  (get-current-instrument-for-player 2 'sax 3 mini))

=>
INSTRUMENT: lowest-written: BF3, highest-written: FS6
lowest-sounding: CS3, highest-sounding: A5
starting-clef: TREBLE, clefs: (TREBLE), clefs-in-c: (TREBLE)
prefers-notes: NIL, midi-program: 66
transposition: EF, transposition-semitones: -9
score-write-in-c: NIL, score-write-bar-line: NIL
chords: NIL, chord-function: NIL,
total-bars: 5 total-notes: 25, total-duration: 20.000
total-degrees: 2920, microtones: T
missing-notes: (BQF3 BQF4), subset-id: NIL
staff-name: alto saxophone, staff-short-name: alt sax,

largest-fast-leap: 999, tessitura: BQF3
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: ALTO-SAX, tag: NIL,
data: NIL
```

SYNOPSIS:

```
(defmethod get-current-instrument-for-player (section player sequence
                                              (sc slippery-chicken))
```

39.28 slippery-chicken/get-event

[slippery-chicken] [Methods]

DESCRIPTION:

Retrieve a specified event object from a slippery-chicken object, giving bar number, event number, and player.

NB: This counts returns event objects, regardless of whether they are notes or rests.

ARGUMENTS:

- A slippery-chicken object.
- An integer that is the number of the bar from which the event object is to be returned.
- An integer that is the number of the event object to be returned from that bar. This number is 1-based and counts all events, including notes, rests, and tied notes.
- A symbol name for the player.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether an error should be signalled if the event doesn't exist.

RETURN VALUE:

An event object.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))
                     (vc (cello :midi-channel 2))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) (e) e+e. 32 (32)))
                                :pitch-seq-palette (((1) 2)))))
        :rthm-seq-map '((1 ((vn (1 1 1))
                            (vc (1 1 1))))))))
```

```
  (get-event mini 2 4 'vn))

=>
EVENT: start-time: 3.750, end-time: 3.875,
duration-in-tempo: 0.125,
compound-duration-in-tempo: 0.125,
amplitude: 0.700
bar-num: 2, marks-before: NIL,
tempo-change: NIL
instrument-change: NIL
display-tempo: NIL, start-time-qtrs: 3.750,
midi-time-sig: NIL, midi-program-changes: NIL,
8va: 0
pitch-or-chord:
PITCH: frequency: 293.665, midi-note: 62, midi-channel: 1
pitch-bend: 0.0
degree: 124, data-consistent: T, white-note: D4
nearest-chromatic: D4
src: 1.122462, src-ref-pitch: C4, score-note: D4
qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
micro-tone: NIL,
sharp: NIL, flat: NIL, natural: T,
octave: 4, c5ths: 0, no-8ve: D, no-8ve-no-acc: D
show-accidental: T, white-degree: 29,
accidental: N,
accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: D4, tag: NIL,
data: D4
**************

written-pitch-or-chord: NIL
RHYTHM: value: 32.000, duration: 0.125, rq: 1/8, is-rest: NIL,
score-rthm: 32.0, undotted-value: 32, num-flags: 3, num-dots: 0,
is-tied-to: NIL, is-tied-from: NIL, compound-duration: 0.125,
is-grace-note: NIL, needs-new-note: T, beam: NIL, bracket: NIL,
rqq-note: NIL, rqq-info: NIL, marks: NIL, marks-in-part: NIL,
letter-value: 32, tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: 32, tag: NIL,
data: 32
```

SYNOPSIS:

```
(defmethod get-event ((sc slippery-chicken) bar-num event-num player
                      &optional (error t) (warn-no-bar t))
```

39.29 slippery-chicken/get-events-from-to

[slippery-chicken] [Methods]

DATE:

22-Jul-2011 (Pula)

DESCRIPTION:

Return a list of event objects for a given player, specifying the region by bar and event number.

ARGUMENTS:

- A slippery-chicken object.
- A player ID.
- An integer (1-based) that is the first bar from which to return events. If nil, it will default to 1.
- An integer (1-based) that is the first event object in the start-bar to return. If nil it will be set to the last bar of the piece.
- An integer (1-based) that is the last bar from which to return events.

OPTIONAL ARGUMENTS:

- An integer (1-based) that is the last event within the end-bar to return.

RETURN VALUE:

A flat list of event objects.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((sax ((alto-sax tenor-sax) :midi-channel 1))
                     (db (double-bass :midi-channel 2))))
        :instrument-change-map '((1 ((sax ((1 alto-sax) (3 tenor-sax)))))
                                 (2 ((sax ((2 alto-sax) (5 tenor-sax))))))
        :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
```

```
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((sax (1 1 1 1 1))
                            (db (1 1 1 1 1))))
                       (2 ((sax (1 1 1 1 1))
                           (db (1 1 1 1 1))))))))
  (get-events-from-to mini 'sax 3 2 5 3))

=>
(
EVENT: start-time: 10.000, end-time: 11.000,
duration-in-tempo: 1.000,
compound-duration-in-tempo: 1.000,
amplitude: 0.700
bar-num: 3, marks-before: NIL,
tempo-change: NIL
instrument-change: NIL
display-tempo: NIL, start-time-qtrs: 10.000,
midi-time-sig: NIL, midi-program-changes: NIL,
8va: 0
pitch-or-chord:
PITCH: frequency: 164.814, midi-note: 52, midi-channel: 1
pitch-bend: 0.0
degree: 104, data-consistent: T, white-note: E3
nearest-chromatic: E3
src: 0.62996054, src-ref-pitch: C4, score-note: E3
qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
micro-tone: NIL,
sharp: NIL, flat: NIL, natural: T,
octave: 3, c5ths: 0, no-8ve: E, no-8ve-no-acc: E
show-accidental: T, white-degree: 23,
accidental: N,
accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: E3, tag: NIL,
data: E3
**************

written-pitch-or-chord:
PITCH: frequency: 369.994, midi-note: 66, midi-channel: 1
pitch-bend: 0.0
degree: 132, data-consistent: T, white-note: F4
nearest-chromatic: FS4
src: 1.4142135, src-ref-pitch: C4, score-note: FS4
qtr-sharp: NIL, qtr-flat: NIL, qtr-tone: NIL,
micro-tone: NIL,
sharp: T, flat: NIL, natural: NIL,
```

```
octave: 4, c5ths: 1, no-8ve: FS, no-8ve-no-acc: F
show-accidental: T, white-degree: 31,
accidental: S,
accidental-in-parentheses: NIL, marks: NIL
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: FS4, tag: NIL,
data: FS4
**************

RHYTHM: value: 4.000, duration: 1.000, rq: 1, is-rest: NIL,
score-rthm: 4.0, undotted-value: 4, num-flags: 0, num-dots: 0,
is-tied-to: NIL, is-tied-from: NIL, compound-duration: 1.000,
is-grace-note: NIL, needs-new-note: T, beam: NIL, bracket: NIL,
rqq-note: NIL, rqq-info: NIL, marks: NIL, marks-in-part: NIL,
letter-value: 4, tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: Q, tag: NIL,
data: Q
**************


EVENT: start-time: 11.000, end-time: 11.500,
[...]
```

SYNOPSIS:

```
(defmethod get-events-from-to ((sc slippery-chicken) player start-bar
                               start-event end-bar &optional end-event)
```

39.30 slippery-chicken/get-events-sorted-by-time

[slippery-chicken] [Methods]

DESCRIPTION:

```
Get all the events of all players between the given bars and then order
them by ascending time.
```

ARGUMENTS:

```
- A slippery-chicken object.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
```

- :start-bar. An integer that is the first bar in which the function is to be applied to event objects. Default = 1.
- :end-bar. NIL or an integer that is the last bar for which the event objects should be returned. If NIL, the last bar of the slippery-chicken object is used. Default = NIL.

RETURN VALUE:

A list of event objects.

SYNOPSIS:

```
(defmethod get-events-sorted-by-time ((sc slippery-chicken)
                                      &key (start-bar 1) end-bar)
```

39.31 slippery-chicken/get-instrument-for-player-at-bar

[slippery-chicken] [Methods]

DATE:

09-Feb-2011

DESCRIPTION:

Get the current instrument for a specified player at a specified bar number in a slippery-chicken object, as defined in the instrument-change-map.

ARGUMENTS:

- The ID of a player in the slippery-chicken object.
- An integer that is the number of the bar from which to get the current instrument.
- A slippery-chicken object.

RETURN VALUE:

An instrument object.

EXAMPLE:

```
(let* ((mini
        (make-slippery-chicken
         '+mini+
```

```
         :ensemble '(((sax ((alto-sax tenor-sax) :midi-channel 1))
                      (db (double-bass :midi-channel 2))))
         :instrument-change-map '((1 ((sax ((1 alto-sax) (3 tenor-sax)))))
                                  (2 ((sax ((2 alto-sax) (5 tenor-sax))))))
         :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
         :set-map '((1 (1 1 1 1 1))
                    (2 (1 1 1 1 1)))
         :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                 :pitch-seq-palette ((1 2 3 4 5)))))
         :rthm-seq-map '((1 ((sax (1 1 1 1 1))
                             (db (1 1 1 1 1))))
                         (2 ((sax (1 1 1 1 1))
                             (db (1 1 1 1 1)))))))
  (get-instrument-for-player-at-bar 'sax 3 mini))
=>
INSTRUMENT: lowest-written: BF3, highest-written: FS6
lowest-sounding: AF2, highest-sounding: E5
starting-clef: TREBLE, clefs: (TREBLE), clefs-in-c: (BASS TREBLE)
prefers-notes: NIL, midi-program: 67
transposition: BF, transposition-semitones: -14
score-write-in-c: NIL, score-write-bar-line: NIL
chords: NIL, chord-function: NIL,
total-bars: 5 total-notes: 25, total-duration: 20.000
total-degrees: 2710, microtones: T
missing-notes: (FQS3 FQS4), subset-id: NIL
staff-name: tenor sax, staff-short-name: ten sax,

largest-fast-leap: 999, tessitura: FS3
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: TENOR-SAX, tag: NIL,
data: NIL
```

SYNOPSIS:

```
(defmethod get-instrument-for-player-at-bar (player bar (sc slippery-chicken))
```

39.32 slippery-chicken/get-last-bar

[slippery-chicken] [Methods]

DATE:

```
September 28th 2020, Heidhausen
```

DESCRIPTION:

```
Get the last bars for each of the players
```

ARGUMENTS:

```
- the slippery-chicken object
```

RETURN VALUE:

```
a list of rthm-seq-bar objects
```

SYNOPSIS:

```
(defmethod get-last-bar ((sc slippery-chicken))
```

39.33 slippery-chicken/get-last-event

[slippery-chicken] [Methods]

DATE:

```
September 28th 2020, Heidhausen
```

DESCRIPTION:

```
Get the last events (of the last bars) for each of the players
```

ARGUMENTS:

```
- the slippery-chicken object
```

RETURN VALUE:

```
a list of event objects
```

SYNOPSIS:

```
(defmethod get-last-event ((sc slippery-chicken))
```

39.34 slippery-chicken/get-nearest-event

[slippery-chicken] [Methods]

DESCRIPTION:

Find the nearest event by time to <reference-event> in the <search-player>.

ARGUMENTS:

- the slippery-chicken object
- the reference event object that we want to find the nearest event to in another player
- the player we want to find the nearest event in (symbol).

OPTIONAL ARGUMENTS:

- if T only return sounding events (pitches/chords) otherwise return the nearest rest or sounding event.
- if T only return struck events i.e. ignore those that are tied to

RETURN VALUE:

An event object.

SYNOPSIS:

```
(defmethod get-nearest-event ((sc slippery-chicken) reference-event
                              search-player &optional
                              (ignore-rests t)
                              (ignore-tied t))
```

39.35 slippery-chicken/get-note

[slippery-chicken] [Methods]

DESCRIPTION:

Get a numbered event from a specified bar of a given player's part within a slippery-chicken object.

NB: Slippery-chicken doesn't have 'note' and 'rest' classes, rather both of these are events. The nomenclature 'note' and 'rest' are thus used here and elsewhere merely for convenience, to distinguish between sounding and non-sounding events.

See also rthm-seq-bar methods for accessing notes by other means.

ARGUMENTS:

- A slippery-chicken object.
- An integer that is the number of the bar from which to get the note (counting from 1).
- An integer that is the number of the note to get within that bar, counting tied notes (counting from 1). This can also be a list of numbers if accessing pitches in a chord (see below).
- The ID of the player from whose part the note is to be retrieved.

OPTIONAL ARGUMENTS:

- T or NIL to indicate whether, when accessing a pitch in a chord, to return the written or sounding pitch. T = written. Default = NIL.

RETURN VALUE:

An event object.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1)))
        :rthm-seq-palette '((1 ((((2 4) (e) e+e. 32 (32)))
                                :pitch-seq-palette (((1) 2)))))
        :rthm-seq-map '((1 ((vn (1))))))))
  (print (data (get-rest mini 1 2 'vn)))
  (print (data (get-note mini 1 2 'vn)))
  (print (data (get-note mini 1 '(2 1) 'vn)))
  (print (data (get-note mini 1 '(2 2) 'vn)))
  (print (is-tied-from (get-note mini 1 1 'vn))))

=>
32
"E."
C4
A4
T
```

SYNOPSIS:

```
(defmethod get-note ((sc slippery-chicken) bar-num note-num player
                     &optional written silent)
```

39.36 slippery-chicken/get-num-sections

[slippery-chicken] [Methods]

DESCRIPTION:

```
Return the number of sections in the given slippery-chicken object, as
defined in e.g. in the set-map. N.B. If the object has subsections these
are counted also, unless the optional argument is NIL.
```

ARGUMENTS:

```
- A slippery-chicken object.
```

OPTIONAL ARGUMENTS:

```
- count-subsections: if T (default) count all sections and subsections. If
NIL just the top-level sections.
```

RETURN VALUE:

```
An integer that is the number of section.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((sax (alto-sax :midi-channel 1))))
        :set-palette '((1 ((e3 fs3 b3 cs4 fs4 gs4 ds5 f5))))
        :set-map '((1 ((a (1 1 1 1))
                       (b (1 1 1 1))))
                   (2 (1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((a ((sax (1 1 1 1))))
                            (b ((sax (1 1 1 1))))))
                        (2 ((sax (1 1 1))))
                        (3 ((sax (1 1 1 1 1))))))))
  (get-num-sections mini))

=> 4
```

SYNOPSIS:

```
(defmethod get-num-sections ((sc slippery-chicken)
                             &optional (count-subsections t))
```

39.37 slippery-chicken/get-phrases

[slippery-chicken] [Methods]

DESCRIPTION:

```
This returns lists of events that make up phrases. Its notion of phrases
is very simplistic but hopefully useful all the same: it's any sequence of
sounding notes surrounded by rests.
```

ARGUMENTS:

```
- the slippery-chicken object
- a single symbol or list of symbols representing the players from the
ensemble. If nil all players will be processed.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :start-bar. An integer bar number where the process should start. If
NIL we'll default to 1. Default = NIL.
- :end-bar. An integer bar number where the process should end. If
NIL we'll default to the last bar. Default = NIL.
- :pad. If a phrase starts or ends mid-bar, pad the first bar with leading
rests and the last bar with trailing rests to ensure we have full bars.
```

RETURN VALUE:

```
A list of sublists, one for each requested player. Each player sublist
contains sublists also, with all the events in each phrase.
```

SYNOPSIS:

```
(defmethod get-phrases ((sc slippery-chicken) players
                        &key start-bar end-bar pad)
```

39.38 slippery-chicken/get-pitch-symbols-for-player

[slippery-chicken] [Methods]

DATE:

```
April 27th 2022, Heidhausen
```

DESCRIPTION:

```
Get a list of all the pitches (as symbols) used in a piece on a
player-by-player basis.
```

ARGUMENTS:

```
- the slippery-chicken object
- the player (symbol) or list of players we want the pitches for. NB if nil,
pitches for all players will be returned.
```

OPTIONAL ARGUMENTS:

```
- start bar number. If nil, this will be 1. Default = NIL.
- end bar number. If nil, this will be the last bar. Default = NIL.
- whether written as opposed to sounding note symbols should be
  returned. Default = NIL.
```

RETURN VALUE:

```
A nested list: each element is itself a two-element list where the first
element is the player symbol and the second is an ascending ordered list of
the pitches, with enharmonic equivalents removed.
```

EXAMPLE:

```
(get-pitch-symbols-for-player +percussion+ 'perc)
((PERC
  (B4 C5 DF5 D5 DS5 E5 F5 GF5 G5 AF5 A5 BF5 B5 C6 CS6 D6 EF6 E6 F6 FS6 G6 AF6
   A6 BF6)))
```

SYNOPSIS:

```
(defmethod get-pitch-symbols-for-player ((sc slippery-chicken) player
                                         &optional start-bar end-bar
                                           written)
```

39.39 slippery-chicken/get-player

[slippery-chicken] [Methods]

DESCRIPTION:

```
Return the player object for the specified player.
```

ARGUMENTS:

```
 - A slippery-chicken object.
 - A player ID.
```

RETURN VALUE:

```
 A player object.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((fl (flute :midi-channel 1))
                     (tp (b-flat-trumpet :midi-channel 2))
                     (vn (violin :midi-channel 3))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((fl (1 1 1 1 1))
                            (tp (1 1 1 1 1))
                            (vn (1 1 1 1 1)))))))
  (get-player mini 'vn))
=>
PLAYER: (id instrument-palette): SLIPPERY-CHICKEN-STANDARD-INSTRUMENT-PALETTE
doubles: NIL, cmn-staff-args: NIL, total-notes: 25, total-degrees: 3548,
total-duration: 20.000, total-bars: 5, tessitura: B4
LINKED-NAMED-OBJECT: previous: (TP), this: (VN), next: NIL
NAMED-OBJECT: id: VN, tag: NIL,
data:
INSTRUMENT: lowest-written: G3, highest-written: C7
...
```

SYNOPSIS:

```
(defmethod get-player ((sc slippery-chicken) player)
```

39.40 slippery-chicken/get-rest

[slippery-chicken] [Methods]

DESCRIPTION:

```
 Retrieve the event object that contains the specified rest in a
 slippery-chicken object by giving bar number, rest number and player.
```

ARGUMENTS:

- A slippery-chicken object.
- An integer that is the number of the bar from which to retrieve the rest event object.
- An integer that is the number of the rest (not the number of the event) within that bar, counting from 1.
- The ID of the player from whose part to retrieve the rest object.

RETURN VALUE:

An event object.

EXAMPLE:

```
(let ((mini
(make-slippery-chicken
'+mini+
:ensemble '(((vn (violin :midi-channel 1))
(vc (cello :midi-channel 2))))
:tempo-map '((1 (q 60)))
:set-palette '((1 ((c4 d4 e4 f4 g4 a4 b4 c5))))
:set-map '((1 (1 1 1)))
:rthm-seq-palette '((1 ((((2 4) (e) e+e. 32 (32)))
:pitch-seq-palette (((1) 2)))))
:rthm-seq-map '((1 ((vn (1 1 1))
(vc (1 1 1))))))))
(get-rest mini 2 1 'vc))

=>
EVENT: start-time: 2.000, end-time: 2.500,
duration-in-tempo: 0.500,
compound-duration-in-tempo: 0.500,
amplitude: 0.700
bar-num: 2, marks-before: NIL,
tempo-change: NIL
instrument-change: NIL
display-tempo: NIL, start-time-qtrs: 2.000,
midi-time-sig: NIL, midi-program-changes: NIL,
8va: 0
pitch-or-chord: NIL
written-pitch-or-chord: NIL
RHYTHM: value: 8.000, duration: 0.500, rq: 1/2, is-rest: T,
score-rthm: 8.0, undotted-value: 8, num-flags: 1, num-dots: 0,
is-tied-to: NIL, is-tied-from: NIL, compound-duration: 0.500,
```

```
is-grace-note: NIL, needs-new-note: NIL, beam: NIL, bracket: NIL,
rqq-note: NIL, rqq-info: NIL, marks: NIL, marks-in-part: NIL,
letter-value: 8, tuplet-scaler: 1, grace-note-duration: 0.05
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: E, tag: NIL,
data: E
```

SYNOPSIS:

```
(defmethod get-rest ((sc slippery-chicken) bar-num rest-num player)
```

39.41 slippery-chicken/get-section

[slippery-chicken] [Methods]

DESCRIPTION:

```
Return the section object with the specified reference ID.
```

ARGUMENTS:

```
- A slippery-chicken object.
- A reference ID.
```

RETURN VALUE:

```
A section object.
```

EXAMPLE:

```
(let ((mini
        (make-slippery-chicken
         '+mini+
         :ensemble '(((sax (alto-sax :midi-channel 1))
                      (db (double-bass :midi-channel 2))))
         :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
         :set-map '((1 (1 1 1 1 1))
                    (2 (1 1 1 1 1)))
         :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                 :pitch-seq-palette ((1 2 3 4 5)))))
         :rthm-seq-map '((1 ((sax (1 1 1 1 1))
                             (db (1 1 1 1 1))))
                         (2 ((sax (1 1 1 1 1))
                             (db (1 1 1 1 1))))))))
  (get-section mini 2))
```

```
=>
SECTION:
RECURSIVE-ASSOC-LIST: recurse-simple-data: NIL
num-data: 2
linked: T
full-ref: (2)
ASSOC-LIST: warn-not-found NIL
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
BAR-HOLDER:
start-bar: 6
end-bar: 10
num-bars: 5
start-time: 20.0
end-time: 40.0
start-time-qtrs: 0
end-time-qtrs: 40.0
num-notes (attacked notes, not tied): 50
num-score-notes (tied notes counted separately): 50
num-rests: 0
duration-qtrs: 20.0
duration: 20.0 (20.000)
```

SYNOPSIS:

```
(defmethod get-section ((sc slippery-chicken) reference &optional (warn t))
```

39.42 slippery-chicken/get-section-bar-nums

[slippery-chicken] [Methods]

DATE:

```
Mon 13 Jan 2020 15:23:17 GMT
```

DESCRIPTION:

```
Get the start and end bar numbers of each section of a slippery chicken
piece, handlily formatted in an assoc list.
```

ARGUMENTS:

```
A slippery chicken object
```

OPTIONAL ARGUMENTS:

```
Keyword args:
:start - the first section to look at. (Default = 1)
:end - the last section to look at. (Default = nil, look to the end.)
```

RETURN VALUE:

```
An assoc list where each key is a section number and its data is a list
containing the start and end bar numbers of that section.
```

EXAMPLE:

```
(let ((mini
         (make-slippery-chicken
          '+mini+
          :ensemble '(((sax (alto-sax :midi-channel 1))))
          :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
          :set-map '((1 (1 1 1 1 1))
                     (2 (1 1 1 1 1))
                     (3 (1 1 1 1 1)))
          :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                  :pitch-seq-palette ((1 2 3 4 5)))))
          :rthm-seq-map '((1 ((sax (1 1 1 1 1))))
                          (2 ((sax (1 1 1 1 1))))
                          (3 ((sax (1 1 1 1 1))))))))
      (get-section-bar-nums mini))

=>
ASSOC-LIST: warn-not-found T
CIRCULAR-SCLIST: current 0
SCLIST: sclist-length: 3, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL,
                     this: NIL,
                     next: NIL
NAMED-OBJECT: id: SECTION-BAR-NUMS, tag: NIL,
data: (
NAMED-OBJECT: id: 1, tag: NIL,
data: (1 5)
**************


NAMED-OBJECT: id: 2, tag: NIL,
data: (6 10)
**************
```

```
NAMED-OBJECT: id: 3, tag: NIL,
data: (11 15)
**************
)
**************
```

SYNOPSIS:

```
(defmethod get-section-bar-nums ((sc slippery-chicken)
                                 &key (start 1) (end nil)
                                   (assoc-list-id 'section-bar-nums))
```

39.43 slippery-chicken/get-section-refs

[slippery-chicken] [Methods]

DATE:

```
07-May-2012
```

DESCRIPTION:

```
Return the reference IDs for all section and subsections of a given
slippery-chicken object.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the first top-level section for which to return the
  reference IDs. As this number refers to the number of top-level sections
  only, any subsections will be contained in these and only count as 1.
- An integer that is the number of consecutive sections to return section
  reference IDs.
```

RETURN VALUE:

```
A list of lists containing the section reference IDs of the specified
range in the slippery-chicken object.
```

EXAMPLE:

```
(let ((mini
```

```
       (make-slippery-chicken
        '+mini+
        :ensemble '(((sax (alto-sax :midi-channel 1))))
        :set-palette '((1 ((e3 fs3 b3 cs4 fs4 gs4 ds5 f5))))
        :set-map '((1 (1 1 1))
                   (2 (1 1 1))
                   (3 ((a (1 1 1))
                       (b ((x (1 1 1))
                           (y (1 1 1))))))
                   (4 ((a (1 1 1))
                       (b (1 1 1))
                       (c (1 1 1 1))))
                   (5 (1 1 1))
                   (6 (1 1 1))
                   (7 (1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((sax (1 1 1))))
                        (2 ((sax (1 1 1))))
                        (3 ((a ((sax (1 1 1))))
                            (b ((x ((sax (1 1 1))))
                                (y ((sax (1 1 1)))))))))
                        (4 ((a ((sax (1 1 1))))
                            (b ((sax (1 1 1))))
                            (c ((sax (1 1 1 1))))))
                        (5 ((sax (1 1 1))))
                        (6 ((sax (1 1 1))))
                        (7 ((sax (1 1 1)))))))
  (get-section-refs mini 2 4))

=> ((2) (3 A) (3 B X) (3 B Y) (4 A) (4 B) (4 C) (5))
```

SYNOPSIS:

```
(defmethod get-section-refs ((sc slippery-chicken) start-section num-sections)
```

39.44 slippery-chicken/get-sequenz-from-section

[slippery-chicken] [Methods]

DESCRIPTION:

```
Get the sequenz object specified by the section, player and sequenz number
given.
```

ARGUMENTS:

```
- a slippery-chicken object
- the player (symbol)
- the sequenz number, counting from 1
```

RETURN VALUE:

```
the respective sequenz object
```

SYNOPSIS:

```
(defmethod get-sequenz-from-section ((sc slippery-chicken)
                                     section-ref player-ref seq-num) ; 1-based
```

39.45 slippery-chicken/get-set-for-bar-num

[slippery-chicken] [Methods]

DATE:

```
July 11th 2020, Heidhausen
```

DESCRIPTION:

```
Get the complete-set object from the set-palette that was used in the
generation of a specific bar.
```

ARGUMENTS:

```
- the slippery-chicken object
- the bar number (integer)
```

RETURN VALUE:

```
a complete-set object or NIL if the bar doesn't exist or the first event of
the bar has no set-ref slot value, for any of the players, for some reason
(e.g. not initialized by make-slippery-chicken)
```

SYNOPSIS:

```
(defmethod get-set-for-bar-num ((sc slippery-chicken) bar-num)
```

39.46 slippery-chicken/get-starting-ins

[slippery-chicken] [Methods]

DESCRIPTION:

Return the instrument object that is the first instrument object used by a specified player in a given slippery-chicken object.

ARGUMENTS:

- A slippery-chicken object.
- A player ID.

RETURN VALUE:

An instrument object.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((sax ((alto-sax tenor-sax) :midi-channel 1))
                     (db (double-bass :midi-channel 2))))
        :instrument-change-map '((1 ((sax ((1 alto-sax) (3 tenor-sax)))))
                                 (2 ((sax ((2 alto-sax) (5 tenor-sax))))))
        :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((sax (1 1 1 1 1))
                            (db (1 1 1 1 1))))
                        (2 ((sax (1 1 1 1 1))
                            (db (1 1 1 1 1))))))))
  (get-starting-ins mini 'sax))

=>
INSTRUMENT: lowest-written: BF3, highest-written: FS6
lowest-sounding: CS3, highest-sounding: A5
starting-clef: TREBLE, clefs: (TREBLE), clefs-in-c: (TREBLE)
prefers-notes: NIL, midi-program: 66
transposition: EF, transposition-semitones: -9
score-write-in-c: NIL, score-write-bar-line: NIL
chords: NIL, chord-function: NIL,
total-bars: 5 total-notes: 25, total-duration: 20.000
total-degrees: 2920, microtones: T
missing-notes: (BQF3 BQF4), subset-id: NIL
staff-name: alto saxophone, staff-short-name: alt sax,
```

```
largest-fast-leap: 999, tessitura: BQF3
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: ALTO-SAX, tag: NIL,
data: NIL
```

SYNOPSIS:

```
(defmethod get-starting-ins ((sc slippery-chicken) player) ; symbol
```

39.47 slippery-chicken/get-tempo

[slippery-chicken] [Methods]

DESCRIPTION:

```
Return the tempo object in effect for a specified bar of a given
slippery-chicken object.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the number of a bar within that slippery-chicken
  object.
```

RETURN VALUE:

```
A tempo object.
```

EXAMPLE:

```
(let ((mini
(make-slippery-chicken
'+mini+
:ensemble '(((sax (alto-sax :midi-channel 1))))
:tempo-map '((1 (q 60)) (5 (e 72)) (7 (q. 176 "prestissimo")))
:set-palette '((1 ((e3 fs3 b3 cs4 fs4 gs4 ds5 f5))))
:set-map '((1 (1 1 1 1 1 1 1 1)))
:rthm-seq-palette '((1 ((((4 4) h q e s s))
:pitch-seq-palette ((1 2 3 4 5)))))
:rthm-seq-map '((1 ((sax (1 1 1 1 1 1 1 1))))))))
(get-tempo mini 6))

=>
TEMPO: bpm: 72, beat: E, beat-value: 8.0, qtr-dur: 1.6666666
qtr-bpm: 36.0, usecs: 1666666, description: NIL
```

```
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: NIL, tag: NIL,
data: 72
```

SYNOPSIS:

```
(defmethod get-tempo ((sc slippery-chicken) bar-num)
```

39.48 slippery-chicken/get-time-sig

[slippery-chicken] [Methods]

DESCRIPTION:

```
Get the time-sig object associate with a specified bar number in a given
slippery-chicken object.
```

ARGUMENTS:

```
- A slippery-chicken object.
- An integer that is the number of the bar for which to the time-sig object
  is to be returned. NB: Although this argument is listed as optional in
  the method definition (due to inheritance), it is actually required.
```

OPTIONAL ARGUMENTS:

```
- (see above).
```

RETURN VALUE:

```
A time-sig object.
```

EXAMPLE:

```
(let ((mini
(make-slippery-chicken
'+mini+
:ensemble '(((sax (alto-sax :midi-channel 1))))
:set-palette '((1 ((e3 fs3 b3 cs4 fs4 gs4 ds5 f5))))
:set-map '((1 (1 1 1)))
:rthm-seq-palette '((1 ((((4 4) h q e s s)
((5 8) q e s s e)
((3 16) s e))
:pitch-seq-palette ((1 2 3 4 5 1 2 3 4 5 1
```

```
2)))))
:rthm-seq-map '((1 ((sax (1 1 1)))))))
(get-time-sig mini 2))

=>
TIME-SIG: num: 5, denom: 8, duration: 2.5, compound: NIL, midi-clocks: 24, num-beats: 5
SCLIST: sclist-length: 2, bounds-alert: T, copy: T
LINKED-NAMED-OBJECT: previous: NIL, this: NIL, next: NIL
NAMED-OBJECT: id: "0508", tag: NIL,
data: (5 8)
```

SYNOPSIS:

```
(defmethod get-time-sig ((sc slippery-chicken) &optional bar-num)
```

39.49 slippery-chicken/get-transposition-at-bar

[slippery-chicken] [Methods]

DATE:

```
24-Mar-2011
```

DESCRIPTION:

```
Return the number of semitones difference between the sounding pitches and
written pitches of a given player's part in a specified bar within a
slippery-chicken object; e.g. bass clarinet = -14.
```

ARGUMENTS:

```
- The ID of the player for whom the transposition value is sought.
- An integer which is the number of the bar for which the transposition
  value is sought.
- A slippery-chicken object.
```

RETURN VALUE:

```
An integer.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
```

```
        '+mini+
        :ensemble '(((sax (alto-sax :midi-channel 1))))
        :set-palette '((1 ((e3 fs3 b3 cs4 fs4 gs4 ds5 f5))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                               :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((sax (1 1 1)))))))
  (get-transposition-at-bar 'sax 2 mini))

=> -9
```

SYNOPSIS:

```
(defmethod get-transposition-at-bar (player bar (sc slippery-chicken))
```

39.50 slippery-chicken/lp-display

[slippery-chicken] [Functions]

DESCRIPTION:

```
This function has exactly the same arguments as write-lp-data-for all and
only differs from that method in that after writing all the Lilypond text
files for the score, it calls Lilypond to render the PDF, which is then
opened automatically from within Lisp (if the value of
(get-sc-config 'lp-display-auto-open) is T).

In order to work properly, you'll need to make sure the value of the
'lilypond-command configuration parameter is set via (set-sc-config ...) to
the full path of your Lilypond command (not the app: it's usually
/path/to/Lilypond.app/Contents/Resources/bin/lilypond on OSX).

NB Only available with CCL and SBCL on OSX. With SBCL on OSX the output
from Lilypond is only printed once the process has exited, so it may take a
while until you see anything.
```

ARGUMENTS:

```
See write-lp-data-for-all
```

RETURN VALUE:

```
An integer: the shell's exit code for the PDF open command, usually 0 for
success. Second returned value is the patch to the PDF file.
```

SYNOPSIS:

```
(defun lp-display (&rest args)
```

39.51 slippery-chicken/make-slippery-chicken

[slippery-chicken] [Functions]

DESCRIPTION:

```
Make a slippery-chicken object using the specified data. This is the
function that will be used most often to "put it all together", and many of
its slots require full objects of other classes rather than just straight
data. These objects, such as rthm-seq-palette, rthm-seq-map etc, are also
documented in detail elsewhere in the robodoc and the user's manual.

NB Usually, when no set pitches are available for an instrument, an error
will be signalled. As of October 2018, you can instead allocate an 'empty
sequence' (rests) in such cases by setting the following before calling
make-slippery-chicken:
(set-sc-config 'pitch-seq-no-pitches-error nil)
```

ARGUMENTS:

```
- A symbol that is the name/ID of the object. The value passed to this
  argument will be made into a global variable, so that the newly created
  slippery-chicken object and the data it contains remain in memory and can
  be accessed and modified after the object is generated.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
NB: Although these arguments are technically optional, the slippery-chicken
    object will only be complete and make musical sense if many of the core
    elements are present.
- :title. A string that will be used as the title of the piece. The value
  given for this object will be used as both the header for the printable
  output as well as the base for any file names generated by the
  write-lp-data-for-all method. Default = "slippery-chicken-piece".
- :subtitle. A string that will be used as the subtitle of the piece. Works
  in Lilypond scores only.
- :instrument-palette. An instrument-palette object. This will be the
  palette of instrument objects available to the players of in the given
  slippery-chicken object's ensemble slot.
  Default = +slippery-chicken-standard-instrument-palette+.
```

- :ensemble. A recursive association list that will be used as the data to create an ensemble object populated with player objects within the slippery-chicken object. The format of this list will be a list of user-defined player IDs each coupled with a list of instrument object IDs from the current instrument-palette and various player object parameters. See the user's manual and robodoc entries on the ensemble and player classes for more detail.
- :set-palette. A recursive association list that will be used as the data to create a set-palette object within the slippery-chicken object. This object is where the collections of possible pitches for any given sequence are defined. The format of this list will be a list of IDs for each set of pitches, each coupled with a list of note-name symbols for the pitches that will be used to make that set. See the user's manual and the robodoc entry on the set-palette class for more detail.
- :set-map. A recursive association list that will be used as the data to create a set-map object within the slippery-chicken object. This is where the order in which the pitch collections defined in the set-palette will be used in the piece. The format of this list will be a list of IDs from the given slippery-chicken object's structure coupled with a list of IDs from those given to the sets in the set-palette. There must be an equal number of sections in this list as there are in the rthm-seq-map, and they must have identical names. There must be an equal number of individual set IDs in each list paired with the section IDs as there are in the corresponding lists of the rthm-seq-map. See the user's manual and the robodoc entries for the set-map and sc-map for more detail.
- :rthm-seq-palette. A recursive association list that will be used as the data to create a rthm-seq-palette object within the slippery-chicken object. This object is where the collections of possible rhythm sequences for any given sequence in the piece are defined. This list will take the format of a list of IDs paired with a list of data for individual rthm-seq objects. These in turn will consist of one or more lists of rhythm data for rthm-seq-bar objects, as well pitch-seq-palettes and marks data for the individual rthm-seq objects to be created. See the user's manual as well as the robodoc entries for rthm-seq-palette, rthm-seq, rthm-seq-bar, rhythm, and pitch-seq-palette for more detail.
- :rthm-seq-map. A recursive association list that will be used as the data to create a rthm-seq-map object within the slippery-chicken object. This is where the order in which the rhythm sequences defined in the rthm-seq-palette will be used in the piece. It will take the format of a list of section IDs, of which there must be an equal number as are given in the set-map, each coupled with a list of player IDs, as defined in the ensemble slot of the given slippery-chicken object. The player IDs in turn are coupled with a list of IDs for rthm-seq objects, as defined in the rthm-seq-palette. Each of these lists must contain the same number of elements as are contained in each of the set-map sections. See the user's manual and robodoc entries for rthm-seq-map and sc-map for more details.

- :snd-output-dir. A string that will be used as the directory path for any output generated by clm-play in conjunction with sound files listed in the sndfile-palette (see below). Default = (get-sc-config 'default-dir).
- :sndfile-palette. A recursive association list that will be used as the data to create a sndfile-palette object within the slippery-chicken object. This is where the list is defined that contains all possible source sound files which may be used in conjunction with output generated by clm-play. This list will take the format of a list of IDs for sound-file groups, coupled with lists of file names and various other parameters associated with the sndfile class. The list of sound-file groups is followed by a list of directory paths where the given sound files are located and an optional list of file extensions. See the user's manual and the robodoc entries on sndfile-palette, sndfile, and clm-play for more detail.
- :tempo-map. A recursive association list that will be used as the data to create tempo objects within the slippery-chicken object. This is one of two options for specifying the sequence of tempo changes for a given piece (also see tempo-curve below). The format will be a list of integers that are measure numbers within the piece, each coupled with tempo indications in the form (beat-unit bpm). See the user's manual as well as the robodoc entry for tempo-map for more detail. NB: This slot cannot be used together with :tempo-curve.
- :tempo-curve. A list of data that will be used to create tempo objects within the slippery-chicken object, based on an interpolated list of break-point pairs. This is one of two options for specifying the sequence of tempo changes for a given piece (also see tempo-map above.) The first item in the list will be the number of bars between each new tempo object. The second item is the beat basis for the tempo objects made. The third and final argument is the list of break-point pairs, of which the first is a value on an arbitrary x-axis and the second is a number of beats-per-minute. See the user's manual and the robodoc entry for tempo-curve for more detail. NB: This slot cannot be used together with :tempo-map.
- :staff-groupings. A list of integers that indicate the placement of group brackets for the printable output. Each number represents a consecutive number of players, in the order they appear in the ensemble object, that will be included in each consecutive group. The sum of the numbers in this list must be equal to the number of players in the ensemble. See the user's manual for more detail.
- :instrument-change-map. A recursive association list that will be used as the data to create an instrument-change-map object within the slippery-chicken object. This will be used to indicate where those players in the ensemble that play multiple instruments will change instruments. The format will be a list of section IDs coupled with a list of player IDs, each of which in turn is coupled with a list of 2-item lists consisting of a sequence number paired with the ID (name) of one of

the instrument objects assigned to that player in the ensemble object. See the user's manual and the robodoc entries for instrument-change-map for more detail.
- :set-limits-high. A recursive association list that will be used to limit the uppermost pitches of either the parts of individual players or of the entire ensemble. The format will be a list of player IDs, as defined in the ensemble object, each paired with a list of break-point pairs that consist of a value on an arbitrary x-axis paired with a note-name pitch symbol. These break-point envelopes are applied to the entire duration of the piece. See the user's manual for more detail.
- :set-limits-low. A recursive association list that will be used to limit the lowermost pitches of either the parts of individual players or of the entire ensemble. The format will be a list of player IDs, as defined in the ensemble object, each paired with a list of break-point pairs that consist of a value on an arbitrary x-axis paired with a note-name pitch symbol. These break-point envelopes are applied to the entire duration of the piece. See the user's manual for more detail.
- :fast-leap-threshold. A number that is the longest duration of a note in seconds that can be followed by a leap of a large interval, as defined in the largest-fast-leaps slot of the instrument objects. Default = 0.125.
- :instruments-hierarchy. A list of player IDs from the given slippery-chicken object's ensemble that will specify the order in which slippery chicken's pitch selection algorithm will choose pitches for the instruments. By default (when NIL) this order follows the order in which the instrument objects appear in the ensemble object. See the user's manual for more detail. Default = NIL.
- :rehearsal-letters. A list of numbers that are measure numbers at which consecutive rehearsal letters will be placed. Since rehearsal letters are technically actually place on the right-hand bar line of the previous measure, measure 1 cannot be entered here. Slippery chicken automatically proceeds consecutively through the alphabet, so only numbers are required here. See the user's manual for more detail. If NIL, no rehearsal letters will be added to the score. Default = NIL.
- :avoid-melodic-octaves. T or NIL to indicate whether two linearly consecutive pitches in the part of a given player may be of the same pitch class but a different octave. T = avoid melodic octaves. Default = T.
- :instruments-write-bar-nums. A list of player IDs above whose parts in the score bar numbers should be written. If NIL, bar numbers will be written above the top player in each group. NB: This slot affects CMN output only. Default = NIL.
- :pitch-seq-index-scaler-min. A decimal number that affects the likelihood that slippery-chicken's pitch selection algorithm will choose pitches for an instrument that have also already been assigned to other players. In general terms, the higher this number is, the more likely it will be that instruments may be assigned the same pitches, though this will of course

also be dependent on other factors, such as the characteristics of those instruments and the pitches in the current set. See the user's manual on pitches and the robodoc entries for pitch-seq for more detail. Default = 0.5.
- :bars-per-system-map. A list of 2-item lists, each of which consists of a measure number coupled with a number of measures to be placed in each system starting at that measure number. NB: This list only affects CMN output. See the user's manual on score layout for more details.
- :composer. A string that will be used for the composer portion of the header on the score's first page in LilyPond output. If NIL, no composer's name will appear in the score. Default = NIL.
- :year. A string that will be concatenated with the composer portion of the header on the score's first page in LilyPond output. If NIL, no year will appear in the score. Default = NIL.
- :rthm-seq-map-replacements. A list of lists in the format '(((1 2 va) 3 2) ((2 3 vn) 4 3)) that indicate changes to individual elements of lists within the given rthm-seq-map object. Each such list indicates a change, the first element of the list being the reference into the rthm-seq-map (the vla player of section 1, subsection 2 in the first example here), the second element is the nth of the data list for this key to change, and the third is the new data. If NIL, no changes will be made. See the robodoc entries for rthm-seq-map for more detail. Default = NIL.
- :set-map-replacements. A list of lists in the format '((1 2 2) (3 3 1)) that indicate changes to individual elements of lists within the given set-map object. Each such list indicates a change, the first element of the list being the reference into the set-map (the section, followed by a subsection if any exist), the second element being the nth of the data list for to change, and the third being the new data. If NIL, no changes will be made. See the robodoc entries for sc-map for more detail. Default = NIL.
- :key-sig. A two-element list indicating starting key signature for the piece, e.g. '(ef minor). Usual note name symbols apply (e.g. ds = d sharp, bf - b flat). Implies nothing beyond the signature, i.e. no conformity to tonality expected. Default '(c major) i.e. no key signature.
- :transposition-curve. An envelope describing over the duration of the piece (but using any arbitrary x-axis range) what transpositions in semitones should be applied to the sets. Default = (0 0 100 0)

(- :warn-ties. This slot is now obsolete, but is left here for backwards compatibility with pieces composed with earlier versions of slippery-chicken. Default = T.)

RETURN VALUE:

T

EXAMPLE:

```
;;; An example using all slots
(let ((mini
       (make-slippery-chicken
        '+mini+
        :title "A Little Piece"
        :composer "Joe Green"
        :ensemble '(((fl ((flute piccolo) :midi-channel 1))
                     (cl (b-flat-clarinet :midi-channel 2))
                     (hn (french-horn :midi-channel 3))
                     (tp (b-flat-trumpet :midi-channel 4))
                     (vn (violin :midi-channel 5))
                     (va (viola :midi-channel 6))
                     (vc (cello :midi-channel 7))))
        :set-palette '((1 ((fs2 b2 d4 a4 d5 e5 a5 d6)))
                       (2 ((b2 fs2 d4 e4 a4 d5 e5 a5 d6)))
                       (3 ((cs3 fs3 e4 a4 e5 a5 e6))))
        :set-map '((1 (2 1 2 3 1 3 1))
                   (2 (1 1 3 2 2 3 1))
                   (3 (2 3 1 3 1 1 2)))
        :rthm-seq-palette '((1 ((((4 4) h (q) e (s) s))
                                :pitch-seq-palette ((1 2 3))))
                           (2 ((((4 4) (q) e (s) s h))
                                :pitch-seq-palette ((2 1 3))))
                           (3 ((((4 4) e (s) s h (q)))
                                :pitch-seq-palette ((3 2 1)))))
        :rthm-seq-map '((1 ((fl (2 3 3 1 1 1 2))
                            (cl (3 2 1 1 2 1 3))
                            (hn (1 2 3 1 1 3 2))
                            (tp (2 1 1 3 3 2 1))
                            (vn (3 1 3 2 1 1 2))
                            (va (2 1 1 1 3 2 3))
                            (vc (1 2 3 1 3 2 1))))
                        (2 ((fl (3 1 3 2 2 1 1))
                            (cl (1 1 2 3 1 3 2))
                            (hn (1 3 2 1 3 1 2))
                            (tp (1 1 1 3 3 2 2))
                            (vn (2 1 3 1 3 1 2))
                            (va (2 2 3 1 1 3 1))
                            (vc (1 3 1 2 2 1 3))))
                        (3 ((fl (1 1 3 2 1 3 2))
                            (cl (2 1 2 3 3 1 1))
                            (hn (3 2 1 1 1 3 2))
                            (tp (3 3 1 1 2 1 2))
                            (vn (3 1 3 2 1 1 2))
```

```
                        (va (3 2 1 1 3 2 1))
                        (vc (1 3 2 1 2 3 1)))))
      :snd-output-dir "/tmp"
      :sndfile-palette '(((sndfile-grp-1
                           ((test-sndfile-1.aiff :start 0.021 :end 0.283)
                            (test-sndfile-2.aiff)
                            (test-sndfile-3.aiff)))
                          (sndfile-grp-2
                           ((test-sndfile-4.aiff :frequency 834)
                            (test-sndfile-5.aiff)
                            (test-sndfile-6.aiff))))
                         ("/path/to/test-sndfiles-dir-1"
                          "/path/to/test-sndfiles-dir-2"))
      ;; :tempo-map '((1 (q 84)) (9 (q 72))) ; ; ;
      :tempo-curve '(5 q (0 40 25 60 50 80 75 100 100 120))
      :staff-groupings '(2 2 3)
      :instrument-change-map '((1 ((fl ((1 flute) (3 piccolo) (5 flute))))))
      :set-limits-low '((fl (0 c5 50 g5 100 c5))
                        (cl (0 c4 50 f4 100 c4))
                        (hn (0 f3 50 c4 100 f3))
                        (tp (0 c4 50 f4 100 c4))
                        (vn (0 e5 50 a5 100 e5))
                        (va (0 c3 50 f3 100 c3))
                        (vc (0 c2 50 f3 100 c2)))
      :set-limits-high '((fl (0 d6 50 a6 100 d6))
                         (cl (0 c5 50 a5 100 c5))
                         (hn (0 f4 50 c5 100 f4))
                         (tp (0 f5 50 c5 100 f5))
                         (vn (0 c6 50 e6 100 c6))
                         (va (0 g4 50 d5 100 g4))
                         (vc (0 c4 50 f4 100 c4)))
      :fast-leap-threshold 0.5
      :instruments-hierarchy '(fl vn cl tp va hn vc)
      :rehearsal-letters '(3 11 19)
      :avoid-melodic-octaves nil
      :instruments-write-bar-nums '(fl cl hn tp)
      :pitch-seq-index-scaler-min 0.1
      :bars-per-system-map '((1 1) (2 2) (3 3) (7 4) (11 5))
      :rthm-seq-map-replacements '(((1 va) 3 1) ((2 fl) 4 3))
      :set-map-replacements '((1 2 2) (3 3 1)))))
(midi-play mini :midi-file "/tmp/mini.mid")
(cmn-display mini)
(write-lp-data-for-all mini))
```

SYNOPSIS:

```
(defun make-slippery-chicken (name &key
                                    rthm-seq-palette
                                    rthm-seq-map
                                    set-palette
                                    set-map
                                    sndfile-palette
                                    tempo-map
                                    tempo-curve
                                    (snd-output-dir (get-sc-config
                                                     'default-dir))
                                    instrument-change-map
                                    instruments-write-bar-nums
                                    bars-per-system-map
                                    staff-groupings
                                    rthm-seq-map-replacements
                                    set-map-replacements
                                    set-limits-low
                                    set-limits-high
                                    instrument-palette
                                    ensemble
                                    rehearsal-letters
                                    (fast-leap-threshold 0.125)
                                    instruments-hierarchy
                                    (title "slippery chicken piece")
                                    subtitle
                                    composer year
                                    (avoid-melodic-octaves t)
                                    (avoid-used-notes t)
                                    (pitch-seq-index-scaler-min 0.5)
                                    defer
                                    ;; MDE Mon Jul  2 16:08:42 2012
                                    (key-sig '(c major))
                                    ;; MDE Thu Sep  6 14:34:14 2018
                                    (transposition-curve '(0 0 100 0))
                                    (warn-ties t))
```

39.52 slippery-chicken/midi-play

[slippery-chicken] [Methods]

DESCRIPTION:

```
Generate a MIDI file from the data of the specified slippery-chicken
object. Note that for events that contain chords, the event's amplitude slot
will be used for all pitches unless each individual pitch object in the
```

chord has its amplitude slot set to a number (in which case the event's amplitude slot will be ignored.)

ARGUMENTS:

- A slippery-chicken object.

OPTIONAL ARGUMENTS:

keyword arguments:
- :midi-file. The name of the MIDI file to produce, including directory path and extension. Default is a filename extracted from the title of the sc piece, placed in the (get-sc-config 'default-dir) directory (default /tmp).
- :voices. NIL or a list of player IDs indicating which of the players' parts are to be included in the resulting MIDI file. If NIL, all players' parts will be included. Default = NIL.
- :players. A synonym for voices. This is the preferred keyword as it aligns better with other classes, but :voices is retained for older code compatibility.
- :start-section. An integer that is the number of the first section for which the MIDI file is to be generated. Default = 1.
- :num-sections. An integer that is the number of sections to produce MIDI data for in the MIDI file. If NIL, all sections will be written. Default = NIL.
- :from-sequence. An integer that is the number of the sequence within the specified section from which to start generating MIDI data. NB: This argument can only be used when the num-sections = 1. Default = 1.
- :num-sequences. An integer that is the number of sequences for which MIDI data is to be generated in the resulting MIDI file, including the sequence specified in from-sequence. If NIL, all sequences will be written. NB: This argument can only be used when the num-sections = 1. Default = NIL.
- :update-amplitudes. T or NIL to indicate whether events should have their amplitude slots updated to ensure that any dynamic marks that have been added (and therefore amplitude slots updated as a side-effect) reflect the subsequent events' amplitudes also. This will override any amplitude slots that have been changed other than by (setf (amplitude ...)) or (add-mark ... 'some-dynamic). Also, if T, then hairpins (cresc, dim) will result in increasing/decreasing amplitudes over their extent (but this will only work if the amplitude/dynamic of the hairpins' starting and ending events are correct, and in that case intervening events' amplitudes will be overwritten no matter how they've been set; also this won't handle cases where there's a hairpin end and a new hairpin beginning on the same event). Default = T
- :force-velocity. Either: an integer between 0 and 127 (inclusive) that is

```
   the MIDI velocity value which will be given to all notes in the resulting
   MIDI file, or a function which takes an event object argument and
   returns a velocity from it (e.g. randomising the existing amplitude
   slightly). Default = NIL.
 - :auto-open. Whether to open the MIDI file once written. Currently only
    available on OSX with SBCL or CCL. Uses the default app for MIDI files,
    as if opened with 'open' in the terminal. Default = Value of
    (get-sc-config 'midi-play-auto-open).
 - :suffix. Add some text to the filename just before .mid?. Default = ""
```

RETURN VALUE:

```
 Returns the path of the file written, as a string.
```

EXAMPLE:

```
;;; An example with some typical values for the keyword arguments. ;
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (hn (french-horn :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1 1 1))
                   (2 (1 1 1 1 1 1 1))
                   (3 (1 1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h (q) e (s) s))
                                :pitch-seq-palette ((1 2 3))))
                            (2 ((((4 4) (q) e (s) s h))
                                :pitch-seq-palette ((1 2 3))))
                            (3 ((((4 4) e (s) s h (q)))
                                :pitch-seq-palette ((2 3 3))))
                            (4 ((((4 4) (s) s h (q) e))
                                :pitch-seq-palette ((3 1 2)))))
        :rthm-seq-map '((1 ((cl (1 2 1 2 1 2 1))
                            (hn (1 2 1 2 1 2 1))
                            (vc (1 2 1 2 1 2 1))))
                       (2 ((cl (3 4 3 4 3 4 3))
                           (hn (3 4 3 4 3 4 3))
                           (vc (3 4 3 4 3 4 3))))
                       (3 ((cl (1 2 1 2 1 2 1))
                           (hn (1 2 1 2 1 2 1))
                           (vc (1 2 1 2 1 2 1))))))))
  (midi-play mini
             :midi-file "/tmp/md-test.mid"
```

```
             :voices '(cl vc)
             :start-section 2))

;;; An example that passes a (lambda) function to :force-velocity. Usually, by
;;; default, event amplitudes between 0.0 and 1.0 will map onto MIDI velocities
;;; of 0 to 127. Here we map them to velocities of 0 to 100 instead.
(midi-play +jitterbug+ :force-velocity
           #'(lambda (event)
               (floor (* (amplitude event) 100))))
```

SYNOPSIS:

```
#+cm-2
(defmethod midi-play ((sc slippery-chicken)
                      &key
                        ;; no subsection refs: use from-sequence instead
                        (start-section 1)
                        ;; these voices are used to get the actual sequence
                        ;; orders i.e. each voice will be appended to <section>
                        ;; when calling get-data.
                        ;; if nil then all voices.
                        (voices nil)
                        ;; MDE Thu Oct 18 17:07:57 2018 -- players is a more
                        ;; appropriate keyword than voices but keep the latter
                        ;; for historical code and make the two equivalent
                        (players nil)
                        ;; add something to the filename just before .mid?
                        (suffix "")
                        (midi-file
                         (format nil "~a~a~a.mid"
                                 (get-sc-config 'default-dir)
                                 (filename-from-title (title sc))
                                 suffix))
                        (from-sequence 1)
                        (num-sequences nil)
                        ;; if nil we'll write all the sections
                        (num-sections nil)
                        ;; MDE Mon Jun 13 12:30:55 2016
                        (update-amplitudes t)
                        ;; MDE Tue Jun  4 19:06:11 2013
                        (auto-open (get-sc-config 'midi-play-auto-open))
                        ;; if this is a 7-bit number we'll use this for all
                        ;; notes
                        (force-velocity nil))
```

39.53 slippery-chicken/next-event

[slippery-chicken] [Methods]

DESCRIPTION:

```
Get the events from a specified player's part within a given
slippery-chicken object one after the other (e.g. in a loop). This method
must be called once with a bar number (or any other non-NIL value,
whereupon we start at bar 1) first in order to reset the counter; doing
this will return NIL. Once the counter has been reset, calling the method
without a bar number will return the events in sequence.
```

ARGUMENTS:

```
- A slippery-chicken object.
- A player ID.
```

OPTIONAL ARGUMENTS:

```
- T or NIL to indicate whether to return only events that consist of
  attacked notes (i.e., no ties or rests). T = return only events with
  attacked notes. Default = NIL.
- T, NIL, or an integer to indicate the first bar from which events are to
  be retrieved. If T, the counter is reset to the first event of the
  player's part. This should be NIL after the first resetting call.
  Default = NIL
- NIL or an integer to indicate the last bar from which events are to be
  retrieved. If NIL, all events will be retrieved from the starting point
  to the last event in the given slippery-chicken object. Default = NIL.
  NB This arg should be used in the second call to next-event.
  E.g.: (next-event sc player nil start-bar)
        (loop for ne = (next-event sc player nil nil end-bar)
              while ne
              etc...
```

RETURN VALUE:

```
An event object
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
```

```
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (hn (french-horn :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h (q) e (s) s))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((cl (1 1 1))
                            (hn (1 1 1))
                            (vc (1 1 1))))))))
  (next-event mini 'vc nil t)
  (loop for ne = (next-event mini 'vc)
     while ne
  collect (get-pitch-symbol ne)))

=> (E4 NIL F4 NIL G4 E4 NIL F4 NIL G4 E4 NIL F4 NIL G4)
```

SYNOPSIS:

```
  (defmethod next-event ((sc slippery-chicken) player
                         &optional
                         (attacked-notes-only nil)
                         ;; could be a number too, whereupon it's the bar
                         ;; number to start at
                         (start-over nil)
                         (end-bar nil)) ; inclusive
```

39.54 slippery-chicken/note-count

[slippery-chicken] [Methods]

DATE:

```
May 20th 2022, Heidhausen
```

DESCRIPTION:

```
for statistical purposes, generate a list of the count of all the notes in
the whole or part of a piece.
```

ARGUMENTS:

```
- the slippery-chicken object
- the player: either a single symbol, a list of players, or if nil, all the
players
```

OPTIONAL ARGUMENTS:

```
- how to sort the result. Default = 'pitch meaning that the pitches will
ascend irrespective of their counts. Anything else (e.g. nil) will print in
the order of lowest to highest note count
- the start bar number. Default = NIL = 1
- the end bar number. Default = NIL = last bar
- T or NIL to indicate sounding (nil) or written (t) pitches should be
examined.  Default = NIL = sounding.
```

RETURN VALUE:

```
a list of (note count) pairs
```

SYNOPSIS:

```
(defmethod note-count ((sc slippery-chicken) player
                       &optional (sort 'pitch) start-bar end-bar written)
```

39.55 slippery-chicken/num-bars

[slippery-chicken] [Methods]

DESCRIPTION:

```
Return the number of bars in the piece.
```

ARGUMENTS:

```
- A slippery-chicken object.
```

RETURN VALUE:

```
An integer that is the number of bars.
```

EXAMPLE:

```
(let ((mini
(make-slippery-chicken
'+mini+
:ensemble '(((fl (flute :midi-channel 1))
(tp (b-flat-trumpet :midi-channel 2))
(vn (violin :midi-channel 3))))
:set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
```

```
:set-map '((1 (1 1 1 1 1)))
:rthm-seq-palette '((1 ((((4 4) h q e s s))
:pitch-seq-palette ((1 2 3 4 5)))))
:rthm-seq-map '((1 ((fl (1 1 1 1 1))
(tp (1 1 1 1 1))
(vn (1 1 1 1 1))))))))
(num-bars mini))

=> 5
```

SYNOPSIS:

```
(defmethod num-bars ((sc slippery-chicken))
```

39.56 slippery-chicken/num-notes

[slippery-chicken] [Methods]

DESCRIPTION:

```
Returns the number of attacked notes in a given slippery-chicken object;
i.e., not including ties or rests.
```

ARGUMENTS:

```
- A slippery-chicken object.
```

RETURN VALUE:

```
An integer,
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((fl (flute :midi-channel 1))))
        :set-palette '((1 ((c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1)))
        :rthm-seq-palette '((1 ((((4 4) - e e e e - - e e e e -)))))
        :rthm-seq-map '((1 ((fl (1))))))))
  (num-notes mini))

=> 8
```

SYNOPSIS:

```
(defmethod num-notes ((sc slippery-chicken))
```

39.57 slippery-chicken/num-players

[slippery-chicken] [Methods]

DESCRIPTION:

Get the number of players in a slippery chicken object's ensemble object.

ARGUMENTS:

- A slippery chicken object

RETURN VALUE:

- An integer.

SYNOPSIS:

```
(defmethod num-players ((sc slippery-chicken))
```

39.58 slippery-chicken/num-seqs

[slippery-chicken] [Methods]

DESCRIPTION:

Return the number of sequences (which may contain multiple bars) in a specified section of a slippery-chicken object.

NB This will return the total number of seqs if there are sub-sections.

ARGUMENTS:

- A slippery-chicken object.
- The ID of the section for which to return the number of sequences.

RETURN VALUE:

An integer.

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((sax (alto-sax :midi-channel 1))))
        :set-palette '((1 ((e3 fs3 b3 cs4 fs4 gs4 ds5 f5))))
        :set-map '((1 (1 1 1 1))
                   (2 (1 1 1))
                   (3 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((sax (1 1 1 1))))
                        (2 ((sax (1 1 1))))
                        (3 ((sax (1 1 1 1 1))))))))
  (num-seqs mini 2))

=> 3
```

SYNOPSIS:

```
(defmethod num-seqs ((sc slippery-chicken) section-ref)
```

39.59 slippery-chicken/player-ambitus

[slippery-chicken] [Methods]

DATE:

```
February 9th 2017
```

DESCRIPTION:

```
Find the highest and lowest events in the (given bars of the) piece,
printing if requested.
```

ARGUMENTS:

```
- the slippery-chicken object
- the player ID (generally symbol)
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
```

```
- :written. T or NIL to indicate if the written (as opposed to sounding)
  pitches of the events should be handled/returned/printed. Default = NIL =
  sounding pitches
- :print. Whether to print results to the Lisp interpreter.
- :start-bar. The bar number to start processing. Default = 1.
- :start-bar. The bar number to stop processing (inclusive). Default = NIL =
  the last bar.
- :attacked-only. Whether only to include attacked notes. Default = T.
```

RETURN VALUE:

```
two values: the lowest event and the highest event.
```

SYNOPSIS:

```
(defmethod player-ambitus ((sc slippery-chicken) player &key written print
                                                          (start-bar 1) end-bar
                                                          (attacked-only t))
```

39.60 slippery-chicken/player-doubles

[slippery-chicken] [Methods]

DATE:

```
02-Apr-2012
```

DESCRIPTION:

```
Boolean test to check whether a specified player plays more than one
instrument.
```

ARGUMENTS:

```
- A slippery-chicken object.
- A player ID.
```

RETURN VALUE:

```
T if the player has more than one instrument, otherwise NIL>
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((sax ((alto-sax tenor-sax) :midi-channel 1))
                     (db (double-bass :midi-channel 2))))
        :instrument-change-map '((1 ((sax ((1 alto-sax) (3 tenor-sax)))))
                                 (2 ((sax ((2 alto-sax) (5 tenor-sax))))))
        :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
        :set-map '((1 (1 1 1 1 1))
                   (2 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((sax (1 1 1 1 1))
                            (db (1 1 1 1 1))))
                        (2 ((sax (1 1 1 1 1))
                            (db (1 1 1 1 1))))))))
  (player-doubles mini 'sax))

=> T
```

SYNOPSIS:

```
(defmethod player-doubles ((sc slippery-chicken) player)
```

39.61 slippery-chicken/players

[slippery-chicken] [Methods]

DESCRIPTION:

```
Return a list of all player IDs from the given slippery-chicken object.
```

ARGUMENTS:

```
- A slippery-chicken object.
```

RETURN VALUE:

```
A list of player IDs.
```

EXAMPLE:

```
(let ((mini
       (make-slippery-chicken
        '+mini+
```

```
        :ensemble '(((cl (b-flat-clarinet :midi-channel 1))
                     (hn (french-horn :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h (q) e (s) s))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((cl (1 1 1))
                            (hn (1 1 1))
                            (vc (1 1 1))))))))
  (players mini))

=> (CL HN VC)
```

SYNOPSIS:

```
(defmethod players ((sc slippery-chicken))
```

39.62 slippery-chicken/players-ambitus

[slippery-chicken] [Methods]

DATE:

```
August 5th 2017
```

DESCRIPTION:

```
Call the player-ambitus method for each player in the slippery-chicken
object.
```

ARGUMENTS:

```
see player-ambitus method
```

RETURN VALUE:

```
the lowest and highest events of all the players
```

SYNOPSIS:

```
(defmethod players-ambitus ((sc slippery-chicken)
                            &key written print (start-bar 1) end-bar)
```

39.63 slippery-chicken/quarter-duration

[slippery-chicken] [Methods]

DATE:

```
23rd August 2023
```

DESCRIPTION:

```
return the duration in quarter notes of a slippery-chicken piece
```

ARGUMENTS:

```
the slippery chicken object
```

OPTIONAL ARGUMENTS:

```
start-bar and end-bar, both integers or if nil will default to 1 and the
number of bars in the piece
```

RETURN VALUE:

```
a float: total number of quarters
```

SYNOPSIS:

```
(defmethod quarter-duration ((sc slippery-chicken) &optional start-bar end-bar)
```

39.64 slippery-chicken/reaper-play

[slippery-chicken] [Methods]

DESCRIPTION:

```
Using the sound files (samples) defined for the given reference (group ID)
in the sndfile-palette slot of the slippery-chicken object, write a reaper
files using the pitch and timing information of one or more players' parts
from the slippery-chicken object.

See clm-play for details of these related methods.

One significant difference between this and clm-play is that reaper offers
the possibility of transposing sound files via playback rate (like clm) and
by pitch shift. Set :do-src 'transposition for the latter.
```

ARGUMENTS:

- A slippery chicken object.
- The ID of the starting section.
- The IDs of the player(s) whose events are to be used to obtain the rhythmic structure (and optionally, pitch content) of the resulting sound file. This can be a single symbol for an individual player, a list of player IDs, or NIL. If NIL, the event from all players' parts will be reflected in the output file. Default = NIL.
- The ID of the sound file group in the sndfile-palette slot of the slippery-chicken object that contains the source sound files from which the new sound file is to be generated.

OPTIONAL ARGUMENTS:

keyword arguments:
- :num-sections. An integer or NIL to indicate how many sections should be generated, including the starting section. If NIL, sound file data will be generated for all sections of the piece. NB If there are sub-sections this will include them in the total, i.e., if section 1 has 3 subsections and :num-sections is 2, we'll generate data from the first two subsections of section 1, not all subsections of main sections 1 and 2. Default = NIL.
- :from-sequence. An integer that is the number of the first sequence within the specified starting section to be used to generate the output file. This argument can only be used when num-sections = 1. Default = 1.
- :num-sequences. NIL or an integer that indicates how many sequences are to be generated, including that specified by :from-sequence. If NIL, all sequences will be played. This argument can only be used when num-sections = 1 and the section has no subsections. Default = NIL.
- :srate. A number that is the sampling rate of the reaper project file (independent of the input files). Default = 48000.
  lead time of a specified number of seconds of silence prior to the sound output.
- :inc-start. T or NIL to indicate whether playback of the source sound files is to begin at incrementally increasing positions in those files or at their respective 0.0 positions every time. If T, the method will increment the position in the source sound file from which playback is begun such that it reaches the end of the source sound file the last time it is 'played'. T = increment start times. Default = NIL.
- :ignore-rests. T or NIL to indicate whether silence should be incorporated into the resulting sound file to correspond with rests in the player's parts. If T, the sound files will play over the duration of rests. However, this is only true on a bar-by-bar basis; i.e., notes at the end of one bar will not be continued over into a rest in the next bar. This implies that rests at the start of a bar will not be turned

into sounding notes. T = ignore resets. Default = T.
- :sound-file-palette-ref2. The ID of a sound file group in the given slippery-chicken object's sndfile-palette slot. If this reference is given, the method will invoke fibonacci-transitions to transition from the first specified group of source sound files to this one. If NIL, only one group of source sound files will be used. Default = NIL.
- :do-src. T, NIL, 'transposition, a number, or a note-name pitch symbol to indicate whether transposition of the source sound files for playback will be calculated such that the perceived fundamental frequencies of those sound files are shifted to match the pitches of the current set or score note. If do-src is a number (frequency in Hertz) or a note-name pitch symbol, the method will use only that pitch instead of the sound files' frequencies when transposing to the events' pitches. If :do-src 'transposition, then reaper's pitch-adjust will be used instead of playback rate (default). NB Whichever is used, after being converted to a sample rate conversion factor, this is always multiplied by the src-scaler (see below). T = match sound files' frequencies to set pitches or score pitches via playback rate (depending on :pitch-synchronous). NIL means no transposition will be performed. Default = T.
- :pitch-synchronous: T or NIL to indicate whether the source sound files are to be transposed to match the pitches of the events in the given players' part. This will only be effective if the given source sound file has a perceptible frequency that has been specified using the sndfile object's :frequency slot in the sndfile-palette. :do-src must also be T for this to work. T = match pitches. Default = NIL.
- :reset-snds-each-rs. T or NIL to indicate whether to begin with the first source sound file of the specified group at the beginning of each rthm-seq. T = begin with the first sound file. Default = T.
- :reset-snds-each-player. T or NIL to indicate whether to begin with the first source sound file of the specified group for the beginning of each player's part. T = begin with the first sound file. Default = T.
- :play-chance-env. A list of break-point pairs that determines the chance that a given event from the source player's part will be reflected in the new sound file. It is determined by random selection but uses a fixed seed that is re-initialized each time reaper-play is called. The following default ensures every note will play: '(0 100 100 100).
- :play-chance-env-exp. A number that will be applied as the exponent to the play-chance-env's y values to create an exponential interpolation between break-point pairs. Default = 0.5.
- :max-start-time. A number that is the last time-point in seconds for which events will be processed for the output file. If a maximum start time is specified here (in seconds), events after this will be skipped. The default value of 99999999 seconds (27778 hours) will result in all events being reflected in the sound file.
- :time-scaler. A number that will be the factor by which all start times are scaled for the output file (in effect a tempo scaler). If

:ignore-rests is T, this will also have an indirect effect on durations. This argument should not be confused with :duration-scaler. Default = 1.0.
- :duration-scaler. A number that is the factor by which the duration of all events in the output sound file will be scaled. This does not alter start times, and will therefore result in overlapping sounds if greater than 1.0. This is not to be confused with :time-scaler. Instead of a simple number, :duration-scaler can also be an envelope (a list of break-point pairs). This means you could smoothly transition from staccato-like to overlapping sounds in a single reaper-play call. Default = 1.0.
- :src-scaler: A number that is the factor by which all sample-rate conversion values will be scaled (for increasing or decreasing the transposition of the overall resulting sound file). Instead of a simple number, :src-scaler can also be an envelope (a list of break-point pairs), similar to :duration-scaler. Default = 1.0.
- :note-number. A number that is an index, representing the the nth pitch of the current set or chord (from the bottom) to be used for the lowest player. Default = 0.
- :duration-run-over. T or NIL to indicate whether the method will allow a sound file event to extend past the end of specified segment boundaries of a sound file in the sndfile-palette. T = allow. Default = NIL.
- :short-file-names. T or NIL to indicate whether abbreviated output file names will be automatically created instead of the usually rather long names. T = short. Default = NIL.
- :output-name-uniquifier. A user-specified string that will be incorporated into the file name, either at the end or the beginning depending on whether short-file-names is T or NIL. Default = "".
- :check-overwrite. T or NIL to indicate whether to query the user before overwriting existing sound files. T = query. Default = T.
- :sndfile-palette. NIL or a file name including path and extension that contains an external definition of a sndfile-palette. This will replace any sndfile-palette defined in the slippery-chicken object. If NIL, the one in the slippery-chicken object will be used. Default = NIL.
- :chords. NIL or a list of lists consisting of note-name symbols to be used as the pitches for the resulting sound file in place of the pitches from the set-map. There must be one chord specified for each sequence. If NIL, the pitches from the set-map will be used. Default = NIL.
- :chord-accessor. Sometimes the chord stored in the palette is not a simple list of data so we need to access the nth of the chord list. Default = NIL.
- :pan-min-max. Each event is panned randomly between two channels. Pan values are in degrees where 0 is channel 1 (left in stereo files) and 90 is channel 2 (right). The values used are chosen from the following list of degrees: (15 25 35 45 55 65 75). However, these can be scaled to be within new limits by passing a two-element list to

:pan-mix-max. E.g. '(40 50) would limit most values to the middle area of stereo space. Note that no matter how many channels a sound file or channel has, the item's panner is set as if it were stereo. Also that, for convenience, :min-channels and :max-channels can set the tracks' channel count, it is up to the user to take care of routing and, eventuallay, panning depending on the number of sound file and track channels used in the mix. Default = '(5 85).

- :pan-fun. If you want to take charge of selecting pan positions yourself, pass a function via this keyword. The function must take one argument: the current event, though of course, it can ignore it completely if preferred. The function should return a degree value: a number between 0 and 90. Default = NIL.
- :min-channels and :max-channels 4. Set the reaper tracks' minimum and maximum channel counts. See these named slots in the reaper-track class definition and the cond in create-tracks for details of how these are handled.
- :snd-selector. By default the sound files in the given group are cycled through, one after the other, returning to the beginning when the end is reached. This can be changed by passing a function to :snd-selector. This function must take three arguments: the circular-sclist object that contains the group's sndfile objects (the sndfiles are in a simple list in the data slot); the pitch of the current event (if the event happens to be a chord, each pitch of the chord will be handled separately i.e. the :snd-selector function will be called for each pitch); and the event object. Of course any of these arguments may be ignored by the :snd-selector function; indeed it is imagined that the event object will be mostly ignored and the pitch object used instead, but it is nevertheless passed for completeness. For example, the following function, declared on-the-fly using lambda rather than separately with defun, makes reaper-play act as a traditional sampler would: Assuming our sndfile-palette group has a list of sound files whose frequencies really correspond to their perceived fundamentals (e.g. see the set-frequency-from-filename method, and the kontakt-to-sfp function) we choose the nearest sound file in the group to the frequency of the current event's pitch:

```
:snd-selector #'(lambda (sflist pitch event)
                  (declare (ignore event))
                  (get-nearest-by-freq
                   (frequency pitch) (data sflist)))
```

  NB If :pitch-synchronous is NIL, then the pitch-or-chord slot of the event will be the pitch selected from the current set, rather than the actual pitch of the instrument in question.
  Default = NIL i.e. use the circular selection method.
- :snd-transitions. Set the transition from sound-file-ref to sound-file-ref

```
   over the course of the call to reaper-play using a custom envelope instead
   of the default fibonacci-transitions method, e.g. '(0 0 100 1). x-axis is
   arbitrary, y-axis should range from 0 to 1. When reading, y-values will be
   rounded.
 - :file. reaper file name, if you don't want this to be auto-generated.
   Automatically generated file names will be placed in the default-dir as
   defined in the sc-config. A full path is expected. Default = NIL =
   automatic.
 - :fade-in :fade-out. The fade times expressed as a percentage of the
   duration of any given event/item. Default = 10%
 - :tracks-per-player. How many reaper tracks to use per player. If > 1
   then the tracks will be used one after the other, in rotation, for each
   event. Particularly useful if there are lots of overlapping events due to
   e.g. :duration-scaler, or chords, of course. Default = 1.
 - :items-processor. A function to call to process the list of reaper-item
   objects just before writing the reaper file. This should take one
   argument, the list of reaper-items, and return a list of reaper-items that
   will be written to the file. Default = NIL = no processing.
 - :markers. Whether to write markers at sequenz and tempo changes. This
   should be a list of two booleans: whether to write sequenz and/or
   tempi. NB This only works (for now) if you're generating the whole
   piece. Default = '(t t)
```

RETURN VALUE:

```
 Total events generated (integer).
```

EXAMPLE:

```
;;; using playback speed to effect the transpositions:
(reaper-play +second-law+ 1 nil 'source-sndfile-grp-1
             :check-overwrite nil :tracks-per-player 2
             :pitch-synchronous t)

;;; using reaper's pitch adjust and keeping playback speed implicitly to 1.0
(reaper-play +second-law+ 1 nil 'source-sndfile-grp-1
             :check-overwrite nil :tracks-per-player 2
             :pitch-synchronous t :do-src 'transposition)
```

SYNOPSIS:

```
(defmethod reaper-play ((sc slippery-chicken) section players
                        sound-file-palette-ref
                        &key
                        sound-file-palette-ref2
```

```
(play-chance-env '(0 100 100 100))
(max-start-time 99999999)
(play-chance-env-exp 0.5)
(time-scaler 1.0)
(from-sequence 1)
(num-sequences nil)
(num-sections nil)
(ignore-rests t)
(time-offset 0.0)
(chords nil)
(chord-accessor nil)
(note-number 0)
(fade-in 10)
(fade-out 10)
(inc-start nil)
;; either a number or an envelope
(src-scaler 1.0)
(do-src t)
(pitch-synchronous nil)
;; either a number or an envelope
(duration-scaler 1.0)
(short-file-names nil)
;; by default we generate filenames according to players
;; and other relevent data but if this is given then
;; we'll use it instead
(file nil)
(check-overwrite t)
(reset-snds-each-rs t)
(reset-snds-each-player t)
(duration-run-over nil)
(srate 48000)
(output-name-uniquifier "")
(sndfile-palette nil)
(tracks-per-player 1)
;; MDE Sat Oct  3 18:45:28 2015 -- for Cameron!
pan-fun
;; MDE Thu Oct  1 21:03:59 2015
(pan-min-max '(5 85))
;; MDE Wed Feb 21 13:21:00 2024, Heidhausen
items-processor
;; MDE Thu Feb 22 11:40:43 2024, Heidhausen
(markers '(t t))
;; MDE Thu Oct  1 19:13:49 2015
snd-selector
(min-channels 2) (max-channels 4)
snd-transitions)
```

39.65 slippery-chicken/rebar

[slippery-chicken] [Methods]

DESCRIPTION:

Go through the vertically simultaneous sequences in all players of the given slippery-chicken object and rebar according to the first one that has the least number of bars (but following the player hierarchy).

If rthm-seqs or sequenzes are created algorithmically and bundled into the slippery-chicken piece slot artificially, bypassing the usual generation structure, it might be difficult to end up with each instrument having the same metric structure when combined vertically. This method goes through the vertically combined sequences and rebars as described above.

NB: See documentation in piece class method. Don't confuse this method with the re-bar method.

NB: This method is used internally and not recommended for direct use.

ARGUMENTS:

- A slippery-chicken object.

OPTIONAL ARGUMENTS:

- A list of player IDs from the given slippery-chicken object, ordered in terms of importance i.e. which instrument's bar structure should take precedence.
- NB: The rebar-fun is not yet used.

RETURN VALUE:

Always T.

SYNOPSIS:

```
(defmethod rebar ((sc slippery-chicken)
                  &optional instruments-hierarchy rebar-fun)
```

39.66 slippery-chicken/sc-combine

[slippery-chicken] [Functions]

DATE:

22nd May 2025

DESCRIPTION:

Combine an arbitrary number of slippery-chicken objects into a new slippery-chicken by referencing the bars and players of the existing objects to copy (clone). This takes a map as the first argument that looks something like:

'((sc-object-num start-bar end-bar result-start-bar player1 player2 ...))

There can be as many sublists as necessary. <sc-object-num> is the 1-based index into the <sc-objects> list (2nd argument). <start-bar> and <end-bar> reference the existing sc-object and are inclusive. If <end-bar> is nil, then it will be set to the last bar of the sc-object. <result-start-bar> is where things should be written in the new slippery-chicken object--this can be anywhere, and even overlap/overwrite other copied-in bars, as long as the metric structure matches. There can be as many players referenced as desired but each must of course be found in the referenced sc-object (see also the double-events and delete-events methods for how to move things around afterwards, if necessary. Players can also be omitted from the map's sublist whereupon all players in that sc-object will be copied.

Note that rest bars will automatically be created for players that have no notes to play and that if there are any bars without notes at all, for any player, then rest-bars will be created using the :rest-time-sig keyword argument as the meter.

Note also that writing into new bars doesn't have to appear in the map in any particular order, and neither do the <sc-objects> E.g. we could write bars 50-100 of the result first, using bars from sc-object 3, then 10-20 afterwards using sc-object 1.

The map can also omit the result-start-bar so that bar ranges are just appended one after the other in the result. In that case do set :follow-on t in order for the result start bars to be automatically calculated. The players can also be omitted here. Note however that the order of the map will determine the order in the output. E.g.

'((sc-object-num start-bar end-bar player1 player2 ...))

ARGUMENTS:

- the map: a list as described above
- a list of slippery-chicken objects

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :max-bars. The maximum number of bars that will be generated. Defaults
  generously to 1000 but can go as high as you like, memory permitting.
- :max-players. The maximum number of players we'll use. The same applies
  here as to :max-bars, as both are used to initialise the array we use as
  the workhorse in remapping. Note that the resultant slippery-chicken
  object will only have as many bars and players as needed by the mapping
  procedure. Default = 50.
- :new-sc-name. A symbol to be used as the global variable for the resultant
  slippery-chicken object, as with make-slippery-chicken.
- :instrument-palette. The instrument palette used in creating the
  <sc-objects>. NB it is a (not very daunting) restriction of this method
  that all listed sc-objects must have used the same palette. Default =
  +slippery-chicken-standard-instrument-palette+
- :rest-time-sig. The time signature to be used when creating rest bars
  where no instrument is in play.
- :add-rehearsal-letters. T or NIL to indicated whether rehearsal letters
  should be placed at every point in the map, i.e. for every copy
  procedure. NB in any case, existing rehearsal-letters will not be copied
  over from the sc-objects in the 2nd arguments. Default = T.
- :follow-on. T or NIL or a number to indicate that bar ranges should just
  be appended after each other (see above). If a number is given, this will
  be the start-bar in the result. Default = NIL.
```

RETURN VALUE:

```
- a new slippery chicken object
```

SYNOPSIS:

```
(defun sc-combine (map sc-objects
                   &key (max-bars 1000) (max-players 50)
                   (new-sc-name '*sc-combine*)
                   (add-rehearsal-letters t)
                   follow-on
                   (instrument-palette
                    +slippery-chicken-standard-instrument-palette+)
                   (rest-time-sig '(2 4)))
```

39.67 slippery-chicken/sc-init

[slippery-chicken] [Methods]

DESCRIPTION:

```
Explicitly initialize the slippery-chicken object. This is usually called
implicitly by initialize-instance (i.e. when you call
make-slippery-chicken) but there could be circumstances (e.g. in subclasses
of slippery-chicken) where you'd like to defer initialization and call this
method explicitly instead. In that case set :defer to t when making the
slippery-chicken object. This can also be called with using :defer, to
regenerate the piece after e.g. calling set-limits-by-section.
```

ARGUMENTS:

```
- the slippery-chicken object

OPTIONAL ARUGMENTS
keyword arguments:
- :regenerate-pitch-seq-map: the pitch-seq-map is generated here for each
instrument using the pitch-seqs in the rthm-seq-palette. By setting this
to T we can force regeneration (e.g. if the rthm-seq-palette has changed
and we want to re-init the sc with different data). Default = T.
- :shorten-large-fast-leaps. T or NIL to call the associated method or
  not. Eg. you might want to call this explicitly elsewhere because the
  tempo-map is set after sc-init is called. Default = T.
```

RETURN VALUE:

```
the now fully initialized slippery-chicken object
```

EXAMPLE:

```
(let ((sc (make-slippery-chicken
           '+mini+
           :ensemble '(((vn (violin :midi-channel 1))))
           :set-palette '((1 ((gs4 af4 bf4))))
           :defer t
           :set-map '((1 (1 1 1)))
           :rthm-seq-palette '((1 ((((4 4) e e e e e e e e))
                                  :pitch-seq-palette ((1 2 1 1 1 1 1 1)))))
           :rthm-seq-map '((1 ((vn (1 1 1))))))))
  (sc-init sc))
```

SYNOPSIS:

```
(defmethod sc-init ((sc slippery-chicken) &key (regenerate-pitch-seq-map t)
                                               (shorten-large-fast-leaps t))
```

39.68 slippery-chicken/section-densities

[slippery-chicken] [Methods]

DATE:

```
January 12th 2019, Heidhausen
```

DESCRIPTION:

```
Calculate the densities of notes or rests in the sections of a
slippery-chicken object. Density is the average number of notes/rests per
bar across a section.
```

ARGUMENTS:

```
- a slippery-chicken object
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :slot. The section slot to use. 'num-score-notes is the default and makes
  sense in that ties indicate a higher level of density than just attacked
  notes by themselves, but further useful slots could be 'num-notes
  (attacks) or indeed 'num-rests
- :percent. Whether to return values as percentages (where 0 would be the
  least dense section and 100 the most). If NIL the actual average notes per
  bar in each section would be returned. Default = T.
```

RETURN VALUE:

```
a list of density numbers, one per section/subsection.
```

SYNOPSIS:

```
(defmethod section-densities ((sc slippery-chicken)
                              &key (slot 'num-score-notes) (percent t))
```

39.69 slippery-chicken/shorten-large-fast-leaps

[slippery-chicken] [Methods]

DESCRIPTION:

Modify the pitches of each part in a slippery-chicken object to avoid large melodic leaps at fast speeds, based on the largest-fast-leap slot of the given instrument object and the fast-leap-threshold slot of the slippery-chicken object.

This method is called automatically at init and as such will most likely seldom need to be directly accessed by the user.

ARGUMENTS:

- A slippery-chicken object.

OPTIONAL ARGUMENTS:

keyword arguments:
- :threshold. A number that is the maximum duration in seconds between two consecutive notes in the slippery-chicken object for which linear intervals greater than the number specified in the given instrument object's largest-fast-leap slot will be allowed. This value is taken from the fast-leap-threshold slot of the given slippery-chicken object by default.
- :verbose. T or NIL to indicate whether to print feedback about the method's operations to the Lisp listener. T = print. Default = T.

RETURN VALUE:

Always T

EXAMPLE:

```
(let ((mini
(make-slippery-chicken
'+mini+
:ensemble '(((vn (violin :midi-channel 1))))
:tempo-map '((1 (q 96)))
:set-palette '((1 ((g3 a5 b6))))
:set-map '((1 (1 1 1 1 1 1)))
:rthm-seq-palette '((1 ((((2 4) e s 32 64 64 e s 32 64 64))
:pitch-seq-palette ((1 5 1 5 1 5 1 5 1 5)))))
:rthm-seq-map '((1 ((vn (1 1 1 1 1 1)))))))
(shorten-large-fast-leaps mini :threshold 0.25))

=>
******* section (1)
Getting notes for VN
```

```
Shortening short, fast leaps...
Shortened 23 large fast leaps
seq-num 0, VN, replacing B6 with G3
seq-num 0, VN, replacing B6 with G3
seq-num 0, VN, replacing B6 with G3
seq-num 0, VN, replacing B6 with G3
seq-num 0, VN, replacing G3 with B6
seq-num 0, VN, replacing G3 with B6
seq-num 0, VN, replacing G3 with B6
seq-num 1, VN, replacing G3 with B6
seq-num 1, VN, replacing B6 with G3
seq-num 1, VN, replacing B6 with G3
seq-num 1, VN, replacing B6 with G3
seq-num 1, VN, replacing B6 with G3
seq-num 1, VN, replacing G3 with B6
seq-num 1, VN, replacing G3 with B6
seq-num 1, VN, replacing G3 with B6
seq-num 2, VN, replacing G3 with B6
seq-num 2, VN, replacing B6 with G3
seq-num 2, VN, replacing B6 with G3
seq-num 2, VN, replacing B6 with G3
seq-num 2, VN, replacing B6 with G3
seq-num 2, VN, replacing G3 with B6
seq-num 2, VN, replacing G3 with B6
seq-num 2, VN, replacing G3 with B6
seq-num 3, VN, replacing G3 with B6
seq-num 3, VN, replacing B6 with G3
seq-num 3, VN, replacing B6 with G3
seq-num 3, VN, replacing B6 with G3
seq-num 3, VN, replacing B6 with G3
seq-num 3, VN, replacing G3 with B6
seq-num 3, VN, replacing G3 with B6
seq-num 3, VN, replacing G3 with B6
seq-num 4, VN, replacing G3 with B6
seq-num 4, VN, replacing B6 with G3
seq-num 4, VN, replacing B6 with G3
seq-num 4, VN, replacing B6 with G3
seq-num 4, VN, replacing B6 with G3
seq-num 4, VN, replacing G3 with B6
seq-num 4, VN, replacing G3 with B6
seq-num 4, VN, replacing G3 with B6
seq-num 5, VN, replacing G3 with B6
seq-num 5, VN, replacing B6 with G3
seq-num 5, VN, replacing B6 with G3
seq-num 5, VN, replacing B6 with G3
seq-num 5, VN, replacing B6 with G3
```

```
seq-num 5, VN, replacing G3 with B6
seq-num 5, VN, replacing G3 with B6
seq-num 5, VN, replacing G3 with B6
```

SYNOPSIS:

```
(defmethod shorten-large-fast-leaps ((sc slippery-chicken)
                                     &key threshold (verbose t))
```

39.70 slippery-chicken/statistics

[slippery-chicken] [Methods]

DESCRIPTION:

Print various information about the given slippery-chicken object to the Lisp listener or other specified stream.

ARGUMENTS:

- A slippery-chicken object.

OPTIONAL ARGUMENTS:

- NIL or a stream to which the information should be printed. If NIL, the method will not print the information to any stream. T = print to the Lisp listener. Default = T.

RETURN VALUE:

A number of formatted statistics about the given slippery-chicken object.

EXAMPLE:

```
(let ((mini
(make-slippery-chicken
'+mini+
:ensemble '(((sax (alto-sax :midi-channel 1))))
:set-palette '((1 ((e3 fs3 b3 cs4 fs4 gs4 ds5 f5))))
:set-map '((1 (1 1 1))
(2 (1 1 1)))
:rthm-seq-palette '((1 ((((4 4) h q e s s))
:pitch-seq-palette ((1 2 3 4 5)))))
:rthm-seq-map '((1 ((sax (1 1 1))))
```

```
(2 ((sax (1 1 1))))))))
(statistics mini))

=>
+MINI+
"+MINI+-piece"
start-bar: 1
end-bar: 6
num-bars: 6
start-time: 0.0
end-time: 24.0
start-time-qtrs: 0
end-time-qtrs: 24.0
num-notes (attacked notes, not tied): 30
num-score-notes (tied notes counted separately): 30
num-rests: 0
duration-qtrs: 24.0
duration: 24.0 (24.000)
```

SYNOPSIS:

```
(defmethod statistics ((sc slippery-chicken) &optional (stream t))
```

39.71 slippery-chicken/transpose-events

[slippery-chicken] [Methods]

DESCRIPTION:

```
Transpose the pitches of event objects in a specified region and a
specified player's part.
```

ARGUMENTS:

```
- A slippery-chicken object.
- A player ID.
- An integer that is the first bar in which to transpose events.
- An integer that is the first event in that bar to transpose.
- An integer that is the last bar in which to transpose events.
- An integer that is the last event in that bar to transpose.
- A positive or negative number that is the number of semitones by which
  the pitches of the events in the specified region should be transposed.
```

OPTIONAL ARGUMENTS:

```
keyword argument:
- :destructively. T or NIL to indicate whether the pitches of the original
  event objects should be replaced. T = replace. Default = T.
```

RETURN VALUE:

```
Returns a list of events.
```

EXAMPLE:

```
;;; Print the pitches before and after applying the method ;
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((sax (alto-sax :midi-channel 1))
                     (db (double-bass :midi-channel 2))))
        :set-palette '((1 ((c2 d2 g2 a2 e3 fs3 b3 cs4 fs4 gs4 ds5 f5 bf5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                :pitch-seq-palette ((1 2 3 4 5)))))
        :rthm-seq-map '((1 ((sax (1 1 1 1 1))
                            (db (1 1 1 1 1))))))))
  (print
   (loop for e in (get-events-from-to mini 'sax 3 2 5 3)
      collect (get-pitch-symbol e)))
  (transpose-events mini 'sax 3 2 5 3 11)
  (print
   (loop for e in (get-events-from-to mini 'sax 3 2 5 3)
      collect (get-pitch-symbol e))))

=>
(EF4 AF4 BF4 EF5 CS4 EF4 AF4 BF4 EF5 CS4 EF4 AF4)
(D5 G5 A5 D6 C5 D5 G5 A5 D6 C5 D5 G5)
```

SYNOPSIS:

```
(defmethod transpose-events ((sc slippery-chicken) player start-bar
                             start-event end-bar end-event semitones
                             &key (destructively t))
```

39.72 slippery-chicken/update-instrument-slots

[slippery-chicken] [Methods]

DATE:

23rd August 2013

DESCRIPTION:

This will go through the generated slippery-chicken object's bar structure and update each players' instruments' total-bars, total-notes, total-duration, and total-degrees slots, for statistical purposes only. This might be called by the user after performing one of the editing routines that deletes or changes notes, etc.

ARGUMENTS:

- The slippery-chicken object.

RETURN VALUE:

T

SYNOPSIS:

```
(defmethod update-instrument-slots ((sc slippery-chicken))
```

39.73 slippery-chicken/update-slots

[slippery-chicken] [Methods]

DESCRIPTION:

Called by initialize-instance and others. Updates timings of events and statistics. Not generally called by the user but can be useful if post-generation editing has changed something fundamental to the structure.

If you want to update the statistics for each instrument in the ensemble, use the update-instrument-slots method.

ARGUMENTS:

- A slippery-chicken object

OPTIONAL ARGUMENTS:

- A tempo-map object (not just as a list). If not given, then the tempo-map from the slippery-chicken object will be used. Default = NIL.
- A number that is the start-time of the first event object in

```
   seconds. Default = 0.0.
 - A number that is the start-time of the first event, in 'quarters' (for
   MIDI timing). Default = 0.0.
 - A integer that is the number of the starting bar. Default = 1.
 - The reference of the current section (for internal recursive use in the
   bar-holder class). Default = NIL.
 - The nth sequence (for internal recursive use in the sequenz class).
   Default = NIL.
 - T or NIL to indicate whether to print a warning to the Lisp listener when
   ties are being used at the beginning of a sequence. This argument is now
   obsolete and ignored, but remains for some backward compatibility.
   Default = T.
```

RETURN VALUE:

```
The duration in seconds of the object; in this class: the whole generated
piece.
```

EXAMPLE:

```
;;; Create a slippery-chicken object and print the start time of one of its
;;; events; call update-slots with a start time of 10.0 and print the start
;;; time of that same event to see the difference
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((vn (violin :midi-channel 1))))
        :tempo-map '((1 (q 60)))
        :set-palette '((1 ((c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((2 4) (s) (s) e e e))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((vn (1 1 1 1 1)))))))
  (print (start-time (get-event mini 4 1 'vn)))
  (update-slots mini nil 10.0)
  (print (start-time (get-event mini 4 1 'vn))))

=
6.0
16.0
```

SYNOPSIS:

```
(defmethod update-slots ((sc slippery-chicken)
                         &optional
```

```
(tempo-map nil)
(start-time 0.0)
(start-time-qtrs 0.0)
(start-bar 1)
(current-section nil)
(nth nil)
(warn-ties t)
(update-write-bar-nums nil))
```

39.74 slippery-chicken/write-antescofo

[slippery-chicken] [Methods]

DESCRIPTION:

```
Write an antescofo~ (Arshia Cont's/IRCAM's score follower MaxMSP external)
score file. This allows you to specify a single player to follow (for now:
no polyphonic following) and which players' events you'd like to be
triggered along with this, in the form of MIDI notes (sent via antescofo's
group action commands). Of course this doesn't imply that you have to use
MIDI with antescofo, rather, that you have something that picks up midi
note data and uses them (or ignores them) somehow or other. In any case,
the MIDI notes sent consist of four messages: the MIDI note (in midi cents
e.g. middle C = 6000), the velocity (taken as usual from the amplitude slot
of the event), the channel, and the duration in beats. Each may be preceded
by a delay, which will be relative to the previous NOTE or group MIDI note.

Rehearsal letters already in the slippery-chicken piece will automatically
be written into the antescofo~ file as labels/cues. E.g. if you have
rehearsal letter A, it will show up in the antescofo~ file as "letter-A".
Further labels/cues can be added to any slippery-chicken events and these
will also be written into the antescofo~ file.

Action messages can also be added to events and thus written into the
antescofo~ file. You can push as many messages as you want into this list;
they'll be reversed before writing out so that they occur in the order in
which you added them. If you prefer you can call the event method
add-antescofo-message. NB For the part we're following, we can add messages
to rests but if it turns out we added messages to rests in other players'
(i.e. group event parts) these won't be written to the antescofo~ file.

Do remember to set the instruments-hierarchy slot of the slippery-chicken
object so that the player you're going to follow is top of the list,
otherwise some strange event ordering might occur in the antescofo~ file.
```

Bear in mind that if you want to write antescofo~ files without having to work within the usual slippery chicken workflow, you could generate events by any method, then put them into rthm-seq-bar objects before then calling bars-to-sc in order to create a slippery-chicken object by brute force (as it were).

A note about grace notes: For the voice we're following, antescofo~ considers that grace notes are 'out of time' so have a duration of 0. But for group notes, which we're triggering, they have to have some duration; we default to the grace-note-duration of the event class (0.05 seconds by default, which will be written as a fraction of the beat at the current tempo, of course). Now this means our grace notes are not 'stealing' time from the previous note, as they do in performed music. This is not ideal, but if we've got a long group of grace notes, doing that might mean we end up with a negative duration for the previous note. So we simply make the grace notes short and allow/hope that the score follower catches up for us on the next recognised note.

ARGUMENTS:

- The slippery-chicken object
- the player who we'll follow (single player for now, as a symbol)

OPTIONAL ARGUMENTS:

keyword arguments:
- :group-players (list of symbols). The players for whom midi-note events will be written in the antescofo file as part of a "group" action. If NIL, then we'll write all players' events except for the player we're following. NB There's no reason why we couldn't include the player we're following in these group commands (for unison playing between live and digital instruments perhaps). The easist way to write group events for all players is to write something like :group-players (players +your-sc-object+) Default = NIL.
- :bar-num-receiver. The MaxMSP receiver name to which bar numbers will be sent for display other other purposes. Default = "antescofo-bar-num"
- :midi-note-receiver. The MaxMSP receiver name to which midi notes will be sent. Default = "midi-note", so a typical group output event could be 0.0 midi-note 6500 12 4 0.6666667
- :file. The name of the file to write. If NIL, then the file name will be created from the slippery-chicken title and placed in (get-sc-config 'default-dir). Default = NIL.
- :group-duration-in-millisecs. Should we write the group notes (i.e. those we generate) as fractions of a beat (NIL) or have antescofo~ convert this to millisec duration according to the current tempo (T)? Default =

```
   NIL.
- :warn. Issue a warning when we write labels on group (rather than NOTE)
   events?  (Because labels attached to group event notes can be written
   and read, but they won't show up in MaxMSP as cues which you can jump
   to.)  Default = T.
- :messages-first. Write antescofo messages immediately before the note
   data sent to the midi-note-receiver (T), or after (NIL)? Default = NIL.
- :round-tempi. Round tempo values to the nearest integer? (If you don't do
   this, then MIDI playback could become out of synch with
   antescofo). Default = T.
- :include-program-changes. If T, we write any programme changes attached
   to an event as "midi-program-change <programme> <channel>". If this is a
   symbol or string then that ID will be used instead of
   "midi-program-change". (Symbols will be made lowercase.) Default = T.
- :makenote-compatible. If T, we'll write midi note information in the
   following order: midi note (rounded to nearest chromatic note i.e. no
   longer midi-cents), velocity, duration, channel. Default = NIL.
```

RETURN VALUE:

```
The number of NOTE plus action events we've written. We also print to the
terminal the number of events and actions separately, which should
then correspond to what Antescofo~ prints when it loads the score in MaxMSP.
```

EXAMPLE:

```
;;; Follow the violin part and generate group events for all other parts ;
(let* ((mini
        (make-slippery-chicken
         '+mini+
         :title "antescofo test"
         :ensemble '(((vn (violin :midi-channel 1))
                      (va (viola :midi-channel 2))
                      (vc (cello :midi-channel 3))))
         :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
         :set-map '((1 (1 1 1)))
         :tempo-map '((1 60))
         :rthm-seq-palette '((1 ((((4 4) { 3 tq tq tq } +q e (s) s)))))
         :rthm-seq-map '((1 ((vn (1 1 1))
                             (va (1 1 1))
                             (vc (1 1 1)))))))))
  ;; Adding a label (probably wouldn't need one in bar 1, but to illustrate)
  (setf (asco-label (get-event mini 1 1 'vn)) "test-label")
  ;; start the (fictitious) vocoder when the first cello note in bar 2 is played
  (push "max-receiver1 start-vocoder" (asco-msgs (get-event mini 2 1 'vc)))
  (write-antescofo mini 'vn :file "/tmp/asco-test.txt"))
```

```
-->
******* section (1)
Getting notes for VN
Getting notes for VA
Getting notes for VC
Shortening short, fast leaps...
Shortened 0 large fast leaps
"/tmp/asco-test.txt"
Antescofo~ score written successfully with 15 events and 34 actions.
49

The generated file will begin something like this:

; antescofo~ score generated by slippery chicken version ;
; 1.0.4 (svn revision 4733 2014-01-15 11:27:10) ;
; at 12:02:06 on Thursday the 8th of May 2014 ;
BPM 60
antescofo-bar-num 1 ;;;
```

```
NOTE 6200 0.6666667
group bar2.1 {
0.0 midi-note 6000 12 2 0.6666667
0.0 midi-note 6400 12 3 0.6666667
max-receiver1 start-vocoder
}
NOTE 6200 0.6666667
...
```

SYNOPSIS:

```
(defmethod write-antescofo ((sc slippery-chicken) follow-player
                            &key group-players file (warn t)
                              (group-duration-in-millisecs nil)
                              messages-first
                              ;; could be T, a string, or a symbol
                              ;; todo: include control changes also?
                              (include-program-changes t)
                              (round-tempi t)
                              makenote-compatible
                              (midi-note-receiver "midi-note")
                              (bar-num-receiver "antescofo-bar-num"))
```

39.75 slippery-chicken/write-csound-score

[slippery-chicken] [Methods]

DATE:

```
2023-02-28, Essen
```

DESCRIPTION:

```
Generate a Csound-score file (.sco) from the data of the specified
slippery-chicken object.
```

ARGUMENTS:

```
- A slippery-chicken object.
- The ID(s) of the player(s) whose events are used to obtain the
  rhythmic structure of the score events (i.e. the onset and duration
  values) and, dependent on the p-fields function, pitch content,
  amplitude etc. This must be a list with one or more symbols.
- The ID(s) (number) or name(s) (string) of the Csound instruments
  according to the resp. instrument definition in the Csound orchestra.
```

```
   Each player of the second argument of this methods needs to be assigned
   one Csound instrument. This must be a list with one or more values.
```

OPTIONAL ARGUMENTS:

```
keyword arguments:
- :suffix. A string to be added just before the .sco of the generated
   csound score-file. Will be overridden when manually setting the path
   via :csound-file. Default = "".
- :csound-file. A string indicating the path and filename to the .sco-file
   to be generated with this method. Default is a filename extracted from the
   title of the sc piece, placed in the (get-sc-config 'default-dir)
   directory (per default /tmp).
- :start-section. An integer that is the number of the first section for
   which the Csound-score is to be generated. Default = 1.
- :num-sections. An integer that is the number of sections to generate the
   score data. If NIL, all sections will be written. Default = NIL.
- :from-sequence.  An integer that is the number of the first sequence
   within the specified starting section to be used to generate the output
   file. Default = 1.
- :num-sequences. An integer that s the number of sequences for which
   event data is to be generated in the resulting .sco-file, including the
   sequence spiecified in from-sequence. If NIL, all sequences will be
   written. NB: This argument can only be used when the num-sections = 1.
   Default = NIL.
- :chords. Either a boolean or an integer which determines how to deal with
   chords in events. When T, all notes of a chord are parsed as
   i-statements in the Csound-score. When NIL, chords will be ignored and
   treated like rests (i.e. nothing will be generated). When an integer is
   given, the number determines the max. amount of pitches being parsed as i-
   statements. They are chosen chronologically from the given chord.
   Default = T.
- :offset. An integer or floating point value to indicate the global
   offset of all score events (in seconds). Default = 0.
- :delimiter. A character which separates each score statement. Csound
   recommends using spaces, though #\tab does also work an might, in some
   cases, improve legibility. Default = #\space.
- :p-fields. This should be either NIL, a list of lists or a function used
   to generate the p-fields starting from p4. The lists can be of arbitrary
   length and are solely limited by Csound's internal parsing restrictions.
   When NIL, no additional p-statements are added to any of the score-
   statements, regardless of the amount of included players (i.e. no further
   p-field is added to any of the players).
   When given a list of lists, each sublist contains a set of constant
   p-values for each player. NB: It is mandatory to pass a sub-list for
   each player chosen for score generation (cf. second argument). It is
```

possible to pass an empty list (i.e. NIL) for a single player, if it is
desired not to add any further p-fields other than instrument-name (p1),
onset (p2), and duration (p3).
When a function is given as argument, it will be used to generate the
p-values based on the event-data and other arguments. This enables dynamic
generation of score content dependent on the data contained in the
specific event used for generating the i-statement (e.g. pitch or
amplitude). The pfield-function needs to accept the following arguments:
- event. A sc event.
- event-num. An integer counting the event index of each player.
- cs-instrument. The csound-instrument name as defined in the third
argument (see above). This comes in handy when the
csound instruments require different sets of p-values.
NB: When chords should be able to be processed (see above), the function
needs to be able to process events containing chords and return a list
of the length of the chord (e.g. via is-chord).
For further detail, take a look at csound-p-fields-simple.
- :comments. A boolean indicating whether additional comments should be
included into the generated score. The comments include the title,
composer and generation date of the sc piece resp. the csound-score.
Additionally, the score sections containing the instrument calls for
each player will be preceded with a reference to the player ID
according to the sc ensemble. Default = T.

RETURN VALUE:

Returns the path of the file written, as a string.

EXAMPLE:

```
;;; An example using the csound-p-fields-simple function
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((pno (piano :midi-channel 1))
                     (vln (violin :midi-channel 2))))
        :set-palette '((1 ((f3 g3 as3 a3 bf3 b3 c4 d4 e4 f4 g4 a4 bf4 cs5))))
        :set-map '((1 (1 1 1 1 1 1 1))
                   (2 (1 1 1 1 1 1 1))
                   (3 (1 1 1 1 1 1 1)))
        :tempo-map '((1 (q 60)))
        :rthm-seq-palette '((1 ((((4 4) h (q) e (s) s))
                                :pitch-seq-palette ((1 (2) 3))))
                            (2 ((((4 4) (q) e (s) s h))
                                :pitch-seq-palette ((1 2 3))))
                            (3 ((((4 4) e (s) s h (q)))
```

```
                                :pitch-seq-palette ((2 3 3))))
                             (4 ((((4 4) (s) s h (q) e))
                                :pitch-seq-palette ((3 1 (2))))))
        :rthm-seq-map '((1 ((pno (1 2 1 2 1 2 1))
                            (vln (1 2 1 2 1 2 1))))
                       (2 ((pno (3 4 3 4 3 4 3))
                            (vln (3 4 3 4 3 4 3))))
                       (3 ((pno (1 2 1 2 1 2 1))
                            (vln (1 2 1 2 1 2 1)))))))
  (write-csound-score mini
              '(pno vln)
              '(1 "fmsynth")
              :chords 1
              :delimiter #\tab
              :comments t))

;;; An example that creates different sets of p-field for each
;;; individual player.
;;; - Player 'vln calls the (hypothetical) Csound instrument 1
;;;   which generates tone of frequency p4 and amplitude p5.
;;;   Here, the standard csound-p-fields-simple function is invoked.
;;; - Player 'vlc calls the Csound instrument 2, which is a simple
;;;   sampler capable of playing one out of ten pre-loaded soundfiles.
;;;   These are identified by an index (1 <= i <= 10) which is expected
;;;   to be p4. p5 determines the amplitude of the playback.
;;;   In this examples, the samples will be selected cyclically, while
;;;   each event of the player is allocated one of the ten samples.
;;;   As there is one soundfile per events, chords will be treated as
;;;   pitches, i.e. every chord results in a list instead of a list of
;;;   lists of p-fields.
(defun csound-p-fields-custom (event event-num cs-instrument)
  ;; generate different p-field sets depending on the
  ;; given Csound instrument-number
  (format t "~%ins: ~a; enum: ~a~%"
          cs-instrument
          event-num)
  (case cs-instrument
    (1 (csound-p-fields-simple event
                               event-num
                               cs-instrument))
    (2 (let* ((amplitude (get-amplitude event))
              ;; num of available samples in Csound
              ;; instrument definition / orchestra
              (num-samples 10)
              ;; get the id of the sample to be played
              ;; for this event
```

```
              (current-sample (1+ (mod (1- event-num)
                                       num-samples))))
            (list current-sample
                  amplitude)))))

(let* ((mini
         (make-slippery-chicken
          '+mini+
          :ensemble '(((vln (violin :midi-channel 1))
                       (vlc (cello :midi-channel 2))))
          :tempo-map '((1 (q 60)))
          :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5)))
                         (2 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5)))
                         (3 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
          :set-map '((1 (1 1 1 1 1))
                     (2 (2 2 2 2 2))
                     (3 (3 3 3 3 3)))
          :rthm-seq-palette '((1 ((((4 4) h q e s s))
                                  :pitch-seq-palette ((1 2 3 4 5))))
                              (2 ((((4 4) q e s s h))
                                  :pitch-seq-palette ((1 2 3 4 5))))
                              (3 ((((4 4) e s s h q))
                                  :pitch-seq-palette ((1 2 3 4 5)))))
          :rthm-seq-map '((1 ((vln (1 3 2 1 2))
                              (vlc (3 1 1 2 2))))
                          (2 ((vln (3 1 1 2 2))
                              (vlc (1 3 1 2 2))))
                          (3 ((vln (1 1 3 2 2))
                              (vlc (2 1 1 2 3))))))))
  (write-csound-score mini
                      '(vln vlc)
                      '(1 2)
                      :chords nil
                      :delimiter #\tab
                      :p-fields #'csound-p-fields-custom
                      :comments t))
```

SYNOPSIS:

```
(defmethod write-csound-score ((sc slippery-chicken)
                               players
                               csound-instruments
                               &key
                               (start-section 1)
                               ;; offset in seconds
                               (offset 0)
```

```
;; add something just before .sco?
(suffix "")
(csound-file
 (format nil "~a~a~a.sco"
         (get-sc-config 'default-dir)
         (filename-from-title (title sc))
         suffix))
;; add a comment sections?
(comments t)
(delimiter #\space)
(from-sequence 1)
(num-sequences nil)
(chords t)
;; either NIL, a list of lists, or a function
;; cf. documentation
(p-fields #'csound-p-fields-simple)
;; when NIL, all sections are considered
(num-sections nil))
```

39.76 slippery-chicken/write-lp-data-for-all

[slippery-chicken] [Methods]

DESCRIPTION:

```
Generate all of the .ly files required by the LilyPond application for
printable output from the musical data stored in the given slippery-chicken
object.

This method produces .ly files for the score as well as the parts for all
individual players in the ensemble (unless otherwise specified by the
user). The files are automatically named based on the value passed to the
TITLE slot of the given slippery-chicken object.

NB: This method only produces the .ly files. These must be rendered by the
    LilyPond application separately for PDF output. See the slippery
    chicken installation web page and the manual page on Output for more
    detail. Bear in mind that SBCL and CCL users on OSX can use the
    lp-display macro to call Lilypond and display the resultant PDF
    automatically.

NB: Many of the arguments for this method pass their values directly to
    LilyPond parameters.

NB: Clefs added to grace-note events will not be rendered in Lilypond.
```

ARGUMENTS:

- A slippery-chicken object.

OPTIONAL ARGUMENTS:

keyword arguments:
- :base-path. A string that is the directory path only for the resulting files. The method will automatically generate the file names and extensions. Default = (get-sc-config 'default-dir).
- :start-bar. An integer that is the first bar of the given slippery-chicken object for which output is to be generated. If NIL, the start-bar will be set to 1. Default = NIL.
- :end-bar. An integer that is the last bar of the given slippery-chicken object for which output is to be generated. If NIL, all bars after the start bar will be generated. Default = NIL.
- :start-bar-numbering: For bar counting in the score only: the bar number that the :start-bar will be counted as (integer). NIL = :start-bar. Default = NIL.
- :players. A list of player IDs or NIL to indicate which players' parts are to be generated and included in the resulting score. If NIL, all players' parts will be generated and included in the score. This can be handy, for example, for excluding the computer part of a piece for tape and instruments. Default = NIL.
- :respell-notes. NIL, T or a list to indicate whether the method should also call the respell-notes method on the given slippery-chicken object before generating the output to undertake enharmonic changes. If a list, then these are the specific enharmonic corrections to be undertaken. If this is T, the method will process all pitches for potential respelling. If NIL, no respelling will be undertaken. See the documentation for the respell-notes method for more. Default = NIL.
- :auto-clefs. T or NIL to indicate whether the auto-clefs method should be called to automatically place mid-measure clefs in the parts of instruments that use more than one clef. T = automatically place clefs. Default = T
- :in-c. T or NIL to indicate whether the full score is to contain written pitches or sounding pitches. NB: Some transposing C instruments still transpose at the octave in C scores, such as double-bass and piccolo. NB: Parts will always be transposed. T = sounding pitches. Default = NIL.
- :page-nums. T or NIL to indicate whether page numbers should automatically be added to each page (not including the start page) of the output. T = add page numbers. Default = T.
- :rehearsal-letters-font-size. A number that indicates the font size of rehearsal letters in LilyPond output. Default = 18.
- :rehearsal-letters-all-players. T or NIL to indicate whether rehearsal letters are to be placed in all parts generated. T = all parts.

Default = T. NB: This must be set to T when the user would like the rehearsal letters in all individual LilyPond parts, but printing with CMN thereafter will result in rehearsal letters in all parts as well.
- :tempi-all-players. T or NIL to indicate whether tempo marks are to be placed in all parts generated. T = all parts. Default = T.
- :all-bar-nums. T o NIL to indicate whether the corresponding bar number should be printed above every measure in the score (not including multi-bar rests). T = add a bar number to every measure. Default = NIL.
- :paper. A string to indicate the paper size for LilyPond output. Only LilyPond's predefined paper sizes are valid here. According to the LilyPond manual, these include: "a4, letter, legal, and 11x17... Many more paper sizes are supported... For details, see scm/paper.scm, and search for the definition of paper-alist." NB: This argument will only adjust paper size, but not margins or line widths, which are adjusted using the arguments below. Default = "a4"
- :staff-size. An integer that indicates the size of the notes and staves in the resulting output. Standard for parts is 20. Default = 14.
- :group-barlines. T or NIL to indicate whether bar lines should be drawn through the whole staff group or just one staff. T = through the whole staff group. Default = T.
- :landscape. T or NIL to indicate whether the paper format should be landscape or portrait. T = landscape. NB: This argument will only adjust paper layout, but not margins or line widths, which are adjusted using the arguments below. Default = NIL.
- :barline-thickness. A number that is the relative thickness of the bar lines. Default = 0.5.
- :top-margin. A number that is the margin at the top of the page in millimeters. Default = 10.
- :bottom-margin. A number that is the margin at the bottom of the page in millimeters. Default = 10.
- :left-margin. A number that is the margin at the left of the page in millimeters. Default = 20.
- :line-width. A number that is the width of each line in centimeters. Default = 17.
- :page-turns. T or NIL to indicate if LilyPond should attempt to optimize page breaks for page turns in parts. T = optimize page breaks. Default = NIL.
- :min-page-turn. A two-item list indicating the minimum rest necessary for the method to automatically place a page turn, in a format similar to that of a time signature; i.e., '(2 1) would mean a minimum of 2 whole rests. Default = '(2 1))
- :use-custom-markup. T or NIL. Set to T when using a number of marks that are specific to LilyPond, such as 'bartok or any of the marks that use eps graphics files (whereupon those graphics files would need to be in the same folder as your lilypond files). Default = T.
- :lp-version. A string that will be added to each .ly file generated in

conjunction with the LilyPond \version command. Default = "2.42.1". NB Lilypond syntax changes every now and then. If you get errors that can't be explained by a user error, please post an issue on GitHub, as it might be that we need to update the generated code to reflect Lilypond's new syntax.
- :process-event-fun. NIL or a user-defined function that will be applied to every event object in the given slippery-chicken object. If NIL, no processes will be applied. Default = NIL.
- :extend-hairpins. If you want hairpin (cresc/dim) to extend beyond the previous barline (or beyond the note) set to T. Default = NIL.
- :min-hairpin. If you're getting Lilypond warnings about hairpins being too short (and then not showing up in the score, or dynamics overlapping), you can set this value to, e.g. 10.
- :stemlet-length. NIL or a decimal number < 1.0 that indicates the scaled length of stems over rests in LilyPond output, should this feature be desired. 0.75 is a recommended value for this. NIL = no stems over rests. Default = NIL. NB: LilyPond can be instructed to extend beams over rests (without stemlets) simply by using the '-' in the definition of the rthm-seq-bar object, as is done with any other note; however, starting/ending a beam on a rest and then trying to generate a score with CMN will fail.
- :footer. A string to add to the footer of every score page. Default = NIL.
- :title. Usually T or NIL to include the title of the piece at the top of the score or not, but could also be a string to specify a new title. Default = T.
- :indent. T or NIL to indicate whether the first system in the score should be indented or not. Default = T.
- :force-bracket. T or NIL to indicate whether tuplet numbers on beams should be forced into a bracket. Default = NIL.
- :two-sided. A two-element list to set the inner and outer margins with layout for two-sided printing and binding. E.g. :two-sided '(30 20) means the left-hand margin will be 30mm and the right-hand will be 20mm. If set, this overrides the :left-margin and :line-width arguments. Default = NIL.
- :dummy-staves. The player name(s) as a symbol or a list for any parts that have been created merely as empty staves to add cross-staff notation.
- :between-system-space. The space in cm between systems (groups of staves). Default = NIL (= Lilypond's default distance).
- :staff-basic-distance. This controls the distance between staffs within a system. It's in Lilypond units (relative to staff size). Setting to 15 creates a good distance between piano staves. Default = NIL (= Lilypond's default distance).
- :fixed-width. Set to a rhythmic value (e.g. 1/32) in order to create fixed width rhythmic spacing. This makes it easier, for some, to read complex rhythms as you can see how long rhythms last by the horizontal

space they occupy. The shorter the value given the more space is allocated, so experimentation might be necessary. Default = NIL (usual engraver's spacing algorithm used, according to context).
- :accidental-style. String. See the lilypond documentation for details of applicable styles and their results. Default = 'modern.
- :links. T or NIL to indicate whether the generated PDF should contain notes that are clickable links to the lilypond source code or not. Default = T.

RETURN VALUE:

The path of the main score file generated.

EXAMPLE:

```
;;; An example with values for the most frequently used arguments ;
(let ((mini
       (make-slippery-chicken
        '+mini+
        :ensemble '(((fl (flute :midi-channel 1))
                     (cl (b-flat-clarinet :midi-channel 2))
                     (vc (cello :midi-channel 3))))
        :staff-groupings '(2 1)
        :tempo-map '((1 (q 84)) (9 (q 72)))
        :set-palette '((1 ((f3 g3 a3 b3 c4 d4 e4 f4 g4 a4 b4 c5))))
        :set-map '((1 (1 1 1 1 1 1 1 1))
                   (2 (1 1 1 1 1 1 1 1))
                   (3 (1 1 1 1 1 1 1 1)))
        :rthm-seq-palette '((1 ((((4 4) h (q) e (s) s))
                                :pitch-seq-palette ((1 2 3))
                                :marks (bartok 1)))
                            (2 ((((4 4) (q) e (s) s h))
                                :pitch-seq-palette ((1 2 3)))))
        :rthm-seq-map '((1 ((fl (1 2 1 2 1 2 1 2))
                            (cl (1 2 1 2 1 2 1 2))
                            (vc (1 2 1 2 1 2 1 2))))
                        (2 ((fl (1 2 1 2 1 2 1 2))
                            (cl (1 2 1 2 1 2 1 2))
                            (vc (1 2 1 2 1 2 1 2))))
                        (3 ((fl (1 2 1 2 1 2 1 2))
                            (cl (1 2 1 2 1 2 1 2))
                            (vc (1 2 1 2 1 2 1 2)))))
        :rehearsal-letters '(3 11 19))))
  (write-lp-data-for-all mini
                         :start-bar 7
                         :end-bar 23
```

```
                          :paper "letter"
                          :landscape t
                          :respell-notes nil
                          :auto-clefs nil
                          :staff-size 17
                          :in-c nil
                          :barline-thickness 3.7
                          :top-margin 40
                          :bottom-margin 60
                          :left-margin 40
                          :line-width 22
                          :page-nums t
                          :all-bar-nums t
                          :use-custom-markup t
                          :rehearsal-letters-font-size 24
                          :lp-version "2.20.0"
                          :group-barlines nil
                          :page-turns t
                          :players '(fl cl)
                          :tempi-all-players t))

=> T
```

SYNOPSIS:

```
(defmethod write-lp-data-for-all
    ((sc slippery-chicken)
     &key
       (base-path (get-sc-config 'default-dir))
       start-bar end-bar (paper "a4") landscape
       ;; MDE Tue May 29 21:34:53 2012
       start-bar-numbering
       ;; if a list, then these are the enharmonic corrections
       (respell-notes t)
       ;; automatically add clefs to instruments who read more than one?
       (auto-clefs t)
       (staff-size 14)
       ;; parts will always be transposed but score can be in in C or not
       (in-c nil)
       (barline-thickness 0.5)
       (top-margin 10)                  ; mm
       (bottom-margin 10)               ; mm
       (left-margin 20)                 ; mm
       (line-width 17)                  ; cm
       ;; if not nil, then in cm
       between-system-space
```

```
;; MDE Tue May  3 12:12:46 2016 -- this one controls the distance
;; between staffs within a system. It's in Lilypond units (relative to
;; staff size). Setting to 15 creates a good distance between piano
;; staves.
staff-basic-distance
(page-nums t)
;; print every bar number unless
;; multi-bar-rest?
(all-bar-nums nil)
;; this has to be T if we're going to get letters in the parts--but CMN
;; printing will have all parts all letters too thereafter
(rehearsal-letters-all-players t)
;; set to t if using bartok pizz and othersigns
(use-custom-markup t)
(rehearsal-letters-font-size 18)
;; "2.16.2") "2.14.2") ;"2.12.3") "2.17.95")  "2.20.0")
(lp-version "2.24.1")
;; 24.7.11 (Pula) barlines through whole staff group or just a stave
(group-barlines t)
;; 5.11.11 set to t if you want lilypond to optimize page breaks for
;; page turns in parts
(page-turns nil)
;; MDE Sat Mar 10 16:52:31 2012
(process-event-fun nil)
;; MDE Mon Apr 16 16:08:36 2012 -- added so that we can write a subset
;; of players into the score (e.g. leave out a computer part). If nil
;; all players will be written. NB the order of these will determine
;; the order in the score (overriding the ensemble order).
(players nil)
(footer nil)
;; minimum rest necessary to do a page turn; something like a time
;; signature e.g. (2 1) would mean we need a min. of 2 whole rests
(min-page-turn '(2 1))
;; MDE Tue May 29 22:58:25 2012
(stemlet-length nil)
;; MDE Thu Jan 9 09:20:33 2014 -- if you want hairpin (cresc/dim) to
;; extend beyond the note set to T
(extend-hairpins nil)
;; MDE Wed Dec 30 18:29:28 2020, Heidhausen
(min-hairpin nil)
;; MDE Wed Jul  8 10:45:02 2015 -- tuplet numbers on beams don't usually
;; include a bracket but we can force this if we like
(force-bracket nil)
;; MDE Thu Mar 26 18:49:46 2015: show the title? could also be a string
;; in order to not use the sc title
(title t)
```

```
;; MDE Thu Apr 29 17:39:40 2021, Heidhausen -- show the composer? also
;; possibly a string
(composer t)
two-sided                        ; MDE Wed Oct 21 18:15:08 2015
;; MDE Sat Dec 24 14:54:34 2016 -- try also "dodecaphonic" if you want
;; accidentals before every pitch
(accidental-style "modern")
;; MDE Thu Mar 26 19:18:03 2015
(indent t)
;; MDE Thu Apr 28 15:17:36 2016 -- specify the player name(s) as a
;; symbol or in a list for any parts that have been created merely as
;; empty staves to add cross-staff notation.
dummy-staves
;; MDE Thu Feb  2 14:45:21 2017 -- fixed width rhythmic notation
fixed-width
;; MDE Sat Sep 30 16:55:40 2023, Heidhausen -- the PDF should contain
;; links to the source code?
(links t)
;; sim to rehearsal letters
(tempi-all-players t)
(ekmelic-style "gost"))
```

39.77 slippery-chicken/write-xml

[slippery-chicken] [Methods]

DATE:

```
March 17th 2017, Edinburgh
```

DESCRIPTION:

```
Write a MusicXML file for import into Sibelius, Finale, Dorico, etc. All
voices for the score and parts are written into one file. As it is assumed
that editing will take place in the notation software of choice, there are
only very basic arguments to this method, some of which may even be ignored
upon import. For example, the :in-c keyword argument of other methods is
missing here, as switching between sounding and transposing scores is
assumed to be the job of the notation software used for MusicXML import.

NB Sibelius 7.5 messes up some transposing instruments, getting octave
transpositions wrong and adding generally unwanted key signatures.

NB Dorico has trouble importing some tuplets. I have some understanding for
this as the music-xml definition of rhythm is complex and confusing. If you
```

generated complex nested tuplets (e.g. via RQQ rhythmic notation) then some bars, upon importing into Dorico, will be considered overfull (you'll probably see these marked in red, unless you have 'Hide Signposts' on. You can click on those marks in dorico and enter the duration of the bar, correcting what Dorico thinks is an error, but bear in mind that this arises from what I consider to be a false interpretation of the optional tuplet-actual xml tag which overrides the more pertinent and non-optional duration tag (which is always correct as far as I can tell).

I prefer to correct such problems (e.g. 3:2 tuplets not spanning the rhythms indicated via tuplet start and stop tags) in software semi-automatically. For instance the following will examine the tuplets slots of rthm-seq-bars and if it identifies problematic values (as identified by trial-and-error via Dorico import) you can correct it with the replacements given. In the following case this is always a similar case of a triplet over e.g. 6 notes being replaced by three triplets over 2-note pairs:

See e.g. the following Dorico form post for more details:
https://forums.steinberg.net/t/xml-import-nested-tuplets-problem/1022697

ARGUMENTS:

- the slippery-chicken object

OPTIONAL ARGUMENTS:

keyword arguments
- :staff-height. The suggested height of the staff, in millimetres. Default = 7.
- :page-width. The page width in millimetres. Default = 210 = A4
- :page-height The page height in millimetres. Default = 297 = A4
- :tenths. A number representing tenths of interline staff space (positive or negative). Both integer and decimal values are allowed, such as 5 for a half space and 2.5 for a quarter space. Interline space is measured from the middle of a staff line. Distances in a MusicXML file are measured in tenths of staff space. Tenths are then scaled to millimeters within the scaling element, used in the defaults element at the start of a score. See https://usermanuals.musicxml.com/MusicXML/Content/EL-MusicXML-scaling.htm for more detail. Default = 40.
- :left-page-margins. A 4-element list specifying for left pages in the score the left, right, top, and bottom margins, all in millimetres. Default = '(20 20 20 20)
- :right-page-margins. Similar to :left-page-margins but for right pages.

- :players. Which players to write to the xml file. Default = NIL = process all players.
- :hide-players. Which players to leave out of the score.
- :start-bar. The bar number to start at. Default = NIL = 1.
- :end-bar. The bar number to end at. Default = NIL = the last bar in the slippery-chicken object.
- :respell-notes. NIL, T or a list to indicate whether the method should also call the respell-notes method on the given slippery-chicken object before generating the output to undertake enharmonic changes. If a list, then these are the specific enharmonic corrections to be undertaken. If this is T, the method will process all pitches for potential respelling. If NIL, no respelling will be undertaken. See the documentation for the respell-notes method for more detail. Default = T.
- :file. The full path of the file to write. Default will be derived from the title slot of the slippery-chicken object and will be written into the default directory e.g. (set-sc-config 'default-dir "~/Desktop")
- :suffix. A string to append to the default file name before the .xml extension. Default = "".

RETURN VALUE:

The file name written, as a string.

SYNOPSIS:

```
(defmethod write-xml ((sc slippery-chicken)
                      &key
                        (staff-height 7)  ; mm
                        (page-width 210)  ; mm = a4
                        (page-height 297) ; mm = a4
                        (tenths 40)       ; see below
                        ;; left right top bottom, all mm
                        (left-page-margins '(20 20 20 20))
                        (right-page-margins '(20 20 20 20))
                        players hide-players start-bar end-bar
                        (respell-notes t)
                        ;; MDE Tue Nov 30 09:01:48 2021, Heidhausen
                        (suffix "")
                        file)
```